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INORGANIC ELECTRONIC SPECTROSCOPY (second edition)

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To Bernice

Preface

The first edition of this book, published in 1968, demonstrated how detailed studies of the electronic spectrum of a molecule can provide a great deal of chemical information. Although a number of excellent advanced monographs, dealing with this subject, had appeared prior to 1968, there was perceived a need for a textbook dealing primarily with electronic spectroscopy at a level understood by the younger research worker or final year undergraduate. Much to the pleasure of the author, the book was well received and apparently fulfilled its objectives. In the 16 years since the appearance of the first edition, no other book, dealing in a similar fashion with electronic spectroscopy, has appeared.

Since 1968 various areas of inorganic spectroscopy have developed to such a degree that the original book can contribute little or nothing. These include inorganic chromophores occurring in metalloproteins and other biological environments, organometallic chromophores, especially carbon monoxide, metal-metal bonded and cluster systems, mixed valence systems, and the general area of charge transfer spectroscopy. Clearly it is time for this second edition wherein each of these areas will be specifically addressed. In addition our general knowledge of the ligand field, d-d, spectra of transition metal ions has greatly improved, especially with regard to lower symmetry complexes and to complexes containing heavier transition metal ions.

There have developed many associated techniques to aid our understanding of energy levels in inorganic molecules: these include other spectroscopies such as magnetic circular dichroism, ultraviolet photoelectron and resonance Raman, the general area of inorganic photochemistry, and the emission properties of excited states. These techniques are referenced here when relevant to the spectroscopic discussion.

The first edition was written:
(a) to provide the theoretical background necessary to understand and work with electronic spectra and to take the reader to a level of theoretical understanding to be able to move on to more advanced monographs;
(b) to illustrate and discuss methods involved in assigning spectroscopic transitions;
(c) to show how an assigned spectrum can be analysed to provide information about metal-ligand bonds, and
(d) to document and explain the spectra observed for all common transition metal ions in their various oxidation states and stereochemistries, so that the research worker may use the text as a reference in elucidating his own spectra.

This second edition has the same general aims. However since 1968 a number of theoretical aspects of this research area have become so commonplace that they now appear in undergraduate curricula. Thus the chapters dealing with atomic spectra, molecular symmetry, and group theory are omitted from this edition in the belief that most readers will have met these topics elsewhere, and

that if they have not, there exist so many other texts dealing with these areas that repetition here would be superfluous. Therefore a general undergraduate knowledge of molecular symmetry and group theory, and atomic structure, is assumed.

Early chapters deal with the theory of electronic spectroscopy, one-electron and multi-electron energy level diagrams, selection rules, band intensities and dichroism and lead in to the later chapters which deal with ligand field and charge transfer spectra and with the special topic areas indicated.

Many detailed examples are included to ensure that the reader will be able to obtain a working knowledge of the theory.

Electronic spectra are reproduced directly from the literature. While this has the disadvantage that the scales of one figure may differ from another, this is hopefully not a serious problem.

No attempt is made to present comprehensive coverage of this enormous field. Rather, examples of all important metal ions, oxidation states and stereochemistries are presented to allow the reader to find an entry to the area of interest. Moreover the immensity of the task has required that a review-like style be employed at times, rather than present every detail. For this reason the experimental data are fully referenced, while the early chapters have key references.

Certain aspects have been omitted, since their inclusion would have increased to impractical size an already large task. These include the spectra of the lanthanides, actinides, minerals and, regretfully, main group spectroscopy.

The various detailed molecular orbital and valence bond theories are also largely omitted since their use by most research workers is precluded by the cost of large scale computer manipulation. On the other hand, the prevalence now of small micro and minicomputers does enable the average researcher to undertake less complex fitting of their electronic spectra. Therefore this book does indicate how these machines can be used to solve problems in this area. Indeed much of the computation reported from the literature, in this edition, has been re-checked by the author on a microcomputer. Errors were indeed found. In addition some computation not carried out by the original author is included. Some of the theoretical work presented here is arranged for simple manipulation on a microcomputer.

Fundamental spectroscopy does not change with time and some material is taken directly from the first edition; however the additions are extensive in the light of additional knowledge.

This manuscript was written using Wordpro 4+© on a Commodore CBM Model 8032 microcomputer. The information was stored on disc and printed directly on a Qume printer as camera ready copy.

The use of a microcomputer and wordprocessing software is revolutionising the production of texts. It is likely that this is one of the first highly technical books to be produced in this fashion, and a few comments are worthwhile.

Firstly the flexibility offered by the microcomputer in producing this book is phenomenal. Over a period of about 3 years,

the information for the tables has accumulated and has been stored on disc for final formatting and printing directly into this text. Chapter writing continued over about 2 years with material being constantly up-dated and edited. Various intermediate versions of the chapters were run-off and distributed to critics for comment. Editing and addition of new material continued to within days of final submission of the manuscript.

There are also less obvious advantages such as saving time by using a couple of code letters for frequently used collections of words, such as journal titles, or authors, in the references, followed by replacing all these by software at a later date.

However there are a few drawbacks which I hope my readers will forgive. The flexibility offered by the wordprocessing software and printer are not unlimited. A custom printwheel containing Arabic and Greek letters, and a square root sign, did enable most material to be printed directly. However the numerals used for superscripts and subscripts are slightly larger than normally used, leading to some crowding, especially in the tables.

I am greatly appreciative of the help and criticism offered by many of my colleagues and by the various individuals who kindly read pre-final versions of the various chapters. In particular I would like to thank Ian Walker and Paul McCarthy for their continued collaboration in inorganic spectroscopy over the years. Special thanks also go to Elaine Dodsworth for reading every word of the manuscript and offering both chemical and typographical criticism. I would also like to express my appreciation to, in alphabetical order, Ivano Bertini, John Donini, Colin Flint, Harry Gray, Brian Hathaway, Bryan Hollebone, Tom Meyer, Harvey Schugar, Ed Solomon and Bill Trogler, for their input, criticism and support. In the final outcome, however, any errors are solely my responsibility.

I would also wish to thank the various authors who gave permission for the reproduction of their spectra, and to their publishers. In particular, I acknowledge copyright permission from the following houses: Academic Press Inc., Acta Chemica Scandinavica, The American Chemical Society, The American Institute of Physics, The Chemical Society of Japan, Elsevier Science Publishing Co., Elsevier Sequoia S.A., Gordon and Breach Science Publishers Inc., Johann Ambrosius Barth Verlag, John Wiley and Sons Ltd., The National Research Council of Canada, North Holland Publishing Co., The Royal Society of Chemistry, Springer-Verlag and Taylor and Francis Ltd.

Thanks are certainly due to my secretary, Carol Francis, for incorporation of material not available on the printwheel, and for the extensive correspondance associated with the production.

Finally I owe my wife, Bernice, and family, a debt for their forbearance during the production of this work which did take up a great deal of my energy for the past several years.

A.B.P.Lever
Toronto, March 1984

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Chapter 1

One-Electron Energy Levels

1.1 Introduction

The general principles behind the calculation of the energies of the various group theoretical terms arising from a d^n configuration will be discussed in the first three Chapters both in the weak field and strong field approximations for cubic and non-cubic molecules. The emphasis will be on the use of these models to assign and calculate spectra, and the chemical information which can be derived thereby. No attempt is made, in the limited space available, to provide intensive discussion of each of the theoretical techniques. Moreover some techniques will only be considered superficially if they have been treated often elsewhere.

1.2 Crystal Field Theory

1.2.i) Energy Hamiltonian [1-6]

The atomic orbitals (s,p,d,f etc) are solutions of the one electron Schrodinger equation:

$$(\delta^2/dx^2 + \delta^2/dy^2 + \delta^2/dz^2)\psi + (8\pi^2m/h^2)(E - V) = 0$$

$$\nabla^2\psi + (8\pi^2m/h^2)(E - V)\psi = 0 \qquad (1.1)$$

where E is the total energy and V is the potential energy. ∇^2 is the Laplacian operator.

Equation (1.1) can be re-written in the form:

$$[V - (h^2/8\pi^2m)\nabla^2]\psi = E\psi$$

or

$$H\psi = E\psi \qquad (1.2)$$

where H is the Hamiltonian operator.

For the hydrogen atom, neglecting spin orbit coupling, the Hamiltonian has the form:

$$H = (-h^2/8\pi^2m)\ \nabla^2 - e^2/r \qquad (1.3)$$

where the first factor expresses the kinetic energy, and the second, the potential energy of the electron. For many electron atoms, the wave equation is still represented as (1.2) but the Hamiltonian has the form:

$$H = (-h^2/8\pi^2 m)\Sigma\nabla^2_i - \Sigma Ze^2/r + \sum_{i>j} e^2/r_{ij} + \Sigma\zeta_i(r_i)l_i s_i \quad (1.4)$$

where Z is the 'effective' nuclear charge, r_i is the distance of the i-th electron from the nucleus, and r_{ij} is the distance between the i-th and j-th electrons. The first term sums the kinetic energy over all the electrons, the second sums the attraction of all the electrons by the nucleus, and the third sums the mutual repulsion between each pair of electrons (interelectronic repulsion). The fourth term concerns the coupling between the spin and orbital magnetic momenta (spin-orbit coupling).

To obtain the total energy of a wave-function, reconsider eqn. (1.2). Both sides are multiplied by ψ and rearranged to give:

$$E = \psi H\psi/\psi^2 \quad (1.5)$$

Since ψ is a function of the spatial coordinates, and E is not, both numerator and denominator of the right-hand side of (1.5) are integrated over all space, thus:

$$E = \int\psi^* H\psi d\tau / \int\psi^*\psi d\tau \quad (1.6)$$

where ψ^* is the complex conjugate of ψ.

> The complex conjugate of a real function is the function itself. The complex conjugate of a imaginary function is the negative of that function.

However $\int\psi^*\psi d\tau = 1$, if the wavefunction is normalised, therefore:

$$E = \int\psi^* H\psi d\tau \quad (1.7)$$

This equation is fundamental to the discussion of the energies of wavefunctions and molecular states (composed of linear combinations of wave functions) since it provides a means of deriving the energy of a wavefunction or state under perturbation by some field defined by the Hamiltonian H.

The first two terms in expression (1.4) are common for all electrons and electron configurations and are therefore 'transparent' since they do not contribute to the **energy differences** commonly measured in an electronic spectrum. The third, interelectronic repulsion, term gives rise to the spectroscopic states. For a d^1 ion, the only possible state is ^2D comprising the ten different arrangements of d^1 (m_l = 2,1,0,-1,-2 each with m_s = +1/2 or -1/2). Since there is only one electron inside the valence shell, interelectronic repulsion with inner core electrons is constant and will not contribute to observed energy differences. This third term will contribute to the energies of multi-electron configurations which will be discussed in Chapter 3. The phenomenon of spin-orbit coupling will also be discussed in more depth in Chapter 3.

When the free ion becomes complexed (bound by an ion or ligand), the energies are found by use of the modified Hamiltonian:

$$H = (-h^2/8\pi^2 m)\nabla^2_i - \Sigma Ze^2/r + \sum_{i>j} e^2/r_{ij} + \xi_i(r_i)l_i s_i + V_i \quad (1.8)$$

which is similar to equation (1.4) except for the last term V_i. V_i is the potential energy in the crystalline field, i.e. the energy of interaction between the electrons and the crystal field.

1.2.ii) Generation of Crystal Field Hamiltonians

Crystal field theory treats a molecule as if it consists of a positive metal ion surrounded by negatively charged ions (ligands). The approach was pioneered by Bethe [1] and Van Vleck [2] and was the basis of the first successful analysis of the electronic spectra of octahedral transition metal ions.

The theory was quite unsuccessful in predicting the absolute magnitude of a transition energy. This failure is not surprising since transition metal complexes undoubtedly involve some degree of covalent character and a simple point charge model is quite inadequate.

Nevertheless inorganic spectroscopy developed rapidly following the first correct identification of a d-d transition in 1951 [3]. This development was due, in no small measure, to the success enjoyed by crystal field theory in predicting, semi-empirically, both the number of d-d transitions, and their relative energies. This success arose because crystal field theory is based firmly upon the geometry, or rather molecular symmetry, of the complex concerned [4-7]. The number of excited states which a given d^n configuration generates, depends solely upon molecular symmetry and is quite independent of any model used to describe the metal ligand bonds. Using molecular symmetry arguments the number of excited states, and therefore, in general the number of electronic transitions, can be derived. Moreover, using crystal field theory empirically, their relative energies can be determined. Although this simple theory cannot generate absolute energies, experimental data can be fitted to the theory to generate semi-empirical radial parameters which, in a simple view, reflect the interaction between metal and ligand. In a poly-electron system, the interelectronic repulsion parameters (Racah B and C) are included in a well defined fashion leading to a complete and generally quite acceptable fitting of the electronic spectrum.

The energy associated with an electron j at a distance r_{ij} from a negatively charged ligand i is given by (Fig.1.1):

$$V_i = Z_i e^2/r_{ij} \quad (1.9)$$

where Z_i is the charge on the ligand placed arbitrarily along the z axis.

Summing the potential over all the ligands allows the determination of the potential experienced by an electron in any molecular environment. The crystal field theory assumes that the ligands are point charges or point dipoles and that the d orbitals

are entirely non-bonding. The additional potential energy of each d electron due to the crystal field is determined and the splitting of the d orbitals so deduced. As will be seen, the splitting can readily be predicted qualitatively from group theory, but determination of the energy separations involved requires detailed calculation of the potentials.

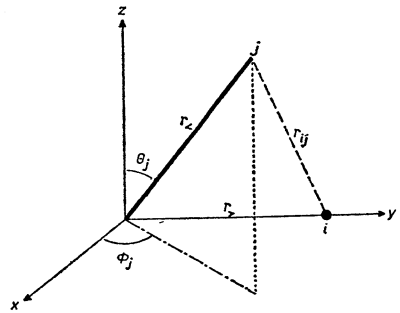

Fig.1.1 Polar coordinates for electron j and ligand i in point charge model.

The potential Ze/r_{ij} can be expanded in terms of spherical harmonics $Y_{lm}(\theta,\phi)$ centred at the origin [7-9]. Since the d orbitals can also be described in terms of spherical harmonics centred at the metal nucleus as origin, it is convenient to centre the harmonics describing $1/r_{ij}$ at the metal nucleus. The potential then takes the form:

$$\sum_i \frac{Ze}{r_i - r_j} = \sum_{k=0}^{\infty} \sum_{q=-k}^{k} \left|\frac{4\pi}{2k+1} \sum_i Ze Y_k^{q*}(\theta'_i,\phi'_i)\right| \frac{r^k}{r'^{k+1}} Y_k^q(\theta_j,\phi_j) \cdot \qquad (1.10)$$

where $Y_k^q(\theta_i,\phi_i)$ refers to the ligand charge, and $Y_k^q(\theta_j,\phi_j)$ refers to the electron. The radial distances r', and r are the shorter and the longer of the radial vectors connecting the origin to the electron and to the charge (ligand). It is usually, but not necessarily, assumed that the electron does not move far from the metal nucleus so that r refers to the electron-nucleus distance.

The first spherical harmonic in (1.10) describes the angular coordinates of the charge (ligand) and is a number which can be derived by using appropriate values of θ_i and ϕ_i for the stereochemistry concerned (see below). The second describes the angular coordinates of the electron and is thus related to the angular part of the wave function of the electron concerned.

The spherical harmonics can be thought of as functions which can describe the standing waves (normal vibrations) set up on a

flooded planet whose ocean is of uniform depth [9]. The harmonics and their components have both magnitude and direction, and by appropriate combination any geometry can be simulated as a standing wave in this ocean. In parallel fashion a set of spherical harmonics can describe the geometry of any molecule.

Using eqn.(1.10) a Hamiltonian operator can be devised, which can be used to determine the energies of an electron configuration within any given crystal field environment. As an example, the Hamiltonian for a square planar D_{4h} environment will be derived. Before beginning this exercise however, note that eqn (1.10) is a summation from k = 0 to k = infinity. To complete this summation might take some time. Therefore consider first if some of the terms will vanish.

Once the form of the Hamiltonian is known, the energy of any given wavefunction will be obtained by evaluating an integral of the type (1.7). Generally we are concerned with the energy differences expressed in an electronic spectrum and, for d^1, need only consider the V_i contribution to H, (neglecting spin-orbit coupling for the present).

Since a wavefunction consists of both a radial part, detailing the extension of the electron from the nucleus, and an angular component, detailing the symmetry of its orbital, so too will the expansion of (1.7) contain a radial component which will be a scalar and an angular component.

The crystal field operator describing a particular geometry is constructed of a linear combination of spherical harmonics **which transform as the totally symmetric representation in the group concerned.** This results because the shape of a molecule is independent of whether it is reflected or rotated and hence all its characters under the operators of the group are unity, i.e it is totally symmetric. This particular observation is very important and will recur several times in this book.

The energy of a function ψ perturbed by the crystal field operator, is given by the integral (1.7). The value (expectation value) of the integral is simply a number which must also transform as the totally symmetric representation of a group. A number does not change if it is rotated or reflected. Thus the direct product of the representations of $\psi \times H \times \psi$ must contain the totally symmetric representation. This imposes strict conditions on the nature of the wavefunctions which can be coupled, and on the nature of the Hamiltonian itself.

For example since the d orbitals are even with respect to inversion, only even harmonics can contribute to the Hamiltonian, i.e. harmonics whose k values are even. Odd harmonics (i.e. those with an odd value of k) are ungerade; the inclusion of such harmonics in the Hamiltonian will make it odd with respect to inversion and therefore unable to couple the even d functions, i.e. the integral would no longer transform as the totally symmetric representation of the group concerned. Moreover as will also be considered in more detail below, functions with k values greater than 4 cannot couple two d orbitals with l = 2. For f orbital

Table 1.1a

SPHERICAL HARMONICS Y_l^m TO l = 4 (Polar)

Harmonic	Polar coordinates
Y_0^0	$(1/\sqrt{2})\sqrt{(1/2\pi)}$
Y_1^0	$\sqrt{(3/2)}(1/\sqrt{2\pi})\cos\theta$
$Y_1^{\pm1}$	$\mp\sqrt{(3/4)}(1/\sqrt{2\pi})\sin\theta e^{\pm i\phi}$
Y_2^0	$\sqrt{(5/8)}(1/\sqrt{2\pi})(3\cos^2\theta - 1)$
$Y_2^{\pm1}$	$\mp\sqrt{(15/4)}(1/\sqrt{2\pi})\sin\theta\cos\theta.e^{\pm i\phi}$
$Y_2^{\pm2}$	$\sqrt{(15/16)}(1/\sqrt{2\pi})\sin^2\theta.e^{\pm 2i\phi}$
Y_3^0	$\sqrt{(7/8)}(1/\sqrt{2\pi})(5\cos^3\theta - 3\cos\theta)$
$Y_3^{\pm1}$	$\mp\sqrt{(21/32)}(1/\sqrt{2\pi})(5\cos^2\theta - 1)\sin\theta.e^{\pm i\phi}$
$Y_3^{\pm2}$	$\sqrt{(105/16)}(1/\sqrt{2\pi})\sin^2\theta\cos\theta.e^{\pm 2i\phi}$
$Y_3^{\pm3}$	$\mp\sqrt{(35/32)}(1/\sqrt{2\pi})\sin^3\theta.e^{\pm 3i\phi}$
Y_4^0	$\sqrt{(9/128)}(1/\sqrt{2\pi})(35\cos^4\theta - 30\cos^2\theta + 3)$
$Y_4^{\pm1}$	$\mp\sqrt{(45/32)}(1/\sqrt{2\pi})\sin\theta(7\cos^3\theta - 3\cos\theta).e^{\pm i\phi}$
$Y_4^{\pm2}$	$\sqrt{(45/64)}(1/\sqrt{2\pi})\sin^2\theta(7\cos^2\theta - 1).e^{\pm 2i\phi}$
$Y_4^{\pm3}$	$\mp\sqrt{(315/32)}(1/\sqrt{2\pi})\sin^3\theta\cos\theta.e^{\pm 3i\phi}$
$Y_4^{\pm4}$	$\sqrt{(315/256)}(1/\sqrt{2\pi})\sin^4\theta.e^{\pm 4i\phi}$

Adapted from ref.[43].

Table 1.1b

SPHERICAL HARMONICS Y_l^m TO l = 4 (Cartesian)

Harmonic	Cartesian coordinates
Y_0^0	$(1/\sqrt{2})(1/\sqrt{2\pi})$
Y_1^0	$\sqrt{(3/2)}(1/\sqrt{2\pi})z/r$
$Y_1^{\pm1}$	$\mp\sqrt{(3/4)}(1/\sqrt{2\pi})(x \pm iy)/r$
Y_2^0	$\sqrt{(5/2)}(1/\sqrt{2\pi})(1/2)(3z^2 - r^2)/r^2$
$Y_2^{\pm1}$	$\mp\sqrt{(5/2)}(1/\sqrt{2\pi})\sqrt{(3/2)}z(x \pm iy)/r^2$
$Y_2^{\pm2}$	$\sqrt{(5/2)}(1/\sqrt{2\pi})\sqrt{(3/8)}(x \pm iy)^2/r^2$
Y_3^0	$\sqrt{(7/2)}(1/\sqrt{2\pi})(1/2)z(5z^2 - 3r^2)/r^3$
$Y_3^{\pm1}$	$\mp\sqrt{(7/2)}(1/\sqrt{2\pi})\sqrt{(3/16)}(x \pm iy)(5z^2 - r^2)/r^3$
$Y_3^{\pm2}$	$\sqrt{(7/2)}(1/\sqrt{2\pi})\sqrt{(15/8)}z(x \pm iy)^2/r^3$
$Y_3^{\pm3}$	$\mp\sqrt{(7/2)}(1/\sqrt{2\pi})\sqrt{(5/16)}(x \pm iy)^3/r^3$
Y_4^0	$\sqrt{(9/2)}(1/\sqrt{2\pi})(1/8)(35z^4 - 30z^2r^2 + 3r^4)/r^4$
$Y_4^{\pm1}$	$\mp\sqrt{(9/2)}(1/\sqrt{2\pi})\sqrt{(5/16)}(x \pm iy)(7z^3 - 3zr_2)/r^4$
$Y_4^{\pm2}$	$\sqrt{(9/2)}(1/\sqrt{2\pi})\sqrt{(5/32)}(x \pm iy)^2(7z^2 - r^2)/r^4$
$Y_4^{\pm3}$	$\mp\sqrt{(9/2)}(1/\sqrt{2\pi})\sqrt{(35/16)}z(x \pm iy)^3/r^4$
$Y_4^{\pm4}$	$\sqrt{(9/2)}(1/\sqrt{2\pi})\sqrt{(35/128)}(x \pm iy)^4/r^4$

Adapted from refs.[6,43]. See Ref.[6], p.93 for l = 5.

functions, it is necessary to extend the Hamiltonian to k = 6.

The summation (1.10) is now reduced, for d orbitals, to k = 0, 2 and 4. The q value represents the projection of the vector moduli k along the z direction, just as the magnetic quantum number m_l is the z component of the angular quantum number l. Which of these q values will contribute to our Hamiltonian ?

In principle all might do so. However consideration of the explicit form of the spherical harmonics (Table 1.1) provides some simplification. Where the principal axis of the point group concerned is even, i.e 2 or 4 for example, the odd values of the component q cannot contribute to the totally symmetric representation. These components only contribute to Hamiltonians involving groups whose principal axis is odd, i.e. most importantly the three-fold groups.

> The above statement is only strictly true where the principal axis is defined as z, the axis along which the functions are quantized. This point is considered further below.

The generation of the Hamiltonian for a square planar (D_{4h}) complex is accomplished by evaluating (1.10) for the harmonics k = 2 (q = -2,0,2) and k = 4 (q = -4,-2,0,2,4). The harmonic with k = 0 need not be considered since it will shift all d levels by the same amount and will not contribute, thereby, to a discussion of electronic transitions within the d orbital set. Using the usual polar coordinate representation, the four ligands lie in the xy plane at 90° to the z axis (i.e. $\theta = 90^\circ$). The ϕ values (angle with respect to the x axis when the ligand is projected into the xy plane) for the four ligands, 1-4, will be 0°, 90°, 180°, and 270° respectively (Fig.1.1).

The general Hamiltonian will have the form:

$$V = \Sigma Ze/r_{ij} = c_2^0.R_2(r).Y_2^0 + c_2^2.R_2(r)(Y_2^2 + Y_2^{-2})$$
$$+ c_4^0.R_4(r).Y_4^0 + c_4^2.R_4(r)(Y_4^2 + Y_4^{-2}) + c_4^4.R_4(r)(Y_4^4 + Y_4^{-4})$$

where

$$c_k^q = (4\pi/2k+1)Ze\ \Sigma Y_k^q(\theta,\phi)$$

$$R_k(r) = r^k/r'^{k+1} \qquad (1.11)$$

and where the contributions from the components -2,+2, and -4,+4 are equivalent and grouped together.

The c_k^q values are evaluated by summation over the ligands as follows (harmonics from Table 1.1):

$$c_2^0 = Ze(4\pi/5)\sqrt{(5/8)}\sqrt{(1/2\pi)}(3\cos^2\theta_i - 1)$$
$$= Ze(4\pi/5)\sqrt{(5/8)}\sqrt{(1/2\pi)}(-4)$$
$$= -4Ze\ \sqrt{(\pi/5)} \qquad (1.12)$$

where the (-4) arises from summation of the four ligands for which cos(90) = 0.

$$c_2^{\pm2} = Ze(4\pi/5)(\sqrt{15}/\sqrt{16})\sqrt{(1/2\pi)}\sin^2\theta_i.e^{\pm2i\phi}$$

summing over the four ligands yields:

Ligand	θ_i	ϕ_i	$c_2^{\pm2}$ $\sin^2\theta e^{\pm2i\phi}$	$c_4^{\pm2}$ $\sin^2\theta e^{\pm4i\phi}$	$c_4^{\pm4}$ $\sin^4\theta e^{\pm4i\phi}$
1	90	0	1	1	1
2	90	90	-1	-1	1
3	90	180	1	1	1
4	90	270	-1	-1	1
Total			0	0	4

(1.13)

where the sums for $c_4^{\pm2}$ and $c_4^{\pm4}$ are also included; however the $c_2^{\pm2}$ and $c_4^{\pm2}$ contributions both sum to zero and therefore these terms do not contribute to the Hamiltonian.

$$c_4^0 = Ze(4\pi/9)\sqrt{(9/128)}\sqrt{(1/2\pi)}(35\cos^4\theta_i - 30\cos^2\theta_i + 3)$$
$$= Ze(4\pi/9)\sqrt{(9/128)}\sqrt{(1/2\pi)}(+12)$$
$$= Ze(\sqrt{\pi})$$

where as before the cos(θ) terms contribute zero. The $c_4^{\pm4}$ contribution follows from (1.13):

$$c_4^{\pm4} = Ze(4\pi/9)\sqrt{(315/256)}\sqrt{(1/2\pi)}\sin^4\theta_i.e^{\pm4i\phi}$$
$$= Ze(4\pi/9)\sqrt{(315/256)}\sqrt{(1/2\pi)}(+4)$$
$$= Ze(\sqrt{\pi})\sqrt{(35/18)}$$

Collecting the terms together and incorporation into (1.10) provides the square planar D_{4h} Hamiltonian as follows:

$$V(Sq.Pl) = Ze(\sqrt{\pi})[-4(1/\sqrt{5})(r^2/a^3)Y_2^0 + (r^4/a^5)(Y_4^0 + \sqrt{(35/18)}(Y_4^4 + Y_4^{-4}))] \qquad (1.14)$$

where the radial terms are included and the in-plane metal-ligand distance is written as (a).

The Hamiltonians for other complexes are generated in exactly the same fashion, by inserting the angular coordinates of the ligands and summing the results. The Hamiltonian for an octahedral (O_h) complex can be obtained by adding two further equivalent ligands, along the z axis, to the square plane above, i.e. at (θ,φ) (0,0) and (180,0). The inclusion of these two additional axial ligands causes the c_2^0 summation to vanish, leaving the terms of order 4, namely:

$$V(Oct) = \sqrt{(49/18)}\sqrt{(2\pi)}(Zer^4/a^5)[Y_4^0 + \sqrt{(5/14)}(Y_4^4 + Y_4^{-4})] \qquad (1.15)$$

If the two added ligands are not the same as those in the plane, and/or lie at different distances, then the complex still

belongs to D_{4h} (assuming retention of 90° angles and equivalent axial ligands) and is a member of a very important and common class of tetragonal trans-ML_4Z_2 species. If these axial ligands with coordinates (0,0) and (180,0) lie at a metal-ligand bond distance of (b), then their addition to the analysis above readily leads to the Hamiltonian:

$$V(D_{4h}) = Ze2\sqrt{(2\pi)}[-\sqrt{(2/5)}((r^2/a^3) - (r^2/b^3))Y_2^0 + (1/\sqrt{72})((3r^4/a^5) + (4r^4/b^5))Y_4^0 + \sqrt{(35/144)}(r^4/a^5)(Y_4^4 + Y_4^{-4})] \quad (1.16)$$

This Hamiltonian collapses to eqn.(1.14) if the terms in (b) are omitted, and collapses to eqn.(1.15) if a = b.

It is often useful to consider a tetragonal six coordinate complex to be a distorted version of an octahedron. Thus eqn.(1.16) can be re-arranged to separate out the octahedral Hamiltonian (1.15) in the fashion:

$$V_{D4h} = V_{Oh} + V_{tetrag}$$

$$V(D_{4h}) = \sqrt{(49/18)}\sqrt{(2\pi)}(Zer^4/a^5)[Y_4^0 + \sqrt{(5/14)}(Y_4^4 + Y_4^{-4})] - 2Ze\sqrt{(2\pi)}[\sqrt{(2/5)}((r^2/a^3) - (r^2/b^3))Y_2^0] - 2Ze\sqrt{(2\pi)}[\sqrt{(2/9)}((r^4/a^5) - (r^4/b^5))Y_4^0] \quad (1.17)$$

Several other important Hamiltonians are included in Table 1.2.

Before demonstrating how to use these Hamiltonians, it is appropriate to ask whether there are any other 'short-cuts' to their generation. Indeed the use of symmetry arguments can be quite beneficial.

The spherical harmonics belong to the spherical group R_3 and span therein the common terms S, P, D, F etc., according to their k (or as usually written, L, their angular momentum) values. In real molecules the harmonics will span representations within a point group. Only those harmonics which span the totally symmetric representation in any point group can contribute to the Hamiltonian.

Thus for an octahedron, the O_h terms correlate with those in R_3 as follows (the parity symbols are dropped since they depend upon whether d or p orbitals etc are involved):

S	L = 0	A_1
P	L = 1	T_1
D	L = 2	$T_2 + E$
F	L = 3	$A_2 + T_1 + T_2$
G	L = 4	$A_1 + E + T_1 + T_2$
etc		

(1.18)

indicating that only the L = 0 and L = 4 spherical harmonics contribute to the octahedral Hamiltonian, in agreement with (1.15)

Table 1.2

EXAMPLES OF SOME CRYSTAL FIELD HAMILTONIANS

Type of Molecule	Crystal Field Hamiltonian
V(Oct) (four fold)	$\sqrt{(2\pi)}\sqrt{(49/18)}(Zer^4/a^5)[Y_4^0 + \sqrt{(5/14)}(Y_4^4 + Y_4^{-4})]$
V(Oct) (three fold)	$-\sqrt{(2\pi)}\sqrt{(98/81)}(Zer^4/a^5)[Y_4^0 + \sqrt{(10/7)}(Y_4^3 + Y_4^{-3})]$
$V(D_{4h})$	$2\sqrt{(2\pi)}\sqrt{(49/72)}(Zer^4/a^5)[Y_4^0 + \sqrt{(5/14)}(Y_4^4 + Y_4^{-4})]$
t-ML_4Z_2	$-2\sqrt{(2\pi)}[\sqrt{(2/5)}Ze((r^2/a^3) - (r^2/b^3))Y_2^0$
	$-2\sqrt{(2\pi)}\sqrt{(2/9)}Ze((r^4/a^5)-(r^4/b^5))Y_4^0]$
$V(D_{4h})$	$2\sqrt{(2\pi)}\sqrt{(1/8)}(Zer^4/a^5)[Y_4^0 + \sqrt{(35/18)}(Y_4^4 + Y_4^{-4})]$
sq. ML_4	$-2\sqrt{(2\pi)}\sqrt{(2/5)}(Zer^2/a^3)Y_2^0$
$V(D_{3d})$	$(12/5)\sqrt{(2\pi)}\sqrt{(5/8)}(Zer^2/a^3)(3\cos^2\theta - 1)Y_2^0$
ML_6	$+ (12/9)\sqrt{(2\pi)}\sqrt{(9/128)}(Zer^4/a^5)(35\cos^4\theta-30\cos^2\theta+3)Y_4^0$
	$+ \sqrt{(35/2)}\sqrt{(2\pi)}(Zer^4/a^5)(\sin^3\theta\cos\theta)(Y_4^3 - Y_4^{-3})$
$V(T_d)$ ML_4	$-(4/9)\sqrt{(2\pi)}\sqrt{(49/18)}Zer^4/a^5[Y_4^0 + \sqrt{(5/14)}(Y_4^4 + Y_4^{-4})]$
$V(D_{2d})$	$(8/5)\sqrt{(2\pi)}\sqrt{(5/8)}(Zer^2/a^3)(3\cos^2\theta - 1)Y_2^0$
ML_4	$+ (8/9)\sqrt{(2\pi)}\sqrt{(9/128)}(Zer^4/a^5)(35\cos^4\theta-30\cos^2\theta+3)Y_4^0$
	$- (8/9)\sqrt{(2\pi)}\sqrt{(315/256)}(Zer^4/a^5)(\sin^4\theta)(Y_4^4 + Y_4^{-4})$
$V(D_{\infty h})$	$\sqrt{(2\pi)}Ze[\sqrt{(8/5)}(r^2/b^3)Y_2^0 + \sqrt{(8/9)}(r^4/b^5)Y_4^0]$

Note the use of rationalised spherical harmonics, described on p.35, which eliminate the $\sqrt{(2\pi)}$ factor and make data manipulation somewhat easier.

(recalling that the L = 0 term was dropped).

In D_{4h}, using correlation tables (see Appendix X), the D term now spans $\mathbf{A_1} + B_1 + \ldots$, while the G term spans $2\mathbf{A_1} + B_1 + \ldots$ No other terms of L = 4 or less span A_1. Thus anticipate the inclusion of an L = 2 term and two L = 4 terms into the D_{4h} Hamiltonian. This is most clearly seen to be the case in (1.17). In the octahedral Hamiltonian, although both Y_4^0 and Y_4^4 are involved they are modified by a common radial term and indeed represent a single A_1 linear combination. In the D_{4h} Hamiltonian there are two separate fourth order terms (see, e.g. Ref.[8] for more detailed discussion of these issues). The actual form of the linear combinations of spherical harmonics which transform as A_1 can be generated by use of a group theoretical projection operator which linearly combines the components of the harmonics to require that they transform as A_1, i.e. for each value of L, projects from these spherical harmonic bases, a linear combination which transforms as A_{1g}. This provides an alternative procedure for generating these Hamiltonians (e.g. see [6,8]) (see section 1.5.)

1.2.iii) Evaluation of Matrix Elements - Cubic systems

Very powerful, and relatively simple, methods exist for the evaluation of crystal field integrals. Some of these are discussed in detail in Chapter 3. Here briefly look at a 'first principles' evaluation.

However it is necessary first to digress and consider the nature of the problem in somewhat more detail and to become familiar with the concept of a 'matrix element'.

Consider a wavefunction ψ_i which could be one member of a basis (ψ_i, i = 1,2,3..n) for describing the functions in some general system. For example the function might be an atomic orbital described by its l (angular quantum number) and magnetic orbital quantum number m_l, e.g. a function described as $|l,m_l\rangle$, such as a d orbital $|2,2\rangle$. It could be the component of a term described by the quantum numbers $|LSM_LM_S\rangle$, e.g. a component of ^3F, say, $|3,1,1,1\rangle$, it may be a real orbital $|xz\rangle$, or indeed may be a linear combination of several of any of these possibilities. For example ψ_i may be:

$$(1/\sqrt{2})[|xz\rangle + |yz\rangle]$$

or, say

$$[\sqrt{(3/5)}|3,1,1,1\rangle - \sqrt{(2/5)}|3,-1,1,1\rangle].$$

Note that small letters l,m_l etc are used for single electrons and capital letters L,M_L etc for the vector sum of all the electrons.

It will prove convenient to use a rather different presentation of (1.7), introduced by Dirac, namely:

$$\int \psi^*_{nlm} V \psi_{n'l'm'} d\tau = C \langle nlm|lm|n'l'm'\rangle \qquad (1.19)$$

where the radial and angular components of the integral on the left have been separated into a scalar (number) C which contains the radial information, and a symbol which contains all the angular information.

Note that the bra $\langle nlm|$ expresses the complex conjugate of a wavefunction with quantum numbers n,l and m, while the ket $|n'l'm'\rangle$ expresses the wavefunction with quantum numbers n',l' and m' (principal, orbital and magnetic orbital respectively). Moreover since the crystal field operator is expressed in terms of spherical harmonics, they may also be defined by their l and m values. The real d orbitals might also be used in (1.19), or the term representations L,M_L etc.

> Note that the Schrodinger expression contained within the left-hand side of (1.19) is expressed in displacement space, while the Dirac symbol on the right exists in momentum space. The numerical value of the Dirac symbol may be calculated by coupling the angular momenta expressed by the angular quantum numbers of the wavefunctions and operator. Note further that the left hand integral form contains the units of energy within the Hamiltonian H, while in the right hand Dirac form the energy is contained within C and the bracketed function is a unitless angular coefficient, i.e. a number. The expectation values of crystal field integrals will be equated to the Dirac formalism matrix element and presented in terms of radial parameters. Moreover, for simplicity the magnitudes of these elements will be presented in a form such as $\langle 2|V|2\rangle = Dq$, which is not a strictly correct presentation since $\langle 2|H|2\rangle$ is simply an unitless number. This is a shorthand notation for $C\langle 2|V|2\rangle = Dq \times 1$.

It is also useful to distinguish at this time a function which is symmetry adapted to a group from one which is not. Thus the real orbital $|x^2-y^2\rangle$ is symmetry adapted to the D_{4h} group (and transforms as b_{1g}), because there are no operations within the group which transform this orbital into another orbital. Similarly the pair of d orbitals described by the wavefunctions $|xz,yz\rangle$ (transforming as e_g) are symmetry adapted to the group D_{4h} because no operator in this group can convert them to other orbitals. However the orbital function $|xz\rangle$ alone is not symmetry adapted to D_{4h} because the C_4 operator converts it into $|yz\rangle$.

If a symmetry adapted function, let us say x^2-y^2, transforms uniquely in a group, i.e there is only one such function with its representation, then its energy in the crystal field, V, is given directly by the value of the function described in (1.7) or (1.19). This may be written:

$$E(x^2-y^2) = \langle x^2-y^2|V|x^2-y^2\rangle.$$

However suppose the C_{2v} group is involved. Reference to the group Character Table reveals that x^2-y^2 (a_1) transforms in the

same way as $z^2(a_1)$. Thus the function $\langle x^2-y^2|V|z^2\rangle$ will also transform as a_1 (recalling that V must transform as a_1) and therefore need not be zero, and will generally not be zero. In this case the crystal field couples the two orbitals and the true bases are not pure d orbitals but are mixed together. To evaluate the degree of mixing and the energies of the resultant orbitals it is necessary to construct a 'secular determinant', viz:

$$\begin{array}{c|cc|} & x^2-y^2 & z^2 \\ x^2-y^2 & \langle x^2-y^2|V|x^2-y^2\rangle - E & \langle x^2-y^2|V|z^2\rangle \\ z^2 & \langle x^2-y^2|V|z^2\rangle & \langle z^2|V|z^2\rangle - E \end{array} = 0 \qquad (1.20)$$

This is effectively a matrix equated to zero. The roots are the two values of E which can be derived and which correspond, in this case, to the energies of the two pairs of mixed orbitals (i.e. x^2-y^2 mixed with some z^2, and z^2 mixed with some x^2-y^2). Re-insertion of the energies into the pair of simultaneous equations expressed within the determinant (cf the Huckel approach to molecular orbital energies) provides the degree of mixing of this pair of orbitals.

If these mixed functions are used as a basis then a 'diagonal' matrix is generated, i.e. only the diagonal elements are non-zero, the off-diagonals vanish (are zero). The values of the diagonal elements are then the roots of the determinant. Indeed this is a procedure for evaluating the determinant roots by rearranging it such that the off-diagonals are zero, a process called diagonalisation.

The functions of the type $\langle\psi_i|V|\psi_i\rangle$ lie on the diagonal of the matrix, and are called diagonal matrix elements, while the functions $\langle\psi_i|V|\psi_j\rangle$ lie off the diagonal, and are called off-diagonal matrix elements.

For example to determine how the complex d orbitals $|2,m_1\rangle$ with $m_1 = 2,1,0,-1,-2$ behave under an octahedral crystal field, it is necessary to calculate the set of matrix elements $\langle m_1|V|m_1'\rangle$ since, in general, these functions are not symmetry adapted to the octahedral field, i.e. will be scrambled by the crystal field operator.

In Dirac notation, the matrix will appear as follows:

$$\begin{array}{c|ccccc|} & 2 & 1 & 0 & -1 & -2 \\ 2 & \langle2|V|2\rangle-E & \langle2|V|1\rangle & \langle2|V|0\rangle & \langle2|V|-1\rangle & \langle2|V|-2\rangle \\ 1 & \langle1|V|2\rangle & \langle1|V|1\rangle-E & \langle1|V|0\rangle & \langle1|V|-1\rangle & \langle1|V|-2\rangle \\ 0 & \langle0|V|2\rangle & \langle0|V|1\rangle & \langle0|V|0\rangle-E & \langle0|V|-1\rangle & \langle0|V|-2\rangle \\ -1 & \langle-1|V|2\rangle & \langle-1|V|1\rangle & \langle-1|V|0\rangle & \langle-1|V|-1\rangle-E & \langle-1|V|-2\rangle \\ -2 & \langle-2|V|2\rangle & \langle-2|V|1\rangle & \langle-2|V|0\rangle & \langle-2|V|-1\rangle & \langle-2|V|-2\rangle-E \end{array} = 0 \qquad (1.21)$$

$$\begin{array}{c|ccccc|} & 2 & 1 & 0 & -1 & -2 \\ 2 & Dq - E & & & & 5Dq \\ 1 & & 4Dq - E & & & \\ 0 & & & -6Dq - E & & \\ -1 & & & & 4Dq - E & \\ -2 & 5Dq & & & & Dq - E \end{array} = 0 \qquad (1.22)$$

from which one may conclude that the complex d functions $|2,1\rangle$, $|2,-1\rangle$ and $|2,0\rangle$ are already symmetry adapted to the octahedral crystal field, and have energies $\langle1|V|1\rangle = \langle-1|V|-1\rangle = -4Dq$ and $\langle0|V|0\rangle = 6Dq$. There are no non-zero elements of the type $\langle1|V|-1\rangle$ or $\langle1|V|0\rangle$ which would mix these functions.

On the other hand the existence of a non-zero value for $\langle2|V|-2\rangle$ requires that the complex d orbitals $|2,2\rangle$ and $|2,-2\rangle$ **are** mixed together. The wavefunctions are $(1/\sqrt{2})[|2,2\rangle + |2,-2\rangle]$ and $(1/\sqrt{2})[|2,2\rangle - |2,-2\rangle]$ with energies corresponding with the remaining two roots of this determinant, of 6Dq and -4Dq respectively.

This last result is obtained by diagonalisation of the sub-determinant:

$$\begin{array}{c|cc|} & 2 & -2 \\ 2 & \langle2|V|2\rangle-E & \langle2|V|-2\rangle \\ -2 & \langle-2|V|2\rangle & \langle-2|V|-2\rangle-E \end{array} = \begin{array}{c|cc|} & 2 & -2 \\ 2 & Dq-E & 5Dq \\ -2 & 5Dq & Dq-E \end{array} = 0$$

For a 2 x 2 determinant, of the form:

$$\begin{array}{c|cc|} & \phi_1 & \phi_2 \\ & H_{11} - E & H_{12} \\ & H_{12} & H_{22} - E \end{array} = 0$$

the energies are:

$$E = H_{11} - H_{12}\cot\theta \quad \text{and} \quad H_{22} + H_{12}\cot\theta$$

and the wavefunctions (eigenvectors):

$$\psi_1 = \sin\theta\,\phi_1 - \cos\theta\,\phi_2 \quad \text{and} \quad \psi_2 = \cos\theta\,\phi_1 + \sin\theta\,\phi_2$$

where $\tan 2\theta = 2H_{12}/(H_{11} - H_{22})$.

In general any problem will require the construction and diagonalisation of one, or more, secular determinants containing a set of matrix elements involving the appropriate operator and a basis set.

It can now be seen that the basis set is in fact the most

complete set of functions which can written down to initiate the solution of the problem. In this case it is the five complex d functions since it might not be known in advance that the $|2,2\rangle$ and $|2,-2\rangle$ are scrambled. Had this fact been known, one might have considered starting with a basis set consisting of $|2,1\rangle$, $|2,0\rangle$, $|2,-1\rangle$, plus the two scrambled functions. A secular determinant would then look like (1.22) but the first and last columns would contain the two scrambled functions. Such a determinant would be 'diagonal', i.e. all the off-diagonals would be zero and one could 'read off' the energies from the diagonal without further computation.

To continue with the development of methods to evaluate the magnitudes of these matrix elements, it is necessary to have a closer look at the analytical expressions for the basis functions and for the operator.

We wish to derive the effect of the octahedral Hamiltonian upon the complex d orbitals i.e. to evaluate the magnitudes of the matrix elements $\langle m|V|m'\rangle$ in (1.21) and (1.22). The detailed evaluation of an element can be considered in two ways. It may be considered as a Schrodinger integral (left-hand side of (1.19)) being an integration over a triple product of functions each of which can be written as a combination of spherical harmonics. Alternatively it may be considered in momentum space in the Dirac formulation (right-hand side of (1.19)), as a coupling of two angular momenta to give a resultant. The former is the more classical approach and will be discussed first. The latter is the more modern approach especially adaptable to computer evaluation, and will be considered in Chapter 3.

The Hamiltonian H incorporates the crystal field Hamiltonians discussed above and recognises that the electric field experienced by the electrons is defined by H = eV. The d orbitals have the analytical forms:

$$d(m_l = 0) = R_{nl}\sqrt{(1/2\pi)}\sqrt{(5/8)}(3\cos^2\theta - 1)$$

$$d(m_l = \pm 1) = \mp R_{nl}\sqrt{(1/2\pi)}\sqrt{(15/4)}\cos\theta\sin\theta.e^{\pm i\phi}$$

$$d(m_l = \pm 2) = R_{nl}\sqrt{(1/2\pi)}\sqrt{(15/16)}\sin^2\theta.e^{\pm 2i\phi} \qquad (1.23)$$

Consider evaluation of the energy of the $m_l = 0$ d orbital, which corresponds with dz^2, in an octahedral field. If the octahedral Hamiltonian is broken down into the Y_4^0 and Y_4^4 components, then the Y_4^0 component may be written:

$$\langle 20|H(Y_4^0)|20\rangle = R_{nl}\sqrt{(1/2\pi)}\sqrt{(5/8)}[\sqrt{(49/18)}\sqrt{(2\pi)}(Ze^2r^4/a^5) \times R_{nl}\sqrt{(1/2\pi)}\sqrt{(5/8)}]\sqrt{(1/2\pi)}\sqrt{(9/128)}\int_0^{2\pi}(35\cos^4\theta - 30\cos^2\theta + 3)(3\cos^2\theta - 1)^2\sin\theta d\theta$$

$$= \alpha_4\,(5/8)\sqrt{(49/18)}\sqrt{(9/128)}(1/\sqrt{2\pi})\int_0^{2\pi}(35\cos^4\theta - 30\cos^2\theta + 3)(3\cos^2\theta - 1)^2\sin\theta d\theta \qquad (1.24a)$$

where, for 3d electrons,

$$\alpha_1 = Ze^2\int_0^\infty R_{3d}[r_<^l/r_>^{l+1}]r^2R_{3d}.dr = Ze^2\overline{r^l/r^{l+1}}$$

$$\text{and } \alpha_4 = Ze^2\int_0^\infty R_{3d}(r_<^4/r_>^5)r^2dr = Ze^2\overline{r^4/a^5} \text{ and } Dq = Ze^2(\overline{r^4/a^5})/6 \qquad (1.24b)$$

In eqn.(1.24a) the long expression represents multiplication of d(m = 0) wavefunction by the Y_4^0 octahedral Hamiltonian component (from 1.15), right multiplied by the m = 0 wavefunction, all multiplied by the analytical expression for the Y_4^0 spherical harmonic from Table 1.1.

The $[Y_4^4 + Y_4^{-4}]$ contribution will be analogous using the appropriate coefficients from (1.15) and the harmonic from Table 1.1. Expression (1.24a)is simplified by collecting together all the radial parts (of wavefunction and Hamiltonian) together in the form of a radial integral, defined in (1.24b). The quantities $r_<$ and $r_>$ are the shorter and longer metal-electron distances almost invariably metal (d) electron and ligand respectively.

Integration of the angular components of (1.24a) can be accomplished by standard mathematical techniques but detailed discussion thereof is omitted here since such evaluations are largely superceded by the more powerful methods discussed in Chapter 3, and because other texts have dealt with this expansion in detail (e.g. [7,8]). The reader is also referred to an article by Companion and Komarynsky which discusses in some detail a procedure, similar to that above, for evaluating the expectation values of matrix elements of crystal field operators [10] (see Appendix II).

Calculations for all the common geometries have, of course, already been published and the results of most are reproduced in this chapter. Irregular geometries can be calculated via use of Appendix II.

The energies of the wavefunctions after perturbation by the crystal field are generally written in terms of the crystal field radial parameters, the fourth order version of which (α_4) is included in (1.24). The parameter r^4/a^5 is regarded as the mean fourth power radius of the metal electron radial function and the denominator and numerator are inseparable. Several special crystal field parameters have been defined, of which Dq (1.24) is the most well known.

For d^1 systems it is only necessary to fit the spectrum to energy differences within the d orbital set and hence derive values of the various radial parameters which can be used as empirical parameters which, however, convey relatively little chemical information. As demonstrated in Chapter 3, the energies of many-electron systems can also be parameterised in terms of the one-electron elements and hence in terms of these various radial integrals.

The expectation values for the matrix elements with the other d orbitals are given by (Following the discussion on the previous page use is made of the Dirac notation here and henceforth since it is much easier to transcribe than the Schrodinger integral form. Recognise that the two approaches are equivalent though the underlying mathematical theory is quite different):

$$\langle 0|V|0\rangle = \alpha_4 = 6Dq$$

$$\langle 1|V|1\rangle = \langle -1|V|-1\rangle = -(2/3)\alpha_4 = -4Dq$$

$$\langle 2|V|2\rangle = \langle -2|V|-2\rangle = \alpha_4/6 = Dq$$

$$\langle 2|V|-2\rangle = (5/6)\alpha_4 = 5\ Dq \qquad (1.25)$$

Note that since the octahedral Hamiltonian components (q) are 0, 4 and -4, then the Hamiltonian can only connect wavefunctions whose m values differ by 0,4 or -4, as shown in (1.25). This explains the previous observation that matrix elements such as $\langle 1|V|0\rangle$ and $\langle 1|V|-1\rangle$ etc necessarily vanish (are zero), when V is the octahedral (four-fold) Hamiltonian.

While the complex d orbitals are usually defined by the 5 m values permitted for l = 2, of more direct use are the real d orbitals which do not contain the complex i, viz:

e_g	x^2-y^2	$(1/\sqrt{2})[\|2\rangle + \|-2\rangle]$
	z^2	$\|0\rangle$
t_{2g}	xy	$(1/i\sqrt{2})[\|2\rangle - \|-2\rangle]$
	xz	$-(1/\sqrt{2})[\|1\rangle - \|-1\rangle$
	yz	$-(1/i\sqrt{2})[\|1\rangle + \|-1\rangle$

(1.26)

and the familiar splitting of the real d orbitals into the t_{2g} (xz,xy,yz) set lying at -4Dq and the e_g (z^2, x^2-y^2) set lying at +6Dq is now appreciated.
Note for example the expansion of the element $\langle d_{xy}|V|d_{xy}\rangle$ to yield - 4Dq.

$$\langle d_{xy}|V|d_{xy}\rangle = (-1/i\sqrt{2})(1/i\sqrt{2})\langle 2 - -2|V|2 - -2\rangle$$

$$= (1/2)[\langle 2|V|2\rangle + \langle -2|V|-2\rangle - 2\langle 2|V|-2\rangle]$$
$$= (1/2)[Dq + Dq - 2x5Dq]$$

$$= -\ 4Dq$$

Remember that with these complex functions, the bra function (left-hand function) must be multipled by -1, since it represents the complex conjugate.

1.2 iv) Evaluation of Matrix Elements - Tetragonal Systems

When the D_{4h} Hamiltonian (1.17) is used, radial integrals are defined to account for the second order terms, and also for the two different metal-ligand distances, indicated by axial(Ax) (distance b) and equatorial(Eq) (distance a), thus:

$$\alpha_2(Eq) = Ze^2\overline{r^2/a^3} \qquad \alpha_2(Ax) = Ze^2\overline{r^2/b^3}$$

$$\alpha_4(Eq) = Ze^2\overline{r^4/a^5} \qquad \alpha_4(Ax) = Ze^2\overline{r^4/b^5} \qquad (1.27)$$

There is generally insufficient information in the spectrum to identify all these parameters, so usually they are grouped together via definition of a new second order radial integral Ds, and a new fourth order integral Dt, defined as:

$$Ds = (2/7)[\alpha_2(Eq) - \alpha_2(Ax)]$$
$$Dt = (2/21)[\alpha_4(Eq) - \alpha_4(Ax)] \qquad (1.28)$$

The (2/7) and (2/21) coefficients are included because these factors occur frequently in the calculations and their inclusion in these definitions simplifies the appearance of the energies. In this fashion the matrix elements of the complex d orbitals in a field of D_{4h} symmetry are given by:

$$\langle 0|V|0\rangle = 6\ Dq - 2\ Ds - 6\ Dt$$
$$\langle 1|V|1\rangle = \langle -1|V|-1\rangle = -4\ Dq - Ds + 4\ Dt$$
$$\langle 2|V|2\rangle = \langle -2|V|-2\rangle = Dq + 2\ Ds - Dt$$
$$\langle 2|V|-2\rangle = 5\ Dq$$

and, for the real d orbitals:

$\langle x^2-y^2\|V\|x^2-y^2\rangle = 6\ Dq + 2\ Ds - Dt$	b_{1g}
$\langle z^2\|V\|z^2\rangle = 6\ Dq - 2\ Ds - 6\ Dt$	a_{1g}
$\langle xy\|V\|xy\rangle = -\ 4\ Dq + 2\ Ds - Dt$	b_{2g}
$\langle xz\|V\|xz\rangle = \langle yz\|V\|yz\rangle = -\ 4\ Dq - Ds + 4\ Dt$	e_g

(1.29)

Since Dq only depends upon (a), which is the in-plane metal-ligand bond distance, the Dq value in a D_{4h} complex refers to the in-plane ligands only. It may be written Dq(Eq) (or Dq(xy)) to indicate it refers to the equatorial ligands. Note however that it possesses exactly the same definition as in the octahedral complexes. Thus it is often found that, in passing from an octahedral ML_6 to a tetragonal trans-ML_4Z_2, the Dq(L) value remains roughly constant. Dt is a difference of two fourth order parameters and by comparison with the definition for Dq it is evident that, in a D_{4h} complex, Dt can be factored into equatorial and axial Dq values, namely:

$$Dt(D_{4h}) = (4/7)[Dq(Eq) - Dq(Ax)]$$

and

$$Dt(C_{4v}) = (2/7)[Dq(Eq) - Dq(Ax)] \qquad (1.30)$$

where the latter definition is for complexes with one axial ligand (ML_5, C_{4v} symmetry) (see section 1.8 for further examples).

1.2 v) Evaluation of Matrix Elements - Trigonal Systems

Many complexes ranging from tris-bidentate systems such as $[M(en)_3]^{n+}$ of octahedral parentage to five coordinate trigonal bipyramidal complexes ML_5 have a three-fold axis of symmetry.

Trigonally distorted octahedral complexes are best approached by using the three-fold axis of an octahedron as the axis of quantization. The appropriate Hamiltonian is shown in Table 1.2. Under such circumstances, the conventional four fold d orbitals are scrambled to yield the new sets [6]:

Transforming as t_{2g}:

Imaginary	Real
$\|0\rangle$	$\|z^2\rangle$
$\sqrt{(2/3)}\|2\rangle - \sqrt{(1/3)}\|-1\rangle$	$\sqrt{(2/3)}\|x^2-y^2\rangle - \sqrt{(1/3)}\|xz\rangle$
$\sqrt{(2/3)}\|-2\rangle + \sqrt{(1/3)}\|1\rangle$	$\sqrt{(2/3)}\|xy\rangle + \sqrt{(1/3)}\|yz\rangle$

Transforming as e_g

$\sqrt{(1/3)}\|2\rangle + \sqrt{(2/3)}\|-1\rangle$	$\sqrt{(1/3)}\|x^2-y^2\rangle + \sqrt{(2/3)}\|xz\rangle$
$\sqrt{(1/3)}\|-2\rangle - \sqrt{(2/3)}\|1\rangle$	$\sqrt{(1/3)}\|xy\rangle - \sqrt{(2/3)}\|yz\rangle$

(1.31)

Distortion along this three-fold axis (compression, elongation, angular twist etc) lifts the degeneracy of the t_{2g} set yielding, generally, e + a_1; however the e_g set remains unsplit under such three-fold distortion. Hence such complexes have three sets of d orbitals (a_1 + 2 e) and there will generally be an off-diagonal element connecting the two doubly degenerate sets. Following Liehr[30] trigonal distortion radial integrals Dσ and Dτ are defined to parallel Ds and Dt [6], leading to the real and imaginary d orbital matrix elements for the trigonal distortion component of the field:

$$\begin{aligned}
\langle \pm 2|V|\pm 2\rangle &= -(2/3)Dq + 2D\sigma - D\tau \\
\langle \pm 1|V|\pm 1\rangle &= (8/3)Dq - D\sigma + 4D\tau \\
\langle 0|V|0\rangle &= -4Dq - 2D\sigma - 6D\tau \\
\langle \pm 2|V|\mp 1\rangle &= \pm(10/3)\sqrt{2}\,Dq
\end{aligned}$$

Table 1.3

ENERGY LEVELS IN THE POINT CHARGE MODEL[a]
GROUND STATE ONE ELECTRON d ORBITAL ENERGIES

Stereochemistry	Point Group	Orbital	α_0^c	α_2	α_4
Cubic	O_h	t_{2g}	8	0	16/27
		e_g	8	0	-8/9
Twisted Cubic	D_{4d}	e_3	8	0	16/27
		e_2	8	0	-4/27
		a_1	8	0	-8/9
Square antiprism (H.S.M)[d]	D_{4d}	e_2	8	21/85	-7/60
		e_3	8	-47/380	21/45
		a_1	8	-21/85	-65/93
Dodecahedron (H.S.M.)	D_{2d}	e	8	8/95	15/79
		a_1	8	15/89	-55/193
		b_1	8	-15/89	7/10
		b_2	8	-15/89	-31/39
Octahedron	O_h	e_g	6	0	1
		t_{2g}	6	0	-2/3
Tetrahedron	T_d	t_2	4	0	8/27
		e	4	0	-4/9
Square Planar	D_{4h}	b_{1g}	4	4/7	19/21
		b_{2g}	4	4/7	-16/21
		e_g	4	-2/7	-2/7
		a_{1g}	4	-4/7	3/7
Trigonal Planar	D_{3h}	e'	3	3/7	3/56
		e"	3	-3/14	-3/14
		a_1'	3	-3/7	9/28
Linear	$D^{\infty}h$	Σ_g^+	2	4/7	4/7
		Π_g	2	2/7	-8/21
		Δ_g	2	-4/7	2/21

			α_0^{Ee}	α_0^{Ae}	α_2^E	α_2^A	α_4^E	α_4^A
Tetragonal	D_{4h}	b_{1g}	4	2	4/7	-4/7	19/21	2/21
		a_{1g}	4	2	-4/7	4/7	3/7	4/7
		b_{2g}	4	2	4/7	-4/7	-16/21	2/21
		e_g	4	2	-2/7	2/7	-2/7	-8/21
Square Pyramid	C_{4v}	b_1	4	1	4/7	-2/7	19/21	1/21
		b_2	4	1	4/7	-2/7	-16/21	1/21
		a_1	4	1	-4/7	2/7	3/7	2/7
		e	4	1	-2/7	1/7	-2/7	-4/21
Trigonal Bipyramid	D_{3h}	a_1'	3	2	-3/7	4/7	9/28	4/7
		e'	3	2	3/7	-4/7	3/56	2/21
		e"	3	2	-3/14	2/7	-3/14	-8/21

a) Dq = (1/6)$\alpha_4(O_h)$ = (1/6)α_4^E (C_{4v}, D_{4h} etc)
b) The coefficients are valid irrespective of whether point charges

or point dipoles are considered.
c) The number in this column is the coordination number
d) Hard sphere model.
e) The numbers in these columns represent the number of equatorial (E) and axial (A) ligands. This section of the Table refers to two different ligands or two identical ligands at different distances.

---

and

$$\begin{aligned} \langle e(t_{2g})|V|e(t_{2g})\rangle &= D\sigma + (2/3)D\tau - 4Dq \\ \langle a_1(t_{2g})|V|a_1(t_{2g})\rangle &= -2D\sigma - 6D\tau - 4Dq \\ \langle e(e_g)|V|e(e_g)\rangle &= (7/3)D\tau + 6Dq \\ \langle e(e_g)|V|e(t_{2g})\rangle &= \sqrt{2}D\sigma - (5\sqrt{2}/3)D\tau \end{aligned} \qquad (1.32)$$

Pryce and Runciman propose an alternate basis [31] in which new parameters (being linear combinations of Dσ and Dτ), are defined:

$$\begin{aligned} \langle e(t_{2g})|V|e(t_{2g})\rangle &= -(1/3)v - 4Dq \\ \langle a_1(t_{2g})|V|a_1(t_{2g})\rangle &= (2/3)v - 4Dq \\ \langle e(e_g)|V|e(e_g)\rangle &= 0 + 6Dq \\ \langle e(e_g)|V|e(t_{2g})\rangle &= v' \end{aligned} \qquad (1.33)$$

which has the advantage that v' is often so small that the splitting then depends only on one variable. Values of v (and v') occur frequently in the discussion of trigonal systems addressed in Chapter 6.

Insertion of the ligand coordinates for a trigonal bipyramid into the crystal field expansion leads to the energies also listed in Table 1.3. Such systems have been studied by various groups, with several different sets of radial parameters employed. Thus, by analogy to the systems discussed above, Ds and Dt may be defined [32,33]: (For trigonal bipyramidal systems)

$$\begin{aligned} Ds &= (1/14)[4\alpha_2(Ax) - 3\alpha_2(Eq)] \\ Dt &= (1/168)[16\alpha_4(Ax) + 9\alpha_4(Eq)] \end{aligned}$$

and hence

$$Dt = (1/28)[16Dq(Ax) + 9Dq(Eq)] \qquad (1.34)$$

These symbols are equivalent to Δ_2 and Δ_1 used by Allen and Hush [34] and are related to another set [35] via $Ds = Q_2/14$ and $Dt = (25/168)Q_4$ [33]. With these parameters the energies of the d orbitals in a trigonal bipyramidal D_{3h} field are represented by [33]:

$$\begin{aligned} \langle 0|V|0\rangle &= \langle z^2|V|z^2\rangle = 2Ds + 6Dt && a_1' \\ \langle \pm1|V|\pm1\rangle &= \langle xz,yz|V|xz,yz\rangle = Ds - 4Dt && e'' \\ \langle \pm2|V|\pm2\rangle &= \langle xy,x^2-y^2|V|xy,x^2-y^2\rangle = -2Ds + Dt && e' \end{aligned} \qquad (1.35)$$

Trigonal bipyramidal systems of the type $ML_3L'Z$ (with L in the equatorial plane) (C_{3v} symmetry) will have the same analysis as regular trigonal bipyramids provided that the angle between axis and plane remains 90°. If this angle deviates from 90° a more complex Hamiltonian, discussed elsewhere [33] is required.

Table 1.3 contains the energies of the real d orbitals in a variety of crystal field environments, expressed in terms of the radial integrals α_2 and α_4. Note that the cubic groups (cubic O_h, octahedral O_h and tetrahedral T_d) all possess essentially the same splitting diagram, a three-fold set t_2 and a two-fold set e, except that the tetrahedral and cubic splitting energy separations are -4/9 and -8/9 that of the octahedron respectively. The negative sign indicates that the splitting is inverted. Since the splitting in a tetrahedral complex is usually designated 10Dq(tet), it is evident that a tetrahedral ML_4 complex is expected to yield a Dq value about 4/9 that of the corresponding octahedral ML_6 species. Despite the simplicity of the approach, this is generally true. For a flattened D_{2d} stereochemistry, see Eqn.8.1.

In summary, the energy relationships shown in Table 1.3 can be used to fit an experimental spectrum (once the transitions have been assigned) and values of the various crystal field radial integrals derived therefrom.

Some considerable space has been devoted [6,7,12,13] to crystal field theory because it provides an excellent basis for the assignment of the spectra of transition metal complexes and because historically it has provided impetus for the development of the entire field of inorganic coordination chemistry.

Crystal field theory is often remarkably successful in:
a) determining the stereochemistry (and coordination number) of metal ions, by assignment of their electronic spectra, and deducing thereby whether the spectrum is consistent with one stereochemistry or another. Generally one might consider several different stereochemistries and assignments and determine which set of numerical values for the radial integrals is most reasonable (based on previous chemical experience),
b) assigning absorption bands to particular transitions and predicting their relative (but not absolute) energies,
c) predicting, through the energy level diagram, the number of unpaired electrons a transition complex may possess,
d) predicting certain thermodynamic phenomena [14] such as the structures of spinels,
e) from a compendium of measured experimental Dq values allowing the prediction of absolute transition energies for hitherto unprepared complexes or for complexes of unknown ligands. This is useful, for example, in metalloprotein chemistry where the nature of the active site may sometimes be deduced from the d-d spectrum of the metalloprotein. Gray and co-workers have been particularly successful in this pursuit by showing that only certain combinations of ligands (derived from amino acids) can give rise to biological chromophores [e.g.15] (see Chapter 8).
f) providing somewhat primitive information concerning the nature

of the metal ligand bond through variation in 10Dq, i.e. the Spectrochemical Series (Chapter 9).
g) to a limited degree, deducing the extent of angular distortion in a low symmetry complex.

However most of this success is really due to the firm basis of molecular symmetry underlying the theory. Once we depart from regular cubic symmetry, the number of radial parameters necessary to solve the spectrum increases yet the chemical significance of these parameters is miniscule. Thus, for example, in a tetragonal complex, the spectrum can be fitted by Dq, Ds and Dt where the sign of Dt, according to (1.28) provides information on the relative magnitudes of the axial and equatorial fields. However little or no chemical information has been derived directly from values of Ds as a function of metal ion or ligand. The radial parameters cannot be calculated 'ab initio' because the basis of the theory, a metal-point charge interaction, is false. The overall fit, in a semi-empirical sense between the predictions of this theory and the experimental data deteriorates as the metal-ligand bond becomes more covalent. Other approaches specifically the use of Normalised Spherical Harmonic Hamiltonians (section 1.5) and the Angular Overlap Model (section 1.6) provide a more chemically useful procedure to manipulate the experimental data. Nevertheless those aspects of crystal field theory which rely wholly on its symmetry basis remain valid and useful. In particular the so-called crystal field splitting diagrams are of great value because they permit one to predict, 'on the back of an envelope', how many electronic transitions to expect, and what kind of magnetism a molecule of any specific geometry should possess. Unfortunately the numerical values of the various radial integrals which can be derived from fitting a spectrum are, with the exception of Dq, of little chemical value (but see (a) above) and their use should be discouraged.

1.3 Crystal Field Splitting Diagrams

The application of the above theory to various stereochemistries yields a d orbital splitting diagram which can be used to predict the electronic spectrum of a transition metal ion in that environment. However such a diagram can usually be obtained by recourse to simple group theoretical arguments if numbers are not required.

Thus in an regular octahedral complex (point group O_h), where there is no distinction between the x, y and z coordinates, the xz, yz and xy orbitals lying between these coordinates must be indistinguishable, and therefore degenerate (of equal energy). They therefore form a basis for a three dimensional representation in the point group O_h, in fact t_{2g}. In the discussion which follows, a general knowledge of group theory is assumed. Readers lacking such knowledge should consult one of the many texts dealing with this subject [4,6,16,17].

The z^2 and x^2-y^2 d orbitals lie along the x, y and z axes. Since these axes are indistinguishable, it follows that these two orbitals are also indistinguishable and therefore form the basis for a two dimensional representation, namely e_g. The two sets of orbitals are distinguishable because the former lies between the axes, and the latter along them. An electron in the e_g set would point directly towards the point charge or ligand and therefore suffer greater destabilisation than an electron in the t_{2g} set which points between the ligands; therefore the e_g set lies at higher energy than the t_{2g} set. This is exactly the result, qualitatively, obtained from a full analytical crystal field treatment, yet obtained in seconds from consideration of the molecular symmetry.

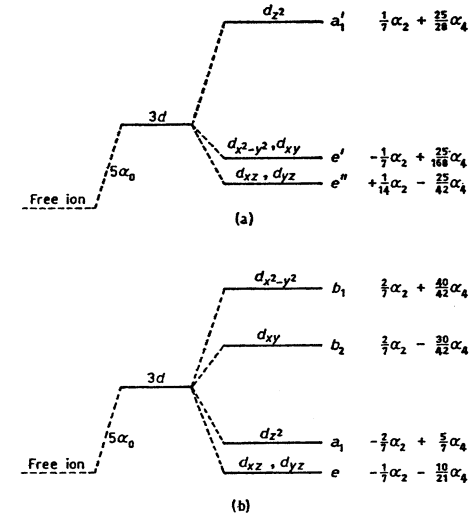

Fig.1.2 Point charge splitting diagrams for (a) trigonal bipyramidal (D_{3h}), and (b) square pyramidal (C_{4v}) five coordinate environments.

Consider now a trigonal bipyramidal molecule ML_5. If the principle axis (the three fold axis) is labelled z, this axis is clearly distinguishable from the x and y axes located in the molecular plane. Thus the z^2 orbital is a unique orbital distinguishable from the other four orbitals (a_1' in D_{3h}). We proceed with a 'gedanken' experiment, useful in dealing with this kind of problem. Choose, arbitrarily, one of the in-plane metal ligand bonds as the x axis; the y axis becomes a vector at 90^o to this x axis but not lying along a metal-ligand bond. Then, relative to these axes the x^2-y^2 and xy orbitals can be defined. Now, in your mind, spin the molecule about the z axis. When the molecule has come to rest it is no longer possible to decide which

metal-ligand bond was the x axis (and hence which was y). Nor is it possible to distinguish where xy and x^2-y^2 were located; therefore they two must represent an indistinguishable pair of orbitals (e' in D_{3h}) lying in the molecular xy plane (and symmetric with respect to reflection therein). The xz and yz orbitals can be distinguished from these last two orbitals (the e' pair) since xz and yz lie above and below the molecular xy plane. However xz and yz cannot be distinguished from each other when, as is the case here, x is indistinguishable from y. Thus the xz and yz orbitals form the basis for a two dimensional representation antisymmetric with respect to reflection in the xy plane, i.e. e" in D_{3h}.

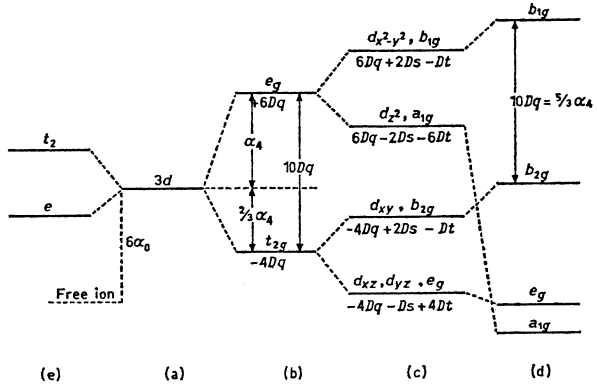

Fig.1.3 Energy levels in the point charge approximation. (a) Spherically perturbed d shell, (b) octahedral perturbation, (c) octahedron with weak tetragonal distortion, (elongation along z axis), (d) strong tetragonal distortion or square planar environment, and (e) tetrahedral perturbation (spherical term $4\alpha_0$).

Therefore in a trigonal bipyramidal field of D_{3h} symmetry, the five d orbitals split into three sets, a', e' and e". Their relative order is less clear but the z^2 orbital will be most destabilised (lie at highest energy) because it points directly at two ligands. The in-plane orbitals should suffer more repulsion than the out-of-plane orbitals, leading to the sequence shown in Fig.1.2. However since there is no group theoretical requirement that the in-plane and out-of-plane metal ligand bonds be of equal length (or even that the molecule should have identical ligands along z and xy) the sequence shown in Fig.1.2 may vary from one molecule to another. Later it is shown that the e" orbitals should have more π-antibonding character than the e' orbitals so that in a covalent molecule where π bonding is important, the sequence may be inverted. Note however that provided the symmetry remains strictly D_{3h}, the d orbitals **must** split into an orbital singlet and two orbital doublets.

Apply the same type of argument to a tetragonal (D_{4h}) complex of type ML_4Z_2 (L in the xy plane, all angles 90^o).

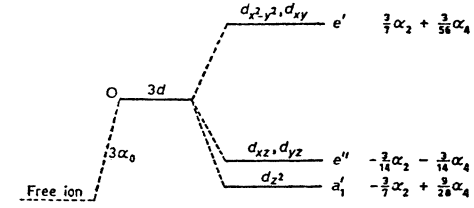

Fig.1.4 Point charge splitting diagram for a D_{3h} trigonal plane

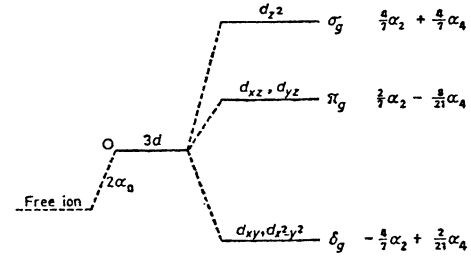

Fig.1.5 Point charge splitting diagram for a linear molecule.

In this 'gedanken' experiment mentally label one M-L bond as x, one as y, and the M-Z bond as z. Throw the molecule (mentally) into the air; catch it and find the axes. Z is found immediately, x and y are known to be in the ML_4 plane along the axes, but they cannot be distinguished. Clearly z^2 is distinguishable from x^2-y^2, and xy is distinguishable from xz,yz. In this case xy is distinguishable from x^2-y^2 (unlike the D_{3h} case) because although the x axis cannot be positively identified, it is known that it lies along a bond vector. Thus xy between the bond vectors is recognisable from x^2-y^2 lying along bond vectors.

In this fashion, the splitting diagrams for a range of different stereochemistries may be obtained. They are displayed in Fig.1.2 - 1.6 together with their calculated crystal field energies in terms of the crystal field radial parameters. These data are also summarised in Table 1.3.

Note that the extreme right hand column of group theoretical Character Tables (see for example Appendix I) contains the representations for the various cartesian coordinates and their products. Products such as xy, xz etc, since they share the same symmetry behaviour as the corresponding d orbitals transform in the same fashion. Therefore the splitting of the d orbitals can be deduced from consideration of the right hand column of the Character Table for the group concerned. For example in the group C_{4v}, appropriate for a square pyramidal five coordinate complex, xz,yz are indistinguishable trans-

forming as the degenerate e representation, while the other three orbitals have different energies since they transform as three different representations, namely xy as b_2, x^2-y^2 as b_1 and z^2 as a_1.

Not surprisingly the splitting pattern for a five coordinate trigonal bipyramidal molecule is similar to that for a trigonal planar three coordinate complex but in the latter case the z^2 orbital will be most stable.

Four coordinate planar, five coordinate square pyramidal and six coordinate ML_4Z_2 complexes can all be approached from the regular octahedron. Begin with an octahedron and replace two trans ligands L, with ligands Z, assumed to have a weaker field strength. The z axis now becomes distinguishable resulting in the loss of degeneracy of z^2 and x^2-y^2; similarly xz,yz can now be distinguished from xy. The initial orbital doublet and triplet of the octahedron will split according to the right hand side of Fig.1.7.

It is only the total charge along an axis which determines the energies of the d orbitals. A five coordinate complex with one Z ligand having twice the field strength of a Z ligand in the six coordinate ML_4Z_2 would have the same first order one-electron d orbital energies. Thus the right-hand side of Fig.1.7 is also appropriate to five coordinate compounds and shows, essentially, what will happpen as the field strength along the z axis diminishes (axial elongation). The energy of the x^2-y^2 orbital is shown to increase, with decrease in z, because in a real molecule as the field along the z axis decreases, the positive charge remaining on the metal increases resulting in increased attraction for the xy ligands. This is not however a prediction of classical point charge crystal field theory. Examples of this effect will be presented in Chapter 9.

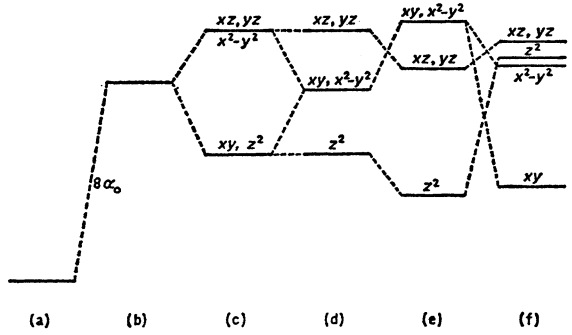

Fig.1.6 Point charge splitting diagram for various eight coordinate environments. (a) Free ion, (b) spherically perturbed d shell, (c) regular cubic (O_h), (d) twisted cube (D_{4d}), (e) square anti-prism (hard sphere, D_{4d}) and (f) dodecahedron (hard sphere D_{2d}) environments.

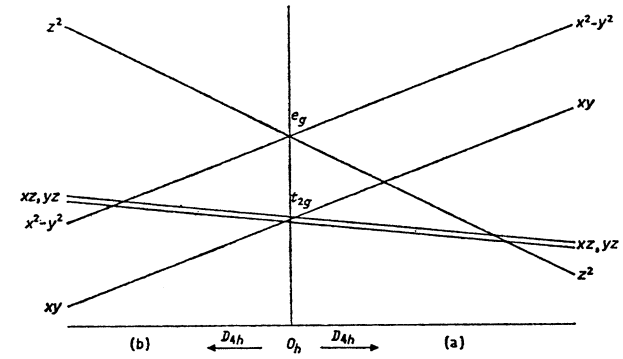

Fig.1.7 Effect of tetragonal distortion upon the energy levels in an octahedral environment. (a) Elongation along the z axis, and (b) elongation in the xy plane.

In the limit of weak field along the z axis, i.e. no field, the square planar situation is achieved, the energy levels for which are those at the extreme right of Fig.1.7.

The left-hand side of Fig.1.7 corresponds to axial compression where the z field is stronger than the xy field either because Z has a stronger field strength than L or the M-Z bond is shorter than the M-L bond (or both). The limit of axial compression approaches a linear molecule. If the xy ligands are weakened to the extent of effectively becoming zero (being removed), the xy and x^2-y^2 orbitals will become degenerate and there results the linear splitting diagram as in Fig.1.5.

In the fashion described above the one-electron energy level diagrams for d orbitals in any environment can be rapidly, but qualitatively, derived.

A one-electron energy level diagram is appropriate for a system in which there is no interelectronic repulsion, namely for d^1 and d^9 species. These diagrams will not be appropriate for multi-electron systems where inter-electronic repulsion causes the generation of additional energy levels. However these one-electron diagrams are crucial for the generation of multi-electron diagrams, usually called term diagrams.

1.4 Ligand Field Theories

In the limit of a point charge approximation, the d orbitals are non-bonding. The general term Ligand Field Theory is applied to developments of crystal field theory recognising that the metal-ligand bond does have some covalent character.

When overlap between metal and ligand orbitals is admitted, then these metal orbitals can, in principle, possess sigma or pi bonding or anti-bonding character.

This can be handled in a qualitative fashion by interpreting the radial parameters which arise from the crystal field, or a related model, in terms of some degree of covalency in the bonds, or can be treated semi-empirically or empirically using a variety of molecular orbital models.

In the following sections an extension of crystal field theory, the normalised spherical harmonic Hamiltonian is discussed, and some molecular orbital approaches are considered.

1.5 Normalised Spherical Harmonic Hamiltonians (NSH Theory) [29]

Suppose the classical ionic crystal field model is used to calculate the energies of an octahedral molecule ML_6 using a standard set of xyz axes passing through the six ligands, the z axis being therefore a four-fold axis. A certain value of Dq, call it ^4Dq, will be derived for a given experimental data set. Now consider that the z axis passes through the face of the octahedron, collinear with a three fold axis, and not now passing through the ligands. The Hamiltonian for the crystal field now looks different, viz:

$$V(Oct) = -\sqrt{(98/81)}\sqrt{(2\pi)}(Zer^4/a^5)[Y_4^0 + \sqrt{(10/7)}[Y_4^3 + Y_4^{-3}] \quad (1.36)$$

More importantly the magnitude of Dq, call it ^3Dq, is related to ^4Dq via $^3Dq = (-2/3)^4Dq$

Within the new framework the same transition energies must be calculated but the actual form of the energies, as functions of ^3Dq and B, will be different from the standard four fold set. The value of Dq has apparently changed. Of course this is artificial, it is simply the definition of Dq which has changed. However it is clear that this can be confusing. Luckily, historically, most researchers used the conventional four fold framework passing through the ligands, so that data from different laboratories were readily compared and contrasted. Imagine the nightmare if several different frameworks were in use.

Nevertheless it is an attribute of conventional crystal field theory, that the magnitudes of crystal field radial parameters may be dependent upon the choice of molecular axes; they are not truly scalar, they do depend upon the geometry and axes employed. For cubic complexes this has not been a problem; it is clearly a problem for lower symmetry complexes where the choice of the 'right' axes to use may not be so obvious. Different laboratories may now choose different axes for the same molecule and publish different sets of radial parameters, obviously a confusing situation. This is one reason why there has been comparatively little development of the analysis and assignment of the spectra of low symmetry metal complexes. Tetragonal systems have been studied in some depth, but there is uniformity among researchers to use invariably the 'obvious' choice $x = y \neq z$.

Proceeding to lower symmetries, such as rhombic, the choice is not necessarily obvious and the radial parameters so derived cannot be readily compared between one system and another.

This is one reason why the NSH approach was developed, since the magnitudes of the NSH parameters are independent of the choice of molecular axes. Moreover as will be seen, the NSH approach allows group theory to be used in a much more powerful fashion both in setting up and solving the secular determinant and in the use of selection rules for the observation of particular observables.

While Dq may, for historical reasons, be continued to be used in the coming years, the use of the lower symmetry crystal field parameters, such as Ds and Dt, should be discouraged. They contain little chemical information (surely the name of the game) and cannot be compared from one stereochemistry to another. The NSH parameters of lower symmetry, DS, DT etc, have specific group theoretical significance and, most importantly, can be compared from one stereochemistry to another. They are ligand field parameters in that no assumption of the type of bonding is made; the only assumption made is that of the molecular symmetry.

Recall that a Hamiltonian describing any geometric environment can be constructed from a suitable linear combination of spherical harmonics. Such a Hamiltonian transforms as the totally symmetric representation of the point group to which the molecule belongs. Moreover the crystal field Hamiltonian is generally constructed by subjecting the set of spherical harmonics (to l = 4 for d orbitals, l = 6 for f orbitals etc.) to the symmetry operators of the group concerned. This procedure selects multiples of the correct linear combination transforming as A_1, but as usually applied, the normalisation relationships are ignored. This neglect of normalisation leads directly to the dependence of the magnitudes of the radial parameters upon choice of molecular framework.

1.5.i) Construction of NSH Hamiltonians [29]

Many six, five or four coordinate complexes can be considered to arise through angular or radial distortion of an octahedron; there are also classes of four coordinate complexes derivable through distortion of a tetrahedron. The octahedron and tetrahedron are the so-called parent or generative groups and the distorted complexes belong to sub-groups of O_h or T_d. The set of symmetry operators common to both the parent group and the sub-group are used to project, from the generative group basis functions, a representation which transforms as A_1 in the sub-group.

Consider a molecule belonging to the point group C_{2v}(III) (the special labelling is discussed below). Using conventional descent in symmetry tables, one can show, for example, that T_{2g} (in O_h) spans $A_1 + B_1 + B_2$ in C_{2v}(III). Since the angular part of the T_{2g} function can be written in spherical harmonics (from R_3), it is evident, that the component transforming as A_1 in C_{2v}(III) can also be written in terms of spherical harmonics from R_3. Indeed since

three functions comprising T_{2g} (in O_h) can be linearly combined in any one of an infinite number of ways, it is certainly reasonable to choose the linear combinations which are eigenfunctions of the C_{2v}(III) operators and which transform as $A_1 + B_1 + B_2$. Indeed it is clear from this development that once this choice is made, these functions are simultaneously eigenfunctions of all the operators, in all the groups, connecting R_3 to C_{2v}(III).

If a Hamiltonian is constructed to transform as A_1 by projecting out a functiom from the complete set of spherical harmonics (to l = 4 for d orbitals), using an A_1 projection operator, then the components of T_{2g} transforming as A_1 in C_{2v}(III) will be included within this set. Indeed the Hamiltonian for C_{2v}(III) is most simply generated by looking at the functions in R_3, or more simply those in O_h, which span A_1 in C_{2v}(III). Again recourse to descent in symmetry tables reveals that in addition to T_{2g}, A_{1g} and a component of E_g span A_1 in C_{2v}(III). These functions may be constructed using normalised spherical harmonics such that they are simultaneously eigenfunctions of R_3, O_h and all the sub-groups down to C_{2v}(III). Hence the Hamiltonian for the group C_{2v}(III) is simply obtained by stringing together the functions in O_h which span A_1 in C_{2v}(III) (some examples are shown below).

The power of this method is recognised by the fact that a single table of functions can be used as basis states wavefunctions for a wide range of point groups, and simultaneously, those which transform as the totally symmetric representation of the group, can be used as the Hamiltonian.

Moreover the symmetry properties and selection rules inherent in the highly symmetric functions of R_3 (and O_h) are essentially retained in the lower groups since the functions are identical. The apparent relaxation of many of these selection rules in the lower groups happens as a consequence of mixing between these functions to an extent that does not occur in the higher groups.

The S, P, D, F and G free ion atomic terms corresponding to L = 0, 1, 2, 3, and 4, are among the more common spectroscopic terms, in R_3, found when dealing with the spectra of many transition metal ions. We require to know which group theoretical terms these functions span in the lower point groups. Some examples are shown here in (1.37), while a more complete set of descent data is shown in Appendix X. Observe:

R_3	O_h	D_{4h}	C_{2v}(I)	C_{2v}(III)
S	$\mathbf{A_{1g}}$	$\mathbf{A_{1g}}$	$\mathbf{A_1}$	$\mathbf{A_1}$
D	T_{2g}	E_g B_{2g}	$B_1 + B_2$ A_2	$A_2 + B_2$ $\mathbf{A_1}$
D	E_g	$\mathbf{A_{1g}}$ B_{1g}	$\mathbf{A_1}$ $\mathbf{A_1}$	$\mathbf{A_1}$ B_1
G	$\mathbf{A_{1g}}$	$\mathbf{A_{1g}}$	$\mathbf{A_1}$	$\mathbf{A_1}$
G	E_g	$\mathbf{A_{1g}}$ B_{1g}	$\mathbf{A_1}$ $\mathbf{A_1}$	$\mathbf{A_1}$ B_1
P,F,G	T_{1g}	E_g A_{2g}	$B_1 + B_2$ A_2	$A_2 + B_2$ B_1
F,G	T_{2g}	E_g B_{2g}	$B_1 + B_2$ A_2	$A_2 + B_2$ $\mathbf{A_1}$
F	A_{2g}	B_{1g}	A_1	B_1

(1.37)

Note that the same group theoretical term can originate from several R_3 terms of different L value, for example T_{1g} from L = 1, 3 and 4. The totally symmetric representations are shown in bold type, because it is these specific components, written in terms of spherical harmonics, which can be strung together to create the Hamiltonian for the point group concerned.

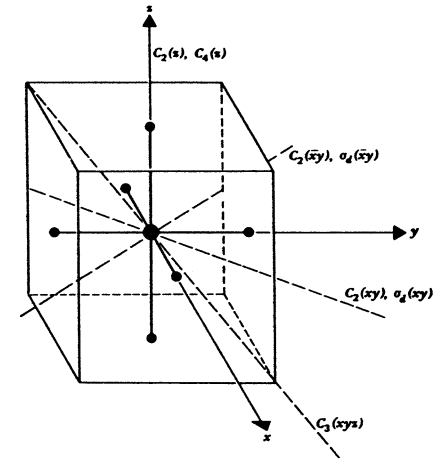

Fig.1.8 The rotation axes of an octahedron

In summary the components of each multi-dimensional representation in O_h are projected from R_3 such that, in a chain of sub-groups, normalised bases are provided for all the representations with which they correlate. This process is similar to symmetry adaptation but the use of a fully defined projection operator provides normalised functions preserving all the phase and coordinate system relationships. The basis functions in the lower

groups can be related directly to the conventions adopted in R_3. The normalised functions retain all the information concerning the magnitudes as well as the directions of the projected vectors.

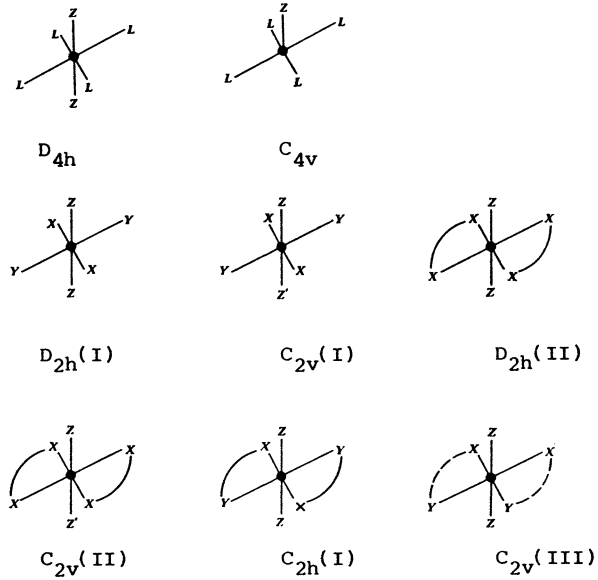

Fig.1.9 Examples of molecules belonging to various octahedral sub-groups.

Four separate chains of sub-groups can be defined by descending along four different rotation axes of the octahedron, namely the axes C_4^z, C_2^z, C_2^{xy}, and C_3^{xyz} (Fig.1.8). The chain of sub-groups is a sequence such that all operators in a given sub-group exist in all previous groups higher in the chain. Fig.1.9 shows examples of lower sub-groups based upon an octahedron, and Fig.1.10 reveals how the chains are constructed. Note that some sub-groups occur in different versions in different chains. For example depending upon which operators of O_h survive, there are three different C_{2v} groups:

$C_{2v}(S)$ $S = I = [E, C_2^z, \sigma_h(xz), \sigma_h(yz)]$

$S = II = [E, C_2^z, \sigma_d(\bar{x}y), \sigma_d(xy)]$

$S = III = [E, C_2(xy), \sigma_h(xy), \sigma_d(xy)]$

and in a similar fashion, there are two D_{2h} groups:

$D_{2h}(S)$ $S = I = [E, 3C_2, i, 3\sigma_h]$

$S = II = [E, C_2^z, C_2(xy), C_2(\bar{x}y), i, \sigma_h(xy), \sigma_d(xy), \sigma_d(\bar{x}y)]$ (1.38)

These chains have some noteworthy properties:

i) The totally symmetric representation in one group is also the totally symmetric representation in all groups below it in the chain, i.e. in all of its sub-groups.

ii) One or more functions transforming as the totally symmetric representation in a group, will derive from functions in higher groups which are not totally symmetric.

iii) Multi-dimensional representations lose their degeneracy in the lower groups.

The projection of basis functions which transform correctly as bases for the various representations in all the groups in a chain, is called **Subduction.** The basis functions are said to be 'subduced' from the generative group bases.

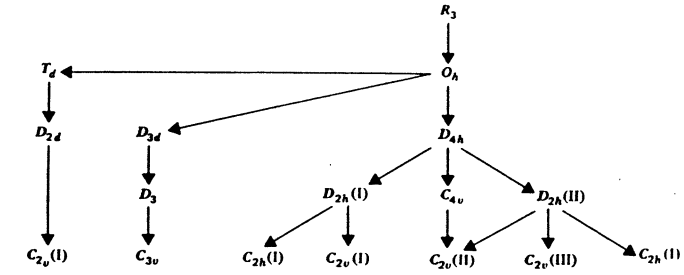

Fig.1.10 Various four-fold, three-fold and two-fold subduction chains (see 1.38).

The basis functions subduced from the generative group may be used to describe both the Hamiltonian and the basis functions of the sub-group. The octahedral basis functions (to l = 4) are shown in Table 1.4 (see footnotes for more details). They are presented in terms of the rationalised spherical harmonics C^l_{ml} which are related to those previously used via:

$$C^l_{ml} = \sqrt{(4\pi/2l + 1)}\, Y_l^{ml}$$

and in terms of the subduction coefficients being the coefficients of the rationalised spherical harmonics for each vector component. Their generation by projection operator techniques from the C^l_{ml} bases was described briefly above and is discussed elsewhere [2,5,15,16] in more detail.

Essentially they can be obtained by using group theoretical projection operators and projecting out the

various representations from the spherical harmonic basis set. Alternatively if, for example, a secular determinant is created by evaluating $\langle 2m|V|2m'\rangle$ where V is the octahedral Hamiltonian, then diagonalisation yields the eigenvectors which will be appropriately adapted bases for $T_{2g} + E_g$. The other functions can be similarly obtained from matrices of L = 3, 4 etc.

Tables 1.5 - 1.7 can be used to determine the parent octahedral state(s) which transform as any desired representation in the lower sub-group of interest. To construct the Hamiltonian add together the octahedral representations which transform as A_1 in the sub-group of interest, using the listing of operator eigenstates shown in Table 1.8 (abstracted from Table 1.4). For an octahedron itself, for d orbitals:

$$H(O_h) = DQ|A_{1g}(i)|^4_{Oh}$$
$$= DQ[\sqrt{(7/12)}C^4_0 + \sqrt{(5/24)}(C^4_4 + C^4_{-4})] \quad (1.39)$$

For an f^n basis extend the Hamiltonian to l = 6 (Appendix VII). The superscript '4' refers to the l value in R_3 from which the representation is subduced.

Referring to (1.37) a tetragonal (D_{4h}) Hamiltonian can be obtained by combining together the representations ($A_{1g} + E_g(\theta)$) which transform as A_{1g} in D_{4h}, viz:

$$H(D_{4h}) = DQ|A_{1g}(i)|^4_{Oh} + DS|E_g(\theta)|^2_{Oh} + DT|E_g(\theta)|^4_{Oh}$$
$$= DQ[\sqrt{(7/12)}C^4_0 + \sqrt{(5/24)}(C^4_4 + C^4_{-4})] + DSC^2_0$$
$$+ DT[\sqrt{(5/12)}C^4_0 - \sqrt{(7/24)}(C^4_4 + C^4_{-4})] \quad (1.40)$$

In this case both l = 2 and l = 4 R_3 sets subduce A_{1g} in D_{4h}. The radial parameters DQ, DS and DT (capital letters) are used to parallel the use of Dq, Ds and Dt in classical crystal field theory.

By comparison of the classical Hamiltonians (1.16, 1.17) with the NSH Hamiltonian (1.40), the following relationships may be deduced:

$$DQ = 6\sqrt{(21)}\ {}^4Dq \text{ (in octahedral molecules)}$$
$$DT = -(7\sqrt{(15)}/2)\ Dt$$
$$DS = -7Ds$$
$$DQ = (6\sqrt{(21)})Dq - (7\sqrt{(21)}/2)Dt \quad \text{(where Dq is actually } Dq^{xy}) \quad (1.41)$$

The classical Hamiltonian scrambles the DQ and DT parameters. Dq is only a measure of the equatorial field, while DQ is a global quantity measuring the totally symmetric contribution to the field, i.e. the average global field strength.

Table 1.4

SUBDUCTION COEFFICIENTS FOR FUNCTIONS SYMMETRY ADAPTED TO IMPORTANT OCTAHEDRAL AXES
SYMMETRY ADAPTED WAVEFUNCTIONS TO L = 4^a

Fourfold Adapted Functions - C_4^z

O_h	D_{4h}	J	M_J	$S(M_J)^2$	$S(M_J+4)^2$	$S(M_J+8)^2$	$(Den)^2$
A_1	A_1	0	0				1
		4	-4	5	14	5	24
A_2	B_1	3	-2	*1	1		2
E	A_1	2	0	1			1
		4	-4	*7	10	*7	24
E	B_1	2	-2	1	1		2
		4	-2	1	1		2
T_1	Ey	1	1	1			1
		3	-3	5	3		8
		4	-3	1	7		8
T_1	A_2	1	0	1			1
		3	0	1			1
		4	-4	1	0	*1	2
T_1	Ex	1	-1	1			1
		3	-1	3	5		8
		4	-1	7	1		8
T_2	Exz	2	1	1			1
		3	-3	*3	5		8
		4	-3	*7	1		8
T_2	B_2	2	-2	*1	1		2
		3	-2	1	1		2
		4	-2	*1	1		2
T_2	Eyz	2	-1	1			1
		3	-1	5	*3		8
		4	-1	1	*7		8

Twofold Adapted Functions - C_2^z

O_h	D_{4h}	J	M_J	$S(M_J)^2$	$S(M_J+2)^2$	$S(M_J+4)^2$	$S(M_J+6)^2$	$S(M_J+8)^2$	$(Den)^2$
A_1	A_1	0	0	1					1
		4	-4	5		14		5	24
A_2	B_1	3	-2	*1		1			2
E	A_1	2	0	1					1
		4	-4	*7		10		*7	24
E	B_1	2	-2	1		1			2
		4	-2	1		1			2

----------cont

Table 1.4 cont----------

Twofold Adapted Functions - C_2^z

O_h	D_{4h}	J	M_J	$S(M_J)^2$	$S(M_J+2)^2$	$S(M_J+4)^2$	$S(M_J+6)^2$	$S(M_J+8)^2$	$(Den)^2$
T_1	Ey	1	-1	1	1				2
		3	-3	5	3	3	5		16
		4	-3	1	*7	7	*1		16
T_1	A_2	1	0	1					1
		3	0	1					1
		4	-4	1				*1	2
T_1	Ex	1	-1	1	*1				2
		3	-3	*5	3	*3	5		16
		4	-3	1	7	7	1		16
T_2	Exz	2	-1	1	1				2
		3	-3	*3	*5	5	3		16
		4	-3	*7	1	1	*7		16
T_2	B_2	2	-2	*1		1			2
		3	-2	1		1			2
		4	-2	*1		1			2
T_2	Eyz	2	-1	1	*1				2
		3	-3	*3	5	5	*3		16
		4	-3	7	1	*1	*7		16

Twofold Adapted Functions - C_2^{xy}

O_h	C_{2v}	J	M_J	$S(M_J)^2$	$S(M_J+2)^2$	$S(M_J+4)^2$	$S(M_J+6)^2$	$S(M_J+8)^2$	$(Den)^2$
A_1	A_1	0	0	1					1
		4	-4	*45	*140	14	*140	*45	384
A_2	B_1	3	-3	*3	5	5	*3		16
E	A_1	2	-2	3	2	3			8
		4	-4	63	*4	250	*4	63	384
E	B_1	2	-1	*1	1				2
		4	-3	*7	*1	1	7		16
T_1	B_1	1	-1	1	1				2
		3	-3	5	3	3	5		16
		4	-3	1	*7	7	*1		16
T_1	A_2	1	0	1					1
		3	-2	*15	2	*15			32
		4	-4	9	7	0	*7	*9	32
T_1	B_2	1	-1	1	*1				2
		3	-3	5	27	*27	*5		64
		4	-3	25	7	7	25		64
T_2	A_2	2	-2	*1		1			2
		3	-2	1	30	1			32
		4	-4	7	*9	0	9	*7	32

cont

Table 1.4 cont----------

Twofold Adapted Functions - C_2^{xy}

O_h	π	J	M_J	$S(M_J)^2$	$S(M_J+2)^2$	$S(M_J+4)^2$	$S(M_J+6)^2$	$S(M_J+8)^2$	$(Den)^2$
T_2	B_2	2	-1	1	1				2
		3	-3	27	*5	5	*27		64
		4	-3	*7	25	25	*7		64
T_2	A_1	2	-2	*1	6	*1			8
		3	-2	1	0	*1			2
		4	-4	*7	4	10	4	*7	32

Threefold Adapted Functions - C_3^{xyz}

O_h	D_{3d}	J	M_J	$S(M_J)^2$	$S(M_J+3)^2$	$S(M_J+6)^2$	$S(M_J+9)^2$	$(Den)^2$
A_1	A_1	0	0	1				1
		4	-3	*10	7	10		27
A_2	A_2	3	-3	2	5	*2		9
E	E	2	-2	*1	2			3
		4	-2	16	4	7		27
E	E	2	-1	2	1			3
		4	-4	*7	4	*16		27
T_1	E	1	1	1				1
		3	-2	*5	1			6
		4	-2	*7	7	4		18
T_1	A_2	1	1	1				1
		3	-3	*5	8	5		18
		4	-3	1	0	1		2
T_1	E	1	-1	1				1
		3	-1	*1	*5			6
		4	-4	*4	7	7		18
T_2	E	2	-2	2	1			3
		3	-2	1	5			6
		4	-2	1	25	*28		54
T_2	A_1	2	0	1				1
		3	-3	1	0	1		2
		4	-3	7	40	*7		54
T_2	E	2	-1	1	*2			3
		3	-1	5	*1			6
		4	-4	28	25	*1		54

The Condon and Shortley phase convention is used. Column 2 in the four and two fold quantized sets shows the representation in the appropriate C_{2v} group at the foot of the chain corresponding to the octahedral representation in column 1. In the three fold quantized chain the representation in D_{3d} symmetry is indicated in column 2. In this case the C_{3v} group at the bottom of the chain still has degeneracy in its E representations subduced by E and components of T_1 and T_2, in O_h. The original publication [29] lists angular

momentum labels in column 2 for all the quantized sets.

Column 3 carries the value of J (or L) and defines whether the representation is subduced by L(J) = 0,(S), 1(P), 2(D), 3(F), 4(G). Higher values are reported in Ref.[29]. Thus if a T_2 representation from an F term is required look for a line initiated with L = 3. The various M_J (M_L) values are obtained by taking the entry in column 4, and modifying it according to the non-zero values in the columns which follow.

Squares of the subduction coefficients are expressed as fractions, the square of the numerator being in the body of the table, and the square of the denominator in the extreme right-hand column. Numbers carrying an asterisk are negative. For example the first five entries in this table expand, using $|L,M_L\rangle$ to:

A_1 A_1 $|00\rangle$ (an S term component)
A_1 A_1 $\sqrt{(14/24)}|40\rangle + \sqrt{(5/24)}[|44\rangle + |4\text{-}4\rangle]$ (G term)
A_2 B_1 $(1/\sqrt{2})[|32\rangle - |3\text{-}2\rangle]$ (an F term component)
E A_1 $|20\rangle$
A_1 $\sqrt{(10/24)}|40\rangle - \sqrt{(7/24)}[|44\rangle + |4\text{-}4\rangle]$

The four-fold quantized functions for L = 1 and L = 3, commonly used in coordination chemistry, are shown in Table 3.5. Note that they are suitable for the four-fold groups. For the two fold groups, as explained in the text, the additional second rank operators mix functions such as $|11\rangle$ and $|1\text{-}1\rangle$ which are unmixed under a four fold axis. Use the apppropriate functions in the C_2^z or C_2^{xy} section of the table.

In $C_{2v}(I)$ complexes, there are three octahedral representations, $A_{1g} + E_g(\theta) + E_g(\varepsilon)$ in O_h which subduce A_1. The complete $C_{2v}(I)$ Hamiltonian is then given by:

$$H(C_{2v}(I)) = DQ|A_{1g}(i)|^4_{Oh} + DS|E_g(\theta)|^2_{Oh} + DT|E_g(\theta)|^4_{Oh} + DU|E_g(\varepsilon)|^2_{Oh} + DV|E_g(\varepsilon)|^4_{Oh}$$

$$= DQ[\sqrt{(7/12)}C^4_0 + \sqrt{(5/24)}(C^4_4 + C^4_{-4})] + DSC^2_0 + DT[\sqrt{(5/12)}C^4_0 - \sqrt{(7/24)}(C^4_4 + C^4_{-4})] + DU[\sqrt{2}(C^2_2 + C^2_{-2})] + DV[\sqrt{2}(C^4_2 + C^4_{-4})] \quad (1.42)$$

Additional labels are added to the octahedral representations to clearly identify which component is involved. Two types of additional labels may be used. A trivial label such as xy or z (or θ etc) which distinguishes the components on the basis of their symmetry, and a label such as 0, 1-, 1+, 2- which is a measure of the surviving angular momentum in the representation and is important where selection rules are

Table 1.5

CORRELATION TABLE FOR THE GERADE IRREDUCIBLE REPRESENTATIONS OF O_h RELEVANT TO CHAINS TERMINATING IN $C_{2v}(I)$ OR $C_{2h}(I)$

O_h	D_{4h}	C_{4h} or	$D_{2d}(I)$ or	$D_{2h}(I)$	$C_{2v}(I)$	or $C_{2h}(I)$
A_{1g}	A_{1g}	A_1	A_1	A_g	A_1	A_g
$E_g(\theta)$	A_{1g}	A_1	A_1	A_g	A_1	A_g
$E_g(\varepsilon)$	B_{1g}	B_1	B_2	A_g	A_1	A_g
A_{2g}	B_{1g}	B_1	B_2	A_g	A_1	A_g
$T_{2g}(xy)$	B_{2g}	B_2	B_1	B_{1g}	A_2	A_g
$T_{1g}(z)$	A_{2g}	A_2	A_2	B_{1g}	A_2	A_g
$T_{2g}(xz)$	$E_g(xz)$	E(xz)	E(xz)	B_{2g}	B_1	B_g
$T_{1g}(y)$	$E_g(y)$	E(y)	E(y)	B_{2g}	B_1	B_g
$T_{2g}(yz)$	$E_g(yz)$	E(yz)	E(yz)	B_{3g}	B_2	B_g
$T_{1g}(x)$	$E_g(x)$	E(x)	E(x)	B_{3g}	B_2	B_g

$D_{2d}(I)$ I = $[E,2S_4,C_2(z),2C'_2,2\sigma_h]$ $D_{2h}(I)$ I = $[E,3C_2,i,3\sigma_h]$
$C_{2v}(I)$ I = $[E,C_2(z),\sigma_h(xz),\sigma_h(yz)]$ $C_{2h}(I)$ I = $[E,C_2(z),i,\sigma_h]$
Note that the transformation properties of a given function, in say the C_{2v} group, will not necessarily be the same as in the right-hand column of a standard character table for C_{2v}. This arises because the operators which survive in a given C_{2v} group may not be the same as those in a standard Character Table.

Table 1.6

CORRELATION TABLE FOR THE GERADE IRREDUCIBLE REPRESENTATIONS OF O_h RELEVANT TO CHAINS TERMINATING IN $C_{2v}(II)$ OR $C_{2v}(III)$

O_h	D_{4h}	C_{4v} or	$D_{2h}(II)$	$C_{2v}(II)$	$C_{2v}(III)$
A_{1g}	A_{1g}	A_1	A_g	A_1	A_1
$E_g(\theta)$	A_{1g}	A_1	A_g	A_1	A_1
$T_{2g}(xy)$	B_{2g}	B_2	A_g	A_1	A_1
A_{2g}	B_{1g}	B_1	B_{1g}	A_2	B_1
$E_g(\varepsilon)$	B_{1g}	B_1	B_{1g}	A_2	B_1
$T_{1g}(z)$	A_{2g}	A_2	B_{1g}	A_2	B_1
$T_{1g}(\tau_1)$	$E_g(\tau_1)$	$E(\tau_1)$	B_{2g}	B_1	B_2
$T_{2g}(\tau_2)$	$E_g(\tau_2)$	$E(\tau_2)$	B_{2g}	B_1	B_2
$T_{1g}(\tau_1')$	$E_g(\tau_1')$	$E(\tau_1')$	B_{3g}	B_2	A_2
$T_{2g}(\tau_2')$	$E_g(\tau_2')$	$E(\tau_2')$	B_{3g}	B_2	A_2

$D_{2h}(II)$ II = $[E,C_2(z),C_2(xy),C_2(\bar{x}y),i,\sigma_h(xy),\sigma_d(xy),\sigma_d(\bar{x}y)]$
$C_{2v}(II)$ II = $[E,C_2(z),\sigma_d(xy),\sigma_d(\bar{x}y)]$
$C_{2v}(III)$ III = $[E,C_2(xy),\sigma_h(xy),\sigma_d(xy)]$. Note that $C_{2v}(III)$ is not subduced by C_{4v}.

Table 1.7

CORRELATION TABLE FOR THE GERADE IRREDUCIBLE REPRESENTATIONS OF T_d AND ITS SUBGROUPS

O_h	T_d	D_{2d}(II)	C_{2v}(II)
$A_{1g}0$	A_1	A_1	A_1
$E_g(\theta)$	E	A_1	A_1
$T_{2g}(xy)$	$T_2(xy)$	B_2	A_1
A_{2g}	A_2	B_1	A_2
$E_g(\varepsilon)$	E	B_1	A_2
$T_{1g}(z)$	$T_1(z)$	A_2	A_2
$T_{1g}(\tau_1)$	$T_1(\tau_1)$	$E(\tau_1)$	B_1
$T_{2g}(\tau_2)$	$T_2(\tau_2)$	$E(\tau_2)$	B_1
$T_{1g}(\tau_1')$	$T_1(\tau_1)$	$E(\tau_1)$	B_2
$T_{2g}(\tau_2')$	$T_2(\tau_2)$	$E(\tau_2)$	B_2

D_{2d}(II) II = $[E,2S_4,3C_2,2\sigma_d]$; C_{2v}(II) II = $[E,C_2(z),\sigma_d(xy),\sigma_d(xy)]$
Note: Three fold chains of O_h and T_d. The A_{1g}, A_{2g} and E_g representations of O_h correlate simply with A_1, A_2 and E down the trigonal chain. Similarly T_{1g} spans A_2 + E, and T_{2g} with A_1 + E.

involved. These last labels are not covered in this volume but their significance is discussed elsewhere [29]. Note that the A_1 representation subduced by A_{2g} in O_h and by F in R_3 cannot be used to form a Hamiltonian since the spherical harmonics in F are odd (with respect to inversion) and cannot couple the even d functions with each other. If d-p configurational interaction was to be studied, then this contribution from F in R_3 would be required.

Descent down the C_3(xyz) axis, yields a chain:

$$O_h \quad D_{3d} \quad D_3 \quad C_{3v}$$

The A_1 representation in these last three groups is subduced by A_{1g} + T_{2g}; thus the same Hamiltonian suffices for all three groups. This case has been especially studied by Hempel [36,37] who coined new radial parameters defined as follows:

$$H = DSIG|T_{2g}0|^2_{Oh} + DQ|A_{1g}0|^4_{Oh} + DTAU|T_{2g}0|^4_{Oh}$$

where the expansion of each component can be obtained from Table 1.4, using the three-fold directed functions. These parameters parallel Dσ, Dq and Dτ used classically (see section 1.8).

Hempel and Miller [38] consider trigonal groups which are not subgroups of an octahedron. The most important are the three cooordinate trigonal planar, the five coordinate trigonal bi-pyramidal and the six coordinate trigonal prismatic complexes and their analogs. The logical parent group is D_{6h} and there are

several possible chains:

$$\begin{array}{llll} D_{6h} & D_{3h} & C_{3v}(I) & C_3 \\ D_{6h} & D_{3h} & D_3 & C_3 \\ D_{6h} & D_{3d} & C_{3v}(II) & C_3 \\ D_{6h} & D_{3d} & D_3 & C_3 \end{array} \qquad (1.43)$$

where the surviving operators in the two C_{3v} groups are I = $[E,2C_3,3\sigma_v]$ and II = $[E,2C_3,3\sigma_d]$. For example C_{3v}(II) comprises an appropriate radial distortion from a D_{3d} trigonal anti-prismatic complex in which the pair of bonding triangles are staggered. C_{3v}(I) arises from a similar distortion of a D_{3h} trigonal prismatic complex, with the bonding triangles eclipsed.

The correlation for D_{6h} to its lower groups, paralleling Eqn.(1.37) is:

D_{6h}	D_{3d}	C_{3v}(II) or D_3	C_3	and	D_{6h}	D_{3h}	D_3 or C_{3v}(I)	C_3
A_{1g}	A_{1g}	**A_1**	**A_1**		**A_{1g}**	**A_1'**	**A_1**	**A_1**
A_{2g}	A_{2g}	A_2	**A_1**		A_{2g}	A_2'	A_2	**A_1**
B_{1g}	**A_{1g}**	**A_1**	**A_1**		B_{1g}	A_1''	**A_1**	A_2	**A_1**
B_{2g}	A_{2g}	A_2	**A_1**		B_{2g}	A_2''	A_2	**A_1**	**A_1**

(1.44)

where the operator eigenstates are NOT those of the R_3 set shown above, but rather are appropriate for D_{6h} [38] namely:

$$\begin{aligned} |A_{1g}|^2_{D6h} &= C^2{}_0; \quad |A_{1g}|^4_{D6h} = C^4{}_0 \\ |B_{1g}|^4_{D6h} &= (1/\sqrt{2})[C^4{}_{-3} - C^4{}_3] \\ |B_{2g}|^4_{D6h} &= i(1/\sqrt{2})[C^4{}_{-3} + C^4{}_3] \end{aligned} \qquad (1.45)$$

where Hempel assumes the use of the radial parameters DS, DT, DA and DB in the sequence cited in (1.45). Note that the Hamiltonian for a D_{6h} complex is equivalent to the infinite axial potential:

$$H_{D6h} = DSC^2{}_0 + DTC^4{}_0 = V = H_{D3h}$$

and hence:

$$\begin{aligned} H &= V + DB|B_{2g}|^4_{D6h} \quad \text{for } G = C_{3v}(I) \\ H_G &= V + DA|B_{1g}|^4_{D6h} \quad \text{for } G = D_{3d}, D_3, C_{3v}(II) \\ H_G &= V + DA|B_{1g}|^4_{D6h} + DB|B_{2g}|^4_{D6h} \quad \text{for } G = C_3 \end{aligned} \qquad (1.46)$$

The Hamiltonians for the various cubic sub-groups may be summarised as follows:

$H_G = V$ for $V = O_h$ and T_d

$H_G = V + V'$ for $G = D_{4h}, C_{4v}, D_{2d}(I), D_{2d}(II), \ldots$

$H_G = V + V' + V''$ for $G = D_{2h}(I), C_{2v}(I), \ldots$

$H_G = V + V' + V'''$ for $G = D_{2h}(II), C_{2v}(II), C_{2v}(III) \ldots$.

$H_G = V^3$ for $G = D_{3d}, D_3 \ldots$

where

$$V = DQ|A_{1g}|^4_{Oh}$$

$$V' = DS|E_g(\theta)|^2_{Oh} + DT|E_g(\theta)|^4_{Oh}$$

$$V'' = DU|E_g(\varepsilon)|^2_{Oh} + DV|E_g(\varepsilon)|^4_{Oh}$$

$$V''' = DM|T_{2g}(xy)|^2_{Oh} + DN|T_{2g}(xy)|^4_{Oh}$$

$$V^3 = DQ|A_{1g}|^4_{Oh} + DSIG|T_{2g}(0)|^2_{Oh} + DTAU|T_{2g}(0)|^4_{Oh} \qquad (1.47)$$

and may be assembled using Table 1.8.

The matrix elements of these NSH operators can be obtained by expanding the Schrodinger triple integral, following the previous crystal field integral discussion. However their high symmetry lends itself to much more facile evaluation using tensor methods to be described in Chapter 3. The results of such evaluation for d^1 matrix elements are shown in Tables 1.9 and 1.10.

They are presented in terms of the matrix elements for the various C_{2v} groups at the foot of the chains. By inserting zero values for some components, the matrix elements for sub-groups to be found higher in the chain are readily obtained. Note that unlike the classical crystal field approach, the splitting of the octahedral states in the lower symmetries is a symmetric function of the NSH radial integrals.

With DS,DT,DU and DV (or DM and DN) all zero, the octahedral solution is obtained. With non-zero DQ,DS and DT, and zero DU,DV (or DM,DN), the fourfold D_{4h} or C_{4v} solution is obtained.

Table 1.9 and subsequent similar Tables contain information which may be used in the following fashion. Where an orbital has a representation which only occurs once in the basis set involved, its energy may be directly read accross a row of the Table. For example the $|x^2-y^2>$ function transforms uniquely as b_{1g} in D_{4h}, and corresponds to the $e_g(\varepsilon)$ component in O_h. Since this representation only occurs once within D_{4h}, its energy is given directly by the matrix element $<E_g(\varepsilon)|V_{D4h}|E_g(\varepsilon)>$ which occurs on the fourth row of Table 1.9. Similarly the $|z^2>$ orbital transforms as a_{1g} ($e_g(\theta)$ in O_h)) and is unique. Its energy is given directly by the magnitude of $<E_g(\theta)|V_{D4h}|E_g(\theta)>$ given by the top row of Table 1.9.

TABLE 1.8 [29]

OPERATOR EIGENSTATES QUANTIZED IN THE POINT GROUP O_h [a]

$$|A_{1g}0(i)|^4_{O_h} = C^0_0$$

$$|E_g0(\theta)|^2_{O_h} = C^2_0$$

$$|E_g2+(\varepsilon)|^2_{O_h} = (\tfrac{1}{2})^{\frac{1}{2}}(C^2_2 + C^2_{-2})$$

$$|T_2 2-(xy)|^2_{O_h} = -i(\tfrac{1}{2})^{\frac{1}{2}}(C^2_2 - C^2_{-2})$$

$$|A_{1g}0(i)|^4_{O_h} = (\tfrac{7}{12})^{\frac{1}{2}} C^4_0 + (\tfrac{5}{24})(C^4_4 + C^4_{-4})$$

$$|E_g0(\theta)|^4_{O_h} = (\tfrac{5}{12})^{\frac{1}{2}} C^4_0 - (\tfrac{7}{24})^{\frac{1}{2}}(C^4_4 + C^4_{-4})$$

$$|E_g2+(\varepsilon)|^4_{O_h} = (\tfrac{1}{2})^{\frac{1}{2}}(C^4_2 + C^4_{-2})$$

$$|T_{2g}2-(xy)^4_{O_h} = -i(\tfrac{1}{2})^{\frac{1}{2}}(C^4_2 - C^4_{-2})$$

$$|T_{1g}0(z)|^4_{O_h} = -i(\tfrac{1}{2})^{\frac{1}{2}}(C^4_{-4} - C^4_4)$$

Quantization along $C_2(xy)$

$$|A_{1g}0(i)|^{L=0}_{O_h} = C^0_0$$

$$|E_g0(\theta)|^2_{O_h} = \tfrac{1}{2} C^2_0 + (\tfrac{3}{8})^{\frac{1}{2}}(C^2_2 + C^2_{-2})$$

$$|E_g0(\theta)|^4_{O_h} = (\tfrac{125}{192})^{\frac{1}{2}} C^4_0 - (\tfrac{1}{96})^{\frac{1}{2}}(C^4_2 + C^4_{-2}) + (\tfrac{63}{384})^{\frac{1}{2}}(C^4_4 + C^4_{-4})$$

$$|T_{2g}0(xy)|^2_{O_h} = (\tfrac{3}{4})^{\frac{1}{2}} C^2_0 - (\tfrac{1}{8})^{\frac{1}{2}}(C^2_2 + C^2_{-2})$$

$$|T_{2g}0(xy)|^4_{O_h} = (\tfrac{5}{16})^{\frac{1}{2}} C^4_0 + (\tfrac{1}{8})^{\frac{1}{2}}(C^4_2 + C^4_{-2}) - (\tfrac{7}{32})^{\frac{1}{2}}(C^4_4 + C^4_{-4})$$

$$|A_{1g}0(i)|^4_{O_h} = (\tfrac{7}{192})^{\frac{1}{2}} C^4_0 - (\tfrac{35}{96})^{\frac{1}{2}}(C^4_2 + C^4_{-2}) - (\tfrac{45}{384})^{\frac{1}{2}}(C^4_4 + C^4_{-4})$$

Table 1.8 [29] cont

OPERATOR EIGENSTATES QUANTIZED IN THE POINT GROUP O_h

Quantization along $C_3(xyz)$ [c]

$$^3|A_{1g}0(i)|^{L=0}_{0_h} = C^0_0$$

$$^3|A_{1g}0(i)|^{4}_{0_h} = (\tfrac{7}{27})^{\frac{1}{2}}\, C^4_0 + (\tfrac{10}{27})^{\frac{1}{2}}(C^4_3 - C^4_{-3})$$

$$^3|T_{2g}0|^{2}_{0_h} = C^2_0$$

$$^3|T_{2g}0|^{4}_{0_h} = (\tfrac{20}{27})^{\frac{1}{2}}\, C^4_0 - (\tfrac{7}{54})^{\frac{1}{2}}(C^4_3 - C^4_{-3})$$

a) The coefficients in this table are subduction coefficients extracted from Table 1.4. The resulting radial parameters have magnitudes independent of the quantization axis provided that both the operator and basis functions are quantized along the same axis. It is therefore convenient, for example, to use functions quantized along the C_3^{xyz} axis for the three-fold groups, and those down C_2^{xy} for groups in chains terminating in C_{2v}(III). The operators quantized along C_2^z can be used for groups which contain a C_4^z axis either alone or collinear with a C_2^z axis. Eqn.1.47 defines the appropriate combinations of operator eigenstates to 'string together' to create Hamiltonians for a wide range of complexes. Not all those listed in this Table are mentioned. However they may arise in other less common situations, e.g. the use of $|T_{1g}z|$ in the group C_4. b) In some earlier publications the phases followed Griffith rather than Condon and Shortley as these do. The principal change is in the sign of the $|E_g(\theta)|$ component which causes a concomitant change in the sign of DT relative to these early papers. c) The functions along C_3^{xyz} are associated with a negative value of DQ.

The review [29] from which this table originated labelled the the functions in Table 1.4 with symbols which reflected the angular momentum residing in each function. Although such symbols are not discussed in depth here, it is pertinent to note that any eigenstate which may be used to form part of a (necessarily) totally symmetric Hamiltonian has zero angular momentum. Its label will usually be (0) though values of (2+) and (2-) were defined for some functions (also having no net angular momentum) in order to create an 'algebra' which could be used to determine which matrix elements would vanish. The reader is referred to [29] for further detail.

TABLE 1.9 [29]
a) NONZERO MATRIX ELEMENTS FOR NORMALIZED SPHERICAL HARMONIC OPERATORS ON THE d^1 BASIS SYMMETRY ADAPTED TO C_{2v}(I), C_{2v}(II), AND C_{2v}(III)

$\langle {}^2a(O_h); {}^2a(C_{2v}(S))|v|{}^2a'(O_h); {}^2a(C_{2v}(S))\rangle$

$a(C_{2v}(S))$											
I	II	III	$a(O_h)$	Operator[a]	$a'(O_h)$	DQ	DS	DT	DU	DV	DM	DN
A_1	A_1	A_1	$E_g(\theta)$	V+V'	$E_g(\theta)$	$(\frac{1}{21})^{\frac{1}{2}}$	$\frac{2}{7}$	$\frac{5}{7}(\frac{1}{15})^{\frac{1}{2}}$				
			$E_g(\theta)$	V''	$E_g(\varepsilon)$				$-\frac{2}{7}$	$\frac{1}{7}(\frac{5}{3})^{\frac{1}{2}}$		
	A_1	A_1	$E_g(\theta)$	V'''	$T_{2g}(xy)$						$-\frac{2}{7}$	$\frac{1}{7}(\frac{5}{3})^{\frac{1}{2}}$
A_1	A_2	B_1	$E_g(\varepsilon)$	V+V'	$E_g(\varepsilon)$	$(\frac{1}{21})^{\frac{1}{2}}$	$-\frac{2}{7}$	$-\frac{5}{7}(\frac{1}{15})^{\frac{1}{2}}$				
A_2	A_1	A_1	$T_{2g}(xy)$	V+V'	$T_{2g}(xy)$	$-\frac{2}{3}(\frac{1}{21})^{\frac{1}{2}}$	$-\frac{2}{7}$	$\frac{20}{21}(\frac{1}{15})^{\frac{1}{2}}$				
B_1			$T_{2g}(xz)$	V+V'+V''	$T_{2g}(xz)$	$-\frac{2}{3}(\frac{1}{21})^{\frac{1}{2}}$	$\frac{1}{7}$	$-\frac{10}{21}(\frac{1}{15})^{\frac{1}{2}}$	$\frac{1}{7}(3)^{\frac{1}{2}}$	$\frac{10}{21}(\frac{1}{5})^{\frac{1}{2}}$		
B_2			$T_{2g}(yz)$	V+V' V''	$T_{2g}(yz)$	$-\frac{2}{3}(\frac{1}{21})^{\frac{1}{2}}$	$\frac{1}{7}$	$-\frac{10}{21}(\frac{1}{15})^{\frac{1}{2}}$	$-\frac{1}{7}(3)^{\frac{1}{2}}$	$-\frac{10}{21}(\frac{1}{5})^{\frac{1}{2}}$		
	B_1	B_2	$T_{2g}\tau_2$	V+V'+V'''	$T_{2g}\tau_2$	$-\frac{2}{3}(\frac{1}{21})^{\frac{1}{2}}$	$\frac{1}{7}$	$-\frac{10}{21}(\frac{1}{15})^{\frac{1}{2}}$			$\frac{1}{7}(3)^{\frac{1}{2}}$	$\frac{10}{21}(\frac{1}{5})^{\frac{1}{2}}$
	B_2	A_2	$T_{2g}\tau_2'$	V+V'+V'''	$T_{2g}\tau_2$	$-\frac{2}{3}(\frac{1}{21})^{\frac{1}{2}}$	$\frac{1}{7}$	$-\frac{10}{21}(\frac{1}{15})^{\frac{1}{2}}$			$-\frac{1}{7}(3)^{\frac{1}{2}}$	$-\frac{10}{21}(\frac{1}{5})^{\frac{1}{2}}$

Table 1.10

NON-ZERO MATRIX ELEMENTS FOR THE NORMALIZED SPHERICAL HARMONIC OPERATOR ON THE d^1 BASIS SYMMETRY ADAPTED TO THE THREE-FOLD AXIS C_3^{xyz}

$\langle\psi_1|H(D_3)|\psi_2\rangle$

C_{3v}	ψ_1	ψ_2	DQ	DM	DN
A_1	T_2	T_2	$(2/3\sqrt{21})$	2/7	$(20/21\sqrt{15})$
E	T_2	T_2	$(2/3\sqrt{21})$	-1/7	$-(2\sqrt{5}/21\sqrt{3})$
	T_2	E		$\sqrt{(2/7)}$	$-(5\sqrt{2}/14\sqrt{15})$
	E	E	$-(1/\sqrt{21})$		

However in the group C_{2v}(I) (but not C_{2v}(II) or C_{2v}(III) - see right-hand three columns of Table 1.9, and Table 1.5), both these orbitals transform as a_1. They will therefore be scrambled by the off-diagonal element $\langle E_g(\varepsilon)|V_{D4h}|E_g(\theta)\rangle$ whose magnitude is given on row 2 of Table 1.9 in terms of DU and DV. It is necessary to construct a 2X2 secular determinant with the previously cited values of the energies of these two functions on the diagonals, and this latter element off-diagonal.

In the groups C_{2v}(II) and C_{2v}(III), the functions $|z^2\rangle$ and $|xy\rangle$ share the same representation (a_1, see right-hand three columns of Table 1.9, and Table 1.6). Their energies must then be obtained from a 2x2 secular determinant set up with the off-diagonal element $\langle E_g(\theta)|V_{C2v(II)}|T_{2g}(xy)\rangle$ in terms of DM and DN.

The d^1 matrix elements described in this chapter, may be utilised for high spin d^4, d^6 and d^9 systems. In the case of d^4 and d^9, the radial parameter values must be multiplied by -1. Other d^n configurations are treated in Chapter 3.

1.6 Some Applications of Molecular Orbital Theory

1.6.i) Introduction

There are many approaches to understanding metal ligand bonds which incorporate some metal-ligand overlap. No attempt is made here to survey these in any detail. In this chapter, dealing with one electron systems, several semi-empirical approaches which have proved of value in the analysis of electronic spectra are considered.

Using standard group theoretical techniques, the symmetries spanned by the various sigma and pi group molecular orbitals on the ligands can be deduced. For example the six sigma lone pairs, Lσ in an octahedral ML_6 species, span $e_g + a_{1g} + t_{1u}$. The orbitals of corresponding symmetry on the metal, namely (x^2-y^2,z^2), s, and (p_x,p_y,p_z), respectively, can, from the viewpoint of molecular symmetry have a non-zero overlap with these ligand orbitals and hence may have sigma character. Since the ligand orbitals lie lower in energy than the metal orbitals, these metal orbitals will be sigma anti-bonding. Similarly a ligand such as chloride has 2 pairs of p orbitals which have pi symmetry with respect to the metal in an M-L bond. In an MCl_6 species, these 12 pairs of pi orbitals span $t_{1u} + t_{2u} + t_{1g} + t_{2g}$. Hence the metal p orbitals (t_{1u}) and metal (xz,yz,xy)(t_{2g}) orbitals will have pi character. In general when the ligand pi oritals are filled they lie below the metal (xz,yz,xy) d orbitals which then become pi anti-bonding. However ligands such as CN^- possess empty pi orbitals which also span t_{2g} (in an octahedral $[M(CN)_6]^{n-}$ molecule) leading to pi bonding character for the metal t_{2g} d orbitals. In an octahedral ML_6 species, the ligand pi t_{1g} and t_{2u} orbitals will remain non-bonding since there are no equivalent orbitals on the metal.

Table 1.11 contains ligand group molecular orbitals for octahedral and tetrahedral molecules with their representations (based upon the framework shown in Fig.1.11).

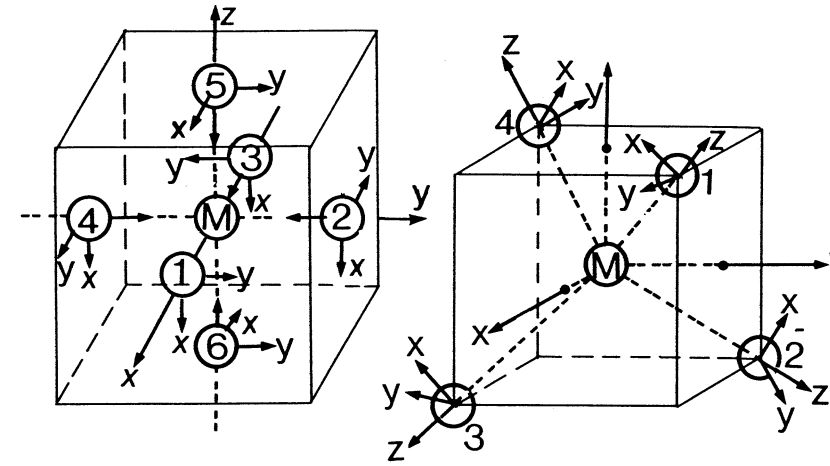

Fig.1.11 Ligand numbering and cartesian coordinate scheme

A generalised molecular orbital diagram for an ML_6 molecule is shown in Fig.1.12 while others appear in Figs.5.12, 5.15 and 5.16. The ML_6 sigma bond framework utilises the z^2, x^2-y^2, s, p_x, p_y and p_z orbitals. Overlap of the z^2 and x^2-y^2 orbitals with ligand sigma orbitals results in a pair ($1e_g$) of bonding orbitals primarily ligand in character, and a pair of anti-bonding orbitals ($2e_g$)

primarily metal in character (Fig.1.12). The metal (xz,yz,xy) orbitals, labelled $2t_{2g}$ (in Fig.1.12) interact with both the ligand pi bonding orbitals ($1t_{2g}$) and pi anti-bonding orbitals ($3t_{2g}$). If the ligands do not possess pi orbitals, the t_{2g} set would remain non-bonding. This would be the case, for example, with ammonia and aliphatic amine ligands.

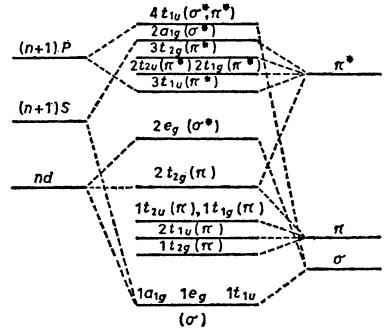

Fig.1.12 A general molecular orbital diagram for an octahedral ML_6 complex with both π and π* orbitals.

In trigonal bipyramidal complexes ML_3Z_2, one can readily visualize that the p_z orbitals on the equatorial ligands would interact in a pi sense with the e" orbitals on the metal. Indeed these three orbitals span a_2" + e". In a similar fashion, any in-plane pi orbitals on the equatorial ligands (spanning a^{2}' + e') have the correct symmetry to interact with the e' metal orbitals.

As a final example, consider a trans complex ML_4Z_2 of D_{4h} symmetry. Assuming the z axis is the principal axis, containing the Z ligands, a pair of pi orbitals on ligands Z will perturb xz,yz which then become pi anti-bonding (or pi bonding if the pi orbitals of Z are empty). These same xz,yz pair will pi interact with the z directed pi orbitals on ligands L. The xy orbital only interacts, in a pi sense, with pi orbitals of L lying in the xy plane. The z^2 orbital is sigma anti-bonding with respect to both the Z ligands and the L ligands (the latter through the 'doughnut' in the xy plane), while the x^2-y^2 orbital is sigma anti-bonding to the L ligands only.

Clearly arguments of this kind can be used qualitatively, as will be shown in a rationalisation of the Spectrochemical Series (Chapter 9), or can be used quantitatively.

Complete ab-initio molecular orbital treatments may, in principle, predict electronic transition energies. However such calculations are expensive and out of the reach of most experimental chemists. Moreover most do not, currently, fare too well in predicting energies of virtual orbitals (i.e. orbitals which are empty until the electronic transition takes place), though the Xα method seems better than most in this respect.

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Table 1.11

GROUP MOLECULAR ORBITALS FOR OCTAHEDRAL AND TETRAHEDRAL MOLECULES

	Metal	σ-orbitals	π-orbitals
Octahedral			
a_{1g}	s	$(1/\sqrt{6})[\sigma_1+\sigma_2+\sigma_3+\sigma_4+\sigma_5+\sigma_6]$	
e_g	x^2-y^2	$(1/2)[\sigma_1-\sigma_2+\sigma_3-\sigma_4]$	
	z^2	$(1/2\sqrt{3})[2\sigma_5+2\sigma_6-\sigma_1-\sigma_2-\sigma_3-\sigma_4]$	
t_{1u}	p_x	$(1/\sqrt{2})[\sigma_1-\sigma_3]$	$(1/2)[-py_2+px_5+py_4-px_6]$
	p_y	$(1/\sqrt{2})[\sigma_2-\sigma_4]$	$(1/2)[py_1+py_5-py_3+py_6]$
	p_z	$(1/\sqrt{2})[\sigma_5-\sigma_6]$	$(1/2)[-px_1-px_2-px_3-px_4]$
t_{2g}	xz		$(1/2)[-px_1+px_5+px_3+px_6]$
	yz		$(1/2)[-px_2+py_5+px_4-py_6]$
	xy		$(1/2)[py_1-py_2+py_3-py_4]$
t_{1g}			$(1/2)[-px_1-px_5+px_3-px_6]$
			$(1/2)[-px_2-py_5+px_4+py_6]$
			$(1/2)[py_1+py_2+py_3+py_4]$
t_{2u}			$(1/2)[-py_2-px_5+py_4+px_6]$
			$(1/2)[py_1-py_5-py_3-py_6]$
			$(1/2)[-px_1+px_2-px_3+px_4]$
	Metal	**Ligand Group Orbitals**	
Tetrahedral			
a_1	s	$(1/2)[s_1+s_2+s_3+s_4]$, $(1/2)[pz_1+pz_2+pz_3+pz_4]$	
e	z^2	$(1/2)[px_1-px_2-px_3+px_4]$	
	x^2-y^2	$(1/2)[py_1-py_2-py_3+py_4]$	
t_2	p_x, yz	$(1/2)[pz_1-pz_2-pz_3-pz_4]$, $(1/2)[s_1-s_2+s_3-s_4]$	
		$(1/4)[px_1+px_2-px_3-px_4 + \sqrt{3}(-py_1-py_2+py_3+py_4)]$	
	p_y, xz	$(1/2)[pz_1+pz_2-pz_3-pz_4]$, $(1/2)[s_1+s_2-s_3-s_4]$	
		$(1/4)[px_1-px_2+px_3-px_4 + \sqrt{3}(py_1-py_2+py_3-py_4)]$	
	p_z, xy	$(1/2)[pz_1-pz_2-pz_3+pz_4]$, $(1/2)[s_1-s_2-s_3+s_4]$, $-(1/2)[px_1+px_2+px_3+px_4]$	
t_1		$(1/4)[\sqrt{3}(px_1+px_2-px_3-px_4) + py_1+py_2-py_3-py_4]$	
		$-(1/4)[\sqrt{3}(px_1-px_2+px_3-px_4) - py_1+py_2-py_3+py_4]$	
		$-(1/2)[py_1+py_2+py_3+py_4]$	

See Fig.1.10 for coordinate frames. Adapted from ref.[44]. The relative positions of the x and y axes on each ligand in the octahedral molecule do not coincide with those in [44], but have been changed to concur with the AOM matrices in Table 1.12.

1.6 ii) A Semi-Empirical Molecular Orbital Approach

McClure [18] and Yamatera [19] were the first to use semi-empirical molecular orbital methods to derive chemical information from electronic spectroscopy.

If lower symmetry complexes retaining a four-fold axis of rotation are considered then the distinction that the z^2 and x^2-y^2 orbitals have sigma symmetry while the xz,yz and xy orbitals have pi symmetry is retained.

Consider a complex ML_5Z with Z on the positive z axis, then following McClure [18] the energy of x^2-y^2 may be written as 4σ because there are contributions from four equivalent sigma bonds in the xy plane. The energy of the z^2 orbital may be obtained, relative to that of the x^2-y^2 by comparing the charge density of the z^2 orbital relative to that of x^2-y^2 along the three cartesian axes. The relative charge densities in each direction may be obtained by squaring the coefficients for each of the ligand orbitals and dividing by the sum of the squares of these coefficients. For the x^2-y^2 orbital (Table 1.11) the contribution from each of the four sigma ligand orbitals is $(1/4)/1 = 1/4$, while the charge density along the z axis is zero. For the z^2 orbital the contribution from each ligand orbital along the z axis is $(1/3)/1 = 1/3$, and from each ligand in the xy plane, $(1/12)/1 = 1/12$. Therefore in the xy plane the z^2 orbital has $(1/12)/(1/4) = 1/3$ times the charge density of the x^2-y^2 orbital. Similarly the charge density of the z^2 orbital along the z axis is $(1/3)/(1/4) = 4/3$ times that of the x^2-y^2 orbital in the xy plane, per bond.

Using these arguments, and defining the bonding of the x^2-y^2 orbital in the xy plane as 1σ, the energy of the z^2 orbital in the ML_5Z complex may be written:

$$(4/3)\sigma(L) + (4/3)\sigma(Z) + 4(1/3)\sigma(L) = (8/3)\sigma(L) + (4/3)\sigma(Z) \qquad (1.48)$$

being the contribution from five M-L bonds and from one M-Z bond.

Similarly the energy of the xy orbital is $4\pi(L)$ there being four equivalent π-bonding contributions from the four in-plane M-L bonds, and the energy of the xz,yz pair is $3\pi(L) + \pi(Z)$.

If the differences in sigma and pi capability of the L and Z ligand are called $d\sigma$ and $d\pi$ respectively, then:

$$d\sigma = \sigma(Z) - \sigma(L) \quad \text{and} \quad d\pi = \pi(Z) - \pi(L) \qquad (1.49)$$

and the energies of the d orbitals relative to the cubic ML_6 molecule are [14]:

	z^2	x^2-y^2	xy	xz	yz
ML_5Z	0	$(4/3)d\sigma$	0	$d\pi$	$d\pi$
trans-ML_4Z_2	0	$(8/3)d\sigma$	0	$2d\pi$	$2d\pi$

	z^2	x^2-y^2	xy	xz	yz
cis-ML_4Z_2	$2d\sigma$	$(2/3)d\sigma$	$2d\pi$	$d\pi$	$d\pi$
sym-ML_3Z_3	$2d\sigma$	$2d\sigma$	$2d\pi$	$2d\pi$	$2d\pi$

(1.50)

In the one-electron case the splitting of the e_g orbitals provides a direct measure of $d\sigma$ and the splitting of t_{2g} a direct measure of $d\pi$ (see section 1.8). If assignments can be made, the sign of these parameters may be obtained. Usually it may be inferred from previous experience concerning the ligating nature of L and Z.

Notice that the energy separations within the e_g and t_{2g} sets are twice as large for trans ML_4Z_2 complexes as for their cis analogs and that the energy levels within these sets are inverted between cis and trans.

The McClure parameters have not been used extensively as a means of assigning and calculating spectra though the $d\sigma$,$d\pi$ parameters are often reported from crystal field analyses where the splitting of e_g and of t_{2g} has been derived. It is clear from this approach that the splitting of the d orbitals in an octahedron is dependent both upon the sigma and pi bonding characteristics of the ligand and that pi acceptor ligands, by stabilising t_{2g} should provide larger values of 10Dq than pi donor ligands which destabilise t_{2g}; this is generally the case (see Chapter 9).

Nevertheless the approach provided impetus for the development of a more analytical procedure, the Orbital Angular Overlap Model.

1.7 Orbital Angular Overlap Model (AOM)

1.7 i) Introduction

Schaffer and Jorgensen [20-23,41] developed an analysis of electronic spectra based upon the ligand orbital sigma and pi interaction with the d orbitals, in a fashion quite similar to that previously suggested by McClure and by Yamatera. This theory is also very solidly based upon the molecular symmetry of the complex. There is necessarily a one-to-one complementarity between crystal field theory and the AOM theory in that the AOM parameters can be used to derive the corresponding crystal field parameters by a simple mapping procedure (or vice versa) i.e. there is a unitary relationship between the two theories. However the AOM parameters, unlike the crystal field parameters, do appear to have chemical significance, or at least, they are more justifiable given that the point charge model is basically inaccurate. Moreover, in principle, the AOM parameters may be calculated from knowledge of metal ligand bonding distances and overlap integrals. Unfortunately this has not yet been very successful. As shown later, the chemical significance of AOM parameters must be treated with some caution.

In parallel with crystal field theory the AOM model may be broken down into radial parameters and angular factors. The radial

parameters describe the interaction between the metal and ligand in sigma or pi bonding or anti-bonding terminology, and the angular factors depend upon the molecular symmetry. Before launching into a mathematical treatment of the coupling of the various metal-ligand bonds, let us obtain a subjective understanding of this approach first.

Consider a diatomic molecule M-L in which the z axes of both metal and ligand cartesian coordinate systems are collinear along the metal-ligand bond. The ligand is assumed to be an axially symmetric linearly ligating ligand. A ligand sigma orbital then perturbs the z^2 orbital on the metal ion. The p pi orbitals on the ligand will perturb the xz and yz orbitals on the metal, while the xy and x^2-y^2 orbitals will have delta symmetry with respect to the ligand. The AOM theory makes the simple assumption that the d orbital, interacting with the ligand orbital, is destabilised by this interaction by an amount $e_\sigma(L)$ as follows:

$$E(z^2) = e_\sigma(L)$$

$$E(xz) = E(yz) = e_\pi(L)$$

$$E(xy) = E(x^2-y^2) = 0 \qquad (1.51)$$

where the magnitude of this parameter is relative to the corresponding delta anti-bonding interaction assumed zero (here) or at least constant.

A slightly different labelling scheme from the original authors is used here. Schaffer and Jorgensen used the parameters e_i' where the prime signified energy relative to the delta anti-bonding component and i is pi or sigma. The prime is omitted since the delta bonding component is never considered.

For a diatomic d^1 (or d^9) molecule the problem is over. Transitions between the d levels are expressed as differences in AOM parameters following (1.51). However more complex molecules are usually being considered. The beauty of the AOM model lies in the fact that the energy levels of a complex molecule ML_n are obtained simply by adding the perturbation of each ligand L upon the five d orbitals of the central metal ion, carefully taking into account their geometric relationship to each other.

The z axis of each ligand is retained collinear with their metal ligand bond (and pointing towards the metal) but obviously the z axis of the metal cannot be collinear with every metal-ligand bond. Having defined one metal-ligand bond to be collinear with the positive metal z axis, the other metal-ligand bonds lie along or between other axes. Taking the simple example of an ML_4Z_2 system with trans M-Z bonds along the z axis, and the M-L bonds collinear with x and y, recognise that the L sigma bonds (along x and y) will each perturb one lobe of the x^2-y^2 orbital. Because of the relative charge densities of the various d orbitals this interaction is $(3/4)e_\sigma(L)$ (rather than $1e_\sigma(L)$) i.e. each of the four ligands

perturbs the x^2-y^2 orbital by $(3/4)e_\sigma(L)$ for a total perturbation of $3e_\sigma(L)$.

These coefficients follow naturally from the development in section 1.6 after recogising that the AOM parameters are normalised to the σ-interaction along the z axis being unity not the xy interaction as used by McClure.

These ligand (L) sigma orbitals lie in nodes of the xz,yz and xy orbitals and therefore do not perturb them at all. However the z^2 orbital has a 'doughnut' of electron density in the xy plane and will be perturbed by the L sigma bonds to the extent of $0.25\ e_\sigma(L)$ per ligand (see section 1.6), for a total perturbation of $1e_\sigma(L)$. Remembering that the perturbation along the z axis is $e_\sigma(Z)$ per Z ligand, the total sigma perturbation, i.e. destabilisation, of the d orbitals is:

$$E(x^2-y^2) = 3e_\sigma(L)$$
$$E(z^2) = 2e_\sigma(Z) + e_\sigma(L)$$
$$E(xz) = E(yz) = E(xy) = 0 \qquad (1.52)$$

Taking into account the p pi orbitals on the ligand (say F^-) and recognising that each lobe of the xy orbital will be perturbed by a ligand (L) pi orbital, that the xz orbital will be perturbed by pi (p_z) orbitals of two ligands L on the x axis, while the yz orbital will be perturbed by two ligands L (p_z) on the y axis, and both xz and yz will be perturbed by two ligands Z (p_x,p_y) along the z axis, the total perturbation energy will be:

$$E(x^2-y^2) = 3e_\sigma(L)$$
$$E(z^2) = 2e_\sigma(Z) + e_\sigma(L)$$
$$E(xz) = 2e_\pi(L) + 2e_\pi(Z)$$
$$E(yz) = 2e_\pi(L) + 2e_\pi(Z)$$
$$E(xy) = 4e_\pi(L) \qquad (1.53)$$

Note that if L = Z , the octahedral solution, the octahedral splitting diagram is recovered, with two sets of orbitals separated by $3e_\sigma(L) - 4e_\pi(L)$ the AOM analog of 10Dq (vide infra). It is difficult to conceive of the d sigma orbitals being anything other than anti-bonding ($e_\sigma(L)$ positive) but since L may be a pi donor or acceptor, the parameter $e_\pi(L)$ can be positive (anti-bonding) or negative (bonding). A negative value of $e_\pi(L)$ implies that L is a pi-acceptor.

Before proceeding to a more rigorous approach, consider one further example of an M-L bond bisecting the x and y axes, i.e. at an angle of 45° relative to the x axis as zero ($\phi = 45°$, $\theta = 90°$). This ligand L now sits in the nodal plane of the x^2-y^2 orbital and will contribute no sigma destabilisation thereto. However it sits on the lobe of xy and will destabilise xy by $(3/4)e_\sigma(L)$. The in-plane (xy) pi orbital of L will interact directly with x^2-y^2 destabilising it by $e_\pi(L)$. The previous interaction of xz (or yz) by $e_\pi(L)$ is now shared with the result that both xz and yz are now (de)stabilised by $0.5e_\pi(L)$. Since the z^2 'doughnut' is symmetric

in the xy plane, its destabilisation, $0.25e_\sigma(L)$ per ligand L, is independent of the angle ϕ. It should be clear that if an angle ϕ, such as 30^o is used where the ligand is not sitting in the nodal plane of either xy or x^2-y^2, the sigma and pi contributions from L to the energies of these two orbitals will be scrambled.

1.7 ii) AOM - Quantitative Approach

In the AOM approach the energy of a d orbital, E_d is treated as a perturbation of the energy of the metal orbital prior to overlap with the ligand, say H_M, caused by overlap of metal and ligand orbitals, the ligand energy being defined, as say, H_L. After overlap the ligand orbital is also perturbed and its new energy is, say, E_l. Further suppose that the overlap integral is S_{ML} and that this is small. Then following the Wolfsberg-Helmholz approximation, and the development in [24,25], the shift in energy due to overlap may be written:

$$E_d - H_M = (H_M + H_L)^2.(S_{ML})^2/(H_M - H_L) = e_\sigma(L)$$

$$E_l - H_L = -(H_M + H_L)^2.(S_{ML})^2/(H_M - H_L) = -e_\sigma(L) \qquad (1.54)$$

For sigma bonding orbitals, $e_\sigma(L)$ is positive and there is a destabilising effect upon the metal orbital and a stabilising effect upon the ligand orbital. For a given metal and ligand, the destabilisation is clearly proportional to the overlap between metal and ligand orbitals. This overlap may be factored into a radial component which depends only upon the metal, ligand, and their distance apart, and an angular term which depends only upon the geometry of the molecule and is independent of both metal and ligand.

For sigma anti-bonding to the z^2 orbital, for example, this factorisation may be written:

$$E(z^2) = e_\sigma(L)F(z^2,p_z)^2 \qquad (1.55)$$

where the F term represents the angular part only of the overlap integral, and the $e_\sigma(L)$ term encompasses all energy terms in (1.7) and the radial part of the overlap integral. For maximum overlap at a given metal-ligand distance, this F term is defined to be unity. In general therefore the energies of the real d orbitals in a fragment M-L are written as above in (1.53).

Consider further the M-L fragment, assume that the M-L bond is the z axis for both metal and ligand frameworks and further that the ligand x and y orbitals parallel the metal x, y orbitals as in Fig.1.11. The atomic orbitals form several types of overlap integral, which may be written [24]:

$$\begin{aligned} \langle d|L\rangle = \langle s|\sigma\rangle &= S(s,\sigma) \\ \langle p_z|\sigma\rangle &= S(p_\sigma,\sigma) \\ \langle p_x|\pi_x\rangle &= S(p_\pi,p_\pi) \end{aligned}$$

$$\begin{aligned} \langle z^2|\sigma\rangle &= S(d_\sigma,\sigma) \\ \langle p_x|\pi_y\rangle &= 0 \\ \langle yz|\pi_y\rangle &= S(d_\pi,p_\pi)_y \\ \langle yz|\pi_x\rangle &= 0 \\ \langle p_z|\pi_x\rangle &= 0 \end{aligned}$$

$$\langle z^2|\pi_x\rangle = 0 \quad \text{etc.} \qquad (1.56)$$

Now suppose the ligand no longer lies exactly along the metal z axis, but has been rotated about the y axis so that it now lies at point j in Fig.1.1, then the overlap with the unrotated d orbitals will be reduced by a geometric factor dependent upon the angle ϕ^o. This may be written [24]:

$$\begin{aligned} \langle d|L\rangle &= \langle s|\sigma\rangle = S(s,\sigma) \\ \langle p_z|\sigma\rangle &= S(p_\sigma,\sigma)\cos\phi \\ \langle p_x|\pi_x\rangle &= S(p_\pi,p_\pi)\cos\phi \\ \langle p_x|\pi_y\rangle &= 0 \end{aligned}$$

$$\begin{aligned} \langle z^2|\sigma\rangle &= S(d_\sigma,\sigma)(\cos^2\phi - 0.5\sin^2\phi) \\ \langle z^2|\pi_x\rangle &= -0.5(\sqrt{3})S(d_\pi,p_\pi)_x\sin(2\phi) \\ \langle yz|\pi_y\rangle &= S(d_\pi,p_\pi)_y\cos\phi \\ \langle yz|\pi_x\rangle &= 0 \\ \langle p_z|\pi_x\rangle &= S(p_\pi,p_\pi)_x.\sin\phi \\ &\quad \text{etc.} \end{aligned} \qquad (1.57)$$

Some overlap integrals are reduced because of the mismatch which now occurs between the metal and ligand orbitals, while other overlap integrals which were zero due to zero overlap, become non-zero because some overlap is now possible in the rotated system.

The destabilisation energy of a d orbital is a summation of perturbation by all the ligand orbitals having non-zero overlap therewith. When the ligand lies along the z axis of the metal, the σ, π and δ components of the bonding along that axis are orthogonal so that energies (shown in (1.53)) are functions only of one type of bonding. However in the general case when mismatch occurs, the energy of a d orbital may have a contribution from any or all of σ, π and δ components, i.e. in general:

$$E(d) = e_\sigma(L)F(d,\sigma)^2 + e_\pi(L)F(d,\pi)^2 + e_\delta(L)F(d,\delta)^2 \qquad (1.58)$$

Delta bonding is usually assumed to be zero; if this is not the case then the sigma and pi parameters should be considered to be referenced against the delta bonding contribution (see [21]). We shall neglect it henceforth.

A basic assumption of the AOM theory is that the total perturbation of any d orbital is a sum of the perturbations due to each ligand, namely, considering the diagonal energy:

$$\langle d_i|V_{OAM}|d_i\rangle = \sum_{n=1}^{n=N} e_\sigma(L_n)F(d_i,L_n)^2 + \sum_{n=1}^{n=N} e_\pi(L_n)F(d_i,L_n)^2 \quad (1.59)$$

Moreover if there are several types of ligands or if the same ligand occurs at different distances from the metal, then the AOM parameters must be further factored to recognise that these will each have different radial parameters e_i.

The metal axis system is considered to be fixed and the problem is to sum the effects of each ligand upon the d orbitals quantized along the metal framework as a function of the position of the ligand with respect to the metal framework. We can imagine placing each ligand on the metal z axis and then rotating it on the surface of a unit sphere to place it in its final position. A ligand atom may reach any position of the surface of this unit sphere first by a rotation of θ^o about the z axis, followed by a rotation of ϕ^o about the y axis. The overlap of this ligand with each of the original orbitals can now be found by deducing the overlap terms, as for example in (1.57), after taking into account these two rotations.

This procedure is the equivalent of creating a new rotated axis system, x',y',z' on the ligands and describing the original d orbitals within the xyz system in terms of the primed system. Schaffer has calculated these relationships and they are shown in Table 1.12 for p and d orbitals [22].

> Table 1.12 applies to a linearly ligating or axial ligand. If the ligand is non-axial, such as water or pyridine, then it may be necessary to take account of the orientation of this ligand with respect to the metal-ligand bond. In Schaffer's original tabulation this was accounted for by inclusion of a third angle ψ. For simplicity this is omitted here. The reader is referred to the original literature [22] for such information.

Consider an octahedral complex ML_6, with coordinate system shown in Fig.1.11. The ligands, labelled as shown, have angular coordinates as follows:

Ligand	1	2	3	4	5	6
θ	90	90	90	90	0	180
ϕ	0	90	180	270	0	0

Consider L_1, leading accross the top of Table 1.12, and inserting $\theta = 90^o$ and $\phi = 0^o$:

$$z^2 = -0.5\ z'^2 + (\sqrt{3})/2\ x'^2-y'^2 \quad (1.60)$$

the other terms in y'z', x'z' etc collapsing to zero. Similarly, for L_1:

Table 1.12

ANGULAR OVERLAP MODEL TRANSFORMATION MATRICES

D^d matrix

	z'^2	y'z'	x'z'
z^2	$(1/4)(1 + \cos2\theta)$	0	$-(\sqrt{3}/2)\sin2\theta$
yz	$(\sqrt{3}/2)\sin\phi\sin2\theta$	$\cos\phi\cos\theta$	$\sin\phi\cos2\theta$
xz	$(\sqrt{3}/2)\cos\phi\sin2\theta$	$-\sin\phi\cos\theta$	$\cos\phi\cos2\theta$
xy	$(\sqrt{3}/4)\sin2\phi(1 - \cos2\theta)$	$\cos2\phi\sin\theta$	$(1/2)\sin2\phi\sin2\theta$
x^2-y^2	$(\sqrt{3}/4)\cos2\phi(1 - \cos2\theta)$	$-\sin2\phi\sin\theta$	$(1/2)\cos2\phi\sin2\theta$

	x'y'	$x'^2-y'^2$
z^2	0	$(\sqrt{3}/4)(1 - \cos2\theta)$
yz	$-\cos\phi\sin\theta$	$-(1/2)\sin\phi\sin2\theta$
xz	$\sin\phi\sin\theta$	$-(1/2)\cos\phi\sin2\theta$
xy	$\cos2\phi\cos\theta$	$(1/4)\sin2\phi(3 + \cos2\theta)$
x^2-y^2	$-\sin2\phi\cos\theta$	$(1/4)\cos2\phi(3 + \cos2\theta)$

D^p matrix

	z'	y'	x'
z	$\cos\theta$	0	$-\sin\theta$
y	$\sin\phi\sin\theta$	$\cos\phi$	$\sin\phi\cos\theta$
z	$\cos\phi\sin\theta$	$-\sin\phi$	$\cos\phi\cos\theta$

See text for use of these matrices. Note discussion of this procedure in refs. [8,21,3-25]. Where orientation of the ligand is important, a third angle, ψ, relating to this orientation, should be used. Complete matrices inclusive of ψ are given in [21].

$$yz = -x'y'$$
$$xz = -x'z'$$
$$xy = y'z'$$
$$x^2-y^2 = (\sqrt{3})/2\ z'^2 + (0.5)x'^2-y'^2 \quad (1.61)$$

Recognise that the functions listed in Table 1.12 are, in fact, the F(p,L) and F(d,L) angular overlap coefficients representing the mismatch of ligand orbitals with the original d orbital set. The interaction energies of the primed orbitals with the d orbitals are the squares of these functions (see 1.55). If a row of Table 1.12 is written in the general form:

	z'^2	xz'	yz'	xy'	$x'^2-y'^2$
d_i	D_{i1}	D_{i2}	D_{i3}	D_{i4}	D_{i5}

then the energies of the d orbitals (diagonal matrix elements) are given by:

$$E(d_i) = \langle d_i|V_{AOM}|d_i\rangle = (D_{i1})^2e_\sigma(L) + (D_{i2})^2e_{\pi x}(L) + (D_{i3})^2e_{\pi y}(L) + (D_{i4})^2e_\delta(L) + (D_{i5})^2e_\delta(L) \quad (1.62)$$

In this case, (using (1.61, 1.62)) the energies of the d orbitals due to perturbation by L_1 are (neglecting delta contributions):

$$E(z^2) = (0.25)e_\sigma(L)$$
$$E(yz) = 0$$
$$E(xz) = e_{\pi x}(L)$$
$$E(xy) = e_{\pi y}(L)$$
$$E(x^2-y^2) = (3/4)e_\sigma(L) \quad (1.63)$$

where the x,y subscripts indicate which ligand pi orbital is involved.

Eqns. 1.61 and 1.63 are more readily understood if one views the original unprimed d orbital set quantized along the metal z axis (Fig.1.11) from the viewpoint of L_1. L_1 sees the 'doughnut' of the z^2 orbital and one lobe of the x^2-y^2 orbital both of which have the same symmetry viewed from L_1 and are therefore scrambled. There will be fractional sigma contributions to the energies of both z^2 and x^2-y^2 in a manner similar to that described above in the McClure-Yamatera treatment. Within the cartesian framework of L_1 (primed) the yz orbital appears like an xy orbital and is therefore x'y' in the L_1 frame. Any energy perturbation to yz from L_1 can then only be delta in nature. The xz orbital lies in the x'z' plane of L_1 and π-interacts with the p_x orbital of L_1. The xy orbital lies in the y'z' plane of the L_1 frame and is perturbed by p_y of L_1 (Fig.1.11). The signs of eqn.1.61 depend upon the choice of initial frame prior to rotation and are an arbitrary initial choice. They are appropriate to the array in Fig.1.11. Other authors may differ (e.g. [44]), and therefore care must be taken when using explicit molecular orbital desciptions (such as Table 1.11) from various sources. Obviously the final energies do not depend upon one's choice of axes.

Now considering a similar treatment for each ligand, the following may be derived.

	$E(z^2)$	$E(yz)$	$E(xz)$	$E(xy)$	$E(x^2-y^2)$
L_1	$(1/4)e_\sigma(L)$	0	$e_\pi(L)$	$e_\pi(L)$	$(3/4)e_\sigma(L)$
L_2	$(1/4)e_\sigma(L)$	$e_\pi(L)$	0	$e_\pi(L)$	$(3/4)e_\sigma(L)$
L_3	$(1/4)e_\sigma(L)$	0	$e_\pi(L)$	$e_\pi(L)$	$(3/4)e_\sigma(L)$
L_4	$(1/4)e_\sigma(L)$	$e_\pi(L)$	0	$e_\pi(L)$	$(3/4)e_\sigma(L)$
L_5	$e_\sigma(L)$	$e_\pi(L)$	$e_\pi(L)$	0	0
L_6	$e_\sigma(L)$	$e_\pi(L)$	$e_\pi(L)$	0	0

(1.64)

where the x,y subscripts are omitted from the π-terms. However if the ligand does not have axial π-bonding (e.g. pyridine), care must be taken to distinguish those pi terms which exist from those which do not.

The total perturbation due to the six ligands (see 1.59) is obtained by summing each column in (1.64) to obtain, for an octahedral ML_6 complex:

$$E(z^2) = E(x^2-y^2) = 3e_\sigma(L)$$
$$E(xy) = E(xz) = E(yz) = 4e_\pi(L) \quad (1.65)$$

The anticipated orbital doublet and triplet are indeed obtained, and the energy separation between these orbitals provides a relationship between the crystal field Dq parameter and the AOM procedure, as previously noted, namely:

$$10\ Dq = 3e_\sigma(L) - 4e_\pi(L) \quad (1.66)$$

Equation (1.66) provides a convenient method of factorising the ligand field into sigma and pi components, and will be further considered in a discussion of the Spectrochemical Series (Chapter 9).

Note a characteristic property of the AOM approach that the sum of the sigma terms equals the number of ligands, and that the sum of the pi terms equals the sum of the number of pi orbitals (12 in this case, for an axially symmetric ligand).

1.7.iii) Tetrahedral species

Using the same approach for a tetrahedral species, inserting the angular positions of the tetrahedrally disposed ligands, yields:

$$E(t_2) = (4/3)e_\sigma(L) + (8/9)e_\pi(L)$$

$$E(e) = (8/3)e_\pi(L) \qquad (1.67)$$

providing the well known observation that the splitting in a tetrahedral complex should be -(4/9) of that in an analogous octahedral complex, a result also obtainable from crystal field theory. Note also that in tetrahedral symmetry, the t_2 set possesses both sigma and pi character.

1.7.iv) Tetragonal Species

Expression (1.64) can be used for any pseudo-octahedral complex even if the ligands are different provided that the angles remain 90°, i.e. the ligands are in the same angular position, and provided that the symmetry is sufficiently high that there are no two metal orbitals with the same symmetry representation, i.e there are no off-diagonals.

Hence a complex such as trans-ML_4Z_2 is easily handled by factoring the $e_\sigma(L)$ parameters into those appropriate for L and for Z. In this fashion, the energies (for simplicity assume ligands 1 and 6 are Z) are simply those previously obtained in a less rigorous manner (1.53).

1.7.v) Trigonal Bipyramidal Complexes

Inserting the angular positions of a regular trigonal bipyramidal molecule (Table 1.12) but distinguishing the axial (L_a) from equatorial (L_e) ligands, yields:

$$E(z^2) = 0.75e_\sigma(L_e) + 2e_\sigma(L_a)$$

$$E(xz) = E(yz) = 1.5e_\pi(L_e) + 2e_\pi(L_a)$$

$$E(xy) = E(x^2\text{-}y^2) = 1.125e_\sigma(L_e) + 1.5e_\pi(L_e) \qquad (1.68)$$

where the pairs of degenerate orbitals were anticipated by our earlier analysis.

1.7.vi) Lower Symmetry Complexes

The process of inserting the ligand angular coordinates into Table 1.12 and deriving the d orbital perturbations is quite standard for all molecules. However when the symmetry is fairly low, several orbitals may have the same symmetry representation. For example in C_{2v}(I) molecules both z^2 and $x^2\text{-}y^2$ transform as a_1. Under these circumstances these orbitals may be scrambled by an off-diagonal AOM matrix element $\langle z^2 | V_{AOM} | x^2\text{-}y^2 \rangle$. The general problem of calculating such off-diagonal elements is now entertained.

Consider a matrix for the perturbation of the d orbitals by ligand L_1 in the octahedral complex. It may be written (combining eqns. 1.60, 1.61):

	z'^2	$y'z'$	$x'z'$	$x'y'$	$x'^2\text{-}y'^2$
z^2	-1/2	0	0	0	√3/2
yz	0	0	0	-1	0
xz	0	0	-1	0	0
xy	0	1	0	0	0
$x^2\text{-}y^2$	√3/2	0	0	0	1/2

(1.69)

The matrix element $\langle z^2 | V_{AOM} | x^2\text{-}y^2 \rangle$ can be evaluated by a simple extension to eqn.(1.62), namely:

$$\langle d_i | V_{AOM} | d_j \rangle = (D_{i1}\cdot D_{j1})e_\sigma(L) + (D_{i2}\cdot D_{j2})e_{\pi x}(L) + (D_{i3}\cdot D_{j3})e_{\pi y}(L) + (D_{i4}\cdot D_{j4})e_\delta(L) + (D_{i5}\cdot D_{j5})e_\delta(L) \qquad (1.70)$$

In other words rather than squaring the coefficients in each row for each orbital, one obtains the product of the coefficients in each column for the two orbitals involved. In this case it is necessary that each column of the z^2 and $x^2\text{-}y^2$ rows should be multiplied together, and the result summed.

Consider this product for L_1, and obtain, from (1.69), $-\sqrt{3}/4\ e_\sigma(L_1)$, (neglecting the delta terms). The corresponding elements for the other ligands are:

$$\begin{array}{ll} L_1 & -\sqrt{3}/4\ e_\sigma(L_2) \\ L_2 & +\sqrt{3}/4\ e_\sigma(L_3) \\ L_3 & -\sqrt{3}/4\ e_\sigma(L_4) \\ L_4 & +\sqrt{3}/4\ e_\sigma(L_5) \\ L_5 & 0 \\ L_6 & 0 \end{array} \qquad (1.71)$$

Clearly if all the ligands are identical, then this sums to zero. All other off-diagonals are also zero.

This lack of off-diagonal elements arises from intelligent choice of the initial framework. To avoid unnecessary mathematical drudgery, care must be taken to choose a framework which will provide least labour. For example if there were chosen for the octahedral ML_6 complex, a coordinate axis system in which the x,y axes passed at 45° between ligands instead of being collinear with them, non-zero off-diagonal elements would have been observed. After full calculation the result would have been the same as before, but it would have required additional effort (see [24] for a worked example).

The sum in (1.71) also goes to zero for the tetragonal case where L_5 and L_6 are Z. If, in the tetragonal case, L_2 and L_4, for example, were defined as Z, then the sum in (1.71) does not go to

zero. There will now be off-diagonal elements but the final result will still be the same as previously determined. This is another example where the injudicious choice of coordinate axis causes additional labour. In this case it is necessary that the four-fold axis, containing the Z ligands, be addressed as the z axis if off-diagonals are to be avoided.

The cis-ML_4Z_2 complex also has no off-diagonal elements if the cis Z ligands are placed in positions 1 and 2 (or 3 and 4) (see 1.71). The final result in this cis case is:

$$\begin{aligned}
\langle z^2|V|z^2\rangle &= E(z^2) = 2.5\, e_\sigma(L) + 0.5\, e_\sigma(Z)\\
\langle yz|V|yz\rangle &= E(yz) = 3\, e_\pi(L) + e_\pi(Z)\\
\langle xz|V|xz\rangle &= E(xz) = 3\, e_\pi(L) + e_\pi(Z)\\
\langle xy|V|xy\rangle &= E(xy) = 2\, e_\pi(L) + 2\, e_\pi(Z)\\
\langle x^2-y^2|V|x^2-y^2\rangle &= E(x^2-y^2) = 1.5\, e_\sigma(L) + 1.5\, e_\sigma(Z)
\end{aligned} \qquad (1.72)$$

This is an example of so-called 'intermediate symmetry' [26] where the molecule behaves as though it had D_{4h} symmetry though the actual symmetry is only C_{2v}. This arises because the metal only senses the total field along an axis, and does not distinguish say, positive x from negative x. Thus the total field along the x and along the y axes is due to (L + Z), while along z it is 2L. The system then behaves as a tetragonal rather than a rhombic molecule in so far as symmetry is concerned.

Now consider the D_{2h} complex $MX_2Y_2Z_2$ where opposite non-adjacent ligands are equivalent and the ligands lie on their named axes. The diagonal elements can be obtained using Table 1.12, or intuitively, they are:

$$\begin{aligned}
\langle z^2|V|z^2\rangle &= E(z^2) = 0.5\, e_\sigma(X) + 0.5\, e_\sigma(Y) + 2e_\sigma(Z)\\
\langle yz|V|yz\rangle &= E(yz) = 2\, e_\pi(Y) + 2\, e_\pi(Z)\\
\langle xz|V|xz\rangle &= E(xz) = 2\, e_\pi(X) + 2\, e_\pi(Z)\\
\langle xy|V|xy\rangle &= E(xy) = 2\, e_\pi(X) + 2\, e_\pi(Y)\\
\langle x^2-y^2|V|x^2-y^2\rangle &= E(x^2-y^2) = 1.5\, e_\sigma(X) + 1.5\, e_\sigma(Y)
\end{aligned} \qquad (1.73)$$

Expression (1.71) does not now collapse to zero, but rather sums to:

$$\langle z^2|V|x^2-y^2\rangle = (\sqrt{3}/2)(e_\sigma(Y) - e_\sigma(X)) \qquad (1.74)$$

In calculating the energies of the z^2 and x^2-y^2 orbitals in D_{2h} complexes of this type, this off-diagonal energy must be taken into account by solving the 2 x 2 secular determinant:

$$\begin{vmatrix} \langle z^2|V|z^2\rangle - E & \langle z^2|V|x^2-y^2\rangle \\ \langle z^2|V|x^2-y^2\rangle & \langle x^2-y^2|V|x^2-y^2\rangle - E \end{vmatrix} = 0 \qquad (1.75)$$

after insertion of appropriate values of these matrix elements according to eqns. (1.73) and (1.74). The other orbitals already

have diagonal energies (no off-diagonal terms connecting to any other orbitals) and their energies can be obtained directly by insertion of values for the AOM parameters into (1.73). A detailed calculation of a more complex molecule is presented in Appendix III.

The analysis described above provides the one-electron energies of the d orbitals in any geometry. The process is readily programmed on a micro-computer and provides rapid access thereby to one-electron energy level diagrams.

However note that the spectroscopic method provides energy differences and not absolute energies of electronic states. With five d orbitals, all excited state energies can be described by means of four one-electron energy level differences plus the necessary inter-electronic repulsion terms. Thus four is also the maximum number of independent radial integrals which can be derived from a d^1 electronic spectrum. Further (non-spectroscopic) experiments would be necessary to obtain further information. This is akin to the fact that in an octahedral complex $10Dq = 3e_\sigma - 4e_\pi$ but e_σ and e_π cannot be determined independently in this symmetry because only one parameter, 10Dq, is available.

The relative magnitudes of the AOM parameters appear to provide both sigma and pi bonding information about the metal ligand bond. However some caution should be exercised in the interpretation of such data. The AOM parameters provide information about the relative energies of the d orbitals. The primary metal ligand bond involves both s and p contributions. One cannot necessarily infer that these bond energies are accurately reflected in d orbital perturbations (e.g. see [27]). AOM data are presented in later chapters.

Finally, although real d orbitals are generally encountered in AOM calculations, there are some situations where the imaginary orbitals are more useful (see Chapter 3). For completion the AOM matrix elements for the complex d orbitals are reported here for a rhombic complex of the type all trans-$MX_2Y_2Z_2$, assuming all X-M-Y, X-M-Z etc type angles are 90^o (D_{2h}):

$$\begin{aligned}
&\langle 2,2|V|2,2\rangle = \langle 2,-2|V|2,-2\rangle = (0.75)(e_\sigma(X) + e_\sigma(Y)) + (e_\pi(X) + e_\pi(Y))\\
&\langle 2,1|V|2,1\rangle = \langle 2,-1|V|2,-1\rangle = e_\pi(X) + e_\pi(Y) + 2\, e_\pi(Z)\\
&\langle 2,0|V|2,0\rangle = 2\, e_\sigma(Z) + (0.5)(e_\sigma(X) + e_\sigma(Y))\\
&\langle 2,2|V|2,-2\rangle = \langle 2,-2|V|2,2\rangle = (0.75)(e_\sigma(X) + e_\sigma(Y)) - (e_\pi(X) + e_\pi(Y))\\
&\langle 2,1|V|2,-1\rangle = e_\pi(Y) - e_\pi(X)\\
&\langle 20|V|22\rangle = \langle 20|V|2-2\rangle = (\sqrt{6}/4)(e_\sigma(Y) - e_\sigma(X))
\end{aligned} \qquad (1.76)$$

Clearly, writing x = y = z provides the imaginary d orbital

octahedral energies in the AOM formalism, while a tetragonal solution is obtained when $x = y \neq z$.

If appropriate sigma and pi values for X, Y and Z ligands are inserted into the secular determinant formed by (1.76) then, of course, the one electron d orbital energies for this particular data set are obtained. It clearly cannot matter which ligands we call X, which we call Y and which we call Z, since the framework is arbitrary. Therefore the same energies are obtained if the values of the X, Y and Z parameters are permuted. The symmetry labels of any specific electronic transition in the spectrum of such a species will depend upon the choice of X, Y and Z, but these too are arbitrary in that they are defined by our choice of axes. Note that this is generally true for any symmetry; permutation of the choice of the ligands on the x, y and z axes does not alter the calculated energies. This is a useful, though not definitive, method for testing AOM computer programs. A program which permutes correctly is not necessary correct. A program which does not permute correctly, is definitely incorrect.

In summary, in Table 1.13 are collected the one electron energy levels, within the AOM formalism, for a range of common geometries. These may be used as a basis for d^1 (and d^4, d^6 and d^9) calculations in these geometries, and, indeed, as shown in Chapter 3, can be used as a basis for other poly-electron calculations.

The AOM approach is basically an additive model in that it assumes that the final energy levels can be obtained by adding contributions from each of the ligands, i.e. the additive AOM approach. This assumption can be obviated by setting up the energy matrices in terms of one-electron energy level differences as discussed in some depth both by Schaffer [23,41] and by Horrocks [28].

Briefly, following Schaffer [23] for an octahedral ML_6 system, the parameters σ and π can be defined via:

$$\Delta_\sigma = 3e_\sigma(L) \text{ and } \Delta_\pi = 4e_\pi(L)$$

further

$$\Delta = \Delta_\sigma - \Delta_\pi = he(O_h) - ht_2(O_h) \quad (= 10Dq) \tag{1.77}$$

where $he(O_h)$ and $ht_2(O_h)$ are the one-electron energies of the e and t_2 octahedral levels respectively. For example in a tetragonal trans ML_4Z_2 complex, the so-called non-additive AOM model deals with the one-electron separations generated by this tetragonal distortion, thus:

$$\Delta(e) = hb_1(D_{4h}) - ha_1(D_{4h}) \quad (= \Delta_3)$$
$$\Delta(t_2) = hb_2(D_{4h}) - he(D_{4h}) \quad (= \Delta_1)$$

and

$$he(O_h) = (1/2)[hb_1(D_{4h}) + ha_1(D_{4h})]$$
$$ht_2(O_h) = (1/3)[hb_2(D_{4h}) + 2he(D_{4h})] \tag{1.78}$$

Table 1.13

ANGULAR OVERLAP MODEL MATRIX ELEMENTS FOR d ORBITALS IN VARIOUS STEREOCHEMICAL ENVIRONMENTS[a]

Matrix Element	ML_6 O_h	t-ML_4Z_2 D_{4h}	ML_4 T_d
$\langle t_2\|V\|t_2\rangle$	$4e_\pi(L)$		$(4/3)e_\sigma(L) + (8/9)e_\pi(L)$
$\langle e\|V\|e\rangle$	$3e_\sigma(L)$		$(8/3)e_\pi(L)$
$\langle z^2\|V\|z^2\rangle$		$e_\sigma(L) + 2e_\sigma(Z)$	
$\langle x^2-y^2\|V\|x^2-y^2\rangle$		$3e_\sigma(L)$	
$\langle xy\|V\|xy\rangle$		$4e_\pi(L)$	
$\langle xz\|V\|xz\rangle$		$2e_\pi(L) + 2e_\pi(Z)$	
$\langle yz\|V\|yz\rangle$		$2e_\pi(L) + 2e_\pi(Z)$	

Matrix Element	$MX_2Y_2Z_2$ D_{2h}	c-ML_4Z_2 C_{2v}
$\langle z^2\|V\|z^2\rangle$	$0.5e_\sigma(X) + 0.5e_\sigma(Y) + 2e_\sigma(Z)$	$2.5e_\sigma(L) + 0.5e_\sigma(Z)$
$\langle x^2-y^2\|V\|x^2-y^2\rangle$	$1.5e_\sigma(X) + 1.5e_\sigma(Y)$	$1.5e_\sigma(L) + 1.5e_\sigma(Z)$
$\langle xy\|V\|xy\rangle$	$2e_\pi(X) + 2e_\pi(Y)$	$2e_\pi(L) + 2e_\pi(Z)$
$\langle xz\|V\|xz\rangle$	$2e_\pi(X) + 2e_\pi(Z)$	$3e_\pi(L) + e_\pi(Z)$
$\langle yz\|V\|yz\rangle$	$2e_\pi(Y) + 2e_\pi(Z)$	$3e_\pi(L) + e_\pi(Z)$
$\langle z^2\|V\|x^2-y^2\rangle$	$\sqrt{3}/2[e_\sigma(Y) - e_\sigma(X)]$	0

Matrix Element	ML_3Z_2 D_{3h}	ML_5Z C_{4v}
$\langle z^2\|V\|z^2\rangle$	$0.75e_\sigma(L) + 2e_\sigma(Z)$	$2e_\sigma(L) + e_\sigma(Z)$
$\langle x^2-y^2\|V\|x^2-y^2\rangle$	$1.125e_\sigma(L) + 1.5e_\pi(L)$	$3e_\sigma(L)$
$\langle xy\|V\|xy\rangle$	$1.125e_\sigma(L) + 1.5e_\pi(L)$	$4e_\pi(L)$
$\langle xz\|V\|xz\rangle$	$1.5e_\pi(L) + 2e_\pi(Z)$	$3e_\pi(L) + e_\pi(Z)$
$\langle yz\|V\|yz\rangle$	$1.5e_\pi(L) + 2e_\pi(Z)$	$3e_\pi(L) + e_\pi(Z)$

Matrix Element	ML_4Z out-of-plane C_{4v} [b]
$\langle z^2\|V\|z^2\rangle$	$(1/4)(1+3\cos2\theta)^2e_\sigma(L) + (3\sin^2 2\theta)e_\pi(L) + e\sigma(Z)$
$\langle x^2-y^2\|V\|x^2-y^2\rangle$	$(3/4)(1-\cos2\theta)^2e_\sigma(L) + (\sin^2 2\theta)e_\pi(L)$
$\langle xy\|V\|xy\rangle$	$4e_\pi(L)$
$\langle xz\|V\|xz\rangle$	$(1.5\sin^2 2\theta)e_\sigma(L) + 2(\cos^2\theta + \cos^2 2\theta)e_\pi(L) + e_\pi(Z)$
$\langle yz\|V\|yz\rangle$	$(1.5\sin^2 2\theta)e_\sigma(L) + 2(\cos^2\theta + \cos^2 2\theta)e_\pi(L) + e_\pi(Z)$

a) Except where shown, all angles are assumed to be 90^o. Each ligand is assumed to be ortho-axial with a pair of pi orbitals perpendicular to the M-L bonding axis. If this is not the case, e.g. pyridine, appropriate care must be taken to remove some pi bonding coefficients. b) θ is the out-of-plane angle. When $\theta = 90^o$ the metal lies in the plane.

There are three one-electron energy separations within the d orbital set of a tetragonal complex and it is clearly possible to define the spectra of such a complex of any d^n configuration in terms of these three variables. This 'non-additive' approach makes no assumptions about how the fields due to each ligand are related, only the final splitting pattern is considered.

In a tetragonal complex, the three relevant parameters are defined as $\Delta(e)$ and $\Delta(t_2)$ as defined in (1.78) and [46]:

$$\Delta = \Delta(d) = 1/2[hb_1(D_{4h}) + ha_1(D_{4h})] - (1/3)[hb_2(D_{4h}) + 2he(D_{4h})]$$

The one-electron energies of the d orbitals may be expressed in terms of these parameters as follows [47]:-

$$\begin{aligned}
\langle x^2-y^2|V|x^2-y^2\rangle &= (3/5)\Delta(d) + (1/2)\Delta(e)\\
\langle z^2|V|z^2\rangle &= (3/5)\Delta(d) - (1/2)\Delta(e)\\
\langle xy|V|xy\rangle &= -(2/5)\Delta(d) + (2/3)\Delta(t_2)\\
\langle xz|V|xz\rangle &= \langle yz|V|yz\rangle = -(2/5)\Delta(d) - (1/3)\Delta(t_2)
\end{aligned} \qquad (1.79)$$

These may be used to parameterize any d^n calculation as, for example, shown by Glerup and co-workers [47], for d^3 or d^8 ions in a D_{2h} (or higher) environment.

The 'non-additive' approach may be related to the 'additive' approach through the following equations (shown for a trans-ML_4Z_2 species):

$$\begin{aligned}
\Delta = \Delta(d) &= e_\sigma(Z) - (4/3)e_\pi(Z) + 2e_\sigma(L) - (8/3)e_\pi(L)\\
\Delta(e) &= 2e_\sigma(L) - 2e_\sigma(Z)\\
\Delta(t_2) &= 2e_\pi(L) - 2e_\pi(Z)
\end{aligned} \qquad (1.80)$$

Since there are only three 'non-additive' parameters (and only three observables for a d^1 species) it is not possible to uniquely define all four AOM 'additive' parameters from the 'non-additive' set. If saturated amines (or similar non-π-bonding species) are used for L, the $e_\pi(L)$ may be set to zero, or, in another sense, the AOM parameters may be defined with respect to $e_\pi(L)$.

One may then write [23,47]:

$$\begin{aligned}
\Delta_\sigma(L) - \Delta_\pi(L) &= 3e_\sigma(L) - 4e_\pi(L) = \Delta(d) + (1/2)\Delta(e) - (2/3)\Delta(t_2)\\
\Delta_\sigma(Z) - \Delta_\pi(L) &= 3e_\sigma(Z) - 4e_\pi(L) = \Delta(d) - \Delta(e) - (2/3)\Delta(t_2)\\
\Delta_\pi(Z) - \Delta_\pi(L) &= 4e_\pi(Z) - 4e_\pi(L) = -2\Delta(t_2)
\end{aligned} \qquad (1.81)$$

Note that all these equations collapse to the expected octahedral solutions if L = Z.

1.8 Inter-Relationships Between the Crystal Field, Ligand Field and Molecular Orbital Models.

Depending upon the problem involved, the kind of spectroscopic and chemical information available, and perhaps the computer programs available, it may prove useful to solve a data set within one particular model and then use the mapping between the several models to derive parameters within the other approaches. Therefore in this section are presented relationships between the various models.

Whether the classical (Dq, Ds, Dt etc), NSH (DQ, DS, DT etc) or AOM ($e_\sigma(L)$ etc) methods are used, the final energies and wavefunctions will be the same. The three methods are simply alternative ways of achieving the same end and are clearly related to one another by unitary transformations. It is possible to write (map) relationships between them such that if one method is used and parameters derived, parameters in the other approaches can be calculated therefrom. First note certain facets of the mapping process.

i) The diagonal elements for a given wavefunction cannot be equated directly from one method to another because the zero point differs for each method. It is necessary therefore, in deriving a map, to equate differences between diagonal energies from one approach to another.

> Note however that the difference between the zero points of two different models can be derived, and will be constant for all energy levels. It can then be used to compare diagonal energies directly, by compensating for the zero point energy difference.

ii) The off-diagonal elements do not depend upon the zero energy and off-diagonal elements can therefore be directly compared from one approach to another. However there is a phase ambiguity in such a comparison and care must be taken to ensure that the phase is consistent throughout the analysis. Changing the sign of an off-diagonal element does not effect the energy if carried out consistently, but will effect the eigenvectors. Inconsistent sign changes in the off-diagonal elements introduce errors.

The maps are quite general and may be used for the poly-electron systems under discussion in later Chapters. For completeness some equations previously encountered in this Chapter are repeated here. Take especial note to distinguish crystal field parameters Dq,Ds and Dt, from the NSH parameters DQ, DS and DT.

a) Octahedral ML_6

$$\begin{aligned}
10Dq &= 3e_\sigma(L) - 4e_\pi(L)\\
DQ &= 6\sqrt{21}\ Dq
\end{aligned} \qquad (1.82)$$

b) Tetragonal trans-ML_4Z_2 (Z on z axis).

$$10Dq(L) = 3e_\sigma(L) - 4e_\pi(L) = 10\,Dq(Eq)$$
$$10Dq(Z) = 3e_\sigma(Z) - 4e_\pi(Z) = 10\,Dq(Ax)$$

$$Ds = -7Ds$$
$$Dt = -(7\sqrt{15})Dt/2 = -(1/3)\sqrt{(5/7)}[DQ(Eq) - DQ(Ax)]$$

$$DQ = (6\sqrt{21})Dq(L) - (7\sqrt{21})Dt/2$$

$$DQ(Ax) = DQ + 2\sqrt{(7/5)}DT$$
$$DQ(Eq) = DQ - \sqrt{(7/5)}DT$$
$$DQ = (1/6)[4DQ(Eq) + 2DQ(Ax)]$$

$$\Delta_1 = -(3/7)DS + (10/7\sqrt{15})DT = \Delta(t_2) = 3Ds - 5Dt$$
$$\Delta_2 = (5/3\sqrt{21})DQ - \sqrt{(5/27)}DT = 10Dq(Eq)$$
$$\Delta_3 = -(4/7)DS - (10/7\sqrt{15})DT = \Delta(e) = 4Ds + 5Dt \qquad (1.83)$$

The above are true generally for all tetragonal complexes. Beware several different definitions of the Δ parameters in the literature, e.g. those for low spin square d^8 (p.113) differ from those here.

$$DQ = (\sqrt{21}/5)[3e_\sigma(Z) + 6e_\sigma(L) - (4e_\pi(Z) + 8e_\pi(L))]$$

$$Ds = (2/7)[e_\sigma(L) + e_\pi(L) - e_\sigma(Z) - e_\pi(Z)]$$
$$= -(1/21)[8d\sigma + 6d\pi]$$
$$DS = -2[e_\sigma(L) + e_\pi(L) - e_\sigma(Z) - e_\pi(Z)]$$

$$Dt = (2/35)[3e_\sigma(L) - 4e_\pi(L) - 3e_\sigma(Z) + 4e_\pi(Z)]$$
$$= -(1/35)[8d\sigma - 8d\pi]$$
$$= (4/7)[Dq(L) - Dq(Z)]$$
$$DT = -(\sqrt{15}/5)[3e_\sigma(L) - 4e_\pi(L) - 3e_\sigma(Z) + 4e_\pi(Z)]$$

$$d\sigma = (3/4)[e_\sigma(Z) - e_\sigma(L)] = -(1/8)[15Dt + 12Ds]$$
$$= (1/28)[6DS + (\sqrt{15})DT] = -(3/8)\,\Delta_3$$
$$d\pi = e_\pi(Z) - e_\pi(L) = (1/2)[5Dt - 3Ds]$$
$$= (1/14)[3DS - 2\sqrt{(5/3)}DT] = -(1/2)\,\Delta_1$$

If L is non-pi bonding, such that $e_\pi(L) = 0$, then:

$$e_\sigma(L) = (5/9)[DQ/\sqrt{21} - DT/\sqrt{15}]$$
$$e_\sigma(Z) = (1/189)[(5\sqrt{21}DQ) + 54DS + (2\sqrt{15}DT)] \qquad (1.84)$$

c) Tetragonal ML_5Z

$$Dt = (1/35)[3e_\sigma(L) - 4e_\pi(L) - 3e_\sigma(Z) + 4e_\pi(Z)]$$
$$= (2/7)[Dq(L) - Dq(Z)]$$
$$Ds = (1/7)[e_\sigma(L) + e_\pi(L) - e_\sigma(Z) - e_\pi(Z)]$$

$$d\sigma = -(1/8)[35Dt + 28Ds]$$
$$d\pi = (1/8)[35Dt - 28Ds]$$

$$10Dq(Eq) = 3e_\sigma(L) - 4e_\pi(L)$$
$$10Dq(Ax) = 0.5[3e_\sigma(Z) - 4e_\pi(Z) + 3e_\sigma(L) - 4e_\pi(L)] \qquad (1.85)$$

d) Tetragonal pyramid ML_4Z (90^o angles)

$$DQ = (\sqrt{21})[1.2e_\sigma(L) + 0.3e_\sigma(Z) - 1.6e_\pi(L) - 0.4e_\pi(Z)]$$
$$Dt = (2/35)[3e_\sigma(L) - 4e_\pi(L) - 1.5e_\sigma(Z) + 2e_\pi(Z)]$$
$$= (2/35)[10Dq(L) - 5Dq(Z)]$$
$$Ds = (1/7)[2e_\sigma(L) + 2e_\pi(L) - e_\sigma(Z) - e_\pi(Z)]$$

$$10Dq(Eq) = 3e_\sigma(L) - 4e_\pi(L)$$
$$10Dq(Ax) = 1.5e_\sigma(Z) - 2e_\pi(Z) = 0.5Dq(Z)$$

If L is non-pi bonding such that $e_\pi(L) = 0$, then:

$$e_\sigma(L) = (5/9)[DQ/\sqrt{21} - DT/\sqrt{15}]$$
$$e_\sigma(Z) = (10DQ/9\sqrt{21}) + (20DT/(63\sqrt{15})) + (4DS/7) \qquad (1.86)$$

e) Cis-ML_4Z_2 (pseudo-tetragonal, all angles 90^o)

$$DQ = (1/5)(\sqrt{21})[6e_\sigma(L) + 3e_\sigma(Z) - 4e_\pi(Z) - 8e_\pi(L)]$$
$$Ds = (1/7)[e_\sigma(Z) - e_\sigma(L) + e_\pi(Z) - e_\pi(L)]$$
$$Dt = (1/35)[3e_\sigma(Z) - 3e_\sigma(L) + 4e_\pi(L) - 4e_\pi(Z)]$$
$$= (2/7)[Dq(Z) - Dq(L)]$$

$$d\sigma = (1/4)[15Dt + 12Ds] = (3/4)[(-4DS/7) - (10DT/7\sqrt{15})]$$
$$d\pi = 3Ds - 5Dt = (-3DS/7) + (10DT/7\sqrt{15})$$

$$10Dq(Eq) = 0.5[10Dq(L) + 10Dq(Z)]$$
$$10Dq(Ax) = 10Dq(L)$$

If L is non-pi bonding such that $e_\pi(L) = 0$, then:

$$e_\sigma(L) = (5/9)[(DQ/\sqrt{21}) + (2DT/\sqrt{15})]$$
$$e_\sigma(Z) = (5DQ/9\sqrt{21}) - (20DT/(63\sqrt{15})) - (4DS/7)$$

(1.87)

Within these tetragonal systems some interesting comparisons can be drawn. Consider pairs of analogous complexes such as ML_4Z_2 and MZ_4L_2. Thus writing Dx for either Ds or Dt (but NOT Dq) note:

$$Dx(\text{trans-}ML_4Z_2) = 2\,Dx(ML_5Z)$$
$$Dx(\text{cis-}ML_4Z_2) = -(1/2)\,Dx(\text{trans-}ML_4Z_2)$$
$$Dx(\text{cis-}ML_4Z_2) = -\,Dx(ML_5Z)$$

and

$$Dx(\text{trans-}ML_4Z_2) = -\,Dx(\text{trans-}MZ_4L_2)$$

Table 1.14

COMPARISON OF PARAMETER VALUES AND TRANSITION ENERGIES IN A SERIES OF CHROMIUM(III) CYANIDE DERIVATIVES

Complex	Observed Energies[a]	State	Calculated Energies	Dq	Ds	Dt	B
$Cr(CN)_5(H_2O)^{2-}$	23,500	^4E	23,480	2640(Eq)	387	309	573
	26,400	$^4B_2$	26,400	2100(Ax)			
	29,000	$^4A_2$	29,620				
	32,100	^4E	31,735				
$Cr(CN)(H_2O)_5^{2+}$	17,500	$^4B_2$	17,500	1750(Eq)	-387	-309	560
	19,200	^4E	19,480	2290(Ax)			
	25,455	^4E	25,000				
		$^4A_2$	26,510				
$c\text{-}Cr(CN)_4(H_2O)_2^-$		$^4B_2$	21,460	2146(Eq)	-387	-309	575
	23,420	^4E	23,815	2690(Ax)			
	30,120	^4E	29,240				
		$^4A_2$	30,780				
$c\text{-}Cr(CN)_2(H_2O)_4^+$	18,520	^4E	18,540	2146(Eq)	387	309	575
	21,460	$^4B_2$	21,460	1605(Ax)			
		$^4A_2$	24,535				
	26,455	^4E	26,620				

a) Data from [39]. With Ds = -387, Dt = -309, $d\sigma$ = 1160, $d\pi$ = -190.

$$Dx(ML_5Z) = -Dx(MZ_5L)$$

and hence

$$Dx(cis\text{-}ML_4Z_2) = Dx(MZ_5L)$$

and finally

$$Dq(Eq)(cis\text{-}ML_4Z_2) = Dq(Ax)(ML_5Z) \qquad (1.88)$$

As a test of some of the relationships shown in (1.83-88), Perumareddi [39] analysed the spectra of a series of chromium(III) cyanide derivatives, though the connection between the radial parameters of the cis complexes and those of the penta-ligand systems was not included. In Table 1.14 are collected calculated and observed energies based upon these relationships. Allowing for the fact that the bands are not always fully resolved, and using band averages in these cases, a single set of Ds and Dt values, with appropriate sign, does appear to reproduce the band energies remarkably well. This is the more remarkable since the rather covalent cyanide ion is involved.

In a procedure slightly different from that used by Perumareddi [39], Dt and Ds were fitted to the spectrum of cis-$[Cr(CN)_2(H_2O)_4]^+$, and then the same values of Ds and Dt, with appropriate sign, used for all the other complexes. The value of Dq (= Dq(eq)) was not obtained by use of the above equations, but was fitted to the observed $^4B_2$ band.

Using the data in Table 1.14, note that the experimental Dq values for the NC-Cr-CN, H_2O-Cr-OH_2 and NC-Cr-OH_2 bond vectors are 2640, 1750 and 2146 cm^{-1} respectively. The average band energies of the $^4T_2$ and $^4T_1(F)$ transitions in mer-$Cr(H_2O)_3(CN)_3$ depend only upon these Dq values, and not upon their breakdown in terms of $e_\sigma(CN)$, $e_\pi(CN)$ etc, using the numbers quoted here, the average band energies of the first two transitions in mer-$Cr(H_2O)_3(CN)_3$ are 21,190 and 28,580 cm^{-1}, (B = 575 cm^{-1}), compared with 21,435 and 27,625 cm^{-1} [39], certainly not unreasonable agreement.

Finally some further relationships:

f) Trigonally distorted octahedra

$$D\sigma = DSIG/\sqrt{70}$$
$$Dq = (1/6)(1/\sqrt{30})DQ + (1/60)\sqrt{(7/6)}DTAU$$
$$D\tau = -(3/10)(1/\sqrt{42})DTAU$$

and [36]

$$DTAU/DQ = [(\sqrt{7}) + R(\sqrt{10})]/[(\sqrt{20}) - R\sqrt{(7/2)}] \qquad (1.89)$$

where

$R = 1.5(1/\sqrt{35})[(35/3)\cos^4\phi - 10\cos^2\phi + 1]/[\sin^3\phi\cos\phi]$ where ϕ is the angle between the threefold symmetry axis and a metal ligand bond. Positive DTAU corresponds to trigonal compression and negative DTAU to trigonal elongation for D_{3d} systems [36]. The Liehr [30] - McFarlane [40] parameters are related to NSH via [37]:

$$Dq = DQ/(6\sqrt{30})$$
$$v = 6DTAU/(3\sqrt{42}) - 3DSIG/(\sqrt{70})$$
$$v' = DSIG/(\sqrt{35}) + DTAU/(2\sqrt{21}) \qquad (1.90)$$

g) Rhombic $MX_2Y_2Z_2$ D_{2h} (I) (all angles 90°) [45]

$$DQ = (1/5)(\sqrt{21})[3(e_\sigma(X) + e_\sigma(Y) + e_\sigma(Z)) - 4(e_\pi(X) + e_\pi(Y) + e_\pi(Z))]$$
$$DS = (2e_\sigma(Z) + 2e_\pi(Z)) - (e_\sigma(X) + e_\pi(X)) - (e_\sigma(Y) + e_\pi(Y))$$
$$DT = (1/5)(\sqrt{15})[3e_\sigma(Z) - 1.5(e_\sigma(X) + e_\sigma(Y)) - 4e_\pi(Z) + 2(e_\pi(X) + e_\pi(Y))]$$
$$DU = (\sqrt{3})[(e_\sigma(X) - e_\sigma(Y)) + (e_\pi(X) - e_\pi(Y))]$$
$$DV = 6(1/\sqrt{5})[(3/4)(e_\sigma(Y) - e_\sigma(X)) + (e_\pi(X) - e_\pi(Y))] \qquad (1.91)$$

The following relationships cannot be solved unless one assumption is made; usually one works with at least one non-π-bonding ligand (this set assumes $e_\pi(Y) = 0$). Note also that there is insufficient information in a d^1 species spectrum to obtain all five NSH parameters unambiguously. Thus at least two assumptions would have to be made in this case. F and P term ions have a greater number of observables; in these cases all five NSH parameters (plus Racah B) can be defined.

$$
\begin{aligned}
e_\sigma(Z) &= (1/9)[5(1/\sqrt{21})DQ + (18/7)DS + (10/7)(1/\sqrt{15})DT \\
&\quad + (6/7)(\sqrt{3})DU + (20/7\sqrt{5})DV + 12e_\pi(Y)] \\
e_\sigma(Y) &= e_\sigma(Z) - (2/7)DS - (5/7)(1/\sqrt{15})DT + (2/7)(1/\sqrt{3})DU \\
&\quad + (1/21)(\sqrt{5})DV \\
e_\sigma(X) &= 2e_\sigma(Z) - e_\sigma(Y) - (4/7)DS - (10/7)(1/\sqrt{15})DT \\
e_\pi(Z) &= (3/14)DS - (5/7)(1/\sqrt{15})DT + (1/14)(\sqrt{3})DU \\
&\quad + (5/21)(1/\sqrt{5})DV + e_\pi(Y) \\
e_\pi(Y) &= 0 \text{ (see text)} \\
e_\pi(X) &= (1/7)(\sqrt{3})DU + (10/21)(1/\sqrt{5})DV + e_\pi(Y)
\end{aligned}
\qquad (1.92)
$$

h) Trigonal Bipyramidal (ML_3Z_2) D_{3h}

$$
\begin{aligned}
DS &= -(\sqrt{70})Ds \\
DT &= 3(\sqrt{70})Dt \\
\\
Ds &= -(1/7)[1.5e_\sigma(L) - 2e_\sigma(Z) + 1.5e_\pi(L) - 2e_\pi(Z)] \\
Dt &= -(4/35)[-(27/32)e_\sigma(L) - 1.5e_\sigma(Z) + (9/8)e_\pi(L) + 2e_\pi(Z)] \\
&= (1/28)[16Dq(Ax) + 9Dq(Eq)]
\end{aligned}
\qquad (1.93)
$$

i) Linear ML_2 ($D_{\infty h}$)

If a = M-L bond length, then:

$$
\begin{aligned}
Ds &= -(6Ze/7)(r^2/a^3) \\
Dq &= Ze/6(r^4/a^5) = (3e_\sigma(L) - 4e_\pi(L))/10 \\
\\
Ds &= -6/7(e_\sigma(L) + e_\pi(L)) \\
e_\sigma(L) &= (1/7)[10Dq - 14Ds/3] \\
e_\pi(L) &= -(1/7)[10Dq + 7Ds/2]
\end{aligned}
\qquad (1.94)
$$

For further detail and energy matrices, see [42].

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Chapter 2

Term Diagrams

Many-Electron Energy Levels

2.1 Introduction

In this chapter are discussed the procedures by which the one-electron energy levels derived in chapter 1 can be used to predict the transition assignments and energies for multi-electron systems [1-7] at least in so far as the construction of term diagrams is concerned.

Historically chemists have discussed electronic spectroscopy within the framework of two limiting viewpoints, the weak field, and the strong field. In the weak field, the coupling of the electrons due to interelectronic repulsion into a set of spectroscopic terms is considered more important than interaction with the external ligand field. The energy levels of the molecule are derived by applying the perturbation due to the ligand field, to the free ion spectroscopic states.

In the strong field limit, the ligand field is considered first to couple the electrons, resulting, for example, in the well known real d orbitals and interelectronic repulsion is then added as a perturbation to the ligand field. With very strong fields (high field ligands and heavy transition elements), interelectronic repulsion can sometimes be neglected.

In this chapter term diagrams are constructed using a correlation between the strong field and the weak field approach.

2.2 Correlation Diagrams

2.2 i) Octahedral Complexes

Consider a d^1 ion in an octahedral environment (see Fig.1.3b). The ground state configuration will be $(t_{2g})^1$ and the total electronic wavefunction may be described by the group theoretical term $^2T_{2g}$. Such terminology is analogous to the ^2D free ion term description. The (intra 3d) excited state configuration is $(e_g)^1$ and the excited term ^2E_g. One absorption band is predicted corresponding to the electronic transition $(t_{2g})^1$----$(e_g)^1$ and represented ^2E_g---$^2T_{2g}$. This information can be written in the form of a correlation diagram as shown in Fig.2.1.

Electronic transitions between electron configurations are generally represented with the ground state written on the left. Transitions between total electronic wavefunctions, i.e. shown occurring between the group

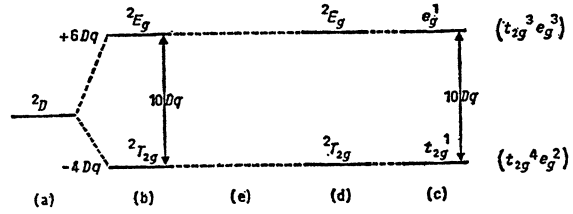

Fig.2.1 Correlation diagram for a d^1 ion in an octahedral field. (a) Spherically perturbed free ion spectroscopic term, (b) group theoretical terms in a weak octahedral field, (c) electronic configurations in a strong octahedral field, (d) group theoretical terms in a strong octahedral field, (e) intermediate field region. The same diagram may also be used for a d^6 ion in an octahedral field. The electronic configurations concerned, which are given in parenthesis to the right of the diagram, will yield spin quintet terms.

theoretical states or free ion terms, are usually written with the excited state on the left and a reverse facing arrow. Electron configurations are written in lower case, while total wavefunctions are written in capitals.

The ions d^1, d^4 ,d^6 and d^9 all have D ground terms. In their high spin state these configurations all yield $T_{2g} + E_g$ with the sense of the splitting depending upon the configuration. Looking at the number of orbital ways of writing each configuration it is easily seen that the T_{2g} term is the ground term for d^1 and d^6, with the inverse splitting, E_g ground term, for d^4 and d^9. The splitting of d^n is the same as d^{n+5} and the opposite of d^{10-n}. These results can be displayed by an 'Orgel diagram' (Fig.2.2) (as first developed by Leslie Orgel [10]).

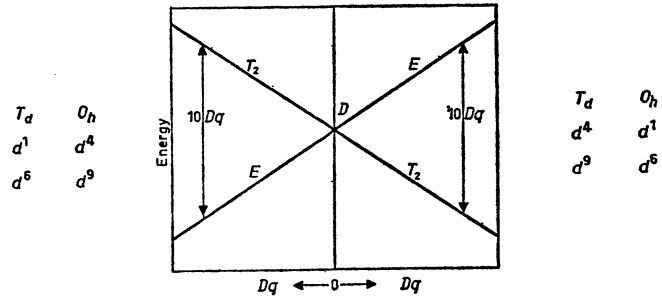

Fig.2.2 'Orgel Diagram' depicting the high spin energy levels for d^1,d^4, d^6 and d^9 ions in both octahedral and tetrahedral fields.

Suppose a d^2 ion is now placed in an octahedral field. The ground state electron configuration in the strong field (see below) $(t_{2g})^2$ would give rise to four terms:

$A_{1g} + E_g + T_{2g} + T_{1g}$ (from the direct product $t_{2g} \times t_{2g}$)

These four terms differ in their interelectronic repulsion energy, but have the same one-electron ligand field energy in the absence of mixing with any other states. There are 6 arrangements for each t_{2g} electron (2 spin and 3 orbital) and 15 arrangements for $(t_{2g})^2$ allowing for the Pauli Exclusion Principle. The total dimension of the four states shown must be 15, but which are spin singlets and which spin triplets ? There are three ways of writing the $(t_{2g})^2$ configuration while maintaining S = 1 (spin triplet) namely $|xz^+,yz^+\rangle$, $|xz^+,xy^+\rangle$ and $|yz^+,xy^+\rangle$ (Fig.2.3), hence the spin triplet has dimension three and MUST correspond either with $^3T_{1g}$ or $^3T_{2g}$. To decide which is correct, it is usual to descend in symmetry [2,7,8]. Thus in D_{4h} symmetry, (xz,yz) transforms as e_g and with an electron in each orbital, $|xz^+,yz^+\rangle$ as $^3A_{2g}$, while the states $|xz^+,xy^+\rangle$ and $|yz^+,xy^+\rangle$ form a degenerate ^3E_g pair. Reference to descent in symmetry or correlation tables shows that:

T_{1g} (in O_h) correlates with $A_{2g} + E_g$ in D_{4h}
T_{2g} (in O_h) correlates with $B_{2g} + E_g$ in D_{4h}

Thus the spin triplet in O_h MUST be $^3T_{1g}$. Hence $(t_{2g})^2$ spans $^1A_{1g} + {}^1E_g + {}^1T_{2g} + {}^3T_{1g}$.

Fig.2.3 The three possible orbital arrangements for the ground state of the d^2 ion ($M_s = 1$) in an octahedral field.

The strong field octahedral excited state configuration, $(t_{2g})^1(e_g)^1$ gives rise to four terms, $^1T_{1g} + {}^3T_{1g} + {}^1T_{2g} + {}^3T_{2g}$ (since the spins of the two electrons are independent of one another (Fig.2.4)) (use group theoretical direct products) and the remaining excited state $(e_g)^2$ to three terms $^1A_{1g} + {}^1E_g + {}^3A_{2g}$. Therefore the $(t_{2g})^2$, $(t_{2g})^1(e_g)^1$, and $(e_g)^2$ configurations, give rise to eleven states and therefore up to ten electronic transitions may be observable. Of these, three are spin-allowed (see chapter 4). The total dimension of these eleven states is seen to be 45.

Clearly the one electron diagrams shown in the previous chapter are not able to predict transitions for many electron species. The creation of term diagrams for many electron species is best presented in the form of correlation diagrams linking the weak

field and the strong field limits.

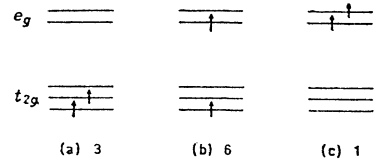

Fig.2.4 Strong field electronic configurations for a d^2 ion (S = 1, M_S = 1) in an octahedral field. (a) Three fold orbitally degenerate ground state, (b) two three fold orbitally degenerate excited states, (c) orbital singlet second excited state.

Consider first that the electrons couple to generate the various spectroscopic terms. In the d^1 case the ten possible orientations (microstates) of the electron according to l = 2, m_l = 2,1,0,-1,-2 and m_s = +1/2, -1/2 have the same energy in the absence of any perturbing environment. These all belong to the ^2D spectroscopic term. The 45 different orientations of the d^2 ion do not all belong to the same spectroscopic term. Specifically because of different measures of interelectronic repulsion for different orientations of the electron vectors, these 45 microstates break down into 5 spectroscopic terms, ^3F (ground term), ^3P, ^1G, ^1D, and ^1S. Within each spectroscopic term each orientation (e.g. the 21 microstates in ^3F) all have the same interelectronic repulsion [5].

> The designation ^nL is used for spectroscopic terms where n is the spin multiplicity, 2S + 1, and L is the total orbital angular momentum where L = 0,1,2,3,4.. is given the symbol S, P, D, F, G etc.

It is no accident that the total dimension of the octahedral group theoretical terms listed above for the d^2 ion is also 45. These 45 terms, generated in the strong field limit by obtaining direct products of the real one electron orbitals must correlate with the 45 microstates contained within the five free ion spectroscopic terms. Standard group theoretical procedures reveal how the free ion terms transform in an octahedral field [8]. The results are shown in Table 2.1. The splitting of the states is independent of their spin multiplicity.

> The group theoretical terms for symmetries lower than octahedral are obtained by using Table 2.1 and a correlation table (see Tables 1.5, 1.6 and Appendix X). For example the D term splits into $E_g + T_{2g}$ in an octahedral field. In a tetragonal field (D_{4h}) the D term transforms as $A_{1g}(E_g) + B_{1g}(E_g) + B_{2g}(T_{2g}) + E_g(T_{2g})$ where the octahedral parents have been included for clarity.

We know how the degeneracy of a term may be lifted, after Table 2.1, but how do we know the relative order ?. This information cannot always be simply obtained but in most cases emerges readily from the strong field approach.

Table 2.1

THE BEHAVIOUR OF S, P ,D, F, AND G TERMS IN CUBIC FIELDS[a]

Free Ion term[b]	Group Theoretical Terms
S (1)	A_{1g}
P (3)	T_{1g}
D (5)	$T_{2g} + E_g$
F (7)	$A_{2g} + T_{2g} + T_{1g}$
G (9)	$A_{1g} + E_g + T_{1g} + T_{2g}$

a) This Table applies to octahedral, cubic (eight coordinate) and tetrahedral fields; in the last case the g subscript should be dropped. Note that if the states are derived from odd electron configurations, e.g. p^n, then the group theoretical terms carry a u subscript (odd).
b) The orbital degeneracy is given in brackets. This will be equal to the sum of the orbital dimensions of the group theoretical terms.

2.2.ii) Correlation diagrams

In Fig.2.1 a simple correlation diagram is shown for d^1 with the weak field free ion terms on the left, and the strong field configurations on the right. The energies in the strong field are readily obtained by summing -4Dq for each t_{2g} electron and +6 Dq for each e_g electron. Interelectronic repulsion energy is ignored in this approximation. WHEN A GROUP THEORETICAL TERM (inclusive of spin designation) APPEARS ONLY ONCE IN AN ENERGY LEVEL DIAGRAM, ITS WEAK FIELD LIMIT ENERGY IS EQUAL TO ITS STRONG FIELD LIMIT ENERGY. This is the case in Fig.2.1 so the energies are readily incorporated into the diagram and the transition energies (10 Dq in this case) readily predicted.
Using Table 2.1 for d^2:

$$\begin{array}{ll} ^3F & ^3A_{2g}\ ,\ ^3T_{2g},\ ^3T_{1g} \\ ^3P & ^3T_{1g} \\ ^1D & ^1T_{2g}\ ,\ ^1E_g \\ ^1G & ^1A_{1g},\ ^1T_{2g},\ ^1T_{1g},\ ^1E_g \\ ^1S & ^1A_{1g} \end{array} \qquad (2.1)$$

In a very strong field, ignoring interelectronic repulsion, the ground state configuration will be $(t_{2g})^2$ lying at an energy -8Dq (Fig.1.3b). The first excited state will be $(t_{2g})^1(e_g)^1$ lying at + 2Dq, and the second excited state is $(e_g)^2$ lying at +12Dq. This provides us with the right hand side of Fig.2.5 (c).

Using the direct products as shown above, these three states split into eleven states when interelectronic repulsion is 'turned on'. These must correlate with corresponding states on the left

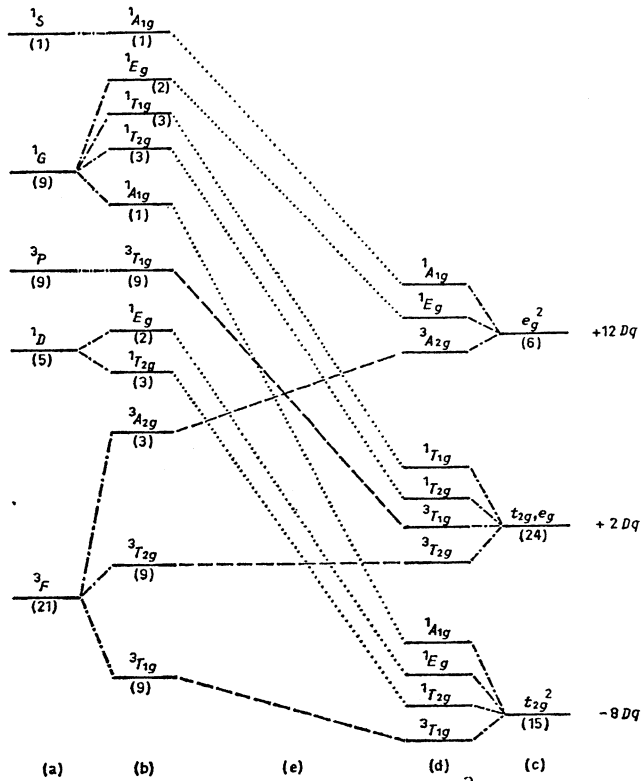

Fig.2.5 Complete correlation diagram for a d^2 ion in an octahedral field. (a) Weak field states, (b,d) Group Theoretical Representations, (c) Strong Field States, (e) Intermediate Field region.

hand side of the diagram.

Fig.2.5 (a) comprises the terms shown in (2.1). The sequence of free ion terms is determined by their interelectronic repulsion energy, shown for a typical case in Fig.2.5. Considering, for the moment, only the high spin terms, the sequence of levels on the left is assumed to be the same as the sequence on the right (this will be proven later). There is a one-to-one correspondance between terms on the left and on the right. Levels of the same symmetry are joined accross the Figure. There is a fundamental rule for the construction of such energy level diagrams that terms of the same symmetry label (spin and orbit) cannot cross one another. This removes the ambiguity concerning the $^3T_{1g}$ levels. The lower $^3T_{1g}$ level on the right correlates with the lower on the left, and the upper on the right with the upper on the left.

This provides the high spin d^2 correlation diagram

unambiguously. Three spin allowed transitions are expected, $^3T_{1g}$---$^3T_{1g}$, $^3A_{2g}$---$^3T_{1g}$ and $^3T_{1g}(P)$---$^3T_{1g}$. Since the $^3T_{2g}$ and $^3A_{2g}$ states occur only once in the diagram, their energies will be + 2Dq and +12Dq respectively. The $^3T_{1g}$ term occurs twice. The wavefunctions for these two terms are mixed together by the crystal field operator so that their energies depend upon the relative size of Dq and B. In the strong field limit, the energies are -8Dq and 2Dq as we have already seen. In the weak field limit they are -6Dq and (15B + 6Dq) relative to the ^3F term energy as zero. Methods for obtaining such numbers in the weak field are presented later.

The transition energies in the weak field limit would be:

$$\begin{aligned} \nu_1 \quad & ^3T_{2g}\text{------}{}^3T_{1g} && = 8Dq \\ \nu_2 \quad & ^3A_{2g}\text{------}{}^3T_{1g} && = 18\ Dq \\ \nu_3 \quad & ^3T_{1g}(P)\text{---}{}^3T_{1g} && = 6Dq + 15B \end{aligned} \qquad (2.2)$$

These are only approximate because the ground state energy, due to mixing of this state with $^3T_{1g}(P)$, will have energy dependent upon both Dq and B. However it is exactly true that the energy separation $\nu_2 - \nu_1 = 10Dq$.

The sequence of spin singlets in Fig.2.5 cannot be judged without recourse to calculation. Given the sequence on the right, and the non-crossing rule, correlation to the corresponding singlets on the left is straightforward.

A correlation diagram for d^3 spin allowed terms is shown in Fig.2.6. Its construction follows that described above for d^2. In this case the strong field ground state is $(t_{2g})^3$ whose high spin term can be shown by direct product to be $^3A_{2g}$ [8].

The wavefunction is described by the determinant $|xz^+,yz^+,xy^+\rangle$ which can readily be shown to transform as A_{2g} under the O_h symmetry operators.

The $^3A_{2g}$ level will lie at -12Dq (3 x -4Dq). The first excited state, namely, $(t_{2g})^2(e_g)^1$ gives rise to $^4T_{1g}$ + $^4T_{2g}$ (plus doublet states) and lies at -2Dq ((2 x -4Dq) + 6Dq), and the second excited state $(t_{2g})^1(e_g)^2$ yields $^4T_{1g}$, at + 8Dq. The representations arising from different configurations are summarised in Table 2.2. Using the non-crossing rule, construction of the correlation diagram for d^3 follows automatically. Three spin allowed transitions are expected from $^4A_{2g}$ to the three excited quartets. The energy of the first transition, $^4T_{2g}$---$^4A_{2g}$ is exactly 10Dq since the two lower states occur only once and have exact energies independent of B.

$$\begin{aligned} \nu_1 \quad & ^4T_{2g}\text{------}{}^4A_{2g} && = 10Dq \\ \nu_2 \quad & ^4T_{1g}(F)\text{---}{}^4A_{2g} && = 18Dq \\ \nu_3 \quad & ^4T_{1g}(P)\text{---}{}^4A_{2g} && = 12Dq + 15B \end{aligned} \qquad (2.3)$$

Because of the 'non-crossing rule' the energy of ν_2 must always be less than ν_3 which is not required in the d_2 case (2.2) where these two transitions may cross. The energies shown in (2.2) and (2.3) are limiting approximations when the field strength is very small

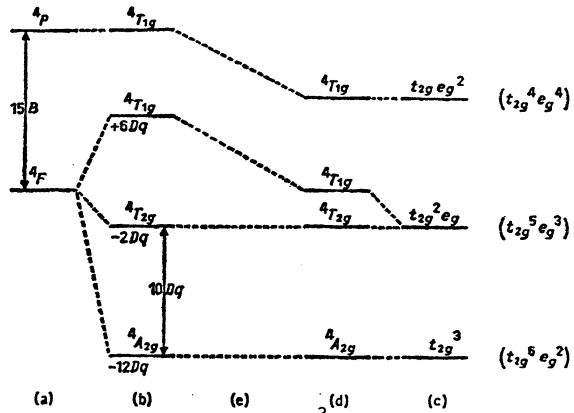

Fig.2.6 Correlation diagram for a d^3 ion (S = 3/2, M_S = 3/2) in an octahedral field. (a) - (e) as in Fig.2.5. This diagram also suffices for the d^8 ion in an octahedral field, in which case the electron configurations concerned give rise to spin triplet terms.

compared with interelectronic repulsion (10Dq << B). These equations should NOT be used for real complexes. Generally the 'intermediate field' region is appropriate, where 10Dq is not so weak that the weak field approximation holds, yet not so strong that B can be ignored. The detailed equations for this region are presented later.

Note that the F term splitting in d^3 is the inverse of that seen in d^2. The F term always splits in the sequence T_{1g}, T_{2g}, A_{2g}, with T_{2g} in the middle. The ground state depends upon the d^n configuration. The d^7 and d^8 ions also possess F ground terms and will split in a similar fashion to d^2 and d^3 respectively.

This is also most readily seen by writing the ground state in the strong field representation, e.g. $(t_{2g})^5(e_g)^2$ for d^7 and asking how many ways the configuration can be written (following Fig.2.3). In this case three orbital representations are possible and hence the ground state for d^7 is $^4T_{1g}$ and the splitting parallels d^2. The high spin correlation diagram for d^7 will be the same as for d^2 (and d^8 the same as d^3). An 'Orgel diagram' appropriate for d^2, d^3, d^7 and d^8 systems is shown in Fig.2.7.

So far the 'high spin' or 'spin free' configurations have been dealt with. With d^4 - d^7, 'low spin' or 'spin-paired' configurations are possible. As shown in Table 2.2 if the four electrons in d^4 are constrained to occupy the t_{2g} sub-shell (spin-paired) $(t_{2g})^4$, then the maximum spin possible is a spin triplet rather than the spin quintet of spin free d^4 $(t_{2g})^3(e_g)^1$. This spin triplet correlates with a spin triplet free ion term which is an excited state in the free ion. A partial diagram for d^4 is shown in Fig.2.8. This diagram clearly shows how the ground state is dependent upon the relative magnitudes of the crystal field strength, determined via Dq, and the interelectronic repul-

Table 2.2
GROUP THEORETICAL TERM REPRESENTATIONS FOR t_{2g} AND e_g CONFIGURATIONS IN CUBIC FIELDS. THE g SUBSCRIPT SHOULD BE OMITTED FOR TETRAHEDRAL STEREOCHEMISTRY IN CUBIC FIELDS[a]

$(t_{2g})^6$ and $(t_{2g})^0$	$^1A_{1g}$
$(t_{2g})^5$ and $(t_{2g})^1$	$^2T_{2g}$
$(t_{2g})^4$ and $(t_{2g})^2$	$^3T_{1g} + {}^1T_{2g} + {}^1E_g + {}^1A_{1g}$
$(t_{2g})^3$	$^4A_{2g} + {}^2E_g + {}^2T_{2g} + {}^2T_{1g}$
$(e_g)^4$ and $(e_g)^0$	$^1A_{1g}$
$(e_g)^3$ and $(e_g)^1$	^2E_g
$(e_g)^2$	$^3A_{2g} + {}^1E_g + {}^1A_{1g}$

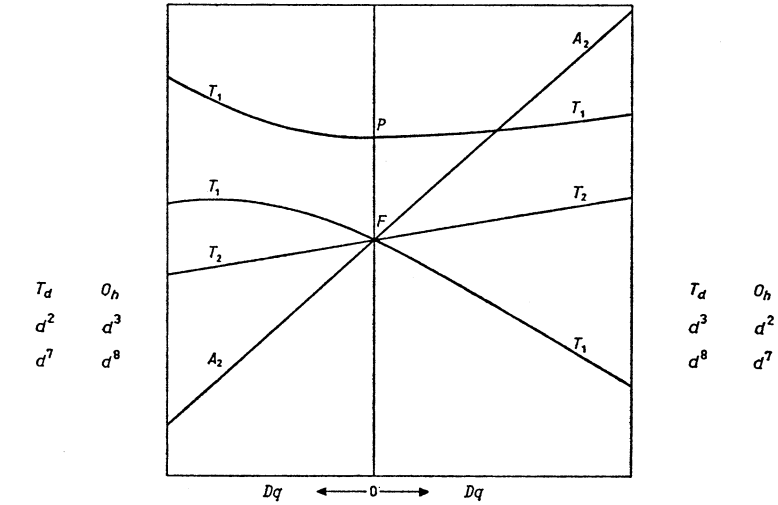

Fig.2.7 'Orgel Diagram' depicting the high spin energy levels for d^2, d^3, d^7 and d^8 ions in both octahedral and tetrahedral fields.

sion energy. Because of the orbital degeneracy of the t_{2g} sublevel, the spin singlet states of $(t_{2g})^4$ can never become ground states in an octahedral (or cubic field). In lower symmetry fields however, spin singlet ground states may be possible. Note that both the spin triplets and especially the spin singlets have a large degree of interelectronic repulsion energy as is obvious from their elevated position on the left-hand side of the diagram. However these states have very negative crystal field stabilisation energy. When Dq is sufficiently greater than B, then the spin triplet can become the ground term as shown in the middle to right of the diagram.

Despite the complexity of this diagram, whose detailed

construction requires more detailed calculation, only one spin-allowed transition is possible for the high spin state, namely $^5T_{2g}$---^5E_g. However in low spin complexes with a $^3T_{1g}$ ground term, there are a larger number of possible spin-allowed transitions to higher excited spin-triplets, some of which are shown in Fig.2.8.

2.2.iii) Tanabe-Sugano diagrams

Correlation diagrams could be drawn for high d configurations, but they are rather clumsy to use. In 1954 Tanabe and Sugano introduced an alternative way of presenting the ground and excited states possible for a d^n configuration in a cubic field [9]. Figs 2.9, 2.10 show such Tanabe-Sugano diagrams (as they are now known) for d^2 and d^3. The complete equations including configurational interactions between states of like symmetry were used to construct this diagram. It is presented as a function of E/B (E is the energy of a state) versus Dq/B. The ground state is drawn as the horizontal baseline, so that a vertical line to an excited state is a direct measure of the transition energy, in terms of Dq and B. An experimental spectrum can be mapped to this diagram and values of Dq and B 'read off'.

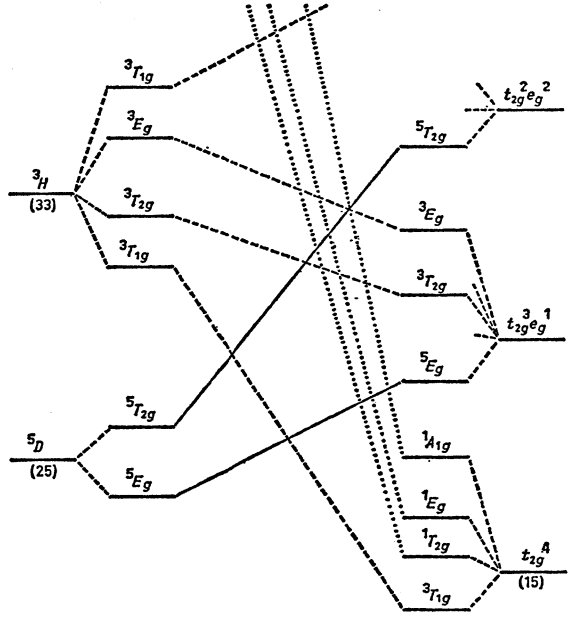

Fig.2.8 Partial correlation diagram for d^4. Not to scale.

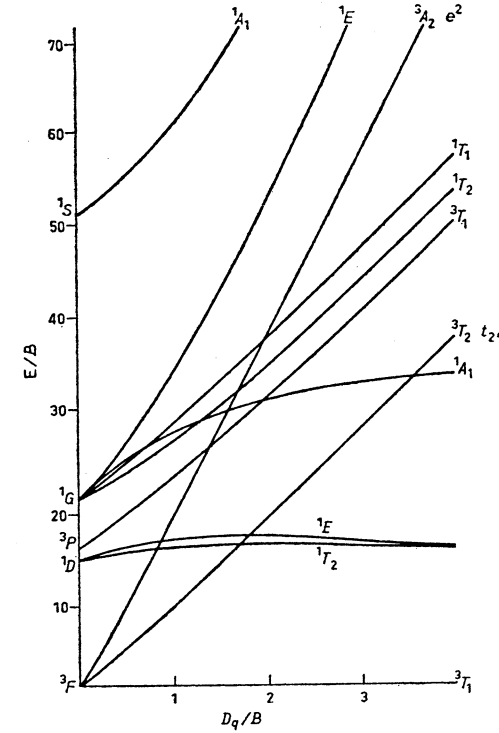

Fig.2.9 Energy level diagram (Tanabe-Sugano) for d^2 ions in an octahedral field (C = 4.42B).

Fig.2.11 for d^4 appears different from the previous diagrams because it displays both the high and low spin ground state possibilities. When Dq/B is large enough, the $^3T_1$ term displaces the ^5E as ground term. Note that $^3T_1$ has a steeply negative slope on the left of the diagram. It has a crystal field energy of -16Dq (4 electrons in t_{2g}) so that it becomes progressively more stable as Dq increases. When it crosses $^3T_1$, it becomes the new ground state. However in so doing, it is drawn, for the purpose of this diagram, as a horizontal line of zero slope. There is therefore an apparent abrupt change in the excited state energies. It is important to recognise that this does not occur in reality, but is an artifact of the method of construction of the diagram. It is true however that the spectra of high spin d^4 species with only one spin-allowed transition, will be very different from the spectra of low-spin d^4 species with many spin-allowed transitions. Note incidentally that transitions to spin singlet excited states are forbidden from both ground states (see Chapter 4).

The d^5 situation is shown in Fig.2.12 where a $^6A_1$ ground state is observed having five unpaired electrons, one unpaired electron in each d orbital. There is only one orbital way of writing this

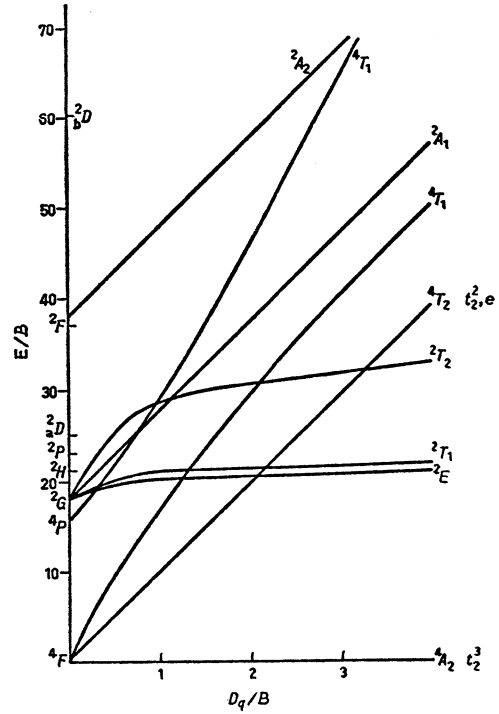

Fig.2.10 Energy level diagram (Tanabe-Sugano) for d^3 ions in an octahedral field (C = 4.5B).

ground state, and hence it is an orbital singlet, and, more interestingly, there are no possible spin-sextet excited states. All d-d excitations necessarily lead to lower spin states. There are therefore no spin allowed transitions in spin free d^5 complexes resulting in only weak d-d absorption due to spin forbidden transitions. This is the reason for the very pale colour of most manganese(II) complexes.

States such as Fig.2.12f are excited states because they have higher spin repulsion energy than Fig.2.12a. However, to first order, they are Dq independent because there is no electron jump between t_{2g} and e_g. They appear as horizontal lines in the Tanabe-Sugano diagram for d^5, Fig.2.13, with energies functions of B and C relative to the ground state. As will be seen in Chapter 4, they generally appear as much sharper transitions than the Dq dependent transitions. The strong field, low spin configuration for cubic d^5 is $(t_{2g})^5$ with S = 1/2. Spin allowed transitions are permitted from this level as illustrated on the right hand side of Fig.2.13.

The configuration d^6 is the hole equivalent of d^4. The same

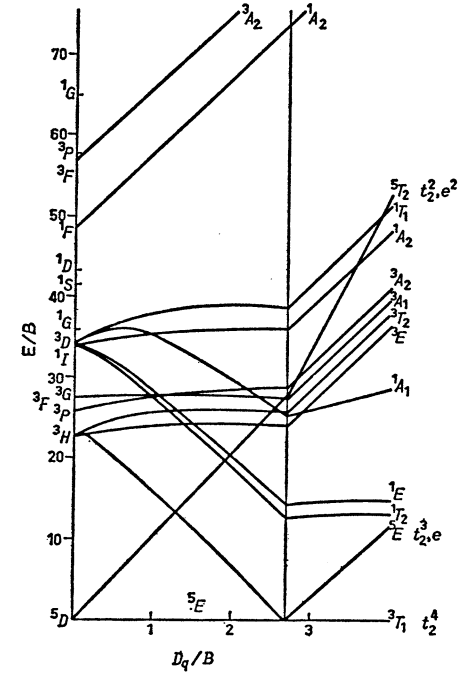

Fig.2.11 Energy level diagram (Tanabe-Sugano) for d^4 ions in an octahedral field (C = 4.61B).

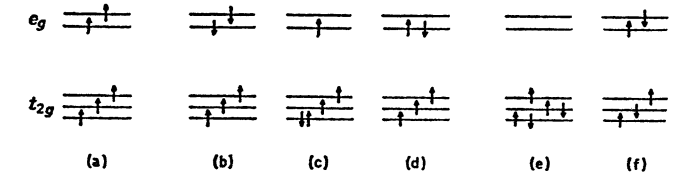

Fig.2.12 Some strong field electronic configurations for the d^5 ion in an octahedral field. (a) High spin ground state (S = 5/2, M_s = 5/2). Excited states (b) S = 5/2, M_s = 1/2, (c) S = 3/2, M_s = 3/2, (d) S = 3/2, M_s = 3/2, (e) S = 1/2, M_s = 1/2 and (f) S = 5/2, M_s = 1/2.

high spin states are obtained but the splitting is inverted. In this case high spin d^6, (^5D) $(t_{2g})^4(e_g)^2$ yields a $^5T_2$ ground term and an ^5E_g excited high spin term again separated by 10Dq. The Tanabe-Sugano diagram in Fig.2.14 illustrates a wealth of excited spin triplets and spin singlets which can also arise from the d^6 configuration. In the low spin state, the completely filled $(t_{2g})^6$

Table 2.3

FREE ION GROUND STATES AND GROUP THEORETICAL GROUND TERMS FOR d^n CONFIGURATIONS IN OCTAHEDRAL AND TETRAHEDRAL FIELDS

Configuration	Free Ion State	Octahedral field ground state		Tetrahedral field ground state
		High spin	Low spin	
d^1	^2D	$^2T_{2g}$		^2E
d^2	^3F	$^3T_{1g}$		$^3A_2$
d^3	^4F	$^4A_{2g}$		$^4T_1$
d^4	^5D	^5E_g	$^3T_{1g}$	$^5T_2$
d^5	^6S	$^6A_{1g}$	$^2T_{2g}$	$^6A_1$
d^6	^5D	$^5T_{2g}$	$^1A_{1g}$	^5E
d^7	^4F	$^4T_{1g}$	^2E_g	$^4A_2$
d^8	^3F	$^3A_{2g}$		$^3T_1$
d^9	^2D	^2E_g		$^2T_2$

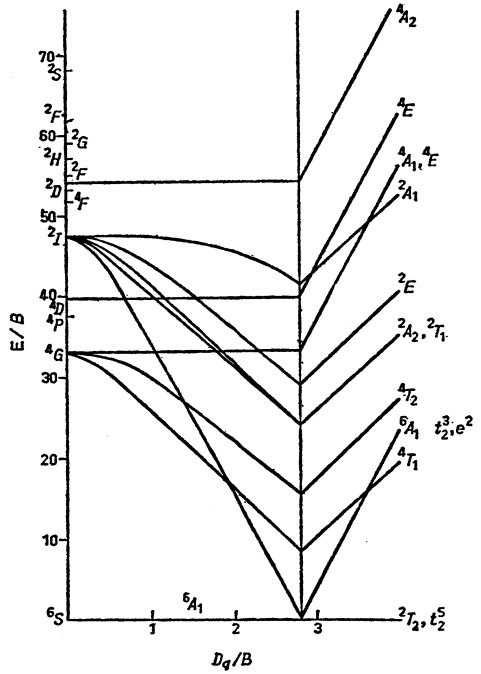

Fig.2.13 Energy level diagram (Tanabe-Sugano) for d^5 ions in an octahedral field (C = 4.477B).

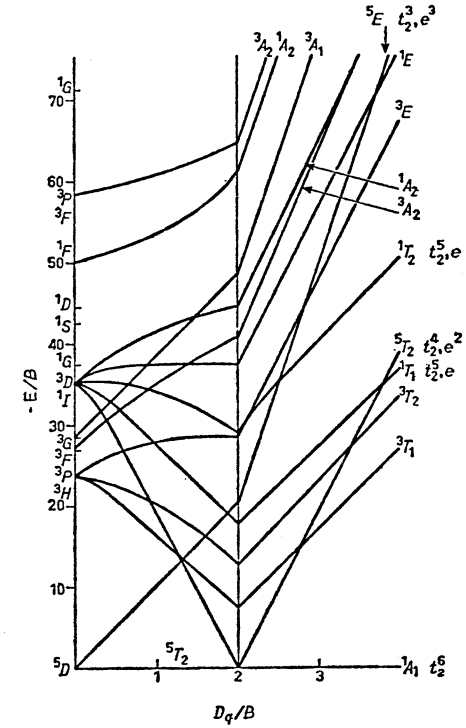

Fig.2.14 Energy level diagram (Tanabe-Sugano) for d^6 ions in an octahedral field (C = 4.8B).

configuration yields the diamagnetic $^1A_1$ ground term. Excitation to the configuration $(t_{2g})^5(e_g)^1$ in which the odd spins in the t_{2g} and e_g configurations are spin opposed, yield several important spin singlet excited states ($^1T_1$, $^1T_2$) to which spin allowed transitions can occur (Fig.2.14). These transitions are clearly seen in many low spin d^6 species such as octahedral cobalt(III) complexes. There will be spin parallel versions of $(t_{2g})^5(e_g)^1$ yielding $^3T_1$ and $^3T_2$. Weak spin forbidden transitions to these spin triplets are often seen especially with the heavier transition elements of high spin-orbit coupling constant (see Chapter 4).

The high spin d^7 configurations are the hole equivalents of d^3. Their correlation diagram would parallel that for d^3 (Fig.2.6) but the F term splitting is inverted. It does, in fact, have the same splitting pattern as does d^2 (Fig.2.5) because it differs from d^2 by 5 d electrons, one in each orbital, and this is a totally symmetric difference. The Tanabe-Sugano diagram is show in Fig.2.15. The low spin state has the configuration $(t_{2g})^6(e_g)^1$ having an odd unpaired electron in the e_g subshell. There are many

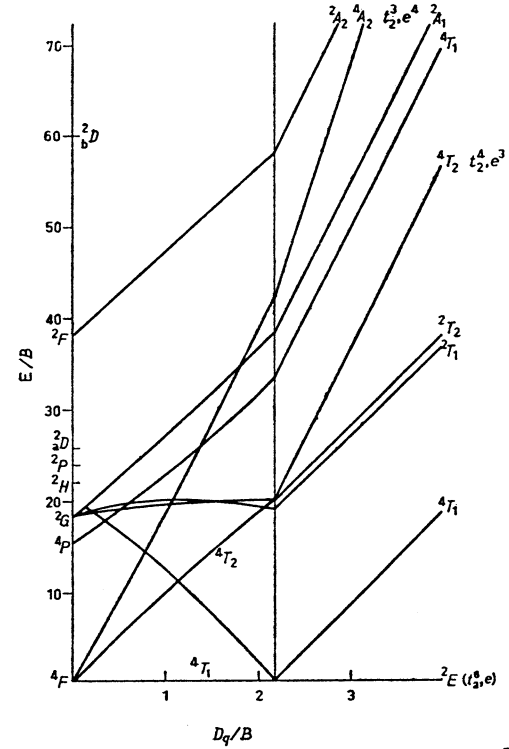

Fig.2.15 Energy level diagram (Tanabe-Sugano) for d^7 ions in an octahedral field (C = 4.633B).

other spin doublet states to which spin allowed transitions may occur in low spin d^7 derivatives (Fig.2.15).

Octahedral complexes with a d^8 configuration can only exist with S = 1, arising from $(t_{2g})^6(e_g)^2$. The correlation diagram is the same as for d^3 (Fig.2.6) and the Tanabe-Sugano diagram is shown in Fig.2.16. The two low lying spin singlets have almost horizontal lines. Their most important configuration is $(t_{2g})^6(e_g)^2$ in which the two spins in $(e_g)^2$ are opposed. They would be horizontal except that they are mixed with higher lying spin singlet states which do depend upon Dq. With d^9 we return to inverse d^1 and a trivial diagram (Fig.2.1) with only two states.

Table 2.3 summarises the ground state group theoretical terms for the d^n configurations.

Tanabe-Sugano diagrams are very useful for obtaining a rapid idea of the transitions to be expected in a given region of the spectrum of an octahedral complex. For a given value of Dq, and B, for example, the spin-allowed and spin-forbidden transition energies can be readily, but approximately evaluated. The approximation arises not because the diagrams are inaccurate, but

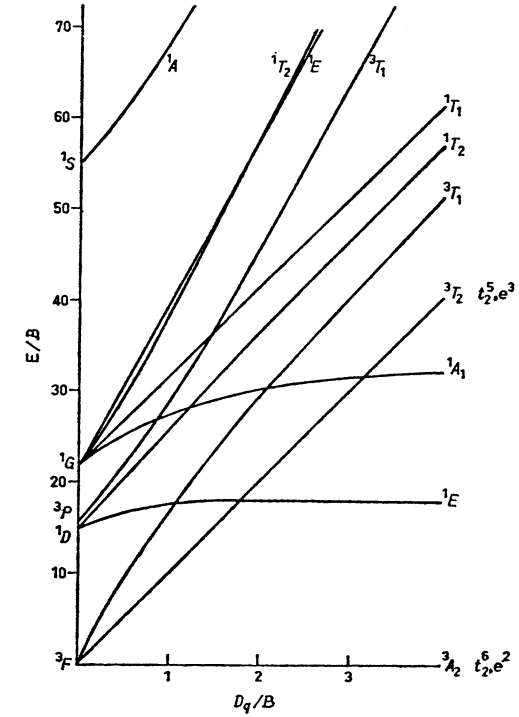

Fig.2.16 Energy level diagram (Tanabe-Sugano) for d^8 ions in an octahedral field (C = 4.709B).

because published versions are almost invariably of small scale. To overcome this problem, at least for high spin ions, in Appendix III are reported Tables of ratios of transition energies versus Dq/B values, which can be used to rapidly solve the energy matrices in an acceptably accurate fashion. This is especially valuable for the student who does not have access to a computer and appropriate program.

2.2.iv) Tetrahedral Species

Recall that the d orbital splitting diagram for a tetrahedral species is the inverse of that for an octahedral species. Therefore tetrahedral d^2 has the same diagram, omitting g subscripts, as octahedral d^2 but with the term splitting inverted. It is therefore possible to use the 'Orgel diagrams' (Fig.2.1, 2.7) for tetrahedral as well as octahedral species. These provide a ready and obvious means of assigning the spin allowed transitions for all high spin cubic d^n ions. For further understanding of the tetra-

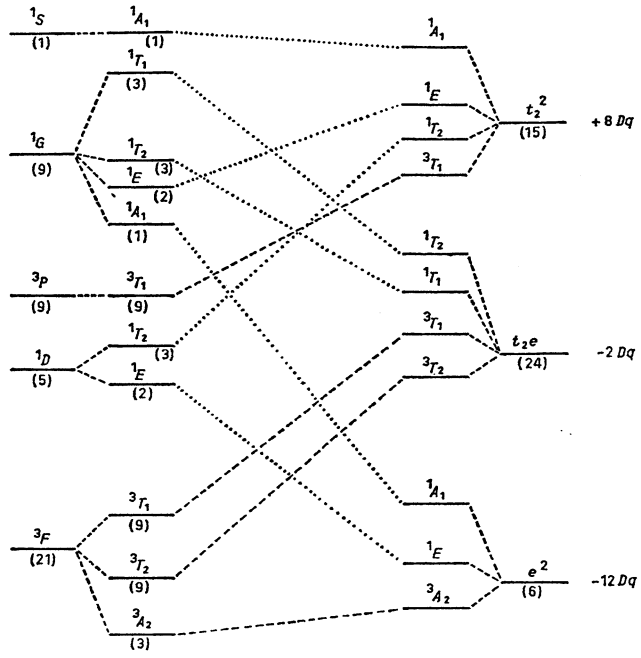

Fig.2.17 Correlation diagram for a d^2 ion in a tetrahedral field,

hedral system, a correlation diagram for d^2 in a tetrahedral field is shown in Fig.2.17.

2.3 Term Diagrams in Lower Symmetries

If a d^1 ion were to be placed into a tetragonal environment the ground state configuration would be $(e_g)^1$ (according to Fig.1.3c) and there would be three possible excited state configurations $(b_{2g})^1$, $(a_{1g})^1$ and $(b_{1g})^1$. By utilising the selection rules (Chapter 4) three possible transitions are expected, expressed, by capitalising the d^1 configurations, as $^2B_{2g}$---^2E_g, $^2A_{1g}$---^2E_g, and $^2B_{1g}$---^2E_g respectively. The energies of these transitions would correspond with the differences in energy between the corresponding one-electron levels as shown in Chapter 1.

The one electron d orbital splitting diagrams can be used as a basis to generate qualitative term diagrams for any multi-electron species in any environment assuming knowledge of group theoretical direct products [7,8]. As an example, a correlation diagram for the high spin (S = 1) levels of d^8 in a D_{4h} environment is derived.

The crystal field diagram appropriate for such species is shown

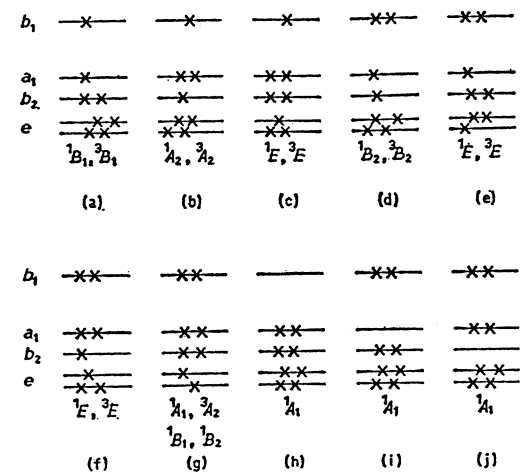

Fig.2.18 Electronic configurations for d^8 ions in a square planar or tetragonal (D_{4h}) field. Neglecting interelectronic repulsion, and assuming that the one electron orbital energies are qualitatively as shown, the configurations (a) to (g) are given in order of increasing energy. (Δ_1 = splitting of t_{2g} level, Δ_3 = splitting of e_g level; $\Delta_1 < \Delta_3 < 10Dq$).

to the right of Fig.1.7. Begin by writing all possible d^8 electron occupations of these orbitals as illustrated in Fig.2.18. The ground state is the spin triplet version of Fig.2.18a; all other configurations are excited states. The representations spanned by each configuration are obtained by generating the direct product of all partially filled orbitals. Filled orbitals are ignored because they span $^1A_{1g}$. Configurations (a) through (g) have two partially filled orbitals whose electrons may be paired, yielding spin singlets, or unpaired yielding spin triplets. The total spin-orbit degeneracy of configurations (a-j) is 45 equal to the number of microstates in d^2.

The corresponding weak field free ion terms may be deduced from Table 2.1 with a suitable correlation Table (see Table 1.5, 1.6, Appendix X):

They are (high spin terms only):

$$\begin{array}{lll} ^3A_{2g} & ^3T_{2g} \text{ and } ^3T_{1g} & \text{in } O_h \\ ^3B_{1g} & ^3E_g + ^3B_{2g} \quad ^3E_g + ^3A_{2g} & \text{in } D_{4h} \end{array} \qquad (2.4)$$

The resultant spin triplet correlation diagram is shown in Fig.2.19. Bearing in mind the 'non-crossing' rule, the correlation is unique. The diagram may be used to predict the spin allowed transitions in D_{4h} d^8 complexes but this is left until selection rules have been dealt with in Chapter 4.

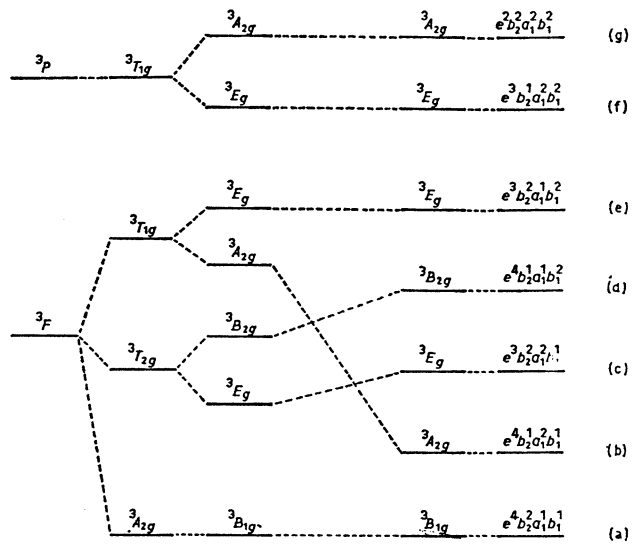

Fig.2.19 Correlation diagram (spin triplets only) for a d^8 ion in a square or tetragonal (D_{4h}) field. The strong field energy order is derived from Fig.2.18. The sense of the splitting of the various orbital triplets is assumed and need not necessarily be as depicted.

Notice that at least in the strong field limit, the components of $^3T_{1g}(F)$ split in the inverse sense to those of $^3T_{1g}(P)$. This is a general result for the splitting of these pairs of T_{1g} states in lower symmetry fields.

Diagrams of this type can provide a powerful diagnostic tool for the assignment of low symmetry electronic spectra. However a secure assignment cannot be obtained without detailed calculation and other techniques such as single crystal polarised light spectroscopy, as detailed in Chapters 3 and 4.

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Chapter 3

Many-Electron Energy Levels Quantitative Considerations

3.1 Introduction

The procedures by which the one-electron energy levels derived in chapter 1 can be used to calculate the energies for multi-electron systems, i.e. to generate quantitatively the term diagrams discussed in chapter 2, are the topic of this chapter.

The 'classical' weak and strong field methods which have been used for many years and which are covered in many prior texts [1-4] are discussed briefly and then we discuss in rather more detail the use of the Wigner-Eckart theorem as a means of solving this problem. We also demonstrate how the Normalised Spherical Harmonic Hamiltonian (NSH) procedure is well suited to the use of the Wigner-Eckart theorem and also consider the use of the AOM model in a many-electron system. Various molecular orbital approaches, such as the X-alpha procedure, are currently under intense development as a means of predicting transition energies and at the same time predicting many other properties of the molecule. It is likely that these will become more prominent in the future as access to the necessary programs and computers becomes more general. Because they are accessible today to only a limited number of researchers, they are not covered in this text.

3.2 Strong Field Analysis [1-6]

Conceptually the strong field approach in which use is made of real orbitals and interelectronic repulsion is subsequently included, is probably the easiest to understand, easier perhaps than visualising a set of weak field wavefunctions. Begin by considering in rather more detail the initial wavefunctions or basis states which initiate the calculation.

It has already been demonstrated that the choice of basis states for a particular calculation is crucial. The final wavefunctions obtained after full calculation will generally be very complex in appearance. Where only energy level information is required, it will probably not prove necessary to derive these final functions. For example, if intensity or magnetic calculations are to be made, then derivation of these wavefunctions will be necessary. However the functions, called basis functions, with which the problem is begun should be the most complete functions which can nevertheless be written down by inspection. In the strong field case they will almost invariably be the real orbitals. In the weak field they may be spectroscopic state wavefunctions ($|LSJM_J\rangle$ or $|LSM_LM_S\rangle$ kets for example), or imaginary d orbital populations. As long as the basis is complete it does not matter in the final outcome where the problem is begun, though the amount of work necessary to complete the calculation will depend upon this choice. We return to this problem later in this chapter.

Earlier the one-electron functions were simply written as $|2\rangle$ or $|xz\rangle$. How are many-electron functions to be written ? Consider two real orbital populations, for example xz^+, and yz^+, where the superscript + implies an electron of spin +1/2 occupies each of these orbitals (these are so-called spinors, spin-orbitals, because they simultaneously define both orbit and spin). It is tempting to write the function (xz^+,yz^+) to represent this two electron wavefunction. However wavefunctions must be anti-symmetric with respect to exchange, are usually normalised to unity, and are orthogonal to one another such that the integral $\langle\psi(A)|\psi(B)\rangle$ is unity if $\psi(A) = \psi(B)$, but zero otherwise.

Using the spinors indicated above, such a function may be written:

$$\psi_i = (1/2)^{1/2}[xz^+(1)yz^+(2) - yz^+(1)xz^+(2)] \qquad (3.1)$$

where dummy electron labels, (1) and (2) are used. If (1) and (2) are interchanged, this function changes sign, i.e. it is anti-symmetric to exchange. Moreover the square root function in front assures that it is normalised to unity, and it is assumed that the functions xz^+ and yz^+ are orthogonal.
Thus function (3.1) is a suitable two electron function. It may be written in a determinantal form, and generalised for n electrons as follows:

$$\psi(A) = 1/(n!)^{1/2} \begin{vmatrix} a_1(1) & a_2(1) & \ldots\ldots & a_n(1) \\ a_1(2) & a_2(2) & \ldots\ldots & a_n(2) \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots \\ a_1(n) & a_2(n) & & a_n(n) \end{vmatrix} \qquad (3.2)$$

Since the magnitude of a determinant changes sign with interchange of columns, this function (3.2) is anti-symmetric with respect to exchange as required. Rather than write out such a function in full, it is generally symbolised by its diagonal element, without specifically citing the normalisation constant which is assumed to be present. Thus function (3.2) can be written - $(a_1,a_2,a_3\ldots a_n)$. The earlier example, (xz^+,yz^+) is therefore considered to be a determinantal function which, when fully expanded, would indeed be (3.1). One of the most important attributes to remember is that, since it is determinantal, $(xz^+,yz^+) = -(yz^+,xz^+)$. In this book, such functions will be written inside round brackets.

As example of a many-electron calculation consider the generation of matrices for d^2 in a tetragonal trans-ML_4Z_2 complex

with the Z ligands along the z axis.

Using the energy level diagram in Fig.1.3c, the strong field orbital populations of the various energy levels are readily described. Considering first only spin-triplet configurations, electrons are simply fed into each energy level to generate bases whose symmetry is obtained by group theoretical direct products.

The bases, B_i, written as determinantal wavefunctions, are obtained very readily:

$$
\begin{array}{ll}
B_1 = (xz^+,yz^+) & {}^3A_{2g} \\
B_2 = (1/\sqrt{2})[(xz^+,xy^+) + (yz^+,xy^+)] & {}^3E_g \\
B_3 = (1/\sqrt{2})[(xz^+,xy^+) - (yz^+,xy^+)] & {}^3E_g \\
B_4 = (1/\sqrt{2})[(xz^+,z^{2+}) + (yz^+,z^{2+})] & {}^3E_g \\
B_5 = (1/\sqrt{2})[(xz^+,z^{2+}) - (yz^+,z^{2+})] & {}^3E_g \\
B_6 = (1/\sqrt{2})[(xz^+,x^2\text{-}y^{2+}) + (yz^+,x^2\text{-}y^{2+})] & {}^3E_g \\
B_7 = (1/\sqrt{2})[(xz^+,x^2\text{-}y^{2+}) - (yz^+,x^2\text{-}y^{2+})] & {}^3E_g \\
B_8 = (xy^+,z^{2+}) & {}^3B_{2g} \\
B_9 = (xy^+,x^2\text{-}y^{2+}) & {}^3A_{2g} \\
B_{10} = (z^{2+},x^2\text{-}y^{2+}) & {}^3B_{1g}
\end{array}
\qquad (3.3)
$$

Write all possible combinations of two electrons in Fig.1.3c, while maintaining S = 1. In the case of the e (xz,yz) level, there is no distinction between placing an electron in xz or in yz, or, in other words, there is a symmetry operation which mixes xz and yz (C_4). Thus in this case linear in-phase and out-of-phase combinations should be used (B_2 and B_3 etc). These coupled linear combinations are appropriately symmetry adapted to the D_{4h} group. Thinking in terms of a correlation diagram for the weak field (e.g. Fig.2.19) all these states must correlate to the two spin triplets, ^3F and ^3P, with a total orbital degeneracy of 10; hence ten real orbital bases are expected.

The one-electron energies of each of these states are also written very readily using, for example, the classical crystal field or the AOM model. The energies of each populated orbital are simply multiplied by the number of electrons therein, one or two, and the results added together. In this procedure note that the crystal field operator is a one-electron operator which operates on one electron at a time. Any other electrons must be 'integrated out', that is separated out as overlap integrals which are, in effect, Kronecker delta functions, e.g.:

$$\langle abc|F|abc\rangle = \langle a|F|a\rangle\langle b|b\rangle\langle c|c\rangle + \langle b|F|b\rangle\langle a|a\rangle\langle c|c\rangle + \langle c|F|c\rangle\langle a|a\rangle\langle b|b\rangle.$$

If the entire element is not to vanish, all of these integrated out overlap integrals must be unity. Since they all multiply one another, the entire function will disappear if one is zero (see Appendix VIII). Hence, using the crystal field data in Fig.1.3c or Eqn.1.29:

$$
\begin{aligned}
\langle (xy^+,z^{2+})|V_{cft}|(xy^+,z^{2+})\rangle &= \langle xy^+|V|xy^+\rangle\langle z^{2+}|z^{2+}\rangle \\
&\quad + \langle z^{2+}|V|z^{2+}\rangle\langle xy^+|xy^+\rangle \\
&= (-4Dq + 2Ds - Dt) + (6Dq - 2Ds - 6Dt) \\
&= 2Dq - 7Dt
\end{aligned}
\qquad (3.4)
$$

where the overlap integrals $\langle xy^+|xy^+\rangle$ and $\langle z^{2+}|z^{2+}\rangle$ are unity.

The expansion of the determinantal functions is carried out as though they were simply products of one electron functions, which, of course, they are. The implicit normalisation constant would provide a factor 0.5; however if the determinantal function is expanded fully as in (3.1), then twice as many integrals would be obtained on the right of (3.4) as now. Since the energy is doubled and then multiplied by 0.5, the result is precisely the same. Thus, in most cases (with one-electron operators), determinantal functions can be expanded simply as products of one electron functions with the normalisation factor 'taking care of itself'. Note that the states must be ordered to ensure that they expand out correctly. If $\langle (xz^+,yz^+)|V_{cft}|(yz^+,xz^+)\rangle$ is expanded to generate $\langle xz^+|V|yz^+\rangle\langle yz^+|xz^+\rangle$ +... etc this leads to zero, because the overlap terms ($\langle yz^+|xz^+\rangle$ etc.) vanish. This particular function must be written as $-\langle (xz^+,yz^+)|V_{cft}|(xz^+,yz^+)\rangle$ and then expanded as in (3.4).

Once an initial ordering is decided it should not be changed without taking account of the sign reversal when (a,b) is changed to (b,a).

Using the classical crystal field model, and the energies indicated in Eqn.1.29, the energies of these basis states are simply:

$$
\begin{array}{lcll}
\langle B_1|V|B_1\rangle &=& -8Dq - 2Ds + 8Dt & {}^3A_{2g} \\
\langle B_2|V|B_2\rangle &=& -8Dq + Ds + 3Dt & {}^3E_g \\
\langle B_3|V|B_3\rangle &=& -8Dq + Ds + 3Dt & {}^3E_g \\
\langle B_4|V|B_4\rangle &=& 2Dq - 3Ds - 2Dt & {}^3E_g \\
\langle B_5|V|B_5\rangle &=& 2Dq - 3Ds - 2Dt & {}^3E_g \\
\langle B_6|V|B_6\rangle &=& 2Dq + Ds + 3Dt & {}^3E_g \\
\langle B_7|V|B_7\rangle &=& 2Dq + Ds + 3Dt & {}^3E_g \\
\langle B_8|V|B_8\rangle &=& 2Dq - 7Dt & {}^3B_{2g} \\
\langle B_9|V|B_9\rangle &=& 2Dq + 4Ds - 2Dt & {}^3A_{2g} \\
\langle B_{10}|V|B_{10}\rangle &=& 12Dq - 7Dt & {}^3B_{1g}
\end{array}
\qquad (3.5)
$$

Where a state of given symmetry designation (spin and orbit) occurs only once in a given basis, its energy can be generated directly as shown, for example, in Eq.(3.5). States of the same symmetry may be scrambled (mixed) by the crystal field operator to an extent dependent upon the magnitudes of the crystal field radial integrals (via off-diagonal matrix elements). Thus this problem will break down into two simple equations for the $^3B_{1g}$ and $^3B_{2g}$ states, a 2 x 2 determinant for the two $^3A_{2g}$ states, and two equivalent 3 x 3 determinants for the ^3E_g states.

The energies in (3.5) are written in terms of the expectation values of the diagonal crystal field matrix elements $\langle B_i|V|B_i\rangle$. Neglecting spin-orbit coupling and kinetic and potential terms common to all electrons (see Chapter 1), the appropriate Hamiltonian for these two-electron functions is $H = V + e^2/r_{ij}$, i.e. the crystal field plus interelectronic repulsion (see eqn. 1.8). Thus an element $\langle B_i|H|B_j\rangle$ can be expanded to $\langle B_i|V|B_j\rangle + \langle B_i|e^2/r_{ij}|B_j\rangle$. Since in the one electron d orbital basis, shown in Fig.1.3c, no two orbitals have the same symmetry, the crystal field operator does not couple any B_i with B_j where i is not j, i.e. there are no non-zero off-diagonal crystal field elements $\langle B_i|V|B_j\rangle$.

This follows because the crystal field operator is a one-electron operator, transforming as the totally symmetric representation of the group concerned. In an off-diagonal element (B_i not equal to B_j), if a pair of identical functions is integrated out to ensure that the overlap integral is unity, two different orbitals are necessarily left, which, in this case must have two different group theoretical representations. For example the coupling of B_2 and B_4 may be written in group theoretical terms as the element $\langle b_2 e|V|a_1 e\rangle$ which can be expanded to $\langle b_2|V|a_1\rangle\langle e|e\rangle + \langle e|V|e\rangle\langle b_2|a_1\rangle$ both terms of which are zero, the first because V cannot couple functions of different symmetry and the second because the overlap integral vanishes. The reader might prove that off-diagonal elements such as $\langle B_2|V|B_3\rangle$ between corresponding pairs of E_g functions, also disappear.

In the strong field limit, the problem could be solved using the energies in (3.5). However in real molecules interelectronic repulsion is generally significant and must be included if a good fit to an experimental spectrum is to be achieved.

3.2.i) Interelectronic Repulsion

In essence it is necessary to evaluate matrix elements of the type:

$$\langle B_i|e^2/r_{ij}|B_j\rangle$$

where i and j both, in this case, take values from 1 through 10. The interelectronic repulsion operator is a two electron operator coupling pairs of electrons to each other.

Interelectronic repulsion energies are described in terms of the Slater-Condon F_k or Racah A, B and C parameters. The latter are probably most useful because the energy separation between terms of maximum spin multiplicity for any d^n configuration, is a function of the Racah parameter B alone, reducing thereby the number of variables in the problem. They are related by the equations:

$$A = F_0 - 49\,F_4$$
$$B = F_2 - 5\,F_4$$
$$C = 35\,F_4$$

Note that the Racah parameters are spherical parameters appropriate to R_3. They are generally incorporated into lower symmetry analyses without any change other than to allow their magnitudes to fall below free ion values. It is likely that in future work, recognition will be made of this anomaly (cf. differential expansion in Cu(II) complexes), also see Chapter 9.

Both diagonal and off-diagonal non-zero elements involving the Racah interelectronic repulsion parameter B may generally be anticipated. The reader is referred to several excellent texts dealing with the detailed evaluation of interelectronic repulsion energies [1-6,20]; the procedures are summarised here. Consider the determinantal functions (a^+,b^+) and (a^+,b^-). The diagonal interelectronic repulsion energies of these functions are obtained by expanding the full form of the determinant in each case:

$$E(a^+,b^+) = \langle(a^+,b^+)|e^2/r_{ij}|(a^+,b^+)\rangle$$

$$= 0.5\langle a^+(1)b^+(2) - b^+(1)a^+(2)|e^2/r_{ij}|a^+(1)b^+(2) - b^+(1)a^+(2)\rangle$$

$$= 0.5\,[\langle a^+(1),b^+(2)|e^2/r_{ij}|a^+(1),b^+(2)\rangle + \langle b^+(1),a^+(2)|e^2/r_{ij}|b^+(1),a^+(2)\rangle - 2\,\langle a^+(1)b^+(2)|e^2/r_{ij}|b^+(1)a^+(2)\rangle]$$

$$= J(a,b) - K(a,b)$$

and

$$E(a^+,b^-) = \langle(a^+,b^-)|e^2/r_{ij}|(a^+,b^-)\rangle$$

$$= 0.5\langle a^+(1)b^-(2) - b^-(1)a^+(2)|e^2/r_{ij}|a^+(1)b^-(2) - b^-(1)a^+(2)\rangle$$

$$= 0.5\,[\langle a^+(1),b^-(2)|e^2/r_{ij}|a^+(1),b^-(2)\rangle + \langle b^-(1),a^+(2)|e^2/r_{ij}|b^-(1),a^+(2)\rangle - 2\,\langle a^+(1)b^-(2)|e^2/r_{ij}|b^-(1)a^+(2)\rangle]$$

$$= J(a,b)$$

where

$$J(a,b) = \langle a(1)b(2)|e^2/r_{ij}|a(1)b(2)\rangle$$
$$K(a,b) = \langle a(1)b(2)|e^2/r_{ij}|b(1)a(2)\rangle \qquad (3.6)$$

where J(a,b) is a **Coulomb Integral** and K(a,b) is an **Exchange**

Integral and the numbers are electron labels.

The determinantal wavefunctions indicated by round brackets within the first matrix element above, are expanded on the right to give two terms in which the round brackets are omitted to signify that the functions are no longer determinantal. The spins are integrated out before the integral is evaluated, i.e.

$$\langle a^+(1)b^+(2)|e^2/r_{ij}|a^+(1)b^+(2)\rangle = J(a,b).\langle +(1)|+(1)\rangle\langle +(2)|+(2)\rangle$$

In both the J and K integrals for (a^+,b^+) the spin integral multiplier is $\langle +(1)|+(1)\rangle\langle +(2)|+(2)\rangle$ which is unity. In the J integral for determinant (a^+,b^-), the spin integral multiplier is $\langle +(1)|+(1)\rangle\langle -(2)|-(2)\rangle$ which is also unity; however in the K integral for (a^+,b^-), the spin overlap multiplier is $\langle +(1)|-(1)\rangle\langle -(2)|+(2)\rangle$ which is zero. The K integral therefore vanishes for pairs of opposed spins.

Following Ballhausen [20] the total energy of an n electron determinant composed of n spinors is given by:

$$W = \sum_n h_{n,n} + \sum_{m<n} J_{m,n} - \sum_{m<n} K_{m,n}.\delta(s_m,s_n) \qquad (3.7)$$

where $h_{n,n}$ is the so-called one electron energy being equal to the kinetic energy plus nuclear attraction of the nth electron (and includes the 'ligand field' energy). The J and K terms reflect the repulsion energy between the nth electron and all other electrons in the system, and the Kronecker delta (δ) function allows the K term to vanish between pairs of electrons of opposed spin. Ballhausen demonstrates [20] that for the special case of a closed shell n orbital determinant (with a pair of electrons in each orbital), the total energy of the system is given by:

$$W = 2\sum_n h_n + \sum_{n,m}(2J_{n,m} - K_{n,m}) \qquad (3.8)$$

where h_n is the one electron energy of an electron in orbital n, and the 2 multiplier reflects the double occupancy of each orbital. In a similar fashion an orbital energy w_n can be defined for each spin-orbital [20], in a closed shell wavefunction:

$$w_n = h_n + \sum_m (2J_{n,m} - K_{n,m}) \qquad (3.9)$$

where this orbital energy reflects the kinetic energy of the spin-orbital, the potential energy of attraction to the nucleus and the repulsion between this orbital and all others (Eq.(1.4)). In photoelectron spectroscopy, it is this orbital energy which is being measured, assuming that Koopman's theorem is valid, i.e. the other electrons do not relax after excitation - not a very good approximation.

Table 3.1

COULOMB (J) AND EXCHANGE (K) INTEGRALS FOR ELECTRONS IN THE REAL d ORBITALS IN TERMS OF RACAH A, B AND C

$J(xy,xy) = J(xz,xz) = J(yz,yz) = J(x^2-y^2,x^2-y^2) = J(z^2,z^2)$
$= A + 4B + 3C$
$J(xy,yz) = J(xy,xz) = J(xz,yz) = J(x^2-y^2,xz) = J(x^2-y^2,yz)$
$= A - 2B + C$
$J(z^2,xz) = J(z^2,yz) = A + 2B + C$
$J(z^2,xy) = J(z^2,x^2-y^2) = A - 4B + C$
$J(x^2-y^2,xy) = A + 4B + C$

$K(xy,xz) = K(xy,yz) = K(xz,yz) = K(x^2-y^2,yz) = K(x^2-y^2,xz)$
$= 3B + C$
$K(z^2,x^2-y^2) = K(z^2,xy) = 4B + C$
$K(z^2,yz) = K(z^2,xz) = B + C$
$K(x^2-y^2,xy) = C$

Table 3.2

MATRIX ELEMENTS $\langle ab|e^2/r_{ij}|cd\rangle$ DIFFERING FROM ZERO (d ORBITALS)

a	b	c	d	$\langle ab\|e^2/r_{ij}\|cd\rangle$
xz	z^2	xz	x^2-y^2	$-2\sqrt{3}$ B
yz	z^2	yz	x^2-y^2	$2\sqrt{3}$ B
yz	yz	z^2	x^2-y^2	$-\sqrt{3}$ B
xz	xz	z^2	x^2-y^2	$\sqrt{3}$ B
z^2	xy	xz	yz	$\sqrt{3}$ B
z^2	xz	xy	yz	$-2\sqrt{3}$ B
z^2	xy	yz	xz	$\sqrt{3}$ B
x^2-y^2	xy	xz	yz	3 B
x^2-y^2	xy	yz	xz	- 3 B

The matrix element remains unchanged under i) inter-change of a and c, ii) interchange of b and d, iii) simultaneous permutation of ab to ba **and** cd to dc, and iv) interchange of ab with cd, thus:
$\langle ab|e^2/r_{ij}|cd\rangle = \langle ad|e^2/r_{ij}|cb\rangle = \langle cb|e^2/r_{ij}|ad\rangle$
$= \langle bc|e^2/r_{ij}|da\rangle = \langle cd|e^2/r_{ij}|ab\rangle$

Finally, consider exciting an electron from a filled orbital a^2 to spin triplet a^+,b^+ and to spin singlet a^+,b^-. What are the energies involved ? The determinant (a^+,a^-) is an adequate description of the ground state, and the determinant (a^+,b^+) describes the spin triplet excited state. However the determinant (a^+,b^-) is not an adequate description of the excited spin singlet state since there is no reason why (a^-,b^+), of equal energy, should not have been chosen. The function $(1/\sqrt{2})[(a^+,b^-) - (a^-,b^+)]$ is chosen as the spin singlet, where the negative sign ensures that the function is anti-symmetric. The energies of these functions follow from Eq.(3.6,3.7):

$$E(a^+,a^-) = 2h_a + J(a,a)$$

$$E(\text{singlet}) = h_a + h_b + (1/2).\langle(a^+,b^-) - (a^-,b^+)|e^2/r_{ij}|(a^+,b^-) - (a^-,b^+)\rangle$$

$$= h_a + h_b + 0.5[\langle a^+(1)b^-(2)|e^2/r_{ij}|a^+(1)b^-(2)\rangle + \langle a^-(1)b^+(2)|e_2/r_{ij}|a^-(1)b^+(2)\rangle + 2\langle a^+(1)b^-(2)|e^2/r_{ij}|b^+(1)a^-(2)\rangle]$$

$$= h_a + h_b + J(a,b) + K(a,b)$$

$$E(a^+,b^+) = h_a + h_b + J(a,b) - K(a,b) \qquad (3.10)$$

where subtraction of the ground state energy $2h_a + J(a,a)$ from the excited state energies yields the transition energies. Note that the energy difference between the excited spin singlet and spin triplet states is $2K(a,b)$, a general result.

> In expanding the interelectronic repulsion component of the excited spin singlet state, eight matrix elements are generated since each determinant breaks into two functions (as in Eq.(3.6)). Only the non-zero terms are shown in Eq.(3.10). The J terms connecting the two different determinants vanish through spin orthogonality, but in this case the corresponding K terms exist.

The J and K parameters for d orbitals are listed, in terms of the real d orbitals, in Table 3.1. Relatively few off-diagonal terms $\langle a,b|e^2/r_{ij}|c,d\rangle$ exist. These are listed in Table 3.2 and are used to evaluate the off-diagonal contributions to the interelectronic energy matrix. Note carefully the relationships shown below Table 3.2.

Using Tables 3.1 and 3.2 the diagonal interelectronic repulsion matrix elements for the bases B_i in (3.3) are:

$$\langle B_1|e_2/r_{2ij}|B_1\rangle = J(xz,yz) - K(xz,yz) = (A - 2B + C) - (3B + C) = A - 5B$$

$$\langle B_2|e^2/r_{ij}|B_2\rangle = 1/2[J(xz,xy) + J(yz,xy) - K(xz,xy) - K(yz,xy) + 2\langle(xz,xy)|e2/r_{ij}|(yz,xy)\rangle]$$

$$= 1/2[A-2B+C + A-2B+C - 3B+C - 3B+C + 0]$$
$$= A - 5B \qquad (3.11)$$

and hence:

$$\begin{aligned}
\langle B_1|e^2/r_{ij}|B_1\rangle &= A - 5B\\
\langle B_2|e^2/r_{ij}|B_2\rangle &= A - 5B\\
\langle B_3|e^2/r_{ij}|B_3\rangle &= A - 5B\\
\langle B_4|e^2/r_{ij}|B_4\rangle &= A + B\\
\langle B_5|e^2/r_{ij}|B_5\rangle &= A + B\\
\langle B_6|e^2/r_{ij}|B_6\rangle &= A - 5B\\
\langle B_7|e^2/r_{ij}|B_7\rangle &= A - 5B\\
\langle B_8|e^2/r_{ij}|B_8\rangle &= A - 8B\\
\langle B_9|e^2/r_{ij}|B_9\rangle &= A + 4B\\
\langle B_{10}|e^2/r_{ij}|B_{10}\rangle &= A - 8B
\end{aligned} \qquad (3.12)$$

The interelectronic repulsion operator transforms as the totally symmetric representation in R_3 and is therefore totally symmetric in all the lower groups. It therefore only connects states of the same symmetry. Off-diagonal elements within the E_g and A_{2g} determinants are all that need be considered. The reader should check that all others vanish.

Considering the set of E_g levels then it is necessary to calculate off-diagonal elements such as $\langle B_2|e^2/r_{ij}|B_4\rangle$ for each of the six E_g levels. Note first that the functions B_2,B_3; B_4,B_5 and B_6,B_7 form pairs and that the off-diagonal element between members of each pair disappear (they are symmetry adapted to the group D_{4h}). There are, in effect, two 3 x 3 E_g determinants each containing one component from each pair.

Consider the off-diagonal element $\langle B_2|e^2/r_{ij}|B_4\rangle$. This may be written out, omitting the positive spin designation and writing ir for the operator, for clarity:

$$\langle(1/\sqrt{2})[(xz,xy) + (yz,xy)]|ir|(1/\sqrt{2})[(xz,z^2) + (yz,z^2)]\rangle$$

$$= 1/2[\langle(xz,xy)|ir|(xz,z^2)\rangle + \langle(yz,xy)|ir|(yz,z^2)\rangle + \langle(xz,xy)|ir|(yz,z^2)\rangle + \langle(yz,xy)|ir|(xz,z^2)\rangle] \qquad (3.13a)$$

Comparison with Table 3.2 reveals that the combination of orbitals in the first two elements do not occur in a single line, and hence these first two elements are zero. The third element may be expanded:

$$\langle (xz,xy)|ir|(yz,z^2)\rangle$$

$$= \langle (1/\sqrt{2})[xz(1)xy(2) - xy(1)xz(2)]|ir|(1/\sqrt{2})[yz(1)z^2(2) - z^2(1)yz(2)]\rangle$$

$$= (1/2)[\langle xz,xy|ir|yz,z^2\rangle - \langle xz,xy|ir|z^2,yz\rangle - \langle xy,xz|ir|yz,z^2\rangle + \langle xy,xz|ir|z^2,yz\rangle]$$

$$= (1/2)[\ -2\sqrt{3}B\ -\ \sqrt{3}B - \sqrt{3}B\ -\ 2\sqrt{3}B]$$
$$= -3\sqrt{3}B \qquad (3.13b)$$

The fourth element in (3.13a) can be expanded:

$$\langle (yz,xy)|ir|(xz,z^2)\rangle$$

$$\langle (1/\sqrt{2})[yz(1)xy(2) - xy(1)yz(2)]|ir|(1/\sqrt{2})[xz(1)z^2(2) - z^2(1)xz(2)]\rangle$$

$$= (1/2)[\langle yz,xy|ir|xz,z^2\rangle - \langle yz,xy|ir|z^2,xz\rangle - \langle xy,yz|ir|xz,z^2\rangle + \langle xy,yz|ir|z^2,xz\rangle]$$

$$= (1/2)[\ -2\sqrt{3}B\ -\ \sqrt{3}B - \sqrt{3}B\ -\ 2\sqrt{3}B]$$
$$= -3\sqrt{3}B \qquad (3.13c)$$

Combining these results with (3.13a) yields $\langle B_2|e^2/r_{ij}|B_4\rangle = -3\sqrt{3}B$.

Take especial care with the relationships shown below Table 3.2. In evaluating each element in (3.13b) and (3.13c), search for the entry which corresponds, permuting according to the rules below Table 3.2, until a match is found. If a match is not found, the element is zero. Note that (ab) and (cd) are NOT Slater determinants. It is NOT correct to write $\langle ab|e^2/r_{ij}|cd\rangle = -\langle ab|e^2/r_{ij}|dc\rangle$, though for some functions it may be true. Note that the data in Tables 3.1 and 3.2 can readily be programmed into a microcomputer lessening the possible confusion which may arise when these matrix elements are evaluated.

Had the off-diagonal element between B_2 and B_5 been chosen, the presence of the negative sign in B_5 would have caused the element to vanish ((3.13b) - (3.13c) instead of (3.13b) + (3.13c)). Thus B_2 and B_4 belong to the same 3 x 3 determinant but B_2 and B_5 belong to different determinants. The remaining component which does not vanish with B_2 and B_4 is found to be B_7, thus $\langle B_2|e^2/r_{ij}|B_7\rangle$ is expanded:

$$\langle (1/\sqrt{2})[(xz,xy) + (yz,xy)]|ir|(1/\sqrt{2})[(xz,x^2-y^2) - (yz,x^2-y^2)]\rangle$$

$$= (1/2)[\langle (xz,xy)|ir|(xz,x^2-y^2)\rangle - \langle (yz,xy)|ir|(yz,x^2-y^2)\rangle + \langle (yz,xy)|ir|(xz,x^2-y^2)\rangle - \langle (xz,xy)|ir|(yz,x^2-y^2)\rangle] \qquad (3.14a)$$

Once again the combination of functions in the first two elements do not occur in a single line of Table 3.2, and hence both are zero.

The third element may be further expanded:

$$\langle (yz,xy)|ir|(xz,x^2-y^2)\rangle$$

$$= (1/2)\langle yz(1)xy(2) - xy(1)yz(2)|ir|xz(1)x^2-y^2(2) - x^2-y^2(1)xz(2)\rangle$$
$$= (1/2)[\langle yz,xy|ir|xz,x^2-y^2\rangle + \langle xy,yz|ir|x^2-y^2,xz\rangle - \langle yz,xy|ir|x^2-y^2,xz\rangle - \langle xy,yz|ir|xz,x^2-y^2\rangle]$$

$$= (1/2)[\quad 0\ +\ 0\ -(-3B) - (-3B)]$$
$$= 3B \qquad (3.14b)$$

The fourth element may be further expanded:

$$\langle (xz,xy)|ir|(yz,x^2-y^2)\rangle$$

$$= (1/2)\langle xz(1)xy(2) - xy(1)xz(2)|ir|yz(1)x^2-y^2(2) - x^2-y^2(1)yz(2)\rangle$$

$$= (1/2)[\langle xz,xy|ir|yz,x^2-y^2\rangle + \langle xy,xz|ir|x^2-y^2,yz\rangle - \langle xz,xy|ir|x^2-y^2,yz\rangle - \langle xy,xz|ir|yz,x^2-y^2\rangle]$$

$$= (1/2)[\quad 0\ +\ 0\ -\ 3B\ -\ 3B]$$
$$= -3B \qquad (3.14c)$$

Combining (3.14a,b,c) yields $\langle B_2|e^2/r_{ij}|B_7\rangle = 3B$.

A similar calculation shows that the remaining off-diagonal element in this determinant is $\langle B_4|e^2/r_{ij}|B_7\rangle = -3\sqrt{3}B$. A complete E_g matrix is then given by:

$$\begin{array}{c|ccc|c} & B_2 & B_4 & B_7 & \\ B_2 & -8Dq+Ds+3Dt-5B-E & -3\sqrt{3}B & 3B & \\ B_4 & -3\sqrt{3}B & 2Dq-3Ds-2Dt+B-E & -3\sqrt{3}B & = 0 \\ B_7 & 3B & -3\sqrt{3}B & 2Dq+Ds+3Dt-5B-E & \end{array} \qquad (3.15)$$

The off diagonals in one matrix are the negative of those in the other since the two components of each e representation are orthogonal. The sign of an

off-diagonal has no effect on the resultant energies. The Racah A parameter is omitted since it shifts all levels to the same degree.

Similarly the off-diagonal for the A_{2g} matrix is $\langle B_1|ir|B_9\rangle$:

$$\begin{aligned}&\langle(xz,yz)|ir|(xy,x^2-y^2)\rangle\\ &=(1/2)[\langle xz,yz|ir|xy,x^2-y^2\rangle - \langle xz,yz|ir|x^2-y^2,xy\rangle\\ &\qquad + \langle yz,xz|ir|x^2-y^2,xy\rangle - \langle yz,xz|ir|xy,x^2-y^2\rangle]\\ &=(1/2)[-3B - 3B + (-3B) - 3B]\\ &=-6B\end{aligned} \tag{3.16}$$

Note that the initial choice of the order of basis states was arbitrary. Had B_1 been written as (yz^+,xz^+) this last element (3.16) would equal + 6B. Since the function $(yz^+,xz^+) = -(xz^+,yz^+)$ this is quite consistent.

The A_{2g} matrix is then:

$$\begin{array}{c|cc|c} & B_1 & B_9 & \\ B_1 & -8Dq-2Ds+8Dt-5B-E & -6B & \\ B_9 & -6B & 2Dq+4Ds-2Dt+4B-E & \end{array} = 0 \tag{3.17}$$

The remaining energy levels are simply obtained by summing the crystal field and interelectronic repulsion energy (in 3.5 and 3.12) to obtain:

$$\begin{aligned}E(^3B_{2g}) &= E(B_8) = 2Dq - 7Dt - 8B\\ E(^3B_{1g}) &= E(B_{10}) = 12Dq - 7Dt - 8B\end{aligned} \tag{3.18}$$

The energies of all ten levels in a d^2 tetragonal ML_4Z_2 complex (D_{4h}) are now available. No statement has been made about which is the ground state. In octahedral symmetry, the ground state is $^3T_{1g}$. In D_{4h} symmetry the degeneracy of this state is lifted to yield $^3A_{2g} + {}^3E_g$. One of these states will be the ground state, i.e. one of the roots of either the E_g or the A_{2g} matrix will have the most negative energy. Transition energies are then obtained by subtracting this energy from all the other energy levels. In advance of the experiment one cannot say which will be the ground state. However it is demonstrated later that knowing the complex under study, chemical experience often advises the most likely ground state. The experimentally observed energy levels are fitted to the energy equations derived above.

The energy levels for a d^8 trans ML_4Z_2 complex (see Fig.2.18) can be obtained directly from these energy level calculations recalling that d^8 is the 'hole' analog of d^2. If Dq, Ds and Dt (but not B) in (3.15, 3.17, 3.18) are multiplied by -1, then these equations now describe the energy levels for high spin d^8. The d^8 ground state in O_h is $^3A_{2g}$ which transforms as $^3B_{1g}$ in D_{4h} symmetry. The state B_{10} is now definitively the ground state in these d^8 complexes. If high spin d^8 complexes are of especial interest, it is convenient to subtract the energy of the $^3B_{1g}$ level (3.18) from all the energy expressions lying on the diagonals in (3.15, 3.17 and 3.18) to obtain transition energies directly after solving the matrices. Clearly the d^8 splitting diagram for these tetragonal systems is the inverse of that for d^2.

That the energy levels for octahedral d^3 and d^7 (high spin) are equivalent to those for d^8 and d^2 respectively, suggests that eqns. (3.15, 3.17 and 3.18) can also be used for D_{4h} high spin complexes of d^3 and d^7, and with the exception of a change in the spin multiplicity label to a spin quartet; this is indeed the case. Note that if Ds = Dt = 0 then Eqns.(3.13 - 3.18) reproduce the octahedral (O_h) solution.

3.2.ii) Low spin states

States of lower spin than the high spin configuration for d^n can be handled in a similar fashion, though choice of suitable basis states is slightly more complex. The weak field method is probably a more useful vehicle for low spin calculations. However note the following development.

If electrons are introduced into the levels in Fig.1.3c, then states such as $S_1 - S_3$ (in 3.19) can be visualised, together with the appropriately generated states $S_4 - S_6$ as described above.

$$\begin{aligned}&S_1 \quad (xz^+,xz^-)\ ,S_2 \quad (yz^+,yz^-)\ ,S_3 \quad (xy^+,xy^-)\\ &S_4\ (1/\sqrt{2})[(xz^+,xy^-) - (xz^-,xy^+)]\\ &S_5\ (1/\sqrt{2})[(yz^+,xy^-) - (yz^-,xy^+)]\\ &S_6\ (1/\sqrt{2})[(xz^+,yz^-) - (xz^-,yz^+)]\end{aligned} \tag{3.19}$$

The symmetries of the singlet states are obtained from the direct product ($t_{2g} \times t_{2g}$) (in O_h) yielding $A_{1g} + E_g + T_{1g} + T_{2g}$. The octahedral spin triplet state is known to be $^3T_{1g}$, leaving the singlets $^1A_{1g} + {}^1E_g + {}^1T_{2g}$, which in D_{4h} symmetry will span $^1B_{1g} + {}^1A_{2g} + {}^1E_g + 2 \times {}^1A_{1g}$. Use of the D_{4h} symmetry operators reveals that the functions $S_4 - S_5$ transform as ^1E_g, S_6 as $^1A_{2g}$, and S_3 as $^1A_{1g}$. However basis states S_1 and S_2 (which by subtraction must span $^1A_{1g} + {}^1B_{1g}$) are not properly symmetry adapted to the point group D_{4h} (e.g. they scramble under C_4). While they could be used, additional non-zero off-diagonal elements would arise making the problem more complex than necessary. In fact the states $(1/\sqrt{2})[S_1 + S_2]$ and $(1/\sqrt{2})[S_1 - S_2]$ will suffice. Ballhausen discusses the projection of appropriately adapted singlet states [2].

Energy matrices for the spin singlets can be obtained by

summing the crystal field and interelectron repulsion energies for all possible spin singlet states and solving the resulting matrices. Note for example, that the diagonal energy of the basis function S_4 in (3.19) is derived as follows:

$$E(S_4) = \langle S_4|V_{cft} + e^2/r_{ij}|S_4\rangle$$

expanding the crystal field term yields:

$$\langle S_4|V_{cft}|S_4\rangle$$

$$= 0.5\ \langle(xz^+,xy^-)|V|(xz^+,xy^-) + 0.5\ \langle(xz^-,xy^+)|V|(xz^-,xy^+)\rangle - \langle(xz^+,xy^-)|V|(xz^-,xy^+)\rangle \qquad (3.20)$$

Recalling that the crystal field operator is a one-electron operator, the first two terms can be expanded as sums of individual crystal field matrix elements multiplied by an overlap integral:

The first term in (3.20) can be expanded as follows:

$$0.5\ [\langle xz^+|V|xz^+\rangle\langle xy^-|xy^-\rangle + \langle xy^-|V|xy^-\rangle\langle xz^+|xz^+\rangle] \qquad (3.21)$$

in which the overlap integrals are unity. The second term expands in an identical fashion except that the spins are interchanged. The third term is zero because the overlap integral will be of the type $\langle xy^-|xy^+\rangle$ which is zero because of spin orthogonality.

Hence, since the crystal field operator does not depend upon spin (from 1.29):

$$\begin{aligned}\langle S_4|V_{cft}|S_4\rangle &= \langle xz|V|xz\rangle + \langle xy|V|xy\rangle \\ &= (-4Dq-Ds+4Dt) + (-4Dq+2Ds-Dt) \\ &= -8Dq+Ds+3Dt\end{aligned} \qquad (3.22)$$

The interelectronic repulsion terms expand as follows:

$$\langle S_4|e^2/r_{ij}|S_4\rangle$$

$$= 0.5\ [\langle(xz^+,xy^-)|ir|(xz^+,xy^-)\rangle + \langle(xz^-,xy^+)|ir|(xz^-,xy^+)\rangle - \langle(xz^+,xy^-)|ir|(xz^-,xy^+)\rangle] \qquad (3.23a)$$

In the first two terms the J integral can be non-zero but the K integral will vanish because of spin orthogonality. However in the third term, the J integral vanishes because the spins are not correctly ordered, but the K integral will now be non-zero as the spins do match. Hence:

$$\begin{aligned}\langle S_4|e^2/r_{ij}|S_4\rangle &= 0.5\ [J(xz,xy) + J(xz,xy)] + K(xz,xy) \\ &= J(xz,xy) + K(xz,xy)\end{aligned} \qquad (3.23b)$$

Summing (3.22) and (3.23b) gives the first order energy of S_4.

However matrices containing all possible spin singlet states should be derived and solved before the spin singlet energies can be calculated since there will generally be more than one state of the same symmetry and scrambling will occur.

Low spin square planar complexes occur frequently especially with d^8 ions. The ground state in square planar D_{4h} systems would be $(e_g)^4(a_{1g})^2(b_{2g})^2(b_{1g})^0$ $(^1A_{1g})$ where the energy sequence indicated here is often assumed. It may however vary from one complex to another (see Chapter 6). Excitation from the three lower lying orbitals into the empty b_{2g} (x^2-y^2) orbital leads to the excited states $^nA_{2g}$, $^nB_{1g}$ and ^nE_g where n may be 1 or 3.

Assume the one-electron energy separations, commonly written:

$$E(b_{1g})-E(b_{2g}) = \Delta_1,\ E(b_{2g})-E(a_{1g}) = \Delta_2,\ E(a_{1g})-E(e_g) = \Delta_3$$

One may also, of course, write these in terms of Dq,Ds,Dt etc.

The transition to $^1A_{2g}$ is represented by the transition:

$$(z^2)^2(xz,yz)^4(xy)^2 \text{ --- } (z^2)^2(xz,yz)^4(xy)(x^2-y^2)$$

The wavefunction for the $^1A_{2g}$ state may be written $(1/\sqrt{2})[(xy^+,x^2-y^{2-}) - (xy^-,x^2-y^{2+})]$, whose interelectronic energy, following (3.19) and (3.23) is $J(xy,x^2-y^2) + K(xy,x^2-y^2)$, plus a repulsion term representing its interaction with all the other electrons. In reaching this transition, a pair of electrons has been broken in $(xy)^2$. The coulombic energy of this pair is simply J(xy,xy) (see eqn.3.6) plus the same interelectronic interaction with all the other electrons. Thus the difference between these two energies, the excitation energy for $^1A_{2g}$---$^1A_{1g}$ is simply $[J(xy,x^2-y^2) + K(xy,x^2-y^2)] - J(xy,xy) = -C$. To this must be added the crystal field energy Δ_1. In similar fashion the energy of the $^3A_{2g}$---$^1A_{1g}$ will be $[J(xy,x^2-y^2) - K(xy,x^2-y^2)] - J(xy,xy) = -3C$ $(+\Delta_1)$. The energies for all the relevant states are listed in (3.24). The reader may derive the other examples for practice.

$$\begin{array}{llll} ^1A_{2g}\text{---}^1A_{1g} & \Delta_1 - C & ^3A_{2g}\text{---}^1A_{1g} & \Delta_1 - 3C \\ ^1B_{1g}\text{---} & \Delta_1 + \Delta_2 - 4B - C & ^3B_{1g}\text{---} & \Delta_1 + \Delta_2 - 12B - 3C \\ ^1E_g\text{---} & \Delta_1 + \Delta_2 + \Delta_3 - 3B - C & ^3E_g\text{---} & \Delta_1 + \Delta_2 + \Delta_3 - 9B - 3C \end{array} \qquad (3.24)$$

These equations do not take into account configurational interaction with other states of the same spin and orbital symmetry; however such interaction is generally ignored. Note that these equations are valid for the definitions of one-electron energies, independently of the exact order of d orbitals, i.e the delta values may be negative.

3.3 Weak Field Analysis

The advantage of the weak field approach lies in the realisation that all microstates within a given spectroscopic term have the same interelectronic repulsion energy. The interelectronic repulsion calculation then becomes essentially trivial. Since interest generally lies in energy differences, it is feasible to write the interelectronic repulsion energy for the ground spectroscopic term as zero and reference all higher energy terms to this zero. The interelectronic repulsion energies of these higher terms are tabulated and can therefore be inserted into the calculation without any further calculation. Such data are listed in Table 3.3 with free ion values for the Racah B and C values for various ions in Table 3.4.

In the crystal field model (no metal-ligand overlap) the magnitudes of the interelectronic repulsion values are those obtained from the analysis of free ion atomic spectroscopy. In real complexes, the interelectronic repulsion parameters usually have values below the free ion limit, especially if the complex is decidedly covalent. We are then in the realm of ligand field theory where metal-ligand overlap is assumed to occur. See the 'Nephelauxetic Series" in Chapter 9, and ref.[20].

The basis states can simply be the $|LM_L SM_S\rangle$ kets of the free ion microstates (Russell Saunders Coupling). If spin-orbit coupling is important they might be the corresponding $|LSJM_J\rangle$ kets.

For example for ^3F, the $|LM_L SM_S\rangle$ are the 21 microstates with L = 3, M_L 3,2,1,0,-1,-2,-3, S = 1 and M_S = 1,0,-1.

Such a starting point has the advantage that the mathematical treatment is concise. However it requires the diagonalisation of very large matrices if such matrices have not been simplified by symmetry adaptation. Where a large computer is available diagonalisation is not a difficulty. However we prefer for three reasons to perform some symmetry adaptation to these bases prior to calculation. Firstly when broken down in this fashion, calculations can generally be completed rapidly on a micro-computer, which is becoming a very common tool; secondly, prior manipulation provides a better pedagogical understanding of exactly what is involved in the calculation, and therefore what chemically can be expected to result. Thirdly, it is easier to locate errors in the calculation (or program) if it is approached in small bites (bytes).

3.3.i) Axes of Quantization and Symmetry Adapted Wavefunctions

The conventional axis of quantization used to study the energy levels in an octahedral complex is the four-fold axis, defined as z, and labelled C_4^z. The symmetry operators of the octahedral group

Table 3.3

THE FREE ION TERM ENERGIES FOR d^2 AND d^3 CONFIGURATIONS

d^2	^3F	A - 8B	$F_0 - 8F_2 - 9F_4$
	^3P	A + 7B	$F_0 + 7F_2 - 84F_4$
	^1G	A + 4B + 2C	$F_0 + 4F_2 + F_4$
	^1D	A - 3B + 2C	$F_0 - 3F_2 + 36F_4$
	^1S	A + 14B + 7C	$F_0 + 14F_2 + 224F_4$
d^3	^4F	3A - 15B	$3F_0 - 15F_2 - 72F_4$
	^4P	3A	$3F_0 - 147F_4$
	$^2H, ^2P$	3A - 6B + 3C	$3F_0 - 6F_2 - 12F_4$
	^2G	3A - 11B + 3C	$3F_0 - 11F_2 + 13F_4$
	^2F	3A + 9B + 3C	$3F_0 + 9F_2 - 87F_4$
	2 ^2D	$3A + 5B + 5C \pm \sqrt{(193B^2 + 8BC + 4C^2)}$	$3F_0 + 5F_2 + 3F_4 \pm \sqrt{(193F_2^2 + 8325F_4^2 - 1650F_2F_4)}$

See Refs [5,9] for a more complete listing.
See Table 2.2 for values of B and C. The Racah data are appropriate for corresponding terms of the d^7 and d^8 configurations, save that A must be modified by the number of pairs of electrons concerned. Further information can be obtained from [5, p.86].

Table 3.4

FREE ION VALUES OF B AND C, ACCORDING TO ELECTRON CONFIGURATIONS, FOR GASEOUS IONS

	Ion	B	C		Ion	B	C
$3d^2$	Ti(II)	718	2629	$4d^3$	Mo(III)	610	
	V(III)	861	4165	$4d^6$	Rh(III)	720	
	Cr(IV)	1039	4238	$4d^7$	Rh(II)	620	4002
$3d^3$	Sc(I)	480		$4d^8$	Pd(II)	683	2620
	V(II)	766	2855	$5d^2$	Os(VI)	780	
	Cr(III)	918	3850	$5d^3$	Re(IV)	650	
	Mn(IV)	1064			Ir(VI)	810	
$3d^4$	Cr(II)	830	3430	$5d^4$	Os(IV)	700	
	Mn(III)	1140	3675	$5d^6$	Ir(III)	660	
$3d^5$	Mn(II)	960	3325		Pt(IV)	720	
$3d^6$	Fe(II)	1058	3901	$5d^8$	Pt(II)	600	
	Co(III)	1100					
$3d^7$	Co(II)	971	4366				
$3d^8$	Ni(II)	1041	4831				

See [5,9] for a more complete listing.

may be used to project a set of bases from the Y_l^m harmonics aligned along (quantized along) C_4^z [2]. The resulting functions will generally be eigenfunctions off i.e. be diagonal with respect

to, C_4^z, and are said to be **symmetry adapted** to the group. Presented in yet another way, this means that the operators of the group (O_h in this case) operating on these functions yield characters of 0,1 or -1, but will not scramble one function with another (except for scrambling within a degenerate representation).

> Functions in this last argument mean the group theoretical representations, e.g. e_g or t_{2g}, not necessarily individual orbitals such as $|1,m_1\rangle = |2,1\rangle$.

Alternatively the octahedral Hamiltonian (1.15) can be used to generate this same set of functions; if the Hamiltonian is used to generate a series of matrix elements within, say, the bases L = 2, the eigenvectors of this matrix will be the wavefunctions symmetry adapted to the Hamiltonian and hence to the group. Since the octahedral Hamiltonian (1.15) (quantized along C_4^z) only contains operator eigenstates of the form C^4_0, C^4_4 and C^4_{-4}, only the angular parts of the wavefunction, $Y_1^{m_0}$, whose m differ by zero or four can be coupled by this operator. The various sets of harmonics are therefore expected to fall into sets of functions $|LM_L\rangle$:

$$
\begin{array}{ll}
L = 1 & |1,1\rangle;\ |1,0\rangle;\ |1,-1\rangle; \\
L = 2 & |2,0\rangle;\ |2,2\rangle + |2,-2\rangle;\ |2,2\rangle - |2,-2\rangle;\ |2,1\rangle;\ |2,-1\rangle; \\
L = 3 & |3,0\rangle;\ |3,2\rangle + |3,-2\rangle;\ |3,2\rangle - |3,-2\rangle; \\
 & |3,3\rangle + |3,-1\rangle;\ |3,3\rangle - |3,-1\rangle;\ |3,1\rangle + |3,-3\rangle; \\
 & |3,1\rangle - |3,-3\rangle; \\
L = 4 \text{ etc} &
\end{array}
\qquad (3.25)
$$

There is no first order coupling by the octahedral field operator, within a set of given L value, between states separated by semi-colons in (3.25). A list of these four-fold adapted functions, to L = 4, is given in Table 1.4. In particular see the footnote to Table 1.4 for a listing of the L = 3 functions excised from the Table.

> The octahedral operator will also couple states of different L value, if their M_L values differ by zero or four; for example $|1,1\rangle$ with $|3,-3\rangle$. This is configurational interaction coupling, in this case, mixing the $^3T_1$ representations of ^3P and ^3F.

Now consider a sub-group, such as D_{2h}, where the four-fold axis is lost, but a C_2^z axis remains. The relevant Hamiltonian for this group possesses operator eigenstates of the type C^2_2, C^2_{-2}, C^4_2, and C^4_{-2} (see Chapter 1, Table 1.8, Eqns.1.47). These will have the effect of coupling Y_1^m whose m values differ by zero or two. Thus states (in 3.25 above) such as ($|3,3\rangle + |3,-1\rangle$) will now be coupled with ($|3,1\rangle + |3,-3\rangle$), as will ($|3,3\rangle - |3,-1\rangle$) and ($|3,1\rangle - |3,-3\rangle$), and $|3,0\rangle$ with ($|3,2\rangle + |3,-2\rangle$) etc. Thus the projected symmetry adapted bases in the C_2^z principal axis groups must include linear combination of this type. For example, from Table 1.4, the $B_1(T_1\ (F))$ state is:

$$\sqrt{(5/16)}|3,-3\rangle + \sqrt{(3/16)}|3,-1\rangle + \sqrt{(3/16)}|3,1\rangle + \sqrt{(5/16)}|3,-3\rangle$$

and, for the d orbital set, $|2,1\rangle$ and $|2,-1\rangle$ must now be combined. A list of these states, symmetry adapted to the C_2^z groups is also included in Table 1.4.

The advantage of using a symmetry adapted basis set for a specific calculation lies in the ease of subsequent calculation. For example, for a d^2 ion in C_{2v} symmetry, the energies of the S = 1 levels could be obtained by diagonalising the 10 x 10 matrix of the 7 orbital states of ^3F and the 3 orbital states of ^3P without using any prior linear combinations of these $|L,M_L\rangle$. However if the symmetry adapted functions are used, the problem immediately breaks down to one simple equation, and three 3 x 3 matrices. If repetitive calculations are necessary to fit an experimental spectrum, the solving of such small matrices is much faster than diagonalizing the 10 x 10, and has the advantage that it can readily be handled by a small desk top computer. Moreover, assigning symmetry labels to the 10 roots of a diagonalised 10 x 10 matrix can be tricky. When the symmetry adapted basis is used the labels are already known.

Now consider sub-groups whose principal axis is not the z axis of the parent octahedron. With the three-fold groups, the set of functions shown in Eq.(1.31) can be used (e.g. see [2]). Consistent with the fact that a Hamiltonian for three-fold groups, quantized along C_3, has harmonics such as C^4_3 and C^4_{-3} which couple states differing by three (Table 1.8), the functions in Eq.(1.31) show scrambling of states such as $|2,1\rangle$ and $|2,-2\rangle$.

It should be clear that if a set of bases adapted to C_2 were used in a problem involving a three-fold group, there would be extensive scrambling of these states because of the coupling of m = 3 with m = 0 etc, and we would essentially be back to handling the full 10 x 10 matrix (for d^2 S = 1). There is therefore a strong computational advantage to starting with the three-fold symmetry adapted functions.

Consider now, for example, a C_{2v} complex in which the principal axis bisects a pair of metal-ligand bonds, rather than being collinear therewith. This axis is considered the C_2^{xy} axis of a parent octahedron. The Y_2^m set (d orbitals) quantized along this C_2^{xy} axis is:

$$
\begin{array}{ll}
t_{2g} & \sqrt{(6/8)}|2,0\rangle - \sqrt{(1/8)}[|2,2\rangle + |2,-2\rangle] \\
 & \sqrt{(1/2)}[|2,1\rangle + |2,-1\rangle] \\
 & \sqrt{(1/2)}[|2,2\rangle - |2,-2\rangle] \\
 & \\
e_g & 1/2\ |2,0\rangle + \sqrt{(3/8)}[|2,2\rangle + |2,-2\rangle] \\
 & \sqrt{(1/2)}[|2,1\rangle - |2,-1\rangle]
\end{array}
\qquad (3.26)
$$

Table 1.4 contains a list of wavefunctions, to L=4, for each of these three quantization axes. A more extensive list can be found in [13].

In the remainder of Chapter 3, Table 1.4 provides bases for all

future examples. Note that an example illustrating how to read Table 1.4 is included as a footnote thereof.

3.3.ii) Octahedral Complexes

We begin by deriving the energies of a d^2 ion in an octahedral environment, by calculating the energies of the representations shown in Table 3.5 (obtained from Table 1.4) for the octahedral components of the ^3F and ^3P terms, quantized along C_4^z.

> The energies along any quantization axis could be calculated; obviously the same result would be obtained, but care must be taken that the Hamiltonian used is quantized along the same chosen axis.

3.3.ii.a Decomposition into one-electron sums

The spherical harmonics define the total orbital angular momentum of a wavefunction and are therefore directly related to the $|LM_L SM_S\rangle$ kets which are components of the free ion terms. Thus the wavefunction defined by the angular component $|Y_3^{-3}\rangle$ for example, is represented by the set of kets $|3,-3,S,M_s\rangle$ where the spin will be determined by the configuration concerned.

Considering the high spin states of d^2, $|Y_3^{-3}\rangle$ is represented by $|3,-3,1,M_s\rangle$. Since all the spin components of a representation have the same energy in the absence of an applied magnetic field, simplification can be achieved by working always with the $M_s = 1$ component. In this case $|Y_3^{-3}\rangle$ is represented directly by the ket $|3,-3,1,1\rangle$ this being one microstate of the ^3F term; it is also part of the $T_1(F)$ and $T_2(F)$ octahedral representations (Table 3.5). Consider the component of T_2 whose wavefunction is:

$$T_2(F) \quad \sqrt{(5/8)}|Y_3^{1}\rangle - \sqrt{(3/8)}|Y_3^{-3}\rangle$$

The interelectronic repulsion energy of this term, in the absence of a configurational interaction with T_2 representations of other terms, is simply the interelectronic repulsion energy of the ^3F state, i.e. -8B (neglecting A). Its crystal field energy can be readily obtained (via 1.25) if it can be written in terms of a sum of one electron d orbitals. This is achieved by writing the free ion kets in terms of one-electron spinors.

There are 45 two-electron microstates associated with the d^2 configuration and 30 of these can be related to the ^3F (spin degeneracy 3, orbital degeneracy 7, hence 21 microstates) and ^3P (spin degeneracy 3, orbital degeneracy 3 hence 9 microstates) terms through the use of angular and spin momentum operators.

> These are the 45 different ways of writing two-electron Slater determinants for the five d orbitals using the m_l values 2,1,0,-1 and -2, and spins +1/2 and -1/2, e.g. $(2^+,1^+)$ for which $M_L = 3$ and $M_S = +1$, $l = 2$ being assumed since d orbitals are being represented. Other examples include $(1^-,0^+)$ ($M_L = 1$, $M_S = 0$), and $(-1^-,-2^-)$ ($M_L = -3$,

Table 3.5

LINEAR COMBINATIONS OF SPHERICAL HARMONICS TRANSFORMING IN THE POINT GROUP O_h [C_4^z Quantization] [2] (Also see Table 1.4)

S		
A_1	Y_0^0	
P		
T_1	Y_1^{-1} Y_1^0 $-Y_1^1$	
D		
T_2	$(1/\sqrt{2})[Y_2^2 - Y_2^{-2}]$ Y_2^1 Y_2^{-1}	E: $(1\sqrt{2})[Y_2^2 + Y_2^{-2}]$ Y_2^0
F		
T_1	$\sqrt{(3/8)}Y_3^1 + \sqrt{(5/8)}Y_3^{-3}$ $\sqrt{(3/8)}Y_3^{-1} + \sqrt{(5/8)}Y_3^3$ Y_3^0	T_2: $\sqrt{(5/8)}Y_3^1 - \sqrt{(3/8)}Y_3^{-3}$ $\sqrt{(5/8)}Y_3^{-1} - \sqrt{(3/8)}Y_3^3$ $(1/\sqrt{2})[Y_3^2 + Y_3^{-2}]$
A_2	$(1/\sqrt{2})[Y_3^2 - Y_3^{-2}]$	
G		
T_2	$(1/\sqrt{2})[Y_4^2 - Y_4^{-2}]$ $\sqrt{(1/8)}Y_4^{-1} - \sqrt{(7/8)}Y_4^3$ $\sqrt{(1/8)}Y_4^1 - \sqrt{(7/8)}Y_4^{-3}$	
A_1	$\sqrt{(7/12)}Y_4^0 + \sqrt{(5/12)}[Y_4^4 + Y_4^{-4}]$	
E	$(1/\sqrt{2})[Y_4^2 + Y_4^{-2}]$ $\sqrt{(5/12)}Y_4^0 - \sqrt{(7/24)}[Y_4^4 + Y_4^{-4}]$	
T_1	$(1/\sqrt{2})[Y_4^4 - Y_4^{-4}]$ $\sqrt{(7/8)}Y_4^{-1} + \sqrt{(1/8)}Y_4^3$ $\sqrt{(7/8)}Y_4^1 + \sqrt{(1/8)}Y_4^3$	

$M_S = -1$). Note that convention dictates a standard order in which the most positive m_l numbers are placed first. Do not forget that permuting two numbers will change the sign (phase) of the determinant, i.e. $(2^+,1^+) = -(1^+,2^+)$. Once a set of determinants has been decided upon to solve a specific problem, do not change the order without recognising this phase change. The reader is invited to write out the 45 microstates before continuing with this section. Russell Saunders coupling is assumed for the present, i.e. the coupling between spin S and orbit L is small and is neglected.

Certain relationships can be written by inspection. For example the ket $|3,-3,1,1\rangle$ is the microstate with the largest negative value of M_L associated with parallel spins ($S = 1$). There is only one such state with $M_L = 3$ and $M_S = 1$ and hence $|3,-3,1,1\rangle = (-1^+,-2^+)$.

Other determinantal wavefunctions which can be uniquely associated with $M_S = 1$ kets are:

$$\begin{aligned} |3,-2,1,1\rangle &= (0^+,-2^+) \\ |3,+3,1,1\rangle &= (2^+,1^+) \\ |3,+2,1,1\rangle &= (2^+,0^+) \end{aligned} \qquad (3.27)$$

The remaining microstates can be related to the two electron determinants through the use of step-down (ladder) angular momentum operators [2,5]. A set of ^3P and ^3F wavefunctions for d^2 are shown in Table 3.5, 3.6. See Appendix VIII for an example of the use of ladder operators.

The energy of one component of the state $^3T_{2g}(F)$ is expanded as follows:

$$^3T_{2g}(E_g)(F) \qquad \sqrt{(5/8)}|31\rangle - \sqrt{(3/8)}|3\text{-}3\rangle$$

$$E(^3T_{2g}(F)) = \langle ^3T_{2g}|V|^3T_{2g}\rangle$$

$$= (5/8)\langle 3,1,1,1|V|3,1,1,1\rangle + (3/8)\langle 3,-3,1,1|V|3,-3,1,1\rangle - 2\sqrt{(15)}/8\ \langle 3,1,\ 1,1|V|3,-3,1,1\rangle \qquad (3.28)$$

Eqn.(3.28) can be expanded by insertion of the two-electron determinants. Each term is expanded and evaluated as follows:

$$(5/8)\langle 3,1,1,1|V|3,1,1,1\rangle$$

$$=(5/8)\langle[\sqrt{(2/5)}(1^+,0^+) + \sqrt{(3/5)}(2^+,-1^+)]|V|[\sqrt{(2/5)}(1^+,0^+) + \sqrt{(3/5)}(2^+,-1^-)]$$

$$=(5/8)[(2/5)\langle(1^+,0^+)|V|(1^+,0^+)\rangle + (3/5)\langle(2^+,-1^+)|V|(2^+,-1^+)\rangle + 2\sqrt{(6)}/5\ \langle(1^+,0^+)|V|(2^+,-1^+)\rangle] \qquad (3.29)$$

Each term on the right of (3.29) can be expanded, omitting spins for clarity:

$$(2/5)\langle(1^+,0^+)|V|(1^+,0^+)\rangle = (2/5)[\langle 1|V|1\rangle\langle 0|0\rangle + \langle 0|V|0\rangle\langle 1|1\rangle + 2\langle 1|V|0\rangle\langle 1|0\rangle]$$

$$= (2/5)[(-4Dq) + 6Dq + 0]$$
$$= (4/5)Dq$$

$$(3/5)\langle(2^+,-1^+)|V|(2^+,-1^+)\rangle = (3/5)[\langle 2|V|2\rangle\langle -1|-1\rangle + \langle -1|V|-1\rangle\langle 2|2\rangle + 2\langle 2|V|-1\rangle\langle 2|-1\rangle]$$

$$= (3/5)[Dq + (-4Dq) + 0]$$
$$= -(9/5)Dq$$

Since the third term of (3.29) is zero, this yields:

$$(5/8)\langle 3,1,1,1|V|3,1,1,1\rangle = (5/8)[(4/5)Dq - (9/5)Dq + 0]$$
$$= -(5/8)Dq$$

The second term of (3.28) is similarly expanded:

$$(3/8)\langle 3,-3,1,1|V|3,-3,1,1\rangle = (3/8)\langle(-1^+,-2^+)|V|(-1^+,-2^+)\rangle$$
$$= (3/8)[\langle -1|V|-1\rangle\langle -2|-2\rangle + \langle -2|V|-2\rangle\langle -1|-1\rangle + 2\langle -1|V|-2\rangle\langle -1|-2\rangle]$$
$$= (3/8)[(-4Dq) + Dq + 0]$$
$$= -(9/8)Dq$$

and the third term:

$$-2\sqrt{(15)}/8\ \langle 3,1,1,1|V|3,-3,1,1\rangle$$
$$= -2\sqrt{(15)}/8\ \langle\sqrt{(2/5)}(1^+,0^+) + \sqrt{(3/5)}(2^+,-1^+)|V|(-1^+,-2^+)\rangle$$
$$= -2\sqrt{(15)}/8\ [\sqrt{(2/5)}\langle(1,0)|V|(-1,-2)\rangle + \sqrt{(3/5)}\langle(2,-1)|V|(-1,-2)\rangle]$$

$$= -2\sqrt{(15)}/8\ [\quad 0 \quad - \sqrt{(3/5)}\langle(2,-1)|V|(-2,-1)\rangle]$$

$$= -2\sqrt{(15)}/8\ [-\sqrt{(3/5)}\langle 2|V|-2\rangle\langle -1|-1\rangle - \sqrt{(3/5)}\langle -1|V|-1\rangle\langle 2|-2\rangle]$$
$$= -2\sqrt{(15)}/8\ [-\sqrt{(3/5)}\ 5Dq\ - 0]$$
$$= +(30/8)Dq \qquad (3.30)$$

Hence

$$E(^3T_{2g}(F)) = -(5/8)Dq - (9/8)Dq + (30/8)Dq$$
$$= 2\ Dq \qquad (3.31)$$

Note, in (3.30) for example, the right-hand determinant in the element $+\langle(2,-1)|V|(-1,-2)\rangle$ is permuted to yield the element $-\langle(2,-1)|V|(-2,-1)\rangle$ so that the element will be in a sequence to be suitably expanded into a sum of two one-electron elements.

Use of the other components of $^3T_{2g}(F)$ would, of course, also yield an energy of 2Dq. The energy of the $^3A_{2g}$ representation (Tables 1.4 and 3.5) is found in a similar fashion (where we omit some steps including the cross-terms in the expansion of the two electron crystal field energies; such cross-terms are always zero) [$^3A_{2g} = (1/\sqrt{2})[\ |3,2,1,1\rangle - |3,-2,1,1\rangle\ ]$:

$$E(^3A_{2g}) = (0.5)[\langle 3,2,1,1|V|3,2,1,1\rangle + \langle 3,-2,1,1|V|3,-2,1,1\rangle - 2\langle 3,2,1,1|V|3,-2,1,1\rangle]$$

$$= (0.5)[\langle(2^+,0^+)|V|(2^+,0^+)\rangle + \langle(0^+,-2^+)|V|(0^+,-2^+)\rangle - 2\langle(2^+,0^+)|V|(0^+,-2^+)\rangle]$$

$$= (0.5)[(Dq + 6Dq) + (6Dq + Dq) + 2(+5Dq)]$$

$$= 12Dq \qquad (3.32)$$

Choosing a component of the $^3T_{1g}$ (F) term, say $|3^0\rangle$ (Table 1.4 or 3.5), then from Table 3.6:

$$E(^3T_{1g}(F)) = \langle 3,0,1,1|V|3,0,1,1\rangle$$

$$= \langle\sqrt{(4/5)}(1^+,-1^+) + \sqrt{(1/5)}(2^+,-2^+)|V|\sqrt{(4/5)}(1^+,-1^+) + \sqrt{(1/5)}(2^+,-2^+)\rangle$$

$$= (4/5)\langle(1,-1)|V|(1,-1)\rangle + (1/5)\langle(2,-2)|V|(2,-2)\rangle + (4/5)\langle(1,-1)|V|(2,-2)\rangle$$

$$= (4/5)(-8Dq) + (1/5)(2Dq) + (4/5)(0)$$

$$= -6Dq \qquad (3.33)$$

The $^3T_{1g}$(P) energy may be readily obtained from, say, Y_1^{-1}:

$$E(^3T_{1g}(P)) = \langle 1,-1,1,1|V|1,-1,1,1\rangle$$

$$= \langle\sqrt{(2/5)}(1^+,-2^+) - \sqrt{(3/5)}(0^+,-1^+)|V|\sqrt{(2/5)}(1^+,-2^+) - \sqrt{(3/5)}(0^+,-1^+)\rangle$$

$$= (2/5)\langle(1,-2)|V|(1,-2)\rangle + (3/5)\langle(0,-1)|V|(0,-1)\rangle - 2\sqrt{(6)}/5\,\langle(1,-2)|V|(0,-1)\rangle$$

$$= (2/5)(-4Dq + Dq) + (3/5)(6Dq - 4Dq) - 0$$

$$= 0 \qquad (3.34)$$

There remains the off-diagonal element connecting the two $^3T_{1g}$ states:

$$\langle^3T_{1g}(F)|V|^3T_{1g}(P)\rangle = \langle 3,0,1,1|V|1,0,1,1\rangle$$

$$= \langle\sqrt{(4/5)}(1^+,-1^+) + \sqrt{(1/5)}(2^+,-2^+)|V|\sqrt{(4/5)}(2^+,-2^+) - \sqrt{(1/5)}(1^+,-1^+)\rangle$$

$$= (4/5)\langle(1,-1)|V|(2,-2)\rangle - (1/5)\langle(2,-2)|V|(1,-1)\rangle + (2/5)\langle(2,-2)|V|(2,-2)\rangle - (2/5)\langle(1,-1)|V|(1,-1)\rangle$$

$$= (4/5)(0) - (1/5)(0) + (2/5)(2Dq) - (2/5)(-8Dq)$$

$$= 4Dq \qquad (3.35)$$

Therefore the $^3T_{1g}$ matrix is:

	$^3T_{1g}$(F)	$^3T_{1g}$(P)	
$^3T_{1g}$(F)	-6Dq - E	4Dq	= 0
$^3T_{1g}$(P)	4Dq	15B - E	

(3.36)

where 15B has been added to the $^3T_{1g}$(P) energy, this being the interelectronic repulsion energy of the ^3P level relative to ^3F as zero (see Table 3.3).

Slightly better agreement between the observed and theoretically calculated spectra can be obtained by use of the Trees' correction. Trees [22] showed empirically that better agreement between theory and experiment could be obtained by adding an energy equal to $\alpha L(L+1)$ to the diagonal energies of each term. α is an empirical quantity generally less than 100 cm^{-1}. Note that the correction is applied to energy differences. Thus comparing the energy difference between a D and a P term, 4α ($6\alpha - 2\alpha$) is added to the diagonal energy of the D term.

Solving the determinant (3.36) provides the octahedral spin triplet d^2 energies:

$$E(^3T_{1g}(F)) = 7.5B - 3Dq - (0.5)[225B^2 + 100Dq^2 + 180DqB]^{1/2}$$
$$E(^3T_{2g}) = 2\,Dq$$
$$E(^3A_{2g}) = 12\,Dq$$
$$E(^3T_{1g}(P)) = 7.5B - 3Dq + (0.5)[225B^2 + 100Dq^2 + 180DqB]^{1/2} \qquad (3.37)$$

Spin-singlet energies may be treated in a similar fashion. For example, the $^1T_{1g}$ representation of ^1G (L=4, S=0, Tables 1.4 and 3.5,3.6 for wavefunctions).

$$^1T_{1g}(G) = (1/\sqrt{2})[|4,-4,0,0,\rangle - |4,4,0,0\rangle]$$

$$E(^1T_{1g}(G)) = (1/2)[\langle 4,4,0,0|V|4,4,0,0\rangle + \langle 4,-4,0,0|V|4,-4,0,0\rangle - 2\langle 4,4,0,0|V|4,-4,0,0\rangle]$$

$$= (1/2)[\langle(2^+,2^-)|V|(2^+,2^-)\rangle + \langle(-2^+,-2^-)|V|(-2^+,-2^-)\rangle - 2\langle(2^+,2^-)|V|(-2^+,-2^-)\rangle]$$

$$= (1/2)[(2Dq) + (2Dq) - 0]$$

$$= 2\,Dq \qquad (3.38)$$

There are no other $^1T_{1g}$ representations for d^2 (see Fig.2.5) so after addition of the interelectronic repulsion energy, relative to ^3F (Table 3.3), there results:- $E(^1T_{1g}(G)) = 2Dq + 12B + 2C$

The energies in (3.37) are also appropriate for d^7, and for both d^3 and d^8 if -Dq is written for Dq. This is the classical way to obtain the octahedral d^n energies, which are summarised in Table

Table 3.6

THE SET OF d^2 ^3F, ^3P, ^1G, ^1D and ^1S WAVEFUNCTIONS IN TERMS OF 2-ELECTRON DETERMINANTAL WAVEFUNCTIONS[a]

^3F

$|3,3,1,1\rangle = -(2^+,1^+)$

$|3,2,1,1\rangle = (2^+,0^+)$

$|3,1,1,1\rangle = -[\sqrt{(3/5)}(2^+,-1^+) + \sqrt{(2/5)}(1^+,0^+)]$

$|3,0,1,1\rangle = \sqrt{(1/5)}(2^+,-2^+) + \sqrt{(4/5)}(1^+,-1^+)$

$|3,-1,1,1\rangle = \sqrt{(2/5)}(0^+,-1^+) + \sqrt{(3/5)}(1^+,-2^+)$

$|3,-2,1,1\rangle = (0^+,-2^+)$

$|3,-3,1,1\rangle = (-1^+,-2^+)$

^3P

$|1,1,1,1\rangle = -[\sqrt{(2/5)}(2^+,-1^+) - \sqrt{(3/5)}(1^+,0^+)]$

$|1,0,1,1\rangle = \sqrt{(4/5)}(2^+,-2^+) - \sqrt{(1/5)}(1^+,-1^+)$

$|1,-1,1,1\rangle = \sqrt{(2/5)}(1^+,-2^+) - \sqrt{(3/5)}(0^+,-1^+)$

^1G

$|4,4,0,0\rangle = (2^+,2^-)$

$|4,3,0,0\rangle = -(1/\sqrt{2})[(2^+,1^-) - (2^-,1^+)]$

$|4,2,0,0\rangle = \sqrt{(3/14)}[(2^+,0^-) - (2^-,0^+)] + \sqrt{(4/7)}(1^+,1^-)$

$|4,1,0,0\rangle = -\sqrt{(1/14)}[(2^+,-1^-) - (2^-,-1^+)] - \sqrt{(3/7)}[(1^+,0^-) - (1^-,0^+)]$

$|4,0,0,0\rangle = \sqrt{(1/70)}[(2^+,-2^-) - (2^-,-2^+)] + \sqrt{(8/35)}[(1^+,-1^-) - (1^-,-1^+)] + \sqrt{(18/35)}(0^+,0^-)$

$|4,-1,0,0\rangle = \sqrt{(1/14)}[(1^+,-2^-) - (1^-,-2^+)] + \sqrt{(3/7)}[(0^-,-1^+) - (0^+,-1^-)]$

$|4,-2,0,0\rangle = \sqrt{(3/14)}[(0^+,-2^-) - (0^-,-2^+)] + \sqrt{(4/7)}(-1^+,-1^-)$

$|4,-3,0,0\rangle = (1/\sqrt{2})[(-1^+,-2^-) - (-1^-,-2^+)]$

$|4,-4,0,0\rangle = (-2^+,-2^-)$

^1D

$|2,2,0,0\rangle = \sqrt{(2/7)}[(2^+,0^-) - (2^-,0^+)] - \sqrt{(3/7)}(1^+,1^-)$

$|2,1,0,0\rangle = -\sqrt{(3/7)}[(2^+,-1^-) - (2^-,-1^+)] - \sqrt{(1/14)}[(1^-,0^+) - (1^+,0^-)]$

$|2,0,0,0\rangle = \sqrt{(2/7)}[(2^+,-2^-) - (2^-,-2^+)] - \sqrt{(2/7)}(0^+,0^-) + \sqrt{(1/14)}[(1^+,-1^-) - (1^-,-1^+)]$

$|2,-1,0,0\rangle = \sqrt{(3/7)}[(1^+,-2^-) - (1^-,-2^+)] + \sqrt{(1/14)}[(0^-,-1^+) - (0^+,-1^-)]$

$|2,-2,0,0\rangle = \sqrt{(2/7)}[(0^+,-2^-) - (0^-,-2^+)] - \sqrt{(3/7)}(-1^+,-1^-)$

^1S

$|0,0,0,0\rangle = \sqrt{(1/5)}[(2^+,-2^-) - (2,^-,-2^+) - (1^+,-1^-) + (1^-,-1^+) + (0^+,0^-)]$

a) These functions follow a Condon and Shortley phase convention that odd positive m_l wavefunctions carry a negative sign. Only the $M_S = 1$ components of the spin triplets are included. Also see [4].

3.7. Since octahedral d^n is equivalent to tetrahedral d^{10-n}, Table 3.7 also provides data for tetrahedral species.

Clearly this technique can be used for any geometry and any d^n configuration by projecting the appropriate wavefunctions and deriving their relationships to the one electron d orbitals. It may also be used for low spin configurations. However both for low spin and low symmetries, this method is rather clumsy.

Note that nevertheless, the use of a microcomputer can simplify the calculations shown above quite significantly. For example the wavefunctions can be set up as a matrix of $|LM_LSM_S\rangle$ kets (10 columns for the ^3P and ^3F terms) with each row being a group theoretical representation e.g. row No.1 of such a matrix could have $(1/\sqrt{2})$ under $|3,2\rangle$ and $-(1/\sqrt{2})$ under $|3,-2\rangle$ and thereby represent the $A_{2g}(F)$ wavefunction. An energy matrix can be generated by calculating the 100 elements $\langle L,M_L,1,1|V|L',M_L',1,1\rangle$ in terms of Dq and B, many of which, however, are zero (some are obtainable from the calculations above, e.g. (3.35)). Multiplication of these two matrices yields a matrix whose solutions are the energies of the wavefunctions. If the 10 x 10 matrix is first simplified into three 3 x 3 matrices ($^3T_{1g}$, $^3T_{1g}$ and $^3T_{2g}$) and a simple equation ($^3A_{2g}$) the calculation becomes facile. This method can readily be adapted to lower symmetries (see below).

An experimental spectrum fitted to the equations above will yield values of Dq and B. Once an assignment has been made (see Chapter 4 for methods of assigning spectra) several procedures exist to fit the spectra to the theoretical equations. Where a rapid and reasonably accurate solution is required, this can be obtained by ratio-ing the energies of the observed spin-allowed transitions and deriving therefrom values of Dq and B. This is explained in full in Appendix V.

3.3.iii) Lower Symmetry Species

Knowing the wavefunctions for lower symmetry species such as trans-ML_4Z_2, their energies can be obtained by the method above, using, in this case, the one electron tetragonal energies (Eqn.1.29). Using the AOM one electron energies (Eqns.1.73 or 1.76), the d^2 AOM energies can be deduced. For example, the wavefunction B_{1g} in D_{4h} subduced by A_{2g} (in O_h), i.e. (3.32), has a diagonal energy in such a trans species as shown in (3.39).

$$
\begin{aligned}
E(^3B_{1g}) &= (0.5)[\langle 3,2,1,1|V|3,2,1,1\rangle + \langle 3,-2,1,1|V|3,-2,1,1\rangle - 2\langle 3,2,1,1|V|3,-2,1,1\rangle] \\
&= (0.5)[\langle (2^+,0^+)|V|(2^+,0^+)\rangle + \langle (0^+,-2^+)|V|(0^+,-2^+)\rangle - 2\langle (2^+,0^+)|V|(0^+,-2^+)\rangle] \\
&= 12Dq - 7Dt \quad \text{(from 1.29)} \\
&= 4\,e_\sigma(L) + 2\,e_\sigma(Z) \quad \text{(from 1.76)}
\end{aligned}
\qquad (3.39)
$$

Table 3.7

ENERGY LEVELS IN OCTAHEDRAL AND TETRAHEDRAL COMPLEXES

d^2	t_2^2	$\mathbf{^3T_1}$	$7.5B - 3Dq - 0.5[225B^2 + 100Dq^2 + 180DqB]^{1/2}$
		^1E	$-8Dq + 9B + 2C - 6B^2/10Dq$
		$^1T_2$	$-8Dq + 9B + 2C - 12B^2/10Dq$
		$^1A_1$	$-8Dq + 18B + 5C - 108B^2/10Dq$
	t_2e	$^3T_2$	$2Dq$
		$^3T_1$	$7.5B - 3Dq + 0.5[225B^2 + 100Dq^2 + 180DqB]^{1/2}$
		$^1T_1$	$12B + 2Dq + 2C$
	e^2	$^3A_2$	$12Dq$
d^3	t_2^3	$\mathbf{^4A_2}$	$-12Dq$
		^2E	$-12Dq + 9B + 3C - 50B^2/10Dq$
		$^2T_1$	$-12Dq + 9B + 3C - 24B^2/10Dq$
		$^2T_2$	$-12Dq + 15B + 5C - 176B^2/10Dq$
	t_2^2e	$^4T_2$	$-2Dq$
		$^4T_1$	$7.5B + 3Dq - 0.5[225B^2 + 100Dq^2 - 180DqB]^{1/2}$
	t_2e^2	$^4T_1$	$7.5B + 3Dq + 0.5[225B^2 + 100Dq^2 - 180DqB]^{1/2}$
d^4	t_2^4	$^3T_1$	$-16Dq + 6B + 5C - 64B^2/10Dq$
		^1E	$-16Dq + 12B + 7C - 82B^2/10Dq$
		$^1T_2$	$-16Dq + 12B + 7C - 208B^2/10Dq$
		$^1A_1$	$-16Dq + 21B + 10C - 436B^2/10Dq$
	t_2^3e	$\mathbf{^5E}$	$-6Dq$
	$t_2^2e^2$	$^5T_2$	$4Dq$
d^5	t_2^5	$^2T_2$	$-20Dq + 15B + 10C - 140B^2/10Dq$
	t_2^4e	$^4T_1$	$-10Dq + 10B + 6C - 26B^2/10Dq$
		$^4T_2$	$-10Dq + 18B + 6C - 38B^2/10Dq$
	$t_2^3e^2$	$\mathbf{^6A_1}$	0
		$^4A_1$	$10B + 5C$
		^4E	$10B + 5C$
		$^4T_2$	$13B + 5C$ +..
		^4E	$17B + 5C$
		$^4T_1$	$19B + 7C$ +..
		$^4A_2$	$22B + 7C$
d^6	t_2^6	$^1A_1$	$-24Dq + 5B + 8C - 120B^2/10Dq$
	t_2^5e	$^3T_1$	$-14Dq + 5B + 5C - 70B^2/10Dq$
		$^3T_2$	$-14Dq + 13B + 5C -106B^2/10Dq$
		$^1T_1$	$-14Dq + 5B + 7C - 34B^2/10Dq$
		$^1T_2$	$-14Dq + 21B + 7C - 118B^2/10Dq$
	$t_2^4e^2$	$\mathbf{^5T_2}$	$-4Dq$
	$t_2^3e^3$	^5E	$6Dq$
d^7	t_2^6e	^2E	$-18Dq + 7B + 4C - 60B^2/10Dq$
	$t_2^5e^2$	$\mathbf{^4T_1}$	$7.5B - 3Dq - 0.5[225B^2 + 100Dq^2 + 180DqB]^{1/2}$
	$t_2^4e^3$	$^4T_2$	$2Dq$
		$^4T_1$	$7.5B - 3Dq + 0.5[225B^2 + 100Dq^2 + 180DqB]^{1/2}$
d^8	$t_2^3e^5$	$^4A_2$	$12Dq$
	$t_2^6e^2$	$\mathbf{^3A_2}$	$-12Dq$
		^1E	$-12Dq + 8B + 2C - 6B^2/10Dq$

cont

Table 3.7 cont

	$^1A_1$	$-12Dq + 16B + 4C - 108B^2/10Dq$
$t_2^5e^3$	$^3T_2$	$-2Dq$
	$^3T_1$	$7.5B + 3Dq - 0.5[225B^2 + 100Dq^2 - 180DqB]^{1/2}$
	$^1T_2$	$-2Dq + 8B + 2C - 12B^2/10Dq$
	$^1T_1$	$-2Dq + 12B + 2C$
$t_2^4e^4$	$^3T_1$	$7.5B + 3Dq + 0.5[225B^2 + 100Dq^2 - 180DqB]^{1/2}$

These data were adapted from ref.[24]. The configurational interaction, indicated by multiples of $B^2/10Dq$ were calculated assuming C = 4B. The spin free ground states are indicated in bold face. To obtain tetrahedral d^n, use d^{10-n} energy levels.

However be conscious that in such low symmetry, many off-diagonal elements can be expected because the unidimensional representations will often occur more than once. Note that matrices for high spin tetragonal d^2 (d^3,d^7,d^8) are shown in Appendix VI.

From a computational point of view, there are simpler procedures available for the derivation of energy levels. Moreover they are more readily extendable to more complex calculations, involving, for example, spin-orbit coupling. The basis of the next procedure to be discussed is the Wigner-Eckart theorem.

3.4 Wigner-Eckart Theorem and 3-j Symbols

By now it should have become clear that both wavefunctions and Hamiltonian operators can be represented by the use of spherical harmonics which themselves are measures of angular momentum. The matrix elements are of the type $\langle Y_a|Y_b|Y_c\rangle$ representing the integration of a product of two angular momenta to form a resultant total angular momentum. There are mathematical techniques available for the coupling of angular momenta and in this section these techniques are applied to the problem of many-electron energy levels.

Brink and Satchler [7] is one of several texts [7-11] which introduces this topic in a manner understandable to chemists. The reader is referred to these texts for detailed discussion of this topic and a less rigorous approach is presented here.

We enter the world of **tensor operators.** A tensor is the general name for the group of functions to which belong scalars (a tensor of rank zero, with only one component), vectors (a tensor of rank 1, with three components) and tensors of higher rank (for example rank 2 with five components, or rank 1 with (2l+1) components). A one-electron tensor operator T^l_m of rank l and component m, has the same angular behaviour as the spherical harmonic of the same rank and component; it may also have radial dependence. A general tensor operator is written T^k_q where k is the rank and q is the component. This operator may couple two functions, say two one-electron functions $|jm\rangle$ where j is a general quantum number

which may represent the orbital quantum number (l), the spin quantum number (s) or the total angular momentum (j), and m is its component. The matrix element may be written:

$$\langle j_1 m_1 | T^k_q | j_2 m_2 \rangle = (-1)^{j1-m1} \begin{pmatrix} j_1 & k & j_2 \\ -m_1 & q & m_2 \end{pmatrix} \langle j_1 || T^k || j_2 \rangle \qquad (3.40)$$

This is the mathematical expression of the **Wigner-Eckart** theorem, and represents the coupling of two angular momenta by an operator.

The matrix element on the right is simply a number representing some observable. In our case it will most usually be the energy of a wavefunction with the tensor operator being a component of one of the ligand field Hamiltonian operators.

The six membered symbol in large brackets is known as a **3-j symbol.** Extensive tabulations exist for this function [7,11]; it is simply a number which can be looked up in a table. The symbol on the right of (3.40) is a **reduced matrix element**, a number independent of the component m, and these too can be tabulated. The evaluation of (3.40) then becomes essentially trivial.

The 3-j symbol is always written with the ranks in the upper row, and the components in the lower row. However tensor operators are written variously in different texts, sometimes with q above k and sometimes with k above q.
The 3-j symbol has a number of properties which should be memorised.

a) It is necessary that in the symbol:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \qquad \begin{aligned} j_1 + j_2 - j_3 &\geq 0 \\ j_1 - j_2 + j_3 &\geq 0 \\ -j_1 + j_2 + j_3 &\geq 0 \end{aligned}$$

This is called the "triangle rule" since the numbers j_1, j_2 and j_3 must form a triangle in their number space. For example 1,3,2 is permitted since a triangle could be constructed therefrom, but a 3-j symbol with the top line 1,1,3 is zero.

b) The components must obey:

$$m_1 + m_2 + m_3 = 0$$

c) Negation of the components leads to the relationship:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j1+j2+j3} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix}$$

d) The symbol is invariant under permutation except for a possible change in s gn (phase factor):

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j1+j2+j3} \begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix}$$

$$= (-1)^{j1+j2+j3} \begin{pmatrix} j_1 & j_3 & j_2 \\ m_1 & m_3 & m_2 \end{pmatrix}$$

e) If $m_1 = m_2 = m_3 = 0$ then $j_1 + j_2 + j_3$ must be odd, otherwise the symbol vanishes.
f) If $m_1 = m_2$, $j_1 + j_2 + j_3$ is odd, and $j_1 = j_2$ then the symbol vanishes.

Previously, for example Table 3.6, we have seen how a many-electron representation can be shown to be a sum of one-electron representations. It is also possible to envisage the reverse of this operation whereby one asks to which many-electron representations, a one-electron state contributes.

For many-electron atoms, the 3-j symbol has the useful property of allowing the coupling of one-electron states (labelled with small letters) into many electron states (labelled with capital letters) or the uncoupling of many-electron states into one electron states, via:

$$|j_1 j_2 JM\rangle = \sum_{m_1 m_2} |j_1 j_2 m_1 m_2\rangle \langle j_1 j_2 m_1 m_2 | j_1 j_2 JM\rangle$$

$$|j_1 j_2 m_1 m_2\rangle = \sum_{JM} |j_1 j_2 JM\rangle \langle j_1 j_2 JM | j_1 j_2 m_1 m_2\rangle \qquad (3.41)$$

The first equation demonstrates which one electron kets contribute to the many electron kets $|JM\rangle$. For example which linear combination of d orbital Slater determinants contributes to a given $|LSM_L M_S\rangle$.

The second equation does the reverse and would, for example, show to which $|LSM_L M_S\rangle$ kets a given Slater determinant contributes.

The capital letters J and M_J represent the total angular momentum J, and component M_J obtained through the coupling of the two vectors. Remember J is still a general or 'dummy' label which can also represent L or S.

The symbol $\langle jjJM | jjmm\rangle$ in (3.41) is called a 'vector addition coefficient' or Wigner or Clebsch-Gordan coefficient. Its square is proportional to the probability that a measurement of the state

$|JM_J\rangle$ will give particular values of $m_j(1)$ and $m_j(2)$ for the components j_{1z} and j_{2z}. In the reverse sense, where the two vectors precess about z, the coefficient expresses the probability that their resultant will be J [see ref.[7] p.32].

This coefficient can be expanded via the 3-j symbols:

$$\langle j_1 j_2 m_1 m_2 | j_1 j_2 JM\rangle = (-1)^{j2-j1-M}(2J+1)^{1/2} \begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & M \end{pmatrix} \tag{3.42}$$

An example shows how this expression (3.42) is used. Consider a two-electron ket belonging to ^3F, say $|3,1\rangle$, namely, $|l_1 l_2; LM_L\rangle = |22;31\rangle$.

The symbol (using the format of eqn.3.41) signifies that a two-electron d state ($l_i = 2$) of $L = 3$ and $M_L = 1$ is to be uncoupled into a sum of one-electron d states. There are two ways in which an M_L value of 1 can be achieved, namely 1,0 and 2,-1. These then are the values of m_1 and m_2 appropriate to (3.41) and (3.42):

$$|22;31\rangle = (-1)^1\sqrt{(7)}\Big[\begin{pmatrix} 2 & 2 & 3 \\ 1 & 0 & -1 \end{pmatrix} |21\rangle|20\rangle + \begin{pmatrix} 2 & 2 & 3 \\ 0 & 1 & -1 \end{pmatrix} |20\rangle|21\rangle$$

$$+ \begin{pmatrix} 2 & 2 & 3 \\ 2 & -1 & -1 \end{pmatrix} |22\rangle|2{-1}\rangle + \begin{pmatrix} 2 & 2 & 3 \\ -1 & 2 & -1 \end{pmatrix} |2{-1}\rangle|22\rangle\Big]$$

$$= -\sqrt{(7)}\Big[\sqrt{(1/35)}|21\rangle|20\rangle - \sqrt{(1/35)}|20\rangle|21\rangle + \sqrt{(3/70)}|22\rangle|2{-1}\rangle$$

$$- \sqrt{(3/70)}|2{-1}\rangle|22\rangle\Big]$$

$$= -\left[\sqrt{(2/5)}(1^+0^+) + \sqrt{(3/5)}(2^+,-1^+)\right] \tag{3.43}$$

where in the last line, the Slater determinant version of the previous line has been inserted. This agrees with the expression shown in Table 3.6. The 3-j symbols were obtained from ref.[12]; a collection of 3-j symbols are also included in Appendix IX.

The expansion (3.42) is readily generalised into more than two electrons, providing a simple means of uncoupling a many-electron wavefunction into sums of one-electron determinantal products. It should now be apparent that using these functions with the Wigner-Eckart theorem provides a ready means of generating the secular determinants to solve many-electron atom spectra.

However if the classical crystal field Hamiltonians are used, the utility of the Wigner-Eckart theorem is reduced. The use of the Normalised Spherical Harmonic Hamiltonians allows the exploitation of the full power of the Wigner-Eckart approach.

3.5. Use of the Wigner-Eckart Theorem with NSH Hamiltonians

There are several alternate procedures in which d^n energy levels can be derived using the Wigner-Eckart theorem and NSH Hamiltonians. Consider the use of unitary tensor operators.

A form of the Wigner-Eckart theorem useful for our purposes may be written (see full development in [13]):

$$\langle j_1 m_1 | u^k_q | j_2 m_2\rangle = (-1)^{j1-m1} \begin{pmatrix} j_1 & k & j_2 \\ -m_1 & q & m_2 \end{pmatrix} \langle j_1 || u^k || j_2\rangle \tag{3.44}$$

where the left-hand integral is a one-electron matrix element, and the right-hand side contains a 3-j symbol and a reduced matrix element called a reduced unitary matrix element. Further:

$$\langle j_1 || u^k || j_2\rangle = 1 \text{ if } j_1 + j_2 \geq k \geq |j_1 - j_2| \tag{3.45}$$

and vanishes otherwise. The unitary operator on the left-hand side of (3.44) is related to the C^k_q operator through (3.46):

$$\langle lm | C^k_q | l'm'\rangle = \langle l || C^k || l'\rangle . \langle lm | u^k_q | l'm'\rangle \tag{3.46}$$

where:

$$\langle l || C^k || l'\rangle = (-1)^l[(2l+1)(2l'+1)]^{1/2} \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix} \tag{3.47}$$

Combining (3.44 - 3.47) provides a simple expression for the evaluation of one-electron matrix elements of C^k_q operators, namely:

$$\langle lm | C^k_q | l'm'\rangle$$
$$= (-1)^{l-m}[(2l+1)(2l'+1)]^{1/2} \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l' \\ -m & q & m' \end{pmatrix} \langle l || u^k || l'\rangle \tag{3.48}$$

Examples of the use of this expression are presented below, but first note that a many-electron reduced unitary matrix element may be defined:

$$U^k_q = u^k_{q1} + u^k_{q2} + \ldots u^k_{qn} = \sum_{i=1}^{n} u^k_{qi} \tag{3.49}$$

where n is the number of electrons of a configuration l^n, and u_i acts on electron i. Further development shows that the Wigner-Eckart theorem for many-electron ions can be written:

$$\langle l^n L M_L | C^k_q | l^n L' M_L' \rangle$$

$$= (-1)^{L-M+1}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} L & k & L' \\ -M & q & M' \end{pmatrix} \langle l^n L || U^k || l^n L' \rangle \qquad (3.50)$$

Eqns.(3.48 and 3.50) are in a suitable form to generate the secular determinants for any d^n electron configuration and any Hamiltonian expressed as a sum of rationalised spherical harmonic terms. Each element is constructed from products of 3-j symbols and reduced unitary matrix elements, obtainable from standard tables or calculated and stored in computer memory. This has been a brief introduction to the use of the Wigner-Eckart theory and the literature should be consulted for more complex applications [7,8,10].

3.5.i) Example of a Calculation for d^1

First, the use of equation (3.48) is illustrated with a complete calculation of the d^1 configuration in D_{4h} symmetry, as an example of a general approach which can be used for any point group. The ^2D state of d^1 has ten states (5 orbital and 2 spin); since the ligand field operator does not influence spin, then, in the absence of spin-orbit coupling it may be ignored. The five orbital states are represented by the five values of $|L,M_L\rangle$ viz:

$$|22\rangle, |21\rangle, |20\rangle, |2\text{-}1\rangle \text{ and } |2\text{-}2\rangle.$$

The brute force solution would solve expression (3.48) for the five states with operator (1.40), thereby generating a 5 x 5 matrix in DQ, DS and DT; diagonalisation would yield the five energy levels.

However since the basis wavefunctions in any group can be referred back to R_3 in the NSH approach, it is simpler to use a basis already symmetry adapted for D_{4h}. Indeed the real orbitals expressed in terms of $|L,M_L\rangle$ would serve as a perfectly acceptable basis. In this fashion eqn.(3.48) is solved to yield 5 simple equations, no matrix diagonalisation being required. In the discussion which follows, the DQ, DS and DT matrices are calculated separately, so that they can be stored on micro-computer file and used for any geometry in future work. This is a time-saving manoeuver if studies of various different geometries are anticipated. In the step by step solution of the d^1 tetragonal problem, first generate the C^2_0, C^4_0, and $[C^4_4 + C^4_{-4}]$ matrices. Thus for C^2_0 the element:

$$\langle 22|C^2_0|22\rangle = \langle 2\text{-}2|C^2_0|2\text{-}2\rangle$$

$$= (-1)^0 .(2(2)+1) \begin{pmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 2 & 2 \\ -2 & 0 & 2 \end{pmatrix} \langle 2||u^2||2\rangle$$

$$= +1 \times 5 \times -\sqrt{(2/35)} \times \sqrt{(2/35)} \times 1$$
$$= -2/7 \qquad (3.51)$$

Note that for in-text simplicity 3-j symbols will be written $(j_1\ j_2\ j_3\ m_1\ m_2\ m_3)$, and as is obvious from the symmetry of the 3-j symbol, for these operators, $\langle 2m|C^k_q|2m\rangle = \langle 2\text{-}m|C^k_q|2\text{-}m\rangle$ if q = 0.

The next element differs from (3.51) only in the phase (power of the (-1) factor) and in the second 3-j symbol. In this case it is (2 2 2 -1 0 1) which is $\sqrt{(1/70)}$. The value of the unitary matrix element for these d^1 bases is always unity. Therefore (3-j symbols are listed in Appendix IX):

$$\langle 21|C^2_0|21\rangle = \langle 2\text{-}1|C^2_0|2\text{-}1\rangle$$

$$= (-1)^1 \times 5 \times -\sqrt{(2/35)} \times \sqrt{(1/70)} \times 1 = 1/7$$

and

$$\langle 20|C^2_0|20\rangle$$

$$= (-1)^2 \times 5 \times -\sqrt{(2/35)} \times -\sqrt{(2/35)} \times 1 = 2/7 \qquad (3.52)$$

The C^4_0 matrix is similarly generated noting that the common 3-j symbol (2 4 2 0 0 0) is $\sqrt{(2/35)}$, thus:

$$\langle 22|C^4_0|22\rangle = \langle 2\text{-}2|C^4_0|2\text{-}2\rangle$$

$$= (-1)^0 \times 5 \times \sqrt{(2/35)} \times \sqrt{(1/630)} \times 1 = 1/21$$

$$\langle 21|C^4_0|21\rangle = \langle 2\text{-}1|C^4_0|2\text{-}1\rangle$$

$$= (-1)^1 \times 5 \times \sqrt{(2/35)} \times \sqrt{(8/315)} \times 1 = -4/21$$

$$\langle 20|C^4_0|20\rangle$$

$$= (-1)^2 \times 5 \times \sqrt{(2/35)} \times \sqrt{(2/35)} \times 1 = 2/7 \qquad (3.53)$$

Matrices of C^k_0 operators are necessarily diagonal, since all off-diagonal elements $\langle lm|C^k_0|lm'\rangle$ are zero (a 3-j symbol vanishes).

Finally for this problem the $[C^4_4 + C^4_{-4}]$ matrices are required. There is only one non-zero element, after noting that $\langle 22|C^4_4|2\text{-}2\rangle = \langle 2\text{-}2|C^4_{-4}|22\rangle$ and (2 4 2 -2 4 -2) is 1/3:

$$\langle 22|C^4_4|2\text{-}2\rangle$$

$$= (-1)^4 \times 5 \times \sqrt{(2/35)} \times (1/3) \times 1 = \sqrt{(10/63)} \qquad (3.54)$$

Using Eq.(1.39) these matrices are now multiplied by the relevant factors in this Hamiltonian and summed to yield matrices for the DQ, DS and DT components of the tetragonal operator. Beginning with the DQ component, and noting that $H_{DQ} = \sqrt{(7/12)}\ C^4_0 + \sqrt{(5/24)}\ [C^4_4 + C^4_{-4}]$:

$\langle 22|H_{DQ}|22\rangle = \langle 2\text{-}2|H_{DQ}|2\text{-}2\rangle$

$= (1/21) \times \sqrt{(7/12)}\ DQ \quad = (1/6)(1/\sqrt{21})\ DQ$

$\langle 21|H_{DQ}|21\rangle = \langle 2\text{-}1|H_{DQ}|2\text{-}1\rangle$

$= -(4/21) \times \sqrt{(7/12)}\ DQ \quad = -(2/3)(1/\sqrt{21})\ DQ$

$\langle 20|H_{DQ}|20\rangle$

$= (2/7) \times \sqrt{(7/12)}\ DQ \quad = (1/\sqrt{21})\ DQ$

$\langle 22|H_{DQ}|2\text{-}2\rangle = \langle 22|H_{DQ}|2\text{-}2\rangle$

$= \sqrt{(10/63)} \times \sqrt{(5/24)}\ DQ \quad$ (from C^4_4 matrix)

$= (5/6)(1/\sqrt{21})\ DQ$ (3.55)

The DT matrix is similarly generated, where $H_{DT} = \sqrt{(5/12)}\ C^4_0 - \sqrt{(7/24)}[C^4_4 + C^4_{-4}]$:

$\langle 22|H_{DT}|22\rangle = \langle 2\text{-}2|H_{DT}|2\text{-}2\rangle$

$= 1/21 \times \sqrt{(5/12)}\ DT = (1/42)(\sqrt{5}/\sqrt{3})\ DT$

$\langle 21|H_{DT}|21\rangle = \langle 2\text{-}1|H_{DT}|2\text{-}1\rangle$

$= -4/21 \times \sqrt{(5/12)}\ DT = -(2/21)(\sqrt{5}/\sqrt{3})\ DT$

$\langle 20|H_{DT}|20\rangle$

$= 2/7 \times \sqrt{(5/12)}\ DT = (1/7)(\sqrt{5}/\sqrt{3})\ DT$

and

$\langle 22|H_{DT}|2\text{-}2\rangle = \langle 2\text{-}2|H_{DT}|22\rangle$

$= \sqrt{(10/63)} \times -\sqrt{(7/24)}\ DT \quad = -(1/6)(\sqrt{5}/\sqrt{3})\ DT$ (3.56)

All other elements are zero. The DS matrix is simply equal to the C^2_0 matrix above. These matrices can be diagonalised to yield one-electron energies in terms of DQ, DS and DT. However the symmetry adapted wavefunctions subduced from R_3 may be used. These are the real d orbitals shown in Eq.(1.26). Their energies in a tetragonal field (H_{D4h} Eqn.(1.40)), here indicated as H_{LF}, are derived simply as follows (Note that the above results are summarised in matrix form in Appendix XI):

$\langle x^2\text{-}y^2|H_{LF}|x^2\text{-}y^2\rangle = 0.5\ [\langle 22|H_{LF}|22\rangle + \langle 2\text{-}2|H_{LF}|2\text{-}2\rangle] + \langle 22|H_{LF}|2\text{-}2\rangle$

$= [(1/6)(1/\sqrt{21})\ DQ + (5/6)(1/\sqrt{21})\ DQ] - (2/7)\ DS + [(1/42)(\sqrt{5}/\sqrt{3})\ DT - (1/6)(\sqrt{5}/\sqrt{3})\ DT]$

$= (1/\sqrt{21})\ DQ - (2/7)\ DS - (3/21)(\sqrt{5}/\sqrt{3})\ DT$ (3.57)

where H_{LF} is a sum of the DQ, DS and DT matrices, i.e. (corresponds to 1.40).
Similarly:

$\langle z^2|H_{LF}|z^2\rangle = \langle 20|H_{LF}|20\rangle$

$= (1/\sqrt{21})\ DQ + (2/7)\ DS + (1/7)(\sqrt{5}/\sqrt{3})\ DT$

$\langle xy|H_{LF}|xy\rangle = 0.5\ [\langle 22|H_{LF}|22\rangle + \langle 2\text{-}2|H_{LF}|2\text{-}2\rangle] - \langle 22|H_{LF}|2\text{-}2\rangle$

$= [(1/6)(1/\sqrt{21})\ DQ - (5/6)(1/\sqrt{21})\ DQ] - (2/7)\ DS + [(1/42)(\sqrt{5}/\sqrt{3})\ DT + (1/6)(\sqrt{5}/\sqrt{3})\ DT]$

$= -(2/3)(1/\sqrt{21})\ DQ - (2/7)\ DS + (4/21)(\sqrt{5}/\sqrt{3})\ DT$

$\langle xz|H_{LF}|xz\rangle = 0.5\ [\langle 21|H_{LF}|21\rangle + \langle 2\text{-}1|H_{LF}|2\text{-}1\rangle] - \langle 21|H_{LF}|2\text{-}1\rangle$

$= -(2/3)(1/\sqrt{21})\ DQ + (1/7)\ DS - (2/21)(\sqrt{5}/\sqrt{3})\ DT$

also

$\langle yz|H_{LF}|yz\rangle = -(2/3)(1/\sqrt{21})\ DQ + (1/7)\ DS - (2/21)(\sqrt{5}/\sqrt{3})\ DT$ (3.58)

The xz and yz orbitals are degenerate in this case, as expected, because the off-diagonal term $\langle 21|H_{LF}|2\text{-}1\rangle$ is zero. These data are summarised in Table 1.9.

Moving to rhombic symmetry, e.g. a complex belonging to the group $C_{2v}(I)$, such as $MX_2Y_2Z_2$ (all trans), it is necessary to add additional terms to the Hamiltonian according to Eqn.(1.42). These involve the operators $(1/\sqrt{2})[C^2_2 + C^2_{-2}]$ and $(1/\sqrt{2})[C^4_2 + C^4_{-2}]$, which couple wavefunctions differing by 2 in their magnetic angular quantum number (m). Considering the real orbitals utilised above, they will couple (z^2) with ($x^2\text{-}y^2$) and will distinguish (xz) from (yz). This also follows from the observation that the former pair of orbitals both transform as a_1 in this group, while the e (in D_{4h}) becomes $b_1 + b_2$. The necessary matrices are shown in Appendix XI, derived by the use of 3-j symbols as shown above.

Consider the off-diagonal element $\langle z^2|H_{LF}|x^2\text{-}y^2\rangle$. Only the two pairs of operators mentioned immediately above can couple

these functions. Recalling the form of these real d orbitals (Eq.(1.26), this element may be expanded as follows:

$$(1/\sqrt{2})\langle 20|C^2_2 + C^2_{-2}|(1/\sqrt{2})(22 + 2\text{-}2)\rangle$$

$$= (1/2)\langle 20|C^2_{-2}|22\rangle + (1/2)\langle 20|C^2_2|2\text{-}2\rangle$$

$$= \langle 20|C^2_2|22\rangle = -(2/7)DU \qquad (3.59)$$

where the first square root $(1/\sqrt{2})$ in the top line of (3.59) comes from the operator, and in line 3 it is recognised that the two terms in line 2 are equal in magnitude. Similarly:

$$(1/\sqrt{2})\langle 20|C^4_2 + C^4_{-2}|(1/\sqrt{2})(22 + 2\text{-}2)\rangle$$

$$= (1/2)\langle 20|C^4_{-2}|22\rangle + (1/2)\langle 20|C^4_2|2\text{-}2\rangle$$

$$= \langle 20|C^4_2|22\rangle = (1/7)(\sqrt{5}/\sqrt{3})\ DV \qquad (3.60)$$

The other elements are similarly calculated and the results also included in Table 1.9. These data parallel the AOM data in (1.76). There is a direct relationship between the one-electron d orbital energy differences which can be derived from Table 1.9, and those derived from (1.76). The diagonal energies cannot be directly equated because the zero point energies for the two models differ. However differences in energies between pairs of orbitals, in the two schemes, may be directly equated to provide a mapping such as that shown in section 1.8.

3.5 ii) A High Spin d^2 Calculation in a Tetragonal D_{4h} Field

The procedure to generate the d^2 energies (or the energies for any d^n configuration) parallels those above very closely. The use of equation (3.49) is illustrated with a complete calculation of spin S = 1 d^2 in a tetragonal field (D_{4h}). Before beginning, consider the steps necessary to solve this problem. There are 30 S = 1 orbital electronic states (from the 30 possible $|L,M_L,S,M_S\rangle$ kets of ^3F and ^3P), forming three equivalent sets of 10 $|L,M_L\rangle$ (for M_S = 1, 0 and -1). Since the ligand field operators do not influence spin, all three sets have the same energy (in the absence of spin-orbit coupling) and as before, only one such set is considered. Again the brute force approach is to set up the 10 x 10 secular determinant of $|L,M_L\rangle$ summing the various terms of Hamiltonian (1.40), diagonalising this matrix and hence generating the necessary energies as a function of DQ, DS, DT and B.

However considerable simplification is possible. The basis set wavefunctions of the D_{4h} representations, subduced from R_3 and O_h, can be written as linear combinations of the $|L,M_L\rangle$ kets. This make use of the NSH approach in which the functions subduced from R_3 are symmetry adapted to all members of a chain from R_3 through

O_h to C_{2v} (Chapter 1). Using these linear combinations as a basis the 10 x 10 matrix blocks out to one 3 x 3 matrix (E_g), a 2 x 2 matrix (A_{2g}) and two simple equations (B_{1g}, B_{2g}).

This time the C^k_q matrices must be constructed for the d^2 $|L,M_L\rangle$ kets. The procedure is identical to that used for d^1 except that the unitary reduced matrix elements are no longer unity. Within the S = 1 manifold for each value of k, there are three such elements:

$$\langle d^2\ ^3F||U^k||d^2\ ^3F\rangle,\ \langle d^2\ ^3F||U^k||d^2\ ^3P\rangle$$

$$\langle d^2\ ^3P||U^k||d^2\ ^3P\rangle$$

whose magnitudes can be obtained from Table 3.8. Consider some representative elements. Within the C^2_0 matrix, for example, and using Eqn.(3.49 and 3.50), the $\langle 31|C^2_0|11\rangle$ element:

$$\langle d^2LM_L|C^2_0|d^2L'M_L'\rangle = \langle 31|C^2_0|11\rangle$$

$$= (-1)^{3-1}\ (5) \begin{pmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix} \langle d^2\ ^3F||U^2||d^2\ ^3P\rangle$$

$$= +1 \times 5 \times -\sqrt{(2/35)} \times \sqrt{(2/35)} \times \sqrt{(24/25)}$$

$$= -(4/35)(\sqrt{6}) = -0.27994 \qquad (3.61)$$

Within the $[C^4_4 + C^4_{-4}]$ matrix the element:

$$\langle d^2LM_L|C^4_4|d^2L'M_L'\rangle = \langle 31|C^4_4|3\text{-}3\rangle$$

$$= (-1)^{3-1}\ (5) \begin{pmatrix} 2 & 4 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 & 4 & 3 \\ -1 & 4 & -3 \end{pmatrix} \langle d^2\ ^3F||U^4||d^2\ ^3F\rangle$$

$$= +1 \times 5 \times \sqrt{(2/35)} \times (1/\sqrt{33}) \times -\sqrt{(11/5)}$$

$$= -\sqrt{(2/21)} = -0.30861 \qquad (3.62)$$

Within the C^4_2 matrix the element:

$$\langle d^2LM_L|C^4_2|d^2L'M_L'\rangle = \langle 11|C^4_2|3\text{-}1\rangle$$

$$= (-1)^{1-1}\ (5) \begin{pmatrix} 2 & 4 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 4 & 3 \\ -1 & 2 & -1 \end{pmatrix} \langle d^2\ ^3P||U^4||d^2\ ^3F\rangle$$

$$= +1 \times 5 \times \sqrt{(2/35)} \times \sqrt{(5/84)} \times -\sqrt{(2/5)}$$

$$= -\sqrt{(5/147)} = -0.18443 \qquad (3.63)$$

Table 3.8

UNITARY MATRIX ELEMENTS $[\langle d^2\ n_L||U^k||d^2\ n_{L'}\rangle]^2$

		U^2 Matrix Elements			
^3P	^3P	*21/25	^1D	^1D	*9/49
	^3F	24/25		^1G	144/245
^3F	^3F	6/25	^1G	^1G	198/49
^1S	^1D	4/5			
		U^3 Matrix Elements			
^3P	^3F	6/5	^1D	^1D	*64/49
^3F	^3F	*3/5		^1G	90/49
			^1G	^1G	99/49
		U^4 Matrix Elements			
^3P	^3F	*2/5	^1D	^1D	16/49
^3F	^3F	*11/5		^1G	110/49
^1S	^1G	4/5	^1G	^1G	143/245

The unitary elements are listed as their square. An asterisk indicates that the negative square root should be used. These elements were obtained from [25] where data for other d^n configurations may be found.

In this fashion the d^2 matrices shown in Appendix XI were generated. By multiplying these matrices by the appropriate factors in the Hamiltonian (Eq. 1.40), the NSH DQ, DS and DT matrices also shown in Appendix XI can be generated, in exactly parallel fashion to the d^1 calculation shown in (3.51-3.54). Finally using the wavefunctions in Table 3.5 and multiplying the DQ, DS and DT matrices therewith, the energies for d^2 in D_{4h} symmetry are obtained.

For example, calculate the A_{2g} (in D_{4h}) matrix, noting that both $^3T_{1g}(P)$ and $^3T_{1g}(F)$ will span A_{2g} in D_{4h} (see 1.37 or Appendix X). The two relevant wavefunctions from Table 3.5 are $|30\rangle$ and $|10\rangle$ respectively, or see Table 1.4.

The diagonal elements are given by the following, noting that 15B must be added to the ^3P wavefunction energy since the ^3P lies 15B above ^3F taken as zero.

$$\langle 30|H_{LF}|30\rangle = \langle 30|H_{DQ}|30\rangle + \langle 30|H_{DS}|30\rangle + \langle 30|H_{DT}|30\rangle$$

$$= -0.21822\ DQ + 0.11429\ DS - 0.18443\ DT$$

and

$$\langle 10|H_{LF}|10\rangle = \langle 10|H_{DQ}|10\rangle + \langle 10|H_{DS}|10\rangle + \langle 10|H_{DT}|10\rangle + \langle 10|H_{ir}|10\rangle$$

$$= 0\ DQ + -0.4\ DS + 0\ DT + 15\ B = -0.4\ DS + 15\ B$$

The off-diagonal element is evaluated:

$$\langle 30|H|10\rangle = [\langle 30|H_{DQ}|10\rangle + \langle 30|H_{DS}|10\rangle + \langle 30|H_{DT}|10\rangle]$$

$$0.14548\ DQ - 0.34286\ DS + 0.12295\ DT \qquad (3.64)$$

The numerical entries to (3.64) and the equations which follow are taken from Appendix XII. Although the calculations shown above provide the expectation values of the matrix elements in fractional form, the computer program generates in decimal form. The calculation is made to ten decimal places but is rounded off to five. In some cases, this roundoff may lead to errors in the fifth decimal place. Computer output numbers are added in their unrounded versions and then rounded off. Note further that the absence of DQ and DT contributions to the P terms diagonal energy should not be surprising since fourth order harmonics cannot couple L = 1 functions ('triangle rule' for 3j symbol).

Hence (3.64) provides the elements for the 2x2 matrix which generates the A_{2g} energies. Extension to evaluate the energies in a rhombic complex such as $MX_2Y_2Z_2$ (of $C_{2v}(I)$ symmetry) is straightforward. The additional components of the Hamiltonian are of the C^2_2 and C^4_2 type, modified by the radial integrals DU and DV respectively (see Eqn.(1.42) and Table 1.8). As far as diagonal terms are concerned, any wavefunction containing components whose magnetic quantum numbers differ by ± 2 may have a contribution from DU and DV in the point group $C_{2v}(I)$. It is clear from this that the functions $|30\rangle$ and $|10\rangle$ transforming as A_{2g} in D_{4h} will not have a contribution from DU and DV in this lower symmetry. However reference to (1.37) (or Appendix X) shows that the B_{2g} state in D_{4h} (from T_{2g} in O_h) now transforms as A_2 in $C_{2v}(I)$, as do indeed the two A_{2g} terms mentioned above. Table 3.5 shows that this third A_2 term is $(1/2)^{1/2}[|32\rangle + |3\text{-}2\rangle]$ and it is obvious that it must couple via DU and DV both to $|30\rangle$ and to $|10\rangle$, but that its diagonal energy will not contain DU and DV, viz:

$$1/2\ \langle 32 + 3\text{-}2|H|32 + 3\text{-}2\rangle = 1/2[\langle 32|H|32\rangle + \langle 3\text{-}2|H|3\text{-}2\rangle] + \langle 32|H|3\text{-}2\rangle$$

$$= \langle 32|H_{DQ}|32\rangle + \langle 32|H_{DS}|32\rangle + \langle 32|H_{DT}|32\rangle + \langle 32|H_{DQ}|3\text{-}2\rangle + \langle 32|H_{DS}|3\text{-}2\rangle + \langle 32|H_{DT}|3\text{-}2\rangle$$

$$= 0.25459\ DQ + 0\ DS + 0.21517\ DT - 0.18185\ DQ + 0\ DS + 0.21517\ DT$$

$$= 0.07274\ DQ + 0.43033\ DT \qquad (3.65)$$

The off-diagonal elements are readily given by:

$$\langle 30|H|(1/\sqrt{2})(32 + 3\text{-}2)\rangle = (1/\sqrt{2})\langle 30|H|(32 + 3\text{-}2)\rangle$$

$$= (1/\sqrt{2})[\langle 30|H|32\rangle + \langle 30|H|3\text{-}2\rangle] = (\sqrt{2})\langle 30|H|32\rangle$$

and similarly

$$\langle 10|H|(1/\sqrt{2})(32 + 3\text{-}2)\rangle = (1/\sqrt{2})\langle 10|H|(32 + 3\text{-}2)\rangle$$

$$= (\sqrt{2})\langle 10|H|32\rangle$$

leading to:

$$(\sqrt{2})\langle 30|H|32\rangle = (\sqrt{2}) \times [(1/\sqrt{2})\langle 30|C^2_2 + C^2_{-2}|32\rangle\ DU + (1/\sqrt{2})\langle 30|C^4_2 + C^4_{-2}|32\rangle\ DV]$$

$$= \langle 30|C^2_{-2}|32\rangle + \langle 30|C^4_{-2}|32\rangle$$

$$= -0.12778\ DU + 0.08248\ DV$$

and

$$(\sqrt{2})\langle 10|H|32\rangle = (\sqrt{2}) \times [(1/\sqrt{2})\langle 10|C^2_2 + C^2_{-2}|32\rangle\ DU + (1/\sqrt{2})\langle 10|C^4_2 + C^4_{-2}|32\rangle\ DV]$$

$$= \langle 10|C^2_{-2}|32\rangle + \langle 10|C^4_{-2}|32\rangle$$

$$= -\ 0.25555\ DU + 0.16496\ DV \qquad (3.66)$$

where there are no contributions from DQ, DS or DT since none of the operators involved can connect $|0\rangle$ with $|2\rangle$.

These energies (3.64 - 3.66) are the necessary entries to the 3 x 3 A_2 matrix in $C_{2v}(I)$, and may be checked against the entries in Table 3.9. The B_1 and B_2 matrices are similarly generated from the functions in Table 1.4 and are listed in Table 3.9. Note that the function $(1/\sqrt{2})[|32\rangle - |3\text{-}2\rangle]$ transforming as B_{1g} in D_{4h} and as A_1 (F) in $C_{2v}(I)$ cannot be coupled by any operator in Eqn.(1.42) and remains a linear expression transforming as A_1. The energy of this state is derivable from (3.65) by changing the appropriate sign in the wavefunction, giving 0.43644 DQ.

Energy matrices for d^2 in other sub-groups are obtainable by the same methods using the appropriate Hamiltonian from (1.47). Tables 3.9 - 3.11 summarise these results. Because of the 'hole' relationships, these matrices may also be used for high spin d^7 complexes. Moreover by multiplying the radial integrals (but not B) by -1, they also generate the energies of maximum spin d^3 and d^8.

Since the d^1 matrices are suitable for high spin d^4, d^6 and d^9, most common complexes are covered. They are also relevant for tetrahedral and cubic complexes.

RECOGNISE CLEARLY THAT TABLES 3.9 - 3.11, TOGETHER WITH TABLES 1.8 - 1.9, PROVIDE THE MATRIX ELEMENTS FOR ALL d^n HIGH SPIN SPECIES (except d^5 which has no high spin transitions) IN ALL FOUR, FIVE OR SIX COORDINATE COMPLEXES WHICH ARE PROPER SUB-GROUPS OF O_h AND T_d. This accounts for the vast majority of complexes of first row transition metal species.

With DT = DS = DU = DV (=DM = DN) = 0, energies of octahedral species are generated. With DQ, DS and DT non-zero, and the remaining parameters zero, four fold distorted group energies (e.g. D_{4h}, C_{4v}) are generated. With DQ,DS,DT and DU and DM, or DU and DV non-zero, various two fold groups are accommodated (Eqn.(1.47)). It is necessary to fit these energies against an experimental spectrum to obtain experimental values for these parameters.

Systems such as trigonal bipyramids which do not fall into this sub-group classification can be related via procedures previously noted (see Eqns. 1.43 - 1.46). Low spin complexes can be treated in exactly the same manner, but the matrix elements are not reported here and must be generated as needed.

3.5 iii) Example of a Low Spin d^2 Calculation

The energy matrices for low spin species are calculated in an exactly similar manner. As one example, the diagonal energies of a singlet state of d^2 can be obtained. The relevant wavefunctions are shown in Table 3.5. For a complete analysis the C^2_0, C^4_0, C^4_4, etc matrices would be needed for all the singlet state $|LM_L\rangle$ functions derived from ^1D, ^1G and ^1S, namely a 15 x 15 matrix, there being considerable mixing. These functions were not needed in the calculation above, because of spin orthogonality, i.e. in the absence of spin-orbit coupling, the spin singlet states cannot couple to the spin triplet states.

The matrix of $|LM_L\rangle$ from the ^1D term can be generated by evaluating the elements $\langle d^2\ ^1D\ 2M_L|H|d^2\ ^1D\ 2M_L\rangle$, where the terms of the relevant Hamiltonian (1.39), are expanded according to (3.50).

For example, for the $|LM_L\rangle = |22\rangle$ state of ^1D, the C^4_0 element:

$$\langle 22|C^4_0|22\rangle =$$

$$= (-1)^{2-2} \times (5) \times \begin{pmatrix} 2 & 4 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 4 & 2 \\ -2 & 0 & 2 \end{pmatrix} \langle d^2\ ^1D||U^k||d^2\ ^1D\rangle$$

$$= 1 \times 5 \times \sqrt{(2/35)} \times \sqrt{(1/630)} \times 4/7 = 4/147 \qquad (3.67)$$

where the unitary matrix element for $\langle d^2\ {}^1D||U^4||d^2\ {}^1D\rangle$ is 4/7 (see Table 3.7 and [25]). In similar fashion all the $|LM_L\rangle$ elements of ^1D can be generated for C^4_0 and for the other operator matrices (see Appendix XII). Using the wavefunctions in Table 3.5 and summing as previously (3.62 - 3.64) yields the diagonal energies:

$$E(^1B_{2g})[^1T_{2g}] = -0.08313\ DQ + 0.12245\ DS + 0.14052\ DT$$
$$= -(16/7)Dq - (6/7)Ds - (4/7)Dt$$

$$E(^1E_g)[^1T_{2g}] = -0.08313\ DQ - 0.06122\ DS - 0.07026\ DT$$
$$= -(16/7)Dq + (3/7)Ds + (16/7)Dt$$

$$E(^1A_{1g})[^1E_g] = 0.12470\ DQ - 0.12245\ DS + 0.10539\ DT$$
$$= (24/7)Dq + (6/7)Ds - (24/7)Dt$$

$$E(^1B_{1g})[^1E_g] = 0.12470\ DQ + 0.12245\ DS - 0.10539\ DT$$
$$= (24/7)Dq - (6/7)Ds - (4/7)Dt \qquad (3.68)$$

representing the energies of the ^1D term in a tetragonal field in the absence of configurational interaction with other spin singlet terms. The classical Dq, Ds and Dt values are also listed in (3.68) derived from the NSH values via relationships (Eq.1.41,1.83).

Note that the NSH values split symmetrically about the baricentre of the T_{2g} and the E_g states (in O_h), while the classical values do not. This is a consequence of the well defined symmetry behaviour of the NSH model. See footnote below (1.41). Note also that the symmetry labels in (3.68) were obtained by descent in symmetry from the octahedral labels in Table 3.5 using the correlation information in (1.37) or the tables in Appendix X. The relationship between the $|LM_L\rangle$ values of ^1D and the symmetry labels in D_{4h} follows the same behaviour as the one electron states.

The results in (3.68) could have been obtained by more classical methods by writing the states in terms of one-electron d orbital states and summing the necessary energies. For example, from Table 3.6, [and ref.2 p.72], the $^1A_{1g}$ state is $^1D\ |20\rangle$ ($|LM_L\rangle$ S = 0). Its energy, expanded as a sum of one electron elements is:

$$\langle 20|H|20\rangle$$
$$= (2/7)[\langle 2|H|2\rangle + \langle -2|H|-2\rangle + \langle 2|H|2\rangle + \langle -2|H|-2\rangle$$
$$+ \langle 0|H|0\rangle + \langle 0|H|0\rangle] + (1/14)[\langle 1|H|1\rangle + \langle -1|H|-1\rangle$$
$$+ \langle 1|H|1\rangle + \langle -1|H|-1\rangle]$$
$$= (8/7)\langle 2|H|2\rangle + (4/7)\langle 0|H|0\rangle + (2/7)\langle 1|H|1\rangle$$
$$= (24/7)Dq + (6/7)Ds - (24/7)Dt \qquad (3.69)$$

Table 3.9

NON-ZERO MATRIX ELEMENTS OF THE $C_{2v}(I)$ HAMILTONIAN FOR THE SPIN TRIPLET STATES OF A d^2 BASIS[a] $[\langle\psi_1|H(C_{2v}(I)|\psi_2\rangle]$

ψ_1	ψ_2	DQ	DS	DT	DU	DV
A_1						
$A_2(B_1)$	$A_2(B_1)$	0.43644				
A_2						
$T_1(A_{2g})P$	$T_1(A_{2g})P$		-0.40000			
$T_1(A_{2g})P$	$T_1(A_{2g})$	0.14548	-0.34286	0.12295		
$T_1(A_{2g})P$	$T_2(B_{2g})$				-0.25555	0.16496
$T_1(A_{2g})$	$T_1(A_{2g})$	-0.21822	0.11429	-0.18443		
$T_1(A_{2g})$	$T_2(B_{2g})$				-0.12778	0.08248
$T_2(B_{2g})$	$T_2(B_{2g})$	0.07274		0.43033		
B_1[b]						
$T_1(E_gy)P$	$T_1(E_gy)P$		0.20000		0.34641*	
$T_1(E_gy)P$	$T_1(E_gy)$	0.14548	0.17143*	-0.06148*	0.29692*	0.10648
$T_1(E_gy)P$	$T_2(E_gxz)$		-0.22131*	-0.14286*	0.12778	-0.08248*
$T_1(E_gy)$	$T_1(E_gy)$	-0.21822	-0.05714*	0.09221*	-0.09897*	-0.15972*
$T_1(E_gy)$	$T_2(E_gxz)$		-0.11066*	-0.07143*	0.06389	-0.04124*
$T_2(E_gxz)$	$T_2(E_gxz)$	0.07274		-0.21517		0.37268*
B_2[b]						

a) Symmetry adapted to $C_{2v}(I)$. In column 1 are the labels in $C_{2v}(I)$ symmetry. The various octahedral representations in columns 2 and 3, grouped according to common labels in $C_{2v}(I)$ have their components in D_{4h} symmetry indicated in parentheses. See Tables 1.4 - 1.7 for further information. b) The non-zero matrix elements for the B_2 states are the same as those of B_1 except for inversion of the sign of the starred* elements. The D_{4h} components for the B_2 functions are $T_1(E_gx)$ and $T_2(E_gyz)$ (see Table 1.5). The repulsion energy 15B must be added to the P term diagonals. This Table may be used to calculate d^2 energies for all groups in the chain from O_h to $C_{2v}(I)$. This Table may also be used for d^7 as is, and for d^3 and d^8 after multiplication of all radial integrals by -1 [Ref.13, Table X].

where the final result in (3.69), obtained from (1.29), agrees with the corresponding entry in (3.68). Certainly the NSH approach is more amenable to machine calculation, using 3-j symbols and unitary reduced matrix elements, where many-electron calculations are involved.

Table 3.10

NON-ZERO MATRIX ELEMENTS OF THE $C_{2v}(S)$ HAMILTONIAN FOR THE SPIN TRIPLET STATES OF A d^2 BASIS FOR S = II AND III.[a]
$[\langle\psi_1|H(C_{2v}(II,III)|\psi_2\rangle$

ψ_1	ψ_2	DQ	DS	DT	DM	DN
A_1, A_1						
$T_2(B_{2g})$	$T_2(B_{2g})$	0.07274		0.43033		
B_2,A_2						
$T_1(E+)P$	$T_1(E+)P$		0.20000		-0.34641*	
$T_1(E+)P$	$T_1(E+)$	0.14548	0.17143	-0.06148	0.19795*	-0.05324*
$T_1(E+)P$	$T_2(E-)$		-0.22131*	-0.14286*	0.25555	0.12372
$T_1(E+)$	$T_1(E+)$	-0.21822	-0.05714	0.09221	-0.02474*	-0.31944*
$T_1(E+)$	$T_2(E-)$		-0.11066*	-0.07143*	-0.03194	0.16496
$T_2(E+)$	$T_2(E-)$	0.07274		-0.21517	0.12372*	0.10648*
A_2, B_1						
$T_1(A_{2g})P$	$T_1(A_{2g})P$		-0.40000			
$T_1(A_{2g})P$	$T_1(A_{2g})$	0.14548	-0.34286	0.12295		
$T_1(A_{2g})P$	$A_2(B_{1g})$				-0.25555	0.16496
$T_1(A_{2g})$	$T_1(A_{2g})$	-0.21822	0.11429	-0.18443		
$T_1(A_{2g})$	$A_2(B_{1g})$				-0.12778	0.08248
$A_2(B_{1g})$	$A_2(B_{1g})$	0.43644				
B_1,B_2[b]						

a) Adapted from Table XI [13] which regretfully contains a few labelling typographical errors. Note also the omission, in Table XI [13] of a star on the second entry in the DM column. b) The non-zero matrix elements for the B_1 states of $C_2v(II)$ are the same as those of B_2 except for inversion of the sign of the starred* elements. The transformation properties of the octahedral representations in D_{4h} symmetry are indicated in parentheses. E+ is a component of E in D_{4h} which transforms as $(1/\sqrt{2})(S_x + S_y)$ (functions which behave like p_x and p_y but which are gerade with respect to inversion); E- is a component of E in D_{4h} which transforms as $(1/\sqrt{2})(yz - xz)$. The B_2 states of $C_{2v}(III)$ are the same as those for A_2 except for inversion of the sign of the starred* elements. The repulsion energy 15B must be added to the P term diagonals. This Table may be used to calculate d^2 energies for all groups in the chain from O_h to $C_{2v}(II)$ or $C_{2v}(III)$. This Table may also be used for d^7 as is, and for d^3 and d^8 after multiplication of all radial integrals by -1.

Table 3.11

NON-ZERO MATRIX ELEMENTS OF THE C_3 HAMILTONIAN FOR THE SPIN TRIPLET STATES OF A d^2 BASIS[a] $[\langle\psi_1|H(C_3(xyz))|\psi_2\rangle]$

ψ_1	ψ_2	B	DQ	DM	DN
A_1					
T_2	T_2	15	$-(1/3\sqrt{21})$	-1/7	$-(2\sqrt{5}/21\sqrt{3})$
A_2					
$T_1(P)$	$T_1(P)$	15		-2/5	
$T_1(P)$	T_1		$(2/3\sqrt{21})$	-8/35	$(5/21\sqrt{3})$
$T_1(P)$	A_2			$-(4/7\sqrt{5})$	$(2/7\sqrt{3})$
T_1	T_1		$(1/\sqrt{21})$	-1/35	$-(2\sqrt{5}/7\sqrt{3})$
T_1	A_2			$(2/7\sqrt{5})$	$-(1/7\sqrt{3})$
A_2	A_2		$-(2/\sqrt{21})$		
E					
$T_1(P)$	$T_1(P)$	15		1/5	
$T_1(P)$	T_1		$(2/3\sqrt{21})$	4/35	$-(\sqrt{5}/42\sqrt{3})$
$T_1(P)$	T_2			$-(4/7\sqrt{5})$	$-(\sqrt{3}/14)$
T_2	T_2		$-(1/3\sqrt{21})$	1/14	$(\sqrt{5}/21\sqrt{3})$
T_2	T_1			$-(1/14\sqrt{5})$	$(2/7\sqrt{3})$
T_1	T_1		$(1/\sqrt{21})$	1/70	$(\sqrt{5}/7\sqrt{3})$

a) The operator is appropriate for D_{3d}, D_3 and C_{3v}.

3.5 iv) Other Axes

The above treatment can in principle be used for any molecule by choosing an NSH Hamiltonian with an appropriate number of terms. However it is useful to take note of the symmetry axes of the molecule. For example, consider a species of $C_{2v}(III)$ symmetry whose principle axis lies along the $C_2(xy)$ axis of the octahedron. The $T_{2g}(xy)$ operator eigenstate is now required. If the z quantized Hamiltonian is used, the imaginary i is introduced. While the problem can be solved in this fashion, life is easier without i. If the Hamiltonian operator eigenstates are re-quantized along $C_2(xy)$, as shown in Table 1.8, then the i does not appear, and calculation proceeds in a simple manner, using a basis symmetry adapted to the $C_2(xy)$ axis (from Table 1.4). In a similar fashion, the three-fold groups should preferably utilise a Hamiltonian quantized along $C_3(xyz)$ (Table 1.8) with appropriately quantized functions (Table 1.4) if diagonalisation of large matrices (of $|LM_L\rangle$) is to be avoided (see Ref.[13] for such functions). Matrix elements for several such chains are shown in Tables 3.9 - 3.11.

3.6 Angular Overlap Model Many-Electron Matrix Elements [26]

In Chapter 1, Table 1.13 presented the one electron matrix elements for the d orbital energies in the AOM formalism. Using the relationships between the d^2 $|LM_L\rangle$ functions and the one electron states (Table 3.5), a set of d^2 $\langle LM_L|H_{AOM}|L'M'_L\rangle$ matrix elements can readily be derived. These are shown for the 90^o case in Table 3.12. They are quantized along the original four fold axis of the octahedron.

The ^3P and ^3F matrices in Table 3.12 correspond within the AOM formalism, with the DQ, DS, DT etc matrices within the NSH formalism. Diagonalisation of these matrices will yield a set of AOM energies. However use can again be made of symmetry adapted wavefunctions. Within this four fold quantized set use can be made of the wavefunctions symmetry adapted to the sub-groups along the four-fold chains.

For example, the A_2 matrix for a $MX_2Y_2Z_2$ species can be evaluated following (3.64 - 3.65, and Table 3.12):

$$\langle(1/\sqrt{2})(32 + 3\text{-}2)|H_{AOM}|(1/\sqrt{2})(32 + 3\text{-}2)\rangle$$

$$= \langle 32|H_{AOM}|32\rangle + \langle 32|H_{AOM}|3\text{-}2\rangle$$

$$= [(5/4)(e_\sigma(x) + e_\sigma(y)) + 2e_\sigma(z) + (e_\pi(x) + e_\pi(y))] + [-(3/4)(e_\sigma(x) + e_\sigma(y)) + (e_\pi(x) + e_\pi(y))]$$

$$= 0.5[(e_\sigma(x) + e_\sigma(y)] + 2e_\sigma(z) + 2(e_\pi(x) + e_\pi(y)) \qquad (3.70)$$

and

$$\langle 30|H_{AOM}|30\rangle$$

$$= (3/10)(e_\sigma(x) + e_\sigma(y)) + 2(e_\pi(x) + e_\pi(y)) + (16/5)e_\pi(z)$$

$$\langle 10|H_{AOM}|10\rangle$$

$$= (6/5)(e_\sigma(x) + e_\sigma(y)) + 2(e_\pi(x) + e_\pi(y)) + (4/5)e_\pi(z) \qquad (3.71)$$

While the off-diagonal elements will be, following (3.66):

$$\langle 30|H_{AOM}|10\rangle = (3/5)(e_\sigma(x) + e_\sigma(y)) - (8/5)e_\pi(z)$$

$$\langle 30|H_{AOM}|(1/\sqrt{2})[32 + 3\text{-}2] = (\sqrt{2})\langle 30|H_{AOM}|32\rangle$$

$$= (\sqrt{2})[\sqrt{(6/5)}(e_\sigma(y) - e_\sigma(x))/4] = \sqrt{(3/20)}[e_\sigma(y) - e_\sigma(x)]$$

$$\langle 10|H_{AOM}|(1/\sqrt{2})(32 + 3\text{-}2)\rangle = (\sqrt{2})\langle 10|H_{AOM}|32\rangle$$

$$= (\sqrt{2})[\sqrt{(24/5)}(e_\sigma(y) - e_\sigma(x))/4]$$

$$= \sqrt{(3/5)}[e_\sigma(y) - e_\sigma(x))] \qquad (3.72)$$

Table 3.12 [26]

d^2 ANGULAR OVERLAP MODEL MATRIX ELEMENTS[a]

$\langle d^2\ LM_L|H_{AOM}|d^2\ L'M'_L\rangle$ Spin Triplets

---

^3P State Diagonal Elements[bc]

$$\langle 11|H|11\rangle = \langle 1\text{-}1|H|1\text{-}1\rangle = E1 + (0.6\ E0) + (0.4\ E2)$$
$$= (3/5)(e_\sigma(x) + e_\sigma(y)) + (6/5)e_\sigma(z) + (7/5)(e_\pi(x) + e_\pi(y)) + 2e_\pi(z)$$

$$\langle 10|H|10\rangle = (0.4\ E1) + (1.6\ E2)$$
$$= (6/5)(e_\sigma(x) + e_\sigma(y)) + 2(e_\pi(x) + e_\pi(y)) + (4/5)e_\pi(z)$$

^3F State Diagonal Elements

$$\langle 33|H|33\rangle = \langle 3\text{-}3|H|3\text{-}3\rangle = E1 + E2$$
$$= (3/4)(e_\sigma(x) + e_\sigma(y)) + 2(e_\pi(x) + e_\pi(y) + e_\pi(z))$$

$$\langle 32|H|32\rangle = \langle 3\text{-}2|H|3\text{-}2\rangle = E0 + E2$$
$$= (5/4)(e_\sigma(x) + e_\sigma(y)) + 2e_\sigma(z) + (e_\pi(x) + e_\pi(y))$$

$$\langle 31|H|31\rangle = \langle 3\text{-}1|H|3\text{-}1\rangle = (0.4\ E0) + E1 + (0.6\ E2)$$
$$= (13/20)(e_\sigma(x) + e_\sigma(y)) + (4/5)e_\sigma(z) + (8/5)(e_\pi(x) + e_\pi(y)) + 2e_\pi(z)$$

$$\langle 30|H|30\rangle = (1.6\ E1) + (0.4\ E2)$$
$$= (3/10)(e_\sigma(x) + e_\sigma(y)) + 2(e_\pi(x) + e_\pi(y)) + (16/5)e_\pi(z)$$

^3P - ^3P Off-Diagonal Element

$$\langle 11|H|1\text{-}1\rangle = (-2\sqrt{6}/5\ ET0) - (0.6\ EN1)$$
$$= -(3/5)(e_\sigma(y) - e_\sigma(x) + e_\pi(y) - e_\pi(x))$$

^3F - ^3F Off-Diagonal Elements

$$\langle 33|H|31\rangle = \langle 3\text{-}3|H|3\text{-}1\rangle = (-\sqrt{(2/5)}\ ET0) + (\sqrt{(3/5)}\ EN1)$$
$$= \sqrt{(3/5)}[-(1/2)(e_\sigma(y) - e_\sigma(x)) + (e_\pi(y) - e_\pi(x))]$$

$$\langle 31|H|3\text{-}1\rangle = (2\sqrt{(6/25)}\ ET0) - (0.4\ EN1)$$
$$= (3/5)(e_\sigma(y) - e_\sigma(x)) - (2/5)(e_\pi(y) - e_\pi(x))$$

$$\langle 33|H|3\text{-}1\rangle = \langle 3\text{-}3|H|31\rangle = -\sqrt{(3/5)}\ EN2$$
$$= -\sqrt{(3/5)}[(3/4)(e_\sigma(x) + e_\sigma(y)) - (e_\pi(x) + e_\pi(y))]$$

$$\langle 30|H|32\rangle = \langle 30|H|3\text{-}2\rangle = \sqrt{(1/5)}\ ET0$$
$$= \sqrt{(6/5)}((e_\sigma(y) - e_\sigma(x))/4$$

---cont

Table 3.12 cont----------

$\langle 32|H|3\text{-}2\rangle = -EN2$
$= -(3/4)(e_\sigma(x) + e_\sigma(y)) + (e_\pi(x) + e_\pi(y))$

$^3F - {}^3P$ Off-Diagonal Elements

$\langle 10|H|30\rangle = 0.8(E2 - E1)$
$= (3/5)(e_\sigma(x) + e_\sigma(y)) - (8/5)e_\pi(z)$

$\langle 11|H|3\text{-}1\rangle = \langle 1\text{-}1|H|31\rangle = (\sqrt{}(6/25)\ EN1) - (0.2\ ET0)$
$= (\sqrt{}6)[((e_\pi(y) - e_\pi(x))/5) - ((e_\sigma(y) - e_\sigma(x))/20)]$

$\langle 11|H|31\rangle = \langle 1\text{-}1|H|3\text{-}1\rangle = \sqrt{}(6/25)(E2 - E0)$
$= -(1/5)(\sqrt{}6)[2e_\sigma(z) - (1/4)(e_\sigma(x) + e_\sigma(y)) - (e_\pi(x) + e_\pi(y))]$

$\langle 11|H|33\rangle = \langle 1\text{-}1|H|3\text{-}3\rangle = (\sqrt{}(3/5)\ ET0) + (\sqrt{}(2/5)\ EN1)$
$= \sqrt{}(18/5)((e_\sigma(y) - e_\sigma(x))/4) + \sqrt{}(2/5)(e_\pi(y) - e_\pi(x))$

$\langle 11|H|3\text{-}3\rangle = \langle 1\text{-}1|H|33\rangle = -\sqrt{}(2/5)\ EN2$
$= -\sqrt{}(2/5)[(3/4)(e_\sigma(x) + e_\sigma(y)) - (e_\pi(x) + e_\pi(y))]$

$\langle 10|H|32\rangle = \langle 10|H|3\text{-}2\rangle = \sqrt{}(4/5)\ ET0$
$= \sqrt{}(24/5)((e_\sigma(y) - e_\sigma(x))/4)$

---

a) Quantized along the z four fold or two fold axes
Two solutions are reported. The first solution provides the energies in terms of the one-electron AOM elements, $\langle 2m|H|2m'\rangle$ where: E0 = $\langle 20|H|20\rangle$, E1 = $\langle 21|H|21\rangle$, E2 = $\langle 22|H|22\rangle$, EN1 = $\langle 21|H|2\text{-}1\rangle$, EN2 = $\langle 22|H|2\text{-}2\rangle$, and ET0 = $\langle 22|H|20\rangle$.
The second result is cast in terms of the AOM parameters for the specific case of a $D_{2h}(I)$ all-trans-$MX_2Y_2Z_2$ system with all angles 90^o. By writing some of these parameters equal to one another, or to zero, this data set can be used for other point groups involving ligands lying along the cartesian coordinates.
b) Recall that $\langle LM_L|H|L'M_L'\rangle = \langle L'M_L'|H|LM_L\rangle$
c) Interelectronic repulsion energy should be added where appropriate.
Note that the matrix elements for the spin singlet states ^1D, ^1G and ^1S can be obtained in a similar fashion by expanding the states in terms of their one electron determinants (Table 3.6), and writing these determinants in terms of their AOM energies (e.g 1.76, Table 1.13).

These calculations provide a means of determining the A_2 energies of d^2 complexes in a rhombic (90^o, four fold and two fold) field. Corresponding calculations lead to the energies of the B_1, B_2 etc representations. However in practice, rather than reproduce such solutions here, it is easier to program the matrix

elements in Table 3.12 and multiply by the matrix of wavefunctions (e.g. in Tables 1.4 or 3.5) to generate the necessary energies. By appropriate choice of wavefunction in Table 3.12, and by multiplying by -1 where necessary, this approach provides the AOM energies for all high spin d^n configurations in sub-groups which retain two or four fold axes. The table should not be used for the three-fold groups or for groups which lack a symmetry axis.

Indeed Table 3.12 also contains the d^2 solutions in a general form, as a function of the d^1 matrix elements [22]. These d^1 elements can be generated and inserted into Table 3.12 to enable calculation of the d-d transition energies, subject again to the restrictions in the paragraph above.

Low spin calculations may be accomplished by uncoupling the many electron functions (e.g. using 3.41) and then calculating the many electron elements via the one electron functions in (1.76) or by using more sophisticated approaches, based on the Wigner-Eckart theorem, and published by Schaffer [17] or by Gerloch [18].

3.7 Spin-Orbit Coupling [2,5,20]

It has been assumed until now that the spin and orbit vectors of the atom are uncoupled. This is a good approximation when the spin-orbit coupling coefficient λ is small, as is the case with the lighter main group and transition series elements. With second, and especially third row transition metal ions, this approximation is not valid and a proper treatment must be set in terms of the $|JM_J\rangle$ basis states rather than the $|LM_L\rangle$ basis states. Spin-orbit coupling can lift the degeneracy of otherwise orbitally degenerate states leading to structure in absorption bands, and can provide a mechanism to mix states of different spin resulting in the borrowing of intensity by spin forbidden bands from spin allowed bands.

Recall that the orbital (L) and spin (S) vectors when dealing with free-ion states labelled $|LSM_LM_S\rangle$, may couple to yield states designated $|LSJM_J\rangle$. These states split to form a multiplet where the energy separation between the J and (J+1) state is $\lambda(J + 1)$ where λ is the Spin-Orbit Coupling Coefficient. For a shell less than half-full, λ is positive, while for more than half-full shells, λ is negative, (see ref.[2,3,6] for further details).

When dealing with group theoretical terms, splitting may also occur as a consequence of spin-orbit coupling.

If spin-orbit coupling is invoked with orbital singlet states, whatever their spin, no first order spin-orbit splitting can occur, though the energy of such states may be shifted by spin-orbit coupling to other states of like symmetry [27].

Splitting of an orbitally degenerate state ψ_i will occur if the matrix element $\langle\psi_i|\ LS|\psi_i\rangle$ is non-zero. The complete form of the spin-orbit operator is totally symmetric and therefore can couple only states of the same spin-orbit symmetry Γ_i. However it is useful to remember that L transforms as a rotation, e.g. T_{1g} in O_h. Considering only the orbital symmetry ψ_o, only states for which the

matrix element $\langle\psi_o|\lambda_L|\psi_o\rangle$ does not vanish, can be split by spin-orbit coupling. Hence in octahedral symmetry T_{1g} and T_{2g} states may split, but E_g cannot. In D_{4h}, however, the rotations transform as $A_{2g} + E_g$ and hence E_g states can be split by spin-orbit coupling.

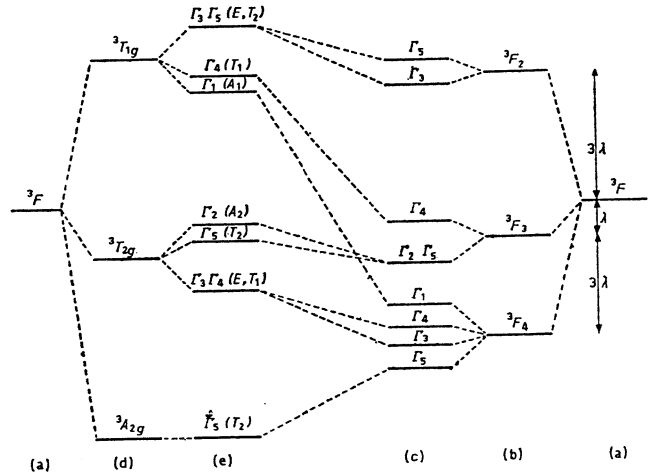


Fig.3.1 A d^8 correlation diagram relating spin-orbit terms with their crystal field and free-ion parents (^3F term only). (a) Spherically perturbed ion, (b) strong spin-orbit coupling with Dq = 0, (c) strong spin-orbit coupling with weak crystal field, Dq small, (d) Dq non-zero with zero spin-orbit coupling, and (e) both Dq and spin-orbit coupling large.

In common with the Racah parameters, the spin-orbit coupling coefficient parameter is an atomic spectroscopy (spherical free ion) parameter. It is incorporated into the theory of lower symmetry species with no modification other than to allow it to vary from the free ion value. A more complete treatment should recognise that in lower symmetry, different configurations might have different values for this coefficient. Note however some examples in Chapter 5 where different charge transfer states do have effectively different values for the spin-orbit coupling coefficient due to different mixing with the ligand.

To determine the total spin-orbit symmetry, it is necessary to form the direct product of the orbital and spin components. Where the spin is integral, this represents no problem. For example, in O_h, considering the degeneracies of the spin states:

S = 0 transforms as A_{1g} (Γ_1)
S = 1 transforms as T_{1g} (Γ_4)
S = 2 transforms as $E_g + T_{2g}$ ($\Gamma_3 + \Gamma_5$)
etc

where the commonly used Bethe nomenclature for spin-orbit states is also shown.

$\Gamma_1 = A_1'$, $\Gamma_2 = A_2'$, $\Gamma_3 = E'$, $\Gamma_4 = T_1'$, $\Gamma_5 = T_2'$, $\Gamma_6 = E_2'$, $\Gamma_7 = E_3'$ and $\Gamma_8 = G'$ (four-fold degenerate). The primes on the conventional symbols imply that they belong to the double octahedral group. The last three symbols occur only in the double group and not in the simple O_h group.

Thus, spin-orbit mixing in the ^1E_g octahedral state leads to the unsplit Γ_3 spin-orbit state, while $^3T_{1g}$ forms the multiplet ($T_1 \times T_1$) $\Gamma_1 + \Gamma_3 + \Gamma_4 + \Gamma_5$ (Fig.3.1). The existence of Γ_3 in both ^1E_g and $^3T_{1g}$ implies that these components can be mixed in second order by spin-orbit coupling. This mixing is responsible for a historically famous study of these states in octahedral nickel(II) spectra in which the spin-singlet gains significant intensity [1,2,19,20].

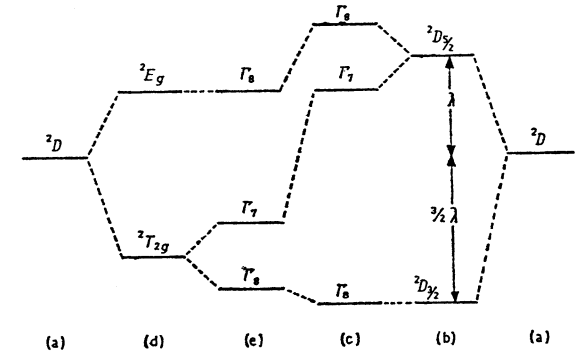


Fig.3.2 A d^1 correlation diagram relating spin-orbit terms with their crystal field and free ion parents. (a) to (e) as Fig.3.1.

There is a simple formula to derive the splitting of octahedral T states, to first order, utilising the special relationship these T states have with P free ion states, such that these octahedral T states have an effective L value of 1 [2,5,20]. Thus in a $^{2S+1}T_1$ or $^{2S+1}T_2$ state, the J levels may be derived via coupling of L = 1 with S. Thus for S = 3/2, J is 5/2, 3/2 and 1/2 leading to the splitting of $^4T_1$ or $^4T_2$ into a triplet with degeneracies (2J +1) of 6, 4 and 2 respectively. The energies of these states are given by [20]:

$$W_{J,S} = (1/2)\,\lambda[J(J+1) - 2 - S(S+1)] \qquad (3.73)$$

In general these $^{2S+1}T_i$ ($i = 1,2$) levels split into three subsets even though group theory may demand more (see above). Further splitting may occur by interaction of the various sub-levels with other spin-orbit states.

It is necessary for a complete treatment to consider the symmetry of half-integral spin states.

Group theoretically spin-orbit coupling introduces some difficulties if the spin is half-integral. In such cases the double groups must be used [15,21]. A brief example is made of the ^2D states of d^1 under the influence of spin-orbit coupling.

The character (chi) for a rotation of θ for a given value of J is determined from:

$$\text{chi}(\theta) = \sin(J + 1/2)\theta/\sin(\theta/2) \qquad (3.74)$$

Since a rotation of 360^o is equivalent to the identity operation, chi(θ) should equal chi($\theta + 2\pi$). This is true for integral J, but for half-integral J:

$$\text{chi}(\theta) = -\,\text{chi}(\theta + 2\pi)$$

hence simple groups cannot be used to handle half-integral J values. Bethe introduced the fiction that a rotation of 360^o called R should be treated as a separate symmetry operation but not as the identity operation.

> The Mobius strip is a good illustration of this idea. Such a strip is constructed by taking a piece of paper, say 12" by 1", twisting one end through 180^o and joining it to the other end. Now draw a line from a point on the surface of the strip, and continue the line until you return to the point of departure. It is found that it is necessary to make TWO complete revolutions of the circle ($2 \times 2\pi$) to reach the point of departure.

Double groups are constructed by multiplying all the symmetry operations of the single group by R. Such double groups are used to determine the symmetry behaviour of half-integral J values. For example, in the double group O', the J = 5/2 and J = 3/2 (and S = 1/2) states of ^2D behave as follows:

	E	R	$8C_3$	$8C_3R$	$6C_2$	$6C_4$	$6C_4R$	$12C_2'$
S = 1/2	2	-2	1	-1	0	$\sqrt{2}$	$-\sqrt{2}$	0
J = 3/2	4	-4	-1	1	0	0	0	0
J = 5/2	6	-6	0	0	0	$-\sqrt{2}$	$\sqrt{2}$	0

(3.75)

which may be reduced by conventional means to:

$$(J = 3/2) = \Gamma_8 \qquad (J = 5/2) = \Gamma_7 + \Gamma_8 \qquad (3.76)$$

In an alternative and perhaps more readily used procedure, the ^2D state is first perturbed by the octahedral field to yield ^2E_g + $^2T_{2g}$, and the spin-orbit representations of the corresponding spinor states obtained by direct product of the orbital part of these representations with the spin part. The latter is obtained from (3.73) and (3.74), namely, $(S = 1/2) = \Gamma_6$. Using the double group (in Appendix I) and the Bethe nomenclature for E_g (Γ_3) and T_{2g} (Γ_5) these products are:

$$\Gamma_6 \times \Gamma_3 = \Gamma_8$$
$$\Gamma_6 \times \Gamma_5 = \Gamma_8 + \Gamma_7 \qquad (3.77)$$

and are summarised in Fig.3.2.

We are concerned briefly with the methods of calculating the energies of spinor states and how they are coupled together. Matrix elements of the spin-orbit coupling operator mix orbit and spin together and couple states of the same spinor symmetry. They have the general form:

$$\langle LSM_L M_S | \lambda L.S | LSM_L M_S \rangle \qquad (3.78)$$

where the operator L.S may be expanded:

$$\begin{aligned} L.S &= 0.5(L_x + iL_y)(S_x - S_y) + 0.5(L_x - iL_y)(S_x + iS_y) + L_zS_z \\ &= 0.5\,L_+S_- + 0.5\,L_-S_+ + L_zS_z \end{aligned} \qquad (3.79)$$

The four 'ladder operators' (abbreviated L_+, L_-, S_+, S_- and described in some detail in Appendix VIII) have the effect of raising or lowering the m_l or m_s value of a function. For example the first bracket operating on the ket $|l,s,m_l,m_s\rangle$ yields:

$$(L_x + iL_y)|l,s,m_l,m_s\rangle = h[l(l+1) - m_l(m_l+1)]^{1/2}|l,s,m_l+1,m_s\rangle \qquad (3.80)$$

If a ket $|l,s,m_l+1,m_s\rangle$ exists in the set, then it is coupled to $|l,s,m_l,m_s\rangle$ by this operator, viz:

$$(l,s,m_l+1,m_s|(L_x + iL_y)|l,s,m_l,m_s\rangle = h[l(l+1) - m_l(m_l+1)]^{1/2} \qquad (3.81)$$

otherwise operation with this component of the operator yields zero. These four 'ladder operators' contribute off-diagonal elements to the spin-orbit matrix. The L_z and S_z operators do not change m (e.g. $L_z\,|m_l\rangle = m_l\,|m_l\rangle$) and therefore generate diagonal elements to the matrix. Given a one electron basis set, the spin-orbit matrix is generated via (3.79) and the resultant matrix diagonalised to yield the spin-orbit energies in terms of the spin-orbit coupling coefficient.

For example within the d orbital set, m_l = 2,1,0,-1 and -2, for a single d electron, m_s = +1/2 or -1/2, then, only the following operations may yield non-zero results [28]:

$$
\begin{aligned}
&\text{i) } L_+|m_l\rangle \text{ is non-zero for } m_l = 1,0,-1,-2\\
&\text{ii) } L_-|m_l\rangle \text{ is non-zero for } m_l = 2,1,0,-1\\
&\text{iii) } S_+|m_l\rangle \text{ is non-zero for } m_s = -1/2\\
&\text{iv) } S_-|m_l\rangle \text{ is non-zero for } m_s = +1/2\\
&\text{v) } L_z|m_l \text{ is non-zero for } m_l = 2,1,-1,-2\\
&\text{vi) } S_z|m_s \text{ is non-zero for } m_s = +1/2,\ -1/2
\end{aligned}
\qquad (3.82)
$$

Following [28] consider the spin-orbit matrix for $d^1\ {}^2T_{2g}$ ground term.

The six spinor states are:

$$
\begin{aligned}
\phi1 &= (1/\sqrt{2})[|2^+\rangle - |-2^+\rangle] & \phi4 &= |1^-\rangle\\
\phi2 &= (1/\sqrt{2})[|2^-\rangle - |-2^-\rangle] & \phi5 &= |-1^+\rangle\\
\phi3 &= |1^+\rangle & \phi6 &= |-1^-\rangle
\end{aligned}
\qquad (3.83)
$$

obtained simply by multiplying each real t_{2g} orbital, written in terms of its m_l value, by positive or negative spin.

Consider the operation $L.S|\phi1\rangle$, then:

$$
\begin{aligned}
L.S|\phi1\rangle &= 0.5L_+S_-|\phi1\rangle + 0.5L_-S_+|\phi1\rangle + L_zS_z|\phi1\rangle\\
&= 0.5L_+S_-|\phi1\rangle + 0 + 0
\end{aligned}
\qquad (3.84)
$$

where the second and third terms will contribute zero elements of the type (3.81) because of statements (3.82 iii) and (3.82 v) respectively, i.e. one cannot raise a spin of +1/2 (S_+) and the effective m_l value of $\phi1$ (and $\phi2$) is zero.

The first term may be expanded as follows:

$$
\begin{aligned}
0.5L_+S_-|\phi1\rangle &= (1/\sqrt{2})[0.5L_+S_-|2^+\rangle - 0.5L_+S_-|-2^+\rangle]\\
&= (1/\sqrt{2})[0.5L_+|2^-\rangle - 0.5L_+|-2^-\rangle]\\
&= (1/\sqrt{2})[\ 0\ - 0.5\sqrt{(6-2)}|-1^-\rangle]\\
&= -(1/\sqrt{2})|-1^-\rangle = -(1/\sqrt{2})|\phi6\rangle
\end{aligned}
\qquad (3.85)
$$

where the first term disappears because $|2\rangle$ cannot be raised by L_+ (3.82 i). It should be clear that the set of six matrix elements:-

$$\langle\phi i|\lambda L.S|\phi1\rangle \quad i = 1 \text{ to } 6$$

all disappear except for:

$$\langle\phi6|\lambda L.S|\phi1\rangle = -(1/\sqrt{2})\lambda$$

Similarly:

$$
\begin{aligned}
L.S|\phi3\rangle &= 0.5L_+S_-|\phi3\rangle + 0.5L_-S_+|\phi3\rangle + L_zS_z|\phi3\rangle\\
&= 0.5L_+S_-|1^+\rangle + 0 + L_zS_z|1^+\rangle\\
&= 0.5L_+|1^-\rangle + 0 + (1 \times 1/2)|1^+\rangle\\
&= (\sqrt{4})0.5\ |2^-\rangle + 0.5\ |1^+\rangle\\
&= |2^-\rangle + 0.5\ |\phi3\rangle
\end{aligned}
\qquad (3.86)
$$

and the set of matrix elements $\langle\phi i|\ L.S.|\phi3\rangle$ are all zero except:

$$
\begin{aligned}
\langle\phi2|\lambda L.S|\phi3\rangle &= (1/\sqrt{2})\lambda\\
\langle\phi3|\lambda L.S|\phi3\rangle &= 1/2\lambda
\end{aligned}
$$

The complete set is [28]:

$$
\begin{aligned}
L.S|\phi1\rangle &= -(1/\sqrt{2})\ |\phi6\rangle\\
L.S|\phi2\rangle &= (1/\sqrt{2})\ |\phi3\rangle\\
L.S|\phi3\rangle &= |2^-\rangle + 0.5\ |\phi3\rangle\\
L.S|\phi4\rangle &= -(1/2)\ |\phi4\rangle\\
L.S|\phi5\rangle &= -(1/2)\ |\phi5\rangle\\
L.S|\phi6\rangle &= |-2^+\rangle + 0.5\ |\phi6\rangle
\end{aligned}
\qquad (3.87)
$$

Finally the complete matrix is:

$$
\begin{array}{c|cccccc|c}
 & \phi1 & \phi2 & \phi3 & \phi4 & \phi5 & \phi6 & \\
\phi1 & 0-E & & & & & -(1/\sqrt{2}) & \\
\phi2 & & 0-E & (1/\sqrt{2}) & & & & \\
\phi3 & & (1/\sqrt{2}) & (1/2)-E & & & & \\
\phi4 & & & & -(1/2)-E & & & = 0\\
\phi5 & & & & & -(1/2)-E & & \\
\phi6 & -(1/\sqrt{2}) & & & & & (1/2)-E &
\end{array}
\qquad (3.88)
$$

which has two roots at $+\lambda$ and four at -0.5λ. Re-insertion of these energies into the matrix yields the eigenvectors which are:

at $+\lambda$

$$
\begin{aligned}
\psi_1 &= (1/\sqrt{3})[\phi2 + \sqrt{2}\phi3]\\
\psi_2 &= (1/\sqrt{3})[\phi1 - \sqrt{2}\phi6]
\end{aligned}
$$

at -0.5λ

$$
\begin{aligned}
\psi_3 &= (1/\sqrt{3})[\sqrt{2}\phi2 - \phi3]\\
\psi_4 &= (1/\sqrt{3})[\sqrt{2}\phi1 + \phi6]\\
\psi_5 &= \phi4
\end{aligned}
$$

$$\psi_6 = \phi 5 \qquad (3.89)$$

which show the extensive mixing induced by spin-orbit coupling, and the splitting of the six-fold degenerate $^2T_{2g}$ into a set of two and a set of four.

Further details of these operators and the theory of spin-orbit coupling are presented elsewhere [2,5]. Note that the Wigner-Eckart theorem may also be used to solve spin-orbit matrices (e.g. [18]).

The theory of spin-orbit coupling is encountered again in Chapter 4 when the concept of intensity stealing by spin forbidden transitions is considered.

3.8 Ground State Energy Space Diagrams

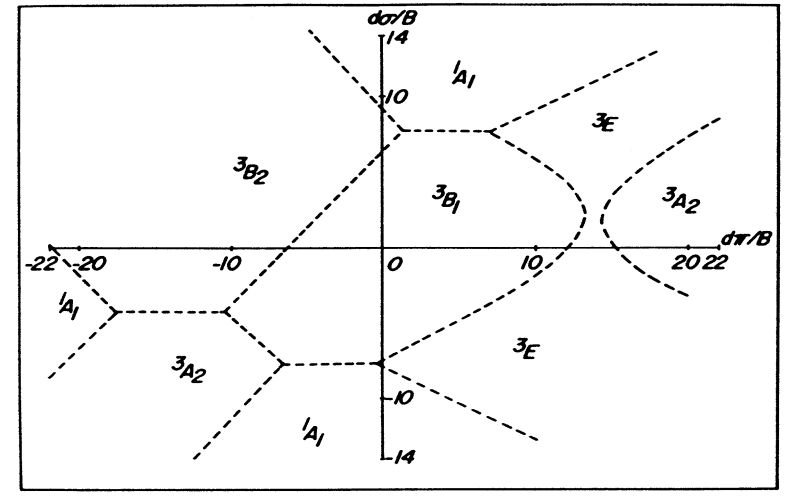
The ground state of a complex is not always readily predicted. This is particularly true of low symmetry complexes where several possible ground states may occur depending upon the distortion present. Moreover different spin states may occur, depending upon the degree of distortion.

Suppose, for example, a set of energy levels in a tetragonal molecule whose energies are determined by the parameters DQ, DS, DT and B. Imagine a five dimensional space in which each parameter is varied, yielding a set of surfaces lying along an energy coordinate. Suppose further that such surfaces are set up along an energy coordinate with the lowest lying state defined as energy zero. Some excited states can never become this energy zero, i.e the ground state, whatever the combination of values of DQ, DS, DT and B. Other states may become the ground state for specific values or ranges of values of any or all of these parameters.

If a plane is cut through this E = 0 level, then regions of parameters are found inside this plane in which various states could become the ground state. Such a diagram is called a Ground State Energy Space Diagram [23].

Because of the constraints of a two dimensional piece of paper, two parameters are usually plotted while maintaining other parameters constant, e.g. as shown in Fig.3.3, though it would certainly be feasible to vary three parameters in a perspective drawing (or computer program). They may be drawn in terms of any of the theoretical models and may have particular chemical utility when drawn in terms of AOM or McClure parameters.

Such diagrams can be generated for all values of the distortion parameters deemed reasonable for the complex under consideration, to provide a ready picture of the ground states which might be accessible to the species. If the actual distortion parameters measured place the complex close to a ground state boundary then there must be a low lying state near to the ground state. Use of these diagrams could provide impetus to synthetic chemistry by showing what kind of distortion is needed to generate a species with a hitherto unknown ground state.



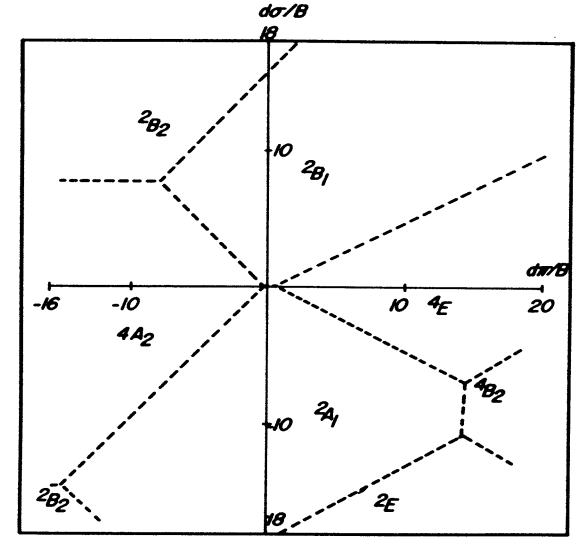
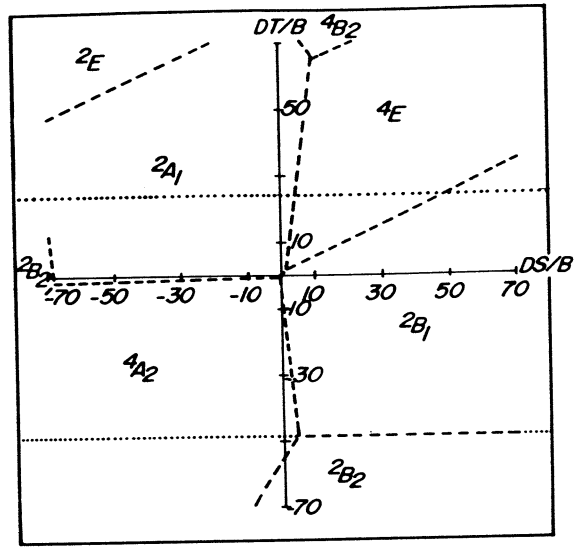


Fig.3.3 (a) D_{4h} ground state energy space dσ,dπ diagram for d^8 ions (DQ = 18,000, B = 800, C = 3000 cm^{-1}). (b) D_{4h} ground state energy space dσ,dπ diagram for d^7 ions (DQ = 40,000, B = 700, C = 3000 cm^{-1}. Also see Fig.6.37 (b).



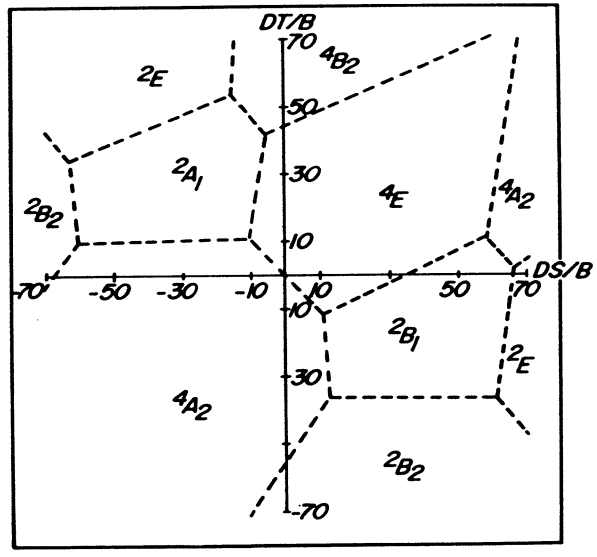


Fig.3.3 (c) D_{4h} ground state energy space DT,DS diagram for d^7 ions (DQ = 40,000, B = 700, C = 3000 cm^{-1}). (b) D_{4h} ground state energy space DT,DS diagram for d^7 ions (DQ = 30,000, B = 700, C = 3000 cm^{-1}.

The d^7 D_{4h} ground state energy space diagrams displayed here exhibit some of the interesting aspects of this procedure. With relatively small global DQ (e.g. Fig.6.37 (b), and Figs.3.3 (d), both the $^4A_2$ and ^4E ground states are accessible depending upon the sign of $d\pi$. Since this can often be deduced from the nature of the ligands involved, this diagram may provide an immediate assessment of the correct ground state in an otherwise ambiguous situation. However the crossover does depend upon B; the exact boundary does not accurately follow the $d\sigma = 0$ coordinate and some caution should be exercised if $d\pi$ is small (Fig.6.37 (b)). Clearly the use of appropriately π-bonding ligands can dictate one ground state or the other.

More interestingly, when stronger field ligands are present such that the global DQ is raised (Figs.3.3 (b,c)), four possible states lie very close together. This implies that any of these four might be the ground state depending upon the nature of the ligands. Moreover whatever the ground state, the other three states may lie very close thereto, leading to unusual magnetic and spectroscopic properties.

Fig.3.3 (a) shows that for a d^8 species, very considerable distortion is necessary before a ground state other than $^3B_1$ can be achieved. However the diagram does predict that very strongly π-donating ligands along the z axis or strongly π-accepting ligands in the plane may produce a $^3A_2$ ground term, while the reverse situation should generate a $^3B_2$ ground term. Several linear environment ground state energy space diagrams are shown in Fig.6.59.

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Chapter 4

SELECTION RULES, BAND INTENSITIES, VIBRONIC COUPLING, DICHROISM AND EMISSION SPECTROSCOPY

4.1 Introduction

If a beam of monochromatic light of intensity I_o passes through a solution of path length l (measured in cm.) containing solute of molar concentration c, then the transmitted light, of intensity I, is related to the incident light intensity, I_o, by:

$$T = I/I_o = 10^{-\varepsilon cl}$$

or

$$E = \log_{10}(I_o/I) = \varepsilon cl \qquad (4.1)$$

where ε is the molar extinction coefficient (in units of l cm^{-1} $mole^{-1}$ or cm^2 $mole^{-1}$) E is the optical density and T is the transmittance (both dimensionless). This is the basis of the Beer-Lambert Law which states that the amount of light absorbed is proportional to the number of absorbing molecules through which the light passes. If a solute is dissolved in a transparent medium, then the absorbance is proportional to the concentration of the solute. The majority of systems obey the Beer-Lambert Law but it will fail if there are phenomena such as aggregation which is dilution dependent, or a dilution dependent reaction with the solvent.

For a given concentration and path length, the actual absorbance depends upon the molar extinction coefficient; this chapter discusses the factors upon which this depends and shows how the measurement of absorption intensities aids assignment, and indeed provides information about the metal-ligand bonds.

Coupling between electronic and vibrational motion plays an important role in electronic spectroscopy. The group theoretical aspects of this coupling are discussed and methods are presented to obtain valuable information about the molecular structure both in the ground state and excited state. Various other topics such as the Jahn-Teller effect, and the techniques of photoselection spectroscopy, circular dichroism, magnetic circular dichroism and resonance Raman spectroscopy are briefly touched upon to provide background for later chapters.

4.2 Electronic Dipole Selection Rules

Transitions may occur through electric dipole, magnetic dipole or electric quadrupole mechanisms [1]. As far as transition metal, d block, complexes are concerned, their spectra usually arise through electric dipole transitions; magnetic and electric quadr-

upole transitions have a very much lower probability of occurrence (i.e. give rise to very low intensity absorption, see below), though the former have been detected. Magnetic dipole transitions are more common in the spectra of the lanthanides and actinides.

An electric dipole transition will be intense if it is accompanied by a large change in dipole moment, i.e. there is a considerable charge translation during the transition, and will be weak (or not observed) if there is a small (or zero) change in the dipole moment. Group theoretically the components of the electric dipole operator transform like translations.

The components of the magnetic dipole operator transform like rotations and those of the electric quadrupole operator as products of cartesian coordinates (such as xy, z^2 etc) with the same selection rules as the Raman effect. Magnetic dipole allowed origins are often seen very weakly when a transition is magnetic dipole allowed, but electronic dipole forbidden. Magnetic quadrupole allowed transitions can be detected in actinide systems where they can be quite strong [25]. The intensities of electric dipole, magnetic dipole, and electric quadrupole transitions [3] are of the order of 6.5×10^{-36}, 8.7×10^{-41} and 4×10^{-42} c.g.s. units respectively.

The molar extinction coefficient defines the molar absorbance of a transition at the peak wavelength of an absorption band. Of more fundamental significance is the oscillator strength f which is a measure of the integrated intensity under the entire absorption profile (plotted as extinction coefficient versus frequency) and is directly related to a theoretically derivable quantity called the dipole strength D of the transition. If the band stretches from ν_1 to ν_2 (in cm^{-1}) then:

$$f = 10^3 \log_e(10)(mc^2/N\pi e^2) \int_{\nu_1}^{\nu_2} \varepsilon d\nu$$

$$= 4.315 \times 10^{-9} \int \varepsilon d\nu$$

$$= \text{ca. } 4.6 \times 10^{-9}\ \varepsilon_{max} \Delta_{1/2} \qquad (4.2)$$

(where e and m are the charge and mass of an electron, c is the velocity of light in vacuo, and N is Avogadro's number). $\Delta_{1/2}$ known as the half-bandwidth, is the width of the band, in cm^{-1} where the molar extinction coefficient has a value half its maximum. Effectively f is a measure of the intensity summed over all the vibrational components. The magnitude of f as measured experimentally using Eqn.4.2 can be related to the probability that the transition will occur, via the following.

The matrix element of the dipole moment operator connecting states ψ_a and ψ_b, usually called the transition moment, P is:

$$P = \langle \psi_a | M | \psi_b \rangle = P = eD^{1/2} \qquad (4.3)$$

and where $D = P^2$ is called the dipole strength.

Note that some authors use the letter D for the transition moment leading to some confusion.

The square of the transition moment is proportional to the transition probability, i.e. the probability that the transition will take place, and is related to the oscillator strength via:

$$f = (8\pi^2 mc/3he^2) G\nu P^2 = 0.470 \times 10^{30}\ G\nu P^2$$
$$= 1.085 \times 10^{11}\ G\nu D \quad \text{(if P is in cm)}$$
$$= 1.085 \times 10^{-5}\ G\nu D \quad \text{(if P is in angstroms)} \qquad (4.4)$$

where h is Planck's constant, ν is in cm^{-1}, and G is a number referring to the degeneracies of the states concerned. G is unity for a transition between unidimensional states. For states of degeneracy d, G = 1/d, and the oscillator strength must be summed over all pairs of possible transitions.

M (Eqn.4.3) is the dipole moment operator with components M_x, M_y and M_z, such that, for electric dipole allowed transitions:

$$M_x = -e\sum_i x_i \qquad (4.5)$$

i.e. the intensity of an absorption band depends on the relative orientation of the electric vector of the incident light, and the symmetry axes of the molecule (vide infra). The term is summed over all electrons i of charge -e. Combining (4.3 - 4.5) provides:

$$f = \text{const. } \nu \langle \psi_a | M | \psi_b \rangle^2$$

$$= \text{const.} \nu [\langle \psi_a | M_x | \psi_b \rangle^2 + \langle \psi_a | M_y | \psi_b \rangle^2 + \langle \psi_a | M_z | \psi_b \rangle^2] \qquad (4.6)$$

Simple group theoretical procedures can decide whether these various elements must be zero or can exist. In general a group theoretical approach is used, qualitatively, to decide the relative intensities of electronic transitions. This approach will also be used here. It is beyond the scope of this book to calculate precise values of the transition moments as a function of the wavefunctions involved, though we shall comment on this calculation in several places below. We note that the classical approach to the evaluation of molecular intensities does have its critics and that Mason, especially, argues for the use of the Ligand Polarisation Model as a more appropriate vehicle for intensity analysis [43]. The reader is also referred to Ballhausen [3] for more detail on intensities.

An element in (4.6) may exist if the direct product of the ground and excited state wavefunctions with the dipole moment operator, contains the totally symmetric representation of the group concerned [4,5]. The element will certainly vanish if this is not true. Generally speaking the representations of the wavefunctions and operators are known and therefore the direct product can be evaluated. Since the dipole moment operator is odd with respect to inversion, it can only couple an odd to an even wavefunction. The selection rule that:

electric dipole transitions between states of equal parity are forbidden

is derived immediately. In other words, d-d, p-p, f-f transitions are forbidden, but d-p, d-f transitions would be permitted by this selection rule. This rule may also be written in the form: $\Delta l = \pm 1$ for orbitally allowed transitions, and is known as Laporte's Rule. Since the dipole moment operator is a one-electron operator, i.e. it operates on one electron at a time, an electric dipole transition can only involve one electron excitation, hence:

transitions involving excitation of two or more electrons are forbidden

The total wavefunction can be factored, to a first approximation, into rotational, vibrational and electronic components such that:

$$\psi = \psi_R \cdot \psi_V \cdot \psi_E \qquad (4.7)$$

This is usually known as the Born-Oppenheimer approximation. Electronic transitions take place in such a short time interval (ca. 10^{-15} s) that the nuclei may be assumed to remain stationary during the transition (Franck-Condon transition).

> At the instant of excitation, the excited state has the same nuclear geometry as the ground state, and may be highly excited, vibrationally. Relaxation to the ground vibrational level of the excited electronic state occurs very rapidly (picoseconds). The equilibrium nuclear configuration of the excited state, may or may not, be the same as that of the ground state. Detailed study of the electronic absorption and emission data can provide information about this excited state geometry.

Consider, for now, only the electronic component of the wavefunction. It will consist of an orbital (o) component, and a spin (s) component:

$$\psi_a = \psi_a^{o} \cdot \psi_a^{s} \qquad (4.8)$$

Since the dipole moment operator does not couple electronic spins, they must be 'integrated out' of the transition moment expression:

$$\sqrt{f} = \text{const.} \langle \psi_a^{o} | M | \psi_b^{o} \rangle \langle \psi_a^{s} | \psi_b^{s} \rangle \qquad (4.9)$$

Expression (4.9) vanishes because of spin orthogonality unless the spin functions of the ground and excited state are the same. Hence:

transitions between wavefunctions of different spin are forbidden

or $\Delta S = 0$ for spin-allowed transitions.

Since the totally symmetric representation of any group can only be obtained from a direct product of the type Γ_i^2, and never from $\Gamma_i \times \Gamma_j$ $(i \neq j)$ it follows that the transition moment will be zero unless the direct product of the representations of the ground and excited state wavefunctions spans a representation of the dipole moment operator.

The actual magnitude of the intensity of a transition depends, inter alia, upon the overlap between the ground and excited state wavefunctions. For example if the ground state function is localised in the plane of a molecule, and the excited state is in an orthogonal orbital perpendicular to the plane, the overlap will be zero, and, in the absence of any second order effects (for example mixing of a third state into ground or excited state), the intensity will also be zero. This remains true even if the group theoretical direct product contains A_1. Also see section 5.5.

4.2.i) Polarisation of Electronic Transitions

Since the transition moment is a tensor (of rank 1 or greater) it has both magnitude and direction as indicated above (4.5,4.6). In fluid media the question of polarisation does not arise since the molecules are randomly oriented with respect to the incident light. No linear polarisation information can be obtained from a solution spectrum to argue the assignment of the transition. If however a beam of polarised light is incident along an extinction axis of a crystal of known crystal structure, information concerning assignments can often be derived. When a crystal absorbs light to a different degree along different crystal directions it often appears a different colour along these various directions; the phenomenon is called dichroism.

> More precisely, when a crystal appears a different colour along two orthogonal axes it is called 'dichroic', and if a different colour along more than two axes, 'polychroic'.

If a beam of plane polarised light is passed perpendicularly through the plane of a uniaxial crystal, transition moments are projected along the symmetry axis, and normal to this symmetry axis.

> Uniaxial crystals include trigonal, tetragonal and cubic crystals. A crystal without a symmetry axis, may exhibit frequency dependent polarisation behaviour [3].

By analysing the relative intensity of absorption as a function of wavelength, direction of polarisation and crystal axis, single crystal polarised light absorption may provide unequivocal information about band assignments and may also, through the use of linear dichroism, decide whether a transition is magnetic dipole or electronic dipole allowed.

For example one may measure an axial spectrum by passing polarised light along the symmetry axis c of a crystal. In this instance both the electric vector E and the magnetic vector H are perpendicular to the optic axis. If plane polarised light is now propagated perpendicular to c, there are two possibilities, i) the

electric vector is perpendicular to c, and the magnetic vector is parallel to c, ii) the electric vector is parallel to c, and the magnetic vector is perpendicular to c. These are termed the σ- and π-spectra respectively. If the c axis of the crystal corresponds with the z axis of the molecule, these spectra correspond with the (x,y) and z polarised spectra respectively, and the symmetry of the electric dipole moment operator will correspond to those of the translations T_x,T_y and T_z respectively. If the axial spectrum corresponds with the σ-spectrum, the transition is electric dipole allowed; if the axial spectrum corresponds with the π-spectrum, it is magnetic dipole allowed.

Beware of some earlier literature where the relationship between σ and π and (xy) and (z) is reversed.

Single crystal polarised light studies therefore provide a powerful means of unequivocally assigning observed electronic transitions on the basis of their polarisation. In the lower groups, such as C_{2v}, the three dipole moment operators, M_x, M_y and M_z, belong to three different representations so that it is theoretically possible to distinguish transitions polarised along x, y, or z. However this is not possible for all space groups.

As will be seen in examples throughout this book (and Fig.4.1, 4.2 below) absorption peaks may occur with the light oriented along one coordinate, but not another, leading sometimes to an unequivocal determination of their symmetry designation, if the point group can be accurately defined. Often ambiguities in the choice of the effective point group prevent unequivocal assignments.

Usually the crystal is cooled to low temperatures, at least to liquid nitrogen temperature, but preferably to liquid helium temperature, as a means of obtaining sharp, highly resolved polarised data (by removing transitions from 'hot' excited vibrational levels of the electronic ground state).

Even in a crystal of known structure where the molecules do not line up perfectly with the crystal axes, the ratios of the intensities of the polarisations of a given band along different crystal directions can provide usable information to define the transitions.

Note that although it is most desirable to know the X-ray structure of the crystal and hence know precisely the orientation of the electric vector with the molecular axes, useful spectra and information may be obtained even without X-ray data. Thus the extinction axes of a crystal may be determined using a polarising microscope, and polarised light propagated along them. If the molecules are lined up in an advantageous manner, highly dichroic spectra may be obtained. Very often the relative energies of the various transitions can be deduced from knowledge of the ligand field strengths of the relevant ligands and essentially unequivocal assignments may follow. Sometimes comparison of polarisation data obtained with room temperature crystals with those obtained with low temperature crystals can provide information concerning temperature dependent structural changes. Many examples of the use of polarisation are presented in Chapter 6.

4.2.ii) Molecules without inversion centres (acentric complexes)

The parity selection rule forbidding d-d transitions is relaxed under certain circumstances, including the situation where the molecules do not possess a centre of symmetry. Under such circumstances parity has no absolute significance, though in practice, intensities are usually low if the distortion of an otherwise centrosymmetric molecules, is small.

In the group C_{4v}, M_x,M_y transform as E, and M_z transforms as A_1. For a transition ψ_a---ψ_b to be orbitally allowed, the product $\Gamma_a . \Gamma_b$ must contain E and/or A_1. Under such circumstances the triple direct product implied in the elements of (4.6) must contain A_1. Therefore:

Transition ψ_b	ψ_a	Direct product $\Gamma_a \times \Gamma_b$	Orbitally Allowed or forbidden	Polarisation
A_2	A_1	A_2	Forbidden	
E	A_1	E	Allowed	x,y
B_2	B_1	A_2	Forbidden	
B_2	B_2	A_1	Allowed	z
E	E	$A_1+A_2+B_1+B_2$	Allowed	z

(4.10)

In the cases cited in (4.10), if the light beam is so arranged that the electric vector is parallel to the z axis, absorption corresponding to B_2---B_2 and E---E transitions could occur, but an E---A_1 transition, for example, could not occur (as an electronic transition). Conversely if the electric vector lies along the x or y axes, the E---A_1 transition could be seen, but neither of the other two cited transitions will occur.

4.2.iii) Molecules with inversion centres (centrosymmetric complexes).

In the octahedral group, the electric dipole moment operator components transform as T_{1u}, and all d-d transition are forbidden. Nevertheless transition metal complexes are coloured. Absorption does occur, even though it is forbidden. However, in general, absorption intensities in centrosymmetric transition metal complexes are rather small, and smaller than similar absorption in acentric transition metal complexes.

There are two common mechanisms by which the parity selection rule is relaxed, namely static or dynamic removal of the centre of symmetry. If any distortion occurs which permanently or temporarily removes the centre of symmetry, then the effective group to which the molecule belongs is no longer O_h. In general such effects do not relax the selection rules greatly, but they do provide a pathway for some absorption intensity. Molecules such as the

tris-acetylacetonates, for example, have an octahedron of oxygen atoms about the metal atom, but if the rest of the molecule is taken into account, the highest symmetry is the acentric D_3. Whether or not the d-d spectra follow O_h or D_3 selection rules depends upon the metal and can only be deciphered by recourse to experiment.

In general the more ionic transition metal complexes in general have smaller absorption intensities than covalent analogs, whatever the symmetry, probably as a consequence of the Laporte Rule.

4.2.iv) Vibronic coupling in centrosymmetric molecules

In practice the most important mechanism for relaxing the Laporte Rule in a centrosymmetric molecule is vibrational coupling. The question of vibronic coupling is one of the more difficult aspects of electronic spectroscopy. This section begins with a simplified approach and then becomes more rigorous.

Returning to (4.7) recognise that there is a vibrational contribution to the wavefunction. If a molecule absorbs a quantum of vibrational energy corresponding to an odd vibration, then the centre of symmetry will be lost. The vibrationally excited molecule is no longer restricted by the parity rule and can undergo an electronic transition. In practice it is assumed that the molecule is electronically excited with simultaneous excitation of an appropriate odd vibration. Such a mechanism is known as vibronic coupling (VIBRational electrONIC). Eqn. (4.9) can now be re-written, excluding spin:

$$\sqrt{f} = \text{const.} \langle \psi_{aV}\psi_a^{o} | M | \psi_{bV}\psi_b^{o} \rangle \qquad (4.11)$$

and the direct product derivation must now include the direct products of the ground and excited vibrational contributions.

> Note that the operator M in (4.11) is not precisely that shown in (4.5) but must include a term which can operate on the vibrational part of the wavefunction. This is the Herzberg-Teller expansion which is discussed below in section 4.2.viii).

Normally in absorption, we deal with molecules in their ground vibrational states, with v" = 0 states which are, in fact, always totally symmetric [5].

> The ground vibrational state, v" = 0 of any vibration, whatever its symmetry, is totally symmetric. Note that the double prime is reserved for vibrational states of the electronic ground state. The single prime, a symbol, v', is used to describe vibrational states of an excited electronic state. A vibrational transition is described with the excited state written first, i.e. a transition from ground to excited state is written v'---v".

Hence ψ_{aV} can be ignored if it is a vibrational ground state (v" = 0) since it will not contribute to the direct product. For

the integral in (4.11) to be non-zero, the direct product of the initial and final electronic wavefunctions with the component of the dipole moment operator must span the representation of an odd vibration of the molecule. Moreover in a molecule ML_n where L is a polyatomic ligand, it is normally, though not exclusively, the vibrations of the metal ligand framework (skeletal vibrations) which may couple to the electronic motion.

Consider for example, an octahedral molecule ML_6; the dipole moment operator spans T_{1u} and the vibrations of the molecule are given in Table 4.1; the permissible odd vibrations are t_{1u} and t_{2u}:

	ψ_b --- ψ_a	Direct Product $\Gamma_j = \Gamma(\psi_b)\Gamma(\psi_a)$	Direct Product of Γ_j with T_{1u} [a]	
1	A_{2g}---A_{2g}	A_{1g}	T_{1u}	Vibronically allowed
2	E_g----E_g	$A_{1g} + A_{2g} + E_g$	$2T_{1u} + 2T_{2u}$	Vibronically allowed
3	A_{2u}---A_{2g}	A_{1u}	T_{1g}	Forbidden
4	A_{1g}---A_{2g}	A_{2g}	T_{2u}	Vibronically allowed
5	T_{2g}---A_{2g}	T_{1g}	$A_{1u} + E_u + T_{1u} + T_{2u}$	Vibronically allowed
6	T_{2g}---A_{2u}	T_{1u}	$A_{1g} + E_g + T_{1g} + T_{2g}$	Orbitally allowed
7	T_{2u}---A_{1g}	T_{2u}	$A_{2g} + E_g + T_{2g} + T_{1g}$	Forbidden

a) T_{1u} is the representation of the dipole moment operator

(4.12)

If the direct product in the penultimate column contains A_{1g} then the transition is orbitally allowed (eqn. 4.9). If it does not contain A_{1g} but does contain an odd vibration of the framework, it is vibronically allowed. If neither is true, it is forbidden, in the absence of any other allowing perturbing mechanism. Of the examples cited only transition 6 is fully allowed and would therefore be expected to have a high intensity (ε, $10^3 - 10^4$ l cm^{-1} mole^{-1}). Transitions 1,2,4 and 5 are all vibronically allowed and would be expected to occur, but with a lower intensity (ε, $1 - 10^2$ l cm^{-1} mole^{-1}). They are said to become allowed by an 'enabling vibration', and occur at an energy equal to the sum of the electronic transition energy plus the energy of the vibrational quantum. Transitions 3 and 7 are both orbitally and vibronically forbidden and should not appear. Notice that if the parity rule were not considered, transition 5 would be orbitally allowed. For this reason it may have a slightly higher intensity than other vibronic transitions especially if the molecule does not unequivocally have a centre of inversion.

Later in this volume examples will be discussed where the equilibrium nuclear configuration of an excited state is not the

Table 4.1

FUNDAMENTAL SKELETAL NORMAL MODES OF VIBRATION OF SOME OF THE MORE COMMONLY OCCURRING CENTROSYMMETRIC AND ACENTRIC MOLECULES

Molecular Skeleton	Group	Normal Modes of Vibration
ML_6	O_h	$A_{1g} + E_g + T_{2g} + 2T_{1u} + T_{2u}$
ML_6	D_3	$3A_1 + 2A_2 + 5E$
ML_5	D_{3h}	$2A_1' + 3E' + 2A_2'' + E''$
ML_5	C_{4v}	$3A_1 + 2B_1 + B_2 + 3E$
ML_4	D_{4h}	$A_{1g} + B_{1g} + B_{2g} + A_{2u} + B_{2u} + 2E_u$
ML_4	C_{4h}	$A_g + 2B_g + A_u + B_u + 2E_u$
ML_4	T_d	$A_1 + E + 2T_2$
ML_4	D_{2d}	$2A_1 + B_1 + 2B_2 + 2E$
ML_4L_2'	D_{4h}	$2A_{1g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$
ML_2L_2'	D_{2h}	$2A_g + B_{1g} + 2B_{1u} + 2B_{2u} + 2B_{3u}$
ML_3L'	C_{3v}	$3A_1 + 3E$
ML_2L_2'	C_{2v}	$4A_1 + A_2 + 2B_1 + 2B_2$

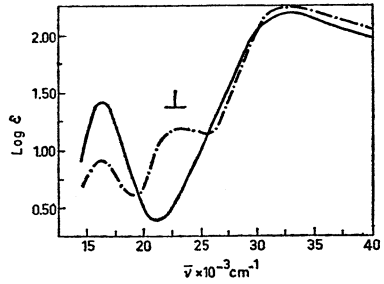


Fig.4.1 The polarised absorption spectra of a crystal of the trans-$[Co(en)_2Cl_2]^+$ ion.

same as the ground state. Recognise that because there is no time for nuclear rearrangement, the excited state when first populated will be in a higher vibrational state whose nuclear configuration is exactly the same as the ground state; only when it relaxes will the new geometry be generated. If the relaxed excited state belongs to a different point group from the ground state molecule, then group theoretical calculations of the type indicated above must use the set of symmetry operators which are common to both symmetry groups.

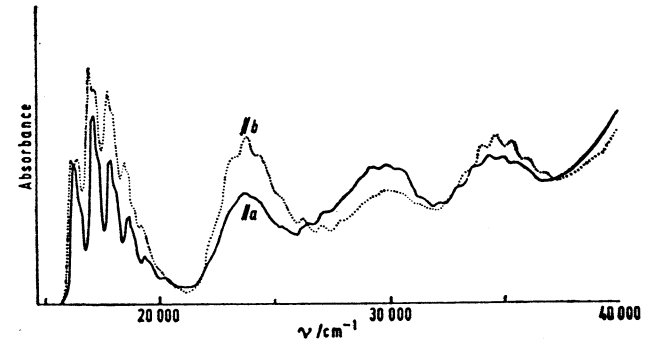


Fig.4.2 The charge transfer spectrum of the $[MnO_4]^{2-}$ ion in a crystal of K_2SO_4 at 4.2 K [44].

4.2.v) Vibronic coupling in acentric molecules

In a centrosymmetric molecule, a formally forbidden electronic transition can often be observed if it is coupled to an enabling vibration. If an allowed electronic transition couples to a vibration it may be allowed or forbidden depending upon the symmetries of the functions concerned. Clearly if a totally symmetric vibration couples to an allowed electronic transition, the transition will still occur. Indeed a progression of totally symmetric vibrations can be co-excited with an electronically allowed transition to give a progression in which the vibrational quantum number increases by unit steps.

As we explore in more detail later, such a progression can be expected only if there is a distortion along that normal coordinate in the electronic excited state. If the ground and excited state have the same equilibrium internuclear distance, then, within the harmonic model, the v" = 0 level is orthogonal to all v' except v' = 0 (consider a vertical transition in Fig.4.3 below).

States which exhibit Jahn-Teller distortion, necessarily degenerate, may also couple to Jahn-Teller activating modes, e_g and t_{2g}, giving rise to observable progressions in these modes. Examples are presented below.

Consider a transition $^2T_2 \text{---} ^2E$ in a tetrahedral molecule. The dipole moment operator transforms as T_2, so this transition is electronically allowed. The direct product ($T_2 \times T_1 \times E$) spans all the representations of the T_d group; thus any vibration may be coupled. Fig.4.2 shows the electronic spectrum of the $[MnO_4]^{2-}$ ion in the charge transfer region. The lowest energy band is assigned $^2T_2 \text{---} ^2E$ and a progression in the totally symmetric, a_1, Mn-O stretching mode is clearly observed (average interval 783 cm^{-1}

compared with a ground state Mn-O stretching vibration of 810 cm^{-1})[14]. A t_2 vibration can also be identified in the vibronic structure of the lower energy band. Note that a progression of the a_1 mode is also observed in the band centred near 35,000 cm^{-1} (and also in the d-d transition (not shown)) [14].

It should also be evident that a vibronically allowed transition may be observed in an acentric molecule, where an orbitally forbidden electronic transition is 'enabled' by a vibration, and that totally symmetric progressions may superimpose on top of an enabled vibronic transition in a centrosymmetric molecule.

In general many vibrational transitions can be co-excited if they are permitted by the selection rules. In room temperature solution spectra they usually result in a broadening of the observed electronic transitions. In low temperature crystal spectra they may be resolved, as in Fig.4.2. Further more detailed dicusssion of vibrational coupling is presented in section 4.3.

4.2 vi) Symmetry Ascent Selection Rules

We briefly refer to selection rules within the framework of the Normalised Spherical Harmonic Hamiltonian recognising that in this model the wavefunctions are octahedral eigenstates symmetry adapted to the lower groups.

As an example consider a d^2 square pyramidal complex, and the transition $E[T_1(F)]---B_1[A_2(F)]$, where the parentage in O symmetry of the states in C_{4v} is indicated in square brackets. Given that the dipole moment operator transforms as $A_1 + E$ in C_{4v}, this transition should be allowed via the matrix element $\langle B_1|E|E\rangle$ which does not appear to vanish (direct product contains A_1). Since the wavefunctions are octahedral eigenstates, the parentage should be considered by utilising the matrix element $\langle B_1(A_2)|E(T_1)|E(T_1)\rangle$. The brackets are used to recognise the parentages of functions and dipole moment operator. If this matrix element is not to vanish, the direct product of the bracketed functions should also contain A_1, and in this case they do not. Thus if the transition $E[T_1(F)]---B_1[A_2(F)]$ is 'pure' in the sense that there is no mixing with other states of like symmetry, the transition moment vanishes, despite the fact that $\langle B_1|E|E\rangle$ contains A_1. In most cases, configurational interaction with other states will occur (e.g. with $E[T_2]$, transitions to which are allowed), but the excitation may well be weaker than expected because of this 'forbiddenness'. Where distortion from one sub-group to another is small, the use of 'symmetry ascent selection rules' and appeal to the symmetry selection rules of the higher group may prove useful. Arguments of this kind can also be applied to vibrational coupling, see section 4.4.

Table 4.2

MOLAR EXTINCTION COEFFICIENTS FOR VARIOUS TYPES OF COMPLEX

Molar Extinction Coefficient ($l\ cm^{-1}\ mole^{-1}$)	Type of Transition	Type of Complex (d-d and CT absorption)
10^{-3} - 1	Spin forbidden Laporte forbidden	Many octahedral and tetrahedral complexes of d^5 ions, e.g. $[Mn(H_2O)_6]^{2+}$.
1 - 10	Spin forbidden Laporte forbidden	Tetrahedral complexes of d^5 ions with intensity stealing e.g. $[MnBr_4]^{2-}$. Spin forbidden bands of transition metal ions.
	Spin allowed Laporte forbidden	Ionic six coordinate molecules, e.g. $[Ni(H_2O)_6]^{2+}$.
10 - 10^2	Spin forbidden Laporte forbidden	Certain of the more covalent tetrahedral complexes of the d^5 ions, e.g. $[FeBr_4]^-$. 3rd row transition metal complexes.
	Spin allowed Laporte forbidden	Six coordinate molecules with intensity stealing - complexes with organic ligands. Some square complexes, e.g. $[PdCl_4^2]^-$, copper complexes.
10^2 - 10^3	Spin allowed Laporte forbidden	Tetrahedral complexes, e.g. $[NiCl_4]^{2-}$. Certain six coordinate molecules of low symmetry e.g. $(2\text{-Picoline})_2Co(NO_3)_2$ Many square complexes particularly with organic ligands.
	Spin allowed Laporte allowed	Some MLCT charge transfer bands in molecules with unsaturated ligands. Some forbidden charge transfer bands.
10^2 - 10^4	Spin allowed Laporte forbidden	Acentric complexes with covalent ligands such as acac, soft ligands P, As etc.
10^3 - 10^6	Spin allowed Laporte allowed	Many charge transfer absorptions. Electronically allowed transitions in organic species.

4.2. vii) Band Intensities

Having learned how to determine whether a transition is electronically or vibronically allowed, or is forbidden, how is such information obtained from an experimental spectrum ? Fully allowed electronic transitions, such as the $\pi - \pi^*$ transitions of aromatic organic compounds have molar extinction coefficients of the order of $10^4 - 10^6$ l cm^{-1} $mole^{-1}$, with dyestuff molecules, such as the phthalocyanines, at the upper end of this range. At the other extreme, Laporte and spin forbidden d-d transitions in centrosymmetric molecules, such as the crystal field spectra of octahedral manganese(II) species (very pale yellow or pink in colour) may have intensities of the order of $10^{-2} - 10^{-3}$ l cm^{-1} $mole^{-1}$. Table 4.2 correlates the observed intensities with the various types of electronic transition.

While group theory can provide a qualitative 'yes' or 'no' answer to the question of the anticipated intensity for a transition, it cannot simply estimate relative intensities of permitted transitions. Clearly the group theoretical approach as presented above, contains no information about the nature of the wavefunctions, be they metal, or ligand, or mixed, but only deals with their symmetry.

Two extensions of the previous discussion permit further understanding of the data presented above. In acentric molecules, both d and p orbitals may have the same symmetry permitting d-p mixing. Although this can provide a means of bypassing the Laporte Rule, it is not the only factor in intensity enhancement. Of greater potential importance, a covalent interaction between metal and ligand orbitals, in a molecular orbital, can mix metal d orbitals with ligand p orbitals, in both ground and excited states, providing a mechanism for their net overlap. Under such circumstances, the observed intensity becomes a qualitative measure of the degree of covalency in a metal-ligand bond (see section 5.5).

Thus spin allowed d-d transitions of octahedral, and in general, most six coordinate, transition metal complexes involving inorganic ligands usually have intensities in the range 1 - 50 l cm^{-1} $mole^{-1}$, transitions in six coordinate complexes of organic ligands are often more intense. In cases where the degree of mixing between metal and ligand orbitals is expected to be small (especially centrosymmetric molecules) the intensity gain is small. However in acentric molecules (even of inorganic ligands) and especially where metal-ligand covalency is significant, bands attributable to crystal field transitions may have intensities of the order of $10^2 - 10^3$ l cm^{-1} $mole^{-1}$, e.g. tetrahedral $[MX_4]^{n-}$ species (X a halogen and many other possibilities), tris-acetylacetonate derivatives, many unsaturated ligands, complexes with heavy ligands such as those containing arsenic, iodide etc.

In any complex, there will arise charge transfer states of odd symmetry, transition to some of which will be strongly allowed. The mechanism of vibronic coupling discussed above, provides a pathway to couple an even d orbital function, via an odd vibration to an odd charge transfer state, thus mixing some odd character into the otherwise even ground state. The Laporte rule is thereby relaxed.

While it may be expected that all odd vibrations (of appropriate symmetry) of a molecule will couple to the electronic motion, these vibronic origins will not all have the same intensity. The observed intensity depends upon coupling of these vibronic wavefunctions to odd, usually charge transfer states, almost invariably occurring at high energy. The degree of mixing depends upon the energy separation between the d ground state wavefunction and the odd excited state, the closer the two states, the greater the degree of mixing. Crystal field states, for this reason, inter alia, generally increase in intensity as they move into the blue, where they can couple with higher energy wavefunctions. This coupling mechanism is commonly called 'intensity stealing'.

4.2.viii) Intensity Stealing in Spin Allowed Transitions

In a centrosymmetric molecule, a d-d transition can gain intensity by coupling to an odd electronic state via a mechanism involving an appropriate odd vibrational mode. Indeed in the previous study of the group theoretical requirements for vibronic coupling, these specific vibrational modes were identified. Thus when a vibration enables an otherwise parity forbidden transition, it does by coupling an even electronic state to one or more odd electronic states thereby introducing 'odd' character into an otherwise even-even transition. In transition metal complexes, these odd electronic states are generally charge transfer in nature.

The Hamiltonian which accomplishes this coupling process is obtained by expanding the crystal field Hamiltonian V_o about an asymmetric (odd) coordinate (Q) such that [4]:

$$\begin{aligned} V &= V_o + Q(dV/dQ)_o + \ldots \\ &= V_o + H_v \end{aligned} \qquad (4.13)$$

Since the product Q(dV/DQ) must transform as A_{1g}, it follows that (dV/dQ) must transform in the same fashion as the odd coordinate Q, and is therefore odd. H_v can therefore mix odd excited state wavefunctions into the ground state. For example, using first order perturbation theory:

$$|a'\rangle = |a_g\rangle + |b_u\rangle\langle b_u|H_v|a_g\rangle/(E_b - E_a) \qquad (4.14)$$

This is known as the Herzberg-Teller mechanism for introducing odd character and hence intensity into a d-d transition.

The mechanism of (so-called) intensity stealing has been considered in some detail by Fenske [6]. Briefly the argument

proceeds as follows. Consider a d-d transition t_{2g}---e_g in a d^1 octahedral complex. There will be odd orbitals, primarily ligand in origin, lying below the metal d levels; generally these will have t_{1u} and t_{2u} symmetry. Assume that the highest such orbital has t_{1u} symmetry, though it makes no difference to the argument if the alternative is chosen. The ground state, call it A, may be written $[(t_{1u})^6(t_{2g})]$. Excited states B, C and D may written in which there is a d-d transition (B) and in which there are transitions from the ligand odd orbitals to e_g (C) and to t_{2g} (D), viz:

$$A = [(t_{1u})^6(t_{2g})] \qquad C = [(t_{1u})^5(t_{2g})(e_g)]$$

$$B = [(t_{1u})^6(e_g] \qquad D = [(t_{1u})^5(t_{2g})^2] \qquad (4.15)$$

where lower lying filled orbitals have been omitted. Nominally the intensity of a transition between A and B is zero since it is a d-d transition. However, an odd vibration of appropriate symmetry (e.g. t_{1u} or t_{2u}) can couple both A and B to C and D. Using first order perturbation theory, the coupled wavefunctions carrying primes for identification may be written (following 4.14):

$$\begin{aligned} |A'\rangle &= |A\rangle + \sum_i |C_i\rangle\langle C_i|H_v|A\rangle/(E_A-E_{Ci}) \\ |B'\rangle &= |B\rangle + \sum_i |C_i\rangle\langle C_i|H_v|B\rangle/(E_B-E_{Ci}) \end{aligned} \qquad (4.16)$$

where the sum ranges over all relevant excited states C_i being C and D in this case, and where H_v is the vibrational component of the Hamiltonian. The transition moment between $|A'\rangle$ and $|B'\rangle$ can now be written:

$$\langle A'|M|B'\rangle = \langle A|M|C_i\rangle\langle C_i|H_v|B\rangle/(E_B-E_{Ci}\rangle + \langle B|M|C_i\rangle\langle C_i|H_v|A\rangle/(E_A-E_C) \qquad (4.17)$$

where the terms $\langle A|M|B\rangle$ and $\langle C_i|M|C_i\rangle$ are zero and are omitted.

Both the dipole moment operator and the vibrational Hamiltonian are one electron operators. Neither can therefore couple B with D since these two wavefunctions differ by more than one orbital. Therefore within this approximation, the excited state represented by D cannot contribute to the intensity of the d-d transition. Rather the d-d transition must gain intensity by coupling to the L---e_g charge transfer band which, in a complex with an incomplete t_{2g} sub-shell may not be the lowest (LMCT) charge transfer band.

Following Fenske we note a more detailed expansion of the terms in (4.17), involving states A,B and C. Each function is designated by its unique orbital the other orbitals being common and integrated out, thus A = $[..t_{1u},t_{2g}]$, B = $[..t_{1u},e_g]$, C = $[..t_{2g},e_g]$:

$$\langle A|M|C\rangle\langle C|H_v|B\rangle/(E_B-E_C) = \langle(t_{1u})|M|(e_g)\rangle\langle(t_{1u})|H_v|(t_{2g})\rangle/(E_C-E_B) \qquad (4.18)$$

and

$$\langle B|M|C\rangle\langle C|H_v|A\rangle/(E_A-E_C) = \langle(t_{1u})|M|(t_{2g})\rangle\langle(t_{1u})|H_v|(e_g)\rangle/(E_C-E_A) \qquad (4.19)$$

Both (4.18) and (4.19) contribute to the intensity of the d-d transition but for two reasons, (4.18) is expected to make the major contribution. Firstly the denominator of (4.18) is smaller than that of (4.19) since (C) is closer to (B), by 10 Dq, than is (C) or (A). Secondly the transition moment component of (4.19) is smaller than that of (4.18) since charge transfer transitions from orbital triplets to the metal t_{2g} are significantly weaker than those to e_g (see Chapter 5). Expression (4.19) can then be neglected, yielding a transition moment:

$$\langle A'|M|B'\rangle = \sum_i \langle A|M|C_i\rangle\langle C_i|H_v|B\rangle/(E_B-E_{Ci}\rangle \qquad (4.20)$$

Combining (4.4) with (4.20) provides an absolute measure of the intensity of a d-d transition coupled to one or more charge transfer states by the vibrational Hamiltonian. However the magnitude of the vibrational Hamiltonian matrix element is not generally known, nor will the charge transfer transition dipole strength always be known. Fenske noted that the relative intensities of several d-d transitions could be obtained by cancellation of some of these terms, resulting in an expression for the relative intensity in terms only of the differences in energy between the d orbital and charge transfer states concerned. Quite acceptable agreement was demonstrated for a series of d^6 species where the charge transfer transition to e_g could be observed. The treatment was extended to the presence of two coupled charge transfer states by assuming that the vibrational Hamiltonian matrix element was the same for both transitions. While a rather more involved analysis is necessary to obtain absolute intensities from molecular orbital coefficient data, this simple treatment rationalises how, in a group theoretical sense, the d-d transition is coupled to the charge transfer transition, and can be used to confirm, or otherwise, proposed d-d assignments, on the basis of calculated relative intensities.

4.2.ix) Intensity Stealing in Spin-Forbidden Transitions

Transitions involving a change in spin are forbidden since the dipole moment operator does not couple spins. However the mechanism of spin-orbit coupling (see Section 3.7) permits the coupling of spin and orbital angular momenta. We have already illustrated that two Russell Saunders states, nominally of different spin, may share the same spin-orbit designation and be mixed. Thus a spin-forbidden transition may occur through, for example, a transition from a singlet state, to the singlet component of a mixed triplet-singlet state. This can be written in a more analy-

tical form using first order perturbation theory. Following Ballhausen [3], consider a ground state spin singlet, ^1A, and an excited state spin singlet, ^1B, transition between which is allowed with dipole strength D_{AB}. Now consider a third and spin triplet state, ^3C, whose symmetry allows spin-orbit coupling with the higher energy singlet, ^1B. The mixed triplet state may be written:

$$|^3C'\rangle = |^3C\rangle + [\langle ^3C|H|^1B\rangle/(E(C) - E(B))].|(^1B)\rangle \qquad (4.21)$$

where H is the spin-orbit coupling Hamiltonian [3]. The dipole strength of the transition from ^1A to the unmixed ^3C, D_{AC}, is zero through spin orthogonality. However, considering the mixed state $|^3C'\rangle$, the dipole strength is given by:

$$D_{AC} = [\langle ^3C|H|^1B\rangle/(E(C) - E(B))].D_{AB} \qquad (4.22)$$

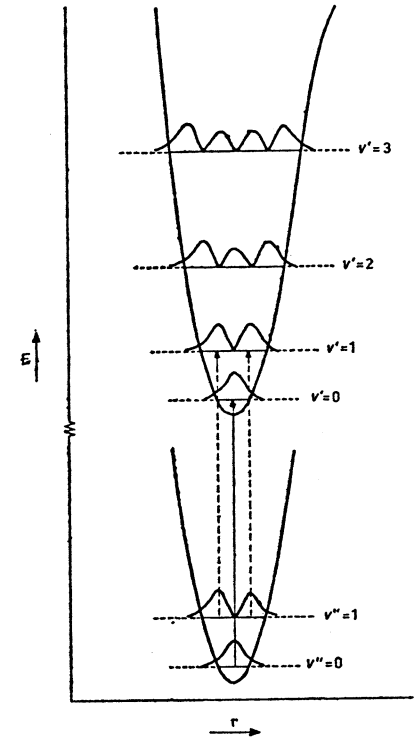
Clearly this provides a general mechanism for mixing two spin states and its importance depends upon the magnitude of the spin-orbit coupling coefficient, and the energy separation of the states being mixed. With heavy transition metals where the spin orbit coupling coefficient is very large, of the order of 10^3 cm^{-1} (e.g. third row transition metal ions), 'spin-forbidden' transition intensities can be quite high and the designation of states having specific spins is largely illusory.

Ballhausen emphasises [5] that the mixing process only perturbs the relevant wavefunction, $|^3C\rangle$ above, to a small degree. Thus if we were to observe vibronic structure on the transition ^3C---^1A, this would be described in terms of the vibronic structure of the ^3C state, not the ^1B state to which it is coupled.

4.3 Vibrational Progressions and Franck-Condon Factors in Electronically Allowed Transitions [3]

A more detailed study is now presented of the vibrational structure of a transition in order to determine how useful information can be derived therefrom. The discussion is also expanded to include vibrational progressions observed in luminescent emission. In absorption the progressions will be in frequencies appertaining to the excited electronic state. However it is often possible to record the luminescent emission from an excited state (see section 4.7) cascading down the vibrational levels of the ground electronic state. Thus vibronic structure in luminescent emission corresponds with ground state vibrational frequencies, and may be related directly with observed infrared and Raman vibrational data.

Incorporating vibrational overlap, the transition moment may be written:



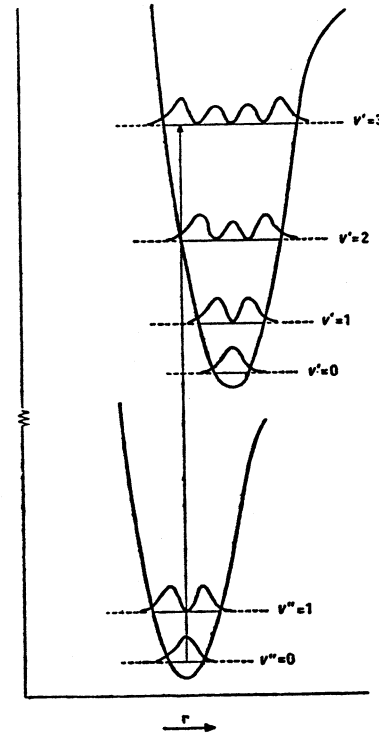


Fig.4.3 (Left) Potential energy versus metal-ligand distance (Morse curve). Equilibrium internuclear distances in the ground and excited state are the same.
Fig.4.4 (Right) Potential energy versus metal-ligand distance (Morse curve). Equilibrium internuclear distance in the ground state is less than in the excited state. Both Fig.4.3 and 4.4 qualitatively represent the discussion in the text but strictly speaking would only apply to diatomic systems.

$$D_{oj} = \langle \psi_j^o|M|\psi_o^o\rangle\langle (v'',\zeta_1)|(v',\zeta_1)\rangle\langle (v'',\zeta_2)|(v',\zeta_2)\rangle .. \\ ..\langle (v'',\zeta_{3N-6})|(v',\zeta_{3N-6})\rangle \qquad (4.23)$$

where the first term, the electronic motion (M corresponding with eqn.(4.5)), is multiplied by a series of vibrational overlap integrals between the ground and excited state vibrational levels with the vibrational overlap summed over 3N-6 terms. The intensity is proportional to the square of the transition moment which,

according to (4.23) will incorporate the squares of these vibrational overlap integrals, commonly called Franck-Condon factors.

Firstly, consider two general situations where the excited state potential surface may have the same equilibrium nuclear distance as the ground state (Fig.4.3), or have a different equilibrium internuclear distance (Fig.4.4). Given that the transition is vertical, i.e. no movement of nuclei is permitted, the second situation (Fig.4.4) may not allow for the observation of 0'-0" transitions but only higher transitions, n'-0". Moreover, in this latter situation, the magnitudes of the vibrations (force constants) are expected to differ from those of the ground state, being reduced or increased depending upon whether the equilibrium inter-nuclear distance in the excited state is greater or smaller than that of the ground state. Thus study of vibronic structure can provide information concerning excited state bond distances, force constants and other bond data (see below).
There are a number of special situations:

i) An electronically allowed transition to an excited state with the same equilibrium metal-ligand distance.

In this case the 0'-0" transition will be observed. A progression of totally symmetric vibrations is not expected since their Franck-Condon factors will vanish in the harmonic limit, i.e. overlap of v"= 0 with v'= n will be zero except for small contributions from anharmonicity. As indicated above, a progression in a totally symmetric frequency M-L is only expected when there is distortion along the M-L coordinate, i.e. in general when the M-L bond length changes.

ii) An electronically allowed transition to an excited state with a different internuclear equilibrium metal-ligand distance.

In this case a vertical transition from v" = 0 must reach a vibrationally excited state of the electronic excited state (Fig.4.4) and the (0'-0") origin may not be observed. Because of the displacement in equilibrium internuclear distance, the excited vibrational intervals may now be different from the ground state frequencies. The vibrational progression will be in a coordinate which generates the change in metal-ligand bond length.

In general the region of the vibrational progression of maximum intensity (maximum value of the Franck Condon factors) will depend, inter alia, upon the difference in equilibrium inter-nuclear distances of the ground and excited state. Therefore analysis of the relative intensities of the members of a totally symmetric vibrational progression can yield a value for this difference in internuclear distance.

Ballhausen [3,15,20] develops this argument and shows that the relative intensities of such a progression are given by the expression [3,16,19,20]:

$$I_n = [0.5\phi(\delta r)^2]^n I_o/n \qquad (4.24)$$

where $\phi = 4\pi^2 \nu_g cM/h$, and (δr) is the change in extension at the minimum of the potential energy surface for the normal coordinate corresponding to ν_g, for a ligand of mass M, assumed monoatomic for a totally symmetric progression. Eqn.4.24 compares the intensity of the n'th line in the progression with the zero'th line. Generally the intensity of a progression goes through a maximum, which, for fairly large values of v', occurs at $v' = 0.5\phi(\delta r)^2$. Thus identifying the v' value of the most intense member of a progression can provide a simple estimate of (δr).

However this analysis assumes that the ground and excited state vibrational frequencies are the same. This is not generally the case for inorganic systems of this type, and a more sophisticated analysis is generally necessary.

Following Gliemann and co-workers [17], the relative intensities of the n'th and zero'th lines are given by:

$$[I(em)^{0,n}/I(em)^{0,0}] = [E_{0,n}/E_{0,0}]^4 [R_{0,n}/R_{0,0}]^2 \text{ (emission)}$$

$$[I(abs)^{m,0}/I(abs)^{0,0}] = [E_{m,0}/E_{0,0}][R_{m,0}/R_{0,0})]^2 \text{ (absorption)}$$

where

$$[E_{0,n}/E_{0,0}]^4 = [E_{0,0} - nh\nu_0"]^4/E_{0,0}^4 \text{ (emission)}$$

$$[E_{m,0}/E_{0,0}] = (E_{0,0} + mh\nu_0')/E_{0,0} \text{ (absorption)} \qquad (4.25)$$

The most useful vibrational functions are:

$$R_{0,0} = [2\delta/\sqrt{(1+\delta^2)}]\exp(-.5\sigma^2)$$

$$R_{n+1,0} = [-2\delta DR_{n,0} - (\sqrt{2n})(\delta^2-1)R_{n-1,0}]/(\delta^2+1)\sqrt{(2n+2)} \qquad (4.26)$$

where $\delta = \sqrt{(\nu_o'/\nu_o")}$, $D = C(\sqrt{m\nu_o"})(\delta r)$ and $\sigma = D/\sqrt{(1+\delta^2)}$
D is defined for emission. For absorption, $\nu_o"$ should be replaced by ν_o'. Where the displacement (δr) in the minimum of the normal coordinate, is expressed in angstroms, the vibrational energy in cm^{-1}, and the masses (monatomic) from the G matrix in amu, then C = 0.1722 [27]. Thus insertion of (4.26) into (4.25) permits evaluation of the relative intensities of a progression with vibrational frequencies differing in the ground and excited states, as a function of the change in distance between the minima in the ground and excited state potential surface of the normal coordinate.

Note that to determine the actual change in length of a metal-ligand bond (an internal coordinate), the relationship between normal and internal coordinates is required. This is dealt with by Wilson, Decius and Cross [21]. Note also that there are several different procedures for analysing vibrational progressions (discussed in [17]); approximately the same result is obtained for each, since they are primarily sensitive to the change in equilibrium bond distance.

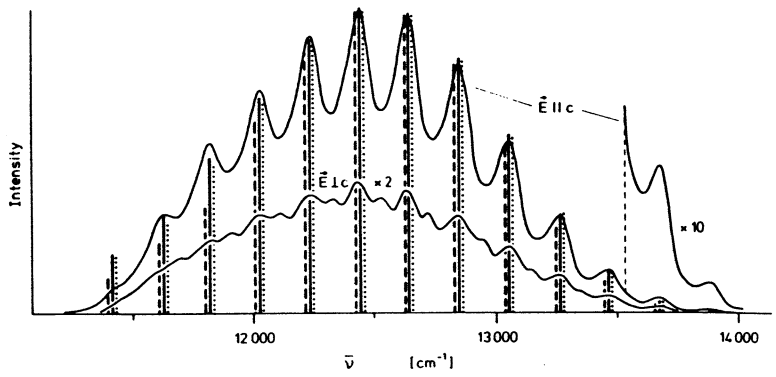


Fig.4.5 Single crystal polarised luminescence spectrum of $K_2[PtBr_4]$ at 5 K. The normalised vibronic intensity ratios were calculated with:-
(a) Solid lines - $[E_{0,n}/E_{0,0}]^1$, $\nu_o'/\nu_o'' = 1.0$, $\delta r = 170$pm.
(b) Dotted lines - $[E_{0,n}/E_{0,0}]^4$, $\nu_o'/\nu_o'' = 1.0$, $\delta r = 175$pm.
(c) Dashed lines - $[E_{0,n}/E_{0,0}]^4$, $\nu_o'/\nu_o'' = 0.83$, $\delta r = 190$pm. In (a,b,c) $E_{0,0} = 13{,}900$, $\nu_o'' = 205$ cm^{-1}; atomic mass is 79.9amu [17].

Indeed it is possible to obtain an estimate of the bond dissociation energy (BDE) using the information contained in the vibrational progression. In an anharmonic oscillator, the overtones crowd closer together with increasing quantum number until they reach the continuum. The energy separation beteween the potential energy minimum at v" = 0, and this continuum is defined as the BDE. Thus if sufficient members of a progression are observable it is possible to extrapolate to where this continuum begins. For an anharmonic oscillator, the frequencies of a progression defined by the vibrational quantum number v are approximated by [36-39]:

$$\nu(v) = v\omega_e + vx_e(2m + d + v) + \ldots. \qquad (4.27)$$

where ω_e is the harmonic wavenumber and x_e is the first (and largest) anharmonicity constant. If the function $\nu(v)/v$ is plotted against v, then a straight line should be generated. The slope of this line is x_e, and the intercept is $\omega_e + x_e(2m + d)$. The quantity d is the degeneracy of the vibration and will be one for a unidimensional vibration; m = 0 if a fundamental rather than a 'hot' band is involved. Thus the intercept of the line is simply $\omega_e + x_e$.

Finally the BDE is related to these quantities via the Birge-Sponer relationship [36,40]:

$$BDE = \omega_e/4x_e \qquad (4.28)$$

The value of the BDE so obtained is only considered approximate for various reasons. Certainly the expression (4.27) is an approximation since it neglects interaction (coupling) with other vibrations.

iii) An electronically allowed transition to an excited state with a different equilibrium configuration (i.e. different point group).

In this case there must be a non-totally symmetric mode which distorts the molecule from the ground state to the excited state. This mode however will be found to be totally symmetric with respect to the set of symmetry operators which are common to both groups involved, and a progression therein can often be observed.

4.4 Vibrational Progressions in Electronically Forbidden but Vibronically Allowed Spectra

This is a very common and important case in transition metal d-d spectra. From previous statements, however, it should now be clear, that we cannot see a totally symmetric vibration based upon the 0'-0" line, since this line is not observed. Rather the transition becomes vibronically allowed by coupling of the electronic motion to a non-totally symmetric vibration such that the matrix element (4.11) exists. The observed energy will be equal to the vertical difference in ground and excited state electronic energy plus the energy of the enabling vibration. Built upon this 'false' origin, progressions in totally symmetric vibrations may be observed. Where more than one non-totally symmetric vibration can induce intensity into the forbidden transition, further totally symmetric progressions may be built. If the crystal is 'hot', excitations of the type 2'-1" may also be observed.

Consider one further example to illustrate the arguments set out above. Flint has studied the luminescent emission of the $[Cr(CN)_6]^{3-}$ ion [23]. The ^2E_g---$^4A_{2g}$ transition is both spin and Laporte forbidden, and becomes weakly allowed by a combination of spin-orbit and vibronic coupling. The dipole moment operator spans T_{1u} in O_h, therefore the vibrations which can enable this transition are spanned by the direct product $E_g \times T_1 \times A_{2g}$, namely $t_{1u} + t_{2u}$.

The $[Cr(CN)_6]^{3-}$ ion has thirteen normal modes of vibration - 2 a_{1g} (Raman), 2 e_g (Raman), t_{1g} (inactive), 4 t_{1u} (ir), 2 t_{2g} (Raman), and 2 t_{2u} (inactive). On the basis of the theory presented above, six independent vibronic origins can be expected with energies corresponding in absorption to the (0'-0") line of ^2E_g---$^4A_{2g}$ plus one quantum of each of these six vibrations (originating in the excited state ^2E_g). In luminescence, these six emissions should occur at E(0'-0") less one quantum of each vibration (originating in the ground state A_{2g}). The luminescence in this region of the spectrum is shown in Fig.4.6. Indeed Flint and Greenough [23] were able to identify the four t_{1u} origins at energies corresponding closely with their energies in the known infrared vibrational spectrum of this species. In addition they

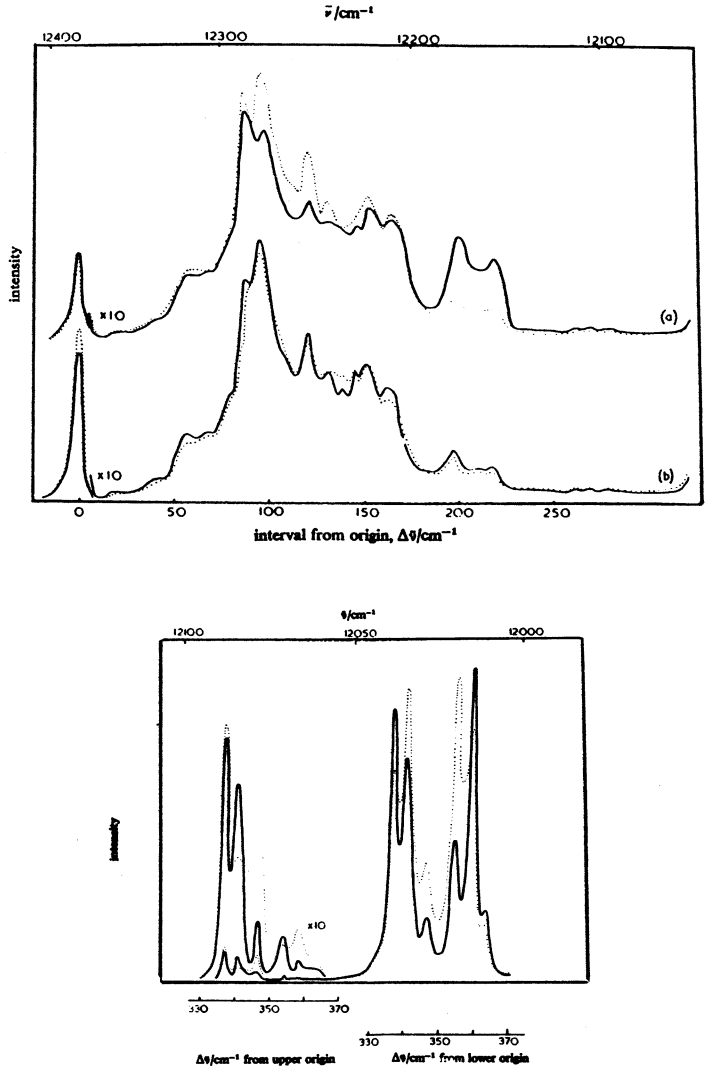


Fig.4.6 (a) (Upper) The low energy region of the 5 K luminescence spectra of $K_3(Co,Cr)(CN)_6$ in several different polarisations [23]. (b) (Lower) The 19 K luminescence spectra of $K_3(Co,Cr)(CN)_6$ in the region of two different vibronic origins in two polarisations. The emission between 12,100 and 12,050 cm^{-1} is from the upper component of the ^2E state and is absent at 5 K [23].

observed the t_{2u} origins enabling them to derive values for these two vibrations which are both ir and Raman inactive. They also observed combination bands involving the two a_{1g} vibrations, i.e. several progressions in each a_{1g} built on the vibronic origins. They specifically did not observe any emission due to coupling with either the e_g or t_{2g} modes in this case. Most of the expected lattice modes could also be identified.

The spectrum was measured by doping Cr(III) into a single crystal of the $[Co(CN)_6]^{3-}$ ion which is distorted very slightly from octahedral symmetry (site symmetry D_{2h}). At 5 K the (0'-0") origin line of the ^2E_g---$^4A_{2g}$ line was observed as a very narrow feature at 12,373 cm^{-1}. As the temperature was raised, a second line appeared at 12,422 cm^{-1} which is believed to be the second component of the ^2E state split in the lower symmetry. As shown in Fig.4.6, vibrations built on this second origin can be seen at higher temperatures. The (0'-0") lines probably occur because inclusive of spin orbit coupling, they are magnetic dipole allowed in O_h (also in D_{2h}).

4.5 Vibronic Coupling, an Alternate View [11,12]

Two limiting stages of vibronic coupling can be envisaged, so-called 'weak' and so-called 'strong'. The weak coupling is typified by a vibronic transition which shows a totally symmetric vibrational progression laid on to the false origin of one electronic plus one (odd) vibrational quantum. Strong coupling is typified by a progression of the odd mode superimposed on the false origin. Since the even harmonics of an odd mode of vibration are themselves even, they should in fact be missing, and, a progression in v = 2 should be seen. In fact, because of anharmonic motion and other second order effects, one might see a progression with alternating relative intensity, weak, strong, weak etc. In the strong coupling limit, the angular momenta associated each with the electronic and vibrational vectors are not conserved, and it is necessary to deal with the resultant angular momentum (i.e. the Born-Oppenheimer approximation is not valid).

Recently Hollebone has published several papers introducing an alternate view of some aspects of strong vibronic coupling [11,12]. A brief review is presented here, but the reader should consult the original texts for a detailed analysis. The idea of symmetry ascent selection rules (see section 4.2.vi) above) is utilised to show in a more explicit manner than previous theories, which vibrations can enable an otherwise forbidden electronic transition.

Following Hollebone [11], we note that a transition moment operator can be expanded into a series of multipoles, viz- dipolar, quadrupolar, octupolar, hexadecupolar etc. It is well known that electronic absorption intensity is allowed utilising the dipolar operator since there is a displacement of charge. The centre of charge is not displaced but since the motion is ungerade, it maps onto a displacment of mass for an uncharged body. In octahedral symmetry the appropriate operator has T_{1u} symmetry (xyz trans-

lation) and is subduced by L = 1 in R_3.

Although intensity in vibrational spectra is usually associated with a dipolar mechanism, Hollebone points out [11] that this cannot be the case since it infers a translation of mass, and such translation is forbidden for a normal mode of vibration. The next term in the multipole expansion is the quadrupole, but this cannot be used since it does not provide an anti-symmetric displacement of nuclear charge which can interact with the photons. The first term which does permit such displacement without permitting motion of the centre of mass, is the octupole. This odd (ungerade) mode is much more effective than the even hexadecupole invoked to explain the circular dichroism of vibronic transitions [13].

In an octahedral complex, to first order, the octupole operator also spans T_{1u} but it is the T_{1u} arising from L = 3, i.e. it is subduced by L = 3 in R_3. Here we begin to see the concept of symmetry ascent being used to provide a distinction between two functions apparently of the same symmetry.

The fundamental proposal in the Hollebone approach is to note that strongly coupled vibronic transitions must follow the same constraints as previously discussed but be allowed by the octupole operator. Hence in addition to the usual requirement in octahedral symmetry, that the ground and excited state be capable of being coupled by a T_{1u} mode, there is the additional requirement that the difference between the total angular momentum of the ground state, and the total angular momentum of the excited state, in an allowed strongly coupled vibronic transition, must be 3. The same restrictions apply to lower sub-groups by appropriate correlation, though mixing of states of the same symmetry may obscur the effect.

The L values for the electronic states are readily obtained by correlation with the free ion terms from which they derive, thus A_{2g} from F would have L = 3. Because of the angular momentum of the octahedral operator, L = 4, 5, 6, 7, 8 etc is re-assigned to L = 0, 1, 2, 3, 0 etc. In a similar fashion, the vibrational wavefunctions are assigned momentum (say V), values. Thus an A_{1g} vibration (with no nodes) correlates with S, hence V = 0. A T_{1u} antisymmetric stretch correlates with P, hence V = 1. E_g and T_{2g} normal modes may clearly correlate with D, V = 2, while a T_{2u} mode and the T_{1u} twist normal mode, because it has more nodes than the anti-symmetric stretch, correlates with F, i.e. V = 3.

The final selection rule may be stated:

$$\Delta S + \Delta L + \Delta V = |3| \qquad (4.29)$$

indicating that the modulus of the total change in spin, orbit and vibrational angular momentum must be three.

Two examples were discussed in some depth by Hollebone [11]. In the $[MnF_6]^{2-}$ ion, the $^4T_{2g}$---$^4A_{2g}$ transition exhibits nicely resolved vibronic structure with a clearly alternating intensity pattern, suggestive of a superimposed ungerade vibration (i.e. strong coupling). In this case S = 0, L = 0 and the required

change in vibrational quantum number must be 3. The standard theory requires that a T_{1u} vibrational mode activate this electronic transition, while the Hollebone approach requires that this t_{1u} vibration have L = 3, and hence be the twisting rather anti-symmetric stretching mode. Indeed on the basis of the vibrational separations, the vibronic structure is associated with the twisting mode with no contribution from the anti-symmtric stretch [11].

In a second example, the MCD spectrum of the vibronically activated ($^2T_{1g}$,$^4T_{2g}$)---$^4A_{2g}$ transition in the $[Cr(H_2O)_6]^{3+}$ ion was considered. The spin allowed component of this spectrum is also enabled by the T_{1u} twisting vibration since the analysis is the same as for the manganese species. In the spin forbidden part of the band envelope, there is a transition to $^2T_{1g}$ (L = 4). Hence ΔL is 1, but there is also a change of spin of $\Delta S = 1$. Hence applying (4.27), $\Delta V = 1$. Thus the enabling vibration in the spin-forbidden component must be the anti-symmetric T_{1u} stretching vibration; this is indeed seen to be the case.

Hoggard and Albin [41] re-analyse the vibrational emission data of $[PtCl_4]^{2-}$, and also show data for $[Pt(bpy)Cl_2]$ which are consistent with the Hollebone strong coupling approach. However the interpretation of the data is not unambiguous.

This treatment has necessarily been brief but should serve to introduce a theoretical model which, if properly tested and found valid, will exercise a profound effect upon our understanding of vibronic spectra of strongly coupled systems.

4.6 Bandwidth and Vibrational Structure

Given a transition from a level of energy E" to a level of energy E', a sharp line of energy E' - E" might be expected in the absorption spectrum. Such a line is the exception rather than the rule, and broad bands of halfwidths of the order of 500 - 2000 cm^{-1} are common. On the other hand, in certain cases, half-bandwidths of less than one wavenumber are observed. Many factors contribute to bandwidth and whether or not vibrational structure can be observed. Each one is discussed briefly here [26].

4.6 i) Inhomogenous Broadening: Where single crystal spectra are concerned, it is naturally impossible to have a perfect crystal in which every metal ion (or group of ions) is in an identical environment. There are certain to be some imperfect sites, where the metal differs from the majority of its neighbours (strained crystal). If the transition concerned is not sensitive to this strain (e.g. a spin forbidden transition with no change in the number of bonding or anti-bonding electrons), the effect is very small; however if the transition is sensitive to environment, this can contribute a band broadening effect of 10 - 15 cm^{-1}.

4.6 ii) Homogenous Broadening: Sequences of transitions involving excitations from 'hot' vibrational levels (e.g. 1'-1", 2'-2") can occur. In a perfectly harmonic oscillator such transitions will have the same energy as the 0'-0" transition. However vibrations generally exhibit some anharmonicity which becomes evident with higher vibrational quantum numbers. Thus this phenomenon will contribute broadening.

4.6 iii) Dispersion and Factor Group Splitting: Where several molecules are present in the same unit cell, coupling between these molecules will lead to an increase in the number of similar energy vibrations which may be coupled. Such coupling will also take place between molecules in adjacent unit cells.

4.6 iv) Heisenberg Uncertainty: Very short lived highly excited states will give broad bands because of the Heisenberg uncertainty principle.

These first four mechanisms cause quite small effects and are responsible for determining the bandwidths of many spin-forbidden transitions independent of DQ.

4.6 v) Continuum Broadening: Every state is embedded in the vibrational continua of the states (including the ground state) which lie below it. Although the coupling of a vibrational level of an excited state to these continua is very small, there are so many vibrations submerged in the continua that the effect is noticeable. This is one important reason why vibrational structure is not so commonly observed, especially at room temperature.

4.6.vi) Variation in Crystal Field Strength: Since DQ is inversely proportional to a^5, the 5th power of the metal-ligand distance, it is very sensitive to changes in this distance a. At ordinary temperatures the metal-ligand bonds will be vibrating so that the metal-ligand distance is constantly fluctuating. If the transition concerned has a dependence upon DQ, then the transition will be observed over a range of energies corresponding with the range of DQ values as the bond vibrates. An energy level will have a slope of dE/d(DQ), where for the configuration $(t_{2g})^p(e_g)^r$ (octahedral field) dE/d(DQ) = 4p - 6r. The greater the value of this slope, the greater the effect of thermal fluctuations upon the energy level. The effect on the transition energy will depend upon the coefficient $d(h\nu)/d(DQ)$ which is at least 8 for spin-allowed transitions of cubic molecules. This phenomenon can be a major contributor to the bandwidth of spin allowed transitions, and is certainly more important than the previous five mechanisms discussed above. Spin forbidden transitions, such as the 'ruby' lines of chromium(III) ($(t_{2g})^3$ M_s = 3/2 to $(t_{2g})^3$ M_s = 1/2) do not involve a change in electron population so that their transition energy is independent of DQ. They will not be broadened by this mechanism and can be very narrow.

4.6.vii) Spin-Orbit Coupling: In the higher groups where orbitally degenerate states can exist, spin-orbit coupling may lift the degeneracy of these states resulting in a broadening or even splitting of an absorption band. In the first row transition series, the spin orbit coupling coefficient is small (expect perhaps for Cu(II)) and this effect is responsible for broadening of up to 1000 cm^{-1}. It is certainly important in the heavier transition series and in the lanthanides where it is more likely to cause resolvable splitting of a degenerate state. Where identifiable it may be used to derive the effective spin orbit coupling coefficient (see Section 3.7).

4.6.viii) Low Symmetry Components to the Crystal Field: By far the majority of six coordinate complexes have an approximately octahedral arrangement of donor atoms around the central ion. The absorption spectrum can often be assigned on the basis of the group O_h, but if the ligands are not all equivalent, as will often be the case, the actual symmetry of the molecule is less than O_h. The degeneracy of E and T terms will necessarily be partially or wholly lifted. This splitting may be small resulting in increased bandwidth, or may result in several absorption peaks. In many low symmetry complexes several transitions occur close together often resulting in broad ill-defined absorption, e.g. in many square planar nickel(II) and copper(II) complexes. Many examples of this situation appear later in this book.

4.6. ix) The Jahn-Teller Effect: An important phenomenon which can distort the more highly symmetric molecules, either in the ground state or in the excited state, is the Jahn-Teller effect. This distortion by lifting the degeneracy of otherwise degenerate levels plays an important role in line broadening. According to the Jahn-Teller theorem [8,9] when the orbital state of a non-linear molecule is degenerate, the nuclear framework will suffer a spontaneous distortion until the ion assumes a configuration of lower symmetry (thus resolving the degeneracy) and of lower energy.

Consider, for example, a d^4 ion (Fig.4.7) in an octahedral environment. The ground state has an energy of -6Dq relative to the spherically perturbed ^5D state. Suppose a splitting of the e level by δ, caused, say, by an axial elongation. The e electron is stabilised by $-\delta/2$ in energy while the total energy of the t_{2g} level electrons remains unaltered (1 x $(2\delta/3)$ - 2 x $(-1\delta/3)$). The system loses energy and therefore the axial elongation should be spontaneous. This is a simple view of the Jahn-Teller effect. Thus for molecules which have such degenerate levels, the high symmetry shape is unstable, and the molecule may oscillate between several structural minima [9]. In practice axial elongation usually occurs, but there is also a reduction of bonding energy which counterbalances the stabilisation of the e electron. Nevertheless the degeneracy of the e level is lost and transitions thereto will be broad or split.

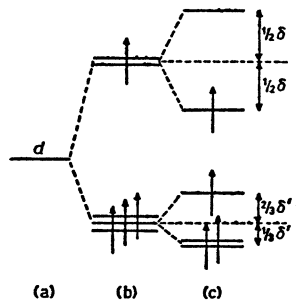


Fig.4.7 The effect of Jahn-Teller distortion upon a d^4 ion. (a) Spherically perturbed d shell, (b) splitting of the d shell in an octahedral field, (c) Jahn-Teller splitting of the d orbitals.

In a more quantitative sense the distortion of a degenerate state δ_i, takes place by the coupling to the electronic state of an even vibrational mode, usually e_g (spanned by the symmetric direct product $\Gamma_i \times \Gamma_i$) resulting in the axial distortion (which may in principle be elongation or compression). When the odd electron giving rise to the degeneracy has σ-anti-bonding character, as is the case here (d^4), and also in six coordinate d^9 complexes, the Jahn-Teller distortion can be quite large with pronounced cryst-allographic distortion and splitting of the e_g level by several thousand wavenumbers. Where non-bonding, or weakly bonding electrons are concerned, e.g. degeneracy within t_{2g}, both the overt crystallographic effect, and the net splitting are much smaller. Generally, pronounced Jahn-Teller effects are seen in E terms but not in T terms. This is also true of excited states. The familiar single peak of the ^2E_g---$^2T_{2g}$ transition of titanium(III) shows a shoulder almost certainly due to Jahn-Teller splitting of the excited ^2E_g state. Similarly many high spin Co(III) species, such as $[CoF_6]^{3-}$ show two components to their ^5E_g---$^5T_{2g}$ transition, probably a similar consequence. However vibrational progressions due to Jahn-Teller coupling have been observed in a T_{2g} state [22,24]. Thus Flint reports that the luminescent emission of the ^2E_g---$^4A_{2g}$ band in the $[Cr(NH_3)_6]^{3+}$ ion (in contradistinction to the $[Cr(CN)_6]^{3-}$ ion discussed briefly above) fails to show a progression in the a_{1g} vibrational modes but does show progressions in an e_g mode [24].

The intensity of the e_g vibrational structure is seen to be inversely proportional to the energy difference ^2E_g---$^4T_{2g}$, in a series of Cr(III) derivatives, and is believed to arise through the coupling of a Jahn Teller distorted (by the e_g mode) $^4T_{2g}$ state with ^2E_g. The matrix element permitting this interaction would have the general form $\langle E_g|T_{1u}|T_{2g}.t_{1u}.e_g\rangle$ which contains A_{1g}. For further detail see discussion of Cu(II) in Chapter 6.

4.6 x) Vibrational Coupling: In the case of an electronically allowed transition, as noted above, if there is a change in equilibrium internuclear distance, one can expect a progression in the totally symmetric mode which distorts along the appropriate coordinate. Clearly this leads to broadening of an otherwise narrow transition. When the measurement is made in fluid medium each of these lines is associated with a rotational subset. If the measurement is made in the solid state, each electronic 'line' will couple to a manifold of (low energy) lattice modes providing another broadening mechanism.

In the case of a vibronic transition, the electronic state will couple with EVERY odd mode of vibration permitted by the selection rules (e.g. in the most symmetric case, the ML_6 octahedron, three vibronic false origins will occur, 2 t_{1u} + t_{2u}). As above, in the solid state, lattice vibrations will also couple. Each of these vibronic origins will, in general, have a progression based thereon.

Consider, for example, $[Cr(NH_3)_6]^{3+}$. The first excited state is $^4T_{1g}$ which will be split (broadened) by spin orbit coupling. There are 11 odd modes of vibration which can enable the $^4T_{2g}$---$^4A_{2g}$ transition. Thus there will be 11 false origins for transition to EACH spinor level of the excited state (not all of the same intensity (probability)). EACH of these multiplicity of transitions may have a progression of totally symmetric a_{1g} Cr-N vibrations based thereon, PLUS progressions of other a_{1g} modes in this molecule, PLUS a progression in a Jahn-Teller distorting e_g mode. One can soon envisage hundreds if not thousands of overlapping lines. Since each 'line' is likely to be 10 - 15 cm^{-1} wide, it is clear that the overlapping of so many components may lead to a broad band with no observable structure. Indeed it is amazing that structure is sometimes seen.

4.7 The Effect of Temperature upon Absorption Spectra

In general two effects can be considered, changes in intensity and changes in band energy. The effect of temperature upon band envelopes has been discussed above with respect to changes in vibrational contributions.

When the temperature is reduced there is evidence [10] that DQ increases, as a consequence of the decrease in vibrational energy causing a small decrease in average bond length as the system cools. Whether this will result in a shift in the energy of the band maximum depends upon the slope dE/d(DQ) of the energy level to which the transition occurs. If the slope is positive there will be a shift to higher energy with decreasing temperature, if it is negative, the shift is to lower energy, and if the slope is zero, there should be no temperature dependence, at least from this cause. Note in passing that M-L stretching frequencies also tend to increase in energy when a sample is cooled while internal ligand vibrations are less affected [42].

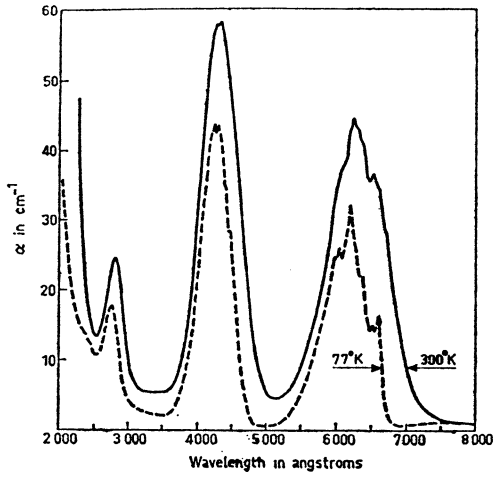


Fig.4.8 The effect of temperature upon the spectrum of (a) K_2NaCrF_6 (with centre of inversion) and (b) emerald (no centre of inversion).

There may be an apparent shift because of loss of vibrational intensity. If the transition is electronically allowed the band will be broad with a long wavelength tail corresponding to 3'-3", 2'-2", 1'-1" sequences. This tail will diminish as the system cools and may result in an apparent shift to higher energy but the shift, if evident, will be small. As already mentioned the oscillator strength does not change because the population of the 0" level increases and hence the intensity of the 0'-0" transition increases with cooling. In vibronic transitions, this long wavelength tail (due to 3'-2", 2'-1", etc transitions) diminishes, but is replaced by the 1'-0" transition. However the contribution to the 'stolen'

intensity by the 3'-2", 2'-1" etc transitions is greater because they are of larger amplitude and induce greater distortion [3]. There is therefore a real reduction in oscillator strength and a real shift to higher energy of the band envelope, with cooling, when a vibronically activated transition is studied.

These effects are illustrated in Fig.4.8. The loss of the long wavelength tails upon cooling is clearly evident in the spectrum of K_2NaCrF_6 where the chromium sits at the centre of symmetry of an octahedron of fluoride atoms. In emerald the environment of the chromium atom is acentric, as indicated by the splitting of the very narrow ^2E transition near 7000 angstroms.

> The intensity of the d-d absorption in emerald is about ten times greater than in the centrosymmetric hexafluoride. The low intensity of the hexafluoride spectrum must be a consequence of the very high energy of the charge transfer band (F to Cr(III)) which would be coupled to the d-d transitions.

In a vibronically allowed transition, the loss in intensity with temperature is given by [3,4]:

$$f = X \coth(h\nu/2kT) \qquad (4.30)$$

where ν is the weighted sum of the frequencies of the activating vibrations, and X is a constant. Therefore a study of the temperature dependence of a vibronically activated transition can provide a value for this sum, which may generally refer to a single dominant enabling vibration. With polarised data, this analysis may give a different weighted sum for different polarisations.

4.8 Emission Spectroscopy

The field of inorganic photochemistry has exploded in the last decade, promoted in part by a major publication [13] and by the work of many excellent researchers. It has also been aided by the development of very short timescale pulsed lasers (in the nanosecond and picosecond regime) which have enabled the detailed study of the emission spectroscopy of many inorganic systems. Emission spectroscopy is of relevance to this book since such data can provide details concerning the nature of the lowest excited states. In the later experimental chapters such emission data are reported, where relevant to the discussion. Here brief fundamental ideas are provided concerning the emission of light from excited states.

After an electronic transition has taken place, the molecule is in an excited state which must relax in some fashion. In general an excited state relaxes rapidly to the lowest excited state having the same spin by a process called 'internal conversion' in which the extra energy is delivered to the environment as heat on a very short time scale (picoseconds). Subsequently this state will further relax to the ground state by:

a) relaxing rapidly to the ground state by delivering energy (as

heat) to its environment ('non-radiative transition'), or
b) relaxing to a lower lying state of different spin ('intersystem crossing') followed by a non-radiative transition to the ground state, or
c) the states in a) or b) may undergo a chemical reaction (photochemistry), or
d) the states in a) or b) may emit energy as light i.e. luminescence, or
e) energy transfer may occur to a lower lying energy level of another molecule.

We are concerned primarily with mechanism d above. The selection rules governing emission are exactly the same as those governing absorption. Thus if a transition is strongly allowed, emission back to the ground state will also be strongly allowed; in other words the excited state lifetime will be very short. Conversely if a transition is forbidden, emission back to the ground state will also be forbidden and the excited state should have a longer lifetime.

Emission from a state of the same spin as the ground state will generally be allowed and rapid. Such emission is called fluorescence. Consider a spin singlet ground state molecule. A spin allowed singlet-singlet transition occurs, but before emission can occur (or before it occurs completely) there may be relaxation to a spin triplet partner of the excited spin singlet (mechanism b). Almost invariably this spin triplet lies at lower energy than the spin singlet, and here we concern ourselves with ligand centered states. Intersystem crossing is slower than internal conversion and depends upon a mechanism for coupling the two excited states. This mechanism is usually spin-orbit coupling. The rate of intersystem crossing is therefore greater with the heavier transition elements where spin orbit coupling is more important. Emission from this spin triplet is spin forbidden and hence the spin triplet excited state may have a fairly long lifetime. The process of emission from the spin triplet to the ground spin singlet (or in general emission between states of different spin) is called phosphorescence. With metal centered emission, the lowest lying excited state may or may not have the same spin as the ground state, depending upon the relative magnitude of DQ and the Racah parameters, and the configuration, d^n. In general, depending upon these factors, either phosphorescence or fluorescence may be observed. At room temperature in fluid media, fluorescent states have lifetimes in the nanosecond to microsecond regime, while phosphorescent states may have lifetimes in the microsecond to millisecond regime. Generally lifetimes increase upon cooling the sample and when the sample is solid rather than fluid.

> There are several mechanisms by which the lifetime of an excited state can be reduced. One involves collisional de-activation which takes place in fluid media but is reduced upon cooling or upon solidification.

Since there is a correlation between the allowedness of a transition and its emission lifetime, and allowedness is reflected in the oscillator strength of the absorption, there should be a correlation between lifetime and oscillator strength. An approximate relationship does exist, namely:

$$t_0 = g_u/(g_l.f.\nu^2) \qquad (4.31)$$

where g_u and g_l are degeneracy numbers for the upper and lower state respectively, f is the oscillator strength and ν the frequency. This equation (4.29) provides a rough estimate of the intrinsic lifetime in the absence of any external quenching effects (see [13] p.65 for some restrictions).

Most excited states live long enough to relax to their equilibrium states. Thus in a forbidden d-d transition, excitation occurs to a higher vibrational state of the excited electronic state, but emission, if observed will descend from the v' = 0 level of the excited electronic state. Thus emission from a state is to the red of absorption to that same state. Moreover if the states are displaced along the metal-ligand coordinate (e.g. Fig.4.4) the excited state may relax to higher excited vibrational states of the ground state, and cascade through them to the ground vibrational state, exhibiting vibrational structure, e.g Fig.4.5, 4.6. We repeat that such emission spectra may contain vibrational information concerning transitions which are vibronically coupled but which are forbidden both in Raman and ir. These spectra therefore provide a useful means of obtaining otherwise unavailable vibrational information for possible force constant analyses. Vibrational structure in emission is usually sharper than in absorption because it is less 'immersed' in the vibrational continuum.

Moreover emission occurs from the lowest lying vibrational state of the excited electronic state into the ground state vibrational regime which is not cluttered by interaction with other vibrational continua. Even if the excited state is Jahn Teller distorted, this complexity will not be seen since emission is only from one state. Thus the vibrational structure in an emission spectrum is generally much more highly resolved and clearly evident than in absorption.

Note that for an allowed band exhibiting the 0'-0" absorption, the absorption and emission will generally be mirror images of one another and will overlap at the position of the 0'-0" line. For a vibronic transition, the 0'-0" line is usually absent, but the 0'-1" line may be observed. Similarly in emission, the 0'-0" line is forbidden. In this case there will be a gap between the emission and absorption spectra corresponding to an energy of twice the quanta of the missing vibration.

The shift between absorption and emission is called the 'Stokes Shift' and is clearly related to the difference in equilibrium internuclear distances of the states concerned. Where emission occurs from a state with the same internuclear distance as the ground state, often phosphoresence from a spin forbidden state with no dependence upon Dq, the Stokes shift is very small and

absorption and emission are almost coincident.

If an excited state has a different equilibrium geometry than the ground state, then its lifetime may be quite long since a geometric change is necessary before it can relax. Examples of such systems are discussed in Chapter 7.

Excitation spectra can also provide electronic and vibrational information and examples will be presented in later chapters.

> In a luminescence experiment, the intensity of luminescence at a given wavelength can be scanned as a function of the exciting wavelength; the result is an excitation spectrum, which may exhibit vibrational structure.

Emission spectra will, in general, be polarised in lower symmetry complexes, obeying the same selection rules as in absorption. Clearly this polarisation data can provide useful information concerning the nature of the emitting states. DeArmond [28,29] has utilised the technique of emission photoselection presented by Albrecht [30] to obtain quite subtle information concerning the nature of emitting states. The polarisation of emission is defined by:

$$P = (I_{vv} - I_{vh})/(I_{vv} + I_{vh}) \qquad (4.32)$$

where I_{vv} and I_{vh} are the measured emission intensities with the analysers vertical(v) and horizontal(h) respectively, and the excitation source is polarised vertically(v). This polarisation is measured as a function of emission or excitation wavelength. The reader is referred to the original literature for further information [28-30].

4.9 Circular Dichroism and Magnetic Circular Dichroism

A brief description of the techniques of circular, and magnetic circular dichroism is presented since data therefrom can provide useful information complementary to those from electronic spectroscopy.

If plane polarised light is passed through a solution containing an optically active substance, the plane will be rotated in a direction which depends upon which enantiomer is present, the rotation being opposite and equal for the two enantiomers. The rotation results from the unequal refraction of left and right circularly polarised light. A spectrum displaying the magnitude of the rotation against wavelength is called Optical Rotatory Dispersion (ORD); such a spectrum shows a dramatic change in sign at absorption maxima, an observation called the 'Cotton Effect'. However such spectra are rarely utilised with inorganic systems.

Rather a spectrum can be obtained by plotting the difference in absorption of left and right circularly polarised light, as a function of wavelength. This is the Circular Dichroism (CD) experiment. The sign of the differential absorption depends upon which enantiomer is present; thus the CD spectrum can provide information concerning the absolute configuration of an enantiomer.

For the purpose of this discussion, consider the electronic and assignment information which is contained within a CD spectrum. The rotational strength depends upon the scalar product of the electric dipole and magnetic dipole transition moments. Allowing for the present that the transition is electronically (or vibronically) allowed, for a CD band to be observed, the magnetic dipole transition moment must be non-zero. This requires that the direct product of the ground and excited state must span a rotation in the point group concerned. Transitions which are both electronically allowed and magnetic dipole allowed will show a strong circular dichroism rotation.

Transitions which are electronically forbidden but which might have a small intensity due to vibronic interactions, may show up quite clearly in the CD spectrum if they have a strong magnetic dipole allowedness. Thus the CD spectrum may contain clear features which are largely absent, or obscured, in the electronic spectrum and therefore provide valuable additional information for deducing the energy levels within a molecule (see examples in Chapters 6,8).

For a given transition, the ratio of the differential intensity ($\Delta\varepsilon$) in the CD spectrum to the intensity (ε) in the electronic spectrum is known as the Kuhn anisotropy (or dissymmetry) factor, $|\Delta\varepsilon/\varepsilon|$.

Measured in terms of band areas, this factor reflects the ratio of rotational strength to the dipole strength of the transition (4R/D) [31-33]. The rotational strength R is related to $\Delta\varepsilon$ by [34]:

$$R = 22.9 \times 10^{-40} \int(\Delta\varepsilon/\nu)d\nu \qquad (4.33)$$

and the dipole strength is related to the extinction coefficient by: (compare 4.2 - 4.4)

$$D = 91.8 \times 10^{-40} \int(\varepsilon/\nu)d\nu \qquad (4.34)$$

where the integral $\int(\varepsilon/\nu)d\nu$ may be approximated by:

$$\ln 2.(\sqrt{\pi})\varepsilon_{max}\Delta_{1/2}/\nu^{o} \qquad (4.35)$$

(ε_{max} is the maximum extinction coefficient, $\Delta_{1/2}$ is the bandwidth at half-height and ν^{o} is the frequency of the band intensity maximum)[35].

Consideration of the magnitude of this dissymmetry factor can provide useful electronic information. For example Solomon and co-workers [33] note that the sulphur to copper charge transfer transitions in copper proteins, fall into two classes. The πS---Cu(II) bands are magnetic dipole allowed and show large dissymmetry factors, while the σS---Cu(II) transitions, though very much more intense in the electronic spectrum, have small dissymmetry factors because they are magnetic dipole forbidden.

While the CD technique is hampered by the need for an optically

active material, it is especially important in metalloenzyme spectroscopy, since natural enzymes are usually obtained in only one enantiomeric form.

If the above experiment is carried out by placing the sample inside a strong magnetic field, a **Magnetic Circular Dichroism (MCD)** spectrum is obtained. In this case it is not necessary to use a chiral material.

The technique depends upon the differential absorption of circularly polarised light by different components of a degenerate state split by the Zeeman effect in a magnetic field. It is necessary that the ground and/or excited state possess non-zero angular momentum (j) such that splitting into components of different m_j will take place in a magnetic field. Thus left circularly polarised light induces a transition in which $\Delta m_j = -1$, while right circularly polarised light induces a transition in which $\Delta m_j = +1$. A transition from a non-degenerate to a degenerate state, split in a magnetic field will have a differential intensity curve as shown in Fig.4.9 (a) known as an **A** term; the shape is independent of temperature.

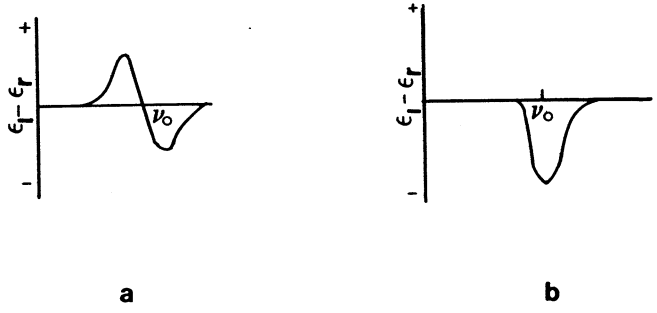


Fig.4.9 (a) An mcd **A** term, and (b) an mcd **C** term.

If the ground state is degenerate, but the excited state is not, there will be a Boltzmann distribution of left and right circularly polarised absorption from the Zeeman levels of the ground state. The result is a **C** term (Fig.4.9 (b)) which will, of course, have a temperature dependent band envelope (inversely proportional to absolute temperature).

Finally if there is a field induced mixing of the various states, a **B** term may be observed. This appears like a **C** term but is temperature independent.

Analysis of MCD spectra provides evidence of the degeneracy of states and hence assists band assignments. Perhaps more importantly, it provides a procedure for elucidating the angular momentum associated with a specific state, a number which can then be compared with, for example, a molecular orbital analysis of the system.

4.10 Resonance Raman Spectroscopy [45]

If a solid or solution is irradiated with monochromatic light, the scattered light contains vibrational information. Thus, for example, the spectrum of scattered light collected at 90^o to a monochromatic beam of frequency, ν incident in a solution of some species, will show strong emission at frequency ν due to simple scattering (Raleigh scattering) plus additional emission corresponding to the stimulation of normal modes of vibration (Stokes emission). This additional emission will occur at a frequency of ν less the frequency of each stimulated normal mode (Stokes lines), plus weaker lines to higher frequency at ν plus the normal mode frequencies (anti-Stokes emission). This is the simple Raman experiment. It parallels the infrared experiment in generating vibrational information but since the selection rules (change in polarisability rather than change in dipole) are different, the data complement the infrared experiment [25,45].

In the resonance Raman experiment [45], the incident light, using a laser, is tuned directly into a specific absorption band of a species. A vibrational spectrum will again be generated, but vibrations which are coupled to the electronic transition which is being irradiated may exhibit enhanced intensities. Generally it is the relevant totally symmetric vibrations which will be enhanced, though other lower symmetry vibrations may also increase in intensity.

This resonance enhancement provides a powerful means for assigning spectra since it can provide a direct indication of the type of electronic transition being irradiated. It has been used extensively in the study of charge transfer spectra where, for example an X-M LMCT transition may show enhancement of M-X stretching vibrations. Examples of the use of resonance Raman spectroscopy are given in Chapters 5-8. Of especial interest is the study of the spectra of oxymetalloproteins (Chapter 8). Such species can show quite complex spectra. Irradiation into a specific absorption band will generally cause enhancement of the intensity of the O-O stretch if the band concerned is O_2-M LMCT in nature.

If the intensity of a specific vibration is scanned as a function of the exciting frequency then a resonance Raman excitation spectrum is generated. If the absorption band concerned involves an excited state whose internuclear M-L distancee is significantly different from the ground state, then the excitation spectrum will often show a long progression in the M-L stretching frequency [45].

In a parallel experiment involving low symmetry complexes, the excitation spectrum may show structure in an absorption band where the simple electronic spectrum does not. In such a situation, e.g. see Fig.5.19 (a), only some of the components of a split state couple to the vibration concerned.

4.11 Isosbestic Points [25]

Finally a note about the experimental observation of isosbestic points. Suppose there is a reaction between two species A and B to yield a product AB such that in a specific spectroscopic region, only A and AB absorb. If the initial concentration of A is held constant, then as the concentration of B varies, the sum of the concentrations of A and AB remains constant. The absorbance of the solution at any specific wavelength is given by:-

$$E = \varepsilon_A[A] + \varepsilon_{AB}[AB] \qquad (4.36)$$

There are likely to be one or more wavelengths where it is true that $\varepsilon_A = \varepsilon_{AB}$. At these wavelengths the absorbance of the solution will remain constant with variation of [B], provided [A] + [AB] is constant. Such constant absorbing points are called isosbestic points.

The existence of isosbestic points in a spectroscopic study of a reaction mixture is a useful guide to the purity of the reaction since such points are unlikely to occur if one or more reacting species are decomposing to other products. Moreover appropriate analysis can allow one to deduce how many species are in equilibrium. Although two species, A and AB, are assumed in equilibrium above, there are conditions where three or more species in equilibrium will yield isosbestic points. Such situations were discussed in some depth by Drago [25].

The conditions for observation of one or more isosbestic points in a mixture containing more than two species may be written:-

In a system containing 2 + N absorbing species, an isosbestic point may arise if there are N independent equations of the type [X]/[Y] = constant where X and Y are two of the species concerned.

For example consider a reaction between a Lewis acid A, and an ambidentate ligand such as thiocyanate. One might imagine that the following equilibria exist:-

$$A^+ + NCS^- \text{ ------ } ANCS$$

$$A^+ + NCS^- \text{ ------ } ASCN$$

where perhaps A^+, and the two adducts each absorb in the region of interest. The equilibrium constants for these reactions may be written:-

$$K_1 = [ANCS]/[A^+][NCS^-] \qquad K_2 = [ASCN]/[A^+][NCS^-] \qquad (4.37)$$

and the desired independent equation is:

$$K_1/K_2 = [ANCS]/[ASCN] \qquad (4.38)$$

Thus if $[A^+]$ is held constant and $[NCS^-]$ is varied, an isosbestic point (or more than one) should be observed. See [25] for further detail.

Coleman, Varga and Mastin [46] have discussed graphical methods for elucidating how many absorbing species there are in a given solution spectrum.

For example if the absorbance of a specific solution, j, at wavelength i, is denoted A_{ij}, then a plot of data for different wavelengths and different solutions can provide useful information. Thus a plot of $A_{i'j}-A_{ij}$, versus this difference at a different wavelength i', will yield a straight line, passing through the origin, if only two absorbing species are present. Three absorbing species in solution will give rise to a straight line plot of $(A_{2j}-A_{2j'})/(A_{1j}-A_{1j'})$ versus $(A_{3j}-A_{3j'})/(A_{1j}-A_{1j'})$ at three wavelengths, 1,2 and 3.

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Chapter 5

Charge Transfer Spectra

5.1 General Introduction

While colour in inorganic compounds is generally associated with the presence of a partially filled d shell, there are many such compounds with d^0 or d^{10} configurations which are intensely coloured. Mercury(II) iodide (brick red, d^{10}), the manganate(VII) ion (purple, d^0), bismuth(III) iodide (orange red, $d^{10}s^2$) and lead iodide (yellow, $d^{10}s^2$) are common examples. In such cases colour arises, at least in part, as a consequence of light absorption due to transfer of an electron from an orbital primarily on the ligand, to one primarily on the metal, or vice versa. Such charge or electron transfer processes are termed 'ligand to metal charge transfer', LMCT, and 'metal to ligand charge transfer', MLCT, respectively. Such a process is not confined to d^0 and d^{10} species and occurs frequently in transition metal complexes. Indeed all polyatomic species must exhibit charge transfer spectra, but only if this absorption occurs in the visible region, will this process give rise to colour visible to the human eye. This area of spectroscopy has grown explosively during the past fifteen years. We begin with a general introduction and develop the theory in a quasi-chronological order from a general approach to a more sophisticated analysis. The chapter then deals specifically with a range of metal configurations and ligands and special topics.

Charge transfer processes frequently lie at the blue end of the visible region, or in the ultraviolet region. However if the metal is easily oxidisable, and the ligand easily reducible, or vice versa, then charge transfer transitions may occur at quite low energies, even in the near infrared. Charge transfer spectra are often intense and may obscure weaker crystal field transitions in the same region. However this is not always the case, weak, i.e. partially forbidden, charge transfer transitions are now more commonly being identified.

5.1.i) Ligand to Metal Charge Transfer Transitions

Jorgensen [70,275,282,419-422,424] pioneered the study of charge transfer spectra with a detailed investigation of the charge transfer spectra of heavy metal halides. His work laid the foundation for later development of this area.

Consider first the heavy metal hexahalides. In general three main regions of absorption are observed in the charge transfer electronic spectra of these species. Firstly a group of narrow bands (half-width, 400 - 1000 cm^{-1} energies in sequence I < Br < Cl) lie approximately between 15,000 and 30,000 cm^{-1} in nd^3, nd^4,

nd^5 but not low spin nd^6 (n = 4,5) complexes. Next, all hexahaloanions of the 4d and 5d elements exhibit broad strong absorption (half-width 2000 - 4000 cm^{-1}) in the region 25,000 - 45,000 cm^{-1}. This often consists of a single band in the case of chlorides, a double band for bromides and a triple band for iodides. Finally in certain cases, it is possible to observe another very strong band lying above 44,000 cm^{-1}.

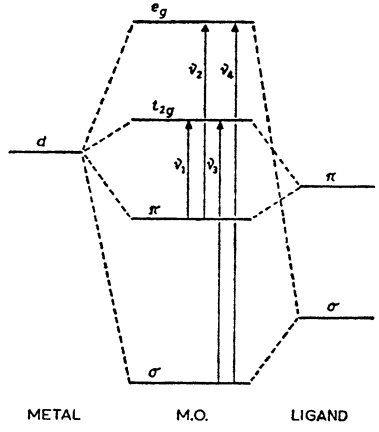


Fig.5.1 A simplified molecular orbital diagram illustrating the potential ligand to metal charge transfer transitions.

Before considering how these transitions vary as a function of metal and oxidation state, consider a very simplified approach to the type of charge transfer spectra which may be anticipated. Consider a single metal-halogen bond, then the energy levels concerned may be depicted, in a simple approximation, as shown in Fig.5.1, in which the lone pairs of the halogen form the basis for the σ and π bonds to the metal atom.

Four basic types of transition may be expected, viz:

$$\begin{array}{ll} \nu_1 \quad \pi L \text{---} Mt_{2g} \ (\pi^*) & \nu_2 \quad \pi L \text{---} Me_g \ (\sigma^*) \\ \nu_3 \quad \sigma L \text{---} Mt_{2g} \ (\pi^*) & \nu_4 \quad \sigma L \text{---} Me_g \ (\sigma^*) \end{array} \qquad (5.1)$$

in order of increasing energy. This model can be used to obtain a gross understanding of the spectra of the d^n hexahalides.

The first set of comparatively low energy bands observed in the hexahalides may be ascribed to ν_1 transitions. They are narrow because they involve transitions between orbitals which are almost non-bonding in character and thus invoke a low probability of vibrational excitation. Clearly these bands must be absent in the low spin nd^6 species since the t_{2g} level is filled. The energy of these transitions, for a given halogen, increases by 6000 - 9000 cm^{-1} in proceeding from $4d^n$ to $5d^n$ ions and also increases with

increasing oxidation state of the central metal. Their energies increase in the sequence I < Br < Cl in parallel with the ionisation energies of the halide ions (See section 5.4, Table 5.6).

The first set of bands in the nd^6 (t_{2g}^6) complexes must necessarily involve transitions to the e_g metal orbitals and are assigned to the ν_2 transition. Since the upper level is anti-bonding, these transitions are expected to be broad. Their energies vary in a fashion similar to the ν_1 transitions.

The very high energy, and very intense, bands are assigned to ν_4. The alternative explanation, ν_3, is eliminated on the basis that they are seen in nd^6 complexes and that ν_3 transitions are expected to be weak [70,419-422]. The presence, in most cases, of multiple bands rather than single bands, in these hexahalide complexes follows from a more complete molecular orbital treatment.

Since we are dealing not with a single metal-halogen bond, but with six in an octahedral array, then the six σ bonds span $a_{1g} + e_g + t_{1u}$, and the twelve π orbitals span $t_{1g} + t_{2g} + t_{1u} + t_{2u}$. Then, for example, the 'single' $\pi L \text{---} t_{2g}$ transition becomes two potentially allowed excitations from t_{1u} and t_{2u} and two Laporte forbidden excitations from t_{1g} and t_{2g}.

Moreover, consider the excitation $t_{1u} \text{---} 2t_{2g}$. The direct product of these representations spans four states, and in a d^0 complex, for example, the excited states, all nominally ascribed to ν_1, span $^1A_{2u}, ^3A_{2u}, ^1E_u, ^3E_u, ^1T_{1u}, ^3T_{1u}, ^1T_{2u}$ and $^3T_{2u}$, though not all transitions are permitted.

Next, the existence of more structure in the spectra of the bromo and iodo complexes is a consequence of their large spin-orbit coupling coefficient which will lift the degeneracy of orbital triplet states leading to an increased number of possible transitions. Finally many charge transfer bands show vibrational structure.

Despite this complexity, this last fifteen years has seen some very elegant work in which detailed and reliable analyses of charge transfer structure has been achieved.

The same basic approach may be used to develop the LMCT spectra of other stereochemistries which will be analysed below.

5.1.ii) Metal to Ligand Charge Transfer (MLCT)

Such absorption is expected when the ligand has low lying empty orbitals; unsaturated ligands with empty π anti-bonding orbitals are prime candidates. For example pyridine, pyridine N-oxide, pyrazine, 2,2'-bipyridine, 2,2'-bipyrazine and o-phenanthroline complexes of the more readily oxidisable transition metal ions (e.g. Mo(0), Re(I), Cu(I), Fe(II), Ti(III), Ru(II) etc) frequently generate fairly low energy charge transfer absorption of the MLCT class. Clearly such absorption must lie at lower energies than the internal $\pi - \pi^*$ transitions of the ligands themselves, or the charge transfer absorption is likely to be obscured.

This type of charge transfer was first identified by Jorgensen

who termed these transitions 'inverted charge transfer' [436] in a series of iridium(III) pyridine complexes.

Ligand to metal (LMCT) and metal to ligand (MLCT) transitions may usually be distinguished by the way in which their energy changes when the metal ion (or its oxidation state) or ligand is altered. Thus for a given metal ion, ligand to metal charge transfer energies decrease as the ligand becomes more readily oxidisable and/or as the metal becomes more readily reducible. Clearly metal to ligand charge transfer transitions decrease in energy as the ligand becomes more reducible, and/or the metal becomes more readily oxidisable.

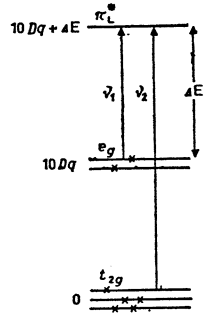


Fig.5.2 A simplified molecular orbital diagram of the potential metal to ligand charge transfer transitions in a d^7 ion. The one electron energies relative to $E(t_{2g}) = 0$ are shown on the left.

This is shown in Table 5.1 by a series of pyridine N-oxide complexes. For example compare the cobalt(II) complexes of 2-methylpyridine N-oxide, pyridine N-oxide and 4-nitropyridine N-oxide, whose MLCT energies decrease in this order as the ligand becomes a better acceptor.

With picolinic acid N-oxide (picO) as a ligand, the MLCT energies decrease in energy in the sequence:

$$Cr(III) >> Cu(II) > Mn(II) > Ni(II) > Co(II) > Fe(II) \qquad (5.2)$$

This is not the order of the ionisation potentials of these ions, nor need it be, since in the complex, the ion will no longer carry net positive charges but will more nearly approach electrical neutrality (Pauling Electroneutrality Principle). It is interesting however that the sequence (5.3) is reflected by the chemical properties of these complexes in that direct chemical oxidation leads to picO complexes of Fe(III), Co(III), Ni(III) and Mn(III) but not Cu(III) or Cr(IV) [423].

However the order is also determined by the fact that these bands do not all belong to the same transition.

Table 5.1

MLCT CHARGE TRANSFER SPECTRA OF PYRIDINE N-OXIDE COMPLEXES

Complex	Configuration	t_{2g}---π^*	e_g---π^*	Ref
$Mn(PicO)_2.2H_2O$	$(t_{2g})^3(e_g)^2$		24,700(280)	423
$Fe(PicO)_2.2H_2O$	$(t_{2g})^4(e_g)^2$	21,050(270)		423
$Co(PicO)_2.2H_2O$	$(t_{2g})^5(e_g)^2$	23,800(125)		423
$Ni(PicO)_2.2H_2O$	$(t_{2g})^6(e_g)^2$	24,400		423
$Cu(PicO)_2.2H_2O$	$(t_{2g})^6(e_g)^3$		26,700	423
$Mn(PyNO)_6^{2+}$	$(t_{2g})^3(e_g)^2$		24,300	298
$Co(PyNO)_6^{2+}$	$(t_{2g})^5(e_g)^2$	25,450		298
$Co(4NO_2PyNO)_6^{2+}$	$(t_{2g})^5(e_g)^2$	21,100		298
$Co(2Me-PyNO)_6^{2+}$	$(t_{2g})^5(e_g)^2$	26,600(400)		298
Co(Pico)Cl (T_d)	$(e)^4(t_2)^3$		30,800	423

See [426,429,430] for further amine oxide CT data. Many tables in this chapter include MLCT data.

This series is a nice example of how MLCT (and certain LMCT) assignments depend upon the relative magnitude of the crystal field splitting and the inter-electronic repulsion energy.

Consider the two possible transitions indicated in Fig.5.2, their energies are given approximately by:

$$E(\nu_1) = E$$
$$E(\nu_2) = E + 10Dq - P \qquad (5.3)$$

where P is the mean pairing energy per electron pair. For a spin free (high spin) complex, P is greater than 10Dq by definition, therefore $E(\nu_2) < E(\nu_1)$ although this does not appear to be the case in Fig.5.2. Such a situation will be true for spin-free compounds when there is an electron pair in the t_{2g} set but not in the e_g set. Thus, using these qualitative arguments, the assignments shown in Table 5.1 are derived. Clearly there will be circumstances where the transition energies are comparable and both bands will lie close together. We return in section 5.3 to a more detailed analysis of the problem of repulsion energy contributing to charge transfer transitions.

5.1.iii) The Effect of Coordination Number and Stereochemistry upon Charge Transfer Energies

The tetrahedral complex Co(picO)Cl has a MLCT band at 30,800 cm^{-1}. The shift from 23,800 cm^{-1} in the octahedral complex is due principally to a decrease in the spherical one-electron energies of

the d orbitals when the number of ligands around the metal is decreased. If bonding is primarily ionic then we may generally expect MLCT transitions to blue shift as the coordination number of the ligand decreases, while for LMCT transitions we should expect a red shift, with decreasing coordination number. However it is well known that bonding tends to become more covalent with decreasing coordination number as the metal must accept more charge from fewer ligands if it is to approach electroneutrality. Thus for example in the d^0 series [243]:

$$[TiCl_6]^{2-}\ (25{,}000) \quad < \quad TiCl_4\ (35{,}600)$$
$$[TiBr_6]^{2-}\ (21{,}800) \quad < \quad TiBr_4\ (29{,}500)$$
$$[TiI_6]^{2-}\ (12{,}100,\ 14{,}300) \quad < \quad TiI_4\ (19{,}600) \qquad (5.4)$$

the six and four coordinate complexes listed here have LMCT X--Ti(IV) energies increasing in the sequence as shown, i.e. increasing with decreasing coordination number. There is no doubt that the Ti-Cl bond is much more covalent in the liquid $TiCl_4$ than in the salt-like $[TiCl_6]^{2-}$ anion. Thus the metal ion will become less positively charged in the sequence (5.4), while the chlorine atom becomes less negatively (more positively) charged. The highest energy chlorine orbital therefore becomes more stable, and the lowest energy metal orbital becomes less stable, in the sequence (5.4). This clearly results in a blue shift in the transition energy as observed. There will also be a crystal field effect upon the acceptor metal orbital (in simple terms, $-4Dq(T_d)$ for the four coordinate system and $-6Dq(O_h)$ for the six coordinate system) which also has the effect of shifting the CT band to the blue in the sequence of (5.4). In this case, the crystal field effect probably represents about 20% of the actual shift. In terms of optical electronegativity (see section 5.3), and correcting for this crystal field contribution, there is a change of about 0.3 optical electronegativity units in sequence (5.4). This should probably not be ascribed to either the metal or ligand alone, but to the net effect upon both.

As discussed in some depth [243], ligand-ligand repulsion will also have an effect upon CT energies in ML_n species. Octahedral complexes have a shorter L-L distance than tetrahedral complexes (by a factor of 1.41:1.63) resulting in a significant destabilisation of ligand orbitals in these octahedral complexes relative to their tetrahedral analogs. This L-L factor probably also contributes to the lower LMCT energies in octahedral species shown in (5.4).

5.1 iv) Solvent Dependence of Charge Transfer Bands

It is generally understood that crystal field (d-d) transitions have energies which are essentially insensitive to the solvent in which the spectrum is obtained.

Beware however situations where the solvent reacts with the species by displacement of another ligand (solvolysis) or breakup of a polymeric structure followed by ligation. In such situations the d-d energies may be solvent dependent to some degree.

It is also usually true that CT bands do show some solvent dependence. In simple terms this is a consequence of the following general situation.

In the ground state, the solvent is organised with its dipole oriented to interact with the ground state dipole of the complex. The dipole of the excited CT state will usually differ markedly (in direction or magnitude) from the ground state dipole. Thus the solvent will no longer be favourably oriented in the Franck-Condon (instantaneous) excited state. Depending upon the nature of the solvent, and the change in dipole, the excited state may be stabilised or de-stabilised. In either case the energy of the transition will depend upon the solvent, and usually upon its polarity.

Such solvent dependence may be quite small, or even effectively non-existent if the molecule is symmetric and has no permanent dipole. However in the special situation that the transition moment of the CT state lies along the ground state dipole and hence will cause a significant change upon excitation, the solvent effect can be very large indeed.

One of the earlier treatments of solvent effects was a report by Saito and co-workers [200] on the solvatochromism of $M(CO)_4(N\text{-}N)$, M = Cr,Mo and W. Although the assignment of a higher energy band has since changed, the paper presents an early interesting discussion. These authors showed that d-d bands did not shift appreciably, that with saturated amines such as ethylenediamine, the solvatochromism of the spectral features was also small, but that with diimine ligands there was marked solvatochromism in the low energy M---diimine MLCT band. In general the MLCT band shifted to the blue with the more polar solvents. This is an example where the transition moment of the MLCT transition does lie along the permanent ground state dipole.

Work by tom Dieck and colleagues with these systems has contributed to our understanding of solvatochromism in carbonyl-containing complexes [145]. They also demonstrated that systems $M(diimine)(CO)_4$ (M = Mo,W) (containing an $MN_2(CO)_2$ plane) show MLCT transitions which lie in the visible region, and whose energies are frequently highly solvent dependent [145,146]. For example with diimine = N,N'-ditert-Bu-1,4-diazabutadiene, and M = Mo, the MLCT band varies from 21,120 cm^{-1} in DMSO to 17,745 cm^{-1} in cyclohexane. Additionally the CO frequencies associated with the in-plane CO groups have vibrational frequencies which are solvent dependent (varying linearly with the MLCT transition), whereas the axial CO frequencies show little variation. In this example, the ground state dipole lies along the z axis (C_{2v} symmetry axis in $N_2M(CO)_2$ plane and bisecting the NMN and CMC angles), with negative charge build up in the carbonyl region. Assuming the MLCT excitation is from metal to diimine, then this transition is in opposition to the ground state dipole and, by the arguments presented above, the

shift to the blue in the more polar solvents as is observed.

However a study of a series of complexes with different in-plane diimines, and with axial CO groups substituted by phosphines reveals several possible solvatochromism patterns [145]. Some of these complexes show no solvatochromism, while others, such as Mo(N,N'-dicyclohexyl-1,4-diazabutadiene)$(CO)_2(PEt_3)_2$, show a reversed solvatochromism in which the band shifts to the red with the more polar solvents. If the CT transition shifts charge in the same direction as the ground state dipole then again using the arguments expressed above, the transition will shift to the red in the more polar solvents. tom Dieck argues that, in these complexes, the CT transition is of the LMCT type from diimine to metal, and hence transports charge in the opposite sense to the di-tert-butyl species discussed above [145]. This seems an over-simplification given that the metal is in the zerovalent state.

Rather we note that any description of the relevant excited state must involve both diimine and the two equatorial (in diimine plane) CO groups, since their π orbitals will overlap through the metal. In both the cases discussed here, with opposing solvatochromism behaviour, charge moves from an orbital mainly localised on the metal, to an orbital localised on both diimine and in-plane CO. In the di-tert-butyl1,4-diazabutadiene case one assumes that most charge moves towards the diimine, while in the cyclohexyl-1,4-diazabutadiene case, one would assume that more charge moves towards the CO groups, facilitated by the electron donor character of the diphosphine substituents. In situations where the solvatochromism disappears, the charge is evenly distributed over both diimine and in-plane CO and the ground state dipole is not affected.

Stufkens and associates have also studied the solvatochromism (and thermochromism) of many $M(CO)_{4-x}L(PR_3)_x$ systems (where x = 0,1,2 and L is a diimine) [114,187,188,190]. In general where the diimine has two accessible π^* states, transitions to both states from the d manifold may be observed. These shift in concert with change in solvent (see section 5.8 vi)).

Burgess and co-workers [201] have summarised the data for various diimine systems (containing, CO, CN^- etc, e.g. $Fe(bipy)_2(CN)_2$) and show that there is often, though not invariably, a linear relationship between band energy and the empirical Reichardt parameter, E_T [202]. In a series of solvents there are often two separate correlations for a given spectroscopic data set, one with hydroxylic solvents and one with non-hydroxylic solvents. This is especially true of correlations against Reichardt E_T, while correlations with Acceptor Number (AN) may give a single line for all solvents [431].

In their extensive study of carbonyl systems, Wrighton and co-workers have noted solvent dependence of many CT bands, especially those terminating on a π^* orbital [152,154,155,158-160].

While many authors have associated the changes in energy as a function of solvent upon properties of the solvent such as Donor or Acceptor number, optical or static dielectric constant, Reichardt

Table 5.2

SOLVENT PARAMETERS APPROPRIATE FOR SOLVATOCHROMIC BEHAVIOUR[a]

Solvent	D_s	D_{op}	μ_g	E_T	Z	DN	AN	β	H_p
Non-Hydroxylic									
n-Hexane	1.90	1.882		30.9			0.0		
iso-Octane	1.94	1.936			60.1				
Benzene	2.27	2.244	0	34.5		0.1	8.2		
CCl_4	2.23	2.129	0	32.5			8.6		
Toluene	2.38	2.232	0.36	32.5					
CS_2	2.64	2.663	0	32.6					
Ethyl ether	4.22	1.828	1.30	34.6		19.2	3.9		
Chloroform	4.7	2.085	1.01	39.1	63.2		23.1		
THF	7.4	1.971	1.63	37.4		20.0	8.0		
Tri-N-Bu Phos	7.8	2.0292				23.7			
CH_2Cl_2	8.9	2.028	1.60	41.1	64.2		20.4		
Quinoline	9.22	2.646	2.29	39.4					
$ClCH_2CH_2Cl$	10.4	2.085	2.06	41.9		0			
Pyridine	12.3	2.271	2.19	40.2	64.0	33.1	14.2		
Cyc-pentanone	13.5	2.0638						0.537	5.50
Acetone	20.7	1.8413	2.88	42.2	65.7	17.0	12.5	0.495	5.59
Benzonitrile	25.2	2.3375	4.98	42.0		11.9	15.5	0.409	
HMPA	29.6	2.1255				38.8	10.6	0.990	8.72
Nitrobenzene	34.8	2.403	4.22	42.0		4.4	14.8	0.393	
Acetonitrile	36.2	1.8066	3.92	46.0	71.3	14.1	19.3		4.2
DMF	36.7	2.0458	3.82	48.3	68.5	26.6	16.0	0.710	6.97
DMA	37.8	2.0609	3.81	41.0		27.8	13.6	0.749	7.44
Nitromethane	38.6	1.9091	3.46	46.3		2.7	20.5		
DMSO	49.0	2.1815	3.96	45.0	71.1	29.8	19.3	0.752	7.21
Propyl. Carb.	65.1	2.0133	4.98	46.6		15.1	18.3		4.53
N-Me formam.	182.0	2.0478	3.83	54.1					6.44
Hydroxylic									
Acetic acid	6.15	1.877	1.74	51.9	79.2		52.9		
1-Decanol	8.1	2.066							
1-Octanol	10.3	2.036							
1-Pentanol	13.9	1.982							
1-Butanol	17.1	1.952	1.66	50.2	77.7				
2-Propanol	18.3	1.890	1.66	48.6	76.3		33.5		
1-Propanol	20.1	1.913	1.68	50.7	78.3				
Ethanol	24.3	1.847	1.69	51.9	79.6	20	37.1		
Methanol	32.6	1.758	1.70	55.5	83.6	19	41.3		
Et Glycol	37.7	2.042	2.28	56.3	85.1				
Water	78.5	1.777	1.85	63.1	94.6	18.0	54.8		

a) Abstracted from Table IX in [52]. See [52] and other sources

such as the 'Handbook of Chemistry and Physics' for additional data and error limits on these numbers. See also V.Gutmann, "The Donor-Acceptor Approach to Molecular Interactions", Plenum Press, NY 1980, and Reichardt, Angew.Chem., **4** (1965) 29. D_s and D_{op} are the static and optical dielectric constants, μ_g is the dipole moment in gas phase, E_T is the Reichardt parameter, Z is due to Kosower [624], DN and AN are the Gutmann donor and acceptor numbers, β is due to Kamlet and Taft [54] and ΔH_p is due to Arnett [55].

parameter etc, it is difficult to extract useful information. One may argue specific interactions between the solvent and a part of a complex such as a polar carbonyl or cyanide group or a potentially hydrogen bonding group such as ammonia, especially if the CT band fails to track the solvent dielectric (e.g. eqn.7.11). For example if there is a correlation with the hydrogen bonding scales of Taft (β) or Arnett, a specific hydrogen bonding interaction may be present [52]. Indeed one may use the solvent dependence to probe solvent characteristics by observing the shift in a CT band as a function of mole fraction in mixed solvents [52,431]. To aid such development, we include here a Table of relevant solvent properties (Table 5.2). It is possible that a charge transfer transition can be treated theoretically [52,146] much like an intervalence transition (Chapter 7) in which case some interesting solvent and molecular characteristics may be derivable from solvatochromic behaviour (see Chapter 9).

5.2 Spectroscopic Terms and Charge Transfer Energies

If a tetrahedral complex d^1 ML_4 complex is considered, the ground configuration may be written $...(1t_1)^6(e)^1$ which transforms as ^2E. The 'first' excited configuration is $(1t_1)^5(e)^2$. Which states are possible for this configuration and what are their relative energies ? This problem has recently been considered in some detail by Vanquickenbourne and Verdonck [247,248].

Crucial to the understanding of this problem are two facts. Firstly, the LMCT states of a d^n configuration may be generated by obtaining the direct product of the symmetries of the states spanned by the metal d^{n+1} configuration with the symmetry of the hole on the ligand. Secondly, since Coulomb integrals of the type J(d,d) (localised on the metal) are larger than those of the type J(d,L) (delocalised between metal and ligand) the relative energies of these states will be primarily determined by the relative energies of the states of the metal d^{n+1} configuration.

Thus, in this specific case, the $^3A_2$, ^1E and $^1A_1$ states of d^2 $(2e)^2$ follow from the Pauli exclusion principle and the direct product e x e (see Table 2.2). The $^3A_2$ state (M_s = 1) can couple with the m_s = 1/2 of $(1t_1)^5$ to yield a spin quartet or spin doublet, both of T_2 symmetry (t_1 x a_2). The spin singlet states ^1E and $^1A_1$ can only yield spin doublets when coupled with m_s = 1/2 of $(1t_1)^5$. They produce the orbital states e x t_1 = t_1 + t_2, and

a_1 x t_1 = t_1 respectively. Thus the 36 microstates follow (Fig.5.3):

$$^3A_2 \times t_1 \qquad ^1E \times t_1 \qquad ^1A_1 \times t_1$$

$$(1t_1)^5(2e)^2 \text{ spans } (^4T_2 + {}^2T_2) + (^2T_1 + {}^2T_2) + (^2T_1)$$

where they are listed in relative order given that the d^2 states lie in the energy order $^3A_2 < {}^1E < {}^1A_1$. Since transitions from ^2E are allowed to all the spin doublet states, a tetrahedral d^1 ML_4 species should exhibit three regions of LMCT absorption (for this particular excitation), the first and last comprising one transition, and the second comprising two closely spaced transitions (Fig.5.3 on lower left). Other states arising from the excitations t_1---2e and t_1---$2t_2$ are shown in Fig.5.3.

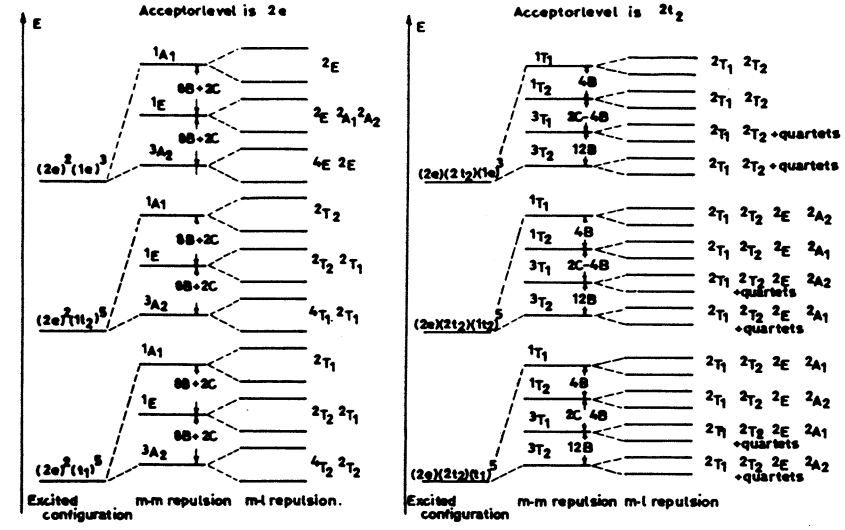


Fig.5.3 Qualitative energy levels diagrams for charge transfer excited states in T_d d^1 systems. The left-hand diagram shows excitation from ligand levels into the metal 2e level, and the right-hand side into metal $2t_2$. Interelectronic repulsion on the metal is indicated by m-m, and between metal and ligand by m-l [247]. The allowed transitions are $^2T_1$, $^2T_2$---^2E.

As a further example consider Fig.5.4 which shows the states possible for a tetrahedral d^7 cobalt(II), $[CoX_4]^{2-}$, species involving excitation from the two upper ligand levels ($1t_1$ and $1t_2$) into the half-full (d)$2t_2$ level ((d)2e is full). The centre column shows the states arising from the d orbital configuration $(e)^4(t_2)^4$ (d^8) (Table 2.2) which are generated by this transition, while the

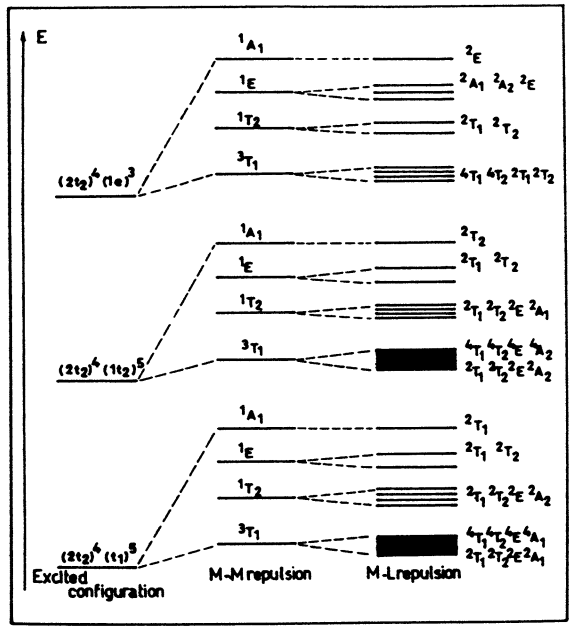


Fig.5.4 Qualitative energy levels diagram for charge transfer excited states in d^7 $[CoX_4]^{2-}$ (X = halogen) [247].

right-hand column shows the final states obtained by coupling these d^8 states to the ligand 'hole'.

Thus, for the lowest configuration, the $^3T_1$, $^1T_2$, ^1E and $^1A_1$ states will be separated to a degree comparable to the corresponding states (in this case) of d^8. For a given metal state, the several states on the right-hand side, such as those spanned by the lowest lying $^3T_1$, will all lie close together (separated only by M-L repulsion terms) and may not be separable. However the four sets of levels on the right-hand side, originating from the four metal states of $(2t_2)^4(1t_1)^5$, should be readily resolvable. Thus, in this fashion, one may distinguish those component states which should lie close together, from those which are separated by repulsion integrals on the metal.

In this specific case, only the transitions $^4T_1$---$^4A_2$ are electric dipole allowed so that only one transition will arise from each configuration. This will not generally be the case however. Table 5.3 surveys d^n configurations to identify those configurations which give rise to several parent metal spectroscopic states having several electric dipole allowed transitions. In these cases, e.g. d^1 in Fig.5.3, the CT states should be separately identifiable, and, where assignment is possible, Dq, B and C values

Table 5.3

CHARGE TRANSFER EXCITED STATE TERM CLASSIFICATIONS [247,248]

Configuration	2e Acceptor	$2t_2$ Acceptor	Example
Tetrahedral-symmetry			
d1	M	M	VCl_4, MnO_4^{3-}, MnO_4^{2-}
d2	S	M	MnO_4^{2-}, FeO_4^{2-}
d3 high spin	M	M	
low spin	S	M	
d4 high spin	M	M	
low spin		S	
d5 high spin	S	S	FeX_4^-
low spin		M	
d6 high spin	S	M	FeX_4^{2-}
low spin		M	
d7		S	CoX_4^{2-}
d8		S	NiX_4^{2-}
d9		S	CuX_4^{2-}
Octahedral-symmetry			
Configuration	e_g Acceptor	t_{2g} Acceptor	Example
d1	M	M	TiX_6^{3-}, ReF_6, $Mo(NCS)_6^{2-}$, VX_6^{3-}
d2	M	M	TcX_6^{2-}
d3	M	S	$Mn(H_2O)_6^{3+}$
d4 high spin	M	M	OsX_6^{2-}
low spin	M	X	
d5 high spin	X	X	$Mn(H_2O)_6^{2+}$
low spin	M	X	IrX_6^{2-}
d6 high spin	M	X	CoF_6^{3-}
low spin	S		IrX_6^{3-}
d7 high spin	M	S	$Co(H_2O)_6^{2+}$
low spin	M		
d8	S		NiX_6^{4-}
d9	S		$Cu(H_2O)_6^{2+}$

M = different electric dipole allowed states arise from different parent metal ion states; S = Only one electric dipole allowed state for each configuration; X = several electric dipole allowed states arise from a single configuration.

of the excited state may be obtainable. Examples of this analysis will be met later in this chapter.

An essentially identical treatment is possible for octahedral complexes as shown in Table 5.3. The relevant repulsion energy matrices for these different parent terms can be obtained via Table 3.6 but are also included in Refs.[247,248]. Data for some hexahalides were re-assigned using this type of analysis. The key point is that the separation of groups of LMCT transitions (for configurations d^n labelled M in Table 5.3) follow the separations in the crystal field spectra of the d^{n+1} ions. The above arguments also apply in an exactly analogous fashion to MLCT transitions.

Inclusion of spin-orbit coupling:- The above analysis excludes spin-orbit coupling. Clearly for the heavier ions (both metal and ligand), the various degenerate states may be further split due to spin-orbit coupling. In general, chloride to metal LMCT presents a much simpler spectrum than does bromide or iodide. In cubic systems, the 'single' bands of chloride LMCT usually become two or three in the case of bromide, and especially iodide, because of the loss of degeneracy of orbital triplet states through spin-orbit coupling.

We follow Schatz and co-workers [380] to provide an example of one procedure for handling this kind of problem. Consider the d^2 $[MoCl_6]^{2-}$ ion. Electron promotion (L---t_{2g}) may occur from the ligand orbitals $(\pi + \sigma)t_{1u}$, πt_{2u} or $(\sigma + \pi)t_{1u}$ into the metal d shell to yield $Lt^5(t_{2g})^3$ type configurations.

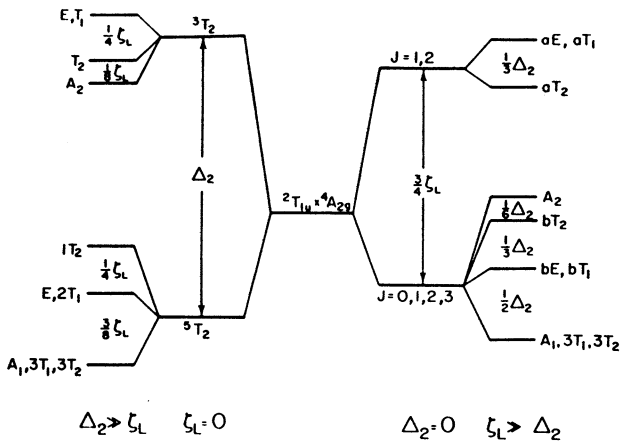


Fig.5.5 (a) Behaviour of the lowest charge transfer states in an octahedral d^2 ion under the influence of spin-orbit coupling and electrostatic interactions. ζ_L is the effective spin-orbit coupling constant for the ligands, including the effect of σ-π mixing. Δ_2 is defined as the electrostatic splitting between $^3T_{2u}$ and $^5T_{2u}$ [380].

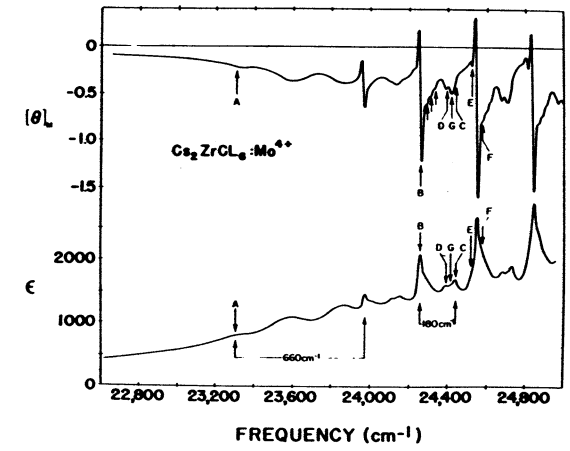


Fig.5.5 (b) High resolution absorption and MCD spectra of the lowest energy region of the spectrum of d^2 $[MoCl_6]^{2-}$. Line A is the first member of a broad progression. Lines B-F have been labelled at their first repeat (in a_{1g}) for clarity, while line G is noted at its first appearance [380].

The metal $(t_{2g})^3$ configuration spans ^4A_2g, (^2E_g, $^2T_{1g}$) and $^2T_{2g}$, in order of increasing energy, each of which may couple with the hole in the various ligand configurations. Thus coupling of the ground (metal) $^4A_{2g}$ state with the three ligand states generates three sets of states as follows:

$$
\begin{aligned}
&^4A_{2g} \times {}^2T_{1u} \quad (\pi + \sigma)t_{1u}^5 \qquad {}^3T_{2u} + {}^5T_{2u} \\
&^4A_{2g} \times {}^2T_{2u} \qquad (\pi)t_{1u}^5 \qquad {}^3T_{1u} + {}^5T_{1u} \\
&^4A_{2g} \times {}^2T_{1u} \quad (\sigma + \pi)t_{1u}^5 \qquad {}^3T_{2u} + {}^5T_{2u}
\end{aligned}
\tag{5.5}
$$

Coupling of the subsequent metal states (^2E_g, $^2T_{1g}$ and $^2T_{2g}$) will obviously generate many more states whose symmetries are also obtained by direct product analysis.

The lowest LMCT excitation from $(\pi + \sigma)t_{1u}$ to t_{2g} generates the configuration $(t_{2g})^3(^4A_{2g}))(t_{1u})^5(^2T_{1u}))$ and the states $^3T_{1u}$ + $^5T_{1u}$, arising from the metal $^4A_{2g}$ configuration (Fig.5.5a). The ground state is $^3T_{1g}$ and in the absence of spin-orbit coupling the only possible allowed electronic transition is $^3T_{2u}$---$^3T_{1g}$, since transition to $^5T_{2u}$ will be spin forbidden. If spin-orbit coupling is small (compared with the exchange terms which separate the spin triplet and quintet), the spin triplet and quintet states will not be greatly mixed but the degeneracy of the $^3T_{2u}$ state will be

lifted (Fig.5.5a, left-hand side). In this specific case the metal originating state being an orbital singlet, only the ligand spin-orbit coupling interaction is relevant. Also in this specific case, the $^3T_{1g}$ ground state splits under spin-orbit coupling with an E spinor state lying lowest. Two transitions are now allowed, $T_1, T_2(^3T_{2u})$---$E(^3T_{1g})$ separated by 1/4 of the ligand spin-orbit coupling coefficient (Fig.5.5a, left, upper).

The right-hand side of Fig.5.5a shows the situation if the spin-orbit coupling is large compared with the exchange energy. Now there will be extensive mixing of spin triplet and quintet and most of the components will lose their spin identity. Four transitions are now possible, to the states marked aT_1, aT_2, bT_1 and bT_2. Transitions to the remaining T states do not occur because they are pure quintet components which do not mix with the spin triplet.

Analysis of the $[MoCl_6]^{2-}$ spectrum by absorption and MCD spectroscopy, in fact reveals seven origins (Fig.5.5b, A-G) corresponding to these transitions but with additional vibrational coupling in some cases. Each of the labelled origins has a progression in the totally symmetric Mo-Cl vibration. Values of $_2$ and were shown to be 515 and 1174 cm^{-1} respectively by fitting the spectrum. The rather large spin-orbit coupling coefficient occurs because coupling of the π and σ t_{1u} states permits this coefficient to be as high as twice the chloride spin-orbit value (587 cm^{-1}). For a t_{2u} state, where such mixing does not occur, the value is limited to 587 cm^{-1} for a chloride complex.

5.3 Optical Electronegativities [418,420,424]

Comparison of the position of the lowest charge transfer band for a series of metal halide complexes led Jorgensen [70,275,280, 282,418-422,424] to note the regularity that the energies in the chloro, bromo and iodo complexes are related to that in the fluoro complex by:

$$F\ \nu_{CT} \qquad Cl\ \nu_{CT}\text{-}28{,}000\ cm^{-1}$$

$$Br\ \nu_{CT}\text{-}34{,}000 \qquad I\ \nu_{CT}\text{-}44{,}000\ cm^{-1} \qquad (5.6)$$

These variations are roughly proportional to the differences in the electronegativities of the ligands. The basic assumption of Jorgensen's treatment is that charge transfer transition energies may be directly related to the electronegativity difference between the donor and acceptor orbital. By suitable analysis of charge transfer spectra, the electronegativity of an orbital may be evaluated. Such a datum is called the 'Optical Electronegativity'. Its principal value lies in the ability to use such information to predict charge transfer energies and hence assign spectra.

The optical electronegativity of the fluoride ion is chosen to be 3.9 to relate it to the Pauling electronegativity scale.

Table 5.4

EVALUATION OF $[\langle S(S+1)\rangle - S(S+1)]$ AS A FUNCTION OF SPIN FOR d^n CONFIGURATIONS

n	S = 0	1/2	1	3/2	2	5/2
2 or 8	+4/3		-2/3			
3 or 7		+1		-2		
4 or 6	+2		0		-4	
5		+4/3		-5/3		-20/3

Then we may write:

$$\nu'_{CT} = 30{,}000[\chi_{opt}(X) - \chi_{opt}(M)]\ cm^{-1} \quad (LMCT) \qquad (5.7)$$

where $\chi_{opt}(X)$ and $\chi_{opt}(M)$ are the optical electronegativities of the donor and acceptor orbitals on the ligand and metal respectively. The constant 30,000 cm^{-1} related this scale to the Pauling scale, and the transition energy ν'_{CT} is corrected for interelectronic repulsion differences between ground and excited states (see below). For metal to ligand charge transfer, the quantity in brackets in expression (5.7) is reversed, i.e.:

$$\nu'_{CT} = 30{,}000[\chi_{opt}(M) - \chi_{opt}(X)]\ cm^{-1} \quad (MLCT) \qquad (5.8)$$

To provide a ready comparison of optical electronegativities for different metals and different stereochemistries, it is usual to calculate the optical electronegativity of the spherically perturbed shell. This is assumed to correspond with the optical electronegativity of the almost non-bonding t_{2g} orbitals (ligand field model) in an octahedral complex, or the almost non-bonding e orbitals in tetrahedral complexes.

It is necessary to deal with the changes in interelectronic repulsion energy, within the d shell, and between the d shell and the ligand electrons, as a consequence of the transition. The assumption is made that all J and K integrals involving electrons on different atoms are zero, or cancel out, so that the interaction between the d shell and the ligand is ignored. The change in interelectronic repulsion energy on the ligand will probably be approximately constant for all complexes of the ligand, and may therefore be incorporated into the optical electronegativity value for the ligand. This last assumption is not very good since the repulsion energy probably depends upon stereochemistry, and more

importantly, upon the number of ligands of a given type in the complex, i.e. ligand-ligand repulsion is important. At this time we choose to ignore this effect, but will recall it later when data are analysed.

The change in interelectronic repulsion within the d shell can be thought of in terms of the spin pairing energy (SPE). Considering the transition

$$L^n t_{2g}^{\ a} e_g^{\ b} \text{ ------ } L^{n-1} t_{2g}^{\ a+1} e_g^{\ b}$$

the transition energy will depend upon the difference in one electron energies of the t_{2g} and L orbitals, and the difference in spin pairing energy of the $t_{2g}^{\ a} e_g^{\ b}$ and $t_{2g}^{\ a+1} e_g^{\ b}$ configurations.

The spin pairing energy (SPE) may be represented by [420-424]:

$$[\langle S(S+1)\rangle - S(S+1)]D \qquad (5.9)$$

where $\langle S(S+1)\rangle$ is the average value of $S(S+1)$ for a given configuration 1^q, equal to:

$$\langle S(S+1)\rangle = [q(q+2)/4] - [(2l+2)q(q-1)/2(4l+1)] \qquad (5.10)$$

(l is the orbital quantum number, S is the spin quantum number, and q is the number of electrons). D is called the spin pairing energy parameter, and for d and p electrons is given by: [424]

$$\text{(p electrons)} \quad D = (15/4)\,F_2$$
$$\text{(d electrons)} \quad D = (7/6)[(5/2)B + C] = \underline{ca}\ 7B \qquad (5.11)$$

Consider as an example, the tetrahedral $[CoCl_4]^{2-}$ ion which has its lowest charge transfer band near 43,000 cm^{-1} assigned as $\pi(Cl)$----$Co(II)(t_2)$, the e metal orbital being full, i.e. $L^n e^4 t_2^{\ 3}$---$L^{n-1} e^4 t_2^{\ 4}$. The spin pairing energy of the d shell is obtained by inserting q = 7 and S = 3/2 into eqns. (5.9) and (5.10); thus SPE = -2D for the ground state. Values for the expression (5.9) are tabulated (Table 5.4).

The excited charge transfer configuration has q = 8 and S = 1; hence SPE = -(2/3)D. The net change in SPE is then + (4/3)D; thus:

$$\nu_{CT}(\text{observed}) = E(t_2) - E(L) + (4/3)D$$

and

$$\nu_{CT}' = E(t_2) - E(L) = \nu_{CT}(\text{observed}) - (4/3)D$$

with B = 670 cm^{-1}, D = 4700 cm^{-1} then:

$$\nu_{CT}' = {}_{CT}(\text{observed}) - 6300 = 36{,}700\ cm^{-1}$$

Using eqn.(5.7) with $\chi_{opt}(Cl) = 3.0$, then $\chi_{opt}(Co(II)t_2) = 1.8$. In view of the approximations involved, optical electronegativity values should not be quoted to more than one decimal place. The value of 1.8 applies to the anti-bonding t_2 sub-shell.

Table 5.5

OPTICAL ELECTRONEGATIVITY DATA FOR METAL IONS AND LIGANDS

Ion	Octahedral	Tetrahedral	Ion	Octahedral	Tetrahedral
Ti(IV)	2.05	1.8	V(III)	1.9	2.1
Cr(III)	1.8-1.9		V(IV)	2.6	
Cr(IV)	2.65				
Mn(IV)	2.7-3.0		Mn(III)	2.0L	
Fe(III)	2.1L 2.5H	2.4-2.5	Fe(II)		1.8
Co(III)	2.3L		Co(II)		1.8 - 1.9
Ni(II)		2.0-2.1	Ni(III)	3.05	
Ni(IV)	3.4		Cu(II)		2.3-2.4
Cu(III)	3.5		Zn(II)		1.2
Hg(II)	1.5		Bi(III)	1.7	

	Octahedral	Square		Octahedral	Square
Nb(V)	1.85		Ta(V)	1.80[a], 1.62[b]	
Zr(IV)	1.6		Nb(V)	1.85	
Mo(III)	1.7		Mo(IV)	1.95	
Mo(V)	2.0		Mo(VI)	2.1	
W(V)	1.95		W(VI)	2.0	
Re(IV)	2.1		Re(VI)	2.0-2.1	
Tc(IV)	2.25				
Ru(IV)	2.45		Ru(III)	2.0-2.1L	
Rh(IV)	2.65		Rh(III)	2.3-2.4L	
Ir(III)	2.25		Ir(IV)	2.3-2.4L	
Os(III)	1.95		Os(IV)	2.2	
Os(VI)	2.6				
Pt(II)		2.3L	Pt(IV)	2.6-2.7L	
Pt(V)	3.0		Pt(VI)	3.2L	
Pd(IV)	2.75		Pd(II)		2.2-2.4L
			Au(III)		2.9L (2.4)[f]
UO_2(VI)	1.8		Ni(II)		2.2L
U(VI)	2.4				
Np(VI)	2.6				
Pu(VI)	2.85				

Ligands	π	σ		π	σ
F^-	3.9	(4.4)	Cl^-	3.0	3.4
Br^-	2.8	3.3	I^-	2.5	3.0
H_2O	3.5		CN^-	2.8	
dtp^-	2.7		$dsep^-$	2.6	
PyNO	0.5 (π^*)		acac	2.7	
O_2^-	(3.2)		SO_4^{2-}	3.2	
N_3^-	2.8		NCO$^-$	3.0	
NH_3		3.3	RNH_2		3.2

cont

Table 5.5 cont----------

Ligands	π	σ		π	σ
ROH	3.1				
R_3P		(2.6)	S^{2-}	2.5	
R_2S	2.9		R_3As		2.5
S_2PPh_2	2.6		SCN^-	2.6	
NCS^-	2.9		$SeCN^-$	2.85	
$NCSe^-$	2.8				
$R(1)S^-$	2.79		$R(2)S^-$	2.87	
$R(3)S^-$	2.95		Py2SH	2.4	

$R(1)S^- = SCH_2CH(NH_2)CO_2Et$; $R(2)S^- = SC_6H_4NO_2$; $R(3)S^- = SC_6F_5$ [302]. L = low spin state H = high spin state. Sources: [70,195,254,420, 424,389,608]. a) [70], b) [254].

To obtain the corresponding value for the 'non-bonding' e shell, ν_{CT}' must be further decreased by 10Dq (3300 cm^{-1}); hence ν_{CT}' = 33,400 cm^{-1} and χ_{opt}(Co(II)e) = 1.9 (tetrahedral environment).

Table 5.5 summarises the OE values for various metal ions in different stereochemistries and spin states. Table 5.5 also includes ligand OE values.

There are many subtleties associated with discussion of optical electronegativities and care should be taken not to interpret OE values too rigidly. Thus in oxide derivatives, for example, the oxide ion will perturb the π metal orbitals to an extent which depends upon the number of oxide ligands. This effect may be about 0.15 OE units per oxide ion. Hence in mixed systems, such as CrO_2Cl_2, $[CrO_3Cl]^-$, the Cl---Cr(VI) transition energies depend upon how many oxide ligands are present, i.e. the OE value appears to change. Jorgensen and co-workers [243] discuss this in considerable detail. The sequence of lowest energy LMCT bands in $[MO_4]^{n-}$ ions follows [243] (see section 5.12 (i)):

Mn(VII) (18,700) < Ru(VIII) (26,000) = Cr(VI) (26,800) < Os(VIII) (33,800) = Tc(VII) (34,600) < V(V) (36,900) < Mo(VI) (43,200) = Re(VII) (43,700) < W(VI) (50,300)

(5.12)

If OE values remained essentially constant, then the corresponding series of $[MS_4]^{n-}$ ions should parallel (5.12) but be displaced a constant amount to lower energy. This is clearly not the case, viz [240,243]:

VO_4^{3-}	36,900	VS_4^{3-}	18,600	difference 18,300
MoO_4^{2-}	43,200	MoS_4^{2-}	21,400	21,800
ReO_4^-	43,700	ReS_4^-	19,800	23,900
WO_4^{2-}	50,300	WS_4^{2-}	25,500	24,800

(5.13)

wherein it has been noted that these differences correlate roughly linearly with the oxide LMCT energies [243]. The explanation probably lies in the considerable covalency in these systems which modifies both metal and ligand orbital energies to varying extents from one species to another.

The spectra of mixed oxide-sulphide systems are discussed in section 5.11. Note in general that for a system such as MX_xY_y independent transitions from X and Y to M (or vice versa) may be expected if the relevant one-electron energies (diagonal energies) of X and Y differ very significantly. If they do not differ significantly then, assuming appropriate symmetry axes, there will be extensive mixing between the orbitals of X and Y, and the transitions will occur at energies roughly the average of those expected for X and Y, i.e. an average environment rule applied to charge transfer transitions. See examples of mixed halides in section 5.5, and sulphido-selenides in refs.[241,243].

5.3 i) Charge Transfer Transitions and 10Dq

The possibility of observing charge transfer transitions to both the t_{2g} and e_g d levels permits the derivation of 10Dq from charge transfer data. However this is not quite so straightforward as has been presumed by some authors.

It is straightforward to identify the energies of LMCT transitions, in octahedral complexes, to t_{2g} and e_g levels, using the theory described immediately above, thus:

LMCT Octahedral complexes

d^n	Transition to t_{2g}	Transition to e_g
0	30(X(X) - X(M))	30(X(X) - X(M)) + 10Dq
1	30(X(X) - X(M)) - (2/3)D	30(X(X) - X(M)) + 10Dq - (2/3)D
2	30(X(X) - X(M)) - (4/3)D	30(X(X) - X(M)) + 10Dq - (4/3)D
3	30(X(X) - X(M)) + 2D	30(X(X) - X(M)) + 10Dq - 2D
4HS	30(X(X) - X(M)) + (7/3)D	30(X(X) - X(M)) + 10Dq - (8/3)D
4LS	30(X(X) - X(M)) + (4/3)D	30(X(X) - X(M)) + 10Dq - (5/3)D
5HS	30(X(X) - X(M)) + (8/3)D	30(X(X) - X(M)) + 10Dq + (8/3)D
5LS	30(X(X) - X(M)) + (2/3)D	30(X(X) - X(M)) + 10Dq - (4/3)D
6HS	30(X(X) - X(M)) + 2D	30(X(X) - X(M)) + 10Dq + 2D
6LS		30(X(X) - X(M)) + 10Dq - D
7HS	30(X(X) - X(M)) + (4/3)D	30(X(X) - X(M)) + 10Dq + (4/3)D
7LS		30(X(X) - X(M)) + 10Dq - (5/3)D
8		30(X(X) - X(M)) + 10Dq + (2/3)D
9		30(X(X) - X(M)) + 10Dq

(5.14)

where the '30' modifier is in kiloKayser (1000 cm^{-1}), and HS and LS are high spin and low spin respectively. These equations provide the energy of the lowest LMCT transition to t_{2g} or e_g in terms of the optical electronegativities of the ligand and metal, and the spin pairing energy D as described above.

The following facts should be noted:

i) For the configurations d^n, n = 0,1,2,5HS, 6HS and 7HS the energy difference between the two LMCT transitions is 10Dq (but see (iii) below).
ii) For the configurations d^n, n = 3,4LS,4HS,and 5LS, a value for 10Dq can be obtained from the energy difference between the two LMCT transitions corrected for the difference in interelectronic repulsion energy.
iii) In some d^n configurations, the inclusion of 10Dq in (5.14) is an approximation. Thus in those situations where more than one LMCT state has the same symmetry, mixing of LMCT states will modify the above equations.
iv) It is not always the case that the transition to t_{2g} lies, in energy, below that to e_g. Thus for d^3, the energy difference LMCT(e_g - t_{2g}) = 10Dq - 4D. If 10Dq > 4D (ca. 14,000 - 24,000 cm^{-1} depending upon B), then transition occurs to t_{2g} before e_g. If 10Dq < 4D, the transition occurs to e_g before t_{2g}. For example, in the $[Cr(NCS)_6]^{3-}$ ion 10Dq is roughly equal to 4D and both transitions are predicted to lie at close to the same energy (see Table 5.6).
v) As discussed in section 5.2, a transition Lπ---t_2, for example, may encompass several excited states separated by differing interelectronic repulsion energy. The analysis reported in Eqn.5.14 may, in some examples, only provide an approximate value for 10Dq. The analysis in section 5.2 should be used if sufficient data are available for a complete calculation; this is especially true of the heavier transition metal ions.

Obviously (5.12) also provides the limits for defining high and low spin behaviour in terms of 10Dq and D. Thus for d^4 and d^5, the spin crossover occurs at 10Dq = 4D, and for d^6, 10Dq = 3D. Low spin complexes, by definition, will exhibit transitions to t_{2g} below those to e_g. Complexes close to the spin crossover should show these LMCT transitions at similar energies.

For MLCT transitions in octahedral complexes, a similar analysis yields:

MLCT Octahedral complexes

d^n	t_{2g}---π^*	e_g---π^*
1	30(χ(M) - χ(X))	
2	30(χ(M) - χ(X)) + (2/3)D	
3	30(χ(M) - χ(X)) + (4/3)D	
4HS	30(χ(M) - χ(X)) + 2D	30(χ(M) - χ(X)) - 10Dq + 2D
4LS	30(χ(M) - χ(X)) - 2D	
5HS	30(χ(M) - χ(X)) + (8/3)D	30(χ(M) - χ(X)) - 10Dq + (8/3)D
5LS	30(χ(M) - χ(X)) - (4/3)D	
6HS	30(χ(M) - χ(X)) - (8/3)D	30(χ(M) - χ(X)) - 10Dq + (7/3)D
6LS	30(χ(M) - χ(X)) - (2/3)D	
7HS	30(χ(M) - χ(X)) - 2D	30(χ(M) - χ(X)) - 10Dq + 2D
7LS	30(χ(M) - χ(X)) - D	30(χ(M) - χ(X)) - 10Dq + D
8	30(χ(M) - χ(X)) - (4/3)D	30(χ(M) - χ(X)) - 10Dq + (5/3)D
9	30(χ(M) - χ(X)) - (2/3)D	30(χ(M) - χ(X)) - 10Dq - (2/3)D
10	30(χ(M) - χ(X))	30(χ(M) - χ(X)) - 10Dq

(5.15)

where, as was observed in the Introduction (section 5.1), there is also the possibility that transitions from t_{2g} to π^* will occur at an energy below that from e_g, related, as described above, to the relative magnitudes of 10dq and D. Indeed high spin d^6 and d^7 ions will always show MLCT transitions from t_{2g} below transition from e_g, because the fact that they are high spin defines the relationship between 10Dq and D.

Finally the analysis can be extended to LMCT transitions in tetrahedral complexes:

LMCT Tetrahedral complexes

d^n	Transition to t_2	Transition to e
0	30(χ(X) - χ(M))	30(χ(X) - χ(M)) + 10Dq
1	30(χ(X) - χ(M)) - (2/3)D	30(χ(X) - χ(M)) + 10Dq - (2/3)D
2	30(χ(X) - χ(M)) - (4/3)D	30(χ(X) - χ(M)) + 10Dq + (5/3)D
3	30(χ(X) - χ(M)) - 2D	30(χ(X) - χ(M)) + 10Dq + 2D
4	30(χ(X) - χ(M)) - (8/3)D	30(χ(X) - χ(M)) + 10Dq + (7/3)D
5	30(χ(X) - χ(M)) + (8/3)D	30(χ(X) - χ(M)) + 10Dq + (8/3)D
6	30(χ(X) - χ(M)) + 2D	30(χ(X) - χ(M)) + 10Dq + 2D
7	30(χ(X) - χ(M)) + (4/3)D	
8	30(χ(X) - χ(M)) + (2/3)D	
9	30(χ(X) - χ(M))	

(5.16)

where similar comments to those above apply.

It is also valid to question whether the value of 10Dq determined in this way, should agree with the value obtained from the d-d spectrum of the same species. The crystal field spectrum provides a measure of 10Dq for the M(n) species bound by L. The excited LMCT state, at equilibrium, represents an energy for the species M(n-1) bound to L(+). In the Franck-Condon excitation, the energy corresponds to a vibronically excited version of M(n-1)/L(+) in which the metal-ligand distance is the same as that of the ground state. Experience shows that 10Dq(d-d) and 10Dq(CT) are, in fact, fairly similar; however it is clear from this analysis that they do not have to be identical.

5.4 Charge Transfer Spectra of Halide Complexes

5.4 i) Octahedral MX_6 Species

Combination of the s and p orbitals on each of the six halide ions yields π orbitals transforming as $t_{1g} + t_{2g} + t_{1u} + t_{2u}$, and σ orbitals transforming as $a_{1g} + e_g + t_{1u}$. The relative energies of these transitions have been the subject of some debate [70].

McCaffery and Schatz [204], in an early important paper, reported the MCD spectra of the $[IrCl_6]^{2-}$ ion (low spin d^5) doped into $(CH_3NH_3)_2SnCl_6$. This work defined, via the sign of C/D, that the $^2T_{1u}$ LMCT state lies below $^2T_{2u}$.

The energy levels, on this basis, fall in the sequence:

$$\pi 1t_{1g} > (\pi + \sigma)2t_{1u} > \pi 1t_{2u} > \pi 1t_{2g} > \sigma 1e_g > (\sigma + \pi)1t_{1u} > \sigma 1a_{1g} \quad (5.17)$$

where it is recognised that the π and σ t_{1u} orbitals will be extensively mixed (Fig.5.6), and where the order, $(\pi+\sigma)$ and $(\sigma+\pi)$ places the dominant contribution first in the bracket.

Following the ideas developed briefly in sections 5.1 and 5.2, excitation from these various levels into $2t_{2g}$ and into $2e_g$ will generate one or more charge transfer states. As indicated in Table 5.6, it has proven possible to identify transitions to dt_{2g} from $\pi 1t_{1g}$ (weak, Laporte forbidden), from $(\pi+\sigma)2t_{1u}$ (usually fairly narrow), from $\pi 1t_{2u}$, and from $\pi 1t_{2g}$ (weak, forbidden). With low field ligands such as bromide and iodide, transitions from $(\pi+\sigma)2t_{1u}$ to $2e_g$ are also identifiable. Transitions from $\pi 1t_{2g}$ (forbidden) or from $(\sigma+\pi)1t_{1u}$ (expected to be very weak) to $2t_{2g}$ have rarely been identified.

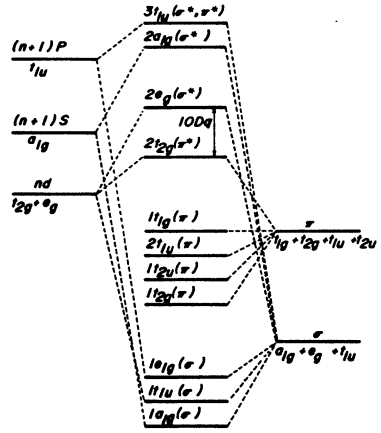


Fig.5.6 Molecular orbital diagram suitable for a six coordinate hexahalide complex.

Identification was initially achieved by simple MO calculations and by comparison of a wide variety of metal complexes with various halides. Recall that the chlorides will give the simplest spectra since the orbital triplet states are not significantly split by spin-orbit coupling. Regions of absorption can then often be associated with individual transitions as indicated in section 5.1.

Table 5.6

CHARGE TRANSFER SPECTRA OF SIX COORDINATE HALIDE AND pseudo-HALIDE COMPLEXES[a]

Complex	d^n	$\pi 1t_{1g}$ --$2t_{2g}$	$(\pi+\sigma)2t_{1u}$ --$2t_{2g}$	$\pi 1t_{2u}$ --$2t_{2g}$	$(\pi+\sigma)2t_{1u}$ --$2e_g$	$(\sigma+\pi)1t_{1u}$ --$2e_g$	Ref
Hexahalides							
$ZrCl_6^{2-}$	0		41,700				271
$NbCl_6^-$	0	28,900	(31,800)	34,500	41,600		254
$NbBr_6^-$	0	(21,500)	23,800	29,000	37,600	45,800	415
			26,700				
MoF_6	0		54,000				274
WCl_6	0	(20,300)	25,800	34,000	44,000		
		(22,400)		(37,400)			414
$TaCl_6^-$	0	34,800	38,000	41,500	49,500		254
$TaBr_6^-$	0	(22,200)	(24,000)	29,400	34,300	44,000	
			25,600	(32,300)	37,400		414
WCl_6	0	(22,400)	26,400	29,900			275
$MoCl_6^-$	1	26,400	30,200	36,300	45,200		414
WCl_6^-	1	25,400	29,400	36,600	45,200		414
WBr_6^-	1	(18,100)	27,000br	31,200	34,000	45,100	414
OsF_6	2	(31,000)	35,700	40,800			276
$MoCl_6^{2-}$	2		24,500	28,300			
			32,800[b]	36,000[b]			380
$ReCl_6^{2-}$	3	(28,000)	(31,600)	36,500			386
			(34,100)	(39,100)			277
IrF_6	3	(31,000)	35,500	42,000			276
OsI_6^{2-}	4		12,300	etc			111
$MoCl_6^{3-}$	3		45,300				422
$OsCl_6^{2-}$	4	23,900	27,000	(29,300)	47,900		
				30,000			279
$OsCl_6^{2-}$	4		26,000	29,000			384
PtF_6	4	20,000	25,000	32,600			276
$RuCl_6^{2-}$	4	19,400	20,700	24,300			387
$RuBr_6^{2-}$	4	14,200	15,800	19,600			387
$RuCl_6^{3-}$	5	(25,600)	28,600	32,400	43,600		279
$RhCl_6^{2-}$	5	(10,000)	12,000	15,600			
				16,700			280
$OsCl_6^{3-}$	5	(32,600)	35,400	38,200			
				39,400			279
OsI_6^{3-}	5		19,100				111
IrF_6^{2-}	5		47,000				281
$IrCl_6^{2-}$	5	(18,000)	19,700	23,500			396
$O_2^+ PtF_6^-$	5		28,600				283
$RhCl_6^{3-}$	6				41,000ca	51,000	284
$RhBr_6^{3-}$	6				28,600	38,600	284
$IrCl_6^{3-}$	6				48,500		284
$IrBr_6^{3-}$	6				37,000		284

cont

Table 5.6 cont

Complex	d^n	$\pi 1t_{1g}$ --$2t_{2g}$	$(\pi+\sigma)2t_{1u}$ --$2t_{2g}$	$\pi 1t_{2u}$ --$2t_{2g}$	$(\pi+\sigma)2t_{1u}$ --$2e_g$	$(\sigma+\pi)1t_{1u}$ --$2e_g$	Ref
$PdCl_6^{2-}$	6				29,400	41,000	284
$PtCl_6^{2-}$	6				37,000	49,200	285
$PtBr_6^{2-}$	6				30,580, 31,540	44,400	285
PtI_6^{2-}	6				20,250	etc	422
Hexa-pseudohalides							
$Ru(N_3)_6^{3-}$	5		22,300		35,700		416
$Ru(NCS)_6^{3-}$	5		18,200				389
$Os(NCS)_6^{3-}$	5		22,000				389
$Ir(N_3)_6^{3-}$	6				38,300		416
$Rh(N_3)_6^{3-}$	6				32,600	42,000[c]	416
$Pt(N_3)_6^{2-}$	6				32,600		416
$Rh(SCN)_6^{3-}$	6	(30,300)[d]			34,800		389
$Ir(SCN)_6^{3-}$	6				(41,000)		389
$Pt(SCN)_6^{3-}$	6				34,600		389

Complex	d^n	$\pi X \rightarrow M$ LMCT[e]			Ref
Mixed Halides					
$Co(NH_3)_5Cl^{2+}$	6	(37,100)(600)	43,900(20,000)	σX→Co	435
$Co(NH_3)_5Br^{2+}$	6	32,000(850)	39,500(18,500)	σX→Co	435
$Co(NH_3)_5I^{2+}$	6	26,100(2700)	34,900(16,500)	σX→Co	435
$c\text{-}Co(en)_2Cl_2^{+}$	6	(32,000)(600)	42,000(19,000)	σX→Co	437
$t\text{-}Co(en)_2Cl_2^{+}$	6	(32,500)(1300)	40,300(27,000)	σX→Co	437
$t\text{-}Co(en)_2Cl(NCS)^{+}$	6	(37,850)	45,800(26,800)	σX→Co	511
$t\text{-}Co(en)_2Br_2^{+}$	6	28,600(1950)	34,600(22,300)	σX→Co	511
$t\text{-}Co(CRH)Br_2^{2+}$	6	26,500(2860)			602
$Rh(NH_3)_5I^{2+}$	6	36,000(3200)	44,300(20,000)	σX→Rh	439
$Rh(NH_3)_5Br^{2+}$	6	40,600	50,000 σX→Rh		439
$Ru(H_2O)_5Cl^{2+}$	5	31,800			422
$Ru(NH_3)_5Cl^{2+}$	5	30,500(1930)			432,438
$Ru(NH_3)_5Br^{2+}$	5	25,100(1920)			432,438
$Ru(NH_3)_5I^{2+}$	5	18,500(2050)			432
$Os(NH_3)Cl_5^{2-}$	5	37,170(4500)			346
$Os(NH_3)_5Cl^{2+}$	5	37,450(2000)			357,432
$Os(NH_3)_5Br^{2+}$	5	31,850(1840)			357,432
$Os(NH_3)_5I^{2+}$	5	24,570(1970)			357,432
$c\text{-}Ru(NH_3)_4Cl_2^{+}$	5	28,400(1640)	32,250(1380)		432
$c\text{-}Ru(NH_3)_4Br_2^{+}$	5	22,600(1670)	27,000(1370)		432
$t\text{-}Ru(NH_3)_4Cl_2^{+}$	5	30,200(4750)			432
$t\text{-}Ru(NH_3)_4Br_2^{+}$	5	25,000(4950)			432
$t\text{-}Os(NH_3)_4Cl_2^{+}$	5	36,900(4810)			357
$t\text{-}Os(NH_3)_4Br_2^{+}$	5	31,500(4690)			357
$t\text{-}Os(NH_3)_4I_2^{+}$	5	23,900(4920)			357
$c\text{-}Os(NH_3)_4Cl_2^{+}$	5	35,200(1680)	39,220(2030)		357,432
$c\text{-}Os(NH_3)_4Br_2^{+}$	5	29,070(1620)	33,560(1600)		357,432

cont

Table 5.6 cont

Complex	d^n	$\pi X \rightarrow M$ LMCT[e]			Ref
$c\text{-}Os(NH_3)_4I_2^{+}$	5	21,280(2070)	26,460(1680)		357,432
$m\text{-}Os(NH_3)_3Cl_3$	5	37,310(4600)			346
$m\text{-}Os(NH_3)_3Br_3$	5	31,450(4200)			346
$m\text{-}Os(NH_3)_3I_3$	5	23,580(3220)			346
$t\text{-}Os(NH_3)_3I_2(N_2)^{+}$	5	17,300			354
$Os(NH_3)Cl_5^{-}$	4	28,410(8350)			346
$c\text{-}Os(NH_3)_4I_2^{2+}$	4	10,960(2480)	13,740(3220)	18,350(1560)	346
$t\text{-}Os(NH_3)_4I_2^{2+}$	4	15,820(11,800)			346
$t\text{-}Os(NH_3)_4Cl_2^{2+}$	4	28,010(11,600)			346
$t\text{-}Os(NH_3)_4Br_2^{2+}$	4	23,570(13,100)			346
$c\text{-}Os(NH_3)_4Cl_2^{2+}$	4	25,000	28,570		346
$c\text{-}Os(NH_3)_4Br_2^{2+}$	4	19,610	24,100		346
$m\text{-}Os(NH_3)_3I_3^{+}$	4	15,670(8760)			346
$m\text{-}Os(NH_3)_3Cl_3^{+}$	4	29,240(9400)			346
$m\text{-}Os(NH_3)_3Br_3^{+}$	4	22,520(7850)			346
$t\text{-}ReCl_4(PMP)_2$	3	17,800			434
$c\text{-}ReCl_4(DPP)_2$	3	21,300			434

a) For third row transition metal ions, spin-orbit coupling is so large that extensive mixing of the various states occurs and assignment of specific bands to specific transitions as indicated above should be treated with caution. Recall also that some excited metal configurations cause quite a wide separation in excited state energies with overlap in energy ranges for the several excitations, e.g. marked (b) above. Many of the transition are very broad with vibrational structure so that energies indicated here are approximate band centres not 0-0 lines. c) Assigned as $1t_{2u} \rightarrow 2e_g$ d) Tentatively assigned as $1t_{1g} \rightarrow 2e_g$. e) For d^n, $n < 6$, transitions are $\pi X \rightarrow 2t_{2g}$, for d^6, transitions are $\pi X \rightarrow 2e_g$, or components thereof. Refer to [422] Table VIII, for an extensive listing of early halide data.

Moreover comparison of d^n, n = 1 to 5, with low spin n = 6-9, reveals those transitions to the $2t_{2g}$ level which must be absent when the $2t_{2g}$ level is full. The energy of the lowest lying LMCT band can also be calculated using the arguments of the optical electronegativity principle as previously discussed (section 5.3).

Recent studies have extensively used MCD spectra to provide definitive identification of the symmetries of the relevant states.

Many groups have studied d^0 (and d^1) species (see Table 5.6) since they are apparently so simple. The hexachlorides often exhibit a weak absorption followed by two much stronger bands which are associated with transitions from $\pi 1t_{1g}$ (very weak), $(\pi+\sigma)2t_{1u}$ and $\pi 1t_{2u}$. Bromo and iodo complexes can then be analysed by detailed comparison with the chloro analogs.

Some care must be used in studying the spectra of halides published in the earlier literature (prior to about 1975). This is

especially true for the elements to the left of the transition series. Most of these complexes are sensitive to water and many authors, not fully recognising this sensitivity, published spectra of mixed or hydrolysed species or failed to recognise the true nature of the species for which they reported data (e.g. see comparative data in [414]). There was also considerable guesswork with respect to assignments, and these should also be treated with caution. Such assignments also had a knack of changing from one publication to another, even by the same authors.

While there is an extensive literature concerning octahalides of the second and third row transition metal ions, little is available for octahedral first row ions. This is due to the comparative scarcity of hexahalo anions of first row ions, in part to the smaller electronegativity of these ions pushing the energy of the CT transitions of interest, into the far uv where they may be obscured by atomic halide transitions. Such data can be obtained by doping metal ions into halide lattices such as lithium or cadmium halides (Table 5.7). Three regions of absorption are generally seen, and these are associated with $\pi L \text{---} 2t_{2g}$, $\pi L \text{---} 2e_g$ and $\sigma L \text{---} 2e_g$ (recall $\sigma L \text{---} 2t_{2g}$ should be very weak). The data shown in Table 5.7 probably under-estimate the complexity of these spectra which should be analysed in more detail.

To provide further understanding, some examples of the analyses of some individual hexahalide species will now be introduced.

5.4 ii) The $[FeCl_6]^{3-}$ Ion

Since Fe(III) is a moderately oxidising ion, many complexes exhibit L---Fe(III) LMCT. In general, both $L \text{---} 2t_{2g}$ and $L \text{---} 2e_g$ transitions may be expected.

Gudel, Schatz and co-workers [7] have reported a detailed study of the $[FeCl_6]^{3-}$ ion using both absorption and MCD, in doped crystals of Cs_2NaYCl_6. Seven transitions were clearly observed between 20,000 - 46,000 cm^{-1} and assigned to charge transfer (Fig.5.7 (a)).

The CT states arise from the three open-shell configurations $(t_{xu})^5(2t_{2g})^{3+n}(2e_g)^{3-n}$ where x = 1,2 and n = 0,1, and where the ligand group molecular orbital may be $2t_{1u}(\pi + \sigma)$, $1t_{2u}(\pi)$ or $1t_{1u}(\sigma + \pi)$. The ground state is $^6A_{1g}$ and only transitions to $^6T_{1u}$ are electric dipole allowed. These workers take the reasonable view to consider the states as arising from a coupling of an effective Fe(II) $(2t_{2g})^{3+n}(2e_g)^{3+n}$ core with the $(Cl_6)^{5-}$ ligand core. The latter is a spin doublet. Only spin quintets of the former can couple with this doublet to yield the required spin sextet. There are only two such states, $^5T_{2g}$ from $2t_{2g}{}^4(^3T_{1g})2e_g{}^2(^3A_{2g})$ and, ^5E_g from $2t_{2g}{}^3(^4A_{2g})2e_g{}^3(^2E_g)$ lying 10 Dq to higher energy. There are therefore six possible transitions from the three ligand orbital triplets to each of these states. These are identified in Fig.5.7 (a) and Table 5.7, as supported by MCD and theoretical analysis, including a full spin-orbit diagonalization of the spin quartets and sextets, using tensor methods. Values for the spin orbit coupling coefficients for iron (320 cm^{-1}, 20 percent reduction from free ion) and for chloride (580 cm^{-1}, usual literature value) were obtained. It is pertinent to note that the lowest energy CT transition is a forbidden $1t_{1g} \text{---} 2t_{2g}$ transition.

The energy interval between bands 2,4, bands 3,5 and bands 6,7 are almost identical and are associated with the energy difference between the metal $2t_{2g}$ and metal $2e_g$ orbital. Following previous discussion this energy interval is estimated to be 10 Dq. The band 2,3 interval is the interval between ligand $2t_{1u}(\pi + \sigma)$ and ligand $1t_{2u}$. A value here of about 3000 cm^{-1} is typical of hexahalide complexes (also see section 5.14).

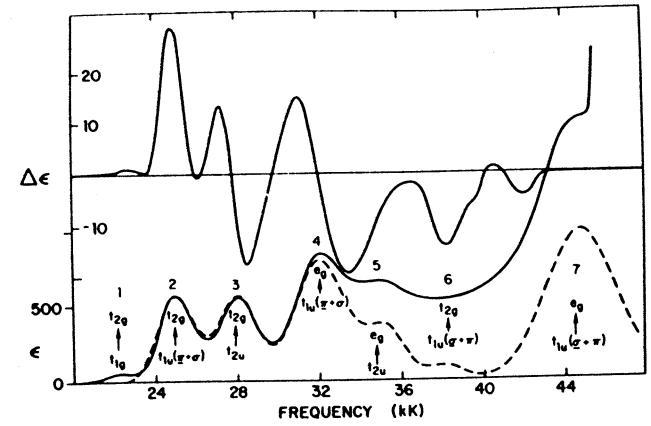


Fig.5.7 (a) Absorption spectrum (at 8 K) and MCD spectrum (at 4.2 K) for $[FeCl_6]^{3-}$ doped into Cs_2NaYCl_6 [7].

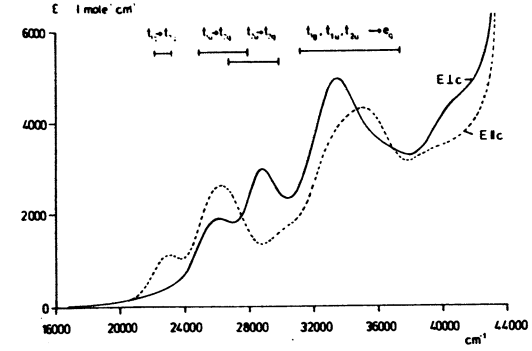


Fig.5.7 (b) Polarised single crystal absorption spectrum (at 12 K) of $[(NH_4)_2In_{1-x}Fe_xCl_5(H_2O)]$ in the charge transfer region [25].

Table 5.7

CHARGE TRANSFER SPECTRA OF FIRST ROW SIX COORDINATE TRANSITION METAL HALIDE AND pseudo-HALIDE DERIVATIVES

Complex	π---$2t_{2g}$	π---$2e_g$	σ---$2e_g$	Ref
$K_2Ti(IV)Cl_6$	30,000	43,800		
	23,400 ($\pi 1t_{1g}$)			
	34,500 ($\pi 1t_{2u}$)			271
$Cr(NCS)_6^{3-}$	31,700[a]	31,700[a]		389
Mn(II)/LiBr	48,700			150
Fe(II)/LiCl	46,600	52,700		150
$Fe(II)/CdCl_2$	42,800			151
$Fe(II)/CdBr_2$	36,700			
	35,900w			151
Fe(II)/LiBr	37,400	42,400	49,800	
		45,600		150
Co(II)/LiCl	39,400	45,000	52,500	150
$Co(II)/CdCl_2$	36,900			151
Co(II)/LiBr	30,600	35,300	42,700	
	32,800 s.o.	39,100 s.o.	45,700 s.o.	150
$Co(II)/CdBr_2$	28,800	ca.34,000		151
$Ni(IV)F_6^{2-}$		32,800	ca 51,000	272
Ni(II)/LiCl		39,600	48,000	150
$Ni(II)/CdCl_2$		37,700		151
Cu(II)/LiCl		28,200	36,200	150
$Ni(II)/CdBr_2$		28,900	36,200	
		(30,000)		151
Ni(II)/LiBr		30,300	38,000	
		(33,900)		150

$Cs_2NaY(Fe)Cl_6$	22,400(73)	$1t_{1g}$---$2t_{2g}$	Cl LMCT	
	24,900(619)	$2t_{1u}(\pi + \sigma)$---$2t_{2g}$		
	27,900(579)	$1t_{2u}$---$2t_{2g}$		
	32,000(845)	$2t_{1u}(\pi + \sigma)$---$2e_g$		
	35,000(644)	$1t_{2u}$---$2e_g$		
	38,100(496)	$1t_{1u}(\sigma + \pi)$---$2t_{2g}$		
	44,800(1740)	$1t_{1u}(\sigma + \pi)$---$2e_g$		7
$(NH_4)_2In(Fe)Cl_5(H_2O)$	23,040(800)[b]	$e(t_1)$---$e(t_2)$	Cl LMCT	
	26,280(2600)	$2t_{1u}$---$2t_{2g}$	Cl LMCT	25
	....			

a) Both these transitions are calculated to occur at approximately the same energy (see section 5.2).

Although not strictly octahedral, it is convenient to discuss, here, the $[FeCl_5(H_2O)]^{2-}$ ion, of C_{4v} symmetry, also studied by Gudel [25]. The sextet-quartet d-d spectra show clear dichroism indicative of splitting of up to 2000 cm^{-1}, in the orbital triplet transitions, due to the lower symmetry (Fig.5.7 (b)). The CT spectra are also clearly dichroic. Given that the optical electronegativity of water is about 0.5 unit more than for chloride, the O---Fe(III) CT should lie at least 15,000 cm^{-1} to the blue of the Cl---Fe(III). Thus the shoulder near 40,000 cm^{-1} may be water---Fe(III) CT but the lower energy strong bands are all Cl---Fe(III) LMCT. The bands below 32,000 cm^{-1} are appreciably narrower than the broad band at 34,000 cm^{-1}. It is therefore reasonable to associate the former with components of the $2t_{1u}(Cl)$, $1t_{2u}(Cl)$---$2t_{2g}(Fe(III))$ and the broad band to corresponding transitions to $2e_g(Fe(III))$. The lowest energy band, at 23,040(800) cm^{-1} is totally polarised along the Fe-O bond axis. This transition is associated with the $1t_{1g}(Cl)$---$2t_{2g}$ transition, observed though forbidden in the $[FeCl_6]^{3-}$ ion, but which would become allowed in C_{4v} symmetry. The representation t_{1g} spans $a_2 + e$, and t_{2g} spans $b_2 + e$. Only the transition $e(1t_{1g})$---$e(2t_{2g})$ is likely to have significant intensity, because of overlap considerations, and it is polarised along Fe-O (z). Thus the 23,040 cm^{-1} band is so assigned (Table 5.6).

5.4 iii) The Cs_2ZrCl_6/Mo(IV) Doped System

This is an interesting system which shows not only the theory necessary to understand the transitions involved, but also provides a lesson in being exceptionally careful to determine the true nature of the species under investigation. Earlier workers assumed, in error, this system to contain the d^1 $[MoCl_6]^-$ ion, but in the later work, the MCD data were shown to be consistent with a mixture of the desired $[MoCl_6]^{2-}$ ion plus impurity d^1 $[MoOCl_5]^{2-}$ ion. Thus the observed spectra are a superimposition of the spectra of the two species.

The low temperature MCD spectra recorded between 5 K and 12 K are readily separated into two components, one which is inversely dependent upon temperature, and one which is independent of temperature. The former is associated with the impurity $[MoOCl_5]^-$ ion, and the latter to $[MoCl_6]^{2-}$.

The $[MoCl_6]^{2-}$ ion is d^2 and possesses a $^3T_{1g}$ ground term split by spin-orbit coupling. The nine spinor components of this state split according to $E < T_2 < T_1 < A_1$. The E component is the ground state lying about 27 cm^{-1} below T_2 (as indicated by a 'complete' calculation, and by analysis of the temperature dependence of the MCD spectra). A preliminary analysis of this system, under the influence of spin-orbit coupling, was presented in section 5.2 (see Fig.5.5). This first complex band system was assigned to the $(\pi + \sigma)1t_{1u}$---$2t_{2g}$ transition. Fitting this band using the Jorgensen optical electronegativity approach (section 5.3), yields

a value for Mo(IV) of 2.1, slightly higher than the value (1.95) suggested by Jorgensen.

A second structured band centred at about 28,200 cm^{-1} was then assigned to the several states originating in the $\pi 1t_{2u}$---$2t_{2g}$ transition ($(2t_{2g})^3(^4A_{2g})(\pi 1t_{2u})^5(^2T_{2u})$).

The separation between the first two sets of bands of about 3500 cm^{-1}, represents the energy separation between the ligand $(\pi + \sigma)2t_{1u}$ and $\pi 1t_{2u}$ levels. This same separation is seen between the next two higher absorption manifolds lying at 31,500 - 34,500 and 34,500 - 38,000 cm^{-1}. These two manifolds are then assigned to the set of states generated by the products $(^2E_g, ^2T_{1g}) \times {}^2T_{1u}$ and $(^2E_g, ^2T_{1g}) \times {}^2T_{2u}$. The authors [380] also note that the separation between the centres of the first set of states associated with the $^4A_{2g}$ and the second set associated with $^2E_g, ^2T_{1g}$, namely ca. 9000 cm^{-1}, corresponds well with the energy separation between these states in the d^3 $[MoCl_6]^{3-}$ ion.

5.4 iv) The Cs_2ZrCl_6/Os(IV) Doped System

While Jorgensen assigned all strong bands in the spectrum of the $[OsCl_6]^{2-}$ species to LMCT [381] a fact supported by later work [383], other workers assigned many transitions as d-d in origin [382]. Detailed MCD and absorption studies [384], however, confirm the LMCT assignment. The first region of d-d absorption occurs between 15,000 - 32,000 cm^{-1} in $[IrCl_6]^{3-}$, 12,000 - 28,000 cm^{-1} in $[RhCl_6]^{3-}$ and 16,000 - 24,000 cm^{1} in $[PtCl_6]^{2-}$, in all cases with intensities of less than 100 l mol^{-1} cm^{-1} [385]. The much greater intensities in this system argue strongly for LMCT, as do the MCD data. The ground state $^3T_{1g}$ state splits into the same spinor levels as for d^2 Mo(IV) above, but with inverted order since $(t_{2g})^4$ is the hole equivalent of $(t_{2g})^2$. Thus the ground state is the spinor A_1, and only transitions to spinor T_1 states are electric dipole allowed. Since Os(IV) is a very heavy ion, the spectra were assigned in the j-j approximation, i.e. assuming spin-orbit coupling is dominant and L and S have lost their significance. The procedure is as above, the hole in the metal configuration $(2t_{2g})^5$ being coupled into the hole in the ligand configuration $Lt^5(2t_1u$ or $1t_{2u})$ using the double octahedral group to generate the representations. The spectrum is simpler than might initially be imagined.

The configuration $(\pi + \sigma)2t_{1u}{}^5(2t_{2g})^5$ comprises 36 spinors which include $4T_1$. However transitions to three of these levels are forbidden because they represent two-electron transitions. Thus only one allowed transition occurs (similar to $[IrCl_6]^{2-}$ discussed below).

> The t_{2g} level spans $u_g' + e_g''$ in the double group, and the ground state A_1 comprises $(u_g')^4(e_g'')^0$. The $2t_{1u}$ ligand level spans $u_u' + e_u'$. Of the various T_1 excited spinor states, transition to $(u_u')^3(e_u')^2(u_g')^4(e_g'')^1$ is clearly a one-electron transition and is allowed. However the T_1 state derived from $(u_u')^4(e_u')(u_g')^3(e_g'')^2$ cannot be achieved from the ground state via a one-electron transition and is therefore forbidden.

A band system centred at 26,000 cm^{-1} is assigned to this single T_1 arising from $(\pi + \sigma)2t_{1u}$---$2t_{2g}$. The $\pi 1t_{2u}$---$2t_{2g}$ transition generates two accessible T_1 levels and these are assigned to two regions of absorption overlapping at 29,000 cm^{-1}. Their separation defines a value for the chloride spin-orbit coupling coefficient.

In addition to weak d-d transitions occurring near 4700 (intraconfigurational $(2t_{2g})^4$), 10,500 and 17,100 cm^{-1}, there are regions of intermediate intensity at 24,000 - 25,200, at 27,000 - 28,600 and 32,000 - 34,000 (ca 1000 l mol^{-1} cm^{-1}) which are too strong to be d-d and are assumed to be (forbidden) vibrationally induced CT transitions. These must plausibly originate in the even ligand orbitals $\pi 1t_{1g}$ and $\pi 1t_{2g}$, and perhaps $\sigma 1e_g$, probably vibrationally induced by the t_{1u} vibration. For further analysis, see ref. [384].

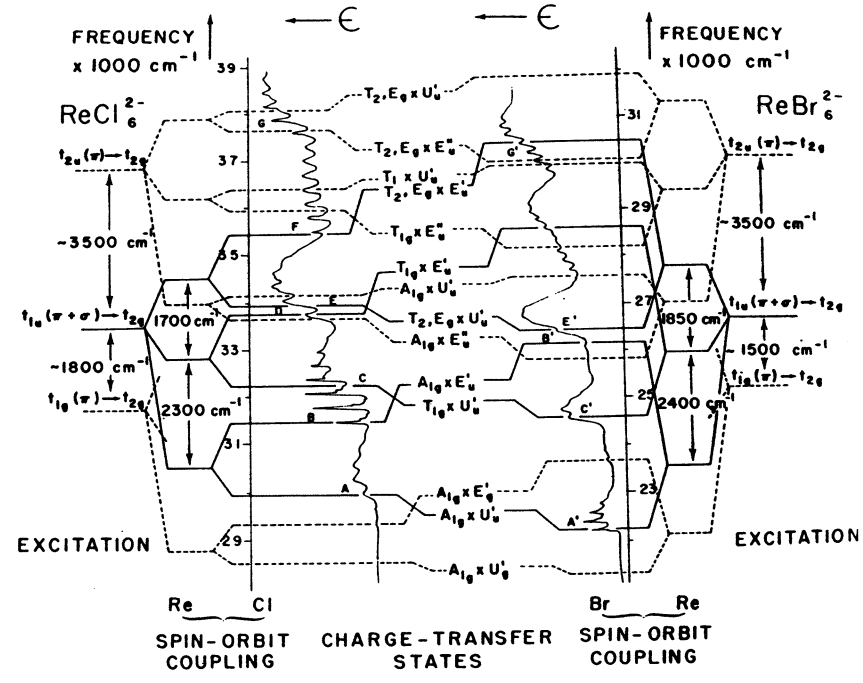


Fig.5.8 Charge transfer transitions in the spectra of Cs_2ZrCl_6:Re(IV) and Cs_2ZrBr_6:Re(IV) showing both the spectra and molecular orbitals involved [386].

5.4 v) The $[ReX_6]^{2-}$ (X = Cl,Br) Systems in Various Crystals

The hosts Cs_2ZrCl_6, Cs_2YCl_6 and Cs_2ZrBr_6 were used to study these $(t_{2g})^3$ ions [386]. The ground state is $^4A_{2g}$ and excited states are generated by combination of $(2t_{2g})^4$ with Lt^5. The lowest state of $(2t_{2g})^4$ is $^3T_{1g}$ which in combination with the hole in the ligand orbitals, gives rise to a plethora of spinor states, transitions to many of which are allowed. Thus the spectra are much more complex than those associated with, for example, Os(IV). While an extensive effort has been made to understand these spectra, only a diagram showing the possible relationship between theory and experiment (Fig.5.8) is presented here and the reader is referred to [386] for detail.

5.4 vi) The $[RuX_6]^{2-}$ (X = Cl,Br) Systems in Various Crystals

The hosts Cs_2ZrCl_6 and Cs_2SnBr_6 were used to study these $(2t_{2g})^4$ ions [387]. Spin-orbit coupling is less important than for Re(IV) and Os(IV) above, but is more important than for a first row ion. Thus intermediate rather than j-j coupling is appropriate. The LS ground state is $^3T_{1g}$. Again spin-orbit coupling will place the A_1 spinor level as the ground state.

> Note that the magnitude of the spin orbit coupling coefficient will determine how 'pure' the ground state spinor A_1 will be, i.e. to what degree higher energy spinor states of the same spinor A_1 symmetry will be admixed therein. For example [387] in the j-j limit the ground state A_1 contains 2/3 of $A_1(^3T_{1g})$ and 1/3 A_1 ($^1A_{1g}$ of $(2t_{2g})^4$). Using the value of the spin-orbit coupling coefficient derived from analysis of the d-d spectrum [388] (about 1044 cm^{-1}), the contribution of A_1 ($^1A_{1g}$ of $(2t_{2g})^4$) in the ground state spinor is calculated to be about 0.02. Thus the j-j limit is poor for this system.

Excitation to $(2t_{2g})^5$ configuration provides only a single metal state, $^2T_{2g}$ to couple to the hole in the ligand Lt^5; thus the problem is apparently more simple than the previous cases. The authors [387] consider the possibility that other higher energy metal configurations such as $^6A_{1g}$ $(2t_{2g})^3(2e_g)^2$ and $^4T_{1g}$ $(2t_{2g})^4(2e_g)$ might generate states to which transitions could occur weakly through spin-orbit mixing. However they are considered not to be significant in the present (small coupling coefficient) case. Once again the analysis is too complex for extensive discussion here. Note that the excitations $(\pi + \sigma)2t_{1u}$ and $\pi 1t_{2u}$ to $2t_{2g}$ generate nine T_{1u} spinor states. In the j-j limit, transitions to only three of these are allowed from the j-j limit ground state A_1. In the intermediate limit, four other such transitions gain intensity because of the change in the nature of the ground state. Naturally transitions to other 'forbidden' levels may also occur through vibronic coupling. Assignments are shown in Table 5.6. The Ru(IV) spin-orbit coupling coefficient has a value of 800 - 1000 cm^{-1} and the ligand value is 600 (Cl) and ca 2500 for Br in the $1t_{1g}$ and $(\pi + \sigma)2t_{1u}$ levels, but only about 1650 cm^{-1} in

$\pi 1t_{2u}$ [387], as deduced from spin-orbit splitting between states.

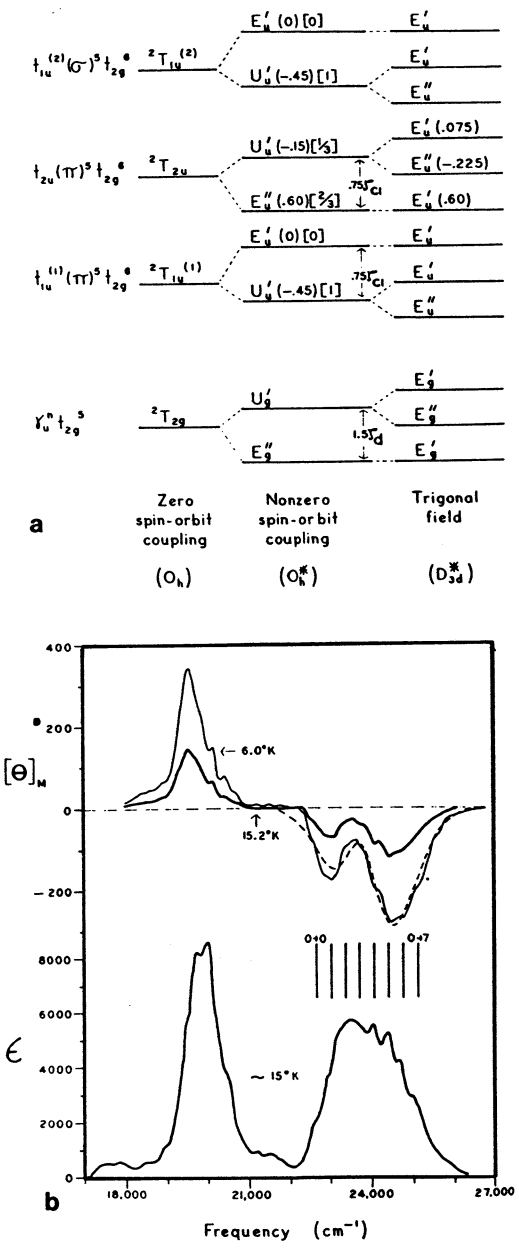
5.4 vii) The $[IrX_6]^{2-}$ Ions (X = Cl,Br) Doped in Various Crystals

These d^5 complexes, in doped crystals, have been extensively studied by Schatz, Stephens and co-workers [394-398]. They offer an opportunity to study a relatively simple CT system in that the accessible LMCT transitions will involve transitions from ligand to $(2t_{2g})^5$ forming a closed sub-shell $(2t_{2g})^6$ $^1A_{1g}$ state with which to couple the hole on the ligand. Thus, in the absence of spin-orbit coupling (which will be small for the chloride), there should be only one electronic transition for each configuration. In general these will split due to spin-orbit coupling but the splitting will be small (Fig.5.9 (a)). The two lower lying allowed transitions will originate from $(\pi + \sigma)2t_{1u}$ and $\pi 1t_{2u}$ and these are shown in Fig.5.9 (see Table 5.6) [397].

Consider Fig.5.9 (c) which shows the region of Fig.5.9 (b) to higher resolution (also different doped crystal). The weak absorption lines labelled 1-5 is associated with the parity forbidden $\pi 1t_{1g}$---$2t_{2g}$(Ir(IV)) transition. The next group of stronger lines is associated with the parity allowed $(\pi + \sigma)2t_{1u}$---$2t_{2g}$ transition. These authors [396] assign line 6 (19,296) to the no-phonon (0-0) spin-orbit origin $U_u'(^2T_{1u})$ (U_u'---E_g''). A progression in a_{1g} Ir-Cl (ca. 316 cm^{-1}) is built theron. Transition to the other spin-orbit origin E_u', E_u'---E_g'' is forbidden in O_h^*. It is assumed that line 9 is a (0-1) transition to E_u' so that the energy difference between lines 6 and 9 corresponds to the sum of the spin-orbit splitting plus the enabling vibration. In this fashion the spin-orbit splitting is estimated to be about 200 - 240 cm^{-1}.

With respect to the $^2T_{2u}$ transition, only the transition E_u''---E_g'' and its associated vibronic satellites can be observed. The authors argue on the basis of the shape of the MCD spectrum and the presence of a shoulder on this last (0-0) transition, that the spin-orbit splitting is very small. A shoulder, at most 6 cm^{-1} from E_u''---E_g'', is tentatively associated with U_u'---E_g''. The spin-orbit splitting is said to be quenched by the Ham effect.

Another region of moderately intense absorption (20,800-22,000 cm^{-1}) is associated with $\pi 1t_{2g}$---$2t_{2g}$(Ir(IV)) rather than d-d absorption. Indeed except for some very weak bands in the 22,000 - 22,600 cm^{-1} no d-d absorption is assigned. This contradicts an earlier study [399] which assigned some CT bands as d-d transitions. The d-d assignment is excluded by the MCD data reported here (sign of C/D etc) and also by the relatively high intensity of the visible absorption. Note [396] that one cannot argue that visible region d-d bands pick up intensity from the nearby dipole allowed LMCT transitions since according to the Fenske argument (section 4.2 viii)), these d-d transitions cannot gain intensity from L---$2t_{2g}$ transitions. The parity forbidden LMCT transitions, however, may pick up intensity by vibronic mixing with these stronger LMCT bands.



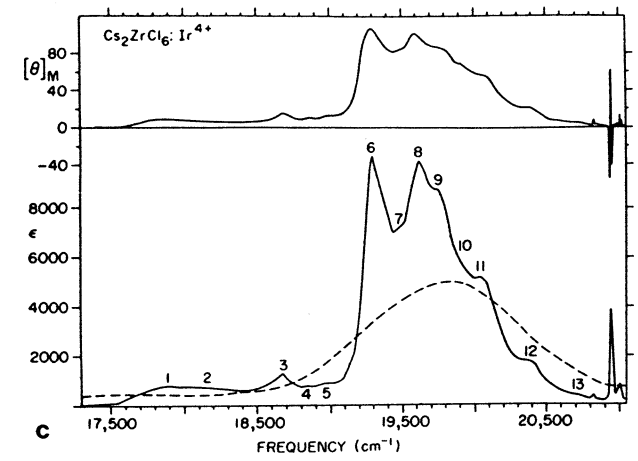


Fig.5.9 (a) (Previous page, upper) Schematic energy level diagram showing the states generated by excitation from octahedral ligand levels into $(t_{2g})^5$, both inclusive and exclusive of spin-orbit coupling [397]. (b) (previous page, lower) MCD (at 6.0 and 15.2 K) and absorption spectra (at ca 15 K) for $[IrCl_6]^{2-}$ doped into $(CH_3NH_3)_2SnCl_6$. The dashed line shows the best fit to the MCD spectrum. The vertical lines show the frequencies of the proposed totally symmetric progression based on a 345 cm^{-1} vibration [397]. (c) (Above) Absorption and MCD spectra of Ir(IV) doped into Cs_2ZrCl_6. The solid line is recorded with the crystal at ca 11 K and the hatched line at room temperature [396]. For significance of numerals, see text.

The $[IrBr_6]^{2-}$ ion has been studied in several different crystalline environments [394,398]. The spin-orbit coupling is much greater for bromide (2460 cm^{-1}) than for chloride (590 cm^{-1}), and greater splitting of the excited orbital triplet states is expected. The absorption spectrum doped into Cs_2ZrBr_6 is particularly striking with highly resolved vibrational structure. The weak absorption (Fig.5.10, lines 1,2) is associated with $\pi 1t_{1g}$, and lines 3-4 to the first parity allowed transition to U_u'---$E_g''(^2T_{1u})$. The other spinor component, E_u'---E_g'' (forbidden in O_h^*), probably lies on the low energy edge of the band system at 17,000 - 19,500 cm^{-1}. This double band structure is associated with the two allowed split components of $^2T_{2u}$ (Fig.5.10). The reader should read the original publication [394] for detail on the vibrational analysis.

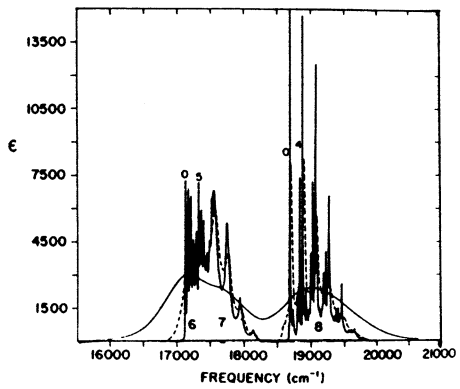


Fig.5.10 Absorption spectra of Cs_2ZrBr_6:Ir(IV) [394].

5.4 viii) Pseudo-Octahedral ML_5X, cis- and trans ML_4X_2 Systems

There are a myriad of six coordinate complexes containing one or more halogen ligands. In general these will show halogen to metal LMCT transitions defined most readily by consideration of the relevant optical electronegativity values. Some examples are included in Table 5.7. Detailed experimental studies of the more subtle aspects of such systems have not, in general, been pursued.

A simple molecular orbital approach to this problem (for low spin d^5 complexes) has been considered by Zwickel and Creutz [323] and by Verdonck and VanQuickenborne [432] using essentially the same procedure, the former dealing with MLCT transitions and the latter with LMCT transitions, but the results (for low spin d^5) are equivalent.

We follow the treatment of Verdonck and VanQuickenborne [432] but extend the treatment to all d configurations. The special case of d^5 is shown in Fig.5.11.

A single halogen in a system such as ML_5X (of symmetry C_{4v}) has a pair (p_x,p_y) of π-orbitals which transform as e. The dt_{2g} orbitals span e + b_2, while de_g spans a_1 + b_1. Transitions from πe(X) to all these levels are possible, depending upon the d configuration. In the low spin d^5 systems shown in Table 5.6 a single transition from πe(X) to the hole in $(2t_{2g})^5$ is observed. In the low spin d^6 systems such as the Co(III) pentammine series, the transition is to z^2 this being the lowest available acceptor orbital. In principle, transitions from the lower lying σX orbital (involved in bonding to M) can be expected at higher energies.

In a trans-ML_4X_2 complex, the four πX orbitals (2 on each X) transform as e_g + e_u. Transitions from πe_g to the d shell will be Laporte forbidden, but all transitions from πe_u will be allowed.

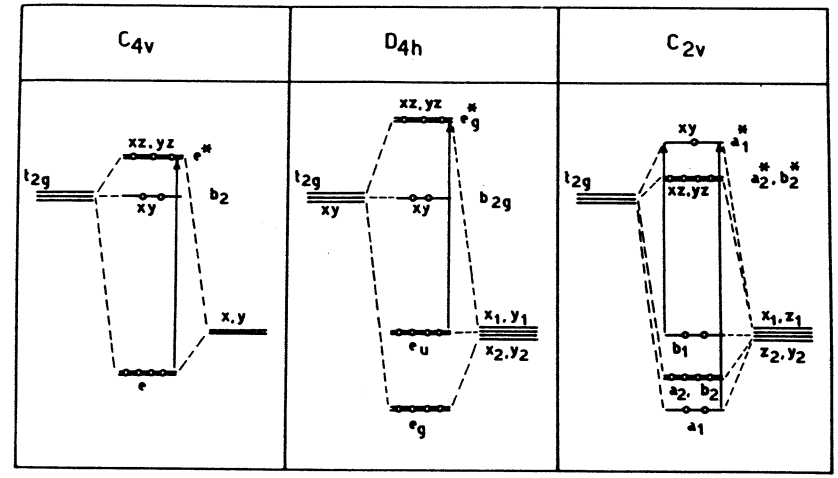


Fig.5.11 Approximate energy level scheme for the orbitals arising from the π interactions between a low spin d^5 metal and X ligands. The left-hand diagram refers to ML_5X, the middle to trans-ML_4X_2 and the right to cis-ML_4X_2 [432].

The possible transitions πe_u---$e_g(xz,yz)$ and πe_u---$a_{1g}(z^2)$ will be strong. However both πe_u--- $b_{1g}(x^2-y^2)$ or πe_u---$b_{2g}(xy)$, will be weak due to poor overlap. In the low spin d^5 complexes shown in Table 5.6, transition from πe_u to the hole in $(2t_{2g})^5$ is responsible for the single πX---M LMCT transition observed at low energies.

The appearance of a strong π-LMCT transition to t_{2g} in a d^5 ML_4X_2 system presumably implies that the hole in t_{2g} lies in e_g, i.e. configuration $(b_{2g})^2(e_g)^3$. If L has no π-orbitals, such as an ammine, then interaction of X with M should indeed destabilise e_g with respect to b_{2g}, as observed. However no doubt situations could arise where the hole would be in b_{2g}, in which case the transition should be much weaker or absent.

A cis-ML_4X_2 complex has C_{2v} symmetry, and the four π orbitals span a_1 + a_2 + b_1 + b_2. In discussing the spectra of these species, Verdonck and VanQuickenborne chose not to use a standard (character Table) coordinate frame for the C_{2v} group, but rather assumed the cis-X_2 ligands lay in the xy (ML_2X_2) plane, along axes x and y. This is more useful where a comparison of the three systems under discussion here is desired (i.e. a common coordinate frame for all three complexes). In this framework, the p π orbitals interact as follows:

(ML_2X_2 defined as xy plane, X(1) along -y, X(2) along +x).

d	Ligand group orbital	Symmetry
xy	y(2) - x(1)	a_1
$1/\sqrt{2}(xz + yz)$	z(1) - z(2)	a_2
$1/\sqrt{2}(xz - yz)$	z(1) + z(2)	b_2
z^2	-	a_1
x^2-y^2	[y(2) + x(1)]	b_1

(5.18)

Simple visualisation of this situation predicts that the ligand a_1 group orbital will overlap the metal xy orbital to a greater degree than overlap of a_2 and b_2 with xz or yz (which are equivalent). The ligand b_1 group orbital has the same symmetry as x^2-y^2 but clearly cannot overlap it (nor will the πa_1 orbital have any net overlap with z^2). Thus the energy sequence is expected to be $a_1 < a_2,b_2 < b_1$(nb).

Interaction of the p π orbitals should destabilise $(a_1)^*$(xy) to a greater degree than $(a_2,b_2)^*$(xz,yz) which remain degenerate ('intermediate symmetry'). In a low spin d^5 complex, the hole will be in $xy(a_1)^*$, and two transitions are predicted, namely a_1---$a_1{}^*$(xy) and b_1---$a_1{}^*$(xy). Of the other two possibilities, a_2---$a_1{}^*$(xy) is electric dipole forbidden, and b_2---$a_1{}^*$(xy) will be very weak because of poor overlap (transition from an out-of-plane to an in-plane orbital) (Fig.5.11). Further transitions are possible for d^n configurations with holes in $(a_2,b_2)^*$, though they will lie close together; well defined experimental examples are not available. Low spin d^6 systems also commonly show two π-LMCT transitions, to z^2. Transitions from πX to x^2-y^2 should be very weak. In fact there are three possible πX transitions to $z^2(a_1)^*$, from πa_1, πb_2 and πb_1, but they are not generally completely resolved.

Zwickel and Creutz [323] were interested specifically in the MLCT transitions in low spin d^6 $Ru(NH_3)_5L$, and cis- and trans-$Ru(NH_3)_4L_2$ complexes where L is a π^* electron acceptor. The number of predicted M---L transitions is the same as found above, 1, 2 and 1 respectively for transitions from t_{2g} to π^* (see section 5.12 for further detail).

Both publications [323,432] include discussion of a simple MO approach to calculating the relative energies of these transitions.

Using simple symmetry arguments and defining:

$$\langle d|H|d\rangle = H_d \quad \langle p|H|p\rangle = H_p \quad \langle p|H|d\rangle = \beta \quad H_d - H_p = \delta$$

The following orbital energies and transition energies (t_{2g} levels) were derived [323,432]:

Energies

ML_5X

$$e \quad (0.5)[H_d + H_p - (\delta^2 + 4\beta^2)^{1/2}]$$

$$e^* \quad (0.5)[H_d + H_p + (\delta^2 + 4\beta^2)^{1/2}]$$

cis-ML_4X_2

$$a_1 \quad (0.5)[H_d + H_p - (\delta^2 + 8\beta^2)^{1/2}] \qquad (a_2,b_2)$$

$$b_1 \quad H_p$$

$$a_1{}^* \quad (0.5)[H_d + H_p + (\delta^2 + 8\beta^2)^{1/2}] \qquad (a_2,b_2)^*$$

trans-ML_4X_2

$$e_g \quad (0.5)[H_d + H_p - (\delta^2 + 8\beta^2)^{1/2}]$$

$$e_u \quad H_p$$

$$e_g{}^* \quad (0.5)[H_d + H_p + (\delta^2 + 8\beta^2)^{1/2}]$$

(5.19)

where, for completion we also note that the form of the energies for the a_2 and b_2 orbitals will be, in this approximation, the same as for a_1 but with different numerical values of the parameters.

These energies are derived by a simple first order molecular orbital calculation. Thus for the ML_5X complex, construct a 2 x 2 secular determinant (A(i,j)) whose roots are the two e energies. One may begin with a basis consisting of a pure d pair (xz,yz) and a pure p orbital pair (x,y) whose energies are H_d and H_p respectively. Then $A(1,1) = H_d - E$, $A(2,2) = H_p - E$ and $A(1,2) = A(2,1) = \langle H_p|H|H_d\rangle = \beta$. Equate the determinant to zero and solve to obtain the energies shown in (5.19). Re-insertion of the energies into this matrix will generate the eigenvectors for the two MOs showing the extent of mixing between d and p. The two a_1 energies in the cis complex are obtained in a very similar fashion except that the basis will now be d, and $(1/\sqrt{2})[p(y2) - p(x1)]$. The matrix (B(i,j)) will then necessarily contain $B(1,1) = H_d - E$, $B(2,2) = H_p - E$ and $B(1,2) = B(2,1) = 2\beta$. It will be evident that the trans-ML_4X_2 system will generate the same matrix B(i,j) (and hence the cis and trans energies are the same (5.19)), once one visualises that the pπ basis necessary to overlap xz is $(1/\sqrt{2})[p(x1) - p(x2)]$ (and correspondingly for yz).

The relevant transition energies, for low spin d^5, are then seen to be:

ML_5X	$e---e^*$	$(\delta^2 + 4\beta^2)^{1/2}$
cis ML_4X_2	$a_1---a_1^*$	$(\delta^2 + 8\beta^2)^{1/2}$
	$b_1---a_1^*$	$(0.5)[\delta + (\delta^2 + 8\beta^2)^{1/2}]$
trans-ML_4X_2	$e_u---e_g^*$	$(0.5)[\delta + (\delta^2 + 8\beta^2)^{1/2}]$

(5.20)

Note that the lower energy transition in the cis complex should equal the energy of the transition in the corresponding trans complex, and if $\beta << \delta$ (as is the case, see below) then the energies of the transitions in the pentammine and trans complex will be very similar as is the case (Table 5.7).

Using data from a set of corresponding complexes (data in Table 5.7), Verdonck and VanQuickenborne calculated values for δ and β, as follows:

Metal	Ligand	$\delta(H_d - H_p)$	β	Calc.Energy
Ru(III)	Cl	28,600	5300	30,400 cm^{-1}
	Br	23,100	5000	25,000
Os(III)	Cl	35,600	5800	37,400
	Br	30,000	5300	31,800
	I	22,500	4900	24,500

(5.21)

where values of δ and β were obtained by fitting the energies of the ML_5X transition and the higher energy transition in cis-ML_4X_2. Thus the calculated energy refers to the lower energy transition in cis-ML_4X_2 and the single transition in trans-ML_4X_2 which should be equal. Comparison with the data in Table 5.7 shows that agreement with the trans complex energy is excellent, but that the cis energy is over-estimated.

Experimentally the lower energy cis transition seems always to be about 2000 - 3000 cm^{-1} lower in energy than the band in the corresponding trans complex. This analysis omits ligand-ligand interactions and their existence may cause some destabilisation of the b_1 level and hence a decrease in the $b_1---a_1^*$ transition energy. Since the X ligands are much further apart in the trans complexes than in the cis complexes, omission of L-L interactions is valid in the former.

The trend in δ is as expected based on the decreasing ionisation potentials of the halides in the sequence Cl > Br > I. The differences between Cl and Br for both Os(III) and Ru(III) are about the same implying that the energy of the ligand basis does not depend upon the metal. Covalency is reflected by the β/δ ratio which is greater for Ru(III) than Os(III), and increases in the anticipated sequence Cl < Br < I.

The above analysis may be used to derive energies for transitions to z^2 if one may estimate the energy separation between z^2 and components of t_{2g}.

Verdonck and VanQuickenborne [432] also comment on the relative intensities of the spectra. It is necessary to derive values for the transition moment $\langle d(MO)|r|X(MO)\rangle$ and square for the dipole strength (see Chapter 4).

A very simple first order analysis notes the work of previous authors [390,433] who show that the main contributions to the transition moments arise from ligand diagonal elements $\langle p|r|p\rangle$. The essential argument is the view, expressed previously (section 4.2), that for a transition to have non-zero intensity there must be some net overlap between ground and excited state wavefunctions. Thus, for example, a CT transition from a pure p orbital on a halogen to a pure d orbital, of different symmetry, on the metal would have zero intensity, even if the transition is group theoretically allowed. The model calculates the contribution of the ligand p orbitals to the 'd' molecular orbital wavefunction and calculates therefrom the relative dipole strengths. In this fashion the intensities of the πLMCT transitions are calculated to decrease in the sequence trans > pentammine > both cis, in the ratio 4:2:1:1. Experimentally the trans system undoubtedly yields the most intense spectra, the ratio for the series being roughly 5:2:1.7:1.6 [432]. See the discussion of the $[CuX_4]^{2-}$ species in section 5.5 for further comment on this approach.

5.5 Charge Transfer Spectra of Square Planar and Tetrahedral Complexes

5.5 i) Square Planar Complexes

Tetrahalocomplexes: An M.O.Diagram appropriate for square planar tetrahalides is shown in Fig.5.12 (a). The relative energies of the ligand group orbitals, and indeed, the d orbitals, may vary slightly from that shown. The following LMCT transitions may be anticipated [69], where the spectroscopic states assume a d^8 diamagnetic $^1A_{1g}$ ground state:

π levels	$b_{2u}---b_{1g}(\sigma^*)$	$^1A_{2u}$	A_{2u}'
		$^3A_{2u}$	$E_u' + (A_{1u}')$
	$e_u---b_{1g}(\sigma^*)$	^1E_u	E_u'
		^3E_u	$E_u' + A_{2u}' + (A_{1u}', B_{1u}', B_{2u}')$
	$a_{2u}---b_{1g}(\sigma^*)$	$^1B_{2u}$	(B_{2u}')
		$^3B_{2u}$	$(B_{2u}') + E_u' + (A_{1u}')$
σ levels	$e_u---b_{1g}(\sigma^*)$	^1E_u	E_u'
		^3E_u	$E_u' + A_{2u}' + (A_{1u}', B_{1u}', B_{2u}')$

(5.22)

where the initial state is effective in the absence of spin-orbit coupling, and this is followed by the spin-orbit states. Assuming a $^1A_{1g}$ ground state, in D_{4h} symmetry, transitions to the E_u' states are allowed in xy, and to A_{2u}' allowed in z; transitions to other states (in parentheses) are electronically forbidden.

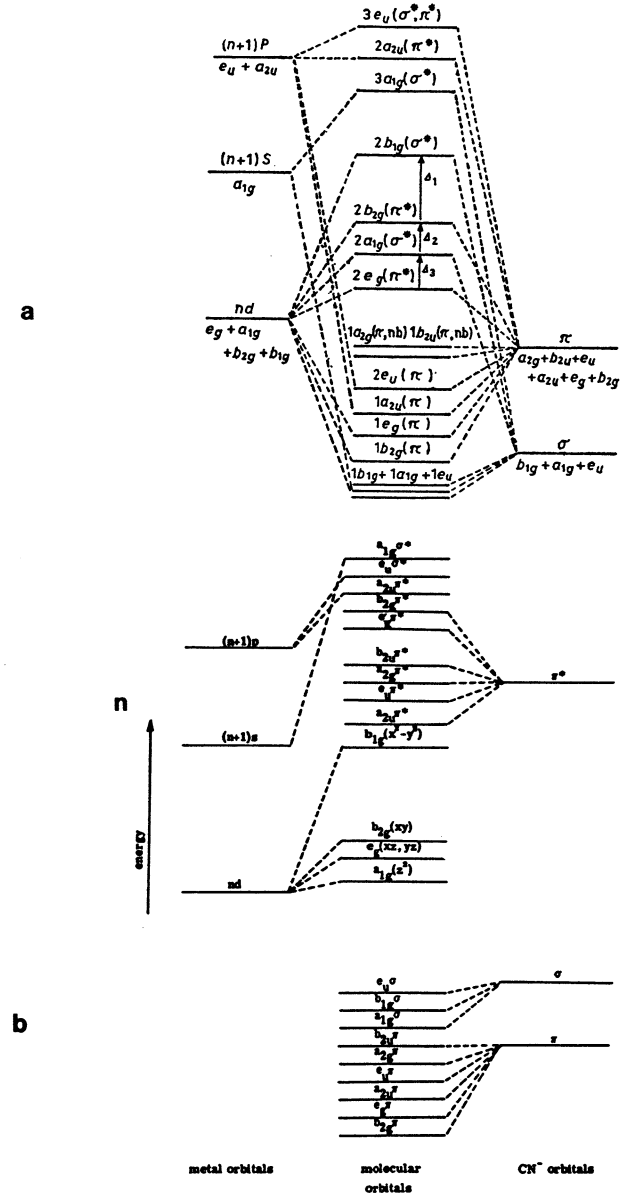


Fig.5.12 (a) (Upper) Relative molecular orbital energies for a square planar (D_{4h}) $[MX_4]^{n-}$ ion (X = halogen). Only the principal correlation lines are shown. Various definitions of the

Δ_i parameters appear in the literature. The z^2 orbital may lie below e_g. (b) (Lower) (previous page) Relative molecular orbital energies for a square planar (D_{4h}) $[M(CN)_4]^{n-}$ ion. Only the principal correlation lines are shown [69] NB the x^2-y^2 level may lie above CN π^* a_{2u} [69].

The spin-orbit approximation is necessary when a heavy metal with a large spin-orbit coupling coefficient, e.g. Pt(II), is involved.

The intensity of the σ---σ^* transition is expected to be much greater than any of the π---σ^* transitions, and within this latter group, the intensity should increase in the order $^1B_{2u} << {}^1A_{2u} < {}^1E_u$. In general the π---σ^* transitions lie some 10,000 - 16,000 cm^{-1} below the σ---σ^* transitions [69].

The configuration d^8 is most prone to form low spin square planar complexes and the discussion of this large group of complexes is left until section 5.12 viii). It is however appropriate to consider the planar $[CuCl_4]^{2-}$ here as an example.

The square planar $[CuCl_4]^{2-}$ ion. This species forms complexes containing either a square planar or a tetrahedral (or distorted tetrahedral) copper(II) site, depending upon the nature of the associated cation (see Chapter 6). It is useful to compare a single species with varying stereochemistry, and the results presented here are compared with the tetrahedral analog in the next section.

Solomon and co-workers have recently presented detailed single crystal polarised light spectroscopy, ultra-violet photoelectron spectroscopy and an X-α calculation of both the square planar and distorted (D_{2d}) tetrahedral $[CuCl_4]^{2-}$ species [609]. Various square planar $[CuCl_4]^{2-}$ species show common CT characteristics involving an essentially wholly xy polarised spectrum below the limit (ca. 40,000 cm^{-1}) of the study. There are two intense xy polarised bands near 26,400 and 35,900 cm^{-1}, both showing some splitting when viewed along x or y.

Only transitions from e_u into b_{1g} are xy polarised; thus the two strong transitions must be associated with $\pi 4e_u$---$3b_{1g}$ and $\sigma 3e_u$---$3b_{1g}$ respectively. A very weak feature near 23,700 cm^{-1} is assigned to the forbidden $1a_{2g}$---$3b_{1g}$ transition. It is probably vibronically activated by an e_u vibration. The absence of any z polarised transition is illuminating since transition from the $1b_{2u}$ πCl orbital is group theoretically allowed along z.

The absence of intensity in the $1b_{2u}$---$3b_{1g}$ transition follows from the following argument. As discussed in Chapter 4, there must be net overlap between ground and excited state wavefunctions for a transition to have non-zero probability. The σp and πp orbitals of the chloride ions are orthogonal to the copper d orbitals, and therefore if the p and d bases are exactly pure, there can be no transition between them. However there is a $\sigma p 1b_{1g}$ orbital on chloride which contributes σp character to the $3d$-$3b_{1g}$ orbital through symmetry mixing. Any other chloride orbital which has σp character can give rise to non-zero net overlap with $3b_{1g}$ if group

theory permits. Calculation [609] shows that the $3e_u$ level has about 70% σp character and hence the transition $\pi 3e_u$---$3b_{1g}$ is strong. The $\pi 1b_{2u}$ orbital has no σp character (there is no σp orbital of b_{2u} symmetry) and hence the transition $1b_{2u}$---$3b_{1g}$ vanishes even though it is group theoretically allowed.

Tetracyano complexes: CT transitions in these complexes tend to blue shift as the metal becomes more oxidising, suggesting that the relevant transitions are MLCT. Considering the two low lying odd π^* CN^- orbitals, the following transitions are possible:

b_{1g}---$a_{2u}(\pi^*)$	$^1B_{2u}$	Forbidden	(refer to 5.22 for spin-orbit states)
	$^3B_{2u}$		
e_g---$a_{2u}(\pi^*)$	^1E_u ^3E_u	Allowed xy	
a_{1g}---$a_{2u}(\pi^*)$	$^1A_{2u}$ $^3A_{2u}$	Allowed z	
b_{2g}---$e_u(\pi^*)$	^1E_u ^3E_u	Allowed xy	
e_g---$e_u(\pi^*)$	$^1A_{2u}$ +.. $^3A_{2u}$	Allowed z	
a_{1g}---$e_u(\pi^*)$	^1E_u ^3E_u	Allowed xy	

(5.23)

In general the first four are seen below 50,000 cm^{-1}. As usual the E_u state are the most intense and may show some splitting. Spin forbidden charge transfer transitions will also be seen (see below), and will become important with the heavier ions such as Pt(II). The polarisation data in (5.23) are relevant in the absence of spin-orbit coupling. Otherwise use the spin-orbit coupling and polarisation relevant to the corresponding states of (5.22). Square planar tetracyanides appear generally to have the d-d energy sequence $a_{1g} > e_g > b_{2g}$. See section 5.6 for a discussion of the use of this information.

5.5 ii) Tetrahedral Complexes

Tetrahalocomplexes: Early work in this area includes major contributions by Bird and Day [390] and Rivoal and Briat [392]. These authors using simple MO theory, and utilising MCD spectra, comfirmed the most probable order of ligand group orbitals (also supported by more recent PES data) as (Fig.5.13):

$$s\sigma 1a_1 < s\sigma 1t_2 < p\sigma 2a_1 < p\sigma 2t_2 < p\pi 1e < p\pi 3t_2 < p(nb)t_1 < d2e < d4t_2 \quad (5.24)$$

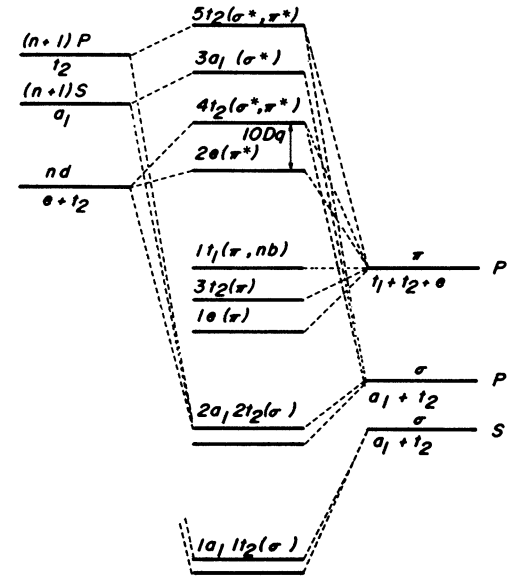


Fig.5.13 Relative molecular orbital energies for a tetrahedral $[MX_4]^{n-}$ ion (X = halogen).

It was originally anticipated that LMCT transitions in the visible and near ultra-violet region will originate from t_1, $\pi 3t_2$ and, in some cases, $\pi 1e$. However a very recent study [609] reveals contributions from the pσCl orbitals in D_{2d} $[CuCl_4]^{2-}$ ion (see below). One may expect transitions from t_1, being a non-bonding orbital, to be narrower than those from bonding orbitals. In addition, through consideration of overlap between ground and excited configurations one predicts that the $\pi 3t_2$---d2e transition should be significantly weaker than the other transitions.

As usual, chloro complexes are expected to show the most simple spectra because their orbital triplet states will not be split significantly by spin-orbit coupling. Complexes of bromide and iodide will show more complex spectra due to spin orbit splitting of the orbital triplets. These expectations are well founded as shown in Fig.5.14 a,b. In general six sets of transitions can be expecte d, by promoting an electron from 1e, $3t_2$ or t_1 into d2e or $d4t_2$. By analysing the symmetries of the states and deriving how many electric dipole allowed transitions are possible, the assignments for many of the spectra follow fairly readily.

Table 5.8

CHARGE TRANSFER SPECTRA OF TETRAHEDRAL TETRAHALIDES AND PSEUDOHALIDES (T_d)

Complex	Absorption Spectra	Assignment	Ref
$MnCl_4^{2-}$	49,400	p-s Cl	393
$MnBr_4^{2-}$	46,100 (51,300)	p-s Br	393
$Mn(NCO)_4^{2-}$	>46,000	πNCO---Mn(II) ?	417
$Mn(NCS)_4^{2-}$	ca.35,000	πNCS---Mn(II) ?	417
$FeCl_4^-$	27,500(7350)	t_1---2e $^6T_2$	390
	31,500	t_1---$4t_2$ $^6T_2$	
	36,700w	$3t_2$---2e $^6T_2$	
	41,200	$3t_2$---$4t_2$ $^6T_2$	234,392
$FeBr_4^-$	21,350(5800)	t_1---2e $^6T_2$	
	23,600 - 27,200	t_1---$4t_2$ $^6T_2$	
	31,000	$3t_2$---2e $^6T_2$	390
	35,600	$3t_2$---$4t_2$ $^6T_2$	234,392
FeI_4^-	14,300	t_1---2e $^6T_2$	
	17,200 - 21,300	t_1---$4t_2$ $^6T_2$	
	(25,600)	$3t_2$---2e $^6T_2$	
	27,000	$3t_2$---$4t_2$ $^6T_2$	400
$FeCl_4^{2-}$	45,500(3840)	t_1---2e	234
$FeBr_4^{2-}$	(39,000) 40,900(6220)	t_1---2e	234
$Fe(NCS)_4^{2-}$	34,200(6200)	t_1---2e $^5T_2$	389
$Fe(NCSe)_4^{2-}$	31,400(9000)	t_1---2e $^5T_2$	389
$CoCl_4^{2-}$ b)	42,735(4320)	t_1---$4t_2$ $^4T_1$	
	46,085	$3t_2$---$4t_2$?	203,234
	51,700(24,900) 57,100	p-s Cl	393
$CoCl_2Br_2^{2-}$ b)	(35,460)	πX---$4t_2$	
	37,735	πX---$4t_2$	203
$CoCl_2I_2^{2-}$ b)	30,030	πX---$4t_2$	
	32,785	πX---4t2	203
$CoBr_4^{2-}$	34,000 - 36,000	t_1---$4t_2$ $^4T_1$	390
	37,000 - 39,000	$3t_2$---$4t_2$ e	392
	47,400(31,000) (52,100)	p-s Br	393
CoI_4^{2-}	25,800 27,800 29,750	t_1---$4t_2$ $^4T_1$	
	33,500 (35,750) (36,500)	$3t_2$---$4t_2$ $^4T_1$	
	38,650	1e---$4t_2$ $^4T_1$	390
	(40,800) 44,100(16,800)	p-s I	234,393
$Co(NCO)_4^{2-}$	42,500(9320)	πNCO---Co(II) ?	417
$Co(NCS)_4^{2-}$	31,000(12,170)	πNCS---Co(II)	
	39,500(3500)		417
$NiCl_4^{2-}$	35,460 38,250	t_1---$4t_2$	390
	(40,400) 42,350	$3t_2$---$4t_2$	
	51,700(22,500)	p-s Cl	234,393

cont

Table 5.8 cont

Complex	Absorption Spectra	Assignment	Ref
$NiBr_4^{2-}$ b)	28,410 30,050 31,200	t_1---$4t_2$	
	33,700 (35,500) 37,100	$3t_2$---$4t_2$	390
NiI_4^{2-}	18,850 - 21,670	t_1---$4t_2$	
	23,450 - 29,000	$3t_2$---$4t_2$	
	31,700 (32,800)	1e---$4t_2$	390
	(39,200) 42,000(13,400)	p-s I	393
$NiBr_4^{2-}$	47,200(22,400) 56,000	p-s Br	234,393
$Ni(NCO)_4^{2-}$	36,300(5890)	πNCO---Ni(II)	
	39,500(2500)	NCO ?	417
$Ni(NCS)_4^{2-}$	26,300(5000)	πNCS---Ni(II)	
	ca.34,000(800)		417
$NiCl_2Br_2^{2-}$ b)	31,950	πX---$4t_2$	203
$CuCl_4^{2-}$	(22,500)	t_1---$4t_2$ $^2A_2(^2T_1)$ c	
	24,400	t_1---$4t_2$ $^2E(^2T_1)$ c	
	(29,200)	$3t_2$---$4t_2$ $^2B_2(^2T_2)$ c	
	34,050	$3t_2$---$4t_2$ $^2E(^2T_2)$ c	390
	41,600	1e---$4t_2$ $^2A_1(^2E)$	392
	53,200(18,600)	p-s Cl	234,393
$CuCl_4^{2-}$ D_{2d}	22,700	$1a_2$---$5b_2$ $^2A_2(^2T_1)$	
	24,730	π5e---$5b_2$ $^2E(^2T_1)$	
	(28,880)	π4e---$5b_2$ $^2E(^2T_2)$	
	33,480	σ3e---$5b_2$ $^2E(^2T_2)$	
	43,000	$\sigma 2a_1$--$5b_2$ $^2A_1(^2A_1)$	609
$CuBr_4^{2-}$	16,500	t_1---$4t_2$ $^2A_2(^2T_1)$ c	
	18,300 - 19,000	t_1---$4t_2$ $^2E(^2T_1)$ c	
	23,500	$3t_2$---$4t_2$ $^2B_2(^2T_2)$ c	
	28,000 - 30,000	$3t_2$---$4t_2$ $^2E(^2T_2)$c	390
	35,800	1e---$4t_2$ $^2A_1(^2E)$	392
	48,700(40,000)	p-s Br	234,393
$Cu(NCO)_4^{2-}$	>46,000	πNCO---Cu(II) ?	417
$Cu(NCS)_4^{2-}$	20,650(2490)	πNCS---Cu(II) ?	
	39,000(3560)	NCS ?	417
$ZnBr_4^{2-}$	48,600	t_1---4s	440
$CdBr_4^{2-}$	48,600	t_1---5s	440
$HgBr_4^{2-}$	35,500	t_1---6s	440

a) All these transitions are LMCT from halogen or pseudohalogen to metal unless otherwise indicated b) Molten salt data c) D_{2d} distortion, tetrahedral parent in parenthesis e) Ref.[390] assigns this to spinor components of t_1---$4t_2$.

The analysis used below for d^n species, is that described earlier where the metal states of the d^{n+1} configuration are coupled with the hole in the ligand group orbital. The ground state is high spin and, usually, the metal d^{n+1} configuration only generates one high spin state with which to couple the ligand hole. Lower spin possibilities will be omitted from the analysis used here. The dipole moment operator spans T_2 in T_d.

The spectra are illustrated by reference to individual complexes.

Titanium(IV) Tetrahalides: These d^0 species represent a relatively simple situation where the states generated by the six possible transitions are as follows (spin triplet states are omitted). The ground state is $^1A_1$ and only transitions to $^1T_2$ are allowed. States to which transitions are orbitally forbidden are enclosed in parentheses. The metal configurations (e) ^2E and (t_2) $^2T_2$ are coupled with the ligand hole to obtain:

Transition	States	Transition	States
i) t_1---2e	$(^1T_1)\ ^1T_2$	ii) t_1---4t_2	$(^1A_2)(^1E)(^1T_1)\ ^1T_2$
iii) 3t_2---2e	$(^1T_1)\ ^1T_2$	iv) 3t_2---4t_2	$(^1A_1)(^1E)(^1T_1)\ ^1T_2$
v) 1e---2e	$(^1A_1)(^1A_2)(^1E)$	vi) 1e---4t_2	$(^1T_1)\ \ ^1T_2$

(5.25)

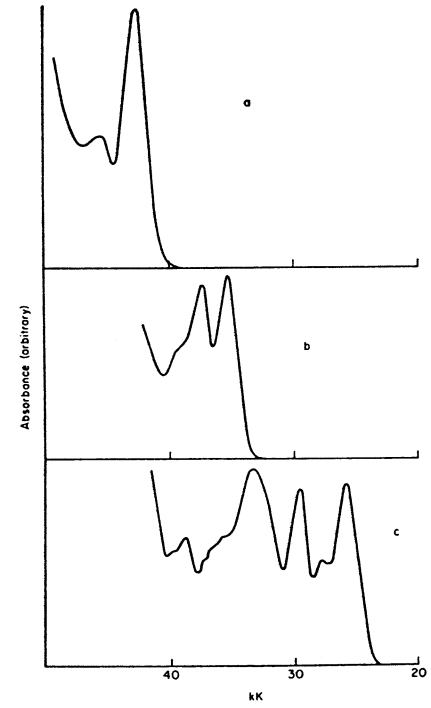
Data are shown in Table 5.8. Two bands are observed, some 8000 cm^{-1} apart. They were assigned to transitions (i) and (ii) so that their separation is 10Dq. Since the t_1 - 3t_2 separation is generally of the order of 4000 - 8000 cm^{-1} (see data below), the second transition could be assigned to (iii). Recent PES and calculations suggest that this is indeed the case and that the second band probably comprises both transitions (ii) and (iii) [410,413]. A low energy shoulder may be a spin-forbidden LMCT band [410].

The TiI_4 species shows a much richer spectrum, the two lowest absorptions of which (Table 5.8) probably constitute transitions (i) and (ii) [411,412].

Vanadium(IV) Tetrahalides: These d^1 species have a ^2E ground state and transitions to both $^2T_1$ and $^2T_2$ are electric dipole allowed. The metal configurations $(e)^2\ ^3A_2$ and $(e)(t_2)\ ^3T_2$ are coupled with the hole on the ligand to obtain (omitting quartets):

Transition	States	Transition	States
i) t_1---2e	$^2T_1$	ii) t_1---4t_2	$(^2A_2)(^2E)\ ^2T_1\ \ ^2T_2$
iii) 3t_2---2e	$^2T_2$	iv) 3t_2---4t_2	$(^2A_1)(^2E)\ ^2T_1\ \ ^2T_2$
v) 1e---2e	(^2E)	vi) 1e---4t_2	$^2T_1\ \ ^2T_2$

(5.26)



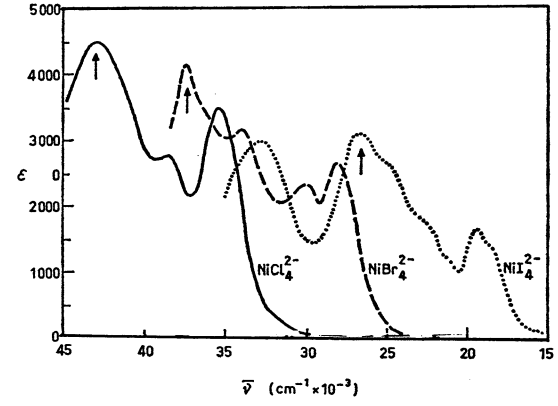
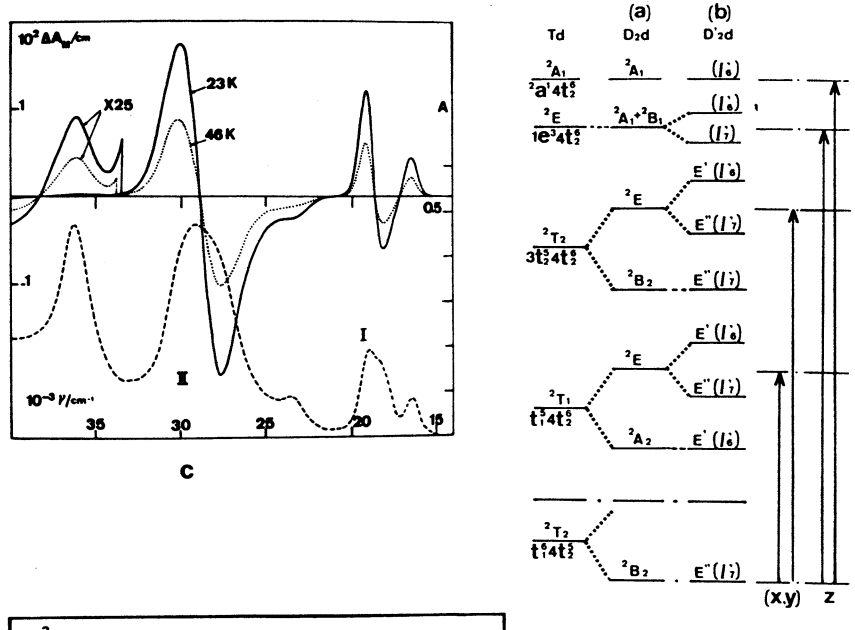


Fig.5.14 (a) (Upper) The charge transfer spectra of a: $[CoCl_4]^{2-}$, b: $[CoBr_4]^{2-}$ and c: $[CoI_4]^{2-}$ at 77 K [390]. (b) (Lower) The absorption spectra of tetrahedral ions as identified in the Figure, recorded as tetra-alkylammonium salts in dichloromethane [234].



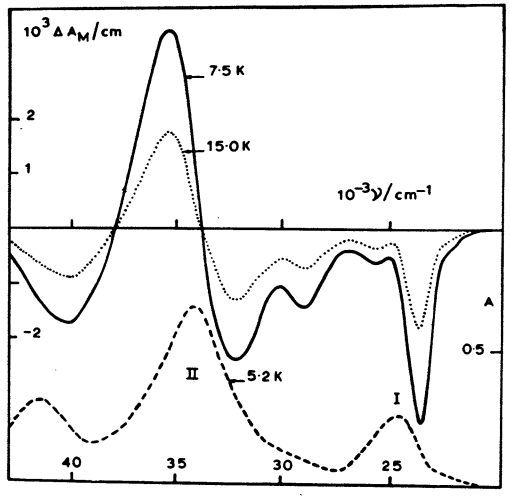


Fig.5.14 (c) (Top left) Low temperature absorption (bottom curve) and MCD (top curves) spectra of $[CuBr_4]^{2-}$ [392]. (d) (lower left) Low temperature absorption (bottom curve) and MCD (top curves) spectra of $[CuCl_4]^{2-}$ [392]. (e) (Top right) Influence of the tetragonal field and spin-orbit splitting on the ground state and various excited states of $[CuX_4]^{2-}$ ions [392].

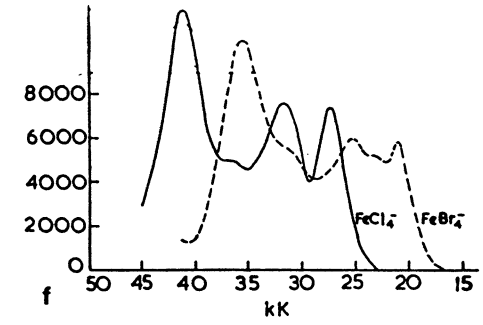


Fig.5.14 (f) The absorption spectra of tetrahedral $[FeX_4]^{2-}$ ions as identified in the Figure, recorded as tetra-alkylammonium salts in dichloromethane [234].

where we note that the higher energy metal state $(e)(t_2)\ ^3T_1$ will generate another set of levels corresponding to (ii), (iv) and (vi) but displaced to higher energy. The species VCl_4 displays a d-d band (10Dq) at 9030, an LMCT band at 24,500 and a more intense LMCT band at 34,840 cm^{-1} [406,408]. The lower energy LMCT band is assigned to (i) and the higher to (both) (ii) [410].

Iron(III) Tetrahalides: The d^5 ion has a $^6A_1$ ground state, and only transitions to $^6T_2$ are electric dipole allowed. Coupling of the metal states $(e)^3(t_2)^3$ (^5E) and $(e)^2(t_2)^4$ ($^5T_2$) with the hole in the ligand group orbital, yields (omitting spin quartets):

Transition		States	Transition		States
i)	t_1---2e	$(^6T_1)\ ^6T_2$	ii)	t_1---$4t_2$	$(^6A_2)(^6E)(^6T_1)\ ^6T_2$
iii)	$3t_2$---2e	$(^6T_1)\ ^6T_2$	iv)	$3t_2$---$4t_2$	$(^6A_1)(^6E)(^6T_1)\ ^6T_2$
v)	1e---2e	$(^6A_1)(^6A_2)(^6E)$	vi)	1e---$4t_2$	$(^6T_1)\ ^6T_2$

(5.27)

The spectrum of the $[FeCl_4]^-$ ion should be especially simple because each transition gives rise to only one electric dipole allowed transition which should not be significantly split. Reference to Fig.5.14 f) reveals four clear regions of absorption, the third being much weaker than the others. If this weak transition is associated with 3t2---2e, then the four regions correspond with the first four allowed $^6T_2$ states. Since the spin-pairing energy contributions to these states are the same (section 5.2), the energy differences between transitions i) and ii), and between iii) and iv), should be the same and should

correspond with 10Dq. This is seen to be true with 10Dq = ca.4000 cm^{-1}. Detailed analysis of the MCD spectra [390,392] is consistent with this assignment (Table 5.8). While certainly more complex, the spectra of the $[FeBr_4]^-$ and $[FeI_4]^-$ ions can also be divided into four regions, and the assignments listed in Table 5.8 are also consistent with MCD data analysis. Resonance Raman excitation into the lowest LMCT band in the halides results in long progressions in the Fe-X a_1 stretching vibration [403].

Cobalt(II) Tetrahalides: Complexes of these deep blue or deep green ions are very well known in coordination chemistry. The ground state for this d^7 ion is $^4A_2$ and only transitions to $^4T_1$ are electric dipole allowed. Considering the six possible transitions, note first that 2e is full, so that an even more simple situation may prevail:

Transition	States	Transition	States
i) t_1---2e	-	ii) t_1---$4t_2$	$(^4A_1)(^4E)(^4T_2)$ $^4T_1$
iii) $3t_2$---2e	-	iv) $3t_2$---$4t_2$	$(^4A_2)(^4E)(^4T_2)$ $^4T_1$
v) 1e---2e	-	vi) 1e---$4t_2$	$(^4T_2)$ $^4T_1$

(5.28)

Thus only transitions (ii) and (iv) might be observed in the visible region, with transition (vi) being allowed but probably beyond the usual instrumental range. Indeed the spectrum of the $[CoCl_4]^{2-}$ ion is very simple with only two bands being observable (Fig.5.14 a)). The first is clearly transition (ii), and the second may be transition (iv), but, compared with the $[FeCl_4]$ ion, the energy separation between t_1 and $3t_2$ seems too small for the latter assignment to be certain. The bromide and iodide complexes also show two regions of absorption, with each region having a multiplet due to spin-orbit splitting (Table 5.8). In the iodide complex these regions are separated by about 8000 cm^{-1} which should correspond with the t_1---$3t_2$ separation.

Nickel(II) Tetrahalides: The d^8 ground state is $^3T_1$ and transitions to $^3A_2$, ^3E, $^3T_1$ and $^3T_2$ are all electric dipole allowed. Thus:

Transition	States	Transition	States
i) t_1---2e	-	ii) t_1---$4t_2$	$^3A_2$ ^3E $^3T_1$ $^3T_2$
iii) $3t_2$---2e	-	iv) $3t_2$---$4t_2$	$(^3A_1)$ ^3E $^3T_1$ $^3T_2$
v) 1e---2e	-	vi) 1e---$4t_2$	$^3T_1$ $^3T_2$

(5.29)

Clearly the greater number of allowed transitions results in a greater complexity in the spectra as shown in Fig.5.14 b). Since all the allowed states for a given configuration arise from a single metal state, the various allowed states will not differ greatly in energy and should overlap one another. It is therefore possible to assign groups of bands to each configuration as shown in Table 5.8. In the bromo and iodo complexes, assignments to individual spinor components have been made [390,392], and the reader is referred thereto for further information. Note that in the nickel tetrahalides, the t_1 - $3t_2$ separation again appears quite small.

The MCD spectrum of the $[NiI_4]^{2-}$ ion has been studied in considerable detail [404] with special emphasis on the ^1G d-d state. It is noted that the effective spin-orbit coupling coefficient in the ground state is only ca. 15 cm^{-1}, an observation tentatively ascribed to covalent interaction with the iodide ligands. The 'quenching' increases with the magnitude of the ligand spin-orbit coupling coefficient and so is more important in the iodide than, for example, in the tetrachloride where the effective ground state spin-orbit coupling coefficient is about -190 cm^{-1} [404]. MCD data in the lower energy LMCT region are also presented but not analysed in detail.

Copper(II) Tetrahalides: The d^9 case should be simple, since the metal excited state configuration is closed shell d^{10}. Thus excitation from 1e, $3t_2$ and t_1 will yield ^2E, $^2T_2$ and $^2T_1$ states respectively. The ground state is $^2T_2$ and excitation to each of these excited states is electric dipole allowed. Therefore three regions of absorption should be seen, and are most clearly exemplified in the spectra of the $[CuCl_4]^{2-}$ ion [390,392,401,609] (Fig.5.14 c,d)).

We discuss first the assignment of Rivoal and Briat [392] which seems to confirm this simple picture. However very recent work by Solomon and co-workers suggests that the reality is far more complex and unexpected.

The $[CuX_4]^{2-}$ ions are subject to tetragonal distortion, flattening, resulting in the more appropriate point group being D_{2d}. Under such circumstances ^2E spans $^2A_1 + {}^2B_1$, while $^2T_2$ spans $^2B_2 + {}^2E$ and $^2T_1$ spans $^2A_2 + {}^2E$. The ground state is $^2B_2$ and excitations to ^2E and $^2A_1$ are electric dipole allowed (Fig.5.14 e)). The first two transitions are then assigned as $^2E(D_{2d})$---$^2B_2$ (clear A terms in their MCD spectra) with lower energy shoulders on each band assigned as transitions to the other (forbidden) components of the orbital triplet states (in T_d) (Table 5.8). The third, fairly intense and z polarised band is then an allowed transition to the $^2A_1$ component of ^2E (in T_d) (as shown in Fig.5.14 e). Subject to additional spin-orbit splitting, the same pattern is seen in the spectrum of the $[CuBr_4]^{2-}$ ion, for which resonance Raman data are also available [402]. Indeed the spin-orbit splitting of the first transition ($1t_1(\pi nb)$- $4t_2$ in T_d) is greater by about a factor of four in the bromide relative to the chloride showing that it is the spin-orbit splitting of the halogen which

dominates this excitation.

The Solomon study [609] has already been alluded to in the discussion of the square planar $[CuCl_4]^{2-}$ ion above. The distorted tetrahedral $[CuCl_4]^{2-}$ ion has D_{2d} symmetry with orbital energies in the sequence (based upon calculation):

$$\sigma sCl(1a_1 < 2e < 2b_1) < \sigma pCl(2a_1 < 3b_2 < 3e) < \pi pCl(1b_1 < 3a_1 < 4e < 4b_2 < 5e < 1a_2) < 3dCu(4a_1 < 2b_1 < 6e < 5b_2 \quad (5.30)$$

where orbitals up to and including 6e are filled, and there is a hole in $5b_2$. The sequence in (5.30) is grouped to reflect the parentage in T_d symmetry, namely following (5.24).

Solomon and co-workers [609] note that the x^2-y^2 orbital is stabilised by about 5000 cm^{-1} in proceeding from square D_{4h} $[CuCl_4]^{2-}$ to D_{2d} $[CuCl_4]^{2-}$, and that transition from $\sigma p3e(Cl)$ in the square planar complex occurs at 35,900 cm^{-1}. If the Rivoal assignment is accepted, the $\sigma p(Cl)$ LMCT bands must lie beyond 45,000 cm^{-1}, which is inconsistent with the stabilisation of the x^2-y^2 acceptor orbital. The ^2E state assigned to the 35,000 cm^{-1} in the D_{2d} complex is undoutedly correct, since it clearly yields an MCD A term. However Solomon and co-workers [609] propose that it represents the $\sigma p3e---5b_2$ transition and therefore that the z polarised band at 43,000 cm^{-1} be the $^2A_1$ state derived through the $\sigma p2a_1---5b_2$ transition.

They then conclude that the weak feature near 30,000 cm^{-1} should be re-assigned as the required $\pi p4e---5b_2$ transition (Table 5.8). These rather unexpected assignments were supported by intensity calculations based upon the X-α analysis.

It is initially strange that both the square planar and D_{2d} ions have CT bands at similar energies even though the acceptor x^2-y^2 orbital is much more stable in the latter. This is a consequence, primarily, of the difference in L-L repulsion in the two systems. Thus in the latter case the ligands are further apart such that the ligand orbitals are generally stabilised with respect to the situation in the square planar example. It is also noted [609] that for states which exhibit a large change in metal-ligand overlap upon distortion from square to tetrahedral, changes in CI and many-electron re-organisation can also contribute markedly to CT energies. Other copper halide LMCT spectra are considered in section 5.12 (also see [405]).

5.6 Charge Transfer Spectra of Hexacyanide Derivatives [109-111,115-118,120-124,287]

In constructing a molecular orbital diagram for a hexacyanide, it is usual to assume that the bonding C atom possesses two π orbitals, and two σ orbitals, the latter being nominally associated with the 2s and $2p_z$ orbitals. This assumption led Gray to the MO diagram shown in Fig.5.15, and a hexacyanide spectro-

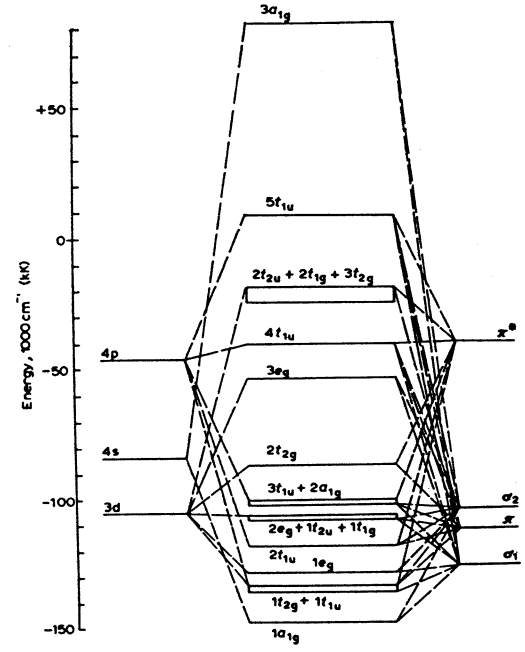


Fig.5.15 Energy levels appropriate for the $[Fe(CN)_6]^{4-}$ ion [121].

scopic study published in several sections [111,115,121]. In general both LMCT and MLCT involving the cyanide groups, may be anticipated. In the discussion which follows the d-d spectra, which may occur at high energies within the general CT manifold, will not be analysed (see appropriate sections of Chapter 6).

The d^6 hexacyanides display the simplest spectra. In view of the large ligand field strength of the cyanide ion, the $t_{2g}-e_g$ interval is generally very large (>20,000 cm^{-1}), and the complexes are low spin $(t_{2g})^6$. LMCT transitions, from lower lying CN states, must terminate on the high energy e_g level (since t_{2g} is full) and such transitions will generally fall beyond 50,000 cm^{-1}. Since the ground state is $^1A_{1g}$, only transitions to $^1T_{1u}$ are fully allowed. There are two such transitions expected, namely $t_{2g}---t_{1u}$ π^* MLCT and $t_{2g}---t_{2u}$ π^* MLCT. These must also possess a moderately high energy and will be observed in the high uv. They are commonly observed [115,121] (Table 5.9).

Alexander and Gray [111] discussed the spectra of other d^n hexacyanides by using a modified Wolfsberg-Helmholz molecular orbital calculation to provide a rough idea of where to expect the various transitions. They also compared different oxidation states and configurations to predict the way in which, energetically, the

various possible LMCT and MLCT transitions should shift.

The ferricyanide ion has a hole in t_{2g} which will permit lower lying LMCT transitions, absent in the ferrocyanide [287]. The MLCT transitions might be expected to shift to the blue. However such an assumption must be used with caution. The effective nuclear charge on the iron atom may not change much in such a covalent species. The prediction ignores any repulsion energy changes, and the t_{2g}-e_g splitting may increase and possibly increase thereby the separation between t_{2g} and ligand acceptor orbital.

In fact, as seen in Table 5.9, the pair of MLCT bands in the ferricyanide ion lie at essentially the same energies as in the ferrocyanide (as with Mn(II) and Mn(III)). However there are four possible LMCT transions from the lower lying $\sigma 3t_{1u}$, $\pi 1t_{2u}$, $\pi 2t_{1u}$ and the low lying $\sigma 1t_{1u}$ to metal $t_{2g}(\pi)$. Significant mixing of the σ and π character of the t_{1u} levels is likely. Note that the e_g level lies about 35,000 cm^{-1} above t_{2g} so that LMCT transitions to e_g are likely in the vacuum uv. The three lower energy possibilities, of these four, are assigned in the ferricyanide case (Table 5.9) [111]. The authors also note that since the final state of the metal ion is $d^6\ ^1A_{1g}$, there should be no multiplet structure to the CT transitions. The lowest lying transition at 23,500 cm^{-1} is relatively weak for a CT transition and its assignment has previously been controversial [122,123]. However its oscillator strength rises slightly with decreasing temperature requiring that it be Laporte allowed (and hence not d-d). The assignments are supported by MCD measurements [109,124]. Data for substituted $[Fe(CN)_5X]^{3-}$ ions (X = NCS, NCSe, N_3^-) [287] have also been reported.

In the case of the isoelectronic $[Mn(CN)_6]^{2-}$ ion, the LMCT transitions might shift to the blue and the MLCT to the red, when compared with ferricyanide, and this is seen to be the situation (Table 5.9). By similar arguments, and with the help of the MO calculation, the other assignments in Table 5.9 were chosen.

Certain general observations could be made as a result of this study. The first LMCT transition shifts monotonically from Cr(III) (>38,600) to Fe(III) (24,000 cm^{-1}). MLCT transitions shift from 38,600 (Cr(III)) to > 50,000 cm^{-1} (Co(III). Interestingly the MLCT bands in the Fe(II), Ru(II) and Os(II) lie at essentially the same energy, as do the lower lying LMCT bands in the Fe(III), Ru(III) [125] and Os(III) cyanides. This is not the case for the corresponding hexahalides (Table 5.9). This is interpreted in terms of much stronger π-back donation to the t_{2g} level in the heavier metal hexacyanides, relative to the hexahalides. Note also that $[Mn(CN)_6]^{3-}$ and $[Mn(CN)_6]^{4-}$ also show their LMCT transitions at roughly the same energy perhaps because the effective nuclear charge on the metal does not change much in the lower oxidation states of cyano complexes.

Table 5.9

CHARGE TRANSFER SPECTRA OF SIX COORDINATE TRANSITION METAL CYANIDE COMPLEXES

Complex	Absorption Spectra	Assignment	Ref
$Cr(CN)_6^{3-}$	38,600(5900)	$^4T_{2u}$ t_{2g}---π^*	111
$Mn(CN)_6^{3-}$	31,100(3670)	$^3T_{2u}$ σt_{1u}---t_{2g} Mn	
	36,700(1170)	$^3T_{1u}$ πt_{2u}---t_{2g} Mn	
	(41,100)(1621)	$^3T_{2u}$ t_{2g}---$t_{1u}\pi^*$	111
$Mn(CN)_6^{4-}$	31,250(2650)	$^2T_{1u}$ σt_{1u}---t_{2g} Mn	
	37,200(12,100)	$^2T_{2u}$ πt_{2u}---t_{2g} Mn	
	41,000(15,050)	$^2T_{1u}$ t_{2g}---$t_{1u}\pi^*$	
	48,000(22,700)	$^2T_{1u}$ t_{2g}---$t_{2u}\pi^*$	111
$Fe(CN)_6^{3-}$	23,500(493)	$^2T_{1u}$ σt_{1u}---t_{2g} Fe	
	33,000(1338)	$^2T_{2u}$ πt_{2u}---t_{2g} Fe	
	38,460(1157)	$^2T_{1u}$ πt_{1u}---t_{2g} Fe	
	44,000(5480)	$^2T_{1u}$ t_{2g}---$t_{1u}\pi^*$	
	50,000(10,780)	$^2T_{1u}$ t_{2g}---$t_{2u}\pi^*$	111
$Fe(CN)_5H_2O^{2-}$	24,600	$^2T_{1u}$ πLMCT	
	27,700	πH_2O---t_{2g} LMCT	
	34,800	$^2T_{2u}$ πLMCT	
$Fe(CN)_5NH_3^{2-}$	25,400	$^2T_{1u}$ πLMCT	110
	26,900	σNH_3---t_{2g} LMCT	
	34,600	$^2T_{2u}$ πLMCT	
$Fe(CN)_5PPh_3^{2-}$	16,000	σPPh_3---t_{2g} LMCT	110
	25,300	$^2T_{1u}$ πLMCT	
	34,200	$^2T_{2u}$ πLMCT	
$Fe(CN)_5(NCS)^{3-}$	19,200(3600)	π(NCS)---t_{2g} LMCT	110
	(25,300)(520)	d-d	
	28,600(995)	σCN---t_{2g} Fe(III)	
	(31,000)(1090)	πCN---t_{2g} Fe(III)	
	(33,300)(1360)	d-d	
$Fe(CN)_5(N_3)^{3-}$	18,000(1580)	$\pi(N_3)$---t_{2g} LMCT	287
	23,900(990)	σ(CN)---t_{2g} Fe(III)	
	(24,700) (33,300)	d-d	
$Fe(CN)_6^{4-}$	45,870(23,700)	$^1T_{1u}$ t_{2g}---$t_{1u}\pi^*$	287
	50,000	$^1T_{1u}$ t_{2g}---$t_{2u}\pi^*$	
$Ru(CN)_6^{4-}$	48,500	$^1T_{1u}$ t_{2g}---$t_{1u}\pi^*$	115
	52,000	$^1T_{1u}$ t_{2g}---$t_{2u}\pi^*$	
$Ru(CN)_6^{3-}$	21,800(3000)	$^2T_{1u}$ σt_{1u}---t_{2g} Ru	115
	28,200(5000)	$^2T_{2u}$ πt_{2u}---t_{2g} Ru	
$Os(CN)_6^{3-}$	24,100(952)	$^2T_{1u}$ σt_{1u}---t_{2g} Os	441
	29,900(1141)	$^2T_{2u}$ πt_{2u}---t_{2g} Os	
	32,400(1218)	$^2T_{1u}$ πt_{1u}---t_{2g} Os	
$Os(CN)_6^{4-}$	47,950(49,300)	$^1T_{1u}$ t_{2g}---$t_{1u}\pi^*$	111
	51,300(25,000)	$^1T_{1u}$ t_{2g}---$t_{2u}\pi^*$	
$Co(CN)_6^{3-}$	49,500(22,900)	$^1T_{1u}$ t_{2g}---$t_{1u}\pi^*$	115
			115

cont

Table 5.9 cont

Complex	Absorption Spectra	Assignment	Ref
$Rh(CN)_6^{3-}$	52,000(2920?)	$^1T_{1u}$ t_{2g}---$t_{1u}\pi^*$	111
$Rh(CN)_5Br^{3-}$	34,800(402)	^1E d-d	
	(42,200)(1700)	πBr---Rh(III)	
	48,800(24,000)	σBr---Rh(III)	113
$Rh(CN)_5I^{3-}$	31,900(1500)	^1E d-d	
	38,800(4200))	πI---Rh(III)	
	45,500(42,000)	σI---Rh(III)	113
$Ir(CN)_6^{3-}$	> 52,000		115
$Ir(CN)_5Br^{3-}$	34,500(70)	^3E d-d	
	38,900(412)	^1E d-d	
	(46,500)(2000)	πBr---Ir(III)	113
$Ir(CN)_5I^{3-}$	31,300(60)	^3E d-d	
	36,400(1400)	^1E d-d	
	(43,500)(3000)	πI---Rh(III)	
	(51,300)(21,800)	σI---Rh(III)	113

The LMCT transitions fall in the sequence:

Ti(III) > V(III) > Cr(III) > Mn(III) = Mn(II) > Fe(III) = Ru(III) = Os(III)

and the MLCT transitions in the sequence

Ir(III) > Rh(III) > Co(III) > Os(II) = Ru(II) > Fe(II) > Cr(III)

For further discussion of cyano derivatives, see the tetracyanides in section 5.5 and 5.12.

5.7 Electronic Spectra of Nitro and Nitrito Derivatives

The nitrite group may be bound through nitrogen (nitro), through oxygen (nitrito, monodentate or bidentate), or act as a bridge in various fashions [362]. Its electronic spectrum can often be used as a guide to the mode of bonding via the presence (or absence) of CT and possible internal ligand modes.

The 8 π-electrons and lone pair of the nitrite ion possess the configuration...$(1a_1)^2(1b_1)^2(b_2)^2(a_2)^2(2a_1^*)^2(2b_1^*)^0$. The nitrite ion itself shows four bands assigned as follows:

$^3B_1$---^{1}A 1, $2a_1^*$---$2b_1^*$ 22,500(0.01) (n--π*)

$^1B_1$---$^1A_1$, $2a_1^*$---$2b_1^*$ 27,000(28) (n--π*)

$^1A_2$---$^1A_1$, a_2----$2b_1^*$ 34,000(9)

$^1B_2$---$^1A_1$, b_2----$2b_1^*$ 46,000(5000)

Both singlet and triplet n---π* transitions exhibit vibrational structure consistent with a progression in an N-O bending mode ($NaNO_2$, 829 cm^{-1}) and an N-O stretching mode ($NaNO_2$ 1018 cm^{-1}) (compared with ground state values of 829 and 1325 cm^{-1} respectively) [379].

The nature of metal-nitrite spectra has been studied in some depth by Caulton and Fenske [366], by Reinen [373,375], by Hitchman [369], by Walker and Lever [372] and by Barnes [367,368] but unequivocal assignments are still elusive. Chemically, nitrite ion can act both as a reducing agent and an oxidising agent leading one to conclude that both MLCT and LMCT transitions might be observed under appropriate conditions. It is useful to begin by considering the spectra of $[M(NO)_2)_6]^{n-}$ species.

Hexanitrometallates containing metals such as the lanthanides which are generally non-redox active, show bands at 46,000, (33,000)w, 28,000(20) and 23,500(1) cm^{-1} [368]. These bands are clearly associated only with the nitro group; in particular the bands at 23,500 and 28,000 cm^{-1} (which show associated vibrational structure) must be the spin-forbidden and spin-allowed n---π* transitions respectively. Similarly weak bands in Tl(I), Cd(II) and Ag(I) nitrite solutions near 22,000 cm^{-1} are assigned as spin forbidden n---π* transitions [442,443]; the intensities, while low, are considerably greater than in the free nitrite ion, probably because of spin-orbit mixing induced by interaction with the heavy metal ion.

Solid hexanitrometallate complexes with redox active metal ions show bands in the visible and near ultraviolet correlated as follows [366-368]:

		(i)	(ii)	(iii)	(iv)	(v)	Ref.
$K_2BaFe(NO_2)_6$	d^6	20,000w	24,400s	31,000s	40,800s	50,000s	366
$Cs_3Co(NO_2)_6$	d^6	21,500w	26,800s	37,600s		48,600s	366
$Na_3Rh(NO_2)_6$	d^6		31,500s	37,400s		>50,000s	366
$K_2BaCo(NO_2)_6$	d^7	23,000w	28,400s	34,400s	39,900	49,500s	366
$K_2CaNi(NO_2)_6$	d^8	13,600w	28,600s	33,000s	(40,200)w	48,500s	366
$M_3Ce(NO_2)_6$	f^1		23,500s	29,000s		46,000br	368
$M_4Cd(NO_2)_6$	d^{10}		21,500s	26,500s			367
$M_4Pb(NO_2)_6$	s^2		21,500s	26,500s			367
$M_3Bi(NO_2)_6$	s^2		21,500s	26,500s			367
							(5.31)

In the first group bands (ii) and (iii) follow a common sequence of:

Fe(II) < Co(III) < Rh(III) > Co(II) = Ni(II)

while bands (iv) and (v) appear to follow an inverse relationship. Note that Caulton and Fenske assume that band (iv) in the Co(III)

complex is hidden by band (iii) (and is therefore lower in energy than in the Fe(II) complex).

On the basis that the Fe(II) complex excitation was lower in energy than the Co(III) complex excitation, bands (ii) and (iii) were assigned [366] as MLCT from t_g (point group T_h) to π^* (NO_2) t_u. Bands (iv) and (v) were then assigned as LMCT from $\pi(NO_2)$ t_u to de_g. This assignment, however, creates many problems, as was first pointed out by Barnes and co-workers [367]. In particular it is surprising that the excitation from Co(II) lies at a higher energy than from Co(III), especially as the former is low spin and has a very unstable electron in z^2.

If the first strong bands are MLCT then the optical electronegativity of the acceptor orbital on NO_2 must be about 1.25 (based upon Co(III) and Fe(II) data). With this value, the Co(II) and Ni(II) complexes should show strong bands very much lower in energy than ca. 28,000 cm^{-1}. Caulton and Fenske considered that a weak band near 13,600 cm^{-1} in the nickel complex (and 15,500 cm^{-1} in the Co(II) complex) might be e_g---π^*. This would be consistent with the Co(III) and Fe(II) data if these transitions can be assigned as MLCT, but then it is necessary to explain why they are so weak.

An LMCT assignment must be considered as an alternative possibility. Using the Fe(II) and Co(III) data and the estimate of 27,400 cm^{-1} for 10Dq in the cobaltinitrite [366], an optical electronegativity value for the donor orbital on NO_2 is calculated to be about 2.4 [509]. Using this value, bands in the 28,000 cm^{-1} region for L---e_g LMCT transitions in the Co(II) and Ni(II) complexes are reasonable. The assignment is also consistent with the fact that Mn(II) doped into $[Cd(NO_2)_6]^{4-}$ shows no intense absorption in the visible region. If MLCT transitions were involved, such absorption must be expected [367].

The data in the second group of complexes in (5.31) are also assumed to be MLCT in origin [367,368] based on the earlier MLCT assignment [366]. While, energetically this is not unreasonable for Ce(III), the relatively low energies for Cd(II) is puzzling, as is the insensitivity between Cd(II), Pb(II) and Bi(III) spectra. At this time it is not clear whether the MLCT or LMCT assignment should accepted for the lower energy bands in the hexanitrito systems. More study is obviously necessary to obtain an understanding of the spectra of these species.

Note in passing that many salts of the type MM'[M"$(NO_2)_6$] show charge transfer transitions, in the solid state, between M" and M or M', somewhat like the intervalence transitions [367] (see Chapter 7).

Consider now complexes with nitrite and other ligands. Chromium(III) complexes show the n---π^* internal nitrite transition as a weak structured band near 28,000 cm^{-1}, and in these cases its assignment is probably secure [363,364]. A similar band is seen near 20,000 cm^{-1} in many nickel(II) nitrogen-bound nitro complexes but not in oxygen-bound monodentate nitrito complexes [369,500-505].

Its intensity is very similar to that of a d-d transition and it falls near the expected position of the $^3T_1(F)$---$^3A_2$ Ni(II) d-d transition. It has therefore often been erroneously so assigned. However its position in a nitro nickel(II) complex is roughly independent of other ligands and of the number of nitro groups present, i.e. it does not obey the average environment rule [504].

In the spectra of cooled crystals, this '20,000 cm^{-1}' band usually shows structure in a ca. 600 cm^{-1} vibration, presumably the bending mode. Such structure may also be seen in room temperature spectra, though not in every case. It is clear that this band is not d-d in origin. In some cases both the $^3T_1(F)$ transition and this 'extra' transition are clearly separated [369,375,503].

In principle this transition could be MLCT, LMCT or internal n---π^* NO_2. Hitchman [369] argues that the reduced value of the bending mode (relative to the ground state) requires that the N-O bond order be reduced which requires the addition of an electron to π^* (MLCT, or internal n--π^*) not loss of an electron (LMCT); this is in agreement with earlier considerations [366,373-375]. Presumably occurring as low as 21,000 cm^{-1}, an n---π^* assignment requires the transition to be spin-forbidden (see 5.30) [370,371]. Its intensity is generally slightly greater than the d-d band intensities which seems inconsistent with a spin-forbidden assignment unless some kind of coupling to the nickel(II) state provides some extra probability. For this reason, Hitchman prefers an MLCT assignment (from e_g to π^*). This assignment also has problems as indicated above in the hexanitrometallate discussion. Nevertheless, from a diagnostic point of view, the presence of this structured band near 20-21,000 cm^{-1} seems indicative of nitro bonding to Ni(II) (including bridging NO_2 groups containing an Ni-N bond).

Monodentate nitrito bound to nickel(II) does not exhibit the '20,000 cm^{-1}' band possibly because the direct bond between metal and oxygen destabilises the π^* orbital and shifts the transition to the blue. However nitrito bound bidentate to nickel generates a weak and structured band centered near 22-23,000 cm^{-1} together with a stronger structured band near 27,000 cm^{-1}. These are assigned as spin forbidden and spin allowed n---π^* internal transitions of the nitrite ion respectively [372] (see Fig.6.51 (c)).

Some aspects of the vibrational structure in these two bands are worth comment. Firstly the vibrational spacing of the bending mode in the 22,000 cm^{-1} band, in the bidentate nitrito complexes, is generally (though not exclusively) greater than the values found in the '20,000 cm^{-1}' band in nitro nickel(II) complexes, ca. 630-680, versus ca. 600 cm^{-1} respectively. Secondly, the N-O stretching mode generally appears in a short progression on the higher energy spin-allowed transition in oxygen bound nitrito nickel(II) complexes. Thirdly, significant progressions in the bending mode appear on the spin allowed band in monodentate oxygen bonded nitrito, and in the nitrite ion (often the counter ion in many nitrite complexes) but they are much less obvious in bidentate

nitrito.

See [362,369,372] for further detailed and recent discussion. Whatever the true assignment, the spectra of nitro and nitrito (monodentate and bidentate) bound to nickel(II) are sufficiently different to allow use as a fingerprint for structure determination; be wary however of mixed spectra from two types of nitrite coordination.

5.8 Metal Carbonyl Charge Transfer Spectra [115,127,137,206,207]

It is convenient to sub-divide this section into binary carbonyls of various configurations, followed by mixed carbonyl ligand systems [610]. In general the CO group behaves much as the CN^- group, with which it is iso-electronic. However although MLCT transitions to the CO group are commonly observed, the energetics of the system are such that LMCT from σ or π levels of CO to the metal ions are only rarely identifiable [115,127,206].

In a hexacarbonyl derivative, the two low lying π^* orbitals are $t_{1u} < t_{2u}$ (Fig.5.16). Transitions from $d(t_{2g})$ to these two levels, generate states as shown in Table 5.10. These states were derived using the procedure outlined in section 5.2, i.e. by obtaining the direct product of all the possible metal states (shown in Table 2.2) with the hole in the ligand group orbital (t_{1u} or t_{2u}).

The d^6 species are expected to have simple spectra since there is only one predicted transition for each excitation. Other d^n species should be more complex. Data are presented in Tables 10-12.

Table 5.10

SYMMETRY SPECIES OF POSSIBLE SPIN-ALLOWED CHARGE TRANSFER BANDS IN d^4, d^5 AND d^6 BINARY CARBONYLS

Excitation	Configuration and ground term d^4 ($^3T_{1g}$)	d^5 ($^2T_{2g}$)	d^6 ($^1A_{1g}$)
t_{2g}---t_{1u}	$4\ ^3T_{2u} + 3\ ^3T_{1u}$ $2\ ^3E_u + {}^3A_{1u}$ $+ (^3A_{2u})$	$3\ ^2T_{2u} + 4\ ^2T_{1u}$ $2\ ^2E_u + {}^2A_{2u}$ $+ (^2A_{1u})$	$^2T_{1u} + (^2T_{2u})$ $(^2A_{2u}) + (^2E_u)$
t_{2g}---t_{2u}	$3\ ^3T_{2u} + 4\ ^3T_{1u}$ $2\ ^3E_u + {}^3A_{1u}$ $+ (^3A_{2u})$	$4\ ^2T_{2u} + 3\ ^2T_{1u}$ $2\ ^2E_u + {}^2A_{2u}$ $+ (^2A_{1u})$	$^2T_{1u} + (^2T_{2u})$ $(^2A_{1u}) + (^2E_u)$

All transitions are orbitally permitted except those shown in parenthesis. These data were obtained from the direct product of the symmetry of the ligand state with the possible metal states as shown in Table 2.2.

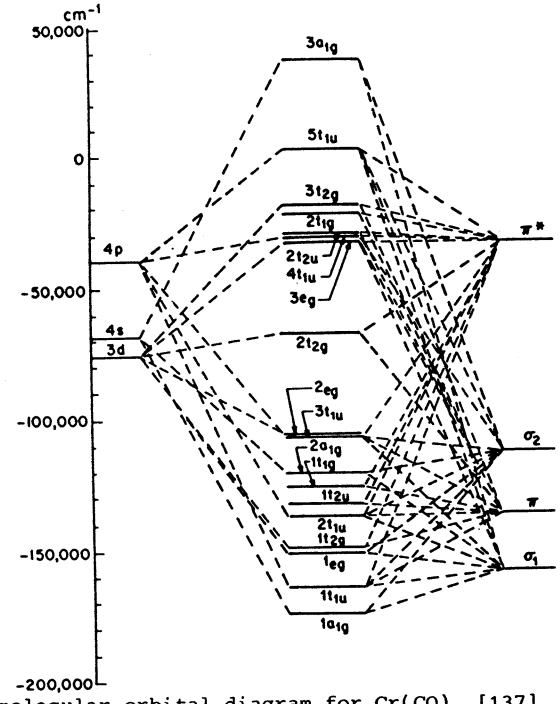
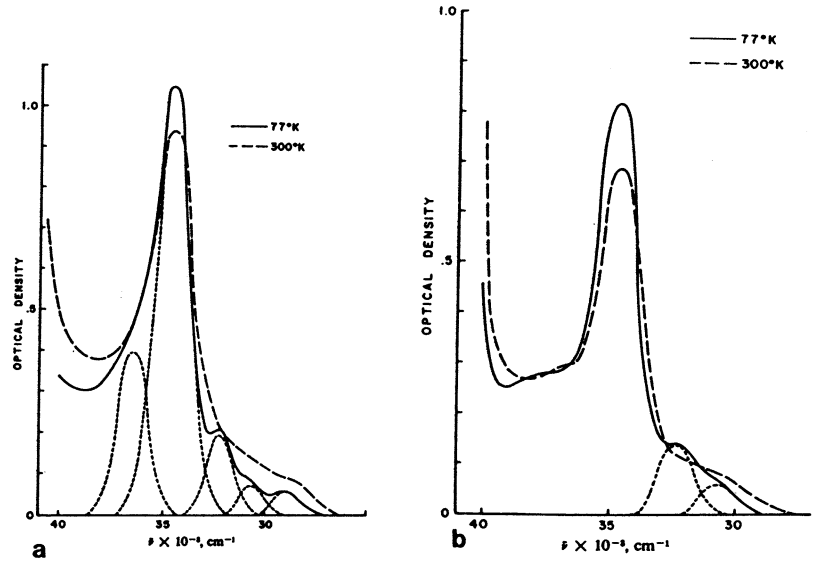
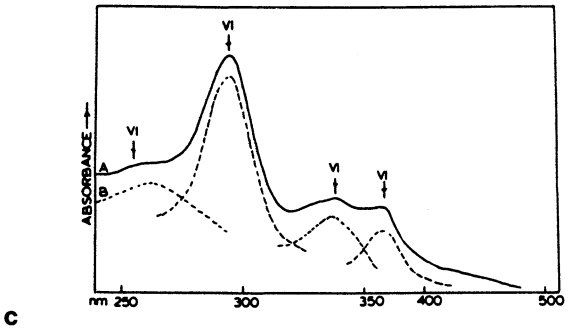


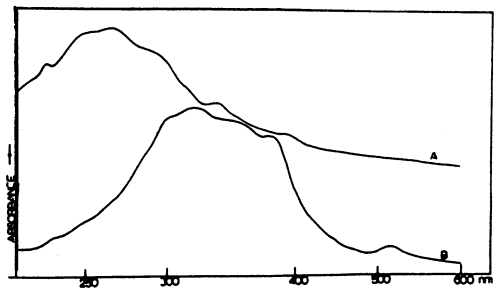
Fig.5.16 A molecular orbital diagram for $Cr(CO)_6$ [137].

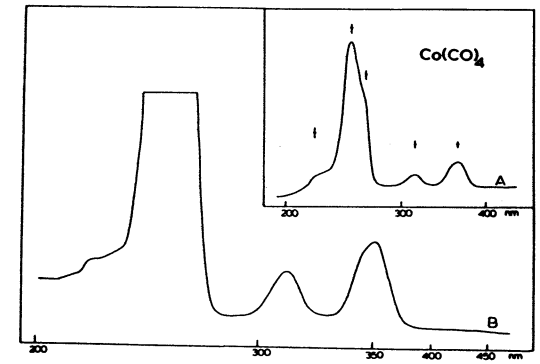
5.8 i) The d^4 and d^5 Binary Carbonyls

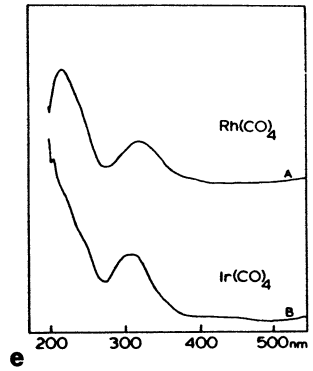
The d^5 $V(CO)_6$ species has broad ill defined absorption peaking near 40,000 cm^{-1} (Fig.5.17) [207]. The ground configuration is $...(t_{2g})^5$ and the first excited CT configuration is $(t_{2g})^4(Lt_{1u})$. Following section 5.2, the lowest energy states of $(t_{2g})^4(Lt_{1u})$ are obtained from $^3T_{1g} \times t_{1u} = {}^2T_{1u} + {}^2T_{2u} + (^2A_{1u}) + (^2E_u)$, since the lowest energy state of $(t_{2g})^4$ is $^3T_{1g}$. Thus the Mt_{2g}---Lt_{1u} excitation contains two electric dipole allowed transitions, which may be split by lower symmetry perturbation, spin-orbit coupling (small in this case) or vibronic interactions. The many higher energy states of $(t_{2g})^4(Lt_{1u})$ (Table 5.10) probably overlap the two low energy states similarly derived from the excitation to $(t_{2g})^4(Lt_{2u})$; thus the spectrum is complex (Fig.5.17). However it is appropriate to assign the centres of the two broad overlapping regions of absorption to the t_{2g}---t_{1u} and t_{2g}---t_{2u} transitions. Note however that a recent calculation [444], while confirming the existence of the transitions cited above, in the near uv, also places the LMCT transition Lt_{1u}---dt_{2g} in the near uv where it probably adds to the complexity of the spectrum.











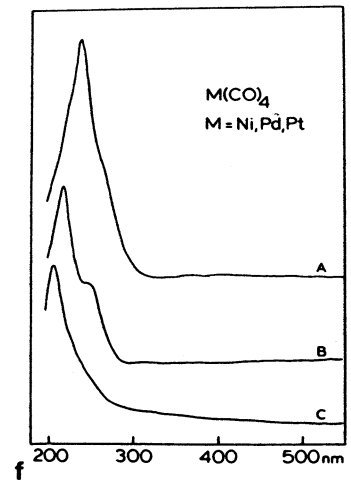


Fig.5.17 Previous page (a) The electronic spectra of $W(CO)_6$ in EPA (0.80×10^{-4} M) at 77 and 300 K [137]. (b) The electronic spectra of $Mo(CO)_6$ in EPA (0.55×10^{-4} M) at 77 and 300 K [137]. (c) (Upper panel) Absorption spectra of $Ti(CO)_6$ formed from Ti/CO co-condensation at 10 K. VI refers to this species [135]. (Lower panel) Absorption spectra of A:$V(CO)_6$ (from V/CO co-condensation at 10 K) and B:$V(N_2)_6$ (from co-condensation of V/N_2 at 20 K) [135]. This page d) Matrix spectrum of the co-condensation product of Co/CO at 12 K. B is a scale expansion of A [139]. e) Matrix spectra of the co-condensation products of Rh/CO and Ir/CO at 12 k [139]. f) Matrix spectra of the co-condensation products of A:Ni/CO, B:Pd/CO and C:Pt/CO at 12 K. All co-condensation reactions were carried out at a ratio of $1/10^4$ M/CO [139].

This latter study [444], comments on the black colour of solid state $V(CO)_6$ (the solutions are yellow). A broad band at 17,250 cm^{-1} is responsible. It is argued that the transition is due to the process:

$$2\ V(CO)_6 \xrightarrow{h\nu} [V(CO)_6^+].[V(CO)_6^-]$$

with an energy of:

$$h\nu = IP[V(CO)_6] - EA[V(CO)_6] - e^2/r(V\text{-}V) \qquad (5.32)$$

where r(V-V) is the vanadium-vanadium separation in the solid state. The calculation is in good agreemment with experiment. Both the ionisation potential and electron affinity of $V(CO)_6$ are fairly low contributing to the low energy of the transition. Mixed solids containing $V(CO)_6$ and another carbonyl such as $Cr(CO)_6$ also show this phenomenon but the absorption is displaced to the blue [444].

The lowest excited configuration of d^4 $Ti(CO)_6$ is $(t_{2g})^3(Lt_{1u})$ and the lowest energy state of $(t_{2g})^3$ is $^4A_{2g}$. Thus the symmetry of the lowest CT state is $^4A_{2g} \times t_{1u} = {}^3T_{2u}$, and this should be well separated from higher energy states of $(t_{2g})^3(Lt_{1u})$. Since only one state is generated, rather than two as with $V(CO)_6$, a simpler spectrum is anticipated. Indeed as seen in Fig.5.17, the absorption is clearly resolved into two fairly narrow regions of absorption [135]. Lower energy absorption in the spectra of these species is d-d in origin.

5.8 ii) d^6 Binary Carbonyls

Hexacarbonyls [115,130,137,207]:- Gray and Beach were the first to clearly identify the transitions to be anticipated for octahedral d^6 hexacarbonyl species [115,137]. Because of the pseudo closed shell $(t_{2g})^6$ configuration, no LMCT transitions are expected below 50,000 cm^{-1}, and the upper part of the MO diagram in Fig.5.17 is relevant. Some representative spectra are shown in Fig.5.16. In parallel with the previous discussion of hexacyanide spectra, assignments for the hexacarbonyls follow readily (Table 5.11).

Wrighton and co-workers [107] investigated a range of d^6 hexacarbonyl anions and identified the $t_{2g}(M)\text{---}\pi^*t_{1u}$ MLCT transition (Table 5.11).

Crystal field d-d spectra were also reported for many of these ions and vibrational structure, probably associated with the M-C stretch, observed on the $^3T_{1g}$ emission of $[Ta(CO)_6]^-$.

The sequence of MLCT energies follows:

Nb(-I) = Ta(-I) < V(-I) < Mo(0) = W(0) < Cr(0) < Mn(I) = Re(I)

Table 5.11

CHARGE TRANSFER SPECTRA OF d^6 BINARY CARBONYL COMPLEXES

Complex	Absorption Spectrum	Assignment	Ref
$V(CO)_6^-$	23,200(300) 25,100	$^1T_{1g}$ d-d	
	28,400(6240)	$V\text{---}\pi^*$ t_{1u}	
	31,100(3300)	$^1T_{2g}$ d-d	
	37,550(60,900)	$V\text{---}\pi^*$ t_{2u}	107,136,137
$Cr(CO)_6$	29,500(700)	$^1T_{1g}$ d-d	
	31,550(2670)	$^1T_{1g}$ d-d	
	35,700(13,1009	$Cr\text{---}\pi^*$ t_{1u}	
	44,480	$Cr\text{---}\pi^*$ t_{2u}	
	51,280w	$t_{2g}(\pi)\text{---}t_{2g}(\pi^*)$	115
$Cr(CO)_5$	18,450(3200)	$^1E\text{---}^1A_1$	
	42,200(39,500)	$d\text{---}\pi^*$	127
$Mn(CO)_6^+$	33,250(600)	$^3T_{1g}$ d-d	
	37,300(1100)	$^1T_{1g}$ d-d	
	39,600(2200)	$^1T_{1g}$ d-d + v	
	44,500(16,000)	$Mn\text{---}\pi^*$ t_{1u}	
	49,900(27,000)	$Mn\text{---}\pi^*$ t_{2u}	130,137
$Mo(CO)_6$	28,850(350)	$^3T_{1g}$ d-d	
	30,150(1690)	$^1T_{1g}$ d-d	
	31,950(2820)	$^1T_{1g}$ d-d	
	34,600(16,800)	$Mo\text{---}\pi^*$ t_{1u}	
	37,680w	$^1T_{2g}$ d-d	
	43,860s	$Mo\text{---}\pi^*$ t_{2u}	
	51,810m	$t_{2g}(\pi)\text{---}t_{2g}(\pi^*)$	107,115
$Mo(CO)_5$	23,000(3300)	$^1E\text{---}^1A_1$ d-d	
	40,800(20,000)	$d\text{---}\pi^*$	142,127
$K(dig)Nb(CO)_6$	22,730(1500)	$^3T_{1g}$ d-d	
	24,390	$^1T_{1g}$ d-d	
	26,660(6000)	$Nb\text{---}\pi^*$ CO	107
$W(CO)_6$	28,300(1000)	$^3T_{1g}$ d-d	
	29,950(1680)	$^1T_{1g}$ d-d	
	31,850(3250)	$^1T_{1g}$ d-d	
	34,650(17,600)	$W\text{---}\pi^*$ t_{1u}	
	37,170w	$^1T_{2g}$ d-d	
	44,640s	$W\text{---}\pi^*$ t_{2u}	
	50,510	$t_{2g}(\pi)\text{---}t_{2g}(\pi^*)$	
	52,630	$t_{2g}(\pi)\text{---}t_{2g}(\pi^*)$	107,115
$W(CO)_5$	22,700(6300)	$^1E\text{---}^1A_1$ d-d	
	42,200(63,500)	$d\text{---}\pi^*$	142,127
$(Ph_4As)Ta(CO)_6$	22,170	$^3T_{1g}$ d-d	
	24,690	$^1T_{1g}$ d-d	
	26,180	$Ta\text{---}\pi^*$ CO	107

cont

Table 5.11 cont

Complex	Absorption Spectrum	Assignment	Ref
$Re(CO)_6^+$	36,850(710)	$^3T_{1g}$ d-d	
	38,500(1500)	$^1T_{1g}$ d-d	
	40,700(2900)	$^1T_{1g}$ + v d-d	
	44,500(20,000)	Re---π^* t_{1u}	
	47,100(4600)	$^1T_{2g}$ d-d	
	51,200(77,900)	Re---π^* t_{2u}	137

In parallel with the hexacyanide derivatives (but not hexahalides) the constancy of band energy down a periodic group implies that π-back donation increases down the group. Beach and Gray also note that the t_{2u} π^* orbital does not interact, for symmetry reasons, with metal orbitals, while t_{1u} π^* will scramble with metal p_z. Thus the difference in energy between these two orbitals, as indicated by the difference in the two MLCT transitions, reflects the degree of p_z interaction. This increases in the order $Ti(CO)_6$ < $V(CO)_6$ < $Cr(CO)_6$ < $Mo(CO)_6$ < $W(CO)_6$ suggesting that np bonding increases down the group, and accross the period. This should also be a measure of π-back donation. Note also that the intensity of the transition to t_{2u} should increase with increasing π-back donation [115]. There is an apparent relative increase in the sequence $Ti(CO)_6$ < $V(CO)_6$ < $Cr(CO)_6$ in agreement with intuitive expectation [207].

Trogler and Solomon [144] studied both the MLCT transitions in $[M(CO)_6]$ (M = Mo,W) in argon matrices at 10 K. The lower energy t_{2g}---t_{1u} π^* transition show considerable vibrational structure associated with the excited state a_{1g} M-C stretching vibration (ca 400 cm^{-1}). In the tungsten complex, there is a further progression in this frequency based upon the C-O a_{1g} stretch, i.e. with an origin of energy E(MLCT) + CO. Analysis of the vibrational structure reveals a substantial lengthening of the M-C bond (ca 15pm) but little change in CO bond length. An X-α calculation confirmed the veracity of the assignments.

Photolysis of many $M(CO)_5L$ systems results in the scission of the M-L bond and formation of $M(CO)_5$ which can be trapped in a matrix at cryogenic temperatures [181-185] (Table 5.11). The electronic spectra are dependent upon the nature of the matrix (e.g. frozen argon, frozen xenon etc) resulting in some controversy concerning whether the inert gas actually occupies a coordination site of the electronically deficient $M(CO)_5$ fragment [194]. For example the d-d and MLCT transitions for $Cr(CO)_5$ fall at 18,800 (d-d) and 42,000(MLCT) in argon at 10 K, and at 20,200(d-d) and 40,800(MLCT) in xenon at 10 K [184]. The d-d transition is nominally from xz,yz to z^2 (^1E) and so will reflect the position of the z^2 orbital and hence any interaction along the 'vacant' axis. When this position is occupied by a genuine ligand such as, for example, trimethylamine, this d-d transition occurs near 23,800 cm^{-1}; thus any interaction with the inert gas matrix is very weak. Substituted electronically deficient fragments can also be prepared, e.g. $Cr(CO)_4NMe_3$ which absorbs at 20,000(d-d), 28,900(d-d) and 40,650(MLCT) in xenon at 10 K [184]. Photolysis of such species results in the formation of several different isomers whose spectra have been discussed [184].

5.8 iii) d^8 Binary Carbonyls

Iron pentacarbonyl, $Fe(CO)_5$, a trigonal monomeric species, is the most important species in this group. Its spectrum consists of a low lying weak d-d band (xz,yz---z^2) and two higher lying transitions of the MLCT type from Fe(0) to π^* CO [141] (Table 5.12). Cobalt(I) complexes, such as $[Co(CNR)_5]^+$ are isoelectronic and have very similar spectra to $Fe(CO)_5$, e.g. $[Co(CNMe_3)_5]^+$, 35,600 d-d 5e'---$5a_1$', 43,700 MLCT 5e'---π^* CO, [205].

5.8 iv) d^9 Binary Carbonyls

Odd d^n configurations usually form binuclear carbonyl derivatives which are discussed in Chapter 7. However under certain conditions, e.g. photolysis in low temperature matrices, monomeric and paramagnetic species can be obtained.

The $M(CO)_4$ (M = Co,Rh and Ir) species are reported [206] to have spectra (Fig.5.17 d,e)) which include relatively low energy bands some 10,000 cm^{-1} below the first bands observed in $[M(CO)_4]$ M = Ni,Pd and Pt (Fig.5.17 f)). These low lying bands are unlikely to be d-d in origin since this would imply a very large tetrahedral d-d splitting. Rather, note that these species are open shell 17 electron systems with a hole in t_2(d). It is therefore likely that these low lying transitions (Table 5.12) are σ---t_2 LMCT from CO to M. The photoelectron spectrum of $Ni(CO)_4$ leads to the suggestion that the σ--t_2 separation in this species is about 42,000 cm^{-1} [208]; in the 17 electron systems, the separation between the states concerned should be smaller because a closed 18 electron shell is formed about the metal ion. The cobalt complex shows significant structure probably due to a C_{3v} distortion [209-211]. The rhodium and iridium features are also rather broad and may encompass several transitions. A very weak low lying feature (Fig.5.17 e)) in the iridium complex is probably spin forbidden. Note that a recent calculation suggests that the LMCT and MLCT assignments may be mis-assigned and require interchanging [611].

5.8 v) d^{10} Binary Carbonyls

The group $M(CO)_4$ (M = Ni,Pd and Pt) are the most well known (Fig.5.17 f)) (Table 5.12). The nickel and palladium complexes show a single strong band in the high uv plus a well developed shoulder some 6000 cm^{-1} to lower energy [206]. The platinum derivative shows a similar high energy band but there is no shoulder.

Table 5.12

ELECTRONIC SPECTRA OF BINARY CARBONYL DERIVATIVES

Complex	Electronic spectra	Assignment	Ref
$Ti(CO)_6$	27,200 29,800	d-d	
	33,900	t_{2g}---t_{1u} MLCT	
	38,000	t_{2g}---t_{2u} MLCT	135
$V(CO)_6$	25,850 30,500	d-d	
	32,150 33,800 37,450	t_{2g}---t_{1u} MLCT	
	40,650 44,250	t_{2g}---t_{2u} MLCT	135
$Fe(CO)_4^-$	33,000		127
$Fe(CO)_5$	35,500(3800) 35,450(3900)	5e'---5a_1' d-d	
	41,500(10,200) 41,670(15,700)	5e'---6e' MLCT	
		5e'---2a_2' MLCT	
	50,000(37,000)	MLCT	141
$Co(CO)_4$	28,200	σ---t_2(d) LMCT ^2E--$^2A_1$	
	(28,900)	σ---t_2(d) LMCT	
	31,720	σ---t_2(d) LMCT $^2A_1$--$^2A_1$	
	(37,200)	d----π^* MLCT	
	39,060	d---π^*	
	44,840	d---π^*	139,140
$Rh(CO)_4$	29,850	σ---t_2(d) LMCT	
	41,700	d----π^* MLCT	
	46,500	d---π^*	139
$Ir(CO)_4$	32,600	σ---t_2(d) LMCT	
	(40,650)	d----π^* MLCT	
	(48,100)	d---π^*	138
$Ni(CO)_4$	(36,650)	t_2(d)---$t_2(\pi^*)$ MLCT	
	41,700	d---π^*	139
$Pd(CO)_4$	(40,000)	t_2(d)---$t_2(\pi^*)$	
	45,900	d---π^*	139
$Pt(CO)_4$	48,310	d--p or d---π^*	139
$Fe(CO)_4NMe_3$	21,050(d-d);36,400, 45,450(d---π^*);	$Fe(CO)_4Py$ 27,800(d---π^*Py); 35,700(d---π^*CO);45,450(d---π^*).	

There have been many MO treatments of $Ni(CO)_4$ [208,212-218], and also PES studies [208]; it is evident that the d(e,t_2) levels are split by about 5000 cm^{-1} while the t_2 σCO bonding orbitals lie about 35,000 cm^{-1} below the d manifold. The low energy shoulder is then assigned to d(t_2)---$t_2(\pi^*)$ while the more intense absorption probably contains several transitions from d(e)---π^* and t_2---π^* [206,219]. The band in the platinum complex may also possess d-p character.

5.8 vi) Binary Di-Nitrogen Complexes

The spectra of $M(N_2)_6$ (M = Ti,V and Cr) are very similar to those of the corresponding $M(CO)_6$ species, but are displaced to lower energy [207,224]. Since N_2 is isoelectronic with CO, the similarity is not surprising. However, the shift to lower energy is initially surprising. The lowest π^* level on N_2 lies at approximately the same energy as in CO [220-222]. PES and vibrational data lead to the conclusion that N_2 is a poorer π-acceptor than CO. Thus the off-diagonal element $\langle d|H|\pi^* \rangle$ is greater for metal carbonyls than for metal di-nitrogen complexes [207,223]. It is this fact which shifts the CO MLCT transitions to higher energy (by a factor of about 1.25) with respect to their di-nitrogen analogs.

As indicated above, the separation between the two π^* levels increases with increasing π-back donation. It is seen to be generally smaller in the di-nitrogen complexes compared with the carbonyl derivatives. Estimated 10Dq values for the hexa-di-nitrogen complexes are about 21,300 - 22,500 cm^{-1} compared with about 28,000 - 33,250 cm^{-1} for the carbonyls [207].

5.8 vii) d^6 Mixed Ligand Carbonyl Derivatives

This represents an enormous field of study since such systems are often particularly stable. Complexes of the general classes $M(CO)_5L$ and $M(CO)_4L_2$ (M= Cr,Mo,W), $M(I)(CO)_5X$ (M = Mn,Re), $M(II)(CO)_4X_2$ (M = Fe,Ru,Os) etc offer rich spectra which will be covered in a general fashion in this section.

In complexes of the type $W(CO)_5L$, the lowest state may be d-d or W---L CT depending upon X (Table 5.13) [22,193]. These systems were thoroughly explored by Wrighton [22,127] among others [191], who showed that with poorly accepting ligands such as piperidine, the lowest two transitions were the weak ^3E followed by a relatively strong ^1E d-d transition. At somewhat higher energies came the first W--L MLCT transition. In general, with varying L, the d-d transitions remained roughly constant in energy while the CT band moved with the accepting power of the ligand L. The piperidine complex has no low lying CT bands so that comparison of its spectrum with that of other ligands led readily to the identification of the W---L CT bands (Table 5.13). In a series of substituted pyridines, the MLCT band were observed to move from about 21,300 cm^{-1} with L = 4-formylpyridine, to 28,800 cm^{-1} with L = 3,4-dimethylpyridine. Since the ^3E and ^1E transitions remain constant at approximately 23,000 and 25,000 cm^{-1} respectively, proper choice of L permits either the CT or a d-d state to lie lowest in energy. The emission and photosubstitution properties of this series of complexes are determined by this consideration [22,193]. Oskam and Rest [149] reported emission data for a range of molybdenum and tungsten pentacarbonyl derivatives in cryogenic media.

Since CO will almost invariably have a stronger ligand field than the ligand L, the octahedral levels in $M(CO)_6$ will split in

$M(CO)_5L$ to provide $z^2(a_1) < x^2-y^2(b_1)$. In configurational terms the first ^1E d-d state must then correspond primarily with the transition $(b_2)^2(e)^4 \text{---} (b_2)^2(e)^3(a_1)$; this is confirmed both by MCD [159,171,186,189] and PES data [169,170]. In the former case the presence of an A term confirms an E transition and its sign distinguishes this transition from the alternative transition to x^2-y^2 (b_1). A weaker transition near 30,000 cm^{-1} in complexes where M---L CT absorption does not interfere (e.g. L an alkylamine) may be $(b_2)^2(e)^4 \text{---} (b_2)(e)^4(b_1)$ $^1A_2 \text{---} ^1A_1$, which would require vibronic activation. Note that when L is a phosphine or similar soft ligand, the MCD results reveal less angular momentum in the excited state indicative of greater covalent interaction between metal d and ligand p states in the phosphine than in the N ligands [171].

Since the t_{2g} level (in O_h) is split in C_{4v}, the CT transition labelled d---π^* in Table 5.13 should, in fact, contain at least two components, nominally b_2---π^* and e---π^*. Stufkens and co-workers have extensively explored systems of this general class via resonance Raman spectroscopy and identify structured resonance Raman excitation spectra lying underneath the CT band in terms of these components [172].

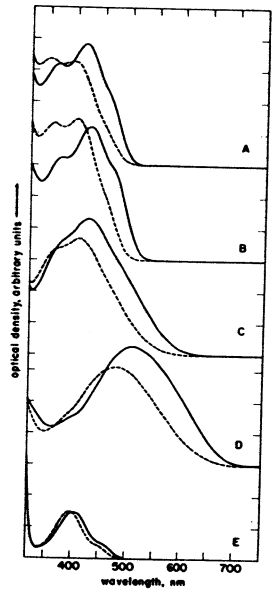


Fig.5.18 Absorption spectra at 298 K in benzene/iso-octane (1:4 v/v) and benzene/iso-octane/ethanol (1:4:5 v/v) (hatched) for cis-$W(CO)_4L_2$ where L is A:3,4-dimethylpyridine, B:pyridine, C:3-benzoylpyridine, D:4-benzoylpyridine and E:tetramethylethylenediamine [155].

Table 5.13

ELECTRONIC SPECTRA OF d^6 CARBONYL DERIVATIVES

Complex	Electronic spectra	Assignment	Ref
$Cr(CO)_5pip$	23,800(3500)	$^1E \text{---} ^1A_1$	
	40,000	d---π^*	127,133
$Cr(CO)_5Py$	25,650(5600)	$^1E \text{---} ^1A_1$?	
	41,300(34,700)	d---π^*	127,133
$Cr(CO)_5Pyz$	21,600	d---π^*(pyz)	
	24,800	d-d	
	34,500	d---π^*	
	41,000	d---π^*	105,106
$Cr(CO)_5PCl_3$	27,600	d-d	
	34,800	d---π^*(CO)	
	39,700	d---π^*(PCl_3)	
	44,100	d---π^*(CO)	105,106
$Cr(CO)_5Br^-$	22,600(840) (28,200)sf	d---σ^* Br,π^*	
	40,320(24,000) 45,450(3000)	d---π^*	
	50,500(30,000)	d---π^*	128
$Cr(CO)_5I^-$	22,850(1000) (28,500)sf	d---σ^* I, π^*	
	39,500(18,000) 44,400(1100)	d---π^*	
	45,900(22,800)	σ---σ^*(I)	
	50,250(17,000)	d---π^*	128
$Cr(CO)_4en^a$	23,600(4900)	$^1A_1, ^1B_2 \text{---} ^1A_1$ d-d	
	29,700(5400) (35,500)	d---π^*	
	40,300(18,200) 46,200(21,900)	d---π^*	126
$Cr(CO)_4bipy^b$	20,100(3700) (24,750)d-d?	d---bipy π^*	
	30,000(6900)	d---π^*	
	33,780(27,500) (34,700)	bipy π---π^*	
	40,900(28,850)	d---π^*	102,126
$Cr(CO)_5imid$	24,900(3200)	^1E d-d	
	39,700(28,000)	d---π^*	114
$Cr(CO)_5thz$	24,600(2650)	^1E d-d	
	(24,900)	d---π^* thz	
	41,700(25,000)	d---π^* CO	114
$Mo(CO)_5pip$	25,100 (32,400) (34,900)		
	39,750(58,000)	d---π^* CO	134
$Mo(CO)_5Cl^-$	24,600(2150) (28,600)d-d sf	d---a_1, σ(MCl)-π^*	
	(31,750)	forbidden transition	
	39,200(25,000)	d---π^*	
	(43,700) 49,500(15,000)	d---π^*	128
$Mo(CO)_5Br^-$	24,600(2150) (32,050)d-d sf	d---a_1, σ(MCl)-π^*	
	(34,500)	forbidden transition	
	39,700(30,000)	d---π^*	
	46,300(28,000)	σ(MBr)---σ^*(MBr)	
	50,600(20,000)	d---π^*	128

cont

Table 5.13 cont

Complex	Electronic spectra	Assignment	Ref
$Mo(CO)_5I^-$	24,500(1500) (32,050)d-d sf	d---a_1, σ(MI)-π^*	
	(34,500)	forbidden transition	
	39,700(30,000)	d---π^*	
	46,300(28,000)	σ(MI)---σ^*(MI)	
	50,500(20,000)	d---π^*	128
$Mo(CO)_4en^a$	25,300(1700)	d-d	
	32,700(10,000)	d---π^*	
	38,400(20,000) (43,400)	d---π^*	126
$Mo(CO)_4bipy^b$	21,600(4800)	d---π^*(Bipy)	
	25,450(3700) (29,100)	d-d	
	(32,600)	d---π^*	
	33,850(28,800) (34,950)	bipy π---π^*	
	38,900(28,800) 40,650(25,700)	d---π^*	
	42,400(25,700)	d---π^*	102,126
$W(CO)_5NH_3$	22,730(600)	^3E---$^1A_1$ d-d	
	24,500(3800)	^1E---$^1A_1$ d-d	130
$W(CO)_5PCl_3$	26,700	d-d sf	
	28,500	d-d	
	34,500	d---π^*	
	41,200	d---π^*(PCl_3)	
	44,400	d---π^*	182
$W(CO)_5Cl^-$	24,000(3100) (26,050)d-d sf	d---a_1, σ(MCl)-π^*	
	(33,300)	forbidden transition	
	40,650(60,000)	d---π^*	
	(46,950) 50,750(10,000)	d---π^*	128
$W(CO)_5Br^-$	24,050(3500) (26,300)d-d sf	d---a_1, σ(MCl)-π^*	
	(33,100)	forbidden transition	
	40,050(70,000)	d---π^*	
	(46,500) 50,760(30,000)	d---π^*	128
$W(CO)_5I^-$	24,600(2750) (29,000)d-d sf	d---a_1, σ(MCl)-π^*	
	30,000(1400)	forbidden transition	
	40,400(47,000)	d---π^*	
	47,200(75,000)	σ(MI)---σ^*(MI)	
	50,750(60,000)	d---π^*	128
$W(CO)_5Pyz$	23,300(500)	^3E---$^1A_1$ d-d	
	24,000	d---π^*(Pyz)	
	30,400	^1E---$^1A_1$ d-d	
$W(CO)_5Et_2O$	21,900	^3E---$^1A_1$ d-d	
	23,900	^1E---$^1A_1$ d-d	127
$W(CO)_5(Et_2NH)$	22,800(730)	^3E---$^1A_1$ d-d	
	24,900	^1E---$^1A_1$ d-d	127
$W(CO)_5PPh_3$	27,500(400)	^3E---$^1A_1$ d-d	
	28,800(2200)	^1E---$^1A_1$ d-d	127
$W(CO)_5ptene$	25,900(550)	^3E---$^1A_1$ d-d	
	27,800(1530)	^1E---$^1A_1$ d-d	
	33,000(7900)	d---π^*(ene) ?	127

cont

Table 5.13 cont

Complex	Electronic spectra	Assignment	Ref
$W(CO)_5acetone$	22,200(910)	^3E---$^1A_1$ d-d	
	24,650(5100)	^1E---$^1A_1$ d-d	127
$W(CO)_5Cl^{-c}$	(21,800)	^3E---$^1A_1$ d-d	
	24,900	^1E---$^1A_1$ d-d	193
$W(CO)_4en^a$	(22,200)(400)	d-d sf	
	25,200(1400)	d-d	
	33,200(8300)	d---π^* or d-d	
	39,200(28,200)	d---π^*	
	45,000(26,300)	?	126
$W(CO)_4bipy^b$	21,000(5100)	d---π^*(bipy)	
	(25,900)d-d 28,900(4000)	?	
	(32,500)	d---π^* (bipy)	
	33,680(27,500) (34,700)	bipy π---π^*	
	39,300(32,350) (40,620)	d---π^* CO	
	(45,900)	? d---π^*	126
c-$W(CO)_4Py_2$	21,600(4740)	$^3A_1$, $^3B_2$ d-d	
	24,200(9370)	$^1A_1$, $^1B_2$ d-d, W--π^*Py ?	
	27,200(7820)	d---π^*(Py)	127,155
c-$W(CO)_4$(4EtPy)	(21,900)	d-d sf	
	24,500(7600)	d-d , W---π^* 4EtPy ?	
	27,800(7280)	W---π^* 4EtPy	155
c-$W(CO)_4$(4CNPy)	(17,850)	W---π^* 4CNPy	
	20,450(7980)	W---π^* 4CNPy, d-d	
	(25,650)	d-d	155
$Mn(CO)_5Me^d$	36,500w	d---σ^*(MX) + π^*(CO)	
	45,700s	d---π^*	
	52,000s	d---π^*	128
$Mn(CO)_5Cl$	26,750(650)	^1E d-d	
	(42,500)(1060)	d---π^*(CO) ?	
	43,800(12,000)	d---π^*	129
	51,000(14,000)	d---π^* ?*	128,130
$Mn(CO)_5Br$	26,100(420)	^1E d-d	
	37,000(1700)	d-d?	
	44,000(15,000)	d---π^* CO	115
	50,500(7000)	d---π^* ?*	128,130
$Mn(CO)_5I$	25,000(360)	^1E---$^1A_1$ d-d	
	33,600(3000)	$^1A_2$---$^1A_1$?	
	36,400(1500)	?	
	42,700(20,000)	d---π^*	
	52,600(88,000)	d---π^*	128,130
$Re(CO)_5H$	37,000	d---a_1 MH* + COπ^*	
	47,300	d---π^*	
	50,630	d---π^*	128
$Re(CO)_5NH_2Me^+$	31,450(1250)	^3E---$^1A_1$ d-d	
	(34,100) 35,340(3270)	^1E---$^1A_1$ d-d	

cont

Table 5.13 cont

Complex	Electronic Spectra	Assignment	Ref
	37,200 (42,550)	d---π*	130
$Re(CO)_5Cl$[e]	29,100(750)	^3E---$^1A_1$ d-d	
	31,100(1940)	^1E---$^1A_1$ d-d	
	33,000(1300)	$^1A_2$---$^1A_1$? d-d	
	36,400(1600) 37,450(1600)	?	130
	44,150(23,000)	d---π*	
	48,600(70,000)	d---π*	
	54,000(100,000)	d---π* ?	128,130
$ClRe(CO)_3$phen	(24,400) 26,530(4000)	d---π* phen	
	37,310(30,620)	π---π* phen	152
$ClRe(CO)_3$2,2BiQ	(20,800) 22,830(1920)	d---π* 2,2BiQ	
	26,670(20,900)	π---π* 2,2BiQ	152
$ClRe(CO)_3(Py)_2$	(29,000)(1000)	d-d ?	
	33,800(5000)	d---π* Py	
	37,900(7000)	π---π* Py	152,163
$ClRe(CO)_3(4PhPy)_2$	33,000(22,600)	d---π* 4PhPy	
	37,300(36,500)	π---π* 4PhPy	163
$IRe(CO)_3(4PhPy)_2$	32,800(21,050)	d---π* 4phpy	
	37,000(38,500)	π---π* Py	163
$MeCNRe(CO)_3phen^+$	27,780(6410)	d---π* phen	162
$Ph_3GeRe(CO)_3phen$	22,200(4940)	σ(Ge-Re)---π* phen	157
$Ph_3SnRe(CO)_3phen$	21,500(4610)	σ(Sn-Re)---π* phen	157
$Fe(CO)_5Cl^+$	38,000s	d---π*	131
$Fe(CO)_4Br^+$	30,300(580)	d-d ?	128
	37,800(20,000)	d---π*	131
$Ru(NH_3)_5CO^{2+}$	27,800(17)	$^1T_1$--$^1A_1$ d-d or sf?	
	36,100(258)	d---π* ?	132

a) Range of diamines, en, pn etc have almost identical spectra. b) Cr(CO)4phen as a similar spectrum. c) $W(CO)5X^-$, X = Cl, Br, I, NCO have almost identical d-d spectra. d) $Mn(CO)_5X$, X = H, Me, Et, CF_3, $SiMe_3$, $GeMe_3$, $SnMe_3$ have very similar spectra. e) Re(CO)5X X= Cl, Br, I have fairly similar spectra. ptene = 1-pentene

Complexes of the general type $M(CO)_5X$ (M = Mn,Re) were extensively studied and compared with isoelectronic $M(CO)_5X^-$ (M = Cr,Mo,W) systems [130]. Relatively weak low lying transitions seen in the Re(I) (and W(0)) but generally not in the Mn(I) (or Cr(0), Mo(0)) complexes are assigned as spin forbidden transitions to ^3E, while at somewhat higher energy, moderately intense transitions to ^1E are observed (Table 5.13). For a given complex, $[M(CO)_5X]^{n-}$, the ^1E transition decreases in energy in the sequence Re(I) > Mn(I) > W(0) > Cr(0). This band appears some 8000 cm^{-1} to the red of the corresponding $^1T_{1g}$ in the octahedral $M(CO)_6$ analog.

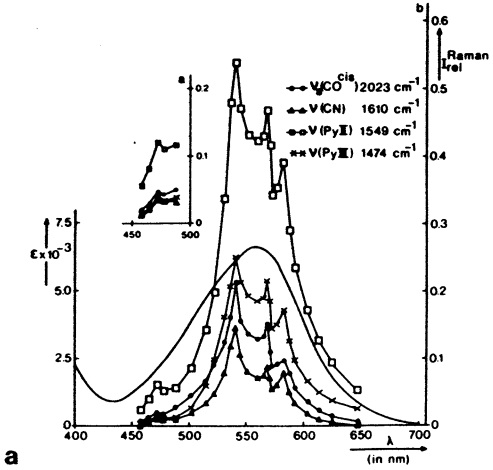
The nature of the ^1E transition is further supported by the similarity of the spectra of isoelectronic species such as $[W(CO)_5X]^-$, $M(CO)_5NH_3$ and $M(CO)_5X$ (X a halogen) where low energy CT transitions do not interfere.

A high energy, high intensity band is evidently a d---π* CO transition, but a lower energy, lower intensity band occurring near 30,000 - 33,000 cm^{-1} in many of these $[M(CO)_5X]^{n-}$ species is less obviously assigned. It is probably another ligand field transition, possibly $^1A_2$---$^1A_1$. It is not likely to involve σX or πX orbitals as has sometimes been suggested, because its energy seems relatively insensitive to the nature of X. The high energy d---π* CO transitions have not been thoroughly assigned in terms of the lower symmetry.

Some authors have assigned relatively low energy transitions involving σM-X (X a halogen) or σ*M-X orbitals. These are cited in Table 5.13 but they have not generally been adequately proven.

In cis-$W(CO)_4X_2$ complexes, the CT band shifts to the red to a greater degree than the LF band such that the CT band lies lowest for most, though not all X [155]. For example in a series of substituted pyridines, most emit from a long lived CT state near 16,000 cm^{-1}, but three of the complexes studied (Table 5.13) emit near 18,000 cm^{-1} from a state previously determined to be ligand field. This shift is also evident in the absorption spectra where the CT band crosses over the LF band as X is changed (Fig.5.18) [155].

Stufkens and collaborators have extensively studied the series cis-$M(CO)_4L_2$ (M = Cr,Mo,W and L often a bidentate nitrogen ligand) [173-179]. Of primary interest were the resonantly enhanced (Raman) vibrations coupled to the M---L MLCT transitions.



a

Fig.5.19 (a) Excitation profile of the diimine system $Mo(CO)_4$(PyCa-Ph) relative to the 758 cm^{-1} band in chloroform [178].

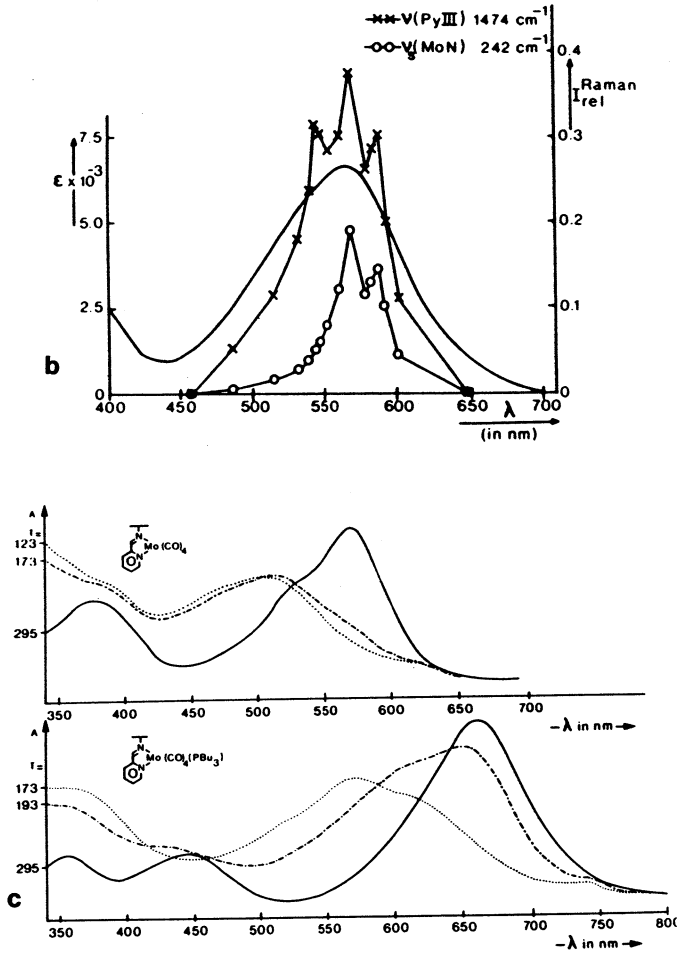


Fig.5.19 (Upper) (b) Excitation profiles of the diimine system $Mo(CO)_4$(PyCa-Ph) relative to the 606 cm^{-1} band of the solvent benzene [178]. (Lower) (c) Thermochromism in the spectra of some diimine molybdenum(0) tetracarbonyl derivatives recorded at temperatures indicated on the left [178].

In these C_{2v} (at highest) compounds, the degeneracy of the t_{2g} level is totally lost and three transitions may lie under the MLCT band envelope. These transitions will be polarised each along a different cartesian coordinate. A study [178,187] of the ligand pyridine 2-carbaldehydeimine in the complex $Mo(CO)_4$PyCA shows

resonance enhancement of several internal pyridine vibrations and also cis-CO stretching. The excitation spectrum shows three clear peaks (Fig.5.19 a,b)) associated with the three possible x,y and z polarised transitions underneath the MLCT transition.

However the Mo-N stretching vibration shows only two peaks in the excitation spectrum. Assuming the molecular plane to be xz, it is proposed that the missing peak is the out-of-plane y polarised electronic transition which would not be coupled to the in-plane Mo-N bonds.

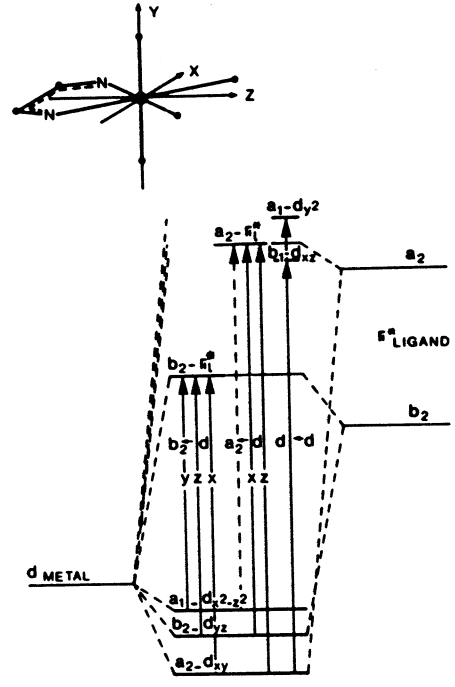


Fig.5.20 (Top) Coordinate framework for the $M(CO)_4$(diimine) species. (Lower) Relevant part of the molecular orbital diagram for these species [188], citing possible transitions and polarisations.

Stufkens and associates also studied the resonance Raman, vibrational, electronic and nmr spectra of the series $Mo(CO)_{4-x}L(PR_3)_x$, where x = 0,1,2 and L is bipyridine, pyridine 2-carbaldehydeimine or diazabutadiene [188]. An appropriate MO diagram is shown in Fig.5.20. These complexes show both solvatochromism and thermochromism (see Fig.5.19 c)). Two broad MLCT transitions to the two different π^* acceptor levels of the diimine encompass several d-d transitions. The a_2---b_2^* transition

should be very weak because of poor overlap between ground and excited state orbitals, while the a_1---a_2^* is orbitally forbidden. With respect to the lowest transition near 15-18,000 cm^{-1}, the strongest transition will be the z polarised b_2---b_2^* transition, and the high energy shoulder which is more evident at low temperatures (or in non-polar solvents) is assigned as the x polarised a_2---b_2^* transition. A very weak low energy shoulder, better seen at low temperatures, should be the weak and y polarised a_2---b_2^*.

With some L, especially bipyridine and bipyrazine [146], the second MLCT band is clearly identifiable about 7000 cm^{-1} higher than the first. Its identity is confirmed by the fact that it shifts in different solvents (solvatochromism) by approximately the same amount as the first MLCT band. This is assigned as a_2---a_2^*. To lower energy a shoulder may be b_2---a_2^*. The difference in energy between these two components is expected to be essentially the same as the corresponding difference in the first MLCT band.

Rhenium(I) forms a well developed class of compounds of general formula [ReX(CO)$_3$LL] which have been extensively studied, especially by Wrighton [152-155,157] and his co-workers. Emission from a low lying CT state is of especial interest in this series. Data are included in Table 5.13. Note that the lowest energy transition is generally the Re(I)---π^* LL MLCT state when LL is an aromatic acceptor ligand. When LL is a substituted o-phenanthroline, the energy of this band shifts to the red in the sequence 5-Me > 5-Cl > 5-Br > 5-NO_2 which is anticipated on the basis of the inductive effects of these substituents [152]. When LL is a poorly accepting ligand such as pyridine, this transition is at much higher energy but no low lying d-d bands are seen. This eliminates the possibility that low lying d-d bands may contribute to the lower energy absorption in systems such as o-phenanthroline. Emission from the triplet MLCT state occurs at about 17,000 - 19,000 cm^{-1} and the energy is dependent upon the medium (so-called rigidochromic effect or luminescence rigidochromic effect).

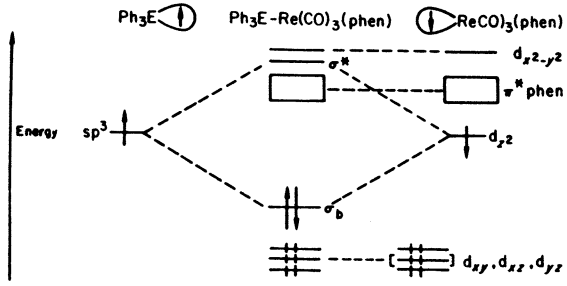


Fig.5.21 One electron scheme for a representative $R_3EM(CO)_3L$ species. (E is P,As etc)[161].

When X is Ph_3Ge or Ph_3Sn, a band is observed near 22,000 cm^{-1} which has been assigned as a transition from σGe-Re (or σSn-Re) to π^* LL, i.e. essentially an intermolecular L to L CT transition [157,161] (Fig.5.21).

The complexes fac-XRe(CO)$_3$(n-benzoylpyridine)$_2$ (n = 3,4) emit both at room and cryogenic temperatures. The room temperature spectra of these species show no structure, but the low temperature of the 3-benzoylpyridine derivative shows marked structure which is due to the superimposition of emission from two different states, namely Re---benzoylpyridine CT, and n---π^* from coordinated 3-benzoylpyridine. These two states do not thermally equilibrate at 77 K perhaps because they have very different geometries. In the case of the 4-benzoylpyridine system, the n---π^* state is too far above the CT state to be involved [156].

In a similar manner, complexes of the type XRe(CO)$_3$(R-Py)$_2$ have Re---π^* R-Py states which lie, in absorption, somewhat below the intramolecular π---π^* state of the substituted pyridine (Table 5.13) [163]. These complexes generally emit both at room temperature and low temperature. In most cases the triplet CT state lies below the triplet π---π^* state and emission therefrom is relatively short-lived and structureless. However with R-Py = 4-phenylpyridine, the triplet π---π^* state lies close to the CT state and emission occurs from both at 77 K. Under such circumstances emission from the triplet π---π^* occurs from a longer lived state and yields structure.

5.9 Di-Oxygen and Di-Sulphur Metal Complexes

5.9 i) Di-Oxygen Complexes [446-464,513]

Di-oxygen complexes form a large group of complexes of interest both for purely spectroscopic reasons and because of their importance in biological systems (see Chapter 8). Charge transfer spectra play an important role in elucidating both structural and electronic aspects of metal-dioxygen binding.

Di-oxygen binding can be divided into two main sections, peroxide and superoxide, and further sub-divided into different binding geometries (Fig.5.22). The peroxide ion carries a formal charge of -2 (O_2^{2-}) and has a longer bond (ca. 144pm) than the superoxide ion (ca. 133pm) which carries a formal charge of -1 (O_2^-). Neutral di-oxygen is supposedly bound to certain manganese complexes [445] but its spectroscopic signature (bound to a metal) is not yet available.

The highest occupied level in the free superoxide ion is the degenerate π_g^* set lying at -0.44 eV [454]. Addition of a proton, to form $HO_2^{\bullet}$, (a model for end-on metal bound superoxide) causes loss of degeneracy of the π_g^* set and stabilises it by about 12 eV. A new transition is observed in the HO_2 radical, at 7100 cm^{-1} [455,456], assigned as a transition between the components of the split π_g^*. The π---π^* transition in this species occurs at 47,620 cm^{-1} [457], implying that the π_u level lies at about -18eV.

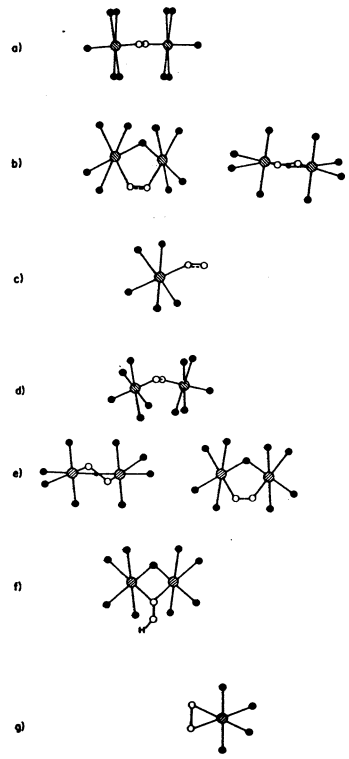


Fig.5.22 Modes of binding for superoxo and peroxo complexes. a) Bridging peroxo, b) doubly bridging peroxo, c) 1:1 end-on superoxo, d) bridging peroxo, e) doubly bridging peroxo, f) bridging hydroperoxo and g) side-on peroxo [446].

The corresponding π_g^* level in peroxide lies above zero (i.e. electrons would spontaneously be lost), but it is stabilised on addition of two protons, to form hydrogen peroxide, and the degeneracy of the π_g^* is lifted. Photoelectron spectra reveal these levels at -11.69 and -12.69 eV [458], a splitting of some 8000 cm^{-1}. The π_u set, now involved in σ bonding to H, lies at about -17.4 eV. This information permits us to generate a qualitative MO diagram for metal-di-oxygen binding as shown in Fig.5.23 and to rationalise the spectra of the various forms of metal di-oxygen binding [446].

Note that the two π^* levels are defined as π_h^* and π_v^* being horizontal and vertical with respect to a plane containing metal and di-oxygen. The π_h orbital, in the M-O_2 plane, will form a

σ-bond to the metal and will be stabilised to a greater degree than π_v^* which can only form a π-bond with the metal.

Superoxide complexes: The following transitions may be expected in a metal complex containing the superoxide ion.

i) One, or more, MLCT transitions from metal orbitals to the hole in π_v^*.
ii) A transition from π_v^* to an acceptor orbital on the metal ($d\sigma^*$).
iii) A transition from π_h^* (σMO) to $d\sigma^*$.
iv) A weak (forbidden) transition between π_h^* (σMO) and π_v^*.
v) A π_u---π_v^* transition at very high energy.

The first three anticipated transitions are observed but have intensities and energies dependent upon the nature of the metal superoxide interaction.

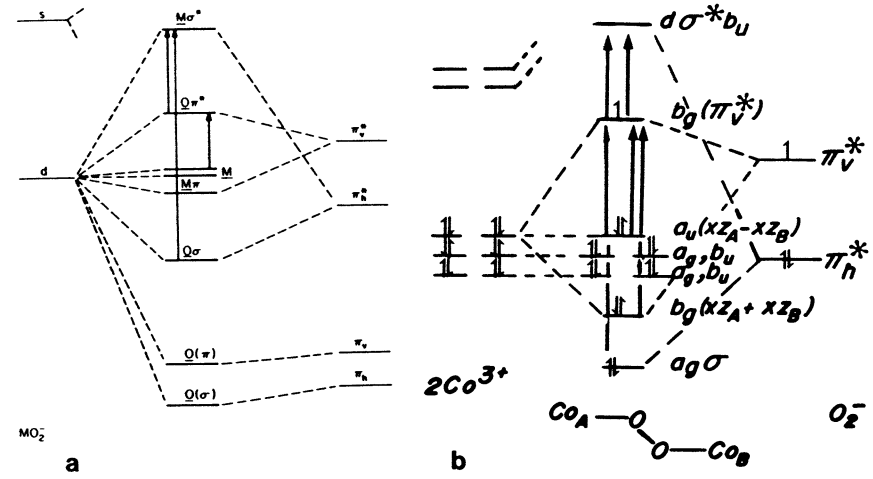


Fig.5.23 Simplified molecular orbital diagrams for superoxometal complexes. If the M-O axis is defined z, then the dσ-metal orbital is dz^2. a) Terminal superoxometal species [446], b) bridging superoxodicobalt(III) species using C_{2h} symmetry [612].

With bridging superoxo M-O_2^--M complexes, a single intense band (intensity 1-3 x 10^4 l mol^{-1} cm^{-1}) in the region 25,000 - 33,000 cm^{-1} [460-464] is assigned [446,464] to the transition from π_h^*---$d\sigma^*$; it is very intense because it is a σ---σ^* type transition. A very weak transition is observed in some systems, near 12,500 cm^{-1} and is assigned as π_h---π_v^* [464] (Fig.5.24).

The presence of a hole in π_v^* permits an MLCT transition ($d\pi$---π_v^*) which is seen in the range 13,000 - 20,000 cm^{-1} with an intensity of ca. 1000 l mol^{-1} cm^{-1}. An alternative assignment for

this band would be π_v^*---$d\sigma^*$. However this is considered improbable because of a comparison of the spectra of mu-superoxo-decaammine cobalt(III) and mu-superoxo-decacyanocobalt(III). In these species bands due to π_h^*---$d\sigma^*$ and π_h^*---π_v^* occur at roughly the same energies. We would then expect a π_h^*---$d\sigma^*$ transition also to be unshifted, whereas the band in question is strongly blue shifted in the decacyanide.

Superoxo bridged systems generally possess a planar M-O_2^--M unit, with a trans configuration about the O-O link. The π_v^*---$d\sigma^*$ transition in such a unit should be very weak (being an in-plane to out-of-plane excitation) but should occur in the CD spectrum. Indeed many of these complexes show a shoulder on the low energy edge of the intense π_h^*---$d\sigma^*$ transition which corresponds to a well developed shoulder in the CD spectrum (Table 5.14, Fig.5.24) (see also the analysis by Solomon of the absorption and CD spectrum of oxyhemocyanin, in Chapter 8). Any d-d transitions in this region should be components of the magnetic dipole forbidden $^1T_{2g}$---$^1A_{1g}$ and are therefore unlikely to give a strongly developed CD band (though perhaps could do so in the lower effective symmetry).

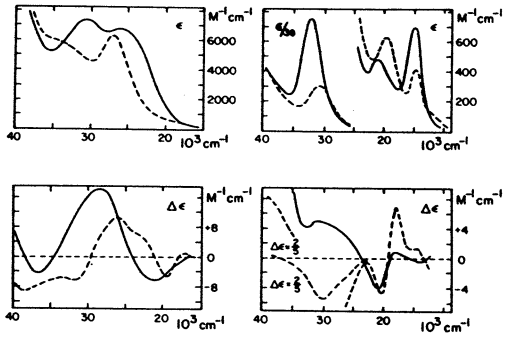


Fig.5.24 Aqueous solution absorption spectra of di-oxygen adducts of Co(II) L-histidine species. Top left:bridging peroxo (solid line) and doubly bridging peroxo (OH^-,O_2^{2-})(hatched). Top-right: bridging superoxo (solid line), and doubly bridging superoxo (OH^-,O_2^-) (hatched). Lower left and right show the corresponding circular dichroism for these species [446,463].

Very recently [612] Miskowski and Gray and co-workers [612] published a single crystal structure and polarised spectra for the superoxy-bridged species $[((NH_3)_5Co)_2O_2](NO_3)_2Cl_3.2H_2O$, being a more complete study than that published very much earlier by Yamada [460]. The planar [Co-O_2^--Co] unit has C_{2h} site symmetry symmetry and the crystal morphology was such that polarisation data could be

obtained almost directly along the O-O bond (defined as σ_2) as well as in the two mutually perpendicular directions (defined σ_1 (in plane),and π (out-of-plane).

The positions and identities of the π_h^*---π_v^* transition (^2A_g---^2B_g, 12,500 cm^{-1}), and MLCT xz---π_v^* (^2A_u---^2B_g, 15,100 cm^{-1}) transition were confirmed. The latter is strongly polarised along σ_2. The intensity along π is much weaker but shows a vibrational progression in a ca. 700 cm^{-1} mode. The intensity along σ_1 is intermediate between that along σ_2 and π, and this shows a progression in 700 cm^{-1} with an apparent sub-progression in ca. 400 cm^{-1} [612].

Most interestingly there is a second progression in π-polarisation, also in a 700 cm^{-1}, but which begins at about 17,200 cm^{-1}. This is believed to be a second MLCT transition most probably (xz+xz)---π_v^*, ^2B_g---^2B_g (Fig.5.23). The splitting of approximately 3000 cm^{-1} between the two MLCT transitions is a measure of the splitting of the metal 't_{2g}' level caused by back-bonding to the superoxo group.

A weak band centred near 21,900 cm^{-1} is associated with the $^1T_{2g}$---$^1A_{1g}$ (in O_h) d-d transition. It displays a developed vibrational progression in a 420 cm^{-1} mode. The intensity decreases with temperature indicative of a vibronic contribution. It appears in all three polarisations but with different intensities, undoubtedly a consequence of the low symmetry.

The crystal also shows a shoulder near 29,000 cm^{-1} which was also discussed above. This band appears weakly in π-polarisation. It was not observed in σ-polarisation but is evident in the isotropic spectrum. The authors therefore argue that there must be a stronger σ-component to this transition which is obscured by the higher energy and σ-polarised LMCT transition. While the CD argument above led to the suggestion that this is the π_v^*---$d\sigma^*$ transition, these authors [612] prefer the d-d assignment [464] to a component of $^1T_{2g}$---$^1A_{1g}$ (in O_h). The π-polarised component is assigned to '$^1B_2$---$^1A_1$' and the presumed more intense σ-component to '^1E---$^1A_1$'. Since the π_v^*---$d\sigma^*$ transition would be π-polarised, the identity of this transition must remain uncertain.

The very strong σ-polarised transition near 30,000 cm^{-1} is consistent with the previous assignments (Table 5.14).

Note that transfer of a π_h^* electron from O_2^- to M 'leaves behind' a neutral di-oxygen molecule in its spin-triplet state, while removal of a π_v^* electron leaves behind a spin singlet dioxygen. It is well known that the spin singlet di-oxygen has significantly higher repulsion energy than spin triplet di-oxygen, perhaps explaining why the two LMCT transitions lie so close together even though there is substantial splitting of the π_g^* set.

Moreover, the addition of an electron to superoxide (2nd electron affinity of di-oxygen) is highly unfavourable, rationalising the relatively high energy of the LMCT transitions.

Di-bridging systems, e.g. with a Co(X)(O_2^-)Co link (X = OH, NH_2 etc) also possess a planar bridge and have spectra not very

different from those described above [461,463,465]. However OH^- and NH_2^- bridged systems will possess an extra band in the 25,000 - 33,000 cm^{-1} region due to the $\pi X \text{---} d\sigma^*$ LMCT transition [446].

Non-bridging terminal 1:1 superoxo-metal species have been investigated [466,467]. The spectra are very similar to those of the bridging variety. The removal of a metal from one end of the superoxide group should presumably destabilise the π^* orbitals with a consequent red shift in the $\pi_h^* \text{---} d\sigma^*$ transition and a blue shift in the $d\pi \text{---} \pi_v^*$ transition; this is generally observed (Table 5.13). Note that the MLCT band shows marked dependence on solvent in both the 1:1 and bridging systems [464,467]. The $\pi_h^* \text{---} \pi_v^*$ and $\pi_v^* \text{---} d\sigma^*$ transitions have not been identified in the 1:1 series.

The complex $[Co(sym\text{-}Me_2en)_2(Cl)(O_2^-)]^+$ shows a Cl---Co LMCT band at 42,400 cm^{-1} consistent with Co(III) rather than Co(II) which is useful information given the on-going argument concerning effective oxidation states and electron distributions in oxygen complexes [452,453].

The free superoxide ion exhibits $\pi_u \text{---} \pi_g^*$ at 40,000 cm^{-1} while the protonated species absorbs at 47,600 cm^{-1}. There seems little doubt that this intraligand transition shifts to the blue in metal complexes, relative to the free ion. It has been observed at 47,200(12,000) cm^{-1} in $[Co(s\text{-}Et_2en)_2MeOH(O_2^-)]^{2+}$ [449].

Note that the LMCT bands do blue shift with decrease in oxidation state. Thus the $Co(II)(O_2)_2$ (see below) and $Co(4,11\text{-dieneN}_4)O_2$ show $\sigma_h^* \text{---} \sigma^*$ at 43,500 and 41,600 cm^{-1} compared with ca. 30,000 - 33,000 cm^{-1} for the Co(III) complexes (Table 5.14). Note also that the $\pi_u \text{---} \pi_g^*$ transition seen in free superoxide ion at about 40,000 cm^{-1}, is blue shifted by the metal interaction to at least 46,500 cm^{-1} [449].

Peroxide Complexes: The peroxide ion is closed shell so that MLCT transitions do not occur at accessible energies. We are left, therefore, with consideration of possible LMCT transitions. These depend upon the binding geometry as shown in Fig.5.22.

In general two classes of spectroscopic behaviour are observed, one type for terminal, or bridging, end-on bound peroxide, and the other for sideways bound peroxide.

While the bridging $M\text{-}O_2^-\text{-}M$ unit is usually planar, the corresponding bridging $M\text{-}O_2^{2-}\text{-}M$ unit is usually, though not exclusively, non-planar. In such non-planar systems, the distinction between π_v^* and π_h^* is lost because there is no longer a plane with which to define them. Transitions from both these orbitals will have some σ-character; both may have significant intensity, but less than the pure $\sigma \text{---} \sigma^*$ transitions in the planar unit.

Thus two closely spaced LMCT transitions are expected, with intensities rather less than observed in comparable superoxide complexes [446,463,464,468,469]. The energy separation between these two components is about the same as observed for the split π_g^* level in hydrogen peroxide.

Table 5.14

ELECTRONIC SPECTRA OF DIOXYGEN CONTAINING COMPLEXES

A. Bridging Superoxodimetal Complexes $[L_xM(O_2^-)ML_x]^{n+a}$

M	L	$d\pi \text{---} \pi_v^*$	$\pi_v^* \text{---} d\sigma^*$	$\pi_h^* \text{---} d\sigma^*$	Ref
Co	NH_3 fhi	15,100(800)	(28,500)3500[j]	33,200(23,960)	
		17,200			612
Co	NH_3 [f]	14,890(935)	(28,900)(3470)	33,100(19,950)	464
Co	en,NH_3	14,100(1120)	(26,700)(2000)	33,100(19,500)	461
Co	trien	14,100(1290)		32,500(19,500)	461
Co	dien,1-pn	14,100(1550)		32,600(22,900)	461
Co	tetren	14,100(1410)	(27,000)(2000)	31,900(19,050)	461
Co	L-histidine	15,100(710)		31,900(22,900)	463
Co	CD	14,200(-0.5)	26,800(+3.5)	31,000(+5.5)	463
Co	CN[g]	20,590(740)		32,200(17,400)	
				37,700(5250)[b]	464
Rh	Cl,pyridine	16,600(2880)		34,720(19,050)[c]	462
Rh	Cl, 4-Pic.	16,370(2700)	(27,000)(2140)	33,900(22,900)[c]	462

B. Double Bridged Superoxodimetal Complexes $[L_xCo(X)(O_2^-)CoL_x]^{n+}$

L	X	$d\pi \text{---} \pi_v^*$	$\pi_v^* \text{---} d\sigma^*$	$\pi_h^* \text{---} d\sigma^*$	Ref
en	NH_2	14,600(435)	(28,500)(1580)	32,500(7400)	
			CD 26,700(-1.05)	30,300(+3.82)	465
NH3	NH_2	14,300(280)		33,000(4070)	461
1-pn	NH_2	14,400(410)	(28,000)(2000)	32,400(6610)	
			CD 26,500(-0.45)	30,300(+5.52)	465
en	OH	14,800(525)	27,800(5010)	ca32,800(500)	461
1-pn	OH	14,900(510)			461
L-His.	OH	14,900(400)		30,700(10,000)	
	CD	14,800(+1.5)	26,800(-3.0)	30,100(-5.5)	463

C. 1:1 Superoxometal Complexes $[L_xXMO_2^-]^{n+}$

M	L	X	$X \text{---} d\sigma^*$	$d\pi \text{---} \pi_v^*$	$\pi_h^* \text{---} d\sigma^*$	Ref
Co	CN	CN		18,870(80)[d]	31,205(2820)[d]	467
Co	s-Me_2en	Cl	42,370	20,790(645)[e]	28,890(5250)[e]	499
Co	s-Et_2en	Cl		20,320[e]	(26,950)[e]	298
Co(II)(4,11-diene)					41,600(24,550)	498

cont

Table 5.14 cont----------

D. Bridging Peroxodimetal Complexes $[L_xM(O_2^{2-})ML_x]^{n+}$

M	L	CD	LMCT Absorption		Ref
Co	CN		(27,000)(5010)	31,750(10,000)	464
Co	en,dien		(27,300)(1820)	33,300(9550)	461
Co	dien, 1-pn			33,200(10,000)	461
Co	tetren		(27,600)(3390)	32,200(12,600)	461
Co	L-histidine	28,100(+1.8)	25,900(6610)	30,600(7250)	463
Co	L-2,3-DMPr	27,550(-2.8)	25,970(3020)	31,750(3470)	468
Co	L-2,4-DMBu	29,240(-0.7)	27,030(2950)	31,850(3890)	468
Co	tetren		27,600(3000)	32,500(16,000)	513
Rh	Cl, 4-Pic		27,020(3980)[c]	32,470(15,850)[c]	462
Rh	Cl, Pyridine		27,780(4265)[c]	32,150(15,150)[c]	462
Cr	H_2O		36,760(1175)[c]	(42,000)(1260)[d]	469

E. Double Bridging Peroxodicobalt Complexes $[L_xCo(X)(O_2^{2-})CoL_x]^{n+}$

L	X	CD	LMCT Absorption		Ref
en	NH_2	27,800(-10.3)	29,800(3890)		461,465
1-pn	NH_2	27,800(-8.61)	29,700(3310)	(33,700)(4075)	465
en	OH		28,000(4900)	36,200(4790)	461
1-pn	OH		28,600(5750)	36,800(5625)	461
L-hist.	OH	25,700(+1.1)	26,900(6165)	33,300(5500)	
		31,500(-0.6)			463
L-2,3DMPr	OH	26,450(-2.2)	27,930(2500)		
		32,890(+2.8)			468
L-1,2dmp	OH	27,030(-6.5)	28,090(2190)		
		32,790(+4.4)			468

a) Differential extinction coefficients for left and right hand polarised light are cited for CD data. b) π_h^*---x^2-y^2. c) Assignment uncertain. d) Acetonitrile solution. e) 20 percent MeOH/EtOH at 10 K. f) π_h^*---π_v^* 12,500(110). g) π_h^*---π_v^* 12,200(12) cm^{-1}. h) Single crystal. i) '$^1T_{1g}$---$^1A_{1g}$' d-d transition observed at 21,880 cm^{-1} with associated vibronic structure at 11 K. 2,3DMPr = 2,3-diaminopropionate. 2,4DMBu = 2,4-diaminobutyrate. 1,2-dmp = 1,2-diaminopropane.

These transitions are magnetic dipole allowed and therefore also give rise to strong CD absorption (Fig.5.23). In the situation where the $M\text{-}O_2^{2-}\text{-}M$ unit is planar, only one LMCT transition (from π_h^*---$d\sigma^*$) is anticipated [461].

Pickens and Martell have offered some interesting linear correlations between the higher energy LMCT band in bridged peroxycobalt(III) complexes and i) the reduction potential of the complex (to bridged $Co(II)\text{-}O_2^{2-}\text{-}Co(III)$), and ii) the oxygenation constant for formation of the complex [513].

Double bridged $M(X)\text{-}O_2^{2-}\text{-}M$ complexes (X usually OH^-), constrain a cis binding to the M-O-O-M link and cause distortion out-of-plane. These systems also show two LMCT bands, but the higher is shifted to the blue where it often appears as an intense shoulder on further rising absorption. This higher energy LMCT component is more clearly identified in the CD spectrum with a rotation which is opposite to that of the lower energy LMCT band [446,468]. Peak intensities in doubly bridged systems invariably appear lower than in singly bridged complexes, though the integrated intensity may not vary significantly. Broad bands probably arise because of the distortion in the bridge link being different in ground and excited states. Since superoxide bridged systems are usually planar, and one of the excited peroxide LMCT states is equivalent to superoxide bound to Co(II), it is likely that this excited state is planar, or at least distorted towards planar.

The best defined terminal peroxide exists in the hydroperoxide complex $[(NH_3)_5Co(O_2H)Co(NH_3)_5]^{5+}$ whose spectrum was reported by Miskowski [470]. The complex contains bound hydroperoxide tilted out of the Co-O-Co plane [471].

An intense peak at 33,330 cm^{-1} is assigned as π_h^*---$d\sigma^*$, and a weaker peak at 27,780 cm^{-1} is probably π_v^*---$d\sigma^*$ [446].

The majority of oxygen complexes contain Co(III), and the theory has developed to explain the significant quantity of Co(III)-oxygen data which are available. Although the principles involved should not be different with other central metals, the fine detail will vary and care should be used in transferring the arguments above to other metals.

Sideways bound peroxide: Many metal ions, especially those of the left-hand transition series, give characteristic colours when treated with hydrogen peroxide [472-479]. Many of these species have been characterised to contain sideways bound peroxide [478,479,484-487] and it is probable that most in fact do so.

Essentially without exception, these sideways bound complexes exhibit a weak absorption band (10^2 - 10^3 l mol^{-1} cm^{-1}) as the lowest identifiable CT feature. Depending upon the metal, this transition may be found in the near infrared, visible or ultraviolet regions, shifting to the red with increasing charge on the metal, hence LMCT [485]. An $M\text{-}O_2$ sideways bound system is necessarily planar, with the 'π_h^*' orbital used to σ-bond to the metal. Thus it is eminently reasonable to associate the weak transition to the π_v^*---$d\sigma^*$ transition. Some complexes show structure on this band, perhaps due to several acceptor d orbitals, but detailed studies have not been undertaken.

Table 5.15

SURVEY OF THE ELECTRONIC SPECTRA OF DIOXYGEN CONTAINING COMPLEXES

Type	Energy Range	Assignment
$LCo(O_2^-)$	27,000 - 32,000 (2500 - 6500)	π_h^*---$d\sigma^*$
	18,000 - 21,000 (100 - 1000)	$d\pi$---π_v^*
$LCo(O_2^-)CoL$	31,000 - 34,000(16,000 - 32,000)	π_{h*}^*---$d\sigma^*$
	26,000 - 29,000(2000 - 3200)[a]	π_v^*---$d\sigma^*$
	14,000 - 15,000(600 - 3200)	$d\pi$---π_{v*}^*
	12,500(100)	π_h^*---π^*
$LCo(O_2^-)XCoL$	30,000 - 33,000(4000 - 10,000)	π_{h*}^*---$d\sigma^*$
	27,000 - 29,000(1500 - 5000)[a]	π_v^*---$d\sigma^*$
	14,000 - 15,000(250 - 500)	$d\pi$---π_v^*
$LCo(O_2^{2-})CoL$	30,000 - 33,000(3000 - 16,000)	π_{a*}^*---$d\sigma^*$
	25,000 - 28,000(2500 - 6500)	π_b^*---$d\sigma^*$
$LCo(O_2^{2-})XCoL$	33,000 - 36,000(3000 - 5500)	π_{a*}^*---$d\sigma^*$
	27,000 - 30,000(2500 - 6500)	π_b^*---$d\sigma^*$
$LM(O_2^{2-})$	13,000 - 35,000(125 - 1300)	π_v---d
	> 40,000vs	σMO--d

In double bridged systems, X is usually OH^-, or NH_2^-; LMCT absorption of the type X---Co(III) probably occurs near 30,000 cm^{-1}. a) May be a d-d band.

The $(\sigma)\pi_h^*$---$d\sigma^*$ transition is blue-shifted and has rarely been identified [486,487]. In some systems there is no further absorption below 50,000 cm^{-1}.

Many metals to the right of the transition series also form 1:1 sideways bound peroxo complexes, e.g. Vaska's oxygen complex [488], and phosphine [489] and arsine [490] complexes. However the π_v^* LMCT transition is rarely seen, being obscured in the uv by other absorption. It has been observed in $Pt(PPh_3)_2(O_2)$ (Table 5.14), at an energy shown to be acceptable by an SCF X-alpha calculation [491].

Many biological complexes containing di-oxygen exist [495], oxygenated hemoglobin being the most well known. In most systems, other strongly absorbing chromophores such as a porphyrin are present, and these partially or totally obscure transitions involving the di-oxygen ligand. Some assignments have, however, been suggested in the porphyrin domain [492]. Complexes such as oxyhemerythrin and oxyhemocyanin do exhibit clearly identified

oxygen to metal LMCT transitions and these are discussed in some depth in Chapter 8. Their assignment follows logically from the discussion presented here; indeed such analysis permits the distinction of one type of oxygen binding from another in systems, such as biological molecules, where the complexity makes structural identification very difficult.

In oxyhemerthyrin, for example, a band at 20,000 cm^{-1} [493] is assigned to π_v^*---$d\pi^*$(Fe(III)) in a complex believed to have peroxide terminally bound to one iron(III) atom. The spectrum is very similar to that obtained by adding hydrogen peroxide to an Fe(III)/EDTA solution [494].

Ozin and Ogden and their co-workers have prepared simple binary complexes of the type $M(O_2)$ and $M(O_2)_2$ in cryogenic matrices [497,498]. These species all exhibit one strong band in the ultra-violet generally with a well developed shoulder to lower energy. A simple group theoretical analysis (based on the treatment described above) leads one to conclude that all these complexes should exhibit a strong σ---σ^* transition and a weaker $d\pi$---σ^* transition, irrespective of whether end-on or sideways bound di-oxygen is assumed. The strong band and shoulder are respectively so assigned [447]. These represent an interesting series of complexes in which electronic spectroscopy can probe electronic structure without interference from other ligands.

The energies of the σ---σ^* transition correlate in an excellent linear fashion with the electron affinity of the M(II) ion, viz [447,450,512]:

	$Ti(O_2)_2$	$V(O_2)_2$	$Fe(O_2)_2$	$Cr(O_2)_2$	$Co(O_2)_2$	$Mn(O_2)_2$
σLMCT (eV)	5.82	5.51	5.46	5.39	5.39	5.21
πLMCT		(4.59)	(3.54)		(3.54)	
EA M(II)	13.6	14.2	16.2	16.5	17.05	15.6

	$Rh(O_2)_2$	$Ni(O_2)_2$	$Pt(O_2)_2$	$Pd(O_2)_2$	$Cu(O_2)_2$
σLMCT (eV)	4.96	4.77	4.68	4.59	4.28
πLMCT		(3.90)		(3.87)	(3.54)
EA M(II)	18.1	18.15	18.6	19.4	20.3

(5.33)

The correlation with the first or third electron affinity is noticeably poorer, which may be taken as evidence that these species are formally bivalent. There is also a continuous change in the O-O stretching vibration which varies from about 1000 to about 1120 cm^{-1} in this series (lower end of superoxide range); indeed there is a good linear correlation between this vibration and the σ---σ^* transition for many of the $M(O_2)_2$ and $M(O_2)$ series which fall on the same correlation line [447].

For example

	$Cu(O_2)_2$	$Pd(O_2)_2$	$Ag(O_2)$	$Ni(O_2)_2$	$Rh(O_2)_2$	$Cu(O_2)$
LMCT (eV)	4.28	4.59	4.51	4.77	4.96	5.39
(O-O) cm^{-1}	1112	1112	1080	1062	1038	1016

(5.34)

These data are interpreted in terms of a continous change in the electron density transferred from metal back to di-oxygen as the electron affinity of the M(II) species decreases.

Table 5.15 summarises the spectra expected for oxygen in different coordination and electronic environments.

5.9 ii) Disulphur Metal Complexes

The bridged disulphur species $[(NH_3)_5Ru(III)\text{-}S\text{-}S\text{-}Ru(III)(NH_3)_5]^{4+}$ exhibits a strong band at 14,000(14,990) cm^{-1} [40]. The species contains a trans-planar Ru-S-S-Ru linkage. Simple M.O. theory delineates the in and out-of-plane π orbitals on the S_2 linkage which can interact with the ruthenium d orbitals. The highest energy and empty orbital is probably a_u, π anti-bonding with respect to the metal S_2 link, and mainly localised on the metal. Although there are several group theoretically allowed transitions from low lying even orbitals to this ungerade empty orbital, the authors, using transition moment theory, argue that only the b_g---a_u transition can have significant intensity. This transition involves good overlap between ground and excited states since it is from an out-of-plane π-bonding orbital to an out-of-plane π-anti-bonding orbital, i.e. π---π^* [40]. Resonance Raman excitation reveals that this transition is strongly coupled to the totally symmetric Ru-S stretching mode. The S-S mode is not enhanced and this is explained by noting that the transition dipole lies along (z) which contains the Ru-S bond. The S-S axis is significantly off-axis and does not couple.

5.10 Bipyridine and Related Diimine Systems

The intense red colour generated by the reaction of iron(II) with diimine ligands such as bipyridine and o-phenanthroline and first studied in depth by Krumholz and his collaborators [324,325], was soon attributed to iron(II) to diimine MLCT [342]. Very many complexes of this general class are known, especially with the d^6 ions Fe(II), Ru(II) and Os(II) which appear to form especially stable derivatives. The stability probably arises through π-back-donation from the filled $(t_{2g})^6$ configuration to the diimine ligand.

Iron(II) complexes ($[Fe(NN)_3]^{2+}$) of various substituted diimines RN=C(R')-C(R')=NR (and asymmetric versions) have multiple bands in the visible spectra peaking at 16,000 - 20,000 (8000-12,000) cm^{-1} usually with shoulders to higher energy [325,505]. A second characteristic absorption is seen near 28,000 cm^{-1} [505].

These transitions are MLCT, from metal t_{2g} to the first two acceptor π^* levels on the diimine ligand. The energy separation between MLCT1 and MLCT2 compares well with the difference between the first two internal π---π^* transitions [325].

Oxidation to the corresponding $[Fe(III)(NN)_3]^{3+}$ species causes a blue shift in the intense visible absorption generating a strong band at 26,000 - 30,000(2000 - 4000) cm^{-1} usually with shoulders to lower energy. A second transition occurs near 30-32,000 cm^{-1}, of slightly lower intensity. Both these transitions are π-diimine to Fe(III) LMCT to the hole in $(t_{2g})^5$ [325]. If each of the LMCT bands in the Fe(III) diimine complex is summed with each of the MLCT bands in the corresponding Fe(II) diimine, the resulting numbers agree well with the two π---π^* internal transitions of the diimine. While a zero order approximation would predict this result, the fact that it works well is surprising [325].

Although the MLCT assignment for the M(II) complexes is clear, a detailed understanding of the polarised and CD spectra of these species is still elusive. Because of its utility in photocatalysis [320], the $[Ru(bpy)_3]^{2+}$ ion has attracted the greatest attention [308-320, 326-334]. Ostensibly the series $[M(bpy)_3]^{2+}$ (M = Fe,Ru and Os) show three strong bands in the visible region (Fig.5.25) (ascribable to the first MLCT transition) but with considerable additional fine structure in the spectra of crystals at low temperatures. These bands are primarily xy polarised, i.e. perpendicular to the molecular C_3 axis. Any model must explain the main features and additional structure, plus answer other questions such as "are the excited states spin singlets or triplets ?", "why do the lifetimes vary as they do (osmium shortest) ?", and "is the 'excited' electron localised in one bipyridine ring, or is it delocalised ?" etc. Since the final answers have not yet been published, this discussion will review current understanding with emphasis on one of the many different approaches to the solution of this problem.

The bipyridine ligand has one low lying π^* orbital of symmetry B_2 and, lying approximately 6000 cm^{-1} to higher energy, two further π^* orbitals of symmetry $A_2 + B_2$. The dominant visible absorption is associated with the lower B_2 orbital.

The three B_2 orbitals (in C_{2v} symmetry) combine, in the D_3 group of the $M(bpy)_3$ species, to yield $(\pi^*)E + (\pi^*)A_2$. The metal t_{2g} orbitals split into $d(A_1) + d(E)$. Thus four possible transitions are expected:

i)	$d(A_1)$---$\pi^*(A_2)$	yielding the states	$^1A_2 + {}^3A_2$
ii)	$d(A_1)$---$\pi^*(E)$		$^1E + {}^3E$
iii)	$d(E)$---$\pi^*(A_2)$		$^1E + {}^3E$
iv)	$d(E)$---$\pi^*(E)$		$^1A_1 + {}^1A_2 + {}^1E$ $^3A_1 + {}^3A_2 + {}^3E$

(5.35)

In the absence of spin-orbit coupling, transitions to the triplet states are forbidden, transitions to the ^1E states are allowed xy, transitions to $^1A_2$ are allowed z, and transition to $^1A_1$ is forbidden. It is argued [328] that the transitions (i) and (ii) should be very weak because the $d(A_1)$ orbital cannot mix with the excited π^* orbitals (wrong state symmetry). However the appearance of three strong xy components seems to eliminate this conclusion perhaps through mixing of $d(A_1)$ with the A_1 component of the second set of π^* levels [316].

The upper $A_2 + B_2$ π^* levels in each ligand, combine in D_3 symmetry to yield $A_1 + A_2 + 2E$.

Kober and Meyer [316] carry out a complete analysis of these four transitions by incorporating the complete spin-orbit matrices, mixing spin singlets and triplets where required. They find that there should be three intense transitions corresponding primarily with the ^1E states of the transitions (ii), (iii) and (iv). In these specific states, mixing with spin triplet E components is small and the three intense features of the absorption spectrum can be regarded primarily to be of the singlet-singlet type. Lower energy and weaker shoulders are deemed to arise primarily via the ^3E states of transitions (ii) and (iv).

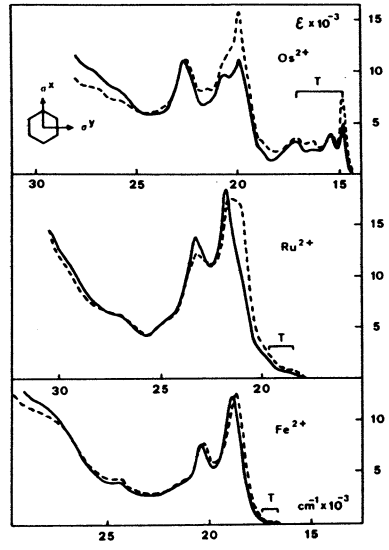


Fig.5.25 Polarised (σ(x) full line, σ(y) hatched line) spectra of Fe(II), Ru(II) and Os(II) doped into $Zn(bpy)_3Br_2.6H_2O$, at 8 K [309]. Note that the wavenumver scales are displaced for comparison of the data set. T marks triplet-singlet transitions.

Table 5.16

CHARGE TRANSFER SPECTRA OF SOME BIPYRIDINE, BIPYRAZINE AND RELATED DIIMINE METAL COMPLEXES

Complex	d---π_1	d---π_2	π---π^*	Ref
$Fe(BPZ)_3^{2+}$	19,670(4270)			
	21,000(3300)			428
$Fe(bpy)_3^{2+}$	19,200(8740)	28,600(6280)	33,560vs	505
	20,500			
$Fe(bpy)(CN)_4^{2-a}$	18,380(3870)	26,670(5360)	33,600vs	505
$Fe(bpy)_2(CN)_2^a$	18,050(6320)	27,320(6170)	33,370vs	505
	19,800			
$Fe(bpy)_2(CN)_2^b$	16,470(10,000)	25,850(10,060)	33,160vs	505
$Ru(BPZ)_3^{2+}$	22,700(13,000)	(29,600)	34,400vs	
	(24,100)		37,455vs etc	1,2
$Ru(BPZ)_2(bpy)_2^+$	21,600(12,000)	26,700(56,000)	34,000vs	
	24,100(9200)	(29,600)		1
$Ru(BPZ)_2(NO_2)^2$	20,660(8700)	28,735(9300)	32,365vs	
		37,450(21,900)	(37,735) etc	336
$Ru(BPZ)_2(NCS)_2$	19,000(9100)	26,740(8900)	32,150vs etc	336
$Ru(BPZ)_2Cl_2$	18,020(10,700)	25,315(8700)	32,050vs etc	336
$Ru(BPZ)_2Cl(CH_3CN)^+$	20,365(10,000)	26,955(7800)	32,785vs etc	2
$Ru(BPZ)_2Br_2$	18,250(9800)	25,510(8100)	31,950vs etc	336
$Ru(BPZ)_2I_2$	18,115(7900)	25,510(7800)	31,450vs etc	336
$Ru(BPZ)_2(oxalate)$	19,380(12,000)	26,740(9300)	32,360vs etc	336
$Ru(BPZ)_2(CO_3)$	20,080(8500)	(26,320)	32,360vs etc	336
$Ru(BPZ)_2(H_2O)_2^{2+}$	20,240(12,900)	(27,780)	32,680vs etc	336
$Ru(BPZ)_2(H_2O)(OH)^+$	18,870(11,000)	26,040(9100)	32,260vs etc	336
$Ru(BPZ)_2(OH)_2$	17,840(11,200)	24,690(9300)	31,550vs etc	336
$Ru(BPZ)(bpy)_2^{2+}$	21,100(9900)	(25,900)	35,300vs	
	24,150(10,000)	(29,150)		1
$Ru(bpy)_3^{2+}$	22,300(14,000)	29,000(6500)		
	23,700(12,000)	31,000(6500)		3,315
$Ru(4CO_2Etbpy)_3^{2+}$	21,200(25,000)	27,600(24,000)		
	22,600(21,000)			315
$Ru(bipym)_3^{2+}$	22,000(8600)	(27,600)		
	23,900(8200)	30,100(17,000)		1
$Ru(bpy)_2Cl_2$	18,280(8900)	26,595(8700)	33,555vs etc	2
$Ru(bpy)_2Br_2$	15,870?	26,810(10,160)	33,440vs	
	18,360(9710)			
	20,580?			505
$Ru(bpy)_2I_2$	15,100?	26,250(9210)	33,410vs	
	18,400(8340)			505
$Ru(bpy)Cl(CH_3CN)^+$	20,790(8700)	28,985(7900)	34,245vs etc	2
$Ru(bpy)_2(NH_3)_2^{2+}$	20,400(9300)	29,000(7600)	34,500vs etc	337
$Ru(bpy)_2en^{2+}$	20,600(9900)	29,100(7600)	34,400vs etc	337

cont

Complex	d---π_1	d---π_2	π---π^*	Ref
$Ru(bpy)_2(AMPy)^{2+}$	21,250(10,000)	29,500(11,000)	33,600vs	
	(23,700)(6300)		41,000vs	337
$Ru(bpy)_2(MPP)_2^{2+}$	23,400(6900)		35,500vs etc	338
$Ru(bpy)_2(P\text{-}P)_2^{2+}$	26,050(5750)	32,400(9300)	35,700vs etc	338
$Ru(bpy)_2(Ph_3P)Cl^+$	22,000(6160)	30,500(7100)	34,250vs etc	338
$Ru(bpy)_2(MPP)Cl^+$	21,500(6160)	30,000(6000)	34,150vs etc	338
$Ru(bpy)_2(Ph_3As)Cl^+$	21,400(6450)	29,700(6300)	34,000vs etc	338
$Ru(bpy)_2(Ph_3Sb)Cl^+$	21,300(7100)	29,400(7200)	34,000vs etc	338
$Ru(bpy)_2(NO_2)_2$	22,850(7470)	30,300	34,770vs	505
$Ru(bpy)_2(CN)_2$[a]	21,740(7730)	31,250	34,710vs	505
$Ru(bpy)_2$(oxalate)	16,250	27,550(9050)	34,090vs	
	19,380(8740)			505
$Ru(bpy)Cl_4^{2-}$[c]	19,690	27,250		505
$Os(bpy)_3^{2+}$	15,620	25,970(9020)		
	17,270(3270)	27,210(9550)		
	20,920(11,100)		34,500vs	505
$Os(bpy)_2(CN)_2$[a]	17,280(2540)	25,810		
	21,050(8790)	27,970(7790)		
	22,780	30,330(7810)	34,400VS	505
$Os(bpy)(en)_2^{2+}$	17,240	23,150(10,000)	34,260vs	
	20,790			505
$Os(bpy)_2Cl_2$	11,880?	21,480		
	17,910	26,110		505
$Cu(bpy)_2^+$	23,000			422
$Cu(o\text{-}phen)^+$	23,000			425
$TiCl_3.bpy^+$	18,000			427
$Co(BPZ)_3^{2+}$	26,400(3280)			428
$Ni(BPZ)_3^{2+}$	27,600(750)			428

Complex	π---t_{2g} LMCT[d]		π---π^*	Ref
$Fe(bpy)_3^{3+}$	16,310(289)?		31,730 etc	597
$Fe(bpy)_2Cl_2^+$	16,950(60)?	27,400(11,150) πCl,πbpy		597
$Fe(bpy)(CN)_4^-$	19,690(104)	24,070(895) πCN	32,260	597
		26,600(1310) πCN		
$Fe(bpy)_2(CN)_2^+$	18,220(194)	25,320(1180) πCN	32,020	597
$Ru(bpy)_2Cl_2^+Cl^-$	(19,570)	(23,810) 26,400(5100) πCl, πbpy?		596,597
$Ru(bpy)Cl_4^-$		24,600(2680) 27,430(3860) πCl		597
$Ru(bpy)Br_4^-$		17,730(792) 23,470(4830) πBr		597
$Ru(bpy)(py)_4^{3+}$	14,730(150)			597
$Ru(bpy)_2(py)_2^{3+}$	14,800(260)	23,900(3600)		597,598
$Ru(bpy)_3^{3+}$	14,790(409)	23,900(3000)		597,598
$Ru(bpy)_2(H_2O)_2^{3+}$	15,100(6020)			597
$Os(bpy)_3$	21,820 (22,220)	(23,810)		595

cont

Table 5.16 Footnotes---
AMPy = 2-aminomethylpyridine MPP = $MePh_2P$ P-P = $Ph_2P(CH_2)PPh_2$. The ML_2X_2 systems with L = bpy, BPZ are almost certainly all cis in configuation. a) Very solvatochromic - data in MeOH. b) Very solvatochromic - data in CH_2Cl_2. c) Prepared only in water. d) πbpy---M except where indicated (but see text).

The spectra were fitted with the following parameters:

	E(1)	E(2)	K	λ
$Fe(bpy)_3^{2+}$	100	-1500	800	440 cm^{-1}
$Ru(bpy)_3^{2+}$	500	-1600	850	1200
$Os(bpy)_3^{2+}$	800	-2100	850	3000

(5.36)

where E(1) is the $d(A_1)$--$d(E)$ splitting, positive with $A_1 > E$, E(2) is the π^* splitting which, being negative, requires $\pi^*(A_2) > \pi^*(E)$, and K is the exchange integral. The magnitudes of these splittings, and their trends, are intuitively reasonable. The observation that the $\pi^*(E)$ level lies below $\pi^*(A_2)$ is at variance with some previous studies [310,318,328] and at variance with simple ligand field theory. Since the d(E) levels must interact with (move upwards) $\pi^*(E)$ and there are no d orbitals to interact with $\pi^*(A_2)$, the order $\pi^*(A_2) < \pi^*(E)$ is predicted. However the d and π^* levels are some 20,000 cm^{-1} apart so that the effect may be small. Much closer lie the 2 E levels of the next set of π^* orbitals and these will depress the $E(B_2)$ state, possibly more than the A_2 state is depressed by the next π^* A_2 (D_3) level [316].

The singlet-triplet exchange energy K is somewhat smaller than found for some platinum complexes (3000-5000 cm^{-1}) [41,74,77,92,98] but since the ligands in the latter cases (CN^-, PR_3 etc) probably have greater metal-ligand mixing than does bipyridine, the K values here should be smaller.

> The greater the degree of metal-ligand orbital mixing, the greater the contribution of one-centre integrals to the total value of the exchange integral K, and the larger the value of K.

In the osmium spectra, the overall result is similar (three strong bands) but the degree of mixing in of spin triplet states is greater by a factor of three. This explains, in part, the shorter lifetimes of these states.

Crosby and co-workers present a very different approach to the tris(bipyridine) systems [310,312,317,318]. The promoted π^* electron is coupled to the $(t_{2g})^5$ core. Complete spin-orbit coupling within the $(t_{2g})^5$ core (in double D_3 group symmetry) leads to three doubly degenerate spinor states. These are then each coupled weakly, with the hole in $\pi^*(A_2)$ or, at higher energy $\pi^*(E)$ to yield the manifold of excited states. Crosby and co-workers report emission data in support of their contention that the $\pi^*(A_2)$ state lies lowest [310]; however this has been disputed in later

studies.

The Crosby model also generates splittings which require $K < 200\ cm^{-1}$ which does seem too small [316]. Finally the fact that spinor states are used as a basis leads this group to propose that it is meaningless to try and distinguish spin singlet from spin triplet states. Clearly this is in contradistinction to the Kober and Meyer approach [316].

Woodruff and co-workers [314,319,335], using the technique of 'time resolved resonance Raman' (TR^3) spectroscopy, revealed that the lowest excited state of the $[Ru(bpy)_3]^{2+}$ ion, at equilibrium, is localised on one of the three bpy ligands. Thus the Raman spectrum of the excited state, 1ns after excitation, appears to be essentially a superimposition of the Raman spectrum of bpy and the Raman spectrum of the isolated $(bpy)^-$ ion. The excited state should then be regarded as $[Ru(III)(bpy)_2(bpy)^-]^{2+}$ with maximum C_2 symmetry.

Reduction of the $[Ru(bpy)_3]^{2+}$ ion can be carried out chemically or electrochemically to generate a series of reduced species. The first reduction product involves addition of an electron to a bipyridine ring (not to Ru(II)), to generate ground state $[Ru(II)(bpy)_2(bpy)^-]^+$. The spectra of this species, and of further reduced species, have been reported [315]. The spectra are very rich and generally include low energy absorption in the near infrared. Such absorption arises, in part at least, from the reduced anions themselves. Electrochemical evidence is introduced [315] to show that the 'odd' electron in the (ground state) first reduced species is also localised on one bpy ring.

Resonance Raman indicates that the Re---bpy excited state of $fac\text{-}ReX(bpy)(CO)_3$ also possesses the characteristics of the bpy anion radical [108]

Many species of the type (cis and trans) $[Ru(diimine)_2XY]$ complexes have been prepared (see Table 5.16). The feature of interest is the effect of changing X and Y upon the energies of the Ru---bpy (and related ligand) MLCT bands. Generally, hard, and π-acceptor ligands raise the potential of the Ru(III)/Ru(II) couple and shift the Ru---diimine MLCT band to the blue. Softer, and π-donor, ligands result in lower potentials and red shifted MLCT bands (Table 5.16).

Ruthenium(III) and related ions form analogous $[Ru(bpy)_3]^{3+}$ species. Data are included in Table 5.16. Note that in the $[Os(bpy)_3]^{3+}$ ion, the splitting of the t_{2g} set (in D_3) symmetry (discussed above) can be directly observed as a d-d transition in the near infrared (4580(450) and 5090(360) cm^{-1}) [341].

Bryant and Fergusson [597] provide absorption data for a wide range of Fe(III), Ru(III) and Os(III) bipyridine species with assignments for LMCT transitions from π-bpy to t_{2g}, and, where relevant, π-halogen---t_{2g} (Table 5.16). The lowest energy CT band in a series of such complexes is shown to correlate in a linear fashion with the M(III)/M(II) redox potential. However the spectra are relatively complex and additional bands with no clearcut assignment are often observed. There is need for further work in this area. The unusual oxidation state, Rh(II), forms bipyridine complexes which are obtained by pulse radiolytic reduction of Rh(III) [103] (Table 5.16). Zelewsky has published a useful compilation of redox and MLCT data for a range of bipyridine analogs with extended π-systems, e.g. bi-quinolyl type ligands and their ruthenium(II) complexes [510]. In general, extension of the aromatic pathway red shifts, as anticipated, the MLCT spectrum, but the band shape usually remains unchanged. In polyene iron(II) macrocycle species, Dabrowiak and Busch have discussed the shift to the blue in the Fe(II)---ene MLCT band in changing from diimine to monoimine [626].

Mason and co-workers discuss the optical rotatory power of a range of tris(bipyridine) and tris(o-phenanthroline) metal complexes [598,599], with special, but not exclusive emphasis on π---π^* transitions. These authors present a useful selection of absorption and CD spectra in their discussion of assignment and absolute configuration. Day and Sanders [327,601], also in an early study, investigate the effect of substituents in phenanthroline ligands upon CT energies in Fe(II), Fe(III) and Cu(II) species.

5.11 Charge Transfer Transitions in Systems of Biological Interest

5.11. i) Copper(II) Charge Transfer of Biological Interest

Because of the interest in the spectra of copper proteins, blue and non-blue, there has been considerable interest in the charge transfer spectra of copper(II) in regimes of biological significance. Specifically these include sulphur, imidazole and tyrosine ligands derived from amino-acids.

Copper sulphur model systems

It is now well known that the blue colour of the 'blue' copper proteins originates in a transition near 16,000 cm^{-1} attributed to σ(S)---Cu(II) LMCT, with a lower energy weaker transition assigned to πS---Cu(II) (see Chapter 8). It is clearly of interest to know how these bands may vary in energy and intensity as a function of other parameters such as the other ligating atoms and the stereochemistry. This has prompted many workers to synthesise sulphur-containing Cu(II) complexes and study their spectra [5,6, 229,230,288,291-295,297,299,606].

One of the most well designed studies was undertaken by Bosnich [6,297], who reported data for a range of different well defined environments. Data are listed in Table 5.17, and some examples are shown in Fig.5.26. Square planar complexes containing thioether sulphur were shown to exhibit a fairly strong band near 30,000 cm^{-1}, which was assigned, because of its intensity, to $\sigma(S)---x^2\text{-}y^2$ Cu(II) LMCT [5,6]. No clear evidence was present of the corresponding π(S)---Cu LMCT transition. Indeed it seems on the basis of the now considerable available evidence that thioether bound to copper(II) does not exhibit a πS---Cu(II) LMCT band in the

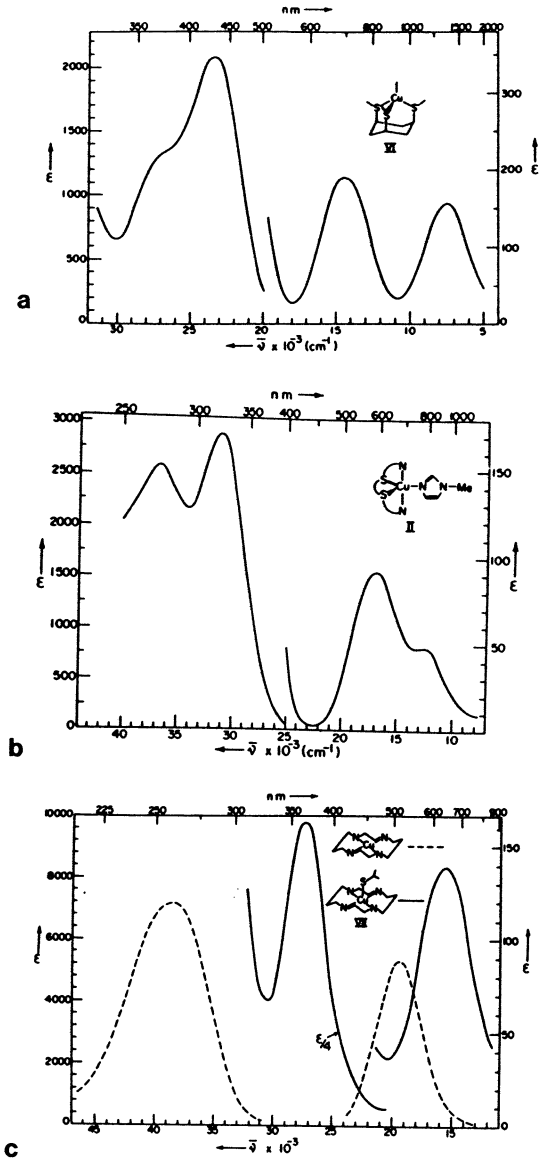


Fig.5.26 (a,b,c) The absorption spectra of the species indicated in a) acetic anhydride, b) water and c) ethanol (solid line), and water (hatched line) [297].

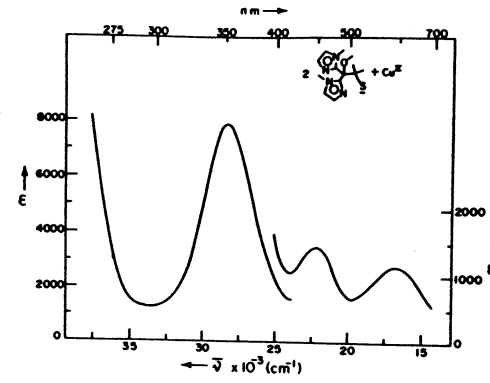


Fig.5.26 (d) The absorption spectrum of the species indicated [297].

electronic spectrum. However this conclusion should be treated with caution since stereochemistry may also be a factor.

Shoulders were seen on the low energy edge of transitions in complexes with more than one thioether S ligand. These may be due to resolution of several σ(S)---Cu(II) transitions or possibly the π(S) transition. There was a rough correlation between intensity and the number of thioether atoms. The trigonal bipyramidal complex (Table 5.17) also showed the LMCT band near 30,000 cm^{-1} while a tetrahedral CuS_3O chromophore displayed this band to somewhat lower energy, near 25,000 cm^{-1}.

Rorabacher and co-workers [291,294,295] studied a wide series of square (or approximately square) $[CuS_4]^{2+}$ chromophores using S_4 macrocycles of varying ring size (Table 5.17). All of these complexes exhibit a strong band near 25,000 cm^{-1} and a weaker, but still relatively strong (1000 - 2000 l mol^{-1} cm^{-1}) band near 16,000 cm^{-1}. These were assigned as σS---Cu(II) and πS---Cu(II) LMCT respectively. However the general absence of πS---Cu(II) transitions in thioether Cu(II) complexes and the lack of any relationship between the energy of this purported πS---Cu(II) transition and the potential of the E(1/2) Cu(II)/Cu(I) couple [294,298] lead to the conclusion that the band is mis-assigned. Since the 25,000 cm^{-1} transition does correlate reasonably well with E(1/2), we conclude [298] that the lower energy band is d-d (of enhanced intensity due to the close proximity of the stronger LMCT band) while the upper energy band is correctly assigned as σS---Cu(II) LMCT.

Ligating thioether is observed in the complex $Cu(MeSCH_2CH_2NH_2)_2 \cdot 2ClO_4$ [229]. This species displays a band at 25,600 cm^{-1} not generated by the analogous $Cu(NH_2CH_2CH_2NH_2) \cdot 2ClO_4$ and methylated analogs (see section 5.12). Given that

σ(N)---Cu(II) falls at 37,000 cm^{-1} in the thioamine derivative and that the ionization potential of CH_3SCH_3 is about 8000 cm^{-1} lower than CH_3NH_2, then it is reasonable to suppose [228] that the 25,600 cm^{-1} band should be assigned σ(S)---Cu(II)(x^2-y^2).

A planar CuS_4 chromophore is purported to exist in $[Cu(BuSCH_2CH_2SBu)_2](ClO_4)_2$, and bands at 17,500 and 22,700 may arise from πS and σS---Cu(II) LMCT from equatorial bound thioether though the former transition, by analogy with the Rorabacher data, may be mainly d-d [229]. A disulphide complex exhibits at least three S---Cu(II) LMCT transitions [6] (Table 5.17).

Thiolate systems (RS^-) and sulphhydryl systems (RSH) do appear to exhibit both σS-- and πS---Cu(II) LMCT. Data for a range of other thiolate bound complexes in various stereochemical environments were reported by Bosnich and co-workers [6] (Table 5.17). Most exhibit a moderately intense band near 22,000 - 25,000 cm^{-1} assigned to πS---Cu(II), and a stronger band near 28,000 cm^{-1} due to σS---Cu(II) (Fig.5.26).

A purported axial thiolate system, [Cu(cyclam)]SR displays a band at 27,500 cm^{-1}, of intensity [6] about 1/3 that for the corresponding transition in equatorially bound thiolate. Certainly axial Cu(II) interactions are expected to be long and should not give rise to strong CT features (poor overlap due to length and orthogonality to acceptor x^2-y^2 orbital). Indeed an intensity as high as in fact observed seems anomalous, and it is possible the species does not have the proposed structure, e.g. could it be trigonal bipyramidal in solution ?

Indeed another apparent axially coordinated thiolate was studied by Schugar and co-workers [230]. The 14-membered macrocycle tetb complex of Cu(II) reacts with o-mercaptobenzoic acid, to yield $Cu(tetb)(SC_4H_4CO_2)$ for which X-ray structural analysis revealed a trigonal bipyramid with a buckled macrocyclic ring. The sulphur atom occupies an equatorial site but the intensity of the σS---Cu(II) transition (Table 5.17) is rather low, indeed similar to the cyclam complex. Assignment was facilitated by comparison of the spectrum of the Cu(II) complex with that of the zinc analog. Note that thioether axially coordinated to Cu(II) does not yield a detectable LMCT transition [300].

The models discussed above contain copper(II) in a square, octahedral or five coordinate environment and do not exhibit an LMCT band as low in energy as in the tetrahedral Cu(II) proteins (Chapter 8). The stereochemistry in the proteins evidently moves the acceptor orbital to quite a low energy. Indeed tetrahedral pyrazolylborate species have recently been reported to contain a CuN_3S chromophore and to exhibit strong bands near 16,000 cm^{-1} (Table 5.17) assigned to σS---Cu(II) LMCT [303,304].

Some model studies have been reported for metals other than copper(II), to mimic the behaviour of copper proteins substituted by metal ions such as Co(II). Such data are also incorporated into Table 5.17.

Table 5.17

CHARGE TRANSFER SPECTRA OF SOME COPPER SULPHUR LIGAND MODEL SYSTEMS

Complex	Structure	$E_{1/2}$	Charge Transfer	d-d	Ref
Thiol models					
Cu(cyclam)SR	SqPy CuN_4S		27,500(2400)σ	16,000(140)	
			38,600(7200)σN		6
CuL(1)(p-S(Cl)Ph)	SqPy CuN4S-		23,400(5310)π		
			28,200(5325)σ		235
Cu(tet b)(o-$SC_6H_4CO_2$)[e]			23,250(800)π	10,900(300)	
	TBP CuN_4S		23,900(1000)π	13,700(1000)	
			27,800(2200)σ	17,000(200)	230
$Cu(BVI)_2$	Oh CuN_4S_2		22,200(1500)π	16,950(1100)	
			28,600(8000)σ		297
$Cu(BVI)_2 + H^+$	Oh CuN_4S_2		24,700(300)π(SH)		
		ca	32,200(1750)σ(SH)		297
Cu(MPG)(H_2O)	Sq $CuNO_2S$		25,000(700)π	16,500(300)	293
Cu(MAH)(H_2O)	Sq CuN_2SO		22,800(970)π	16,700(830)a	292
Cu(pyrazbor)SR[b]	Td CuN_3S		17,000(3900)σ		303
Cu(pyrazbor)SR[c]	Td CuN_3S		14,700(2000)σ		303
Thioether models					
$Cu(Rmi1)_2(ClO_4)_2$	Sq t-CuN_2S_2	+.33	24,150σ	16,200	5
$Cu(Rmi2)_2(ClO_4)_2$	Sq t-CuN_2S_2	+.37	25,200σ	16,000	5
$Cu(Rmi3)_2(ClO_4)_2$	Sq t-CuN_2S_2	+.34	25,250σ	16,500	5
$Cu(Rmi4)_2(ClO_4)_2$	Sq t-CuN_2S_2	+.36	28,600σ	14,950	5
$Cu(dtn)^{2+}$	Sq c-CuN_2S_2		(28,040)(3060)	17,800(400)	
			31,090(5680)σ		
			40,650(2870)Nσ		6
$Cu(dat)^{2+}$	SqPy CuN_3S_2		(27,600)	17,800(180)	
			29,350(5680)σ		
			40,200(2500)Nσ		6
Cu(mtc)NMeIm	TBPy CuN_2S_2N		31,250(3000)σ	12,650(50)	
				17,150(100)	6
Cu(tmtc)(Ac_2O)	Td CuS_3O		23,500(2100)σ	7500(150)	
			(26,000)	14,500(200)	6
$Cu(idh)(ClO_4)_2$	Sq c-CuN_2S_2	+.23	28,500(4250)σ	15,100	5,6
Cu(NSN)NMeIm	Sq CuN_3S		31,170(2700)σ	16,500(150)	6
$Cu(MeSCH_2CH_2NH_2)_2$	Sq CuN_2S_2		25,650σS	18,200σ? d-d	
			37,000σN		229
$Cu(BuSCH_2CH_2SBu)_2$	Sq CuS_4			17,500(1050)	
			22,700(11,240)σ	+ πCT?	
			28,900(790) 36,600(2400)		
			44,000(2000)		229

cont

Table 5.17 cont

Complex	Structure	$E_{1/2}$	Charge Transfer	d-d	Ref
$[Cu(BuSCH_2CH_2SBu).2Cl]_2$			(17,550)π 22,300σ 25,850 34,250πCl? 41,700πCl?	13,300(460)	229
$Cu(EDDA).H_2O$	Oh	CuO_4S_2	30,800(2600)σ 36,500(4000)(CO_2)?	12,800(62)	229
$Cu(Py_2S_2\text{-}555)$[f]	Sq	CuN_2S_2	+.71 29,100(2300)σ		288
$Cu(Py_2S_2\text{-}565)$	Sq	CuN_2S_2	+.75 27,800(3430)σ		288
$Cu(Py_2S_2\text{-}656)$	Sq	CuN_2S_2	+.84 27,800(5560)σ		288
$Cu(Py_2S_2\text{-}666)$	Sq	CuN_2S_2	+.89 26,300(5020)σ		288
$Cu(12\text{-}aneS_4)^{2+}$	Sq	CuS_4	+.72 25,800(6000)σ	14,800(2000)	295
$Cu(13\text{-}aneS_4)^{2+}$	Sq	CuS_4	+.67 25,600(6000)σ	16,000(1800)	295
$Cu(14\text{-}aneS_4)^{2+}$	Sq	CuS_4	+.69 25,600(3100)σ	17,500(1900)	295
$Cu(15\text{-}aneS_4)^{2+}$	Sq	CuS_4	+.78 24,100(8000)σ	17,700(1140)	295
$Cu(16\text{-}aneS_4)^{2+}$	Sq	CuS_4	+.80 22,700(6100)σ	16,600(800)	295
$Cu(15\text{-}aneS_5)^{2+}$	Sq	CuS_4	+.85 24,150(7000)σ	17,700(2000)	295
$Cu(20\text{-}aneS_6)^{2+}$	Sq	CuS_4	+.80 23,800(7100)σ	18,300(1400)	295
$Cu(Et_2TTU)$	Sq	CuS_4	24,400(6800)σ	16,300(1100)	295
Disulphide model					
$Cu(Me_4\text{-cystam})$		CuN_3S_2	22,500 29,850 37,000	16,300	297

The symbols σ and π refer to σS---Cu(II) and πS---Cu(II) LMCT transitions except where indicated otherwise. a) Assigned as σS---Cu(II) by authors [292]. b) R = $SC_6H_4NO_2$. c) R = $SCH_2CH(NH_2)CO_2Et$. d) Electrochemical data for this series obtained from ref. [294]. e) TBP with equatorial Cu-S. f) N_2S_2 macrocycles with ring sizes as indicated by numerals. Pyrazborate is the pyrazoylborate ligand $HB(3,5\text{-}Me_2pyr)_3$. Rmi = (4-(alkylmercaptomethyl)imidazole; Rmi1 R = n-Pr, Rmi2 R = t-Bu, Rmi3 R = CH_2Ph, Rmi4 R = Ph. L(1) = 3,9-dimethyl-4,8-diazaundeca-3,8-diene-2,10-dione dioxime. See additional data in [231,232].

Imidazole model systems

The binding of imidazole to metal ions is of especial interest because of the importance of histidine binding in metalloproteins. Great effort has been expended, especially by the Schugar, Krogh-Jespersen group, in identifying the spectroscopic fingerprint of imidazole bound to Cu(II) [228,231,232,296]. These transitions are of the LMCT type, from the higher lying orbitals of imidazole to Cu(II). In addition to the σ-lone pair, there are two high lying

filled π-type orbitals on imidazole (and related molecules such as pyrazole), labelled π_1 and π_2.

PES data reveal that these orbitals lie at [228,239]:

	π_1	π_2	σ
Imidazole	8.78	10.3	10.3 eV
Pyrazole	9.15	9.88	10.7

We expect [232] that tetragonal copper-imidazole complexes will show three LMCT transitions, namely σ(im)---$\sigma^*(x^2\text{-}y^2)$(Cu(II)), π_2(Im)---σ^* and π_1(Im)---σ^* near 45,500, 38,500 and 30,300 cm^{-1} respectively. In general the σ(N)---Cu(II) transition was sharply defined, but the π(Im)---Cu(II) transitions were often only poorly defined shoulders.

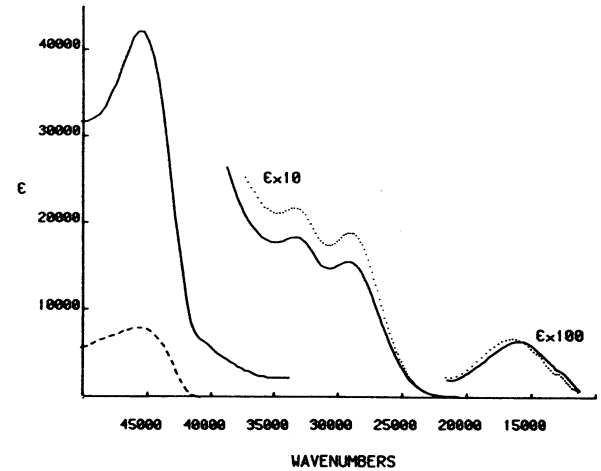


Fig.5.27 Absorption spectra of 0.0035 M $Cu(II)(ClO_4)_2.6H_2O$ in methanol, containing 0.014 M (solid line) and 0.028 M (....) 4,5-dimethylimidazole. The free ligand spectrum (0.014 M) is also shown (----) [231].

This was interpreted as being due to rotation of the imidazole about the Cu-N bond, resulting in broadening of the π---Cu(II) transitions and hence loss of their definition. When this rotation is inhibited by use of substituted imidazoles these two π---Cu(II) transitions become more clearly defined [231,296] (Fig.5.27, Table 5.18). Tetragonal complexes were prepared by treatment in solution of Cu(II) ions with excess of substituted imidazole. Pyrazole forms similar complexes, and the $M(pyrazole)_4$ chromophore is most readily obtained through use of pyrazolylborate and pyrazolylgallate ligands [231,236,237]. The intense σN---σ^* transition is generally overlapped by the internal ligand π---π^* transition. The two

π---Cu(II) transitions are only about 4000 cm^{-1} apart in the imidazole and simple pyrazole systems and are not always separately resolved. The separation is somewhat larger in the pyrazolylborates (5000 - 7000 cm^{-1}). In tetrahedral species there is a red shift of about 10,000 - 12,000 cm^{-1} due primarily to the marked stabilisation of the acceptor d orbital relative to the tetragonal case. A blue shift of about 5000 cm^{-1} occurs when Cu(II) is replaced by Co(II) or Ni(II) (Table 5.18). The lower energy π_1---Cu(II) transition has been observed in some copper(II) protein derivatives (see Chapter 8). Schugar also recently studied the Ru(III)-imidazole system [506] based upon preparative work of Taube [507,508], providing support for the validity of the Cu(II)-imidazole interpretation discussed here. The excited L^+d^{10} LMCT configuration of Cu(II) produces only one state, as does $L^+(t_{2g})^6$ of Ru(III); thus the spectra are similar and relatively simple.

Comparison of the spectra of imidazole-containing Ru(III) complexes with systems bereft of imidazole, indicates the presence of two extra charge transfer transitions in the visible and near uv (in $[(NH_3)_5Ru(Im)]^{3+}$ species). These transitions shift substantially to the red in corresponding imidazolate anion complexes (the π-levels are destabilised) providing evidence that they are LMCT, π_1,π_2---Ru(III), in nature. Krogh-Jespersen and Schugar so assign them [506] (Table 5.18) and note that they shift to the red when the imidazole is C-substituted with alkyl groups.

Taube also characterised some interesting C-bonded linkage isomers of the type $[L(NH_3)_4(Im\text{-}C)]^{3+}$ where L may be NH_3 or other ligands [507,508]. The electronic structure of these complexes is very similar to those of the N-bonded isomers with the low energy LMCT band (from π_1) shifted somewhat to the red of the corresponding N-bonded isomer. Further C-substitution causes a further red shift. The higher energy π_2---Ru(III) is absent. The separation between π_1 and π_2 is greater in the C-bonded than in the N-bonded isomers, so the π_2---Ru(III) transition may be obscured in the uv. However π_2 for the C-bonded isomer has a node at the C atom which would be likely to provide a low oscillator strength for this transition. An Fe(III)-imidazole study has very recently been published [623].

Imidazolato (Im^-) groups exist in such proteins as superoxide dismutase (see Chapter 8) and there is obvious interest concerning the possible use of electronic spectroscopy to distinguish bound imidazole from bound imidazolato. Schugar [228] reports absorption near 26,700 cm^{-1} which appears associated with Im^----Cu(II) LMCT, probably $\pi_1(Im^-)$---Cu(II) [228].

5.11 ii) Tyrosine-Metal Charge Transfer

Copper: Tyrosine is the only amino acid with a pendant hydroxy group which, when bound to a redox active metal atom, can give rise to charge transfer in or close to the visible region. The binding of this amino acid to such a metal can therefore be detected via the charge transfer transition.

Table 5.18

CHARGE TRANSFER IN MODEL SYSTEMS OF BIOLOGICAL INTEREST (excluding Cu-S)

Species	Absorption spectra	Assignment	Ref
Tyrosine Models			
$[FeL_4]^{2-}$			
2,3,5,6-trimethylphenolate	23,700(9440)	$t_1(O)$---e	
	31,700(14,300)	$t_1(O)$---t_2	
	35,600(25,900)		56
2,4,6-trichlorophenolate	22,900(9830)	$t_1(O)$---e	
	30,300(11,000)	$t_1(O)$---t_2	
	34,500(29,200)		56
2,3,5,6-trimethylthiophenolate	22,300(7230)	$t_1(S)$---e	
	29,100(6880)	$t_1(S)$---t_2	
	33,900(14,300)		56
$FeCl_4^-$	27,480(7350)	$t_1(Cl)$---t_{2g}	
	31,500(7600)		233,234
Fe(4Me-phenolato)$(H_2O)_5^{2+}$	16,400		58
Fe(EDDHA)$^-$	21,050(4000)	πO---t_{2g}	
	(31,750)(3000)	πO---e_g	60
Sulphur amino acid models (excluding copper(II))			
$Co(SPh)_2$	23,800	S---Co(II)	289
$Co(thiourea)_2Cl_2$	31,250	S---Co(II)	290
$Co(Py2SH)_4(ClO_4)_2$	22,720	πS---Co(II)	391
$Co(Py2SH)_2Cl_2$	25,060	πS---Co(II)	391
$Co(Py2SH)_2I_2$	24,150	πS---Co(II)	391
$(Et_4N)_2[Co(S_2\text{-o-xyl})_2]$	28,170(3450)	πS---Co(II)	46
$Ni(Py2SH)_2Cl_2$	26,300	πS---Ni(II)	391
$Zn(Py2SH)_4(ClO_4)_2$	26,850	πS---Zn(II)	391
$Cd(Py2SH)_2Cl_2$	27,550	πS---Cd(II)	391
$Hg(Py2SH)_2Cl_2$	27,700	πS---Hg(II)	391
$Bi(Py2SH)_2Cl_3$	20,890	πS---Bi(III)	391
$Sn(Py2SH)_2Cl_4$	26,600	πS---Sn(IV)	391
$Co(2\text{-}Pic\text{-}6SH)_4(BF_4)_2$	22,370	πS---Co(II)	391
$Co(2\text{-}Pic\text{-}6SH)_2Cl_2$	24,400	πS---Co(II)	391
$Co(SC(CH_3)_2CH_2NH_2)_2$	27,000	πS---Co(II)	
	35,700	*S---Co(II)	301
$Co(HB(3,5Me_2pyz)_3)SR$ R^a	29,400	S---Co(II)	302
R = $SC_6H_4NO_2$	31,450	S---Co(II)	302
$Ru(NH_3)_5SEt^{2+}$	19,700(1900)	S---Ru(III)	509
Histidine models			
Imidazole	48,500(6900)	π---π^* Im	231

cont

Table 5.18 cont

Species	Absorption spectra	Assignment	Ref
$Cu(Im)_4 \cdot 2NO_3$	19,000	d-d	
	34,000	π_1---σ^* Cu(II)	
	40,000	π_2---σ^*	296
4,5-Et_2-imidazole (4,5EtIm)	45,500(8000)	π---π^*	231
4:1 4,5-EtIm/Cu(II)	16,100(67)	d-d	
	29,000(1550)	π_1---π^* Cu(II)	
	33,000(1770)	π_2---π^* Cu(II)	
	45,200(41,000)	π---π^* + σN---σ^*	231
1,4,5-trimethylimidazole	44,800(4570)	π---π^*	296
$Cu(tmp)_4 \cdot 2ClO_4$	16,800(50)	d-d	
	28,700(1880)	π_1---σ^* Cu(II)	
	32,800(2060)	π_1---σ^*	
	44,600(25,100)	π---π^*	296
1,2-dimethylimidazole	48,100(4300)	π---π^*	296
$Cu(1,2\text{-dmp})_4 \cdot 2ClO_4$	16,800(50)	d-d	
	28,900(1920)	π_1---σ^* Cu(II)	
	32,500(1860)	π_2---σ^*	296
$Ni(1,2\text{-dmp})_4 \cdot 2ClO_4$	22,200(130)	d-d	
	35,100(440)	π_1---Ni(II)	296
Pyrazole (Pyz)	48,100(6300)	π---π^* Pyz	231
4:1 Pyz/Cu(II)	13,900(41)	d-d	231
	32,100(430)	π_1---σ^* Cu(II) + π_2---σ^* Cu(II)	
	47,400(34,300)	π---π^* + σN---σ^*	231
3,5-dimethylpyrazole (3,5-dmp)	46,700(5700)	π---π^*	231
4:1 3,5-dmp/Cu(II)	14,100(40)	d-d	
	28,600(685)	π_1---σ^* Cu(II)	
	32,300(890)	π_2---σ^* Cu(II)	
	45,900(41,200)	π---π^* + σN---σ^*	231
4:1 1,2-MeIm/Co(II)[b]	17,000(570) 17,500(550)	$^4T_1(P)$---$^4A_2$	
	18,900(330)	π_1---Co(II)	228
	33,900(400)	π---π^*	
$Zn(H_2B(Pyz)_2)_2$	45,900(24,000)	π---π^*	231,236
	(46,500)(23,000)		
$Cu(H_2B(Pyz)_2)_2$	18,500(68)	d-d	
	30,500(1000)	π_1---Cu(II)	
	37,700(1000)	π_2---Cu(II)	
	49,300(20,000)	π---π^* + σ(N)---Cu(II)	231,236
$Co(H_2B(Pyz)_2)_2$[b]	35,700(1700)	π_1---Co(II)	
	41,300(5000)	π_2---Co(II)	
	48,100(18,000)	π---π^*	231,236

cont

Table 5.18 cont

Species	Absorption spectra	Assignment	Ref
$Cu(Me_2B(Pyz)_2)_2$	20,000(33) (17,000)	d-d	
	31,600(1140)	π_1---Cu(II)	231,237
$Cu(Me_2Ga(Pyz)_2)_2$	15,400(93) (12,500)	d-d	
	29,600(1190)	π_1---Cu(II)	231,237
$Cu(Me_2Ga(35diMe\text{-}1\text{-}Pyz)_2)_2$[b]	37,700(2790)	σN---σ^* Cu(II)	
	24,700(1900)	π_2---σ^* Cu(II)	
	20,200(1970)	π_1---σ^* Cu(II)	
	(12,000) 9380(160)	d-d	231,237
$Ni(1,2\text{-}Me_2\text{-imidazole})_4^{2+}$	35,100(440)	π_1---Ni(II)	
	22,200(130)	d-d	228
$Ni(1,2\text{-}Me_2\text{-imidazole})_2Cl_2$	37,000(440)	π_2---Ni(II)	
	30,800	π_1---Ni(II)	
	16,700 18,200	d-d	228,238
$Ni(H_2B(Pyz)_2)_2$	36,100(1400)	π_1---Ni(II)	
	21,200(72)	d-d	228,236
$Ni(Me_2B(Pyz)_2)_2$	37,310(1740)	π_1---Ni(II)	
	21,370(65)	d-d	228,237
$Ru(NH_3)_5(Im)^{3+}$	23,300(250)	π_1---Ru t_{2g}	
	33,400(1880)	π_2---Ru t_{2g}	506
$Ru(NH_3)_5(NMe\text{-}Im)^{3+}$	23,700(186)	π_1---Ru t_{2g}	
	32,200(2310)	π_2---Ru t_{2g}	506
$Ru(NH_3)_5(4Me\text{-}Im)^{3+}$	21,300(267)	π_1---Ru t_{2g}	
	33,000(2020)	π_2---Ru t_{2g}	506
$Ru(NH_3)_5(4,5\text{-}DiMe_2Im)^{3+}$	19,300(168)	π_1---Ru t_{2g}	
	32,100(2190)	π_2---Ru t_{2g}	506
$Ru(NH_3)_5(Im\text{-}N^-)^{2+}$	18,000(350)	π_1---Ru t_{2g}	
	27,700(2990)	π_2---Ru t_{2g}	506
$Ru(NH_3)^5(4Me\text{-}Im\text{-}N^-)^{2+}$	16,700(472)	π_1---Ru t_{2g}	
	27,000(3000)	π_2---Ru t_{2g}	506
$Ru(NH_3)_5(4,5\text{-}DiMe_2Im\text{-}N^-)^{2+}$	13,300(750)	π_1---Ru t_{2g}	
	25,500(2830)	π_2---Ru t_{2g}	506
$Ru(NH_3)_5(Im\text{-}C)^{3+c}$	20,700(890)	π_1---Ru t_{2g}	506
$Ru(NH_3)_4(OH)(Im\text{-}C)^{2+}$	23,530(1750)	π_1---Ru t_{2g}	508
$Ru(NH_3)_4(NCS)(Im\text{-}C)^{2+}$	19,800(1850)	π_1---Ru t_{2g}	508
$Ru(NH_3)_4(H_2O)(4,5\text{-}DiMe_2Im\text{-}C)^{2+}$	16,600(5100)	π_1---Ru t_{2g}	506
$Ru(NH_3)_5(Im\text{-}C^-)^{2+}$	17,000(1300)	π_1---Ru t_{2g}	506

a) R = $SCH_2CH(NH_2)CO_2Et$. b)Tetrahedral. c) Im-C is a carbon bonded imidazole.

Bosnich reported the spectrum of a Schiff base complex of Cu(II) based upon o-aminomethylphenol [6]. Bands at 25,500 and 30,500 cm^{-1} are not readily associated either with d-d or

intra-ligand bands. The copper glycinate complex shows no CT absorption below 40,000 cm^{-1} so that these phenoxy bands cannot be σO---Cu(II) LMCT. The 8-hydroxyquinoline complex of Cu(II) also shows these two bands but displaced by about 2000 cm^{-1} to higher energy. Since the quinoline ring allows more delocalisation than the benzene ring, these transitions are likely to be MLCT not LMCT. Bosnich therefore assigns them to the two lowest d(Cu(II)---π^* transitions of phenoxide [6]. These complexes should also exhibit πO(phenoxide)---Cu(II) LMCT but, based upon the positions of these transitions observed in iron(III) complexes (discussed below), such transitions, to the much less stable Cu(II) x^2-y^2 orbital, will be displaced into the ultraviolet.

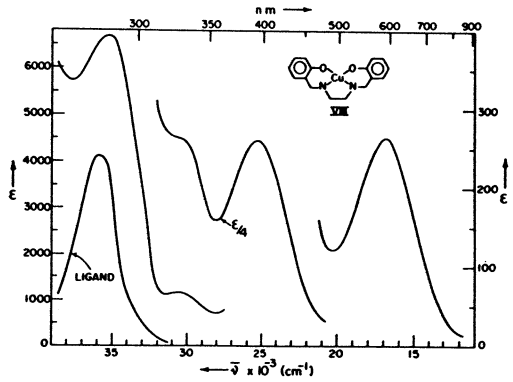


Fig.5.28 The absorption spectrum of system VIII (as indicated) in methanol, and that of the free ligand in methanol containing 2 equiv. of NaOH [297].

Iron: The reaction of ferric chloride with phenols has been recognised since last century as a test for the phenol functionality. Thus Fe(III) ion forms a purple colour with many phenols, a colour due to a charge transfer state. Many proteins contain an Fe(III) tyrosine residue (see Chapter 8). Koch has recently reported some model FeO_4 systems containing the phenol unit [56]. Redox and spectroscopic data were reported for $[Fe(OR)_4]^{2-}$ complexes of 2,3,5,6-tetramethylphenol and 2,4,6-trichlorophenol [56]. These contain a tetrahedral Fe(phenoxide)$_4$ unit and exhibit intense absorption in the visible region ascribed to π-oxygen (phenol) to iron(III) LMCT (Table 5.18). The two bands terminate on the e and t_2 orbitals and are therefore separated by 10Dq. It is a little disturbing to this writer that the CT bands occur at essentially the same energy while the E(1/2) values for the (supposed) Fe(III)/Fe(II) couple occur at -1.32 and -0.45V respectively (see Chapter 9).

In discussing the spectra of lactoferrin, Ainscough and

Table 5.19

MISCELLANEOUS CHARGE TRANSFER DATA

	A	B	C	L	Absorption Spectra kK	Ref
$Ti(S_2CNR_2)_4$	0	4	8	S	25.3, 29.2, 36.9	517
$TiCl_4$	0	4	4	Cl	35.6	422
$TiCl_4$	0	4	4	Cl	34.84	572
$TiBr_4$	0	4	4	Br	27.75, 29.4	422
TiI_4	0	4	4	I	19.46, 24.88, 27.62etc	587
TiI_6^{2-}	0	4	6	I	(9.5)pf, 12.7, 18 etc	422
$TiCl_6^{2-}$	0	4	6	Cl	25	422
$TiBr_6^{2-}$	0	4	6	Br	(17.7)pf, 21.7, 24.9etc	422
$TiCl_3Me$	0	4	4	Cl	43	572
$TiMe(OiPr)_3$	0	4	4		33	572
$TiCl_3(NMe_3)_2$	1	3	5	Cl	30, 37, 44.9	569
$TiBr_3(NMe_3)_2$	1	3	5	Br	27.7, 35.6, 44.8	569
TiF_6^{3-}	1	3	6	F	47	539
$TiCl_3$	1	3	6	Cl	37.1	573
$TiCl_3$	1	3	6	Cl	18.9, 27.5	576
$TiBr_3$	1	3	6	Br	15.5, 23.3	576
$Ti[N(SiMe_3)_2]_3$	1	3	6	N	28.6	588
$V(S_2CNR_2)_4$	1	4	8	S	31.7, 34.8, 44.1	517
VCl_6^{2-}	1	4	6	Cl	21.05	514
VCl_4	1	4	4	Cl	24.5e, 34.84t2	515,407
VF_6^{2-}	1	4	6	F	34,37	531
$V(OBut)_4$	1	4	4	O	28	516
$V(NMe_2)_4$	1	4	4	N	22	516
VF_6^{3-}	2	3	6	F	42.2	539
$V(NCS)_6^{3-}$	2	3	6	NCS	24.8?,31.9	526
$V(S_2CNR_2)_3$	2	3	6	S	24.0, 30.5 40.7	517
$VCl_3(NMe_3)_2$	2	3	5	Cl,N	37.85, 44.05	569
$VBr_3(NMe_3)_2$	2	3	5	Br,N	30.1, 34.4, 40.8, 45.5	569
$VCl_3(SMe_2)_2$	2	3	5	Cl	37.74, 41.2	569
$VCl_3(SMe_2)_2$	2	3	6	Cl	28.6, 37.8, 45	569
$VBr_3(SMe_2)_2$	2	3	5	Br	30, 33.8	569
$VBr_3(SMe_2)_2$	2	3	6	Br	23.5	569
$V(CN)_6^{4-}$	3[a]	2	6	CN	36.4, 38.0, 47.4	567
$V(S_2PPh_2)_3$	2	3	6	S	22.5	518
$V(S_2AsPh_2)_3$	2	3	6	S	21.8	520
$V[N(SiMe_3)_2]_3$	2	3	3	N	24.7, 28.1	588
$V(Py)_4Cl_2$	3	2	6	P	22.0,31,8	528
$V(NCS)_6^{4-}$	3	2	6	NCS	25?, 32	529
$VO(PyNO)_4^{2+}$	1	4	5	N-O	22	514

cont

Table 5.19 cont

	A	B	C	L	Absorption Spectra kK	Ref
$VO(S_2CNR_2)_2$	1	4	6	S	22.5, 32.8, 45.0	517
$VO(CN)_5^{3-}$	1	4	6	CN	37.4, 45.6	567
$VO(H_2O)_5^{2+}$	1	4	6	=O	ca.33-34	568
$VO(CH_3CO_2)_2$	1	4	5	=O	41.7	519
$VOCl_3$	0	5	4	Cl	29	407
$VOCl_4^{2-}$	0	5	5	Cl	16.1f, 21.5, 29.2	571
$VOCl_5^{2-}$	0	5	6	Cl	15.9f,21.3, 29.4	571
$VO(O_2)_2(NH_3)^{2-}$	0	5	6	O_2?	45.5	577
$VOCl_4$	1	4	5	Cl	31.1f, 37.25, 41.3, 46	571
$VO(dca)_2$	1	4	5	O	40, 47.3	574
$VO(tca)_2$	1	4	5	O	40.8, 44.4	574
$Cr[N(SiMe_3)_2]_3$	3	3	3	N	25.3, 31.4	588
$CrCl_3(NMe_3)_2$	3	3	5	Cl	37.5, 45.6	569
$Cr(OBu^t)_4$	2	4	4	O	37,45	592
CrF_6^{3-}	2	4	6	F	29.6, 37	531
CrO_4^{2-}	1	5	4	O	16-18,24-27	270,578
CrO_4	0	6	4	O	26.316π,34	581-583
CrO_3F^-	0	6	4	O	22.5	582
CrO_3Cl^-	0	6	4	O	22.75	582
CrO_3Br^-	0	6	4	O,Br	22.2, 27.9, 34.2, 38.4	585
CrO_3I^-	0	6	4	O,I	22.2, 27.4, 34.5	586
CrF_6^{3-}	3	3	6	F	>52	539
$Cr(NCS)_6^{3-}$	3	3	6	NCS	31.3	526
$Cr(NCSe)_6^{3-}$	3	3	6	NCSe	28.7	527
$Cr(BipyNO)_3^{3+}$	3	3	6	N-O	26.45	532
$Cr(2,6\text{-}LutNO)_6^{3+}$	3	3	6	N-O	>28	533
MnO_4^-	0	7	4	O	18.32	582,583
MnO_3F	0	7	4	O	15.00	582
MnO_3Cl	0	7	4	O	16.00	582
MnO_2Cl_2	1	6	4	Cl	21,5, 32	579
MnO_4^{3-}	2	5	4	O	33.0	589
MnF_6^{3-}	4	3	6	F	43.51f, >50	539
MnF_6^{2-}	3	4	6	F	39.02	535
$Mn(CN)_6^{2-}$	3	4	6ls	CN	25.5, 34.7	536
Mn[macl]Cl_2	4	3	6	Cl	36.1	542
Mn[macl]Br_2	4	3	6	Br	34.80	542
$MnCl_5^{2-}$	4	3	5	Cl	23.9	545
Mn(acen)Cl	4	3	5	acen	22.4, 26.2, 34.2	544
Co(t-[14]dien)$^+$	8	1	4	C=N	14.4	549
Co(t-[14]dien)$^{2+}$	7	2	4	C=N	29.8 (34.5)	549
Co([14]teteneN4)$^+$	7	2	6	C=N	18.4	553
Co(Me6[14]dien)$^{2+}$	7	2	6	C=N	30.0	553

cont

Table 5.19 cont

	A	B	C	L	Absorption Spectra kK	Ref
$Co(Et_4D)_3^{2+}$	7	2	6	S	23	551
$Co(Se_2PPh_2)_2$	7	2	4	Se	23.4	525
$Co(Ph_2P(Se)S)_2$	7	2	4	(Se)S	24.6	242
$Co(N_3)_6^{3-}$	6	3	6	N_3	27	593
Co(dptpi)$_3$	6	3	6	S	22.2 27.7	82
Co(dtp)$_3$	6	3	6	S	24.6	82
CoF_6^{3-}	6	3	6	F	39.6	539
t-Co(CRH)N_3Cl^+	6	3	6	N_3	29.5	602
t-Co(CRH)$(NCS)_2^+$	6	3	6	NCS	31.3	602
t-Co(CRH)$(NO_2)_2^{2+}$	6	3	6	NO_2	28.1	602
Rh(dtp)$_3$	6	3	6	S	31.2	82
Rh(dptpi)$_3$	6	3	6	S	28.4 35.6	82
Ir(dtp)$_3$	6	3	6	S	37.4	82
Ir(dptpi)$_3$	6	3	6	S	33.6 39.1	82
$ZrCl_6^{2-}$	0	4	6	Cl	41.7	422
$ZrBr_6^{2-}$	0	4	6	Br	(34.5), 38.9 (40.8)	422
$NbCl_6^-$	0	5	6	Cl	(31.5)pf,35.5,43.5	414,422
$NbCl_6$	0	5	6	Cl	28.9sh,31.8sh,34.5,41.4	584
$NbBr_6$	0	5	6	Br	(21.5)pf,(23.8),26.7,29.0	422
$NbBr_6^-$	0	5	6	Br	17.0,21.3,22.8,27.2,35.7	584
MoF_6	0	6	6	F	54	422
$MoCl_6^-$	1	5	6	Cl	26.4pf?, 30.2, 36.3	414
$MoBr_6$	1	5	6	Br	18.9pf?, 23.5, 26.3	414
$MoCl_6$	1	5	6	Cl	24.1, 27.8,31.7	414
$Mo(en)_3^{3+}$	3	3	6	N	40	537
$Mo(CN)_4^{4-}$	2	4	8	CN	42.7, 45.5sh	522
$Mo(NCS)_6^{3-}$	3	3	6	NCS	31.4	526,538
$Mo(NMe_2)_4$	2	4	4	NMe_2	19.6 21.7	566
$MoPy_3Cl_3$	3	3	6	Py	27.2, 33.6	543
Mo(o-phen)Cl_4^-	3	3	6	o-phen	18.9, (22.2)?, 26.3	543
Mo(bipy)Cl_4^-	3	3	6	bipy	18.2, (25)	543
Mo(bipy)Cl_4	3	3	6	bipy	25.6, 33.2	546
$TaCl_6^-$	0	5	6	Cl	30.8pf?, 36.4, 42.4	414,422
$TaCl_6$	0	5	6	Cl	34.8, 38.0, 41.5, 49.5	584
$TaBr_6$	0	5	6	Br	22.2pf?, 24.0, 25.6 etc	414
$TaBr_6^-$	0	5	6	Br	22.7, 26.0, 29.8sh, 32.0 33.6, 41.7	584
WF_6	0	6	6	F	57.1	422
WCl_6	0	6	6	Cl	(22.4)pf, 26.4, 29.9	422
WOX_2	2	4	4?	O	31, 37-38	580

cont

Table 5.19 cont

	A	B	C	L	Absorption Spectra kK	Ref
$WOCl_2$	2	4	4?	Cl	45, 47.4	580
$WOBr_2$	2	4	4?	Br		580
WOI_2	2	4	4?	I	44.4, 26	580
$W(CN)_4^{4-}$	2	4	8	CN	40.8,43.8sh	522
$W(oxine)_4$	2	4	8	oxine	14.3, 15.3, 25.6	534
ReO_4^-	0	7	4	O	44.00	582,583
ReO_3F	0	7	4	O	40.00	582
ReF_6^{3-}	0	5	6	F	35.1?, 43	523
ReO_4^{2-}	1	6	4	O	34, 37.0 ,40.5	521
$Re(DPE)_2Cl_2$	5	2	6	P	16.7, 22.2, 25(Cl)	548
$Re(DPE)_2I_2$	5	2	6	P,I	14.3, 18.2	548
TcO_4^{2-}	1	6	4	O	31.5, 33.6, 35.1, 38.85 40.5	521
$TcOBr_4^-$	2	5	5	Br	28 ?, 33.3, 39.7	524
$TcOCl_4^-$	2	5	5	Cl	33.9	524
$Tc(NCS)_6^{2-}$	3	4	6	NCS	20	540
$Tc(NCS)_6^{3-}$	3	3	6	NCS	(22.73) 24.82 (35.7)	540
FeO_4^{2-}	2	6	4	O	17.8, 19.6	591
FeF_6^{3-}	5	3	6	F	(42.6)lf,47.6	541
$Fe[N(SiMe_3)_2]_2$	5	3	3	NSi	25.3, 29.7	590
$Ru(CN)_6^{3-}$	5	3	6	CN	21.05	550
$Ru(N_3)_6^{3-}$	5	3	6	N3	22.3, 35.7	552
$Ru(dtc)_3$	5	3	6	S	20.8(ML), 25.6	
$Ni(dtc)_2$	8	2	4	S	(23.0)of, 25.6 MLCT	557
$Ni(Me_4D)_3^{2+}$	8	2	6	S	23.8	551
$Ni(Et_4D)_3^{2+}$	8	2	6	S	23.0	551
$Ni(dpdo)Cl_2$	8	2	4	P	26.7, (32.4)	556
$Ni(dpdo)Br_2$	8	2	4	P	25.5, 40.5	556
$Ni(CN)_4^{2-}$	8	2	4	CN	34.36, 34.66, 35.84, 36.50 36.70	558
Ni(1,4-dienemac)	8	2	4	C=C	35.6	559
$Ni(PMe_3)_4^{2+}$	8	2	4	P	35.7sh 43.4	560
$Ni(PMe_3)_2Br_2$	8	2	4	Br	26.3 - 29.4, 36.4	560
$Ni(PMe_3)_2Cl_2$	8	2	4	Cl	28.2 - 31.5, 36.0	560
$Ni(PMe_3)_2I_2$	8	2	4	I	23.3, 26.0, 28.5	560
$Ni(dmma)_2$	8	2	4	S	23.5 26.3MLCT 40.53LMCT	562
$Ni(PMe_3)_3I_2$	8	2	5	I	23.7, 25.4, 29.6, 30.0	560
$Ni(tas)Cl_2$	8	2	5	Cl,As	30.2, 32.3	555
$Ni(tas)Br_2$	8	2	5	Br,As	28.3, 32.3	555

cont

Table 5.19 cont

	A	B	C	L	Absorption Spectra kK	Ref
$Ni(tas)I_2$	8	2	5	I,As	26.0, 32.3	555
$Ni(dap)Br_2$	8	2	5	Br,P	27.4, 33.7	555
$Ni(dap)I_2$	8	2	5	I,P	25.0, 33.7	555
$NiCl_2(PMe_3)_3$	8	2	5	Cl	30.3(π), 34.5(σ)	554
$NiBr_2(PMe_3)_3$	8	2	5	Br	28(π), 34(σ)	554
$NiI_2(PMe_3)_3$	8	2	5	I	23.8(π), 29.5(σ)	554
$NiX_2(PMe_3)_3$	8	2	5	P	40	554
$Ni(macrocycle)^+$	9	1	4	C=C	20	594
$Ni(Ph_2P(Se)S)_2$	8	2	4	(Se)S	28.1	242
$PdBr_4^{2-}$	8	2	4	Br	26.99sf 30.2(^1E_u) 30.9($^1A_{2u}$) 40.4(σ)	561
$PdCl_4^{2-}$	8	2	4	Cl	35.72(^1E_u) 37.40($^1A_{2u}$) 44.98(σ)	561
$Pd(dtaa)_2$	8	2	4	S	19.84 20.49 MLCT 33.44 46.30 LMCT	562
$Pd(dptpi)_2$	8	2	4	S	(27.8) 32	82
$Pt(dptpi)_2$	8	2	4	S	33.9	82
$Cu(NCS)_4^{2-}$	9	2	4	NCS	23.5, 26.45	563
$Cu(CN)_4^{2-}$	9	2	4	CN	35.7	563
Ag(mac)	9	2	4	N	28.75, 35.7 LMCT	564
$Ag(dipic)_2(H_2O)_4$	9	2	6	dipic	27.0, 31.0	565

A: d^n, B: Oxidation state, C: Coordination number, L: Ligand presumably involved in CT band. The reader should consult the original literature for details on the data in this table. a) May be $[V(III)(CN)_7]^{4-}$.

colleagues [57] have studied the electronic spectra of a wide range of Fe(III) complexes bound to phenolate oxygen. Within the range of complexes studied, from FeO_6 to $FeO_2N_2O'_2$ (where O' is carboxylate or water etc), a band appears between 17,750 - 24,800 cm^{-1}, which is probably associated with the π(O, phenol)---Fe(III) CT transition though this is not proven in all cases. There appears to be a general trend that increasing the number of phenolate oxygen atoms in the coordination sphere causes a ca 2000 cm^{-1}/phenolate blue shift in the spectrum. The intensity appears to be about 2000 l mol^{-1} cm^{-1}/phenolate but perhaps reaches a maximum at 2 phenolate residues [57-60] (see Transferrin, Chapter 8). Addition of imidazole into the coordination sphere also appears to shift the O---Fe(III) LMCT band to higher energy by some 1000-2000 cm^{-1}/imidazole [57].

5.12 d^n Charge Transfer Spectra

This section discusses, under a d^n classification, a number of topics which were not so readily included in previous sections. It makes no attempt to survey all known d^n CT systems.

Relatively few researchers study the CT spectra of metal complexes for their specific interest. However very many researchers report electronic spectra for newly prepared complexes. Bands are frequently assigned as CT sometimes for no other reason than that they are not readily assigned as d-d or intraligand. These spectra represent a useful data base for studies in CT spectra. During the preparation of this book, some of the more useful such data have been accumulated and are reported in Table 5.19. However the validity of the assignments in this table (quoted in kiloKaysers, 1000 cm^{-1}) is not guaranteed and the reader should read the original publications with care.

5.12 i) The d^0 Charge Transfer Spectra

Ti(IV), Zr(IV), Hf(IV), V(V), Nb(V), Ta(V), Cr(VI), Mo(VI), W(VI)
Mn(VII), Re(VII), Tc(VII), Fe(VIII), Ru(VIII), Os(VIII)

The oxidising power of these species increases from left to right across the Periodic Table. Ti, Zr and Hf have little oxidising character in their (+IV) states so that LMCT transitions occur towards the blue end of the visible region and into the ultraviolet (unless the ligand is very readily oxidisable). Since the oxidising character of a given oxidation level decreases down the transition blocks, Cr(VI), Mn(VII) and Fe(VIII) are much more oxidising than their later transition metal congeners. Thus for a given complex, such as $[MO_4]^{n-}$, the LMCT bands should blue shift as we proceed from first to second to third row transition metal ion.

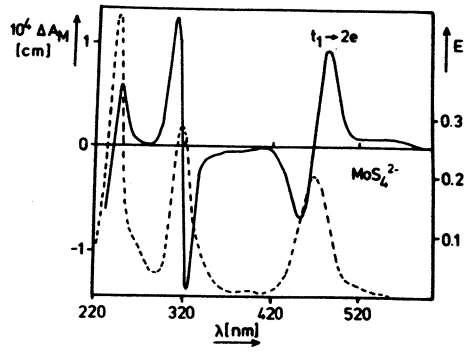


Fig.5.29 The absorption (hatched) and MCD spectra of the $[MoS_4]^{2-}$ ion [240].

Because of the high oxidation state, virtually all the complexes are tetrahedral or low symmetry versions thereof. The crystal field splitting is small so that LMCT transitions to e and t_2 (or their components) may generally be expected. Obviously MLCT in the visible and near UV region is impossible.

An MO diagram for $[MX_4]^{n-}$ species (X = O,S,Se) is shown in Fig.5.13. The sequence of MO levels can be shown to be $..[(\sigma)2a_1 < (\sigma)2t_2 < (\pi)1e] < (\pi)3t_2 < (\pi)1t_1 < (d)2e < (d)4t_2$, with the last two d orbitals being empty in the ground state. Extensive PES data generally support this conclusion though the sequence of the orbitals grouped in square brackets may vary [285,307]. It is generally expected that the lowest, long wavelength, feature will be the $1t_1$---2e LMCT transition ($^1T_2$---$^1A_1$), a conclusion supported by MCD data (Fig.5.29) [243]. The excited configuration $(t_1)^5e^1$ yields four states $^3T_1$, $^3T_2$, $^1T_1$ and $^1T_2$ but these all appear closely spaced (and only the last is electric dipole allowed) and cannot generally be distinguished. In an extended series of complexes, this t_1---2e transition shifts in a fashion consistent with the optical electronegativity of the central ion (Fig.5.30) [240]. Thus this transition shifts to the red as the oxidation state of the central metal becomes more positive; compare the purple colour of the manganate(VII) ion with the colourless vanadate(V). The left-hand data show the variation with central ion, while the data on the right show variation with common ligand (Fig.5.30).

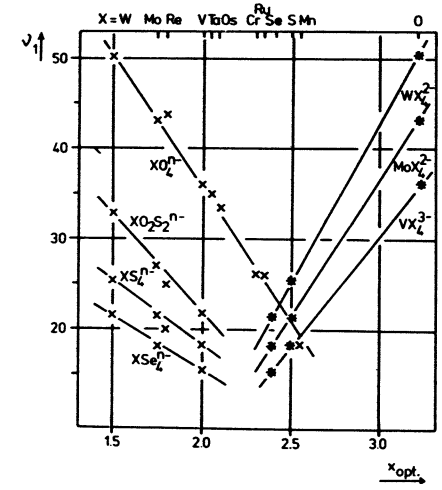


Fig.5.30 The t_1---2e transition energies of species of T_d symmetry as a function of the optical electronegativity of the central atom, or of O, S or Se (Energy data in units of 1000 cm^{-1} (kilokaysers) [240].

The excited $^1T_2$ state should be Jahn-Teller active and resonance Raman studies with the $[MoS_4]^{2-}$ ion confirm this expectation [259]. Indeed some structure due to the excited state JT effect is observed in the absorption spectrum on a crystal cooled to 14 K. The excited state Mo-S distance is calculated to be about 7pm longer than the ground state distance [259].

Frequently one or two further higher energy transitions are readily identified. Their assignment is less certain. The next two transitions to be expected would be $1t_1$---$4t_2$ and $3t_2$---2e. On the basis of analysis of MCD data comparing calculated and observed angular momenta of the excited states, it is shown that the best aggreement is obtained by assigning ν_2 and ν_3 to these latter two transitions respectively [243]. This would make the energy difference between ν_1 and ν_2, 10Dq. These authors reject this assignment arguing that 10Dq should be near 20,000 - 30,000 cm^{-1} and therefore choose the reverse assignment with $_3$ assigned to $1t_1$---$4t_2$. This author sees no justification in assuming such a large 10Dq value in a covalent tetrahedral species of the first or even second transition series and prefers the initial assignment which best fits the MCD data. However it is clear that the assignment probably varies from one complex to another. Moreover the assignment for a given compound varies from one publication to another even by the same authors (e.g. see data for $[MoS_4]^{2-}$ in [241,243,246]).

There is little to gain by a detailed discussion of the higher energy transitions since their assignments are so insecure. The reader is referred to the original literature for such discussion.

Many complexes of the type $[MX_3Y]^{n-}$ have been prepared. In these C_{3v} species, $1t_1$ spans a_2 + e but 2e is unsplit. The transition $a_2(1t_1)$---2e generates ^1E, while $e(^1T_1)$---2e spans $^1E + {}^1A_2$. Alternatively note that the $1t_1$---2e transition creates the $^1T_1$ and $^1T_2$ states, transition to the former being forbidden. However in C_{3v} symmetry (of $[MO_3X]$ etc.) transition to ^1E of $^1T_1$ would be allowed. Thus three transitions may be expected, namely $^1E(^1T_1)$, $^1A_2(^1T_2)$ and $^1E(^1T_2)$. The two components of $^1T_2$ evidently lie close together since in general, only two transitions are observed (Table 5.20). However as discussed below these three transitions are resolved in $[CrO_3X]^-$ and $[OsO_3N]^-$ [344].

It transpires, as seen in Table 5.20, that the first band of $[MOS_3]^{n-}$ essentially corresponds with the first band of the corresponding $[MS_4]^{n-}$ species, and the second band of $[MOS_3]^{n-}$ coincides with the first band of $[MO_3S]^{n-}$. This arises [243] because the set of sulphur orbitals in $[MOS_3]^{n-}$ constituting the a_1 molecular orbital, is the same as that constituting $1t_1$ in $[MS_4]^{n2}$ Note that resonance Raman data for $[WOSe_3]^{2-}$ show enhancement of W-Se modes but not W-O modes when irradiated into the a_2 transition, a fact fully consistent with the assignment since this molecular orbital has no W-O component [245].

As indicated earlier, mixed sulphido-selenido species have spectra which obey a type of average environment rule for LMCT spectra. Species such as MO_2Y_2, (Y=S,Se) as shown in Table 5.20,

show a band at almost the same wavelength as the e component of the corresponding mono-oxo and tri-oxo species [243,244]. Oxide has an optical electronegativity value about 0.6 to 0.8 units higher than sulphide.

The series of species $[CrO_3X]^-$ (X = I,Br,Cl and F) represent an interesting data set [245]. The lowest three transition energies are almost independent of X, indicating that they must be primarily oxygen to Cr(VI). Thus bands near 22,000, 28,000 and 34-35,000 cm^{-1} were assigned as π(O)---Cr(VI), the last having π(I)---Cr(VI) character in the iodo complex, while the other halides show π(X)---Cr(VI) near 38,000 (Br) and 42,000 (Cl) cm^1.

These systems, and the $[OsO_3N]^-$ ion were later studied in more detail [344]. As indicated above, three transitions correspond with the $1t_1$---2e transition in $[MO_4]$ and these are in fact seen (Table 5.20) with progressions in the Cr-O a_1 mode superimposed thereon. The excited state frequencies were found to be 790, 780 and 897 cm^{-1} respectively for $[CrO_3F]^-$, $[CrO_3Cl]^-$ and $[OsO_3N]^-$, corresponding with ground state frequencies of 912, 913 and 1024 cm^{-1} respectively. Not unexpectedly there is a close similarity between the bands observed in $[CrO_3X]^-$ and the $^1T_1$ and $^1T_2$ levels of $[CrO_4]^{2-}$ [344].

Data for the MnO_3X series are also included in Table 5.20. Ions of the type $[MO_{4-n}S_n]^{x-}$ coordinate readily to divalent ions to form complexes of the type $[M'(MO_{4-n}S_n)_2]^{m-}$ and $[LM'(MS_4)_n]^{2-}$ in which the M' metal is coordinated by oxido or sulphido ligands of the central anion [240]. These complexes have a rich electronic spectrum often with low lying S---M' LMCT bands.

This section is continued with some specific studies. Note that titanium and vanadium tetrahalides were discussed in section 5.5 iii). Dijkgraaf and Rousseau [510-512] have reported a collection of data involving $TiCl_xL_{4-x}$ (x = 0-4, L = alkyl, phenoxy, alkoxy etc). In general the Cl---Ti transition shifts to the blue in these series.

The Manganate(VII) Ion

The purple permanganate $[MnO_4]^-$ ion has been the subject of innumerable investigations [109,259-261,269, 607 and refs. therein]. The spectrum is shown in Fig.5.31 [261]. In this early detailed study it was considered that 4 distinct electronic transitions could be observed in the energy region studied. More recent studies, incorporating resonance Raman [260] confirm this and also reveal some additional structure. The vibrational structure arises from the excited Mn-O totally symmetric stretch which varies only slightly from one state to another (764 - 777 cm^{-1}) (ground state 838.9 cm^{-1}, from resonance Raman).

These four singlet states comprise states arising from the four possible configurational excitations (all π(O)---Mn(VII)):

$(3t_2)^6(1t_1)^6$----$(3t_2)^6(1t_1)^5(2e)$ $^3T_1 + {}^3T_2 + {}^1T_1 + {}^1\mathbf{T_2}$

$(3t_2)^6(1t_1)^6$----$(3t_2)^6(1t_1)^5(4t_2)$ $^3A_2 + {}^3E + {}^3T_1 + {}^3T_2$
$^1A_2 + {}^1E + {}^1T_1 + {}^1\mathbf{T_2}$

$(3t_2)^6(1t_1)^6$----$(3t_2)^5(1t_1)^6(2e)$ $^3T_1 + {}^3T_2 + {}^1T_1 + {}^1\mathbf{T_2}$

$(3t_2)^6(1t_1)^6$----$(3t_2)^5(1t_1)^6(4t_2)$ $^3A_1 + {}^3E + {}^3T_1 + {}^3T_2$
$^1A_1 + {}^1E + {}^1T_1 + {}^1\mathbf{T_2}$

(5.38a)

where the only electric dipole allowed transitions (shown bold) are the $^1T_2$---$^1A_1$ transitions.

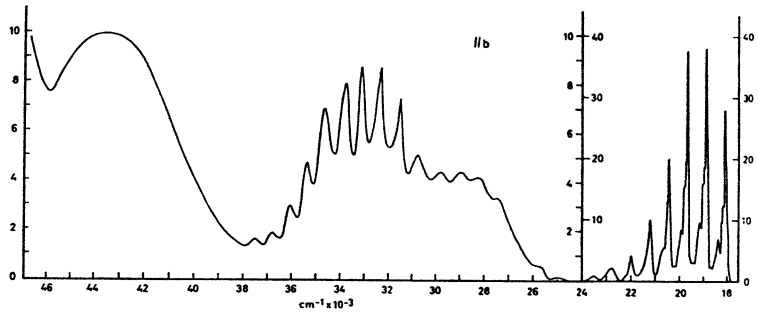


Fig.5.31 The absorption spectrum of $KMnO_4$ dissolved in $KClO_4$ at liquid helium temperature [261].

The currently accepted assignments incorporate the four possible electric dipole allowed transitions and include some electric dipole forbidden features; these are for tetraethylammonium MnO_4, citing (0-0) transitions (not the intensity maxima):

$1t_1$---2e	$^1T_1$	(forbidden)	15,100 cm^{-1}
	$^1T_2$		17,700
	$^3T_2$	(forbidden)	18,000
$3t_2$---2e	$^1T_1$	(forbidden)	24,700
	$^1T_2$		30,300
$1t_1$---$4t_2$	$^1T_2$		29,500
$3t_2$---$4t_2$	$^1T_2$		44,400

(5.38b)

which provide an average ($4t_2$---2e) separation of about 13,000 cm^{-1}. These authors argue [260] that this figure should be corrected by about 2500 cm^{-1}, to yield a 10Dq value of 10,500 cm^{-1} because of the difference in the Franck-Condon maxima for 2e and $4t_2$ levels. The resonance Raman spectrum of the lowest electric dipole allowed transition (the first main absorption band) shows not only a progression in the totally symmetric Mn-O stretch, but also in ν_3 indicating a Jahn-Teller distortion in the excited $^1T_2$ state [259]. The equilibrium Mn-O bond length in the lowest excited $^1T_2$ state is observed, from resonance Raman studies to be 9pm longer than the ground state [286].

The Chromate(VI) Ion

This yellow ion has also been extensively studied [109,260,262-264,613 and refs. therein]. Containing a less oxidising central ion than in manganate(VII), the low energy $^1T_2$ state ($1t_1$---2e) occurs as a structured band between 25,800 (0-0) and 31,700 cm^{-1} with (Cr-O) = 730 cm^{-1} [613] (Fig.5.32). A weak band system near 22,200 cm^{-1} (with the same vibrational mode) is probably the electric dipole forbidden $^1T_1$, by analogy with permanganate. Absorption between 35,000 and 42,000 cm^{-1} (maximum at 38,000 cm^{-1}) may contain both $3t_2$---2e and $1t_1$---$4t_2$.

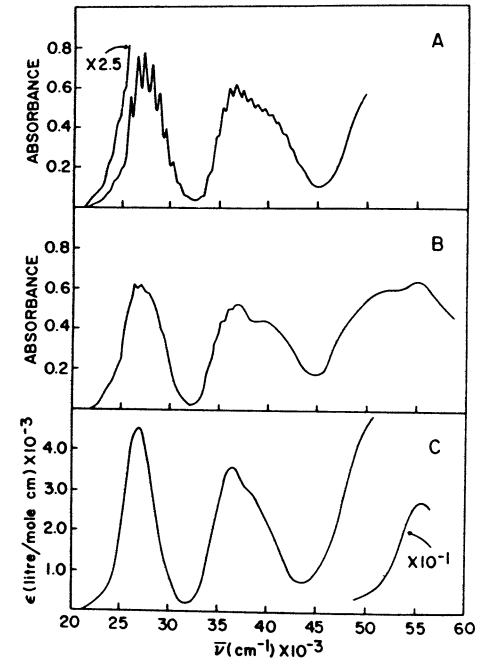


Fig.5.32 The absorption spectra of Cr_2CrO_4. A: Thin film at 77 K. B: Thin film at 300 K. C: Aqueous solution at 300 K (2 x 10^{-3}M) [613].

Table 5.20

CHARGE TRANSFER SPECTRA OF d^0 SPECIES

Complex	Absorption spectra			Other Bands	Ref
	$1t_1$--2e	(a)	(b)		
$[MX_4]^{n-}$ species					
TiI_4	19,600	27,800	25,000		249
$TiCl_4$	35,400	42,800	42,800		249
$TiBr_4$	29,500	37,000	35,800		249
VO_4^{3-}	36,200(8000)	45,000(5700)			242
VO_4^{3-}	39,500(^1E)[c]				
	(41,000)($^1B_2$)[c]				360
VS_4^{3-}	18,600(5800)	25,400(3500)	28,500(4800)		
				37,500(1100)	242
VSe_4^{3-}	15,600				242
CrO_4^{2-}	22,200e	33,300f	50,000		241,243
	25,800e		55,560		613
MoO_4^{2-}	42,700	47,600			241
MoS_4^{2-}	21,300	31,300	41,200		241
WS_4^{2-}	25,300	35,200			241
MnO_4^-	18,900	31,200	28,400		250
TcO_4^-	34,600(2100)	40,500(5700)	53,200(1500)		243
ReO_4^-	43,700(3600)	48,300(6100)			243
ReS_4^-	19,800(9600)	32,000(18,000)	44,000s		243
RuO_4	26,000	32,300	40,000w		243
OsO_4	34,000	39,200	47,200		241

Complex	Absorption spectra			Ref
	(π)a_2---2e	(π)e---e	Other bands	
$[MX_3Y]^{n-}$ species				
VOS_3^{3-}	(19,200)	21,800	30,800 33,900	243
$VOSe_3^{3-}$	15,600	19,200		243
CrO_3F^-	22,500	28,500	36,400	251,252,256
CrO_3F^-	22,200[d]	25,160[d]($^1A_1$)	38,130	
		28,750[d](^1E)		344
CrO_3Cl^-	22,750	28,400	35,600 41,400	251,256
CrO_3Cl^-	23,400[d]	25,170[d]($^1A_1$)	33,560d	
		28,480[d](^1E)		344
CrO_3Br^-	22,300	27,600	34,300 38,400	255
MoO_3S^{2-}		25,400	34,700s	243
$MoOS_3^{2-}$	21,500(2300)	25,500(8700)	32,000(6600)	243
WO_3S^{2-}		30,600	41,000s	243
WO_3Se^{2-}		27,000	38,000	240
WOS_3^{2-}	(26,700)	29,900(11,300)	37,000(7200)	243

cont

Table 5.20 cont

Complex	Absorption spectra				Ref
	(π)a_2---2e	(π)e---e	Other bands		
$[MX_3Y]^{n-}$ species					
MnO_3F	15,000	22,400	33,200	39,400	251
MnO_3Cl	16,000	20,800	30,500		253
ReO_3S^-		28,600	33,600s		243
$ReOS^{3-}$	19,800	25,500	32,300		243
OsO_3N^-	28,000	31,500(1000)	35,800	44,000	257
OsO_3N^-	27,280[d]	30,520[d]($^1A_1$)			
		35,580[d](^1E)	40,720		344

Complex	Absorption spectra			Ref
$[MO_2Y_2]^{n-}$ species				
$VO_2S_2^{3-}$	21,800	27,800	32,800	243
$MoO_2S_2^{2-}$	25,400(3000)	31,400(6000)	34,700(3000)	243
$MoO_2Se_2^{2-}$	22,000	28,500	32,000	243
$WO_2S_2^{2-}$	30,600(4000)	36,600(6900)	41,000(3900)	243
$WO_2Se_2^{2-}$	(25,000)	29,100	36,000 etc	243

Many more examples are given in refs.[240,241,243,256]. (a) Possible assignment $3t_2$---2e. (b) Possible assignment $1t_1$---$4t_2$. However assignments to (a) and (b) are interchanged in various publications even by the same authors. c) Doped into Ca_2PO_4Cl in D_{2d} site. d) Doped crystal at 4 K. The (0-0) lines are quoted. e) 0-0 Lines for the $^1T_1$ (lower energy) and $^1T_2$ states. f) 0-0 Line.

The Vanadate(V) Ion

This colourless species shows a strong absorption at about 37,000 - 43,000 cm^{-1} which clearly originates in $1t_1$---2e [360,361]. In the Ca_2PO_4Cl crystal the site is D_{2d} distorted and the polarised spectrum at 4.2 K shows structure. Group theoretical analysis shows that, in this site and crystal orientation, the $^1B_2$ transition is not seen when E is parallel to the a direction, while ^1E should be seen in all crystal orientations, though with differing intensities. On this basis the lower energy component of the absorption is assigned to $^1E(^1T_2)$ and the upper to $^1B_2(^1T_2)$ [360] (Table 5.20).

5.12 ii) Charge Transfer Spectra of d^1 Complexes

The manganate(VI) ion is one of the more interesting d^1 species which has been studied in some depth [265-267]. The ground state is ^2E, $(3t_2)^6(1t_1)^6(2e)$.

The lowest excitation will be to the configuration $....(3t_2)^6(1t_1)^5(2e)^2$. The two (2e) electrons transform as $^3A_2$ + ^1E + $^1A_1$ which can couple with the hole in $1t_1$ to generate the states $2(^2T_1)$ + $2(^2T_2)$ + $^4T_2$. Since the lowest energy state of $(2e)^2$ is $^3A_2$, the lowest CT state of the excited configuration is expected to be $^2T_2$. All these transitions except that to $^4T_2$ are electric dipole allowed. Following the discussion in section 5.3, terms arising from $^3A_2$, ^1E and $^1A_1$ are separated by repulsion energies based upon the J and K integral values for the d^2 ion. These are estimated by Day and co-workers [265] who show that these terms should be separated ($^2T_2...^2T_2$, $^2T_1...^2T_1$) by about 20,000 cm^{-1}. Day proposes that the next excitation will be to the $(3t_2)^6(1t_1)^5(2e)(4t_2)$ configuration which produces many states; however only transitions to $^2T_1$ are permitted since it transpires that transitions to $^2T_2$ differ by more than one spin-orbital from the ground state.

Absorption bands in the $[MnO_4]^{2-}$ ion, e.g. doped into K_2SO_4 or Rb_2SO_4, are centred near 18,000, 24,000, 27,000, 30,000 and 35,000 cm^{-1} (see Fig.4.2). The transition $(1t_1---2e)$ $^2T_2$ is clearly identified as the lowest transition (near 18,000 cm^{-1}) but the higher transitions are less clearly assigned. The next lowest transitions are expected to be $(1t_1---2e)$ $^2T_1$, $^2T_2$ but calculation places components of $(1t_1---4t_2)$ close in energy thereto.

The spectrum of this ion doped into Rb_2SO_4 at 4.2 K is very similar to that doped into K_2SO_4 and previously displayed (Fig.4.2). In Rb_2SO_4, the site symmetry is C_s which splits the $^2T_2$ into 2 ($^2A'$) + $^2A''$. The former is allowed from $^2E(x^2-y^2)$ along the b axis of the crystal, while the latter is allowed when light is polarised along a or c. The complex structure in the polarised spectrum can be interpreted in terms of these three origins plus progressions in the totally symmetric Mn-O stretch, combined, in some cases, with an additional t_2 vibration. The ligand field transition will be discussed in Chapter 6.

Data for the $[MO_4]^{2-}$ ions (M = Tc,Re) have also been reported [268]. The 10Dq values are reported to vary as Mn(VI) 11,000, Tc(VI) 17,000 and Re(VI) 19,000 cm^{-1} [268].

Simo and Holt [270] report the single crystal spectrum of the $[CrO_4]^{3-}$ doped into Ca_2PO_4Cl wherein the site symmetry is close to D_{2d}. The excited d state $^2T_2$ splits into $^2B_2$ + ^2E and Holt assigns these two transitions (respectively) to two closely spaced origins at 9912 and 10,204 cm^{-1}. A band at 17,100 cm^{-1} (^2E state) is then assigned as a component of the first LMCT state assumed to be $^2T_1$. At 28,000 cm^{-1} there exists a much stronger band With a clear progression in Cr-O; Holt assigns this to the next LMCT state $^2T_2$.

This assignment is criticised [265] on the basis that LMCT from oxygen to Cr(V) ought to lie at a significantly higher energy than oxygen to Mn(VI); this isnot the case with the Holt assignment. These authors [265] prefer to assign the 17,100 cm^{-1} band as the $^2B_2$ component of the ligand field band, with the two closely spaced origins as components of $^2E(^2T_2)$. The 28,000 cm^{-1} band then becomes the first LMCT transition ($^2T_2$).

The splitting of the ligand field state is exceptionally large in this assignment (weighted average 10Dq = 12,400 cm^{-1}) due supposedly to distortion in the host lattice.

In a later study, Borromei, Oleari and Day [343] investigated the spectrum of the $[CrO_4]^{3-}$ doped into $Sr_5(PO_4)_3Cl$. The ligand field band extends from about 10,000 - 17,000 cm^{-1} with a baricentre resulting in 10 Dq = 13,500 cm^{-1}. The lowest LMCT band is a complex structure spanning 26,000 - 31,000 cm^{-1}. The Cr-O a_1 mode is evident (ca. 800 cm^{-1}) together with a lattice mode. The assignment is probably $1t_1---2e$ ($^2T_2$).

5.12 iii) Charge Transfer Spectra of d^2 Manganate(V)

Day and co-workers have studied the $[MnO_4]^{3-}$ ion doped into various crystals [278,284]. The ground state is $..(1t_1)^6(2e)^2$, $^3A_2$. Doped into a crystal of $Sr_5(PO_4)_3Cl$, at 4.2 K, (C_{3v} effective symmetry), ligand field transitions are seen at 10,000 - 12,500 $^3E(^3T_2)---^3A_2$, 14,250 - 15,000 $^3E(^3T_1)---^3A_2$, and 16,245 cm^{-1} $^3A_2(^3T_1)---^3A_2$, which were fitted with values of 10Dq = 11,000, B = 500 and a C_{3v} distortion parameter of 600 cm^{-1} [278]. Spin singlet states were also discussed. Strong absorption at 30,000 - 35,000 cm^{-1} was assumed to be LMCT, presumably, $1t_1---2e$ but no detailed analysis was reported [278,306].

5.12 iv) Charge Transfer Spectra of d^3 and d^4 Complexes

Detailed studies of these systems appear not to have been undertaken. However there are many isolated examples of species of these configurations included in Table 5.19.

5.12 v) Charge Transfer Spectra of d^5 Complexes

Iron(III): Perhaps the single most well known iron(III) CT band is associated with the intense red colour which occurs when Fe(III) is mixed with thiocyanate ion. This is clearly an LMCT transition from the readily oxidisable thiocyanate to the fairly oxidising Fe(III).

Octahedral and tetrahedral iron(III) halides were discussed previously in this chapter (sections 5.4, 5.5), diimine systems in section 5.10, and phenoxides in section 5.11.

Iron sulphur proteins are briefly discussed in Chapter 8 but since they are cluster complexes with many iron and sulphur atoms, their spectra are extremely complex. Holm has extensively investigated these species and has also reported CT data for some model Fe(II) and Fe(III) tetrahedral MS_4 chromophores [46]. The $[Fe(S_2\text{-o-xyl})_2]^-$ ion shows fairly strong absorption at 14,530 (1670), 15,625(1600), 20,570(6240), (22,200)(4300) and 28,250 (8300) cm^{-1}, all of which are probably associated with S---Fe(III) LMCT (Table 5.21). This spectrum is also similar to oxidised rubredoxin which is believed to possess a similar chromophore [46].

Table 5.21.

LMCT DATA FOR d^5 COMPLEXES

Complex	Absorption Spectrum	Assignment	Ref
$Ru(NH_3)_5SMe_2^{3+}$	22,100(300)	$\pi S \rightarrow t_{2g}$	345
$Ru(NH_3)_5SeMe_2^{3+}$	20,500(150)	$\pi Se \rightarrow t_{2g}$	345
$Ru(NH_3)_5TeMe_2^{3+}$	16,700(60)	$\pi Te \rightarrow t_{2g}$	345
$Ru(NH_3)_4(glycine)^{2+}$	34,700(1900)	$\pi O \rightarrow t_{2g}$?	359
$Ru(NH_3)_4(glyc\text{-}NH)^{2+}$	27,800(3900)	$\pi N^- \rightarrow t_{2g}$?	359
$Ru(NH_3)_4(glyc\text{-}NEt)^{2+}$	24,600(3800)	$\pi N^- \rightarrow t_{2g}$?	359
$Ru(NH_3)_4(glyc\text{-}NR')^{2+}$	25,800(3700)	$\pi N^- \rightarrow t_{2g}$?	359
$Ru(NH_3)_4(glyc\text{-}NR^-)^{+}$	24,600(3600)	$\pi N^- \rightarrow t_{2g}$?	359
$Ru(NH_3)_5(4\text{-}NH_2Py)^{3+}$	19,800(3235)	$\pi NH_2 \rightarrow t_{2g}$	347
	31,150(4790)	$\pi Py \rightarrow t_{2g}$?	347
$Ru(NH_3)_4Cl_2^{+}$	30,200(5270) (34,500)(900)	$\pi Cl \rightarrow t_{2g}$	614
$Ru(en)_2Cl_2^{+}$	29,150(3850) (34,250)(800)	$\pi Cl \rightarrow t_{2g}$	615
	36,760(900)	$\pi N \rightarrow t_{2g}$?	616
$Ru(1\text{-}pn)_2Cl_2^{+}$	28,900(3460) 34,400(1380)	$\pi Cl \rightarrow t_{2g}$	615
$Ru(2,3,2\text{-}tet)Cl_2^{+}$	28,650(3430) 33,670(1160)	$\pi Cl \rightarrow t_{2g}$	615
$Ru(3,2,3\text{-}tet)Cl_2^{+}$	28,500(2560) 33,100(1230)	$\pi Cl \rightarrow t_{2g}$	615
$Ru(3,3,3\text{-}tet)Cl_2^{+}$	28,300(2610) 32,800(1450)	$\pi Cl \rightarrow t_{2g}$	615
$Ru(cyclam)Cl_2^{+}$	28,000(2260) 32,050(1250)	$\pi Cl \rightarrow t_{2g}$	615
$Ru([15]aneN_4)Cl_2^{+}$	27,850(2410) 31,550(1350)	$\pi Cl \rightarrow t_{2g}$	617
$Ru([16]aneN_4)Cl_2^{+}$	27,800(2230) 31,400(1320)	$\pi Cl \rightarrow t_{2g}$	617
$Ru(teta)Cl_2^{+}$	26,450(2310) 32,050(1700)	$\pi Cl \rightarrow t_{2g}$	614
$Ru(tetb)Cl_2^{+}$	26,300(2340) 32,250(1820)	$\pi Cl \rightarrow t_{2g}$	614
$Ru(cyclam)Br_4^{+}$[a]	23,350 (24,570) 28,170)	$\pi Br \rightarrow t_{2g}$	614
$Ru(cyclam)I_2^{+}$		$\pi I \rightarrow t_{2g}$ b	618
$[Ru(NH_3)_6]Cl_3$	34,000(255)	$\pi Cl \rightarrow t_{2g}$ b	620
$[Ru(NH_3)_6]Br_3$	32,700(300)	$\pi Br \rightarrow t_{2g}$ b	620
$[Ru(NH_3)_6]I_3$	25,000(252) 31,250(251)	$\pi I \rightarrow t_{2g}$ b	620
$(Et_4N)[Fe(S_2\text{-}o\text{-}xyl)_2]$	14,530(1670)	$\pi S \rightarrow Fe(III)$	
	15,625(1600)	$\pi S \rightarrow Fe(III)$	
	20,575(6240)	$\pi S \rightarrow Fe(III)$	46
$Fe(das)_2Cl_2^{+}$	18,100	$\pi As \rightarrow xz,yz$ c	
	27,100	$\pi As \rightarrow z^2$	
	31,500	$\pi As \rightarrow x^2\text{-}y^2$	621
	40,000	internal das	
$Fe(das)_2Br_2^{+}$	16,500	$\pi As \rightarrow xz,yz$ c	
	24,400	$\pi As \rightarrow z^2$	
	30,800	$\pi As \rightarrow x^2\text{-}y^2$	621
$Fe(das)_2I_2^{+}$	13,000	$\pi As \rightarrow xz,yz$ c	
	21,100	$\pi As \rightarrow z^2$	
	30,500	$\pi As \rightarrow x^2\text{-}y^2$	621
$Fe(das)_2BrI^{+}$	14,300	$\pi As \rightarrow xz,yz$ c	
	23,000	$\pi As \rightarrow z^2$	621

a) Data in water; system is moderately solvatochromic [614].

b) Outer-sphere charge transfer in an ion-pair. c) In the $[Fe(das)_2X_2]^+$ series, the transition from πAs will be mixed with halogen content through symmetry. Glyc-NH is $[H_2NCH_2CONH]^-$; glyc-NEt is $[H_2NCH_2CONEt]^-$; glyc-NR' is $[H_2NCH_2CON(CO_2H)]^-$; glyc-NR$^-$ is $[H_2NCH_2CONH(CO_2)]^{2-}$. See Tables 5.6 and 5.16 for further data.

Zink and co-workers [621] published an interesting study of the spectra of $[Fe(das)_2XY]^+$ species (das = diarsine), where X and Y are Cl,Br, and I. These complexes showed up to three clear LMCT transitions below an internal das transition at 40,000 cm^{-1} (Fig.5.33). The relatively small dependence of the energies of these transitions upon the nature of the halogen (Table 5.21) led Zink to propose that they are all πAs---d(Fe(III)) transitions, necessarily mixed through symmetry with some halogen content. These complexes are low spin with a configuration $(b_2)^2(e)^3(a_1)^0(b_1)^0$ which permits three LMCT transitions, namely πL---xz,yz(e), $\pi L \rightarrow z^2(a_1)$ and $\pi L \rightarrow x^2\text{-}y^2(b_1)$, and it is reasonable therefore to associate the three observed bands with these three transitions. Using the AOM model and correcting for changes in spin-pairing energy (parameter D) associated with the LMCT transitions (section 5.3), the relative energies of the d orbitals are [621 b].

$$\begin{aligned} E(x^2\text{-}y^2) &= 3e_\sigma(As) - (4/3)D \\ E(z^2) &= 2e_\sigma(X) + e_\sigma(As) - (4/3)D \\ E(xz,yz) &= 2e_\pi(X) + 2e_\pi(As) + (2/3)D \\ E(xy) &= 4e_\pi(As) + (2/3)D \end{aligned} \qquad (5.39a)$$

Unfortunately there is insufficient energy information in the CT spectra to solve for these parameters unambiguously. However Zink argues that the lowest d-d transition seen near 5000 cm^{-1} corresponds approximately with the xy---xz,yz transition providing an extra piece of information. If the zero point for the energies shown in (5.38) were to be taken as the arsenic donor level, then the CT energies can be equated directly with (5.38) and, with an estimate of 4000 cm^{-1} for D, the AOM parameters can be evaluated. In this fashion Zink obtained [621 b)]:

$[Fe(das)_2Cl_2]^+$ $e_\sigma(Cl) = 10,100$ $e_\sigma(As) = 12,300$ cm^{-1}
$e_\pi(Cl) = 5100$ $e_\pi(As) = 2600$

$[Fe(das)_2Br_2]^+$ $e_\sigma(Br) = 8800$ $e_\sigma(As) = 12,000$
$e_\pi(Br) = 4700$ $e_\pi(As) = 2200$

$[Fe(das)_2I_2]^+$ $e_\sigma(I) = 7300$ $e_\sigma(As) = 11,900$
$e_\pi(I) = 3900$ $e_\pi(As) = 1300$ (5.39b)

which data also explain the CT energies in the mixed $[Fe(das)_2XY]^+$ species in an acceptable fashion. However the true zero energy will be more negative by an amount W being the energy difference between

the unperturbed d orbitals and the arsenic donor level. The transition energy to x^2-y^2, for example, is then $3e_\sigma(As) + W$, and the data in (5.39) must be appropriately modified by an indeterminate amount. Judging from the position of the internal das transition, W probably does not exceed 8000 cm^{-1}. Despite the uncertainties, this represents a most useful system for evaluating AOM parameters from CT energies.

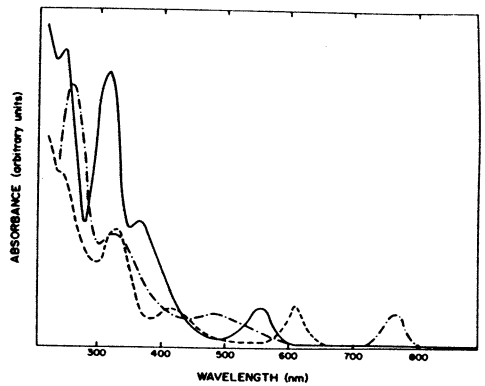


Fig.5.33 Charge transfer spectra of $[Fe(das)_2)X_2]^+$ species. X = Cl (solid line), X = Br (---) and X = I (-.-.) [621].

Ruthenium(III): The spectra of Ru(III) systems have been studied in some detail, especially by the Taube group (Table 5.21). The hole in the low spin $(t_{2g})^5$ configuration permits relatively low LMCT bands with reducing-type ligands. This is, of course, in contradistinction to Ru(II) which commonly displays MLCT transitions. Comparison of corresponding Ru(II) and Ru(III) complexes usually leaves no doubt as to the nature of CT transitions. Imidazole systems were discussed previously in section 5.11, and some halides in section 5.4.

Substituted glycinamide derivatives show the expected trends in LMCT energies when the N substituent ($H_2NCH_2CONR^-$) is modified. These N,N' ligands show a band near 24,500 - 28,000 cm^{-1} due to N^- to Ru(III) LMCT, a transition which shifts to lower energies with more electron donating R groups. Glycinamides which bind through the amidic oxygen (N,O) show a corresponding O---Ru(III) LMCT band near 31,500 - 34,700 cm^{-1} which also shifts predictably with changing R [359].

The ligand 4-aminopyridine bound to Ru(III) (via pyridine nitrogen) shows a peak attributable to NH_2---Ru(III) which disappears upon protonation of the complex (at $-NH_2$ group). However the position of this LMCT band is profoundly dependent upon the nature of the anion in solution. The energy varies from 20,000 cm^{-1} for fluoride, through 19,450 cm^{-1} for chloride to 18,900 cm^{-1} for

hexafluorophosphate or perchlorate [347]. It is suggested that the anion stabilises (to varying degrees) the positive charge on the nitrogen atom (of NH_2) when the electron is promoted to Ru(III), i.e. a kind of solvation effect.

Walker and Taube [617] noted that in a series of trans-$[Ru(macrocycle)Cl_2]^+$ species, two $\pi Cl \text{---} t_{2g}(Ru(III))$ transitions could be seen, rather than the expected one (section 5.4 viii)). They attributed this observation to the splitting of the πCl e_u level due to interaction of the chloride ions with hydrogen atoms attached to bulky in-plane nitrogen ligands. This study was extended in depth by Poon [614,619] who noted that the energy of the lower energy band, in the series trans-$[RuN_4Cl_2]^+$, decreased in the sequence (Table 5.21):

NH_3 > en > 1,3-propylenediamine > 2,3,2-tet > 3,2,3-tet > cyclam

> [15]aneN_4 > [16]aneN_4 > teta > tetb (5.37)

This was explained in terms of increasing steric hindrance, in sequence (5.37) causing some slight elongation of the Ru-Cl bond. This is expected [614] to cause a red shift in the LMCT transition energy because decreasing overlap results in a smaller destabilisation of the $d\pi^*$ (acceptor) orbitals. This decrease in overlap should also cause a decrease in intensity, in the same sequence, and this is evidently observed (Table 5.21). However the logical extension of this argument is that ion-pair outer-sphere charge transfer in systems such as $[Ru(NH_3)_6]Cl_3$ should occur at even lower energies, but this is not the case (Table 5.21). Such systems [620] absorb at higher energies, and with the expected very low intensities. One must also take into account that as the halide moves further away from the ruthenium(III), it comes into greater contact with the solvent such that solvation phenomena become important.

The Poon study [614] also noted the existence of the second, higher energy, but weaker $\pi Cl \text{---} t_{2g}$ (Ru(III)) component reported initially by Walker and Taube. This transition also shifted roughly in the order (5.37), but its intensity increased in this sequence. Evidently increasing steric hindrance causes a greater degree of distortion favouring an increase in intensity even if the bond length is increasing. The energy separation between the two transitions tends to increase with increasing steric hindrance, i.e. roughly in sequence (5.37) which fact is consistent with the distortion interpretation.

Two bands are also observed with some bromide complexes but with a smaller separation [614,618]. No splitting is seen with iodo complexes; evidently the interaction is dependent upon the ability of the halide ion to hydrogen bond, which decreases in the sequence Cl > Br> I. Poon [614] did consider that the second and higher energy CT transition might be $\sigma(Cl) \text{---} t_{2g}$ but this seems unlikely in that the difference in optical electronegativity values of the σ and π-orbitals of chloride corresponds with a 12,000 cm^{-1} energy

difference. Data for many Ru(III) systems are 'buried' in the studies of intervalence Ru(II)-L-Ru(III) systems studied, principally, by the Taube group [351-353 and refs therein]. Often the LMCT bands in the Ru(III) fragment, do not differ greatly from analogous monomeric systems.

Osmium(III) halides were noted previously (Table 5.6). An interesting study [346] compares the spectra of mono-, di- and tris-halides in different configurations (see section 5.5). Di-nitrogen complexes of osmium(III) have also been reported [354].

5.12 vi) Charge Transfer of d^6 Metal Complexes

Low spin d^6 systems are good pi donor ions and often display relatively low energy MLCT transitions, depending, of course, upon the metal ion. Johnson and Shepherd [8] have reviewed the MLCT spectra of $[MX_5L]^{n-/+}$. Complexes [9-22] such as $[(CN)_5ML]^{3-}$ (M = Fe,Ru), $[(NH_3)_5ML]^{2+}$ (M = Ru,Os) and $W(CO)_5L$ exhibit low energy (15,000 - 30,000 cm^{-1}) MLCT transitions when L has low lying π^* acceptor orbitals. Many examples are shown in Table 5.22.

Using substituted pyridines [14], a roughly linear correlation is demonstrated between the Hammett σ substituent constants and the MLCT transition energy. There is also a good correlation between the MLCT energy and the reduction half-wave potential for the ligand, since the latter reflects the LUMO energy of the ligand, to which the MLCT transition occurs [21,23,24].

A plot of the MLCT energies of the data set versus those of the corresponding $[(CN)_5FeL]^{3-}$ series, shows roughly the same slope for the ruthenium pentacyano (slope 0.87), and pentammine (0.78) and tungsten pentacarbonyl (0.87) series, but a much lower slope (0.42) for the osmium series. This indicates that the $[(CN)_5FeL]^{3-}$ series shows greatest sensitivity in MLCT energy to L, and osmium the least. It is concluded that the mixing (back-bonding) in the osmium series is much greater than with the other complexes, and that the donor orbital has significant ligand character [8].

Both LMCT and MLCT (to π^* CN) transitions were observed in the $[Co(CN)_5X]^{n-}$ species whose CT spectra were compared with analogous $[Co(NH_3)_5X]^{n+}$. The σ,π X---Co(III) LMCT transitions in the cyanides are blue shifted but similar to those in the ammine [625].

Where L is a bridging ligand capable of binding another metal, then bimetallic systems such as $[(CN)_5Fe(pyz)Rh(NH_3)_5]$ can be prepared [47,49,61]. These clearly will also display M(II) to bridge MLCT perturbed, though not greatly, by the other metal atom. The remote metal atom acts as a Lewis acid shifting the MLCT transition to lower energy and shifting the M(III)/M(II) redox couple more positively with respect to A_5ML [49]. The addition of Lewis acids such as H^+, Me^+ and BF_3 to the bifunctional ligand also usually results in a red shift in the MLCT transition.

The solvent dependence of CT bands has already been discussed in some depth. Meyer and co-workers [52] have investigated the solvent dependence of the MLCT and LMCT transitions in Ru(II) and Ru(III) amine derivatives respectively [52].

Table 5.22

CHARGE TRANSFER DATA FOR MX_5L SPECIES[a]

Ligand L	$(NH_3)_5OsL^{2+}$	Ref.	$(NH_3)_5RuL^{2+}$	Ref.	$(CN)_5FeL^{3-}$	Ref
Me_2S			27,900w[b]	345		
			38,800[b]	345		
Me_2Se			26,450w[b]	345		
			36,350[b]	345		
Me_2Te			31,450[b]	345		
Me_2SO			31,750[b]	358		
Py	23,300	9,10	24,600	12	27,600	13,14
2MePy					27,600	258
4MePy			25,100	12	28,100	13,14
$2NH_2Py$					25,850	258
$4NH_2Py$			27,900	11	31,300	14
2ClPy					(29,000)	258
3ClPy			23,500	104		
4ClPy			23,800	13,14	26,100	13,14
4HCOPy			18,350	104		
$4CO_2HPy$			20,100	12		
$4CO_2$-Py			21,900	12	24,100	13,14
$4NH_2COPy$	19,700	9,10	20,900	12	23,000	13,14
$4CF_3Py$			22,000	104		
2CNPy					26,000	258
3CNPy[c]					27,030	61
3CNPy[d]					24,150	61
4CNPy[c]			23,500	49	24,700	61
4CNPy[d]			19,600	16	21,000	21,61
$4CNPyH^+$			18,800	49		
$4CH_3COPy$					20,800	21
4HCOPy			18,300	12	19,800	13,14
Pyz	21,700	9,10	21,200	12	22,000	13,14
$PyzH^+$	23,400	9,10	18,900	12	16,000	20
$PyzMe^+$			18,400	13,14	15,100	13,14
2MePyz			21,400	15	22,300	18,19
4,4bpy					23,200	47
$4,4bpyMe^+$					19,200	47,49
isn			21,400	104		

Ligand L	$(CN)_5RuL^{3-}$	Ref.	$(CO)_5WL$	Ref.	$(NH_3)_4XOsL^{n+}$	Ref
Py	31,600	8	28,200	22		
4MePy	32,300	8	28,500	22		
$4CO_2HPy$	28,000	8				
$4CO_2$-Py	28,700	8				

cont

Table 5.22 cont

Ligand L	$(CN)_5RuL^{3-}$	Ref.	$(CO)_5WL$	Ref.	$(NH_3)_4XOsL^{n+}$	Ref
$4NH_2COPy$	27,500	8				
4CNPy	26,500	8	22,000	22		
$4CH_3COPy$			22,700	22		
4HCOPy	24,400	8	21,300	22		
Pyz	27,200	20			21,500[e]	20
$PyzH^+$	20,400	20			23,250[e]	20
$PyzMe^+$	19,100	8				
2MePyz	27,600	8				
4,4bpy			27,030	48		
$4,4bpyaW(CO)_5$			28,820	48		
$4,4bpyeW(CO)_5$			22,200	48		
$4,4bpyW(CO)_5$			22,800	48		

a) All data are MLCT except where indicated b) Probably LMCT c) Fe-NCPy d) Fe-PyCN X = Cl, n = 2. Many of these data were abstracted from Table I, ref. [8] which contains additional data of this type. In general see listed refs for additional data. Also NOTE that most of these spectra are solvent dependent and that the data reported here cannot all be in the same solvent. Care should therefore be taken in comparing one datum with another.

They demonstrate that complexes such as $[Ru(NH_3)_5L]^{2+}$, $[Ru(LL)_x(NH_3)_{6-2x}]^{2+}$ (x=1,2) etc, show Ru(II)---L (or LL) MLCT bands whose energies vary linearly with the Gutmann Donor Number [53] of the solvent (hydroxylic solvents excluded), (but no correlation with other common solvent parameters except those noted below). MLCT transitions have a negative slope in a plot of transition energy versus DN, while LMCT have a positive slope. Moreover the slope increases with the number of coordinated ammine groups. There are also good correlations with the Taft-Kamlet beta parameter [54] which is a composite scale of five different indices of a solvents ability to hydrogen bond as a Lewis base, and with the Arnett enthalpy parameter. This parameter [55] measures the enthalpy of hydrogen bond formation to p-fluorophenol in various solvents. The authors note that a plot of MLCT band maximum against mole fraction in a mixed solvent can be a useful probe for preferential solvation effects. For example in mixed nitromethane/DMSO, the band maximum approaches that seen in DMSO in as low as 1/10 DMSO in nitromethane [52].

Complexes of the type trans-$[Ru(NH_3)_4LL']^{2+}$ are also of especial interest. If L and L' both have acceptor orbitals of appropriate energy, will two MLCT bands be seen, or only one of some median energy ? Since the two ligands in a trans complex lie on a symmetry axis (assuming maximal D_{4h} symmetry) their acceptor orbitals should combine in and out-of-phase to yield two new orbitals of opposite parity. Thus in the limiting situation, only

Table 5.23

MLCT CHARGE TRANSFER SPECTRA OF BINUCLEAR d^6 COMPLEXES

Complex	Absorption Spectrum	Assignment	Ref
$(CN)_5Fe(4,4\text{-bpy})^{3-}$	23,200	Fe(II)---4,4bpy	47
$(CN)_5Fe(4,4bpy)Co(CN)_5^{5-}$	22,000	Fe(II)---4,4bpy	47
$(CN)_5Fe(4,4bpy)Fe(CN)_5^{6-}$	22,500	Fe(II)---4,4bpy	49
$(CN)_5Fe(4,4bpy)Rh(NH_3)_5$	20,800	Fe(II)---4,4bpy	47
$(CN)_5Fe(4,4bpy\text{-}Me+)^{2-}$	19,200	Fe(II)---4,4bpyMe+	47
$(CN)_5Fe(pyz)Rh(NH_3)_5^-$	17,500	Fe(II)---pyz	47
$(CN)_5Fe(4CNPy)Co(CN)_5^{5-}$	23,300	Fe(II)---4CNPy	49
$(CN)_5Fe(4CNPy)Rh(NH_3)_5^-$	25,000	Fe(II)---4CNPy	49
$(CN)_5Fe(4CNPy)Co(NH_3)_5$	20,400	Fe(II)---4CNPy	61
$(CN)_5Fe(Py4CN)Co(NH_3)_5$	18,975	Fe(II)---Py-4CN	61
$(CN)_5Fe(3CNPy)Co(NH_3)_5$	23,150	Fe(II)---3CNPy	61
$(CN)_5Fe(Py3CN)Co(NH_3)_5$	24,390	Fe(II)---Py-3CN	61
$(NH_3)_5Ru(pyz)Rh(NH_3)_5^{5+}$	21,900	Ru(II)---pyz	47
$(NH_3)_5Ru(pyz)Ru(NH_3)_5^{5+}$	17,700	Ru(II)---pyz	49,50
$(NH_3)_5Ru(pyz)Ru(NH_3)_5^{4+}$	18,300	Ru(II)---pyz	49,50
$(NH_3)_5Ru(4CNpy)Rh(NH_3)_5^{5+}$	20,500	Ru(II)---4CNPy	47
$(NH_3)_5Ru(4,4bpy)Ru(NH_3)_5^{5+}$	19,000	Ru(II)---pyz	49,51
$(NH_3)_5Ru(4,4bpy)Ru(NH_3)_5^{4+}$	19,200	Ru(II)---pyz	49,51
$(NH_3)_5Ru(N(CN)_2)Ru(NH_3)_5^{4+}$	44,850(25,100)	Ru(II)---$N(CN)_2$	353
	31,000(3160)	Ru(II)---$N(CN)_2$?	353

one parity allowed transition will be seen. However if the symmetry is not strictly centrosymmetric a second, and probably weaker, transition should occur.

Moreover this model will be apposite only when the energies of the two orbitals are similar - if they differ significantly, two transitions, being effectively two independent MLCT transitions may be expected. This analysis is also valid when L = L', which returns us to the discussion of section 5.8 viii). Experimentally [52,104,105,322,323], the second and higher energy transition is not observed, when L = L', except when some distortion mechanism is operative, e.g. the Ru(III) complexes of the previous section. However when the two ligands are different, the complex is no longer centrosymmetric, the D_{4h} model may no longer be appropriate, and the second transition is observed (Table 5.24). Thus whether, or not, the second transition is seen depends upon whether L and L' differ sufficiently to make the system effectively acentric.

It is relevant that when one of the two ligands is modified, e.g. substitution or protonation, both of the transitions are affected because of this coupling mechanism (and also because the charge on the metal may be altered). Note that in systems such as $[Ru(NH_3)_5L]^{2+}$ and $[Ru(NH_3)_4LL']^{2+}$, the nature of L' can have a

Table 5.24

CHARGE TRANSFER SPECTRA OF RUTHENIUM(II) TETRAMMINE DERIVATIVES

Complex	Absorption Spectrum	Assignment[a]	Ref
$Ru(NH_3)_4(bpy)^{2+}$	19,050(3950)	Ru(II)---bpy	52
$t\text{-}Ru(NH_3)_4(Py)(4CNPyNMe)^{3+}$	19,080(20,200)	Ru(II)---4CNPyMe$^+$	
	29,600(7290)	Ru(II)---Py	52
$t\text{-}Ru(NH_3)_4(3ClPy)(4CNPyNMe)^{3+}$	19,380(21,000)	Ru(II)---4CNPyMe$^+$	
	28,250(5900)	Ru(II)---3ClPy	52
$t\text{-}Ru(NH_3)_4(Py)(4,4bpyNMe)^{3+}$	17,920(14,700)	Ru(II)---4,4bpyNMe$^+$	
	26,750(4830)	Ru(II)---Py	52
$t\text{-}Ru(NH_3)_4(3ClPy)(4,4bpyNMe)^{3+}$	18,215(11,000)	Ru(II)---4,4bpyNMe$^+$	
	25,450(3100)	Ru(II)---3ClPy	52
$t\text{-}Ru(NH_3)_4(Py)(CH_3CN)^{2+}$	26,450(10,000)	Ru(II)---Py	104
$c\text{-}Ru(NH_3)_4(Py)(CH_3CN)^{2+}$	26,700(7100)	Ru(II)---Py	104
$t\text{-}Ru(NH_3)_4(CychexNC)_2^{2+}$	41,000(27,000)	Ru(II)---NC ?	349
$t\text{-}Ru(NH_3)_4(isn)(C_7H_7NC)^{2+}$	27,900(7000)	Ru(II)---isn	349
$t\text{-}Ru(NH_3)_4(isn)(CH_3CN)^{2+}$	23,400(8700)	Ru(II)---isn	104
$c\text{-}Ru(NH_3)_4(isn)(CH_3CN)^{2+}$	23,250(7250)	Ru(II)---isn	104
$c\text{-}Ru(NH_3)_4(2\text{-}NH_2CH_2Py)^{2+}$	24,150(6300)	Ru(II)---2NH62CH$_2$Py	
	40,300(5800)	π---π^* L	105
$c\text{-}Ru(NH_3)_4(2\text{-}NHCHPy)^{2+}$	19,200(6100)	Ru(II)---2NHCHPy	
	26,450(4400)	Ru(II)---2NHCHPy	
	37,450(7500)	π---π^*	105
$c\text{-}Ru(NH_3)_4(2CH_3COPy)^{2+}$	16,100(5400)	Ru(II)---2CH$_3$COPy	
	25,900(4000)	Ru(II)---2CH$_3$COPy	
	34,250(16,000)	π---π^*	105
$t\text{-}Ru(NH_3)_4(Py)_2^{2+}$	23,650(17,000)	MLCT Ru---Py	
	40,800(8900)	π---π^*	322
$c\text{-}Ru(NH_3)_4(Py)_2^{2+}$	24,400 26,600	MLCT Ru---Py	340
$t\text{-}Ru(NH_3)_4(Py)(4\text{-}AcetylPy)^{2+}$	19,700(16,200)	MLCT1 (4AcPy)	
	27,300(3000)	MLCT2 (Py)	
	36,900(3600) etc	π---π^*	322
$t\text{-}Ru(NH_3)_4(Pz)_2^{2+}$	21,100(18,600)	MLCT1 Ru---Pz	
	28,700(2500)	MLCT2 Ru---Pz	
	39,800(9800)	π---π^*	322
$c\text{-}Ru(NH_3)_4(Pz)_2^{2+}$	21,300 25,100	MLCT Ru---Pz	323
$t\text{-}Ru(NH_3)_4(Pz)(PzH)^{3+}$	17,500(20,900)	MLCT1 Ru---PzH+	
	26,700(1290)	MLCT2 Ru---Pz	
	36,600(8300) etc	π---π^*	322
$t\text{-}Ru(NH_3)_4(PzH)_2^{4+}$	16,000(28,800)	MLCT1 Ru---PzH+	
	22,700(850)	MLCT2 Ru---PzH$^+$	
	35,500(8900) etc	π---π^*	322
$c\text{-}Ru(NH_3)_4(isn)_2^{2+}$	21,000 24,200	MLCT Ru---isn	323

a) In mixed ligand complexes, the assignment of two MLCT bands to different ligands is a limiting assignment. Certainly each transition involves excitation to both ligands to some degree.

profound effect upon the Ru(II)-L MLCT band (viewed as separate transitions). For example compare the Ru(II) --isonicotinamide MLCT bands in $[Ru(NH_3)_5isn]^{2+}$ (Table 5.22) and $[Ru(NH_3)_4(isn)(RNC)]^{2+}$ (Table 5.24), where the acceptor nature of the isonitrile shifts the Ru(II)---isn band to much higher energies.

Many of the tetrammine spectra are strongly solvatochromic.

Data for some $[Ru(NH_3)_4(XY)]^{n+}$ species (XY an asymmetric bidentate ligand) are also included in Table 5.24. Note that bidentate species such as 2-NH_2CH_2Py show a single band near 24,000 cm^{-1} assigned to Ru(II)---Py MLCT. Where the 2-substituent is unsaturated and conjugated to the pyridine, two bands are seen, the lower energy being associated with the presence of the unsaturated X=C.C=N chromophore [105], though obviously some coupling must be involved.

Generally speaking, the lowest MLCT transition in trans-$[RuX_4LL']$ falls at a lower energy than the corresponding transition in $[RuX_5L]$, and is approximately twice as intense.

While trans-$[Ru(II)(NH_3)_4L_2]$ systems generally only exhibit one strong Ru(II)---L LMCT transition, the cis analogs show two closely spaced transitions (Table 5.24)[323]. This follows from the above treatment through coupling of the π^* orbitals on each ligand. However, in this C_{2v} situation, both transitions are allowed (in orthogonal polarisations). The splitting is of the order of 2000 - 4000 cm^{-1}. The lowest energy LMCT transition in cis-$[Ru(NH_3)_4L_2]$ falls at slightly higher an energy than in the trans analog. Though most of the data supporting the above analysis stems from ruthenium(III) tetrammine derivatives, the approach must be general for all analogous systems.

5.12 vii) Charge Transfer Spectra of d^7 Complexes

Many examples of cobalt(II) have appeared previously in this Chapter, with special reference to tetrahedral halides, CoS_4 systems etc. The tetrahedral CoS_4 chromophore is of especial interest since cobalt is frequently used to replace zinc or copper in natural enzymes and can then be used as a probe for the metal site environment. Holm reports data for the tetrahedral CoS_4 in the $[Co(S_2\text{-}o\text{-}xyl)_2]^-$ ion [46] (Table 5.18), while the reader is referred to Tables 8.2 and 8.8 for data for cobalt in various copper protein environments. In general the Co-S LMCT absorption appears, in tetrahedral species, near 28,000 cm^{-1}. Lever quotes an extensive CoS_4 data set in [391].

5.12 viii) Charge Transfer Spectra of d^8 Complexes

Nickel: The $[Ni(CN)_4]^{2-}$ ion has been the subject of significant study and the assignment of the observed bands has varied with time [65,69,72,74,84,119]. A polarised spectrum taken with a crystal at 5 K is shown in Fig.5.34, and the currently accepted assignment in Table 5.25. In general, it is now accepted that the spectra of

tetracyanides arise from d---π^* MLCT transitions where π^* is the lowest available CN π^* orbital, namely a_{2u}. The combination of MCD data [69] and dichroism [84] provide a convincing argument for most of the assignments. The two intensely $(\pi)(z)$ polarised bands must be assigned to the spinor A_{2u}' states as originally suggested, to explain the MCD data [69]. The lower energy transition is assigned to the MLCT A_{2u}' state of $^1A_{2u}$ mixed with some A_{2u}' from ^3E_u, while the higher energy is to A_{2u}' of ^3E_u mixed with some $^1A_{2u}$ (see section 5.5).

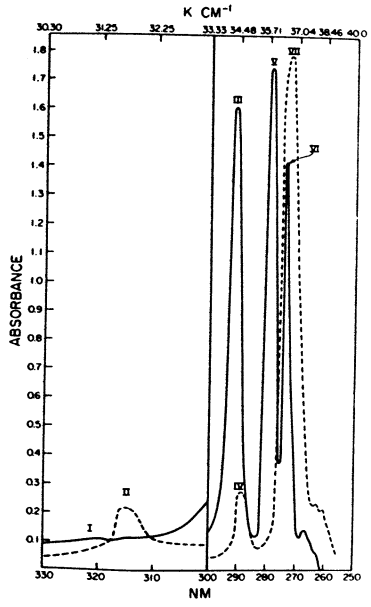


Fig.5.34 The polarised single crystal spectra of $(n\text{-}Bu_4N)_2Ni(CN)_4$ at 5 K. The solid line is the π-spectrum and the dashed line the σ-spectrum [84].

The first $(\sigma)(xy)$ polarised band is weak and is unusual in being exceptionally broad. Energy arguments lead to the conclusion that the d-d states, $^1B_2$, $^1A_2$ and ^1E may be in this region. As discussed in some depth in Chapter 6 (cf square planar nickel(II)), the excited D_{4h} states $^1B_{2g}$, $^1A_{2g}$ and ^1E_g are unstable with respect to distortion to D_{2d}, while $^1A_{2g}$ is stable. Transitions to $^1B_{2g}$ and ^1E_g will be broad because of the consequent vibrational excitation of these states. Since the band in question is broad and (σ) polarised it is best assigned to ^1E; the shoulder may be due to some splitting of this state.

The highest energy $(\sigma)(xy)$ polarised band is readily assigned

to the e_g---a_{2u} MLCT transition. MCD data define a degenerate state near 35,000 cm^{-1} and the $(\sigma)(xy)$ polarised band at 34,660 cm^{-1} is then best assigned to E_u' from the various states from which this can derive (Table 5.25) (section 5.5). Finally the band at 36,500 cm^{-1} is exactly 2140 cm^{-1} above the lower A_{2u}' transition and this corresponds with the a_{1g} C=N stretching vibration [84].

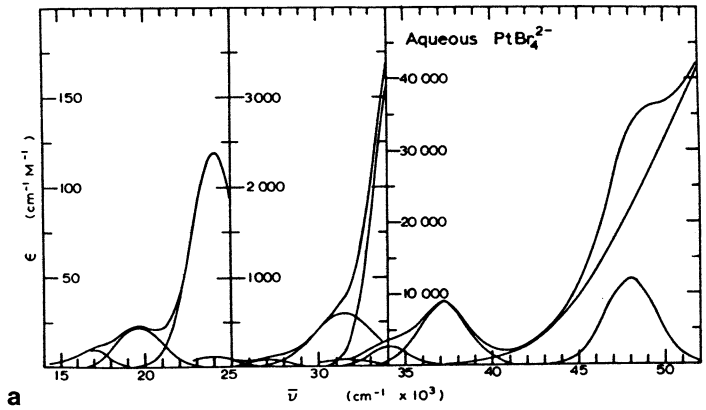
Maleonitriledithiol (mnt) forms square planar complexes with d^8 metals and the complexes $[M(mnt)_2]^{2-}$ (M = Ni(II),Pd(II) and Pt(II)) have rich spectra which must contain both S---M LMCT and M---mnt MLCT [90,100]. An early MO treatment [100] led to an assignment shown in Table 5.25. In comparing the series Ni(II), Pd(II) and Pt(II), the two types of CT band behave in opposite senses hence providing a procedure for identification. Thus the LMCT bands shift to higher energies in the sequence Pd(II) < Ni(II) < Pt(II), while the MLCT bands decrease in the same sequence. There is also an internal mnt π---π^* transition whose energy is almost invariant of the metal ion. Note also that LMCT transitions are generally more intense than MLCT transitions, as is the case here.

Octahedral nickel(II) species exhibit Ni-halogen CT at relatively high energy (Table 5.7) while tetrahedral systems are at significantly lower energy (Table 5.8). Les Dartiguenaves studied [554,560,565] both square planar and five coordinate halogen complexes also containing phosphines, and identified the LMCT transitions (Table 5.19). Other five coordinate halo complexes exhibit accessible halogen LMCT [555] (Table 5.19). There is little sensitivity in LMCT energies to the detailed stereochemistry in these low spin four and five coordinate complexes.

Palladium(II), Platinum(II) and Gold(III) Complexes: Following the previous analysis of the spectra of square planar halo complexes (section 5.5), the gross spectra of these species follow in a fairly straightforward manner. However the situation is complicated in the Pt(II) series by the expectation that d-p transitions will occur at comparable energies to the LMCT transitions. Thus a clear distinction between these two types of transition may not be possible. Absorption may be to states which are an admixture of LMCT and d-p. In the case of Pd(II) and Au(III), the separation between d-p and LMCT transitions is such that, below 50,000 cm^{-1}, d-p transitions are not expected.

These complexes were the subject of early debate [65-73,86-88,96,97]. Data are presented in Table 5.25, Figs.5.35, 6.61. The sequence of odd ligand levels appears to be $\pi b_{2u} > \pi e_u > \pi a_{2u} > \sigma e_u$.

Following the earlier discussion (section 5.5), two closely spaced πX---σ^*M transitions ($^1A_{2u}$, $^1E_u(1)$) should be followed by a more intense σX---σ^*M ($^1E_u(2)$). Transition from the non-bonding πX a_{2u} to the b_{1g} σ^* level yields an orbitally forbidden $^1B_{2u}$ state which may appear weakly. These transitions are clearly seen in a recent absorption and MCD study of $[AuX_4]^-$ species [64], following Mason and Gray [69]. A detailed MCD study [64] was supportive of the assignments shown in Table 5.25.



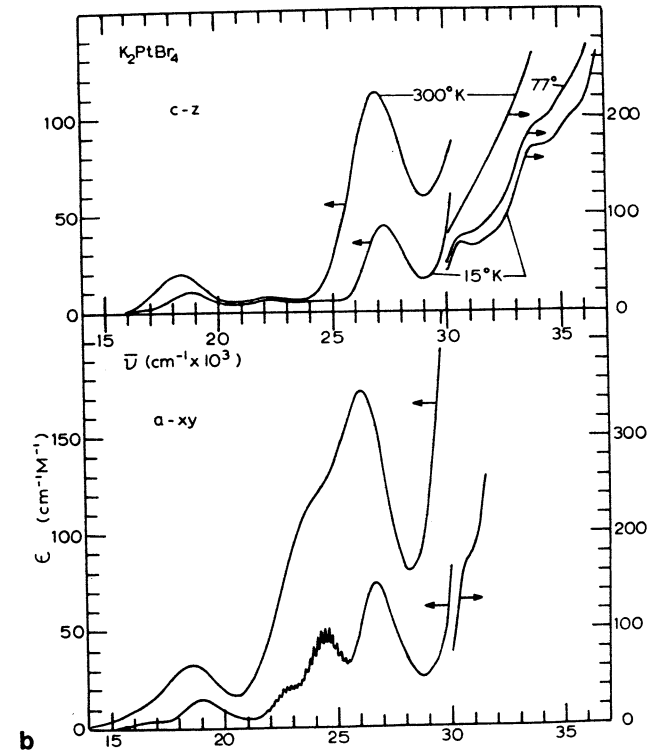


Fig.5.35 a) (Upper) The spectrum of K_2PtBr_4 in aqueous solution. The spectrum is shown deconvoluted into its component absorption peaks [32]. b)(Lower) A polarised single crystal spectrum of K_2PtBr_4 at ambient and low temperature as indicated [32].

Table 5.25

CHARGE TRANSFER SPECTRA OF d^8 COMPLEXES

Complex	Absorption spectrum	Assignment	Ref
$[Ni_2(S_2\text{-o-xyl})_2]_3^-$	20,570(3760)	πS---Ni(II)?	46
$(Bu_4N)_2Ni(CN)_4$	31,100(530)	$^1B_2$ d-d (π)	
	31,650(840)	^1E d-d (σ)	
	(31,850)	split ^1E ? (σ)	
	34,360(6200)	$A_{2u}(^1A_{2u}, ^3E_u)$(π) CT	
	34,660	Eu (^3E_u, $^3B_{1u}$, $^3A_{2u}$, ^1E_u)	
	35,840(5200)	$A_{2u}(^3E_u, ^1A_{2u})$(π) CT	
	36,500	A_{2u} + a_{1g} C=N	
	36,700(15,200)	^1E_u (e_g---a_{2u})(σ) CT	69,74,84
$Ni(mnt)_2^{2-}$	11,690(30)	x2-y2---xy d-d	
	17,500(570)	xz---xy d-d	
	19,250(1250)	d---π* MLCT	
	21,000(3800)	d---π* MLCT	
	26,400(6600)	πL---xy LMCT	
	31,300(30,000)	π---π* L	
	37,000(50,000)	σL---xy LMCT	100
$Pd(en)Cl_2$	38,900(9300)[a]	π_h(Cl)---xy	
	44,600(62,900)[a]	σ(Cl)---xy	26,164
$(Bu_4N)_2Pd(CN)_4$	42,920(1200)	$^1B_{1u}$ b_{2g}--a_{2u}* MLCT	
	44,310(5700) 45,090(9800)	^1E_u e_g--a_{2u} π* MLCT	
	46,190(9200)	$^1A_{2u}$ a_{1g}--a_{2u}* π* MLCT	
	49,000(10,400)	^1E_u a_{1g}--e_u π* MLCT	
$Pd(CNMe)_4^{2+}$	42,800(1280)	$E_u(^3A_{2u})$ a_{1g}---a_{2u}π*	69
	45,650(6600)	$A_{2u}(^1A_{2u})$ a_{1g}---a_{2u}	
	51,650(13,720)	$E_u(^1E_u)$ e_g---a_{2u}	77
$Pd(Ph_2PS_2)_2$	(27,800) 32,000(20,000)	πS---σ* LMCT ?	82
$Pd(N_3)_4^{2-}$	30,800(15,350)	πN_3---σ* ?	
	41,900(17,000)	σN_3---σ* ?	83
$PdCl_4^{2-}$	35,720(9330)	^1E_u πe_u---b_{1g}	
	37,400(388)	$^1A_{2u}$ πb_{2u}---b_{1g}	
	40,780(900)		
	44,980(25,800)	^1E_u σe_u---b_{1g}	85

cont

Table 5.25 cont

Complex	Absorption spectrum	Assignment	Ref
$PdBr_4^{2-}$	26,990(3700)	^3E_u πe_u---b_{1g}	
	30,200(10,600)	^1E_u πe_u---b_{1g}	
	30,900(550)	$^1A_{2u}$ πb_{2u}---b_{1g}	
	36,500(5400)	^3E_u σe_u---b_{1g}	
	37,000	Br---σ^*	
	40,400(28,900)	^1E_u σe_u---b_{1g}	
	43,200	Br---σ^*	85
$Pd(mnt)_2^{2-}$	15,700(64)	x2-y2---xy d-d	
	22,700(5700)	πL---xy LMCT	
	35,800(2840)	d---π^* MLCT	
	30,800(20,200)	π---π^* L	
	33,900(47,000)	σL---xy LMCT	100
$(Bu_4N)_2Pd_2Cl_6$	30,000(2100)	[Cl(t)π---dσ^*]	
	34,700(2260)	[Cl(b)π---dσ^* LMCT]	
	41,980(31,200)	[Cl(t)σ---dσ^* LMCT]	
	ca.50,000(70,000)	[Cl(b)σ---dσ^* LMCT]	69
$(Bu_4N)_2Pd_2Br_6$	24,270(3470)	[Br(t)π---dσ^* LMCT]	
	30,720(4470)	[Br(b)π---dσ^* LMCT]	
	36,760(36,000)	[Br(t)σ---dσ^* LMCT]	
	46,300(44,700)	[Br(b)σ---dσ^* LMCT]	69
$Pt(en)Cl_2$	35,100(55,100)[a]	dz^2---p_z	
	49,200(6700)	LMCT ? d-p ?	26,165
cis-$Pt(Py)_2Cl_2$	27,250(28)	x^2-y^2---xy	
	33,300	x^2-y^2---π^*	
	34,500 36,400?	n---π^*	
	38,000(7500)	xz,yz---π^*	
	39,200(7870) 40,300(9240)	π---π^*	
	43,500(10,600)	d--p_z	27
trans-$Pt(Py)_2Cl_2$	27,550(21)	xy---x^2-y^2	
	32,300(3310)	xz,yz---π^*	
	35,700(3640)	n---π^*	
	37,600(9760)	z^2---π^*	
	38,800(10,700) 39,400(10,900)	π---π^*	
	43,900(6260)	d--p_z	27
K_2PtBr_4	35,300(15)	3B1u xy--p_z	
	37,200(8500)	π(eu)---x^2-y^2	
		xz,yz---p_z	
	48,100(11,800)	π(b2u)---x^2-y^2	
		z^2---p_z	32

cont

Table 5.25 cont

Complex	Absorption spectrum	Assignment	Ref
$(Bu_4N)_2PtBr_4$	33,330(3860)	$^1A_{2u}$ πBr---Pt ?	
	36,040(8400)	^1E_u πBr---Pt ?	
	47,380(55,400)	^1E_u σBr--Pt,d-p	69,166
$PtCl_4^{2-}$	17,800(3) 21,000(17)	sf d-d	
	25,600(56) 30,200(59)	d-d	
	37,900(400) 43,400(7200)	πCl---dσ^*c	
	46,500(9050)		622
$(Bu_4N)_2Pt(CN)_4$	35,700(1270)	$E_u(^3A_{2u})$ a_{1g}--a_{2u} π^*	
	39,000(9000) 39,250(15,460)	E_u,A_{2u} (^3E_u) e_g--a_{2u}	
	41,300(1300)	? see [74]	
	46,190(18,700)	$E_u(^1E_u)$ e_g--a_{2u},π^*	
		$A_{2u}(^1A_{2u})$ a_{1g}--a_{2u}	
		etc.	69,74
$Pt(NH_3)_4^{2+}$	50,760(11,070)	d-p e_g--a_{2u}	69
$Pt(NH_3)Cl_3^-$	48,400(8050)	[πCl---dσ^*]	167,168
$Pt(CNMe)_4^{2+}$	36,750(2080)	E_u ($^3A_{2u}$) a_{1g}---$a_{2u}\pi^*$	
	39,400(7490)	$A_{2u}(^1A_{2u})$ a_{1g}---a_{2u}	
	41,600(7060)	E_u (^3E_u) e_g---a_{2u}	
	43,700(3400)	E_u (^3E_u) + CN	
	46,500(6350)	$A_{2u}(^3E_u)$ e_g---a_{2u}	
	48,700(18,600)	E_u (^1E_u) e_g---a_{2u}	77
$Pt(Ph_2PS_2)_2$	33,900(30,000)	πS---σ^* LMCT ?	82
$Pt(N_3)_4^{2-}$	40,500(19,800)	πN_3---σ^* ?	83
c-$Pt(PEt_3)_2Me_2$	32,600(622)	^{3}[z^2---π^*]	
	36,100(5640)	^{1}[z^2---π^*]	
	36,600(5640)	^{3}[xz,yz---π^*]	
	40,600(2870)	^{3}[x^2-y^2---π^*]	91
c-$Pt(PEt_3)_2Et_2$	(31,000)(400)	^{3}[z^2---π^*]	
	(34,000)(2060)	^{1}[z^2---π^*]	
	(36,400)(2670)	^{3}[xz,yz---π^*]	
	(40,800)(3870)	^{3}[x^2-y^2---π^*]	
	49,300(24,000)	^{1}[xz,yz---π^*]	91
c-$Pt(PEt_3)_2Cl_2$	31,300(370)	d-d	
	38,100(1960)	^{3}[z^2---π^*]	
	41,400(7400)	^{1}[z^2---π^*]	
		^{3}[xz,yz---π^*]	
	45,600(10,900)	πCl---σ^*, d-p	91

cont

Table 5.25 cont

Complex	Absorption spectrum	Assignment	Ref
c-$Pt(PEt_3)_2Br_2$	(29,500)(450)	d-d	
	33,700(1660)	$\pi Br \rightarrow \sigma^*$ MLCT	
	(37,800)(1620)	$^3[z^2 \rightarrow \pi^*]$	
	(40,500)(6900)	$^1[z^2 \rightarrow \pi^*]$	
		$^3[xz,yz \rightarrow \pi^*]$	
	(44,000)(16,200)	$\sigma Br \rightarrow \sigma^*$, d-p	
		$^1[xz,yz \rightarrow \pi^*]$	91
c-$Pt(PEt_3)_2MeCl$	(35,000)	$^3[z^2 \rightarrow \pi^*]$	
	40,900(7500)	$^1[z^2 \rightarrow \pi^*]$ + ^{3}xz,yz	
	(45,000)(8900)	$^3[x^2-y^2 \rightarrow \pi^*]$	
	48,200(16,500)	$^1[xz,yz \rightarrow \pi^*]$	91
t-$Pt(PEt_3)_2MeCl$	34,100(1040)	$^3[z^2 \rightarrow \pi^*]$	
	37,200(2500)	$^3[yz \rightarrow \pi^*]$	
	39,500(5800)	$^1[z^2 \rightarrow \pi^*]$	
		$^1[yz \rightarrow \pi^*]$	91
t-$Pt(PEt_3)_2Br_2$	23,000(20)	^{3}d-d	
	30,300(455)	^{1}d-d	
	35,500(7780)	$\pi Br \rightarrow \sigma^*$	
		$^3[yz \rightarrow \pi^*]$	
	(38,900)(4830)	$^1[z^2 \rightarrow \pi^*]$	91
t-$Pt(PEt_3)_2I_2$	29,900(2620)	$\pi I \rightarrow \sigma^*$	
	33,500(3370)	$^3[yz \rightarrow \pi^*]$	
	(40,000)(12,200)	$^1[yz \rightarrow \pi^*]$	
	44,300(48,800)	$\sigma I \rightarrow \sigma^*$	
	48,300(28,600)	d-p ? $^1[z^2 \rightarrow \pi^*]$	91
t-$PtCl_2(CO)(Py)$	15,700 18,700	d-d	
	24,400 29,700	d-d	
	34,970(1679)	Pt $\rightarrow \pi^*$ Py	
	37,300 38,300 39,100	$\pi \rightarrow \pi^*$ Py ?	192
t-$PtCl_2(C_2H_4)(Py)$	22,300 25,000 28,200	d-d	
	32,030(925)	Pt $\rightarrow \pi^*$ Py	
	38,170(3825)	$\pi \rightarrow \pi^*$ Py	192
$Pt(mnt)_2^{2-}$	14,410(49)	$^3[x^2-y^2 \rightarrow xy]$	
	18,500(1220)	$^1[x^2-y^2 \rightarrow xy]$	
	15,650(56)	$^1[xz \rightarrow xy]$	
	18,500(1220) 21,100(3470)	$d \rightarrow \pi^*$ MLCT	
	29,700(15,600)	$\pi \rightarrow \pi^*$ L	
	32,300(13,400)	$\pi L \rightarrow xy$ LMCT	
	43,800(43,500)	$\sigma L \rightarrow xy$ LMCT	100

cont

Table 5.25 cont

Complex	Absorption spectrum	Assignment	Ref
Rh(P-P)$_2$Cl^b	21,400(124)	E_u ($^3A_{2u}$) $z^2 \rightarrow a_{2u}$	
	24,800(4600)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	28,490(9100)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	
	29,580	?	
	31,900(9400)	?E_u (^1E_u) $z^2 \rightarrow e_u$	41,42,197
t-Rh(CO)Cl(PPh$_3$)$_2$	27,700(3100)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	35,100(15,000)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	41,99
[RhBr$_2$(CO)$_2$]$^-$	29,800(2700)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	37,000(5400)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	41
Rh(CNEt)$_4^+$	23,000(260)	E_u ($^3A_{2u}$) $z^2 \rightarrow a_{2u}$	
	26,300(8400)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	30,000(3450)	E_u (^3E_u) $e_g \rightarrow a_{2u}$	
	32,500(24,500)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	
	(35,500)(2100)	E_u ($^3B_{1u}$) $b_{2g} \rightarrow a_{2u}$	41,77
Rh(bpy)$_2$(H$_2$O)$_2^+$	18,000 (19,455)	$d \rightarrow \pi^*$ MLCT	101
Rh(en)(cod)$^+$	26,400(1120)	$^1[z^2 \rightarrow \pi^*$(diene)]	196
Rh(bipy)(cod)$^+$	20,600(790)	$^1[z^2 \rightarrow \pi^*$(bpy)]	196
Rh(bpz)(cod)$^+$	17,700(524)	$^1[z^2 \rightarrow \pi^*$(bpz)]	196
Ir(P-P)$_2$Cl^b	19,200(700)	E_u ($^3A_{2u}$) $z^2 \rightarrow a_{2u}$	
	22,800(3900)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	26,500(4100)	E_u (^3E_u) $e_g \rightarrow a_{2u}$	
	31,800(8600)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	41,42,197
Ir(2=phos)(cod)+	16,800(505)	$^3[z^2 \rightarrow a_{2u}]$ phos	
	19,800(2790)	$^1[z^2 \rightarrow a_{2u}]$ phos	
	(20,400)(2010)	?	
	24,800(3380)	$^1[z^2 \rightarrow a_{2u}]$ cod + ?	197
IrCl(CO)(PPh$_3$)$_2$	23,100(530)	E_u ($^3A_{2u}$) $z^2 \rightarrow a_{2u}$	
	26,100(2780)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	29,800(2370)	E_u (^3E_u) $e_g \rightarrow a_{2u}$	
	36,500(5500)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	41,99
IrCl$_2$(CO)$_2$	26,000(700)	E_u ($^3A_{2u}$) $z^2 \rightarrow a_{2u}$	
	28,500(2380)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	31,600(2040)	E_u (^3E_u) $e_g \rightarrow a_{2u}$	
	38,600(4700)	E_u (^1E_u) $e_g \rightarrow a_{2u}$	
	42,300(4200)	$xy \rightarrow a_{2u}$?	41
Ir(CN)$_2$(CO)$_2^-$	21,700(1570)	E_u ($^3A_{2u}$) $z^2 \rightarrow a_{2u}$	
	24,900(2960)	A_{2u}($^1A_{2u}$) $z^2 \rightarrow a_{2u}$	
	28,900(1980)	E_u (^3E_u) $e_g \rightarrow a_{2u}$	

cont

Table 5.25 cont

Complex	Absorption spectrum	Assignment	Ref
$Ir(CN)_2(CO)_2^-$	34,500(5030)	E_u (^1E_u) e_g---a_{2u}	
cont	(36,000)(4450)	xy---a_{2u} ?	41
$Ir(CNEt)_4^+$	21,400(4600)	E_u ($^3A_{2u}$) z^2---a_{2u}	
	23,800(5500)	A_{2u}($^1A_{2u}$) z^2---a_{2u}	
	27,100(4170)	E_u (^3E_u) e_g---a_{2u}	
	32,500(6670)	E_u (^1E_u) e_g---a_{2u}	
	34,550(3850)	A_{2u}(^3E_u) e_g---a_{2u}	41,77
$Ir(en)(cod)^+$	21,000(196) 21,700(165)	^{3}[z^2---π^*(diene)]	196
	24,200(1060) 24,900(997)	^{1}[z^2---π^*(diene)]	196
$Ir(bpy)(cod)^+$	17,300(734)	^{1}[z^2---π^*(bpy)]	
	22,300(1080)	^{1}[z^2---π^*(diene)]	196
$(Bu_4N)AuCl_4$	31,000(5800)	$^1A_{2u}$, ^1E_u(1) πCl--M	
	(38,000)(880)	$^1B_{2u}$ nCl---M	
	44,100(44,800)	^1E_u(2) σCl---M	64
$(Bu_4N)AuBr_4$	(21,000)(1230)	E_u($^3A_{2u}$, ^3E_u)	
	25,400(4900)	$^1A_{2u}$, ^1E_u(1) πBr--M	
	(33,000)(660)	$^1B_{2u}$ nBr---M	
	39,200(52,500)	^1E_u(2) σBr---M	
	50,500(17,400)	CTTS Br^-	64,69
$(Bu_4N)Au(CN)_2Cl_2$	34,200(1060)	$^1B_{3u}$, $^1B_{1u}$ πCl---M	
	45,500(21,400)	$^1B_{2u}$ σCl---M	
	(49,000)(13,700)	$^1B_{2u}$ πCN---M	64
$(Bu_4N)Au(CN)_2Br_2$	28,000(650)	$^1B_{3u}$ πBr---M	
	30,800(840)	$^1B_{1u}$ πBr---M	
	41,500(29,000)	$^1B_{2u}$ σBr---M	
	47,800(13,600)	$^1B_{2u}$ πCN---M	
	49,500(14,100)	CTTS Br^-	64
$(Bu_4N)Au(CN)_2I_2$	(22,500)(920)	$^1B_{3u}$ πI---M	
	26,700(5800)	$^1B_{2u}$ πI---M	
	36,700(41,700)	$^1B_{2u}$ σI---M	
	42,200(19,400)	$^1B_{2u}$ CN---M	
	(47,000)(21,700) 49,000(33,900)	CTTS I^-	64
$(Me_4N)Au(CN)_2ClBr$	29,000(400)	πX---M	
	32,700(675)	πX---M	
	42,500(18,500)	σX---M	
	47,800(12,000)	CN---M	64

cont

Table 5.25 cont

Complex	Absorption spectrum	Assignment	Ref
$(Me_4N)Au(CN)_2BrI$	26,800(1840)	πX---M	
	38,900(25,000)	σX---M	
	41,800(15,000)	CN---M	
	47,200(16,800) 49,000(23,500)	CTTS X^-	64
$Au(N_3)_4^-$	30,000(31,300)	πN_3---σ^*	
	45,200(15,530)	σN_3---σ^*	83

a) Crystal extinction coefficient. For comparison with solution spectrum, divide by 3. b) Data for the unsaturated diphosphine, PPh_2.CH=CH.PPh_2 are almost identical. c) Original publication proposed MLCT assignment. This publication [622] also contains data for many aquo species of Pt(II) and Pd(II). The bromo and chloro derivatives of Rh(I) and Ir(I) generally have very similar spectra.

The weak $^1B_{2u}$ state is only tentatively identified. It did not prove necessary to be concerned about spin-orbit splitting of terms in the bromide complex. Mixing of the spin singlets and triplets seems small and it appears that mixing of the σ and π e_u levels serves, together with a large separation between σ--σ^*, n and π---σ^* levels, to nullify a strong spin-orbit perturbation of the LMCT spectra in gold complexes.

Schwartz [67] has also reported detailed MCD spectra for $[AuCl_4]^-$ doped into Cs_2YCl_6. Considerable fine structure was noted in the lower energy LMCT transition when the crystals were at cryogenic temperatures. Vibrations due to the $[AuCl_4]^-$ molecule and the lattice were observed.

Data for the $[PdX_4]^{2-}$ (X = Cl,Br) ions are included in Table 5.25 [85]. The pattern of transitions outlined above is readily observed and supported both by polarisation [85] and MCD data [39].

The square complex K_2PtBr_4 was studied by Martin and co-workers [32,33]. The molecular orbital scheme shown in Fig.5.12 a)is appropriate. Thus, in addition to LMCT transitions, the complex is expected to exhibit d--p type transitions, as is relatively common for platinum complexes. The spectrum is shown in Fig.5.35. The solution spectrum shows three strong uv bands at 34,200, 37,200 and 48,000 cm^{-1}. Previous workers discussing $[PtX_4]^{2-}$ ions (X = Cl,Br) [34-39] have proposed various assignments involving both d--p and LMCT transitions. The assignments proposed in Table 5.25 by Martin and colleagues [32] are based upon MCD identifying degenerate states, upon comparison of chloro and bromo data, and upon polarisation data. This last shows that the two lower energy bands are x,y polarised while the upper energy band is z polarised and must be associated with A_{2u}. There is extensive mixing between

certain d---p and LMCT states such that it is unrealistic to choose one pure transition over the other.

Data for $[Au(CN)_2X_2]^-$ species [64] are also shown in Table 5.25. By comparison of X = Cl,Br,I, it was facile to identify halogen dependent bands as halogen to gold LMCT. A band whose energy varied little with halogen, was then attributed to $\pi CN \text{---} \sigma^* Au$. At higher energy CTTS transitions from halide to solvent were observed. The $[Au(CN)_4]^-$ anion does not absorb below 52,000 cm^{-1} [66], and the red shift in the mixed system occurs because the x^2-y^2 orbital (σ^*) is more stable due to a lower net ligand field. The mixed $[Au(CN)_2XY]^-$ complexes show LMCT bands at energies roughly midway between those expected for X and Y.

Unlike the $[Au(CN)_4]^-$ ion, the series $[M(CN)_4]^{2-}$, M = Ni,Pd,Pt do exhibit CT spectra below 50,000 cm^{-1}. Since they are red-shifted relative to Au(III) it is evident that they must be MLCT [74-81]. Indeed these complexes exhibit the various d---π^* transitions discussed above (section 5.5) (Table 5.25). Also unlike the gold complexes, spin-orbit coupling appears essential in understanding these complexes, even the light nickel complex. Thus both $[M(CN)_4]^{2-}$, M = Ni(II), Pt(II) show three MCD positive A terms indicative of excited E_u states. The simple analysis in section 5.5 cannot yield three E_u states in the absence of spin-orbit coupling.

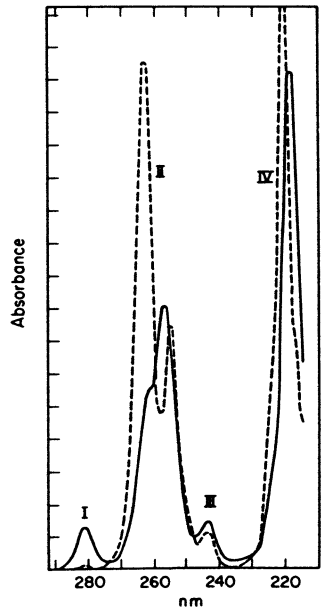


Fig.5.36 The polarised single crystal spectra of $(n\text{-}Bu_4N)_2Pt(CN)_4$ at 5 K. The solid line is the σ-spectrum and the dashed the π-spectrum [75].

Thus the earlier assignments [69] which neglected spin orbit coupling, require revision [74]. Based on MCD data, relatively strong bands in the $[Pt(CN)_4]^{2-}$ ion previously assigned to spin singlet transitions, were re-assigned to the E_u components of spin triplet transitions.

Subsequently a single crystal study of $(Bu_4N)_2Pt(CN)_4$ at 5 K, revealed polarisation data fully consistent with the previous MCD study. Thus three xy polarised bands at 35,520w, 38,880s and 45,700s cm^{-1} are assigned to the E_u components of $^3A_{2u}$, ^3E_u and ^1E_u respectively. Two z polarised bands at 38,020s and 45,290s cm^{-1} are assigned to the A_{2u} components of ^3E_u and $^1A_{2u}$ respectively (Fig.5.36) [75]. The weak feature at 41,070 cm^{-1} appears in both polarisations and may involve $b_{2g} \text{---} a_{2u}$ transition, or is possibly a vibrational satellite of ^3E_u (with a_{1g} C=N stretch).

Isci and Mason have discussed the spectra of isonitrile complexes [41,92]. While the earlier paper [41] considered the possibility that higher energy transitions were to the second π^* of CNR, the more recent and complete MCD study led to the conclusion that all states involved transitions to the lowest a_{2u} π^* CNR level [77]. Complete spin-orbit coupling is necessary to understand the data (Table 5.25). The d level sequence is the same as for the cyanides, namely $z^2 > xz,yz > xy$. The z^2 level is considered essentially non-bonding so that the stabilisation of the xz,yz level reflects π-bonding to the CNR group. This stabilisation is some 5-7000 cm^{-1} for $[Pt(CN)_4]^{2-}$, but in the $[M(CNR)]^{2+}$ complexes ranges from ca. 7000 cm^{-1} for Pt(II) and Ir(I), to 13,000 cm^{-1} for Pd(II) and Rh(I). Evidently CNR is a stronger π-acid than cyanide, consistent with a general blue shift for MLCT bands when comparing the two series of complexes.

Anex and Peltier have considered the quartz ultraviolet spectra of $M(en)Cl_2$ (M = Pd,Pt) [26], using the technique of polarised single crystal specular reflection spectroscopy [28-31]. Assignments are shown in Table 5.25. Because of crystal stacking high energy LMCT transitions red shift in a manner similar to the Magnus Green Salt (MGS) systems discussed in Chapter 7.

Patterson and co-workers present luminescence, absorption and MCD data for cis and trans-$Pt(Py)_2Cl_2$. Time resolved luminescent data show the existence of two emitting π^* states only 240 cm^{-1} apart. In comparison with the ammine analogs (see Chapter 6), the d-d bands lie at slightly lower energies but there exist a manifold of π^* levels of Py near the high energy empty d orbital. Thus in trans-$Pt(Py)_2Cl_2$ the $xy \text{---} x^2-y^2$ d-d transition occurs at 27,550 cm^{-1}, but at 32,600 cm^{-1} is observed a $xz,yz \text{---} \pi^*(Py)$ transition (Table 5.25). In general LMCT transitions in the trans isomer lie at slightly lower energy than corresponding transitions in the cis isomer [27].

Geoffroy and co-workers [91] have studied the electronic and MCD spectra of the series cis- and trans-$Pt(PEt_3)_2XY$. Fig.5.37 shows the anticipated molecular orbital sequence for these complexes assuming that z is perpendicular to the molecular plane.

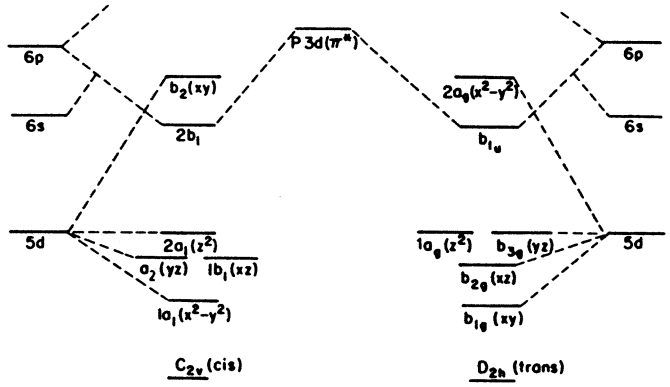


Fig.5.37 One electron molecular orbital energy level diagrams for planar complexes of C_{2v} and D_{2h} symmetry, involving a phosphine π^* acceptor ligand [91].

MLCT transitions to the low lying 3d(π^*) state are expected to occur below d-d transitions. Typical spectra are also shown in Fig.5.38. Indeed these workers studied many compounds with X,Y ranging from alkyl groups to halide to hydride and combinations thereof. Only selected data are shown in Table 5.25. Assignment was based on analogy to the $[Pt(CN)_4]^{2-}$ assignment, upon MCD data, and upon the supposition that only one phosphorus π^* state is involved. There is considerable consistency in the assignment. Thus for the series of complexes (X,Y) = (H,Cl), (H,Br), (Me,Cl), (Me,Br), (Et,Cl), (Et,Br), (Cl,Cl), (Br,Br) and (I,I), the following transitions fall within the ranges indicated.

trans series

$^3[z^2 \text{---} \pi^*]$	32,400 - 33,700	$^1[z^2 \text{---} \pi^*]$	35,000 - 39,700 cm^{-1}
$^3[yz \text{---} \pi^*]$	33,000 - 36,500	$^1[yz \text{---} \pi^*]$	38,800 - 41,800
$^3[xz \text{---} \pi^*]$	43,500 - 44,800	$^1[xz \text{---} \pi^*]$	46,000 - 47,600

cis series

$^3[z^2 \text{---} \pi^*]$	32,100 - 37,900	$^1[z^2 \text{---} \pi^*]$	36,600 - 41,600 cm^{-1}
$^3[xz,yz \text{---} \pi^*]$	36,600 - 41,600	$^1[xz,yz \text{---} \pi^*]$	46,300 - 49,000

(5.40)

The largest deviations occur with the dihalides such that variations within the mixed halide/hydride and alkyl/halides are much smaller. The halo complexes also show intense absorption near 50,000 cm^{-1} which does not occur in the di-alkyl derivatives; this is presumably π-halide to Pt LMCT, perhaps mixed with d-p ($z^2 \text{---} 6p_z$?). The distinction between $xz \text{---} \pi^*$ and $yz \text{---} \pi^*$ is made on the basis that where, for example, the P-Pt-P axis is defined as x, the xz orbital will lie below yz through π-back donation to P, and

hence the former transition should lie at higher energy. While the position of the π^* level is assumed to be invariant with changing X,Y, the d orbital energies, and hence the donor d orbitals involved in the MLCT transition, will depend upon the ligand field strength of X,Y. Thus in this fashion, the spectrochemical series is seen to be H,Me,Et >PEt_3 > Cl > Br > I [91]. With X = Y = Cl,Br and I, the high energy empty d orbital lies below π^* and hence ligand field transitions are seen below the lowest MLCT band.

The authors argue that since the z^2 orbital is almost non-bonding, the difference between the two MLCT transitions $z^2 \text{---} \sigma^*$ and $d\pi \text{---} \pi^*$ provides a measure of the extent of π-stabilisation (back donation) by the ligands concerned. This difference is 7000 cm^{-1} for $[Pt(CN)_4]^{2-}$ compared with 9000 cm^{-1} for $[Pt(CNEt)_4]^{2+}$. The values for the cis and trans $[Pt(PEt_3)XY]$ depend upon X and Z but generally fall between 7000 - 9000 cm^{-1}. Thus PEt_3 has a π-acceptor strength between CN^- and CNR [91]. Stufkens has reported data for a range of trans-$PtX_2L(RPy)$ where L may be CO or ethylene, and RPy are a series of substituted pyridines [192]. Corresponding palladium complexes have been studied but in much less detail [92,93].

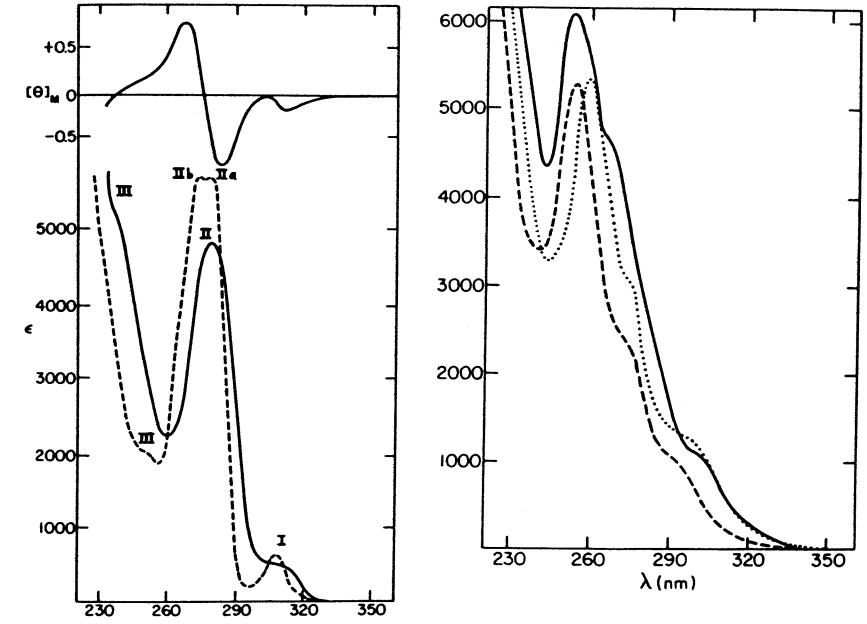


Fig.5.38 (a) Absorption (lower curves) and MCD spectra of cis-$[Pt(PEt_3)_2Me_2]$ at 298 K in acetonitrile (solid line) or at 77 K in 7/4 methylcyclohexane/pentane glass (hatched line) [91]. (b) Absorption spectra of trans-$[Pt(PEt_3)_2(H)Cl]$ (solid line), trans-$[Pt(PEt_3)_2(Me)Cl]$ (---), and trans-$[Pt(PEt_3)_2(Et)Cl]$ (...) in acetontrile solution [91].

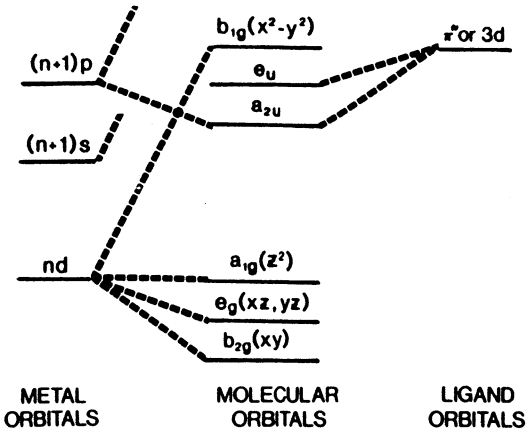


Fig.5.39 Simplified molecular orbital energy level diagram for planar complexes containing π^* acceptor ligands [41].

Square Planar Rhodium(I) and Iridium(I) Complexes [41-44]: These species have been extensively studied partly because of their rich spectra and perhaps because they are generally luminescent. Complexes with π-acceptor ligands are unusually stable and a molecular orbital diagram appropriate for their analysis is shown in Fig.5.39. In complexes such as $[M(diphosphine)_2]^+$, MLCT transitions of the type z^2 or xz,yz to a_{2u} and e_u, and xy---e_u are allowed. Because of large spin-orbit coupling for these ions, one may expect that spin forbidden transitions may gain intensity, and that degenerate states might show spin-orbit splitting.

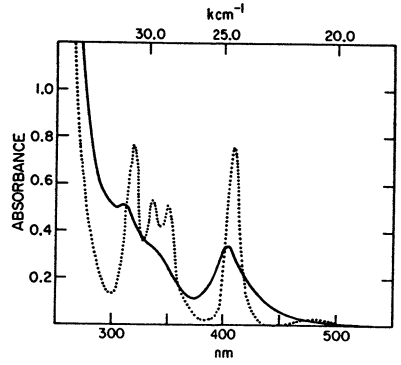


Fig.5.40 The absorption spectra of $[Rh(2{=}phos)_2]Cl$ in EPA solution at room temperature (solid line) and 77 K (...) [43].

Table 5.26

CHARGE TRANSFER SPECTRA OF SOME COPPER(II) DERIVATIVES[a]

Complex	Structure[a]	Charge Transfer	d-d	Ref
$Cu(cyclam)_2^{+}$	Sq CuN_4	38,600(7200)σN	19,500(90)	6
$Cu(salenH_4)$	Sq c-$CuN_2O'_2$	25,500(1000)O'[a]	17,500(260)	
		30,500(1000)O'[a]		6
$Cu(glycinate)_2$	Sq t-CuN_2O_2	40,000σN,σO		6
$Cu(en)_2^{2+}$	Sq t-CuN_4[b]	43,860(6025)σNH_2		45
$Cu(en)_2^{2+}$	Sq t-CuN_4	40,050(6450)σNH_2		44
$Cu(s\text{-}Me_2en)_2^{2+}$	Sq t-CuN_4	38,280(6025)σNHMe		45
$Cu(as\text{-}Me_2en)_2^{2+}$	Sq t-CuN_4	(40,820)σNH_2	35,550(4700)σNMe_2	45
$Cu(s\text{-}Et_2en)_2^{2+}$	Sq t-CuN_4	39,260(5000)σNHEt		45
$Cu(as\text{-}Et_2en)_2^{2+}$	Sq t-CuN_4[b]	39,670(3550)σNH_2		
		33,330(3300)σNEt_2		45
$Cu(en)_2Cl_2$	Tet t-CuN_4Cl_2	36,770σNH_2		44
$Cu(en)_2Br_2$	Tet t-CuN_4Br_2	36,060σNH_2		44
$Cu(s\text{-}Me_2en)_2Cl_2^{2+}$	Tet t-CuN_4Cl_2	35,170σNHMe		44
$Cu(N\text{-}Me\text{-}pn)_2^{2+}$	Sq CuN_4	41,200σN	16,700CD	
			18,700CD	44
$Cu(NN\text{-}Me_2\text{-}pn)_2^{2+}$	Sq CuN_4	33,000	15,000CD	
		40,600	19,200CD	44
$Cu(NN\text{-}Et_2\text{-}pn)_2^{2+}$	Sq CuN_4	33,000	14,000CD	
		38,000	18,500CD	44
$Cu(NN\text{-}Pr_2\text{-}pn)_2^{2+}$	Sq CuN_4	33,900	14,400CD	
		41,000	17,500CD	
			21,200CD	44
$[Cu(s\text{-}Me_4en)Cl_2]_2$	SqPy CuN_2Cl_3	25,550(1020)πCl	(11,800)	
		34,000(6980)πCl	14,500(145)	
		36,300σN		229
$[Cu(s\text{-}Me_4en)Br_2]_2$	SqPy CuN_2Br_3	19,600(595)πBr	(11,200)(120)	
		(24,400)(2000)πBr	14,200(240)	
		30,600(6540)πBr		
		35,700σN		229
$CuCl_2$(gaseous)	Lin $CuCl_2$	19,000πCl		603
Cs_2CuCl_4	Td $CuCl_4$	23,000 24,800πCl	29,000 -	
		34,500σCl		600
$(MeNH_2)_2CuCl_4$	Sq $CuCl_4$	24,000πCl 33,300br	38,500σCl	600
$(dienH_3)CuCl_5$	D3h $CuCl_5$	24,200πCl 28,170	37,040σCl	604

a) Where intensity data are reported, the solvent, usually water, probably coordinates at a longer distance along axis. b) MeOH solution. R_2-pn is $RR'NCH(CH_3)CH_2NH_2$ (optically active). O' is phenolic oxygen. The O'* CT state is MLCT Cu(II) to π^*(phenol). $SalenH_4$ is the reduced version of the Schiff base from condensation of salicylaldehyde with ethylenediamine. See Tables 5.7, 5.17 - 5.19 for additional data.

Thus individual transitions to the A_{2u} and E_u spinor states of $^3A_{2u}$, ^3E_u and $^3B_{1u}$ may be observed. This should be particularly true for the heavier Ir(I) compared with Rh(I).

A typical spectrum is shown in Fig.5.40, and assignment in Table 5.25. The assignments follow fairly readily from consideration of the molecular orbital diagram, and were supported by excitation polarisation [42], MCD data [43], and emission spectra [44].

5.12 ix) Charge Transfer Spectra of d^9 Complexes

Copper: Copper(II) is relatively easy to reduce to copper(I) such that many ligands give rise to LMCT transitions. Thus even simple amine complexes exhibit $\sigma \text{---} \sigma^*$ transitions from the Cu-N bond to the empty x^2-y^2 orbital in a tetragonal or square complex. Transitions from $-NH_2$ groups occur near 40,000 cm^{-1}, while -NHR and $-NR_2$ groups, with somewhat longer and weaker Cu-N bonds, show this transition at lower energy [44,45,229] (Table 5.26). CD data have been reported for this LMCT transition [44], whose energy is linearly related to the estimated first ionisation potential of the amine ligand [339]. Complexes with several types (lengths ?) of Cu-N bond show several σN---Cu(II) CT transitions [45].
Halogen to copper(II) LMCT occurs at relatively low energy (e.g. the black colour of anhydrous copper bromide). It also appears relatively independent of stereochemistry. The range of $[CuCl_4]^{2-}$ ions of different stereochemistry, linear $CuCl_2$ and trigonal bipyramidal $[CuCl_5]^{3-}$ all have grossly similar chloride to Cu(II) LMCT [600,603,604]. A range of simple halo complexes are included in Table 5.26. Quite low energy transitions seem to have no other provenance than halogen to Cu(II) LMCT [229]. Recall the work of Solomon [609] dealing with square planar versus D_{2d} distorted $[CuCl_4]^{2-}$ ions in section 5.5.

Note that most Cu(II) complexes are six coordinate tetragonal with long axial Cu-L bonds. In general one does NOT see CT absorption from such axially ligated ligands. Thus strong CT absorption in a tetragonal Cu(II) complex is an indicator of equatorial binding for the ligands concerned.

Sulphur, imidazole and phenoxide copper(II) CT has been dealt with in section 5.11 and in Chapter 8.

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Chapter 6

ELECTRONIC SPECTRA OF d^n IONS

6.1 General Introduction

In this chapter the electronic spectra of d^n ions will be discussed as a function of n, and of stereochemistry. The discussion will include first, second and third row ions, though the majority of data in the literature refer to first row transition metal ions.

Electronic spectra may be recorded under various conditions; historically they were obtained by photographic methods, more recently by manual photoelectric spectrophotometers where each point is obtained individually. Both methods are now largely superceded by the use of automatically recording spectrophotometers. Indeed in recent years microprocessor controlled spectrometers have appeared. These have a number of advantages including the ready processing of spectroscopic data and the acquisition of automatic baselines. The latter eliminates the need to use matching cells since the spectrometer memorises the spectrum of solvent plus cell and subtracts it from solvent plus cell plus solute.

A sample for spectroscopic study may be investigated in many ways, viz:

a) Solution spectroscopic data, where the sample is dissolved in a suitable solvent. The spectrum is restricted to the wavelength region in which the solvent is transparent. Aromatic solvents, for example, cannot be used in the ultraviolet region. Because of vibrational overtones it is difficult to use solvents in the near infrared beyond 1600nm.

b) Diffuse reflectance spectra (drs), where light is reflected from the powdered sample. This method [1-6] suffers mainly from the fact that it is difficult to obtain any information about the intensity of absorption. Techniques however do exist to extract such information [7].

Even though one band may appear stronger than another, its oscillator strength may not in fact be greater. Apparent band intensities vary appreciably according to the state of subdivision of the sample, the degree of scattering (wavelength dependent) and whether the sample is mixed, or not, with an inert diluent. Band positions, however are rather less dependent upon these variations.

Drs spectra are useful as a means of determining whether a solid species undergoes chemical change when dissolved in a solvent. If the drs spectrum in the solid is essentially identical

with the solution spectrum then the inference is that the species has dissolved unchanged. However some care must be utilised. Similar species may have very similar spectra, e.g. $[PtI_4]^{2-}$ and $[Pt_2I_6]^{2-}$ [8].

c) Crystal spectra arising from the absorption of oriented, or unoriented, crystals. Considerably more information can be obtained from crystal studies than from solution or drs. The use of polarised light with an oriented crystal of known crystal structure allows the absolute determination, in suitable cases, of the symmetry of excited states. Data can be collected over a wide spectroscopic range, though absorption in the ultraviolet region is generally so great that even the mechanically thinnest crystal usable is too thick. It is more simple to obtain low temperature data, down to liquid helium temperatures, than with solutions (freezing problems). Such low temperature data are important because they can reveal, by enhanced resolution, fine details in the spectrum (weak electronic bands, vibronic structure etc). Even where the crystal structure is unknown the use of polarised light and a crystal oriented along its extinction axes can provide well resolved and useful data.

A modification of this method involves isomorphic substitution. For example if vanadium(III) is substituted for aluminium(III) in a crystal of otherwise transparent $CsAlCl_4$, the observed spectrum is consistent with that expected for vanadium(III) in a tetrahedral environment [9]. However some care must be exercised since examples are known where the dopant ion does not take up the same structure as the host molecule [10].

d) Mull spectra for which the sample is mulled into a solvent in which it is insoluble. This method is rarely employed since it offers no obvious advantages to methods (a) and (b). However it has been used in the near infrared region (10000 - 3500 cm^{-1}) where solvents are unreliable. Hexachlorobutadiene, and liquid paraffin are generally used. When organic ligands are present, the vibrational spectrum of the ligand generally appears as fine structure on the d-d transitions making analysis uncertain. The transmission spectra obtained by this method often differ appreciably from diffuse reflectance spectra especially below 5000 cm^{-1}. Transmission spectra may also be recorded using KBr discs, compressing the sample into powdered KBR under high pressure. Under such conditions the bromide ion may substitute into the complex.

e) Fused salt spectra for which the sample is dissolved into the molten salt [11 -14]. Many transition metal ions when dissolved into a molten eutectic mixture of KCl and LiCl form complex ions of the type $[MCl_n]^{m-}$ where n is generally 4 or 6. There is often a temperature dependent equilibrium between tetrahedral (n = 4) and octahedral (n = 6) forms. Under such conditions, stereochemistries unobtainable by other routes are often stabilised. Many other melts have been studied, including fluorides, nitrates and thiocyanates

and low temperature melts such as tetramethylphosphonium bromide.

6.2 Crystal Field Spectra

The discussion which follows is subdivided in terms of the d^n configuration, metal ion, and stereochemistry. It is not intended to be comprehensive but will strive to provide characteristic examples of all known transition metal systems. Special systems such as bio-inorganic chromophores, carbonyls, metal-metal bonded systems, including solid state phenomena such as occurring in the Platinum blues, and mixed valence complexes are reserved for Chapters 7 and 8.

Note also that because of the incredible amount of information in the literature it has sometimes been necessary to adopt a review-like presentation and direct the reader to the literature without providing detailed discussion here.

The terms octahedral, tetrahedral etc will be reserved for complexes clearly belonging to the O_h and T_d point groups etc, and the terms six coordinate, four coordinate, pseudo-tetrahedral etc used where coordination number only or distorted stereochemistry is implied. Since the electronic spectra of d^0 species are not d-d in nature, such data are collected in Chapter 5. The charge transfer spectra of other d^n configurations have already been dealt with in Chapter 5 and, except in passing, such CT data are not reported here. The reader should be aware therefore that the Tables of data reported in this Chapter do not include the charge transfer absorption which may occur at energies above the d-d absorption, and cross reference with Chapter 5 is desirable. However, where charge transfer absorption occurs at energies below or within the range of observable d-d absorption, it is usually included here. Detailed discussion of crystal field and nephelauxetic data derived from d-d spectra is also generally excluded from this Chapter since such discussion would tend to be repetitious. Chapter 9 deals with the whole body of such data obtained from the studies outlined in this Chapter.

The Tables do not include information on the solvent used to obtain the spectrum except in rare instances where there is a pronounced solvent effect; the reader should obtain such information from the original literature. In some cases, the spectra may reflect a solvated species. Emphasis is placed on single crystal low temperature studies since these are usually of very high quality, and are based on a system of accurately known structure. Shoulders are reported in parentheses in the various Tables. Minerals are not covered in this volume. Where space permits, the microstoichiometry of each species is included in the various Tables.

6.2. (i) d^1 Sc(II), Ti(III), Zr(III), Hf(III), V(IV), Nb(IV), Ta(IV), Cr(V), Mo(V), W(V)

The free ion ground term is ^2D, and the octahedral ground term is

Table 6.1

SIX COORDINATE TITANIUM(III) COMPLEXES

Complex	Absorption Spectrum		ΔE	Chromophore	Ref
Ti^{3+}/Al_2O_3	17,870	(17,720)	150[a]	O_6	210
Ti^{3+}/LiF	15,630[b]			F_6	32
$Ti^{3+}/Al(urea)_6^{3+}$	16,950	18,700	1750	O_6	368
$\alpha TiCl_3$	14,300	18,900CT?	4500	Cl_6	111
$\alpha TiBr_3$	12,500	15,500CT?	3000	Br_6	111
$Ti(H_2O)_6^{3+}$	20,100(5.7)	(17,400)	2700	O_6	16,17
$Ti(ROH)_6^{3+}$	16,800(4.3)	(15,200)	1600	O_6	17,36
$Ti(Urea)_6^{3+}$	17,500(10.6)	(16,000)	1500	O_6	18,19
$Ti(Urea)_6^{3+a}$	18,530	16,640	1850	O_6	33
TiF_6^{3-}	18,900	(16,100)	2800	F_6	29
TiF_6^{3-}	19,310	16,210	3100	F_6	113
$TiCl_6^{3-}$	13,000	10,000	3000	Cl_6	24
$Ti(MeUrea)_{6c}^{3+}$	17,400	15,350	2050	O_6	35
$TiCl_3(tht)_{2c}$	15,300	11,800	3500	Cl_3S_3	62
$TiBr_3(tht)_{2c}$	13,800(13)	(11,900)	1900	Br_3S_3	62
$TiI_3(tht)_2$	12,600(13)	(10,800)	1800	I_3S_3	62
$TiCl_3(SMe_2)_{2c}$	15,600(25)	12,050	3550	Cl_3S_3	62
$TiBr_3(SMe_2)_2$	14,300(12)	(11,800)	2500	Br_3S_3	62
$Ti(MeUrea)_5Cl^+$	17,500	13,600	3900	O_5Cl	35
$Ti(DMeUrea)_4Cl_2$	17,200	12,550	4650	O_4Cl_2	35
$Ti(C_2O_4)_3^{3-}$	24,000			O_6	20
$Ti(acac)_3$	16,900(1500)			O_6	28
$TiCl_3$	13,800	(12,000)	1800	Cl_6	23
$TiCl_3(Py)_3$	16,670(b)			Cl_3N_3	25
$TiBr_3(dioxan)_3$	14,100	(12,500)	1600	Br_3O_3	26
$TiCl_3(CH_3CN)_3$	17,200(31)	(14,700)	2500	Cl_3N_3	27,34
$TiCl_3$.bipyridyl	15,750	13,500	2250	Cl_3N_3	31
$TiCl_2(CH_3CN)_4^+$	20,400	16,800	3600	Cl_2N_4	34
$Ti(H_2O)_4F_2^+$	23,300[d]	15,200[d]	8100	O_4F_2	307
$Ti(H_2O)_4Cl_2^+$	19,220[d]	14,970[d]	4250	O_4Cl_2	229
$NH_4ZnTiF_6.6H_2O$	22,124	14,535	7590	F_6?	454

a) Single crystal at 20 K (NB assignments not as shown). b) Fluorine to Ti(III) charge transfer reported at 48,080, 51,550 and 64,940 cm^{-1} [32]. c) measured in sulphur ligand as solvent, six coordinate complex. d) see text for assignment; shoulder at 21,860 cm^{-1} in 10 K spectrum (chloride). The $[Ti(CN)_6]^{3-}$ ion absorbs at 18,900 and 22,300 cm^{-1} [20] but the species is suspect. All bracketed data in this Table and those which follow refer to shoulders. The molar intensity (in 1 mol^{-1} cm^{-1}) is reported in parentheses immediately following an energy given in wavenumbers. The absence of an intensity value may infer that the measurement was carried out with the sample in the solid state. In some cases it is omitted

because of lack of space in the Table. To save space, the solvents used to obtain solution data are not generally identified. See the original reference for further information. A query (?) sign means uncertainty in assignment not in energy. Abbreviations used in these and the following Tables include CT = charge transfer, br = broad, c- = cis, t- = trans, sf = spin forbidden. x,y,z are polarisations.

$^2T_{2g}$. The excited state is ^2E_g and will be subject to Jahn Teller distortion. Assuming a tetragonal distortion (to D_{4h}), the excited state will split into $^2A_{1g}$ + $^2B_{1g}$ (see discussion for Ti(III) below).

SCANDIUM(II) d^1

The spectrum of scandium(II) in an axially distorted environment has been observed [15] through the bombardment of scandium(III) doped in calcium fluoride, with 2MeV electrons. Four main absorption bands were observed between 15,100 and 42,400 cm^{-1}, but were not assigned.

TITANIUM(III) d^1

Six Coordinate Complexes: A discussion of the spectrum of titanium(III) is of historic importance since Ilse and Hartmann [16] first assigned the weak absorption band observed in solutions of the octahedral hexaaquotitanium(III) ion at 20,100 cm^{-1} (Fig.6.1a) to a crystal field transition between the split levels of the ^2D term, namely, ^2E_g---$^2T_{2g}$. Most six coordinate titanium(III) complexes have an absorption band in this region, and are generally purple in colour. On the lower energy edge of the absorption band in the hexaaquo ion spectrum there is a shoulder attributed to Jahn-Teller splitting of the excited ^2E_g state. This is a common feature of titanium(III) spectra. If the usual e_g mode of vibration leading to four short equatorial bonds and two longer axial bonds is assumed to be responsible for the lifting of degeneracy of the ^2E_g state, then the energy sequence will probably be $z^2(a_{1g}) < x^2\text{-}y^2(b_{2g})$. In this case the energy of the upper component will be approximately 10Dq (depending upon the extent and sense of splitting of t_{2g}) (Fig.1.3 (c)). However the sense of the splitting of ^2E_g cannot be assured without further measurement and will probably vary from one complex to another; unless the authors have obtained definitive information relating to the splitting of ^2E_g, such information is not included in Table 6.1. A rough assessment of the crystal field splitting energy could be ascertained from averaging the transition energies. Note that if the $^2T_{2g}$ term splits with ^2E_g (in D_{4h}) as ground term, neither of the transitions is exactly equal to 10 Dq (see Chapter 3). In general the splitting of the ground term is of the order of 420-800

cm^{-1}, [16] while that of the excited state is of the order of 1000-4500 cm^{-1} (Table 6.1).

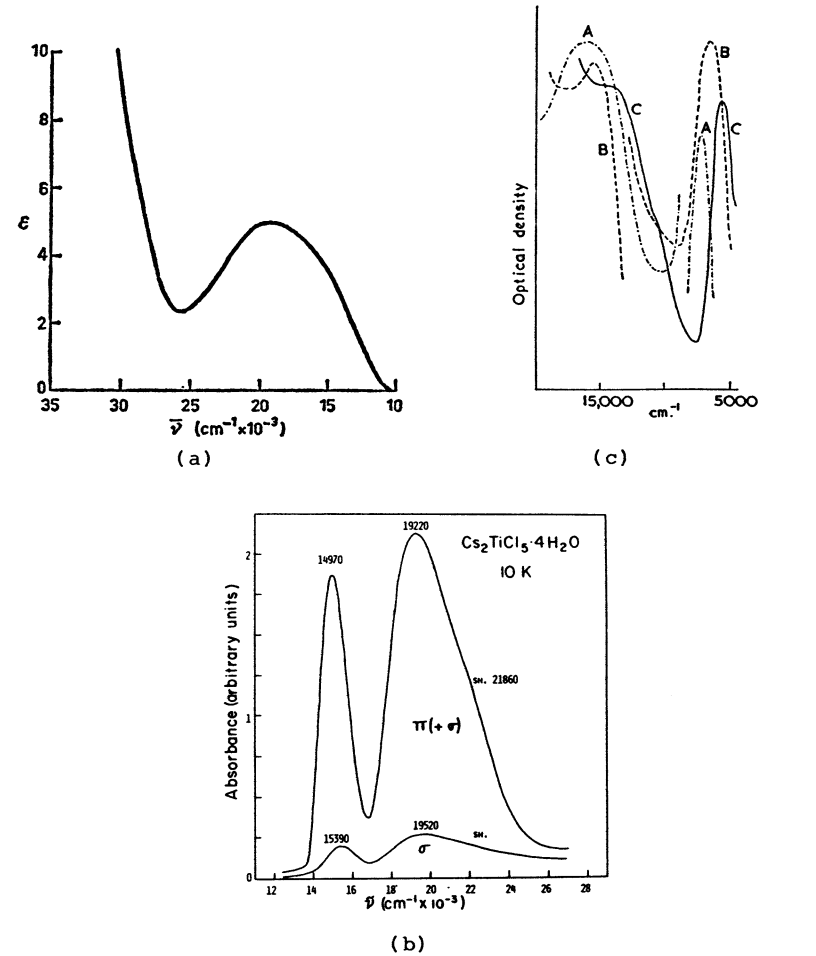


Fig.6.1 (a) The electronic spectrum of the octahedral $[Ti(H_2O)_6]^{3+}$ ion. (b) The polarised crystal spectra at 10 K of $Cs_2TiCl_5.4H_2O$ containing the trans-$[Ti(H_2O)_4Cl_2]^+$ cation. The weaker spectrum is σ, and the stronger 78.5 π and 21.5 σ. (c) The electronic spectra of five coordinate $TiX_3.2$(2-picoline), A, X = Cl, B, X = Br, C, X = I.

Recall that the splitting of the E_g state (in O_h) in the AOM model for an ML_4Z_2 species (Z on z axis) would be

given by $2(e_\sigma(Z) - e_\sigma(L))$ being positive with $E(A_{1g}) > E(B_{1g})$. Thus the sense of the splitting is readily predicted if it is known whether, for an ML_6 complex, axial compression (positive splitting) or axial elongation (negative splitting) occurs. For a tetragonal system, ML_4Z_2, the sense of the splitting probably follows from the relative magnitudes of the e_σ values of the ligands concerned.

Six coordinate complexes, formally of lower symmetry than octahedral, also show a single absorption band, with a shoulder or separate peak (Table 6.1). In most of these lower symmetry complexes there is no need to invoke the Jahn-Teller effect to split the ^2E_g level, since the lower symmetry provides a sufficient mechanism. In general the magnitude of the splitting of the ^2E_g level is rather greater (Table 6.1) [35,36]. However the sense of the splitting will depend upon the relative magnitudes of the fields of the contributing ligands. Six coordinate complexes with non-axial ligands, such as ethylenediamine [38], acetylacetone [28] and urea [33] show characteristics of D_3 symmetry. In this group the degeneracy of the $^2T_{2g}$ ground term in O_h symmetry is lifted to yield $^2A_1 + {}^2E$, while the excited state remains ^2E. The presence of two transitions whose separation decreases with temperature but whose absolute intensity does not change significantly (electronic selection rules) is consistent with a Jahn-Teller splitting of this ^2E (in D_3) excited state [33].

The species trans-$[Ti(H_2O)_4F_2]^+$ has been studied both by esr and electronic spectroscopy and exhibits one of the largest splittings of the e_g set yet observed (8100 cm^{-1}) consistent with the fluoride ion being a much better sigma donor than water [307] if the upper state is assumed to be $^2A_1$. The single crystal spectrum of $NH_4ZnTiF_6.6H_2O$ also exhibits similarly large splitting [454] (Table 6.1). In these two cases there is probably a large tetragonal distortion not necessarily directly associated with the Jahn Teller effect. McCarthy and Richardson [229] report the single crystal data for trans-$[Ti(H_2O)_4Cl_2]^+$ (Fig.6.1(b)). The observation of vibronic absorption for the components of ^2E_g (in O_h) in both σ and π polarisation is only consistent with a ^2E_g (in D_{4h}) ground state leading to $e_\pi(H_2O) > e_\pi(Cl)$. The splitting of ^2E_g (in O_h) is also unexpectedly large (4250, $d\sigma$ = 1594 cm^{-1}), but the sense of the splitting was not determined. Utilising the 'average environment rule' where necessary, the spectrochemical series (see Chapter 9) for titanium(III) can be expressed as follows:

Br (10,500) < Cl (13,000) < ROH (16,800) = acetylacetone (16,900) < urea (17,500) = dioxan (17,700) < F (18,900) < H_2O (20,100) = pyridine (20,340) < bipyridyl (21,250) < acetonitrile (21,400) < cyanide (22,300) (6.1)

Five Coordinate Complexes: Fowles and co-workers have investigated five coordinate and trigonal bipyramidal amine complexes of the type TiX_3L_2, where L is trimethylamine or 2-picoline, and X is Cl or Br [37,39,40]. Their electronic spectra in the visible region are similar to octahedral analogs but they are readily distinguishable therefrom by the appearance of a d-d band in the near infrared region at 5000-8000 cm^{-1} (Fig.6.1.(c)). The assignments are clearly identified, using Fig.1.2(a) as shown in Table 6.2. Some ill-defined binuclear five coordinate titanium(III) complexes have been reported [41].

Table 6.2

FIVE COORDINATE TITANIUM(III) COMPLEXES

Complex	$^2E' \leftarrow {}^2E''$	$^2A_1' \leftarrow {}^2E''$	Ref
$TiCl_3(NMe_3)_2$	6100(88)	15,100(17)	37
$TiBr_3(NMe_3)_2$	5100	14,100	37
$TiCl_3(2\text{-Pic})_2$	7500	15,900	37
$TiBr_3(2\text{-Pic})_2$	6500	15,800	37
$TiI_3(2\text{-Pic})_2$	5700	14,700	37

Four Coordinate Complexes: Few tetrahedral derivatives, and no square planar derivatives, of Titanium(III) have been characterised. Titanium trichloride dissolved in molten LiCl/KCl eutectic has absorption bands at 13,000 and (10,000) cm^{-1} at 400°C attributable to the $[TiCl_6]^{3-}$ ion. As the temperature is raised there is evidence for the equilibrium:-

$$TiCl_6^{3-} \rightleftharpoons TiCl_4^- + 2\ Cl^- \qquad (6.2)$$

shifting to the right. Bands are observed near 6000 and 8000 cm^{-1} in this presumed tetrahedral species. This same species can be observed in molten $CsGaCl_4$ at 750°C [13,24].

Other Complexes: The oxalate group appears to form both seven and eight coordinate bidentate and bridging complexes with titanium(III) [42]. X-ray data show the presence of a pentagonal bipyramidal structure in the former. The d-d transitions are observed near 9300 - 10,500 and near 12,500 cm^{-1}. The cyclopentadienyl group (Cp) forms a range of pseudo tetrahedral complexes of the type Cp_xTiCl_{3-x}. Apparently the d and Cp orbital energies are comparable in energy and complex spectra involving both d-d and charge transfer transitions are seen [43-45] (see Chapter 7).

Tri-coordinate titanium(III) is observable in the spectrum of vapor phase $Ti(III)Cl_3$ and exhibits bands at 14,300 and 12,000 cm^{-1} [46]. On the other hand the three coordinate $Ti[(N(SiMe_3)_2]_3$ shows

d-d absorption at 4,800(10) and 17,400(122) cm^{-1} [136].

VANADIUM(IV) d^1

Many vanadium(IV) compounds contain the oxovanadium(II) (VO^{2+}) group; the section which follows is subdivided into simple vanadium(IV) and oxovanadium(IV) species.

Six coordinate complexes: The spectra of these species are similar to those of Titanium(III). Earlier workers [e.g. 49,50] reported a considerably greater apparent splitting of the ^2E_g excited state in many of the complexes studied (Table 6.3). However low energy charge transfer transitions may be expected because of the higher oxidation state of the metal. Caution should be exercised in assigning bands above about 20,000 cm^{-1} to d-d transitions without seriously considering the charge transfer option. The $[VCl_6]^{2-}$ ion has an absorption band [47] at 21,050 cm^{-1} (in acetonitrile) whose intensity (1040 l $mole^{-1}$ cm^{-1}) is sufficiently large that it is probably charge transfer in origin, and a charge transfer assignment seems likely for bands in this region [83], (see Chapter 5 for full discussion). Complexes such as $V(acac)_2Cl_2$ and $[V(acac)_3]^+$ are all intensely red or violet through high intensity low energy charge transfer absorption [455].

Five coordinate complexes: The unusual complex, $[PCl_4]^+.[VCl_5]^-$ has been reported [52]. This exhibits weak, presumed d-d transitions at (6200), 8100 and 16,000 cm^{-1}. The spectrum has been analysed both in terms of a square pyramid and a distorted trigonal bipyramid [53,54]. It is trigonal bipyramidal [1377].

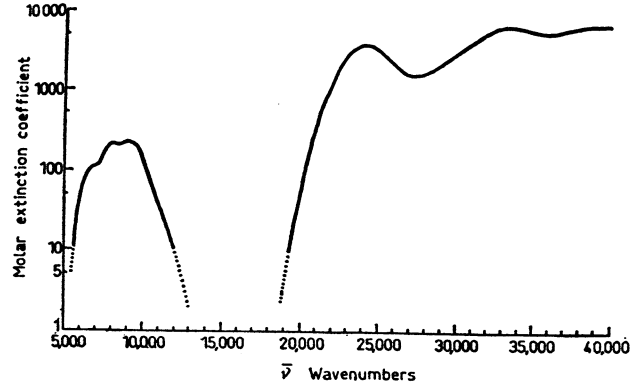


Fig.6.2 The absorption spectrum of VCl_4.

Four coordinate complexes: A band observed [51] at 9000 cm^{-1} ($f = 2 \times 10^{-3}$) (Fig.6.2) in the vapour phase spectrum of tetrahedral VCl_4 is assigned to the $^2T_2 \text{---} ^2E$ transition. The absorption band is clearly resolved into three components at 6600, 7850 and 9250 cm^{-1}. These may arise through vibrational fine structure, separate electronic transitions due to a lowering of symmetry, spin-orbit coupling, or operation of the Jahn-Teller effect. Calculations [51,55] favour the Jahn-Teller effect. Xα calculation [93] confirms a 10 Dq value of about 9000 cm^{-1} for this species. The spectrum also exhibits prominent chlorine to vanadium charge transfer [56-58] (see Chapter 5), also predicted [93]. The species VL_4 (L = Bu^tO, Me_2N, and Et_2N) are also believed to be four coordinate with distorted tetrahedral structures [59,60] (Table 6.3). These green or blue compounds show two d-d transitions assignable as $^2E \text{---} ^2B_1$, and $^2B_2 \text{---} ^2B_1$ in D_2d symmetry. The energy separation between these two transitions is in good agreement with the photoelectron spectra thereof [79]. These species exhibit quite low energy charge transfer bands as anticipated [59,60].

Other vanadium species: The red-brown dodecahedral complex $V(S_2CNEt_2)_4$ [61] exhibits a weak d-d band at 13,600 cm^{-1}, followed by stronger, charge transfer and intraligand absorption from 31,700 cm^{-1}. Other octacoordinate dithiocarbonate derivatives have similar spectra [312].

VANADYL COMPLEXES d^1

Vanadium(IV) species containing the V=O grouping [63] have spectra which are quite distinct from other vanadium(IV) derivatives, and present several controversial features. Evidently the strong axial perturbation by the axial V=O group plays an important role in determining the spectrum. The most symmetrical vanadyl(IV) complex, VOL_5 (all ligands equivalent) would have C_{4v} symmetry in which the metals transform as:

$$\begin{array}{ccc} x^2\text{-}y^2 \text{---} b_1 & z^2 \text{---} a_1 & xy \text{---} b_2 \\ xz,yz \text{---} e & s,\ pz \text{---} a_1 & p_x,p_y \text{---} e \end{array} \qquad (6.3)$$

Whatever the relative order of these orbitals, there are three potential intra d shell transitions, although not all are necessarily allowed. In general three bands are observed below 30,000 cm^{-1}, with intensities of the order of 5 - 100 l $mole^{-1}$ cm^{-1} (Fig.6.3)(Table 6.4), but their assignments have long been the subject of controversy [64] .

In C_{4v} symmetry, the metal d orbitals in a OVL_4 species would be perturbed as follows:

$$z^2(a_1) \quad \sigma^*(O); \quad x^2\text{-}y^2(b_1) \quad \sigma^*(L);$$
$$xy(b_2) \text{ in-plane } \pi^*(L); \quad xz,yz(e) \quad \pi^*(O), \text{ out-of-plane } \pi^*(L)$$

Table 6.3

VANADIUM(IV), NIOBIUM(IV), MOLYBDENUM(V) AND TUNGSTEN(V) DERIVATIVES

i) Eight coordinate Vanadium(IV) derivatives

Complex	d-d transitions	Charge Transfer	Ref
$(Diars)_2VCl_4$	13.2		310
$(S_2CNEt_2)_4V$	15.6		311
$(Dta)_4V$	(12.8)(60) (19.2)(316)	(21.7)(5755)	312
$(Dtpa)_4V$	(12.4)(41) (13.3)(5) (18.9)(288)	(21.3)(1380) (27.0)(3310)	312

ii) Six Coordinate Vanadium(IV), Molybdenum(V) and Tungsten(V) Derivatives

Complex	Peak Positions		Chromophore	Ref
$[VCl_6]^{2-}$	(13,800)	15,300(70)	Cl_6	
		21,200(330)CT		47,83
$V(OCH_3)Cl_5$	13,500(3)	14,500(3.4)	Cl_5O	
		26,100(1250)		83
$[VF_6]^{2-}$	20,120		F_6	48,253
$(Pyridine)_2VCl_4$	15,750		Cl_4N_2	47
$(4\text{-}NO_2\text{-}PyNO)_2VCl_4$	14,600	22,000CT	Cl_4O_2	112
$(DPK)_2VCl_4$	14,300	20,400CT	Cl_4O_2	49
$(Dioxan)VCl_4$	17,400	22,200CT	Cl_4O_2?	49
$(MeCN)_2VCl_4$	ca.20,000	ca.22,000CT	Cl_4N_2	49
$(Bipy)VCl_4$	17,400	21,300CT	Cl_4N_2	50
$(o\text{-}Phen)VCl_4$	18,000	21,700CT	Cl_4N_2	50
$[MoCl_6]^-$	23,830-25,700[a]		Cl_6	110
$[MoCl_6]^-$	26,700br		Cl_6	99
$[MoBr_6]^-$	18,100 CT?		Br_6	122
$[WCl_6]^-$	25,400 CT?		Cl_6	122
$[WBr_6]^-$	23,300 CT?		Br_6	122

a) Doped into a single crystal of Cs_2ZrCl_6; spectrum measured at 4 K shows extensive vibronic structure.

iii) Four Coordinate Vanadium(IV) and Niobium(IV) Derivatives

Complex	Peak Positions	Chromophore	Ref
VCl_4	6600,7850,9250	Cl_4	51
$V(OBu^t)_4$	10,930sh, 13,900(107)	O_4	59
$V(NMe_2)_4$	13,300sh, 17,900br	N_4	59

cont

Table 6.3 cont

Complex	Peak Positions	Chromophore	Ref
$V(NEt_2)_4$	13,300sh, 17,400(270)	N_4	59
	13,120sh, 17,065sh[b]		60
$Nb(NR_2)_4$	18,200, 20,400	N_4	89

b) using $Ti(NEt_2)_4$ as a reference.

Ballhausen and Gray [65], in a molecular orbital study of the $[VO(H_2O)_5]^{2+}$ ion, suggested that the d orbital energy order might then be expected to be:

$$b_2 < e < b_1 < a_1$$

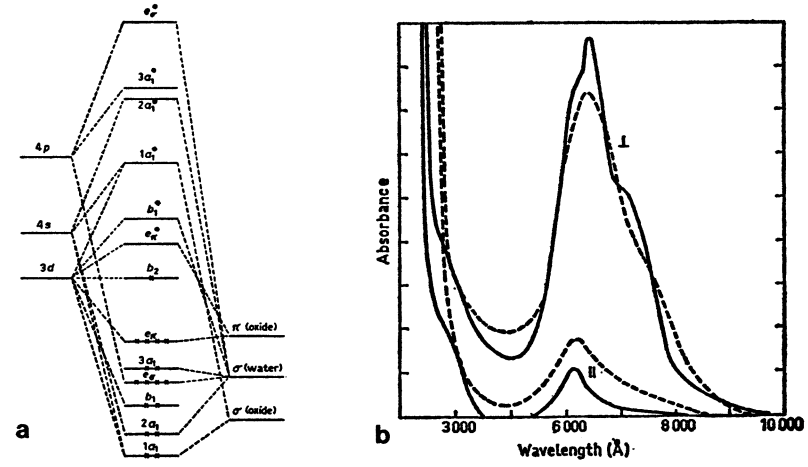


Fig.6.3 a) A molecular orbital diagram for the $[VO(H_2O)_4]^{2+}$ ion. b) The polarised spectra of the $[VOCl_5]^{3-}$ ion substituted in $K_3TiCl_6.2H_2O$ (full curve -196° C, broken curve 25°C).

The interaction of xz,yz with ligand L may possibly be π acceptor in character in which case the relative order of the b_2 and e orbitals could be inverted [70]. This possibility can usually be checked via an esr experiment. Moreover distortion from C_{4v} symmetry could occur especially where LL is a bidentate ligand in common complexes of the type $VO(LL)_2$. In this case the highest possible symmetry is C_{2v} and the degeneracy of the e orbitals is lifted [71]. An additional d-d transition may then be seen. Considering the data in Table 6.4, the three transitions may correspond to:

$$\begin{array}{lll} \text{I} & (xy)\ b_2 \text{---} e^* & {}^2E \text{----} {}^2B_2 \\ \text{II} & \quad b_2 \text{---} b_1^* & {}^2B_1 \text{---} {}^2B_2 \\ \text{III} & \quad b_2 \text{---} 1a_1^* & {}^2A_1 \text{---} {}^2B_2 \end{array} \qquad (6.4)$$

The electronic dipole selection rules in C_{4v} symmetry are:

b_2----e	xy---xz,yz	allowed xy
b_2----b_1	xy---x^2-y^2	forbidden
b_2----a_1	xy---z^2	forbidden

A band observed near 25,000 cm^{-1} could be transition III but is often too intense for a transition which is electronically forbidden. Indeed various groups have proposed that the transition xy---z^2 should be placed at 50,000 - 70,000 cm^{-1} [73-78]. On the other hand Selbin suggested that all the transitions are close to Band I ('cluster' assignment) and that the band near 25,000 cm^{-1} is a spin or orbitally forbidden charge transfer transition [69,72].

Within the AOM model, the energies for transitions I-III in $[VOL_4L']^{2-}$ (L' trans to V=O) are represented by:

$$\begin{array}{ll} \text{I} & e_\pi(O) + e_\pi(L') - 2e_\pi(L) \\ \text{II} & 3e_\sigma(L) - 4e_\pi(L) \\ \text{III} & e_\sigma(L) + e_\sigma(L') + e_\sigma(O) - 4e_\pi(L) \end{array} \qquad (6.5)$$

The V=O axial bond is exceptionally short relative to the equatorial bonds, and the axial oxygen AOM parameters may be quite large. There are insufficient data in the spectra of these complexes to assign all these parameters unequivocally. If band III is assumed to lie above 50,000 cm^{-1} then the value of $e_\sigma(O)$ is required to be between 40,000 - 50,000 cm^{-1} about 7 - 9 times higher than the required value of $e_\sigma(L)$. This is unreasonable. If band III occurs below 30,000 cm^{-1} then more reasonable values of $e_\sigma(O)$, are deduced [81].

In some early work, Wentworth and Piper [73] studied a crystal containing the $[VOCl_5]^{3-}$ ion, with polarised light, and showed the presence of a perpendicularly polarised, temperature independent, band at 15,500 cm^{-1} reasonably assigned to ${}^2E \text{---} {}^2B_2$. Under this band, at 16,200 cm^{-1} they detected a parallel polarised band, whose intensity was temperature dependent, and which they assigned to the vibronically activated ${}^2B_1 \text{---} {}^2B_2$ transition.

The single crystal spectrum of the vanadyl ion doped into calcium tartrate has been reported and all three bands, I, II and III(?) reported; however the quality of the spectrum is rather poor [90]. Rather better, and polarised, spectra were reported [91] for VO^{2+} doped in potassium oxalate at 77 K. Band III was again assigned and some interesting vibronic structure observed on band II (ca. 760 cm^{-1}, excited V=O stretch ?). The authors comment that the polarisation data are only consistent with C_2 symmetry. Interestingly they note that transition I splits into two components at 11,337 and 11,723 cm^{-1}. The relative intensities of Bands I and II are the reverse of what might be expected, possibly

a consequence of the lower symmetry. Yet another doped spectrum, in zinc cesium sulphate tetrahydrate [94] shows a more typical spectrum but little sign of band III is seen except for a weak shoulder at 30,480 cm^{-1}, assigned thereto. Polarised single crystal data are also reported [92] for $(NEt_4)_2VOCl_4$ at 77 K interpreted with the transitions I and II interchanged. A low energy transition III was tentatively assigned.

Robbins and co-workers [82] have investigated the MCD spectra of $[VO(H_2O)_5]^{2+}$ and $[VO(NCS)_5]^{3-}$ ions and confirm the presence of the expected A term with the correct sign in the lowest energy band confirming its original assignment [65] as a transition to ${}^2E$. In the aquo ion a C term is found under the shoulder of this band, in agreement with transition II gaining intensity through second order spin-orbit coupling therewith (Table 6.4).

Transition III will most likely be observed near 25 - 30,000 cm^{-1} if it is not obscured by charge transfer absorption. A very low transition energy for transition III is rejected because it would imply only a very weak interaction with the double bonded oxygen, and the model which places the band at 50 - 70,000 cm^{-1} is rejected because this would make the interaction much too strong [114].

In the spectra of $[VOCl_4]^{2-}$ species Drake and co-workers [80] have identified a weak band near 25,000 cm^{-1} which they do associate with band III (in 6.4). Very recently, an elegant low temperature single crystal polarised light study of several vanadyl species has appeared to resolve many of the uncertainties in this area [141]. Transitions I and II were assigned on the basis of their variation with different ligands, their polarisation behaviour, and the nature of their allowing vibrations as deduced from their, often rich, vibronic structure. Replacement of in-plane Br^- by Cl^- and by NCS^- caused a marked shift in the ${}^2B_1 \text{---} {}^2B_2$ transition (II) (but note different nomenclature in ref. [141]) because of their increasing field strength (Table 6.4, Fig.6.3). In five coordinate complexes, a richly structured band near 21 - 24,000 cm^{-1} is assigned to transition III, ${}^2A_1 \text{---} {}^2B_2$. Considering, for example, the pair of complexes, $VOCl_2(tmu)_2$ and $VOBr_2(tmu)_2$ this band shows a vibrational progression in about 800 cm^{-1}, in both polarisations, attributable to the excited V=O stretch implicating an orbital involving V=O bonding. In addition, in x,y polarisation, vibrations at 380 and 350 cm^{-1} respectively, are believed to be enabling V-halogen vibrations. The fact that their energies differ little from their ground state values, is taken to indicate that this state does not involve a transition which materially alters V-halogen bonding. A halogen-vanadium charge transfer state is thereby dismissed in favour of transition (III) (6.4). Moreover it is clearly demonstrated that in six coordinate vanadyl species, where the V=O bond length does not differ much from five coordinate species [141] the presence of the sixth, and axial, ligand, shifts transition III to the blue, probably above 30,000 cm^{-1} (Table 6.4).

Bis(acetylacetonato)oxovanadium(IV) is five coordinate, square

Table 6.4

CRYSTAL FIELD SPECTRA OF OXOVANADIUM(IV) COMPLEXES

Complex	I	II	III	Ref
$VO(H_2O)_5^{2+}$	13,100(16)	16,000(7.5)		49
$VO(H_2O)_5^{2+}$	12,930(16.5)E	15,630(7.5)B_1		82
VO^{2+}/Ca Tar[e]	12,340	18,345	19,875	90
VO^{2+}/K Ox[f]	11,337E	14,224	17,893	
	11,723E	14,992 15,740 16,501		91
$VOCl_4^{2-}$[g]	13,500E	11,600B_1	15,500	92
$VOCl_4^{2-}$[h]	13,097E	11,909B_1	21,383A_1	141
VOF_5^{2-}[a]	10,300E	15,550B_1		141
$VO(NCS)_4^{2-}$[h]	14,060E	17,720B_1	21,030A_1	141
$VO(NCS)_5^{3-}$	13,500	17,200	23,900	39
$VO(NCS)_5^{3-}$	13,370(120)E	17,360(39.6)B_1		82
$VO(NCS)_4(OH_2)^{2-}$	12,450E	17,400B_1		141
$VOCl_2(tmu)_2$	14,280E[h]	13,090B_1	21,943	141
$VOBr_2(tmu)_2$	14,020E[h]	12,783B_1	21,875	141
$VO(CN)_5^{3-}$[a]	14,400(13.8)	23,900(12.5)		
		38,900(1650)CT		84
$VO(oxalate)_2^{2-}$	12,600(25)	16,500(9)	29,400(170)	49
$VO(acetate)_2$	11,700	18,000	27,000	
			41,700CT	110
VO^{2+}/ $ZnCsSO_4$[i]	13,365E	16,390B_1	30,480	94
$VO(DMSO)_3Cl_2$	14,000(38)		(25,300)(9)	
			26,700(10)	50
$VOCl_2(PhAsO)_4$	13,400(32)	(15,400)	(25,300)(29)	50
$VO(bipy)Cl_2$	13,300	16,000	26,800	39
$VO(EDTA)^{2-}$	13,030[b]	17,450	25,630	39
	12,990[c]	17,220	23,800	51
	14,820[d]	16,800	25,000	51
$VO(dsp)_2$	10,300	14,300sh(16)	17,100(58)	
			24,100sh(22)	107
VO(sal(+)tnCH_3)	13,160(-0.26)[j]	17,390(-0.45)	21,280(-1.18)	394
VS(salen)[k]	9680(45)	13,700(71)	(19,495)	393
VS(acen)[k]	10,225(50)	14,815(28)	(20,160)(490)	393

a) These species are six coordinate in the solid state. b) in MeOH. c) in pyridine. d) in chloroform. e)Calcium Tartrate tetrahydrate at 80 K. f) Potassium oxalate at 77 K., also CT band at 28,900 cm^{-1}. g) Single crystal at 77 K. h) single crystal at 5 K contains rich vibronic structure. i)Zinc cesium sulphate hexahydrate. j) CD spectrum on the complex with N,N'-bis(salicylidene)-(S)-(+)-butane-1,3-diamine. k) Thiocarbonyl derivatives, assignment uncertain.

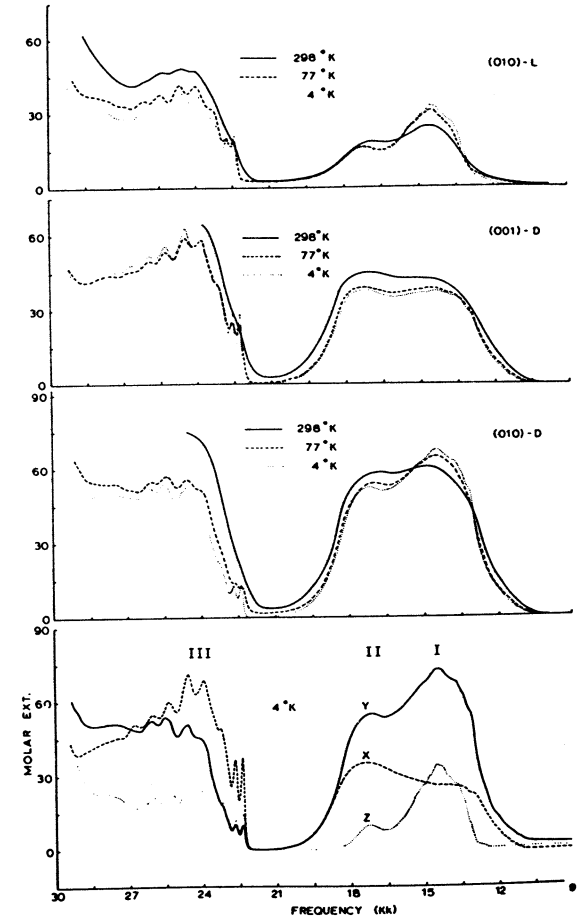


Fig.6.4 Polarised single crystal spectra of the (001) and (010) faces of bis(acetylacetonato)oxovanadium(IV). Spectrum shown along the molecular axes [145].

pyramidal, and dissolves in a wide range of solvents to give spectra which are markedly solvent dependent [133], the solvent presumably occupying the axial site trans to the V=O bond. The energy separation between transitions I and II varies between 1000 and 6000 cm^{-1}. This energy separation has been used as a measure of the extent to which the solvent interacts with the vanadium(IV). The solvents lie in the sequence: H_2O >> MeOH > pyridine > n-propylamine > EtOH > DMSO > DMF > MeCN > $MeNO_2$ > $CHCl_3$ > CS_2 >

benzene > CCl_4. Belford and co-workers [145] have reported a detailed single crystal polarised light study of this species at low temperature. Band I (Fig.6.4) is assigned to transition I. It is noted that the onset of the vibronically induced z polarised absorption begins significantly higher in energy than the onset of the xy polarised, electronically allowed absorption. Band II (Fig.6.4) is the only band whose intensity increases with temperature in all three polarisations; it is presumably, though not necessarily, electronically forbidden.

Very low symmetry oxovanadium(IV) complexes such as those containing tartrate, lactate, mandelate or malate groups [134,135] give rise to four well defined absorption bands between 10,000 and 26,000 cm^{-1}. In these complexes the e level apparently splits, and in the lower symmetry, all four bands become allowed. However further work is necessary on these species to define them adequately. Many Schiff base oxovanadium species are known (e.g. [394]). The CD spectrum of an optically active Schiff base (Table 6.4) provides clear evidence for the three anticipated d-d bands though assignment of band III is still uncertain. Farmer and Urbach [394] note that, on the basis of X-ray data, the loss of a donor molecule from the axial site trans to the V=O bond results both in a shortening of the V=O bond and the shift thereof more out-of-plane. This suggests that in comparing donor (axial site coordinated) and non-donor (axial site empty) solvents, band I should shift to higher energies and band II should shift to lower energies as the donicity of the solvent decreases. There is evidence that this indeed occurs [394].

Thiovanadyl systems (Table 6.4) have been reported [393]. They exhibit similar bands to the oxo species, but shifted to lower energy as anticipated. Since all three bands are so shifted, this may be evidence for the correct assignment of band III at energies below 30,000 cm^{-1}.

NIOBIUM(IV) AND TANTALUM(IV) d^1

The complex K_3NbF_7 is isostructural with $(NH_4)_3ZrF_7$ in which the M(IV) ion is coordinated by seven fluoride ions in a slightly distorted pentagonal bipyramid [85]. The spectrum of K_3NbF_7 is shown in Fig.6.5 and is interpreted in terms of a distorted D_{5h} structure [86].

The higher energy band at 28,570 cm^{-1} is assigned to the transition to a_1', whilst the lower energy pair of bands are assigned to the split components of e_2'. Similar spectra were observed for Nb(IV) in fluoride melts [87].

Relatively weak bands observed near 20,000 cm^{-1} in the spectra of presumed six coordinate $[MCl_6]^{2-}$ (M =Nb, Ta) were assumed to be d-d in origin and were followed by more intense ultraviolet charge transfer absorption [88]. The four coordinate (presumed distorted tetrahedral D_{2d}) $Nb(NR_2)_4$ (R = alkyl) complexes [89] exhibited a weak (ca 200 l mol^{-1} cm^{-1}) transition at 18,000 - 21,000 cm^{-1} with a lower energy shoulder, plus more intense, presumed CT absorption

above 25,000 cm^{-1}. The two bands are considered to arise from the splitting of the allowed b_1---e transition.

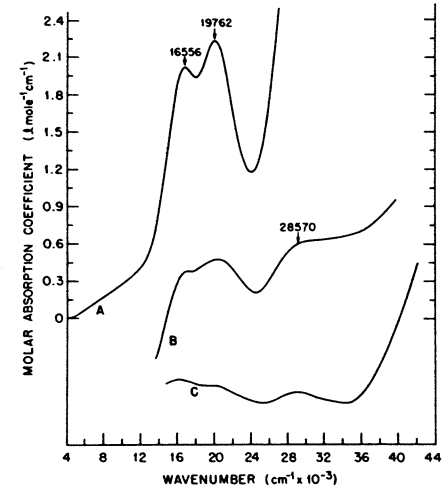


Fig.6.5 Electronic spectrum of K_3NbF_7 at 25°C. a) Polycrystalline plate, b) diffuse reflectance spectrum and c) Kel-F mull.

CHROMIUM(V), MOLYBDENUM(V), AND TUNGSTEN(V) d^1

Absorption of d-d character can be observed in many cases, especially with highly electronegative ultra violet transparent ligands. Charge transfer spectra can occur at very low frequencies, especially with Cr(V) and Mn(VI).

Eight coordinate complexes: The spectrum of the dodecahedral tetrakis(diethyldithiocarbamato)molybdenum(V) has been reported [115,116]. Three d-d bands are reported to lie below 16,000 cm^{-1}. Charge transfer transitions (probably MLCT) lie to higher energies. Both Mo(V) and W(V) form octacyanide complexes [117,118]. Extensive discussion concerns the possible presence in solution of dodecahedral (of D_{2d} symmetry) or tetragonal anti-prismatic (of D_{4d} symmetry) structures [119]. Esr, visible, mcd and vibrational spectroscopy have all been used to decipher these systems. From the conflicting assignments it seems evident that the stereochemistry may vary from solid state to solution, and from one solvent to another. Moreover the actual stereochemistry in any particular situation may be distorted from ideal. In D_{4d} symmetry only two transitions are expected. A mull spectrum of the square-antiprismatic (D_{4d}) $Na_3[W(CN)_8]$ shows a single strong peak at 22,000 cm^{-1}, the second peak presumably being obscured in the ultra-violet.

Table 6.5

CRYSTAL FIELD SPECTRA OF OXO SPECIES OF CHROMIUM(V), MOLYBDENUM(V) AND TUNGSTEN(V)[a]

Compound	xy----xz,yz	xy---x^2-y^2[b]		Ref
$CrO(HSO_4)_3(H_2SO_4)_2$?	13,700	18,100(40)		96
$CrOCl_4^-$	12,480(17)	17,900(40)	25,050(1475)	106
$CrOF_4$	(e)	16,530(0.8)	22,150(17)[f]	106
$CrOCl_5^{2-}$	13,800	23,500		97
$CrOCl_4$[-c]	13,400xyE[d]	18,100xyCT	(21,000)	
			(23,000)	
			(25,000)h	103
$MoOCl_4^-$	14,000(12)	22,500(14)		109
$MoOCl_4$[-c]	15,400xyE	22,500[g]xy	26,500z	
			26,830xy	95
$MoOCl_4^-$[c]	15,500xyE[dh]	22,600B_2[d]		108
$MoOCl_4(H_2O)^-$[c]	13,400xyE[d]	22,800xy		105
$MoOCl_5^{2-}$	13,800	23,000		97
$MoOBr_5^{2-}$	14,300(7)	21,300(560)		98
$MoO(H_2PO_4)_4^-$	13,100(21)	26,000(16)		108
$MoOCl_3(Ph_3PO)_2$	13,500(19)	22,300(11)		99
$MoOCl_3(MeCN)_2$	13,700	19,000		100
$MoOCl_3(bipy)$	13,600	18,800,	23,000	101
$MoOBr_3(dbm)^-$	14,100(41)	18,200(550)	(21,800)	104
$MoO(NCS)_5^{2-}$	12,800	19,200		102
$WOCl_4(H_2O)^-$	12,400[di]	14,700[di]	25,500	126
$WOCl_3(PPh_3O)_2$	12,600[di]	14,800[di]	25,800	126

a) a more extensive tabulation of the absorption spectra of oxomolybdenum(V) species can be found in ref.102. b)Some of the bands listed in this column may be low energy charge transfer bands. c) polarised light single crystal study d) vibronic structure. e) weak bands are seen at 77 K in a Nujol mull at 8290 and 8410 cm^{-1}; these are supposed to correspond with transition I. f) Many bands observed to higher energy. g)Authors assign as charge transfer but this is probably incorrect. h) Some resolution into two components due to spin-orbit splitting. i) Presumed to be spin orbit components of one transition, ^2E---$^2B_2$.

In D_{2d} symmetry, three transitions are expected:- b_2---b_1, a_1---b_1 and e---b_1, and were so assigned, in this sequence of increasing energy, by Perumareddi, Adamson and Liehr [189]; this sequence also implies the order of increasing energy of the d orbitals. The highest energy E transition was split indicative of a lower symmetry perturbation.

Six coordinate complexes: The $[MoCl_6]^-$ ion has been investigated by several workers [99,110,121-123,453]. The solid state absorption reveals signs of Jahn-Teller splitting [120,121]. However when doped into single crystals of $CsZrCl_6$ a cubic spectrum with beautifully resolved fine vibronic structure was observed at 4 K [110,453] (Table 6.3). A single sharp transition ($\Gamma_8(^2T_{2g})$---$\Gamma_8(^2E_g)$) was associated with vibrational structure in the range 23,830 - 25,697 cm^{-1}. There is a major progression in the totally symmetric mode (292 cm^{-1}) based upon the false vibronic origins due to excitation of the odd t_{1u} lattice mode (49 cm^{-1}), and the t_{1u} Mo-Cl stretching mode (170 cm^{-1}) [453].

However in a more recent study of the MCD and electronic spectra of this latter system, Schatz and co-workers [1375] propose that the doped crystal contains a mixture of d^1 $[MoOCl_5]^{2-}$ and d^2 $[MoCl_6]^{2-}$ and does not contain d^1 $[MoCl_6]^-$. The main visible region absorption mentioned above is then re-assigned to the lowest lying charge transfer state of $[MoCl_6]^{2-}$ (see section 5.4).

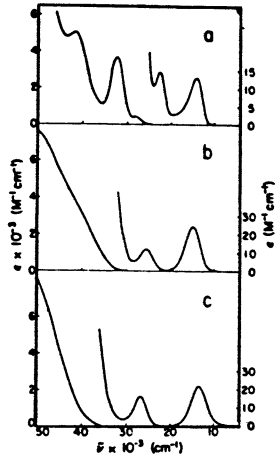
Walton and co-workers [122] reported the electronic spectra of several $[MoX_6]^-$ and $[WX_6]^-$ species (X = Br,Cl). They assigned weak absorption near 15,000 cm^{-1} as d-d in origin whilst bands above about 18,000 cm^{-1} were accorded parity forbidden charge transfer status, on the basis of earlier related work [123-125]. Certainly the band near 24,000 cm^{-1} in the $[MoCl_6]^-$ ion is a vibrationally activated forbidden transition [110].

CHROMYL(V), MOLYBDENYL(V) AND TUNGSTENYL(V) d^1

Both chromium(V) and molybdenum(V) exhibit OM(V) spectra analogous to vanadyl(IV) spectra. Two low energy, low intensity, absorptions may be observed, ascribed to the xy---xz,yz and xy---x^2-y^2 transitions (Table 6.5) [97]. Since charge transfer transitions are observed at lower energies than in vanadyl(IV) species, the third transition, xy---z^2 has never been observed. Various single crystal studies [95,103,105,108,1334] provide clear evidence for the assignment of the first transition as ^2E---$^2B_2$. Garner and co-workers [95,105] assign the second low intensity transition in the $[(H_2O)MoOCl_4]^-$ ion to a e(Mo=O pi)---b_2(xy) LMCT transition, for several reasons including the fact that its polarisation is xy rather than z as anticipated for a xy---x^2-y^2 transition, and because its energy is insensitive to changes in the ligands cis to the oxo group. Polarised data reported for $MoOCl_3(OP(NMe_2)_3)_2$ [1334] show a band at 13,000 cm^{-1}, assigned to xy---xz,yz with vibrational structure at 40 K having an interval of 750 cm^{-1} assumed to arise from Mo-O stretching and reduced from the ground state value of 996 cm^{-1}. As previously, a band at 22,500 cm^{-1} is tentatively assigned as the oxygen to xy CT band because of its polarisation (xy), and because of the earlier work cited above.

However Winkler and Gray [108] report a detailed polarised light study of the $[MoOCl_4]^-$ ion at 4 K. They show that the second band is indeed xy---x^2-y^2 , and that the polarisation arises through this forbidden transition being vibronically activated via

an e vibration (Fig.6.6, 6.7). A z polarised vibrational component is associated with a symmetric Mo-Cl stretch expected for this transition since the x^2-y^2 level is σ^* antibonding with respect to the $MoCl_4$ plane. A corresponding absorption in the $[CrOCl_4]^-$ ion spectrum [103] is probably also $xy---x^2-y^2$ rather than oxygen or halogen to chromium charge transfer despite several SCF MO calculations [103,166] which must under-estimate these charge transfer energies.



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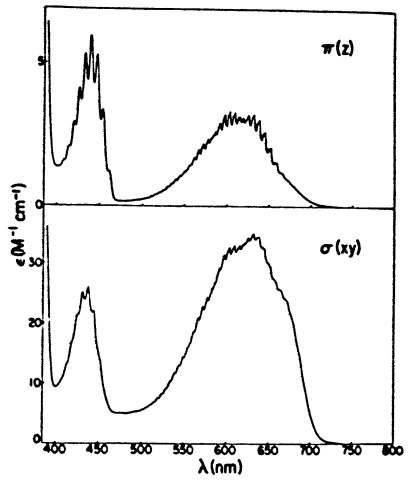


6.7

Fig.6.6 Electronic spectra of molybdenyl ions at 25°C. a) $[MoOCl_5]^{2-}$ in 12M HCl, b) $[MoO(HSO_4)_4]^-$ in 17M H_2SO_4, and c) $[MoO(H_2PO_4)_4]^-$ in 13M H_3PO_4 [108].

Fig.6.7 Single crystal polarised electronic spectra of $(Ph_3As)MoOCl_4$ at 5 K.

Oxy species such as $[MoO(HSO_4)_4]^-$ do not exhibit intense absorption below 35,000 cm^{-1} (Table 6.5) which implies that π(Mo=O)---d(Mo) LMCT transitions should not be anticipated below this energy. Absorption near 28,000 and 32,500 cm^{-1}, in the chloro species, is probably π(Cl)---d(Mo) LMCT in character [108].

The strength of the M=O bond increases in the sequence Cr(V) < Mo(V) < W(V) and is reflected in the increasing energy of the $^2E---^2B_2$ transition energy (see orbital angular overlap model for vanadyl(IV) above). Since oxidising power decreases in this sequence, the LMCT transitions shift to the blue in this sequence. For this reason the band at 18,100 cm^{-1} in $[CrOCl_4]^-$ might involve chlorine to chromium charge transfer (see Chapter 5, section 5.12).

Note that the e(πCl)---xy LMCT transition is expected to be weak because of poor overlap.

Mixed ligand complexes, e.g. $MoOX_3L_2$ exhibit absorption near 13,000 and 23,000 cm^{-1} [99] probably assignable to the two low energy d-d transitions. The five coordinate $(Ph_3PS)MoOCl_3$ has a structure midway between square pyramidal (with oxide at the apex) and trigonal bipyramidal [1323]. A band at 14,900 cm^{-1} is assigned as $^2E---^2B_2$ and is followed by bands at 22,700 and beyond, which are charge transfer in nature.

Little is known of tungstenyl(V) spectra; they appear similar in character to OCr(V) and OMo(V) but with transitions displaced to the blue [108,126]. However Milstein and co-workers [128] do report data for $[WOCl_4(H_2O)]^-$ and $WOCl_3(PPh_3O)_2$ at 5 K. Rich structure between 10 - 17,000 cm^{-1} in the spectra of these species is assigned to spin orbit and vibronic components of the $^2E---^2T_2$ d-d transition. The vibrational interval (ca 850 - 900 cm^{-1}) compares well with the expected value of the excited W=O stretching vibration. Bands observed near 25,500 cm^{-1} in these complexes are probably d-d in origin, perhaps $^2B_1---^2B_2$, since charge transfer in tungsten(V) should be at fairly high energy [108].

Four coordinate Complexes: Chromium(V), in concert with other high oxidation state first row ions, forms a tetrahedral oxy-anion species $[CrO_4]^{3-}$. Such species are usually highly coloured by intense charge transfer absorption occurring in the visible region. However d-d bands can be observed at lower energies. In the tetrahedral species $[MO_4]^{n-}$, the ground state configuration is $t_1^6e^1t_2^0(^2E)$. The (presumed) lowest excited state is the $(^2T_2)$ $t_1^6e^0t_2^1$, arising by d-d excitation. The t_1 level lies on the ligand (oxygen) atoms and there are ligand to metal charge transfer excitations to $t_1^5e^2t_2^0$ and $t_1^5e^1t_2^1$ giving rise to a series of $^2T_1$ and $^2T_2$ states.

Holt and co-workers have studied the chromate(V) ion doped into hosts such as phosphates and vanadates(V) [127,128]. Although the chromate(V) ion is tetrahedral, dichroic spectra were observed because of lower symmetry distortion in the crystal. Thus low energy and broad structured absorption near 9500 - 14,000 cm^{-1} is assigned to the $^2T_2---^2E$ transition (in T_d) split by lower symmetry components and by vibronic coupling. Bands near 17,000 and 26,000 cm^{-1} are then assigned as oxygen to chromium(V) charge transfer. However DiSipio and co-workers [132] reassign the 17,000 cm^{-1} band to a split component of the d-d transition, (see Chapter 5).

MANGANESE(VI), TECHNETIUM(VI) AND RHENIUM(VI) d^1

Manganese(VI) also forms tetrahedral oxy and chlorospecies. The polarised spectra of manganate(VI) doped into various alkali metal sulphates have been recorded at 4.2 K [128,131,132]. The lowest energy band is assigned to the $^2T_2---^2E$ transition also split by low symmetry components, and exhibiting several vibrational

progressions. According to these workers the splitting of the $^2T_2$ d-d excited state is much smaller for the manganate(VI) ion than for the chromate(V) ion. However Milstein and co-workers [128] do assign a band near 15 - 19,000 cm^{-1} as a component of this d-d transition. This seems unlikely. Doped into $BaSO_4$, the manganate(VI) ion has dichroic properties consistent with an effective symmetry of C_{2v} [167]. In this symmetry all three components of $^2T_2$ are electronically allowed, but only two components of $^2T_1$ are allowed. Kosky and co-workers [167] noted that the d-d absorption near 10,000 cm^{-1} consisted of three distinct origins, one in each polarisation consistent with the $^2T_2$ parentage. Near 16,000 cm^{-1} several regions of absorption could be detected having band origins and polarisations consistent with several closely spaced $^2T_1$ and $^2T_2$ charge transfer states. Extensive vibronic structure was observed. Comparison of the baricentres of the d-d bands yields 10Dq values of 12,405 and 11,177 cm^{-1} for chromate(V) and manganate(VI) respectively [132]. The very highly structured charge transfer transitions [132] are discussed in Chapter 5.

The manganese(VI) species MnO_2Cl_2 shows weak absorption at 16,200 (100) cm^{-1} which is probably $^2T_2$---^2E (in T_d). More intense absorption at 21,500(1100) cm^{-1} and 32,000(1200) cm^{-1} is oxygen to manganese(VI) charge transfer [129] and parallels that reported above for chromate(V) (see Chapter 5). Charge transfer spectra, and an X-α calculation were also reported for this species, by Jasinski and Holt [180].

The free ion spin orbit coupling coefficients for Mn(VI), Tc(VI) and Re(VI) are 540, 1700 and 4200 cm^{-1} respectively; thus the technetate(VI) and rhenate(VI) oxyanions are expected to show significant spin-orbit coupling effects. Spin orbit coupling will split the T_2 state into two levels Γ_8 at -0.5λ and Γ_7 at $+\lambda$. The E state, on the other hand, is not split to first order by spin-orbit coupling in T_d. Diffuse reflectance spectra of $(Me_3N)_2MO_4$ (M = Tc,Re) (T_d site symmetry) shows such split components of the $^2T_2$---^2E d-d transition with additional vibronic structure on the Γ_8 level. The 10Dq values are 11,000, 17,000 and 19,000 for M = Mn(VI), Tc(VI) and Re(VI) respectively [181].

The octahedral d^1 ReF_6 species has a t_{2g}^1 configuration which is split by spin orbit coupling into two states labelled U'_g (four fold degenerate) and E''_g (two fold degenerate). Transitions between these levels are seen at about 5000 cm^{-1} [182,358]. Since 10Dq in this case is about 30,000 cm^{-1}, these 'intraconfigurational crystal field transitions' are the only d-d transitions which can be observed. A magnetic dipole origin is accompanied by electric dipole transitions vibronically activated by single t_{1u} and t_{2u} quanta. Progressions of two or three quanta of an upper state t_{2g} mode are superimposed on both the magnetic and vibronic origins, providing evidence for a Jahn-Teller distortion.

6.2 (ii) d^2 Ti(II), V(III), Cr(IV), Mo(IV), W(IV), Mn(V), Tc(V), Re(V), Fe(VI), Ru(VI), Os(VI)

For most of the first row transition metal d^2 ions, the ground term is ^3F in all compounds, except in some eight coordinate systems where spin pairing gives rise to a spin singlet ground term. Eight coordinate and diamagnetic complexes are common with the heavier d^2 transition ions.

In an octahedral environment the ground term is $^3T_{1g}$. Three transitions are expected, namely:

$$\begin{array}{ll} \text{I} & {}^3T_{2g}\text{------}{}^3T_{1g} \\ \text{II} & {}^3A_{2g}\text{------}{}^3T_{1g} \\ \text{III} & {}^3T_{1g}(P)\text{---}{}^3T_{1g} \end{array} \qquad (6.6)$$

Transition II is a 'two electron' transition. Since the two $^3T_{1g}$ states are fairly far apart energetically, mixing between them is small, and the 'two electron' transition is expected to be weak (see Chapter 4 for details). Note that depending upon the relative values of DQ and B, transition II may be higher in energy than transition III (see Fig.2.2).

The corresponding transitions for tetrahedral d^2 ions are:

$$\begin{array}{ll} \text{I} & {}^3T_2\text{------}{}^3A_2 \\ \text{II} & {}^3T_1\text{------}{}^3A_2 \\ \text{III} & {}^3T_1(P)\text{---}{}^3A_2 \end{array} \qquad (6.7)$$

Since the two $^3T_1$ states are now close in energy, the 'two electron' transition, II, will be relatively stronger in intensity.

Low lying spin singlet states are anticipated as shown in the relevant Tanabe-Sugano diagram (Fig.2.9).

Many of the heavier ions form oxy or nitrido species in which pi bonding either through oxygen or nitrogen causes sufficient splitting of the t_{2g} (in O_h) set (xz,yz >> xy) that diamagnetism occurs through the configuration $(xy)^2$.

TITANIUM(II) d^2

Few compounds of titanium(II) have been characterised. The spectrum of black titanium(II) dichloride dissolved in molten aluminium chloride has been reported [137]. The spectrum, presumably derived from the $[TiCl_6]^{4-}$ ion has peaks at 7600(9.1) and 14,500(29.4) cm^{-1}. The lower energy band is presumably the $^3T_{2g}$---$^3T_{1g}$ transition, while the higher energy band is probably $^3T_{1g}(P)$---$^3T_{1g}$. The third transition, $^3A_{2g}$---$^3T_{1g}$ is the expected weak 'two electron transition'. Assuming this assignment Dq = 845, DQ = 23235 and B = 517 cm^{-1}, and the third and weak transition is predicted to lie near 16,000 cm^{-1}.

More recent studies confirm the presence of an octahedral titanium(II) complex [46,142] but it likely involves aluminium as well as chloride [46]. Titanium(II) doped into NaCl yields a green

octahedral species whose spectrum [143,144] shows well defined vibrational structure. The Laporte forbidden d-d spectra are assumed to arise through vibronic coupling to modes of the impurity centre. In this medium Dq = 850, DQ = 23370 and B = 615 cm^{-1}.

Tetrahedral Ti(II) is reported to exist in CdTe [259], CdS [260] and CdSe [382] lattices. Bands are observed at 5000 ($^3T_1(F)$) and 9200 ($^3T_1(P)$) in CdTe, at 3300 ($^3T_2$), 5700 ($^3T_1(F)$) and 9840 cm^{-1} ($^3T_1(P)$) in CdS, and 3100 ($^3T_2$), 4850,5300 ($^3T_1(F)$, $^1E(D)$) and 9500 cm^{-1} ($^3T_1(P)$) in CdSe. The lower energy band in the CdTe lattice exhibits structure due to an overlapping triplet-singlet transition and perhaps spin orbit splitting.

VANADIUM(III) d^2

Six coordinate complexes: The spectra of a considerable number of vanadium(III) complexes, usually green in colour, have been reported (Fig.6.8, Table 6.6). Since DQ is larger for a trivalent ion than for a divalent ion, transition II is expected to lie at much higher energies than that calculated for titanium(II), and will generally occur above transition III. It is often obscured by charge transfer or ligand absorption, though some authors have claimed to identify it (Table 6.6).

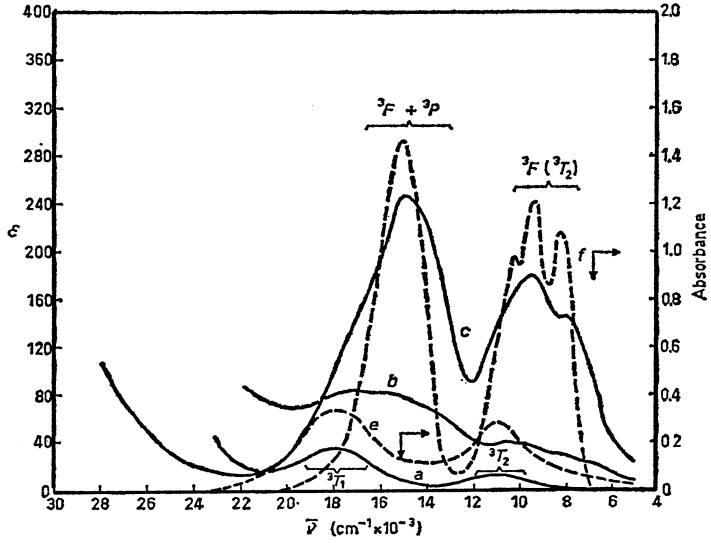


Fig.6.8 The spectrum of the V(III) species in various media. a) Octahedral $[VCl_6]^{3-}$ in LiCl/KCl eutectic at 400°C, b) octahedral-tetrahedral $[VCl_6]^{3-}$ - $[VCl_4]^-$ equilibrium in LiCl/KCl eutectic at 1000°C, c) tetrahedral $[VCl_4]^-$ in $KAlCl_4$ at 700°C, e) octahedral $[VCl_6]^{3-}$ in Cs_3ScCl_6 at room temperature and f) tetrahedral $[VCl_4]^-$ in $CsAlCl_4$ at room temperature. The right-hand absorbance scale should be used for (e) and (f).

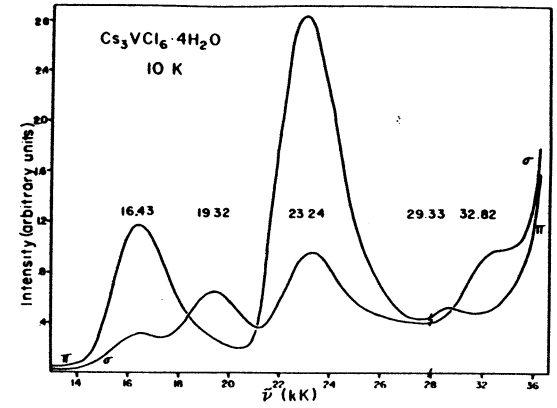


Fig.6.9 The single crystal polarised spectra of $Cs_3VCl_6.4H_2O$ at 10 K. The species is the tetragonal $[VCl_4(H_2O)_2)]^+$ ion [199].

Low symmetry vanadium(III) complexes have not been extensively studied. Tetragonal systems, $[VL_4X_2]^+$ show some resolution of the first 'octahedral' band but few detailed high resolution studies have been carried out. McCarthy and co-workers [199] have studied the $[V(H_2O)_4Cl_2]^+$ ion in $Cs_3VCl_6.4H_2O$ by single crystal polarised spectroscopy at cryoscopic temperatures. Although the splitting is not extensive, the spectra are strongly dichroic as shown (Fig.6.9).

The odd vibrations of the VO_4Cl_2 framework span $a_{2u} + b_{2u} + e_u$. The $^3T_{1g}$ (in O_h) ground term of d^2 will split into $^3E_g + ^3A_{2g}$, but it is not obvious which will be the new ground term. Using simple crystal field theory, since water has a stronger field than chloride, $^3A_{2g} < ^3E_g$ would be anticipated. However the splitting depends primarily on the pi bonding characteristics of these ligands and the $^3A_{2g}$ ground term should not be taken for granted.

The vibronic selection rules may be summarised as follows, the enabling vibrations for each possible transition are inserted:

$^3A_{2g}$ ground term to:	(xy)	(z)	^3E_g ground term to:	(xy)	(z)
B_{2g}	e_u	b_{2u}	B_{2g}	$a_{2u} + b_{2u}$	e_u
E_g	$a_{2u} + b_{2u}$	e_u	E_g	e_u	$a_{2u} + b_{2u}$
A_{2g}	e_u	a_{2u}	A_{2g}	$a_{2u} + b_{2u}$	e_u
B_{1g}	e_u	---	B_{1g}	$a_{2u} + b_{2u}$	e_u

(6.8)

With the $^3A_{2g}$ ground term, all transitions are permitted in both

Table 6.6

CRYSTAL FIELD SPECTRA OF VANADIUM(III) COMPOUNDS

i) Six Coordinate Vanadium(III) (ground term $^3T_{1g}$)

Compound	$^3T_{2g}$	$^3T_{1g}$(P)	10Dq	DQ	B	Ref
$V(H_2O)_6^{3+}$	17,800(3.5)	25,700(6.6)	16,900	46,470	614	67,68
$VBr_3.6H_2O$	14,800	22,100	15,960	43,880	564	203
$VCl_3.6H_2O$	15,100	22,900	16,310	44,845	601	203
$V(urea)_6^{3+}$	16,200	24,200	17,400	47,840	614	67,70
$V(CN)_6^{3-}$	22,200(27)[g]	28,600(50)	23,850	65,575	550	60
$V(NCS)_6^{3-}$	16,700(3.3)	24,000(7200)[a]				67
$V(NCS)_6^{3-}$	16,400(43)	24,800(2200)				196
VF_6^{3-}	14,800	23,000	16,100	44,270	627	69,353
VCl_6^{3-} b	11,000	18,020	12,000	32,995	536	11
$VCl_3.3CH_3CN$	14,400	(20,000)				25
V^{3+}/Al_2O_3	17,420	24,930				63
	17,510	25,310				
$V(H_2O)_6^{3+}$	18,520(3.6)	26,180(7.8)	19,950	54,855	644	71
	18,690(4.4)	27,400(5.6)				
$V(CH_3OH)_4Cl_2$	14,800	22,300	16,050		553	
	(10,450)					146
$V(S_2PPh_2)_3$	12,350(40)	17,400(180)	13,200	36,295	395	107
$V(S_2AsPh_2)_3$	12,200	(17,250)	13,000	35,745	397	168
$V(CH_3SO_2)_3$	14,925	22,990	16,165		620	486
$V(H_2O)_4Cl_2^{+}$	16,430	23,240(^3E_g)	17,100	47,015		
		25,000($^3A_{2g}$)				
		32,820($^3B_{1g}$)[f]				199

Footnotes: a) Probably charge transfer in origin. b) in LiCl/KCl eutectic at 673K. c) 18,870(7.5) 14,700(5.7) d) 8800,9700 ($^1E,^1T_2$), 21,020 ($^1A_1$). e) Diluted in a crystal of a colourless aluminium compound. f) Single crystal polarised cryogenic study. Assignments in D_{4h} symmetry; fitted by DQ = 46672, DS = -1675, DT = -292, B = 580, C = 2870. g) May not be six coordinate, seven ?

ii) Five Coordinate Vanadium(III) (ground term $^3A_2'$)

Compound	$^3A_1"$,$^3A_2"$	$^3E"$	$^1E'$	$^3E'$,$^3A_2'$(P)	Ref
$VCl_3(NMe_3)_2$	4900(76)	7100(16)	11,200	19,900	37,147
$VBr_3(NMe_3)_2$	<4500	5900	(10,800)	18,700	37
$VCl_3(2\text{-Pic})_2$	ca.4500	7300	8600	(19,000)	37
$VBr_3(2\text{-Pic})_2$	<4500	6300	8100	(17,900)	37

cont

Table 6.6 iii) Four Coordinate Pseudo-Tetrahedral Vanadium(III) (ground term $^3A_2$)

Complex	$^3T_2$	$^3T_1$(F)	$^3T_1$(P)	B	Ref.
VCl_4^-	-	8300	15,030(260)		
		9250			
		10,000			225
VCl_4^-	5520	9080	15,100	505	226
$V(Py)_4^{3+}$	9500	16,200	25,000	785	228
		(18,400)			
$V(bipy)_2^{3+}$	-	14,700	20,500		228
		(17,500)			

polarisations except for the transition to $^3B_{1g}$ which is only permitted in (xy) (sigma).
With ^3E_g as ground term, all transitions are vibronically activated. Experimentally the transition at 32,820 cm^{-1} is only seen in the sigma polarisation. Calculation of the data shows that this band is reasonably assigned to the B_{1g} level, thereby confirming the $^3A_{2g}$ ground term.

> Care should be taken in using the simple vibronic arguments presented here. Some vibrations may not 'enable' an electronic transition, even when group theory would permit it; for example the charge transfer state which is vibrationally coupled to the d state, may lie at too high an energy (see Chapter 4).

The assignments are summarised in Table 6.6. The spectra have several interesting aspects. While most of the bands decrease in intensity upon cooling as expected for the vibronic mechanism, that at 19,320 cm^{-1} increases in intensity with cooling. Moreover within the tetragonal model, there should not be an additional electronic transition in this position. Deuteration causes a shift in frequency of this band to 18,800 cm^{-1}, thus providing convincing evidence that it is the first (and only) member of a vibrational progression of the a_{1g} OH stretch of the coordinated water ligands based upon a component of the $^3T_{2g}$ (in O_h) excited state. The peak at 29,330 cm^{-1} is shown to increase when iron is added to the crystal, and in fact originates from an iron impurity. It is likely a Cl-Fe(III) charge transfer band of a substituted $[Fe(H_2O)_4Cl_2]^+$ species. It is exactly polarised along the Cl-Fe-Cl vector.

The polarised crystal spectra of tris(diethyldithiophosphato)-vanadium(III) doped into the colourless indium analog have been reported by Palmer [157]. The spectra are dichroic with polarisation consistent, at least at room temperature, with D_3 symmetry, even though the site symmetry is C_1. At cryogenic temperatures the polarisation essentially disappears and C_1 appears to become a more appropriate point group.

Detailed studies of the spectrum of vanadium(III) in corundum

(Al_2O_3) have been reported [21,138-140,237]. The vanadium(III) ions are present as impurity ions in the alumina lattice and replace aluminium(III) ions. The site symmetry is close to C_{3v}, but the symmetry experienced by the impurity ion is apparently C_3. The results of McClure's studies [139,140] are shown in Fig.6.10. The ground state degeneracy of the $^3T_{1g}$ level (in O_h) is lifted in C_{3v} to give $^3A_2$ + ^3E with the former lower in energy in this case. The first excited state ($^3T_{2g}$ in O_h) is scarcely split by the lower symmetry but shows distinct vibrational structure (at 77 K) occurring as a series of bands with mean separation ca 180 cm^{-1}.

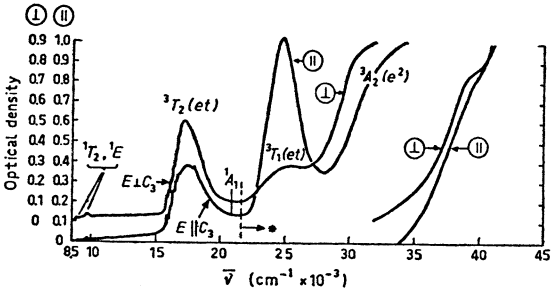


Fig.6.10 The polarised spectra of V(III) in corundum (Al_2O_3) at -196°C; the electric vector is parallel or perpendicular to the C_3 axis.

The band appears in both perpendicular and parallel polarisation yet the transition to $^3A_1$ component of $^3T_{2g}$ is forbidden in C_{3v} symmetry. Its appearance is taken as evidence of the effective symmetry being only C_3 where this transition corresponds with the allowed ^3A---^3A transition. The degeneracy of the $^3T_{1g}(P)$ (in O_h) level is lifted by 380 cm^{-1} with the $^3A_2$ component lying at lower energy. A weak shoulder observed at 31,240 cm^{-1} could be a transition to the $^3A_2$ level ($^3A_{2g}$ in O_h). Variable temperature data were also presented for this system [139]. Apparently above 700 K the ^3E component of the $^3T_{1g}$ (in O_h) ground term becomes thermally populated. Using these data the splitting of the ground state was calculated to be 960 cm^{-1} agreeing well with the figure of 1200 cm^{-1} determined [151] using spin-orbit coupling arguments. A detailed analysis of the spectrum [152] yielded 10 Dq = 18,000, (DQ = 49,490) B = 610, C = 2500, and ζ = 933 cm^{-1}

Dingle, McCarthy and Ballhausen [227] also reported the polarised spectrum of the $V(urea)_6^{3+}$ ion which has D_3 symmetry. The ground state $^3T_{1g}$ (in O_h) splits with the $^3A_2$ (in D_3) as the new ground state. Transitions to the $^3A_1$, and to the ^3E components of the first two orbital triplets are seen in z and xy polarisation respectively; however the transition to $^3A_2$ (of $^3T_1$) is forbidden in both polarisations and is seen only weakly (Table 6.6, Fig.6.11).

Distorted octahedral spectra were also observed when

vanadium(III) ions were doped into beryl [154], scandium and indium chlorides [155] and various chalcogenide semi-conductors [156].

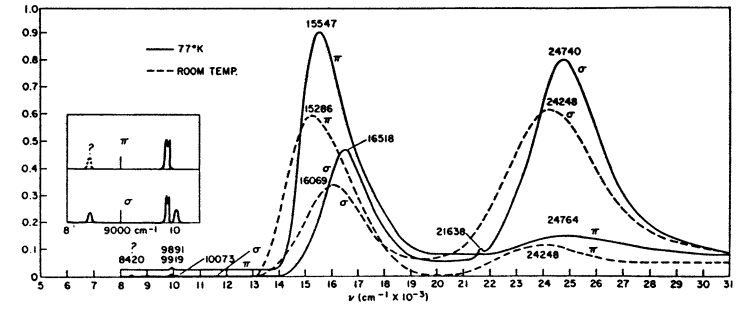
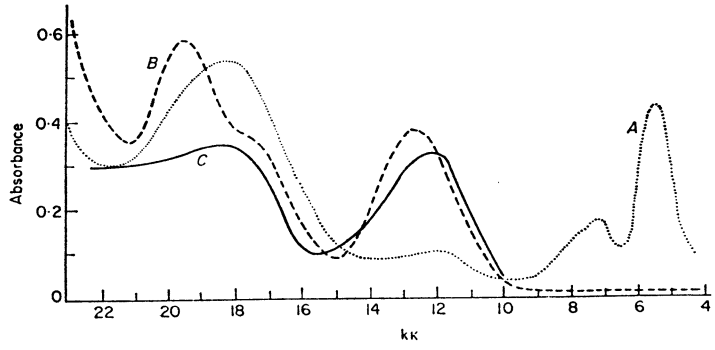


Fig.6.11 The polarised crystal spectrum of $[V(urea)_6](ClO_4)_3$ at 295 and 77 K [227].

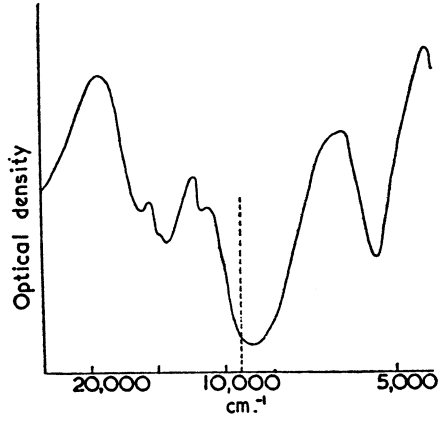
Five Coordinate Complexes: Trigonal bipyramidal complexes of vanadium(III) have been reported (Fig.6.12, Table 6.6). The analysis of such complexes is usually based on Wood [159] or Ciampolini [160]. Three states are expected to lie in the near infrared below 10,000 cm^{-1} and such low energy absorption is observed, clearly distinguishing these complexes from their six coordinate analogs. Assignments in the visible region seem uncertain [37,147].

Four coordinate Complexes: No square planar complexes of vanadium(III) have been reported, and few tetrahedral complexes are known. The spectrum of vanadium(III) in $CsAlCl_4$, at room temperature, is reported to consist of three bands at 8000, 9500 and 10,500 cm^{-1} [153] due to transitions to the components of $^3T_2$ of a distorted tetrahedral $[VCl_4]^-$ ion (Fig.6.8). A further band at 15,000 cm^{-1} probably contains transitions to the components of $^3T_1$. A similar spectrum is observed in $KAlCl_4$ at 973 K. In LiCl-KCl eutectic at 673 K the principle species is the octahedral $[VCl_6]^{3-}$ ion (which is also present in Cs_2ScCl_6/V^{3+} at room temperature), but at 1273 K the spectrum indicates the presence of an octahedral - tetrahedral equilibrium. The tetrahedral $[VX_4]^-$ ions ($X = Br^-$, Cl^-) have been reported as alkylammonium and alkylarsonium salts and have electronic spectra [150] similar to (though not so 'clean'as) the doped spectra. These latter spectra show ultraviolet absorption near 20,000 cm^{-1} not evident in the doped spectra. It is unlikely that charge transfer spectra would be observed at so low an energy (see Chapter 5) and the data may be

erroneous. Indeed other workers do not report such bands [225,226]. Konig [228] has reported an interesting series of tetrahedral amine complexes with a VN_4 chromophore (Table 6.6).



(a)



(b)

Fig.6.12 a) Electronic spectra of $VCl_3.2C_4H_8S$ in benzene solution (1.06×10^{-2} g.mol/l) (A), C_4H_8S solution (2.14×10^{-2} g.mol/l) (B) and diffuse reflectance (C). b) Diffuse reflectance spectrum of $VCl_3.2NMe_3$.

Other complexes of Vanadium(III): Vanadium(III) forms a seven coordinate pentagonal bipyramidal complex, the $[V(CN)_7]^{4-}$ ion. The splitting diagram will be essentially the same as that for the D_{3h} case (and for cyclopentadienyl derivatives) but, especially with cyanide, the overall splitting will be much greater. The spectra have been reported by several workers [158,161-164], and analysed

in terms of a strong field D_{5h} model [158].

Three coordinate V(III) occurs in trigonal $V[N(SiMe_3)_2]_3$ showing d-d bands at 12,000(60), 15,900(150) and at 19,200(268) cm^{-1} assigned to transitions anticipated for a d^2 ion in a D_{3h} environment [136].

CHROMIUM(IV), MOLYBDENUM(IV) AND TUNGSTEN(IV) d^2

Few chromium(IV) complexes have been studied. The tetrahedral alkoxides, $Cr(OR)_4$ were initially reported [169] to have a rather broad band with maximum in the range 14,800 - 15,400 (550-600). More recent studies reported three regions of absorption centered at 9100(10), 15,200(560) and 25,000(500) associated with the three spin allowed transitions of tetrahedral $Cr(IV)(OBu^t)_4$ [261]. The $[CrF_6]^{2-}$ ion is reported to have d-d transitions at 11,400 (^1E_g, $^1T_{2g}$), 20,200 ($^3T_{2g}$) and 27,500 ($^3T_{1g}$) [253,520].

A few paramagnetic six coordinate Mo(IV) species have been investigated. Two relatively low energy bands were attributed to d-d transitions, and are followed by more intense charge transfer transitions. The original assignments of the low energy bands to d^2 transitions I and III may be erroneous since they fall anomalously close together. While the secular determinant for d^2 in O_h can be solved, the value for B obtained will be very small. For example the $[MoCl_6]^{2-}$ ion (Table 6.7) is solved with B = 290 cm^{-1}.

Four coordinate complexes: The $[Mo(NMe_2)]_4$ D_{2d} species, reported by Chisholm and co-workers [1333], exhibited electronic and PES data interpreted in terms of a $(x^2-y^2)^2$ diamagnetic ground state, with a d orbital sequence $x^2-y^2 << z^2 < xz,yz < xy$. A shoulder at 10,500(30) cm^{-1} is assigned as $x^2-y^2 \text{---} z^2$ ($^1B_1 \text{---} ^1A_1$), which is orbitally forbidden. A strong shoulder at 14,300(300) cm^{-1} is probably the orbitally allowed $x^2-y^2 \text{---} xz,yz$ ($^1E \text{---} ^1A_1$) transition, with CT absorption to higher energies [1333].

Other complexes: There has been much discussion concerning the electronic spectra of the diamagnetic octacyanomolybdate(IV) and octacyanotungstate(IV) species, following that referred to above for the corresponding Mo(V) and W(V) species. These complexes also show four relatively low intensity transitions, between 20,000 and 40,000 cm^{-1} attributed to d-d transitions [187]. If the molecule possesses D_{4d} symmetry, only two spin allowed transitions (to two doubly degenerate levels) should be observed, while three spin allowed transitions are anticipated for a D_{2d} structure. These authors [187] propose a distortion to D_2 symmetry, thereby removing all degeneracy and requiring four spin allowed transitions. Single crystal spectra at 4.2 K are reported and comparison of ambient and low temperature data leads to the suggestion that the deviation from dodecahedral symmetry is smaller at lower temperature. The aqueous solution spectra [117] of these $[M(CN)_8]^{4-}$ species are quite different from the crystal spectra [187], with an apparently quite different sequence of energy levels.

Table 6.7

ABSORPTION SPECTRA OF MOLYBDENUM(IV) DERIVATIVES

Complex	$^3T_{2g}$---$^3T_{1g}$	$^3T_{1g}(P)$---$^3T_{1g}$	C.T.	Ref
$MoCl_6^{2-}$	22,200	25,800	28,400	
			33,900	
			36,760	
			41,300	99
$MoCl_4.2PPh_3$	15,600dd?	25,000		
	20,000	27,800 CT?		178
$MoCl_4.2Py$	(21,200)	24,700		178
$MoCl_4.2Ph_3PO$	(20,800)	27,000		178
$MoCl_4.(bipy)$	18,300(690)	(25,200)(800)	27,200(1200)	
			33,100(12,000)[a]	179
			41,700(14,000)[a]	179

a) π---π^*

Eight coordinate dithiocarboxylate complexes of Mo(IV), $Mo(dtb)_4$ show intense absorption accross the visible region, from 10,000 cm^{-1} probably of mixed d-d and charge transfer origin [309]. Eight coordinate tungsten (IV) oxine derivatives show d-d absorption at 9000 - 11,000 cm^{-1} lying below CT absorption beginning near 14,000 cm^{-1} [313]. Lippard and Russ [192] report data for various hydrolysis products of the octacyanides containing both cyanide and oxygen ligands.

Both Mo(IV) and W(IV) form eight coordinate complexes with bidentate ligands, of the type $M(LL')_4$, where, for example, LL' is a dithiocarbamate, dithiocarbonate, 8-hydroxyquinoline, pyridine or pyrazine carboxylate etc. Often these compounds are readily oxidisable to Mo(V) and W(V) analogs and in keeping with this observation, many of the $M(LL')_4$ systems show low lying MLCT transitions [186](chapter 5). Although some of the complexes are known to have a dodecahedral structure (e.g. [183]) there is the same problem as discussed above for the octacyanides that solution structures may vary from crystal structures, which, in any event, may not be available. The MoS_8 and WS_8 chromophores show d-d absorption from 10,000 - 17,000 cm^{-1}, followed by charge transfer (Mo to S) between 17,000 and about 30,000 cm^{-1} [184,185].

MANGANESE(V), TECHNETIUM(V) AND RHENIUM(V) d^2

In common with other high oxidation state first row transition metal ions, Mn(V) is commonly known in a tetrahedral, or distorted tetrahedral environment.

The essentially tetrahedral manganate(V) ion is observed in

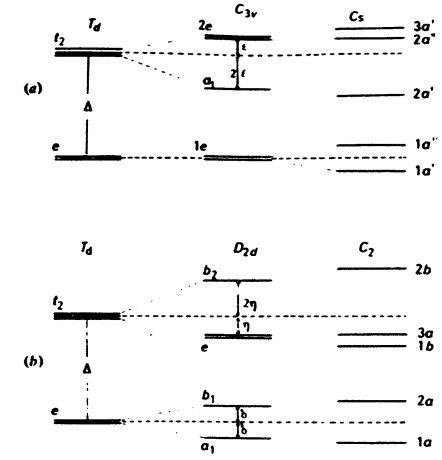


Fig.6.13(a) The splitting of t_2 and e molecular orbitals in (a) C_{3v} and C_s sites, and (b) D_{2d} and C_2 sites [172,174].

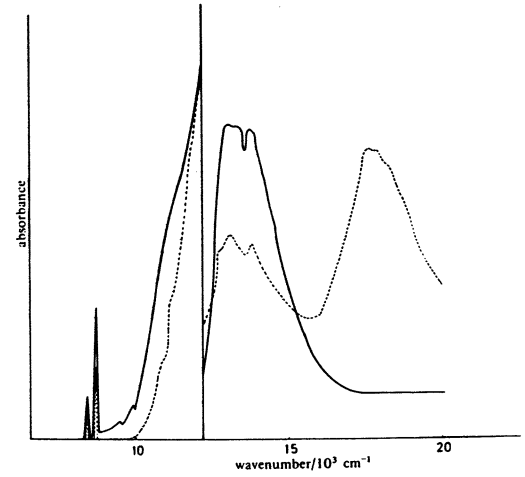


Fig.6.13 (b) The polarised absorption spectrum of a high concentration of $[MnO_4]^{3-}$ in Ca_2PO_4Cl at 4.2 K. Solid line perpendicular to a axis, hatched line parallel to a axis [172,174].

12M NaOH [170] and in single crystals of Li_3PO_4 doped with Mn(V) [128,171]. In T_d symmetry, only transitions of the type $^3T_1$---$^3A_2$ are electric dipole allowed. Early assignment [128] of the spectra of these species included a low lying charge transfer transition in the visible region. In the light of the low symmetry data reported immediately below, this now seems unlikely. Assignment of the tetrahedral species is deferred until after the low symmetry data which follows.

Day, Oleari and co-workers have undertaken some elegant single crystal studies of the manganate(V) ion doped into $Sr_5(PO_4)_3Cl$ and into Ca_2PO_4Cl crystals [172-174]. The former site is primarily trigonal C_{3v}, and the latter primarily distorted tetrahedral D_{2d}. The data show extensive polarisation and the study represents an interesting example of the value of polarisation in determining band assignments.

In both C_{3v} and D_{2d} symmetry, the E states originating from T_1 and T_2 (in T_d) retain their degeneracy. However in both these crystals, there is observed some splitting of the E states consistent with further slight distortion to C_s and C_2 respectively. The energy levels under these circumstances are shown in Fig.6.13a. In general, both crystals show sharp absorption in the near infrared (8000 - 9000 cm^{-1}), highly dichroic and structured absorption between 10,000 - 20,000 cm^{-1}, intense absorption at 29,000 - 36,000 cm^{-1} (which has to be charge transfer in origin), and, in the case of $Ca_2PO_4Cl/Mn(V)$, five relatively sharp peaks in the region 21,000 - 25,000 cm^{-1} (Fig.6.13b).

In the earlier investigation [172], the crystal field parameters derived for the two crystals differed markedly. In the more recent study [174], it was argued that distortion of the $[MnO_4]^{3-}$ ion should not affect the overall tetrahedral crystal field splitting energy, 10Dq, and the Racah parameter, B, to such an extent as had been observed. The lowest energy transition in a tetrahedral species is T_2---A_2 which is electric dipole forbidden. In the lower symmetry fields, the intensity of transitions to components of T_2 will depend upon the extent of mixing of these components with states of the same symmetry but deriving from an electric dipole allowed transition (specifically mixing with components of the T_1 state) (cf. symmetry ascent selection rules, Chapter 4).

In the earlier study a band near 11,500 cm^{-1}, observed by some authors [128,176] had not been detected by Day and co-workers necessitating the assignment of the near infrared absorption to components of T_2 and hence yielding a small overall crystal field splitting parameter. Further study with more heavily doped crystals allowed detection of this band as a weak shoulder (Fig.6.13b)[174].

The undistorted tetrahedral $[MnO_4]^{3-}$ spectrum shows a weak absorption near 10,000 cm^{-1} reasonably associated with $^3T_2$---$^3A_2$, and structured absorption centred near 16,000 cm^{-1} now reasonably associated with spin-orbit and vibrational components of $^3T_1(F)$---$^3A_2$. This assignment yields 10Dq = 10,000 and B = ca 800 cm^{-1}.

Jasinski and Holt have measured the gas phase spectrum of $MnOCl_3$ [308], in the region 12,000 to 52,000 cm^{-1}. Very weak absorption at about 15,000 cm^{-1} is assigned to components of the first ligand field band in C_{3v} symmetry. A series of charge transfer transitions follow into the ultraviolet. Two progressions in $\nu_1(a_1)$ (682 cm^{-1}, ca 34 redn. from ground state) and $\nu_3(a_1)$ (154 cm^{-1}, ca 7 redn. from ground state) shows both chlorine to manganese and oxide to manganese CT character in the 25,000 cm^{-1} region. The assignments were supported by Xα calculations.

$Cs[ReF_6]$ exhibits a very weak absorption between 4000 and 6000 cm^{-1} assigned as a transition within the spin-orbit components of the ground term $^3T_{1g}$ of octahedral d^2. In addition a weak shoulder at 28,000 cm^{-1} is assigned as $^3T_{2g}$---$^3T_{1g}$ and a strong band at 35,100 cm^{-1} to $^3T_{1g}(P)$---$^3T_{1g}$ possibly mixed with charge transfer [188]. However the assignment to the 28,000 and 35,100 cm^{-1} bands is unlikely to be correct since these bands are too close together [177].

The $[Tc(NCS)]_6^-$ ion shows only a strong charge transfer band in the visible region, at 19,880 cm^{-1} [190]. The $[TcOX_4]^-$ ions (X = Cl,Br) are apparently tetragonal and show at least four d-d transitions between 11,000 and 28,000 cm^{-1} [191], followed by charge transfer. The $[TcOCl_5]^{2-}$ and $[TcO_2(en)_2]^+$ ions have been analysed in terms of C_{4v} or D_{4h} symmetry to yield one-electron separations similar to corresponding rhenium compounds [452].

IRON(VI), RUTHENIUM(VI) AND OSMIUM(VI) d^2

The ferrate(VI) ion shows extensive charge transfer (Chapter 5) with absorption near 12,720 cm^{-1} ascribed to a d-d transition [176,231,232]. Although some authors ascribe this to the $^3T_1$---$^3A_2$ transition, data reported in a more recent publication [274] lead to an alternative assignment. Thus in 9M KOH, the ferrate(VI) ion exhibits bands at 12,600(400), (17,200), 19,800(1100) and 29,400 (1100) cm^{-1}. Single crystal data in a host crystal of K_2CrO_4 at 4.2 K show several vibronic origins with polarisation characteristics consistent with a C_s distortion. Progressions in a_1 and t_2 M-O modes are observed. The 12,600 cm^{-1} band is assigned (in T_d) $^3T_2$---$^3A_2$. These authors ascribe the double peak near 18,000 cm^{-1} to both the $^3T_1$ transitions. However these two bands cannot lie this close together. We prefer to assign the split band near 18000 cm^{-1} to low symmetry components of $^3T_1(F)$ and the shoulder near 29,400 to $^3T_1(P)$. These data are fitted by Dq = 12,600, DQ = 34,645 and B = ca 630 cm^{-1}.

Both ruthenium(VI) and osmium(VI) form nitrido complexes of the type $[MNX_4]^-$, where X = Cl,Br, whose electronic spectra have been studied in depth by Mabbs [204] and by Gray and co-workers [205,206]. These species form extremely short M-N bonds (ca 160 pm) and have C_{4v} symmetry, the metal-nitrogen bond forming the axial vector. Simple ligand field theory suggests an energy sequence xy << xz,yz < x^2-y^2 << z^2. The complexes are diamagnetic with a configuration $(xy)^2$, $^1A_1$. Polarised low temperature data are shown

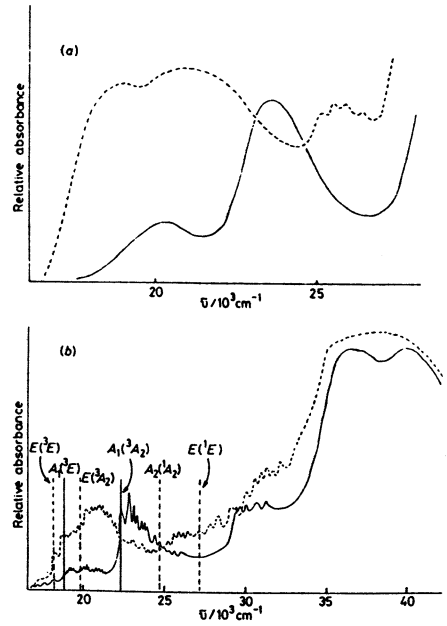


Fig.6.14 The polarised single crystal spectra of $(AsPh_4)[OsNCl_4]$ at (a) 298 K, and (b) 5 K. Solid line z, and hatched line xy polarisation [204].

in Fig. 6.14.

In the absence of spin-orbit coupling, electric dipole selection rules predict only one allowed transition, $^1E---^1A_1$ (xy---xz,yz). Clearly the spectra are more complex. Certainly with these metals, spin-orbit coupling is important. Referring to Chapter 3, S = 1 transforms as T_{1g} in O_h, and hence $A_2 + E$ in C_{4v}, and S = 0 transforms as A_1. Thus the spin-orbit representations of the various states lead to the following transitions and polarisations [204]: (Spinors in the double group carry primes)

	Transition	C_{4v}' Polarisation z	xy
xy---z^2			
$^3B_2---^1A_1$	$B_1'---A_1'$	F	F
	E'	F	A
$^1B_2---$	B_2'	F	F
xy---x^2-y^2			
$^3A_2---$	A_1'	A	F
	A_2'	F	F
$^1A_2$			
xy---xz,yz			
$^3E---$	A_1'	A	F
	A_2'	F	F
	B_1'	F	F
	B_2'	F	F
	E'	F	A
$^1E---$	E'	F	A

(6.9)

There should therefore be four distinct electronic dipole permitted transitions in xy polarisation and two in z polarisation. An examination of the polarised spectrum of $AsPh_4[OsNCl_4]$ shows three distinct regions of z polarised intensity, and four regions of xy polarised intensity. The extra z intensity absorption is ascribed to a vibronically allowed transition [204]. Most of the bands in the osmium complexes show rich vibronic structure. Comparison with ground state (infrared) vibrational spectra allows the recognition of excited state totally symmetric Os-N, and Os-X vibrations. Assignment of the bands is based partly on calculation of their energies based upon the one-electron sequence indicated above, and partly on the identification of these vibrational progressions. For example the low energy band starting at 17,800 cm^{-1} in xy polarisation possesses a progression in a vibration of ca 800 cm^{-1} ascribed to an Os-N bond slightly elongated in the excited state. The electronic transition therefore probably involves an excitation in which the Os-N bond order is reduced. It is therefore assigned to $(xy)^2---(xy)^1(xz,yz)^1$ $(E(^3E)---A^1(^1A_1))$ where the xz,yz orbitals are pi anti-bonding with respect to the Os-N bond. Similarly the band beginning at 19,830 cm^{-1} in xy polarisation has vibrational structure of ca 140 cm^{-1} associated with the A_1 Os-Cl deformation, observed in the ground state at 184 cm^{-1}. Thus the electronic transition involves a reduction in bond order of the Os-Cl bond. It is assigned as the transition $(xy)^2---(xy)^1(x^2-y^2)^1$ $(E(^3A_2)---A_1(^1A_1))$, where the x^2-y^2 orbital is sigma antibonding with respect to the Os-Cl bond. In this fashion, all the d-d transitions are assigned (in a manner slightly different from Gray's earlier work [205,206]). The data are parameterised in terms of the Slater-Condon and spin-orbit coupling coefficients, and the one-electron energy separations between the three sets of d energy levels [204,207]:

	xy - xz,yz	xy - x^2-y^2	xy - z^2	F_2	F_4	λ(M)
$OsNCl_4^-$	31,000	24,200	50,000	1107	41	2800
$OsNBr_4^-$	27,400	23,900	50,000	680	42.5	2700
$OsNI_4$	23,800	21,500	50,000	456	19	2100
$RuNCl_4^-$	36,800	28,600	60,000	1144	104	800
$RuNBr_4^-$	28,700	23,700	60,000	553.5	61.5	1000

(6.10)

The xy---z^2 energy is fixed to place these transitions under the lowest charge transfer band. The decreasing values in the first column reflect an overall decrease in pi bonding down the halogen sequence; similarly the decrease in the second column values (= 10Dq) shows a decrease in in-plane sigma bonding. Using the relative intensities of the Os-N progression, Gray calculates an Os-N bond lengthening of about 17 pm in the chloro anion [205] (see Chapter 4 for method).

6.2 (iii) d^3 V(II), Cr(III), Mo(III), W(III), Mn(IV), Tc(IV), Re(IV), Fe(V), Ru(V), Os(V), Rh(VI), Ir(VI)

The free ion ground term is ^4F in most cases. As with d^2, 7 and 8 coordinate transition metal ions are spin-coupled, as are some of the lower symmetry heavier metal species.

In an octahedral environment, the ground term is $^4A_{2g}$, and three spin-allowed transitions are predicted:

$$\begin{array}{lll} \text{I} & {}^4T_{2g} \text{------} {}^4A_{2g} & (= 10Dq) \\ \text{II} & {}^4T_{1g} \text{------} {}^4A_{2g} & \\ \text{III} & {}^4T_{1g}(P) \text{---} {}^4A_{2g} & \end{array} \qquad (6.11)$$

In this case the 'two electron' transition is scrambled due to the fairly close proximity of the two T_{1g} states; thus all three transitions will be seen unless obscured by charge transfer or other absorption.

In the few tetrahedral d^3 complexes, the two-electron transition (II) will be weak, viz:

$$\begin{array}{lll} \text{I} & {}^4T_2 \text{------} {}^4T_1 & \\ \text{II} & {}^4A_2 \text{------} {}^4T_1 & E(\text{II} - \text{I}) = 10Dq \\ \text{III} & {}^4T_1(P) \text{---} {}^4T_1 & \end{array} \qquad (6.12)$$

VANADIUM(II) d^3

A survey of the spectra of vanadium(II) species, often violet in colour, is shown in Table 6.8. The spectrum of octahedral $[VCl_6]^{4-}$, in $AlCl_3$ melt at 227°C is shown in Fig.6.15. All three spin-allowed transitions are clearly seen. Hauser and Gudel have studied the single crystal spectra of the one-dimensional Heisenberg systems, $CsVX_3$, X = Cl,Br and I [208]. Highly dichroic spectra were recorded due to a significant trigonal distortion of the $[VX_6]^{4-}$ chromophore. Extensive vibrational structure led to excited state V-X frequencies of 256, 160 and 118 cm^{-1} for Cl,Br and I respectively. Spin-forbidden transitions were exceptionally intense due to exchange interactions occurring in the solid (see also [239]). These same authors have also reported emission (and further absorption data) for $CsVCl_3$ and for V(II) doped into $CsMgCl_3$, in a more recent publication [1164].

Table 6.8

THE ABSORPTION SPECTRA OF VANADIUM(II) COMPLEXES

i) Six Coordinate Vanadium(II) (Ground term $^4A_{2g}$ in O_h)

Compound	$^4T_{2g}$ [a]	$^4T_{1g}$	$^4T_{1g}(P)$	$^2T_{1g}$, ^2E_g	B	Ref
$V(H_2O)_6^{2+}$	12,350(4.1)	18,500(6)	27,900(3)	(13,100)	680	200
$V(H_2O)_6^{2+}$	12,100	18,000	(28,000)		650	209
VCl_6^{4-}	7200b[b]	12,000	19,000		800	13
VCl_6	9000b	14,000	21,500		600	50
VCl_2[c]	9300	14,220	22,240	11,830		570
				12,300		234
$VCl_2/AlCl_3$	8000b	13,100(12)	20,300(20)		735	61
VBr_2	8600[d]	13,330	20,330		530	235
VI_2	7870[d]	12,270	18,920		510	235
$V(en)_3^{2+}$	15,500	21,400	(32,000)		581	202
$V(CN)_6^{4-}$	22,300(116)	27,700(147)	36,700CT(7300)		484	84
$V(NCS)_6^{4-}$	15,500	22,000	25,000 CT		670	236
$V(NH_3)_6^{2+}$	14,800	21,200	(32,000)		660	203
$V(dipy)_3^{2+}$	15,930	25,000CT?		12,270[e]		201
$V(imid)_6^{2+}$	15,300(23)	21,800(31)				258
$V(2MeIm)_6^{2+}$	13,650	20,000				1364
$V(py)_4Cl_2$	(13,300)[c]	22,000 CT	-		-	
	(16,400)[c]					
	(19,500)[c]					233
$V(py)_4I_2$	9800[f]	(21,800)CT	-		-	
	(14,500)[f]					
	(19,500)[f]					233
$V(pz)_4Cl_2$	11,600[f]	16,250[g]				
	15,800[f]	20,200[g]				1364
$V(2meIm)_4Cl_2$	9800[f]	13,000[g]				
	11,400[f]	15,200[g]				1364

ii) Four Coordinate Vanadium(II) (Ground term $^4T_1$ in T_d)

Compound	$^4T_2$	$^4A_2$	$^4T_1(P)$	Dq	B	Ref
V(II)/ZnS	4200	9200	11200	490	485	156
V(II)/CdTe		6650	9900	356	495	259

Footnotes: a) = 10 Dq. b) In LiCl/KCl eutectic at 400°C. c) Single crystal at 22 K d) spectrum at 5 K. e) Fluorescence. f) Components of $^4T_{2g}$ in tetragonal symmetry. g) Components of $^4T_{1g}$ in tetragonal symmetry. See also ref [208,210,1364] for further data.

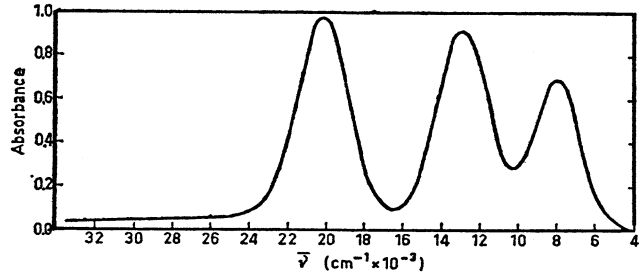


Fig.6.15 The octahedral $[VCl_6]^{4-}$ species in $AlCl_3$ at 227°C.

The data were interpreted in D_{3d} local symmetry. A very sharp spike (halfbandwidth 5.5 cm^{-1}) was observed at 8299 cm^{-1} in $CsVCl_3$ (and 8178 cm^{-1} in $CsMgCl_3$:V(II)) coincident in both emission and absorption. This is assigned to the magnetic dipole allowed $^4A_{1g}$---$^4A_{2g}$. Built on this origin was a progression in the V-Cl stretching vibration; in addition six vibronic origins were built upon this magnetic dipole origin. Though too many for a molecular system of D_{3d} symmetry this is consistent with a factor group analysis of AMX_3 systems which predicts six enabling modes in D_{6h} symmetry. Moreover a progression of the a_{1g} V-Cl mode is built on each of these origins. Using the intensities of such an a_{1g}, Hauser and Gudel calculate a V-Cl bond lengthening of 17pm [1164].

The spectra of VCl_2 doped into sodium chloride [144], and into other main group chlorides [237] have been reported. Spin forbidden transitions are clearly identified and fitted to conventional theory. Highly resolved spectra were also reported for the three halides VX_2 (X = Cl, Br, I) at 5 K [224,235]. Of especial interest were the appearance of very sharp, narrow lines (half-width < 5 cm^{-1}) ascribed to zero phonon (i.e. 0'-0") pure electric dipole (but spin forbidden) transitions allowed through some magnetic coupling mechanism in the solid state [235]. A very broad, low intensity transition was observed at an energy corresponding to twice that of $^4T_{2g}$—$^4A_{2g}$ and is indeed believed to be a two centre, and hence strongly forbidden, transition. It becomes partially permitted through a magnetic interaction between a neighboring pair of ions in the crystal whereby a single photon facilitates two one-electron transitions. The intensity of this transition increases with cooling allowing its identification in the presence of the vibronically induced transitions which decrease in intensity upon cooling [235]. A study of V(II) doped into

several fluoride lattices discusses anti-resonance effects where a sharp spin forbidden line appears inside a broad spin allowed band resulting in interference between the absorptions and the appearance of a dip in the peak (rather than a spike) [238]. Solutions of vanadium(II) halides in a range of organic solvents [223] and a series of interesting tetragonal $V(amine)_4X_2$ [233] and $V(H_2O)_4X_2$ [1230] species have been studied.

Few tetrahedral vanadium(II) species have been studied. Tetrahedral sites are involved for V(II) doped into several sulphide and selenide lattices [156,382]. Thus, for example, V(II)/CdSe exhibits bands at 3500 ($^4T_2$), 6900 ($^4A_2$) and 10,600cm^{-1} ($^4T_1$(P)) (10Dq = 3400 and B = 520 cm^{-1})[382].

Linear two coordinate vanadium(II) is known in the gas phase spectrum of VCl_2 [675] and has been extensively discussed [378,675,676]. A broad structured band is observed at 14,500 cm^{-1} with lower energy peaks at 9600 and 10,700 cm^{-1}.

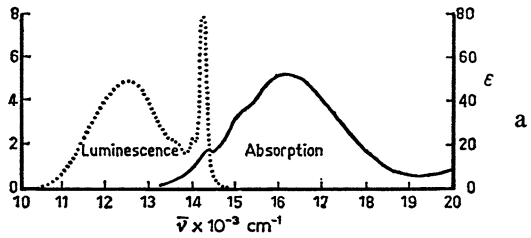
CHROMIUM(III) d^3

Although almost totally restricted to six coordinate complexes, the electronic spectra of chromium(III) complexes have played a special role in the development of inorganic electronic spectroscopy. This is ascribed primarily to the kinetic inertness of Cr(III), such that many varied complexes can be isolated, and to the ready observation of all three spin allowed transitions and a number of clearly defined spin forbidden transitions. The data are surveyed in Table 6.9.

All three spin allowed transitions (I - III) can be observed if ultra-violet transparent ligands are used. Since Cr(III) is neither a good reducing nor a good oxidising agent, charge transfer bands tend not to obscure d-d absorption. They are to be found in the ultra-violet. The $^4T_{1g}$ transition is observable but tends to be weaker than the other d-d transitions.

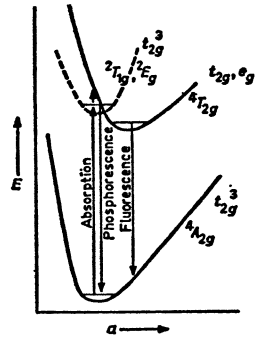
Of particular historical importance, was the study of the so-called 'Ruby' lines, very narrow spin-forbidden (quartet-doublet) transitions observed in most chromium(III) complexes but studied extensively in the spectrum of ruby, a gemstone composed of alumina doped by nature with Cr(III) ions. These lines are extremely narrow and very weak (f = 2 x 10^{-6} compared with spin allowed bands with intensities up to 10^3 times greater, i.e. molecular extinction coefficients of 0.1 rather than about 50 - 100 l mol cm^{-1}). The very narrow ruby lines arise from a 'spin flip' within the t_{2g} set and, to first order, have no dependence upon DQ (see below).

Many chromium(III) complexes are luminescent with emission occurring from the lowest quartet and/or doublet states [211-217]. The existence of a body of both absorption and emission data for very well identified transitions, provided a good test for theoretical inorganic spectroscopy in its early development. Fig.6.16a shows some typical data. Note that the quartet absorption and emission (the broad bands) overlap at the energy of the forbid-



a

Fig.6.16 (a) The absorption (25°C) and emission (- 195°C) spectra of 0.03M $Cr(urea)_6(ClO_4)_3$.



b

Fig.6.16 (b) Potential energy diagrams for the $^4A_{2g}$, $^4T_{2g}$, ^2E_g and $^2T_{2g}$ states of an octahedral d^3 species.

den 0'-0" transition (these are vibronically activated bands) and that the doublet emission and absorption (the very narrow bands) are equi-energetic, i.e. no Stokes shift (see Chapter 4). There is essentially no change in equilibrium bond distance during excitation of the spin-forbidden line (as expected for a spin flip) but there is a marked change in bond distance for the spin allowed transition (as expected since a sigma antibonding e_g^* orbital is being occupied)(Fig.6.16b).

The vibronic structure of the lowest energy quartet-doublet transitions of octahedral Cr(III) has been extensively studied. Subtle differences exist among different CrL_6 systems which provide interesting insights into the detailed nature of the chromium(III) ligand bond, both in the ground and excited states. In these studies, and parallel studies of lower symmetry Cr(III) species to be discussed below, the frequencies of all the ground state normal modes of vibration can often be obtained through combination of infrared and Raman spectroscopy and electronic luminescence spectroscopy. Recall that vibrational structure in luminescence provides ground state frequencies (Chapter 4). Frequencies of vibrations which are forbidden in both the infrared and Raman can sometimes be obtained from the luminescence study.

Table 6.9

SIX COORDINATE PSEUDO-OCTAHEDRAL CHROMIUM(III) COMPLEXES
(ground term $^4A_{2g}$)

Complex	$^2T_{2g}$, ^2E_g	$^4T_{2g}$	$^4T_{1g}$(F)	$^4T_{1g}$(P)	B	Ref
CrF_6^{3-}	15,700[c] (16,400)	14,900	22,700	34,400	896	200
$Cr(NCS)_6^{3-}$		17,700(160)	23,800(130)	32,400 (25,000)d	585	200
$CrCl_6^{3-}$	13,180	18,700				246
$Cr(CN)_6^{3-}$		26,700(80)	32,600(55)		529	200
$Cr(H_2O)_6^{3+}$	15,000(2)	17,400(13)	24,600(15)	37,800(4)	728	200
$Cr(NH_3)_6^{3+}$	15,300	21,550(30)	28,500(27)		657	200
$Cr(NCSe)_6^{3-}$	12,900	18,000	23,000		461	198
$Cr(MeNH_2)_6^{3+}$		20,960(72)	27,540(62)		619	252
$Cr(EtNH_2)_6^{3+}$	15,120[i]	20,910(72)	27,550(62)		626	252
$Cr(acac)_3$	12,950	17,860	23,260		505	357
$Cr(HFac)_3$	12,560	17,450	21,680		383	357
$Cr(dpm)_3$	12,960	17,700	23,260		523	357
$Cr(dma)_6^{3+}$		14,880(60) 15,310(61)	21,550(47)		697	248
$Cr(tmm)_3^{3+}$		17,100	23,500		627	249
$Cr(acam)_6^{3+}$		16,450(53)	22,940(41)		646	248
$Cr(hmpa)_6^{3+}$		16,200	23,000		693	293
$Cr(dmso)_6^{3+}$		15,770(33)	22,520(35)		692	251
$Cr(BipyNO)_3^{3+}$		18,150 19,530	20,575			302
$Cr(BipyNO_2)_3^{3+}$		14,950(75) 15,850(77)	20,820(93) (22,300)(70)		616	303
$Cr(TMNO)_6^{3+}$		16,530	24,690		900	305
$Cr(PyNO)_6^{3+}$		15,385(83)	22,548(126)		763	251
$Cr(4MePyNO)_6^{3+}$		15,270 (16,670)	22,730		690	306
$Cr(4ClPyNO)_6^{3+}$		15,750	22,220		653	306
$Cr(4MeOPyNO)_6^{3+}$		16,130	22,990		701	306
$Cr(2,6\text{-}LutNO)_6^{3+}$		16,860	(25,320)		942	304
$c\text{-}Cr(CN)_4(H_2O)^-$		23,420	30,120		621	914
$CrCl_3$	13,150	13,700	18,900		512	200
$CrBr_3$	13,900	13,400	17,700	19,200	406	247
$Cr(OMe)_3$		17,400	24,200		675	314
$Cr(SMe)_3$		16,670	25,000		925	314
$NaCrS_2$		13,350(3.2) 14,600(10)	(18,000)		447	949
$Cr(en)_3^{3+}$	(h)	22,300(A_1) 22,350(E)	28,670(E) 28,875(A_2)		598	244
$Cr(pn)_3^{3+}$	14,950[i]	21,830(83)	28,500(74)		625	252
$Cr(NO_3)_6^{3-}$		17,000	23,000		579	282

cont

Table 6.9 cont

Complex	$^2T_{2g}$, ^2E_g	$^4T_{2g}$	$^4T_{1g}$(F)	$^4T_{1g}$(P)		Ref
Cr/Al_2O_3 [ab]	14,430(5)	18,000(40)	24,600(60)	(39,000)	643	200
$Cr(EtOCS_2)_3$	13,330	16,120	20,500		402	250
$Cr(ETS)_3$	(13,100)[f]	14,300(380)	18,600(270)			200
	(13,600)		19,100(265)		429	254
$Cr(ETSe)_3$ [f]	(12,600)	13,750(430)	17,650(340)		360	245
	(13,100)					
$Cr(S_2AsMe_2)_3$		13,900(290)	18,350(250)	(29,500)	430	168
$Cr(Ph_2PS_2)_3$		13,650(333)	18,100(179)		422	254
$Cr(Et_2NCS_2)_3$		15,500(380)	20,400(360)		462	255
$Cr(CS_3)_3^{3-}$		16,000(300)	21,500(500)		527	256
$Cr(EtSCS_2)_3$		15,700	19,400		334	257
$Cr(diphos)_3^{3+}$		21,000	26,200	(42,800)	472	283
$Cr(CH_3)_6^{3-}$		20,800	26,500	(43,800)	524	283,284
$Cr(CH_3SO_2)_3$	14,285	15,750	22,125		640	486
$CrCl_3$(QP)		16,300(1060)	21,200(520)		457	679
$CrBr_3$(QP)		16,300(920)	19,600(510)		293	679
$Cr(NH_3)_5CN_2^+$		22,200(43)	28,800(38)			1368
c-$Cr(NH_3)_4(CN)_2^+$		22,900(49)	29,200(38)			1369
t-$Cr(NH_3)_4(CN)_2^+$		22,700(43)	29,100(41)			1370
c-$Cr(en)_2(CN)_2^+$		23,050(69)	29,500(62)			1371
t-$Cr(en)_2(CN)_2^+$		23,100(50)	29,700(43)		1371	1372
t-Cr(cyclam)$(CN)_2^+$		24,150(62)	30,500(62)			1373

a) Ruby b),c),e),g),h) $^2T_{2g}$---$^4A_{2g}$ bands at 21,030(5), (22,000), 19,200 and (18,700), 22,000 respectively. d) internal NCS transition + charge transfer f) + high intensity charge transfer bands. h) ^2E at 14,883 and 14,901; $^2T_1$ at 15,427, 15,439, 15,517 cm^{-1}. i) vibronic origin in emission [378]. See Ref [1343] for additional extensive data.

Excited state vibrational frequencies can be obtained from structure in the absorption spectra. The information derived from the changes in frequency of a particular mode in the ground and excited electronic states, and, inter alia, the bond distance information which can be obtained from the intensities of progressions, yields a sophisticated view of the molecule.

Luminescence in an octahedral chromium(III) species is usually due to the transition $^4A_{2g}$---^2E_g (phosphorescence), which gains intensity primarily through spin-orbit coupling with the higher energy $^4T_{2g}$ state, enabled by the t_{1u} and t_{2u} vibrational modes.

In the system $Cr(NH_3)_6^{3+}$, there are seven t_{1u} modes, some of which involve NH stretching and deformation. All seven are seen in the phosphorescence spectrum of this species, i.e. there are seven vibronic origins occurring at frequencies corresponding with the electronic 0'-0" energy less (luminescence) each of the vibrational

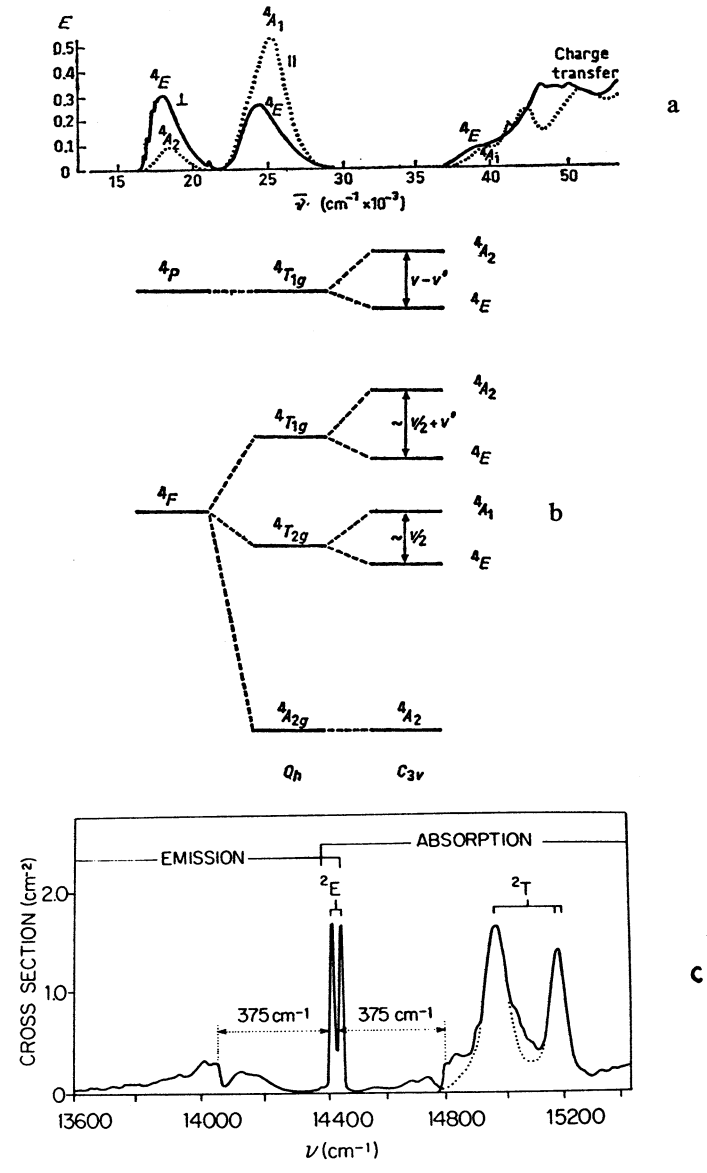


Fig.6.17 (a) The polarised absorption spectra of Cr(III) in corundum at -196°C.(b) Energy level diagram for a d^3 ion in octahedral and trigonal (C_{3v}) environments. (c) Absorption and phosphorescence spectra of ruby at 77 K. The 'R' lines near 14,000 cm^{-1} are the split origin of the ^2E---$^4A_2$ transition.

frequencies. Thus not only skeletal modes, but intramolecular modes of the ligand NH_3, are coupled to the electronic motion. This argument is also supported by a parallel study of $Cr(ND_3)_6^{3+}$ [240], and was recognised much earlier in an absorption study by Adamson and Dunn [221]. A more recent interpretation involves a vibronic model in which the lone pair motion is considered [545]. There are four t_{2u} modes and some, though not all, were positively identified. Clearly not all the vibronic origins have the same intensity. Variations in intensity may reflect the ability of the odd vibration to couple a higher excited odd charge transfer state (e.g. its energy is so high its contribution is very small), or, in the Hollebone mechanism, the adherence to the octupole rule (see section 4.4). No progressions in the totally symmetric Cr-N a_{1g} stretch were observed, nor are they expected in this spin-flip transition where there is no change in the internuclear equilibrium distance.

The $^4T_{2g}$ state, which confers intensity to the emission, involves excitation to $t_{2g}{}^2e_g$ and should have a longer Cr-N distance. Moreover it is theoretically subject to Jahn-Teller distortion via an e_g or t_{2g} mode. Flint [240] argues that to the extent the ^2E_g and $^4T_{2g}$ states are mixed (by spin-orbit coupling), these vibrations might be revealed in the emission spectra. Indeed the emission spectra show combination bands involving one quantum of t_{1u} (or t_{2u}) plus one quantum of e_g. Combinations with a_{1g} (Cr-N) might also arise through this mechanism but are not observed. In an independent analysis, Wilson and Solomon showed that the absorption band $^4T_{2g}$---$^4A_{2g}$ shows vibrational structure arising from a Jahn-Teller distortion via the e_g mode [241] (see Chapter 4 for some additional discussion on this system).

A similar study of the $Cr(NCS)_6^{3-}$ ion, in both absorption and emission identified the vibronic origins of 6 t_{1u} and 3 t_{2u} modes [242]. Following the hexammine example, the totally symmetric progressions are absent and combination bands involving e_g can be seen.

The emission spectrum of $[Cr(CN)_6]^{3-}$ doped into $[Co(CN)_6]^{3-}$ has also been studied by Flint [243]. It differs from those discussed above in that a progression in the a_{1g} Cr-C stretching vibration is observed, while a progression in e_g is absent. This absence is ascribed to the fact that coupling to the $^4T_{2g}$ state is greatly reduced since it is energetically much further from ^2E_g in the hexacyanide relative to the ammine and thiocyanate cases (and coupling is proportional to the square of the energy difference). The presence of the a_{1g} progression implies that indeed the internuclear Cr-C distances in ^2E_g and $^4A_{2g}$ are different. From consideration of the intensities of the progression components, this difference is calculated to be 4 - 5 pm (see Chapter 4). Flint speculates that the metal-ligand distance in the ^2E_g state decreases so as to increase pi delocalisation into the cyanide orbitals and that this is more important than in the other cited cases because such delocalisation would be less efficient and because, with their smaller DQ values, coupling to the $^4T_{2g}$ state

(with its longer M-L bond) would be greater and would tend to cancel the reduction in length. Note that, to first order, the energy of ^2E_g---$^4A_{2g}$ is 9B +3C, and that this transition lies about 2000 cm^{-1} to lower energy in the hexacyanide compared with the other cited complexes. This is consistent with a greater degree of covalency. Oelkrug and co-workers [315] report both Cr-C and C-N symmetric stretching frequencies in the luminescence spectrum of the hexacyanide chromium(III) anion [315].

The elpasolite salts Cs_2NaMCl_6 contain octahedral MCl_6 complexes and are useful for single crystal studies into rigidly octahedral sites. The $[CrCl_6]^{3-}$ ion has been studied in this fashion [395,397]. In this case, DQ is small (DQ = 32,995, 10Dq = 12,000 cm^{-1}) and the $^4T_{2g}$ state lies below ^2E_g. Thus luminescence is fluorescence from the spin quartet rather than phosphorescence from the spin doublet. The $^4T_{2g}$ state splits, under spin-orbit coupling, into E', U'(5/2), U'(3/2) and E". In the elpasolite single crystal spectrum at 8 K, the four magnetic dipole allowed origins (from $^4A_{2g}$ to these four states) are observed at 11,882, 11,890, 11,904 and 11,916 cm^{-1} [395]. The luminescence from this transition (from the lowest lying spin-orbit state) shows false origins due to coupling of the electronic motion with t_{1u} (120 cm^{-1}) and t_{2u} (187 cm^{-1}) vibrational modes of the $[CrCl_6]^{3-}$ ion. In addition there are progressions in the a_{1g} (298 cm^{-1}) and e_g (240 cm^{-1}) modes, the former dominating. When the spin-orbit energy separations within the $^4T_{2g}$ state are calculated in terms of DQ, B, C and λ, a good fit cannot be obtained whatever value of the spin orbit coupling coefficient is considered. The presence of the e_g progression is clear evidence for the existence of a Jahn-Teller distortion in the $^4T_{2g}$ state and when this is taken into account, the spin-orbit splitting can be evaluated in terms of the Ham effect.

> In Jahn-Teller distorted states the spin-orbit splitting is often less than anticipated. Quenching occurs by a mechanism known as the Ham effect [398] as elaborated by Sturge [399]. Consideration of the reduction in spin orbit coupling by this mechanism can lead to a value for the Jahn-Teller energy.

Low symmetry six coordinate Chromium(III) - Three-fold distortion: Specific absorption and emission data for Ruby are seen in Fig.6.17. The transition ^2E_g---$^4A_{2g}$ (in O_h) is responsible for the sharp peak, with an energy, correct to second order of $9B + 3C - (90B^2/10Dq)$ (Chapter 3). Evidence for low symmetry is seen in the polarised data in Fig.6.17 which show dichroism. The absorption spectrum of ruby [139,217-219,1305] has been analysed by McFarlane [220] on the basis of a trigonal distortion of the octahedral field. The high spin energy levels are indicated in Fig.6.17b in the group C_{3v}. Excellent agreement between calculated and observed energy levels was obtained, viz: (10Dq = 18,100; B = 650; v' = 400; v = 800 cm^{-1}). There has been extensive study of the Ruby system and space precludes discussion here. Note however study

of the pairwise interaction of Cr(III) ions [294] and the absorption spectra of excited Ruby [295].

McCarthy and Vala [244] report the single crystal spectra of the D_3 distorted $[Cr(en)_3]^{3+}$ ion. The deviation from regular octahedral symmetry is very small (see Table 6.9). Some 80 vibrational components were identified between 14,700 and 18,900 cm^{-1} at 77 K. An anti-resonance effect ('dip') was observed through interference between overlapping spin allowed and spin forbidden bands. Several other studies of this ion have been reported [396,400-403].

Geiser and Gudel report single crystal absorption and CD data for the $[(+)_D\text{-}Cr(en)_3]^{3+}$ ion doped into $Ir(en)_3Cl_3$ [396]. From the similarity in the sigma and axial spectra, an electric dipole mechanism is demonstrated, though from the small temperature dependence of the intensity, there is also a vibronic contribution. Considering the $^4T_{2g}$---$^4A_{2g}$ absorption, the difference between the sigma and pi components of the absorption spectra, is quite small (50 - 150 cm^{-1}) which may reflect the D_3 splitting of $^4T_{2g}$, spin-orbit and Jahn-Teller effects. Nevertheless these authors assign the sigma intensity to the transition to $^4E(^4T_{2g})$ assuming it is static in origin. There is a 450 cm^{-1} difference between the CD maximum and the axial absorption maximum, the former with no temperature dependence of intensity. The implication is that the CD of the vibronic part of the intensity is smaller than that of the static.

The $^4T_{1g}$ state occurs near 29,000 cm^{-1} and is polarised predominantly sigma. It exhibits only very weak CD. These authors [396] also report extensive vibrational data for the spin doublet states both in absorption and emission. The ^2E state is split into two spin orbit components by the D_3 field and spin orbit coupling.

Piper and Carlin [661] studied the single crystal spectra of the $[Cr(oxalate)_3]^-$ ion [661]. The $^4T_2$ splits by 304 cm^{-1} in the D_3 field, with $^4A_1$ lying lower than ^4E. Well resolved spin forbidden transitions also showed vibrational satellites (Cr-O). Dingle [1165] reported the single crystal polarised spectrum of trans-$K[Cr(oxalate)_2(H_2O)_2].3H_2O$, analysed in D_{2h} symmetry. Resolution of the orbital triplets was observed but an unequivocal assignment thereof did not prove possible.

The tris(diketonato)Cr(III) derivatives have, at best, D_3 symmetry, and have been extensively studied [357,360,361]. In general the first two spin allowed transitions can be observed, together with some spin forbidden transitions, and MLCT transitions to the pi levels of the diketone ligand. The DQ values do not depend markedly upon the substituent in the diketone ligand; however the β_{55} value (see Chapter 9), a function of the degree of pi delocalisation, does depend upon the substituent. Indeed both the nephelauxetic functions are greatly reduced for these species, comparable indeed to sulphur ligands (see Table 6.9). Presumably the Cr-O bond is quite covalent and delocalisation to the pi framework is significant.

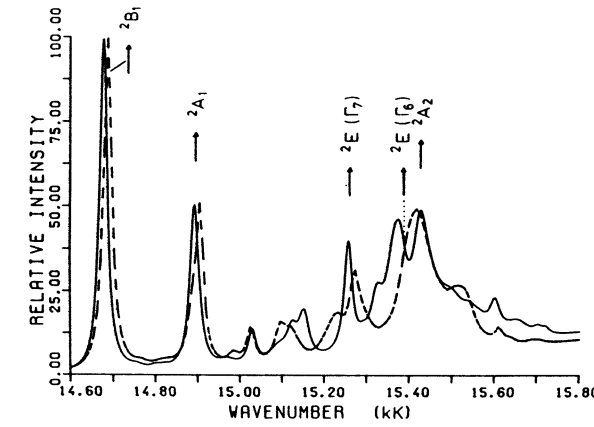


Fig.6.18 The absorption spectrum of $[Cr(NH_3)_5(NCO)](NO_3)_2$ (solid line) and the corresponding deuterated species (hatched line) in transparent KBr discs at 25 K [278].

The tris(diethyl-dithiophosphate)chromium(III) derivatives also have polarised spectra indicative of the D_3 group [157]. β_{55} is greater than β_{35} inferring that sigma delocalisation (sigma Cr-S bonding) is more important than pi bonding in these ligands.

Low Symmetry Six Coordinate Chromium(III) - Four-fold distortion:

The extensive spectra available for four fold distorted Cr(III) systems have contributed greatly to the development of our understanding of low symmetry complexes.

Further, spin orbit coupling, in these four fold groups, will lift the degeneracy of the E states. Some of the key discussions of this area can be found in [269,276,278-280,1343], where a detailed review of tetragonal systems is valuable [1343]. Well defined splitting of the spin allowed orbital triplets is not generally observed in these systems. Thus splitting of the spin doublets is expected to be small. Using high resolution absorption and emission spectra, various authors assign bands to $^2A_1(^2E)$ and $^2B_1(^2E)$ varying from a few tens [276] to a few hundreds of wavenumbers [278,280] (Fig.6.18). The splitting of the $^2T_1$ state is of the order of 100 cm^{-1} and the spin orbit splitting of $^2E(^2T_{1g})$ a few tens of wavenumbers [278]. Flint [280] presents arguments, supported by Schmidtke [278] that conventional ligand field theory cannot satisfactorily account for these splittings (and those of the spin allowed states) even when a complete calculation with all configurational interaction and spin-orbit coupling is included. Gudel [276] challenges this view.

It is certainly true that the Racah parameters obtained through analysis of the spin allowed bands do not usually satisfactorily

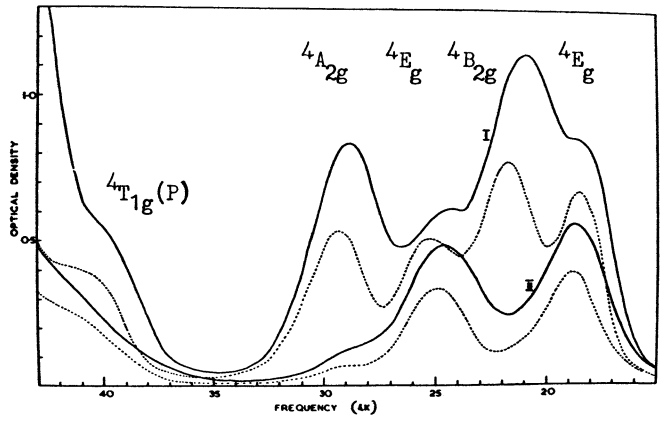


Fig.6.19 The polarised spectra of $[Cr(en)_2F_2]^+$ [270]. The electric vector in I and II lie approximately along xy and z respectively. The solid and dotted lines represent room temperature and 4 K spectra respectively.

allow prediction of the energies of all the spin forbidden transitions. Following discussion in Chapter 3, there are 9 coulombic and exchange integrals involving the d orbitals. These are generally all handled within the framework either of a single value of B and C, or of two sets of values of B and C, one for the spin allowed bands and one for the spin forbidden bands. Even this however fails to accomodate all the transition energies. The resolution of the data obtained with these chromium systems is so good, that the experiment certainly challenges the theory. It is probable that a complete analysis must allow for so-called differential radial expansion (i.e. different B values for different coulombic pairs of orbitals), inclusion of vibronic coupling, and covalency.

The spin forbidden transitions in CrL_5Z systems often display rich vibronic structure [278-280]. For example the Cr-O symmetric stretch (at 570 cm^{-1}) is readily observed in the spin forbidden spectra of $[Cr(NH_3)_5OH]^{2+}$ [276]. The spin allowed bands in these CrL_5Z systems are not generally fully resolved (but see $[Cr(NH_3)_5OH]^{2+}$ in [276]). In the trans CrL_4Z_2 series, nominally of D_{4h} symmetry, splitting of the two lower orbital triplets into pairs of bands is observed (Table 6.10, Fig.6.19). Single crystal spectra using polarised light, have confirmed the assignments for several of these species [e.g. 273,289]. In general however assignments are obtained by comparison of a large body of compounds searching for a consistent set of parameters, recently within the AOM formalism [e.g. 269] (though in earlier classic work via crystal field theory [286-288,292]). When L is a saturated amine, $e_\pi(N)$ is considered zero, and the four components of the $^4T_{2g}$ and $^4T_{1g}$ (P) triplets can be defined by the three AOM parameters $e_\sigma(L)$, $e_\sigma(Z)$ and $e_\pi(Z)$, plus the Racah B. An energy of 10Dq(L) is associated with $^4B_{2g}$ ($^4T_{2g}$) essentially constant for a series trans-CrL_4Z_2, if there is little interaction between the Cr-L and Cr-Z bonds. This provides a benchmark to assign one of the four transitions. Moreover the sense of the splitting of the $^4T_{2g}$ triplet is usually known from the relative DQ values of L and Z (see Chapter 3). Thus the principle uncertainty is the sense of the splitting of $^4T_{1g}$ and this has caused some controversy in the past, especially where fluoro derivatives were concerned.

Discussion of the spectrochemical series obtained from analysis of these systems is left until Chapter 9 when a complete discussion of all relevant systems is presented. Here note that the evaluation of both sigma and pi AOM parameters permits a two dimensional spectrochemical series to be generated for Cr(III). Based upon work by Glerup and co-workers [281], Barton and Slade [365] and Lever and co-workers [269], this may be written:

$$e_\sigma \; I^- < Br^- < Cl^- < OAc^- < dmso < dmf < H_2O < Py < NH_3 = en = 1,3pd < OH^- < F^-$$

$$e_\pi \; Py\ (-ve) < NH_3\ (=0) << I^- < Br^- < OAc^- < Cl < H_2O < OH^- < F^- \qquad (6.13)$$

The species $[Cr(H_2O)_4Cl_2]^+$, like its vanadium analog, shows an additional absorption band due to coupling to an OH mode [289]. Many of these tetragonal systems show extensive vibrational structure which has been analysed. The luminescence spectra of $[Cr(tmd)_2Cl_2]^+$ show all the ungerade vibrations permitted in the infrared absorption spectrum [290].

The study of trans-$[Cr(Py)_4F_2]^+$ by Flint and Matthews [272] is of especial interest. In addition to a useful discussion of the effects of low symmetry upon the splitting of the lower lying doublet states, there is an analysis of the effective vibrational frequencies of various modes in different electronic excited states. For example the ^2E_g (in D_{4h}) components of $^2T_{1g}$ and $^2T_{2g}$ (in O_h) are mixed such that there is a shift of electron density away from the axis towards the xy plane, in the lower lying ^2E_g, and an opposite shift in the higher lying ^2E_g, if the mixing factor $(-3Ds + 5Dt)(= d\pi/2)$ is positive, and the reverse if the mixing factor is negative [301]. In this example $d\pi$ is positive, and the fluoride ion is a stronger pi donor than pyridine. Since the t_{2g} electrons are pi anti-bonding, if pi density is transferred from z to xy, then the Cr-F z axis bond is strengthened (shortened) in the lower lying ^2E_g state, due to lower pi electron repulsion, and lengthened in the upper ^2E_g state. Because of the change in bond length, progressions in Cr-F and Cr-N frequencies are anticipated and are observed. By analysis of the vibronic structure of these states in absorption, these workers showed that the frequencies for the a_{1g} Cr-F stretching vibration was 558 and 512 cm^{-1} in the lower and upper ^2E_g states respectively, consistent with this model.

Table 6.10

LOW SYMMETRY CHROMIUM(III) COMPLEXES

i)

Six Coordinate Tetragonal Chromium(III) Complexes (ground term $^4A_{2g}$ in O_h)[269].

Complex	^4E_g	$^4B_{2g}$	$^4A_{2g}$	^4E_g	$^4T_{1g}(P)$	Ref
$Cr(en)_2F_2^+$	18,500	21,700	29,300	25,300	(41,000)	270
$Cr(en)_2Cl_2^+$	17,700	22,800	25,400	26,000		265,273
$Cr(en)_2Br_2^+$	17,000a	22,500	23,900	26,300		263,273
$Cr(en)_2I_2^+$	15,770	(20,960)c				365
$Cr(en)_2(H_2O)_2^{3+}$	20,000b	22,285	29,300	27,500		262,273
$Cr(en)_2(OH)_2^+$	18,250	19,920	30,300	25,190		262
$Cr(1,3pd)_2F_2^+$	19,260	22,030	30,670	(24,270)		365
$Cr(1,3pd)_2Cl_2$	17,570	22,220	25,510			365
$Cr(1,3pd)_2Br_2$	16,490	23,040	23,800	26,670		365
$Cr(1,3pd)_2I_2$	14,950c					365
$Cr(1,3pd)_2(dmf)_2^{3+}$	19,530	23,360	(29,000)	26,650d		365
$Cr(1,3pd)_2(OAc)_2^+$	20,240	22,520	27,930			365
$Cr(1,2pd)_2F_2^+$	19,690	22,370	29,940	25,190		365
$Cr(chd)_2F_2^+$	19,530	22,700	29,760	25,380		365
$Cr(daud)_2F_2^+$	19,800	22,570	30,450	25,910		365
$Cr(H_2O)_4F_2^+$	16,300	-	25,500	22,700		263
$Cr(H_2O)_4Cl_2^+$	16,630	16,630	21,760	23,040	35,130	289
$Cr(NH_3)_4F_2^+$	18,230	21,110	28,200	24,950		264
$Cr(CN)_5(H_2O)^{2-}$	23,500	26,400	29,000	32,100		914
$Cr(CN)(H_2O)_5^{2+}$	19,200	17,500	-	25,455		914
$c\text{-}Cr(CN)_2(H_2O)_4^+$	18,520	21,460	-	26,455		914
$Cr([14])Cl_2^+$	17,640	24,750	27,472			285
$Cr([15])(H_2O)_2^{3+}$	18,520	22,025	26,525			277
$Cr([15])I(H_2O)^{2+}$	17,795	21,645	25,380			277
$Cr([15])Br(H_2O)^{2+}$	17,860	21,740	25,445			277
$Cr([15])Cl(H_2O)^{2+}$	17,730	21,645	25,445			277
$Cr([15])Me(H_2O)^{2+}$	?	21,370	26,670 ?		38,760 ?	277
$Cr([15])Et(H_2O)^{2+}$	?	21,415	26,110 ?		37,880 ?	277
$Cr(py)_4F_2^+$	17,000	18,500	29,900	24,300		272
$Cr(NH_3)_5(OH)^{2+}$	19,700	-	29,400	25,450		275,276
$Cr(NH_3)_5Br^{2+}$	19,000	21,500	26,500			267

ii)

b) Five Coordinate Complexes

Complex	Absorption Spectrum		Ref
$Cr(Me6tren)Cl^{2+}$	10,800(86)	14,000(32)	411
$Cr(NP_3)Cl^{2+}$	13,800(510)	17,000(470)	412
$Cr(NP_3)Br^{2+}$	13,300(530)	16,000(415)	412

cont

Table 6.10 cont

ii) Five Coordinate Complexes

Complex	Absorption Spectrum			Ref
$Cr(NP_3)I^{2+}$	13,150(520)	15,600(380)		412
$Cr(N_3P)Br^{2+}$	11,600	14,100		412
$CrCl_3.2NMe_3$	10,100	17,600	(30,200)	147

The CrL_4Z_2 examples are all of trans stereochemistry except where noted.

a) Spin forbidden bands at 20,945 ($^2B_{2g}$), 21,040 (^2E_g) b) spin forbidden bands at 21,860 ($^2B_{2g}$) and 22,285 (^2E_g) cm^{-1}. c) Higher energy bands obscured by CT absorption. d) Preferred assignment, but higher two states may be interchanged; however this yields a rather larger value for pi(dmf) than seems likely. See Ref. [1343] for extensive additional data.

Indeed the vibrational frequencies of important Cr-F and Cr-N modes were determined for all the lower lying spin doublet and quartet states from a combination of absorption amd luminescence spectroscopy. Quite significant variations were seen and justified by arguments such as that presented above. The work is a specially beautiful example of how spectroscopy can be used to obtain important structural and electronic binding information.

Various cis-CrL_4Z_2 complexes are known. The splitting of the $^4T_{2g}$ term in such complexes is the inverse of that anticipated for the corresponding trans complex, and of approximately one half the magnitude (Chapter 1). It is this last observation which results in the spectra of cis complexes rarely exhibiting structure and having bands at energies roughly the average of those of their trans analogs [285,291].

Other Coordination Numbers: Cubic eight-fold distorted Cr(III) is found in Cr(III)/CdF_2 [296]. Tetrahedral Cr(III) is observed in $PCrCl_8$ ($[PCl_4]^+[CrCl_4]^-$) [298] and tungstochromic acid (8300(20), 16,000(105) cm^{-1})[297]. A five coordinate, probably trigonal bipyramidal species has been reported [678]. Weaks bands at 10,100, 13,000, 17,600 and 23,200 cm^{-1} are almost certainly d-d in origin, but their assignment is a matter of dispute [147,159]. Better defined five coordinate trigonal bipyramidal complexes are included in Table 6.10 [411,412]. The three coordinate $Cr[N(SiMe_3)_2]_3$ exhibits d-d transitions at 11,800(100) and 14,800(540 cm^{-1} and higher energy more intense transitions in the ultraviolet.

MOLYBDENUM(III) AND TUNGSTEN(III) d^3

The spectra of some six coordinate Mo(III) (and W(III)) derivatives are indicated in Table 6.11 [316-319,322-324,370].

Table 6.11

ABSORPTION SPECTRA OF MOLYBDENUM(III), TUNGSTEN(III), MANGANESE(IV), TECHNETIUM(IV) AND RHENIUM(IV) DERIVATIVES

Complex	^2E_g, $^2T_{1g}$	$^2T_{2g}$	$^4T_{2g}$	$^4T_{1g}$ $^4A_{2g}$	B	Ref
$Mo(H_2O)_6^{3+}$			26,110(16)	31,750(25)	505	325
$MoCl_6^{3-}$	9650	14,800	19,200	24,000	436	96,101
$MoBr_6^{3-}$	9500(4)	14,500 (5.5)	18,300(33)	23,200(CT)		97
MoI_6^{3-}	9200	13,800	16,650	20,050	303	97
$Mo(urea)_6^{3+}$			25,900	31,000	453	100
$Mo(ox)_3^{3-}$		16,600	(27,000)			
		18,400				370
$Mo(acac)_3$			(23,300)	27,000(5000)		95
$Mo(en)_3^{3+}$			23,810	31,745	756	
				40,000s CT?		331
$Mo(en)_2(NCS)_2^+$			23,250	30,305	660	
				31,250s 40,000sCT?		331
$MoCl_3(CH_3CN)_3$	(8400)	13,500	25,200	28,900(CT)		96
	8750					
$MoPy_3Cl_3$	8800	14,500	27,200CT			370
$MoPy_3Br_3$	8800	14,700	27,200CT			370
$Mo(o\text{-phen})Cl_4^-$	9300	13,500	18,900CT			370
$Mo(SMe)_3$		13,500	19,750	30,350		
				35,350 ?		314
$W(SMe_3)$		13,000	30,500	35,150 ?		314
MnF_6^{2-}	16,021	25,707	21,801	28,090	583	102,326
	17,300					327
$MnCl_6^{2-}$	12,020		18,240	25,500	725	77,374
$MnB(OH)_4$	(17,855)?		15,920	19,230	295	103
$Mn(CN)_6^{2-}$	14,890		(31,950)			329
TcF_6^{2-}	11,000	17,600	28,400	34,400	536	426,427
	11,500					
ReF_6^{2-}	9100	17,700	32,800	(37,500)?	409	426,427
	10,100[a]	19,000[a]				
	10,900					

a) spin orbit components

The crystal field transitions parallel those of Cr(III) though the spin allowed bands fall at higher energies since DQ is larger. Generally only the transitions within the F shell are observed, the transition to $^4T_{1g}(P)$ falling well into the uv where it is often obscured. The (t_{2g}) intrashell spin forbidden transitions lie at lower energies than for Cr(III) since the larger size of these ions

confers smaller values for the Racah parameters B and C. The rather smaller optical electronegativity of Mo(III) (1.7) versus Cr(III) (1.9) gives rise to lower energy MLCT transitions which may obscure d-d transitions.

The luminescence spectra of $[MoX_6]^{3-}$ (X = Cl, Br) have been reported [333,594] doped into various octahedral transparent lattices. Emission occurs from and between the ground state and the three lowest spin doublet states, split by spin-orbit coupling. In $Cs_2NaScCl_6/[MoCl_6]^{3-}$ these states have been located as follows:

$^2T_{2g}$	Γ_7	14,667	Γ_8	14,412 cm^{-1}
$^2T_{1g}$	Γ_6	9556	Γ_8	9387
^2E_g	Γ_8	9239		
$^4A_{2g}$	Γ_8	0 Ground term		

(6.14)

with the corresponding bands occurring in the bromide at $\Gamma_8(^2E_g)$ 9115, $\Gamma_8(^2T_{1g})$ 9206, $\Gamma_6(^2T_{1g})$ 9423 and $\Gamma_8(^2T_{2g})$ 14,081 cm^{-1} [333].

At 10 K transitions were observed from the Γ_8 level of $^2T_{2g}$ to all the lower lying levels. In addition the transition from Γ_8 of ^2E_g to the ground state was observed. Each transition generated a vibronic origin by coupling to the three odd vibrational modes of $[MoX_6]^{3-}$ (whose magnitude varied slightly from state to state) and were further coupled to internal and lattice modes of the host lattices. Large variations in the intensities of the several vibronic origins were not immediately explainable [333].

The seven coordinate $[Mo(CN)_7]^{4-}$ ion exists as a pentagonal bipyramid (D_{5h}) in aqueous solutions but as a monocapped trigonal prism (C_{2v}) in the solid state (potassium salt, dihydrate). The crystal field splitting energies of the latter are naturally greater than the former and this is readily seen in the relevant spectra. Thus in solution, the lowest energy d-d band occurs at 24,900 cm^{-1} (e_1''---e_2'), but in the solid, bands occur at 16,100, 20,900 and at 22,500 cm^{-1} [371].

MANGANESE(IV), TECHNETIUM(IV) AND RHENIUM(IV) d^3

Relatively few Mn(IV) systems have been studied. Several six coordinate Mn(IV) complexes are included in Table 6.11. The spin allowed transitions occur at higher energies than the isoelectronic Cr(III) because of the higher oxidation state. The Racah parameters are reduced to a greater degree (nephelauxetic effect) probably because of increased covalency in this ion.

The $[MnF_6]^{2-}$ ion has been extensively studied [326,327, 330,457]. The spectrum (Fig.6.20) consists of weak structured spin forbidden absorption near 16,500 cm^{-1}, a quite highly structured $^4T_{2g}$---$^4A_{2g}$ transition near 22,000 cm^{-1}, a rather less structured $^4T_{1g}(F)$----$^4A_{2g}$ transition near 28,000 cm^{-1}, and a fluoride to manganese LMCT band near 39,000 cm^{-1} (Table 6.11). The structure on the $^4T_{2g}$ band shows a progression of about 500cm^{-1} undoubtedly associated with the totally symmetric Mn-F stretching vibration, since this transition involves excitation of an electron to an Mn-F

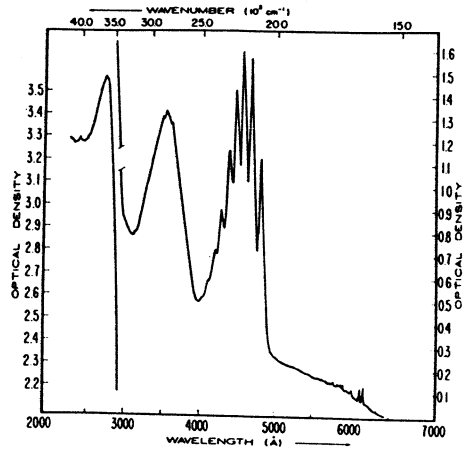


Fig.6.20 The absorption spectrum of a Cs_2MnF_6 mull at -196°C [326].

anti-bonding e_g orbital. Indeed three discrete progression are observed built upon vibronic origins of a spin-orbit split $^4T_{2g}$ state. Both the Racah B parameter and the spin orbit coupling coefficient are reduced to about 60δ of their free ion values.

Flint and co-workers have studied the ^2E_g---$^4A_{2g}$ emission of the $[MnF_6]^{2-}$ ion in various hexafluoride lattices. In each case the spectrum is analysed in terms of a) a magnetic dipole allowed transition, b) phonon sidebands characteristic of the lattice, c) several bands involving two lattice phonons, d) three strong 'odd' vibronic origins, e) progressions in up to three quanta of three even fluoride modes based upon the vibronic origins, and f) bands involving combination of an internal molecular mode with lattice modes [327].

Four coordinate Mn(IV) exists in tungstomanganic acid exhibiting a band at 7000(18) cm^{-1} [300].

The d-d spectra of Tc(IV) and Re(IV) species are rarely observed, being of high energy (large DQ) and usually being obscured by charge transfer absorption [176,200,336,337]. Low lying doublet states within the $(t_{2g})^3$ configuration can be observed through luminescence [334,335], and also through absorption in the near infrared and low energy visible regions. Note that the higher spin orbit coupling coefficient will result in significant mixing of these states. The $[ReX_6]^{2-}$ species have been extensively studied both as pure systems and doped into other crystals [335,362-364, 406-410,456,529]. For example Flint and Paulusz [335] report very sharp structured emission for $[ReBr_6]^{2-}$ in the near infrared (4000

- 15000 cm^{-1}). Transitions from the spin orbit states of $^2T_{2g}$ and $^2T_{1g}$, ^2E_g to the $^4A_{2g}$ ground term are observed as is low energy emission from $^2T_{2g}$ to the lower lying doublets. Extensive vibrational structure is again observed, and the t_{1u} and t_{2u} enabling modes identified. In the spectrum of $[Re(OH)Br_5]^{2-}$, d-d bands are found in the region 4000 - 16,000 cm^{-1}, and gave a derived 10Dq value of 29,650 cm^{-1} (B = 420 cm^{-1}) [362].

Similar intra-configurational data are reported for the polymeric $[ReCl_5]^-$ ion [405].

Rich, fairly narrow absorption spectra (d-d) and many CT bands are seen in the Re(IV) species, $[ReX_5(PPh_3)]^-$ (C_{4v}) and $[ReX_4(PPh_3)_2]$ (D_{4h}) [404].

The d-d and CT spectra of the series $[TcX_6]^{2-}$ (X = F, Cl, Br and I) have been briefly discussed [526]

RUTHENIUM(V), OSMIUM(V) AND IRIDIUM(VI) d^3

Peacock and Sleight report [338] the absorption spectrum of RuF_5 (octahedral in the solid state) in which they observe the lower energy spin forbidden transitions to the spin-orbit components of ^2E_g, $^2T_{1g}$ at 8500 - 9100 cm^{-1}, and to $^2T_{2g}$ at 14,300 - 14,400 cm^{-1}.

The spectrum of the $[OsF_6]^-$ ion has been reported [522,526,528] and transitions to the various spin-orbit states of the d^3 configuration were assigned [526,528].

Structured narrow spin forbidden absorption has also been reported for IrF_6 [339]. This species parallels ReF_6 in that only intraconfigurational transitions within $(t_{2g})^3$ are seen. However the presence of three t_{2g} electrons rather than one as in ReF_6 naturally makes the spectrum considerably more complex. Transitions from $^4S'$ to components of $^2D'$ and $^2P'$ are seen near 6200 - 8800 and 12,300 - 15,200 cm^{-1} respectively. As with ReF_6, most of the intensity is vibronically induced, and evidence for Jahn Teller effects are again observed [359].

The vapour phase spectrum of IrF_6 [359] shows electronic origins from Γ_8 of $^4A_{2g}$ to $^2E_g(\Gamma_8)$ at 6260, $^2T_{1g}(\Gamma_8)$ at 8330, $^2T_{1g}(\Gamma_6)$ at 8860, $^2T_{2g}(\Gamma_7)$ at 12,330 and $^2T_{2g}(\Gamma_8)$ at 15,160 cm^{-1}. No evidence was found for Jahn Teller distortion of the ground term, but the Γ_8 components ^2E_g and $^2T_{2g}$ did show evidence of such distortion.

6.2 (iv) d^4 Cr(II), Mo(II), W(II), Mn(III), Tc(III), Re(III), Fe(IV), Ru(IV), Os(IV), and Pt(VI)

The ground term is ^5D in all high spin complexes but is ^3H for low spin complexes. The ground term in a regular high spin octahedral environment is ^5E_g. In this progressive study of the d^n configurations, d^4 offers two new features. In the first case, as indicated above, there is the possibility of obtaining either high or low spin complexes whose spectra will differ greatly. Secondly the degeneracy of the ^5E_g ground term will necessarily be lifted by the Jahn Teller effect, so that regular octahedral d^4 complexes

Table 6.12

ABSORPTION SPECTRA OF CHROMIUM(II) AND MANGANESE(III) COMPLEXES

i) Six Coordinate Complexes

Complex	$^5A_1 \leftarrow {}^5B_1$	$(^5T_{2g}) \leftarrow {}^5B_{1g}$	Chromophore[e]	Ref
$Cr(en_3)^{2+}$	8300	16,000	N_6	372
$Cr(en)_2Br_2$	13,200	17,900	N_4Br_2	372
$Cr(1,2mpn)_3^{2+}$	7700	16,300	N_6	366
$Cr(1,3pd)_2Cl_2$	(12,600)	17,600	N_4Cl_2	366
$Cr(1,3pd)_2Br_2$	(14,200)	18,000	N_4Br_2	366
$Cr(1,3pd)_2I_2$	(15,300)	18,300	N_4I_2	366
$Cr(s\text{-}Me_2en)_2Cl_2$	(11,400)	16,900	N_4Cl_2	366
$Cr(as\text{-}Me_2en)_2Br_2$	17,700		N_4Br_2	366
$Cr(as\text{-}Me_2en)_2I_2$	16,500		N_4I_2	366
$Cr(H_2O)6^{2+}$	(9500)(1.8)	14,000(5)	O_6	110
CrF_2	11,500	14,700	F_6	110
$CrCl_2$[d]	8750	12,000	Cl_6	110
Cr(II)/AgCl	5700	10,250	Cl_6	375
$CrCl_2.2DMSO$[d]	(8500)	12,000	Cl_4O_2?	110
$Cr(4\text{-}MePy)_4Cl_2$	9700	(17,000)	N_4Cl_2	233
$Cr(4\text{-}MePy)_4Br_2$	10,800	17,000	N_4Br_2	233
$Cr(4\text{-}MePy)_4I_2$	12,000	17,500	N_4I_2	233
$Cr(4\text{-}MePy)_2Cl_2$	(11,700)	14,600	N_2Cl_4?	233
$Cr(4\text{-}MePy)_2Br_2$	(11,500)	14,200	N_2Br_4?	233
$CrBr_2.2Py$	(10,300)	14,300	N_2Br_4?	109
$CrBr_2.6Py$	11,600	17,000	N_4Br_2?	109
$CrI_2.6Py$	12,750	17,000	N_4I_2?	109
CrF_6^{4-}	20,200	29,600CT	F_6	253
$CrCl_6^{4-}$	10,200(9.2)		Cl_6	61
$Cren_2^{2+}$	18,300(25)		N_4?	118
$Cr(N_2H_4)_2Cl_2$	17,500	22,000(CT?)	N_4?	112
$CrCl_2.2CH_3CN$	(9800)	13,300	N_2Cl_4?	119
$CrBr_2.2CH_3CN$	(9600)	13,200	N_2Br_4?	119
$CrI_2.2CH_3CN$	10,000	16,200	N_2I_4?	119
$Cr(pz)_4Cl_2$	(12,500)	18,200	N_4Cl_2	380
$Cr(pz)_4Br_2$	(13,300)	18,500	N_4Br_2	380
$Cr(im)_4Cl_2$	(13,300)	17,500	N_4Cl_2	380
$Cr(im)_4Br_2$	(16,600)	19,200	N_4Br_2	380
$Cr(NMeim)_4Cl_2$	(13,300)	17,700	N_4Cl_2	380
$Cr(NMeim)_4Br_2$	(14,000)	17,700	N_4Br_2	380
$Cr(NMeim)_4I_2$	(15,500)	17,700	N_4I_2	380
$Cr(MeCyclam)Cl^{+}$[b]		14,815(110)	N_4Cl	355
$Cr(MeCyclam)Br_2$[a]		15,270(122)	N_4Br_2	355
$Cr(Me_6[14]aneN_4)Cl_2$	14,300	19,700	N_4Cl_2	356
$Cr(Me_6[14]aneN_4)Br_2$	15,400	19,600	N_4Br_2	356
$Mn(H_2O)_6^{3+}$		21,000(5)	O_6	114,123

cont

Table 6.12 cont

Six Coordinate Complexes

Complex	$^5A_1 \text{---} {}^5B_1$	$(^5T_{2g}) \text{---} {}^5B_{1g}$	Chromophore[e]	Ref
$MnF(H_2O)_5^{2+}$	13,100(9)	18,600(19) 23,200(24)	O_5F	123
$Mn(dmso)_6^{3+}$	14,810	16,670($^5B_2$) 20,000(^5E)	O_6	430
$Mn(dmf)_6^{3+}$	15,380	17,860($^5B_2$) 20,830(^5E)	O_6	430
$Mn(antipyrine)_6^{3+}$		17,090($^5B_2$) 20,620(^5E)		430
$Mn(PyNO)_6^{3+}$		16,000($^5B_2$) 19,610(^5E)	O_6	430
MnF_6^{3-}	9000	(17,800) 19,400		353,354,1304
$MnCl_6^{3-}$	17,540	22,400CT	Cl_6	79
$Mn(OAc)_3$	11,400	22,000	O_6	381
$Mn(tart)_2(H_2O)_2^{5-}$	12,100(60)	17,240(129) 22,800	O_6	431
$Mn(ox)_3^{3-}$	9100	(19,050) 22,400	O_6	381
$Mn(tdpm)_3$	9450(84)	13,580(350)($^5B_2$) 14,800(370)(^5E)	O_6	428
$Mn(acac)_3$	9520(110)	17,900(200)($^5B_2$) 21,500(^5E)	O_6	429
$Mn(5,14[14]aneN_4)Cl_2$	10,900(9)	(24,100)(^5E) 24,400CD	(26,500)($^5B_2$) (31,200) CT? 27,200CD[c] 30,800CD	367 367
$Mn(5,14[14]aneN_4)Br_2$	11,700(17)	(23,200)(^5E) 22,400CD	(27,000)($^5B_2$) (30,800) CT? 26,900CD[c] 30,000CD	367 367
$Mn(CN)_6^{3-}$ ls.	20,540, 20940 ($^3E_g \text{---} {}^3T_{1g}$?)			328

ii) Five Coordinate Complexes

	Absorption Spectra				Ref
$[Et_4N]_2(In,Mn)Cl_5$	11,500($^5A_1$)	15,000($^5B_2$)	16,000(^5E)	23,900CT	390
$[Et_4N]_2MnCl_5$	13,300	17,400	23,800CT		391
$[BipyH]_2MnCl_5$	13,700	16,900	18,200	23,900CT	390
Mn(acen)Cl	16,400(200)	22,400(1700)CT			386
$MnCl_3(PyNO)_2$	(12,000)(355)	16,000(835)	27,000(5200)CT		392
MnCl3(4EtPyO)2	(12,000)(380)	16,100(1040)	(24,400)(3900)CT		392

a) cis-octahedral. b) Probably trigonal bipyramidal. c) (+)[Mn((-)5,7,7,12,12,14-hexamethyl-1,4,8,11,-tetraazacyclotetradecane]X_2^+. d) Extensive spin forbidden data presented. e) Subject to ground state Jahn-Teller distortion.

will rarely occur (see Chapter 4). Thus the single $^5T_{2g}$---^5E_g transition in O_h symmetry is likely to split into at least two transitions, e.g. $^5T_{2g}$ transforms in D_{4h} as $^5B_{2g}$ and ^5E_g. Moreover quite large distortion is expected (since the configuration $(t_{2g})^3(e_g)$ contains a sigma antibonding electron) and the splitting of the ^5E_g ground term may be large enough to yield a transition therein in the near infrared or visible region. Reference to Fig.2.11 suggests that a cluster of spin forbidden quintet-triplet transitions can be anticipated in the visible region. Since all spin allowed transitions occur within ^5D, no inter-electronic repulsion energy information is derivable. Such information can, in principle, be obtained from a thorough analysis of the spin forbidden bands.

The low spin species will have a $^3T_{1g}$ ground term which is not seriously subject to the Jahn Teller effect; octahedral low spin complexes are anticipated. Reference to the d^4 Tanabe-Sugano diagram (Fig.2.11) suggests that the lowest spin allowed transition will be ^3E_g---$^3T_{1g}$ and will occur at relatively high energies.

Many of these species form binuclear M-M bonded systems (due to coupling of the odd e_g electron) but study of their spectra is deferred until Chapter 7.

CHROMIUM(II) d^4

The spectra of chromium(II) complexes are notoriously difficult to obtain because of the ease of oxidation to chromium(III). Nevertheless their spectra are well known.

> In recording the spectra of air sensitive compounds, it is desirable to record the spectra of the oxidation products. These can be compared with the spectra of the reduced species in order to exclude the possibility of oxidation products contributing to extra features in the spectra. The analogous technique should be used with highly water sensitive species.

Six Coordinate Complexes: The spectra of most Cr(II) complexes consist of a broad band spread from about 13,000 - 17,000 cm^{-1} which can often be resolved (e.g. at low temperature) into three spin allowed components [340-350](Table 6.12). In addition some complexes exhibit a shoulder or peak on the low energy side of the main absorption band [23,341,347]. In parallel with many copper complexes, the Jahn Teller distortion results in an energy level diagram as displayed in Fig.6.21, though the sequence of splitting can be expected to vary from one complex to another. High spin d^4 manganese(III) complexes also display a low energy transition plus two higher energy transitions, and since more work has been carried out on these systems, their interpretation seems more secure. Based on such studies (outlined below), the first transition in a chromium(II) high spin spectrum is expected to be $^5A_{1g}$---$^5B_{1g}$ (assuming axial elongation, the reverse for axial compression). Subsequently, as shown in Fig.6.21, there will be two transitions

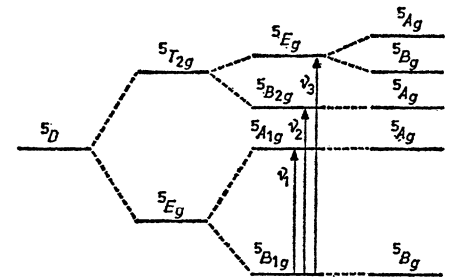


Fig.6.21 Energy level diagram for a d^4 ion in an octahedral, tetragonal (elongated along z axis) and rhombic (C_{2h}) environment (not to scale).

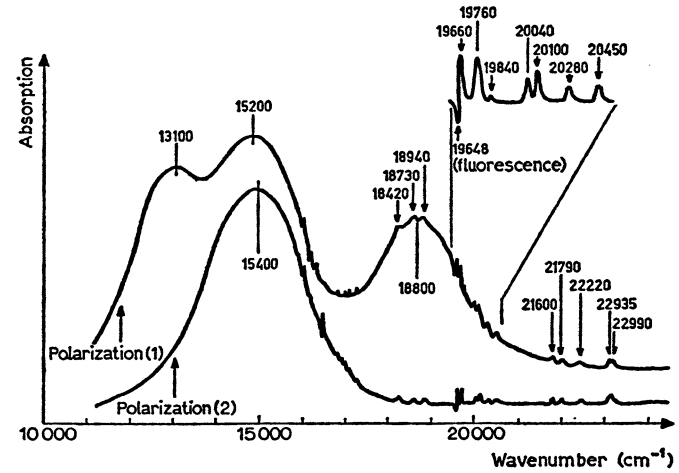


Fig.6.22 The polarised crystal spectrum of $CrCl_2.4H_2O$ at -196°C. The species is a trans-$CrCl_2(H_2O)_2.2H_2O$ chromophore with two long Cr-H_2O vectors along the z axis. The polarisations are uncertain but in (1) the electric vector may be parallel to the short O-Cr-O axis.

to the components of $^5T_{2g}$. Such a spectrum is anticipated for 'octahedral' complexes such as the $[Cr(H_2O)_6]^{2+}$ ion [341] and species such as $[Cr(en)_3]^{2+}$ [365,372]. Lower symmetry complexes, such as CrL_4X_2 will also show such split spectra but due to the lower symmetry field rather than the Jahn-Teller effect (See Table 6.12). In Fig.6.22, is shown the polarised spectrum of trans-$CrCl_2(H_2O)_4$ clearly exhibiting the three bands expected [344].

The splitting of the $^5T_{2g}$ state is often much smaller than the

splitting of the ^5E_g state. In the CrL_6 systems, the transition within the ^5E_g state is often seen below 10,000 cm^{-1} while the visible band to both components of $^5T_{2g}$ is often unresolved unless low temperature or polarised light studies are undertaken. In the trans-CrL_4X_2 series, two bands are seen in the visible region. It is reasonable to suppose that the lower energy band is $^5A_{1g}$---$^5B_{1g}$ and the upper is the unresolved transition to $^5T_{2g}$. Recall that the energy separation of the two components of $^5T_{2g}$ (Fig.6.21) will be $-2d\pi$ (Chapter 1) and that of ^5E_g, $-(8/3)d\sigma$. There is no difficulty in anticipating a fairly large $d\sigma$ for both the CrL_6 (with J-T distortion) and the CrL_4X_2 series, but, for Cr(II), pi bonding is expected to be small (Chapter 9) and hence the splitting of $^5T_{2g}$ should be small as observed.

Note that in a series of complexes trans-CrL_4X_2 the lower energy transition within the components of ^5E_g, should increase in energy as the the crystal field of X decreases, i.e. as the difference between the field strengths, or more exactly, the difference between the e_σ values of L and X increases. This is clearly seen to occur in, for example, the N-Methylimidazole complexes shown in Table 6.12.

The splitting of the main absorption band into B_{2g} + E_g is especially evident in fluorochromium(II) species such as $MCrF_3$, and M_2CrF_4 [351,377]. The crystal field analysis and assignments [377] lead, in the AOM model to large values of $d\sigma$ (-3300 - -3600 cm^{-1}) and quite large values of $d\pi$ (-1465 - -1675 cm^{-1}) implying that fluoride ion does enter into a pi interaction with Cr(II). Note that the sign of $d\pi$ is consistent with an axial elongation, the fluoride ion being a pi donor. The axial ligand field strengths are very low in these compounds (Chapter 9).

Note that many spin-forbidden bands can be anticipated for a d^4 species (see Tanabe-Sugano diagram Fig.2.11) and are often observed (e.g. Fig.6.22). Thus, for example, the species $RbCrCl_3$ and $CsCrCl_3$ show rich, narrow, spin forbidden spectra in the range 15,000 - 19,000 cm^{-1}. These species are linear chain molecules in which the octahedra are formed by octahedra sharing faces in one dimension, with little interaction between adjacent chains. An interesting feature of the spectra of these molecules is the presence of a 'double exciton' band at 22,000 cm^{-1}, corresponding to the pairwise excitation of two adjacent Cr(II) ions to the $^5T_2$---^5E transition. This particular study involved both polarisation and variable temperature data [381].

Five-Coordinate complexes: The quadridentate ligand tris(2-dimethylaminoethyl)amine (Me_6tren) forms five coordinate trigonal bipyramidal Cr(II) complexes of formula [Cr(Me_6tren)X)]X where X is a halogen [373]. There is a broad asymmetric band in the region 10,000 - 15,000 cm^{-1} which contains at least two strong transitions, possibly to the two ^5E states arising from ^5D in C_3, the higher energy transition being weaker and sometimes only observed as a shoulder. Mixed phosphine-amine trigonal bipyramidal chromium(III) complexes have also been reported [412].

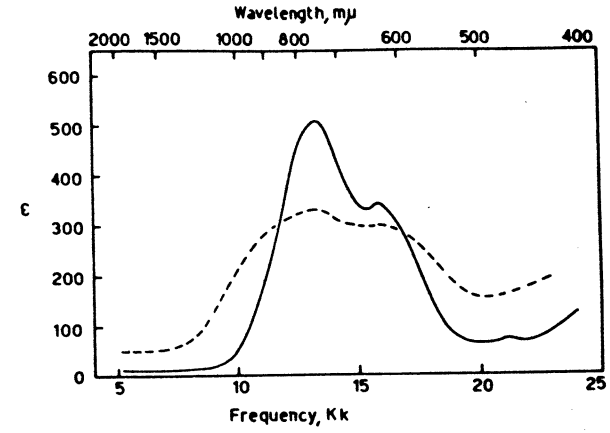


Fig.6.23 Reflectance (hatched line) and solution (CH_2Cl_2) spectra of trigonal bipyramidal [Cr(NP_3)]I [412].

Their spectra are similar to those of Me_6tren but are significantly more intense and are shifted to higher energy, though with similar band shape (Fig.6.23). A folded macrocyclic complex of Cr(II) is also regarded as five coordinate [356], and has a similar spectrum. Note these bands are not so high in energy as generally seen for tetragonal complexes of similar ligands [356].

Four-Coordinate complexes: Rigorously square planar Cr(II) complexes are, with the exception of the porphyrins and phthalocyanines, unknown, though six coordinate tetragonally distorted systems may approach square planarity. The distorted tetrahedral $[CrCl_4]^{2-}$ ion exhibits a broad band at 9800 cm^{-1}, in lithium chloride/potassium chloride eutectic above 670 K attributed to ^5E---$^5T_2$ [13]. The corresponding $[CrBr_4]^{2-}$ ion may exist in acetonitrile solutions of chromous bromide containing excess bromide [350]. Such solutions display an intense absorption band near 10,000(225) cm^{-1}. CdSe [382] provides tetrahedral sites for Cr(II) impurity ions yielding a single peak at 5000 cm^{-1} assigned to ^5E---$^5T_2$. Similar spectra are recorded in ZnS [383], CdS [383], ZnSe [384,385,513] and CdTe [259].

Other complexes: The gas phase spectrum of the linear triatomic $CrCl_2$ have been analysed [378]. Two transitions are observed at 5400 and 9000 cm^{-1} [379]. Within a linear field (Fig.1.4), in the absence of polarisation behaviour (impossible to obtain in the gas phase) we are led to consider two alternate ground states (σ_g^+ and π_g) and four possible assignments for the two transitions. All can be fitted to the AOM model, but only one yields chemically acceptable values of the AOM parameters (both positive, and e_σ(Cl)

> $e_\pi(Cl)$).

The ground state is found to be $^5\Sigma_g^+$, with the two transitions assigned to $^5\Pi_g$ and $^5\Delta_g$ respectively. The AOM parameters are found to be $e_\sigma(Cl) = 4500$ and $e_\pi(Cl) = 1800$ cm^{-1} [378].

MOLBYDENUM(II) d^4

Mononuclear molybdenum(II) species have not been seriously investigated via electronic spectroscopy. However the hepta-coordinate $[Mo(CNR)_7]^{2+}$ species do show a band near 24,400 cm^{-1} assigned to the transition $b_2(yz)$---$a_1(z^2)$ in addition to higher energy charge transfer bands which are observed [693,694].

MANGANESE(III) d^4

Following earlier arguments, and considering Fig.2.11, many spin forbidden transitions (those having a negative dependence upon Dq) in manganese(III) complexes should occur at lower energies than their counterparts in chromium(II) because the higher oxidation state will confer a higher value of DQ. The splitting pattern shown in Fig.6.21 will also apply to manganese(III) d^4, but the increased charge should serve to increase both e_σ and e_π values. The McClure $d\pi$ may now have a higher value than with Cr(II), and the splitting of the $^5T_{2g}$ state may now be resolved (Table 6.12). The manganese(III) ion is moderately oxidising in nature and ligand to metal CT will occur at lower energies than for chromium(II) and may obscure d-d transitions. Because of the destabilisation of the z^2 electron, metal to ligand, MLCT, transitions also occur at fairly low energies with suitable acceptor ligands.

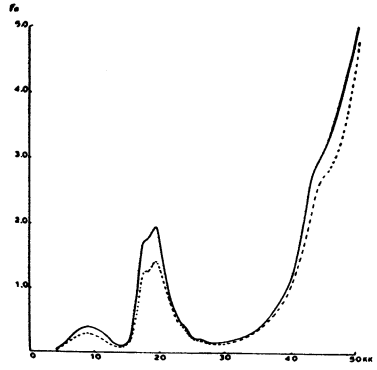


Fig.6.24 The diffuse reflectance spectra of potassium hexafluoromanganate(III) at room temperature (solid line) and -196°C (hatched line) [353].

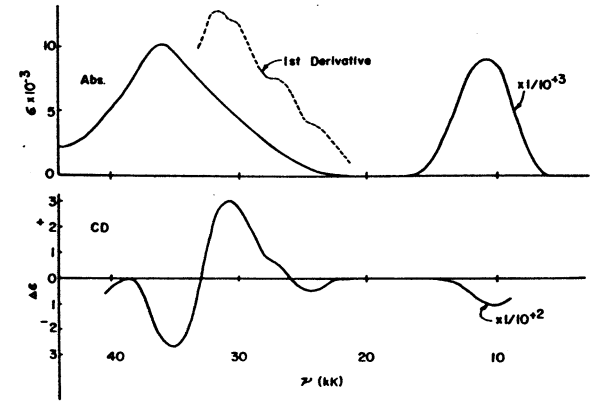


Fig.6.25 The absorption, first-derivative, and circular dichroism spectra of (+)[Mn(III)((-)-5,14-[14]aneN$_4$)Cl$_2$]PF$_6$ in acetonitrile [367].

Six Coordinate complexes: The general principles discussed above for chromium(II) also apply to Mn(III). There will be a low lying absorption associated with a transition within the split components of ^5E_g and a higher energy broad absorption (generally lying between 16,000 - 21,000 cm^{-1})[381,429,433-436] due to transition to the, possibly unresolved, components of $^5T_{2g}$. This assignment was first seriously discussed by Davis, Fackler and Weeks [381] who explored several possible assignments for the lower energy band (spin forbidden transition, forbidden CT band etc) and concluded that the assignment here presented was most likely. Because of the higher oxidation level of Mn(III) versus Cr(II), covalency should be greater, and the e_π values might well be greater leading to more commonly observable splitting in the higher energy transition to the components of $^5T_{2g}$.

Although the intensity of the spin allowed crystal field band in manganese(III) alum $[Mn(H_2O)_6]^{3+}$ is low (5 l mol^{-1} cm^{-1}), manganese(III) complexes generally give rise to fairly intense crystal field bands (See Table 6.12). This probably results from a combination of factors, namely a fairly covalent metal ligand bond, low symmetry components to the ligand field, and, fairly low lying charge transfer bands.

The spectrum of the $[MnF_6]^{3-}$ ion (Fig.6.24, Table 6.12) shows three bands attributed to the three expected transitions [353,377,1304]. The McClure parameters $d\sigma$ = -3375 and $d\pi$ = -1100 cm^{-1} are consistent with an elongated tetragonal distortion in which fluoride to manganese pi bonding is important (Chapter 9).

In an effort to unravel the spectra of tetragonal Mn(III) complexes, Bryan and Dabrowiak synthesized a racemic Ni(II)

macrocycle complex, resolved it, removed the now optically active macrocycle, and bound it to Mn(III). In this fashion they were able to obtain both absorption and CD spectra [367] (Fig.6.25). The band near 36,000 cm^{-1} is certainly too strong to be a d-d transition and is assigned as charge transfer LMCT (halogen to Mn(III)). Following the arguments presented earlier the band near 10,000 cm^{-1} is likely to be the $^5A_{1g}$---$^5B_{1g}$ transition whose energy is 4Ds + 5Dt = (-8/3)dσ (Chapter 1). The simple CD spectrum in this region seems to preclude a spin forbidden assignment for this band, since such would probably give a more complex CD spectrum in this low symmetry. Shoulders resolved between 24,000 - 31,000 cm^{-1} are then assumed to be the remaining transitions to the components ^5E_g + $^5B_{2g}$ of $^5T_{2g}$, which are split by 2dπ. Bryan and Dabrowiak favour the lower energy band at 24,400 in CD (chloride complex) to be $^5B_{2g}$---$^5B_{1g}$ (= 10 Dq). However note that in the AOM model, the energies of these two transitions will be $^5B_{2g}(3e_\sigma(N) - 4e_\pi(N))$ and $^5E_g(3e_\sigma(N) - 2e_\pi(N) - 2e_\pi(X))$. The e_π value for the macrocycle must be zero or close thereto, while that of the halogens will be positive. Thus the $^5B_{2g}$ state should lie to higher energy in these molecules. We prefer the reverse assignment, as shown in Table 6.12, to that reported by Bryan and Dabrowiak [367]. This leads to Ds = ca 1214, Dt = ca 1209 cm^{-1} and dπ (= e_π(Cl)) = ca 1200 cm^{-1}, for the chloride, and Ds = ca 1128, Dt = ca 1438 and dπ (= e_π(Br)) = ca 1900 cm^{-1}, for the bromide. This is a nice example of the utility of the AOM approach in the choice of several possible assignments (however the treatment should not be regarded as proof).

Many pseudo octahedral complexes exhibit both the $^5B_{2g}$ and ^5E_g components of a supposedly tetragonally distorted (Jahn Teller) octahedron (Table 6.12). If the assignments as shown are supposed correct (and few have been unequivocally so defined), then the McClure dσ and dπ values can be derived. While this will be discussed in more detail in Chapter 9, it is interesting that all the complexes shown in Table 6.12 with the necessarily resolved data, exhibit very low axial crystal fields and, negative dσ and dπ values (except for macrocycle systems with positive dπ). Since the axial sigma and pi components must clearly be less than those of the equatorial plane, in an axially elongated complex, this requires that all the ligands shown in Table 6.12 with resolved $^5T_{2g}$ transitions, have positive e_π(L) values, i.e. be π-donors towards Mn(III). This is not an unreasonable result. Note that if the complexes are assumed to be axially compressed, and the calculation repeated, the equatorial field strengths are too small, and the axial field strengths determined to be too large (See Chapter 1 for details of calculation). Thus use of the AOM method supports the view that the distortion must be axial elongation.

Many Schiff base complexes of Mn(III) have been studied [386-389]. They usually exhibit strong charge transfer absorption in the visible region (Mn(III) to imine MLCT) but often have lower lying weaker d-d absorption.

Few low spin complexes have been studied. Consideration of the Tanabe-Sugano diagram (Fig.2.11) leads to the conclusion that the first spin allowed transition, ^3E_g---$^3T_{1g}$ might occur at moderately high energies, but that spin forbidden bands may occur at fairly low energies. The $[Mn(CN)_6]^{3-}$ ion has been studied and two bands reported (Table 6.12)[328,437].

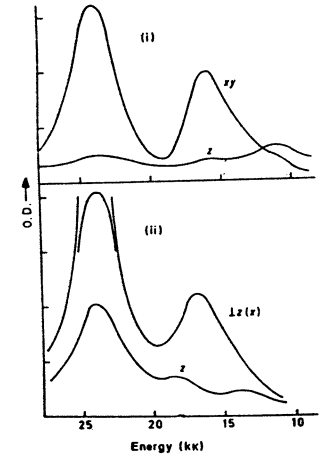


Fig.6.26 The crystal spectra of five coordinate (i) $(Et_4N)_2(In,Mn)Cl_5$ parallel and perpendicular to the C_4 (z) axis, and (ii) $(BipyH_2)MnCl_5$ [390].

Five Coordinate complexes: Manganese(III) shows quite a strong preference to form five coordinate complexes, presumably because the presence of an anti-bonding sigma electron in high spin d^4 species weakens interaction along the axis. The most well known simple example is the $[MnCl_5]^-$ ion which has a square pyramidal structure. Bellitto, Tomlinson and Furlani have studied the electronic spectra of single crystals (polarised light) containing this ion, and also of Mn(III) doped into the five coordinate $[InCl_5]^{2-}$ matrix. In both cases the metal lies above the basal plane. In the indium matrix the apical Mn-Cl bond is shorter than the equatorial bond (242 versus 246 pm resp.) while in $[BipyH]_2[MnCl_5]$, the apical bond length is greater (253 versus 234 pm)[390]. Moreover although the indium system has C_{4v} symmetry, there is angular distortion in the pure complex which has only C_{2v} symmetry. The spectra are shown in Fig.6.26. The solution intensities of these absorption bands are in the range 200 - 400 l mol^{-1} cm^{-1} for the bands below 20,000 cm^{-1} and nearly 1200 l mol^{-1} cm^{-1} for the 23,900 cm^{-1} band which is evidently charge transfer, chloride to Mn(III) LMCT, being too strong for a d-d band. The xy polarised band at 16,000 cm^{-1} is evidently the transition to the ^5E (allowed in xy, forbidden in z, electronic selection rules). With respect to the Mn(III)/In(III) doped system, Bellitto and co-workers reasonably assign the lowest energy weak band (Table

6.12) to the $^5A_1$ state (in C_{4v}) and the weak shoulder on the low energy edge of the xy polarised spectrum to $^5B_2$, both these latter transitions being vibronically activated. Transition to $^5B_2$ is forbidden in z polarisation both electronically and vibronically. Similar data were obtained in a more recent study of $[MnCl_5]^{2-}$ systems [390].

The five coordinate Schiff base complexes $[Mn(acen)X]^+$ (acen = N,N'-ethylenebis(acetylacetone imine)) also show strong absorption near 25,000 cm^{-1} attributed to Mn(III) to imine MLCT. However weaker absorption near 16,000 cm^{-1}, whose energy is independent of the nature of the axial ligand X, is ascribed to the in-plane $xy---x^2-y^2$ transition [386].

Other complexes: Seven coordinate Mn(III) is reported to occur in some pyridine ring containing macrocycles [461]. A pair of electronic transitions (d-d) is seen near 8000 - 10,000 cm^{-1}, and a further d-d transition at about 17,000 cm^{-1}, which were analysed in terms of a simple crystal field approach for a C_{2v} $MA_3B_2C_2$ system [461].

IRON(IV), RUTHENIUM(IV), OSMIUM(IV), IRIDIUM(V) AND PLATINUM(VI) d^4

These ions are also dominated by charge transfer absorption (see Chapter 5 for detail [200])(e.g. Fe(IV) diarsine complexes [438]) but $(t_{2g})^4$ intraconfigurational data can be obtained through absorption and emission. The $[Os(IV)X_6]^{2-}$ ions have been thoroughly studied [413-420]. Where X = Cl, Br and I, the visible spectra are dominated by charge transfer bands. However the hexafluoride does exhibit d-d absorption to at least 42,000 cm^{-1}.

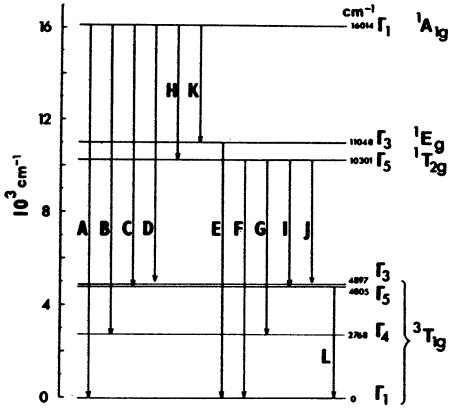


Fig.6.27 Energy level diagram from $Cs_2SnBr_6/[OsBr_6]^{2-}$. The corresponding levels in Cs_2TeCl_6:$OsCl_6$ are: $\Gamma_1(^1A_{1g})$ 17,025; $\Gamma_3(^1E_g)$ 11,952; $\Gamma_5(^1T_{2g})$ 10,520; $\Gamma_3(^3T_{1g})$ 4983; $\Gamma_5(^3T_{1g})$ 4884 and $\Gamma_4(^3T_{1g})$ 2786 cm^{-1} [422].

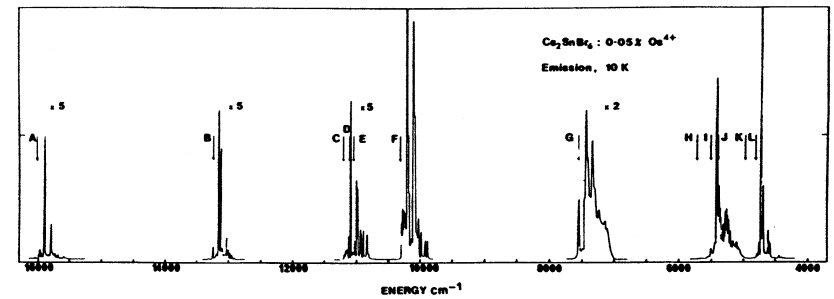


Fig.6.28 The 514.5nm excited luminescence spectrum of $Cs_2SnBr_6/[OsBr_6]^{2-}$ at 10 K [422].

The low spin $(t_{2g})^4$ configuration gives rise to $^3T_{1g}$ (ground term) and $^1T_{2g}$, ^1E_g and $^1A_{1g}$. There is extensive spin-orbit mixing of these states. The $^3T_{1g}$ term splits into four levels (A_{1g} (ground term), T_{1g}, T_{2g} and E_g). Transitions within this set of four levels are seen in the range 3000 - 5700 cm^{-1} in the fluoride (in absorption)[413], and between 2600 - 5200 cm^{-1}, in the other three halides (in their electronic Raman spectra)[419]. All the transitions in the 25,500 - 34,000 cm^{-1} spectra of $[OsF_6]^{2-}$ are attributed to transitions between the A_{1g} spin-orbit ground term of $(t_{2g})^4$ and T_{1g} and T_{2g} spinor components of $(t_{2g})^3(e_g)$. The MCD spectra are characterised by a series of A terms thereby excluding transitions to the A_{1g}, A_{2g} or E_g spinor components of $(t_{2g})^3(e_g)$. All these transitions are 'spin-allowed' in the sense that, transition is between the spin triplets in the limit of zero spin orbit coupling. Analysis of the many sharp vibrational components in this region of the spectrum shows vibronic activation by t_{1u} (both Γ_3 and Γ_4) and t_{2u} (Γ_6) frequencies upon which vibronic origins are built progressions in a_{1g} (Os-F stretching ground state frequency 690 cm^{-1}, and excited state frequency 660 cm^{-1}).

Extremely sharp line luminescence has been observed [422, 423,746,747,946] through excitation of the $[OsBr_6]^{2-}$ and $[OsCl_6]^{2-}$ ions. Emission is observed from each of the spinor states of the $(t_{2g})^4$ configuration (see Figs.6.27 & 6.28) [422]. Transitions from the three 'spin singlet' components of $(t_{2g})^4$ are clearly observable in this luminescence experiment, though they are very weak in absorption [413]. The sequence of spinor states derived from $^3T_{1g}$ has been studied in the tetragonal site present in $K_2SnCl_6/Os(IV)$ by detailed analysis of the vibrational selection rules in this lower symmetry [946].

The $Os(IV)/Cs_2ZrCl_6$ system shows absorption at ca. 4700, 10,500 and 17,100 cm^{-1} respectively assigned to the $T_{2g}(^3T_{2g})$, $T_{2g}(^1T_{2g})$ and $A_{1g}(^1A_{1g})$ states of the $(t_{2g})^4$ configuration. Each has associated vibrational structure [416].

The spectrum of the $[IrF_6]^-$ ion [522,528] shows absorption below 20,000 cm^{-1} due to spin orbit-components of the various states arising from $(t_{2g})^4$ [526,528] [4,000 ($^3T_{1g}$), 6550 ($^3T_{1g}$), 12,900 ($^1T_{2g}$,^1E_g), 19,800 ($^1A_{1g}$), 24,200 (^5E_g), and (33,000) $(t_{2g})^4$---$(t_{2g})^3e_g$ cm^{-1}] but higher energy d-d transitions are less well defined [526].

The spectrum of PtF_6 has been reported [339]; the usual intraconfigurational sharp narrow bands are observed. Allen and Warren [526] suggest these are transitions to $^3T_{1g}(\Gamma_4)$ (at 3340), $^3T_{1g}(\Gamma_5,\Gamma_3)$ (at 5240,5420), $^1T_{2g}(\Gamma_5)$, $^1E_g(\Gamma_5)$ (at 12,000) and $^1A_{1g}(\Gamma_1)$ (at 16,000 cm^{-1}).

6.2 (v) d^5 Cr(I), Mn(II), Re(II), Fe(III), Ru(III), Os(III), Co(IV), Rh(IV), and Ir(IV)

The ground term is the orbital singlet ^6S in high spin complexes. It cannot be split by a crystal field of any symmetry. Since the high spin d^5 ion is its own 'hole equivalent', the same energy level diagram may be used for all cubic groups. The absence of any other spin sextet terms requires that all transitions in high spin d^5 complexes are spin forbidden, as well as Laporte forbidden - thus they will generally be very weak.

As a consequence of the sharpness of many bands, especially at low temperatures, it is possible to obtain very accurate measurements of electronic transition energies with d^5 complexes. Such spectra therefore provide a good test for the theory. Spectroscopic assignments for many of the complexes investigated in the literature [439-448] are shown in Table 6.13. The assignments are based upon the early work by Tanabe and Sugano [449] and by Orgel [450]. Transitions from ^6S to spin doublet levels are highly forbidden (change in spin of 2) and are rarely to be observed. The spin quartets arise from the configurations $(t_{2g})^4(e_g)$, $(t_{2g})^3(e_g)^2$ and $(t_{2g})^2(e_g)^3$, with strong field octahedral crystal field energies of -10 Dq, 0 and +10 Dq relative to ^6S.

In an octahedral field, the $(t_{2g})^4(e_g)$ and $(t_{2g})^2(e_g)^3$ each give rise to $^4T_{1g}$ and $^4T_{2g}$ states. The $(t_{2g})^3(e_g)^2$ configuration gives rise to the spin quartets - $^4A_{1g}$, $^4A_{2g}$, $^4T_{1g}$, $^4T_{2g}$ and 2 ^4E_g. It follows that the $^4A_{1g}$, $^4A_{2g}$ and ^4E_g states have energies independent of any crystal field, and will appear as horizontal lines in the Tanabe-Sugano diagram (Fig.2.13). In the weak field limit, $^4A_{1g}$ and one ^4E_g originate from ^4G, $^4A_{2g}$ from ^4F, and the remaining ^4E_g from ^4D. Hence the transitions to these levels in the spectra of d^5 complexes provide direct evaluation of the energies of the ^4G, ^4F and ^4D terms. The $^4T_{1g}$ and $^4T_{2g}$ levels will be scrambled between the three configurations and will have energies dependent upon the ligand field.

In practice transitions to such lines are generally very narrow and can often be readily picked out from a spectrum. In general acentric complexes will have more intense spectra than centrosymmetric complexes providing one means for distinguishing stereochemistries in this d^5 area.

CHROMIUM(I) d^5

Data for chromium(I) complexes are sparse. However the spectra of some nitrosyl derivatives have been reported [458,459,510]. In the systems $[Cr(I)L_5NO]^{n-}$ (L = CN,NH_3), the sequence of energy levels is purported to be:

$$e(xz,yz)+\pi(NO^+) < b_2(xy) < e(\pi^*NO^+) < b_1(x^2-y^2) << a_1(z^2)$$

where there is an important interaction between the xz,yz orbitals of the metal and the pi orbitals of the nitrosyl group, assumed to be NO^+.

For L = CN, three bands are seen at 13,700, 22,200 and 27,320 cm^{-1}, while for L = NH_3, the bands occur at 17,480, 22,200 and 28,170 cm^{-1}. Upon cooling the cyanide complex, the first two bands show no change in intensity, while the third decreases in intensity. In the case of the ammonia derivative, the first peak does not change in intensity upon cooling, but the second and third bands do. In both cases the 22,200 cm^{-1} band shows significant vibrational structure with vibrational spacings of 390 - 520 cm^{-1}. The ground state, in C_{4v} symmetry, is the low spin ($e^4b_2^1$) and the three transitions are assigned, in the case of L = CN, in increasing order of energy:

^2E----$^2B_2$ [$e(xz,yz)$---$b_2(xy)$]	allowed in C_{4v} symmetry.
^2E----$^2B_2$ [$e(xz,yz)$---$e(\pi^*(NO^+))$]	allowed in C_{4v} symmetry.
$^2B_1$---$^2B_2$ [$b_2(xy)$---$b_1(x^2-y^2)$] = 10 Dq	forbidden in C_{4v} (vibronic).

The third band energy of 27,320 cm^{-1} (L = CN) which is formally equal to 10 Dq, is close to that observed for the $Cr(CN)_6^{3-}$ ion (10 Dq = 26,600 cm^{-1}).

In the case of the ammonia derivative, these authors [458] assign the structured 22,200 cm^{-1} band to the $^2B_1$---$^2B_2$ transition since the energy is similar to 10 Dq for $Cr(NH_3)_6^{3-}$, and since, unlike the cyanide derivative, the intensity of this band decreases with cooling. However the very close similarity in shape and structure of the 22,200 cm^{-1} bands in the two complexes argues strongly for the same assignment in both.

Two macrocycle containing chromium(I) nitrosyl derivatives have been reported by Busch and co-workers [615]. The sequence of energy levels is proposed to be:

$$e(xz,yz)+\pi(NO^+) < b_2(xy) < a_1(z^2) < b_1(x^2-y^2) < e(\pi^*(NO^+))$$

and three bands near 15,300, 22,200 and 26-28,000 are assigned to the $^2A_1$---$^2B_2$, $^2B_1$---$^2B_2$ and ^2E---$^2B_2$ transitions respectively.

Manganese(II) d^5

Manganese(II) high spin complexes are very weakly coloured, often appearing pink. A typical manganese(II) spectrum, that of MnF_2, is shown in Fig.6.29. Two of the bands correspond with the $^4A_{1g}(G), ^4E_g(G) \leftarrow\!\!- {}^6A_{1g}$ (this pair of transitions is degenerate in octahedral symmetry) and the $^4E_g(D) \leftarrow\!\!- {}^6A_{1g}$ transition, whose energies are 10B + 5C and 17B + 5C respectively. Variations in these band energies as a function of ligand (and d^5 metal) provide a simple measure of their nephelauxetic ratios.

The assignments in Table 6.13 are obtained by fitting the observed spectrum to the Tanabe Sugano diagram. The three lowest energy bands may be assigned to the $^4T_1(G)$, $^4T_2(G)$ and $^4E, ^4A_1(G)$ transitions, and values of Dq, B and C computed therefrom. These values are then used to calculate the higher energy transitions. A reasonable fit to within a few hundred wavenumbers can usually be obtained by this method. The overall agreement can be improved by a least squares fitting of all the bands. Any misassignment should show up as a fairly large rms deviation for one or more bands. The spectrum of $MnCl_2.2H_2O$ was fitted in this manner [448] and the values B = 630, C = 3600 and Dq = 800 cm^{-1} reproduce all the bands in the spectrum (with the exception of the $^4T_{1g}(P)$) with an accuracy of 2δ. Somewhat better agreement is obtained by including the spin-orbit coupling coefficient in the calculation [451].

Manganese(II) complexes are generally fairly ionic, in part due to their zero crystal field stabilisation energy. The molar extinction coefficients of octahedral species are in the range 10^{-2} - 10^{-1} l mol^{-1} cm^{-1}. Tetrahedral and other acentric species have somewhat higher intensities but usually less than 10 l mol^{-1} cm^{-1}, except when low lying charge transfer bands, or heavy atom ligands provide an intensification mechanism. Note also that intensification can also occur through cooperative exchange processes in polynuclear species such as demonstrated in the polarised spectrum of the trigonally distorted octahedral $RbMnBr_3$ [582].

It is rarely possible to identify many, if any, of the d-d bands of a manganese(II) complex of an organic ligand since even the very weak tail of uv organic absorption tailing into the visible can obscure them.

Similar low temperature crystal data were reported for the $[MnCl_4]^{2-}$ and $[MnBr_4]^{2-}$ ions [472]. Both spin-orbit and vibrational structure were observed, again with the totally symmetric Mn-X mode being evident (Fig.6.30). The spectra could be fitted with Dq = 330, B = 650 and C = 2990 (chloride), and Dq = 310, ,DQ 9075, B = 630, and C = 3024 cm^{-1} (bromide). In both the halide and MnS_4 chromophore (and in manganese(II) spectra in general) luminescence could be observed from the lowest lying state (zero phonon line of $^4T_1 \leftarrow\!\!- {}^6A_1$), a characteristic green emission for tetrahedral Mn(II). Some very weak absorption observed in this study may be highly forbidden sextet-doublet transitions.

Table 6.13

ABSORPTION SPECTRA OF MANGANESE(II) DERIVATIVES

	MnF_2	$MnCl_2$	$MnBr_2$	$Mn(H_2O)_6^{2+}$	$Mn(en)_3^{2+}$	$MnCl_2.2CH_3CN$	$MnBr_4^{2-}$
$^4T_{1g}$	19,440	18,500	18,450	18,700	15,000	19,000	[b]21,140
$^4T_{2g}$	23,500	22,000	21,650	23,120	19,900	22,400	22,100
$^4A_{1g}, ^4E_g(G)$	25,190	23,590	23,084	24,960	23,600	23,800	22,755
	25,300		23,825		23,550		25,235
	25,550						
$^4T_{2g}(D)$	28,120	26,750	26,520	27,980	26,500	27,200	25,890
	28,370						
$^4E_g(D)$	30,230	28,065	27,505	29,750		28,550	26,300
$^4T_{1g}(P)$	33,060	36,500	29,950	32,960		31,200	27,345
$^4A_{2g}(F)$	39,000	38,400	34,800	40,820b		35,000	34,300
$^4T_{1g}(F)$	41,400	40,650	37,400			37,700	35,700
$^4T_{2g}(F)$	42,370	38,750					36,150
Ref.	441	440	440	439	570	447	472

a) This may also be a $^4T_{1g}(P)$ transition b) Since the relative magnitudes of 10 Dq differ for octahedral and tetrahedral derivatives, the relative order of the terms for such species are not necessarily the same.

Related data are presented in refs. [474-482,587].

Interestingly, excitation spectra have been reported for this emission [469,471,472], and these reproduce the absorption spectra generally very well; indeed the resolution of the excitation spectra can be better than the absorption spectra (compare data in [470,471]).

> Excitation spectra are obtained in a dual monochromator spectrometer, where the intensity of emission of a single emitting wavelength, is monitored as a function of the exciting wavelength.

Note that [442,470] the spectra of tetrahedral manganese(II) complexes fall into three well defined regions because of the fairly low Dq values involved. Thus transitions to the components of the ^4G term lie between 19,000 - 25,000 cm^{-1}, those to ^4P and ^4D between 25,000 - 29,000 cm^{-1}, and those to ^4F lie between 35,000 - 38,000 cm^{-1}.

Single crystal absorption, excitation and emission spectra were also obtained for the D_3 six coordinate $[Mn(OMPA)_3]^{2+}$ ion [473]. The D_3 distortion is revealed in the splitting (at 5 K) of the $^4A_1$, ^4E transitions which are accidentally degenerate in a strictly octahedral environment. The excitation and absorption data may be

(Table 6.13 cont)

AND IRON(III) DERIVATIVES

	$MnCl_4^{2-}$	$Mn(NCO)_4^{2-}$	$Fe(H_2O)_6^{3+}$	FeF_6^{3-}	$Fe(urea)_6^{3+}$	$FeCl_4^-$
$^4T_{1g}$	21,250	20,400	12,600	14,200	12,500	15,600
$^4T_{2g}$	22,235	22,400	18,500	19,700	17,100	16,300
$^4A_{1g}, ^4E_g$	23,020	23,200	24,300	25,400	23,100	18,800
			24,600		23,400	
$^4T_{2g}(D)$	26,080			28,800		20,100
$^4E_g(D)$	26,710			30,200		22,400
$^4T_{1g}(P)$	27,770					27,400CT
						31,800CT
$^4A_{2g}(F)$	33,300	35,800				41,100CT
$^4T_{1g}(F)$	34,500		42,000CT			
$^4T_{2g}(F)$	36,650	36,700				
	472	446	176	176	176,353[c]	176

c) There are some differences in band energies between these two reports.

fit by Dq = 956, DQ = 26,290, DSIG = -2000, DTAU = 2000, B = 670 and C = 3620 cm^{-1}.

The octahedral MS_6 chromophore has been studied using the tetra-alkyl dithiooxamides. The tetraethyl derivative, Et_4D formed a complex $Mn(Et_4D)_3$ exhibiting d-d absorption at 18,520(0.15) ($^4T_{1g}(G)$) and 22,220(0.04) ($^4T_{2g}(G)$) [483].

Five coordinate Mn(II) systems have been studied by Ciampolini and co-workers [485]. The series $MnL_4(ClO_4)_2$ where L is Ph_3PO, Ph_3AsO etc is five coordinate C_{4v} with one coordinated perchlorate group. Complexes such as $[Mn(Me_6tren)X]^+$ (D_{3h}) were also studied. The effect of both C_{4v} and D_{3h} distortion on the d^5 energy levels was calculated and the spectra fitted.

Although polynuclear species are considered elsewhere (Chapter 7), note the interesting species $(Me_3NH)_3Mn_2Cl_7$ which contains both octahedral and tetrahedral Mn(II) sites. The former exhibits significant C_{3v} distortion [571]. Single crystal data confirm the C_{3v} symmetry through identification of appropriately polarised components while emission occurs from both sites (^4E---^6A in the six coordinate (C_{3v}) site and $^4T_1$---^6A in the tetrahedral site). Similar data are reported for the C_{3v} distorted Me_4NMnCl_3. Note that the polynuclear species also exhibits triboluminescence.

Triboluminescence is luminescence caused by the application of mechanical stress to a crystal; it is moderately common and gives rise to luminescence which is usually the same as that observed by conventional excitation.

Reference to the d^5 Tanabe Sugano diagram (Fig.2.13) shows that low spin complexes might have rich spin allowed d-d spectra, if the d-d bands are not obscured by charge transfer. However few low spin Mn(II) complexes are known, and those which are, invariably involve redox active ligands. Thus the $[Mn(CN)_6]^{4-}$ ion shows [487,488] a d-d band at 27,420 cm^{-1} ($^2T_{1g}, ^2A_{2g}$---$^2T_{2g}$) followed by charge transfer to higher energy.

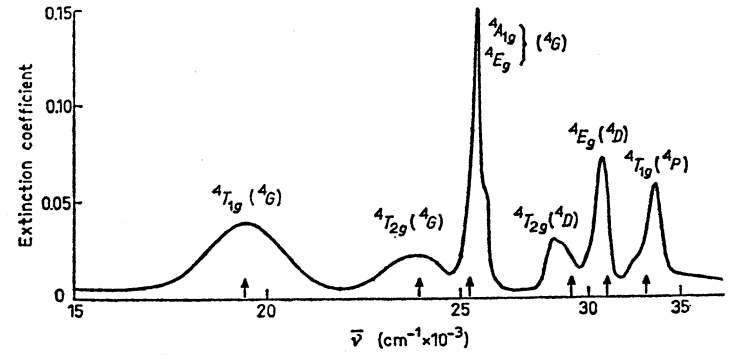


Fig.6.29 The absorption spectrum of octahedral MnF_2 at room temperature. The short arrows indicate calculated energies.

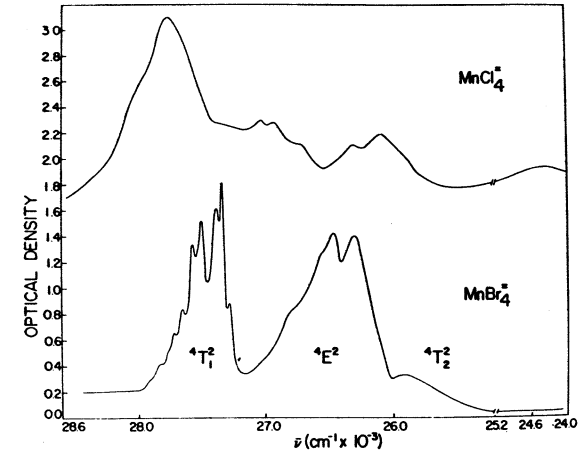


Fig.6.30 Segments of the single crystal spectra of the tetrahedral $((CH_3)_4N)_2MnCl_4$ and $((CH_3)_4N)_2MnBr_4$ at ca. 6 K [472].

RHENIUM(II) d^5

Spectra for this species have rarely been reported. Morazzoni, Cariati and Valenti report data for the low spin, tetragonally distorted $Re(DPE)_2X_2$ [497] (DPE = 1,2-bis(diphenylphosphino)ethane, and X = Cl,I) with S = 0.5. Bands in the region 9500 - 10,500(27-44) are assigned as d-d transitions within the split $(t_{2g})^5$ configuration. Charge transfer from phosphorus to Re(II) is deemed to occur as low as 14,300 cm^{-1}. Rhenium(II) cluster complexes are discussed in Chapter 7.

IRON(III) d^5

Because of the greater oxidising power of Fe(III) relative to Mn(II), ligand to metal charge transfer bands often obscure the very low intensity d-d absorption. In general where observed, the d-d band intensities are somewhat larger than those of Mn(II), especially in acentric systems (presumably because of increased metal ligand bond covalency). Some data are shown in Table 6.13.

Bands whose energies depend solely upon B and C should lie lower in Fe(III) relative to Mn(II) analogs. The Dq value for Fe(III) will be higher than for analogous Mn(II) systems. The relative band positions will depend upon whether they have a positive or negative dependence upon Dq. Note for example, that the forbidden transitions $^4T_1, ^4T_2 \text{---} ^6A_1$ have a strong negative dependence upon Dq and therefore tetrahedral FeL_4 species will exhibit these transitions at greater energies than octahedral FeL_6 species, the reverse of the usual situation, e.g. the $^4T_1 \text{---} ^6A_1$ transition for $Fe(III)O_6$ and $Fe(III)O_4$ chromophores should lie near 11,000 and 22,000 cm^{-1} respectively [1345].

Kambli and Gudel [1347] report the single crystal spectra of species containing the $[FeCl_5(H_2O)]^{2-}$ ion (see Fig.5.7). Broad sextet-quartet transitions ($^4T_{1g}, ^4T_{2g} \text{---} ^6A_{1g}$, intensities ca 0.25 l mol^{-1} cm^{-1}) show a clear dichroism and splitting (ca 2000 cm^{-1}) as a consequence of the C_{4v} field, while the spin flip ^4E_g, $^4A_{2g}$ are less obviously affected though their intensity is undoubtedly enhanced through exchange interactions in the solid state.

The single crystal spectra of both $[FeCl_4]^-$ and $[FeBr_4]^-$ have been reported by Vala and co-workers [489-491], and compared with their manganese analogs. In general, while the manganese tetrahedral anions show totally symmetric vibrations built upon spin-orbit origins, the iron species exhibit significant Jahn Teller distortion as shown by the coupling of non-totally symmetric t_2 modes upon spin-orbit origins whose separation had been quenched by the Ham effect. The $Fe(III)S_4$ tetrahedral chromophore is reported to exhibit the $^4T_1 \text{---} ^6A_1$ transition at 13,400 cm^{-1} [1346].

Fatta and Lintvedt [492] report the transition energies ($^4T_{1g}(G)$ and $^4T_{2g}(G)$) for a series of six coordinate acetylacetonate derivatives. The intensities are about 0.4 and 1.2 l mol^{-1} cm^{-1} respectively. The Dq parameter is not very sensitive to the substituents on the acetylacetone, but the Racah parameter B

is, providing a nephelauxetic sequence (see Chapter 9 for more detail):

H > t-Bu,t-Bu > CH_3,H > CH_3,CH_3 > CH_3,CF_3 > CF_3,Ph = CH_3,Ph > CF_3,CF_3 > Ph,H > Ph,Ph.

The d-d absorption in six coordinate FeS_6 species is rarely observed since the strong Fe-S CT absorption will obscure such weak bands. However weak infrared absorption reported for $Fe[(Ph_2P(S)S)]_3$ at 9000 and 12,000 cm^{-1} is presumably crystal field in origin [493].

The five coordinate species $[Fe(dtc)_2X]^{2+}$ have spin quartet ground terms and show in the near infrared (6500 - 8400 cm^{-1}) weak, sharp peaks which are probably low lying spin forbidden states. Other bands are obscured in these cases by charge transfer bands [509]. The trigonal bipyramidal $FeCl_3(4\text{-cyanopyridine})_2$ [1202] exhibits spin forbidden bands between 5000 and 20,000 cm^{-1} and presumed charge transfer absorption between 27,000 - 33,000 cm^{-1}.

The three coordinate $Fe[N(SiMe_3)_3]_3$ exhibits bands at 16,100(400) ($^4A_1'', ^4A_2'' \text{---} ^6A_1'$) and 20,000(450) ($^4E' \text{---} ^6A_1'$) cm^{-1} [136] which are remarkably intense for d-d transitions of Fe(III), but are so assigned because they surely are at too low an energy to be charge transfer in origin. It is evident however that they must be significantly mixed with CT absorption beginning as low as 25,300 cm^{-1}.

Low spin d^5 has a ground state of $^2T_{2g}$ (in O_h) arising from $(t_{2g})^5$. The spin orbit coupling coefficient for iron(III) is too small for intraconfigurational transitions to be seen. The $(t_{2g})^4(e_g)$ configuration gives rise to $^4T_{1g}$, $^4T_{2g}$, $^2A_{1g}$, $^2A_{2g}$, 2 x ^2E_g, 2 x $^2T_{1g}$ and 2 x $^2T_{2g}$.

Note that there are 60 spin-orbit ways of writing the configuration $(t_{2g})^4(e_g)$ and that the total dimension of the group theoretical states is also necessarily 60.

Low spin Fe(III) ($^2T_{2g}$ in O_h) is uncommon but is represented by the intensively studied ferricyanide ion [487,498-500,508,512]. The absorption features in the spectra are primarily charge transfer in origin (Chapter 5) but various weak features in a crystal of KCl (or NaCl) doped with $[Fe(CN)_6]^{3-}$ are assigned as d-d transitions to doublet and quartet states (in KCl) at 20,000 ($^4T_{1g}$), 28,736 ($^4T_{2g}$), 30,960 ($^2A_{2g}, ^2T_{1g}$) and 36,500 (^2E_g) cm^{-1}, (see Fig. 2.13). The substituted series $[Fe(CN)_5L]^{n-}$ are also well known [508,588]. However since the d-d bands generally lie as weak shoulders on charge transfer absorption in these various cyano species, the crystal field assignments are not yet unequivocal [498,508,512].

RUTHENIUM(III) AND OSMIUM(III) d^5

Ruthenium(III) complexes are almost invariably six coordinate unless doped in special lattices. Charge transfer absorption often occurs at relatively low energies while the crystal field parameter

Table 6.14

ABSORPTION SPECTRA OF LOW SPIN IRON(III), RUTHENIUM(III), COBALT(IV), RHODIUM(IV) AND IRIDIUM(IV)[a]

Complex	Spin Forbidden	$^2A_2, ^2T_1$	$^2T_2$	^2E etc	Ref
$Fe(CN)_6^{3-}$	23,500CT	(30,700)(876)	33,000CT	(35,000)(864)	487
$Fe(CN)_6(NCS)^{2-}$	19,200CT	(25,300)(520)?	CT	(33,300)?	588
RuF_6^{3-}	10,000($^4T_{1g}$)	20,000		(26,500)	425
	15,400($^4T_{2g}$)			34,000	
$RuBr_6^{3-}$		(16,500)(104)			501
$RuCl_3$/LiCl/KCl[b]	8420($^4T_{1g}$)	16,000(26)	17,870(38)	19,230(23)	
	12,020($^4T_{2g}$)			21,730(44)	
				23,560(34)	503
$Ru(H_2O)_6^{3+}$	16,670($^4T_{1g}$)	25,510(40)		30,000–	565
				35,000	504
$Ru(NH_3)_6^{3+}$	23,000($^4T_{1g}$)	30,000(50)			506
$Ru(NH_3)_5Cl^{2+}$		36,200(530)			511
$Ru(NH_3)_5NCO^{2+}$	29,000CT[c]	35,300(231)	44,000(331)		511
$Ru(NH_3)_5NCS^{2+}$	20,200CT[d]	31,100(401)	36,000(209)		511
$Ru(NH_3)_5OAc^{2+}$	33,800CT[e]		45,000(436)		511
$Ru(NH_3)_5NCSe^{2+}$	12,000CT[f]	22,000(325)	30,300(811)		511
	17,100CT[g]				
$Ru(ox)_3^{3-}$	15,850(11)	26,600(350)		34,700(320)	
	20,400(28)				505
$Ru(dtc)_3$		16,900		20,800	494
CoF_6^{2-}	6400($^4T_{1g}$)	17,300	18,900	21,400	
	10,300($^4T_{2g}$)			24,400?	
				32,200?	520
RhF_6^{2-}	11,600($^4T_{1g}$)	19,200	21,200		
	16,000($^4T_{2g}$)			(27,600)	
				(32,800)	425
IrF_6^{2-}[i]	6650[h]	30,000br		38,000br	
	19,800($^4T_{1g}$)[i]				
	24,900($^4T_{2g}$)				520
$IrCl_6^{2-}$	5250[h], 17,400($^4T_{1g}$,CT)				413
$IrBr_6^{2-}$	4700[h]				413

a) The assignments in this Table are those proffered by the relevant authors and are subject to change. b) Bands obtained by Gaussian analysis. c) ε(2650) d) ε(3620) e) ε(1440) also probably overlaps $^2T_{1g}$ transition. f) ε(942) g) ε(2080) h) intraconfigurational $(t_{2g})^5$. i) see also ref [420].

is quite large; thus d-d bands are frequently obscured. Though the spin-orbit coupling coefficient is greater than for iron(III),

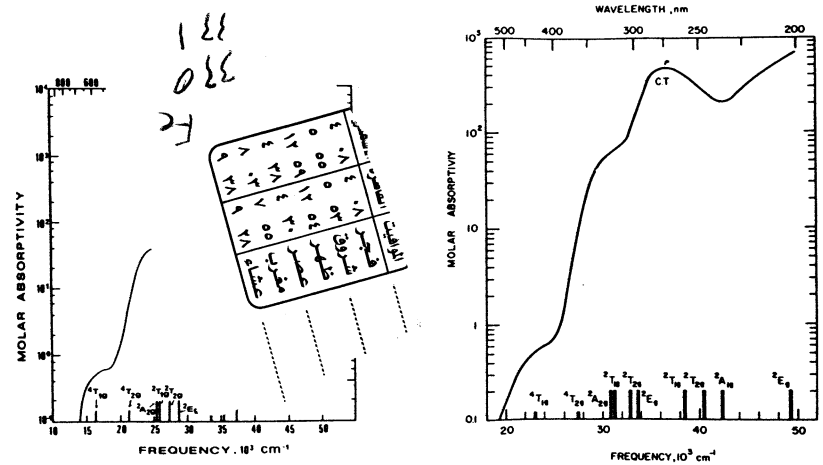


Fig.6.31 (left) The absorption spectrum of $[Ru(H_2O)_6]^{3+}$ in 1 M CF_3SO_3H. The stick diagram is calculated with Dq = 2860, B = 640 cm^{-1} and C/B = 4.0 [504]. (Right) The absorption spectrum of $[Ru(NH_3)_6](ClO_4)_3$ in water [506].

intraconfigurational transitions within $^2T_{2g}$ will still lie at too low an energy to be conveniently identified.

The two lowest absorptions, the spin forbidden $^4T_{1g}$---$^2T_{2g}$ and $^4T_{2g}$---$^2T_{2g}$ can frequently be observed (often as shoulders) with intensities of about 10 l mol^{-1} cm^{-1}. Transitions to the lower set of spin doublets arising from $(t_{2g})^4(e_g)$ can be seen with ligands which do not give rise to low energy charge transfer (Table 6.14). The approximate energies of these states are shown in Table 3.7. Since there are some uncertainties in the detailed assignments, the calculated crystal field parameters derived from the spectra (often based on the assumption that C = 4B) should be treated with some caution (see Chapter 9).

The spectra of the $[Ru(NH_3)_6]^{3+}$ and $[Ru(H_2O)_6]^{3+}$ ion [504,506] are shown in Fig.6.31, together with proposed assignments. The many states arising from $(t_{2g})^4(e_g)$ are also seen in a gaussian analysis of the $RuCl_3$ spectrum in a LiC-KCl eutectic [503]. They are perhaps most clearly seen in K_3RuF_6 [425] (Table 6.14). Low symmetry ruthenium(III) complexes are quite common but their spectra have rarely been clearly scrutinised. Complexes such as $RuCl_3(CH_3OH)(AsPh_3)_2$ and the trigonal bipyramidal $[RuCl_3(AsPh_3)_2]$ have quite rich spectra whose low lying bands have been assigned as d-d in origin [502]. However since the intensities of these bands range from 600 to 6200 l mol^{-1} cm^{-1}, some caution should be exercised in accepting these assignments. Nevertheless, in the case of the trigonal bipyramidal system, the d-d band energies have been calculated and can be fitted. The presence both of an acentric

Table 6.15 cont

Six Coordinate Complexes

Chromophore	^5E_g---$^5T_{2g}$ [a]	-(8/3)dσ	Ref
$Fe(Py)_4I_2$	N_4I_2	5950	11,300	5350	567
$Fe(Py)_4(NCO)_2$	N_6	9430	12,580	3150	567
$Fe(Py)_2(NCO)_2$	N_6	9570	7410	-2160	567
$Fe(Py)_4(NCS)_2$	N_6	9880	11,460	1580	567
$Fe(Py)_2(NCS)_2$	N_6 [e]	12,660	7300	-5360	567
$Fe(Py)_4(NCSe)_2$	N_6	(9900)	11,360	1460	567
$Fe(Py)_2(NCSe)_2$	N_6 [e]	12,820	7090	-5730	567
$Fe(IQ)_4Cl_2$	N_6	N_4Cl_2	(9100)	10,950	1850	538,539
$Fe(IQ)_4Br_2$	N_4Br_2	7800	11,450	3650	538,539
$Fe(IQ)_4I_2$	N_4I_2	5900	11,750	5850	538,539
$Fe([Me_6[14]eneN_4])phen$ [b]	N_6	11,100	19,600	8500	541
$Fe([13]aneN_4])(NCS)_2$ [b]	N_6	9100	12,000	2900	611
$Fe([15]aneN_4])(CH_3CN)_2$	N_6	9630(3.6)	14,060(3.9)	4430	611
$Fe([15]aneN_4])(NCS)_2$	N_6	10,420(6.1)	12,590(5.2)	2170	611
$Fe([16]aneN_4])(CH_3CN)_2$	N_6	9300(3.1)	12,340(4.4	3040	611
$Fe([16]aneN_4])(NCS)_2$ [b]	N_6	9120(1.3)	11,080(4.6)	1960	611
$Fe([15]aneN_4])(NO_2)_2$	N_6	11,100(10)	20,700(1600)CT		611
$Fe([ms\text{-}CRH])(N_3)_2$	N_6	11,700(55)			543
$Fe([ms\text{-}CRH]OAc)^{+2}PF_6^{-}$ [b]	N_4O_2	11,600(10)	(18,500)		543
$Fe([13]aneN_4)(NCS)_2$ (cis)	N_6	8780	11,460	-2680	611
$Fe([16]aneN_4)(NCS)_2$	N_6	9120(1.3)	11,080(4.6)	1960	611
$Fe([15]aneN_4)(CH_3CN)_2^{2+}$	N_6	9630(3.6)	14,060(3.9)	4430	611
$Fe([16]aneN_4)(CH_3CN)_2^{2+}$	N_6	9300(3.1)	12,340(4.4)	3040	611
$Fe(NP_3)(NCS)_2$	N_3P_3	17,300(395)	24,100(483)	CT?	660
Fe(II)/orthopyroxene		5400	11,000[c]		572,573
K_3CoF_6	F_6	11,400	14,500	3100	540
$Ba_3(CoF_6)_2$	F_6	10,200	16,700	6500	540

ii)

Five Coordinate Complexes--Square Pyramidal

Complex	$^5A_1$---$^5T_2$	$^5B_1$---$^5T_2$	*E	Ref
$[Fe(ms\text{-}CRH)Cl]^+ PF_6^-$	4900(3)	10,800(4)	5900	543
$[Fe(ms\text{-}CRH)Br]^+ PF_6^-$	4500(4)	11,000(5)	6500	543
$[Fe(ms\text{-}CRH)I]^+ PF_6^-$	4500(5)	11,400(5)	6900	543
$[Fe(Me_6[14]diene)Cl]^+ClO_4^-$	?	(12,800)		593

iii)

Five Coordinate Complexes--Trigonal Bipyramidal--(High spin)

Complex	Chromophore	$^5E'$---$^5E''$	$^5A_1'$---$^5E''$	Ref
$Fe(Me_6tren)Br^+$	N_4Br	3100	9800	620
$Fe(NP_3)Cl^+$	NP_3Cl	<5000	8900(66)	660

cont

Table 6.15 iii) cont

Five Coordinate Complexes--Trigonal Bipyramidal--(High spin)

Complex	Chromophore	$^5E'$---$^5E''$	$^5A_1'$---$^5E''$		Ref
$Fe(NP_3)Br^+$	NP_3Br	<5000	8700(59)		660
$Fe(NP_3)I^+$	NP_3I	<5000	8300(58)		660
$Fe(SNNMe)Cl_2$	SN_2Cl_2	(5820)	8400		706
$Fe(SNNMe)Br_2$	SN_2Br_2	5620	8470		706
$Fe(SNNMe)(NCS)_2$	$SN_2N'_2$	5740	10,320		706
$Fe(MeTPyA)Cl^+$	N_4Cl	5400(68)	9000(30)		738
$Fe(MeTPyA)Br^+$	N_4Br	5400(69)	8700(27)		738
$Fe(CyNP_3)Cl^+$	NP_3Cl	<5000	9300(60)		740
$Fe(CyNP_3)Br^+$	NP_3Br	<5000	9100(67)		740
$Fe(CyNP_3)I^+$	NP_3I	<5000	8500(49)	16,900(10)?	740
$Fe(CyNP_3)(NCS)_2$	$NP_3N'_2$	<5000	8100(64)		740

iv)

Four Coordinate Complexes

Complex	Chromophore	$^5T_2$---^5E [d]		Ref
$FeCl_4^{2-}$	Cl_4	4050(80)		507
$FeBr_4^{2-}$	Br_4	3000		534
$Fe(NCS)_4^{2-}$	N_4	4650(125)	5780(142)	535
$Fe(NCSe)_4^{2-}$	N_4	4750	6050	532
$Fe(Ph_3AsO)_2Br_2$	O_2Br_2	5000(48)		536
$Fe[(Me_2PS)_2N]_2$	S_4	3235(100)	4190(100)	547
$Fe(S_2\text{-}o\text{-}xyl)_2$	S_4	5560		1350
$Fe(dts)_2^{2-}$	S_4	5800		1351
$Fe(SPh)_4^{2-}$	S_4	5880		1352
Rd(red)	S_4 [f]	(5000)	6250	1353
$Fe(HMPA)_4^{2+}$	O_4	6950		597
$Fe(Py)_2I_2$	N_2I_2	5760		567
$Fe(L^+)Cl_3$	NCl_3	4000, 4910		580
$Fe(Qn)_2Cl_2$	N_2Cl_2	4770, 7150		580
$FeBr_2(HMPA)_2$	O_2Br_2	4650		597
$FeCl_2(TMO)_2$	O_2Cl_2	5600br		598
Fe(II)/ZnSe	Se_4	3150		589
Fe(II)/CdSe	Se_4	2400		590

a) All the six coordinate complexes, except where noted, are assumed to have a trans distorted tetragonal configuration though this has not always been proven unequivocally. In these trans six coordinate complexes, where it is clear that the axial field has a weaker sigma strength than the equatorial field, the sequence of energies is $E(^5A_1) < E(^5B_1)$ and vice versa. b) Believed to be a Cis complex. c) pyroxene der. Broad band at 2350 cm^{-1} may be a transition within the split $^5T_{2g}$ (split in C_{2v}) term. d) Assignment for regular tetrahedral species. e) These complexes have anomalously high splitting and may not have a six coordinate

structure with bridging anion. The lower energy peak can be quite sharp and may in fact be a vibrational overtone. f) reduced Rubredoxin.

Thus:

$$E(^5B_{1g} \text{----} ^5B_{2g}) = 10\ Dq$$
$$\text{and} \quad E(^5A_{1g} \text{----} ^5B_{2g}) = 10\ Dq + (8/3)d\sigma = 10\ Dq - 4\ Ds - 5Dt \qquad (6.15)$$

The value of 10 Dq cannot be derived directly from the spectrum if ^5E_g is the ground term (except by approximating it to the average energy of the split components), but note:

$$E(^5B_{1g} \text{----} ^5E_g) = 10\ Dq - 2\ d\pi = 10\ Dq + 3\ Ds - 5Dt$$
$$\text{and} \quad E(^5A_{1g} \text{----} ^5E_g) = 10\ Dq - 2\ d\pi + (8/3)d\sigma$$
$$= 10\ Dq - Ds - 10\ Dt \qquad (6.16)$$

The single crystal spectra of $Fe(thiourea)_4Cl_2$ has been described by several groups [616-618]. Sharp, narrow, vibrational overtones of the thiourea ligand are seen near 7000 cm^{-1}, and a broad polarised band near 8000 - 10,000 cm^{-1} ascribed to the two components of ^5E_g. The ground state is uncertain, there being disagreement in the various reports. McCarthy and Walker [617] report the unusual observation that the freshly prepared crystals are emerald green in colour while older crystals, or crystals which have been heated, are colourless (yet still retaining essentially the same spectrum below 12,000 cm^{-1}). The green colour arises as a consequence of two intense bands, at 16,240 and 24,660 cm^{-1}, which disappear when the crystal is heated to 150°C, and are not present in solution. They are ascribed to charge transfer arising from some small amount of impurity disulphide iron(II) complex in the crystal [617].

Spin forbidden transitions are anticipated for high spin d^6 but have rarely been assiduously studied. The $MFeX_3$ linear chain systems are of interest in this respect since exchange interactions cause an increase in intensity in the spin forbidden quintet-triplet transitions resulting in some detailed assignments [578].

Low Spin Six Coordinate Complexes: A wide variety of high crystal field octahedral and pseudo octahedral species of iron(II) have a low spin d^6 configuration; many examples are presented in Table 6.16. In the absence of obscuring charge transfer absorption, the two transitions to the $^1T_{1g}$ and $^1T_{2g}$ states are observed, often with weak lower lying absorption to the corresponding triplets. The energy separation between the two states is approximately 16B while the energy of the $^1T_{1g}$ state is 10Dq - C. If the usual assumption is made that C = 4B, then the two band energies allow estimation of 10Dq and B.

Table 6.16

ABSORPTION SPECTRA OF LOW SPIN SIX COORDINATE IRON(II), RUTHENIUM(II), COBALT(III), RHODIUM(III), IRIDIUM(III), PALLADIUM(IV) AND PLATINUM(IV) COMPLEXES

Complex	$^3T_1$	$^3T_2$	$^1T_1$ [a]	$^1T_2$	Ref
$Fe([13]aneN_4)(CN)_2$			18,800(18)	23,000(78)	611
				27,900(71)	
$Fe(Me_6[14]aneN_4(MeCN)_2$			17,900(44)	26,600(47)	544
$Fe(CN)_6^{4-}$	23,700(2.1)		31,000(345)	37,040	553,554
					487
$Fe(CNO)_6^{4-}$	(20,000)?		25,870(322)	32,000(604)	555
$Fe(CN)_5CO^{3-}$			32,790(380)		566
$Fe(das)_3^{2+}$	14,700		21,400(230)	26,600(350)	546
$c\text{-}FeCl_2(ArNC)_4$			17,400	ca(22,900)	549
$c\text{-}FeCl(SnCl_3)(ArNC)_4$			(19,850)	23,600	549
$Ru(aniline)_6^{2+}$			22,000	27,300	627
$Ru(en)_3^{2+}$			27,000(40)	33,200(91)	557
$Ru(CN)_6^{4-}$			31,000		487
$Ru(H_2O)_6^{2+}$			18,870(21)	25,640(31)	556
$Ru(dien)_2^{2+}$			27,600(95)	33,900(76)	557
$Ru(NH_3)_5(NO)^{3+}$	22,000(15)		32,800(56)		595
$Ru(NH_3)_4(NO)Cl^+$	22,700(17)		30,300(264)	CT	595
$Ru(NH_3)_4(NO)OAc^+$	23,000(22)		30,500(131)		
			33,400(162)		595
$Co(CN)_6^{3-}$			32,400(200)	39,000(140)	200
$Co(CN)_6^{3-}$			31,800(138)	38.700(109)	487
$Co(N_3)_6^{3+}$			13,880(315)	(19,500)	1099
$Co(en)_3^{3+}$	13,700 (17,500)		21,550(88)	29,600(78)	200
$Co(NH_3)_6^{3+}$	13,000(0.2)		21,200(56)	29,550(46)	200
$Co(H_2O)_6^{3-}$	8000 12,500		16,500	24,700	558
$Co(NO_3)_6$			15,200(84)	22,500(100)	
				25,500	1160
$Co(NO_3)_3$			16,400	23,150	1160
$Co(P(OMe)_3CMe)_6^{3+}$			30,800(340)	35,700(2000)?	559
$Co(dtp)_3$			13,550(450)	(19,000)	560
$Co(dtc)_3$			15,400(100)	20,900(600)	494
$Co(CNO)_6^{3-}$			25,000(266)	31,500(1100)	555
$Co(F_2PS_2)_3$			13,580	(18,490)	843
$Co(Ph_2PS_2)_3$			13,000(600)	18,200	673
$Co((C_2H_5O)_2PS_2)_3$			13,550(400)	(19,000)	256
$Co((C_2H_5)_2NCS_2)_3$			15,500(100)	20,800(600)	674
$Co(H_2NCS_2)_3$			15,700(280)	20,700(630)	674
$Co(C_2H_5OCS_2)_3$			16,200(270)	20,800(360)	674
$Co(PhCH_2CH_2CS_2)_3$			16,950(280)	22,400(2800)	674
$Co(CS_3)_3^{3-}$			16,000(300)		256

cont

Table 6.16 cont

Complex	$^3T_1$	$^3T_2$	$^1T_1$ [a]	$^1T_2$	Ref
$Co(ox)_3^{3-}$			16,600(125)	23,750(155)	200
$Co((pic)_3tach))_3^{+b}$			21,600(125)		599
$c\text{-}Co(NH_3)_4(NO_2)_2^+$			22,470(148)	30,865(3980)	640
$c\text{-}Co(NH_3)_4(ONO)_2^+$			20,000(91)	28,090(478)	640
$c\text{-}Co(en)_2Cl_2^+$			18,870(78)	26,320(69)	609
$c\text{-}Co(en)_2(NO_2)_2^+$			22,885(178)	31,055(3800)	640
$c\text{-}Co(en)_2(ONO)_2^+$			20,830(123)	30,305(933)	640
$c\text{-}Co(en)_2(ox)^{+2}$			20,080(114)	28,090(142)	607
$c\text{-}Co(en)_2F(H_2O)^+$			20,000(85)	27,800(54)	550
$c\text{-}Co(cyclen)Cl_2^+$			17,860(185)	25,640(165)	608
$c\text{-}Co(cyclen)(NO_2)^{2+}$			22,470(355)	31,250(4460)	608
$c\text{-}Co(2,2,2\text{-}tet)(ox)^+$			20,120(179)	28,010(179)	601
$c\text{-}Co(2,3,2\text{-}tet)(ox)^+$			20,000(140)	27,780(175)	604
$c\text{-}Co(3,2,3\text{-}tet)(ox)^+$			19,530(118)	27,550(177)	604
$c\text{-}Co(2,2,3\text{-}tet)(ox)^+$			20,620(125)	28,170(160)	601
$c\text{-}Co(TTP)(NCS)_2^+$			18,520(840)	23,800?	610
$Co(en)_2(SO_2\text{-}NH_2)_2^{2+}$			23,150(220)	34,725CT	671
$Co(en)_2(OSO\text{-}NH_2)_2^{2+}$			19,530(134)	30,675(4100)	671
$Rh(H_2O)_6^{3+}$			25,500(47)	32,800(55)	552
$Rh(NH_3)_6^{3+}$			32,800(134)	39,200(101)	562
$Rh(NH_3)_5Cl^{2+}$			29,000(101)		707
$Rh(NH_3)_5Br^{2+}$		23,500(24)	28,000(124)		707
$Rh(NH_3)_5I^{2+}$			24,200(286)		
			(26,000)(250)	$^1A_2$ or lf?	707
$Rh(NH_3)_5(H_2O)^{3+}$			31,645(104)	38,020(89)	577
$t\text{-}Rh(NH_3)_4I^{2+}$			21,400(304)		707
$t\text{-}Rh(NH_3)_4Cl_2^+$			24,400(72.5)	34,500(91.0)	707
$t\text{-}Rh(NH_3)_4Br_2^+$			23,200(136)		707
$Rh(CN)_5(H_2O)^{2-}$			37,700(605)^{1}E		632
$Rh(NH_3)_5(CH_3CN)^{3+}$			33,220(158)	39,525(126)	576
$Rh(CN)_6^{3-}$		(39,000)?	44,500(527)		562
$Rh(CN)_5Cl^{3-}$			36,100(425)^{1}E		632
$RhBr_6^{3-}$			18,100(190)	22,200(180)	552
$RhCl_6^{3-}$		14,700(3)	19,300(102)	24,300(82)	552
$Rh(N_3)_6^{3-}$			20,500(893)	24,500(1145)	680
$Rh(ox)_3^{3-}$		19,200(8)	25,100(290)	30,000	552
RhF_6^{3-}		15,500 (16,500)	21,300	27,800	562
$Rh(dtc)_3$			19,800	23,600	494
$Rh(dsep)_3$			20,300(1960)		245
$Rh(en)_3^{3+}$			33,200(210)	39,600(190)	200
$Rh(Me\text{-}en)_3^{3+}$			32,050		950
$t\text{-}Rh(aMe_2\text{-}en)_2Cl_2^+$			26,525(260)	32,050(200)	950
$t\text{-}Rh(sMe_2\text{-}en)_2Cl_2^+$			23,925(72)	33,335(171)	950
$c\text{-}Rh(sMe_2\text{-}en)_2Cl_2^+$		20,410(20)	27,780(670)	(35,840)(1900)	950
$t\text{-}Rh(en)_2(H_2O)_2^{3+}$			29,240(66)	36,760(125)	682
$t\text{-}Rh(en)_2(OH)_2^{2+}$			29,590(136)	34,480(136)	682

cont

Table 6.16 cont

Complex	$^3T_1$	$^3T_2$	$^1T_1$ [a]	$^1T_2$	Ref
$t\text{-}Rh(en)_2Cl_2^+$			24,630(75)	34,965(130)	641
$c\text{-}Rh(en)_2Cl_2^+$			28,410(155)	33,900(180)	641
$t\text{-}Rh(en)_2Br_2^+$			23,530(100)	36,230(1800)	641
$c\text{-}Rh(en)_2Br_2^+$			27,625(210)	(36,230)(900)	641
$t\text{-}Rh(en)_2I_2^+$			21,650(322)	CT	682
$c\text{-}Rh(en)_2I_2^+$			26,670(1200)		641
$t\text{-}Rh(en)_2ClBr^+$	20,580(18)		23,870(94)		682
$t\text{-}Rh(en)_2FH_2O^{2+}$			28,900(80)	35,710(131)	682
$t\text{-}Rh(en)_2(NO_2)_2^+$			(33,335)(590)	(39,215)(2400)	641
$c\text{-}Rh(en)_2(NO_2)_2^+$			34,480(660)	40,815(3500)	641
$t\text{-}Rh(en)_2(NH_3)Cl^{2+}$			29,240(95)	36,365(120)	641
$c\text{-}Rh(en)_2(NH_3)Cl^{2+}$			29,240(150)	36,230(195)	641
$t\text{-}Rh(en)_2(N_3)_2^+$			26,665(780)	CT	641
$c\text{-}Rh(trien)Cl_2^+$			28,410(250)	34,130(200)	641
$c\text{-}Rh(cyclen)Cl_2^+$			27,770(528)	33,900(367)	548
$c\text{-}Rh(cyclam)Cl_2^+$			28,170(224)	33,335(365)	548
$t\text{-}Rh(cyclam)Cl_2^+$			24,630(78)	(32,260)(80)	642
$c\text{-}Rh(cyclam)(CN)_2^+$			35,590(465)	45,870(1830)	1354
$t\text{-}Rh(cyclam)(CN)_2^+$			37,450(270)	45,870(1620)	1354
$t\text{-}Rh(cyclam)(NO_2)_2^+$			(31,250)(535)	(38,460)(2100)	642
$t\text{-}Rh(cyclam)(H_2O)_2^{3+}$			28,410(65)		642
$c\text{-}Rh(cyclam)(H_2O)_2^{3+}$			33,785(249)	39,840(230)	642
$t\text{-}Rh(cyclam)(H_2O)(OH)^{2+}$			29,240(118)	36,495(222)	642
$t\text{-}Rh(cyclam)(OH)^{2+}$			29,325(101)	36,100(163)	642
$c\text{-}Rh(cyclam)(OH)_2^+$			30,210(226)	35,970(202)	642
$Ir(N_3)_6^{3-}$	(15,400)(57) 19,400(232)		25,000){935)	27,600(1050)	680
$IrCl_6^{3-}$	16,300(7.5) 17,900(10)		24,100(76)	28,100(64)	552
$IrBr_6^{3-}$	16,800(27)		22,400(232)	25,800(198)	552
$Ir(dsep)_3$	(19,600)(400)		24,500(200)	(26,600)	245
$Ir(CN)_5(CH_3CN)^{2-}$	(41,700)(250)^{3}E		46,500(720)^{1}E		632
$Ir(CN)_5(H_2O)^{2-}$	(37,000)(180)^{3}E		42,600(790)^{1}E		632
$Ir(NH_3)_6^{3+}$	(31,800)(14)		39,800(92)	46,800(160)	563
$Ir(NH_3)_5(H_2O)^{3+}$	(30,000)(12)		38,800(86)	47,000(128)	564
$Ir(NH_3)_5Cl^{2+}$	27,400(10)		34,965(73)	44,050(333)	579
$Ir(NH_3)_5Br^{2+}$			33,100(84)		707
$Ir(NH_3)_5I^{2+}$			29,900(360)		707
$t\text{-}Ir(en)_2Cl_2^+$	24,000(7)		29,200(48)	36,000(36)	707
$t\text{-}Ir(en)_2Cl_2^+$	23,365		28,990(^{1}E)	36,400(1B2)	947
$c\text{-}Ir(en)_2Cl_2^+$	26,525		31,250($^1B_1$)[c]	39,370($^1A_2$,$^1B_2$)[c]	
			34,250($^1A_2$,$^1B_2$)[c]		947
$t\text{-}Ir(en)_2Br_2^+$	22,500(9)		27,600(68)	34,600(30)	707
$t\text{-}Ir(en)_2I_2^+$	20,300(12)		25,300(165)		707

cont

Table 6.16 cont.

Complex	$^3T_1$	$^3T_2$	$^1T_1$ [a]	$^1T_2$	Ref
$Ir(en)Cl_4^-$	20,365		28,320($^1A_2$,$^1B_2$)[c]		
			32,000($^1A_1$)[cd]		947
$mer\text{-}Ir(en)(enH)Cl_3^+$	21,550		27,780($^1B_2$)[c]	32,785($^1B_1$)[c]	
			36,500($^1B_1$)[c]		947
$PdCl_6^{2-}$			(20,800)		685
PtF_6^{2-}	(22,500)	24,400	31,500	36,400	687,688
$PtCl_6^{2-}$	22,100		28,300		686,687
$PtCl_6^{2-}$	(20,700)(27)		(26,400)(200)	28,300(294)	
	21,500(32)				708
$PtBr_6^{2-}$	(19,100)		(23,000)		686,687
$PtBr_6^{2-}$	19,000(69)		(22,500)(1000)		
	20,600(72)				708
$PtCl_4(OH)_2^{2-}$	20,200		26,700(^1E)		1112
$PtCl_4(H_2O)_2$	21,800		28,300(^1E)		1112
$Pt(SeCN)_6^{2-}$			(19,600)(1480)	(21,700)(3400)	
					708
$Pt(N_3)_6^{2-}$	(20,400)(248)				680,708
$Pt(NH_3)Cl_5^-$	21,600(35)		29,100(540)		1110
$Pt(NH_3)_5Cl^{3+}$			35,400(180)		707
$Pt(NH_3)_5Br^{3+}$			31,100(256)		707
$Pt(NH_3)_5I^{3+}$			25,800(840)		
			(28,700)(610)	$^1A_2$ or sf?	707
$Pt(NH_3)_4I_2^{2+}$	18,000(51)				707
$Pt(NH_3)_4Cl_2^{2+}$	(26,600)(18)		29,900(91.4)		707
$Pt(NH_3)_4Br_2^{2+}$	(23,200)(19)		26,700(175)		707
$t\text{-}Pt(en)_2Cl_2^{2+}$	(27,000)(20)		30,300(101)		707
$t\text{-}Pt(en)_2Br_2^{2+}$	(23,500)(20)		(27,000)(172)		707
$Pt(CN)_5Cl^{2-}$			38,400(222)		707
$t\text{-}Pt(CN)_4Cl_2^{2-}$	(26,500)(30)		(30,500)(180)		707
$t\text{-}Pt(CN)_4Br_2^{2-}$	(23,000)(39)				707

a) The lowest energy spin allowed d-d band (under the $^1T_{1g}$ column) in the trans ML_4Z_2 species is more properly assigned to ^1E_g (in D_{4h}) in the majority of cases where Dq(Z) < Dq(L). b) N,N',N"-tris(2-picolyl)-cis-cis-1,3,5-triaminocyclohexane - complex is probably trigonal prismatic.
c) Labelled in C_{2v} symmetry d) Believed, through calculation, to overlap $^1B_1(^1T_{2g})$

The crystal spectra of sodium nitroprusside, $Na_2Fe(CN)_5(NO).2H_2O$ has been studied by Gray and co-workers [621-623]. The sequence of energy levels was proposed to be:

$$e(xz,yz) < b_2(xy) < e(\pi^*(NO^+)) < b_1(x^2-y^2) < a_1(z^2) < e(\pi^*(CN))$$

and a series of absorption bands lying between 20,080 and 50,000 cm^{-1} were assigned as d-d, and charge transfer (iron(II) to nitrosyl).

The single crystal spectrum of $Fe(bipy)_3^{2+}$ is dominated by charge transfer bands (see Chapter 5) but weak absorption at 11,500 cm^{-1} is assigned to a d-d spin triplet [624].

Many low symmetry derivatives of low spin iron(II) have been reported (Table 6.17). Extensive studies of low spin tetragonally distorted macrocyclic iron(II) complexes have been reported, especially by Busch and his co-workers. In D_{4h} symmetry $^1T_{1g}$ spans $^1E_g + {}^1A_{2g}$, while the higher energy $^1T_{2g}$ spans $^1E_g + {}^1B_{2g}$. The splitting energies are given:

$^1T_1$	$^1A_2$---$^1A_1$	10Dq - C
	^1E----$^1A_1$	10Dq - C - (35/4)Dt
$^1T_2$	$^1B_2$---$^1A_1$	10Dq + 16B - C - 4Ds - 5Dt
	^1E----$^1A_1$	10Dq + 16B - C + 2Ds - (25/4)Dt

(6.17)

Configurational interaction is not taken into account. For weaker field ligands this may introduce some errors (e.g. see data in [947]). Note however that the ^1E mixing element $\langle {}^1E^a|V|{}^1E^b\rangle = -\sqrt{3}(Ds - (5/4)Dt)$ [288].

These equations may be used for tetragonal complexes of the type ML_4Z_2, ML_5Z etc and also cis-ML_4Z_2, making use of the relationships shown in Chapter 1.

Complexes of the type $[LFe(CN)_5]^{n-}$ are well known for various L [1355-1360]. Some examples are shown in Table 6.17. Variations in the energies of the components of $^1T_{1g}$ have been investigated with special reference to replacement of L by amino acids [1355]. Note that Dt = 0.628[Dq(CN) - Dq(L)] is assumed [1355,1361].

In low spin trans-$Fe(II)L_4Z_2$ species, the splitting of the $^1T_{1g}$ state is commonly observed. However the splitting of the $^1T_{2g}$ state is rarely resolved (Table 6.17). If the usual assumption is made that C = 4B, then values of Dq, Dt and B, may be deduced from the energies of the split components of $^1T_{1g}$ and the average energy of $^1T_{2g}$ (see Chapter 9). Since Dt is proportional to the difference in equatorial and axial ligand field strengths (see Chapter 1), values for Dq(L) and Dq(Z) can be deduced [288] but no information concerning the relative sigma and pi bonding strengths can be obtained in the absence of detectable splitting of $^1T_{2g}$. If Dq(Z) < Dq(L) then $E(^1E(^1T_{1g})) < E(^1A_2(^1T_{1g}))$. The variations in field strength in non-cubic iron(II) complexes are discussed in Chapter 9. Note that in a low spin trans-FeL_4Z_2 complex, the $^1A_{2g}$

Table 6.17

ABSORPTION SPECTRA OF LOW SPIN AXIALLY DISTORTED IRON(II), RUTHENIUM(II), OSMIUM(II), COBALT(III), RHODIUM(III) AND IRIDIUM(III) COMPLEXES.

Complex[a]	^1E_g---$^1A_{1g}$	$^1A_2$---$^1A_{1g}$	($^1T_2$)	Ref
$Fe(CN)_5(H_2O)^{3-}$	22,520(444)	30,300(100)		1355
$Fe(CN)_5(NH_3)^{3-}$	25,250(450)			1355
$Fe(CN)_5$(S-methionine)$^{3-}$	25,575(270)	30,770(100)		1355
$FeCl_2(ArNC)_4$	21,000	24,200		549
$Fe(SnCl_3)_2(ArNC)_4$	21,700	23,800		549
$Fe([14]aneN_4)(CH_3CN)_2$	18,510(65)		27,200(140)	611
$Fe([13]aneN_4)(CN)_2$	23,030(78)	18,780(18)	27,900(71)	611
$Fe([14]aneN_4)(CN)_2$	21,900	(17,700)	27,900	611
$Fe([15]aneN_4)(CN)_2$	19,570(27)	15,120(6.2)	25,690(64)	611
$Fe(Me_2[14]teteneN_4)Py_2^{2+}$	12,420(579)	20,000(493)		613
$Fe(Me_2[15]teteneN_4)Py_2^{2+}$	13,200(23.2)	19,600(320)		613
$Fe(Me_2[16]teteneN_4)Py_2^{2+}$	13,500(20)	19,000(133)		613
$FeCl_2$(diars)$_2$	13,500	22,400		956
$FeBr_2$(diars)$_2$	12,900	22,500		956
FeI_2(diars)$_2$	12,700	22,500		956
Ru(aniline)$_4Br_2$	18,400	24,200	28,500	627
$RuCl_2$(Me-S-S-Me)$_2$	21,400(54)	29,000(126)		568
$RuBr_2$(Me-S-S-Me)$_2$	20,200(64)	28,650(153)		568
$RuCl_2$(Ph-S-S-Ph)$_2$	20,700(175)	24,400(320)		568
$RuBr_2$(Ph-S-S-Ph)$_2$	19,400(216)	24,400(255)	(31,700)?	568
$RuCl_2$(Me-P-P-Me)$_2$	24,000(56)	34,000		569
$RuBr_2$(Me-P-P-Me)$_2$	23,000(64)	33,000		569
$Os(NH_3)_5(N_2)^{2+}$	22,720[e]	25,640[e]	32,260[e]	672
$Co(NH_3)_4(CN)_2^+$	24,000(66.3)	20,300(15.4)	31,400(67.6)	668
$Co(NH_3)_5I^{2+}$	17,250			288,628
$Co(NH_3)_4Cl_2^+$	15,900	21,000	24,940	288,629
$Co(NH_3)_4Br_2^+$	14,900	(21,000)		633
$Co(NH_3)_3(H_2O)Br_2^{2+}$	14,550	18,400		633
$Co(NH_3)_3(TCA)_3$[c]	17,500	19,600	26,750	634
$Co(NH_3)_3Cl_3$[c]	15,800	(17,500)		634
$Co(NH_3)_4$(propionate)$_2^+$	18,120	21,800	27,360	288,630
$Co(CN)_5(H_2O)^{2-}$	26,315(280)	(32,300)(110)		164,735
$Co(CN)_5(Cl)^{3-}$	25,500(200)	(31,700)(97)		164,735
$Co(CN)_5(Br)^{3-}$	25,300(170)			164,735
$Co(CN)_5(I)^{3-}$	23,000(200)			164,735
$Co(CN)_5(NCS)^{3-}$	27,600(500)			164,735
$Co(CN)_5(SCN)^{3-}$	26,500(191)			164,735
$Co(CN)_5(SO_3)^{4-}$	29,900(430)			164,735

cont

Table 6.17 cont

Complex[a]	^1E_g---$^1A_{1g}$	$^1A_2$---$^1A_{1g}$	($^1T_2$)	Ref
$Co(CN)_5(NO_2)^{3-}$	(29,000)(490)			164,735
$Co(NH_3)_5(NO_2)^{2+}$	30,865(1660)[d]	21,835(100)		640
$Co(NH_3)_5(ONO)^{2+}$	20,325(71)		27,700(269)	640
$Co(NH_3)_5F^{2+}$	19,450	21,470	28,270	288,628
$Co(NH_3)_5Cl^{2+}$	18,720	21,350	27,500	288,628
$Co(NH_3)_5Br^{2+}$	18,230	21,950		288,628
$Co(MeNH_2)_5Br^{2+}$	17,330(43)	19,820(21)		638
$Co(EtNH_2)_5Br^{2+}$	17,010(59)	19,560(25)		638
$Co(n\text{-}PrNH_2)_5Br^{2+}$	16,900(63)	19,330(32)		638
$Co(n\text{-}BuNH_2)_5Br^{2+}$	16,880(68)	19,160(47)		638
$Co(MeNH_2)_5Cl^{2+}$	17,870(49)	20,230(19)	26,500	639
$Co(EtNH_2)_5Cl^{2+}$	17,590(49)	19,930(22)	25,770	639
$Co(n\text{-}PrNH_2)_5Cl^{2+}$	17,590(55)	19,860(18)	25,680	639
$Co(n\text{-}BuNH_2)_5Cl^{2+}$	17,480(69)	19,780(32)	25,850	639
$Co(Py)_2(CO_3)_2^-$	18,300(89.1)		23,400(35.5)B_{2g} 26,500(57.5)E_g	666
$Co(en)_2(NO_2)_2^+$	29,500(4170)[d]	23,255(219)		640
$Co(en)_2F_2^+$	17,200	22,620	27,600	288,629
$Co(en)_2Cl_2^+$	16,200(34)	22,000(24)	28,170(54)	200
$Co(en)_2Br_2^+$	15,210	21,680		288,629
$Co(en)_2(H_2O)_2^{3+}$	18,200	22,500	29,000	288,631
$Co(en)_2F(H_2O)^{2+}$	17,540(27)	23,000(31)	28,170(54)	550
$Co(en)_2(OAc)_2^+$	18,500	23,000	27,600 28,100	756
$Co(en)(Py)_2(OAc)^{2+}$	18,400	21,000		756
$Co(cyclam)Cl_2^+$	15,700(31)	23,200(41)		551
$Co(2,3,2\text{-tet})Cl_2^+$	16,000(35)	23,530(35)	-	603
$Co(2,3,2\text{-tet})(OAc)_2^+$	18,415(70)	22,575(29)	27,700(82)	602
$Co(3,2,3\text{-tet})Cl_2^+$	15,625(34)	22,220(45)	-	603
$Co(3,2,3\text{-tet})(OAc)_2^+$	18,180(70)	21,505(46)	27,175(93)	602
$Co(2,2,3\text{-tet})Cl_2^+$	15,265(35)	22,220(40)	-	601
$Co(2,2,3\text{-tet})(OAc)_2^+$	18,350(67)	21,980(40)	27,400(120)	601
$Co([13]aneN_4)Cl_2^+$	16,670(33)	23,700(144)		614
$Co([13]aneN_4)Br_2^+$	15,390(22)	(22,220)		614
$Co([14]aneN_4)Cl_2^+$	16,200(35)	21,820(24)	25,860(56)	614
$Co([14]aneN_4)(N_3)_2^+$	17,400(430)			614
$Co([15]aneN_4)Cl_2^+$	15,630(27)	20,410(38)	24,100(62)	614
$Co([15]aneN_4)Br_2^+$	14,390(43)	(19,800)		614
$Co([16]aneN_4)Cl_2^+$	15,110(41)	19,610(35)	23,870(76)	614
$Co([16]aneN_4)Br_2^+$	13,790(38)	(18,520)		614
$Co(Me_2[14]1,3\text{-dieneN}_4)^{3+}$	21,300(120)	23,600(125)	(27,800)	612
$Co(Me_6[14]4,11\text{-dieneN}_4)^{3+}$	21,300(58)	24,100(82)	(27,000)	612
$Co([14]tet\text{-}enN_4)Br_2^+$	15,800(87)	25,600(3470)	CT	798
$Co([14]tet\text{-}enN_4)(MeCN)_2^{3+}$	21,300(56)	25,600(118)	(30,300)	798
$Co(TAAB)Br_2$	15,200	21,700	28,000CT?	605

cont

Table 6.17 cont

Complex[a]	^1E_g---$^1A_{1g}$	$^1A_2$---$^1A_{1g}$	($^1T_2$)	Ref
$Co(TAAB)(NO_3)_2$	16,300(350)	22,000(2500)	28,600(24,000) CT?	605
$Co(TTP)I_2^+$	15,630	20,410	30,490?	610
$CoCl_2(diars)_2^+$	16,400			956
$CoBr_2(diars)_2^+$	15,700			956
$CoI_2(diars)_2^+$	15,200			956
$Co(dtp)_3$	D3 13,600	13,080	18,500	157
$Rh(CN)_5I^{3-}$	31,900(1500)			632
$Rh(NH_3)_5I^{2+}$	24,100(270)	25,800(230)		552
$Rh(Py)_2Cl_4^-$	23,200(92)	20,000(39)		552
$Rh(CN)_5Cl^{3-}$	36,100(425)			632
$Rh(CN)_5Br^{3-}$	34,800(402)			632
$Rh(ox)_2Cl_2^{3-}$	21,300(54)	25,100(38)		552
$Rh(cyclam)I_2^+$	(19,415)(64)	21,460(204)		642
$Rh(cyclam)Br(H_2O)^{2+}$	21,365(37)	24,815(63)	(32,260)(106)	642
$Rh(cyclam)I(OH)^+$	22,575(153)	25,445(158)		642
$Rh(cyclam)BrI^+$	20,120(151)	21,785(152)		642
$Rh(cyclam)(N_3)I^+$	(21,505)(283)	23,980(654)		642
$Rh(cyclam)(NCS)I^+$	23,040(393)	24,270(413)		642
$Ir(CN)_5(NCCH_3)^{2-}$[b]	(46,500)(720)			632
$Ir(CN)_5OH^{3-}$[b]	43,500(630)			632
$Ir(CN)_5H_2O^{2-}$[b]	42,600(790)			632
$Ir(CN)_5Cl^{3-}$[b]	40,800(370)			632
$Ir(CN)_5Br^{3-}$[b]	38,900(412)			632
$Ir(CN)_5I^{3-}$[b]	36,400(1400)			632

a) All ML_4Z_2 complexes reported here are assumed to have a trans configuration. b) A weak spin forbidden band (^3E) is also observed. c) Trans complex of C_{2v} symmetry; the column heading assignments are not appropriate for this group. d) Probably overlapping ^1E_g and $^1T_{2g}$. e) These are very intense and may be charge transfer rather than d-d in origin.

transition should occur at essentially the same energy as the unsplit $^1T_{1g}$ transition in the octahedral FeL_6^{2+} complex (6.17). Thus in a series of species FeL_4Z_2 with common L but differing Z, this transition should remain at almost constant energy. This provides a useful means of assigning this transition (see Co(III) below).

Five Coordinate Iron(II) Complexes: These species are known both as square pyramids and as trigonal bipyramids (Table 6.15). In the former case the spectra are reminiscent of the tetragonally distorted octahedra but generally with a large splitting of the ^5E state. High spin trigonal bipyramidal molecules generally give two

bands in the near infrared spectrum (Table 6.15) assigned to the expected one electron transitions (based on the configuration $(e")^3(e')^2(a_1')$ in Fig.1.3(a)). However low spin trigonal bipyramidal molecules are also possible.

Thus the ligand tris(o-diphenylphosphinophenyl)phosphine, QP, forms the low spin trigonal bipyramid $[Fe(QP)NO_3]^+$ with configuration $(e")^4(e')^2$. Three transitions to the configurations $(e")^3(e')^3$, $(e")^4(e')(a_1')$ and $(e")^3(e')^2(a_1')$ are seen centered at ca. 9000(800), ca 18,000(4000) and ca 30,000(9000) cm^{-1} [625,948]. The intensities are exceptionally high for d-d transitions. However such intensities are fairly normal for such acentric species with highly covalent ligands such as phosphorus or arsenic. These authors [625] report calculations for the energy levels anticipated for low spin trigonal bipyramidal Fe(II) (and other species).

The tri-imine, a,a',a"-Tri-imine,2-6,-(Dibenzothiazol-2-yl)-pyridine (NNN) forms five coordinate complexes with iron(II) of formula $Fe(NNN)X_2$ where X is Cl^-, Br^-, I^- and NCS^- [705]. Before the onset of charge transfer absorption at above 17,000 cm^{-1}, two d-d bands are seen at 8500 - 9600 and 15,300 - 17,000 cm^{-1} at somewhat higher energies than those shown in Table 6.15.

Another tridentate ligand, 6-methylpyrid-2-yl-N-(2'-methyl-thiophenyl)methyleneimine (SNNMe) forms five coordinate iron(II) complexes of formula $Fe(SNNMe)X_2$ [706]. A trigonal bipyramid is assumed to be the more likely configuration. Similar data are reported for tris(2-dicyclohexylphosphinoethyl)amine [740].

Low and intermediate spin trigonal bipyramidal iron(II) species are formed by ligands such as the hexaphenyl-1,4,7,10-tetra-phosphadecane (P4), a linear ligand with four coordinating phosphorus donor atoms. Complexes of the type $Fe(P4)X^+$ have a singlet ground state but, for certain X, show a low lying spin triplet [736], or may have a ground state triplet [1328]. Transitions near 7600 - 8500, 11,700, 15,000 and 23,000 cm^{-1} have been assigned both in tetragonal [736] and in C_{2v} symmetry [1328]. In the latter case an AOM analysis is attempted.

Four Coordinate Iron(II) Complexes: Square planar iron(II) complexes generally exhibit intermediate spin with S = 1. Busch [611,613] and Konig [626], among others, have studied these systems (Table 6.18). There is a low lying absorption band in the near infrared, probably due to a transition to the low lying z^2 orbital, and another absorption in the red at 11,000 - 15,000 cm^{-1} prior to the onset of charge transfer absorption.

Gillespite is a mineral which contains high spin square planar Fe(II) in an oxide environment ($BaFeSi_4O_{10}$). The ground state is $^5A_{1g}$ with $^5B_{2g}$ lying at 8300 and $^5B_{1g}$ at 20,000 cm^{-1} [1322,1323]. The d orbital sequence is $z^2 < xz,yz < xy < x^2-y^2$, and the lowest lying band, to ^5E_g was not observed.

The ground state configuration is $(z^2)^2(xz,yz)^2(xy)^1-(x^2-y^2)^1$. This gives rise to the $^5A_{1g}$ ground term once it is recognised that two unpaired electrons in (xz,yz) will transform as a_2; descent in symmetry can be used to prove

Table 6.18

ABSORPTION SPECTRA OF SPIN TRIPLET IRON(II) COMPLEXES

Ligand	Solvent	Absorption Spectra			Ref
Me_2[14]en(2-)	Tol	6210(10.5)	14,000(21)	+ CT	613
	Py	7930(32)	12,300(33.6)	+ CT	613
Me_2[15]en(2-)	Tol	4800(14)	11,800(26)	+ CT	613
	Py	6760(13)	12,000(46)	+ CT	613
Me_6[15]en(2-)	Tol	4800(11)	11,800(28)	+ CT	613
	Py	6400(14)	11,800(93)	+ CT	613
Me_2[16]en(2-)	Tol	5920(14.8)	15,400(119)	+ CT	613
	Py	6100(11)	+ CT		613
Me_4[16]en(2-)	Tol	5850(8)	14,500(88)	+ CT	613
	Py	5900(9)	+ CT		613
([15]aneN_4)$(NO_2)_2$			11,100(10)	+ CT	611
$(Phen)_2Ox$			11,620(6)	+ CT	626
$(Bipy)_2$mal			11,900(5)	+ CT	626

this point. Thus the states lie in the sequence $E_g < B_{2g} < B_{1g}$.

Hitchman and co-workers have analysed the magnetism and spectrum of this species in terms of the AOM model [1321]. They note that a reasonable fit can only be obtained if the z^2 orbital is stabilised by some 5000 cm^{-1} by interaction with the 4s orbital.

Gillespite undergoing a phase transition above 26 kbar of pressure, is converted to a highly flattened tetrahedral form. The two bands shift to 7150 and 17,200 cm^{-1} [1324]; these data have been considered in the AOM formalism [1321].

Tetrahedral iron(II) species generally exhibit a single spin allowed transition at low energies, generally between 3000 - 7000 cm^{-1}, and often split into a multiplet. Complexes of the type FeL_3L' (C_{3v} symmetry) and $FeL_2L'_2$ (C_{2v} symmetry) will show additional structure due to the loss of degeneracy of the $^5T_2$ state [580]. Data are presented in Table 6.15. Higher energy quintet-triplet transitions may also be present.

Other complexes: Eight coordinate iron(II) occurs when Fe(II) is doped into CdF_2. The $^5T_{2g}$---^5E_g transition is seen as a dynamically Jahn Teller split transition near 5000 cm^{-1} [643]. Seven coordinate pentagonal biypramidal iron(II) has spectra [644] little different from tetragonally distorted six coordinate iron(II). Two coordinate iron(II) exists in the gas phase as $FeCl_2$ [675]. The spectrum shows two bands at 4600 and 7140 cm^{-1} assigned to the transition $^5\Pi$---$^5\Delta$ and $^5\Sigma$---$^5\Delta$ [676]. This yields AOM values of $e_\sigma(Cl)$ = 3570 and $e_\pi(Cl)$ = 2300 cm^{-1} [378].

RUTHENIUM(II) AND OSMIUM(II) d^6

Charge transfer absorption generally obscures d-d bands in these species. This is especially true of unsaturated amine ligands for which a very large body of CT data exists (see Chapter 5). However there are quite a few examples (Tables 6.16, 6.17) of six coordinate low spin ruthenium(II) species where the d-d transitions to $^1T_{1g}$ and $^1T_{2g}$, or their low symmetry components, can be observed. These data are little different from those of low spin iron(II) reported above, though shifted to higher energy for comparable ligands because it is a second row transition metal.

COBALT(III) d^6

Cobalt(III) complexes have been very extensively studied, for reasons similar to those for chromium(III), i.e. the range of materials which can be prepared is very large because of the kinetic inertness of this ion. With the exception of some fluorides (Table 6.15), six coordinate complexes of cobalt(III) (which make up over 99% of known cobalt(III) complexes) are invariably low spin and diamagnetic ($^1A_{1g}$ ground term). Thus their behaviour parallels low spin iron(II), once again shifted to higher energies for comparable ligands, because of the higher oxidation state. Data are presented in Tables 6.16 and 6.17 and detailed analysis of the trends is again left to Chapter 9. The high spin hexafluorides, shown in Table 6.15 show the expected Jahn Teller distortion of the excited ^5E_g state [540].

The $[Co(NH_3)_6]^{3+}$ and $[Co(ND_3)_6]^{3+}$ ions show extensive vibrational structure when cooled to 4.2 K [645]. Various progressions in the Co-N stretching vibration are seen built upon several vibronic origins. Evidence for a coupled Jahn Teller active e_g mode is equivocal. However see [1379].

Both a_{1g} and e_g progressions have also been observed in the emission spectrum of the $[Co(CN)_6]^{3-}$ ion [315,653,654] due to the transition $^3T_{1g}$---$^1T_{1g}$. Fitting of the emission data reveals that the M-C bond is about 10pm longer in the excited $^3T_{1g}$ than in the ground state [653]. These authors [653] discuss the relative energies of the absorption, emission and zero-zero transitions in this example and comment on their relevance to photochemical studies. The absorption spectrum of the excited $^3T_{1g}$ state has been reported [669].

Various single crystal polarised light studies have been reported. These include the highly dichroic, D_3, $Co(dtp)_3$ [157], and the much less dichroic D_3, $[Co(en)_3]^{3+}$ ion [646-648]. In this last case intensity primarily arises through coupling with the t_{1u} and t_{2u} modes despite the static threefold distortion. Also with respect to this last case, note that the CD spectrum permits resolution of the components of $^1T_{1g}$ in D_3 [655,656,1324]. Indeed CD spectra have been used extensively in cobalt(III) spectroscopy as a means of identifying transition assignments (e.g. Fig.6.33 and [1107,1325] and refs. therein).

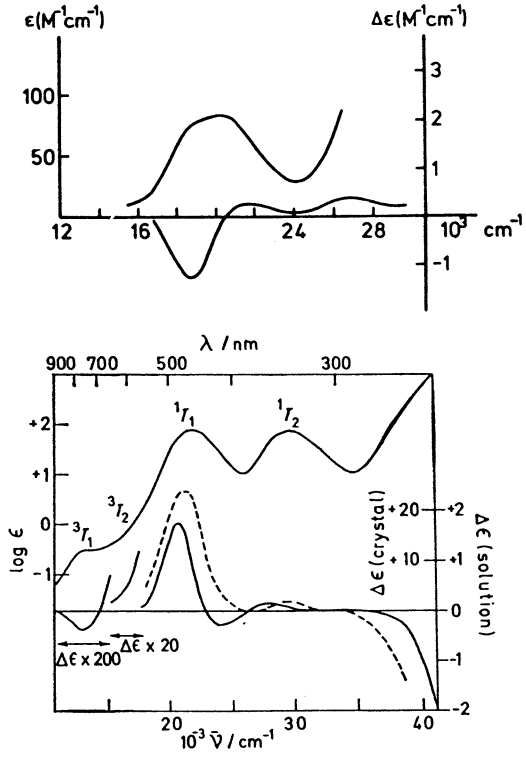


Fig.6.33 (Upper) Absorption and CD spectra of the $(-1)_{589}$ antipode of all-cis-$[Co(L\text{-his})(D\text{-his})]^+$ isomer [658]. (Lower) Absorption and CD spectra of $\Lambda(+)[Co(en)_3]^{3+}$ in aqueous solution (solid lines) and in a single crystal of $2[\Lambda(+)Coen_3Cl_3].NaCl.6H_2O$ for radiation propagated along the optic axis (hatched) [655].

Data have also been reported for Co(III) doped into GASH [649]; a small trigonal splitting was observed in both the $^1T_{1g}$ and $^1T_{2g}$ states. Trigonal splitting of the $^1T_{1g}$ state was also observed when Co(III) was doped into $NaMgAl(C_2O_4)_3.9H_2O$ [661], and into alumina [662,663] where splitting of the $^1T_{2g}$ level was also observed.

Cis and trans $[CoL_4Z_2]^{n+}$ species can usually be readily distinguished on the basis of intensity and band position. Thus cis complexes, being acentric almost invariably exhibit more intense transitions than trans species. Provided that L and Z differ sufficiently in ligand field strength, the $^1T_{1g}$ state will be split in the trans complex, but is rarely split in the cis complex.

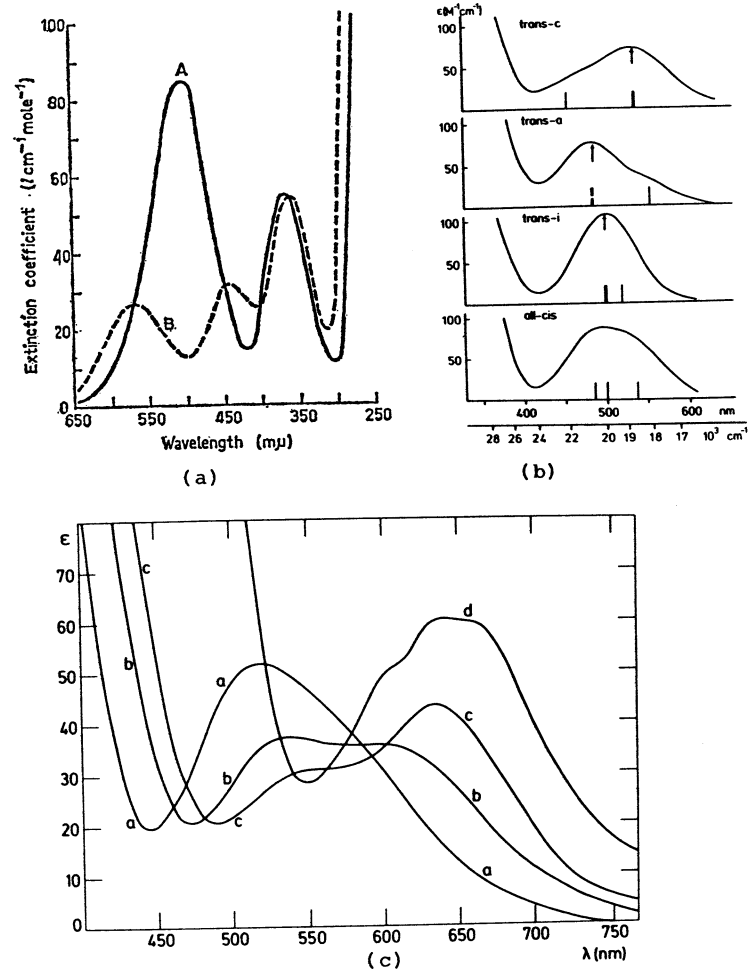


Fig.6.34 (a) Absorption spectra of (A) cis-$[Co(en)_2F(H_2O)]^+$ and (B) trans-$[Co(en)_2F(H_2O)]^+$. (b) Absorption spectra of the four known geometrical isomers of the bis(histidinato)cobalt(III) complex. Vertical lines indicate resolution of electronic states [658]. (c) Absorption spectra of the supposedly meridional isomers a:$[Co(Py)_3(H_2O)_3]^{3+}$, b:$[Co(Py)_3Cl(H_2O)_2]^{2+}$, c:$[Co(Py)_3Cl_2(H_2O)]^+$, and d:$[Co(Py)_3Cl_3]$ in 6 M $HClO_4$, 4 M $HClO_4$, 4 M HCl and chloroform respectively [657].

Appropriate equations were discussed above in the discussion of

low spin Fe(II). However because of kinetic inertness, there have been far more studies of cis and trans Co(III) than of Fe(II).

While one ($^1A_{2g}$) of the two components of the $^1T_{1g}$ state in trans$[CoL_4Z_2]^{n+}$ lies at the same energy as the $^1T_{1g}$ level in $[CoL_6]^{n+}$, this is not the case for the cis species. The single low energy absorption band corresponding, in cis-$[CoL_4Z_2]^{n+}$, to the transition from $^1A_1$ to the (usually) unresolved components of $^1T_{1g}$ will lie at an energy which is roughly the average of the split components seen in the trans species, when Dt is positive (axial field weaker than equatorial field) (see relevant discussion in Section 1.8). Hence in such a pair of equivalent cis and trans-$[CoL_4Z_2]^{n+}$ species, the trans complex will exhibit the lower spin allowed transition. When Dt is negative the lower energy band in the cis complex may fall below that of the corresponding trans complex. This is of course also true for the other low spin d^6 ions. See Table 6.16 and 6.17 for examples. The sense of the splitting of the $^1T_{1g}$ state is readily determined from the sign of Dt which can usually be deduced on the basis of the relative field strengths of L and Z (Eqn.1.81).

In the cis-$[ML_4Z_2]^{n+}$ complex the energy of the $B_1(T_{1g})$ level (corresponding with $A_2(T_{1g})$ in the trans derivative) also has an energy 10Dq - C, but Dq is now defined, as in Chapter 1, as the average Dq for L and Z. Polarised data have also been reported [650-652,756] for a range of trans-$[Co(en)_2Z_2]^{n+}$ species which serve to confirm the sense of the splitting of the $^1T_{1g}$ transition, predicted by simple ideas on relative field strengths. The transitions to ^1E_g are primarily polarised parallel to z (but often have a perpendicular component), while those to $^1A_{2g}$ are primarily polarised perpendicular to z. The rhombic t-$Co(OAc)_2(Py)_2(en)$ has unusual polarisation properties with the transition to ^1E_g state (assuming D_{4h} symmetry) polarised perpendicular to the O-Co-O axis, and the transition to $^1A_{2g}$ polarised parallel to this axis. The assignment is tentative [756].

Schmidtke and co-workers [670] have reported an interesting study of trans-$[Co(NH_3)_4(CN)_2]^+$ (and its deutero analog) at 4.2 K. The $^1A_{2g}$---$^1A_{1g}$ transition in the single crystal spectrum showed a progression in a single totally symmetric Co-N stretching mode, with no contribution from Co-C. Analysis of the progression showed that the C-N distance in the molecular plane is increased 9.6 pm in the excited $^1A_{2g}$ state relative to the ground state. Mixing with the totally symmetric Co-CN mode was shown to be very small and the absence of any elongation in the z direction is consistent with the lack of any superimposed progression due to this mode.

The complexes $[CoL_5Z]^{n+}$ are tetragonal and will behave much as the $CoL_4Z_2^{n+}$ with splitting of $^1T_{1g}$ if L and Z differ sufficiently. Since the field along the z axis will be the summation of L + Z, a greater difference between L and Z is necessary, than in the D_{4h} series, before resolution of $^1T_{1g}$ is observed. The series $[Co(CN)_5L]^{n-}$ is especially interesting and has been studied intensively by Gray and co-workers (Table 6.17, [164,735]). The lowest energy spin allowed transition is ^1E---$^1A_1$ which lies in the

sequence of L, $-CN^-$ > $-SO_3^{2-}$ > $-NO_2^-$ > $-NH_3$ > $-NCS^-$ > $-OH_2$ = $-SCN^-$ > Cl^- > Br^- > I^- essentially parallel to the spectrochemical series (Chapter 9).

Similar arguments can be made to predict the spectra of $CoL_3L'_3$ species and distinguish the various possible isomers. There are two isomers. In the meridional (mer) isomer, each set of three ligands lies in one plane, the two planes are orthogonal, and along the three coordinates lie LL, LL' and L'L'. In the facial (fac) isomer, L' is trans to L along each coordinate, and each set of three ligands lies in a face of the octahedron. The fac species has C_{3v} symmetry but is pseudo octahedral in which the field strength is the average of L + L'. It will appear to have an octahedral spectrum ('intermediate symmetry'). The mer species has C_{2v} symmetry but, within the concept of intermediate symmetry will behave as though it had D_{2h} symmetry [e.g. 633,634,657-659] (see Chapter 3). In D_{2h} symmetry, the $^1T_{1g}$ state should split into three components. Schaffer and his co-workers [657] reported the spectra of the series of complexes $[Co(Py)_3(H_2O)_3]^{3+}$, $[Co(Py)_3Cl(H_2O)_2]^{2+}$, $[Co(Py)_3Cl_2(H_2O)]^+$ and $[Co(Py)_3Cl_3]$ [657] whose spectra are shown in Fig.6.34c. The initial and final members of the series are believed to have mer symmetry and their spectra clearly show three components for the $^1T_{1g}$ state (in O_h).

The intermediate systems can exist in one fac form and and up to three meridional isomeric forms and it is not possible, on the basis of the data in Fig.6.34c, to distinguish them. However it is likely that they are meridional.

In this contribution [657], the energies of the components of $^1T_{1g}$ in D_{2h} are expressed in terms of the field strengths of the ligands and specifically on the orientation of the pyridine ligands which will determine their pi interaction with the d orbitals. An AOM matrix is set up for the t_{2g} orbitals in terms of ψ, the angle between the pyridine planes and the XY plane of the molecule. In some tetrakis(pyridine) systems, ψ is often revealed to be 45° by X-ray analysis, while Schaffer has determined a value of 38.1° for some tetragonal chromium(III) species [281] by electronic spectroscopy. In the specific case where ψ = 0° for all three pyridine ligands, the splitting between each of the three components of t_{2g} is equal to one half that of the t_{2g} splitting in the tetragonal trans $M(Py)_4Z_2$.

Another interesting study of this type of complex was reported by Jensen and Bagger [658,659]. Bis(Histidinato)cobalt(III) contains the tridentate histidine ligand with ammino nitrogen, imidazole nitrogen and carboxylic oxygen donor atoms. Two such ligands must bind in a meridional fashion. However there are five different orientations for such binding, four having been successfully isolated. As shown in Fig.6.34b, all four isomers show a more or less well resolved triplet whose energies were analysed in terms of the AOM model in terms of the expected energies of each isomer. CD spectra were reported for the chiral isomers [659]. The spectra of the bis(aspartato)cobalt(III) isomers [664] and similar systems [665-667] have been similarly analysed. The reader should

also see [1196] for an extensive review of the CD spectra of cobalt(III) tetradentate amine complexes.

Five Coordinate Cobalt(III) Complexes: The spectra of five coordinate $CoX_3(R_3P)_2$ complexes have been discussed [677]. $CoCl_3(PEt_3)_2$ has strong absorption at 18,000(6900) cm^{-1} which, though intense, may well be d-d in origin.

RHODIUM(III) AND IRIDIUM(III) d^6

Data for six coordinate rhodium(III) and iridium(III) are also shown in Tables 6.16. and 6.17. See refs [552,632,641,682-684,707] for especially pertinent discussions. Once again with these heavier elements charge transfer spectra often obscure d-d transitions. All complexes are low spin, and will, in principle display transitions to $^1T_{1g}$ and $^1T_{2g}$, at energies shifted to higher energy relative to cobalt(III). Low symmetry species are known, e.g. trans-$[RhL_4Z_2]^{n+}$ but, rather curiously, splitting of the $^1T_{1g}$ state is not very frequently observed (see Table 6.17 for some exceptions). According to the work of Blanchard and Mason [707], in the species trans-ML_4Z_2 (M = Rh(III), Ir(III), and Dq(Z) < Dq(L)) the splitting of $^1T_{1g}$ is large. They observed the low energy component ^1E_g and assumed that the high energy component, $^1A_{2g}$ is obscured at significantly higher energy. In some cases however it may have been observed (see Table 6.16). In this study (Table 6.16) [707] one can readily see the crystal field trends shifting energies to the red in the sequence Cl > Br > I, and also see the regular increase in intensity in both spin forbidden and spin allowed d-d bands in the sequence Cl < Br <I.

An emission study [681] of $[Rh(NH_3)_6]^{3-}$ and its deutero analog revealed phosphorescence from the $^3T_{1g}$ state with a progression in an e_g mode indicating substantial Jahn Teller distortion in this spin triplet state.

Gray and colleagues [632] investigated the spectra of a wide series of $[M(CN)_5L]^{n-}$ (M = Rh(III), Ir(III)) complexes, generally identifying the lowest ^1E transition for Rh(III), and the ^3E and ^1E states of Ir(III) plus relevant CT data.

In the iridium(III) series, all but the lowest lying spin triplet transition may be obscured by charge transfer absorption [552]. Their intensities are rather high for spin forbidden bands, because the large spin orbit coupling coefficient causes significant mixing of spin states. Note also that since B is small for Ir(III), the $^1T_{1g}$ and $^1T_{2g}$ states lie surprisingly close together (Table 6.16). Thus covalency in these systems is unusually high (Chapter 9).

PALLADIUM(IV) AND PLATINUM(IV) d^6

These systems have not been extensively studied, but a few important studies have been made and are included in Table 6.16. Since the first charge transfer band is ligand to metal into the empty e_g level, and since 10 Dq is very large in these systems, the charge transfer bands are generally displaced to relatively high energy, allowing observation of several d-d bands.

This is, of course, especially true of the hexafluoride [690,691]. Patterson and co-workers have studied [689] the absorption and emission spectra of $[PtF_6]^{2-}$ doped into a Cs_2GeF_6 type host lattice at cryogenic temperatures. All four low lying spin 'triplets' and spin 'singlets' can be observed and each displays astonishingly well developed vibrational structure. Moreover each individual spin orbit state was identified. The progressions were based on three vibronic 'false' origins involving two t_{1u} and one t_{2u} vibrations of the $[PtF_6]^{2-}$ molecule. It was initially [689] assumed that the progression was due to the a_{1g} (Pt-F) stretching vibration, but a more recent study [737] re-assigns the progression to an e_g Jahn Teller mode. From analysis [737,738] of the intensities, the configuration $(t_{2g})^5(e_g)$, has a Pt-F bond length about 19 pm longer than in the ground state.

Emission data have also been reported for the other hexahalides [692].

6.2 (vii) d^7 Co(II), Rh(II), Ni(III), Pd(III) and Pt(III)

For high spin d^7 systems, the ground term is ^4F with a low lying ^4P excited term. As a consequence the spectra are reminiscent of analogous d^2 and d^3 complexes. The most important d^7 species is undeniably cobalt(II) known in all coordination numbers from 8 through 2; however coordination numbers 6 and 4 dominate.

Six coordinate octahedral or pseudooctahedral species will exhibit three transitions:

$$\begin{array}{ll} \nu_1 & {}^4T_{2g} \text{------} {}^4T_{1g} \\ \nu_2 & {}^4A_{2g} \text{------} {}^4T_{1g} \\ \nu_3 & {}^4T_{1g}(P) \text{---} {}^4T_{1g} \end{array} \qquad (6.18)$$

none of which correspond in energy with 10 Dq; however the energy difference $\nu_2 - \nu_1$ is exactly equal to 10 Dq. Unfortunately the ν_2 transition is usually very weak (formally two electron) and rarely unequivocally observed.

Tetrahedral species also exhibit three transitions:

$$\begin{array}{lll} \nu_1 & {}^4T_2 \text{------} {}^4A_2 & (= 10\ Dq) \\ \nu_2 & {}^4T_1 \text{------} {}^4A_2 & \\ \nu_3 & {}^4T_1(P) \text{---} {}^4A_2 & \end{array} \qquad (6.19)$$

and all three can be observed, though ν_1 lies at low energy (generally between 2500 - 6000 cm^{-1}) in the near infrared. Both octahedral and tetrahedral species display rich spin forbidden transitions, whose intensity is often enhanced by intensity stealing. As a consequence the identification of the spin allowed bands can be equivocal. The ground term for low spin d^7 is ^2G.

COBALT(II) d^7

Cobalt(II) (and Nickel(II)) played an important role in the development of coordination chemistry because cobalt(II) complexes had spectra which were indicative of their stereochemistry and were easy to prepare usually without special precautions. In the early literature octahedral cobalt(II) was reported to be pink while tetrahedral cobalt was green or blue. Such generalisations should be treated with extreme caution and assignments made on the basis of detailed assignment not visual colour. For example cobalt orthosilicate and anhydrous cobalt chloride are blue, while anhydrous cobalt bromide is green - yet all are octahedral. Some tetrahedral derivatives, for example Co(DPM)2, are pink!

The literature for cobalt(II) electronic spectra is enormous - we browse through some of the more important aspects.

Six coordinate complexes (high spin): In general, two principal regions of absorption are observed [695-701]. A band near 8000 - 10,000 cm^{-1} of intensity 1 - 10 l mol^{-1} cm^{-1} can be assigned to the $^4T_{2g}$---$^4T_{1g}$ transition. In addition a multiple structured band, assigned to $^4T_{1g}(P)$---$^4T_{1g}$, is seen in the visible region near 20,000 cm^{-1} with intensity 5 - 20 l mol^{-1} cm^{-1} (see Table 6.19). The multiple structure arises from a number of causes, but primarily to admixture of spin forbidden transitions to doublet states mainly derived from ^2G and ^2H (see Fig. 2.15). The transition to $^4A_{2g}$ is very weak often appearing as a shoulder. Many authors have assigned it incorrectly to a component of the structured visible absorption. Reference to Appendix V shows that the energy ratio ν_2/ν_1 falls within 1.9 - 2.2 for the range of Dq/B values appropriate for cobalt(II); such information can be used to determine whether a weak feature in the spectrum might be the $^4A_{2g}$ transition (e.g. see Table 6.19).

Structure on the visible region may also be due to spin-orbit coupling, vibrational or low symmetry components, making a total analysis of six coordinate cobalt(II) complexes rather difficult. Ferguson and co-workers investigated the low temperature spectra of simple cobalt(II) salts [695,702,703]. Fig.6.35 illustrates their recorded spectrum of anhydrous cobalt chloride [695]. The shape of the asymmetric band in the visible region is typical of the spectra of octahedral (and pseudo-octahedral) cobalt(II) complexes, the shoulders on the high energy side being assigned to spin forbidden transitions. The $^4A_{2g}$ transition appears as a very weak feature near 12,000 cm^{-1}.

As in this case, it is often possible to observe spin forbidden transitions lying both to low and high energy of the visible region absorption. The lower energy bands originate in ^2G, while the high energy bands derive from ^2G, ^2H, ^2D and ^2P. With the more electronegative ligands (quasi ionically bound complexes), the positions of all these bands accord well with the four parameter Dq, B, C and λ theory. As in Fig.6.35, the effects of spin orbit coupling may be seen at low temperature but will often be obscured at ambient temperature.

Table 6.19

SIX COORDINATE COMPLEXES OF COBALT(II)

Complex	$^4T_{2g}$	$^4A_{2g}$	$^4T_{1g}(P)$	ν_2/ν_1	Ref
$KCoF_3$	7150	15,200	19,200	2.12	695
$CoCl_2$	6600	13,300	17,150		
			17,350	2.12	695
$CsCoCl_3$	6300	13,000	17,000	2.06	790
$CoBr_2$	5700	11,800	16,000		
			17,500	2.07	733
$Co(Py)_6^{2+}$	9800		20,400		715
$Co(en)_3^{2+}$	10,000		21,000	709	750
$Co(H_2O)_6^{2+}$	8100(2)	(16,000)	19,400(4.6)	1.975	200
$Zn(Co)(H_2O)_6^{2+}$	8680	18,400	19,200	2.12	
			19,500		763
$Co(acrylamide)_6^{2+}$	8000	(16,000)	19,400,	2.0	
			(21,000)		596
$Co(NH_3)_6^{2+}$	9000(3)		21,100(8)		200
$Co(MeOH)_6^{2+}$	8470	(15,700)	19,900	1.85	712
$Co(Bipy)_3^{2+}$	11,000		22,000		778
$Co(DMSO)_6^{2+}$	7407(3)	14,600(0.2)	18,700(11.5)	1.97	444
$Co(DTHO_2)_3^{2+}$	8300	(16,800)	19,000	2.02	
			20,500		
			21,400		791
$Co(PyO)_6^{2+}$	(7250)	17,700?	(18,590)	2.28?	
	8300		(20,410)		718
$Co(4\text{-}MePyO)_6^{2+}$	9000(15)		19,050(33)		
			19,840(32)		
			26,400(360)CT		699
$Co(butanone)_6^{2+}$	7570	(15,750)	18,890	2.08	
			(21,440)		761
$Co(acetaldehyde)_6^{2+}$	8680	(16,400)	19,650	1.89	
			(21,280)		762
$Co(benzaldehyde)_6^{2+}$	8290	16,050	19,550	1.94	762
$Co(OMPA)_3^{2+}$	6750	13,800	18,380	2.04	
	6800		19,050		711
$Co(NNEU)_6^{2+}$	8000	(15,800)	19,500	1.97	
			(21,000)		808
$Co(Et_4D)_3^{2+}$	7020(24)	(14,840)(67)	15,550(96)	2.11	
			(16,980)(83)		619
$Co(NC_2H_5)_6^{2+}$	8900		19,800		734
$Co(P(Py)_3)_2^{2+}$	11,000(4)		21,800(25)		754
$Co(PccBF)^{+2}$	8130(3.6)[a]		(22,000)(230)		754
$Co((Py)_3tach)^{2+}$	9380(8.1)		(20,400)(70)		754
Co_2SiO_4		ca.13,000	ca.18,000		700
$Co(Ph_3PO)_2(NO_2)_2$	7400	12,700	17,950	1.72?	717
$Co(L\text{-}hist)_2(H_2O)_2$	10,000	22,400	20,800	2.24	716

cont

Table 6.18 cont.

Complex	$^4T_{2g}$	$^4A_{2g}$	$^4T_{1g}(P)$	ν_2/ν_1	Ref
$Co(2\text{-}MePy)_2(NO_3)_2$	(7486)		18,460(125)		
	8764(15)				699
$Co(Py)_2(Cl_2A)_2$	8620(5)	(17540)?	19,690(35)	2.04	
			(20,730)		701
$Co(Py)_4(Cl_2A)_2$ +Py	9260		19,900 (19,600)		701
$Co(bpy)_3^{2+}$	11,000		22,000		710
$Co(opd)_3^{2+}$	9800	18,000	20,200	1.84	714
$Co(opd)_2Br_2$	9500	(18,600)	20,600	1.96	714
CoO/MgO	8500	17,200	19,600	2.02	200

For a more complete listing, see ref. [719]. a) Trigonal prismatic

In the octahedral polymeric $MCoCl_3$ systems, the spin forbidden bands are greatly intensified through exchange processes [790].

A range of lower symmetry cobalt(II) complexes have been discussed. The interpretation of the data is often complicated by uncertainty concerning the symmetry of the ground state, since the sense of the splitting of the $^4T_{1g}$ ground state in O_h may not be obvious.

The $[Co(en)_3]^{2+}$ ion has D_3 symmetry and single crystal absorption and CD studies [750] show a small trigonal splitting of the orbital triplet states (viz: ^4E 10,400; $^4A_1$ 10,500; ^4E 21,100; $^4A_2$ 21,800 cm^{-1}). The polarisation data lead to the supposition that $^4A_2$ (of $^4T_{1g}$) is the ground term.

Palmer [711] has also studied single crystals of the tris(octamethylpyrophosphoramide)cobalt(II) cation which also possesses D_3 symmetry. Polarised data reveal absorption at 6750(5.6)(xy,^4E), 6800(4.4)(z,$^4A_1$), 18,380(26.7)(xy,^4E), and 19,050(5.12)(z,$^4A_2$) cm^{-1} [711]. The strong polarisation of the components of the $^4T_{2g}$ band requires that the ground state be $^4A_2$ since a ^4E ground term would lead to transitions permissible in both polarisations (i.e. it should be less dichroic). Upon cooling the crystal, the pi (z) component of the spectrum tends to lose intensity relative to the xy component suggesting that a low lying ^4E state may be losing population. MCD data have also been reported for this system [1327].

Wentworth [599,753] and Holm [754] and their co-workers have discussed the spectra of six coordinate trigonal prismatic derivatives. Such systems have D_{3h} symmetry and energy level diagrams for d^7 have been published [754]. The twist angle which differentiates such species from regular octahedral systems may lie between 0^o (trigonal prism) and 60^o (trigonal antiprism) leading to systems whose spectra are often very much like regular octahedral species.

The $Co(H_2O)_6^{2+}$ ion has been studied by doping Co(II) in a crystal of chiral $Zn(H_2O)_6SeO_4$ [763]. All the bands in the visible

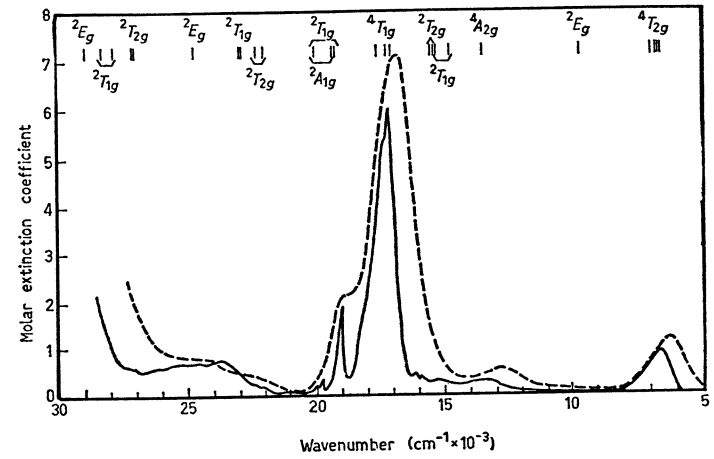


Fig.6.35 The absorption spectra of octahedral $CoCl_2$ at room temperature (hatched) and -253^oC (solid line). The calculated energies are Dq = 690, B = 780, C = 3432 and λ = -140 cm^{-1}, indicated by vertical lines.

and near infrared regions have Kuhn anisotropy factors suggestive of magnetic dipole allowed bands. Several of the spin forbidden transitions are clearly identified, and, at low temperature the CD spectrum shows vibronic structure with a totally symmetric progression built on several false origins [763].

Oleari and Day [1161] report some beautiful low symmetry octahedral data for Co(II) doped into single crystal MWO_4 (M = Zn,Mg) at temperatures from ambient to 4.2 K. The data are essentially temperature insensitive (static distortion, electronic selection rules) with splittings primarily determined by the local C_{2v} site symmetry, with a minor C_2 perturbation. With the electric field aligned along x,y or z of the molecular axes, at most only one transition should be electronically allowed within each 'octahedral' orbital triplet, e.g. $^4A_1$ or $^4A_2$ or $^4B_1$ of $^4T_2$, (see discussion of six coordinate C_{2v} Ni(II) complexes below for more detailed analysis). With the electric field aligned along the crystal a,b or c axes, all three transitions may occur depending upon the direction cosines of x,y and z with respect to a,b, and c. Since these are different for each axis, and are known, it follows that measuring the variation of the intensity ratios for each absorption, along each direction, provides a means of assigning the spectra. This method is utilised by Oleari and Day in their analysis. The procedure is certainly effective in providing a guide to the correct assignments but must be used with caution since, as indicated by these authors, vibronic contributions to the intensity will generally occur rendering the analysis of predicted band ratios much more tiresome.

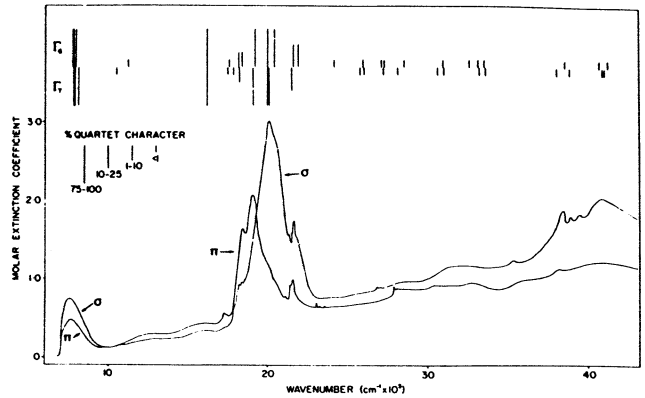


Fig.6.36 The polarised absorption spectra of K_2CoF_4 with energy levels calculated with Dq = 825, Ds = 195, Dt = 10 and ζ = -490 cm^{-1} [751].

Four fold distorted six coordinate cobalt(II) has also been fairly extensively studied, though research here has been much less productive than with analogous nickel(II) systems because of the ground state degeneracy and the complexity due to spin forbidden transitions. The appropriate energy level diagram can be deduced from Fig.2.21 with the F term splitting inverted. The sequence of splitting of the various orbital triplets can only be proven by experiment, though some predictions, as discussed below, can be made.

Various salts M_2CoF_4 show tetragonal distortion of the $[CoF_6]^{4-}$ octahedral array. Single crystal studies [751] reveal dichroic spectra (Fig.6.36) which, depending upon M, arise from axially elongated or compressed octahedra. Generally in tetragonal systems, the sign of Dt can be deduced from the relative magnitudes of Dq(xy) and Dq(z); in such cases as this where the ligands along xy and z are the same, such a choice cannot be made in the absence of structural information. The magnitudes and signs of the parameters obtained in this study were obtained on the basis of the relative polarisations of the magnetic dipole allowed spinor components of $^4T_{2g}$ [751]. The sign of Dt determines the relative magnitudes of Dq(z) and Dq(xy) (Eqn.1.81). Where structural information was available the spectroscopic and structural data were in agreement. For example for the compressed species $K_2Mg(Co)F_4$), Dq(F_z) = 818 and Dq(F_{xy}) = 800 cm^{-1}.

In this last cited case, the authors were unable to deduce the sign of Ds which they assumed was the same as for Dt, i.e. negative. If the AOM model is used in this specific example, a positive Ds leads to negative dσ and

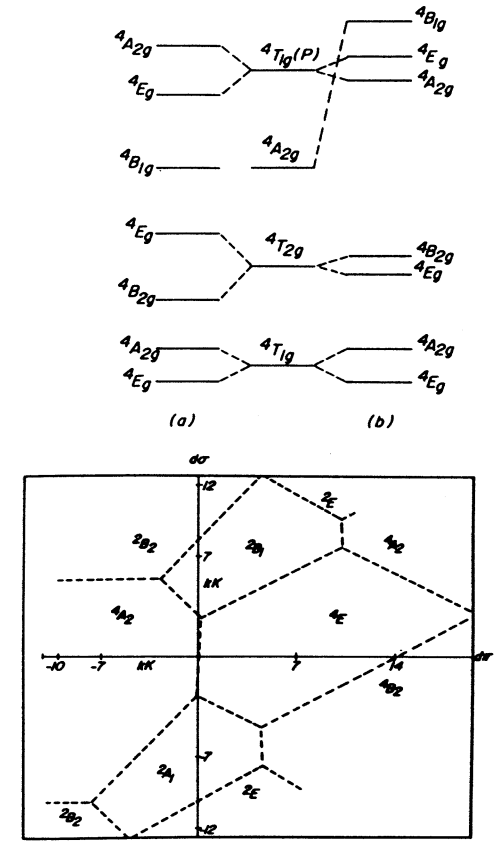


Fig.6.37 (Upper) Splitting diagram for tetragonal (D_{4h}) d^7 appropriate for (a) $Co(s\text{-}Et_2en)_2X_2$, X = Cl, Br etc., and (b) $Co(s\text{-}Me_2en)_2(NO_3)_2$ [749]. (Lower) Ground state energy space diagram for tetragonal (D_{4h}) d^7 with DQ = 30,000, Dq = 10,900, B = 700 and C = 3000 cm^{-1}.

dπ, while a negative Ds leads to positive dσ and dπ. Only the positive values of dσ and dπ are consistent with the known compressed structure of K_2MgF_4, and therefore the authors were probably correct in choosing a negative value for Ds.

Ferguson also studied hydrated cobalt(II) chlorides which have a tetragonally distorted structure. Polarised data for the hydrate were compared with the deuterate [702,703] (see also [704])(see Table 6.20). An axially elongated site was observed for trans-$Co(H_2O)_4Cl_2$ $2H_2O$ (Dq 810, DQ 22,270, Ds -640, Dt 30 cm^{-1}).

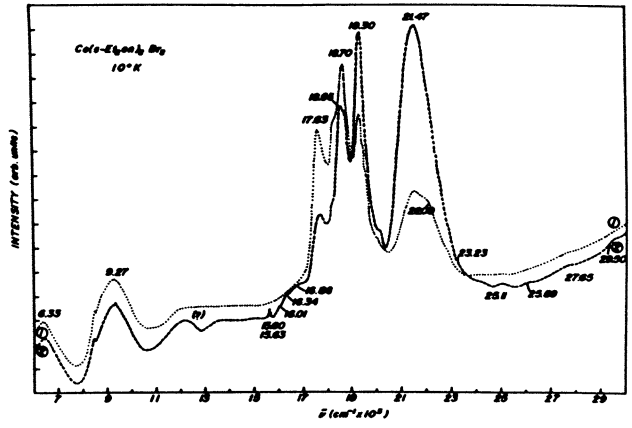


Fig.6.38 The single crystal spectra of $Co(s\text{-}Et_2en)_2Br_2$ at 10 K [749].

In addition, in parallel to hydrate studies reported earlier in this Chapter (e.g. V(III)), an anomalous crystal field band was observed, considerably to higher energy than the principal d-d absorption. The intensity of the band increased with decreasing temperature and the energy separation from the next lower lying d-d bands was approximately 3200 cm^{-1} in the hydrate and 2340 cm^{-1} in the deuterate. It is evident that these 'anomalous' bands involve coupling of an OH stretching vibration to the electronic transition.

Several authors have treated the spectra of trans-CoN_4X_2 species, formally of D_{4h} symmetry [e.g.637,748,749,752,783,784]. Single crystal data were reported [749] for a series trans-$Co(s\text{-}Et_2en)_2X_2$ (Table 6.20). Up to six spin allowed transitions are possible (Fig. 6.37 (upper)) leading to the possibility of deriving significant chemical information from the spectrum (see Chapter 9). The resolution is sufficiently good (Fig.6.38) that all the spin allowed bands (except the lowest between the split components of the ground state $^4T_{1g}$) can be observed. In this case the sign of Dt was deduced from the known relative fields of amine and X and a fit to the spectrum obtained. 'Goodness' of the fit was evaluated not only in terms of the agreement between calculated and observed band energies, but also on the similarity between AOM parameters obtained for parallel pairs of, usually isomorphous, cobalt(II) and nickel(II) complexes.

> Agreement between observed and calculated band energies is not itself a good criterion for accuracy in band assignments, except perhaps where the number of observables significantly exceeds the number of variables.

Table 6.20

TETRAGONAL COMPLEXES OF COBALT(II)

i) Six coordinate---(high spin)

Complex	$^4T_{2g}$ $^4B_{2g}$	^4E_g	$^4A_{2g}$ $^4B_{1g}$	$^4T_{1g}(P)$ ^4E_g	$^4A_{2g}$	Ref
$Co(H_2O)_4Cl_2$	8330	8060	16,450	18,600	22,250	702
$Co(s\text{-}Et_2en)_2Cl_2$ [a]	7180	10,020	19,370	20,050	21,840	749
$Co(s\text{-}Et_2en)_2Br_2$ [a]	6325	9270	17,630	19,000	21,470	749
$Co(s\text{-}Et_2en)_2(NCS)_2$ [a]	10,355	10,355	21,720	20,840	20,560	749
$Co(s\text{-}Me_2en)_2(NO_3)_2$ [b]	7710	10,000	19,867	20,490	23,600	749
$Co(4\text{-}Pic)_4Cl_2$	7300	8800	[18,500	19,200	20,200]	748
$Co(4\text{-}Pic)_4Br_2$	6050	8700	(15,400)	[17,500	18,000	
				18,900	20,600]	748
$Co(4\text{-}Pic)_4I_2$	5400	8300	(14,900)	[17,400	18,500]	748
$Co(dan)_2Cl_2$	(8000)	(9500)	(16,500)	[17,700	19,700]	758
$Co(vinim)_4(SiF_6)_2$	5000	9500	(15,800)	19,200	24,800	
				20,200		931
$Co(vinim)_4Cl_2$	7000	9100	?	18,200	21,200	
				19,700		931

ii) Five Coordinate---(high spin square pyramidal)

$Co(Ph_2MeAsO)_4(NO_3)^+$	<5000	6900	11,300	16,600	20,500	
				17,000		785
$Co(Et_4dien)Cl_2$ [cd]		11,000	15,300	18,300	20,200	
				18,900		786
$Co_2(bz)_4(quin)_2$		8100	13,500	17,400	22,100	
				18,800		787
$Co(dacoda)(H_2O)^{2+c}$		8050	14,000	18,200	22,000	788

a) Ground term ^4E_g b) Ground term $^4A_{2g}$ c) Assignment tentative. The groupings in square brackets probably encompass the assignments indicated but they were not specifically assigned by the authors. d) This complex has a structure intermediate between square pyramidal and trigonal bipyramidal [958].

iii) Five Coordinate---(low spin square pyramidal)

Complex	Absorption Spectra			Ref
$Co([14]aneN_4)/H_2O$	10,400(55)	-		767
$Co(Me_6[14]4,11\text{-}dieneN_4)/H_2O$	13,300(2.5)	22,600(118)		767
$Co([14]tet\text{-}eneN_4)(ClO_4)_2$	7100(39)	23,800(257)	28,200(1836)	
	34,700(1440)			798

cont

Table 6.20 cont

Five Coordinate---(low spin square pyramidal)		
Complex	Absorption Spectra	Ref
$Co([14]tet\text{-}eneN_4)(CH_3CN)^{2+}$	7800(76) 17,400(38) 24,400(258) 29,600(2823) 34,500(2500)	798
$Co([14]tet\text{-}eneN_4)Cl^{2+}$	7100(30) 22,200(1265) (28,500) (37,000)	798
$Co([14]tet\text{-}eneN_4)Br^{2+}$	6800(11) (19,600) (31,300) 37,000(6670)	798
$Co(CN)_5^{3-}$	10,350(233) 16,200(7) 23,300(65) 31,700(527) (38,100)	793
$Co(CNMe)_5^{2+}$	14,280(300) (22,200) 30,300(10) 37,800	793
$Co(CNPh)_5^{2+}$	15,150 (22,200) 31,700	793
$Co(Me_2NPF_2)_3Br_2$	6000 14,050 15,700 21,650 25,400	797
$Co(Me_2NPF_2)_3I_2$	5700 13,150 14,800 20,500 23,700	797
$Co(P_4)Cl^{+2}$	8330 10,500 14,000 (18,200) 22,100 (25,600)	736
$Co(P_4)Br^{+}$	8060 10,000 13,340 (18,200) (22,230) (26,350)	736
$Co(P_4)I^{+}$	7940 9800 12,900 (17,600) (22,200)	736
$Co(P_4)NCS^{+}$	7360 9600 13,050 18,890 23,140	736
$Co(sacsac)_2PPh_3$	6800(17) (8300) 11,000(24) (15,200) 17,700(4330) 21,800(3790) (23,900)	799
$Co(AP)_2Cl$	6900 11,000 12,500 21,900 28,500	799
$Co(AP)_2Br$	6800 11,000 (12,400) 14,000 21,600 28,200	799
$Co(AP)_2I$	6750 11,000 (12,000) 15,700 20,750 22,820 28,600	799
$Co(ttp)Cl_2$	7490(36) (14,300)(86) (17,900)(327) 22,300(1178) 29,900(6900)	794
$Co(ttp)Br_2$	7460(37) (13,900)(78) (17,200)(319) 21,700(1191) (29,400)(4140)	794
$Co(ttp)I_2$	6900(38) 12,800(77) 15,900(332) 21,100(1640) (27,000)	794

iv)

Four Coordinate---(low spin square planar)		
Complex	Absorption Spectra	Ref
$Co(acac)_2(-)pn^a$	3970(96) (4300) (4870) 5550(16) 8500(20) (20,600)(1100) (22,000) 26,800(6500)	772
$Co(aba)_2(-)pn^a$	5100(18) 6250(22) 6900 11,050(73) (11,900) (16,670)(370) 18,900(5000)	772
Co(salen)	3900(65) 8300(16)	768
$Co(trans[14]diene)^{2+}$	6300(35) 22,400(130) 29,800(2500)CT	593

a) diketone Schiff bases

No assumption was made about the splitting of the $^4T_{1g}$ ground term. In fact, for most of the complexes, a good fit could only be obtained for a ^4E_g ground term, i.e. splitting $^4E_g < {}^4A_{2g}$. This is consistent with the ground state energy space diagram approach (see section 3.10) which shows (Fig.6.37 (b)) that for reasonable values of DQ and B, a ^4E_g ground term will arise with positive $d\pi$, and a $^4A_{2g}$ ground term with negative $d\pi$. In these amine cobalt halide complexes, $d\pi$ is reasonably expected to be positive since the amine has no pi capability.

Palmer studied the crystal spectrum of $Co(L\text{-histidine})_2(H_2O)_2$ [757] (Fig.6.39). The absence of significant temperature dependence in the intensity suggested that the transitions are electronically allowed in the point group C_{2v}. The electric dipole rules allow transitions to A_1, A_2 and B_1 when the electric vector is parallel to x, y, and z respectively (ground term B_1 in C_{2v}(III), see Chapter 3). Transition to B_2 is forbidden. Since T_2 (near ir band) spans $A_1 + A_2 + B_2$ there should be no z component to the absorption; in fact there is a z component (perhaps vibronically allowed) but it is weak (Fig.6.39). The visible region transition (T_1) spans $A_2 + B_1 + B_2$ and there should be no x component; again there is presumed vibronic intensity or some low symmetry component to the field. The strong band at 22,400 cm^{-1} (in z) is assigned to $^4B_1(^4A_2)---{}^4B_1$, and that at 18,700 cm^{-1}, also in z, to $^4B_1(^4T_1)---{}^4B_1$. These two transitions are magnetically dipole forbidden and support for their assignment stems from their absence in the CD spectrum of a solution of this species [757].

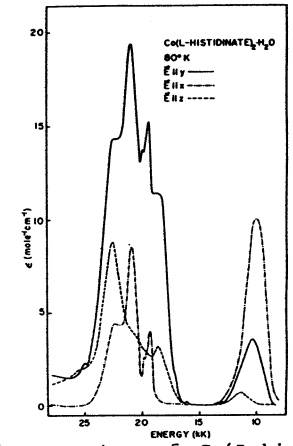


Fig.6.39 The molecular spectra of $Co(L\text{-his})_2.H_2O$ calculated from the polarised spectra obtained at 80 K [757].

Low Spin Six Coordinate Complexes: These species possess a $(t_{2g})^6(e_g)^1$ configuration and should be Jahn Teller distorted. Axially distorted complexes commonly possess an unpaired electron in the z^2 orbital (ground state $^2A_{1g}$) or in the x^2-y^2 orbital (ground state $^2B_{1g}$). The right-hand side of Fig.2.15 is relevant but recognise that the orbital degeneracies of the states will be lifted by the axial distortion, if present. Kida and co-workers have extensively studied low spin Co(II) complexes with especial interest in square planar (vide infra) and six coordinate systems [764,765]. The energy level diagram in Fig.6.40a is appropriate for both these types of low spin system [764] (axial elongation).

The Co(II)salen system has also been exhaustively studied because of its relevance to oxygen absorption [768-772]. Many low spin four and six coordinate Co(II) species show low intensity absorption near 8500 cm^{-1}, initially assigned as a spin forbidden transition [769] but more likely it is spin allowed. In some cases it may be the transition within the split components of ^2E_g (in O_h); however these systems often display additional absorption of similar intensity, at energies between 3000 - 6000 cm^{-1} [768,772]. Clearly care must be taken in assigning these spectra since so many transitions are possible.

The $[Co(NO_2)_6]^{4-}$ ion absorbs at 15,500 cm^{-1} ($^2E_g(^2T_{2g})$---$^2A_{1g}$, and at 18,000 cm^{-1} (transition to components of $^2T_{1g}$)[764]. There is also absorption at 8000 cm^{-1} assigned to $^2B_1$---$^2A_1$, the transition within the split components of ^2E_g (in O_h) [773] (also see Chapter 5.7). Axially compressed systems often have broad ill defined spectra [765,766].

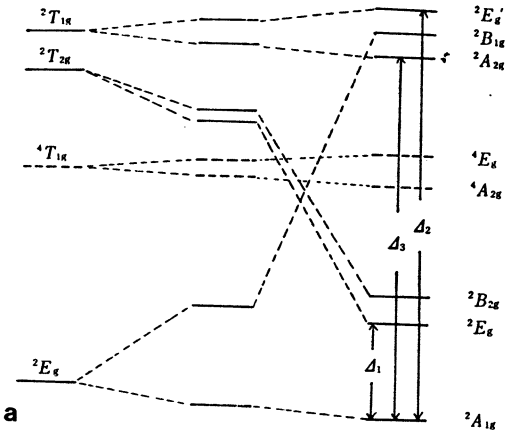


Fig.6.40 (a) Splitting diagram for low spin d^7 states in tetragonal symmetry showing small and large tetragonal distortion [764].

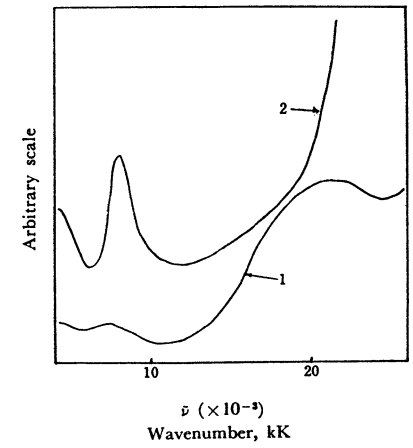


Fig.6.40 (b) The diffuse reflection spectra of 1:$Co(en_2)(AgI_2)_2$ and 2:Co(sals) [764].

There is a close similarity between these spectra and those of low spin five coordinate square pyramidal, and low spin four coordinate square planar complexes to be discussed below. In particular note that the energy of the z^2 level will become more negative (i.e. more stable) as the coordination number changes from six coordinate to five coordinate to four coordinate. In this six coordinate case, the lowest energy transition is expected to be z^2---x^2-y^2, i.e. within the split components of e_g [805]. However in the five and four coordinate systems, the z^2 level may lie close enough to components of t_{2g} that transitions (z^2---t_{2g}) become very low in energy.

Five Coordinate Cobalt(II) Complexes: A wide variety of such species are known, both high and low spin, and both primarily trigonal bipyramidal or square pyramidal.
High spin trigonal bipyramidal - The ^4F term splits, in D_{3h} symmetry, into (in ascending order of energy) $^4A_2' < {}^4A_2'' < {}^4A_1'' < {}^4E'' < {}^4E'$ while ^4P yields $^4A_2' < {}^4E''$ [774-776,789] (Fig.6.41). Such a sequence seems appropriate for a regular trigonal bipyramidal CoL_5 species. However if there is angular distortion [774], or if the species is irregular, e.g. CoL_4L' or $CoL_3L'_2$ the sequence may be slightly modified. For example if $Dq(L_{ax}) < Dq(L_{eq})$ then the splitting of the ^4P state may be inverted [776]. The degeneracy of the $^4A_2''$, $^4A_1''$ pair of states is lost if the axial-equatorial ligand angle deviates from 90° [774,776]. Transitions from the $^4A_2'$ ground state generally occur both in the near infrared, and in the visible region. Indeed the spectra of these species are quite rich (Table 6.21) (Fig.6.42) [738,740,760,

Table 6.21

FIVE COORDINATE COMPLEXES OF COBALT(II)

Complex	High spin trigonal bipyramidal (D_{3h}) (C_{3v}) $^4A_1"$, $^4A_2"$	$^4E"$	$^4E'$	$^4A_2'(P)$ $^4A_2(P)$	$^4E"(P)$ $^4E(P)$	Ref
$Co(Me_6tren)(NCS)^+$	<4000	5800	14,800	16,500	21,000	775
$Co(Me_6tren)Br^+$		5700	12,300	15,600	19,500	
				16,100		979
$Co(Me_6tren)Cl^+$		5800	12,600	15,500	20,200	
				16,610		979
$Co(Me_6tren)I^+$		5700	12,100	16,100	18,600	
					19,300	
					(21,600)	979
$Co(Me_6tren)ClO_4^+$		5900	13,500	15,900	20,600	
				16,400		979
$Co(NS_3)Br^+$		5200	11,800	13,800	18,800	776
$Co(NP_3)Br^+$	9000	5700	10,000	13,300	17,500	760,776
$Co(Et_4dien)(N_3)_2$ [a]			12,400	15,400	19,100	
					20,150	957
$Co(Et_4dien)(NCS)_2$			13,100	15,900	20,000	
					21,000	957
$Co(Et_4dien)Cl_2$			10,500	15,200	19,200	957
$Co(Et_4dien)Br_2$			10,500	15,200	19,000	957
$Co(Et_4dien)I_2$			10,100	14,700	17,170	
				(15,500)	18,100	957
$Co(S\text{-}tan)Br^+$		5600	12,100	16,300	19,100	
					20,100	779
$Co(MeTPyA)Br^+$	(5900)	7100	11,500	17,100	18,700	780

Molar intensities are of the order of 20 - 100 in the near ir, and 50 - 500 l mol^{-1} cm^{-1} in the visible region. a) see Footnote in previous Table concerning these Et_4dien complexes. See [411] for further data.

774-781,899]. Polarised spectra [760,775,776] and CD data [779] provide assurance that the assignments shown in Table 6.21 are basically correct. Many of these complexes however are distorted so that assignment is more appropriate in a lower group such as C_{3v}. Indeed the spectrum of the trigonal bipyramidal $[Co(PyNO)_5]^{2+}$ ion [781,782] can only be analysed in terms of C_{2v} symmetry with a two-fold axis bisecting the equatorial plane. Two bands at 10,500 and at 12,500 cm^{-1} are totally polarised, one in z and one in y in the equatorial (yz) plane, an observation which can only arise if the two directions in this plane are not equivalent (hence C_{2v}).

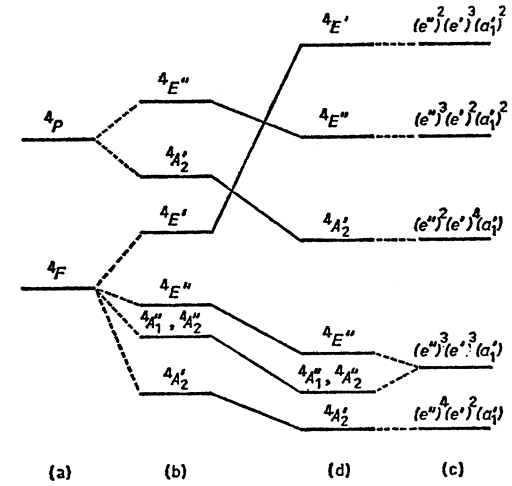


Fig.6.41 Correlation diagram for a d^7 ion in a high spin trigonal bipyramidal (D_{3h}) environment. (a) Spherically perturbed ion, (b) weak field terms, (c) strong field configurations neglecting interelectronic repulsion and (d) including interelectronic repulsion. Only the S = 3/2 states are considered. Not to scale.

These bands are assigned to the split components of $^4E'(F)$ in D_{3h}. A broad band at 18,000 cm^{-1} probably contains the three transitions to the ^4P level in C_{2v} symmetry.

Low spin trigonal bipyramidal - Venanzi and co-workers have calculated the d^7 energy levels in a strong trigonal bipyramidal field [625,998]. The ground configuration, $(e")^4(e')^3$ derives from ^2G. The early theory predicted one fairly low energy transition, probably in the near infrared, and a grouping of closely spaced high energy transitions in the visible (to components of ^2G, ^2H and ^2P). The spectrum shown in Fig.6.43 reveals this pattern. Other complexes show similar patterns [792]. However more recent analysis by the same group [952] led to a slightly different interpretation. The ground state is $^2E'$ and is subject to Jahn-Teller distortion. The weak band in the near infrared is now regarded to be a transition within the Jahn-Teller split $^2E'$ state. The lower energy visible band then becomes a transition from a component of e' to a_1' and the higher energy visible absorption is a transition between the split components of e" and e'. The visible region transitions are remarkably intense, in the range 1000 - 5000 l mol^{-1} cm^{-1}.

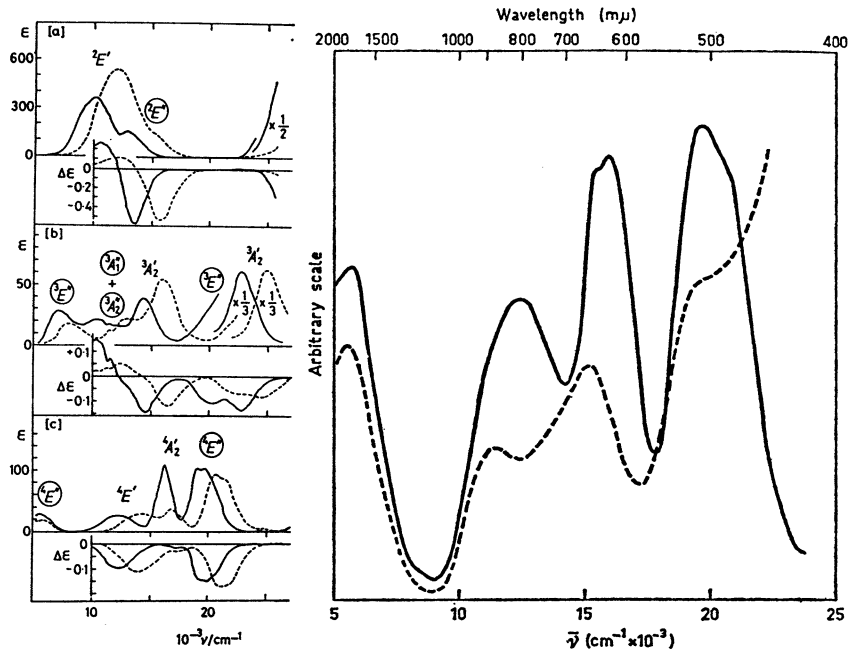


Fig.6.42 (left) Absortion and CD spectra of $[MBr(s\text{-tan})]ClO_4$ (solid lines) and $[M(NH_3)(s\text{-tan})]ClO_4$ (hatched lines) in nitromethane; (a) Cu(II), (b) Ni(II) and (c) Co(II). The circles indicate magnetic dipole allowed transitions under D_{3h} symmetry [779]. (Right) The reflectance spectrum of five coordinate trigonal bipyramidal $[Co(Me_6tren)Cl]^+$ (solid line) and bis(N-methylsalicylaldimino)-cobalt(II) (hatched line).

High spin square pyramidal - These species are best regarded as analogs of the tetragonally distorted six coordinate cobalt(II) with a very weak axial field. The diagram in Fig. 6.38 [789] is appropriate; some data are presented in Table 6.20. The gross spectra (e.g. Fig. 6.44) do not differ so markedly from high spin trigonal bipyramidal systems since both yield near infrared and visible absorption. However the square pyramidal systems do generally have lower intensities (often less than 100 l mol^{-1} cm^{-1}) than the trigonal bipyramidal complex spectra. The magnetic properties may also offer a criterion for distinction. Note that the high spin $Co(Et_4dien)X_2$ species are known to have a structure intermediate between square pyramidal and trigonal bipyramidal [958] and their spectra have been interpreted in both symmetries [786,957].

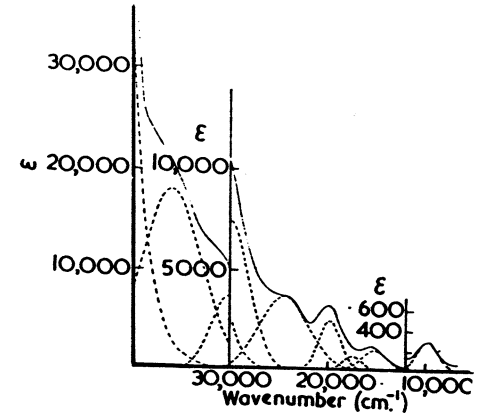


Fig.6.43 Absorption spectrum of low spin trigonal bipyramidal $[Co(o\text{-}Ph_2P(C_6H_4)_3P)NO_3]^+$ in dichloromethane. The dotted lines represent gaussian analysis [625].

Several studies with polarised light have confirmed the assignments shown in Table 6.20 [785,787]. Gerloch and co-workers [785] discuss the complex $[Co(Ph_2AsO)_4]^{2+}$ ion in detail considering the effect of variations in factors such as crystal field strength and bond angle, upon the electronic energy levels. In this study, little dichroism was observed, a fact rationalised by the assumption that the splitting of the $^4T_{1g}$ ground term is small in this system.

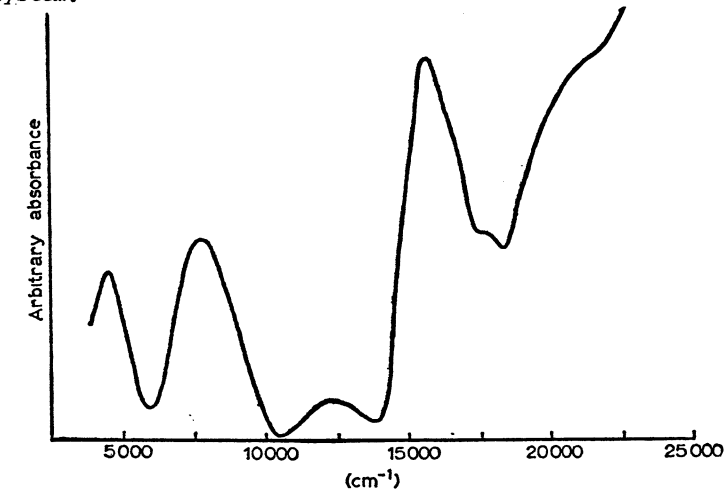


Fig.6.44 The diffuse reflectance spectrum of five coordinate square pyramidal β-Co(paphy)Cl_2.

Similar five coordinate square pyramidal complexes of cobalt(II) with Me_3AsO, [1001], $Ph_3P(As)O$ [1002] and paphy (pyridine2-aldehyde 2'-pyridylhydrazone) [1041] have been studied.

Low spin square pyramidal - These systems parallel low spin six coordinate, and low spin square planar cobalt(II) complexes. Their spectra have been extensively discussed [793-801]. Data are presented in Table 6.20.

The ground state configuration, in order of increasing energy, is generally taken to be $(xy,(b_2))^2(xz,yz,(e))^4(z^2,(a_1))$. Since these are strong field complexes, many authors have chosen to use a strong field approach by considering the possible orbital transitions and then incorporating the necessary interelectronic repulsion term [760,770,793-795,797,802]. Recalling that the energy of the z^2 orbital will depend upon coordination number (above) relatively low energy transitions should occur, specifically $xz,yz \rightarrow z^2$ ($^2E \rightarrow {}^2A_1$) and $xy \rightarrow z^2$ ($^2B_2 \rightarrow {}^2A_1$). Moreover in a series of complexes ML_4Z, these two transitions should shift to the red as X becomes a weaker ligand, e.g. in the series Cl, Br and I. Indeed consideration of the data in Table 6.20 shows that the two lowest energy bands do often shift in this fashion suggesting they are the two transitions indicated above. Detailed assignments for the higher energy bands are not generally available in this complex system, where spin forbidden transitions can also be expected.

The $[Co(CN)_5]^{3-}$ ion is square pyramidal with a strong axial interaction. The preferred assignment places $^2B_1 \rightarrow {}^2A_1$ at 10,350 cm^{-1} and $^2E \rightarrow {}^2A_1$ at 23,300 cm^{-1} [1000].

Note that macrocyclic complexes in this group, e.g. [798] are expected to be close to regular square pyramidal, but that more flexible ligands may give rise to structures which are hybrids of the square pyramidal and trigonal bipyramidal stereochemistries, with concomitant 'mixed' spectra.

Four Coordinate Tetrahedral and Pseudotetrahedral Cobalt(II) Complexes: A wide variety of such species exist, commonly intense blue or green in colour, though other colours, such as the pink dipivaloylmethane derivative, are known [721-725,744,809-834 and refs. in Table 6.22]. The $^4T_1(F) \rightarrow {}^4A_2$ and $^4T_1(P) \rightarrow {}^4A_2$ transitions appear as multiple absorption in the near infrared and visible regions respectively (Table 6.22) (Fig.6.45). The intensities of these bands are generally of the order of $10\text{-}10^2$ for the former, and $10^2\text{--}2 \times 10^3$ l mol^{-1} cm^{-1} for the latter, and allow a fairly clearcut distinction to be made between tetrahedral and octahedral cobalt(II) derivatives.

The bandwidths, in CoL_4 species, ranging between 1500 and 2500 cm^{-1} are too large to be accounted for by spin orbit coupling (theoretically, to first order, 6λ and 4λ $^4T_2$ and $^4T_1$ respectively), although such a mechanism may contribute [820]. Low temperature studies [822] in the solid state suggest that low symmetry fields, and vibrational structure may be responsible.

Table 6.22

FOUR COORDINATE COMPLEXES OF COBALT(II)

Complex	$^4T_1(F)$		$^4T_1(P)$		Ref
Co(II)/CdSe	3000 ($^4T_2$)				
	5500		13,000		382
$CoCl_4^{2-}$	4780	5100	14,300	14,900	
Bu_4N salt	5400	5920	15,750	(16,230)	224
$Co(MORPO)_4^{2+}$	6150		16,340		598
$Co(MORPO)_2Cl_2$	5670		15,720		598
$Co(MORPO)_2Br_2$	5390		15,380		598
$Co(MORPO)_2I_2$	5360		14,930		598
$Co(MORPO)_2(NCS)_2$	6730		16,230		598
$Co(Ph_3AsO)_4^{2+}$	5710(66)	6490(72)	15,480(415)	(16,050)	
	6490(72)		16,670(395)	(17,300)	
			17,950(260)		217
$Co(Ph_3AsO)_2Cl_2$	5240(60)	5780(65)	14,810(536)	(16,000)	
	6560(67)		(16,950)		722
$Co(Ph_3AsO)_2(NCS)_2$	(6450)	7140	15,630(835)	16,390(825)	
	(7750)		(17,400)		217
$Co(DPM)_2$	7000(45)	(10,750)	18,000(41)		229
$Co(NCNPh)_4^{2-}$	7550 (220)		15,380(880)	15,570(825)	
			16,250(980)	17,240(555)	
			15,580(535)		836
$Co(Me_4pn)Cl_2$	6000(26)	7300(65)	15,300(466)	17,300(236)	
	10,000(18)		(18,000)		916
$Co(2\text{-Pic})_2(NCSe)_2$	8340(172)		16,200(757)	16,470	
	9300(155)				230
$Co(4\text{-EtPy})_2Br_2$	5955(73)	6905(85)	15,360(955)	(15,820)	
	9265(71)		16,690(502)		230
$Co(2\text{-EtPy})_2Br_2$	5670(63)	6690(73)	15,130(674)	(15,700)	
	(8280)		(16,690)		230
$Co(2\text{-EtPy})_2(NCS)_2$	7365(129)	8165(129)	16,020(956)	(16,580)	
	9205(147)		(17,710)(416)		725
$Co(2\text{-MeIz})_4^{2+}$	8800		17,700	(18,300)	635
$Co(4\text{-DMP})_2Cl_2$	6100	7080 9460	15,450	15,920 16,780	745
$Co(4\text{-DMP})_2I_2$	6020	7020 8900	14,490	15,270 16,130	745
$Co(Me_4en)Cl_2$	(6000)	7250	15,400	17,200	
	10,000				726
$Co(TDPME)Cl^+$	6940(83)		14,710(454)	15,380(370)	
			17,090(296)	20,410(46)	759
$Co(MImT)_4^{2+}$	6665		14,492		739
$Co[(Ph_2PS)_2N]_2$[b]	6615		14,260		547
$Co[(Me_2PS)_2N]_2$[b]	6575		14,400		547
$Co[(Ph_2PO)_2N]_2$[b]	6875		17,375		547
$Co[(Ph_2PNH)_2N]_2$[b]	8180		17,650		547

cont

Table 6.22 cont.

Complex	$^4T_1$	$^4T_1(P)$	Ref
$Co(NH_3)_4^{2+}$	5900($^4T_2$) 10,000($^4T_1$)	16,000 18,000 19,200	755
$Co(tu)_4^{2+}$	7220(172)	13,890(632) 14,580(690) (15,600)	728
$Co(tu)_2Cl_2$	5300(106) 6300 7500(70)	13,950(396) 15,000(462) 16,200(427)	807
$Co(S_2PF_3)_2$	6000 (7000) 8410 (9300)	14,900 15,310 (16,000) (17,200) 18,600	843
$Co(dtt)_2Cl_2$	5970 7465	13,930 14,880 16,340	806
$Co(dtt)_2Br_2$	5850 7220	13,910 14,770 15,455	806
$Co(dtt)_2I_2^{2+}$	5585 6825 7720	13,460 14,265	806
$Co(Py_2SH)_4^{2+}$	6800 8410	14,200 (14,630) (15,240) (19,050) (19,760)	804
$Co(etu)_4^{2+}$	6500(123)	13,875(639) 14,600(726) 15,720(504)	216
$Co(su)_4^{2+}$	6000 7600	13,200 (14,800)	803
$Co(su)_2Cl_2$	5300 6200 (7500)	13,000 15,000 (15,800)	803
$Co(su)_2Br_2$	5800 6700 7800	13,300 14,200 15,000	803
$Co(su)_2I_2$	4800 6500 7400	13,400 14,900	803
$Co(dietu)_2Cl_2$	5600 7100 7400	14,000 15,400 16,400	729
$Co(esu)_2Cl_2^{2+}$	5815	12,500 15,265 16,000	731
$Co(Me_3PSe)_4^{2+}$	5600 7000	13,300 13,800 14,600	730
$Co(2MePy_6SH)_4^{2+}$	7330	13,500 14,240 15,470	804
$Co(PPh_3)_2Cl_2$	6400 8000	13,600 15,800 16,700	729
$Co(PPh_3)_2Br_2$	6200(110) 7700(95) 10,200(80)	13,100(610) 14,900(970) 15,600(740)	732
$Co(L^+CH_3)Cl_3^{-a}$	4700($^4A_2$) 7700(^4E)	15,400(256)	580
$Co(L^+CH_3)Cl_3^-$	4500($^4A_2$) 7700(^4E)	15,380(^4E) 15,750(44E) 16,690($^4A_2$)	848
$Co(L^+CH_3)Br_3^{-a}$	4300($^4A_2$) 7400(^4E)	15,200(371)	580,847
$Co(L^+CH_3)I_3$	4200($^4A_2$) 6900(^4E)	13,800(645)	580

a) $^4E(^4T_2)$ 3200 cm^{-1}. b) weighted band averages. For a more extensive listing of tetrahedral and pseudo tetrahedral Co(II) systems, see ref. [719,1313].

Moreover, in the visible region there are many spin forbidden quartet-doublet transitions which can gain intensity by interaction with the visible region spin allowed band. A recent highly detailed study [836] outlines the contributions of these various mechanisms. Certainly, as detailed further below, low symmetry pseudo-tetrahedral complexes, CoL_3X and CoL_2X_2 usually have even wider bandwidths.

For computational purposes, the transition energies, to calculate Dq and B, are based on the centres of intensity of these

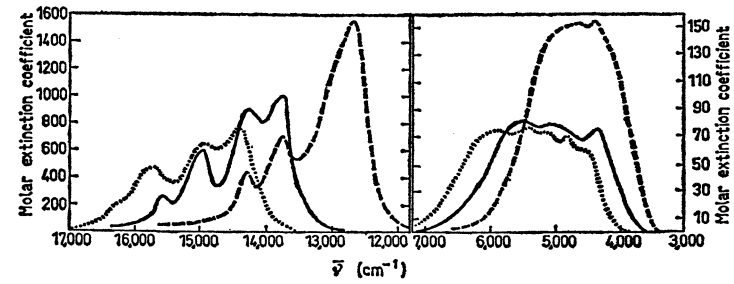


Fig.6.45 The absorption spectra of tetrahedral $(n\text{-}Bu_4N)_2CoX_4$ salts in solution containing an excess of $n\text{-}Bu_4NX$; X = Cl (hatched), X = Br (solid), X = I (dotted).

bands. In Chapters 4 and 9, the relationship between oscillator strength and nephelauxetic parameter is discussed. Generally B increases and f, the oscillator strength decreases, with increasing ionicity.

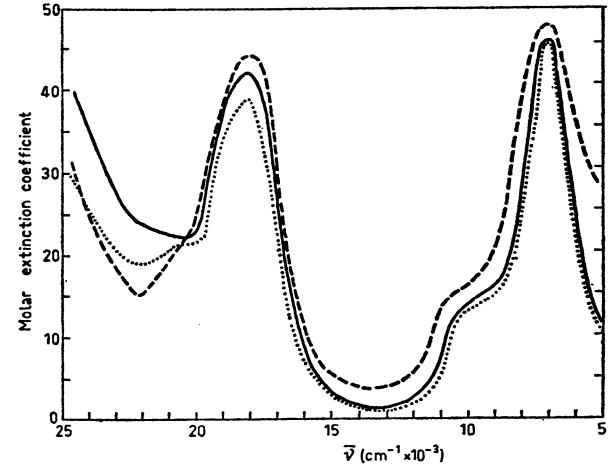


Fig.6.46 The absorption spectra of tetrahedral bis(dipivaloylmethanido)cobalt(II), 0.141 M in benzene (solid), 0.0141 M in benzene (hatched) and diffuse reflectance (dotted).

This point was nicely demonstrated with a series of tetrahedral cobalt(II) complexes by Cotton and Soderberg [816],and later by Csaszar and Ban [845]. In this connection, the ionic diketone cobalt(II) derivatives such as bis(dipivaloylmethane)cobalt(II) are of interest (Fig.6.46). Their spectra are dissimilar to both octahedral and the usual tetrahedral cobalt(II) species. Their low intensity (and colour) reflect the fact that these systems are much more ionic than most other tetrahedral cobalt(II) derivatives. Hollebone has discussed a relationship between spin orbit coupling, as derived from MCD and absorption intensities, and the global DQ parameter, in a closely related series of tetrahedral arylcyanamide derivatives [836]. Moreover there is an inverse linear relationship between DQ and the Racah B value, perhaps due in part to a decrease between the energy of the d-d band, and that of the CT band from which it derives intensity, as DQ increases [836], e.g. see [846].

Bertini and co-workers studied [851] the single crystal polarised spectra of the elongated tetrahedral CoN_4 D_{2d} chromophore bis(N-t-butylpyrrole-2-carbaldimino)cobalt(II). A tentative assignment was proposed and AOM parameters calculated.

CoL_3X Complexes of C_{3v} Symmetry- Several series of such species have been extensively studied. For example Quagliano and co-workers [580,847] considered the complexes $[Co(L^+)X_3]$ where (L^+) is a quaternised 1,4-diazabicyclo[2-2-2]octane. Since of the various F term states, the $^4T_2$ is most sensitive to low symmetry effects (see Chapter 3), it is the transition to this state which broadens or splits to the greatest degree when the T_d symmetry is reduced (Table 6.22). In the $[Co(L^+)X_3]^-$ systems, two well defined transitions are seen in the near infrared, assigned to $^4E + {}^4A_2$ from $^4T_1(F)$, in C_{3v}. The visible band transition to $^4T_1(P)$ is not greatly affected by the low symmetry distortion. However by using single crystal data with polarised light, the individual transitions to $^4E + {}^4A_2$ of $^4T_1(P)$ were assigned [847,848]. Similar data have been reported for mixed halide analogs of these systems [854].

Quagliano [847] discussed the energies of these complexes within the crystal field formalism (as does Furlani [853]) and derived values for the trigonal second and fourth order distortion parameters. Gerloch [848] used the angular overlap model to obtain AOM parameters for these systems (see Chapter 9). The spectra of $[Co(quinoline)Br_3]^-$ have been similarly treated [849,850]; the parameter set used to define the optical transition energies simultaneously fits the magnetic and esr data [850]; indeed in the absence of such magnetic data, the AOM parameter set was not exceptionally well defined by the electronic spectra (also see [852]).

In general, in these systems, and in the CoL_2X_2 systems which follow, if L and X differ significantly in their binding, then spin orbit coupling is less important a perturbation than the lower symmetry field. In such circumstances, the orbital triplets split to give multiplets which are more readily understandable in terms of the lower symmetry (i.e. 2 and 3 components respectively from $^4T_1(F)$ in C_{3v} and C_{2v} respectively) without recourse to arguments involving spin orbit coupling.

This is exemplified further by the spectrum of the [tris-(3,5-dimethyl-1-pyrazolyl)ethylamine]cobalt(II) cation, which has effective C_{3v} symmetry [1193]. In the region 5000 - 25,000 cm^{-1} the two T_1 triplets split into 4 main bands, 2 x ^4E (12,600 and 17,300 cm^{-1}) and 2 x $^4A_2$ (7100 and 21,500 cm^{-1}) polarised in the opposite sense to each other as expected. The electronic and esr data for this species were fitted by a Dq value of 1500 cm^{-1} which is a remarkably high value. The authors suggest that this might arise because of a rather short Co-N bond (197pm) [1193].

CoL_2X_2 Complexes of C_{2v} Symmetry- The number of such species which have been reported is legion (for example [729,745,845,854-860] and refs. in Table 6.22), and only a broad survey is attempted here (see Table 6.22). In dealing with solution spectra most authors note that the Dq values obtained from calculations involving the average band energies of the $^4T_1(F)$---$^4A_2$ and $^4T_1(P)$---$^4A_2$ transitions reflect the average Dq values for L and X. If L and X differ significantly in field strength, then three bands are usually seen in the near infrared, presumably corresponding in most cases with the three components of $^4T_1(F)$ and with an overall bandwidth significantly greater than for the regular CoL_4 species. Indeed the overall bandwidth of the split orbital triplets is usua lly taken, qualitatively as a guide to the degree of distortion from tetrahedral symmetry.

In a study of $Co(x\text{-}R\text{-}Py)_2X_2$ systems, Lever and co-workers [855] showed that the highest energy component of the near infrared band was very senstitive to the position of the R substituent, while other transitions were much less sensitive.

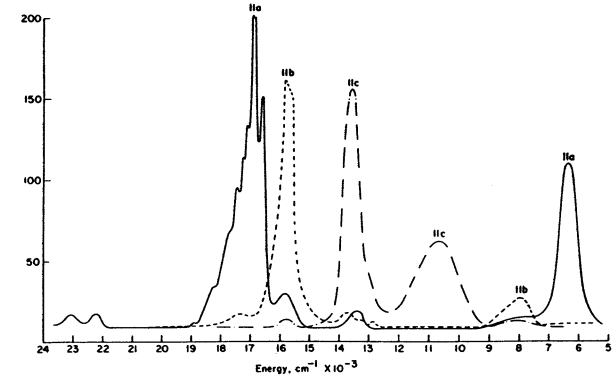


Fig.6.47 The polarised crystal spectrum of pseudotetrahedral $Co(PPh_3)_2Cl_2$ at 80 K [860].

In particular this band (near 10,000 cm^{-1}) shifted by as much as 1000 cm^{-1} (to lower energy) when a 3-substituted pyridine was replaced by a 2-substituted pyridine. Moreover in a series of complexes with X = Cl, the frequency of this transition correlated with the Co-Cl stretching vibration.

An early single crystal study [860] of $Co(Ph_3P)_2Cl_2$ showed that single crystal polarised light spectroscopy could apparently distinguish the various components of the T_1 (in T_d) terms, in C_{2v} symmetry. Absorption clearly polarised along three coordinate directions was observed (Fig.6.47) but no attempt was made at that time to fit the spectra. The polarised absorption data were later extended by doping Co(II) into $Zn(PPh_3)_2X_2$ (X = Br,I) [729] when similar highly resolved and highly polarised data were obtained. The sequence of splitting of the T_1 terms was inverted, i.e. $B_2 < A_2 < B_1$ in $T_1(F)$ and the inverse in $T_1(P)$.

Note that the labels depend critically on the choice of axes for each ligand and differ from one study [729] to another [1043] leading to some confusion. Unfortunately there are also typographical label errors (e.g. the lower B_2 and B_1 entries on p.352 of [729] should be inverted and the second entry of Table 1 [860] should be $^4A_2$).

Several attempts have, of course, been made to reproduce the sequence and energies in the spectra of CoL_2X_2 species. Furlani and co-workers were the first to attempt detailed analysis of the CoP_2X_2 chromophore [729,853] using a conventional crystal field analysis and discussed the difficulties inherent in reproducing both the correct sequence of splitting of the T_1 states, and the correct energies. While they were able to reproduce the correct sequence of energies using empirical values for the crystal field parameters Dq, Ds and Dt, later authors [856] were unable to reproduce fully the earlier calculations.

In this latter study [856] the point was made that in such covalent compounds, a simple crystal field analysis could not be expected to work; they therefore augmented the crystal field approach to permit the charge on the ligands to vary. In this fashion they claim much better agreement between calculated and observed data for a wide range of CoL_2X_2 species. However they failed to obtain satisfactory agreement between their calculation and the experimental work of Simo and Holt [860] with the $Co(PPh_3)_2X_2$ species.

Generally speaking the three regions of absorption in a pseudotetrahedral cobalt(II) complex, i.e transitions to $^4T_2$, $^4T_1(F)$ and $^4T_1(P)$ are well separated. For example, the gap between the transitions to $^4T_1(F)$ and $^4T_1(P)$ is often greater than 6000 cm^{-1}. However in the $Co(PPh_3)_2X_2$ species, the separation essentially disappears. It has been suggested that the distortion in these complexes may be so great that the split components of each of the orbital triplets overlap; if this is the case, then the early assignment to specific components of the orbital triplet states may be incorrect [856]. However Gerloch and co-workers reinvestigated these systems and included a detailed magnetic study

[1043]. Using the AOM model they were able to achieve excellent correspondance between theory and experimental data of the earlier workers [729,860]. Interestingly the phosphine ligand has a large negative e_π value which may be interpreted in terms of significant π-acceptance.

In the development of models for copper proteins and their cobalt analogs, Schugar and co-workers [858] considered the single crystal polarised light spectrum of $Co(SCMe_2CH_2NH_2)_2$ which is quite dichroic both in the near infrared ($^4T_1(F)$) and visible regions ($^4T_1(P)$).

Mason and co-workers have studied the CD and absorption spectrum of several chiral epimers of sparteine in the complexes Co(sparteine)Cl_2 [861,862]. Components of the $^4T_2$ transition were observed in the 2000 - 7000 cm^{-1} region, with large Kuhn dissymmetry factors consistent with the fact that, in T_d symmetry, transition to this state is magnetic dipole allowed but electric dipole forbidden. Curiously, transitions to components of the $^4T_1$ states had low Kuhn factors implying little mixing between the components of $^4T_2$ and $^4T_1$ in the lower field.

Four Coordinate Square Planar (Low Spin) Cobalt(II) Complexes: The spectra of these complexes parallel the low spin five coordinate derivatives discussed above. Low energy, near infrared, transitions from $(xz,yz)^4(xy)^2$ to z^2 are expected if the z^2 orbital still lies above xz,yz and xy, e.g. [768,770,871-875].

Note that for bidentate ligand square planar complexes, $Co(LL)_2$, the x and y axes bisect the bidentate ligand. In this case, the highest (and empty) orbital will be xy, and the low lying three orbitals will be xz,yz and x^2-y^2.

If these systems are rigorously square planar with no interaction at all along the z axis (which is not such a common situation), the z^2 orbital will drop below one or more of the components of t_{2g} (in O_h). Electron spin resonance can generally identify the nature of the orbital containing the odd unpaired electron, thus clarifying this situation [863,864]. Kida and co-workers [863,1047] have reported the spectra of square planar cobalt(II) complexes of macrocycles with varying numbers of cyclic chain lengths (14 to 16 membered), and here the ground state does change depending upon the size of the macrocyclic ring.

The four low lying d orbitals generally lie close together and their relative order will depend upon the system at hand. In the case of the salen systems, for example, the symmetry is so low that xz and yz are not degenerate leading to further complexity. Urbach and co-workers [772] report very low lying transitions (Table 6.19) in square planar acetylacetone Schiff Base complexes presumably involving transitions from xz,yz and x^2-y^2 to z^2 which, according to their esr analysis, contains the unpaired electron. These low spin square planar cobalt(II) complexes exhibit several transitions in the region 3900 - 7000, a complex multiplet of three bands at

8300 - 11,000, and several transitions in the visible region at 14,000 - 20,000 cm^{-1}, for a total of nine observable transitions, which they fit within the strong field formalism [772].

The magnetic and spectroscopic properties of the trans-dimesitylbis(diethylphenylphosphine)cobalt(II) species have been reported [1042]. In this rigidly square planar complex the two Co-P bonds are perpendicular to the plane of the mesityl groups. Low spin square planar cobalt commonly shows a weak band in the 8000 - 10,000 cm^{-1} range [1047,1070]. In this mesityl derivative it is observed at 8000(8.5) cm^{-1} and is assigned to two transitions, $xy---z^2$ and $xz---z^2$. Most of the remaining d-d transitions lie under the onset of CT absorption above 16,000 cm^{-1}. The authors also note that weak transitions between 4000 - 6000 cm^{-1} also appear in the isomorphous nickel complex and must therefore be presumed to be vibrational in origin.

> In this low energy region it is often difficult to distinguish electronic from vibrational (overtone and combination) bands of the ligands. Comparison with other metals in which there can be no electronic transitions in this region is highly desirable.

The esr data were fitted in the AOM model and yielded a parameter set consistent with the limited absorption data. The average e_σ(P) and e_σ(mesityl) AOM parameter is about 6000 cm^{-1} but cannot be accurately defined. The phosphine ligands appear to be π-acceptors as expected. The z^2 orbital has an energy much lower than predicted by the simple AOM theory and is probably stabilised by s-d mixing (see square planar Fe(II), Ni(II) and Cu(II)).

Other Cobalt(II) Complexes: Eight coordinate cobalt(II) exists in the form of the tetranitratocobaltate(II) and some similar tetracarboxylato derivatives. The single crystal polarised spectra of $[Co(NO_3)_4]^{2-}$ as a pure crystal, and doped into the zinc analog, have been reported [865]. In D_{2d} symmetry, polarised transitions were assigned as follows: $^4E---^4B_1$ (xy) at 19,600, 13,200 and (less securely, very weak) 8700 cm^{-1}, and $^4A_2---^4B_1$ (z) at 18,200 and 11,100 cm^{-1}. In this quasi dodecahedral complex, Dq is expected to be twice as great as in an analogous tetrahedral species; thus the appearance of an E component of $^4T_2$ at 8700 cm^{-1} is not unreasonable. Since transition to this level is forbidden in T_d symmetry, its weakness is not unanticipated. The $[Co(O_2CCF_3)_4]^{2-}$ ion also possesses D_{2d} symmetry but there are four short Co-O bonds (200 pm) and four longer Co-O bonds (311pm) so that the system has spectra more closely similar to those of a tetrahedron [866]. Nevertheless the single crystal polarised data are consistent with electronic selection rules for D_{2d} symmetry. Only transitions to $^4T_1$(F) and $^4T_1$(P) were observed, namely $^4E---^4B_1$ (xy) at 18,900, 17,600 and 7600 cm^{-1} and $^4A_2---^4B_1$ (z) at 16,400 and (8600) cm^{-1}. A series of quartet-doublet transitions in the 21,000 - 27,000 cm^{-1} region were also satisfactorily assigned in D_{2d} symmetry. Naphthyridine forms eight coordinate complexes with cobalt(II) [869], e.g. $Co(naphy)_4(ClO_4)_2$ has absorption at 9420(34),

18,500(239) and (19,400)(195) cm^{-1} also reminiscent of a tetrahedral species, though this does not appear to be required [870]. Co(II) doped into the eight coordinate site of CaF_2 shows bands at 3350, 6670 and 18,180 cm^{-1} [876].

Vallee and co-workers [1326] report MCD data for octahedral, pentacoordinate and tetracoordinate cobalt(II) complexes. They report that the MCD 'signatures' of each series are sufficiently different for MCD to be used as a tool to distinguish between them. However each type of coordination gives rise to an MCD spectrum which is comparatively insensitive to further distortion.

Seven coordinate cobalt(II) in a pentagonal bipyramidal macrocyclic environment has been described by Nelson and co-workers [644,867]. A characteristic feature is weak low energy absorption near 7000 cm^{-1} in addition to visible absorption near 21,000 - 25,000 cm^{-1}. Gerloch and Hanton have reported a single crystal magnetic and electronic spectroscopic study of a pentagonal bipyramidal macrocyclic cobalt complex based on o-phenanthroline [1161]. The complex is axially symmetric but reveals exceptionally strong bonding (both sigma and pi donor) from the phenanthroline segment to cobalt(II). An AOM analysis yielding an impressive array of data was reported. However the spectra were generally poorly resolved (absorption near 5100, 12,500 and >17,000 cm^{-1}) and the AOM data should be treated with some caution. The Gerloch group also report single crystal magnetic and spectroscopic data for the open chain pentadentate ligand 2,6-diacetylpyridine bis(semi-carbazone) which forms seven coordinate cobalt(II) complexes [1162]. Absorption is noted near 15,600 and 18,000 cm^{-1} and AOM analysis leads to the suggestion of a relatively strong pi donor interaction from the keto groups.

$CoCl_2$ exists in the gas phase as two coordinate cobalt(II) [868] and exhibits a strong structured band at 14,200 cm^{-1} with weaker absorption at 10,400 and 18,200 cm^{-1}. The assignments are uncertain [378,676,868] and it is possible that the band at 14,200 cm^{-1} arises from an impurity.

RHODIUM(II) d^7

Rhodium(II) complexes occur rarely in mononuclear form; there are however many metal-metal bonded systems which will be considered in Chapter 7.

NICKEL(III) AND PLATINUM(III) d^7

Nickel(III) and platinum(III) complexes, commonly low spin, generally only exhibit charge transfer spectra which obscure any d-d absorption [877-880]. With high field ligands the complexes are presumed to be square planar and can be expected to exhibit a transition from low lying orbitals into z^2; however this transition is narrow and weak and has not been observed in these systems. With weaker field ligands, e.g. fluoride, nickel(III) forms high spin six coordinate complexes.

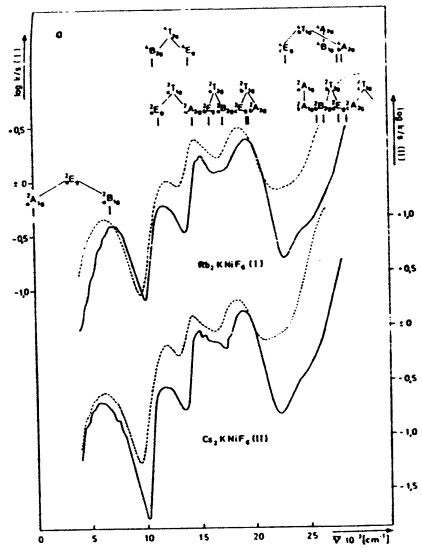


Fig.6.48 Ligand field spectra of tetragonal Rb_2KNiF_6 and Cs_2KNiF_6 at room temperature (dotted) and 4.2 K (solid) [951].

Reinen and co-workers have reported the spectra of a series of Ni(III) hexafluoride complexes with the elpasolite lattice structure [951]. These complexes, such as Rb_2KNiF_6, are low spin. Nominally they have the ^2E_g ground term but this is strongly Jahn-Teller distorted with a splitting of some 7000 - 8000 cm^{-1}. The tetragonal distortion results in axial elongation. The spectra are rich as the example in Fig.6.48 portrays. The high spin ground state ($^4T_{1g}$) lies quite close to the low spin ground state in many of these complexes (at about 1000 cm^{-1}). In one case, it is sufficiently close to the ground state that some 'hot' transitions therefrom are observed.

Our calculation with low spin Rb_2KNiF_6 reveals field strengths of Dq(Eq) = 1388 and Dq(Ax) = 490 cm^{-1}. The reader should also refer to earlier studies [520,1238] one of which [520] includes a detailed Tanabe-Sugano type diagram for tetragonal low spin d^7.

Oxidation of the divalent nickel complex $[Ni(mnt)_2]^{2-}$ yields $[Ni(mnt)_2]^-$, apparently a complex of nickel(III) [1073]. A weak band at 8330(329) cm^{-1} is assigned to the transition xy---z^2. Charge transfer (S---Ni(III)) begins as low as 11,790 cm^{-1}. Additional weak bands at 16,700 and 18,350 cm^{-1} may be d-d in origin (see discussion of d^8 Ni(II)).

6.2 (viii) d^8 Co(I), Ni(II), Pd(II), Pt(II), Cu(III), Ag(III) and Au(III)

The atomic ground state is ^3F and the ground term in an octahedral field is $^3A_{2g}$. Three spin allowed transitions parallel those expected for chromium(III). However this ground state is only observed for nickel(II). Only Co(I) and Ni(II) appear to yield paramagnetic species. The remaining metals are almost invariably low spin and diamagnetic.

COBALT(I) AND RHODIUM(I) d^8

These species are so strongly reducing that their d-d transitions are usually obscured by charge transfer [1100,1103,1104]. Pseudo-tetrahedral Co(I) complexes, $CoX(PPh_3)_3$ (X = Cl,Br and I) have spectra reminiscent of their Ni(II) analogs (see below) but shifted slightly to the red. Bands are observed near 5000, 9000 and 13,500 cm^{-1} [1105,1106]. The presumably square planar macrocycle complex $[Co(teta)]^+$ [593] has three (probable) d-d bands at 14,100(755), 19,600(125) and 23,200sh cm^{-1} which may be the three anticipated d-d transitions for a square planar d^8 species (see Ni(II) below)).

Rhodium(I) forms purported trigonal bipyramidal low spin complexes with quadridentate tripod ligands [952]. Three very intense transitions are seen between 20,000 - 30,000 (9000-14,200) cm^{-1} which could be d-d transitions (mixed with CT) in view of the generally high intensity of spectra of comparable species, such as Co(II) where d-d character is probably assured.

NICKEL(II) d^8

The nickel(II) species are of especial interest existing in all coordination numbers from 7 through 2. There probably exist more electronic spectroscopic data for nickel(II) than for any other metal ion.

Six coordinate octahedral and pseudo-octahedral Nickel(II): The three spin allowed transitions from $^3A_{2g}$ to $^3T_{2g}$, $^3T_{1g}$ and $^3T_{1g}(P)$ generally fall within the ranges 7000 - 13,000, 11,000 - 20,000 and 19,000 - 27,000 cm^{-1} respectively, with intensities less than 30 l mol^{-1} cm^{-1} in regular octahedral systems (Table 6.23) (Fig.6.49). In addition two spin forbidden bands are usually quite prominent (see Fig.2.16), one, the transition to ^1E_g near the second spin allowed transition (for common systems with Dq/B near unity), and the second, primarily $^1T_{2g}$ between the second and third spin allowed band. Indeed with Dq/B ca unity, the ^1E_g state lies so close to $^3T_{1g}$ that extensive mixing takes place leading to observation of a doublet band where the spin forbidden transition has stolen intensity from the spin allowed transition (Fig.6.49).

Table 6.23

SIX COORDINATE NICKEL(II), COPPER(III) AND SILVER(III) COMPLEXES

Complex	$^3T_{2g}$	^1E_g	$^3T_{1g}$	$^3T_{1g}$(P)	B	Ref
$KNiF_3$	7250(6)	15,440 (0.4)	12,530(11)	23,810(23)	843	884
$NiCl_2$	7700	13,640	12,700	22,600	818	961
$CsNiCl_3$	6700	12,900	11,400	21,750	838	
	7200		12,000			942
$NiBr_2$	7540	[12,300--13,300]		20,800	700	961
$CsNiBr_3$	6250	11,700	10,500	20,250	777	
	6750		11,000			942
NiI_2	7375	11,165	12,200			
	8000CT	11,260				962
$CsNiI_3$	6500		11,000?			965
	8000CT	[962]				
NiO/MgO	8600	14,900	13,500	24,600	865	200
$Ni(en)_3^{2+}$	11,700(7)		18,350(7)	29,000(9)	813	906
$Ni(Py)_6^{2+}$	10,150(4)		16,500(5)	27,000(10)	862	1298
$Ni(Im)_6^{2+}$	10,150	13,090	17,600	28,150	944	913
	10,300					
$Ni(his)_2^{2+}$	10,700(7)		18,000(8)	28,000(10)	871	1020
$Ni(NMA)_6^{2+}$	7520	(14,580)	12,660	23,700	919	248
$Ni(etu)_6^{2+}$	8350	14,700	13,500	24,800	909	808
$Ni(pyz)_6^{2+}$	11,100	13,500	17,500	28,100	823	918
$Ni(NH_3)_6^{2+}$	10,750(4)		17,500(5)	28,200(6)	881	906
$Ni(H_2O)_6^{2+}$	8500(2)	(15,200)	13,800(2)	25,300(5)	930	885
$Ni(bpz)_3^{2+}$	12,900(12)		(20,000)	27,600CT		886
$Ni(OMPA)_3^{2+}$	7400(^3E)	14,700	12,700(^3E)	24,000	951	
	7450($^3A_1$)		13,000($^3A_2$)			711
$Ni(MeCN)_6^{2+}$	10,700	(13,000)	17,400	27,810	855	447
$Ni(MeOH)_6^{2+}$	8430(2)		14,225(4)	25,000(6)	915	883
$Ni(detu)_6^{2+}$	8060		13,700	(20,000)	562	881
$Ni(DMSO)_6^{2+}$	7730(4)	(14,285)	12,970(4)	24,040(10)	921	251
$Ni(PyNO)_6^{2+}$	8400(7)		14,125(6)	25,840 CT?		251
$Ni(bipy)_3^{2+}$	12,650(7)	(11,500)	19,200(12)	26,000CT		200
$Ni(NC_2H_5)_6^{2+}$	10,500(7)		17,100(7)	27,500(11)	857	734
$Ni(MeNH_2)_6^{2+}$	10,000(6)	(12,987)	16,780(8.5)	27,320(15)	907	882
$Ni(acetam)_6^{2+}$	8240	(14,700)	13,370	24,510	900	248
$Ni(iPrNH_2)_6^{2+}$	9480		15,265	26,040	871	882
$Ni(Py)_2(Cl_2A)_2$	9615(6)	(13,390)	15,650(8)	25,900(16)	843	701
$Ni(bipy)(DMF)_4^{2+}$	10,155(7)		16,530(5)			945
$Ni(bipy)_2(DMF)_2^{2+}$	10,990(8)		17,920(7)			945
$Ni(etu)_2Cl_2$	7600	(13,500)	12,200	23,000	859	808
$NiQxCl_2$	7580	(11,900)	13,040	(23,640)	908	887
$Ni(dtp)_2(Py)_2^{2+}$	9000(20)		14,000(18)	(22,700)	660	888
$Ni([9]aneN_3)_2^{2+}$	12,500	11,500	19,800	30,800	853	940

cont

Table 6.23 cont

Complex	$^3T_{2g}$	^1E_g	$^3T_{1g}$	$^3T_{1g}$(P)	B	Ref
$Ni([10]anN_3)_2^{2+}$	12,500	11,500	19,500	30,600	834	940
$Ni([11]anN_3)_2^{2+}$	11,200		18,000	28,600	852	941
$Ni([18]dieneN_4)(NCS)_2$	12,690(3)		17,095(8)	27,395(17)	434	1198
$Ni((py)_3tach)^{2+}$	12,000	11,000	19,000			754
$Ni((py)_3tame)^{2+}$	12,400	(11,400)	(19,500)			754
$Ni((py)_3tren)^{2+}$	12,550	11,630	18,300			754
$Ni(acac)_2Py_2$	10,420(7.5)		17,390(8)			944
$Ni(acac)_2(IQ)_2$	10,050(7)		16,130(13)			944
$Ni(Me_4D)_3^{2+}$	(7910)		13,790	23,810CT		
	9010					619
$Ni(Et_4D)_3^{2+}$	(7140)		13,070	22,990CT		
	8480					619
$Ni([12]macS_4)^{2+}$	(9500)(139)		17,200(67)	26,300	825	
	10,500(181)					1064
$Ni([12]macS_3)^{2+}$	10,300(12)		17,500(31)	26,300	716	
	11,200(18)					1064
$Ni([13]macS_4)^{2+}$	(10,300)(120)		19,000(238)	28,200	881	
	11,400(156)					1064
$Ni(dth)_3^{2+}$	(10,000)(22)		17,400(33)	26,700	796	
	11,100(37)					1064
K_3CuF_6	14,100(204)	9600	20,400(189)			1238
Cs_2KAgF_6	18,400(384)	6300	23,400(377)			
		12,900				1238

For an extended list of Ni(II) data, see [1299]. B is calculated from transitions to $^3T_{2g}$ and $^3T_{1g}$(P). Many complexes show spin singlet bands at 18,000 - 24,000 cm^{-1}.

It is not properly correct to assign one component of the doublet to ^1E_g and the other to $^3T_{1g}$; they are scrambled, by the mechanism of spin orbit coupling; for discussion of this problem see [345,905-907], and an interesting and detailed plot of the $^3T_2$---$^3A_2$ and ^1E---$^3A_2$ transitions for a wide range of nickel(II) complexes [1374].

The $[NiF_6]^{4-}$ chromophore (in for example NiF_2, $KNiF_3$, Ni^{2+}/MgF_2 etc) has been extensively studied [884,908-910]. The first spin allowed band has two lines at 6699 and 6867 cm^{-1} (in $KMg(Ni)F_3$) ascribed to the spin orbit components, Γ_3 and Γ_5, of the $^3T_{2g}$ state. These are observed in the same positions in both absorption and fluorescence, and cannot be vibronic in origin. They are in fact magnetic dipole allowed transitions, a conclusion ratified by single crystal measurements [908-910]. Spectra for the tetragonally distorted octahedral M_2NiF_4 and $MNiF_3$ lattices have also

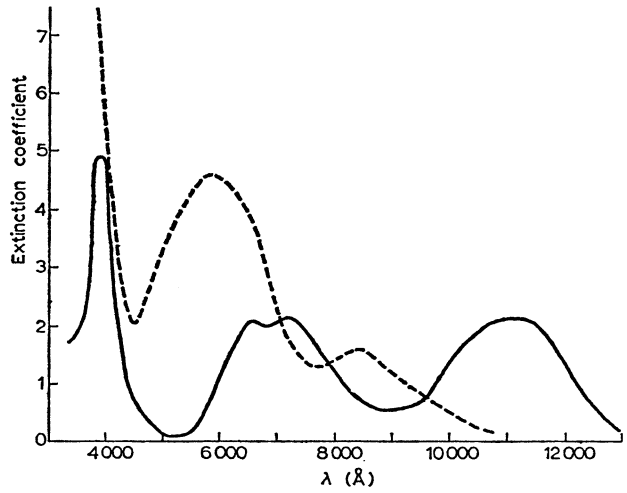


Fig.6.49 The absorption spectra of aqueous octahedral $[Ni(H_2O)_6]^{2+}$ (solid line) and square Ni(2,5-dimethylpyrazine)$_2Br_2$ (hatched line) (solid state reflectance, absorption intensities arbitrary).

been reported [955].

The electronic spectra of the NiX_2 halides (and of other simple halides) have been reviewed [960]. There have been many studies of these systems, with the work of Holt and co-workers [961] being of especial value obtained at 5 K. These investigators also included study of Ni^{2+} doped into CdX_2 (X = Cl,Br). Significant vibrational structure was observed together with structure due to cooperative exchange processes (the latter is present in the magnetically concentrated NiX_2 but not in the magnetically dilute CdX_2 doped systems).

Crystals of NiI_2 [962,963] and $CsNI_3$ [963] have also been studied. In general transition metal iodides are often intensely coloured due to low lying charge transfer absorption. At 5 K the primary CT absorption edge in NiI_2 falls at 13,000 cm^{-1} compared with 15,000 (FeI_2)[962], 25,000 (VI_2)[964], and 33,000 cm^{-1} (MnI_2)[965]. The first two spin allowed transitions and the spin forbidden transition to ^1E_g (^1D) are observable in the spectrum of NiI_2 prior to the onset of CT absorption. Both NiI_2 and $CsNiI_3$ show an anomalous weak band at 8000 cm^{-1}, whose oscillator strength is about the same as that of the transition to $^3T_{2g}$. Unlike this spin allowed transition however, the oscillator strength shows no temperature dependence, leading one to conclude that it is not a vibronically induced band. Rosseinsky and Dorrity argue [962] that this is an intervalence transition in pairs of nickel atoms of the type [2 Ni(II)I(-1)----Ni(I)I(-1) + Ni(II)I(0)].

Schreiner and Hamm [969] report a highly detailed study of the $[Ni(NH_3)_6]^{2+}$ ion using both electronic and MCD spectroscopy. Data were collected at liquid nitrogen temperature and revealed fine details including a progression (on the $^3T_{1g}(F)$ transition) in the a_{1g} Ni-N stretching fundamental at 330 cm^{-1} (excited state, compares with 362 cm^{-1} in the Raman spectrum). A series of very sharp lines were assigned to the spin forbidden $^1A_{1g}$---$^3A_{2g}$ transition, being a summation of the forbidden (and absent) 0"-0' transition coupled to two enabling vibrations and a progression in a_{1g} Ni-N. This transition was enabled both by the t_{1u} skeletal bending fundamental (215 cm^{-1}), and the t_{1u} skeletal stretching fundamental (334 cm^{-1}), but not by t_{2u}. A subsequent more detailed study [1329], by the same authors and which also employed the $[Ni(ND_3)_6]^{2+}$ ion revealed extensive vibrational structure on all but one of the spin allowed and spin forbidden bands. In these data, the t_{2u} and e_u skeletal modes and the symmetric N-H stretch were also identified.

Other less recent single crystal studies include the nickel hexa-aquo ion in various crystals e.g. $NiSiF_6.6H_2O$ [993], and $NiSO_4.6H_2O$ [994], and a range of salicylaldimine systems [995]. Data for many octahedral, and pseudo-octahedral Ni(II) derivatives are shown in Table 6.23.

Lever, Paoletti and Fabbrizzi [1201] showed that for a series of amine substituted octahedral and pseudo-octahedral species $[Ni(H_2O)_x(amine)_{6-x}]^{2+}$, there was a good linear relationship between the energy of either of the first two spin allowed transitions, and the number of amine nitrogen atoms. Moreover it was possible to relate these band transition energies with the heats of formation of these complexes relative to the hexa-aquo ion. The theory was developed within the AOM formalism (see Chapter 9).

Six Coordinate Tetragonally Distorted Derivatives (High Spin) (D_{4h}): This group of complexes, of general formula trans-NiL_4Z_2 has provided a wealth of information and has certainly contributed in a major way to the development of the theory of low symmetry complex spectra [539,637,911-913,918,927-931,966-974,976,990,996, 1063,1089-1091]. The relevant energy level diagram for the spin-allowed transitions was shown in Fig.2.21. All the transitions are vibronically allowed with the transitions to the doubly degenerate E_g states expected to be the most intense. In general the transitions to the $^3A_{2g}$ states are least intense. Since the Ni(II) ion is neither especially oxidising or reducing, the complete d-d spectrum can be observed for a wide range of L and Z without problems due to overlapping CT absorption. Data for many of these complexes are presented in Table 6.24 (with analysis of data in Chapter 9).

If L and Z differ sufficiently in sigma or pi bonding properties, the orbital triplets (in O_h) will show partial or complete resolution into their single and doubly degenerate components. Under ideal conditions, six spin allowed transitions

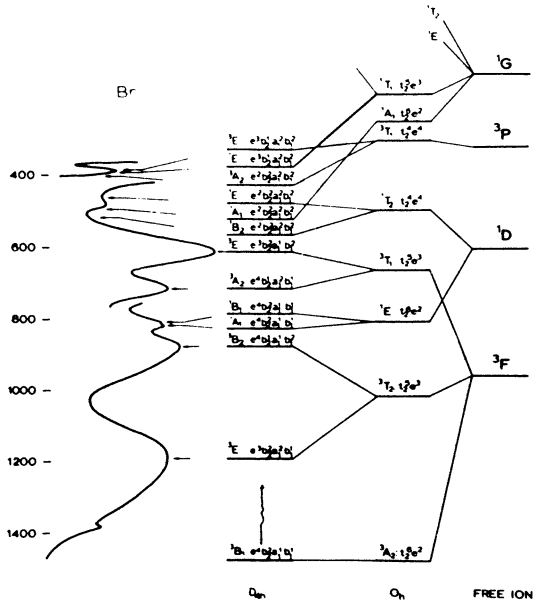


Fig.6.50 The absorption spectrum of tetragonal $Ni(py)_4Br_2$ in Kel-F at -196°C with calculated energies [969].

may be observed to arise from the orbital singlet ($^3B_{1g}$) ground term. Most significant contributions in this area have come from studies on single crystals at cryogenic temperatures; however if L and Z differ significantly, room temperature crystal or mull data can be useful. Although it is desirable to know the X-ray structure, single crystal polarised light studies using light projected along extinction axes (identified by polarisation microscopy) can also provide useful resolved data [927,966] (Fig.6.50).

The sense of the splitting of the $^3T_{2g}$ state depends primarily upon the sign of Dt, and hence upon the relative magnitudes of the axial and equatorial field strengths (see Eqns. 1.81, 1.82). For most L,Z ligand combinations this is known from previous experience; thus the assignment of $^3E_g < {}^3B_{2g}$ or vice versa is usually readily derived. The sense of the splitting of $^3T_{1g}(F)$ depends, to first order, upon $-(2d\sigma + 2d\pi)$[539,967]. It is perhaps unfortunate that for virtually all the trans-NiL_4Z_2 complexes which have been studied, the axial field is smaller than the equatorial field, pi bonding is very small, and hence $d\sigma$ is almost invariably negative. Thus for most complexes studied, as shown in Table 6.24, $E(^3A_{2g}) < E(^3E_g)$ and the $^3A_{2g}$ state is generally seen as a shoulder on the low energy edge of the stronger (because it is doubly degenerate) ^3E_g state. The splitting of the upper $^3T_{1g}(P)$ state depends, to first order, only on $d\pi$, which is invariably found to be very small in high spin six coordinate tetragonal nickel(II) complexes (Chapter 9). Thus the splitting of this band is rarely observed as more than an ill-defined shoulder.

Table 6.24

FIVE AND SIX COORDINATE TETRAGONAL NICKEL(II) COMPLEXES[a]

Complex	^3E_g	$^3B_{2g}$	$^3A_{2g}$	^3E_g	$^3A_{2g}$, $^3E_g(P)$	Ref
i) Six Coordinate-High spin						
$Ni(Py)_4Cl_2$	8995	11,680	15,265	16,810	27,175	913
$Ni(Py)_4Br_2$	8430	11,550	14,105	16,370	25,910	913
$Ni(Py)_4(H_2O)_2^{2+}$	9750	11,780	15,760	17,390	28,000	913
$Ni(Im)_4(H_2O)_2^{2+}$	9340	11,830 14,620	14,815	17,510	28,300	913
$Ni(Im)_4Cl_2$	7220b	11,170	15,630	16,530	26,320	1063
$Ni(Im)_4Br_2$	7270b	10,810	16,810	17,090	21,510?	1063
$Ni(Im)_2Cl_2$	11,900	7350b	13,260		22,470	1063
$Ni(Im)_2Br_2$	10,930	7220b	12,220		21,280	1063
$Ni(pyz)_4Cl_2$	8000	10,950	13,100	16,400	26,500	918
$Ni(pyz)_4Br_2$	7240	10,900	12,000	15,900	25,800	918
$Ni(5Mepyz)_4Cl_2$	8560	11,400	13,680	16,710	26,460	930
$Ni(5Mepyz)_4Br_2$	7650	11,050	12,220	16,200	24,170 25,980	930
$Ni(vinIm)_4(SiF_6)_2$	8000	12,500	(14,500)	17,300	28,500	931
$Ni(vinIm)_4Cl_2$	7800	11,300	13,500	16,200	26,400	931
$Ni(sEt_2en)_2(NCS)_2$	11,310	13,050	16,650	18,115	(28,285) 28,460	927
$Ni(sEt_2en)_2Cl_2$	8600	11,915	14,670	17,285	(25,480) 27,365	927
$Ni(sEt_2en)_2Br_2$	8215	11,770	13,810	17,380	(24,875) 27,030	927
$Ni(sEt_2en)_2(H_2O)Cl^+$	9795	12,525	16,000	18,240	(27,030) 29,000	927
$Ni(sEt_2en)_2(H_2O)Br^+$	9195	12,505	15,440	17,815	28,415	927
$Ni(sEt_2en)_2(H_2O)I^+$	9245	12,620	15,405	18,375	28,700	927
$Ni(sMe_2en)_2(NCS)_2$	11,135	11,990	18,410	18,525	29,110 29,250	927
$Ni(en)_2(NCS)_2$	9600	12,000	16,000	17,900	27,700 28,300	928
$Ni(sMe_2en)_2(NO_3)_2$	9130	12,590	14,365	18,570	28,895	927
$Ni(m\text{-}stil)_2(OAc)_2$	10,010	12,190	16,890	18,180	28,980	966
$Ni(sMe_2en)_2(TCA)_2$	10,150	12,305	16,160	18,025		966
$Ni(sMe_2en)_2(DCA)_2$	10,060	12,600	16,225	18,225		966
$Ni(sEt_2en)_2(TCA)_2$	9900	12,500	16,595	17,940		966
$Ni(sEt_2en)_2(DCA)_2$	9860	12,485	16,130	18,120		966
$Ni(sBz_2en)_2(TCA)_2$	9725	12,250	15,925	17,785		966
$Ni(en)_2(NO_2)_2$	12,200	11,400	19,800	19,800		928
$Ni(NH_3)_4(NCS)_2$	10,750	10,750	17,500	17,350	27,900(^3E_g) 28,000($^3A_2$)	929
$Ni(NH_3)_4(NO_2)_2$	11,200	12,000	19,900	20,350[b]		929
$Ni(urea)_4Cl_2$	8200	7100	13,400	13,700	22,300	996

cont

Table 6.24 cont

Complex	^3E_g	$^3B_{2g}$	$^3A_{2g}$	^3E_g	$^3A_{2g}$,^3E_g(P)	Ref
Ni([14]aneN$_4$)Cl$_2$ [c]	8600	14,750	15,200	19,400	(27,250)	1089
Ni([14]aneN$_4$)(NCS)$_2$	11,080	14,150	(17,800)	20,225		1089
Ni([14]'aneN$_4$)Cl$_2$	8265	13,335	12,990	18,650	28,170	1197
Ni([14]'aneN$_4$)(NCS)2	10,420	13,515	(17,240)	19,415	(28,925)	1197
Ni([14]aneN$_4$)Br$_2$	7850	14,850	13,700	19,330	(27,100)($^3A_{2g}$) 29,300(^3E_g)	1089
Ni([15]aneN$_4$)Cl$_2$	8665	12,390	14,420	17,600	(26,100)($^3A_{2g}$) 27,300(^3E_g)	1089
Ni([16]aneN$_4$)Cl$_2$	7780	11,155	13,455	15,760	25,620(^3E_g)	1089
Ni(TAAB)(N$_3$)$_2$	10,000	14,700	17,000	20,000		1089

a) $^3B_1$ ground term. Most of these data were obtained through single crystal analyses and in most cases many spin forbidden bands were also identified. b) These values seem anomalously low; they may be vibrational in origin. c) Some of the data reported here in ref.[1089] were obtained by Gaussian analysis. This reference contains a large selection of tetragonal macrocyclic nickel(II) data.

ii) Five Coordinate-Square Pyramidal-High spin

Complex	^3E	$^3B_2$	$^3A_2$	^3E	$^3A_2$,^3E(P)	Ref
Ni(RAsO)$_4$(NO$_3$)$^+$	<5000	9300	8200	11,900	19,000($^3A_2$) 22,900(^3E)	932
Ni(ClSalEt)$_2$	7500	9900	12,600	16,500	obs	933
Ni(2-MeIm)$_4$Br$^+$	5150	11,600	10,300	15,300	22,800($^3A_2$) 24,800(^3E)	911
Ni(2-MeIm)$_4$Cl$^+$	5500	11,500	10,550	15,150	23,000($^3A_2$?) 26,000(^3E?)	911
Ni(TMC)N$_3$	5900	10,800	11,800	14,800	23,400(^3E) 26,200(^3E) 24,500($^3A_2$)	934
Ni(Et$_4$dien)Cl$_2$a	?	10,200	11,700	13,100	18,900b 22,500	786

a) Five coordinate in equilibrium with square planar in solution b) Probable square planar Ni(Et$_4$dien)Cl$^+$ absorption. Also see [1158].

There have been several studies on pairs of systems such as Ni(Py)$_4$X$_2$ and Ni(Py)$_2$X$_2$ [968], and also Ni(Im)$_4$X$_2$ and Ni(Im)$_2$X$_2$ [1063] where the NiL$_2$X$_2$ is tetragonal six coordinate with bridging X groups and trans Ni-L bonds. In these pairs of complexes the splitting of the T states is inverted and, ideally, they possess the same values of Ds and Dt, but with inverted sign (Table 6.24).

In common with other nickel(II) complexes, spin forbidden transitions are readily seen in these tetragonal spectra (e.g. Fig.6.50). They may also be fitted to the tetragonal Hamiltonian. In general weak features on the edge of a spin-allowed band may arise through low symmetry splitting of an orbitally degenerate state, but may also arise from spin forbidden transitions. The expected positions of the spin forbidden components will follow from an analysis of the parameters which fit the spin allowed transitions (see Chapter 3 for methods).

Note that Hitchman [911] provides the matrix elements for both spin allowed and spin forbidden transitions in terms of the one-electron energies of the d orbitals. These expressions will suffice for tetragonal d^8 and d^3, and by changing the sign of all the one-electron energies, for tetragonal d^2 and d^7, (also see Appendix VI and [637,987]).

If all six spin allowed transitions are observed, then it may seem that up to six parameters may be defined. However this does not necessarily follow. The energy levels of an trans-ML$_4$Z$_2$ complex are defined, in the NSH formalism by four parameters, DQ,DS,DT and B. In the AOM formalism, they are defined by two AOM parameters for each ligand type, four in all, plus B, i.e. five parameters. Since the two formalisms are related by a unitary transformation, the fifth parameter in the AOM formalism must be redundant. In other words one cannot define unequivocally the four AOM parameters for these systems. As a consequence,most studies have considered systems in which L is a non-pi bonding ligand, thus reducing the AOM parameter problem to three, which is unequivocally soluble if enough absorption bands are resolved. Where this is not the case, only the McClure type parameters, $d\sigma$ and $d\pi$ can be defined. Some authors have chosen to assume that one non-zero parameter can be 'borrowed' from another complex and assumed to be unchanged - this is a dangerous ploy, and rather defeats the objective of the game, which is surely to see how one metal ligand bond, in a low symmetry complex, will influence another. The interested reader should study both the tetragonal Ni(II) and Cr(III) literature for further analysis of this problem (see also high spin five coordinate square pyramidal complexes below). The development of the subject might be judged by a series of papers each of which dealt with the Ni(Py)$_4$X$_2$ species with steadily improving experimental quality [529, 913,968,969]. The MCD data in [969] are of especial interest.

One of the classic papers in this field is that by Hare and Ballhausen presenting the polarised spectra of trans-Ni(NH$_3$)$_4$(NCS)$_2$ and trans-Ni(NH$_3$)$_4$(NO$_2$)$_2$ [929]. The former shows only slight dichroism since the species does have a NiN$_6$ chromophore. The polarisation, though small, is consistent with D_{4h} symmetry (Table 6.24). The nitro derivative shows considerably greater tetragonal distortion since the field due to the nitro group is significantly greater than that of NH$_3$. The dichroism is also greater; however the d-d energies cannot be adequately calculated in this case by the simple d-d transition theory, because of the contribution of Ni-NO$_2$ charge transfer to the d states (see Chapter 5).

Hare and Ballhausen also reported the polarised spectra of $Ni(urea)_4Cl_2$ [996]. The molecular symmetry is no higher than C_{4v} and the site symmetry C_4. The polarisation is inconsistent with C_4 (or S_4 or C_{4v}) with additional intensity in the (z) direction. Although this is an acentric molecule, there is appreciable vibronic intensity contribution. Gerloch [1166] has reported a more detailed study of the single crystal magnetism and electronic absorption spectra of this species, and presents a set of parameters which provides an acceptable fit to both.

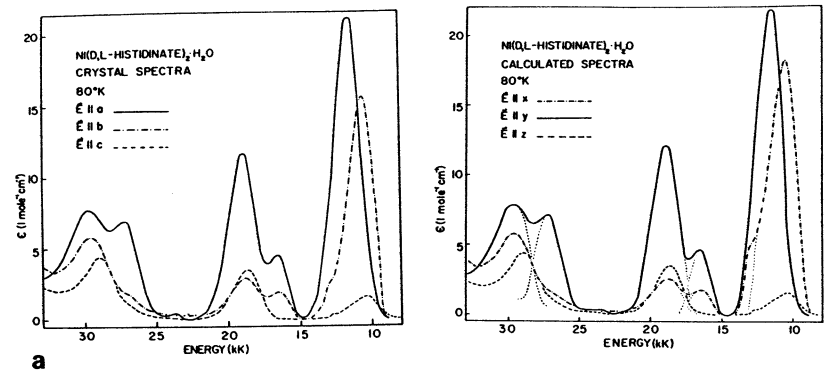
Martin, Sperati and Busch published a most interesting study [1089] of some 29 macrocyclic complexes of general formula trans-$Ni(mac)X_2$ varying the ring size, ring substituent and degree of unsaturation (Table 6.24). A polarised spectrum of $Ni([14]aneN_4)Cl_2$ was also reported. They observed that the 15 membered ring was most appropriate for the nickel ion with a Dq value similar to an open chain amine nickel(II) complex. Larger rings yielded lower values of Dq, and smaller rings yielded a higher value of Dq, due to a shorter Ni-N bond in those cases, such as the $[14]aneN_4$ macrocycle, where a trans complex was obtained. They also provided a significant body of data to show an interaction between the axial and equatorial fields, such that in a series of related derivatives, as the axial field weakens, the equatorial field strengthens, and vice versa.

> Note that low spin six coordinated tetragonal nickel(II) complexes are virtually unknown. If the equatorial field strength is very large, or the axial field strength very low, four coordinate square planar diamagnetic complexes are generally obtained. Thus if the difference in axial and equatorial field is sufficient for the complex to become low spin, it will almost always lose the two axial ligands. This is not the case in analogous copper(II) complexes where the ligands move to greater distances but are not usually lost. Where the equatorial ligands are sterically hindered, tetragonal nickel(II) complexes, if they remain high spin, will exhibit unusually low axial field strengths due, presumably, to axial elongation.

Some penta-aza macrocycle nickel(II) complexes [1378] are anomalous in that the lowest energy d-d is the most intense.

Six coordinate derivatives of C_{2v} symmetry- Meredith and Palmer [1020] report the single crystal spectra of the rhombically distorted $Ni(DL\text{-histidinato})_2.2H_2O$, containing 2 Ni-O(water), 2 Ni-N(imidazole) and 2 Ni-N(amine) bond vectors. The highly polychroic spectra are shown in Fig.6.51(a).

> Note that in general, and as is the case here, the crystal a,b and c axes, are not necessarily collinear with the molecular x,y and z axes. When the crystal structure is known, the true spectra along x,y and z can be calculated via the direction cosines [1067]. Such calculated spectra were reported in this study [1020].



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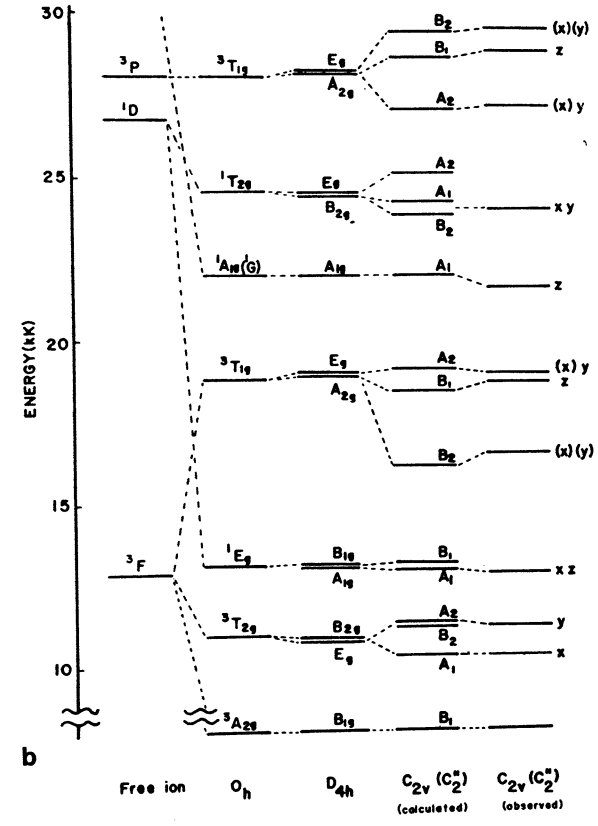

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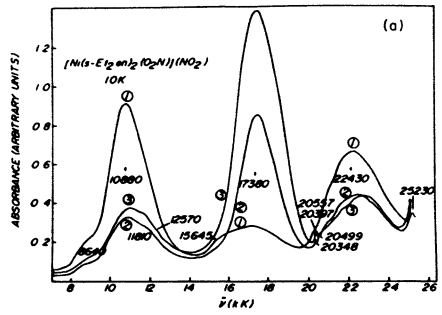

c

Fig.6.51 (a) (Previous page, top), Left - Polarised crystal spectra of $Ni(D,L\text{-his})_2.H_2O$ at 80 K, Right - Molecular spectra of $Ni(D,L\text{-his})_2.H_2O$ [1020]. (b) (Previous page, bottom) Calculated and observed energy levels for $Ni(D,L\text{-his})_2.H_2O$ with free-ion and symmetry parentage shown. Electronic selection rules are shown by x,y and z, and vibronic selection rules by (x),(y) and (z) [1020]. (c) (Above) The single crystal polarised spectra of $cis\text{-}[Ni(s\text{-}Et_2en)_2(ONO)_2]NO_2$ At 10 K [1019].

The orbital states in O_h transform in C_{2v}(III) as follows:

O_h	C_{2v}(III)
A_2	B_1
T_2	$A_1 + B_1 + B_2$
T_1	$A_2 + B_1 + B_2$

(6.20)

and the electronic selection rules are ($^3B_1$ ground level):

	A_1	A_2	B_1	B_2
x	Yes	f	f	f
y	f	Yes	f	f
z	f	f	Yes	f

(6.21)

If the selection rules were strictly obeyed, six transitions should be observed in the energy region investigated (to A_1 and B_1 of T_2, and to A_2 and B_1 in each T_1). In fact these transitions were observed, but a further six transitions were seen, i.e. there was non-zero intensity along coordinates corresponding to transitions which were electronically forbidden (Fig.6.51(b)). Allowing for possible spin forbidden transitions, these extra bands were tentatively assumed to be vibronically activated. In support of this assumption, Meredith and Palmer noted that the electronically allowed bands had only slightly temperature dependent intensities (5-10%), while the vibronically induced bands had much more temperature dependent intensity (30-55%). These authors used a complete crystal field analysis, to try and fit nine spin allowed

transitions to an accuracy of 100 cm^{-1}. They also noted that the splitting of ^3E_g(in D_{4h}), from $^3T_{2g}$, was noticeably less than the splitting of ^3E_g(in D_{4h}) from either $^3T_1$ state.

The single crystal spectra of $[cis\text{-}Ni(diamine)_2(X)]^+ X^-$ (X = NO_3^-, ONO^-) have been reported [1019]. In particular the polarisation behaviour (Fig.6.51(c)) is quite different from that commonly observed in analogous trans systems [927,966]. These complexes possess a bidentate nitrito or nitrato group with a small bite angle (ca 60°). There were analysed in C_{2v}(III) symmetry, with ground state $^3B_1$. Calculation reveals significant rhombic distortion, with overall splitting of all the orbital triplets (in O_h) of some 2000 cm^{-1}. However since transition to ($^3B_2$) in each triplet is forbidden, the observed spectra appear more tetragonal than rhombic. In parallel with the histidine derivative above, the splitting of the ^3E_g (in D_{4h}) from $^3T_{2g}$ is much smaller than the corresponding splitting in the $^3T_1$(F,P) states.

Trans-$Ni(sym\text{-}Et_2en)_2(CCl_3CO_2)_2$ has a typical tetragonal spectrum with splitting of the state $^3T_{2g}$ into two components. The analogous species trans-$Ni(asym\text{-}Et_2en)_2(CCl_3CO_2)_2$ is significantly rhombically distorted having three bond vectors Ni-O, $Ni\text{-}NH_2$ and $Ni\text{-}NEt_2$ where this last bond is much longer than normal due to steric hindrance. The electronic spectrum at 10 K [1021] reveals splitting of $^3T_{2g}$ into three components illustrative of the rhombic field. Curiously the spectra to the upper states of these two complexes are almost identical, which suggests that the distortion may be 'smeared out' in the upper levels.

Note that the transition energies to the three components of the $^3T_{2g}$ triplet, in these rhombic systems, are defined closely by Dq(x), Dq(y) and Dq(z) and are almost totally insensitive to the individual values of $e_\sigma(x)$, $e_\pi(x)$, $e_\sigma(y)$ etc which comprise Dq. The splitting of the upper $^3T_1$ states is however sensitive to the values of the individual AOM parameters.

A more recent paper [1072] shows that the $e_\sigma(N)$ values for Ni-N bonds vary approximately linearly with Ni-N bond length [1072] (see Chapter 9).

Fereday and Hathaway [1065] report the single crystal polarised spectra of the complex bisdi(2-aminoethyl)aminenickel(II) chloride ($[Ni(den)_2]^{2+}$ $2Cl^-$) at room temperature. This NiN_6 chromophore is distorted with three trans pairs of distinctly different Ni-N bonds (ca 206, 212 and 218 pm in length). The spectra consist of the three expected 'octahedral' d-d bands centred at 11,700, 18,600 and 28,800 cm^{-1}. The polarised data however show some splitting of the F term triplets, along the x,y and z coordinates. The polarised data are not consistent with the vibronic selection rules for C_{2v} or D_2 symmetry; rather there is agreement with the expectations for D_{2h} symmetry.

Oleari and Day [1162] have also reported the spectra of Ni(II) doped into MWO_4 (M = Zn,Mg,Cd), at ambient and low temperatures. Using the procedures outlined for the corresponding Co(II) system (discussed previously), complete assignments of the low symmetry

(C_{2v}) data were obtained. Octahedral triplet splittings were of the order of 1000 cm^{-1} or less, and electronic selection rules dominated.

Six coordinate trigonally distorted derivatives (high spin)- There has been much less interest in this group of complexes. Studies include $[Ni(bipy)_3]^{2+}$ [976], $[Ni(en)_3]^{2+}$ [977], $[Ni(OMPA)_3]^{2+}$ [711], $Ni(II)/Al_2O_3$ [997], and complexes of small tri-nitrogen macrocyclic rings [940]. In general such weakly trigonally distorted octahedral complexes have spectra which look very much like the regular octahedral series. However the fit to the usual octahedral equations, in terms of Dq and B, may be poorer [978,988]. The polarised data reported [711] for $[Ni(OMPA)_3]^{2+}$ show only very slight energy differences between the sigma and pi polarised spectra; however the intensities in the two polarisations have markedly different temperature dependencies.

Five coordinate nickel(II) complexes: These are known in both square pyramidal and trigonal bipyramidal, and both high and low spin forms [774,990,992].

Five coordinate high spin square pyramidal complexes (C_{4v})- These complexes are clearly related to the tetragonal six coordinate complexes through removal of one ligand from an octahedral array.

Their spectra [911,932-93,1041] are basically similar to the six coordinate tetragonal systems (Table 6.24) (Fig.2.21, 6.52). The axial distortion will necessarily be larger resulting in a greater splitting of each of the orbital triplet (in O_h) states.

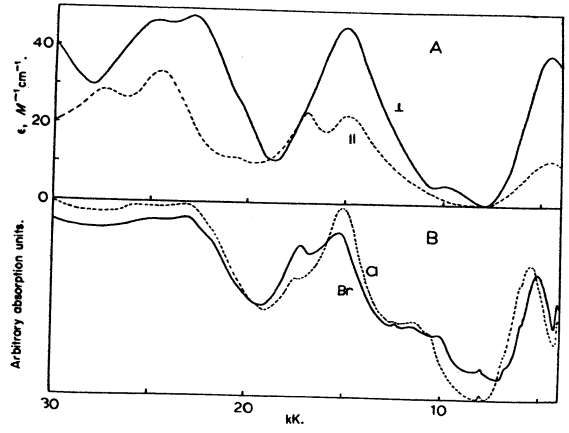

Fig.6.52 (A) The single crystal polarised spectra of high spin five coordinate $[Ni(2\text{-MeIm})_4Br]Br$ at 293 K with vectors aligned as indicated with the molecular z axis. (B) The mull spectra at 80 K of $[Ni(2\text{-MeIm})_4Br]Br$ (solid line) and $[Ni(2\text{-MeIm})_4Cl]Cl$ (dotted line) [911].

The ^3E state will now definitively lie below $^3B_2$ (except in the uncommon situation that one axial ligand has twice or more the field strength of the equatorial field). Indeed the low energy transition will now generally be found in the near infrared. The $^3A_2$ level is also most likely to fall below $^3E(^3T_1(F))$. Indeed the splitting of the $^3T_1(F)$ state may be so large that the lower component ($^3A_2$) lies below $^3B_2(^3T_2)$ (see Table 6.24). In these five coordinate square pyramidal complexes, the $^3T_1(P)$ state will generally split to a sufficient degree for both components to be observed.

> Note that the metal need not lie in the plane of the square pyramid and will generally not do so. An estimate of the angle L(Ax)-M-L(Eq) can be obtained through a crystal field or AOM analysis utilising this angle as a variable (see Chapter 1 and [987]).

Moreover electronic rather than vibronic selection rules may be operative. Thus the transitions to the E_g states are electronically allowed in (xy), while all transitions are permitted vibronically except the transition to $^3B_2$ in (z). Thus, according to Hitchman [911], the three transitions to ^3E should be considerably more intense in (xy) than in (z), while the two transitions to $^3A_2$ should appear weakly in both (xy) and (z), and the transition to $^3B_2$ should appear weakly only in (xy). These selection rules are generally obeyed for the $[Ni(2\text{-MeIm})_4X]^+$ species (Fig.6.52). The electronically allowed transitions to ^3E have intensities in (xy) of about 40-50 l mol^{-1} cm^{-2}, significantly greater than usually seen in tetragonal D_{4h} nickel(II) systems. In the (z) direction, the intensities are more comparable to the centrosymmetric analogs. The other vibronically allowed transitions are generally weak, though there are some puzzling exceptions suggesting that the selection rules are rather more subtle than indicated above [911].

The complex $[Ni(TMC)(N_3)]^+ ClO_4^-$ (TMC = 1,4,8,11-tetramethyl-1,4,8,11-tetraazacyclotetradecane) is square pyramidal with the metal significantly above the macrocyclic plane [934]. The polarisation data cannot be completely understood in C_{4v} symmetry, there being some C_s perturbation.

Gerloch and co-workers have reported a thorough analysis of the polarised spectra of the square pyramidal $[Ni(Ph_2MeAsO)_4(NO_3)]^+$ ion [932]. They reported extensive calculations of the effect of varying axial and equatorial Dq, and the apical angle upon the energy levels. They attempted to fit both the details of the polarised spectroscopic data and the magnetic anisotropy of this system.

Five coordinate low spin square pyramidal derivatives (C_{4v})- There are relatively few square pyramidal low spin nickel(II) complexes. Preer and Gray [1003] discuss the spectra of the series $[Ni(diarsine)_2X]^{n+}$ (X = Cl, Br, I, NCS, thiourea and As). The d orbital sequence should be $xy < xz,yz < z^2 < x^2-y^2$ with a diamagnetic ground state of $(b_2)^2(e)^4(a_1)^2$. Three transitions should be observed, in contradistinction to the low spin trigonal

Table 6.25

TRIGONAL BIPYRAMIDAL NICKEL(II), PALLADIUM(II) AND PLATINUM(II) COMPLEXES[a]

Complex	D_{3h} / C_{3v}	$^3E''$ / ^3E	$^3A_1''$, $^3A_2''$ / $^3A_2$, $^3A_1$	$^3A_2'$ / $^3A_2$	$^3E''(P)$ / $^3E(P)$	$^3A_2'(P)$ / $^3A_2(P)$	Ref
High Spin							
$Ni(DPEA)Br_2$		8300	14,000			22,600	980
$Ni(Me_6tren)(NCS)_2$		7700	13,200	16,200	24,600	21,000	989
$Ni(Me_6tren)(NO_3)^+$		9300	(12,000)	15,600	24,600		979
$Ni(Me_6tren)Cl^+$		7100	(11,000)	14,600	23,300		979
$Ni(Me_6tren)Br^+$		7100	10,500br	14,500	23,000	(24,500)	933
$Ni(Me_6tren)I^+$		7200	10,400	13,800	24,300		979
$Ni(Me_6tren)ClO_4^+$		(8200)	(11,900)	15,300	24,300		979
$Ni_2(bdhe)_2^{2+}$		7000	11,300	15,200	21,200	24,000	
		9000	12,000				936
$Ni(s\text{-}tan)Br^+$		7080	10,500	14,400	20,100	23,000	779
$Ni(s\text{-}tan)NH_3^{2+}$		8060	12,000	16,000	21,300	25,200	779

The assignments in this Table are tentative. Intensities are generally of the order of 20 - 100 l mol^{-1} cm^{-1}. For further data see [411,794,981,984-986].

Low Spin

Complex	$^1E'$----$^1A_1'$	$^1E''$----$^1A_1'$	Ref
$Ni(PMe_3)_5^{2+}$	19,200(3000)	(27,800)	937,1069
$NiCl_2(PMe_3)_3$	15,100($^1A_1$)	23,200($^1B_2$)	
	17,200($^1B_1$)		937,1069
$NiBr_2(PMe_3)_3$[a]	13,800($^1A_1$)(650)	21,600($^1B_2$)[b]	
	17,200($^1B_1$)(950)		937,1069
$NiI_2(PMe_3)_3$[a]	13,700($^1A_1$)(1200)		
	16,700($^1B_1$)(2300)		937,1069
$Ni(CN)_2(PMe_3)_3$	23,400($^1A_1$)		
	25,700($^1B_1$)		1069
$Ni(NCS)_2(PMe_3)_3$	18,200($^1A_1$)	25,000	
	18,900($^1B_1$)		1069
$Ni(CN)_2(PhP(OMe)_2)_3$	(24,400)(800)	(29,000)(1700)[c]	1011
$Ni(CN)_5^{3-}$	23,800		1011
$Ni(Cl)(QP)^+$	17,200(4500)	(26,500)	1013
$Ni(Br)(QP)^+$	16,500(4400)	(27,300)	1013
$Ni(I)(QP)^+$	16,000(3650)	22,200(1240)	1013
$Ni(Cl)(QAs)^+$	16,000(4420)	21,700(285)	1013
$Ni(I)(PAs_3)^+$	16,670(1760)		938
$Ni(Br)(PAs_3)^+$	17,210(2140)		938

cont

Table 6.25 cont

Complex	$^1E'$----$^1A_1'$	$^1E''$----$^1A_1'$	Ref
$Ni(Cl)(PAs_3)^+$	17,790(2330)		938
$Ni(CN)(PAs_3)^+$	22,880(4540)		938
$Ni(I)(AsAs_3)^+$	15,670(1560)		938
$Ni(Br)(AsAs_3)^+$	16,050(1847)		938
$Ni(Cl)(AsAs_3)^+$	16,560(1975)		938
$Ni(CN)(AsAs_3)^+$	21,740(3803)		938
$Ni(I)(SbAs_3)^+$	15,950(1420)		938
$Ni(I)(TSeP)^+$	14,815(1362)	(19,230)	939
$Ni(Br)(SbAs_3)^+$	16,130(1695)		938
$Ni(Cl)(SbAs_3)^+$	16,400(1800)		938
$Ni(CN)(SbAs_3)^+$	21,300(3580)		938
$Ni(Br)(TSeP)^+$	14,925(1697)	20,200(198)	939
$Ni(Cl)(TSeP)^+$	15,150(1860)	20,300(220)	939
$Ni(NCS)(TSeP)^+$	15,875(2585)	20,410(693)	939
$Ni(PL)(TSeP)_2^+$	18,570(2306)	25,640(660)	939
$Ni(TSP)I^+$	15,060(1100)	(19,500)(250)	1045
$Ni(TSP)Br^+$	15,150(1170)	20,750(278)	1045
$Ni(TSP)Cl^+$	15,390(1260)	20,960(303)	1045
$Ni(TSP)NCS^+$	16,100(1840)	20,830(900)	1045
$Ni(TSP)tu^{2+}$	16,850(2470)	(24,100)(1015)	1045
$Ni(TSeP)_2^{2+}$	14,225(267)	22,220(2150)?	939
$PdCl(QAS)^+$	26,400(3380)	(38,200)	1117
$PdCl(QP)^+$	(20,600)	(32,300)	
	22,050(6650)		1117
$PdI(QP)^+$	19,400(6220)	26,800(6850)	1117
$PdCl(AsAs_3Ph)^+$	21,400	27,400	1116
$PdBr(AsAs_3Ph)^+$	20,400	24,600	1116
$PdI(AsAs_3Ph)^+$	20,800	28,600	1116
$Pd(SCN)(AsAs_3Ph)^+$	20,700	24,400	1116
$PdCl(As_3SbPh)^+$	20,750(540)	27,900(6800)	1115
$PdBr(As_3SbPh)^+$	21,215(1800)	24,110(2390)	1115
$PdI(As_3SbPh)^+$	21,275(4500)	29,000(7600)	1115
$Pd(SCN)(As_3SbPh)^+$	21,140(3800)	24,275(4600)	1115
$PtCl(QAS)^+$	27,800(7400)	(37,900)	1117
$PtCl(QP)^+$	(23,600)	(30,800)	
	25,950(5500)		1117
$PtI(QP)^+$	(22,200)	(31,800)	
	22,800(4610)		1117

XAs_3 are $X[CH_2CH_2CH_2AsMe_2]_3$ where X = P, As and Sb. Data reported here for this series were obtained in solution. An additional lower energy shoulder was reported in the solid state nujol data for many of these species.

a) With excess PMe_3; Effective symmetry C_{2v}. b) solid state, very weak. For further low spin data see [625,998,1013]. In this last

set of complexes where only one band is reported, gaussian analysis [1010] generally revealed three (see text). c) In view of the relatively intense nature of this band, it may be mixed with Ni-CN charge transfer.

bipyramidal species (below) which only exhibit two. Of these three, namely $^1B_1$---$^1A_1$, ^1E---$^1A_1$ and $^1A_2$---$^1A_1$, only that to ^1E is orbitally allowed. This is associated with the strong visible region feature in the spectra of the $[Ni(diarsine)_2X]^+$ species. This band sharpens upon cooling with a dramatic increase in the molar extinction coefficient; however when the band narrowing is taken into account, the oscillator strength is seen not to vary consistent with the assignment to an electronically allowed transition. The narrowing of the band permits the observation of the much weaker and forbidden transitions to $^1B_1$ and $^1A_2$ [1003].

Some mixed sulphur-phosphine ligands supposedly form low spin five coordinate square pyramids of the type Ni(tridentate)X_2 and $[Ni(bidentate)_2X]^+$; their spectra [1004] show a strong d-d band near 20,000 cm^{-1} with little sign of the expected structure. However the mixed system $[Ni(tridentate)(bidentate)]^{2+}$ does show three clear transitions near 16,000(ca 100-400), 21,000(500 - 1300) and 27,000(5000 - 13,500) cm^{-1} which may be the expected three transitions, though the last is exceptionally intense (Table 6.24). Data for similar complexes are reported in [1005-1009,1014,1015]. Note that four d-d bands may be observed in distorted square pyramids whose effective symmetry is no higher than C_s [1014,1015].

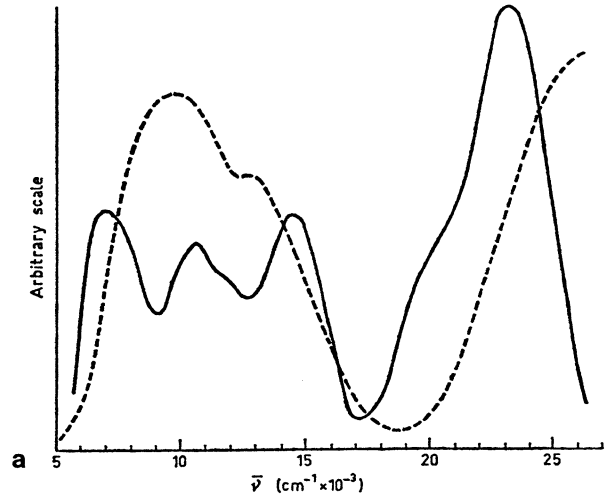

Fig.6.53 (a) The reflectance spectra of [Ni(trenMe)Cl]Cl (solid line) and [Cu(trenMe)Br]Br (dotted line).

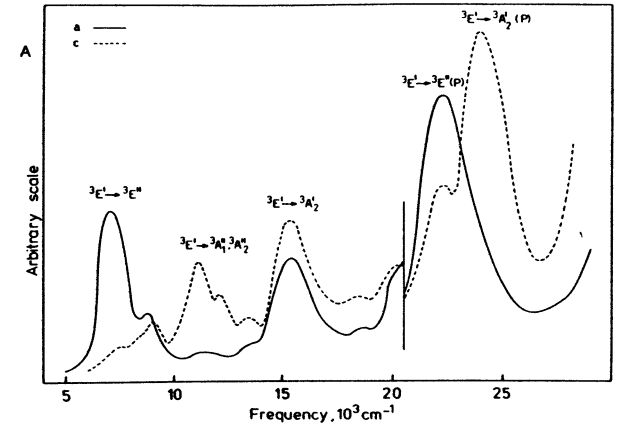

Fig.6.53 (b) The single crystal polarised spectra of the trigonal bipyramidal $[Ni_2(BDHE)_2](ClO_4)_2$ at 4.2 K with radiation incident on the (010) face. The electric vector is parallel to a (solid line) and c (dotted line) [936].

Five coordinate high spin trigonal bipyramidal derivatives (D_{3h})- Ciampolini first reported a detailed theoretical study of five coordinate high spin trigonal bipyramidal (and square pyramidal) nickel(II) [774,987]. Because of the accidental degeneracy of the $^3A_1'$ and $^3A_2''$ states (in rigorous D_{3h} symmetry) only five spin allowed transitions are expected. In general these can be observed (Table 6.25, Fig.6.42, 6.53).

Bertini et al. [989] studied the single crystal spectrum of the archetypical trigonal bipyramidal $[Ni(Me_6tren)(NCS)]^+$ ion. As usual in these systems the effective symmetry may be C_{3v} or C_{2v} rather than the regular D_{3h}. Bertini point outs that an electronically allowed spectrum in C_{3v} symmetry would be predominantly polarised in (xy), with only transitions to the ^3E levels being permitted in both (xy) and (z). In D_{3h} symmetry the spectrum should be quite dichroic with transitions to $^3E''$ allowed in (z), to $^3A_2'$ in (xy), while transitions to $^3A_2''$,$^3A_1''$ are forbidden. In this NiN_5 chromophore, D_{3h} seems to be the effective symmetry. These workers also investigated the effect of the variation of some parameters (axial and equatorial field strength, apical angle etc) upon the energy levels. They also note that a C_{3v} perturbation lifts the accidental degeneracy of the pair of A'' levels. Their results are shown in Fig.6.54.

An elegant piece of work was reported by Gatteschi and co-workers [936]. The binuclear trigonal bipyramidal system bis(N,N-bis(2-(diethylamino)ethyl)((2-hydroxyethyl)amino-O)) di-nickel(II)) behaves from the viewpoint of its electronic spectrum

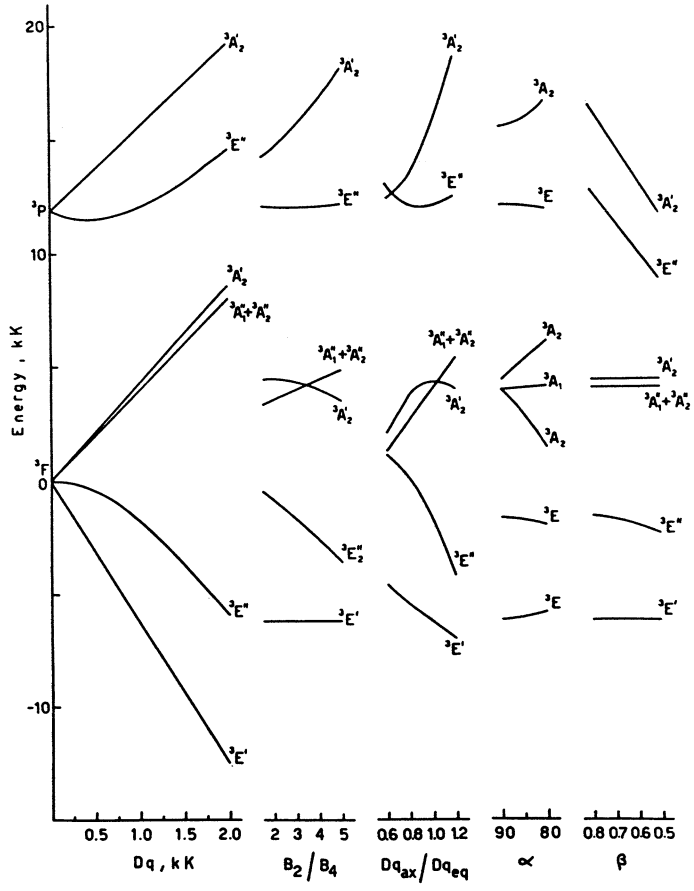

Fig.6.54 Energy level diagrams for Ni(II) in a D_{3h} trigonal bipyramidal field, (left-hand diagram); B_2/B_4 = 3.0, beta = 0.75°, Dq(Ax) = Dq(Eq); subsequent diagrams show the effect of changing B_2/B_4, the Dq(Ax)/Dq(Eq) ratio, the apical angle alpha and the nephelauxetic parameter beta [989].

as a mononuclear complex. In addition to a detailed esr study, these authors report a highly dichroic electronic spectrum (Fig.6.53b). There is evident splitting of the $^3E''$ state and the polarisation is clearly different from that predicted (above) for D_{3h} symmetry. A lower symmetry perturbation is obviously present. An AOM analysis suggested values of $e_\sigma(O)$ of 3500 cm^{-1}, and $e_\sigma(N)$ values (for different N) of 3700 and 4200 cm^{-1}.

Utsuno and co-workers [779] studied the CD spectra of some chiral trigonal bipyramidal nickel(II) complexes providing some support for the assignments as shown (Fig.6.42).

Nemiroff and Holt studied two systems, bis(salicylidene-iminopropyl)aminonickel(II), (Ni(saldipa)) and bis(salicylidene-iminopropyl)methylaminonickel(II), (Ni(salmedpt)). Both complexes are five coordinate with X-ray structural data showing a distorted five coordinate structure between trigonal bipyramidal and square pyramidal [982]. The polarisation data favour effective D_{3h} symmetry for Ni(saldipa) and C_{2v} symmetry for Ni(salmedpt).

Bis(dimethylphosphorodithioato)-2,9-dimethylphenanthroline Ni(II) also possesses a distorted trigonal bipyramidal structure [983]. The single crystal spectrum reveals eleven components (at 4.2 K) with three being assigned to spin forbidden transitions. The remainder are assigned to spin allowed transitions in C_{2v} symmetry. The polarisation data lead one to conclude that the ground state is the $^3B_2$ component of $^3E'$ (in D_{3h}).

Five coordinate low spin trigonal bipyramidal derivatives (D_{3h})- Certainly the most common low spin (diamagnetic) five coordinate nickel(II) complexes are trigonal bipyramidal or a distorted version thereof. They will have an $(e'')^4(e')^4(a_1')^0$ configuration. Most of these complexes possess a soft heavy ligand such as arsenic, phosphorus, sulphur or selenium. Much of the early spectroscopic work was completed by the research groups of Meek, Sacconi and Venanzi. Some typical spectra are shown in Fig.6.55. They are typified by exceptionally high intensities for transitions which are primarily d-d in origin (due probably to a very covalent bond to very soft ligands) and by the presence of two principle d-d bands, the lower energy being the more intense.

In the strong field model, there will be two transitions, from e' and from e" to a_1', and the transitions are so assigned ($^1E'---^1A_1'$ and $^1E''---^1A_1'$ in order of increasing energy) [625,998]. The second transition is forbidden in D_{3h} symmetry and so is seen to be much weaker. Its presence probably arises partly through a vibronic mechanism and partly because these systems do not generally possess strict D_{3h} symmetry. In C_{3v} symmetry, probably a better approximation for these systems, the transition will be allowed. The broadness of the bands may be a consequence of Jahn Teller splitting of the ^1E excited states (derived primarily from the free ion ^1D). An energy level diagram for these systems was reported in [998].

While the pair of anticipated bands is clearly enunciated in Fig.6.55, this is not always the case. Thus Benner and Meek [1010] report data for complexes $[Ni(TAS)X]^+$ of tris(3-dimethylarsino-propyl)arsine, with four coordinating arsenic atoms, exhibiting a single broad structured band centred at 15,000 - 22,000 cm^{-1} depending upon X. The band energies, in complexes of this type, shift to higher energy in the spectrochemical sequence $I^- < Br^- < Cl^- = NO_3^- = ClO_4^- < NCS^- <$ etu < tu < PPh_3 < PPh_2Me < NO_2^- < CN^-.

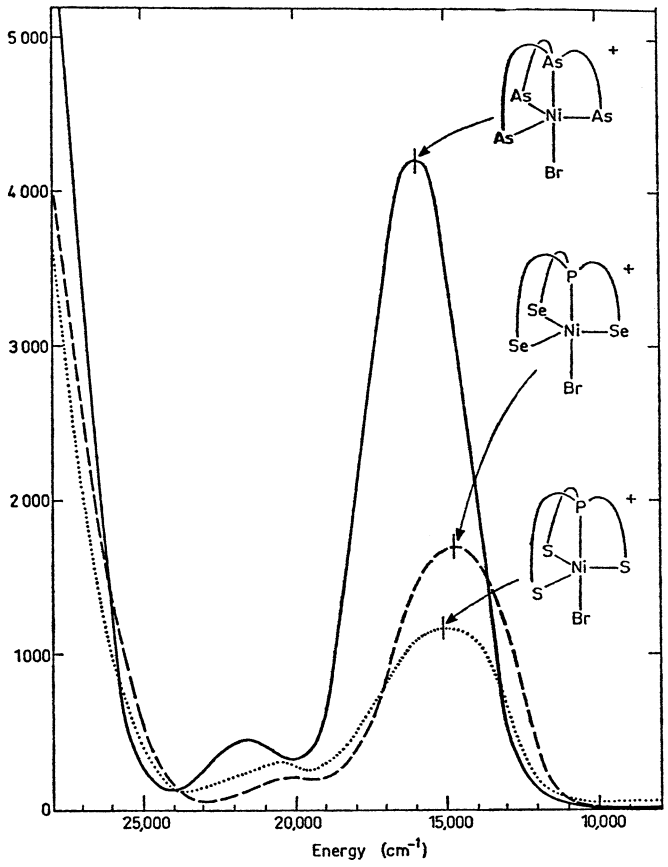

Fig.6.55 The absorption spectra of some low spin trigonal bipyramidal (diamagnetic) nickel(II) complexes as indicated.

Gaussian analysis reveals a strong central band and a weak shoulder some 4000 cm^{-1} to higher energy; presumably this represents the expected pair of transitions. However there is also a lower energy weaker component which has an intensity of 10^2 -10^3 l mol^{-1} cm^{-1}. While this could be a spin forbidden transition, its rather high intensity mitigates against this. It may be a Jahn-Teller component of the $^1E'$ excited state.

Complexes of the type $Ni(PMe_3)X_2$ (e.g. X = Cl, Br, I and CN) also possess a low spin basically trigonal bipyramidal structure [937]. However there is sufficient angular distortion to split both the $^1E'$ and $^1E''$ excited states. In C_{2v} symmetry these states transform as $^1A_1 + {}^1B_1$, and $^1B_2 + {}^1A_2$. The transition to the last of these four states is forbidden in C_{2v}. In the room temperature spectrum the three orbitally allowed transitions are observed, while at 77 K, the fourth and forbidden band also appears (Fig.6.56). Complexes with X = CN appear more regularly D_{3h} with a typical spectrum [937,1011]. The latter work [1011] discusses these complexes in molecular orbital terminology, and compares their spectra with square planar $NiP_2(CN)_2$ analogs. In these five coordinate examples the second transition ($^1E''$) exhibits enhanced intensity and may be mixed with Ni-CN charge transfer absorption.

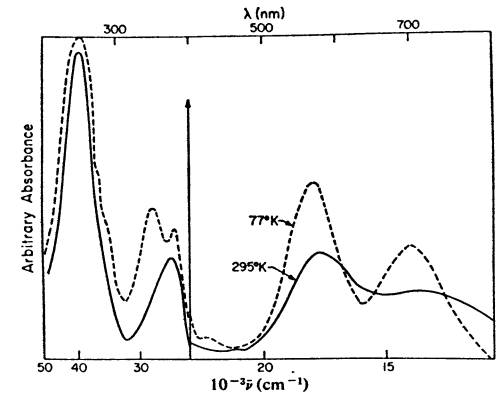

Fig.6.56 The temperature dependence of the solid state spectrum of $NiBr_2(PMe_3)_3$ [937].

Many tripod type ligands have been used to prepare low spin nickel(II) complexes. The lowest band exhibits an energy order in the sequence [1016] QP > TAP > QAS > TAA > TSP > TSeP (see List of abbreviations, but note the key letters refer to the elements arsenic (AS and A), phosphorus (P), sulphur (S) and selenium (Se)).

An interesting temperature effect has been observed in these systems [1016,1017]. It has already been noted that the low energy transition to $^1E'$ sometimes shows doubled structure suggestive of the Jahn-Teller effect. Gray and co-workers have shown that when solutions of these complexes (containing Ni(II), Pt(II) and other ions) are cooled, this band sharpens and often loses the double structure., i.e. the two components move closer together. The effect is much less marked in the solid state.

For the $^1E'$ state to be split by a Jahn-Teller vibration (see Chapter 4), the vibration must span a representation of the symmetric direct product E' x E'. In fact an e' vibration fulfills this requirement and may cause such a splitting. In C_{3v} complexes, an e vibration will suffice. Gray and co-workers present arguments

to show that the energy separation between the two Jahn-Teller components is temperature dependent (function of coth(C/T)) and will actually increase with increasing temperature. Thus the band separation is greater at room temperature than at low temperature. The same authors propose an alternative explanation in which the solution phase is assumed to contain an equilibrium mixture of different stereochemical forms, perhaps intermediate between trigonal bipyramidal and square pyramidal, and influenced by solute-solvent interactions. If this explanation is true, then upon cooling the system, a more symmetric structure is preferred. It is not clear which of these arguments is the more likely. However to some degree they are complementary in that a Jahn-Teller distortion will cause changes in stereochemistry. Species in the solid state which show little temperature effect are probably distorted in their ground state.

The low energy band is also very pressure sensitive, shifting to higher energies as the sample is subjected to high pressures [1018]. The effect is about twice as large for low spin trigonal bipyramids as for low spin square pyramids.

Four coordinate tetrahedral and pseudo-tetrahedral derivatives:
Tetrahedral (T_d) nickel(II) complex spectra are similar in intensity and band energies to those of cobalt(II) (Fig.6.57). However the second spin allowed transition (generally in the near infrared region) terminates on the orbital singlet, $^3A_2$ (rather than the triplet $^4T_1$ in d^7) and therefore does not exhibit the spin-orbit and/or low symmetry structure often seen in the cobalt(II) systems. The visible band ($^3T_1(P)$---$^3T_1(F)$) does exhibit multiple structure with intensity about 10^2 - 10^3 l mol^{-1} cm^{-1}, comparable to (though usually somewhat less than) similar complexes of cobalt(II). Transitions to spin singlet states are often observable in the regions between the spin allowed bands (Fig.6.57). Much of the early work was carried out by Goodgame, Cotton and Drago and their co-workers (Table 6.26).

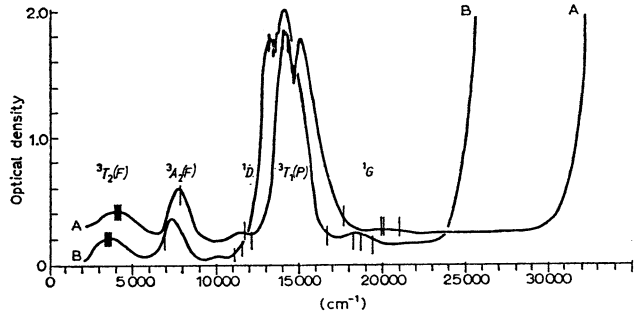

Fig.6.57 The absorption spectra of tetrahedral $[NiX_4]^{2-}$ species in Cs_2ZnX_4 at room temperature. A = Cl, Dq = 385, B = 725, C = 3262; B = Br, Dq = 330, B = 695 and C = 3127 cm^{-1}.

Table 6.26

FOUR COORDINATE TETRAHEDRAL AND PSEUDOTETRAHEDRAL NICKEL(II) COMPLEXES ($^3T_1$ GROUND TERM)

Complex	$^3A_2$	$^1E^a$	$^3T_1(P)$	$^1T_2$	Ref
Ni(II)/CdSe	4200($^3T_2$)		11,500		
	8000($^3A_2$)				382
Ni(II)/ZnSe	4050($^3T_2$)		11,250-11,809		
	8840($^3A_2$)				1023
$NiCl_4^{2-}$	6550(21)	11,630(6)	14,250(160)	(19,800)	
	(8700)		15,240(160)		
			(15,950)		893
$NiBr_4^{2-}$	6995(30)	10,680(5)	13,230(241)	21,190(5)	
			14,140(250)		
			(15,200)	(16,700)	893
NiI_4^{2-}	7040(55)		(11,480)	18,870CT	
			(12,800)		
			14,030(242)		
			(14,530)		893
$Ni(NCO)_4^{2-}$	9460(24)	12,800(6)	15,600(142)		
			16,800(142)		446
$Ni(OMPA)_4^{2+}$	8000(9)	13,175(8)	(14,000)	22,200(17)	
			15,800(91)		
			17,200(77)		892
$Ni[(Ph_2PS)_2N]_2$	8700	10,500	12,700 - 15,000		547
$Ni(Me_4tn)Cl_2$	(9090)	10,700(40)	12,000(35)	(18,880)	
	11,500(34)		19,550(84)		916
$Ni(Ph_3PO)_2Cl_2$	7290	12,120	14,290	22,470	
			16,260		893
$Ni(Ph_3PO)_2Br_2$	7250(19)	11,930	(14,290)	20,490(22)	
			15,580(101)		893
$Ni(Ph_3PO)_2I_2$	7080(18)	11,950	(13,680)	(18,620)	
			15,100(137)		893
$Ni(PPh_3)_2Cl_2$	10,000	(13,500)	15,000(600)	21,500CT?	
			(18,500)		1022
	4500($^3A_2$)	8000($^3A_1$)	11,200($^3A_2$)	17,800($^3A_2$)	
	10,200($^3B_1$)		18,100($^3B_1$)		1043,1044
$Ni(2,3Me_2Py)_2Br_2$	10,050	(11,560)	15,530		890
$Ni(2,3\text{-lut})_2Cl_2$	10,130		16,330	18,080	917
$Ni(2,3\text{-lut})_2Br_2$	10,150		16,670	18,180	917
$Ni(TDPME)(ClO_4)_2$	12,500 (8)		15,750(40)		759
$Ni(MQN)I_2$	(7200)($^3T_2$)		(14,290)	17,000	
	9660($^3A_2$)				1035

cont

Table 6.26 cont----------
Trigonal NiL_3Z[d]----------

	$Ni(Ph_3P)I_3^-$	$Ni(Ph_3P)Br_3^-$	$Ni(L+)Br_3$	$Ni(L+)Cl_3$
$^3E(T_1(F))$	0	0	0	0
$^3A_2(T_1(F))$	7100	5800		
$^3A_1(T_2(F))$	7800			
$^3E(T_2(F))$	5600	5800	5100	5260
$^3A_1(T_2(F)))$		7100	6410	6410
$^3A_2(A_2(F))$	9400	9500	8700	9090
$^3E(P)$	13,400	14,700	[15,000-	16,000
$^3A_2(P)$	(14,800)	15,600	16,200]	17,240
Ref.	915,935	935	847,848	848

a) Data in this column may also be components of other low symmetry spin triplet components. b) ^3E ground term. c) average of split components. d) Also see [850]. For further data see [845, 917,1022,1024,1036-1040].

Early single crystal studies [1026-1028] in oxide and sulphide lattices were difficult to interpret fully because of lattice phonon absorption.

Koester and Dunn [1025] investigated the spectra of $(Et_4N)_2[NiCl_4]$ and of Ni(II) doped into Cs_3ZnCl_5 at 2.2 K. At this temperature, only the lowest spinor component of the $^3T_1$ (in T_d), namely A_1 (in this section labels which do not carry spin designations are spinor labels appropriate to the double group) is populated. The only electronically allowed transitions will be to T_2 . However there is a tetragonal distortion such that the site symmetry is D_{2d}, and transitions are electronically permitted to B_2 and E. Transitions to every possible state except $^1A_1(^1S)$ were observed and with excellent resolution. In addition to electronically allowed transitions to B_2 and E, vibronic transitions to all other states were observed. The e and b_2 enabling vibrational intervals were identified, and progressions in the a_1 Ni-Cl stretch (270 cm^{-1}) were observed. A complete and most elegant analysis of the absorption spectra was provided. In addition, a sharp line emission was observed for Ni(II) doped into $(Et_4N)_2[ZnCl_4]$; the emission originates from ^1G and terminates in the various spin-orbit states of the ground $^3T_1$ term. A vibrational interval of 825-835 cm^{-1} was ascribed to a vibration of the Et_4N^+ ion suggesting exciplex formation between the excited $[NiCl_4]^{2-}$ site and the cation [1025].

The $[NiCl_4]^{2-}/Cs_3MgCl_5$ system was studied previously by Couch and Pedro Smith [1029], who obtained similar but less well resolved data over a smaller wavelength range. Other workers [1031,1032] obtained less resolved data at higher temperatures [1025]. Brown and co-workers [1030] investigated the effect of temperature on the $^3T_1(P)$ band envelope in $Ni(II)/Cs_2ZnCl_4$ with a view to better understanding of the structure of this band. An energy level calculation has also been presented for this system [1033].

There are very many pseudo-tetrahedral systems of general formula NiL_2X_2 [845,890,893,917,1034-1041,1043,1044], especially with L being a nitrogen or phosphorus ligand. In solution these have spectra not very different from regular T_d complexes though the overall splitting of the triplets may be larger. However the presence of structure due to low symmetry and structure due to spin forbidden transitions may make unequivocal assignment difficult [1035,1041]; to add to this complexity many pseudo-tetrahedral nickel(II) complexes are in equilibrium, in solution, with their square planar isomeric forms [1037,1038]. Phosphine containing tetrahedral nickel(II) complexes may also show charge transfer absorption above 25,000 cm^{-1} or so.

The $Ni(PPh_3)_2Cl_2$ species exhibits [1044] one weak band at ca 8000 cm^{-1} in x polarisation, two bands in z at 10,200 and 18,000 cm^{-1}, and four bands in y at <4800, 11,200, 12,000 and 18,000 cm^{-1}; in addition there is a sharp spin forbidden band at 10,800 cm^{-1}. In C_{2v} symmetry, T_1 spans $A_2 + B_1 + B_2$, T_2 spans $A_1 + B_1 + B_2$ and $A_2(T_d)$ spans A_2. On the basis of a $^3B_1$ ground term (chosen because it fits the observed data) the transitions $^3A_1$---$^3B_1$ (in x), $^3A_2$---$^3B_1$ (in y), and $^3B_1$---$^3B_1$ (in z) should be seen. Transitions to $^3B_2$ are electronically forbidden. This provides the ready assignment - $^3A_1$ (8000), $^3A_2$ (<4800, 11,200) and $^3B_1$ (10,200, 18,000 cm^{-1}), where the appearance of 'extra' bands such as that at 18,000 cm^{-1} in y, probably results from vibronic activation.

Gerloch and co-workers also reported an analysis of $Ni(PPh_3)_2X_2$ [1043]. On the basis of absorption spectra alone, there is a range of AOM parameters which will provide an acceptable fit; however by including magnetic data, a more restricted AOM data set becomes available. Thus there was little dependence upon $e_\sigma(Cl)$ in the range 3400 - 4600 cm^{-1}; there is a degree of correlation between $e_\sigma(P)$ and $e_\pi(P)$ (and also between $e_\sigma(P)$ and $e_\pi(Cl)$) in that acceptable fits are obtained for $e_\sigma(P)$ ranging from at least 3400 - 5000 cm^{-1} if $e_\pi(P)$ is allowed to vary its range from -3500 to -1500 cm^{-1}. Since $Dq(P) = 0.1[3e_\sigma(P) - 4e_\pi(P)]$ the observed correlation is tantamount to stating that the fit is sensitive to Dq, but not to individual values of e_σ and e_π. This is not an unusual situation and care should be taken to explore such parameter variations rather than assume that the first data set giving a good fit is necessarily correct. In this report [1043] it is argued however that for chemically reasonable values of $e_\sigma(P)$ and $e_\sigma(Cl)$, the parameter $e_\pi(P)$ will definitely be negative, and $e_\pi(Cl)$ definitely positive in agreement with chemical expectation.

Cruse and Gerloch [1036] report a combined magnetic and spectroscopic study of bis(N-isopropylsalicylideneiminato)-nickel(II) having local C_{2v} NiO_2N_2 microsymmetry. The parameters which fit the magnetism also yield an acceptable fit to the spectrum with bands near 6000 (components of $^3T_2$), 9000 ($^3A_2$) and 14,000 - 16,000 cm^{-1} (components of $^3T_1(P)$).

Data for some interesting trigonally distorted pseudo-tetrahedral systems, $[NiLX_3]^-$ have been reported (Table 6.26). These may be analysed in terms of C_{3v} symmetry. Gerloch and co-workers have published an interesting series of papers on complexes of this type. Their strategy is to vary a set of AOM parameters over as wide a range as possible in order to find a data set which provides a satisfactory fit to both the spectra and the magnetism of the complexes.

Bertini and co-workers [915] report the spectra of $[Ni(PPh_3)X_3]^-$ (X = Br,I), while Gerloch and co-workers report the magnetic data and analyse both the spectroscopic and magnetic data within the AOM framework. The band intensities are temperature dependent with a dramatic increase in intensity for some peaks as the temperature is lowered. The intensity mechanism is undoubtedly electronic due to the asymmetric crystal field (intensities of the order of 10^2 - 10^3 l mol^{-1} cm^{-1}) and the increase must be due to changes in Boltzmann distributions in the degenerate (^{3}E) ground term. The assignments are not unequivocal and there are some differences between the assignments of the two sets of workers. In particular the assignment of a narrow feature at 7100 cm^{-1} to a water overtone (presumably seen because of beam imbalance in the instrument) by the Bertini group, is contradicted by the Gerloch group on the basis that no such band is expected near 7100 cm^{-1}, and because in the Gerloch analysis, a spin-allowed d-d band is expected in that region. It is noted that all reasonable sets of AOM parameters do indeed lead to a splitting of the $^3T_1$ (in T_d) ground term to yield ^{3}E as the ground term in C_{3v}. Although polarisation data were obtained [915] the complexity of low symmetry plus spin-orbit effects makes an unequivocal use of the polarisation to confirm the assignment unreliable. The assignments shown in Table 6.26 are due to Gerloch and were fitted acceptably well by the AOM data set (see Chapter 9) which also fitted the magnetism of these species.

Rather lower quality polarised data were also reported for $Ni(L^+)X_3$ where L^+ is the N-ethyl-1,4-diazabicyclo[2,2,2]octonium cation [847] (Table 6.26). In this case a standard crystal field approach was used, and values of the trigonal parameters Dσ and Dτ derived. However Gerloch and co-workers remeasured these data, obtained the magnetic properties and fitted both to the AOM model [848] (Chapter 9). A relatively rich spectrum was reported for the $[Ni(quinoline)Br_3]^-$ ion cooled to 4 K [850]. Principle bands were fitted near 4000, 6000, 9000 and 15,000 cm^{-1} using the AOM model. Quinoline seems to be a weak pi acceptor (see Chapter 9).

Four Coordinate Square Planar Derivatives: The configuration d^8 is especially prone to form four coordinate diamagnetic planar derivatives, especially with stronger field ligands, or where steric hindrance impedes high coordination numbers [875,887,888, 890,899, 896,900-904,1053,1074,1077-1086, and refs. in Table 6.27]. Such complexes typically have a single band at 18,000 - 25,000 cm^{-1} of intensity generally near 50 - 500 l mol^{-1} cm^{-1}, and are often

Table 6.27

FOUR COORDINATE SQUARE PLANAR NICKEL(II) COMPLEXES[a]

Complex	ν_1	ν_2	ν_3	Ref
$Ni(CN)_4^{2-}$	31,100($^1B_1$)	31,650(^{1}E)		1053
$Ni(en)_2^{2+}$		21,550		875
$Ni(dmg)_2$	(21,050)	(24,390)	27,030(1400)	900
$Ni(dpm)_2$	16,000	18,690(60)	20,000	904,1069
$Ni(mesoStien)_2^{2+}$		23,450(66)		902
$Ni(C,C'\text{-tetraMeen})_2^{2+}$		23,050(64)		903
$Ni(3,4Me_2Py)_4^{2+}$		23,390		890
$Ni(as\text{-}Et_2en)_2^{2+}$		21,860		1051
$Ni(daco)_2^{2+}$		22,500(115)		901
$Ni(daco)_2^{2+b}$	21,350	22,300	23,780	1046
$Ni(Q)_2I_2$	11,000w	15,430	(21,750) 25,000	891
$Ni(Qx)_2I_2$	11,000w	16,160	(22,730)(23,530)	887
$Ni(25Me_2Pz)Br_2$	11,650w	17,100	26,000	899
$Ni(25Me_2Py)_2Cl_2$		18,020		890
$Ni(25Me_2Py)_2Br_2$		17,390		890
$Ni(etu)_4^{2+}$		16,500		881
$Ni(dtp)_2$	14,500(75)	19,100(90)	26,100(800)	888
$Ni(dtc)_2$	15,750($^1B_1$)	17,000($^1B_{2g}$)	19,000($^1B_{3g}$)	
		21,000($^1B_{1g}$)		1052
$Ni(dtc)_2$	15,750(70)	20,500(100)		888,1052
$Ni(S_2CNH_2)_2$	15,600(63)	20,400(150)	(23,500)	
			26,000(3300)	888
$Ni(exan)_2$	15,500(50)	20,800(1500)	23,900(2700)	
			(25,700)	888
$Ni(mnt)_2^{2-}$	17,950($^1B_{3g}$)	18,380 L-L*		1068
$Ni(WS_4)_2^{2-}$	14,300($^1A_{2g}$)	19,000(1B1g)		1048,1050
$Ni(WOS_3)_2^{2-}$	14,500($^1A_{2g}$)			1049,1050
$Ni(WO_2S_2)_2^{2-}$	15,150($^1A_{2g}$)	22,200?		1050
$Ni(CS_3)_2^{2}$	(16,200)(10)	(18,500)(100)	20,100(5000)	256
$Ni(macS_4)^{2+}$	20,200(268)			1348
$Ni(dap)Cl^+$		22,800(1050)		1014
$Ni(dpdo)Br_2$	17,000(200)	25,500(4050)		1037
$Ni(dpo)(NCS)_2$		(22,700)	26,200	1038
$Ni(CyNP_3)Cl^{+2}$		21,100(1450)		740
$Ni(CyNP_3)Br^+$		20,400(1218)		740
$Ni(CyNP_3)I^+$		19,400(1005)		740
$Ni(PEt_3)_2Cl_2$		20,410(450)	27,030(14,200)	896
$Ni(PEt_3)_2Br_2$		18,450(350)	25,000(6200)	896
$Ni(PMePh_2)_2Cl_2$	(18,200)	21,750	26,200	895
$Ni(PEtPh_2)_2Br_2$	(18,000)	(20,600)	24,300	895
$Ni(Ph_2POEt)_2(CN)_2$		29,400		1011
$Ni(PMe_3)_4^{2+}$	16,600($^1B_{1g}$)	20,400(^1E_g)	25,600($^1A_{2g}$)	1069
$Ni(PMe_3)_2Cl_2$	15,150(^1A_g)	20,200($^1B_{2g}$,$^1B_{3g}$)		1069

cont

Table 6.27 cont

Complex	ν_1	ν_2	ν_3	Ref
$Ni(PMe_3)_2Br_2$	14,500(^1A_g)	18,500($^1B_{2g}$,$^1B_{3g}$)	21,800($^1B_{1g}$)	1069
$Ni(PMe_3)_2I_2$	13,900	19,600		1069
$Ni([14]\text{-}N_4mac)^{2+}$		19,000(500)	(22,400)	1047
$Ni([15]\text{-}N_4mac)^{2+}$		18,000(500)	(21,800)	1047
$Ni([13]\text{-}macN_4Lb)^{2+}$		22,470		1055
$Ni([14]\text{-}macN_4Lc)^{2+}$		22,220		1055
$Ni([14]\text{-}macN_4Ld)^{2+}$		21,550(77)		1056
$Ni([13]\text{-}macN_4)^{2+}$		24,270(110)		1057
$Ni([12]\text{-}aneN_4)^{2+}$ (cyclen)		23,255		1058,1060
$Ni([13]\text{-}aneN_4)^{2+}$		21,460		1058
$Ni([14]\text{-}aneN_4)^{2+}$ (cyclam)		22,520		1058
$Ni([15]\text{-}aneN_4)^{2+}$		21,550		1058
$Ni([18]dieneN_4)^{2+}$		21,275(56)		1198
$Ni([12]\text{-}cyclenmodN_4)^{2+}$		24,900		1061
$Ni([14]\text{-}1,4\text{-}dienemacN_4)^{2+}$		22,520(66)		1062
$Ni(2Miz)_4^{2+}$		22,300		635
$Ni(2\text{-}2\text{-}2\text{-}tetN_4)^{2+}$		22,520		1058
$Ni(2\text{-}3\text{-}2\text{-}tetN_4)^{2+}$		22,420		1058
$Ni(3\text{-}2\text{-}3\text{-}tetN_4)^{2+}$		21,785		1058
$Ni(3\text{-}3\text{-}3\text{-}tetN_4)^{2+}$		21,050		1058
$Cu(13\text{-}macN_4Lb)^{3+}$		28,570		1055

a) Transitions listed under the same column do not necessarily have the same assignment b) Gaussian analysis of single crystal spectrum. c) Also see [1082].
See individual references for details on other macrocycles. CyNP3 is tris(2-dicyclohexylphosphinoethyl)amine with one non-bonded phosphine in this case. For additional data see [1054,1071,1200].

orange, yellow or red. A second, more intense band may be seen near 23,000 - 30,000 cm^{-1} which is often charge transfer in origin. Square complexes differ from those previously discussed in that absorption below 10,000 cm^{-1} is not seen (except perhaps for some vibrational structure), the crystal field strength in a square planar complex being very high.

Diamagnetism is a consequence of eight electrons being paired in the four lower lying d orbitals. The upper orbital is $x^2\text{-}y^2$ (b_{1g}) in monodentate NiL_4 D_{4h} systems where the x,y cartesian coordinates pass through the Ni-L bonds. However note that if bidentate ligands are used, the x,y coordinates are usually written to bisect these bidentate ligands. Under such circumstances the upper orbital is xy (b_{2g}). The four lower orbitals are often so close together in energy, that individual transitions therefrom to

the upper d level, cannot be distinguished - hence the single absorption band.

Clearly there are several possible d orbital sequences. Assuming, for example, the upper level is $x^2\text{-}y^2$ (see also Chapter 1):

$$xy < z^2 < xz,yz \quad I \qquad xz,yz < z^2 < xy \quad II$$
$$xy < xz,yz < z^2 \quad III \qquad xz,yz < xy < z^2 \quad IV$$
$$z^2 < xz,yz < xy \quad V \text{ and } z^2 < xy < xz,yz \quad VI \qquad (6.22)$$

In addition a D_{2h} distortion will split xz,yz to the extent that they may overlap other levels.

Recognising that a square planar ML_4 complex has a_{2u}, b_{2u} and e_u skeletal vibrations, and that the transition moment operator transforms as $a_{2u} + e_u$, three spin allowed transitions may be vibronically induced:

$z^2 \text{---} x^2\text{-}y^2$	$^1B_{1g} \text{---} ^1A_{1g}$	Polarisation xy, z
$xy \text{---} x^2\text{-}y^2$	$^1A_{2g} \text{---} ^1A_{1g}$	Polarisation xy
$xz,yz \text{---} x^2\text{-}y^2$	$^1E_g \text{---} ^1A_{1g}$	Polarisation xy, z

(6.23)

where the energy order will depend upon the relative magnitudes of the one-electron splitting energies between these orbitals and the inter-electronic repulsion terms (see Chapter 3).

In the various papers quoted here for NiL_4 systems, some authors have assumed the empty orbital is $x^2\text{-}y^2$ (b_{1g}), and others xy (b_{2g}). For clarity, for monodentate ligands L, the empty orbital is assumed to be b_{1g}, irrespective of the original choice of the paper under discussion. Thus E_g, B_{1g} and A_{2g} are the three excited states. If xy is the empty orbital, the state label B_{1g} is replaced by B_{2g}.

The energy sequence of the lower four orbitals (which is also relevant for discussion of planar Pd(II), Pt(II), Cu(II) etc) must probably vary from one complex to another, and has been the subject of much debate [1052,1073-1077,1084,1148]. In particular note the detailed molecular orbital study of Gray and Ballhausen [1084] (Fig.5.12). Generally if there is no interaction whatsoever along the z axis, the z^2 orbital should be very stable. The e_g orbitals have pi symmetry and their energy relative to the other two orbitals will depend upon the pi nature of the ligands. The xy (or $x^2\text{-}y^2$) orbital will also have pi symmetry and its energy will depend upon whether the ligands have pi character in the xy plane.

There are many examples where three transitions, often in the form of a weak shoulder or peak on each side of the main stronger absorption band, are observed (Table 6.27). A low energy band seems characteristic of sulphur-containing systems (see Table 6.27 and, e.g. [1054]). Many complexes of the type NiL_2X_2 belong formally to D_{2h}, and may show four vibronically induced transitions (Table 6.27) due to splitting of xz,yz.

Various single crystal polarised light studies have been

undertaken [1046,1052,1053,1078,1080,1082,1083,1086]. Unfortunately while of interest, they have not always provided unequivocal results. Consider, for example, $[(1,5\text{-diazacyclooctane})_2Ni]^{2+}$. This is of especial importance since the ligand is 'innocent', in the Jorgensen sense (i.e. other than sigma bonding, does not interact with the metal), and the structure shows a rigid square planar D_{4h} configuration. Room temperature and low temperature crystal data have been reported (Fig.6.58) [1046,1086], and three bands lying under a single broad envelope were determined by gaussian analysis. The central and most intense peak is much more prominent in xy than in z, and the two outer absorptions are weak; other than these observations, little unambiguous information is derivable.

The data are expected to be consistent with vibronic selection rules in D_{4h} symmetry. The only transition forbidden in these vibronic selection rules (see above) is $^1A_{2g}---^1A_{1g}$ in (z), which fact identifies the central main peak in this complex spectrum to this transition. However it was not possible unequivocally to determine the assignments of the outer absorption. Hitchman and Bremner [1086] consider both possibilities and use AOM theory to see if one assignment is greatly favoured over the other. The two possibilities are:

	A	B
22,370 cm^{-1}	$^1B_{1g}---^1A_{1g}$	$^1E_g---^1A_{1g}$
23,480	$^1A_{2g}---^1A_{1g}$	$^1A_{2g}---^1A_{1g}$
24,540	$^1E_g---^1A_{1g}$	$^1B_{1g}---^1A_{1g}$
e_σ(Ni-N)	10,870	9430 cm^{-1}
e_π(Ni-N)	1885	795

(6.24)

The assignments for the copper analog [1046] are more secure leading to $e_\sigma(Cu\text{-}N) = 7800$ and $e_\pi(Cu\text{-}N) = 1300$ cm^{-1}. The ratio e_σ/e_π for the copper system is most closely similar to assignment A of the nickel(II) system. Given that the M-N bond length in the copper(II) complex will be longer than in the nickel(II) complex, assignment A is preferred (d sequence IV (6.22)). The rather high value for the $e_\pi(N)$ parameter is disturbing since ideally it should be zero for this saturated amine. If proven, the non-zero values of e_π may arise through an electrostatic contribution to the splitting of xy - xz,yz [1087].

In connection with the selection rules for square planar complexes, it is pertinent to note an early analysis which considered possible distortion in the excited states [1083]. These Authors note that if there is excitation to the sigma anti-bonding x^2-y^2 orbital this might 'unbalance' the square planar configuration via a D_{2d} distortion leading ultimately to a tetrahedral excited state. They calculated the energies, using an MO model, in both the square D_{4h}, and tetrahedral T_d geometries and constructed a correlation diagram connecting the two possibilities.

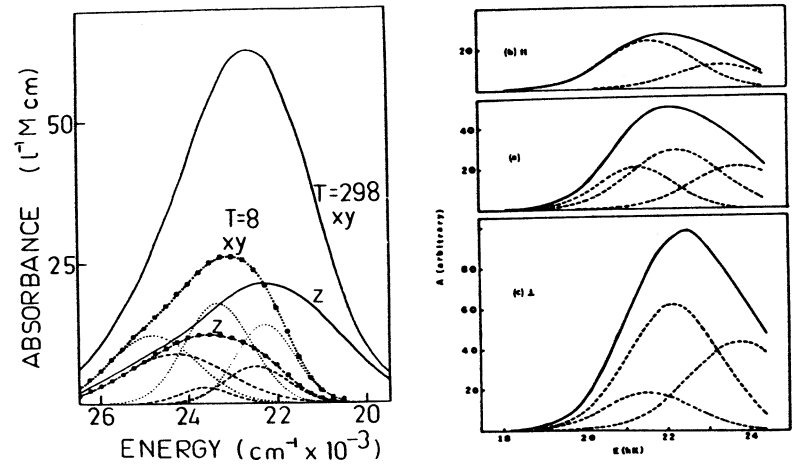

Fig.6.58 (Left) The polarised spectra of the (100) crystal face of $[Ni(daco)_2]^{2+}$ measured at 8 K and 298 K. Best fit Gaussian curves are indicated [1086]. (Right) The single crystal spectra of $[Ni(daco)_2](ClO_4)_2$ - a) non-polarised, b) - parallel to z axis, c) perpendicular to z axis [1046].

The $^1B_{1g}$ and ^1E_g states are more stable in the tetrahedral than the planar form, and therefore distort along a coordinate leading to a tetrahedral configuration. The $^1A_{2g}$ state, on the other hand, has a lower energy in the planar form and should resist distortion.

Thus transitions to the $^1B_{1g}$ and ^1E_g states should terminate on a highly vibrationally distorted state of the D_{2d} surface (symmetry $^1B_2$ and ^1E in D_{2d} respectively), and under appropriate circumstances, a progression in the distorting vibration may be seen. Recall that in considering selection rules, those symmetry elements common to both D_{4h} and D_{2d} must be used (in effect since D_{2d} is a sub-group of D_{4h}, this former group is used), viz:

Transition in D_{4h}	Transition in D_{2d}	Polarisation Electronic	Polarisation Vibronic
		(D_{2d})	(D_{4h})
$^1A_{2g}---^1A_{1g}$	$^1A_2---^1A_1$	forbidden	xy,z
$^1B_{1g}---^1A_{1g}$	$^1B_2---^1A_1$	z	xy,z
$^1E_g---^1A_{1g}$	$^1E_1---^1A_1$	xy	xy,z

(6.25)

where note that the transition moment transforms as $b_2 + e$ in D_{2d}, and the skeletal vibrations are shown in Table 4.1.

Thus if the distortion of the excited state is ignored then the transition $^1B_{1g}---^1A_{1g}$ is allowed vibronically in D_{4h} in both (xy) and (z). Allowing for distortion, however, the transition is now electronically allowed, but only along (z).

Careful experiment should distinguish these two possibilities (see discussion below). Evidence for such distortion in 'planar' excited states has been presented. Thus care should be exercised in using only vibronic selection rules in these D_{4h} complexes.

Numerous authors have investigated the spectra of the planar $[Ni(CN)_4]^{2-}$ ion [163,1053,1076,1083-1085,1088,1092,1340]. Most interest has centred on the charge transfer spectra which were considered in Chapter 5. Here we are concerned with the d-d spectra, which occur as weak features below 30,000 cm^{-1}.

In addition to crystal field absorption, this ion can exhibit Ni-CN MLCT and LMCT, and internal CN^- absorption, leading to a complicated situation. Moreover the stacking of the planar $[Ni(CN)]_4^{2-}$ ions in fairly close proximity (in the solid state) can produce additional interesting phenomena.

Indeed a very weak band near 22,500(2) cm^{-1} has an energy dependent upon the metal-metal distances in the solid state; it is polarised with the electric vector in the xy plane. Since the magnetic vector will now be along the z axis, an earlier study suggested this absorption was due to the magnetic dipole allowed $xy---x^2-y^2$ ($^1A_{2g}---^1A_{1g}$) transition [1092]. A more recent study [1053] proposes that it is in fact a spin forbidden transition enhanced in intensity because of the close proximity of the Ni(II) ions [1053] (it is not seen in solution). The issue is not resolved [1098].

In the species $(Et_4N)_2Ni(CN)_4$, a much stronger xy polarised band is seen at 31,650(840) cm^{-1}. The strength of this band, and its polarisation, provides convincing evidence that it is the electronically allowed $^1E---^1A_1$ transition to the D_{2d} distorted excited state, an assignment consistent with the MCD data [1088]. Finally, a band at 31,100(530) cm^{-1} appears only in (z) and is reasonably assigned to the electronically allowed D_{2d} transition $^1B_2---^1A_1$. This last transition is seen at 23,000 cm^{-1} in the salt $BaNi(CN)_4$ [1083].

> Note that this band should be seen in both polarisations assuming vibronic selection rules based on D_{4h} symmetry. Its observation in only one polarisation provides support for the excited state distortion.

In an early study Shupack and co-workers [1073] studied the maleonitriledithiolate derivatives of Ni(II), Pd(II) and Pt(II), the square planar $[M(mnt)_2]^{2-}$ ions (NiS_4^{2-} chromophore) (x,y axes bisecting ligand, hole in b_{1g}). A complex spectrum is expected to arise from d-d, LMCT and MLCT and internal L transitions (see Chapter 5). In this case there should be significant D_{2h} distortion and the degeneracy of $xz(b_{2g})$ and $yz(b_{3g})$ will be lost. Indeed calculation [1073] leads to the conclusion that ligand filled pi levels lie between these two components. A weak transition at 11,690(30) cm^{-1} is assigned to x^2-y^2---xy ($^1B_{1g}---^1A_{1g}$) while a shoulder, at 17,500(570) cm^{-1}, on the rising charge transfer absorption is assigned to $xz---xy$ ($^1B_{3g}---^1A_g$).

> Note that the energy of the transition x^2-y^2---xy is 10Dq - C. This is the same as the energy of the low spin d^6 $^1T_{1g}---^1A_{1g}$ transition. The energy of this band in square planar nickel complexes $Ni(II)L_2$ is similar to that observed in octahedral low spin $Co(III)L_6$ species with common L [176].

The bis(diethyldithiocarbamato)nickel(II), $Ni(dtc)_2$, species has a similar structure with a square D_{2h} NiS_4 chromophore. In parallel with the Shupack study [1073], Dingle [1052] assigned the lowest energy transition at 15,750(70) cm^{-1} in a chloroform solution of $Ni(dtc)_2$, to the x^2-y^2---xy ($^1B_{1g}---^1A_g$) transition. He noted that such a transition should have an energy which is solvent insensitive, which is indeed the case; higher energy transitions were shown to be solvent dependent. In the crystal spectrum of this species, Dingle noted four absorptions below 22,000 cm^{-1} each polarised only in one direction. The low energy x^2-y^2---xy transition is polarised along the x coordinate (matrix element $\langle A_g|B_{3u}|B_{1g}\rangle$ which spans B_{2u}) requiring a b_{2u} vibration to activate it. A strong x polarised shoulder at 21,000 cm^{-1} is assigned to z^2---xy ($^1B_{1g}---^1A_g$). Between these two bands are two weaker absorptions, one at 19,000 cm^{-1} polarised along z, which must surely be $xz---xy$ ($^1B_{3g}---^1A_g$) ($\langle B_{3g}|B_{1u}|A_g\rangle$ spans b_{2u}), and a less strongly polarised band at 17,000 cm^{-1} assigned to the remaining transition $yz---xy$ ($^1B_{2g}---^1A_g$). Note that this last transition, which is the weakest, cannot be enabled by b_{2u}. It appears that all the expected d-d transitions can be observed and that they are primarily enabled by a b_{2u} vibration.

Some caution should be exercised in assigning the spectra of planar NiS_4 chromophores in that they can exist in different (formal) oxidation states. In the more highly oxidised species the observed absorption may be LL^*, or CT in character and d-d transitions are obscured thereby [1174,1175].

Dingle notes that the high energy absorption above about 23,000 cm^{-1} is observed along all three directions assuring that if other vibrations were active, they ought to have been seen. The analysis does not exclude the possibility that one (or more) of the weak bands are symmetry (or spin) forbidden charge transfer bands. Moreover the analysis is based on the assumption that the lowest transition is x^2-y^2---xy (rather than z^2---xy), which assumption however is fairly secure. The observation of a single enabling vibration (or at least a single representation - possibly two different b_{2u} vibrations are involved) is striking (see Table 4.1). The observed d-orbital sequence, based on the Dingle analysis, is: $z^2 < xz < yz < x^2-y^2 < xy$, the same as found by Shupack et al. [1073], except for the inversion of xz and yz.

The Shupack d-orbital sequence was also derived by Latham et al. [1093] in their study of bis(dithiooxalato)nickel(II), $[Ni(dto)_2]^{2-}$. Once again the lowest transition, a shoulder at 17,700(1300) cm^{-1}, is assigned to x^2-y^2---xy; however in view of its relatively large intensity, it is probably mixed with charge transfer character (also see a resonance Raman study [1094]).

In the absence of strongly binding counter anions, the nickel(II) ion commonly forms square complexes when surrounded by

four nitrogen atoms. While systems such as the macrocyclic N_4 ligands readily form such square complexes, even ethylenediamine will form a square $[Ni(en)_2]^{2+}$ species if a weakly coordinating counter-ion is used (e.g. $[AgI_2]^-$ [875]). Thus square nickel(II) complexes can often be converted to octahedral species by the addition of strongly binding counter-ions (especially thiocyanate) or by dissolution in strongly binding donor solvents, such as water. Conversely, octahedral NiN_4X_2 species can often be converted to square NiN_4 species by removal of the X^- groups, achievable, for example, by adding excess salt to an aqueous solution, or by heating [1058,1095]. Under appropriate conditions, equilibria can be observed between octahedral and square planar nickel(II) complexes (and between square planar and tetrahedral) and due note should be taken of this possibility in that an observed spectrum might represent a mixture of species.

As far as the ready preparation of square planar complexes is concerned, the optimum macrocyclic ring size for square planar nickel(II) appears to be 14-membered (e.g. cyclam). This provides a higher energy absorption band maximum than larger rings (Table 6.27), reflecting the ligand field strength. It is more difficult to make square planar complexes with smaller rings, folding of the ring to generate a cis-octahedral species often occurring (e.g. [561,1058]). However by using salting out or similar procedures, data for square planar complexes of 13 and 12-membered rings may be obtained. These have increased values of the band maximum due to a compressed Ni-N bond length [1058] (Table 6.27). It is often possible to obtain crystal field parameters for the same ligand in both square and octahedral environments about nickel(II). In general Dq(xy) in an octahedral (or tetragonal high spin) nickel(II) complex is about 50 - 65% of the value of the square planar band maximum.

The Busch group has contributed a great deal to our understanding of macrocyclic nickel(II) (and other metal) complexes (e.g. [1200] and refs. therein). In this recent contribution the relationships between d-d band maxima, half-wave potentials, and Hammett constants are explored [1200]. For an extensive series of substituted macrocycles, the observed d-d band maximum (for a square planar complex) increases with increasing electron withdrawing strength of the substituent.

In the many cases where it is not possible to resolve the band envelope in the spectrum of a square planar nickel(II) complexes, one may tentatively assume the band maximum approximately represents the $xy \text{---} x^2\text{-}y^2$ transition whose energy is 10Dq-C. The sequence of energy of this band maximum follows:

$$NiS_4 < NiN_2O_2 < NiN_2X_2 < NiO_4 < NiP_2X_2 < NiN_4$$

while for sulphur ligands, the sequence [1093] is:

$$mnt^{2-} < (CF_3)_2C_2S_2^{2-} < dtp^- < exan^- < dtc^- < dmp^{2-} = dto^{2-}$$

Other complexes of nickel(II): Both open chain pentadentate [1163] and macrocyclic [1161] seven coordinate pentagonal bipyramidal complexes of nickel(II) are known. Single crystal magnetic and absorption spectra led to a fairly detailed AOM analysis. The macrocyclic system, based on o-phenanthroline showed poorly resolved absorption at <5000, 7500, 12,500 and >17,500 cm^{-1}. The open chain complex absorbed at 13,900, 15,600 and >22,000 cm^{-1} with additional bands calculated to lie near 9500 and 4000 cm^{-1} [1163].

Three coordinate nickel(II) derivatives are known. When Ni(II) is dissolved in molten $CsAlCl_4$ and CsCl is added, the final spectrum is consistent with that expected for the $[NiCl_4]^-$ ion. However prior to formation of this ion, when a lesser quantity of CsCl has been added, a unique spectrum attributed to the planar $[NiCl_3]^-$ ion is obtained [1096]. Two bands are observed in the visible region at 13,100 and 17,700 cm^{-1}. No absorption was observed between 5000 cm^{-1} and the visible absorption.

Gaseous nickel(II) chloride is two coordinate and exhibits two relatively intense bands at 13,000 and 21,700cm^{-1} which are almost certainly spin allowed transitions, plus a sharp band at 11,700 cm^{-1}, probably spin forbidden, and a broad plateau between 4000-7000 cm^{-1} which may contain both spin allowed and spin forbidden transitions [1097]. Depending upon the magnitude of Dq and Ds, several possible ground states are possible for a d^8 linear species as illustrated in Fig.6.59. This shows a plot of r^2/r^4 (G) versus Dq, where G = -7Ds/144Dq [378]. A computer program scanned for spin allowed transition energies near 13,000, 21,700 and below 4000 cm^{-1} with the constraint that there should be no other spin allowed bands in the this range. To minimise computer time no attempt was made to fit the spectra with ground states in the lower left quadrant of Fig.6.59. Such a fit would infer that $e_\pi(Cl) > e_\sigma(Cl)$ which is highly improbable (see Eqn.1.91). In fact four sets of parameters yielded fits to these data [378], all with a $^3\pi_g$ ground term. Two sets were excluded because the $e_\sigma(Cl)$ values were very much higher than those found for other MCl_2 systems studied. Of the remaining two data sets, one was chosen as most probable in that it predicted the bands positions of weak spin forbidden features in the spectrum more accurately than the other, viz:

Transition	Obsd	Calcd	Transition	Obsd	Calcd
$^3\Sigma_g^-(F)$		879	$^1\Delta_g(G)$	~16,000	15,466
$^3\Delta_g(F)$		2,454	$^1\Delta_g(D)$	~18,000	18,236
$^3\Phi_g(F)$	4,000-6,000	5,853	$^3\Sigma_g^-(F)$	21,700	21,701
$^1\Sigma_g^+(D)$	10,000-11,000	10,601	$^1\Pi_g(G)$		26,740
$^1\Pi_g(D)$	11,727	11,727	$^1\Sigma_g^+(G)$		27,234
$^3\Pi_g(P)$	13,000	12,984	$^1\Gamma_g(G)$		28,825
$^1\Phi_g(G)$	~14,000	14,717	$^1\Sigma_g^+(S)$		55,433

(6.26)

This assignment, which agrees with earlier work by DeKock and Gruen [379] and by Smith [676], but not Hougen [1097], generates $e_\sigma(Cl) = 4798$ and $e_\pi(Cl) = 3098$ cm^{-1} (see Chapter 9).

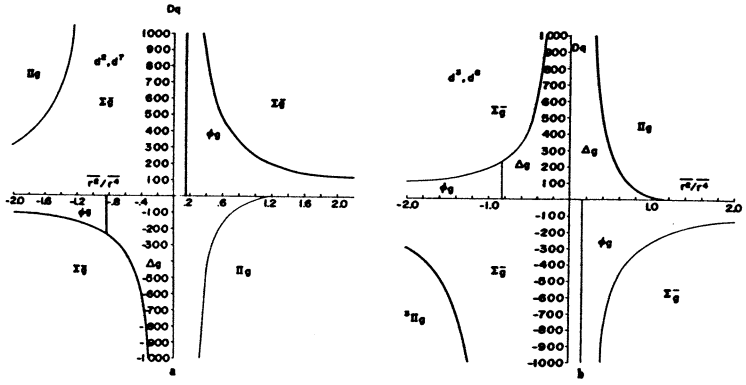

Fig.6.59 Ground state energy space diagrams for (a) d^2, d^7 ions, and (b) d^3, d^8 ions (B = 500 cm^{-1}) [378].

PALLADIUM(II) AND PLATINUM(II) d^8

These species commonly form low spin diamagnetic square planar ML_4 species, though a range of five coordinate diamagnetic complexes are are also known.

Five coordinate palladium(II) and platinum(II) complexes: Tripod ligands form low spin trigonal bipyramidal complexes of the type $[PdL_4X]^+$ [1115-1120,1125]. In parallel with their low spin trigonal pyramidal Ni(II) analogs, these species exhibit two intense peaks assignable, in the absence of marked distortion from D_{3h} symmetry, to the transitions to $^1E'$ and $^1E''$. The band energies generally shift to lower values as a function of X, in the normal spectrochemical series; however note that in the complexes $PdLX^+$ with $L = (o\text{-}Me_2AsPh)_3L'$ (L' = As,Sb), I > Br probably because of compression along the z axis (see Chapter 9) (Table 6.25). Intensities are very large for d-d transitions due to mixing with charge transfer states and perhaps intra-phenyl transitions [1116]. Low spin square pyramidal complexes have spectra not so dissimilar from their four coordinate analogs [1121,1124] (see below).

Four coordinate square planar palladium(II) and platinum(II): Generally the spectra of these square planar derivatives are similar to those of square planar nickel(II). However spin orbit coupling is much more important in these heavier metal complexes

and must be considered in more highly resolved spectra. Moreover many complexes reveal rich vibrational structure at low temperatures. Indeed some of the most elegant single crystal studies so far published utilise these systems (for an early review, see a report by Don S. Martin Jr. [1122], one of the major contributors in this field).

Before discussing these data in more detail, consider the selection rules when spin-orbit coupling is important. It is necessary to work in the D_{4h} double group, thus, following [1122]:

Vibration	Excited State and Polarisation				
	A_1'	A_2'	B_1'	B_2'	E'
a_{2u}	z	f	f	f	xy
b_{2u}	f	f	z	f	xy
e_u	xy	xy	xy	xy	z

(6.27)

and where the the various states (and transitions since the ground state is $^1A_{1g}$) transform as:

$xy{---}x^2\text{-}y^2$ $\quad {}^1A_{2g}\ A_2'$ $\quad {}^3A_{2g}\ A_1' + E'$

$z^2{---}x^2\text{-}y^2$ $\quad {}^1B_{1g}\ B_1'$ $\quad {}^3B_{1g}\ B_2' + E'$

$xz,yz{---}x^2\text{-}y^2$ $\quad {}^1E_g\ E'$ $\quad {}^3E_g\ A_1' + A_2' + B_1' + B_2' + E'$

(6.28)

Thus the transitions to ^1E_g and $^1B_{1g}$ should appear in both (xy) and (z), while $^1A_{2g}$ should appear in (xy) only (in fact the same selection rules as previously obtained (6.23)). The spin triplet transitions will have the polarisation of the spin singlet from which they steal intensity [1123]. Thus, for example, there is a spin triplet observed in the spectrum of the $[PtX_4]^{2-}$ ions (see below) which is polarised completely (xy). The only singlet state polarised totally (xy) is $^1A_{2g}$ (A_2'), and the only triplet with a component A_2' is ^3E_g. Thus this triplet should be ^3E_g [1123].

> The ^3E_g state could couple with all of the spin singlets since it possesses spinor components in common with all of them. It would then appear in both (xy) and (z). The fact that it does not appear in (z), must imply that coupling to other spin singlets, while allowed, is unimportant.

In this section, it is appropriate to discuss individual compounds in some depth. Note however some very early studies [1146-1151].

The tetrachloro and tetrabromopalladate(II) ions [1102,1111,1123, 1126-1128]:- The polarised single crystal spectra of K_2PdCl_4 are shown in Fig.6.60a at ambient and 15 K. The data are summarised in Table 6.28. The band at 21,700 cm^{-1} only appears in (xy) and must be $^1A_{2g}$.

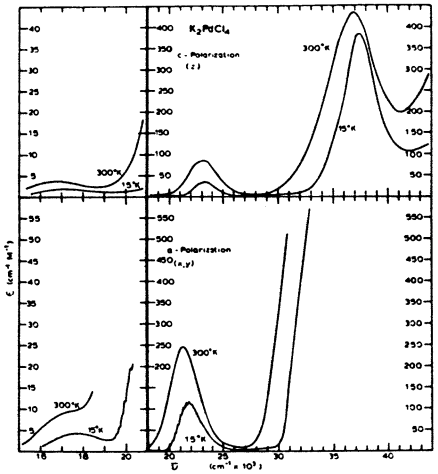

Fig.6.60 (a) (Upper) The polarised single crystal spectra of K_2PdCl_4, at 15 K and 300 K [1111]. (b) (Lower) Microphotometer tracing of the vibrational structure of the $^1A_{2g}$ transition [1111].

Assuming d-orbital sequence V (6.22), generally presumed to be relevant to rigidly square planar complexes, the band at 23,200 cm^{-1} in both (xy) and (z) is assigned to ^1E_g. This assignment is also supported by an MCD study which reveals this band to possess degeneracy (A term) [1129]. A feature seen [1111] in a doped crystal at 2 K, at 28,910 cm^{-1}, assigned to the transition to $^1B_{1g}$, not observed in Fig.6.60a.

Spin forbidden transitions in palladium(II) complexes will be less intense than those in platinum(II) complexes because of the lower spin-orbit coupling coefficient. Because of strong mixing (intensity stealing) with ^1E_g, their intensity increases the closer they are to the ^1E_g transition. Thus, on this basis, the $^3B_{1g}$ transition (E' component) should be most intense [1123]. For this reason, the weak feature at 17,700 cm^{-1} (Fig.6.60a) is assigned [1111,1123] to $^3B_{1g}$ (differs from initial assignment [1102]).

Patterson and co-workers [1111] studied the $[PdCl_4]^{2-}$ ion with different counter ions, and in different lattices, at 2 K. Two spin-allowed transitions ($^1A_{2g}$ and $^1B_{1g}$) reveal considerable vibrational structure, ^1E_g does not. Fig.6.60b shows the vibrational structure of the $^1A_{2g}$ transition. Three distinct progressions are observed, each based on a vibronic origin. This transition is enabled solely by e_u. There are two skeletal e_u modes (Table 4.1), which must be associated with two of these origins, the third must be a lattice e_u mode. The average separation between members of each progression is 271 cm^{-1} corresponding with the a_{1g} Pd-Cl stretching mode of the $^1A_{2g}$ excited state. Interestingly, the corresponding progression in the $^1B_{1g}$ excited state has a separation of 290 cm^{-1}, while the Cl--Pd LMCT ^1E_u state at 32,710 cm^{-1} has a progression frequency of 220 cm^{-1} decreased due to a 35pm increase in Pd-Cl bond length (calculated from the Franck-Condon intensities in the progression) [1111].

Using the data for Pd(II)/Cs_2HfCl_6, the one-electron energy separations between the d orbitals are shown [1111] to be ($xy--x^2-y^2$) 21,491, ($z^2--x^2-y^2$) 33,095 and ($xz,yz--x^2-y^2$) 24,968 cm^{-1} with B = 850 and C = 1750 cm^{-1}.

Data for the $[PdBr_4]^{2-}$ ion parallel those for the chloride and are included in Table 6.28 [1102,1126]. Similar vibrational structure was observed and Pd-Br stretching frequencies and bond elongation were determined.

The tetrachloro and tetrabromoplatinate(II) ions [1076,1108,1109, 1123,1129-1135]:- Under low resolution the spectra of these species are quite similar to those of their palladium analogs, though shifted to higher energies (Table 6.28) (Fig.5.34, 6.61). In both systems, the two lower energy spin singlet transitions are clearly seen, the lowest, $^1A_{2g}$ being observed only in (xy). As anticipated, the spin triplet transitions have much greater relative intensity than their palladium(II) analogs. Vibrational progressions built upon several vibronic origins are again observed. In the $^1A_{2g}$ state, the Pt-Cl a_{1g} stretching vibration has wavenumber 295 cm^{-1} [1131]. On the basis of polarisation and calculation, spinor transitions [1133] were assigned [1131] as follows:-

Obs. Energy $[PtCl_4]^{2-}$	Polarisation	Assignment	Calc. Energy
17,000		E' (^3E_g)	17,039 cm^{-1}
18,100	xy	B_2' (^3E_g)	18,113
20,735	xy,z	E' ($^3A_{2g}$)	20,723
		A_1' ($^3A_{2g}$)	21,198
23,970	xy,z	E' ($^3B_{1g}$)	23,959
26,050	xy	A_2' ($^1A_{2g}$)	26,048
29,575	xy,z	E' (^1E_g)	29,569
36,500	xy,z	B_1' ($^1B_{1g}$)	36,503

(6.29)

where the calculated data are shown in Table 6.29, and the observed energies were averages of those seen in (xy) and (z) by Mortensen [1133]. More detailed polarisation data were reported [1131].

Patterson and co-workers also reported the luminescence spectrum of Cs_2PtCl_4 in Cs_2ZrCl_6 [1131,1132]. The emission is assigned $^3B_{1g}(E') \text{----} ^1A_{1g}(A_1')$, and shows extensive highly resolved vibrational structure. The gross structure was assigned [1131] as a series of progressions in ground state a_{1g} (Pt-Cl) originating from those vibronically enabled states of $^3B_{1g}$. Specifically these were vibronic states involving $a_{2u} + e_u$ (stretch) + e_u (in-plane bend). No coupling to b_{2u} was observed. Under high resolution, individual vibronic components were split further by from 2 cm^{-1} at the high energy end of the emission spectrum, up to 19 cm^{-1} at the low energy end. These authors argue that this extra splitting is due to isomeric $[PtCl_4]^{2-}$ ions involving ^{35}Cl(75%) and ^{37}Cl(25%).

The a_{1g} Pt-Cl stretching frequency depends solely upon the mass of chlorine. The frequency separation between Pt-^{35}Cl and Pt-^{37}Cl is given by $E(^{35}Cl)[1 - \sqrt{(35/37)}] = 9.2$ cm^{-1}. However there are five isomers involving different stereochemistries possible for the various isotopic substitution patterns. Allowing for these various isomers, the separation between each isomer peak would be 2.3 cm^{-1}. However if n quanta of a progression are excited, then the separation between components can be shown to be 2.3n. Where n = 9, then the separation is approximately 19 as observed [1132].

The spectra of the $[PtBr_4]^{2-}$ ion are similar to those reported above. The weak structured peak near 22,700 cm^{-1} [1108] is only seen in (xy) and must evidently be the $^3B_{1g}$ transition. The lower energy weak transitions are assigned to the states ^3E_g and $^3A_{2g}$ but their precise assignments are less secure (as is evident from comparison of [1108] with [1109] by the same authors); there is obvious complication due to the several spinor components and the fact that there is no strongly preferred polarisation (Fig.5.34). The excited state Pt-Cl a_{1g} vibration has a frequency of about 170 cm^{-1} in both the $^1A_{2g}$ and $^3B_{1g}$ states.

Gliemann et al. [1155] propose an alternate assignment for these, and other, species. These workers identify the $^1B_{1g}$ transition at much lower energy (ca. 19,000 cm^{-1} in $[PtCl_4]^{2-}$).

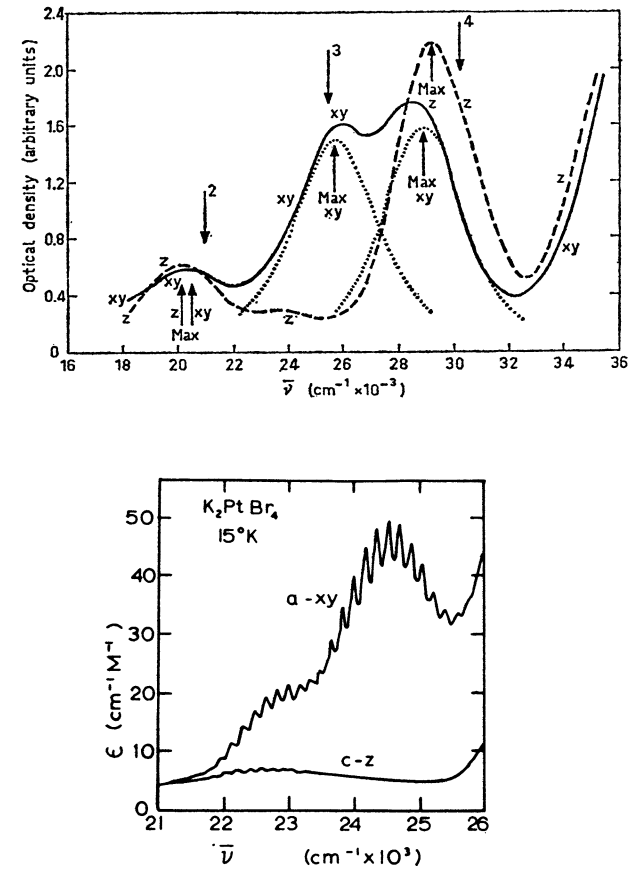

Fig.6.61 (a) (Upper) The polarised absorption spectra of K_2PtCl_4, with gaussian components. (b) (Lower) Vibrational section of the crystal spectrum of K_2PtBr_4 at 15 K [1108].

They propose that the highest filled orbital is z^2 and assign to a singlet transition a band others have assumed to be triplet. Such an assignment leads to a number of inconsistencies and to unacceptably low values for the inter-electronic repulsion parameters. Although this last assignment seems unlikely to be correct, and has been criticised accordingly [1123], it is clear that there are still some features in these spectra which are controversial (see [1123] for a recent analysis).

Table 6.28

FOUR COORDINATE SQUARE PLANAR PALLADIUM(II), PLATINUM(II) AND GOLD(III) COMPLEXES. Ground Term $^1A_{1g}$

(A) Single Crystal and Other Assigned Data

Complex	$(nBu_4N)_2PtCl_4$	$(nBu_4N)_2PtBr_4$	c-$PtCl_2(NH_3)_2$	t-$PtCl_2(NH_3)_2$
^3E_g	17,860(3.3)	16,210(5.5)	23,700MCD[c]	24,600MCD[b]
$^3A_{2g}$	20,510(9.4)	19,010(16.9)	27,000(25)[c]	26,900(30)[b]
$^3B_{1g}$			30,500MCD[c]	29,000MCD[b]
$^1A_{2g}$	25,190(33.4)	23,580(84.6)	32,500(170)[c]	31,700(75)[b]
^1E_g	29,850(40.8)	26,810(204)	35,500(110)[c]	36,700(100)[b]
$^1B_{1g}$	(37,000)(427)	(30,580)(588)		
Ref.	1076,1108	1076	1159	1159

Complex	K_2PtBr_4[a]	$KPt(NH_3)Cl_3$	$[Pt(NH_3)_4]Cl_2$	$Pt(en)Cl_2$
^3E_g		24,000(20.5)		27,300(31)[c]
$^3A_{2g}$	19,100(15)	20,800(6)[c]		
$^3B_{1g}$	22,700(13)	28,900(109)	34,800	24,900(12)[c]
$^1A_{2g}$	24,400(46)			33,200(226)[c]
^1E_g	26,800(73)	33,400(71)	42,100[c]	36,900(94)[c]
$^1B_{1g}$	33,800(39)		46,080(560)	
Ref.	1108, 1109	1110,1142	1076,1114,1143	1152

Complex	$(ncetylMe_3N)_2Pt(N_3)_4$	$(Et_4N)_2Pt(mnt)_2$	$[Pd(NH_3)_4]Cl_2$
^3E_g		(15,650)(56)[b]	
$^3A_{2g}$	20,400(92)	(14,410)(49)[b]	(27,500)
$^1A_{2g}$	26,700(755)	(18,500)(1220)[b]	33,000
^1E_g	(30,300)(820)		35,800
$^1B_{1g}$	40,500CT(19,800)		
Ref.	680	1068,1073	1114

Complex	$(nBu_4N)_2Pd_2Br_6$	K_2PdCl_4[a]	K_2PdBr_4	$(nBu_4N)_2Pd_2Cl_2$
$^3B_{1g}$		17,000(1)	16,960(4.8)	
$^3A_{2g}$	14,600(15.1)		15,700(<1)	16,290(11.9)
$^1A_{2g}$	16,950(74.8)	21,700(118)	20,200(177)	22,420(291)
^1E_g	19,420(481)	23,200(36)	21,730(33)	25,160(403)
$^1B_{1g}$	24,630CT(5120)	28,910[e]		30,580CT(3310)
Ref.	1076	1102,1123	1102,1123	1076

cont

Table 6.28 cont

Complex	$(ncetylMe_3N)_2Pd(N_3)_4$	$(Et_4N)_2Pd(mnt)_2$	$Pd(en)Cl_2$
^3E_g			23,600(16)[c]
$^3A_{2g}$			
$^1A_{2g}$	(20,000)(458)	15,700(64)[b]	26,700(292)[c]
^1E_g	(24,200)(1730)		29,400(300)[c]
$^1B_{1g}$	30,800CT(1530)		34,300(100)[c]
Ref.	680	1068,1073	1113

Complex	$(nBu_4N)AuCl_4$	$(nBu_4N)AuBr_4$	$KAu(SCN)_4$
$^3B_{1g}$?	21,930(17.3)	18,500(119)	
$^1A_{2g}$	26,520(319)	21,300(1090)	19,960(375)
^1E_g	-	-	25,180(534)
CT	30,530CT(6740)	24,940CT(4460)	31,250CT(25,400)
Ref.	1076,1101,1156	1076,1101	1101

Complex	$(nEt_4N)Au(N_3)_4$	(2-Pic)$AuBr_3$	Et_3PAuBr_3
$^1A_{2g}$	(20,600)(258)?	19,230	19,600
^1E_g	30,000CT(31,300)	22,500	24,400
CT		25,250(1800)	26,450(5000)
Ref.	680	1362	1362

a) Single crystal 15 K. b) actually assigned in D_{2h} symmetry. c) actually assigned in C_{2v} symmetry. d) The d orbital sequence is not the same in all these complexes. Hence the energy sequences are not all in regular ascending order. The first column identifies the transition in D_{4h} symmetry with x^2-y^2 as the least stable orbital. However some of the complexes, as indicated, belong to a different point group and were assigned accordingly in the original publication. This should be read to ascertain the correct labels. e) Pd^{2+} doped into Cs_2HfCl_6 [1111].

B) General Data

Complex	Absorption Spectra	Ref
$PdI(TP)^+$	(24,000) 26,400(3120)	1117
$Pd(Ph_2urea)_4^{2+}$	25,000(410)	1120
$Pd(NH_3)_4^{2+}$	33,550(212)	1076
$PdCl_2$(1-pn)	22,500(19) 26,200(234) 28,700(182)	1143
$Pd(Al_2Cl_8)$[f]	20,000	

cont

Table 6.28 cont----------

Complex	Absorption Spectra			Ref
$Pd(dtaa)_2$	18,120(447)	19,840(1778)		1145
$PtI(TF)^+$	30,700(5250)			1117
$PtCl_2(l\text{-}pn)$	27,300(36)	33,100(234)	37,000(126)	1143
$Pt(l\text{-}pn)_2^{2+}$	35,700(45)	41,600(10)	44,600(479)	1143
$Au(NH_3)_4^{3+}$	(40,320)(500)			1076
$Au(l\text{-}pn)_2^{3+}$			43,500CD	1143

f) melt phase. Also see [1153].

The amminotrichloroplatinate(II) ion [1110,1136,1142]: This species belongs to the group C_{2v}. The d-orbital sequence (V) (6.22) is appropriate but with the e level split into $xz(a_2)$ and $yz(b_1)$. There are two relatively strong bands clearly associated with spin allowed d-d transitions at 28,900 cm^{-1} (seen in x, absent in z) and 33,400 cm^{-1} (observed in z) which must be associated with $^1B_2$ ($^1A_{2g}$ in D_{4h}) and $^1A_2$,$^1B_2$ (^1E_g in D_{4h}), the latter showing no evident resolution. Indeed electronic selection rules show that the $^1B_2$ is allowed only in (x), while of the components of ^1E_g, transition to $^1A_2$ is electronically forbidden, while $^1B_1$ is allowed only in (z). Unlike the $[MX_4]^{2-}$ species, whose spectra are temperature dependent (vibronic), these spectra are essentially temperature independent (static distortion, electronic) (though there is some temperature dependence for $^1B_1$) [1110]. No vibrational structure was observed even though the crystals were cooled to 15 K. Possibly the presence of 4 a_1 modes in this MNX_3 species causes so much overlap that resolution is poor [1110].

These crystals are generally yellow (Yellow Cossa's salt). However there is a modification which is red (Red Cossa's salt). This contains some impurity Pt(IV). The resulting inter-valence transitions generate the colour (see Chapter 7).

Less highly resolved data with more tentative assignments have been reported for $Pd(en)Cl_2$ [1113] and for other ammine Pd(II) and Pt(II) complexes [1076,1114,1136-1139,1143,1152-1154]. Data for various sulphur and other soft ligand systems have been reported [1068,1073,1121,1140,1145]. Using the energy of the presumed $xy \rightarrow x^2\text{-}y^2$ transition as a probe, the energies (10Dq) of PdS_4 chromophores increase in the sequence:

mnt(17,800) < dtaa(20,200) < dpma(22,380) < dtma(22,510)

< dmma(22,800) < dtp(23,900) < dtc(24,300 < dto(28,100)

In systems such as the mnt series, both LMCT and MLCT transitions generally interfere with the observation of the weaker d-d transitions [1068,1073].

Weakly coupled systems, such as the $K_2M_2Cl_6$ complexes, and more strongly coupled systems such as Magnus' Green salt, are considered in Chapter 7. Recall that sd mixing is probably important for regular square planar complexes and should therefore be included in any analysis [1123] (see Chapter 1).

COPPER(III), SILVER(III) AND GOLD(III) d^8

In these higher oxidation state species, charge transfer spectra again tend to obscure d-d transitions [1157]. The hexafluoride anions of copper(III) and silver(III) have been discussed. The spectrum of the former is readily interpretable for octahedral high spin d^8 with transitions at 9600(^1E_g), 14,100($^3T_{2g}$), 16,400($^1A_{1g}$), and 20,400($^3T_{1g}$(F)) cm^{-1} [1238]. The spectrum of the latter shows evidence for the higher spin-orbit coupling coefficient of silver(III) relative to copper(III). Some d-d transitions can be detected on the low energy tail of the CT absorption in simple gold(III) complexes [1076,1101,1156]. In the earlier assignments of $[AuX_4]^-$ species [1076,1101], the first weak feature was tentatively assigned to the first spin allowed transition to $^1A_{2g}$ (assuming d-sequence (V) (6.22)), and the second and stronger feature to ^1E_g. Brown and Ewen Smith [1156] reported a polarised spectrum of $NaAuCl_4$ showing that the second feature (near 25,000 cm^{-1}) was polarised exclusively (xy). It is therefore likely that the weak band is indeed a spin forbidden transition, and the second feature is $^1A_{2g}$ (Table 6.28). Further low temperature studies are in order for these species.

6.2 (ix) d^9 Ni(I), Cu(II), Ag(II) and Au(II)

The dominating ion in this group is clearly copper(II), there being only a handful of reports on these other species. The d^9 ion is characteristically stereochemically flexible. While strong field ligands generally yield square planar or six coordinate tetragonal complexes, rather weak fields give rise to almost any stereochemistry. The presence of nine electrons ensures that some electrons are sigma anti-bonding and it is perhaps this feature, together with the Jahn-Teller effect in appropriate cases which give rise to a range of distorted stereochemistries of uncertain description. Thus while, for example, the spectrum of a nickel(II) species is usually characteristic of a specific stereochemistry, this is rarely true for copper(II). The four coordinate complexes may be tetrahedral, square or a distorted mixture thereof, and the five coordinate square pyramidal, trigonal bipyramidal or a mixed version thereof. In the six coordinate species, the structures vary from compressed tetragonal through regular octahedral to elongated tetragonal but also include examples where the bond angles are clearly not 90°. All of this makes the spectroscopy of these species rather uncertain.

If this is not enough, the various theoretical calculations

which work moderately well for other species, often break down for copper(II). This may be ascribed to several effects. Firstly of the first row ions, copper(II) has the largest spin orbit coupling coefficient, and this must probably be taken into account; it rarely is. All transitions are spin allowed and occur within the ^2D term; they should all have the same inter-electronic repulsion energy which should therefore be irrelevant. In fact there is some evidence of differential radial expansion necessitating allowance for inter-electronic repulsion. At least in the regular square complexes, sd mixing, is important.

NICKEL(I) d^9

Few coordination complexes of nickel(I) are known. Reduction of the commonly available macrocyclic nickel(II) complexes can yield nickel(I) when unsaturated macrocycles are involved, e.g. [1167]. However only charge transfer absorption near 20,000 cm^{-1} was reported. Sacconi and Midollini reported [1106] a pseudo tetrahedral nickel(I) complex with a tripod triphosphine ligand; a structured band was observed at 10,000 cm^{-1} with low and high energy shoulders (Table 6.30).

COPPER(II) d^9

A survey of copper(II) electronic spectroscopy presents difficulty because of the lack of definitive statements which can be made to relate spectra with structure. Moreover because of the general ease with which copper(II) complexes can be made, there is an extraordinary amount of spectroscopic information available in the literature.

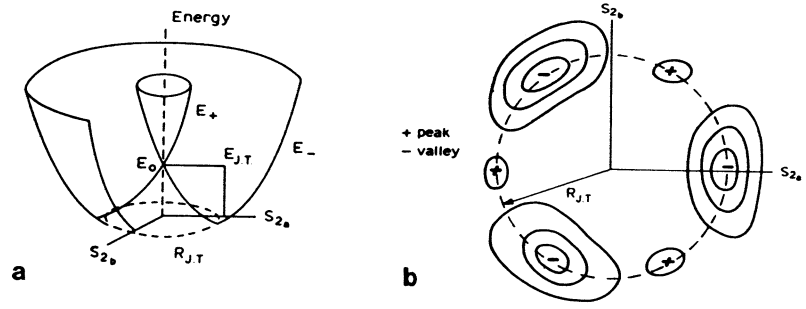

Fig.6.62 (a) (Left) The potential energy surface (Mexican Hat) with normal coordinates S_{2a} and S_{2b}. (b) (Right) Projection of the energy level surface (in (a)) warped by inclusion of non-linear terms and viewed down the principal axis with R_{JT} being the radius of the minimum potential [adapted from 1173,1221].

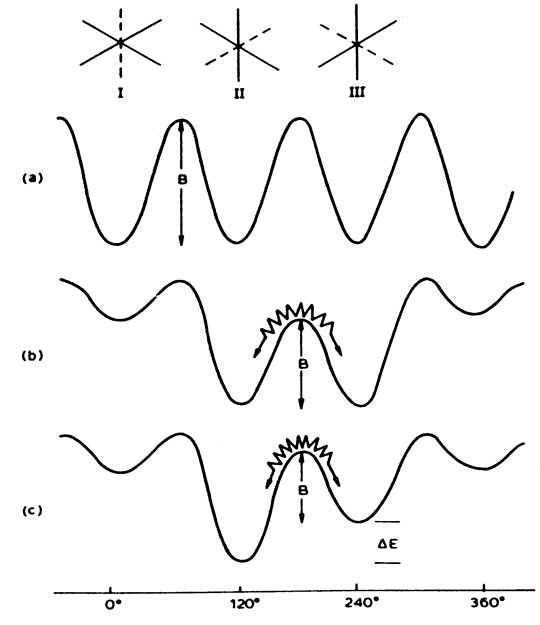

Fig.6.62 (c) Circular cross sections of the warped potential energy surfaces for (a) three wells of equal energy, (b) two wells of equal energy, and (c), one low energy well [1173,1221].

By far the bulk of this consists of complexes with a single broad poorly resolved absorption band in the visible region. Finally for reasons commented on above, copper(II) complexes frequently exist in highly distorted stereochemistries [1173,1335]. However, in parallel with nickel(II), copper(II) systems with square planar or close to square planar stereochemistry, will have no electronic absorption below 10,000 cm^{-1}. Similarly tetrahedral or close to tetrahedral copper(II) will absorb primarily in the red and near infrared. Other somewhat less secure statements will be presented below.

Since the ground state in an octahedral field is the Jahn-Teller unstable ^2E_g, (or ^2E in a trigonal system such as $[Cu(en)_3]^{2+}$), very few regular octahedral copper(II) complexes exist [345,906,1171-1173]. Following Hathaway [1173] consider the special problems associated with identifying the stereochemistries of copper(II) complexes.

Consider first the complexes CuL_6. Because of the Jahn-Teller effect [1336] operating on the ^2E_g ground term of an octahedron,

distortion will occur. It is useful to define [1171,1172,1316] a tetragonality factor **T** (do not confuse with temperature T) where **T** = mean in-plane Cu-L distance / mean out-of-plane Cu-L distance. In general **T** may be less than one (elongated) or greater than one (compressed); for the regular octahedron **T** = 1. The even e_g mode of vibration can couple to the electronic motion generating a surface of three fold symmetry [1221,1222] which can permit elongation (or compression) along any one of the three cartesian coordinates ('Mexican hat' [1220]). Fig. 6.62, reveals the three potential wells [1255].

If the coupling is relatively weak (cubic and quadratic terms important only), all three Wells may have the same energy and be equally populated. If B < kT then the molecule would appear to be regular octahedral (**T** = 1) but would actually be suffering a dynamic Jahn-Teller effect with the distortion averaged over the three cartesian directions. The cubic species $Cs_2PbCu(NO_2)_6$ and the trigonal $[Cu(en)_3]SO_4$ are examples of this behaviour [1173]. However if B > kT then such a situation could give rise to a static disordered situation in which all three distortions may occur simultaneously, with their elongation axes misaligned in three mutually perpendicular directions.

Fig.6.62 shows however that if the Jahn-Teller coupling is strong, higher order terms may warp the surface to yield one Well of higher energy than the other two equi-energetic Wells, or three Wells of different energy. In the former case a tetragonal system will be generated. If B < kT then the molecule will be fluxional with interconversion of two elongated axes at each copper site. Such a system would appear to be compressed octahedral as far as the bond lengths are concerned. Since this situation is dynamic, Hathaway prefers to label it pseudo-compressed octahedral. If B > kT then at any one crystallographic site, a copper atom may be elongated along any one of two orthogonal axes, the third axis being unaffected. Crystallographically an average of the two mixed axes would be seen. This is referred to as pseudo-compressed tetragonal octahedral [1173] or just compressed rhombic octahedral. Note that crystallography cannot readily distinguish these last two cases, B < kT and B > kT, since the former yields a time averaged distortion and the latter a statistically averaged distortion. Molecules such as Ba_2CuF_6 and $Cu(dien)_2(NO_3)_2$, among many others, appear to fall into this class.

In the final case, with three Wells of different energy, then if B > kT, a static distorted, temperature independent, tetragonal stereochemistry will be observed, generally with **T** = 0.80 - 0.85; most copper(II) complexes fall into this classification. If B < kT then the structure will be fluxional with the value of **T** dependent upon temperature [1255] (e.g. $(NH_4)_2Cu(OH_2)_6(SO_4)_2$). In general the Jahn-Teller radius (R_{JT}, Fig.6.62) is near 30pm and is available from X-ray crystallography. The barrier height B = 100 - 1000 cm^{-1} is available from electron spin resonance (esr), and the Jahn-Teller energy, E_{JT} (of the order of 2000 cm^{-1}) is available from electronic spectroscopy.

Table 6.29

ASSIGNED POLARISED DATA FOR A SELECTION OF COPPER(II) AND SILVER(II) COMPLEXES

(4-Fold groups including elongated octahedral)

Complex	Coord. No.	Transition to x^2-y^2 from z^2	xz,yz	xy	Ref
$(NH_4)_2Cu(H_2O)_6(SO_4)_2$	$6\text{-}D_{4h}$	6400	11,540 12,350	10,650	1234
$Na_4[Cu(NH_3)_4]$ $-[Cu(S_2O_3)_2]_2.NH_3$	$6\text{-}D_{4h}$	13,600	17,400		1171
$Cu(NH_3)_4(NO_2)_2$	$6\text{-}D_{4h}$	13,400	17,100	16,200	1171
$Cu(NH_3)_4(SCN)_2$	$6\text{-}D_{4h}$	14,300	17,500	15,700	1171
$K_2BaCu(NO_2)_6$	$6\text{-}D_{4h}$	8600	16,500	16,500	1172
$Cu(H_2O)_4(UO_2)_2$. $(AsO_4)_2.2H_2O$	$6\text{-}D_{4h}$	13,000	15,000	12,000	1172
$Cu(H_2O)_4(HCO_2)_2$	$6\text{-}D_{4h}$	9200	13,200	11,200	1172
$Cu(H_2O)_4TiF_6.NH_4F$	$6\text{-}D_{4h}$	11,100	16,000	13,500	1235
$Cu(NH_3)_4(CH_3CO_2)_2$	$6\text{-}D_{2h}$	17,500	17,800	15,000	1171
$Cu(EtOCH_2CO_2)_2(H_2O)_2$	$6\text{-}D_{2h}$	11,300	13,300 13,700	--	1242
$Ba_2Cu(HCO_2)_6.4H_2O$	$6\text{-}D_{2h}$	8400	13,100 13,500	10,600	1172
$Cu(ClCH_2CO_2)_2(2MePy)_2$	$6\text{-}D_{2h}$	18,000	17,700 18,700	15,000	1219
$Cu(Pyz)_2(NO_3)_2$	$6\text{-}D_{2h}$	17,400	17,200 17,600	14,800	1241
$Cu(OAc)_2(NH_3)_2$	$6\text{-}D_{2h}$	-	17,500 17,800	15,000	1219
$Cu(en)_2(BF_4)_2$	$6\text{-}D_{2h}$	18,000	19,000	(18,500- 19,000)	1256
$Cu(en)_2Cl_2.H_2O$	$6\text{-}D_{2h}$	15,000	18,100	(15,000- 18,000)	1256
$Cu(en)Cl_2$	$6\text{-}D_2$	14,000	ca.15,000	13,000	1172
$Cu(dien)_2Br_2.H_2O$	$6\text{-}D_2$	8800	15,400 15,900	9900	1172
$Zn(Cu)(dien)_2.Br_2$	$6\text{-}D_2$	9400	15,200 15,400	10,300	1236
$Cu(imidazole)_6^{2+}$	$6\text{-}D_{4h}$	12,500	16,200	10,700	1210
$[Cu(bipy)_2NO_3]^+ NO_3^-$	$6\text{-}C_{2v}$	13,400	12,200 13,600	11,400	1228
$K[Cu(NH_3)_5](PF_6)_3$ [b]	$5\text{-}C_{4v}$	11,400	15,500		1171
$Cu(NH_3)4H_2O.SO_4$	5		17,000		1171
$Cu(NH_3)_4.H_2O.SO_4$	$5\text{-}C_{4v}$	-	17,500	16,000	1172
$AgZrF_6$	$6\text{-}D_{4h}$	8340	15,230 17,340	13,080	1244

cont

Table 6.29 cont

Complex	Coord. No.	Transition to x^2-y^2 from z^2	xz,yz	xy	Ref
$Cu(acac)_2$.quinoline	5-C_{4v}	10,750	16,000	14,300	1209
$Cu(2\text{-pic})_2(NO_3)_2$	5-C_{2v}	15,500	15,800	14,000	
			17,400		1241
$(cat^+)_2CuCl_4$ [a]	4-D_{2h}	17,000	14,050(xz)	12,500	
			14,450(yz)		1184
$CuCl_4$	4-	8300	14,150	12,500	
			14,400		1205
$Pt(NH_3)_4CuCl_4$	4-D_{4h}	10,900	14,300	13,100	1213
$-[Cu(S_2O_3)_2]_2.H_2O$	4-D_{4h}	18,400	19,200		1171
$CaCuSi_4O_{10}$	4-D_{4h}	18,800	15,800	12,900	1172
$Cu(bzacac)_2$	4-D_{2h}	14,200	18,100	15,600	1215
$Cu(N\text{-Prsal})_2$	4-D_{2h}	19,500	14,800	-	
			16,400		1216

a) cat^+ = methylphenethylammonium. In some cases the coordinate axes bisected the in-plane Cu-L bonds with the result that xy and x^2-y^2 should be interchanged. b) Not ammonium salt as in literature (B.J.Hathaway private communication).

(3-Fold groups, compressed and cis octahedral)

Complex	Coord. No.	Transition to z^2 from x^2-y^2	xz	yz	xy	Ref
$Cu(bipy)_2I$ [+a]	5-C_{2v}	13,800	10,800	9300	12,700	1172
$Cu(dien)_2(NO_3)_2$	6-C_2	9200	15,600	15,600	11,700	1172
$Cu(Meoxyac)_2.2H_2O$	6-D_{2h}	9800	12,300	12,300	16,100	1172
$Cu(bipy)_2(ONO).NO_3$	6-C_{2v}	14,600	15,000	(14,600)	9500	1232
$Cu(phen)_2(H_2O)^{2+}$	5-D_{3h}	13,000	15,000	15,000	12,000	1203
$Zn(Cu)(F_6acac)(Py)_2$	6-C_{2v}	13,000 - 15,000			8500	1211

a) Data may be alternatively assigned in the point group D_2.

(2-Fold groups)

Complex	Coord. No.	Transition to xy from z^2	xz	yz	x^2-y^2	Ref
$CaCu(CH_3CO_2)_4.6H_2O$	8-D_{2d}	12,700	14,400	14,400	15,800	1172
$Cu(6\text{-}NH_2hex)_4(ClO_4)_2$	8-D_{2d}	16,000	17,500	17,500	13,600	1237
$Cu(H_2edta)H_2O$	6-D_2	8200	12,600	14,600	13,100	1232
$Cu(hfacac)_2bipy$	6-D_2	9400	13,700	14,600	-	1232
$Cu(phen)_3(ClO_4)_2$	6-D_2	8000	15,000	15,000	-	1172
$Cu(trien)(SCN)^+$	5-C_{2v}	14,100	16,800	16,800	-	1218
$Cu(3\text{-Meacac})_2$	4-D_{2h}	19,200	15,800	(14,000 - 15,500 16,000)		1172

cont

Table 6.29 (2-Fold groups)

Complex	Coord. No.	Transition to xy from z^2	xz	yz	x^2-y^2	Ref
$Cu(acac)_2$	4-D_{2h}	14,500	15,600	18,000		1178
$Cu(3\text{-Phacac})_2$	4-D_{2h}	15,400	19,000	20,600	16,900	1179
$Cu(dpm)_2$	4-D_{2h}	15,600	16,400	20,000	18,200	1180
$Cu(dpm)_2$	4-D_{2h}	15,900	16,400	18,100	16,300	1082
Cs_2CuCl_4	4-D_{2d}	9050	5550	4800	7900	1172
$Cu(dpoe)Cl_2$	4-D_{2d}	11,500	8800	9100	11,500	1245
$Cu(dtc)_2$	4-C_{2v}	--	14,480	18,600	--	1233
$Cu(Ph_3PO)_2Cl_2$	4-C_2	10,400a	7200	5500a	8500a	1240
$Ag(dtc)_2$	4-C_{2v}	--	16,000	--	--	1233

a) In C_2 symmetry the levels noted (a) all transform as a and will be mixed. The assignment suggested shows the predominant configurations obtained by comparison with the tetrahedral $[CuCl_4]^{2-}$ spectrum.

For equivalent ligands in CuL_6, and if E_{JT} is greater than kT, then the splitting of the e_g state due to the Jahn Teller splitting is approximately $4E_{JT}$. The Jahn Teller radius is given by $R^2_{JT} = \sum_{i=1}^{i=6} \Delta d_i^2$ [1173,1221].

Copper(II) complexes of lower coordination number also appear to exist in a wide range of stereochemistries making it difficult to use electronic spectroscopy alone as a definitive tool for identifying structure. Certainly, in the absence of X-ray crystallography, esr should be regarded as essential to complement electronic spectroscopic data. Although there is an enormous body of copper(II) electronic spectra in the literature much of it is published without full knowledge of the detailed structure of the complex concerned, and is therefore often ambiguous. For this reason in the discussions which follow, use is made almost exclusively of spectroscopic data obtained from compounds of known structure.

For example the lowest g value of a compressed octahedron is generally closer to 2.00 than in an elongated octahedron. Moreover the value of R $((g_2-g_1)/(g_3-g_2))$ is greater than unity for compressed and less than unity for an elongated octahedron [1172].

Table 6.29 summarises single crystal polarised and assigned data for a range of structures, while Table 6.30 contains information also from compounds of known structure but where assignments are less secure. Much of the data originates from the Hathaway group [1171-1173, and Refs. in Tables 6.29, 6.30].

Note that in the discussion which follows the assignments presented are those of the relevant authors who obtained them through use of specific selection rules. These rules were based

upon a particular choice of point group and on a decision concerning the presence of electronic or vibronic transitions and whether or not spin-orbit coupling effects should be included. The results are not always unambiguous and an intuitive or subjective choice may have been made. Where the symmetry is relatively low, the choice of appropriate point group may not be clearcut. It is evident in Table 6.29 that different authors sometimes have different assignments for the same set of data, and it is not necessarily obvious to the reader which is the correct one. Moreover the d orbital labels will only be truly relevant in high symmetry where there are no off-diagonal elements connecting several d orbitals of the same symmetry transformation. In the lower groups some d orbitals may be mixed (e.g. z^2, and x^2-y^2 in C_{2v}, both a_1) and the labels are then rather misleading, and state labels (e.g. $^2A_1$) should rather be used. Since most authors have used one electron d orbital labels, we will also do so. It is clear that in this area of spectroscopy, caution should be exercised in accepting a particular assignment of data. There is much more to learn of copper(II) spectroscopy.

We have also chosen in this case to present an overview of the various spectroscopic signatures identified in copper(II) spectroscopy without spending too much space defending each one. Note also that because the factors which determine structure can be so subtle and certainly include crystal lattice forces, a given cationic or anionic species may have a structure which is variable and depends upon the counter ion. The $[CuCl_4]^{2-}$ ion, for example, may be planar, tetrahedral, a distorted hybrid thereof, or elongated octahedral (in the solid state through polymerisation), depending upon the associated cation [1224]. Thus counter-ions are generally included in Tables 6.29, 6.30 and in the ensuing discussion. Note studies [1317,1318] which use the average environment rule to predict copper(II) spectroscopic band centres.

Six coordinate tetragonal copper(II) complexes: The t_{2g}--e_g separation in a regular octahedral copper(II) complex varies from about 13,000 cm^{-1} for CuO_6 to about 18,000 cm^{-1} for CuN_6. In the distorted versions thereof, several absorption bands may be expected in these regions, corresponding to transitions from components of t_{2g} to x^2-y^2. In general there will be a marked tetragonal splitting of the e_g level such that z^2 lies from 6000 - 15,000 cm^{-1} below x^2-y^2 and will give rise to absorption at those energies (Table 6.29). This absorption band can generally be unequivocally identified with z^2---x^2-y^2 either on the basis of low energy and/or its polarisation (along z). In general the greater the tetragonality, e.g. the longer the axial bond, the greater will be the energy of the z^2---x^2-y^2 transition [1257,1258]. The usually broad visible band envelope will contain xy---x^2-y^2, and xz,yz---x^2-y^2 (the latter perhaps split into two transitions) and analysis of the polarisation data may provide identification of the correct assignment. Indeed as the axial bond lengthens, the xy---x^2-y^2 and xz,yz---x^2-y^2 transitions shift to the blue because

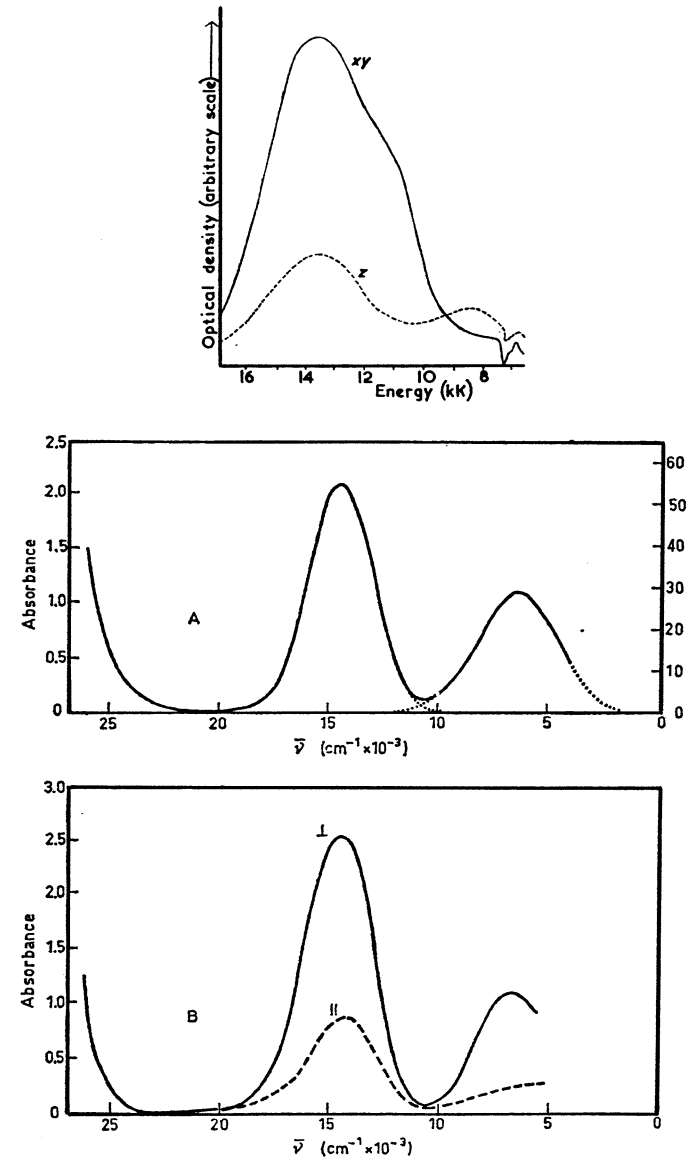

Fig.6.63 (a) (Upper) Polarised crystal spectra (100 face) of six coordinate (D_{2h}) $Ba_2Cu(HCO_2)_6.4H_2O$ at -180°C [1223]. (b) (Lower) The crystal spectrum of trigonal A - $Cu(bpy)_3Br_2.H_2O$, and B - $Zn(Cu)(bpy)_3SO_4.7H_2O$ at 27°C.

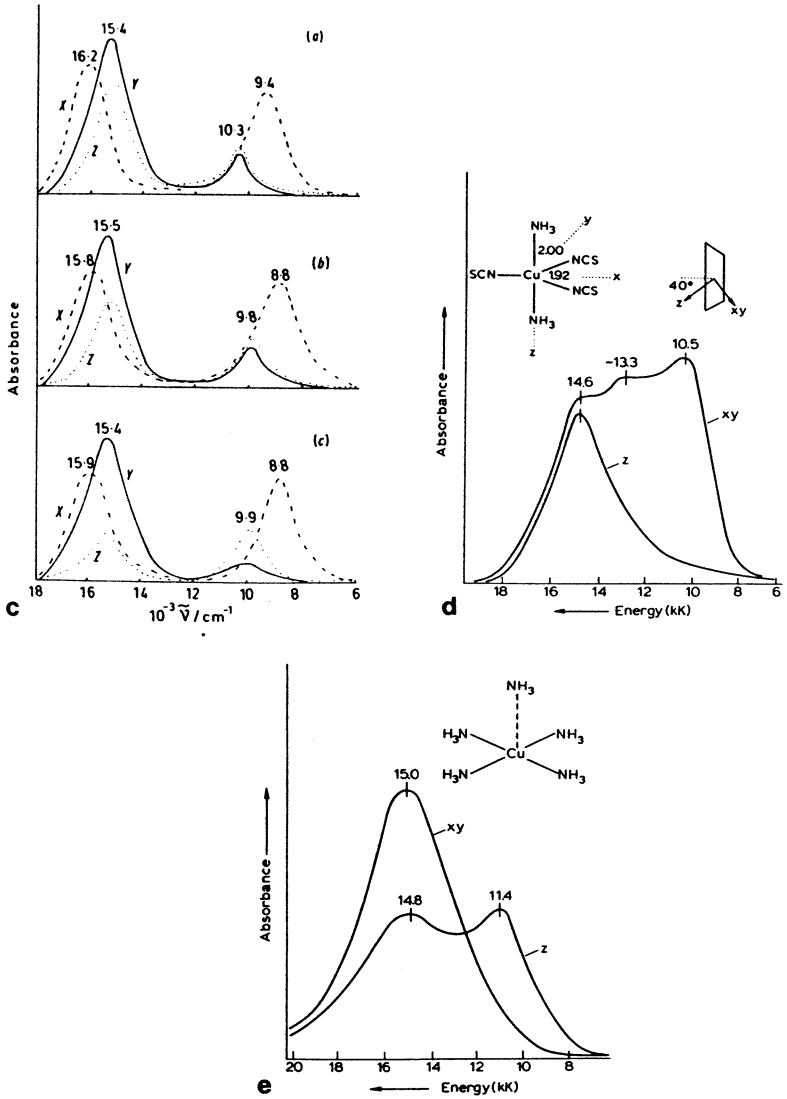

Fig.6.63 (c) (Upper left) Polarised single crystal spectra [1236] of (from top to bottom) Cu(II) doped $[Zn(dien)_2]Br_2.H_2O$, Cu(II) doped $[Zn(dien)_2](NO_3)_2$ (monoclinic type I), and $[Cu(dien)_2]Br_2.H_2O$; (d) (Upper right) Polarised spectra and structure of $[Cu(NH_3)_2(NCS)_3]$ moiety [1171,1190]; (e) (Bottom) Polarised spectra of $K[Cu(NH_3)_5](PF_6)_3$ [1171].

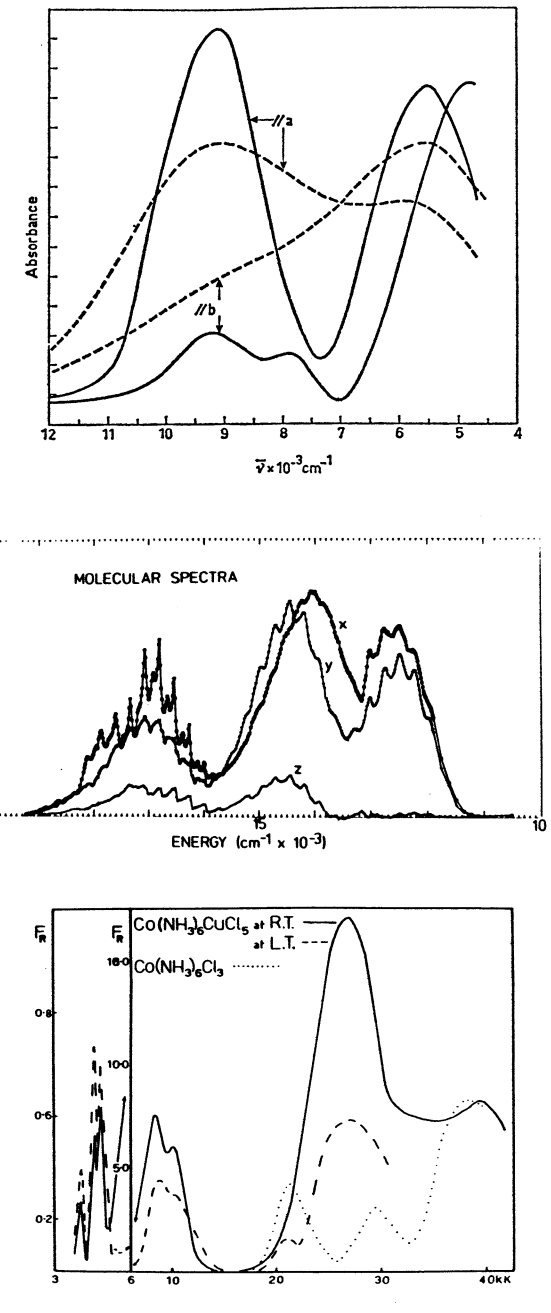

Fig.6.63 (Previous page) (f) (Top) Polarised absorption spectra of tetrahedral Cs_2CuCl_4 (010 face) in the near infrared, at room temperature (hatched) and 20 K (solid line) [1297], (g) (Middle) Molecular spectra of the planar $[CuCl_4]^{2-}$ ion in $(amph)_2CuCl_4$ [1184], and (h) (Bottom) Reflectance spectra of pentacoordinate $[CuCl_5]^{3-}$ at ambient and low temperature [1214].

of a synergic effect. As the axial bond weakens, the copper atom becomes more positive, and attracts the in-plane ligands more strongly [1259,1262]. Thus the x^2-y^2 orbital becomes more anti-bonding. As the axial field is removed, the xz,yz orbitals will become more stable. Hence the entire band envelope shifts to the blue [1258,1261]. Some examples of polarised copper(II) spectra are shown in Fig.6.63, and in Tables 6.29,6.30.

Note that in monodentate ligand complexes CuL_6, or four fold CuL_5 or CuL_4, the cartesian coordinates will lie along the Cu-L bond vectors and the highest, and half-filled, d orbital is the sigma anti-bonding x^2-y^2. With bidentate ligands such as ethylenediamine, and a square complex such as $[Cu(en)_2]^{2+}$, the cartesian coordinates, following the point group D_{2h}, bisect the in-plane Cu-L vectors such that xy is now the highest and sigma anti-bonding d orbital. With lower symmetries, the choice of where to place the cartesian framework may be equivocal such that either xy or x^2-y^2 (or possibly another orbital) may contain the unpaired electron. In comparing data from one laboratory to another the reader should pay attention to the choice of coordinate frame. Note however that where esr data are available and the directions of the g factors have been identified, these may be considered to define the cartesian coordinates. Sometimes these occur in directions which simple group theory would not have predicted. Note that for compressed octahedral CuL_6, the highest orbital will be z^2.

In general in tetragonal six coordinate molecules, three transitions can be expected though they may not always be resolved. Careful single crystal studies may show that one of these transitions, xz,yz---x^2-y^2 is split into two (i.e. absorption along x has a slightly different energy than along y). This may be a consequence of spin-orbit coupling and/or lower symmetry distortion. Thus the simple $[Cu(H_2O)_6]^{2+}$ ion [763,1234] ($\mathbf{T}$ = 0.86-0.90) depending upon specific complex) clearly shows these three bands. The z^2--x^2-y^2 transition appears at a relatively low energy (Table 6.29) indicative of the splitting of the e_g (in O_h) caused by the tetragonal distortion (Jahn-Teller).

Unfortunately the electronic spectra can often be quite insensitive to the detailed nature of the distortion in copper(II) complexes. For example a series of $[Cu(dien)_2]^{2+}$, CuN_6 chromophores observed in various salts and in a copper(II) doped zinc complex, exhibit a range of elongated, compressed and fluxional behaviour.

Table 6.30

A SELECTION OF COPPER(II), SILVER(II) AND NICKEL(I) COMPLEX ABSORPTION SPECTRA

Complex	Chromophore[a]	Absorption Spectra	Ref
Tetragonal			
$Cu(NH_3)_6^{2+}$	$N_4N'_2$	(11,200)(z^2--x^2-y^2) 16,400(xy,xz,yz--x^2-y^2)	1181
$Cu(NCS)_4^{2-}$	N_4	12,660(xy--x^2-y^2) 16,670(xz,yz,z^2--x^2-y^2)	1204
$Cu(imidazN_4)_2$	N_4	15,400 (17,800)	1229
$Cu(imidazN_2S_2)_2$	N_4S_2	14,700 20,000	1229
$Cu(thiapy)_2Cl_2$	$N_2S_2Cl_2$	(10,360)(200) 12,345(209)	1182
$K_2Cu(CN)_4$	$(CN)_4$	18,180(220) 35,715(3900)CT	1212
$Cu(Et_4dien)Cl^+$	N_3Cl	15,600(232) 18,500(140)	786
$Cu(Et_4dien)Br^+$	N_3Br	15,200(226) (17,500)	786
$Cu(Et_4dien)I^+$	N_3I	16,800(340)	786
$CsCuCl_3$	$Cl_4Cl'_2$	8300(z^2--x^2-y^2 10,000(xy--x^2-y^2) 12,930(xz,yz--x^2-y^2)	1225
$(EtNH_3)_2CuCl_4$	$Cl_4Cl'_2$	11,300(z^2--x^2-y^2) 12,390(xy--x^2-y^2) 13,300(xz,yz--x^2-y^2)	1224
$[Ph(CH_2)_2NH_2Me]_2CuCl_4$	$Cl_2Cl'_2$	12,500(xy--x^2-y^2) 14,150, 14,500(xz,yz) 16,900(z^2--x^2-y^2)	1228,1229
Cu(AcAcen)	N_2O_2	16,400(x^2-y^2--xy) 18,400(z^2--xy)	1251
$Cu(daco)_2^{2+}$	N_4	20,300(337)	1246
$Cu(dach)_2^{2+}$	N_4	19,800(172)	1246
$Cu(dach)_2Cl^+$	N_4Cl	16,200(209)	1246
$Cu(dach)_2H_2O^+$	N_4O	18,500(121)	1246
$Cu(3\text{-}2\text{-}3N_4)^{2+}$	$N_4(O_2)$	18,850(151)	1252
$Cu(3\text{-}3\text{-}3N_4)^{2+}$	$N_4(O_2)$	16,700(171)	1252
$Cu(2\text{-}2\text{-}2N_4)^{2+}$	$N_4(O_2)$	17,400(150)	1252
$Ag(dipic)_2.4H_2O$	N_2O_4	11,000(500) 17,400(800)	1239
$Ag(dipicH)(terpy)^+$	N_4O_2?	15,200(240)	1240
$Ag(terpy)_2^{2+}$	$N_4N'_2$	15,600 21,600(1380)CT?	1243
Cis-distorted octahedral			
$Cu(bipy)_2(ONO)^+$	$N_2N_2'O_2$	9500(z^2--x^2-y^2) 14,900(xz,yz,xy---x^2-y^2)[d]	1206
Square pyramidal			
$CuCl_5^{3-}$[a]	Cl_4Cl'	11,050 (23,260)CT 27,030CT	1183
$CuCl_4Br^{3-}$[b]	Cl_4Br	11,240 (22,470)CT 27,030CT	1183
$CuClBr_4^{3-}$[b]	$ClBr_4$	11,430 (19,610)CT 22,220CT	1183
$CuCl_3Br_3^{3-}$[b]	Cl_3Br_2	11,320 (22,220)CT 26,040CT	1183
$CuCl_2Br_3$	Cl_2Br_3	11,360 (21,740)CT 25,640CT	1183
$KCu(NH_3)_5(PF_6)_3$	N_5	(11,000) 15,300	1192

cont

Table 6.30 cont.

Complex	Chromophore[a]	Absorption Spectra	Ref
Trigonal bipyramidal and distorted versions thereof.			
$Cu(NH_3)_2Ag(SCN)_3$	$N_2N'_3$	12,800 14,500 26,300CT	1190
$Cu(tren)NCS^+$	$N_3N'N''$	11,900 14,700 29,400CT	1190,1207
Cu(tetb)(S-bz)	N_4S	10,870(200) 13,700(9200 (17,000)(200)	1188
$Cu(bipy)_2OAc^+$	N_4O	10,860 15,150	1189
$Cu(tren)Br_2$	$N_3N'Br$	11,200 (13,800)	1192
$Cu(tren)I_2$	$N_3N'I$	11,100 (14,500)	1192
$Cu(tren)NH_3^{2+}$	$N_3N'N''$	11,400 (15,200)	1192,1207
$Cu(Me_6tren)Br_2$	$N_3N'Br$	10,400 (13,400)	1192
$Cu(Me_6tren)I_2$	$N_3N'I$	9800 (13,000)	1192
$Co(NH_3)_6CuCl_5$	Cl_5	8200 10,400 27,000CT	1190
$Rh(NH_3)_6CuCl_5$	Cl_5	9000 10,990	1214
$dienH_3CuCl_5$	Cl_5	(11,360) 12,990	1214
$Rh(NH_3)_6CuBr_5$	Br_5	8600	1207
$Cu(bpy)_2CN\ NO_3$	N_4C	12,820 14,920	1365
$Cu(phen)_2CN\ NO_3$	N_4C	13,070[f]	1365
$Cu(bpy)_2Cl\ ClO_4$	N_4Cl	10,470 13,240	1365
$Cu(bpy)_2Cl\ Cl$	N_4Cl	12,500[f]	1365
$Cu(bpy)_2NCS\ BF_4$	N_4N'	11,170 14,120	1365
$Cu(bpy)_2OH_2\ S_2O_6$	N_4O	12,450[f]	1365
$Cu(bpy)_2OH_2\ S_5O_6$	N_4O	10,075 14,120	1365
$Cu(phen)_2OH_2(NO_3)_2$	N_4O	12,500[f]	1365
$Cu(phen)_2OH_2(BF_4)_2$	N_4O	13,400[f]	1365
Pseudo-tetrahedral			
$Cu(MPA)_2$	N_4	8500(51) (12,500) 15,900(737) +CT	1185
$Cu(bipyam)Br_2$	N_2Br_2	9000(xz,yz---xy) 13,000 (z^2,x^2-y^2---xy)	1186
$Cu(bipyam)_2(ClO_4)_2$	N_4[e]	10,400 13,500 15,700 18,400	1338
Cs_2CuCl_4	Cl_4	4800 5550 7900 9050	1226
$Cu(dmso)_4^{2+c}$	O_4	12,195(33)	1187
$Cu(dmso)_3Cl^{+c}$	O_3Cl	11,015(89)	1187
$Cu(dmso)_2Cl_3^-$	OCl_3	9525(103)	1187
$(BzNMe_3)_2CuCl_4$	Cl_4	5920 8850	1227
$Cu(spart)_2Cl_2$	N_2Cl_2	12,820(173)	1248
$Cu(spart)_2Br_2$	N_2Br_2	12,050(336)	1248
$Zn(Cu)(phen)Cl_2$	N_2Cl_2	11,630 13,900	1249
$K_2Cu[N(CN)_2]_4$	N_4	13,160(120)	1250
Ni(P_3ethane)Cl	P_3Cl	(7000) 9800(140) (15,000)vw	1106

a) In some cases, solvent molecules will also be coordinated. b) Square pyramidal versions with cation N-(2-ammoniumethyl)piperazinium cation. c) No. of dmso groups is uncertain. d) Tentative assignment, fluxional molecule. e) Polarised data, assignment uncertain. f) Higher energy unresolved shoulder. For additional useful data see [1252-1254,1259,1260].

However the electronic spectra, though clearly indicative of a significant tetragonal distortion, are almost identical [1236]; it is through use of esr and X-ray studies that the diversity of structure is revealed.

As an example of the use of selection rules consider the spectrum of $Ba_2Cu(HCO_2)_6.4H_2O$ (Fig.6.63 (a)), the data are explained in terms of vibronic selection rules in D_{2h} symmetry [1223], in which the CuO_6 skeleton spans $b_{1u} + b_{2u} + b_{3u}$, not a_u.

The vibronic selection rules for transitions to the hole in x^2-y^2 (a_g) are as follows:-

Electric dipole operator	Symmetry of modes permitting vibronic transition to x^2-y^2 (a_g) from:			
	$z^2(a_g)$	$xy(b_{1g})$	$xz(b_{2g})$	$yz(b_{3g})$
x b_{3u}	b_{3u}	b_{2u}	b_{1u}	a_u
y b_{2u}	b_{2u}	b_{3u}	a_u	b_{1u}
z b_{1u}	b_{1u}	a_u	b_{3u}	b_{2u}

(6.30)

Using these selection rules z^2---x^2-y^2 should be seen in x,y and z, xy---x^2-y^2 should be seen in xy, but not z, while the components xz,yz should be seen in x,z and y,z respectively. Comparison with other systems suggested that the b_{1u} vibration was more effective in enabling a vibronic transition than the other vibrations. This is probably a consequence of a low lying charge transfer band of appropriate symmetry. Thus although the z^2---x^2-y^2 is allowed in x,y and z, it is much stronger in z than in x,y. Consideration of Fig.6.63 (a) shows that the assignment in Table 6.29 is consistent with these selection rules.

Tetragonal copper complexes CuN_4X_2 frequently exhibit thermochromism whereby a blue quasi octahedral form interchanges with a red quasi planar form as a function of temperature [1309-1312]. Usually the square form is favoured at lower temperature, but this is not exclusively true [1309]. The shift in band envelope to the blue as the complex becomes more tetragonal may be due either to a diminished interaction along the z axis [1310], or, to a change in conformation in the CuN_4 in-plane field [1311,1312] depending upon the complex concerned.

Six coordinate trigonal copper(II) complexes: In general, highly trigonally distorted six coordinate copper(II) complexes exhibit two band envelopes, one in the visible and one in the near infrared [906,1172,1232,1268,1286,1287,1337]. Recall that a tris(bidentate ligand)copper(II) complex, such as $[Cu(bipy)_3]^{2+}$ with formal D_3 symmetry will also suffer Jahn-Teller distortion in the ^2E (in D_3) ground term. Such complexes often have D_2 symmetry with tetragonal distortion, whose magnitude however, is likely to be constrained

since the 'long bond' is part of a bidentate ligand bound in the 'equatorial plane'. Thus distortion may be smaller than many tetragonal copper(II) complexes with monodentate ligands. This will result in a smaller splitting of ^2E and hence a smaller transition energy for z^2---xy. This is indeed the case and explains why such species have absorption in the near infrared (z^2---xy) (z polarised). The visible band envelope (primarily, but not exclusively, x and y polarised) contains two of the remaining transitions (Fig.6.63 (B), Table 6.29), the position of x^2-y^2---xy being uncertain. It may also occur in the near infrared. In D_2 symmetry the z^2---xy and x^2-y^2---xy transitions are each z polarised (electronic selection rules), while the transitions from xz and yz are x and y polarised respectively [1337].

The division of copper complexes into trigonal and tetragonal distortion is rather artificial since the former also suffer a tetragonal perturbation. However it is useful to consider the spectra in terms of a tetragonally distorted trigonal $Cu(LL)_3$ system. Compare for example the spectra of the $[Cu(dien)_2]^{2+}$ systems [1236] with the trigonal systems discussed here; they are very similar (Fig.6.63 (C)).

Five Coordinate Copper(II) Complexes: Clearly there are two extreme structures, the square pyramid and the trigonal bipyramid; not unexpectedly many distorted intermediates are known. An essentially regular trigonal bipyramidal CuN_5 chromophore is seen in $Cu(NH_3)_2Ag(SCN)_3$ with peaks extending from 10,500 - 14,600 cm^{-1} (Fig.6.63 (D)), and greater absorption intensity to lower energy. The square pyramidal CuN_5 chromophore in $K[Cu(NH_3)_5](PF_6)_3$ (Fig.6.63 (E)) has a similar band envelope with peaks extending from 11,400 - 15,000 cm^{-1}, but with the greater intensity absorption to higher energy. The polarisation is totally different with the z polarised absorption occurring to higher energy in the former complex, and to lower energy in the latter complex. Where polarised data are available the spectra of regular square pyramids can probably safely be distinguished from regular trigonal bipyramids, using polarisation and intensity information. While the reflectance spectra of trigonal bipyramidal copper(II) complexes usually exhibit a strong low energy band followed by a weaker band or shoulder to higher energy, this is not invariably true (e.g. [1293]) and the data should be interpreted with care [979,1292, 1294,1295].

Many five coordinate chromophores have been studied (e.g. [1183,1188,1190,1192,1209,1214,1217,1218, 1263-1267,1269,1292-1295, 1339,1363,1365]. There is certainly a general trend that as distortion proceeds from regular square pyramidal to regular trigonal bipyramidal via distorted intermediates, the band intensity maximum in the absorption envelope shifts to lower energy, while the z polarised intensity shifts to higher energies (e.g. [1265]). For a given set of ligands the energy of the overall band envelope does not shift appreciably. These observations are well represented by the spectra of the square pyramidal $[CuCl_5]^{2-}$ ion in [1183] compared with spectra of the trigonal bipyramidal stereoisomer in [1214], and also by the comparison of the spectra of $[Cu(tren)NH_3]^{2+}$ with $[Cu(NH_3)_5]^+$ in [1192]. Some intermediate stereochemical forms exhibit twin peaks [1266,1269].

With respect to square pyramidal complexes, the tetragonal distortion caused by a single axial ligand will generally be greater than that caused by two axial ligands in a six coordinate complex. Thus for comparable five and six coordinate complexes, the z^2---x^2-y^2 transition should occur at lower energy in the six coordinate complex. Indeed the band should shift to higher energy as the coordination number decreases from 6 to 5 to 4 (square planar) (cf Co(II) analogs).

A range of $[Cu(bipy)_2X]^+$ systems (X in trigonal plane) show different in-plane NCuN angles in a distorted trigonal bipyramidal geometry [1365]. The distortion from the regular 120° angles may give rise to the twin-peaked d-d spectra sometimes seen in such systems. It is present when the two in-plane angles XCuN(1) and XCuN(2) differ by more than a few degrees [1365].

Electronic and CD data have been presented for tetragonal six coordinate and square pyramidal five coordinate Cu(II) complexes of a range of 12, 13, 14, 15, 16 and 18-membered macrocycles [1363].

Four coordinate tetrahedral and pseudo-tetrahedral copper(II) complexes: A regular tetrahedral copper(II) complex (in the absence of spin orbit splitting) should yield a single transition (^2E---$^2T_2$) at relatively low energy. However in common with the other coordination numbers, 'tetrahedral' copper(II) complexes are almost invariably distorted. Since a four coordinate square planar complex is strongly favoured for the d^9 configuration, the usual distortion is a flattening of the tetrahedron along a two-fold axis. This will result in a structure of D_{2d} symmetry which would retain the degeneracy of xz,yz and lead to three possible transitions, namely from the ground state $^2B_2$ to ^2E, $^2B_1$ and $^2A_1$ [1297]. Commonly four transitions are observed. This may reflect further distortion to D_2 or C_s, and/or the splitting of the e level in D_{2d} by spin orbit coupling. Indeed Ferguson [1297] has studied the single crystal polarised spectra of Cs_2CuCl_4 and reports four transitions below 10,000 cm^{-1} (Fig.6.63 (f)) as indicated in Table 6.29, ascribed both to a moderately high spin orbit coupling coefficient and some additional distortion to C_s.

A regular tetrahedral $[CuCl_4]^{2-}$ species will have a dihedral angle of 90° between the two Cl-Cu-Cl planes. A regular planar complex will have a value of 0° for this angle. Many different $(cat^+)_2CuCl_4$ species are known where the dihedral angle (as measured by X-ray spectroscopy) varies widely between these two limits. Generally as the molecule is flattened (dihedral angle approaches zero, the band envelope shifts to higher energies. Several authors have demonstrated a smooth (though not necessarily linear) relationship between band maxima and the value of this angle [1272-1274,1278]. In the planar $[CuCl_4]^{2-}$ ion [1205,1224,

1279] the absorption has shifted to 12,500 - 16,900 cm^{-1}. The $[CuBr_4]^{2-}$ ion has also been extensively studied [1280-1283], as have mixed halo-anions, e.g. $[CuCl_2Br_2]^{2-}$ [1284].

With stronger field ligands, such as amines, the absorption for 'tetrahedral' copper(II) complexes will naturally lie towards the blue (Table 6.29,6.30). Thus the tetrahedral CuN_4 chromophore is obtained via use of dipyridylamine type ligands. With angles close to a regular tetrahedron, absorption occurs near 7000-8000, 10,000-11,000, 13,000 to 15,000 and 15,000 to 16,000 cm^{-1} [1185, 1270,1271,1330]. These spectra have been assigned to transitions within the spinor components of ^2D in a D_{2d} field [1271]. Examples of complexes with variously compressed tetrahedral CuN_4 chromophores have been studied and their spectra shown to correlate with the dihedral angle [1185,1270,1271,1338], shifting to the blue as the degree of compression increases. Other CuL_4 chromophores such as the salicylaldimines also show spectra which shift to higher energy according to the extent of compression towards the planar structure [1275-1277].

The spectra of deprotonated amide N ligands have been reported in tetragonal [1331] and pseudotetrahedral [1330] systems.

Four Coordinate Planar Copper(II) Complexes: The square planar CuN_4 chromophore is known with ligands such as daco, various N_4 macrocycles, linear tetradentate tetrammines, and with more simple ligands such as ethylenediamine in the presence of weakly coordinating anions, e.g. $Cu(en)_2(AgI_2)_2$ [875]. Such complexes are usually red and have absorption showing little structure (in the absence of polarised data) between 18,000 and 21,000 cm^{-1} (Tables 6.29, 6.30). Square planar CuN_2O_2 chromophores (of D_{2h} symmetry) are well known in Schiff base complexes such as the salicylaldimines. They also may absorb as high as 20,000 cm^{-1} but commonly exhibit a broad structured band from about 13,000 to 20,000 cm^{-1} (see Tables 6.29, 6.30).

Hitchman and Cassidy have reported [1184] highly vibrationally resolved spectra for the planar $[CuCl_4]^{2-}$ ion (Fig.6.63 (g)). Polarisation data are clearly in agreement with selection rules for D_{2h} symmetry, with assignments as shown in Table 6.29. All three transitions show vibrational progressions in the a_{1g} Cu-Cl mode and were analysed to demonstrate a lengthening in the Cu-Cl bond of 8.9pm in the $^2B_{1g}$ and $^2B_{3g}$ states and 10.5pm in the ^2A_g state. transitions to the two former states were enabled in y polarisation by the out-of-plane b_{3u} vibration of frequency ca. 105 cm^{-1} (from temperature dependence measurements). The ^2A_g state, in xy polarisation, was coupled to the e_u in-plane Cu-Cl bending mode (188 cm^{-1}) plus three low energy lattice modes. Each of these four vibronic origins has a superimposed progression in a_{1g}. The d orbital sequence is (V) (6.22). The data were analysed in terms of the AOM model but with additional stabilisation of the z^2 orbital through mixing with 4s.

The planar CuO_4 chromophore was extensively studied as the $Cu(acac)_2$ species (and substituted versions thereof) in a series of

early papers [1082,1284,1285]. The $Cu(acac)_2$ complex also absorbs in a broad structured band from about 13,000 to 20,000 cm^{-1}. Hitchman and Belford studied the bis(benzoylacetonato)copper(II) complex and commented on the intensity mechanism [1215]. Many of these complexes show marked polarisation along y. These authors comment that it is not appropriate to argue that only one vibration is effective at enabling the vibronic spectra. Rather they conclude that there is one charge transfer band (of symmetry $^2B_{3u}$), lying at an energy significantly below the others and which supplies most of the intensity to the d-d transitions. The d orbital sequence is believed to be (IV) (6.22).

Yokoi [1303] has rationalised the electronic spectroscopic and esr data for copper diketonates and has demonstrated that the intensity of the d-d absorption varies in a linear fashion with the pK_a of the diketone. This is consistent with a diketone to metal charge transfer state being primarily involved in the vibronic intensity mechanism. It is also recognised [1303] that the intensity of planar copper complexes generally increases in the sequence:

$$CuO_4 < CuO_2N_2 < CuN_4 \qquad (6.31)$$

provided that the ligands concerned are not bound strongly via pi bonds to the copper. However there are likely to be many exceptions.

The planar CuS_4 chromophore is represented by the diethyldithiocarbamate system, $Cu(dtc)_2$ [1233]. The species exhibits d-d bands at energies close to those observed in the acac systems; polarised data provide the assignments shown in Table 6.29. CT absorption begins at about 20,000 cm^{-1}.

Hathaway [1319] reports the regions in which CuN_x chromophores of various types absorb, as a function of their stereochemistry. These data are reproduced in Fig.6.64, but are best applied to a closely related series of compounds.

Nikolov [1308] has commented on the mixing of orbitals, and modification of spectra which arise as the bite angle in a $Cu(LL)_2$ type complex is varied.

Other complexes of Copper(II):- Two coordinate copper(II) is known via $CuCl_2$ in the gas phase [378,379,676,1097]. Two bands are observed at 9000 and 18,000 cm^{-1}. On the basis of AOM calculations and the perceived need to keep the AOM parameters in line with those found for other 2 coordinate MCl_2 species, Lever and Hollebone [378] argue that the 9000 cm^{-1} band is the $^2\Delta_g \text{---} ^2\Sigma_g^+$ d-d transition, while the 18,000 cm^{-1} band is charge transfer (Cl to Cu(II) [379].

Eight coordinate copper is known in complexes such as $CaCu(OAc)_4.6H_2O$ [1315] and tetra-(6-aminohexanoic acid)copper(II) diperchlorate [1237]. Polarised data were reported for these systems (Table 6.29).

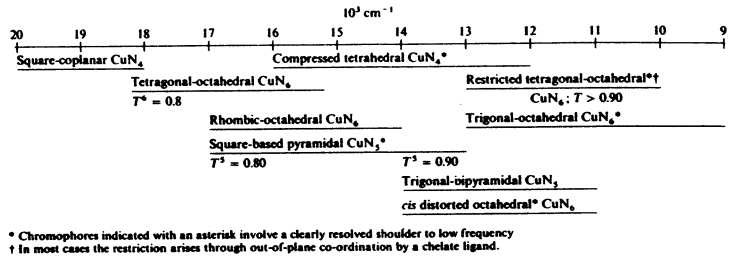

Fig.6.64 Summary of energy ranges for closely related CuN_x chromophores with different stereochemistries [1319].

SILVER(II) d^9

Relatively few data are available for silver(II) complexes. However it does appear that these species can exist in both tetragonal six coordinate (e.g. the $[AgF_6]^{4-}$ chromophore) or square planar stereochemistry (e.g. $[AgF_4]^{2-}$ chromophore) and that their spectra parallel those of copper(II) but are shifted to higher energies (Tables 6.29, 6.30) [1231,1307].

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Chapter 7

Metal-Metal Bonding, Mixed Valence, and Cyclopentadienyl Systems

7.1 Metal-Metal Bonding [1]

Bimetallic complexes $(ML_x)_2$ may possess a metal-metal interaction. If the metal-metal interaction is very weak, it may be revealed by magnetic measurements, but the electronic spectrum will generally appear as those of the isolated ML_x fragments. In polymeric chains $(ML_x)_n$ with weak exchange interactions, this may still be generally true, but spin forbidden transitions may acquire additional intensity.

Slightly stronger interactions between the metals will result in coupling between orbitals on adjacent metals. This can be so small as to be barely detectable or can give rise to characteristic new spectra. It is not necessary for there to be a formal bond between the metal atoms, i.e. binuclear metal complexes with a formal bond order of zero may yet display spectra characteristic of a metal-metal interaction.

In recent years there have been many detailed studies of the electronic spectra of such bimetallic (and polymetallic) complexes, primarily, though certainly not exclusively, from the laboratories of Cotton [1] , Gray, and Martin.

Consider the approach of two equivalent eclipsed planar ML_4 species along the axis (z) perpendicular to the ML_4 plane [2]. Direct interaction of the z^2 orbitals on each M along the bonding axis, will generate a σ-pair of orbitals transforming as a_{1g}, and σ^* transforming as a_{2u} in the group D_{4h}.

> Of key importance in this development is the creation, from 'even' d orbitals, of 'odd' molecular orbitals as a consequence of the metal-metal interaction, and hence the possibility of d-d type Laporte allowed transitions.

Interaction between the pair of xz,yz orbitals will generate a pair of π-orbitals, the bonding set transforming as e_u and the anti-bonding set as e_g. Finally the xy orbitals will interact in the δ-bonding sense to generate a bonding orbital, b_{2g}, and a δ^* anti-bonding interaction, b_{1u}. The x^2-y^2 and s orbitals are not usually considered because they interact strongly with the ligand orbitals. However inclusion of the p_z orbital into this analysis results in two non-bonding sigma type orbitals (mixed with z^2) (a_{1g}, a_{2u}) lying along the z axis but pointing outwards from the metal-metal interaction.

Note that if the two ML_4 moieties twist 45° into the staggered configuration, overlap between the xy orbitals is lost. The sequence of energy levels shown in Fig.7.1 is not invariant but will depend upon the coordination number of the mononuclear fragment (especially whether or not there is an axial interaction)

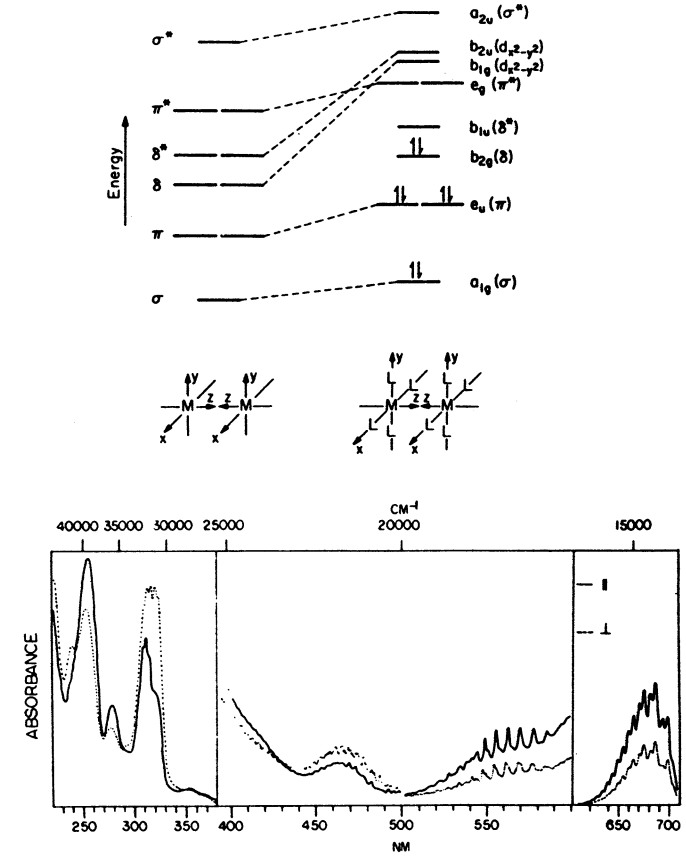


Fig.7.1 (a) (Upper) Relative energies of the d molecular orbitals of the M_2 (left) and M_2L_8 fragments with electron occupation shown for a d^4-d^4 species [2]. (b) (Lower) The single crystal polarised electronic spectrum of $(nBu_4N)_2Re_2Cl_8$ at 5 K (outer spectra) and 15 K (inner spectra). Three different spectra were used to obtain the spectra [2].

and on the distance apart of the two metal centres. For longer distances, and especially for the higher d^n configurations, only the sigma (z^2, z bonding axis) interaction may be important. Assuming the sequence $\sigma < \pi < \delta$ then configurations with a total number of electrons of seven (for example d^3-d^4), eight or nine, will have at least one electron in δ and at least one hole in δ^*. Under such circumstances, the lowest transition is expected to be $\delta - \delta^*$, which is electronically allowed and polarised along the

axis. This transition has been observed in a wide range of complexes and has been the subject of much experimental and theoretical study. It is useful to discuss the data in terms of the configurations of the metal involved.

7.1 i) d^3-d^3 Species

The W_2X_6 Systems [216,217]: Complexes of this group with Mo(III) and W(III) have been prepared but detailed electronic spectroscopic data do not appear to have been published.

7.1 ii) d^3-d^4 Species

The $[Mo_2(SO_4)_4]^{3-}.3.5H_2O$ System: Ultraviolet irradiation of $K_4Mo_2(SO_4)_4$ (see below) leads to the production of this mixed valence Mo(II)-Mo(III) species [4], with configuration $\sigma^2\pi^4\delta$ ($^2B_{2g}$). A band in the near infrared at 6900 cm^{-1} shows a pronounced vibrational progression with spacing 350 cm^{-1} attributable to the totally symmetric Mo-Mo stretch. The transition might arise through an intervalence transition (see section 7.3) or a $\delta - \delta^*$ ($^2B_{1u}$---$^2B_{2g}$) transition. The former is considered unlikely since intervalence transitions rarely exhibit vibrational structure. Data obtained at 15K show five origins in this transition, each with a vibrational progression built thereon. Some of these origins are associated with crystallographically non-equivalent molybdenum sites in the crystal.

Data have also been presented, but not assigned, for the $[Mo_2Br_6]^-$ ion [30].

7.1 iii) d^4-d^4 systems.

Two of the most studied groups of d^4-d^4 complexes are the $[Re_2X_8]^{2-}$ and $[Mo_2X_8]^{4-}$ systems. The energy level diagram assumed appropriate for such species is shown in Fig.7.1 where the bonding σ, π and δ levels lie below the corresponding set of anti-bonding levels. The set of bonding orbitals derived from the d orbitals can accomodate 8 electrons to give, formally, a bond order of four. This perhaps explains the high stability of d^4-d^4 binuclear complexes having this set of bonding orbitals filled (see Fig.7.1). The lowest lying transition is expected to be the δ--δ^* transition and this is usually correct. However this was not always considered to be the case since this interaction gives rise to some unusual spectroscopic properties.

The δ--δ^* transition ($^1A_{2u}$---$^1A_{1g}$ in these d^4-d^4 complexes) is electronically allowed in the group D_{4h}, and is expected to be polarised along the M-M (z) axis. As anticipated for an electronically allowed transition its intensity shows little temperature dependence [3].

Data for many of these species are presented in Table 7.1. Proof for the validity of the δ--δ^* assignment for the lowest prominent, but often not very intense, transition in these d^4-d^4 complexes has been presented by various groups. Thus in an early study Cowman and Gray studied the spectrum of the $Re_2Cl_6(PEt_3)_2$ complex of C_{2h} symmetry [3]. In this particular case the δ--δ^* transition is allowed in xz (C_2=y). The alternate possibility that the low intensity, low energy transition corresponds with δ--σ_n (as once thought) is eliminated in this case because such a transition would be polarised along y and no absorption is seen in this direction with this complex.

> Note that d-d transitions are generally vibronic and therefore are expected to show temperature dependence. The lack of temperature dependence for a d-d transition in a presumed polynuclear species can be taken as strong evidence for the presence of a metal-metal interaction. Of course one must be sure that a d-d transition is being studied, rather than, for example, an allowed, but weak, charge transfer transition.

The $[Re_2X_8]^{2-}$ Systems: The 5K spectra of the $[Re_2Cl_8]^{2-}$ ion was reported by Cowman and Gray in an early publication [3]. In this paper a band at 14,183 (0-0 origin) was assigned to $\delta - \delta^*$ refuting an earlier assignment [5,168]. Several vibrational progressions are built upon this origin. A progression in the totally symmetric Re-Re stretch (248 cm^{-1} compared with the ground state 274 cm^{-1}), is seen. In addition, a pseudo-origin occurs 115 cm^{-1} to higher energy, and a further progression in 245 cm^{-1} is based thereon. The 115 cm^{-1} corresponds with the ground state frequency of the totally symmetric ReReCl bending mode, which is presumably coupled to this excitation. The integrated intensity of the band is temperature independent, implying an electric dipole allowed transition as expected. Moreover the band is z polarized as required for this assigmment [6,7]. Similar data were reported for the $[Re_2Br_8]^{2-}$ ion (see Table 7.1) [3,6]. The $\delta\delta^*$ singlet excited state exhibits emission and rich photoredox chemistry [8,65]. The excited state singlet lifetime is surprisingly long (140 ns in acetonitrile). Indeed it is suggested that emission comes from an excited singlet which has twisted to become non-eclipsed (D_4 or D_{4d}) since emission from $\delta\delta^*$ in these points groups is forbidden. The triplet $\delta\delta^*$ state was not identified but arguments were presented to infer that it might lie at very low energy [65,66] (Fig.7.1 (b)).

Resonance Raman excitation into the $\delta - \delta^*$ transition of these halides [9] yields progressions in the a_{1g} Re-Re stretch, coupled also to the a_{1g} Re-X stretch (X = Cl,Br). Harmonic frequencies and anharmonicity functions were derived from these progressions (see section 4.3), leading to values of the bond dissociation energies [10] (Birge-Sponner [63]).

These two halide species exhibit strong bands at 30,800 cm^{-1} (Cl) and 23,830 cm^{-1} (Br) which exhibit A terms in their MCD spectra and are evidently charge transfer from halide to rhenium, ^1E_u---$^1A_{1g}$ [6,11]. Between the $\delta - \delta^*$ and LMCT transitions there lie up to four weak transitions (intensities 100 - 500 l mol^{-1} cm^{-1}) each with a vibrational progression.

Table 7.1

SPECTROSCOPIC DATA FOR COUPLED METAL–METAL SYSTEMS
(Containing 0–10 d electrons)

Complex	r(M-M)[a]	δ--δ*	π*--d*	π---d*	
d^3-d^4					
$Tc_2(2OP)_4Cl$	2.095	12,194	17,000	18,000	26
$Mo_2Br_6^{3-}$		13,900?	(21,050)	23,800	30
Tc_2Cl_8		5900(630)g	15,700(172)	13,600(35)	24
$Mo_2(SO_4)_4^{3-}$		6900 350			4,116

Complex	r(M-M)[a]	δ - δ*	Spacing	Other Bands[b]	Ref
d^4-d^4		Group A (ε< 200)			
$Cr_2(OAc)_4.2H_2O$	2.836	21,000			13
$Mo_2(L\text{-leuc})_4Cl_2^{2+}$	2.11	21,594(z)(0-0)	347		
		21,851(xy)(0-0 + 256)			
		22,036(xy)(0-0 + 445)			15
$Mo_2(glycH)_4^{4+}$	2.115	21,510(z)(0-0)	343		
		21,790(xy)(0-0 + 280)			
		21,930(xy)(0-0 + 420)			
		22,624(95)f			16
$Mo_2(OAc)_4$	2.093	21,700(z)(0-0)	375		
		21,975(xy)(0-0 + 275)			
		22,345(xy)(0-0 + 545)			
		26,500 (^1E_g---$^1A_{1g}$?)			17
$Mo_2(OC_2H)_4$	2.091	21,870(z)(0-0)	350		
		22,270(xy)(0-0 + 400)			
		22,660(xy)(0-0 + 790)			17,18
$Mo_2(O_2CCF_3)_4$	2.09	22,070(z)(0-0)	360		
		22,325(xy)(0-0 + 255)			
		22,581(xy)(0-0 + 511)			18,19
$Mo_2(SO_4)_4.2H_2O$	2.111	19,400			20
$Re_2(O_2CEt)_4Cl_2$		20,120			158
$Re_2(O_2CEt)_4Br_2$		19,650			158
$Re_2(O_2Cn\text{-}Pr)_4Cl_2$	2.235	19,700			158
$Re_2(O_2Cn\text{-}Pr)_4Br_2$	2.234	19,700			158
$Re_2(O_2CPh)_4Cl_2$	2.235	18,750			158
$Mo_2(O_2CCF_3)_4.2Py$	2.129	21,460(z)(0-0)	331		
		21,690(xy)(0-0 + 230)			
		22,130(xy)(0-0 + 340)			19

cont

Table 7.1 cont

Complex	r(M-M)[a]	δ - δ*	Spacing	Other Bands[b]	Ref
d^4-d^4		Group B (ε250-1200)			
$Mo_2(H_2N(CH_2)_2NH_3)_4^{4+}$		19,800			159
$Mo_2(OAc)_2Cl_4^{2-}$	2.086	20,200			160
$Cr_2(CH_3)_8^{4-}$	1.98	22,000(700)[ef]			21
$Cr_2(mhp)_4$	1.889	21,060(0-0)	320	(29,600)	157
$Re_2F_8^{2-}$		17,900			12
$Re_2(OAc)_4Cl_2$	2.209	15,750			161
$Re_2(O_2Cn\text{-}Pr)_4I_2$		19,530			158
$K_4Mo_2Cl_8$	2.15	17,897	351		3
$K_4Mo_2Cl_8$	2.15	18,083(z)	336	31,400(xy) ^1E_u---$^1A_{1g}$	59
$Mo_2Br_8^{4-}$	2.135	19,400			162
		Group C (ε> 1200)			
$Re_2Cl_8^{2-}$	2.224	14,183	248		3
		14,298[c]	255	32,600(LMCT)	10
$Re_2Br_8^{2-}$		13,597	255		
		13,694	255	23,630(LMCT)	10
		13,765	255		3
$Re_2I_8^{2-}$		13,000		14,800(LMCT)	12
$Re_2(NCS)_8^{2-}$		10,000			163
$Tc_2Cl_8^{2-}$	2.117	14,400			164
$W_2(CH_3)_8^{4-}$	2.26	16,900			22,61
$Mo_2(NCS)_8^{2-}$	2.162	14,500			165
$Mo_2(CH_3)_8^{4-}$	2.148	19,500(1500)			21
$Re_2(CH_3)_8^{4-}$	2.178	18,600(1500)			21
$W_2Cl_4(PMe_3)_2$	2.262	15,210			60
$Mo_2Cl_4(PMe_3)_2$	2.13	17,900		28,800sf?	60
$Mo_2Cl_4(PnBu_3)_4$		16,340	320		65
$Mo_2(mhp)_4$	2.065	19,690(0-0)	400	24,700(12,000)	157
		19,920(0-0 + 230)etc			157
$W_2(mhp)_4$	2.161	18,290(0-0)	320	19,800(18,500)	157
		etc			
$Mo_2(chp)_4$		19,310(0-0)	390	24,700(7800)	157
		etc			
d^5-d^4					
$Re_2Cl_4(PPr_3)_4^+$		6653	275	(13,072)δ*-σ*	27
				14,025 LMCT etc	
$Re_2Cl_5(PMePh_2)_3$		7400		13,700vw 15,900vw	
				(21,150) 26,450	28

cont

Table 7.1 cont

	r(M-M)	δ^*--σ^*	δ^*--π^*	π--σ^* (?)	Ref
d^5-d^5					
$Re_2Cl_4(P(n\text{-}Pr)_3)_4$	2.232	13,790	15,460	etc	27
$Re_2Cl_4(PEt_3)_4$		?	15,385	(20,435)	28

	r(M-M)	σ---σ^*	Other bands	Ref
d^5-d^5				
$\pi Cp_2Mo_2(CO)_6$	3.235	25,460(23,200)	19,530(2300)	117,122
$\pi Cp_2Mo_2(CO)_4C_2H_4$	2.984	27,700(7400)	18,760(850)	117
$\pi Cp_2W_2(CO)_6$	3.222	27,590(21,000)	20,410(2850)	117
$\pi Cp_2W_2(CO)_4C_2H_4$	2.987	28,370(6000)	19,050(900)	117
$\pi Cp_2Cr_2(CO)_6$	3.28	22,320(5000)	16,550(270) (27,000)	244
$\pi Cp_2Cr_2(CO)_4$	2.20	25,600(7100)	16,000(240) 34,350s	244
$\pi Me_5Cp_2Cr_2(CO)_4$	2.28	24,800(12,700)	16,550(300) 33,300s	244

		δ--δ^*	Other absorption	Ref
d^5-d^6				
$Ru_2(OAc)_4Cl$	2.287	9000	(19,000)(z)	
		10,400?	17,900(xy) 22,500(xy)	33
			21,000 (z)	67
$Ru_2(OAc)_4(D_2O)_2^+$	2.287	10,000(60)	23,500(700)	62

a) General distance for M-M bond not necessarily associated with any cation shown. b) Weak, but often important features may not be reported in this Table. See text and original reference. c) 5 K.

Gray and Cotton in a joint publication [10] present elegant group theoretical arguments to assign these weak transitions. In view of their pedagogical value they will be discussed in some depth.

A strongly parallel polarised progression (in 225 cm^{-1}) begins at 17,062 cm^{-1} in the chloro-complex. A less intense progression begins at 17,452 cm^{-1} possibly vibronically based on the former. A perpendicularly polarised progression of the same frequency, is built on a vibronic origin at 17,117 cm^{-1}. This 225 cm^{-1} vibration is also attributed to the a_{1g} Re-Re excited stretch, which may be compared with the values cited above. A maximum in the progression peaks at the third or fourth vibration compared with the first vibration in the case of the δ - δ^* progression. The lower frequency and change in intensity clearly indicates that the Re-Re bond suffers greater distortion in this excited state, than in the δ - δ^* excited state (see section 4.3). The mixed polarisation, presence of several vibronic origins and low intensity, suggest that this is an electric dipole forbidden, vibronically activated transition. A similar system is seen in the bromide at 17,475 cm^{-1}. This transition could in principle be π - π^*, δ - π^* or π - δ^*. The first is seen in $(nBu_4N)_2Re_2Cl_8$ at 39,215 cm^{-1} [6]. Since the π-level has substantial halogen character [6], the π - δ^* transition might be expected to have more halogen dependence than is shown by this transition. Hence the transition is attributed to δ - π^* (^1E_g---$^1A_{1g}$).

A second solely perpendicularly polarised band is seen in the chloride at 20,940 cm^{-1} with a progression in ca.400 cm^{-1} built thereon [10]. A significant halogen effect is seen, the corresponding band in the bromide occurring at 19,150 cm^{-1}. The progression frequency is consistent, not with the Re-Re stretch, but with a a_{1g} Re-Cl mode seen at 359 cm^{-1} in the ground state. These two facts suggest that this transition must involve a halogen dependent orbital . The assignment δ - x^2-y^2 is suggested. The x^2-y^2 orbitals will combine in the binuclear molecule to generate b_{1g} and b_{2u} linear combinations, and therefore $^1A_{2g}$ and $^1A_{1u}$ excited states after promotion of a δ electron. Transitions to both these states are forbidden. Transition to the former can be activated in both perpendicular and parallel polarisation; transition to the latter can be enabled in the perpendicular direction by e_g but not in the parallel direction due to the absence of any a_{2g} vibrational mode in the M_2X_8 D_{4h} unit. Thus this band is assigned as δ - x^2-y^2 ($^1A_{1u}$---$^1A_{1g}$).

A third band system begins at 23,645 cm^{-1}, and is perpendicularly polarised with a progression in 185 cm^{-1}. The intensity is low and parallel components cannot be ruled out. The 185 cm^{-1} frequency is attributable to the a_{1g} ReReCl bending mode which requires that the transition involve halogen in some fashion.Thus the transition is tentatively ascribed to π - δ^* (^1E_g---$^1A_{1g}$). The corresponding band in the bromide is obscured by a charge transfer absorption.

The fourth, and rather stronger, band system consists of two sharp parallel polarised peaks at 27,000 and 28,100 cm^{-1} in the chloride, and 21,290 and 22,080 cm^{-1} in the bromide. X* calculations predict [6] two closely spaced π(Cl)---δ^* transitions (^1E_u---$^1A_{1g}$) and these are attributed to the bands at 30,800 (doublet) and 23,800 cm^{-1} in the chloride and bromide respectively. Thus this weak fourth band system probably represents two spin forbidden charge transfer bands (a,b^3E_u---$^1A_{1g}$). The parallel polarisation requires that they be the A_{2u} spinor components of ^3E_u.

These data may be summarised as follows:

$[Re_2X_8]^{2-}$ X =	F	Cl	Br	I
$\delta - \delta^*$ (z)	17,900	14,183	13,597	13,000
$\delta - \pi^*$ (xy,z)		17,062	17,475	
$\delta - x^2-y^2$ (xy)		20,940	19,150	
$\pi - \delta^*$ (xy, z?)		23,645		
$\pi(X)---\delta^*$ (a^3E_u) (z)		27,000	21,290	
$\pi(X)---\delta^*$ (b^3E_u) (z)		28,100	22,080	
$\pi(X)---\delta^*$ (a^1E_u) (xy)		30,800	23,800	14,800
$\pi---\pi^*$ (z)		39,200	26,000	19,400

(7.1)

where data for the fluoro and iodo complexes are also included [12]. Resonance Raman excitation into the $\pi(Cl)---\delta^*$ ^1E_u gives resonance enhancement of the Re-X stretch as expected [12].

However, the best laid plans of mice and men, etc..in a more recent analysis, one of these authors suggests some reconsideration of the results shown in (7.1), with respect to the weak bands observed above the $\delta--\delta^*$ transition [151]. A more highly resolved spectrum obtained at 5 K reveals that a) the band near 17,000 cm^{-1} actually involves two envelopes, suggesting two electronic transitions are present, and b) the band near 20,000 cm^{-1} has a progression in 200 cm^{-1} (attributed to Re-Re), rather than 406 cm^{-1}, finally c) the band near 23,645 cm^{-1} could not be observed. In this latest paper [151] it is argued that both the $\pi--\delta^*$ and $\delta--\pi^*$ transitions occur near 17,000 cm^{-1}, and that because of the error in progression frequency, it is preferable to assign the 20,000 cm^{-1} region absorption to the triplet $\pi--\pi^*$ ($^3A_{2u}---^1A_{1g}$).

This latter discussion [151] also includes a relativistic molecular orbital calculation which does not accurately calculate the position of the singlet $\delta--\delta^*$ transition (which involves an electron 'hopping' from one rhenium atom to the other) but should accurately calculate the triplet $\delta--\delta^*$ transition which only involves a spin flip. This is calculated to lie near 3600 cm^{-1}.

In early publications [158,214], Cotton and co-workers reported extensive data for binuclear rhenium carboxylates, included in Table 7.1.

A very recent ab initio study suggests further consideration of some of the assignments [172].

The $[Cr_2(O_2CR)_4.L_2]$ Systems: Solomon and co-workers [13] have studied the rather famous red binuclear chromium(II) acetate. This complex has the usual eclipsed D_{4h} structure with water bound axially. However there are some distortions in the crystal such that the effective symmetry is closer to D_{2h}. Spectra resolved along the x, y and z axes are shown in Fig.7.2. The splitting of the x-y spectra is about 600 cm^{-1} with greater intensity along x,y than z for both bands I and II. Evidently they must both be transitions to E states (in D_{4h}, split by the lower D_{2h} symmetry), polarised perpendicular to the Cr-Cr axis. The splitting is the same in both bands suggesting that the same e molecular orbital is

involved in both transitions. Assignments are still uncertain with this species. However combination of these data with a recent SCF-X$_\alpha$ study [14], leads to the suggestion that band I is probably $\delta - \pi^*$ while band II is a charge transfer from oxygen p-π orbitals to π^* (i.e. the same acceptor e level on the metal in both transitions). The latter is unusually weak for a charge transfer transition because of the very small overlap between the orbitals concerned. Band A (Fig.7.2) may be the spin triplet $\delta - \delta^*$ transition, while band C could be the spin allowed analog. The study was expanded to other axially coordinated derivatives and it was shown that the energies of bands I and II vary approximately linearly with the Cr-Cr bond distance.

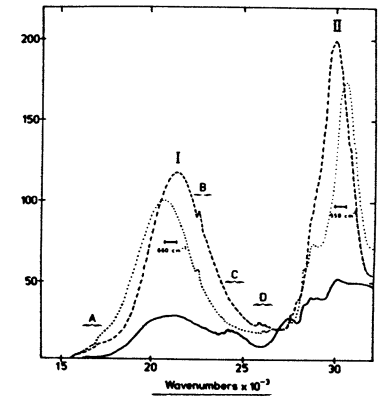


Fig.7.2 Polarised spectra of chromium(II) acetate, x-dotted, y-dashed and z-solid lines [13].

The $[Mo_2(O_2CR_2)_4]$ (R = H, CH_3, CF_3, $[CH_2NH_3]^+$, and Leucine derivatives) System: Complexes of this type [15-19] exhibit a structured peak near 21,500 - 22,500 cm^{-1} which is now generally agreed to originate from the $\delta - \delta^*$ transition (Table 7.1) though it was not always so assigned. The difficulty lay in the observation of a weak z polarised absorption (along the bond axis) plus two xy polarised progressions of only slightly lower intensity. The $\delta - \delta^*$ transition is expected to be z polarised, but if this band is so assigned, why are xy polarised bands of similar intensity seen ? It was eventually recognised [17] that the structured transition is indeed $\delta - \delta^*$ ($^1A_{2u}---^1A_{1g}$), but the transition moment is so low that a pair of vibronically induced xy polarised transitions (coupled via an e_g vibration) have comparable intensity.

The $\delta - \delta^*$ transition involves excitation of an electron from the d xy orbital on one atom to the xy orbital on the other. These orbitals are orthogonal to the M-M axis so the intensity of the transition is low despite being electronically allowed. There are

five e_g modes associated with the Mo-O-C framework but apparently only two are capable of being coupled to the $\delta - \delta^*$ transition.

The vibrational spacings are essentially identical in all three progressions (Table 7.1) and correspond with an Mo-Mo stretch. Photoelectron spectroscopy provides a means of directly 'visualizing' the energies of the δ, π and σ orbital in these systems [154,155].

The $[Mo_2(OAc)_4Cl_4]^{4-}.2L$ System (L = water, methanol) [160]: A binuclear five coordinate complex is formed by two trigonal bipyramidal Mo(II) atoms linked apically via carboxylate bridges, and with equatorial chloride groups. The presumed δ---δ^* transition is observed at 20,200(490) cm^{-1} with two progressions 79 cm^{-1} apart in the Mo-Mo frequency (354 cm^{-1}, L = water). A shoulder is observed at 29,500 cm^{-1}, before the strong uv absorption.

The $[Mo_2(SO_4)_4]^{4-}$ System: This species exhibits a weak (170 l mol^{-1} cm^{-1}) but prominent feature at 19,400 cm^{-1} assigned [20,162] to the $\delta - \delta^*$ transition.

The $K_4[Mo_2Cl_8].2H_2O$ System: This species was originally studied by Gray [3] and subsequently by Cotton and co-workers [59], Norman and Kolari [38] and Resonance Raman data reported by Clark and Franks [64,145]. A structured band with its origin at 18,083 cm^{-1} is assigned to the anticipated $\delta - \delta^*$ transition, with a progression in a_{1g} Mo-Mo of 336 cm^{-1} (ground state 345 cm^{-1}), and polarised along the Mo-Mo axis. Clark and Franks report absorption at 6250 cm^{-1} which they assign to the spin triplet $\delta - \delta^*$ transition. Though it has been argued [59] that this lies at much too low an energy relative to the spin allowed transition to be the triplet, we have earlier noted experimental and theoretical evidence for a large separation between $\delta\delta^*$ singlet and triplet states [65,66]. A weak peak at 28,800 cm^{-1} appears to be a static electric dipole allowed transition, dominant in z polarisation. Its assignment is uncertain [59]. Data for the $[Mo_2(NCS)_8]^{4-}$ ion are also included in Table 7.1 [165].

The $M_2Cl_4(PR_3)_4$ (M = W,Mo) Systems: These complexes have also been extensively discussed [60,61,65]. With R = n-butyl, M= Mo) the $\delta - \delta^*$ absorption begins (0-0) at 16,340 cm^{-1} and there is mirror image emission (from the $\delta\delta^*$ singlet) at 16,210 cm^{-1} (lifetime 16ns at room temperature in 2-methylpentane) [65]. The vibrational progressions in the a_{1g} Mo-Mo stretch in the absorption and emission are 320 (excited state) and 340 cm^{-1} (ground state) respectively. The authors suggest that the corresponding triplet may be much closer to the ground state and decay non-radiatively. This argument is supported by X calculation. Similar data are reported for $M_2Cl_4(PMe_3)_4$, M = Mo,W (Table 7.1).

The $[M_2(CH_3)_8]^{n-}$ Systems: A series of complexes $[M_2(CH_3)_8]^{4-}$, with M = Cr(II), Mo(II) and W(II), and $[Re_2(CH_3)_8]^{2-}$ have been studied. All exhibit a well defined absorption peak between 17,000 - 22,000 cm^{-1} assignable to $\delta - \delta^*$. Indeed there is a good correlation between the frequency of this transition and the M-M bond length [21,22].

The $[Mo_2((CH_2)_2P(CH_3)_2)_4]$ System: The $\delta - \delta^*$ transition in this complex appears at 19,268 cm^{-1} (0-0 origin) with an a_{1g} Mo-Mo progression built thereon (345 cm^{-1}) [23]. There are however four other origins to higher energy (labelled a,b,c,d respectively), also with concomitant progressions in the same frequency. The lowest energy 0-0 origin is about 3/4 as intense as the a-1 pseudo-origin but by the third or fourth member of the progression, this intensity ratio is reversed; there are in fact two sets of intensities displayed in these progressions. The data are explained by the theory outlined in section 4.3. The a, c and d origins all display similar intensity behaviour and are vibronically activated (presumably in xy) by an e_g vibration ($^1A_{2u} \times E_u \times {}^1A_{1g} \times e_g$ spans A_{1g}). The series labelled b has a similar intensity behaviour to the primary origin and is believed to be built therefore upon an a_{1g} vibration. From analysis of the data the excited state Mo-Mo distance appears to be 92pm greater than the ground state ($^1A_{1g}$) [23].

Deprotonated 6-Methyl-2-Hydroxypyridine (mhp) Systems: Manning and Trogler have reported some elegant data for a series of $M_2(mhp)_4$ systems with M = Cr, Mo and W. A fairly intense δ---δ^* transition is observed at 18,000 - 22,000 cm^{-1} (Table 7.1), showing the anticipated progression in the M-M excited stretch. At higher energies (Table 7.1) a more intense transition is assigned to the allowed δ---π^* transition. Emission data are also reported for these complexes. The $\delta\delta^*$ emission is not a mirror image of the absorption because of torsional distortion in this excited state [157].

Reference [157] also includes a useful survey of d^4-d^4 δ---δ^* data with especial comment on intensity. This transition is observed in different systems with a wide variation of intensity, from less than 200 to 18,000 l mol^{-1} cm^{-1}. Indeed in Table 7.1 these data are organised in the fashion of Manning and Trogler, classified according to their intensity. Apparently there is a general trtrend to increasing intensity as the energy of the δ---δ^* transition decreases. As is also evident from Table 7.1, this is not attributable to a simple variation in M-M bond length.

Consideration is given to two extreme situations, a valence bond model limit with overlap S = 0, and a molecular orbital description limiting with overlap S = 1. The δ and δ^* orbitals yield four states, namely δ^2 ($^1A_{1g}$), $\delta\delta^*$ ($^1A_{2u}$, $^3A_{2u}$) and δ^{*2} ($^1A_{1g}$). In the valence bond limit, $^3A_{2u}$ is of low energy and covalent, while $^1A_{2u}$ is ionic (because the two electrons reside on the same metal [166]) and of high energy. As overlap between the δ

orbitals on each metal increases, configurational interaction causes the relative energy of the $^1A_{2u}$ state to decrease and the state becomes more covalent. Configurational interaction has previously been shown, by theory [66,166], to be very important in considering transitions of this type. Since intensity is proportional to the square of overlap, increasing overlap should provide greater intensity to the δ---δ^* transition, which in this model will also decrease in energy as is observed (Table 7.1). Finally although some authors have noted a correlation between M-M bond length and $\delta\delta^*$ energy, this is likely only to be true for closely related systems where variation in bond length is directly related to variation in overlap.

7.1 iv) d^5-d^4 Species.

The $[Tc_2Cl_8]^{3-}$ System: This complex has the well known eclipsed D_{4h} structure. The configuration is believed to be $\sigma^2\pi^4\delta^2\delta^*$. The spectrum is shown in Fig.7.3 [24]. Based upon polarisation and SCF X α calculations [25] the feature whose absorption begins near 6000 cm^{-1} is assigned to the δ - δ^* transition ($^2B_{1u}$---$^2B_{2g}$). The structure is a progression in a 320 cm^{-1} attributed to the excited state Tc-Tc stretch. The evident complexity in the progression probably arises from Davydov splitting due to several molecules in the unit cell. The peak near 15,700 cm^{-1} may be the δ^*---π^* transition which is calculated to lie in this region [25], and which has no counterpart in the d^4-d^4 systems. Possible assignments for the higher energy bands are discussed [24].

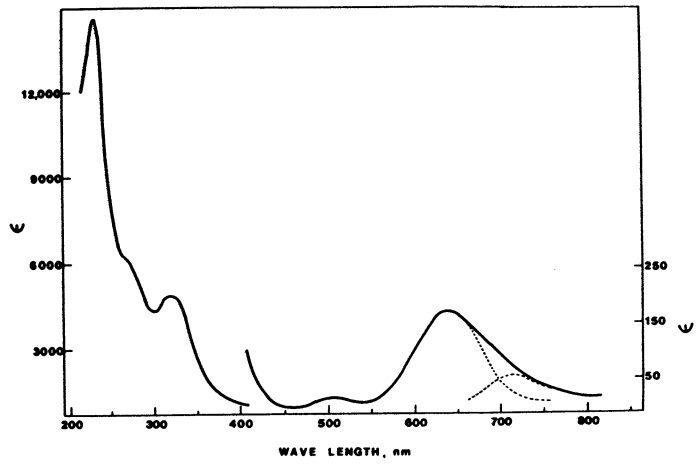


Fig.7.3 (a) The absorption spectrum of the $[Tc_2Cl_8]^{3-}$ ion [24].

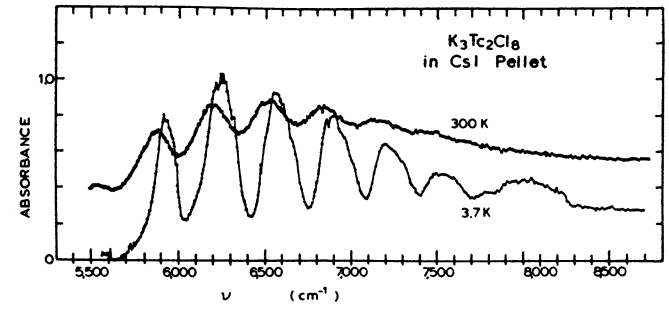


Fig.7.3 (b) The near infrared absorption band of $[Tc_2Cl_8]^{3-}$ ion recorded at room temperature and ca. 5 K [24].

The $[Tc_2(2\text{-oxopyridine})_4Cl]$ System: This complex is eclipsed with the chloridee bound along the Tc-Tc axis; the latter have a formal oxidation state of 2.5, and the configuration is ($\sigma^2\pi^4\delta^2\delta^*$, $^2B_{1u}$). The δ - δ^* transition at 12,194 cm^{-1} [26] shows extensive vibrational structure (both in z and xy) involving participation of 11 vibrations. The z polarised spectrum shows spacings of 264 and 298 cm^{-1} assigned to Tc-O and Tc-N (not necessarily respectively) and 339 cm^{-1} (Tc-Tc, reduced from ground state 383 cm^{-1}). Progressions based on combinations of 298 and 339 cm^{-1} are also seen. The xy polarised spectrum is presumably vibronically induced (see discussion of dimolybdenum(II) systems above). If D_{4h} symmetry were effective, then the transition moment $\langle^2B_{1g}|E_u|^2B_{1u}\rangle$ would span e_g. The transition is electronically forbidden in xy, but could be induced by an e_g vibration.

The $[Re_2Cl_4(PR_3)_4]^+$ Systems: Oxidation of $Re_2Cl_4(PR_3)_4$ yields cations with a hole in δ^*. These also exhibit a structured band near 6650 cm^{-1} attributed to δ - δ^*. The average Re-Re stretch (excited state $^2B_2$ in D_{2d}) is 275 cm^{-1}. A detailed assignment was deduced on the basis of a relativistic X -SW calculation [27]. See Table 7.1 for examples.

7.1 v) d^5-d^5 Species

The $Re_2Cl_4(PR_3)_4$ Systems: A ground state configuration $\sigma^2\pi^4\delta^2\delta^{*2}$ gave rise to rich spectra involving transitions from δ^{*2} to various anti-bonding levels [27]. Assignment was aided by relativistic X -SW calculations. Walton has also reported data for systems of this type [28-31] (Table 7.1). Note that these complexes are generally very air sensitive and if appropriate care is not taken yield near ir absorption which is, in fact, due to the oxidised d^5-d^4 species noted immediately above.

Table 7.2

SPECTROSCOPIC DATA FOR M-M SYSTEMS
(12-18 Electron systems)

Complex	π--σ*	σ---σ*	σ---π* ?	Ref
d^5-d^7				
$Cp(CO)_3MoFe(CO)_2Cp$	18,400(1340)	25,100(12,300)	30,600	238
$Cp(CO)_3WFe(CO)_2Cp$	18,700(1150)	25,800(12,100)	31,250	238
d^7-d^7				
$Mn_2(CO)_{10}$ [a]	(25,640)	29,240(21,400)	(33,000)	118,120
$Mn_2(CO)_9PPh_3$	(23,255)	27,780(24,200)		118,120
$Mn_2(CO)_8(PPh_3)_2$	(22,220)	26,810(23,700)		118
$Tc_2(CO)_{10}$		32,400(26,600)	35,500	119,120
$Re_2(CO)_{10}$		32,800(24,000)	36,000	118,120
$ReMn(CO)_{10}$		31,950(20,100)	36,000	118,120
$Mn_2(CO)_{10}$(phen)	17,420(11,800)CT	28,320(13,500) (26,310)		121
$Re_2(CO)_{10}$(phen)	18,940(7300)CT	(28,570)(5200)		121
$CpMo(CO)_3Mn(CO)_5$ [b]		26,810		237
$CpMo(CO)_3Re(CO)_5$		27,400		237
$CpW(CO)_3Mn(CO)_5$		27,620		237
$[MnCr(CO)_{10}]^-$	24,300(5300)	29,300(13,800)		120,123
$[MnMo(CO)_{10}]^-$	26,900	30,400		123
$[MnW(CO)_{10}]^-$	26,600	30,700		123
$[Cr_2(CO)_{10}]^{2-}$	(24,000)(5300)	27,200(13,500)	30,300CT	120
$[Mo_2(CO)_{10}]^{2-}$				
$[W_2(CO)_{10}]^{2-}$				
$Cp_2Ru_2(CO)_4$	23,000(1280)	30,300(13,900)[c] 37,750(>10,950)[d]		246
$Cp_2Fe_2(CO)_4$	19,450(710) 24,390(1870)	28,900(9190)[d]		246
$Cp(CO)_3MoCo(CO)_4$ [e]	19,100(510)	28,000(14,300)		238
$Cp(CO)_3WCo(CO)_4$	19,600(560)[e]	29,150(12,300)		238
$Pt_2(pop)_4I_2^{4-}$	22,990(15,930)	29,590(42,930)		36
$Pt_2(pop)_4Cl_2^{4-}$	28,980(8190)	35,460(48,840)		36
$Pt_2(pop)_4Br_2^{4-}$	29,410(11,780)	32,790(55,420)		36

Complex	r(M-M)	Absorption spectra	Ref
d^8-d^8			
$K_4Pt_2(pop)_4$(aq)	2.925	22,100(120) ($^3A_{2u}$) 27,200(33,500) ($^1A_{2u}$) 33,000(849) 37,000(1360)	55,56

cont

Complex	r(M-M)	Absorption spectra	Ref
$(n\text{-}BuN)_4Pd_2Cl_6$		16,300($^3A_{2g}$) 22,420($^1A_{2g}$) 25,160(^1E_g) 30,580(LMCT πCl_b) 35,210(LMCT πCl_t) 40,980(LMCT σCl_b) 50,000(LMCT σCl_t)	57
d^8-d^8			
$(n\text{-}BuN)_4Pd_2Br_6$		14,600($^3A_{2g}$) 16,950($^1A_{2g}$) 19,420(^1E_g) 24,630(LMCT πBr_b) 30,490(LMCT πBr_t) 35,330(LMCT σBr_b) 46,300(LMCT σBr_t)	57
$Rh_2(13dcp)_4$		14,925(z) $E_u(^3A_{2u})$ Rh-Rh 150 cm^{-1} 18,080(14,500) $^1A_{2u}$---$^1A_{1g}$	47
		31,450(31,500) ^1E_u---$^1A_{1g}$	
d^9-d^9			
$Co_2(CO)_8$		23,470(980)[c] πd---σ^* 26,460(2200)[d] πd---σ^* 28,570(5500)[c] δ---δ^* 35,460(14,400)[d] σ---δ^* (29,000) (32,100)	32
$Co_2(CO)_6(PPh_3)_2$		(22,200)(5300) πd---δ^* 25,450(23,800) δ---δ^*	32
$Co_2(CO)_6(P(OMe)_3)_2$		(20,800) πd---δ^* 27,780(24,000) δ---σ^*	32
$Co_2(NCC_6H_3Me_2)_8$		35,460 σ---σ^*	239
$Pt_2(dppm)_2Cl_2$	2.652	23,640 (200) 28,000(2600) πd--σ^* 31,250(7800) σ--σ^*	148
$Pd_2(dppm)_2Cl_2$		24,100(7800) 28,820(17,000) πd--σ^* 34,250(25,900) σ--σ^*	148
$Pd_2(dppm)_2Br_2$	2.699	23,470(9500) 27,780(16,200) πd--σ^* 33,100(21,000) σ--σ^* (38,900)	148
$Pt_2(dppm)_2(SnCl_3)_2$		(22,200)(15,700) Pt--Sn 22,830(16,800) 27,780(24,600) 33,000(26,700) σ--σ^* (36,230)(30,800) Pt--P?	148
$Pt_2(CNCH_3)_6^{2+}$		31,250(7600) πd--σ^* (41,700) 45,900(25,600) σ--σ^*	235
$Cu_2(OAc)_4.2H_2O$	2.64	14,600(230)(xy) 14,000(180)(z) d-d 27,770(210)(z) 30,770(30)(xy) LMCT	101
$Cu_2(vinylOAc)_4Py_2$		13,440 d-d (26,810) LMCT	218
$Cu_2(vinylOAc)_4MeOH_2$		14,080(92) d-d - 39,680(3430)	218
$Cu_2(vinylOAc)_4(Et_2O)_2$		14,730(328) (26,880) 38,760(9050)	218

a) Higher energy bands reported for these complexes in [120].

b) Also see [32]. c) Non-bridged isomer. d) Bridged isomer. e) 14-Electron system. X_t and X_b are terminal and bridging halogens, X = Br,Cl.

Chromium(I), molybdenum(I) and tungsten(I) form binuclear carbonyls such as $Cp_2Cr_2(CO)_4$ and $Cp_2Cr_2(CO)_6$ [122,244,245]. In the former there is believed to be a Cr=Cr triple bond, and in the latter a Cr-Cr single bond. The former 10 electron complex probably has the configuration $\pi^4\delta^2\delta^{*2}\sigma_b^2$ where the δ_b is the M-M bonding orbital. A prominent band at 25,000 cm^{-1} is assigned as σ_b--σ^* in $Cp_2Cr_2(CO)_4$ which may be compared with the corresponding transition in $Cp_2Cr_2(CO)_6$ observed at 22,300(5000) cm^{-1}. Although the σ_b---σ^* transition intensifies and sharpens upon cooling the sample, as it is expected to do, the effect is small leading to some uncertainty about the correctness of the assignment. Possibly the 25,000 cm^{-1} transition is due to $\pi(d)$---$\pi^*(d)$ and the σ_b---σ^* may then be associated with a shoulder at 33,300 cm^{-1} and hence a 10,000 cm^{-1} shift to higher energy when passing from a single to a triple bond (Table 7.1). A low energy transition appears like the $d\pi$---σ^* transition seen in many M-M systems. However calculation [245] suggests this transition should lie above σ---σ^* in these complexes. The energy of this band increases as the Cr-Cr bond length increases and the authors [244] suggest δ^*---x^2-y^2 as a possibility.

Complexes of the type $Cp_2M_2(CO)_4$ (M =Fe(I),Ru(I)) also show the same weak band - strong band pattern and may be assigned in a similar fashion (Table 7.2). Some of these complexes exist in isomeric forms with and without CO bridging, resulting in markedly different σ---σ^* transition energies [246]. This behaviour is similar to that described below for dicobalt octacarbonyl.

7.1 vi) d^5-d^6 Species.

The $[Ru_2(OAc)_4Cl]$ System: This complex is eclipsed with the chloride bound (bridging) along the Ru-Ru axis; the Ru atoms have a formal oxidation state of 2.5, and a spin quartet configuration (probably $\sigma^2\pi^4\delta^2\delta^{*1}\pi^{*2}$) [33-35,62]. Data are shown in Table 7.2, but the assignments are uncertain. The visible absorption is attributed to LMCT $\pi(Cl)$-π^* M, rather than δ - δ^* because it is too strong, and because an X calculation [34] places the $\delta\delta^*$ state below 10,000 cm^{-1}. A very weak band near 10,000 cm^{-1} may be δ - δ^*. Resonance Raman data [35,67] using excitation into the visible absorption confirm z polarisation and show enhancement of both the Ru-Ru and Ru-O stretches (and a progression therein), as expected for such an LMCT transition. Aqueous solution electronic and Resonance Raman spectra for a number of binuclear ruthenium complexes have been recorded [67].

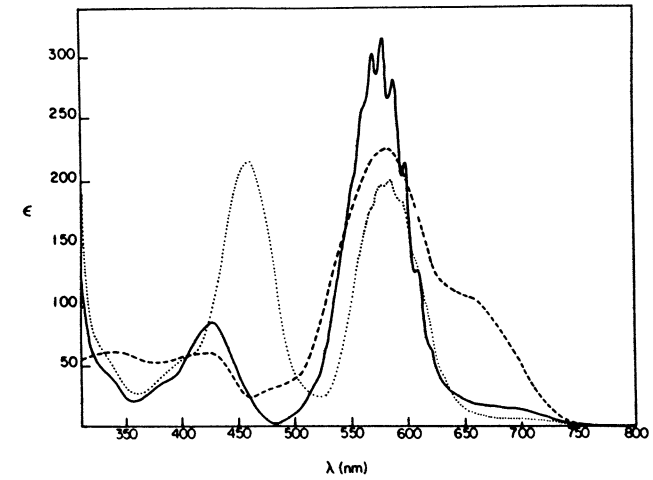


Fig.7.4 The polarised spectra of $Rh_2(OAc)_4$ [40].

7.1 vii) d^7-d^7 Systems

$K_4Pt_2(pop)_4X_2$: The ligand $(pop)^{2-}$ (pop = $(HO)OP\text{-}O\text{-}PO(OH)^{2-}$) forms a binuclear Pt(II) species bridged, in an eclipsed fashion, by four ligand molecules (see below 7.1 viii)). Oxidation with halogen yields binuclear Pt(III) complexes described here, the chloride having a Pt-Pt distance of 269pm. The six coordinate complexes have a ground state in which a high lying z^2 orbital on each metal overlaps to form a sigma bond. The corresponding anti-bonding level is empty ($.....(1a_{1g})^2(1a_{2u})^0$, $^1A_{1g}$). The spectra [36] of these species (Table 7.2) exhibit a pair of bands, the weaker lying at lower energy. The higher energy more intense transition is assigned as the $d\sigma$--$d\sigma^*$ transition ($1a_{2u}$---$^1A_{1g}$, $1a_{1g}$---$1a_{2u}$). It shifts in the sequence Cl>Br>I implying some mixing of halide character into the Pt-Pt bond. Note that the Pt-Pt stretching vibration also varies in the same sequence.

The $[Rh_2(O_2CR)_4L_2]$ System: Rhodium(II) also forms a series of carboxylate dimers which have been studied extensively both theoretically [34,38,153] and spectroscopically [39,40,62]. The molecule also possesses the eclipsed $Rh\text{-}O_4$ binuclear carboxylate molecule with axial bound ligands L. Early studies considered the ready substitution of ligand L, and the effect thereof on spectra [70-73]. Fig.7.4 shows the spectrum of $Rh_2(O_2CCH_3)_4.4H_2O$ (D_{4h}) with water along the axis. Unfortunately the crystal is not ideal for polarised light study since there is Davydov splitting due to

the presence of two molecules per primitive cell and a more useful effective symmetry might be D_{2h} rather than D_{4h}. Fig.7.4 shows the polarised spectrum of this species which exhibits two principle absorption bands in the 15,000 - 30,000 cm^{-1} region. Norman and Kolari [38] assigned the two bands (in solution at 17,100 (I) and 22,700 cm^{-1} (II)) to the Rh-Rh $e_g(\pi^*)$---Rh-Rh $a_{2u}(\sigma^*)$ and to Rh-Rh $e_g(\pi^*)$---Rh-O $b_{1u}(\sigma^*)$ respectively, on the basis of energy computation and the fact that band (I) is more blue shifted by axial donors than band (II). This assignment for band (I) (primarily xy polarised, and therefore consistent with ^1E_u---$^1A_{1g}$) is also accepted by Martin and co-workers [39] who observe a vibrational progression thereon of 297 cm^{-1} for the excited state Rh-Rh vibration, compared with a ground state (Raman) value of 320 cm^{-1}. Several possible transitions may contribute to the higher energy transition which does seem composite.

However Miskowski, Gray and co-workers [40] considered that a frequency of 320 cm^{-1} seemed too high for ground state Rh-Rh stretching when the Rh-Rh bond length is 238.6pm. If this Raman band is erroneously assigned then the structure seen on band (I) may not be Rh-Rh, in which case the assignment was suspect. Moreover a reduction of only 20 cm^{-1} in stretching frequency when passing from the ground state with a formal bond order of 1, to the excited state of formal bond order 0.5, seemed too small. These workers provide convincing evidence that the 'true' Rh-Rh stretch should lie below 200 cm^{-1} and that the Raman band at 320 cm^{-1} must be Rh-O(carboxylate). They then reassign band I to ^1E_u---$^1A_{1g}$ due to Rh-Rh $e_g(\pi^*)$--- Rh-O $b_{1u}(\sigma^*)$. In an elegant and more sophisticated single crystal study, they derive the molecular spectra along each cartesian coordinate. Additional features reported by these workers include a weaker z polarised absorption some 2000 cm^{-1} below band (I) which is assigned as the Rh-Rh δ^*---Rh-O b_{1u*} transition (note that the alternative Rh-Rh δ^*----Rh-Rh σ is forbidden (b_{1u}---a_{2u})). There is weak absorption near 13,000 - 14,000 cm^{-1} which also exhibits the same 320 cm^{-1} spacing and is assigned to the spin triplet components of the π^*,δ^*--- Rh-O σ^* transitions. A splitting of band (I) along x and along y by about 150 cm^{-1} must be due to the lower symmetry. Band (II) is considered to be a transition to Rh-O σ^* from a lower lying Rh-O e_u π level. Even higher energy transitions are discussed and the reader is referred to the paper [40] for further detail.

The 14 Electron Binuclear Carbonyls: Many binuclear carbonyls fall into this classification. The most important are $[M_2(CO)_{10}]$, M = Mn,Tc and Re, and their substituted analogs. These are non-bridging and may be regarded as being composed of two $M(CO)_5$ fragments linked via overlap with their z^2 orbitals each containing an electron, the other orbitals having little mutual interaction.

Thus the z^2-z^2 overlap generates a filled and strongly stabilised a_{1g} σ-bonding orbital which probably lies below the 'non-interacting' d orbitals (e_g,b_{2g}) and an empty a_{2u} σ^* orbital which lies above. Two low lying electric dipole allowed transitions would be (xz,yz, e_g)---a_{2u} ($d\pi$--σ^*) (relatively weak) and a_{1g}---a_{2u} (σ---σ^*) (very strong).

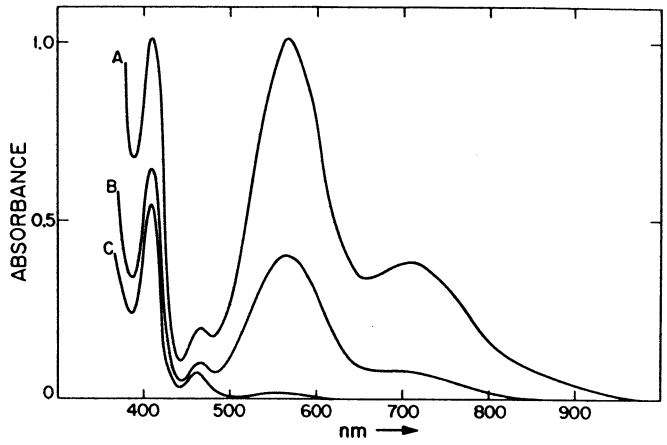
In these complexes a strong band, reasonably assigned to σ---σ^*, is generally observed near 30,000 cm^{-1} [118-123]. A weaker lower energy band, sometimes seen only as a shoulder is then associated with $d\pi$---σ^* (Table 7.2). These bands are polarised along and perpendicular to the M-M axis as required [119]. Indeed many homonuclear and heteronuclear carbonyl M-M' fragments show this type of spectrum including d^5-d^7 systems such as $Cp(CO)_3MoFe(CO)_2Cp$ and d^5-d^9 species such as $Cp(CO)_3MoCo(CO)_4$ (Table 7.2), together with d^9-d^9 species discussed below.

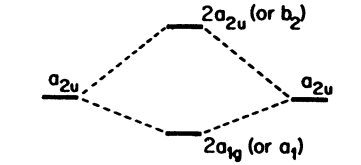
It is characteristic of these species that the σ---σ^* transition will generally sharpen and intensify when the solution is cooled. In addition, the σ---σ^* transition for mixed LM-M'L' systems is often close to the average for the LM-ML and L'M'-M'L' systems. One might conclude from this observation that the mixed heterobinuclear systems have little ionic character.

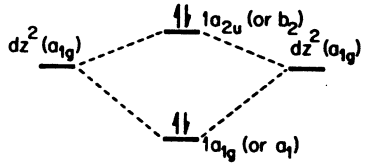
Note also that complexes such as $[M_2(CO)_8(phen)]$ not only show these transitions at the usual energy, but a low lying and medium intensity transition assigned as σ---π^* (phen) MLCT [121], and that substitution of CO by other ligands usually results in a red shift in both π---σ^* and πd---σ^* transitions, probably because of a weakening in the M-M bond.

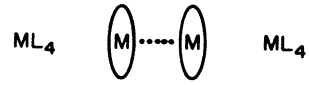
7.1 viii) d^8-d^8 Complexes

Rhodium(I)-Rhodium(I) Systems: In 1975 Gray and co-workers [41] reported that monomeric systems of the type $[Rh(CNPh)_4]^+$ oligomerize in solution to give dimers, trimers and higher oligomers. Fig.7.5 shows the concentration dependence of such a system. Bands at 27,700 (not shown), 24,330 and 21,370 cm^{-1} are attributed to the monomer while new bands at 17,600 and 13,750 cm^{-1} arise from the oligomers. The 17,600 cm^{-1} band grows in first, with increasing concentration. Its intensity is proportional to the square of the intensity of the 24,330 cm^{-1} band confirming that it is due to a dimeric species. Following Fig.7.1, this d^8-d^8 species will have molecular levels filled up to σ^*. In the monomeric species, the bands at 21,370 and 24,330 cm^{-1} are attributed to spin forbidden and spin allowed Rh---π^*(CNPh) charge transfer (MLCT) transitions into an a_{2u} orbital primarily on the ligand (but mixed with metal p_z) (see Chapter 5). As shown in Fig.7.5, in the dimeric unit, this a_{2u} orbital will combine in and out-of-phase to yield two new levels of a_{1g} and a_{2u} symmetry. The band at 17,600 cm^{-1} is then assigned to the electronically allowed $1a_{2u}$---$2a_{1g}$ transition (in Fig.7.5). A similar compound involves a bridged dicyanide (1,3-diisocyanopropane)which forms a binuclear complex $[Rh_2(bridge)_4]^{2+}$. This exhibited an analogous transition at 18,080(14,500) cm^{-1} [42-44] and was assigned similarly. Excitation into this MLCT transition yields emission of very short lifetime (<2ns) occurring at 15,240 cm^{-1} and attributed to fluorescence from









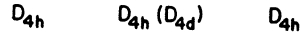


Fig.7.5 (a) (Upper) Electronic absorption spectra of $[Rh(CNPh)_4]PF_6$ in acetonitrile at 25°C. A, 5.7 x 10^{-2} M (0.06mm pathlength), B, 2.7 x 10^{-2} M (0.06mm) and C, 6.3 x 10^{-3} M (0.75mm) (b) (Lower) Relative energies in binuclear D_{4h} (or D_{4d}) $[Rh(CNPh)_4]_2^{2+}$ [41].

the excited MLCT state.

Excitation of these dimers also yields relatively long lived (microseconds) bleaching of the 'dimer' band and transient

absorption near 22,220 cm^{-1} attributed to the triplet MLCT state [45]. Indeed a single crystal spectrum of the $[Rh_2(bridge)_4](BPh_4)_2$ species showed absorption at 14,900 cm^{-1} due to transition to the $^3A_{2u}$ state. This latter state has two spin-orbit components, $E_u + A_{1u}$, and the 14,900 cm^{-1} absorption is attributed to $E_u(^3A_{2u})$---$^1A_{1g}$ which gains intensity from an ^1E_u state lying at 31,450 cm^{-1}. As shown in Fig.7.5, this transition occurs only along z, and has significant vibrational structure. The excitation involves removal of an electron from a σ^* Rh-Rh orbital and placement into the $1a_{1g}$ orbital which has bonding character; thus the Rh-Rh bond order should change, formally, from zero to at least 1/2 [44,46]. Since there is a change in Rh-Rh bond order upon excitation, it is reasonable to associate the structure on the absorption band with the Rh-Rh totally symmetric stretch in the excited $^3A_{2u}$ state [47]. The observation that the frequency of this vibration (150 cm^{-1}) is greater than the ground state Rh-Rh frequency (85 cm^{-1}) is attributed, of course, to the increase in bond order in the excited state. These assigmments were supported by resonance Raman studies [48]. Analysis of the intensities of the vibrational progression led to an Rh-Rh bond distance of 294 pm compared with 324 pm in the ground state.

The 'bridge' compound is of especial interest in view of the redox photochemical activity exhibited by the $^3A_{2u}$ state [42,43, 46,49,50]. The spectra of higher oligomers has been discussed [51]. Rh_m^{n+} chains involving d^8d^8, d^8d^7 and d^7d^7 systems exhibit σ--σ^* transitions (due formally to Rh(II)-Rh(II) interactions) whose energies decrease with increasing chain length as the transition is delocalised. This paper [51] presents an interesting study of the use of band intensities versus [Rh] as a means of elucidating the degree of oligomerisation.

The $[Pt_2X_6]^{2-}$ Systems: These planar molecules involve two square PtX_4 fragments linked through halide bridges [52-54]. Thus, unlike the $[Re_2X_8]^{2-}$ systems, both platinum atoms and all six halide ions are in the same plane in molecules belonging to D_{2h}. Assuming the Pt-Pt axis is z, and the molecule plane yz, then there will be two major interactions between the two platinum atoms. One will be through the yz orbitals overlapping through the halogen bridges, producing the two levels, $(1/2)^{1/2}(yz_1 + yz_2)$ b_{3g} and at higher energy $(1/2)^{1/2}(yz_1 - yz_2)$ b_{2u}. These empty orbitals will be contained by the PtX_4 framework and will be strongly sigma anti-bonding with respect to the terminal halogen atoms. Also contained by the molecular plane, but lying between the Pt-X bonds will be the y^2-z^2 orbital, which is expected to be the HOMO, by analogy with the $[PtX_4]^{2-}$ analogs. Overlap of this pair of orbitals yields a bonding (a_{1g}) and an anti-bonding (b_{1u}) combination (Fig.7.6 a)), both of which are filled in these 16 d electron species. Two transitions are expected (as shown in Fig.7.6), both being y polarised, in the molecular plane.

Note that the net Pt-Pt bond order is formally zero. It is therefore interesting to know if the electronic spectra provide any

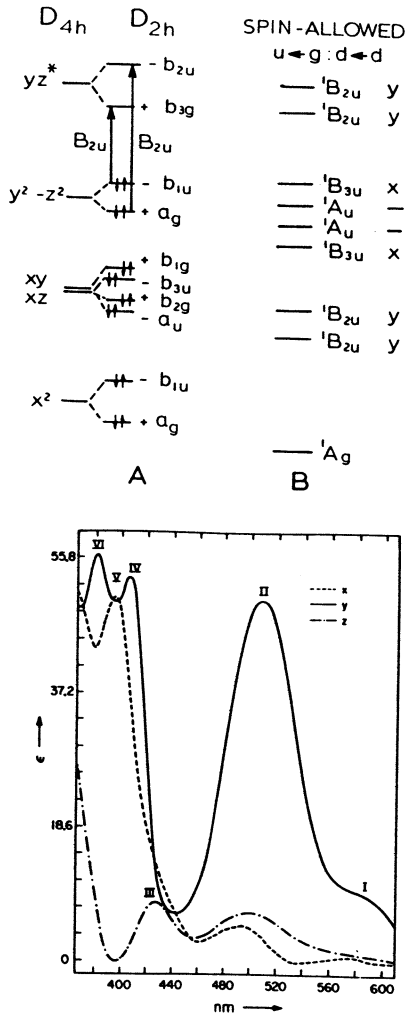


Fig.7.6 (a) (Upper) A, D_{2h} symmetry levels generated from two D_{4h} Pt atoms, and their electron occupation. B, Symmetries and polarisation of spin-allowed and dipole allowed electronic transitions in D_{2h} symmetry [54]. (b) (Lower) Calculated absorption profiles for the D_{2h} $[Pt_2Cl_6]^{2-}$ ion along the molecular axes. X coincides with Pt-Pt and z is out-of-plane [52].

evidence of the Pt-Pt interaction. If this is absent the splitting generated by the interaction will be absent and a single vibrationally activated transition will be seen. Moreover, in this circumstance the spectrum should be very similar to that of the corresponding $[PtX_4]^{2-}$ species. In detailed studies [53,54] of $[Pt_2Br_6]^{2-}$, only a single y^2-z^2--yz transition was observed, at an energy approximately identical to that of the corresponding transition ($^1A_{2g}$) in $[PtBr_4]^{2-}$. However the intensity was some 37 times greater in the former than in the latter. This may be taken as evidence for a Pt-Pt interaction injecting some Laporte allowed character into the transition. In addition the halogen to platinum LMCT transitions are shifted about 8000 cm^{-1} to lower energy in the binuclear complex, probably because the smaller number of halide ions shared by the two platinum atoms makes the platinum more positive.

It is pertinent to question which halogen would give rise to the lower LMCT transition, a bridging or a terminal halide ? Gray has assigned, without detailed argument, lower energy absorption to the bridging halide [57], while Martin assigns the lower energy absorption to the terminal halide [54]. It seems reasonable to this author that a bridging halide would carry more positive charge than a terminal halide, since the former is bonded to two Pt atoms; thus we prefer the terminal halide to Pt LMCT to occur at lower energy (see Chapter 5).

Gray and co-workers [52] studied the $[Pt_2Cl_6]^{2-}$ ion whose Pt-Pt distance of 341.8pm is significantly shorter than that (355pm) in the binuclear bromide. Thus a greater interaction is expected. Indeed a doubled transition (bands IV and VI), split 1650 cm^{-1} and polarised along y, was observed (Fig.7.6 b)) (both $^1B_{2u}$---$^1A_{1g}$), providing a direct indication of the Pt-Pt interaction. Bands I,II and III (Fig.7.6 b)) are spin forbidden transitions. Note that the second triplet spin-forbidden transition II has also become quite intense in the binuclear molecule probably as a consequence of overlap between the xz orbitals on each platinum (xz--xy transition - note that the coordinate axis in Fig.7.6 b) [52] is not the same as assumed in the previous study [54]). Band V is believed to correspond to the same transition as bands IV and VI (x^2-y^2---xy in this framework), but to be vibronically activated along x.

$K_4Pt_2(pop)_4.4H_2O$: The Pt(II)-Pt(II) distance is 292pm. At this distance only the z^2 orbitals interact to any significant extent, yielding sigma bonding and anti-bonding orbitals, both of which are filled ($...(1a_{1g})^2(1a_{2u})^2$). The lowest lying empty orbital will be the non-bonding $2a_{1g}$ level generated through the p_z-p_z interaction. Both the Crosby [55] and Gray [56] groups have studied this system which is characterised by two strong and two weaker bands (Table 7.2) and by emission involving both fluorescence and phosphorescence. The first intense band is assigned $^1A_{2u}$---$^1A_{1g}$ ($1a_{2u}$---$2a_{1g}$) and its weaker lower energy companion as the corresponding spin triplet ($^3A_{2u}$---$^1A_{1g}$) ($1a_{2u}$---$2a_{1g}$). Phosphor-

escence occurs out of $^3A_{2u}$ at 19,400 cm^{-1} (in water) and fluorescence out of $^1A_{2u}$ at 24,800 cm^{-1}. The low temperature phosphorescence spectrum reveals a progression in the Pt-Pt (ground state) stretch at 118 cm^{-1} [55] (Raman 115 cm^{-1}). The excitation spectra also reveal vibrational structure at low temperature. Most interestingly the excited state resonance Raman of the $^3A_{2u}$ state shows a progression in the Pt-Pt stretch for this excited state with a value of 156 cm^{-1} proving that the M-M bond is stronger in the $^3A_{2u}$ state than in the $^1A_{1g}$ ground state. This is anticipated since the excited state contains one less anti-bonding σ-electron than the ground state. The Crosby group also studied the temperature dependence of the phosphorescence lifetime and concluded that the $^3A_{2u}$ state is split by spin-orbit coupling into $A_{1u} < E_u$ components separated by 49.6 cm^{-1}. The higher energy bands may correspond to $1a_{1g}$---$2a_{2u}$ but have not yet been specifically assigned. See [257] for a very recent analysis.

The $[Pd_2X_6]^{2-}$ system [57]: These complexes, also linked in a planar binuclear array in parallel with their platinum analogs, exhibit d-d spectra similar to those expected for the monomeric $[PdX_4]^{2-}$ complexes. Two pairs of LMCT transitions are observed arising from the σ and π levels of the terminal and bridging halide into the empty metal x^2-y^2 σ^* orbital (Table 7.2).

7.1 ix) d^9-d^9 Species

The Copper Acetate system: Copper(II) also forms an eclipsed binuclear acetate $[Cu_2(OAc)_4L_2]$, in parallel to the analogous complexes of Cr(II), Mo(II) etc., and indeed was the prototypical system explored many years ago. There was great debate in the '50s and early '60s as to the nature of the interaction between the two copper atoms [218-232]. It is now generally agreed that the 'odd' electron in the x^2-y^2 orbital of each copper atom interact to form a weak delta bond. Coupling is incomplete with, 2J = ca 286 cm^{-1} and the complex is paramagnetic at room temperature. At 290 K the population of the spin singlet is 57δ, while at 77 K it is 99δ [101]. This is a weakly coupled system and the absorption spectrum (at least in moderate resolution) is expected to show spectra of the individual ions.

The optical spectrum consists of a broad, predominantly xy polarised band near 14,285 cm^{-1} (band I) and a second (z polarised [103]) band near 27,030 cm^{-1} (band II) [101-105]. In solution a third intense band (III) is observed near 38,000 cm^{-1} (Table 7.2). Some authors have reported a weak fourth band near 11,000 cm^{-1} [225,226]. It was rapidly concluded that band I is a simple d-d transition essentially localised independently on each ion. Band I is primarily xz,yz---x^2-y^2 with a low energy shoulder due to z^2---x^2-y^2 [101].

There has been much controversy concerning band II which seems at too high an energy to be a simple d-d band in Cu(II). Many binuclear Cu(II) complexes are known and many have absorption near

27,000 cm^{-1} (cf. Laccase, for example in Chapter 8). It was often suggested therefore that such absorption was 'characteristic of binuclear copper'.

Detailed analysis revealed [101] the presence of two bands in the 27,000 cm^{-1} region. In z polarisation there appears a band at 27,780 cm^{-1}, and in xy polarisation at 30,770 cm^{-1}. The intensity of the 27,780 band approximately doubles between room temperature and 77 K, and remains roughly constant down to 4 K. Evidently therefore this transition originates in the spin singlet. The 77 K intensities are 210 and 30 l mol^{-1} cm^{-1} respectively. The intensity of band I remains roughly constant on cooling, while the behaviour of the 30,700 cm^{-1} band is not clearcut.

Eliminating some of the earlier assignments, such as multiple electron transitions, the most logical assignment for the near uv band would be LMCT from oxygen p π-orbitals to the hole in x^2-y^2 [101,104,105]. The 'obvious' transition is from a p_x orbital in the xy CuO_4 plane to x^2-y^2.

Combination of the x^2-y^2 orbitals on each atom (CuO_4 local xy plane) in and out-of-phase yields $b_{1g} + b_{2u}$. Combination of the p_x π-orbitals of the four oxygen atoms in the CuO_4 plane generates a set of group orbitals spanning $a_{1u} + b_{1u} + e_g$. Excitation from the oxygen e_g group orbital into b_{2u} provides an LMCT transition with xy polarisation, and reasonable strength. The other similar transitions are orbitally forbidden. This is the proposed assignment for the 30,770 cm^{-1} band. Similarly the four p_y π-orbitals of the four oxygen atoms in CuO_4 span $a_{2u} + b_{2u} + e_g$. The transition $b_{2u}(p_y)$---b_{1g} (x^2-y^2) provides an allowed LMCT transition with z polarisation and is assigned to the 27,780 cm^{-1} absorption. Thus this analysis by Dubicki [101] appears to explain the essential features of the structure of band II. Dubicki argues that it is the low energy of this transition which makes it a feature of binuclear copper though clearly the subtle features of this band system also reflect the presence of a binuclear copper unit.

As discussed in some depth in Chapter 9, Cu(II) complexes display structural and electronic flexibility such that in a tetragonal complex, as the axial field increases, the equatorial field decreases, and vice versa. This effect is seen very clearly in a series of binuclear copper vinylacetate complexes which also possess the same structure as copper(II) acetate. These complexes are soluble in a wide range of solvents which undoubtedly bind axially to the binuclear framework. The position of band I in copper vinylacetate axial ligand complexes varies in the sequence:-

PPh_3(13,150) < 4-Methylpyridine(13,280) < pyridine(13,440) < pyridine N-oxide(13,830) = DMSO(13,850) < methanol(14,080) < ethanol(14,160) = pyrazine(14,180) < quinoline(14,310) < benzene(14,490) < acetone(14,600) < THF(14,710) = ether(14,730) < methylene dichloride(14,860) < dioxan(14,930) < aniline(15,230)

The order is approximately that of decreasing axial interaction

resulting in the stabilisation of the xz,yz and z^2 orbitals and hence a blue shift in the transition. There is probably a simultaneous destabilisation of x^2-y^2 as the in-plane Cu-O bond strengthens, further enhancing the blue shift.

Neither band II nor band III reveal much solvent dependence. This is consistent with the LMCT assignment for band II. It is probable that band III is also LMCT, probably $\sigma(O)---x^2-y^2$.

Dimeric Palladium(I) and Platinum(I) Systems [148,233-235]: Various series of complexes of the general type $M_2L_4X_2$, with L a disphosphine, and M_2L_6 with L an isonitrile, are known with M = Pd(I) and Pt(I) (Table 7.2). The complexes are square planar linked along one M-X-M axis via a strong metal-metal bond. In and out-of-phase coupling of the x^2-y^2 orbitals on each atom results in a very stable even σ-orbital (filled) and an unstable odd and empty σ^* level. All other d levels will be filled in these 18 electron systems.

In addition to weak d-d transitions, two lower energy and weaker transitions are assigned as $\pi^*--\sigma^*$, while the anticipated very strong $\sigma--\sigma^*$ occurs in the ultraviolet (Table 7.2).

Bis(diphenylphosphino)methane metal complexes also containing $[PtCl_3]^-$ as a ligand show additional CT absorption centered near 23,000 cm^{-1} (Table 7.2). RR excitation into this absorption band enhances internal diphosphine modes but the M-M stretch is enhanced only very weakly.

The complexes $[M_2(NCCH_3)_6]^{2+}$ show a moderately strong band near 30,000 cm^{-1} and a strong and structured band near 40,000 cm^{-1}. RR excitation into the lower energy band of the Pd complex, causes significant Pd-Pd stretching vibration enhancement; thus this band is assigned $\sigma--\sigma^*$ [235]. The higher energy bands probably involve MLCT.

The energies of these bands are remarkably solvent dependent [235]. In the sequence CH_3NO_2 < acetone < acetonitrile < DMSO, the low energy band shifts to the blue, while the high energy band shifts to the red. The primary effect of axial coordination will be to destabilise the z^2 orbital whose symmetry permits mixing into the stable σ-orbital. Thus the $\sigma---\sigma^*$ transition contains a contribution from z^2. The pronounced red shift of the higher energy absorption then leads to the inference that it probably contains significant $\sigma---\sigma^*$ character.

The 18 Electron Binuclear Carbonyls: Probably the best known member of this class is $Co_2(CO)_8$. In the solid state the bridged system contains both bridging carbonyl groups and a metal-metal interaction. However in solution, the complex is in equilibrium between carbonyl bridged and unbridged forms, both of D_{3d} symmetry [236]. When the solution is cooled, the equilibrium shifts to the bridged isomer [237]. Thus it is easy to distinguish in the electronic spectrum which feature arises from each isomer by comparing room and low temperature spectra. By analogy with other d^9-d^9 species and with the $Mn_2(CO)_{10}$ species discussed above, two

prominent features are expected, namely a fairly intense $\sigma---\sigma^*$ transition and at lower energy a weaker $\pi d---\sigma^*$ transition. Thus the room temperature spectrum of $Co_2(CO)_8$ shows a prominent feature at 28,570 cm^{-1} which is absent from the cold solution spectrum and is therefore assigned to $\sigma---\sigma^*$ for the unbridged isomer. The low temperature spectrum shows a prominent feature at 35,460 cm^{-1} which is assigned to the corresponding transition in the bridged isomer. Both spectra show lower energy shoulders assigned to the $\pi d---\sigma^*$ transitions (Table 7.2). Substituted cobalt carbonyls such as $Co_2(CO)_8(PPh_3)_2$ show similar transitions (Table 7.2) [32]. Mixed systems such as $Co(CO)_3(PPh_3)Mn(CO)_5$ show peaks at the expected average energies (Table 7.2), providing evidence for the veracity of the cobalt assignments.

Although we have not discussed the mixed systems in any depth, a contribution by Abrahamson and Wrighton [238] is interesting. These workers present data for many mixed M-M' systems containing CO and Cp groups (Table 7.2) and discuss their photochemistry.

7.2 Oxo and Nitrido Bridged Binuclear Complexes

There is a fairly extensive series of complexes of general formula $[X_5MOMX_5]^{n-}$ where the central oxide bridge may also be nitride. Frequently the MOM bridge is linear though this is not obligatory for all X. The linear bridged systems are believed to have a common molecular orbital description and all show a band whose behaviour is characteristic of the bridge.

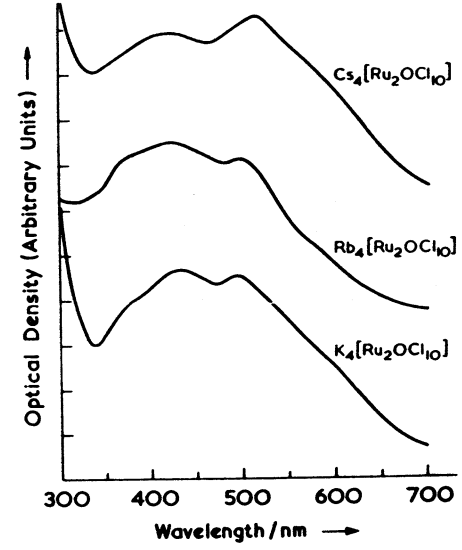


Fig.7.7 (a) Diffuse reflectance spectra of $[Ru_2OCl_{10}]^{4-}$ species [139].

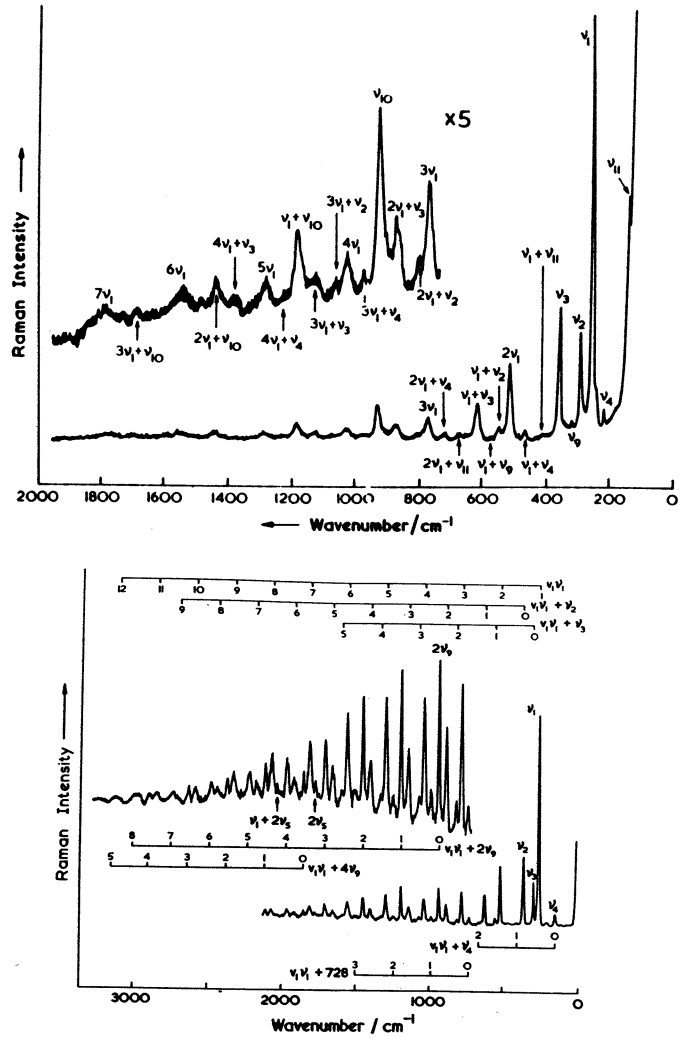


Fig.7.7 (b) (Upper) Resonance Raman spectrum of $K_4[Ru_2OCl_{10}]$ excited at 514.5 nm [139]. (c) (Lower) Resonance Raman spectrum of $K_4[Ru_2OCl_{10}]$ excited at 488.0nm [139].

Thus Dunitz and Orgel [142] considered the linking of two octahedral fragments (C_{4v}) along a linear M-O-M bridge, assumed z, to generate the D_{4h} binuclear molecule.

Table 7.3

SPECTROSCOPIC DATA FOR OXO AND NITRIDO BRIDGED BINUCLEAR COMPLEXES

Complex	Absorption spectra	Ref
$[Ru_2OCl_{10}]^{4-}$	14,000(60) 20,900(5200) 26,700(10,000) 39,400(22,200)	137,139,140
$[Ru_2OBr_{10}]^{4-}$	20,300(4900) 24,600(4300) 32,300(19,200) 35,800(30,000)	140
$[Ru_2NCl_8(H_2O)_2]^{3-}$	(24,800) 31,200	146
$[W_2OCl_{10}]^{4}$	(18,700)(>3800) (33,300)(4500) (37,300)(10,600) 41,000(17,200) (43,400)(16,000)	140
$[W_2OBr_{10}]^{4}$	17,800	137,144
$[Os_2OCl_{10}]^{4-}$	13,900(200) 17,900(140) 25,300(21,500) 30,100(12,600) 32,900(14,400) (36,100)	140
$[Re_2OCl_{10}]^{4-}$	15,400w (19,100) 22,900 (27,000)	137
$[Re_2OCl_{10}]^{3-}$	15,400w 17,000w 19,900s 23,200s 26,200m	138

The e_g---e_u (π--π^*) transition is underlined.

The x^2-y^2 and z^2 orbitals are strongly sigma anti-bonding and generally of high energy and empty. The xy orbitals combine in and out-of phase to generate b_{2g} + b_{2u}. A pair of xz,yz orbitals on each metal combine in and out of phase to generate e_u + e_g. Finally the p(π) orbitals on the oxygen (or nitride) bridge transform as e_u.

Thus the oxygen e_u orbitals can combine with the metal e_u orbitals to generate a bonding and anti-bonding pi interaction. The sequence of energy levels is then generally considered to be $e_u^b < [e_g, b_{2g}, b_{2u}] < e_u^*$, with the 'non-bonding' d orbitals of roughly equal energy lying between the bonding and anti-bonding e_u orbitals. In species such as $[Ru_2OCl_{10}]^{4-}$ the configuration will be $(e_u)^4(e_g)^4(b_{2g})^2(b_{2u})^2$ and two possible electric dipole allowed transitions may occur, namely, e_g---e_u ($^1A_{2u}$---$^1A_{1g}$, z polarised) and b_{2g}---e_u (^1E_u---$^1A_{1g}$, xy polarised). The first transition is of the type π--π^* and can be quite intense. The second will be extremely weak because of poor overlap between ground and excited states. In practice the spectra generally show a strong band with several weaker and lower lying shoulders. The strong band (shown underlined in Table 7.3) shows resonance Raman enhancement of a progression in the totally symmetric M-O-M bridge stretching vibration as would be expected [137-139,145,146]. Indeed the RR experiment is invaluable in identifing this e_g---e_u transition. Note that some species do not contain exactly 8 d electrons to fill the 'non-bonding' set. Thus, for example the $[Re_2OCl_{10}]^{4-}$ ion has the configuration $(e_u)^4(e_g)^4(b_{2g})^2$. Resonance Raman studies [137] show that for the complexes studied (Table 7.3), the transition is

z polarised and it is identified as the e_g---e_u transition, though the term description will vary. Note also that the RR experiment reveals odd enabling modes being enhanced. Thus there are vibronically active electronic transitions possibly contributing to the rather broad absorption spectra envelopes. The RR experiment is illustrated in Fig.7.7 showing the power not only of identifying an electronic transition but also of obtaining significant structural and electronic information through analysis of the vibrational structure [145].

The lighter elements such as Fe(III) and Cr(III) also readily form M-O-M systems in which high spin metal ions are exchange coupled through the oxygen bridge. Schugar, Gray and colleagues have discussed the Fe(III) systems in some depth [169-171]. Complexes such as $enH_2[(FeHEDTA)_2O].2H_2O$ show weak bands below about 25,000 cm^{-1} which are similar in position to those of mononuclear high spin Fe(III) complexes and are assigned to spin forbidden transitions to spin quartet states (see Chapter 6). They are, however about 10 times more intense in the binuclear complexes, as a consequence of exchange coupling. Above 25,000 cm^{-1} a series of fairly narrow and intense bands are seen (intensities ca 10^3 l mol^{-1} cm^{-1}) (Fig.7.8).

The exchange coupled ground state has a spin manifold covering S = 0,1,2,3,4 and 5. The excited state has a spin manifold comprising S^* = 0,1,2 and 3. A transition between two states where $S = S^*$ is spin allowed [169,171]. The more intense uv region bands are then assigned to simultaneous pair electronic transitions (SPE) where there is excitation of two transitions, one on each Fe(III) site, by a single photon. Thus the energy corresponds to the sum of the two individual excitations, e.g.:

$Na_4[(FeEDTA)_2O].12H_2O$ (Ground state $^6A_1$)

11,300(3.1)	$^4T_1$ (^4G)		
18,400(50)	$^4T_2$ (^4G)	29,100	$^4T_1 + {}^4T_2$
21,000(35)	$^4A_1$,^4E (^4G)	32,400	$^4T_1 + {}^4A_1$,^4E
(21,600)(2)			
24,700(100)	$^4T_2$(^4D)		
		36,800	$^4T_2 + {}^4T_2$
		etc	

These data are of relevance to biologically important Fe-O-Fe systems such as hemerythrin (see Chapter 8).

Similar Cr(III) systems have been studied. Where the coupling is weak, the exchange interaction may permit the partial population of several spin states. Thus with binuclear S = 3/2 Cr(III), there exist the total spin states S= 0,1,2, and 3. With small J all of these states may have some population depending upon the Boltzman temperature. Both the ground and excited electronic states will possess a manifold of these spin states and transitions between them will be governed by the spin selection rules. Using the sharp quartet-doublet transitions of Cr(III) it is possible to 'see'

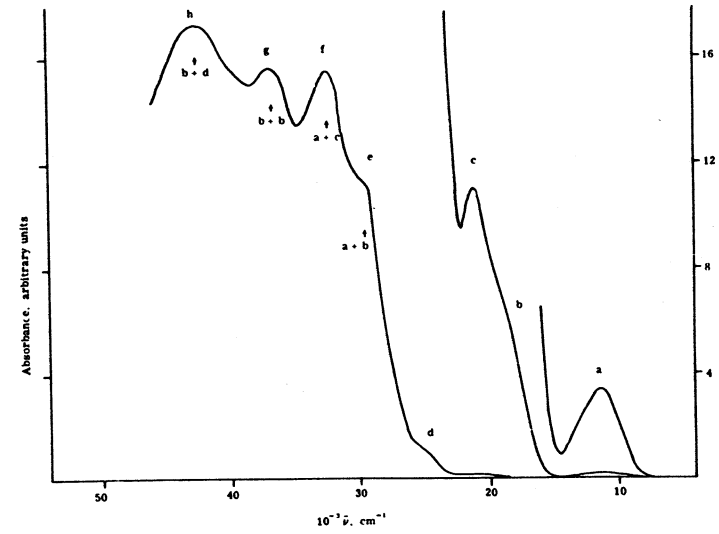


Fig.7.8 Absorption spectrum of 0.20 F aqueous solution of $enH_2[(HEDTAFe)_2O].6H_2O$ at 296 K. Calculated energies of SPE transitions are indicated by arrows [171].

individual transitions from these various spin states. Gudel and co-workers have been especially successful in this pursuit [201-204,213].

A series of hydroxy bridged species, such as binuclear $[(NH_3)_5CrOHCr(NH_3)_5]^{5+}$ have been studied [201-204,213]. Use of polarised light permits the observation of individual components of quartet-doublet transitions involving different spin states. By varying the temperature, generally 6 K - 150 K, the transitions arising from individual spin states vary their intensity in proportion to their Boltzmann populations [202,203]. These complexes also show SPE transitions.

The complexes $[(NH_4)_4Cr(OH)_2Cr(NH_3)_4]^{4+}$ and its en analog are of interest [200]. There is weak exchange coupling across the double hydroxy bridge. Single crystal cryogenic data showed the usual spin quartet transitions of six coordinate Cr(III), but polarisation data revealed that these are split into three components due to the lower symmetry (effective C_{2v}). For example in the en complex, three components of $^4T_{2g}$ are split by 900 and 1700 cm^{-1}. The intensity is much stronger along x than along y and z. Using the Fenske argument (see section 4.2 viii)) it was shown that (using O_h labels for simplicity) the t_{1u}---t_{2g} LMCT states could not contribute intensity to the t_{2g}---e_g d-d excitation because the configuration $t_{1u}{}^5t_{2g}{}^4$ cannot couple to $t_{1u}{}^6t_{2g}{}^2e_g$ via a one electron operator. Rather these spin allowed d-d transitions

are coupled to the t_{1u}---e_g LMCT state. Only one of these is permitted and it is polarised along x.

The exchange nature of the system is evident from the usual appearance of SPE excitations involving the spin quartet-doublet transitions near 29,000 and 35,000 cm^{-1}. It is well known that weak exchange interactions have little effect upon the spin allowed transitions.

In a later paper Decurtins and Gudel [213] have thoroughly analysed the enormous fine structure associated with the transition $^2E_{2g}\,^4A_{2g}$---$^4A_{2g}\,^4A_{2g}$ on these bridged dihydroxides. Although this is only a single electronic transition (not an SPE), there is considerable fine structure because of the pairwise interaction present in both the ground and excited states.

The ground state is coupled through the exchange parameter J(a,b) which is a composite of the several $J(a_i,b_j)$ parameters where i,j are the various t_{2g} orbitals on each Cr(III) atom. Ground state data cannot delineate individual values for these parameters, but excited state information can.

Thus interaction of the ground state quartets via antiferromagnetic exchange results in S = 0 ($^1A_{1g}$), S = 1 ($^3B_{1u}$), S = 2 (^5A_g) and S = 3 ($^7B_{1u}$) where in parentheses are indicated the symmetries of these spin states in D_{2h} symmetry (Fig.7.9).

The pair excited state $^2E_g\,^4A_{2g}$ also couples to yield a set of spin states, eight in all (doublet x quartet), which span, in D_{2h}, $^3B_{2u}$, $^3B_{3g}$, $^3B_{1u}$, ^3A_g, $^5B_{2u}$, $^5B_{3g}$, $^5B_{1u}$ and ^5A_g. The relative energies of these states depend upon the excited state exchange integrals $J(a_i,b_j)$, (i,j 1-3 label the t_{2g} orbitals) i.e. are determined by the pairwise interaction of the t_{2g} orbitals of the pair of Cr(III) atoms, in the excited state.

Thus the fine structure observed in the spectra arises from the possible transitions between the various ground state spin levels and these excited state spin levels.

Two primary mechanisms exist for supplying intensity to spin forbidden transitions in an exchange coupled ion. The single ion mechanism is based on the odd parity crystal field in conjunction with spin-orbit coupling. The pair mechanism has its origin in the exchange coupling [213,214].

Utilising the single ion mechanism for the Cr(III)-Cr(III) pair it can be shown that the excitations may follow both S = 0 and S = 1 spin selection rules, depending upon the symmetries of the states involved. Gudel includes a detailed selection rule Table (Table VI in [213]) delineating all possible transitions between the ground and excited state spin levels of the $^2E_g\,^4A_{2g}$---$^4A_{2g}\,^4A_{2g}$ transition, together with their polarisation.

The exchange governed pair mechanism follows S = 0, and few inter-spin transitions are allowed, in fact only four, namely the transitions to $^3B_{3g}$ (y) and ^3A_g (z) from $^3B_{1u}$ (ground state S = 1), and $^5B_{2u}$ (y) and $^5B_{1u}$ (z) from ^5A_g (ground state S = 2). Thus no pair mechanism transitions are allowed from the ground state S = 0 or S = 3 levels.

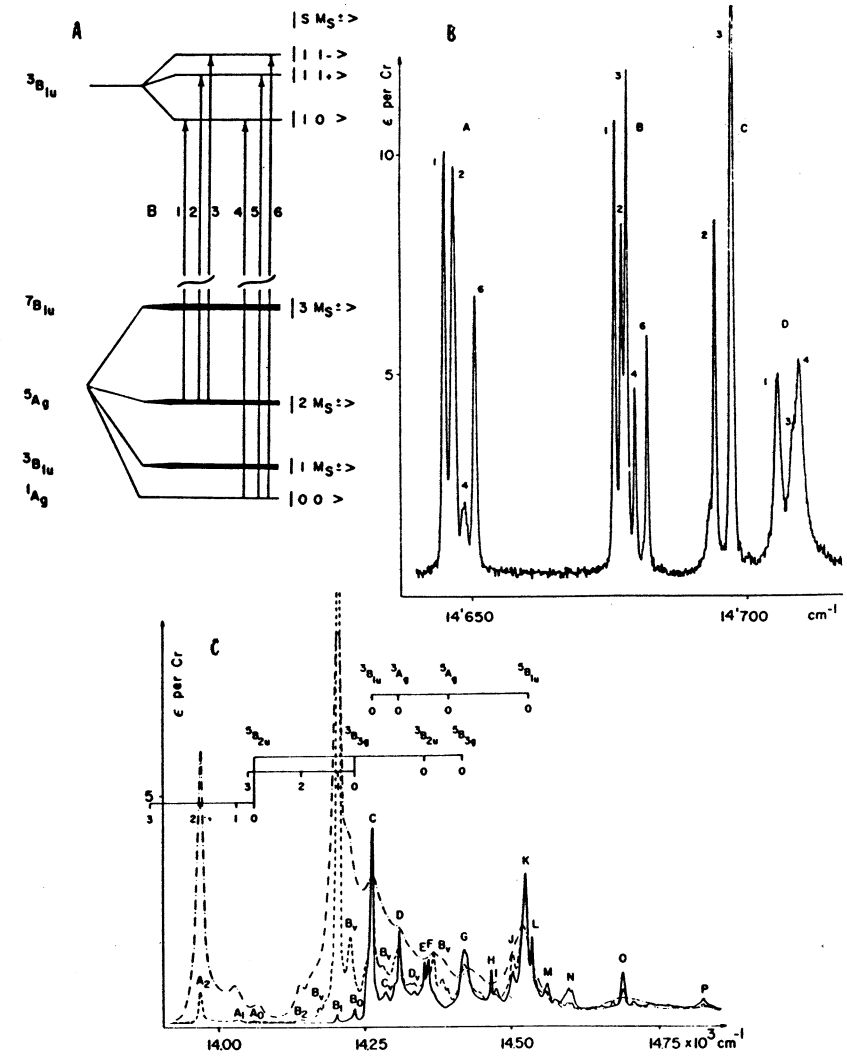


Fig.7.9 (a) (Upper left) Schematic representation of the ground and excited state splitting leading to the fine structure in band B. (b) (Upper right) High resolution absorption spectrum of the four lower energy pair excitations A,B,C and D in binuclear $[(NH_3)_4Cr(OH)_2Cr(NH_3)_4]Cl_4$. (c) (Lower) Y-polarised absorption spectra of $[(en)_2Cr(OH)_2Cr(en)_4]Br_4$ at 6 K (solid line), 19 K(---) and 43 K (-.-.) [213].

Transitions following the single ion mechanism should be most intense in x polarisation because, as indicated above, they derive ultimately from coupling with an x polarised CT state.

Near 14,000 - 15,000 cm^{-1} there are found the transitions $^2T_{1g}$, ^2E_g---$^4A_{2g}$ which overlap in energy. However to lower energy there are some bands arising solely from ^2E_g and these are analysed in detail in this publication [213] (Fig.7.9).

Considering the en complex, at 6 K the spectrum is most intense along x and is almost entirely of the single ion type; line widths are of the order of 1 cm^{-1}. At this temperature only the S = 0 component of the ground state is populated and no pair exchange mechanism transitions can occur. At higher temperatures two y polarised narrow line absorptions (B_1,A_2 in Fig.7.9) originating in the S = 1 (labelled B_1) and S = 2 (labelled A_2) ground levels grow and eventually dominate the spectrum (Fig.7.9). These transitions exist weakly in x and z and are clearly associated with the two allowed pair mechanism transitions $^3B_{3g}$---$^3B_{1u}$ (B_1) and $^5B_{2u}$---^5A_g (A_2).

Peaks A,B,C,D,E,F,H and L (Fig.7.9) are considerably narrower than peaks G,I,J,K,M.. Since transitions to $^2T_{1g}$ are usually broader than transitions to ^2E_g, the fomer set is associated with the ^2E_g $^4A_{2g}$ transition and the latter to $^2T_{1g}$ $^4A_{2g}$. The extensive overlap prevents a complete analysis of the spectrum at this time. However band C is of interest. It is relatively very strong in the 6 K spectrum and must therefore arise from ground state S = 0, ^1A_g. Using the selection rules (Table VI, ref.[213]) only two possible transitions exist for C, namely $^3B_{1u}$---^1A_g or $^3B_{2u}$---^1A_g. On the basis of constraints presumed to determine the relative magnitudes of the J parameters, and hence the energies of these states, C is assigned to the former transition. This allows calculation of the excited state J parameters which are found to be:

$[(en)_2Cr(OH)_2Cr(en)_2]Br_4$ Ground state $J(a,b)_g$ = -15 cm^{-1}.

J(yz,zx) = -145; J(yz,yz) = 33; J(xy,xy) = -5; J(yz,xy) = 22 cm^{-1}

$[(NH_3)_4Cr(OH)_2Cr(NH_3)_4]Cl_4$ Ground state $J(a,b)_g$ = -0.63 cm^{-1}

J(yz,zx) = -65; J(yz,yz) = 25; J(xy,xy) = -8; J(yz,xy) = 20 cm^{-1}

$[(NH_3)_4Cr(OH)_2Cr(NH_3)_4]Br_4$ Ground state $J(a,b)_g$= -0.3 cm^{-1}

J(yz,zx) = -55; J(yz,yz) = 17; J(xy,xy) = -8; J(yz,xy) = 20 cm^{-1}

Of especial interest is the magnitude of J(yz,zx). This parameter must reflect super-exchange through the OH bridge, via overlap of a pair of orthogonal p orbitals on the OH^- bridge with the 90° oriented orbitals on the two metal atoms. It is greater than the direct exchange associated with, for example, J(yz,yz). The excited state $J(a,b)_e$ is almost identical to the ground state value, but this is one of the constraints of the analysis. However

the authors note that the importance of super-exchange follows directly from the fact that $^5B_{2u}$ and $^3B_{3g}$ are the two lowest components of the excited state, and is not determined only by the detailed analysis. The additional fine structure arises from transitions between ground state S levels and excited states M_s levels split by spin-orbit and zero field effects (Fig.7.9). The latter arise through single ion anisotropy; spin Hamiltonian values of D and E are reported. This publication is an elegant example of the enormous amount of information which can be derived from a highly resolved optical spectrum.

7.3 Mixed Valence Complexes

There is considerable interest in complexes which contain two identical metals in different oxidation states. Many such systems are known and they are often intensely coloured. They range from minerals with mixed valences, through species such as the intense blue Prussian Blue [$KFe(III)Fe(II)(CN)_6$], mixed Cu(I),Cu(II) chlorocuprates, the Platinum Blues [130] and unusual valence systems such as 'Au(II)', 'Sb(IV)', 'As(IV)', 'Pb(III)' etc which contain 'mixed' atoms in their more conventional valences (e.g. 'Sb(IV)' = Sb(III) + Sb(V)).

The 'Creutz-Taube' ion, $[(NH_3)_5Ru\text{-pyz-}Ru(NH_3)_5]^{5+}$ [81] (pyz = bridging pyrazine) was one of the earlier and more important examples. There is clearly an ambiguity for complexes of this kind. Should, for example, this species be regarded to contain a Ru(III) and an Ru(II) ion localised in each site, or, at the other extreme, does it contain two identical (delocalised) Ru(2.5) ions ?. Expressed in a different fashion, should the odd electron remain localised on one atom of ruthenium, or does it hop from one atom to the other (and how fast ?). It is not our intent to discuss such systems in great detail since the area has been extensively reviewed [82-87] and many techniques can be utilised to attempt to answer such questions (often different techniques giving different answers). Instead the utility of electronic spectroscopy is emphasized.

The theory is developed below following [82-87]. Consider a binuclear molecule AB in which A_p and B_q represent two metals in two different oxidation states, p and q. Further, for simplicity, assume q-p=1. A simple ground state wavefunction may be written:

$$\psi_a = \psi(A_p)\psi(B_q) \qquad (7.3)$$

Electron transfer from A to B generates an excited state wavefunction:

$$\psi_b = \psi(A_q)\psi(B_p) \qquad (7.4)$$

If these two functions possess the same symmetry, they can be mixed by an off diagonal matrix element $\langle\psi_a|V_{ab}|\psi_b\rangle$, say V_{ab}. This has the effect of mixing some excited state into the ground state to

generate a new ground state whose wavefunction may be written:

$$\psi_g' = c_a\psi_a + c_b\psi_b \qquad (7.5)$$

Using conventional molecular orbital theory and assuming that overlap between ψ_a and ψ_b is negligible, i.e. $\langle\psi_a|\psi_b\rangle = 0$, the energy of the ground state becomes:

$$W_g = 0.5[E_{ab} - (E_{ab}^2 + 4V_{ab}^2)^{1/2}] \qquad (7.6)$$

with respect to the energy of the unmixed function ψ_a. The quantity E_{ab} is the energy difference between the unmixed functions ψ_a and ψ_b.

Zero overlap is assumed to make the mathematics simple. In reality if any interaction occurs between functions ψ_a and ψ_b, there must be non-zero overlap. Thus this theory is relevant for a small degree of mixing between these functions. Where mixing is very extensive, overlap should be considered.

The coefficients in (7.5) are then given by:

$$c_b/c_a = W_g/V_{ab} = 0.5[R - (R^2 + 4)^{1/2}]$$

where $c_a^2 + c_b^2 = 1$ and $R = E_{ab}/V_{ab}$ (7.7)

The ratio (c_b^2/c_a^2) is zero (metals completely localised) if V_{ab} is zero, and non-zero otherwise. Thus the value of V_{ab} determines the degree of mixing between ψ_a and ψ_b. If E_{ab} is zero, then, from (7.6), $W_g = -V_{ab}$, $R = 0$, and $c_b/c_a = 1$. Under these circumstances the ground state is a 50:50 mixture of the two states ψ_a and ψ_b. Complete delocalisation is occurring and the metals should be regarded as having an average oxidation state (p+q)/2. Clearly these represent two extreme situations.

The excited state too must be mixed with the ground state if V_{ab} is not zero. The energy of the excited state, relative to the unmixed ground state and the coefficients in (7.5) are given by:

$$W_e = 0.5[E_{ab} + (E_{ab}^2 + 4V_{ab}^2)^{1/2}]$$

$$c_b/c_a = W_e/V_{ab} = 0.5[R + (R^2 + 4)^{1/2}] \qquad (7.8)$$

Clearly there can be an electronic transition between these two mixed states. Such a transition is termed an 'intervalence' IT transition, and represents a (Franck-Condon) hopping of an electron (or possibly more than one where p-q >1) between the two metal centres. These new transitions (in general there may be more than one) will obviously not be seen in the spectra of isolated A and B, and are therefore fairly diagnostic of a mixed valence system. In this simple model the energy of an IT transition is given by $E_{IT} = W_e - W_g$; it may occur anywhere from very low energies in the near infrared to very high energies in the ultraviolet.

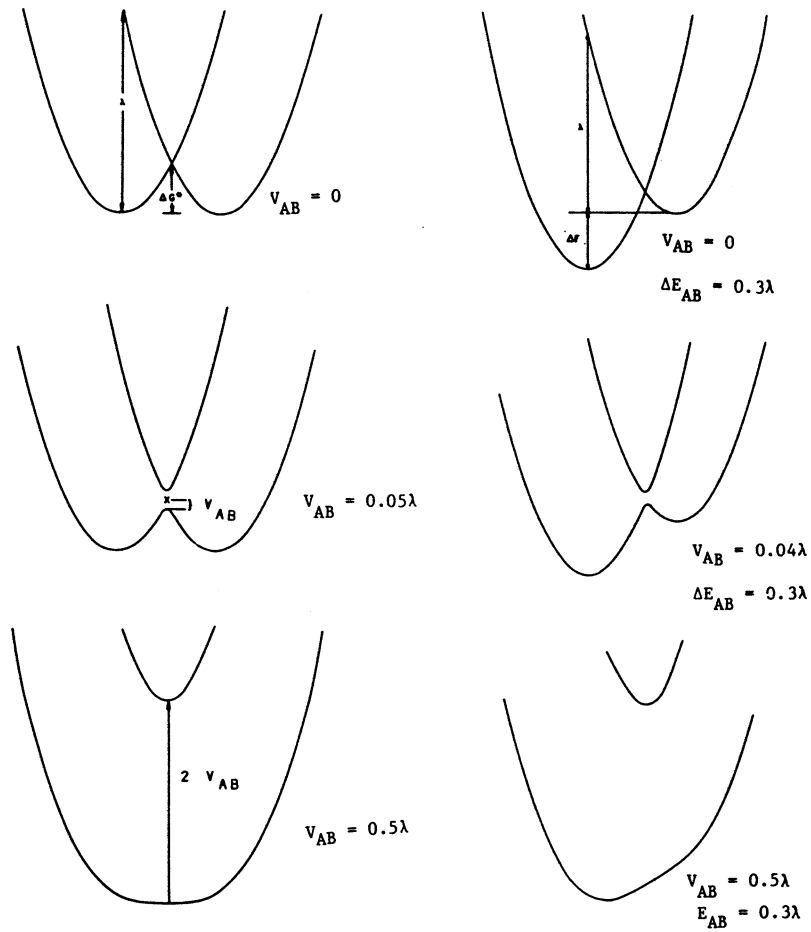


Fig.7.10 Potential energy versus nuclear configuration diagrams for (left) a symmetric mixed valence complex, and (right) an asymmetric mixed valence complex. (adapted from [87]).

In a symmetric mixed valence complex, the transition may be thought of as a transition from, say LM(II)-LL-M(III)L to LM(III)-LL-M(II)L. The free energy for net electron transfer is zero since clearly the initial and final products are the same. However the IT transition is a Franck-Condon transition which will not have zero energy. Thus the M(II)-L distance is different from the M(III)-L distance, and the solvation spheres around the two ions will be different. During the electronic transition there is no time for nuclear motion. Thus after electron transfer, the M(II)

ion finds itself in an M(III) environment (hence higher energy) and the M(III) ion finds itself in an M(II) environment, also of higher energy. There is therefore an activation energy for the optical electron transfer and hence a non-zero value for the IT transition.

In effect the barrier to electron transfer is vibratonal in nature. Vibrational trapping is associated both with intramolecular vibrations and with intermolecular vibrations involving the solvent interaction which is different in ground and excited state.

Robin and Day [85] proposed a classification scheme which presents a useful framework for handling these complexes. Totally localised systems were called Class I, totally delocalised Class III, while intermediate systems were in Class II. A sub-classification of Class III designates clusters as Class III-a and infinitely delocalised solids as Class III-b. These can be summarised as follows [85]:

Class I	Class II	Class IIIa	Class IIIb
A and B very different, ions in different ligand fields etc.	A and B in similar but not totally equivalent surroundings.	A and B not distinguishable in pairs or clusters.	Infinitely delocalised solids, all A and B equivalent.
Valences Trapped	Valences distinguishable but non-zero delocalisation.	Valences delocalised in molecular units.	Valences delocalised totally throughout solid
IT transitions at high energy Rarely observed.	IT transitions observable often in visible or near ir region.	Transitions observable often in visible or near ir region.	Absorption edge. Metal reflectivity
Optical spectra those of constituent ions.	Optical spectra those of constituent ions, with possible minor modification.	Optical spectra of constituent ions not distinguishable	Optical spectra of constituent ions not distinguishable.

(7.9)

where the transitions observed with Class IIIa species is characteristic of the mixed valence system but is not strictly inter-valence in that both (all) metals involved have the same valency (which may be fractional).

A number of useful relationships relating the IT transition with fundamental properties of the system follow. Proof for, and more detailed discussions of these relationships can be obtained from [83,85-89,97,125-129]. In general they are only true for essentially localised systems, i.e. where α^2 is very small. In fact

for many class II complexes this is the usual situation.

Fig.7.10 (left, top to bottom) represent the three extremes of Class I, Class II and Class III behaviour respectively for a symmetrical binuclear species. In Class II behaviour the barrier to thermal electron transfer is $\Delta E(th)^*$, i.e. if this quantity of thermal energy is provided, the electron may hop thermally between the two redox sites. Provided that $\Delta E(th)^* = \Delta G(Th)^*$, and V_{ab} is small, then the following statements are true (subject to certain assumptions in the mathematical development) [87]:

$$E_{IT} = 4\Delta E(th)^* = 4\Delta G(th)^*$$

$$\Delta_{1/2} = [16kT\ln(2)\epsilon_{max}]^{1/2}$$

$$= [2312\ \epsilon_{max}]^{1/2} \quad \text{at 300 K} \tag{7.10}$$

where $\Delta_{1/2}$ is the half-bandwidth of the intervalence transition of maximum intensity ϵ_{max}. These two equations permit evaluation of the thermal electron transfer from the IT energy, and estimation of band-width from band energy. It is useful to compare the calculated half-bandwidth against the observed half-bandwidth, since agreement, or lack of, permits one to identify whether the complex is well behaved (or not) towards the Hush approach.
Further we note:

$$E_{IT} = E_{in} + E_{out}$$

$$E_{in} = n[2f_pf_q/(f_p + f_q)][d_p - d_q]^2$$

$$E_{out} = e^2[(1/2a_p) + (1/2a_q) - 1/r][(1/D_{op}) - 1/D_s]$$

$$= e^2[(1/a) - 1/r][(1/D_{op}) - 1/D_s] \quad \text{(for symmetric ion)} \tag{7.11}$$

where d_p,d_q, f_p,f_q, are the metal ligand bond lengths and force constants for the p and q valency ions, a_p and a_q are the radii of the p and q coordination spheres, and r is the separation between metal centres. In the limit of small V_{ab}, the IT transition energy reflects E_{in} and E_{out}, the inner sphere and outer sphere (solvation energy) and internal energy contributions to the optical barrier for electron transfer. D_{op} and D_s are the optical and static dielectric constants of the solvent, q-p = 1, and n is the number of metal-ligand bonds per metal centre. D_{op} may be taken to be the square of the refractive index of the solvent.

In principle contributions to Eqn.(7.11) can occur from any intramolecular vibration which changes in equilibrium coordinate between the two states; in practice it is the metal-ligand modes which seem most important.

The Eqns (7.11) reveal that a mixed valence IT transition will have an energy which is solvent dependent. This is generally the case because the two different ions will have different solvation energies. If the IT transition energy is plotted versus $[(1/D_{op})$ -

$1/D_s$], then the slope provides a measure of E_{out} and the intercept a measure of E_{in} [87-89,125,127]. Note that a true Class III species will show no solvent dependence because in the totally delocalised situation, both metal ions suffer the same degree of solvation. Thus solvent dependence of the IT transition is an indicator of Class II behaviour.

The magnitude of V_{ab} is given by:

$$V_{ab} = 2.05 \times 10^{-2} [(\varepsilon_{max} \Delta_{1/2})/(E_{IT})]^{1/2}(E_{IT}/R) \quad (7.13)$$

Fig.7.10 (right) define the situation where the two sites differ slightly by an amount ΔE_{AB}. Then:

$$E_{IT} = [16kTln(2)]^{-1} (\Delta_{1/2})^2 + \Delta E_{ab} \quad (7.14)$$

Consider an LCAO description of the ground state of the molecule allowing mixing between A_p and B_q (7.3). The usual expression may be written:

$$\psi_a = (1 - \alpha^2)^{1/2} A_p + \alpha B_q$$

then, ignoring the overlap integral S_{AB}, the transition moment for the IT transition is given by [83]:

$$M_{01} = e\alpha(r_d - r_a)$$

where r_d and r_a are the position vectors of the donor and acceptor nuclei. This may be simplified to:

$$|M_{01}| = e\alpha R$$

with R the donor-acceptor distance. Recalling Chapter 4, the transition moment is related to the intensity via the dipole strength D, the oscillator strength f, or the molar intensity via:

$$|M| = eD^{1/2}$$

$$f = 1.085 \times 10^{-5} \bar{\nu} D$$

$$f = 4.6 \times 10^{-9} \varepsilon_{max} \Delta_{1/2} \quad (7.15)$$

from these equations one may explicitly write the delocalisation coefficient α^2 via [83]:

$$\alpha^2 = 4.24 \times 10^{-4} [(\varepsilon_{max} \Delta_{1/2})/(E_{IT} \times R^2)] \quad (7.16)$$

where E_{IT}, and the half-bandwidth, here and elsewhere are in wavenumbers, and R is in angstroms. The band is assume to have a gaussian shape. The valences are assumed to be 'trapped' if

$\alpha < 0.25$, which is frequently the case for Class II complexes.

For example biferrocene picrate [191,192] shows an IT band at 5260 cm^{-1}, with ε_{max} = 550 l mol^{-1} cm^{-1} and a band-width of 5000 cm^{-1}; hence $f = 1.3 \times 10^{-2}$, $D = 2.2 \times 10^{-1}$ A^2, and assuming R = 5 angstrom, derived from consideration of the geometry, the delocalisation factor α is 0.09.

Since these equations are derived via first order perturbation theory they are only valid for small delocalisation, i.e trapped valences.

Clearly the rate of electron transfer is an important factor. If the electron transfer occurs 10^6 times per sec., then the electronic spectrum which has a time scale of about 10^{-15} s will see a localised class II system. If however the electron transfer occurs 10^{15} times per sec, then electronic spectroscopy would 'see' a delocalised Class III system, while ESCA for example, with a time scale of 10^{-17}s would still see a Class II system. Both thermal and electronic electron transfer rates can be derived from IT spectra.

Meyer and co-workers [97] in a widely ranging paper discuss the effect of solvent on electron transfer and upon the energetics of IT transitions. They note that the electronic electron transfer rate in a symmetrical complex ($\Delta E_{ab} = 0$) can be deduced via:

$$k_{et} = (2\pi/h)V_{ab}^2[\pi/(kTE_{IT})]^{1/2}exp(-E_{IT}/4kT) \quad (7.17)$$

while in an asymmetric complex, the corresponding equation is:

$$k_{et} = (2\pi/h)V_{ab}^2[\pi/(kt(E_{IT} - E_{ab})]^{1/2} exp[-E_{IT}^2/((E_{IT} - E_{ab})4kT] \quad (7.18)$$

However, as pointed out by Meyer, these eqns.(7.17,7.18) are limited in scope. Contributions from high frequency modes must be treated explicitly. This treatment assumes validity of the Born-Oppenheimer approximation which is valid at the optical transition occurring from thermally equilibrated ground state to excited vibrational levels of the product state, both far from the intersection region. Vibrational levels of both reactants and products will overlap near the intersection, breakdown of the Born-Oppenheimer relationship occurs and the thermal transfer rate will be faster than that predicted by eqns.(7.17,7.18). Indeed Eqn.(7.18) is only valid in the limit that electron transfer is slow on the vibrational timescale.

Alternatively, the electron transfer rate may be derived from:

$$k_{et} = \nu_n \mathbf{k} exp[-(\Delta G_{out}^* + \Delta G_{in}^*)/RT] \quad (7.19)$$

where ν_n is a nuclear frequency factor (1-10 x 10^{12} sec^{-1}), and $\mathbf{k}$ is the adiabaticity factor. The two free energy terms may be derived from eqn.7.11, or their sum equated to $E_{IT}/4$ if the entropy term $T\Delta S$ is assumed negligible, and V_{ab} is small.

The reaction may be assumed to be adiabatic ($\mathbf{k} = 1$) if [87]:

$$4.6 \times 10^{-3} V_{ab}^{2}(E_{IT})^{-1/2} > 1$$

while if this is not true, **k** is derived from:-

$$\mathbf{k} = 2[1-\exp(W)]/[2-\exp(W)]$$

with

$$W = [-2.3 \times 10^{-3} V_{ab}^{2}]/(E_{IT})^{1/2} \qquad (7.20)$$

Creutz [87] calculates values for these various parameters showing that V_{ab} varies from about 50 - 1000 cm^{-1}. For an IT band occurring near 10,000 cm^{-1}, the adiabaticity factor is 1 if V_{ab} is greater than about 200 cm^{-1}. The nuclear frequency factor is often approximated to be 5×10^{12} though if sufficient data are available it may be calculated (see [87] for full discussion).

Care should be taken in using any of the equations discussed here. There are many subtleties in mixed valence spectra as evidenced in recent reviews [86,88,106]. For example an IT transition $Ru(II)t_{2g}$---$Ru(III)t_{2g}$ is essentially a transition to a vibrationally excited state; it is not a true electronic transition. The above equations do not explicitly deal with the vibrational aspect of this problem, a fact treated by Wong and Schatz [86].

If the symmetry is not totally cubic then the t_{2g} level will split, and the above cited transition could involve several transitions $Ru(II)t_{2g}$---Ru(III)xz, or xy or yz. At the very least this will generate a band whose half-bandwidth is greater than calculated. A plot of band centre versus solvent dielectric function would give an erroneous intercept etc. Quite generally the half bandwidth of a transition is greater than predicted on the basis the eqn. in (7.11). Such an increase is typical of Class II systems [88]. The reader is urged to read the cited reviews before seriously analysing his own IT data.

Several IT transition are possible in principle. For example in the Ru(II)-Ru(III) system, a higher energy (by 10Dq) IT transition could be $Ru(II)t_{2g}$---$Ru(III)e_g$ which involves both electronic and vibrational excitation.

Note finally that IT transitions rarely if ever show vibrational structure and do not generally lead to emission, since radiationless relaxation to the ground state is very favourable. However with metals such as osmium where the spin-orbit coupling coefficient is large, spin-orbit splitting of the intervalence bands into several components may be observed [254,255].

Charge transfer transitions within a simple molecule such as ML_6, may be thought of as a type of intervalence transition in which the metal and ligand L change their redox level. In this case the energy separation between the two states, ΔE_{ab}, may be large, but the principle is the same. Thus some preliminary attempts have been made to understand simple charge transfer spectra, using the theory developed for intervalence transitions [252,253,256].

The field is still subject to much controversy.

Table 7.4

INTERVALENCE TRANSITION ENERGIES

Complex	IT transition	Ref
$[Ru(bpy)_2Cl]_2pyz^{3+}$	7700(450)	89
$[Ru(bpy)_2Cl]_2pym^{3+}$	7350(40)	89
$[(NC)_5Fe]_2pyz^{5-}$	8300(2200)	90
$[Ru(NH_3)_5]_2pyz^{5+}$	6370(5000)	91
$[Ru(NH_3)_5]_2(CN)_2^{5+}$	6995(410)	92
$[Ru(NH_3)_5]_2(N_2)_2^{5+}$	9800(1400)	93
$[Ru(NH_3)_5]_2(imQ)_2^{5+}$	10,530 7380(1040) 5850(900)	94
$[Ru(NH_3)_5]_2(tmq)_2^{5+}$	10,290(6)	95
$[Ru(NH_3)_5]_2Fum^{5+2}$	10,260(150)	173
$[Ru(II)(bpy)_2ClpyzRu(III)(NH_3)_5]^{4+}$	10,400(530)	96,211
$[Ru(II)(bpy)_2ClpyzOs(III)(bpy)_2Cl]^{3+}$	11,000	97
$[(NH_3)_5Co(III)NCRu(II)(CN)_5]^{-}$	26,700	247
$[(NC)_5Fe(II)CNFe(III)(CN)_5]^{6-}$	7700(32)	98
$[(NC)_5Fe(II)CNFe(III)(CN)_4NH_3]^{5-}$	7700	99
$[(NC)_5Fe(II)pyzRu(III)(NH_3)_5]^{30}$	7300	100
$K_3[Pt(II)(Py)Cl_3][Pt(IV)(Py)Cl_5]$	20,800	107
$K_3[Pt(II)(Py)Br_3][Pt(IV)(Py)Br_5]$	15,600	107
$K_2[Pt(II)(NH_3)Br_3][Pt(IV)(NH_3)Br_5]$	16,800	107
$K_2[Pt(II)(NH_3)I_3][Pt(IV)(NH_3)I_5]$	11,000	107
$[Pt(II)(etn)_4][Pt(IV)(etn)_4Cl_2]Cl_4.4H_2O$	21,000(17,500)	111,114
$[Pt(II)(etn)_4][Pt(IV)(etn)_4Br_2]Br_4.4H_2O$	18,250(16,500)	111,114
$[Pt(II)(etn)_4][Pt(IV)(etn)_4Br_2]Br_4$	23,600	111
$[Pt(II)(etn)_4][Pt(IV)(etn)_4I_2]I_4$	20,600	111
$[Pt(II)(en)_2][Pt(IV)(en)_2Br_2]Br_4$	18,000	108
$[Pt(II)(1,2dp)_2][Pt(IV)(1,2dp)_2Br_2]Br_4$	14,400	108
$[Pt(II)(1,2dab)_2][Pt(IV)(1,2dab)_2Br_2Br_4$	14,600	108
$[Pt(II)(en)I_2][Pt(IV)(en)I_4]$	14,400	109
$K_2[Pt(II)(NH_3)_2(NCS)_2]$- $[Pt(IV)(NH_3)_2(NCS)_2I_2]$	18,200	109
$Cs_2[Pt(II)NO_2(NH_3)Br_2]^{-}$ $[Pt(IV)(NO_2)(NH_3)Br_4]$	19,000	112
$Cs_2[Pt(II)NO_2(NH_3)I_2]$- $[Pt(IV)(NO_2)(NH_3)I_4]$	12,800	112
$[CpFe(II)Cp-CpFe(III)Cp]^{+}$	5000-5630[a]	125
$[CpFe(II)Cp.C{\equiv}C.CpFe(III)Cp]^{+}$	6170-7575[a]	125
b) Ion Pairs		
$[Ru(III)(NH_3)_5Py]_4[Fe(II)(CN)_6]_3$	10,930(40)	126
$[Ru(III)(NH_3)_6]_4[Ru(II)(CN)_6]_3$	17,800	126

cont

Table 7.4 cont

Complex	IT transition	Ref
$Fe(III)/[Ru(II)(CN)_6]^{4-}$	18,000(8400)	199
$Cu(II)/[Ru(II)(CN)_6]^{4-}$	25,200(930)	199
$U(VI)O_2/[Ru(II)(CN)_6]^{4-}$	29,100(5750)	199
$V(IV)O/[Ru(II)(CN)_6]^{4-}$	31,500(3500)	199
$[Cu(phen)_2][Mo(CN)_8].4H_2O$	17,000	249
$Cu_2Mo(CN)_8$	20,200	249
$Fe(III)/[Mo(IV)(CN)_8]^{4-}$	12,500(1300)	250
$Cu(II)/[Mo(IV)(CN)_8]^{4-}$	20,200(750)	250
$U(VI)O_2/[Mo(IV)(CN)_8]^{4-}$	24,650(385)	250
$V(IV)O/[Mo(IV)(CN)_8]^{4-}$	29,000(860)	250
$[Ru(III)(NH_3)_5L]/[Fe(II)(CN)_6]$		
L = pyridine	10,990(33)	255
L = imidazole	12,690(28)	255
$[Ru(III)(NH_3)_5L]/[Ru(II)(CN)_6]$		
L = pyridine	15,550(38)	255
$[Ru(III)(NH_3)_5L]/[Os(II)(CN)_6]$		
L = pyridine	15,290(40)	255

a) Cp units linked directly together through single or double bonds. imQ = NH quinone See [255] for many more examples.

7.3 i) Examples of Intervalence Systems

Several reviews present extensive experimental data [83,85,87] and therefore the listing here will be brief.

In Table 7.4 are listed examples of various intervalence complexes. By far the most intensively studied are the LRu(II)-LL-Ru(III)L' complexes as recently reviewed [87]. A species with orthophenylenedimine as the bridging LL is particularly noteworthy exhibiting at least three low energy IT bands perhaps to different d levels of a low symmetry Ru(III) acceptor [94]. Fig.7.11 shows an example of a mixed valence species $[(NH_3)_5Ru(pyrazine)RuCl(bipy)_2]^{3+}$ compared with spectra of its component parts. It is clear that the mixed valence systems show an admixture of the spectra of the component plus the IT absorption [211].

Fumaronitrile and TCNE bridged Ru(II) complexes have also been studied [173]. The fumaronitrile species are well behaved Class II complexes with IT transitions near 10,000 cm^{-1} whose solvent dependence obeys eqn.7.11. The degree of delocalisation is calculated to be 3×10^{-4}, i.e. an essentially localised system. The TCNE complexes, on the other hand, yielded no clear IT band and appear to be strongly delocalised.

Clearly if two different oxidation states are linked by a saturated linkage, electron transfer should be difficult if not impossible. This possibility was explored by Meyer and co-workers

[211] who showed that mixed valence complexes with two Ru centres linked by trans-1,2-bis(4-pyridyl)ethylene exhibit an IT absorption (Table 7.4), while the analogous complex linked by 1,2-bis-(4-pyridyl)ethane shows no IT absorption.

Taube also reports a series of bridged osmium complexes [156]. The complex $[Os(NH_3)_5]_2pyz^{5+}$ is extremely stable against disproportionation (comproportionation constant 10^7) and shows a unique spectrum not attributable to a summation of Os(II) and Os(III). This Class III species does not exhibit the usual intraconfigurational near ir bands for Os(III), but does have a pair of strong (10^3 - 10^4 l mol^{-1} cm^{-1}) bands at 3000 and 3400 cm^{-1}. These peaks are not influenced by deuteration of the amines.

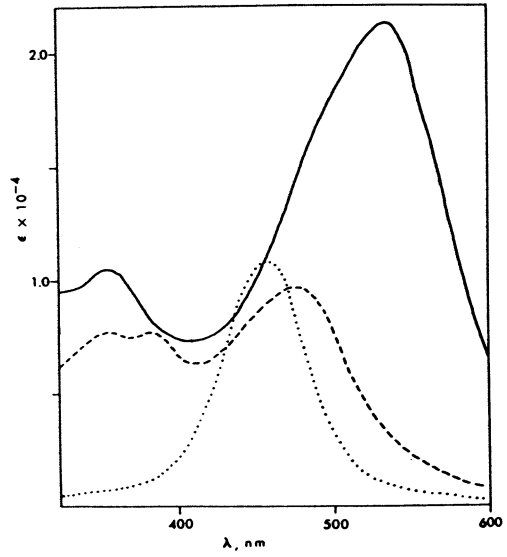


Fig.7.11 The absorption spectra of $[(NH_3)_5Ru(pyz)RuCl(bpy)_2]^{3+}$ (solid line), $[(NH_3)_5Ru(pyz)]^{2+}$ (hatched line) and $[(bpy)_2ClRu(pyz)]^+$ (dotted line) in acetonitrile [211].

Clark and co-workers [106-112,145] and Day [113] have studied the Resonance Raman spectra of anion bridged complexes of the general type $[Pt(II)L_4][Pt(IV)L_4X_2]$. These highly coloured (metallic sheen and semi-conducting) solids have a square $[Pt(II)L_4]$ unit linked by anion bridges to a planar $[Pt(IV)L'_4]$ unit. The Pt(II)-L and Pt(IV)-L distances are different so that this system is Class II rather than Class III. They all show very intense parallel (axis) polarised absorption in the visible region. The transition can be assigned as Pt(II) z^2---Pt(IV) z^2 [111]. Clark has reported beautiful series of vibrational data for many of these complexes. The dominant Resonance enhanced vibration is the

symmetric X-Pt(IV)-X stretching vibration whose overtones can sometimes be detected as far as v = 18. In addition other vibrations may be singly excited and have progressions of the X-Pt(IV)-X vibration built thereon. Substantial changes in axial Pt-X bond distances can be calculated from progression intensity data (Chapter 4). Anharmonicity and other bond properties were also derived. Combination bands involving equatorial groups also gave RR vibrations. The excitation profiles tended to maximise on the low energy side of the IT transition [115].

A recent paper [152] deals with complexes of the type $[M(II)(N{-}{-}N)_2][M(IV)(N{-}{-}N)_2X_2]X_4$, (M = Pd,Pt) in which the N--N bidentate ligand is 1,2-diaminocyclohexane (dach) or 1,2-diaminocyclopentane (dacp). These complexes can exist in several optically active or racemic isomeric and several geometric isomeric forms each of which has characteristic IT bands and progressional RR frequencies. These authors note that the trends in the IT band maximum, excitation profile maximum (to low energy side of IT maximum) and progression frequency all lie in the same sequence, namely Cl > Br > I, Pt > Pd, cis > meso > racemate > trans, and cis-dach > cis-dacp.

Vanadium forms a mixed valence V(IV)-V(V) species with nitriloacetate [150] and related ligands. The vanadyl(IV) component of the complexes exhibit d-d bands similar to monomeric OV(IV) systems (see Chapter 6), but there is an additional IT transition near 10,000 cm^{-1}.

Prussian Blue is the trivial name for salts such as $KFe(III)[Fe(II)(CN)_6]$ which are intensely blue in the solid state due to IT transitions between the two iron atoms. The Fe(III) is octahedrally surrounded by the nitrogen atoms of the cyanides bound to Fe(II). Bands centered near 15,000 and 25,000 cm^{-1} are believed to arise from transitions from the t_{2g} orbitals of the $[Fe(II)(CN)_6]$ fragment to the t_{2g} and e_g orbitals, respectively of the high spin $[Fe(III)(NC)_6]$ fragment [85,240].

Note that ion pairs can also give rise to IT transitions, rather like the situation with Prussian Blue, but occurring in solution. For example metal salts of the $[M(CN)_6]^{4-}$, M = Fe,Ru,Os etc., and $[Mo(CN)_8]^{4-}$ ions etc, often have additional low energy IT absorption [83,199,249-251,255], between anion and cation (Table 7.4). In general these ion pairs show strong solvent dependence which is linear with respect to the solvent dielectric parameter in Eqn.(7.11) [199,250]. Detailed analysis [255] reveals that the energy of the intervalence transition in these ions pairs is related to the energy of electron exchange between the components of the ion pair, as would be expected. Intervalence ferrocene derivatives are discussed below (section 7.4).

Various authors have investigated the ability of an electron to transfer over long distances by synthesizing intervalence complexes with 3 or more metal atoms strung together via bridges [205,207-210]. It is evident that electron transfer can proceed through metal pathways though it does not occur in all such complexes.

Gagne and co-workers have investigated macrocyclic complexes which can contain two metals [248]. For example a species containing a Cu(I)-Cu(II) pair displays a solvent dependent IT band whose energy varies from 8000 - 11,000 cm^{-1}(80).

7.4 Bis(Cyclopentadienyl) (Cp) Metal Complexes [188]

Cyclopentadiene complexes have been of significance for many years with intense interest in their magnetism, esr, PES, optical spectroscopy and bonding characteristics. In this section the familiar bis(cyclopentadienyl) sandwich complexes, $M(Cp)_2$ will be discussed, but the large range of other species, such as $CpM(CO)_x$ will be excluded. The effective point group for $M(Cp)_2$ species is D_{5h} or D_{5d} depending upon whether the two Cp rings are respectively, eclipsed or staggered. However if the rings are presumed to be rotating it is probably more appropriate to use the group $C_{\infty v}$.

Coupling together the π M.O.s of two Cp rings yields 4 orbitals capable of taking the 12 electrons of the two Cp^- units. Their symmetries are such as to yield the M.O. diagram shown in Fig.7.12 [174], where Cp electrons fill up to $1e_{1u}$.

In point group symmetry D_{5d}, the d orbitals span z^2 (a_{1g},σ), xz,yz (e_{1g},π) and xy,x^2-y^2 (e_{2g},δ). The δ orbitals are strongly bonding through interaction with the ligand $2e_{2g}$ orbital, the σ level less so, and the π levels somewhat antibonding, through interaction with filled ligand π e_{1g} levels. This provides the level sequence $\delta < \sigma << \pi$ ($e_{2g} < a_{1g} < e_{1g}$).

It has long been recognised that simple ligand field theory suffices to predict the d-d energies of many Cp metal complexes, and various approaches have been used. The one-electron energies may be written [174,175]:-

$$E(\sigma,a_{1g}) = 2Ds - 6Dt \quad E(\pi,e_{1g}) = Ds + 4Dt \quad E(\delta,e_{2g}) = -2Ds - Dt$$
or
$$E(\sigma,a_{1g}) = 0 \quad E(\pi,e_{1g}) = \Delta_2 \quad E(\delta,e_{2g}) = \Delta_1 \qquad (7.21)$$

while other d^n configuration d-d energies are shown in Table 7.5 [174]. Note that $Ds = (1/7)(\Delta_2 - 2\Delta_1)$ and $Dt = (1/35)(4\Delta_2 - \Delta_1)$.

The splitting ($1e_{2g}$---$2a_{1g}$) is fairly small (4000-7000 cm^{-1}) compared with the pairing energy so that d^3 species are high spin. However the splitting ($2e_{1g}$---$2a_{1g}$) is quite large (17,000 - 25,000 cm^{-1}) and, as a consequence, low spin d^4, d^5 and d^6 species are usually formed. Note however that $Mn(Cp)_2$ is high spin.

The dd and CT states of the various $^2d^n$ configurations are obtained by generating the usual direct products. The group D_{5d} is rarely encountered and it is worth noting here that for the degenerate direct products:-

Table 7.5

ENERGY LEVELS IN d^n CYCLOPENTADIENYL COMPLEXES [174,178]

d^3	a^4E_{1g}---$^4A_{2g}$	$\Delta_2 - 0.5\Delta_1 + 4.5B - 0.5W$
	b^4E_{1g}	$\Delta_2 - 0.5\Delta_1 + 4.5B + 0.5W$
	$^4E_{2g}$	$\Delta_2 - \Delta_1 - 3B$
d^6	a^1E_{1g}---$^1A_{1g}$	$\Delta_2 - 0.5\Delta_1 + 6B - C - 0.5X$
	b^1E_{1g}	$\Delta_2 - 0.5\Delta_1 + 6B - C + 0.5X$
	$^1E_{2g}$	$\Delta_2 - \Delta_1 - 9B - C$
	a^3E_{1g}	$\Delta_2 - 0.5\Delta_1 - B - 3C - 0.5Y$
	b^3E_{1g}	$\Delta_2 - 0.5\Delta_1 - B - 3C + 0.5Y$
	$^3E_{2g}$	$\Delta_2 - \Delta_1 - 9B - 3C$
d^5	a^2A_{1g}---$^2E_{2g}$	$\Delta_1 + 20B$
	b^2A_{1g}	$\Delta_2 - \Delta_1 - 3B - C$
	$^2A_{2g}$	$\Delta_2 - \Delta_1 - 3B - C$
	a^2E_{2g}	$\Delta_2 - 0.5\Delta_1 + 5B - C - 0.5Z$
	b^2E_{2g}	$\Delta_2 - 0.5\Delta_1 + 5B - C + 0.5Z$
	a,b,c $^2E_{1g}$ [a]	$\Delta_2 + 7B - C$
		$\Delta_2 - \Delta_1 + 3B - 2C$
		$\Delta_2 - \Delta_1 - 3B - 3C$

$W = [(\Delta_1 + 3B)^2 + 216B^2]^{1/2}$ $\quad X = [(\Delta_1 + 6B)^2 + 384B^2]^{1/2}$

$Y = [(\Delta_1 + 16B)^2 + 96B^2]^{1/2}$ $\quad Z = [(\Delta_1 + 4B)^2 + 192B^2]^{1/2}$

a) Does not include configurational interaction. The off-diagonal elements are $H_{ab} = 4\sqrt{3}B$, $H_{ac} = 2\sqrt{6}B$ and $H_{bc} = 3\sqrt{2}B$ Complete strong and weak field d^3 matrices are published in [177].

$e_1 \times e_1 = a_1 + a_2 + e_2$; $e_1 \times e_2 = e_1 + e_2$; $e_2 \times e_2 = a_1 + a_2 + e_1$

The d^6 species ferrocene, $Fe(Cp)_2$, is perhaps the most well known cyclopentadienyl derivative, and the species for which most studies have appeared. The ground state is $^1A_{1g}\ (1e_{2g})^4(2a_{1g})^2$. The excited states are generated as follows:

$$\begin{matrix} 2a_{1g}\text{---}2e_{1g} & {}^1E_{1g} + {}^3E_{1g} \\ 1e_{2g}\text{---}2e_{1g} & {}^1E_{1g} + {}^3E_{1g} + {}^1E_{2g} + {}^3E_{2g} \end{matrix} \qquad (7.22)$$

Ferrocene, and in general other d^6 bis(cyclopentadienyl) complexes display the three anticipated spin allowed d-d transitions (to the three spin singlets in 7.22)[174,179,180]. In some cases the very weak spin forbidden transitions can also be seen. In general ferrocene-like derivatives show two closely spaced d-d transitions first identified clearly in cryogenic and single crystal studies by Gray and co-workers [174] (Fig.7.13) (see also MCD data [180]).

Vibrational structure [174,181] on the d-d bands shows a progression in 240-300 cm^{-1} associated with an excited state totally symmetric ring-metal vibration (ground state 309 cm^{-1}). Observed and calculated data are shown in Table 7.6. The Racah parameter is generally quite small, indicative of considerable covalent character.

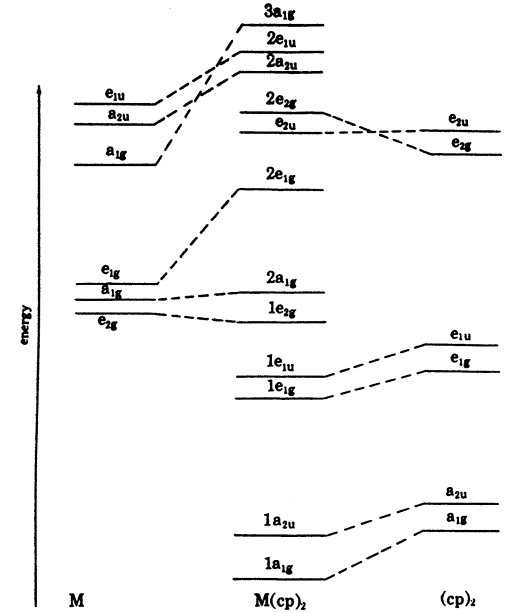


Fig.7.12 A molecular orbital diagram for bis(cyclopentadienyl)-complexes [174].

Warren [175,188,215] has derived energy level diagrams (Tanabe-Sugano type diagrams [188,215]) (as have Pavlik and co-workers [177]) for many of the $M(Cp)_2$ system inclusive of spin-orbit coupling and has reported spectra and assignments for many species [186,187], together with critical analyses of PES data [215]. Optical data for many of these species are presented in Table 7.6 and shown in Fig.7.14. Energy level diagrams for d^n species in a $C_{\infty v}$ field were generated by Warren [188] and an example is displayed in Fig.7.15. In view of the similar energy of the δ and σ d orbitals, several ground states are possible for many species, and a thermal equilibrium may also result (e.g. $Mn(MeCp)_2$). This is clearly demonstrated in the d^4 example displayed in Fig.7.15. Often esr and magnetic data can be used to discriminate between the several possible ground states more readily than optical spectroscopy [188,189].

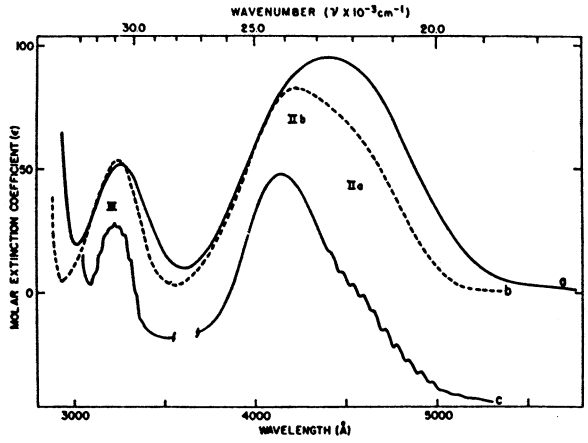


Fig.7.13 Visible absorption spectra of ferrocene at various temperatures. (a) EPA solution at 300 K, (b) EPA glass at 77 K, and (c) single crystal at 4.2 K. Intensities are not to scale [174].

However considering d^4 for example, the diagram is generated on the basis that from previous experience, Dt/Ds is usually near 0.55. Previous experience also shows that Dt/B is often in the range 4.5 - 6.5; thus if this holds true for d^4, the ground state must be to the right, i.e. $^3\Delta$ ($^3E_{2g}$).

CT bands in metallocene derivatives usually occur above 25,000 cm^{-1}. Thus one may assume that weak transitions below about 25,000 cm^{-1} are, in most cases, associated with d-d transitions (but see ferricenium below). Very weak transitions are seen in a few examples and are attributed to spin forbidden d-d transitions. Note for interest the spectra of high spin $Mn(Cp)_2$ and low spin $Mn(Me_5Cp)_2$ (Fig.7.14) which are clearly very different. The sharp spike in the former is obviously a spin forbidden sextet-quartet transition.

Recently the spectra of decamethylcyclopentadienyl complexes have been reported (Table 7.6). Their spectra do not differ greatly from the unsubstituted derivatives, though their intensities are significantly greater. Intensity data for the Ni(IV) species seem anomalously high for d-d transitions. If real, mixing must be very considerable. Where sufficient d-d data exist, radial parameters are calculated and included in Table 7.6. The ratio Dt/Ds seems to remain fairly constant at a value near 0.55 [188]. For a wide range of metallocenes, the values of Δ_1 and Δ_2 do not vary so widely. Thus it is possible to predict the rough energies of all the d-d transitions in a complex whose spectrum may be too poorly resolved to allow a definitive analysis. The d-d assignments in Table 7.6 are therefore fairly secure. A more complete discussion

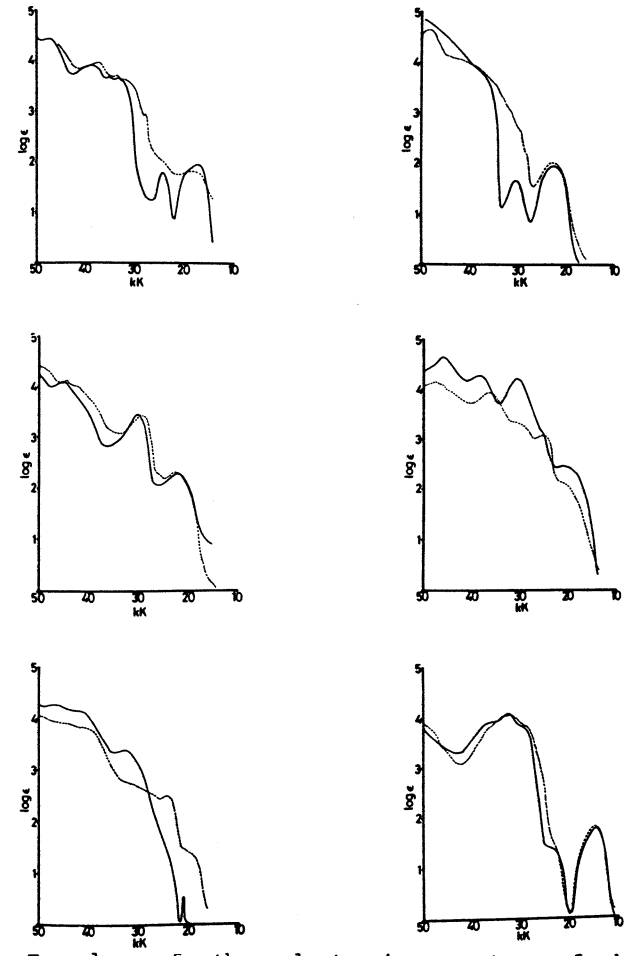


Fig.7.14 Examples of the electronic spectra of bis(cyclopentadienyl)complexes in isooctane solution. The solid lines are the $M(Cp)_2$ species, and the dotted lines the $M(MeCp)_2$ species. Top left - VCp_2 and $V(MeCp)_2$. Middle left - $Cr(Cp)_2$ and $Cr(MeCp)_2$ Bottom left - $Mn(Cp)_2$ and $Mn(MeCp)_2$ Top right - $Fe(Cp)_2$ and $Fe(MeCp)_2$. Middle right - $Co(Cp)_2$ and $Co(MeCp)_2$. Bottom right - $Ni(Cp)_2$ and $Ni(MeCp)_2$ [186].

may be found in [188].

$M(Cp)_2$ complexes show rich CT absorption, usually in the ultra-violet region. Consideration of Fig.7.12, suggests that the following CT transitions may be expected in the usual uv/visible region, depending upon d orbital populations:

LMCT $1e_{1u}$---$1e_{2g}$ $1e_{1u}$---$2a_{1g}$ $1e_{1u}$---$2e_{1g}$
MLCT $2e_{1g}$--- e_{2u} $2a_{1g}$---e_{2u} $1e_{2g}$---e_{2u}

(7.23)

Gordon and Warren have discussed [186] the states and polarisations to be expected from these various excitations in the different d^n configurations, and the possible states are summarised in Table 7.7. For the earlier transition elements with a hole in $1e_{2g}$ and/-or $2a_{1g}$, low energy LMCT transitions to these levels may be anticipated. For the later species these levels are filled and only transitions to $2e_{1g}$ may be anticipated. Such species with an electron in $2e_{1g}$ may also exhibit MLCT transitions. MLCT transitions from the $2a_{1g}$ and $1e_{2g}$ levels will generally be outside the usual range of study.

Comparison with the decamethylcyclopentadienyl derivatives is of value in that LMCT bands should be red shifted in the decamethyl derivatives compared with the unsubstituted derivatives due to the inductive effect of the methyl groups, and this is generally seen (Table 7.8) [186]. Similarly, MLCT transitions should blue shift, and do. It is also worth noting that in-plane (short axis) (x,y) polarised LMCT transitions will probably be weaker than axial (long axis) (z) polarised transitions.

In general the CT assignments shown in Table 7.8 follow fairly systematically from consideration of the shift upon decamethylation, and the number of allowed transitions to be anticipated for a given one electron excitation. For example, considering $Cr(Cp)_2$, this d^4 complex shows peaks at 29,590 and 44,640 cm^{-1} with a shoulder at 40,000 cm^{-1}. Reference to Table 7.7, and following [186], reveals that the anticipated lowest LMCT excitation, $1e_{1u}$---$1e_{2g}$, generates a single CT transition. This is then associated with the band at 29,590 cm^{-1}. The transition $1e_{1u}$---$2a_{1g}$ provides the excited state configuration $(1e_{1u})^1$-$(1e_{2g})^3(2a_{1g})^2$ which spans $^3E_{1u} + {}^3E_{2u}$ (Table 7.7), transitions to both of which are allowed. Thus the bands at 40,000 and 44,640 cm^{-1} are assigned to these two states respectively, the latter more intense band being assigned to $^3E_{2u}$ because this state is allowed in xy, and z, while $^3E_{1u}$ is only allowed in xy.

The group theoretical transition moments for these two transitions may be written as $\langle {}^3E_{2g}|M|{}^3E_{1u}\rangle$ and as $\langle {}^3E_{2g}|M|{}^3E_{2u}\rangle$. The operator r spans $A_{2u}(z) + E_{1u}(xy)$. In the former element, the ground and excited state direct product spans $E_{1u} + E_{2u}$; hence the transition is allowed in xy. In the latter element, the ground and excited state direct product spans $A_{1u} + A_{2u} + E_{1u}$; hence this transition is allowed in both xy, and z.

The energy difference between the $1e_{1u}$---$1e_{2g}$ and $1e_{1u}$---$2a_{1g}$ transitions should be similar to Δ_1 since the repulsion energy contribution to the two transitions should be fairly similar. That the difference is significantly greater than (the probable value of) Δ_1 is a little disquieting, but may be a consequence of configurational interaction separating these states.

Table 7.6

ELECTRONIC SPECTRA (d-d) OF SOME CYCLOPENTADIENYL COMPLEXES

d^3 (ground state $^4A_{2g}$)

	$V(Cp)_2$	$V(Me_5Cp)_2$	$Cr(Cp)_2^+$	$Cr(Me_5Cp)_2^+$
$^2E_{1g}$	8960(.09)	10,500(1.8)		
$^2A_{1g}, {}^2A_{2g}$	13,000(.06)	14,500(5.7)		
a^4E_{1g}	17,330(58)	18,700(23)	17,860(270)	(20,400)(1386)
$^4E_{2g}$	20,240(46)	20,600(25)	21,980(210)	(23,100)(2450)
b^4E_{1g}	24,500(66)	28,200(1037)	27,030(630)	
Δ_2	16,425	18,700	16,570	
Δ_1	-4930	-3800	-6760	
Ds	3755	3760	4300	
Dt	2020	2245	2090	
B	420	630	510	
Ref	175,178,186	176	176	176

d^4 (ground state $^3E_{2g}$)

	$Cr(Cp)_2$
d-d	21,970(186)[c]

d^5 Species (ground state $^2E_{2g}$)

	$Fe(Cp)_2^+$	$Mn(Cp)_2$[d]	$Mn(MeCp)_2$
$^2E_{1u}$	16,200(450)LMCT		
a^2E_{1g}	17,700(250)	19,610(3)[cd]	19,530(22)[c]
a^2E_{2g}	19,100(190)		
b^2E_{1g}	21,400(150)		
b^2A_{1g}	26,300(350)		
Ref	174,182	186	186

d^6 species (ground state $^1A_{1g}$)

	$Fe(Cp)_2$	$Ru(Cp)_2$	$Co(Cp)_2^+$	Fe(Ph-Cp)(Cp)
a^3E_{1g}	18,900(7)	26,000(5)	21,800(7)	17,500(5)
a^1E_{1g}	21,800(36)	29,500(120)	24,300(140)	20,800(160)
$^1E_{2g}$	24,000(72)	32,500(160)	26,400(120)	23,000(320)
b^1E_{1g}	30,800(49)	36,600(150)	33,300(1200)	29,500(850)
Δ_2	22,000	29,800	24,400	21,000
Δ_1	-7100	-6600	-7200	-6900
Ds	5170	6145	5540	4970
Dt	2720	3595	2995	2600
B	390	260	400	370
Ref	174	174	174	174

Table 7.6 cont----------

	$Fe(Me_5Cp)_2$	$Co(Me_5Cp)_2^+$	$Ni(Me_5Cp)_2^{2+}$
a^3E_{1g}		12,700(0.2)	
$^3E_{2g}$		18,500(0.8)	
b^3E_{1g}		21,300(8)	
a^1E_{1g}	23,500(121)	23,800(330)	22,500
$^1E_{2g}$	30,500(180)	29,500(1430)	31,500
b^1E_{1g}	34,500(2970)	40,000(1170)	42,500
Δ_2	23,100	24,100	21,500
Δ_1	-11,200	-14,100	-19,000
Ds	6500	7470	8500
Dt	2960	3160	3000
B	420	630[a]	690
Ref	176	176	176

d^7 (ground state $^2E_{1g}$) d^8 (ground state $^3A_{2g}$)

	$Co(Cp)_2$		$Ni(Cp)_2$	$Ni(Me_5Cp)_2$
d-d	(15,150)	1A1g	11,740(4)	
d-d	(16,390)	a3E1g	14,380(62)	15,900(99)
d-d	(17,860)	3E2g	16,900(23)	18,500(58)
d-d	21,190(302)	b3E1g	23,450(26)	25,000(3250)[b]
Ref	179,186	Δ_2	13,920	15,400
		Δ_1	-4600	-4800
	$Ni(Cp)_2^+$	Ds	3300	3570
d-d	16,400	Dt	1720	1900
Ref	194	B	570	580
		Ref	177,186	176

a) Calculated from spin singlets. Spin triplets yield B = 680 and Δ_1 = -13,900 cm^{-1}. b) Estimated. c) Several d-d bands overlap here ??. d) High spin ground state $^6A_{1g}$; d-d transitions are all spin forbidden.

There are some other difficulties. For example ferrocene displays a strong band at 50,000 cm^{-1} which blue shifts upon decamethylation and is therefore regarded as LMCT. In addition this band seems allied to a band at 38,000 cm^{-1} in the isoelectronic $[Co(Cp)_2]^+$ and at > 51,300 cm^{-1} in $Ru(Cp)_2$. The shift in this band (designated band VI in [174]) is also consistent with LMCT. For a low spin d^6 species, the only possible LMCT excitation is $1e_{1u}$---$2e_{1g}$ which generates the spin allowed states shown in Table 7.7, transition to only one of which is also orbitally allowed. Thus the single intense transition is associated with $^1A_{2u}$---$^1A_{1g}$. However there are two lower lying strong bands in both ferrocene and ruthenocene and no obvious allowed assignment for them. They

are however quite weak compared with band VI and could arise from a forbidden LMCT excitation (red shift on decamethylation [186]).

Cobaltocene (d^7) has at least three clear LMCT bands. The excitation $1e_{1u}$---$2e_{1g}$ generates 4 states, transitions to all of which are orbitally allowed. The isoelectronic $[Ni(Cp)_2]^+$ also exhibits three bands obviously related to the cobaltocene bands but, as required for LMCT transitions, shifted to the red [195].

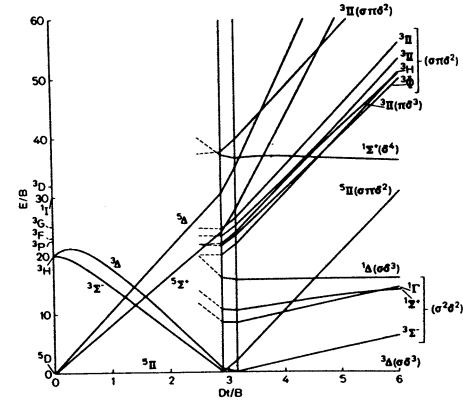


Fig.7.15 A Tanabe-Sugano type energy level diagram for d^4 bis-(cyclopentadienyl)complexes in $C_{\infty v}$ symmetry [188].

The ferrocenium cation possesses the $(1e_{1g})^3(a_{1g})^2$ $^2E_{2g}$ ground term, and gives rise to a rich spectrum [174,182-184]. The hole in $1e_{2g}$ permits the existence of a fairly low LMCT transition from $1e_{1u}$. Indeed, the lowest energy transition, at 15,700 cm^{-1} has an intensity which increases upon cooling, thereby eliminating a d-d type transition. Gray and co-workers [174], and Prins [185] assigned this transition to the low energy LMCT band. Prins noted the band energy shifts significantly upon ring substitution thereby providing evidence for direct ring involvement. Gray demonstrated highly complex vibrational structure [174,182] indicative of splitting of this $^2E_{1u}$ state into two Kramer's doublets. The ring breathing frequency is essentially the same as that in ferrocene.

To higher energy there lie four shoulders whose energies are largely independent of ring substitution; they are assigned to d-d transitions [174]. Theoretical analysis using the equations in Table 7.5, do not accurately reproduce the energies of these states, but the assignments shown in Table 7.6 seem reasonable [174]. Detailed esr, optical spectroscopy etc. studies of cobaltocene [179] and a theoretical analysis of nickelocene [177, 190] have appeared. The redox behaviour of nickelocene in melts has also been studied [194]. Complexes of Ni(II), Ni(III) and Ni(IV) are obtained and spectra reported.

Table 7.7

PREDICTED SPIN ALLOWED d-d AND CT STATES IN d^n BIS(CYCLOPENTADIENYL) COMPLEXES (D_{5d})

	$d^1(^2E_{2g})$	$d^2(^3A_{2g})$	$d^3(^4A_{2g})$
Excitation			
d-d (Laporte forbidden)			
$1e_{2g}$---$2a_{1g}$	$^2A_{1g}$	$^3E_{2g}$	--
$1e_{2g}$---e_{1g}	$^2E_{1g}$	$^3E_{1g}$ + $^3E_{2g}$	$^4E_{1g}$ + $^4E_{2g}$
$2a_{1g}$---e_{1g}	--	--	$^4E_{1g}$
2 electron		$^3A_{2g}$ + $^3E_{1g}$	$^4E_{2g}$ + $^4A_{2g}$
CT			
$1e_{1u}$---$1e_{2g}$	$^2E_{1u}$	$^3E_{1u}$ + ($^3E_{2u}$)	$^4E_{1u}$ + ($^4E_{2u}$)
$1e_{1u}$---$2a_{1g}$	$^2E_{1u}$ + $^2E_{2u}$	$^3E_{1u}$	$^4E_{1u}$
$1e_{1u}$---$2e_{1g}$	($^2A_{1u}$) + ($^2A_{2u}$)	$^3A_{1u}$ + ($^3A_{2u}$)	$^4A_{1u}$ + ($^4A_{2u}$)
	$^2E_{1u}$ +2 $^2E_{2u}$	($^3E_{2u}$)	($^4E_{2u}$)
$2a_{1g}$---e_{2u}			($^4E_{2u}$)
$1e_{2g}$---e_{2u}	$^2E_{2u}$	$^3E_{1u}$ + ($^3E_{2u}$)	$^4A_{1u}$ + ($^4A_{2u}$)
			$^4E_{1u}$
	$d^4(^3E_{2g})$	$d^5(^2A_{1g})$	$d^5(^2E_{2g})$
CT			
$1e_{1u}$---$1e_{2g}$	$^3E_{1u}$	--	$^2E_{1u}$
$1e_{1u}$---$2a_{1g}$	$^3E_{1u}$ + $^3E_{2u}$	$^2E_{1u}$	--
$1e_{1u}$---$2e_{1g}$	($^3A_{1u}$) + ($^3A_{2u}$)	($^2A_{1u}$) + $^2A_{2u}$	2 $^2E_{2u}$ + $^2E_{1u}$
	$^3E_{1u}$ +2 $^3E_{2u}$	($^2E_{2u}$)	($^2A_{1u}$) + ($^2A_{2u}$)
$2a_{1g}$---e_{2u}	($^3A_{1u}$) + ($^3A_{2u}$)	($^2E_{2u}$)	$^2E_{1u}$ + ($^2A_{1u}$)
	$^3E_{1u}$		($^2A_{2u}$)
$1e_{2g}$---e_{2u}	$^3E_{1u}$ +3 $^3E_{2u}$	($^2A_{1u}$) + $^2A_{2u}$	3 $^2E_{2u}$ + $^2E_{1u}$
		$^2E_{1u}$	
	$d^5(^6A_{1g})$	$d^6(^1A_{1g})$	$d^7(^2E_{1g})$
CT			
$1e_{1u}$---$1e_{2g}$	$^6E_{1u}$ + ($^6E_{2u}$)	--	--
$1e_{1u}$---$2a_{1g}$	$^6E_{1u}$	--	--
$1e_{1u}$---$2e_{1g}$	($^6A_{1u}$) + $^6A_{2u}$	($^1A_{1u}$) + $^1A_{2u}$	3 $^2E_{1u}$ + $^2E_{2u}$
	($^6E_{2u}$)	($^1E_{2u}$)	
$2a_{1g}$---e_{2u}	($^6E_{2u}$)	($^1E_{2u}$)	$^2E_{1u}$ + $^2E_{2u}$
$1e_{2g}$---e_{2u}	($^6A_{1u}$) + $^6A_{2u}$	($^1A_{1u}$) + $^1A_{2u}$	$^2A_{1u}$ + $^2A_{2u}$
	$^6E_{1u}$	$^1E_{1u}$	2 $^2E_{1u}$ + $^2E_{2u}$
$2e_{1g}$---e_{2u}	($^6A_{1u}$) + ($^6E_{2u}$)	--	$^2E_{2u}$

cont

Table 7.7 cont

	$d^8(^3A_{2g})$	$d^9(^2E_{1g})$
CT		
$1e_{1u}$---$2e_{1g}$	$^3A_{1u}$ + ($^3A_{2u}$)	$^2E_{1u}$
	($^3E_{2u}$)	
$2a_{1g}$---e_{2u}	($^3E_{2u}$)	$^2E_{1u}$ + $^2E_{2u}$
$1e_{2g}$---e_{2u}	$^3A_{1u}$ + ($^3A_{2u}$)	$^2A_{1u}$ + $^2A_{2u}$
	$^3E_{1u}$	+2 $^2E_{1u}$
$2e_{1g}$---e_{2u}	$^3E_{1u}$ + ($^3E_{2u}$)	$^2A_{1u}$ + $^2A_{2u}$
		2 $^2E_{2u}$ + $^2E_{1u}$

Transitions to states shown in parentheses are orbitally forbidden. In the D_{5d} group $a_1 \times e_n = a_2 \times e_n = e_n$, n = 1,2. The mapping between D_{5d} and $C_{\infty v}$ is:

$C_{\infty v}$	D_{5d}	$C_{\infty v}$	D_{5d}
Σ^-	A_{2g}	Σ^+	A_{1g}
Π	E_{1g}	Φ,Δ	E_{2g}

Metallocenes have been used to quench electronically excited benzil and tris(bipyridine)ruthenium(II) [242]. Quenching occurs via energy transfer to the lowest accessible metallocene triplet state. Ruthenocene emits (ca 17,000 cm^{-1}) [242,243] from this triplet but with a Stokes shift of some 10,000 cm^{-1}. Evidently the excited state is very distorted. The emission shows two vibrational progressions separated by 160 cm^{-1} with spacings of 327 and 340 cm^{-1} (possibly identical to within experimental error) associated with the ring-metal stretch (330 cm^{-1} Raman).

Mixed valence ferrocene complexes have also been studied [191-193,205,206]. Two ferrocene units linked by a C-C bond form biferrocene which can be oxidised by one electron to form biferrocene[Fe(II)Fe(III)] picrate [192]. This exhibits absorption bands similar to those of its constituent parts, ferrocene and ferricenium, but with significantly greater intensities. However there is also a very broad IT band centered at about 5250 cm^{-1} (in ethanol) . The broadness may be due to the flexibility of this molecule whose Fe-Fe distance may vary from about 380 - 510pm. Using the theory outlined in section 7.3, the IT band position and intensity yield values of the activation barrier for thermal electron transfer of 3.78 KCal/M, and an intramolecular electron transfer rate constant of 1.3 x 10^{10} s^{-1}.

A mixed valence biferrocene involving a C≡C bridge has also been studied [125]. This possesses an IT band at 7575 cm^{-1} (in acetone). Both biferrocene derivatives have IT bands which are solvent dependent, and yield fairly linear plots of E_{IT} versus the solvent function in Eqn.7.11 [125]. Data for biferrocene linked by two acetylene bridges have also been presented with E_{IT} near 9000 cm^{-1} [198]. Indeed strings of ferrocenes may be linked together [206] to yield inter-valence systems. Mixed Ru(III) ferrocene systems also yield IT bands near 12,000 cm^{-1} [205].

Table 7.8

ELECTRONIC SPECTRA (CT) OF SOME CYCLOPENTADIENYL COMPLEXES [174,186,194] (Intensities in Log units)

	d^n	LMCT $1e_{1u}$---$1e_{2g}$	LMCT $1e_{1u}$---$2a_{1g}$	LMCT $1e_{1u}$---$2e_{1g}$	MLCT $2e_{1g}$---e_{2u}
$V(Cp)_2$	d^3	33,330(3.65)	38,920(3.91)	47,390(4.42)	
$Cr(Me_5Cp)_2^+$	d^3	29,000(3.38)	32,000(4.20)		
$Cr(Cp)_2^+$	d^3	?	36,630(4.11)	44,640(4.40)	
$Cr(Cp)_2$	d^4	29,590(3.43)	40,000(3.55)	>50,000(4.25)	
			44,640(4.08)		
$Mo(Cp)_2$[e]	d^4	25,100	32,800		
$W(Cp)2$[e]	d^4	25,600	30,950		
$Mn(Cp)_2$	d^5[a]		33,000(3.34)	42,370(4.13)	
				46,950(4.25)	
$Mn(Me_5Cp)_2$	d^5[b]	24,390(2.49)		39,220(3.75)	
		(31,250)(2.72)		42,920(3.88)	
				50,000(4.75)	
$Fe(Cp)_2$	d^6			(38,170)(3.85)[c]	
				(41,670)(4.13)[c]	
				50,000	
$Ru(Cp)_2$	d^6			42,000(3.30)[c]	
				46,100(3.62)[c]	
				>51,300	
$Co(Cp)_2^+$	d^6			38,000(4.58)	
$Ni(Me_5Cp)_2^{2+}$	d^6			31,500(4.78)	
$Co(Cp)_2$	d^7			27,030(3.46)	
				30,770(4.24)	
				38,460(4.28)[d]	46,080
$Ni(Cp)_2^+$	d^7			25,700	
				27,900	
				33,300	
$Ni(Cp)_2$	d^8			29,500(3.79)	
				32,630(4.05)	
				36,100(3.89)	>50,000

a) High spin. b) Low spin. c) Forbidden transitions ?, see text. d) Assigned as $2e_{1g}$---e_{2u} in [174]. CT assignments for $[Cr(Cp)_2]^+$, $[Cr(Me_5Cp)_2]^+$ and $[Ni(Me_5Cp)_2]^{2+}$ are tentatively proposed by this author. e) The assignments for $Mo(Cp)_2$ and $W(Cp)_2$ (data from [212]) are speculative.

UV photolysis of $M(Cp)_2H_2$ or $M(Cp)_2C_2H_4$, M = W,Mo, yield the corresponding $M(Cp)_2$ species in an argon matrix [212]. $W(Cp)_2$ shows a weak band at 3240 cm^{-1} which though in the IR region, is almost certainly an electronic transition since it has no obvious vibrational origin and is too broad to be a vibration. It is probably a transition within the spin-orbit components of the (presumed) $^3E_{2g}$ ground state. Both $M(Cp)_2$ exhibit two intense, probable LMCT, transitions in the near uv at energies lower than the corresponding band in $Cr(Cp)_2$ (Table 7.8).

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Chapter 8

Bio-Inorganic Chromophores

8.1 Metalloproteins and Enzymes - Introduction

Metals clearly have an important role to play in biological chemistry and inorganic chemists have long been fascinated by the coordination environment of metal ions in metalloproteins and enzymes. While stereochemical regularity is the norm in conventional coordination chemistry, Nature appears to have gone out of her way to provide irregular distorted environments for metal ions in biological niches.

Proteins have molecular weights generally lying in the hundreds of thousands making X-ray structural techniques arduous even if, as is rarely the case, crystals can be obtained. Thus spectroscopic analysis provides an enticing method for analysing metal sites if the metal ion is 'coloured'. Iron, copper, nickel, molybdenum and, to some extent, manganese have been studied and attempts made to deduce environments from purported d-d and CT spectra. For those enzymes, such as zinc, whose native metal is not coloured, techniques have been developed to replace the zinc by visible light absorbing chromophores such as Cu(II), Ni(II), Co(II), Mn(II), OV(IV) etc.

Clearly care must be taken to ensure that the new metal goes to the same site vacated by the native metal, and that the site is not modified by the incoming metal ion. Generally speaking these synthetic enzymes do not have the same biological activity as the natural enzymes, and some modification of the active site can certainly take place. Such experiments must therefore be interpreted with caution.

One general approach in this field is to synthesize model compounds designed to have metal environments closely akin to the protein environment. Comparison of the electronic spectra of the models with those of the proteins can then lead to some understanding of the latter. Many of these model systems have been discussed in this book, in Chapters 5 and 6, and, in reading this chapter, the reader should refer to parallel systems documented in earlier chapters.

Here we discuss a range of coloured proteins but exclude the porphyrins because their colour arises primarily from the organic chromophore; moreover there have been several recent reviews of their spectroscopy [1,2].

8.2 The Copper Proteins [4,198,217,220]

While most proteins are colourless, many (but not all) copper-containing proteins are deep blue and evoked thereby the interest of inorganic chemists early in the development of bio-inorganic chemistry. Several classes of copper protein exist depending upon the number of copper atoms per unit of protein, viz:

1 copper/protein unit - e.g. Azurin, Stellacyanin, Plastocyanin etc (blue), Galactose Oxidase (GOase) (non-blue)

2 copper/protein unit - e.g. Hemocyanin, Tyrosinase

4 copper/protein unit - e.g. the Laccases (contain a 'blue copper' chromophore)

8 or more copper/protein - e.g. Ceruloplasmin, Ascorbate Oxidase (both contain 'blue copper')

An arsenal of spectroscopic techniques has been used to elucidate the local structure around the copper atom; these include CD, MCD, uv/vis (including polarised light studies), IR, PES, EXAFS, ESR etc. The development of our understanding of these systems has an interesting history but space constraints preclude full discussion of this here, and emphasis is placed on the current understanding of these systems, and of the problems which wait to be fully resolved.

As developed in detail below, there do appear to be at least three types of copper(II) which can be found in the copper proteins, viz:

Type 1 Copper - Blue copper(II), tetrahedral paramagnetic, as in Stellacyanin, Azurin etc.

Type 2 Copper - Tetragonal paramagnetic (non-blue) copper(II), probably in Dopamine β-Hydroxylase, Galactose Oxidase and the Amine Oxidases.

Type 3 Copper - Epr non-detectable coupled pairs as in Hemocyanin and Tyrosinase.

Mixtures of several types of copper(II) can be found in some of the polycopper proteins such as the laccases (see below).

Since copper(II) complexes are often blue, the colour of the blue copper proteins was initially attributed to d-d absorption with intensity enhanced due to low symmetry. The blue colour arises from a band near 16,600 cm^{-1} with an intensity of ca. 6000 l mol cm^{-1}. This is too great to arise from a d-d transition, however distorted the complex, and is now unequivocally attributed to a charge transfer transition.

Proteins being composed of amino acids are quite limited in the

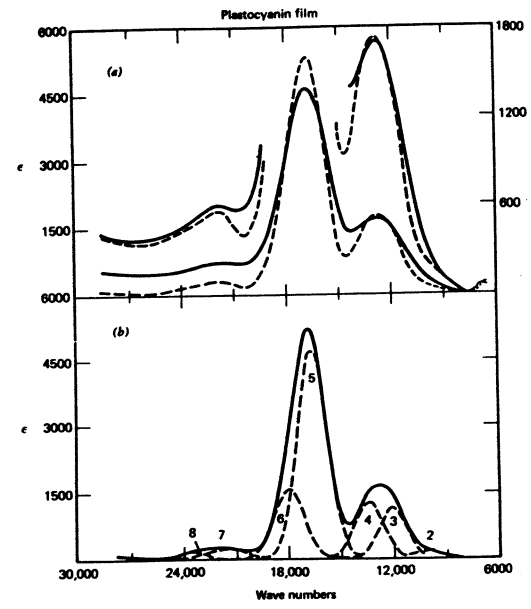


Fig.8.1 Visible and near uv spectra of plastocyanin a) Absorption spectrum at 270 K (solid line) and 35 K (hatched line). b) Gaussian resolution of the 35 K spectrum [4].

ligating atoms which bind to the metal atom. Specifically these are:

N from imidazole (histidine) or an amide linkage

O from water, a carboxylic acid residue, amidic CO linkage or phenol (tyrosine)

S from cysteine, cystine, or methionine

The charge on each of these species may be zero or -1 depending upon their source.

Consideration of optical electronegativities (see Chapter 5) exclude Cu-O or Cu-N charge transfer occurring as low as 16,600 cm^{-1}.

Recall however that CT band energies are dependent upon coordination number and the extent of d orbital destabilisation. Although it is improbable that 4,5 or 6 coordinate copper(II) complexes could exhibit Cu-O or Cu-N CT absorption as low as 16,600 cm^{-1}, complexes of lower coordination number may conceivably do so, and

especially if the O or N ligand is charged.

Cu-S is a prime candidate and copper-sulphur CT is indeed the source of this absorption. Since there are no low lying acceptor orbitals on sulphur, and copper(II) is easy to reduce but difficult to oxidise, the blue absorption band must arise from sulphur-copper LMCT. It is clearly of interest to know what kind of sulphur is bound to copper(II), i.e. from methionine, cysteine or cystine, charged or uncharged ? As demonstrated below, detailed spectroscopic study of the proteins, and of model systems, including single crystal studies with polarised light, have been exceptionally successful in unravelling the structures of these species. Indeed, extensive studies of synthetic copper-sulphur (and copper-imidazole) derivatives, have been spawned specifically to solve the copper protein problem (see Chapter 5).

8.2.i) Azurin, Plastocyanin and Stellacyanin (1 Cu/protein)

These three metalloproteins have similar spectra though there are some subtle differences [3,4]. All possess a distorted tetrahedral chromophore as indicated by characteristic low energy electronic absorption bands in the near infrared. X-ray structural analysis has confirmed a tetrahedral CuN_2S_2 environment both for plastocyanin [5] and azurin [6].

The nitrogen atoms originate from imidazole (histidine), one sulphur belongs to a cysteine residue and the other to a methionine residue, this last at an exceptionally long distance.

The spectroscopic study of these proteins was pioneered by Harry B.Gray and co-workers who were the first to predict a tetrahedral stereochemistry. The spectrum of Stellacyanin at ambient and low temperatures [7], shown in Fig.8.1 may be broadly divided into three overlapping regions:

i) d-d absorption (6000 - 12,000 cm^{-1})

ii) Copper-sulphur LMCT (12,000 - 21,000 cm^{-1})

iii) Copper-imidazole LMCT (21,000 - 25,000 cm^{-1})

Azurin and plastocyanin show similar absorption. The weak absorption below 12,000 cm^{-1} is associated with d-d absorption arising from a distorted tetrahedral copper(II) ion, a conclusion supported by mathematical analysis of the data, by CD and MCD techniques, and because the cobalt analog (replacement of copper(II) by cobalt(II)) yields a spectrum fully characteristic of tetrahedral cobalt(II)[9].

Thus Gray and co-workers [8,120] analysed these data in terms of a flattened tetrahedron of D_{2d} symmetry, and derived a distortion angle of 60-61° (ϕ) (Table 8.1).

Table 8.1

TRANSITION ENERGY ASSIGNMENTS IN PLASTOCYANIN, AZURIN, STELLACYANIN[a] [8]

	Plastocyanin	Azurin	Stellacyanin
d-d			
$^2E--^2B_2$	5000(100)[b]	5800	5250 cm^{-1}
$^2B_1--^2B_2$	9350(200)	ca 10,000	8800
$^2A_1--^2B_2$	11,810(1162)	[12,410][c]	11,110(565)
CT			
$\pi S--x^2-y^2$	13,330(1289)	12,830(779)	12,680(789)
$\sigma S--x^2-y^2$	16,500(4364)	15,850(3798)	16,220(3550)
?	18,100(1163)	17,650(504)	17,490(1542)
$\pi N(Im)-x^2-y^2$	22,200(300)	20,790(198)	22,210(942)
$\pi N(Im)-x^2-y^2$	23,340(100)		
Circular dichroism data [219][e]			
	12,800(-3.78)	12,500(-5.9)	12,800(-5.0)
	16,500(+4.08)	16,100(+6.5)	16,500(+3.6)
	19,000(+0.4)	19,000(+1.2)	19,000(+0.75)
	21,200(-1.32)	21,400(-1.8)	22,400(-7.35)
	24,000(+1.6)	24,500(+0.3)	

a) The d-d data are from the absorption spectrum at 35 K; the CT data are from Gaussian analyses At 270 K. Extinction coefficients in parentheses. b) Observed in MCD. c) Calculated. d) Methionine S or cysteine S to Cu(II) ?. e) Differential extinction coefficient in parentheses. The parameter sets derived from equation (8.1) are: Plastocyanin $\phi = 60°$, Ds = 691, Dt = 465; Azurin $\phi = 61°$, Ds = 700, Dt = 480; Stellacyanin $\phi = 60°$, Ds = 765, Dt = 444 cm^{-1}.

The d electron hole energies are given by:

$$W(^2A_1, z^2) = -12(3\cos^2\phi - 1)Ds - 3\delta\, Dt$$
$$W(^2B_1, xy) = 12(3\cos^2\phi - 1)Ds + 1/2(35\sin^4\phi - \delta)Dt$$
$$W(^2E, xz, yz) = -6(3\cos^2\phi - 1)Ds + 2\delta\, Dt$$
$$W(^2B_2, x^2-y^2) = 12(3\cos^2\phi - 1)Ds - 1/2(35\sin^4\phi + \delta)Dt$$

where $\delta = 35\cos^4\phi - 30\cos^2\phi + 3$.

(8.1)

The goodness of the fit was evaluated in terms of agreement with observed data, and the prediction of a reasonable value (near 21,000 cm^{-1}) for the square planar limit where $\phi = 90°$.

The replacement of copper(II) in these proteins by cobalt(II), also proved useful in identifying the charge transfer bands exhibited by these species. All three species also yield very similar cobalt(II) spectra supporting the view based upon the copper(II) spectra that the sites in all three proteins are very similar. Thus the copper(II) complexes exhibit their most intense visible region band near 16,600 cm^{-1}, but with a prominent lower energy shoulder near 12,000 cm^{-1}. According to the analysis above, this shoulder is not associated with the d-d absorption and is presumably another CT transition. Indeed in the cobalt complexes these bands are absent but are replaced by two new absorptions at 26-27,000 and 30-32,000 cm^{-1} (Table 8.2) whose energy separations parallel those found in the copper analogs [10]. The shift to higher energy of these bands upon replacement of copper(II) by cobalt(II) is definitive evidence for an LMCT transition (since Co(II) is more difficult to reduce than Cu(II), cf Chapter 5).

Since the short Cu-S bond is associated with cysteine (S^-) and since S^- is expected to be more readily oxidisable than the neutral S atom in methionine, it is clear that the 16-17,000 cm^{-1} band must be associated with charge transfer from cysteine S to Cu(II).

The cysteine S-atom possesses 3 lone pairs, one of which must be involved in sigma bonding with copper(II), while the remaining two will have pi symmetry with respect to the Cu-S bond. If the Cu-S-C- angle is not 180° (it is in fact 107°) the two pi orbitals are not related by symmetry and two π---Cu(II) LMCT transitions must be expected. If both the sigma and pi electron pairs are p orbitals, then the pi orbital is expected to be less stabilised by interaction with the metal, than the sigma orbital, and hence the π---M LMCT should lie at lower energy than the σ---M LMCT. Moreover in this case the acceptor orbital on the metal can only be x^2-y^2 which has sigma anti-bonding symmetry. Hence there should be a σ(S-cysteine)---σ^*(Cu(II)) LMCT strong absorption with a weaker and lower energy π(S-cysteine)---σ^*(Cu(II)) LMCT band; these are reasonably associated with the strong 16-17,000 and weaker 12,000 cm^{-1} bands respectively. This assignment is also supported by the Kuhn anisotropy factors observed for these two transitions (see Chapter 4). The π---σ^* transition is both magnetic and electric dipole allowed, while the σ---σ^* transition is only electric dipole allowed. The latter is therefore expected to have a much smaller Kuhn anisotropy factor than the former and this is indeed observed [8].

The three lone pairs comprise two 3p orbitals and a 3s orbital. If there is little hybridisation between these orbitals, one of the lone pairs will be mainly 3s in character and will be much more stable. If there is extensive mixing between 3s and 3p then transition from this orbital might occur near the other cysteine-Cu(II) CT absorption. Several sulphur ligands show PES spectra 3p excitation at 8.5 - 10.5 eV followed by a 3s excitation some 2 eV to higher energy [11]. Thus the second π---M LMCT may well be mainly 3s(S-cysteine)---σ^*(Cu(II)) in character and occur at much higher energies (in the uv ?). On the other hand the PES

spectra of the free ligands may not be good guides to the 3s-3p separation existing in the complexed species.

There remain for assignment a weak shoulder under the high energy edge of the 16,000 cm^{-1} intense LMCT band, and one or more weak bands between 21-24,000 cm^{-1}. Given that the environment of the plastocyanin and azurin Cu(II) ion site is known to be Cu(N-histidine)$_2$(S-cysteine)(S-methionine), the candidates for this absorption are S(methionine)---Cu(II) LMCT, Cu(II)---Imidazole CT, and perhaps the second π(S-cysteine)---σ^*(Cu(II)) LMCT.

On the basis of optical electronegativity arguments the methionine-Cu(II) LMCT transition may be expected to occur near the cysteine absorption, and the weak absorption near 18,000 cm^{-1} was so assigned by McMillin and Morris [11]. However the Cu-S(methionine) bond is 290 pm long and must involve little or no Cu-S overlap. Moreover the sigma Cu-S orbital is almost exactly at 90° to the acceptor x^2-y^2 orbital. For both these reasons, the intensity of a σ(S-methionine)---σ^*(Cu(II)) LMCT transition must be very small, perhaps vanishingly small. The corresponding π(S) transition must also be very weak. This argument was advanced by Solomon and co-workers [12], to dispute the methionine assignment. In addition these workers analysed the crystal spectra of plastocyanin with polarised light and demonstrated that the polarisation of the 18,000 cm^{-1} band (17,870 cm^{-1} in the crystal) was not consistent with the involvement of methionine.

Moreover despite the fact that all three of these proteins show similar bands, stellacyanin (whose structure is currently unknown) does not possess a methionine residue. Possibly the methionine S (-C-S-C-) is replaced in stellacyanin by a cystine S [13,14] (-C-S-S-C-) which may give rise to similar crystal field and charge transfer spectra.

The case for the 18,000 cm^{-1} band being assigned to the second πS(cysteine)---σ^*(Cu(II)) transition is strengthened by the observation [15] that the copper substituted analog of horse liver alcohol dehydrogenase (a zinc enzyme in its natural state) is blue and exhibits the 18,000 cm^{-1} weak absorption. It does not possess any thioether sulphur but has a Cu(N-histidine)(S-cysteine)$_2$(water) chromophore (see Section 8.4 iii)).

Once the existence of coordinated imidazole (histidine) was known, it was clear that imidazole-Cu(II) CT should be anticipated. Extensive studies of model systems carried out by Schugar and his co-workers [16] (see Chapter 5) indicate that a series of imidazole to copper(II) LMCT transitions are possible. Two π---σ^*(Cu(II)) transitions (π_1, π_2) separated by ca. 4000 cm^{-1} should occur in tetrahedral copper(II) species in the region 20-30,000 cm^{-1}. It is reasonable to assign the double band at 21-24,000 cm^{-1} in plastocyanin (and related species) to such π(imidazole)---σ^*(Cu(II)) absorption.

Note however that the polarisation of these two bands is quite different [12] suggesting that they arise from one π-level on each of the two differently located histidine units, rather than from two π-levels on one histidine. If they arise from π_1, then the

Table 8.2

CHARGE TRANSFER DATA FOR COPPER(II) PROTEINS AND TRANSITION METAL SUBSTITUTED ANALOGS[a]

	Plastocyanin	Azurin	Stellacyanin
πS-M(II)			
Cu(II)	13.3	12.8	12.7
Ni(II)		17.8, 20.0	
Co(II)	26.0	24.5, 26.7	27.4
σS-M(II)			
Cu(II)	16.5	15.85	16.2
Ni(II)		22.8	
Co(II)	30.0	30.3	32.3
?			
Cu(II)	18.1	17.65	17.5
Ni(II)		25.5	
πN(Im)-M(II)			
Cu(II)	22.2	20.8	22.2
Ni(II)		28.2	

a) This Table is based on refs.[8,13] but the data herein have been modified slightly in the light of more recent and accurate measurements and assignments.

transitions from π_2 must lie in the uv region. If they are construed to arise from π_2, then there is the possibility that the 18,000 cm^{-1} band might involve the histidine π_1 levels. Schugar's model studies favour the former interpretation.

Tennent and McMillin [13] compared CT spectra of cobalt(II), copper(II) and nickel(II) substituted azurin and observed the anticipated shifted S-M(II) LMCT transitions (Table 8.2) (in absorption and CD). The three lower energy transitions are clearly associated with the three expected cysteine to metal charge transfer bands. The fourth band, for example in cobalt stellacyanin, may be an imidazole to cobalt(II) CT band. Fig.8.2 shows the spectrum of nickel(II) stellacyanin with d-d bands around 16,600 cm^{-1} and CT bands around 25,000 cm^{-1} [4]. Possible assignments are shown in Table 8.3.

Various $Co(N_3)SR$ and $Co(N_2S)SR$ systems (and Cu(I) and Cu(II) analogs) have been reported [90-93] as models for the cobalt substituted copper proteins. X-ray data and spectroscopic information were reported.

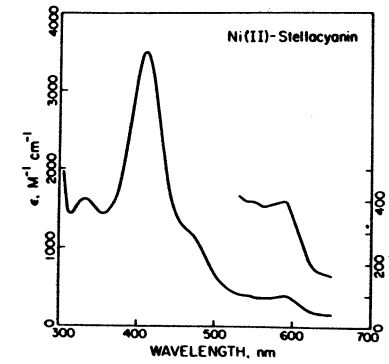


Fig.8.2 The absorption spectrum of nickel(II) substituted stellacyanin [4].

Table 8.3

NEAR ULTRAVIOLET AND VISIBLE SPECTROSCOPIC DATA FOR NICKEL(II) SUBSTITUTED STELLACYANIN AND AZURIN [17]

Ni(II) Stellacyanin	Ni(II) Azurin	
16,900(400)	17,700(320)	d-d
18,200(400)	18,500(350)	d-d
(21,300)	(20,000)	σS(Cys)---Ni(II)
24,400(3500)	22,700(3300)	σS(Cys)---Ni(II)
(27,000)	(25,000)	?
29,800(1600)	28,200(1530)	π(His)---Ni(II)

8.2.ii) Mono-Copper Oxidases - Galactose Oxidase [23,49,132,133,157]

Galactose oxidase catalyses the oxidation of a specific hydroxyl function on certain sugars. It contains one copper atom/protein (68,000 Daltons) and is not 'blue'. There is still an on-going controversy concerning the state of the copper in the active enzyme, though majority opinion would seem to agree that it is probably an example of an isolated type 2 copper(II). The Cu(II) is probably coordinated by two imidazole nitrogen atoms, solvent water and probably tyrosine. At least five electronic transitions are seen through combination of electronic, CD and MCD studies (Fig.8.3, Table 8.4).

Table 8.4

ELECTRONIC SPECTRA OF GALACTOSE OXIDASE [25]

Absorption Spectrum	Circular Dichroism
31,800(1370)	31,800(+ 18,900)
22,500(1155)	25,300(+ 3000)
	20,000(+ 1500)
15,900(1015)	16,400(- 8200)
12,900(905)	12,660(+ 2600)

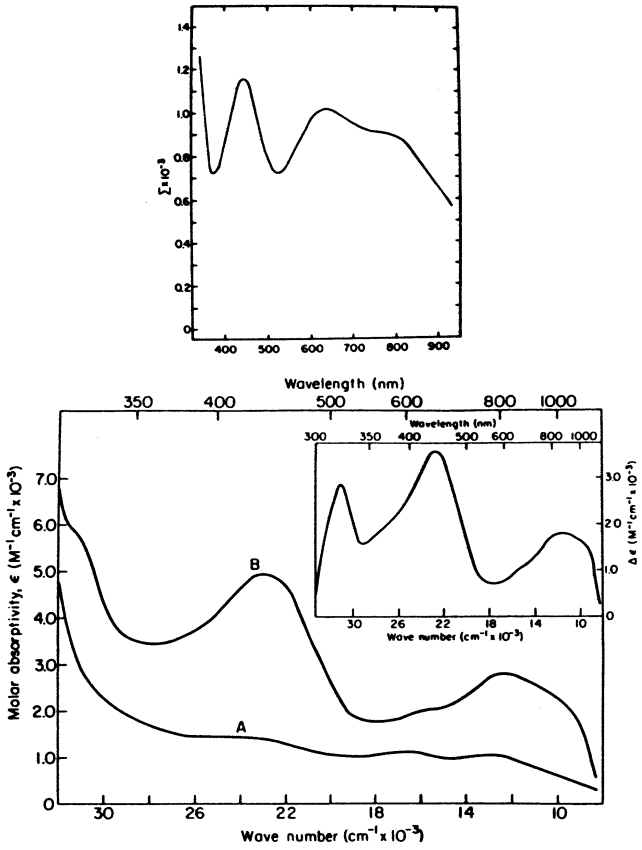


Fig.8.3 a) (Upper) The Cu(II) GOase spectrum in 0.1 M sodium phosphate [25]. b) (Lower) The spectra of resting (A) and oxidised (B) GOase in 0.10 M sodium phosphate [23]. The insert is the difference spectrum between (A) and (B).

The d-d absorption centred at 13-16,000 cm^{-1} clearly arises from a non-'blue' tetragonal copper(II) chromophore, though the intensity is certainly higher than observed in other type 2 copper systems such as the laccases (see below). The strong band at 22,500 cm^{-1} is likely to be a CT band, perhaps histidine to Cu(II). A further CT band is seen at 31,800 cm^{-1} (see Fig.8.3), possibly copper to phenolate MLCT [84]. All five bands disappear upon reduction of the copper(II). The intensities of the d-d bands are unusually high for a tetragonal species; curiously they decrease in intensity when a GOase enzyme substrate is added [24]. Note however that Bereman and co-workers argue that since the addition of cyanide or azide ion to the enzyme causes shifting (to the blue) of all the bands of energy below 30,000 cm^{-1}, but not the 31,800 cm^{-1} band, all of these lower energy bands should be considered d-d in origin [49]. Azide binding leads to an intense absorption band at 26,300 cm^{-1} which is likely azide---Cu(II) LMCT, with the high intensity requiring that the azide bind equatorially to copper(II) (see Chapter 5,6). Indeed exogenous ligands appear to bind in an equatorial site [133].

Oxidation of this enzyme with ferricyanide etc appears to lead to enhanced activity and the loss of the epr signal leading Hamilton to suggest that a formal Cu(III) species might be involved [23]. The electronic spectrum changes as shown in Fig.8.3, with a shift of the Cu(II) d-d bands to lower energy. Otherwise little change in band envelope is seen. On the basis of the small spectrscopic change, it seems unusual that oxidation of Cu(II) has occurred. The involvement of Cu(III) is disputed [25,49].

Addition of horseradish peroxidase (HRP) also leads to an increase in intensity in all the transitions and an apparent shift of the d-d bands to lower energy yielding a spectrum [156] much like that reported by Hamilton for oxidation. It is believed that O-O-dityrosine is generated through reaction of two GOase tyrosines, one or both being bound to Cu(II).

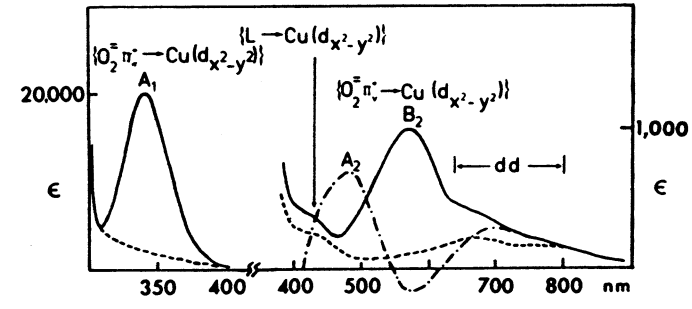


Fig.8.4 The absorption spectrum of oxy (solid line) and met (....) hemocyanin at 15 K (1:1 sucrose glass), and CD spectrum of the oxy species (.-.-.) [198].

8.2.iii) Hemocyanin, Tyrosinase, Dopamine β-Hydroxylase and Amine Oxidases (2 Cu/Protein) [198]

The hemocyanins are proteins which reversibly bind oxygen and are found in the arthropods (lobster, crab etc) and molluscs (snail, octopus etc). Tyrosinase is a mono-oxygenase and also functions as a two-electron oxidase. Both these species contain two copper atoms per protein sub-unit. In the divalent state, the copper atoms are strongly coupled resulting in a diamagnetic species. The reduced species, so-called deoxyhemocyanin and deoxytyrosinase, contain Cu(I) units which bind oxygen to produce, formally, [Cu(II)-O_2^{2-}-Cu(II)], i.e. peroxide bound to binuclear Cu(II). These are referred to as oxyhemocyanin and oxytyrosinase, and, as developed below, contain tetragonal rather than tetrahedral Cu(II). Dopamine β-hydroxylase hydroxylates dopamine, an intermediate in the conversion of tyrosine to adrenaline. It also contains two copper atoms but they appear independent of one another.

a) Oxyhemocyanin and oxytyrosinase: These compounds exhibit intense absorption near 17,550 cm^{-1}(ca 1000) and 29,000 cm^{-1}(ca 20,000) (Fig.8.4) which have no counterpart in the reduced enzyme, are not observed in the met enzyme ([Cu(II)-Cu(II)] without oxygen) and have no obvious relationship to conventional Cu(II) d-d spectra. They are evidently charge transfer in origin, and presumably involve the peroxide unit. These oxy complexes also show a strong band at 20,830 cm^{-1}, absent from the electronic absorption spectrum but seen in the CD spectrum.

> The appearance of these bands in the oxy complexes, and their absence in the deoxy and met complexes does not unequivocally require that they involve oxygen. The complexity of protein chemistry is such that it is possible that, upon oxygenation, some other protein site binds the metal, a site that is not involved in the deoxy and met forms.

Resonance Raman (RR) experiments provided evidence for the detailed assignment of these strong bands. When the excitation is centred in the 17,550 cm^{-1} band, a Raman active ^{16}O-^{16}O stretching vibration is enhanced at 749 cm^{-1}, a frequency corresponding to a peroxide link. The use of oxygen isotopes both confirmed the assignment and indicated that the binding of the peroxide was symmetric, i.e. both oxygen atoms are chemically equivalent [7]. Thus, on the basis of the discussion in Chapter 5, this band is assigned as (O_2)---Cu(II) LMCT. Excitation into the 29,000 cm^{-1} band enhances vibrations at 267 and 226 cm^{-1}, frequencies not affected by oxygen isotopic substitution, but perturbed by both copper and deuterium isotopic substitution; they are therefore copper-ligand, probably Cu-N modes [18].

Despite the resonance Raman result, histidine-Cu CT is an unlikely assignment for the 29,000 cm^{-1} band because there is no resonance Raman enhancement of internal histidine (imidazole) modes.

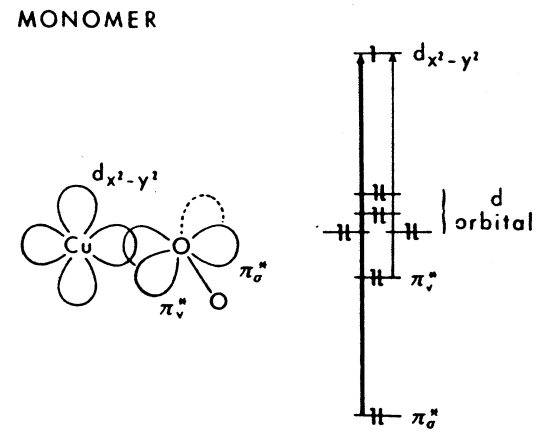


Fig.8.5 Energy level diagram for the monomeric Cu(II) dioxygen interaction [198].

Simultaneous pair excitation seems excluded by the high intensity of this transition. Solomon and co-workers prefer to assign both the 17,550 and 29,000 cm^{-1} to (O_2)---Cu(II) LMCT using the following argument [19,198].

Consider a monomeric Cu-O_2 interaction (Fig.8.5). Two LMCT transitions originating from π_v^* and π_σ^* (see Chapter 5, but note slightly different nomenclature used by the several different authors discussing this topic) are permitted, the latter with much higher intensity than the former because of better overlap with the copper x^2-y^2 acceptor orbital. When a binuclear molecule is constructed by adding an extra Cu(II) atom, the two resulting pairs of transition moments will couple in a symmetric and anti-symmetric fashion, resulting in [198]:

	Monomer	Dimer	Selection Rules
π_v^*---x^2-y^2	A_2	A_2	R_z
		B_2	y,R_x
π_σ^*---x^2-y^2	A_1	A_1	z
		B_1	x,R_y

The 17,550 cm^{-1} band is then B_2 and the intense 29,000 cm^{-1} band A_1 or B_1 with the second component obscured by higher energy protein absorption. The A_2 transition is only permitted in CD and is clearly seen at 20,830 cm^{-1} (Fig.8.4). The B_2 transition involves excitation of a π_v electron localised strongly on oxygen

and hence this band shows strong resonance Raman enhancement of the O-O stretching vibration. The 29,000 cm^{-1} band involves the σ^* electron which is more closely associated with copper; thus resonance Raman enhancement of Cu-L rather than O-O modes is seen [19,198].

The high intensities of these transitions require that the peroxide group be part of the equatorial plane of the copper rather than lie on the axis (see Cu(II) in Chapter 6). Consideration of the relative intensities of the LMCT bands and their energy separation led Solomon to conclude that the peroxide must be bound through both oxygen atoms, Cu(II)-O-O-Cu(II) (though not necessarily linear), rather than one oxygen bridging two copper atoms, and the remaining oxygen uncoordinated as in bridging carbonyl derivatives, i.e. 8.2(a) not 8.2(b), a result consistent with the resonance Raman experiment.

Following Solomon [19,198,220] the intensity ratio of the pairs of coupled transitions A and B is given by the square of the ratio of the vector sum to the vector difference of the transition dipoles, M_1 and M_2 on the monomeric species, viz:

$$I_A/I_B = |M_1 + M_2|^2/|M_1 - M_2|^2$$

The energy splitting between the two transitions is given by the interaction energy between two transition dipoles of magnitude M, at a distance R:

$$E(A - B) = (-2|M|^2/R^3)(3\cos^2\theta - \cos\phi)$$

where θ is the orientation of M with respect to the Cu-Cu axis, and ϕ is the angle between the two dipoles.

(a)

(b)

(8.2)

The group R is part of the protein and is required for several reasons including providing a pathway for the magnetic interaction; it is present even when the peroxide link is absent.

Absorption due to d-d transitions is observed on the low energy edge of the 17,550 cm^{-1} band but is obscured elsewhere. The shoulder at 23,530 cm^{-1} is seen both in oxyhemocyanin and met-hemocyanin and must arise from a L---Cu(II) CT band, where L is one of the protein coordination sites, phenolate ?. The data are summarised in Table 8.5.

Table 8.5

ELECTRONIC TRANSITIONS IN HEMOCYANIN, TYROSINASE AND AMINE OXIDASE DERIVATIVES[a]

Species	d-d Absorption	CT Absorption	Intervalence
Hemocyanin			
Met apo			
N_3	14,085(200)	23,810(2500)	
		18,830 CD[b]	
NCS	12,820	24,390(<300)	
		18,520(<300)	
Met		23,530(390)[c]	
H_2O	15,385(200)[b]		
	11,000(60)[b]		
N_3	13,890(200)[b]	26,200(1180)[b]	
Half-met			
N_3	12,500(200)	20,410(1000)	6450(1200)
xs N_3		20,410(1000)	
		24,390(1500)	
NO_2	14,705(100)		
ca	(11,000)		
OAc	13,890(100)		
	11,000(50)		
F	13,795		
Cl	14,925(500)		10,900(400)
Br	14,085(500)		10,100(600)
I	13,515(510)	23,255(500)	8400(1200)
		19,230(170)	
Oxy[d]	13,800(230)	17,500(1000)	
		20,250(250)	
		20,835 CD	
		(23,000)(360)	
		28,850(20,000)	
De-oxy CO		32,260(1800)[e]	
Tyrosinase			
Half-met			
N_3	15,835(565)[b]	12,445(440)[b]	
Met			
N_3	14,490	27,770(2100)	
		23,810(1400)	

cont

Table 8.5 cont

Species	d-d Absorption	CT Absorption	Intervalence
Tyrosinase (cont)			
Oxy	13,650 CD	16,950	
		19,230 CD	
		22,530[f]	
		28,985(16,500)	
Pig Kidney Diamine Oxidase			
Met		(25,000)[g]	
		(20,800)[g]	
+ N_3	15,400(300)	24,400(6300)	
+ NCS		(20,800)[g]	
		27,400(3000)	

a) Several different species of hemocyanin are incorporated here. See original literature for differences between these species. Data collected from refs [4,19,20,198,216]. b) measured off published spectra - approximate. c) phenolate (protein)---Cu(II) CT. d) Busycon canaliculatum. e) Emission 18,180 cm^{-1}. f) Low temperature spectrum. g) Probably not directly associated with Cu(II).

The 17,550 cm^{-1} band involves the π_v electron primarily primarily on the oxygen molecule and therefore excitation into this band leads to marked (O-O) stretching resonance Raman enhancement. Solomon argues that the 28,985 cm^{-1} band involves the π_σ electrons which are more closely associated with the copper atom. Thus resonance Raman enhancement of Cu-L modes is observed, but enhancement of (O-O) is minimal.

To ascertain the detailed nature of the site, Solomon and co-workers [19,20,198] undertook some especially beautiful studies in which they derived hemocyanins with [Cu(II)Cu(II)] (methemocyanin) bound to various ligands such as azide, cyanide, nitrate, halides etc, and also generated hemocyanins containing only one Cu(II) atom [Cu(II)-] (met-apo hemocyanin), and some containing [Cu(II)Cu(I)] (half-met hemocyanin). By combining esr, visible spectra, EXAFS etc, a picture of the active site is emerging.

Addition of azide ion to the [Cu(II)-] species provides an electronic spectrum (Fig.8.6) (and esr spectrum) consistent with a tetragonal stereochemistry, but with a band maximum rather lower in energy than anticipated (Chapter 6) for a regular square planar system. A five coordinate or six coordinate tetragonal stereochemistry is implicated. Fig.8.6 also shows a strong band at 23,810 cm^{-1} due to azide (πN_3)---Cu(II) CT. The CD spectrum shows two resolved transitions due to the splitting of the otherwise two-

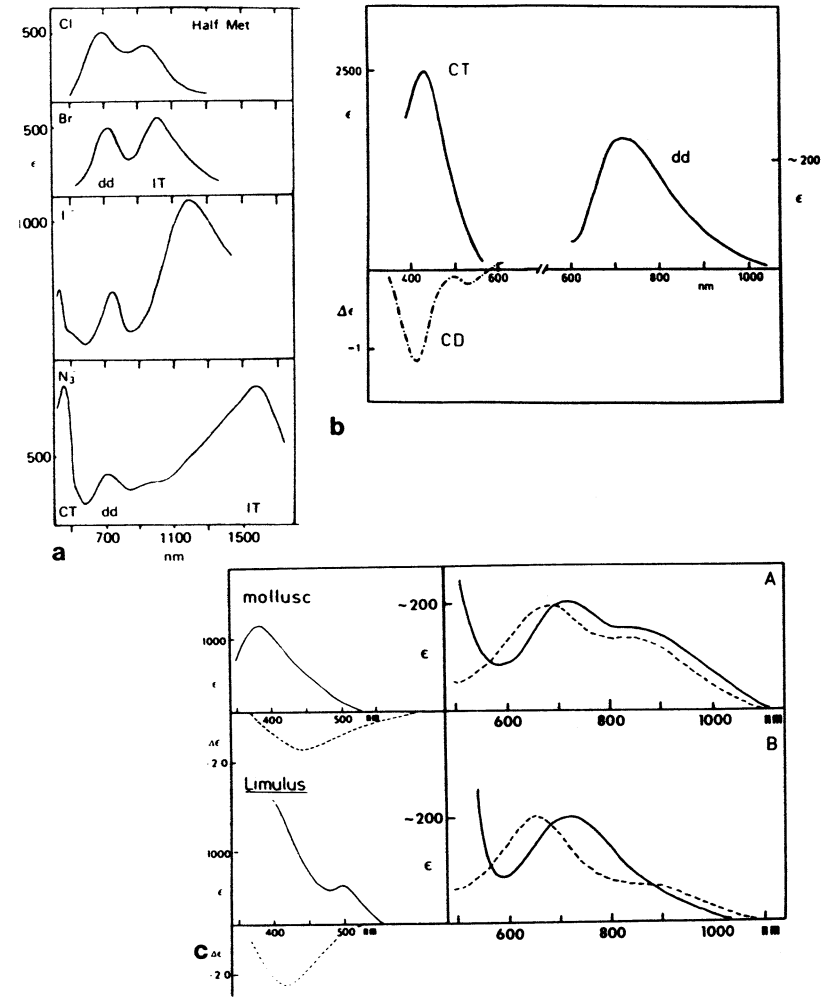


Fig.8.6 a) (Upper left) The absorption spectra (ca 15 K) of the half-Met-L Hemocyanin derivatives (Busycon), with L as indicated. b) (Upper right) The absorption (solid line) and CD (-.-.-) of Met Apo-N_3^- Hemocyanin recorded at ambient temperature (CT region) and ca 15 K (d-d region) [198]. c) (Lower, left) The absorption (solid line) and CD (---) charge transfer region spectra of the met-azide derivatives of the hemocyanines indicated. (Lower, right) The absorption spectra of the met-azide (solid line) and met-aquo (---) derivatives of the hemocyanines indicated [198].

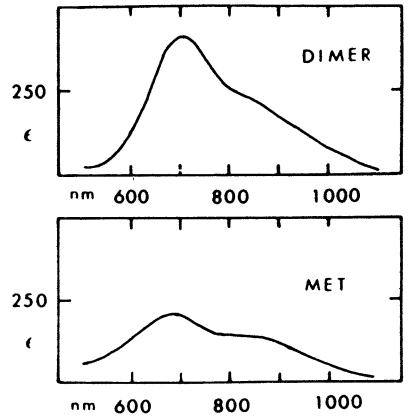


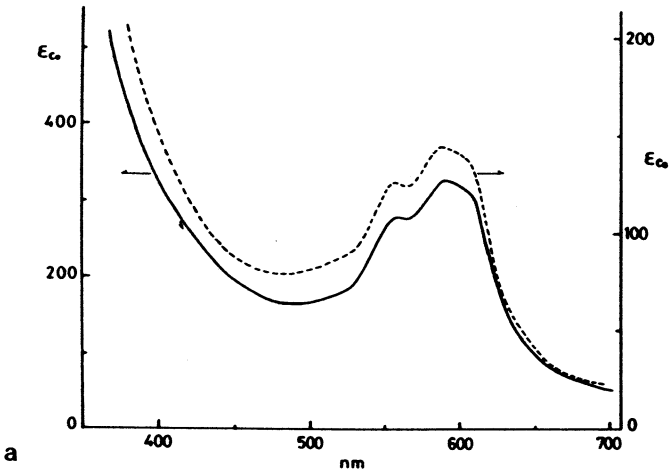
Fig.8.7 The absorption spectra (7 K) in the ligand field region of Dimer and Met Hemocyanin (Busycon) [198].

fold degenerate π orbitals of azide. The intensity of this transition requires the azide ion to be binding in an equatorial site (and hence serves to show that there is a labile equatorial site present in the protein).

The [Cu(II)Cu(I)] half-met species exhibits an additional band in the near infrared attributable to an intervalence transition (see Chapter 7)(Table 8.5). This mixed valence species exhibits class II intervalence behaviour (see Chapter 7). Exogenous (external to protein) ligands will also bind to the half-met species. Ligands such as nitrite, acetate, halides bond only one ligand very strongly (more so than to the met-apo complex), and are referred to here as the group 1 ligands. Ligands such as azide, cyanide and thiocyanate bind two ligands (group 2). Analysis (Chapter 7) of the energy and half-bandwidth of the intervalence band of the mono-azide complex shows a Cu(II)-Cu(I) distance of about 510 pm. Evidently the azide group is bridging the two metals. This bridging azide exhibits a lower energy CT band (20,400 cm^{-1}(1000)) than the terminal azide in the met-apo derivative (Table 8.5). When excess azide is added, a second band attributed to terminal azide CT is seen at 24,390 cm^{-1}.

The group 1 ligand complexes show an intervalence band consistent with an intercopper distance of only 400 pm. Apparently the endogenous (bound to protein) bridge is retained in the group 1 complexes but broken in the group 2 complexes allowing thereby binding of a second exogenous ligand in the latter case, viz:

(8.3)



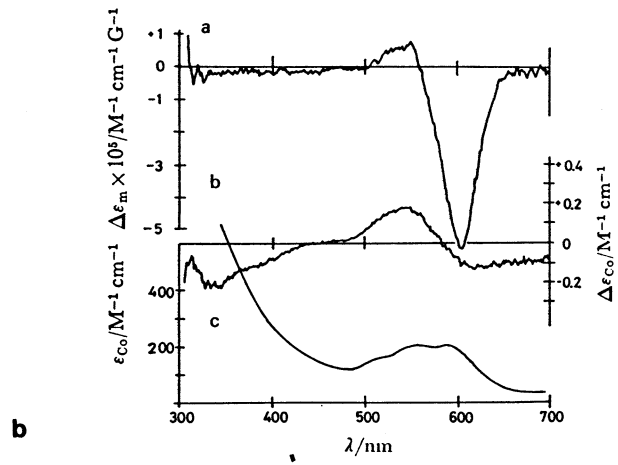


Fig.8.8 Absorption spectrum of half-apo Co(II) hemocyanin (solid line) and excess-Co(II) hemocyanin (hatched line) at ambient temperature [80]. b) MCD (upper), CD (middle) and absorption (lower) spectra of T.gigas Co(II) hemocyanin [135].

The [Cu(II)Cu(II)] met enzyme is also diamagnetic and has a typical tetragonal Cu(II) spectrum (Fig.8.7). This figure also shows the electronic spectrum of the so-called dimer, or esr detectable [Cu(II)Cu(II)] which is paramagnetic. Esr evidence shows an intercopper distance of about 600 pm. Azide binds in different

fashions to different hemocyanins. In some strains, three CT bands are observed, a fact explainable by azide binding to two copper atoms [21,76].

The absence of any intense visible region bands in these Cu(II) containing species precludes sulphur as a donor atom. As far as the other binding (protein) ligands are concerned, information from other techniques leads to the supposition that several histidines are involved [4].

The de-oxy [Cu(I)Cu(I)] has no visible absorption. However it does form a carbonyl derivative which absorbs at 32,250 cm^{-1} (1800), and luminesces at 18,180 cm^{-1}, probably both arising from Cu(I)---CO CT.

There are some marked differences between hemocyanins obtained from different natural sources. The interested reader should refer to the detailed literature on this subject [4]. Recently Co(II) substituted hemocyanins have been prepared [80,135]. Absorption, CD and MCD data were reported for a half-apo [Co(II)-] and a met [Co(II).Co(II)] system. One cobalt atom appears to occupy a tetrahedral site, as is evident in Fig.8.8.

Tyrosinase behaves in a fashion very similar to that of the hemocyanins [198,220] and close structural similarity is assured despite their marked difference in biological function. In this case the 28,985 cm^{-1} (π^*O_2)---Cu(II) band shows somewhat greater resonance Raman enhancement of the (O-O) stretch together with more obvious Cu-L mode enhancement [21] (see Table 8.5). Apotyrosinase has been re-metallated with cobalt(II). Two cobalt(II) ions are inserted and a tetrahedral chromophore is generated (bands at 8500(30), 10,400(15) (to $^4T_2(F)$?), 15,750(469)sh, 16,500(670), 17,750(630) and 19,000(465) cm^{-1}) indicative of a significant number of high field ligands, presumably nitrogen, around Co(II) [142].

Little is yet known about the copper in dopamine β-hydroxylase [143], though it seems evident that isolated type 2 copper ions are involved [27].

Amine oxidases are pink or yellowish in colour due both to copper and protein absorption in the visible region. They catalyse the oxidation of secondary amines to aldehydes and possess one copper in a subunit of about 85,000 Daltons, the enzyme containing two such units. The copper is divalent in the resting enzyme and shows broad absorption between 20-25,000 cm^{-1} [26], in a tetragonal environment. However this band is apparently not associated directly with the copper chromophore [41].

Dooley and Golnik [134] have studied the action of monodentate binding ligands upon pig kidney diamine oxidase and its activity. Both azide and thiocyanate bond to copper and generate LMCT bands at 24,400(6300) and 27,400(3000) cm^{-1} respectively. The high intensity is indicative of binding to an equatorial site. The azide spectrum also shows a weak peak at 15,400 cm^{-1} surely d-d in origin. As in many studies of this type, binding constant data for

the exogenous ligand were obtained from the slope of a plot of intensity of the LMCT band against the added ligand concentration.

8.2.iv) Poly-Copper Proteins - Laccases, Ceruloplasmin, Ascorbate Oxidase

Poly-copper proteins, those with more than 2 Cu/sub-unit, generally possess several different types of copper(II), classified as above.

The blue copper oxidases under discussion here have copper content as follows [4]:

	Type 1	Type 2	Type 3	Total
Laccases	1	1	2	4
Ceruloplasmin	2	1	4	7
Ascorbate Oxidases	3	1	4	8

where some uncertainty still exists concerning the copper content of the latter two species, e.g. see [113].

Absorption spectra occurring at energies lower than 25,000 cm^{-1} seem associated almost exclusively with the type 1, 'blue' chromophore whose intense absorption obscures the weaker contribution from type 2. A band near 30,300 cm^{-1} is clearly associated with the type 3 binuclear chromophore and also shows some perturbation when the exogenous ligands binding type 2 copper are changed.

Extensive evidence [28,29,42,128,130,131,158] shows that the blue site in the various laccases is, in fact, very similar to the blue site in the mono-copper blue proteins discussed above. Some data are shown in Table 8.6.

Malmstrom, Gray and co-workers [22,130,131] have extensively studied the absorption, CD and MCD spectra of both tree and fungal laccases. Some representative spectra, at cryogenic temperatures, are shown in Fig.8.9. The similarity with the blue proteins is quite striking. Weak absorption below 13,000 cm^{-1} is evidently d-d in origin from a distorted tetrahedral site, and the peak and shoulder near 16,000 cm^{-1} are sulphur to copper CT. In parallel with hemocyanin and tyrosinase, the absorption near 29,000 cm^{-1} must be associated with type 3 copper pairs. In agreement with these similarities, the MCD spectra of plastocyanin, azurin, ceruloplasmin and both laccases are strikingly similar [22]. However these workers note that the detailed CD spectra of tree and fungal laccase do differ in the sign of some of the features.

Little is known of the type 3 copper absorption except that it is responsible for the band near 30,300 cm^{-1} (ca 2800 l mol^{-1} cm^{-1}), which disappears if the type 3 copper(II) is reduced. When ligands known to bind type 2 copper are added, this band is perturbed implying either that type 2 copper also absorbs near 30,000 cm^{-1} or that the type 2 copper(II) is structurally close to the type 3 binuclear pair and perturbs its environment [130,131].

Native laccase does not contain fully oxidised copper. It is estimated that the intensity of the 30,300 cm^{-1} absorption in fully oxidised laccase would be about 3600 ± 900 l mol^{-1} cm^{-1} [215].

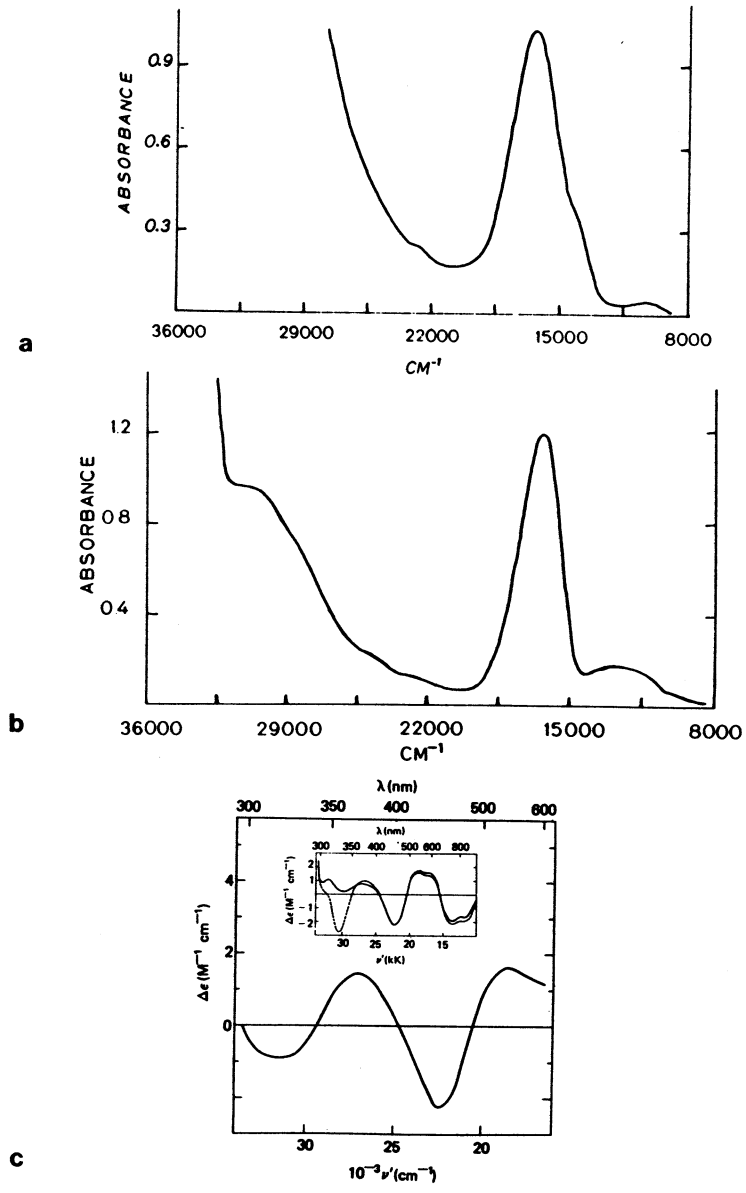


Fig.8.9 a) Absorption spectrum of fungal laccase on a Plexiglas disc at 20 K [29]. b) The absorption spectrum of tree laccase on a quartz disc at 20 K [29]. c) The reaction of partly reduced laccase with oxygen. CD spectrum of product obtained from reaction of oxygen with laccase anaerobically reduced with ascorbate. The insert shows the CD spectrum of native laccase (solid line) and of 'peroxy-laccase' (see text) obtained from reaction of laccase with hydrogen peroxide [37].

It has proven possible [83,85,107] to selectively remove the type 2 chromophore and hence to monitor the other copper ions in a type 2 depleted laccase (T2D laccase). As initially formed T2D laccase has fully reduced type 3 copper with no absorption near 30,000 cm^{-1}. T2D laccase containing reduced type 3 copper provides a clear spectrum of type 1 copper which shows a band at 16,300(5700) cm^{-1} [220]. Oxidation with hydrogen peroxide restores the absorption (30,300 cm^{-1}, (2000)), together with weak bands at 25,000 and 13,420 cm^{-1} [83] but has no effect on the esr spectra [220]. New features (Table 8.7) are associated with the type 3 site. The difference in intensity at 30,300 cm^{-1} between original laccase and re-oxidised T2D laccase may require that type 2 copper absorb in this region. This region is associated, in hemocyanin and tyrosinase, with imidazole to copper CT, while a shoulder at 23,800 cm^{-1} may be phenol (tyrosine) to copper CT [83,84]. Data for a half-met T2D laccase (type 3 Cu(II).Cu(I)) have also been reported [85]. Spectroscopic data reveal that, in contrast to half-met hemocyanin, exogenous ligands do not appear to bridge the two coppers in the type 3 site.

Note that when type 2 copper is removed selectively from laccase, the 'blue' chromophore changes slightly (loss of some intensity) which may lead to the conclusion that the two types of copper are fairly close together in the enzyme [107].

Azide binding to oxidised T2D laccase yields a CT band at 22,220 cm^{-1} but with a fairly small binding constant (K = 200 M^{-1}). With native laccase however, the binding is much stronger (K = 60,000 M^{-1}) and bands are seen at 20,000(500) and 24,390 (630) cm^{-1} [220,221]. As the type 1 and 2 esr are unaffected, these transitions are associated with the type 3 site. At high azide concentration, an additional band at 25,000(1900) cm^{-1} due to N_3^---Cu(II) LMCT appears and the type 2 esr signal changes. Resonance Raman data using these bands show enhancement of the azide stretch but isotopic data reveal that the binding is asymmetric [86]. The data indicate asymmetric azide binding to the type 3 site with strong enhancement of the bonding when type 2 copper is present.

Re-oxidised met-T2D laccase has an electronic spectrum, (after exclusion of the contribution due to type 1 copper(II)), very similar to met-hemocyanin sharing weak, structured d-d absorption centred near 13,000 cm^{-1} and further weak absorptionx near 23,800 cm^{-1} probably due to RO(tyrosine)---Cu(II) LMCT. However met-T2D laccase shows an intense feature at 30,000 cm^{-1} (Table 8.7) (more diffuse in met-hemocyanin) which may be his---Cu(II) LMCT [216]. EXAFS excludes a contribution from S---Cu(II) LMCT [219].

Indeed the type 3 binuclear unit in some laccases can mimic hemocyanin. If fully reduced Rhus laccase reacts with oxygen, the

30,300 and 16,125 cm^{-1} absorption recovers immediately, but the type 2 copper is reoxidised more slowly. Indeed during this re-oxidation, a new spectroscopic feature at 27,780 cm^{-1} (1000-3000)(with negative CD), attributed to an oxygen adduct of laccase appears (Fig.8.9c). This feature is lost when the type 2 copper is fully re-oxidised. Possibly the oxygen atom receives three electrons rapidly (from types 1 and 3), and remains on the laccase unit until the fourth (and kinetically slower) electron is received from the type 2 copper [37]. This same transition is present under steady state oxidising conditions and can also be formed when oxidised laccase reacts with hydrogen peroxide. Note that the intensity is rather lower than observed with oxyhemocyanin. Extensive studies of this oxygen intermediate have been undertaken [37-40,216].

The most recent work [216] suggests that native laccase contains about 25 reduced type 3 sites may be re-oxidised by hydrogen peroxide; however it is proposed that peroxo binding does not occur. This last conclusion follows from the absence of any intense additional features caused by addition of hydrogen peroxide to native or T2D laccase and which would be associated with peroxo---Cu(II) LMCT [216].

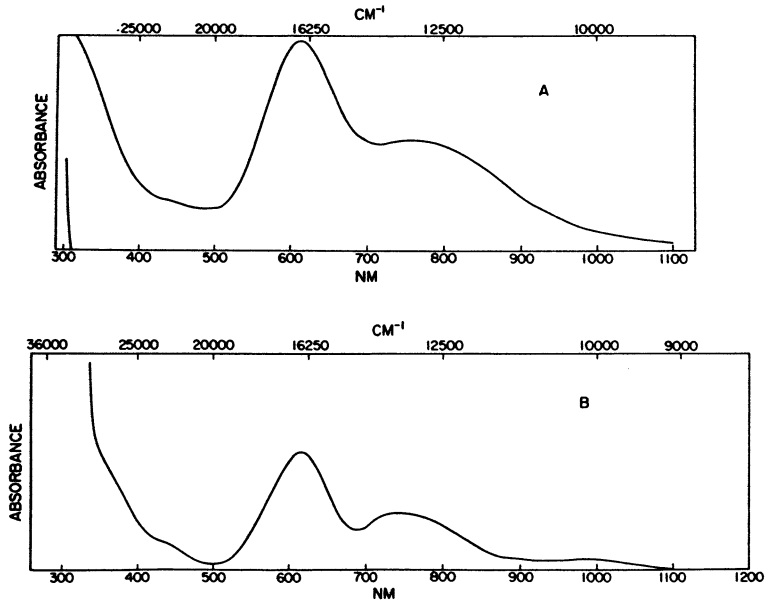


Fig.8.10 The absorption spectra of ascorbate oxidase. A:room temperature in solution (intensity at 610 nm is 9600 l mol^{-1} cm^{-1}), and B:thin film at 20 K [113].

Table 8.6

ELECTRONIC TRANSITIONS IN TREE AND FUNGAL LACCASES [22], CERULOPLASMIN [42] AND ASCORBATE OXIDASE [113][a]

	Energy(intensity)		Kuhn Factor	Assignment
Tree Laccase	6000(100)	0.70	(0.0070)[b]	^2E---$^2B_2$
	9500(200)	0.25	0.0012	$^2B_1$---$^2B_2$
	11,700(600)	2.5	0.0042	$^2A_1$---$^2B_2$
	13,000(600)	2.7	0.0045	πS---x^2-y^2
	16,125(5700)	1.5	0.0003	σS---x^2-y^2
	23,000(530)	2.6	0.0049	πN---x^2-y^2
Fungal Laccase	7250(100)	0.45	(0.0045)[b]	^2E---$^2B_2$
	11,000(170)	0.6	0.0035	$^2B_1$---$^2B_2$
	13,000(830)	2.0	0.0024	$^2A_1$---$^2B_2$
				πS---x^2-y^2
	16,000(4900)	3.2	0.0006	σS---x^2-y^2
	19,000(950)	1.4	0.0015	d-d
	22,650(800)	1.3	0.0016	πN---x^2-y^2
Ceruloplasmin	6100	+2.0		^2E---$^2B_2$
	10,000	-0.9		$^2B_1$---$^2B_2$
	13,750	-9.0		πS---x^2-y^2
	16,400	+3.4		σS---x^2-y^2
	18,350	+5.0		σS^*---x^2-y^2
	22,000	-4.8		πN---x^2-y^2
	28,000	+0.8		Not assigned
	30,750	-0.5		Not assigned
N_3^- added				
	8940	+1.4		^2E---$^2B_2$
	12,000	-1.6		$^2B_1$---$^2B_2$
	15,300	-6.0		πS---x^2-y^2
				σS---x^2-y^2
	19,250	+2.6		σS^*---x^2-y^2
	22,500	-6.7		πN---x^2-y^2
	27,000	+8.0		N_3^---x^2-y^2
NCS^- added				
	13,000	-7.8		πS---x^2-y^2
	16,400	-2.1		σS---x^2-y^2
	18,750	+0.4		σS^*--x^2-y^2
	22,000	-10.6		πN---x^2-y^2
	25,600	+8.4		NCS^----x^2-y^2

cont

Table 8.6 cont

	Energy	Kuhn Factor	Assignment
Ascorbate Oxidase	5800	+1.6	$^2E---^2B_2$
	9100	-0.2	$^2B_1---^2B_2$
	12,000		$^2Z_1---^2B_2$
	13,250	-17.0	$\pi S---x^2-y^2$
	16,530	+7.3	$\sigma S---x^2-y^2$
	18,200	+4.6	$\sigma S^*---x^2-y^2$
	21,275	-4.9	$\pi N---x^2-y^2$
	23,810	+0.7	Not assigned
	30,300	-2.4	Type 3 Copper

a) Collection of absorption and CD data. S* refers to methionine sulphur. Note that these publications contain both electronic absorption and CD data and the two sets of energies do not generally exactly agree. See the original publications for details of absorption data not reported here. b) Estimated.

Table 8.7

ABSORPTION AND CD DUE TO TYPES 2 AND 3 COPPER IN LACCASES [22]

	Absorption	Circular Dichroism	Ref
Tree Laccase	(16,700), 30,000	(16,000), 18,500, (20,000) 24,000	22
Fungal Laccase	(16,700), 30,000	15,000, 19,500, 22,500 (28,500)	22
Met-T2D Laccasea	13,400(150)b 23,800(175) 30,300(2000)		216

a) Type 1 region CT 16,300(5400); CD 13,600(-3.0), 16,300(+1.5), 18,900(+1.4), 22,100(-2.8) [219].

Somewhat less is known of ceruloplasmin and ascorbate oxidase perhaps because the larger number of copper atoms leads to a more complex situation. However it does appear that these enzymes have copper sites very similar to those previously discussed.

Gray and co-workers have presented electronic, CD and MCD data for ceruloplasmin [42] (Table 8.6). The 'blue' spectra (two type 1 copper atoms) are very much like those of the 'blue' proteins (but with doubled intensity), and the site must be very similar. Azide and thiocyanate treated ceruloplasmim yield the extra CT bands (Table 8.6) as anticipated and the data are interpreted in terms of a greater distortion than for the one-copper blue proteins.

The treatment of native ceruloplasmin with ascorbate results in the reduction of the type 1 copper, and loss of bands at 6100,

10,000 and 11,500 cm^{-1}. Absorption and CD data were reported for the remaining type 2 and 3 copper. These data are very similar to those for monomeric tetragonal copper(II) [42].

Ascorbate oxidase has also been intensively studied by the same school [113] using electronic absorption, CD and MCD spectroscopy. The results, summarised in Table 8.6 are remarkably similar to those of ceruloplasmin. In this case the intensities are consistent with two to three type 1 copper(II) atoms. Aside from absorption near 30,000 cm^{-1}, no type 2 or type 3 copper was detected, though it is surely present (Fig.8.10).

8.2.v) The Cuprozinc Proteins

Superoxide dismutase (SOD) catalyses the disproportionation of superoxide ion to oxygen and peroxide. The mammalian enzyme consists of two equivalent units each of which contains copper and zinc sharing a common imidazolato bridge [109,111, 112,116]. The copper, also bound to three other imidazole (histidine) groups is type 2 with a d-d band centred near 14,700(300) cm^{-1} [108]. The tetrahedral zinc site has two other histidine groups and an aspartic carboxylate oxygen [116,174].

Valentine and co-workers [126,136,189,190] have discussed the spectra of native SOD and some of its derivatives in some detail. They have also recorded some CD data and have shown close similarity of these spectra with those of the mono-copper proteins. Near UV absorption can be associated with the several expected imidazole to Cu(II) LMCT (see Chapter 5) and details are presented in Tables 8.8 and 8.9.

The site appears more tetragonal than the blue site in stellacyanin etc; it is also more tetragonal in the $[Cu(II)_2 -]$ species than in the native species. The UV spectra of the $[Ag(I)_2Cu(II)_2]$ species is quite different from that of the native enzyme and it is likely that Cu(II) occupies the zinc site. Indeed the d-d bands are significantly red-shifted in accordance with a much more tetrahedral site [109]. Carboxylate to Cu(II) LMCT may occur [126].

Addition of azide ion results in a shift of the Cu(II) d-d band to 15,150 cm^{-1} while a strong azide---Cu(II) LMCT band appears at 26,810(2700) cm^{-1} [105,173], an energy consistent with equatorial binding of the azide ion. A titration with azide appears to require two azide ligands per molecule of protein. Thiocyanate, cyanate, fluoride and thiosulphate, however, have little effect upon the spectrum. Cyanide ion however causes a loss of the 15,000 cm^{-1} band, and the appearance of a new (d-d ?) band at 18,870 cm^{-1}. Combining esr and optical data, Bertini and colleagues [188] conclude that the anion replaces a histidine in an equatorial site. Moreover the cyanide complex is presumed to be square planar while other anions form five coordinate complexes with an apical water molecule. Some anion data are included in Table 8.8 [173].

Addition of cobalt(II) to the apoprotein generates $[- Co(II)_2]$ with (tetrahedral) cobalt(II) occupying the zinc sites. Subsequent

Table 8.8

TRANSITION ASSIGNMENTS IN BOVINE ERYTHROCUPREIN (SOD) DERIVATIVES [126]

Metallop	Absorption Spectrum	CD	Assignment
Cu(I	39,000(3300)[a]	+4.9	π(imH)---Cu
	35,000(1000)	+1.9	π(imH)---Cu
	29,500(250)	+0.79	π(imH)---Cu
	24,000(110)	+0.23	im⁻---Cu
	18,800	+0.21	Cu d-d
	16,400(155)	+0.70	Cu d-d
	13,000	-0.86	Cu d-d
E (pH 3.6)	39,000	+6.9	π(imH)---Cu
	32,000(225)	+0.69	π(imH)---Cu
	17,000	+0.15	Cu d-d
	13,500(131)	-0.47	Cu d-d
- BE (pH 6.7)	31,000(225)	+0.54	π(imH)---Cu
	18,000	+0.06	Cu d-d
	16,200(124)	+0.31	Cu d-d
	<12,500	<-0.44	Cu d-d
$Ag(I)_2.Cu(II)_2$ BE	>40,000(>2000)		
	33,800(1800)	+1.94	π(imH)---Cu[b]
(pH 6.2)	26,500(250)	+0.21	π(imH)---Cu
	12,000(100)	-1.47	Cu d-d
	10,000	+1.64	Cu d-d
	8000	+0.38	Cu d-d

a) Intensity per Cu(II). b) Also may contain carboxyl to Cu LMCT.

addition of copper(II) re-inserts copper into its native site. This species clearly shows spectra attributable to type 2 copper(II) plus tetrahedral Co(II) (Table 8.8) [106,110].

Addition of Cu(II) to the apoprotein replaces this metal into its original site. However the spectrum of Cu(II) differs slightly from the native enzyme with a clear peak at 28,900 cm^{-1} in the CD spectrum. The Co(II) spectrum is also slightly different in the $[Co(II)_2$ -] species relative to $[Co(II)_2.Cu(II)_2]$. Treatment of a copper free (but zinc-containing) protein with Co(II) in phosphate buffer (which is necessary) generates the $[Co(II)_2Zn_2]$ SOD with Co(II) apparently occupying the Cu(II) sites. However the spectrum is clearly that of tetrahedral Co(II) (Table 8.8). The band energies seem rather too low to be appropriate for a tetrahedral

Table 8.9

TRANSITION ENERGIES IN SOME METALLOENZYMES

Metalloprotein	d-d Transitions/CT	Ref
Vanadium Derivatives		
VO(II)CA	12,900(20) 17,200(9)	69
$[V(III)]_2TF$	15,700(130) (22,500) 27,000(2000)	82
	15,700CD 22,500CD 26,300CD	82
Chromium Derivatives		
$[Cr(III)_2]LF$	16,340(280) 22,625(520)	33
Manganese Derivatives		
$[Mn(III)_2]LF$	/ (15,625) (19,230) 22,990(9620) / (29,410)[d]	33
Mn(III)AP	/ 19,420(2460)[b]	36
Mn(III)SOD	/ 21,140(400)	70
Sweet potato AP Mn(III)	/ 19,400(2460)	102
Iron Derivatives		
Fe(III)TF	/ 21,275[b]	32
Purple Uteroferrin Fe(III)	/ 18,180	99,100
	/ CD 18,700(+18) 20,750(+16) 26,200(-16)	100
Pink Uteroferrin	/ 19,700	99,100
Sweet potato AP Fe(III)	/ 19,400	94
Fe(III) PrCat	/ 21,740	94
Fe(III) Cat	/ 22,000	94
Beef spleen AcP Fe(III)	/ 18,200(2000)	94
reduced pink version	/ 19,800	94
Cobalt Derivatives		
Co(II)CPA[c]	17,480 - 18,020	51
Co(II)CA	8200(47) (10,500) 15,600(100)	
pH = 5.9	16,200(135) 18,100(250) 19,200(180) 19,200(180)	53
Co(II)CA[e]	8200(80) 11,000(18) 15,600(260)	
pH = 8.0	16,200(280) 18,100(380) 19,200(280)	53
Co(II)CA + acetate	8300(7) 10,200(6) 14,100(9) 18,000(110) (19,400) 21,200(100)	53
Co(II)CA + aniline	8900(100) 10,500(55) (17,400) 17,900(580) 18,900(460)	53
$[Co(II)_2]LF$	/ 24,690(10,340)[b]	33
Co(II)PGM	(5715) 6450 7195 8695 (15-30) 16,130 16,810 17,545 18,690 20,620 (150-210)	39

cont

Table 8.9 cont

Metalloprotein		d-d Transitions/CT	Ref
Co(II) AP		15,625(260) 16,530(220) 18,000(380) (19,600)(335)	147
Co(II) YAD		14,100(600) 15,200(800) 16,100(800)	147
Co(II) TL		18,000(90) (20,000)	88
Co(II) NP		(19,050) 18,000(50-100) (21,050)	147
Co(II) RNAP		14,200(670) 17,100(500)	147
Co(II) YA		18,870(150) (20,400)	147
Co(II) YE		18,500(30-40) (20,400)	147
Co(II) Mtn[g]	S_4	7850(165) 8700(155) 13,460(435) 14,450(420) 16,670(245) / (25,000)(2170) 31,250(2845)	151
[- $Co(II)_2$] SOD	N_3O?	17,150(400) (17,730) (18,520)	106
[$Co(II)_2$ -] SOD	N_4?	16,530(330) 17,400(320) (18,700)	175
+ water	N_4O_2?	19,200(100) 21,700(60)	175
[$Ag(I)_2Co(II)_2$] SOD	N_3O	17,150(440) 17,800(435) (18,900)[k]	109
[$Co(II)_2(c)Zn_2(n)$] HLAD[i]	S_2NO	15,385(1200) 19,230(250) / 28,570(7000)	71,72,74
+ NADH		14,750	74
+ NADH + chromophore[h]		14,800 20,900	74
[$Zn_2(c)Co(II)_2(n)$] HLAD[j]	S_4	13,500(250) 15,240(1000) / 29,240(6500)	71,72
Nickel Derivatives			
Ni(II)CPA	N_2O_4?	9430(3) 14,600(7) 24,250(24)	44
Ni(II)PGM	O_6?	7690(5) 13,700(7) 24,390(23)	39
Ni(II)CA	N_3O_2?	15,600(20) 25,600(50)[f]	78,81
Ni(II) Mtn	S_4	(13,300)(300) (17,850)(580) / (23,250) / (29,400)(6560)	151
Copper Derivatives			
Cu(II)CPA	N_2O_2?	12,580(124)	48
Cu(II)CPA-pp	N_2O_2?	11,400(180)	48
Cu(II)CA	N_3O?	13,900(110)	38
		13,400(100) (11,500)(50)	54
- H_2O		13,000(130)	55
- I^-		12,700(400) / 22,500(2800) 28,200(3200)	55 55
- Br^-		12,700(300) / (25,000)	55
- N_3^-		13,600(330) / 25,000(2700)	55
- NCS^-		13,600(150) / 27,800(900)	55
- CN^-		14,500(190)	55
[$Cu(II)_2Zn_2$] SOD		14,700(120) / (28,600)	110,173
- I^-		/ (28,600)	173
- CN^-		18,000(110)	105,173

cont

Table 8.9 cont

Metalloprotein				d-d Transitions/CT	Ref
- N_3^-				14,900(120) / 26,700(1200)	173
				15,150 / 26,800(2700)	105
- NCS^-				14,900(120) / 28,600(800)	173
- NCO^-				14,900(80)	173
[$Cu(II)_2$ -] SOD				14,300 (pH = 6.3)	136
				12,500 15,000 (pH = 10.1)	136
[$Cu(II)_2Cu(II)_2$] SOD				12,500 15,000 / 22,200	114,136
[$Cu(II)_2Co(II)_2$] SOD				14,300(Cu) 16,900 17,600 18,700	106,110
[$Cu(II)_2$]LF[a]				14,770(620) / 22,830(4800)[b]	33
Cu(II) HLAD	77 K	N_2SO		13,200 / 17,200 (19,200) 23,250 / 27,400	15
Cu(II) HLAD	296 K	N_2SO		? / 16,100[l] ? 22,600[m] / 26,500[n]	15
+ pyrazole	77 K	N_3S ?		14,500 / 20,400 24,900	15
Cu(I) HLAD		N_2SO		/ 36,400	15

Charge transfer transitions in this table appear to the right of a slash (/). Transitions without a slash, or to the left of a slash are presumed to be d-d in origin. Note that in the multi-metal enzymes, some authors quote intensities per mole of metal, while others quote per mole of protein.
AP = alkaline phosphatase; AcP = acid phosphatase; CA = carbonic anhydrase; CPA = carboxypeptidase; PGM = phosphoglutamase; LF = lactoferrin; TF = transferrin; PrCat = protocatechuate 3,4-dioxygenase; Cat = catechol 1,2-dioxygenase; YAD = Yeast Alcohol Dehydrogenase; TL = Thermolysin; NP = Neutral Protease B cereus; RNAP = RNA Polymerase E. coli; YE = Yeast Enolase; YA = Yeast Aldolase; Mtn = Metallothionein; HLAD = Horse Liver Alcohol Dehydrogenase. Consult original literature for source of protein.
a) All lactoferrin derivatives show a strong band near 35,335 cm^{-1} attributed to deprotonated tyrosine. b) Tyrosine-metal charge transfer. c) Spectra highly dependent upon conditions; see more recent examples cited in text. d) All bands are strong and probably manganese-tyrosine CT. e) See ref.[53] for an extensive data set on Co(II) CA. f) Additional CD data have also been reported [81]. g) Rabbit liver origin. h) The chromophore is an enzyme substrate trans-4-(N,N-dimethylamino)-cinnamaldehyde. i) 'Blue hybrid'. j) 'Green hybrid'. k) Spectrum almost identical with [$Cu(I)_2Co(II)_2$] SOD. l),m),n) Intensity 2000, 500, and 600 l mol^{-1} cm^{-1} per g atom Cu(II) respectively.

CoN_4 chromophore (cf data for carbonic anhydrase). Removal of the phosphate buffer by dialysis caused a drastic change in the electronic spectrum which became typical of octahedral (or possibly five coordinate) Co(II) [175].

Several other mixed metal enzyme synthetic species have been

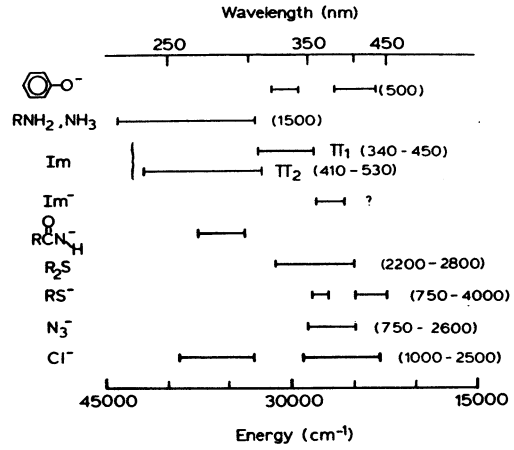


Fig.8.11 Ranges of CT transition energies for various copper(II) enzyme and enzyme model systems [220].

reported (Table 8.8). In the $[Cu(II)_2.Cu(II)_2]$ species, for example, spectra indicative of the usual copper site plus low energy absorption due to Cu(II) in a tetrahedral zinc site, are observed (Table 8.8) [114,136]. The species $[Ag(I)_2Co(II)_2]$ and $[Cu(I)_2Co(II)_2]$ have very similar spectra [109].

In an interesting pH study of a zinc free enzyme $[Cu(II)_2$ -] (Cu(II) in its native site), Valentine and co-workers [136] show that the band maximum at 14,300 cm^{-1} shifts to 12,500 cm^{-1} with increasing pH suggestive of a migration of copper from its native site to a zinc site in a neighbouring molecule. This process is reversible. The native enzyme and the $[Cu(II)_2.Cu(II)_2]$ species show no such pH behaviour. At low pH the zinc binding site has little affinity for any metal as indicated by optical data [189,190].

Table 8.9 should be scrutinised for data not specifically described in the text. Before leaving copper(II) enzymes, note Fig.8.11 which is a useful compilation of the CT energies to be anticipated for copper(II) bound to various amino acid and exogenous ligand species [220].

8.3 Non-Heme Iron Proteins

8.3 i) Transferrins [94,153]

There exist a group of proteins which bind iron very strongly and play a role in its transport in the blood. In verterbrates this role is played by a series of species commonly called the transferrins. Other names such as lactoferrin (from milk) and egg white conalbumin (ovotransferrin), uteroferrin etc are used to describe closely related proteins from non-serum sources [153].

In general these proteins contain two iron(III) atoms per sub-unit [30] and have binding sites which appear to contain both histidine and tyrosine. Some uncertainty exists concerning whether two or three tyrosine residues are bound by water. A water molecule may also occupy one site, while there also appear to be two histidine groups [94] in the site.

The iron(III) atoms have been replaced, for spectroscopic study, by various other metal ions including Cu(II), Cr(III), Mn(III), OV(IV), V(III) and Co(III) [31-33,119]. Metal binding is accompanied by bonding of an equivalent amount of bicarbonate ion, without which no complexation occurs.

All the metallotransferrins show enhancement of intensity of a band seen near 35,335 cm^{-1} in the apoprotein. This is attributed to deprotonation of one or more tyrosine residues upon complexation. The trivalent ions Mn(III), Fe(III) and Co(III) (but not Cr(III)) all show intense absorption between 20,000 and 25,000 cm^{-1} (ca 2000 - 3000 l mol^{-1} cm^{-1}) reasonably assigned to tyrosine(O)---M(III) charge transfer (LMCT), an observation supported by resonance Raman studies [95,96]. The resonance Raman of Fe(III)-serotransferrin exhibits a progression in a 1000 cm^{-1} vibration attributed to the C-O tyrosine stretch, when the protein is irradiated into the 21,250 cm^{-1} absorption [97].

The optical electronegativity of Cr(III) is such that the corresponding band in the Cr(III) transferrin should appear near 42,000 cm^{-1} where it is obscured by protein absorption [34]. The V(III) complex exhibits a band at 27,000(2000) cm^{-1} [82] which the authors suggest is too low in energy to be tyrosine to V(III). However our calculation suggests that a tyr(O)---t_{2g}(III) transition may well lie near 27,000 cm^{-1}. Table 8.9 collects some of these data.

The carbonate ion appears to function by binding to iron (Lewis base) and interacting with protein. Other anions such as thioglycolate, salicylate, glycinate etc appear able to perform both these functions resulting in a shift of the Fe-O LMCT band from about 19,800 cm^{-1} for thioglycolate to 24,390 (sh) for acetoacetate [103].

The electronic spectrum of $Fe(EDDHA)^-$ (ligand is ethylenediaminedi(o-hydroxyphenylacetate)) shows bands at 21,795 cm^{-1} (broad and strong) and 31,745 cm^{-1} (sh) ascribed to phenolic oxygen to Fe(III) LMCT into the t_{2g} and e_g levels respectively. Corresponding features and assignment are identified with the proteins. The similarity leads to the suggestion that two tyrosine residues are involved in the transferrin active site [35].

Several authors have studied model systems incorporating tyrosine-iron(III) interactions [88,89]. A key observation is an increase in energy of the CT band as the number of phenol residues bound to iron increases. Addition of imidazole ligands to the Fe(III) ion also raises the Fe-O LMCT band energy probably by decreasing the net positive charge of Fe(III).

The best model spectroscopic fit to the natural proteins appears to derive from FeO_3N_3 or FeO_2N_4 chromophores (see Chapter 5 for further detail).

Uteroferrin [94,153] is obtained from the allantoic fluid of midpregnant pigs and the uterine flushings of pseudopregnant pigs. The oxidised protein is purple with strong absorption near 18,200 cm^{-1}. Reduction with ascorbate etc provides a pink reduced protein with absorption near 19,700 cm^{-1}, and having acid phosphatase activity. The iron(III) is not reduced in this process; possibly cleavage of a disulphide linkage is involved [118,121]. Resonance Raman data [99-101] also exhibit enhanced tyrosine vibrations when irradiated into the major visible absorption of either the oxidised or reduced protein, leading to the supposition that, in both cases, tyrosine to iron CT is involved. CD data [100] reveal additional absorption to the blue of the main absorption peak, in both the purple and pink forms. These data have been interpreted [100] to require three tyrosine residues bound to iron, one orthogonal to the ferric centre (CD maxima at 18,700 and 32,800 cm^{-1}) and the other two acting as a pair with exciton coupling (CD maxima 20,750 and 26,200 cm^{-1}). However there is not general agreement as to the interpretation of these data [94].

8.3 ii) The Siderophores [154,159,160]

In the inverterbrate world, there is a group of microbial iron transport compounds called the siderophores (ferrichromes); these can be classified into three special groups, namely i) the hydroxamates, ii) the thiohydroxamates, and iii) the catecholates. These compounds all bind Fe(III) extremely strongly and exhibit LMCT (oxygen---Fe(III) and also S---Fe(III) in case (iii)) as broad absorption between 20,000 - 25,000 cm^{-1} [154]. The spectra are very similar to simple iron(III) catechol and iron(III) oximate models. Electronic absorption and CD spectra have been reported for chromium(III) complexes, such as Cr(enterobactin) to provide information, from d-d spectra (not observable in the Fe(III) systems because of the strong CT absorption), about the binding site. These appear typical CrO_6 chromophores [154,160].

8.3 iii) Catechol Dioxygenases [94,117]

There exists an extensive group of dioxygenases which catalyse the cleavage of dihydroxybenzene rings and which contain iron(III). The species have an absorption band near 20,000 - 22,000 cm^{-1} attributed to an O(tyrosine)---Fe(III) LMCT transition, again confirmed by resonance Raman studies [122]. Indeed two distinct tyrosine residues may be detected [123,124], and, most interestingly, the excitation profiles peak at widely different energies, namely 20,000 and 15,400 cm^{-1} [123,124]. The presence of two different (bound) tyrosine residues appears a common feature of this type of enzyme [94]. Water is also identified at the metal site which appears similar to that observed in the Transferrins.

When catechol binds to the enzyme it produces a substrate bound species (ES) in which catechol is probably bound to iron. Thus additional electronic absorption is seen as a shoulder near 14,700 cm^{-1} (catechol to Fe(III) CT) and resonance Raman exciting this transition shows enhancement of both tyrosine and catechol vibrations [94,122] (Table 8.8). Addition of oxygen to such a species (E = protocatechuate 3,4-dioxygenase, S = 3,4-dihydroxyphenylacetate) forms an ESO_2 ternary species absorbing at 19,230 cm^{-1} [125], also showing resonance enhancement of tyrosine modes.

Other reagents, such as carboxylates, phenols and thiophenol bind to the enzymes causing shifts in the tyrosine to Fe(III) absorption. These have been modelled by comparing them with similar shifts seen in the spectra of Fe(salen)X species (which also contains two phenolic to iron bonds) [94] (see also [98]). Que also discusses the spectra of various oxygenated enzyme complexes [94].

8.3 iv) Iron-Sulphur Proteins [137]

A range of proteins such as rubredoxins, ferredoxins, nitrogenases, hydrogenases, sulphite and nitrite reductase, aconitase, xanthine oxidase etc [137], contain Fe-S cluster molecular units. These generally possess 2, 3 or 4 iron atoms per cluster, e.g. $[Fe_4S_4(SR)_4]^{n-}$, composed of FeS_4 tetrahedral sub-units. The molecular binding is highly covalent and simple crystal field models are quite inadequate. The spectra of these species are very complex and can only be usefully analysed in terms of molecular orbital theory.

The richness of the spectra is especially evident in MCD studies, a complexity generally blurred in the room temperature absorption spectra because of the large number of possible transitions. In general weak absorption below about 18,000 cm^{-1} is probably d-d in origin, while more intense absorption to higher energy is Fe-S charge transfer.

These complexes are involved in electron transport and therefore can exist in several oxidation states, formally containing mixtures of Fe(II) and Fe(III). Studies of reduced species reveal near infrared absorption (4000 - 6000 cm^{-1}) which may be spin allowed d-d states of high spin d^6 $Fe(II)S_4$ units [139-141]. Recall that transitions of the type $^4T_1$, $^4T_2$---$^6A_1$ in Fe(III) complexes have a strong negative dependence upon Dq. Thus tetrahedral species show these transitions at higher energies than octahedral species. Rawlings and co-workers [141] identify such a transition for an $Fe(III)S_4$ unit at 13,400 cm^{-1} [141].

The spectra can be used as fingerprints to distinguish one type of cluster from another but no detailed discussion is presented here [137,138].

Table 8.10

ABSORPTION SPECTRA OF HEMERYTHRIN AND ITS DERIVATIVES, AND RIBONUCLEOTIDE REDUCTASE

	Absorption Spectra(intensity)[a]				Ref
Methemerythrin[b]					
Ligand	**A**	**B**	**C**	**D**	
O_2^{2-}	(13,300)	20,000(2200)[c]	(27,780)	30,300(6800)	171
N_3^-	14,700(190)	22,400(3700)	(26,300)	30,700(6750)	171
SCN^-	14,800(200)	22,100(5100)	(27,200)	30,600(7200)	171
Cl^-	15,200(180)	(20,400)(750	26,300(6000)	30,400(6600)	171
NCO^-	15,400(166)	(20,800)(700)	26,500(6500)	29,900(6550)	171
CN^-	14,400(140)	20,300(770)	26,700(5300)	30,300(6400)	171
F^-	(16,800)(200)	(20,800)(400)	27,600(5000)	31,500(5600)	171
OH^-	16,750(160)	(20,800)(550)	27,600(5900)	31,250(6800)	171
H_2O	(17,200)(200)	(20,800)(600)	28,200(6400)[d]	40,000[e]	171
SH^-		19,600(4000)	(27,800)(5000)	30,300(7000)	171
Br^-	14,800(165)	(19,800)(950)	25,800(5400)	30,200(6500)	171
O_2^{2-}-f		19,720(2100)		31,050(6000)	214
N_3^--g		22,300(3900)		30,400(6900)	214
N_3^-		20,500		29,000	
		(22,700)			217
Semi-methemerythrin(r)					
Ligand					
azide		21,300(2400)[e]			169
water		(20,400)(200)	(27,400)(3000)		167
Semi-methemerythrin(o)					
water ?			(27,000)(3000)		167
Ribonucleotide Reductase					
	24,400(2700)	(25,800)(3800)	25,800(5100)	(30,100)(6700)	196

Assigned d-d spectra [184] (Ground state $^6A_1$)

	$^4A_1, ^4E(^4G)$	$^4T_2(G)$	$^4T_1(G)$
Met-hemerythrin			
azide		14,700(184)	9900(10)[h]
hydroxide	20,800(604)	16,400(146)	10,100(8.0)[i]
Semi-methemerythrin(r)			
azide		13,700(80)	11,000(20)
hydroxide	20,400(360)	14,900(80)	10,050(8)
Deoxyhemerythrin		$^5E---^5T_2$	
--	11,700(9.6)	9000(4.2)	

a) Intensity per dimeric Fe(III) unit. Column A: d-d. Columns B and C: d-d where intensity is less than 800, and ligand to Fe(III) LMCT where intensity is greater than 800. Column C is oxo bridge to Fe(III) LMCT [179]. c) Polarised data show two bands here. In parallel polarisation there is a weak feature at 20,200 cm^{-1} also seen in the polarised spectrum of the fluoride and presumably a d-d band, plus a stronger perpendicular (to Fe-Fe axis) polarised band at 19,700 cm^{-1} [187]. d) This is surely too low an energy to be water to Fe(III) LMCT and must presumably be a somewhat lower lying oxo bridge to Fe(III) [179]. e) Ligand to Fe(III) LMCT. f) Lingula reevii, CD bands at 12,600(-0.4), 19,300(-3.5), 29,400(-5.3). g) Lingula reevii, CD bands at 20,000(-4.1), 27,250(-12.0). h) Gaussian resolved 10,300(5) 9500(5.6). i) Gaussian resolved 10,700(4.2) 9600(5.2) cm^{-1}. (r), (o), Prepared via reductive and oxidative pathways (see text).

8.3 v) Hemerythrin [161-169,176-184,214,216-218,220].

Hemerythrin is an oxygen transport protein found in the blood of various marine organisms. The basic unit is a dimeric iron moiety which exists in single, double or quadruple (octameric) form in different proteins. The binuclear unit is very similar in proteins from different sources, usually having a molecular weight near 13,500 Daltons. Despite the name, there is no heme unit.

X-ray structural data for the met protein [Fe(III).Fe(III)], reveal two Fe(III) atoms linked by an oxygen bridge. One Fe(III) is six coordinate with three histidine N atoms, two bridging carboxylate oxygen atoms and the oxygen bridge. The other Fe(III) atom is five coordinate, lacking one of the histidine units [180].

Extensive spectroscopic data have been reported for deoxyhemerthyrin, i.e. [Fe(II).Fe(II)], various methemerythrins, and two types of semi-methemerythrins, i.e. mixed valence [Fe(II).Fe(III)] (Table 8.10). In this last case, different versions are obtained depending upon whether the species is obtained by partial oxidation of deoxy species, or partial reduction of met species [167,170,186].

Deoxyhemerythrin contains two Fe(II) atoms and spectra very similar to isolated Fe(II) high spin species, i.e. a doublet transition due to $^5E---^5T_2$ (Table 8.9) [184]. Reaction with oxygen yields oxyhemerythrin whose spectrum (Fig.8.12) is very similar to simple anionic complexes of methemerythrin (Fig.8.12).

Indeed early resonance Raman data [185] and single crystal data [187] showed that the oxygen is bound as peroxide to Fe(III) and that the peroxo group is bound asymmetrically, probably to only one Fe(III) atom, i.e. behaving as a simple anionic ligand, which can be replaced by many anions to yield anion methemerythrin [Fe(III)Fe(III)] derivatives.

One of the most detailed studies of the spectra of these complexes was published by Garbett, Darnall, Klotz and Williams [179] who reported data for a wide range of hemerythrin complexes,

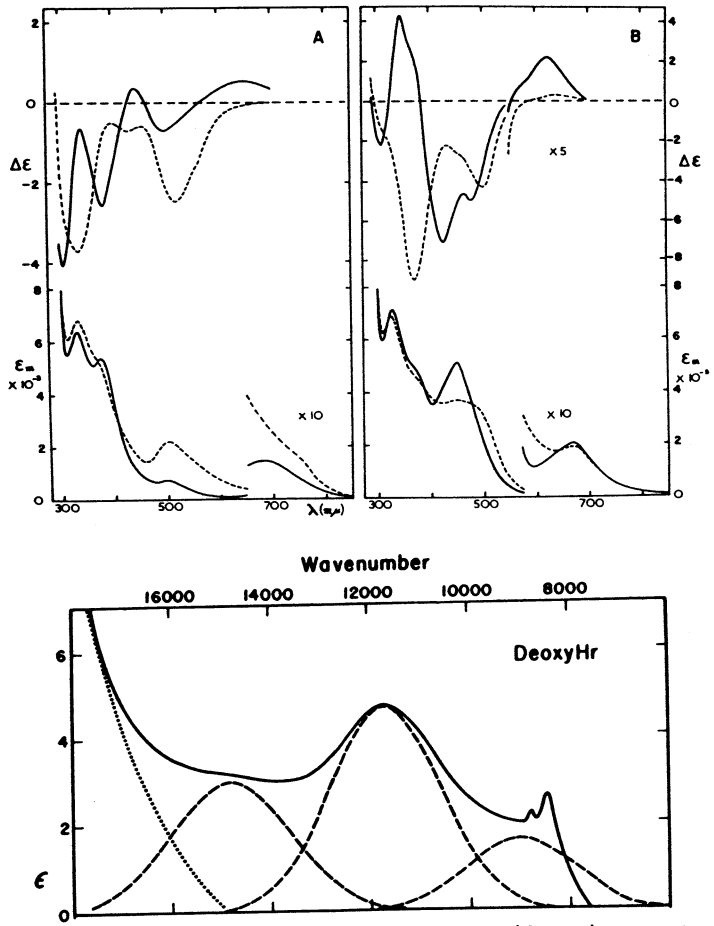


Fig.8.12 The CD (upper) and absorption (lower) spectra of Hemerythrin complexes. A:Cyanomethemerythrin (solid line) and oxyhemerythrin (---), and B:isothiocyanatomethemerythrin (solid line) and azidomethemerythrin (---) [179]. b) (Lower) The Gaussian resolved d-d spectra of deoxyhemeryrthrin [184].

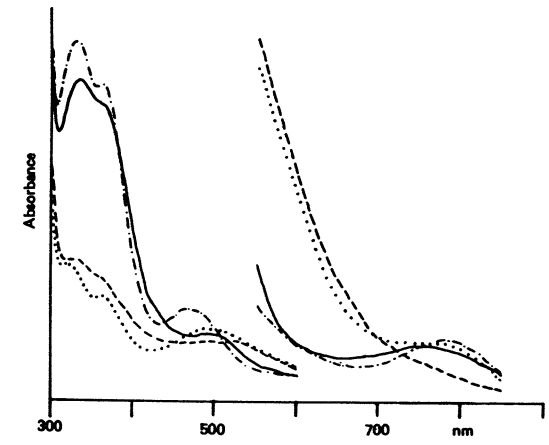
and compared them with many model systems such as $[Fe(III)(H_2O)_5X]^{2+}$ ions, and oxo bridged complexes. CD data were also reported, revealing much structure (Fig.8.12). A linear correlation was noted between the X---Fe(III) LMCT transitions in the models versus the methemerythrin complexes.

Many anionic complexes of the [Fe(III)Fe(III)] methemerythrin display charge transfer transitions to the fairly oxidising Fe(III)

atom. Anions will displace peroxide to form two general classes of complex as classified spectroscopically. Thus [181] class I ligands bind one or both Fe(III) atoms and affect the uv/vis region of the spectrum. They are further subdivided into class I-A having a strong LMCT band in the visible region (intensity 2-5 x 10^3 l mol^{-1} cm^{-1}); these include azide, thiocyanate, selenocyanate, peroxide, sulphide and NCN^{2-}. Class I-B, on the other hand, are less reducing anions and yield slightly more intense LMCT bands further into the ultraviolet, allowing for the observation of fairly weak visible region bands. These include the halides, cyanide, cyanate, nitrito, carboxylato, hydroxo and aquo. Class II anions do not bind Fe(III), but interact at some protein site. These include perchlorate, phosphate, borate, nitrate, carbonate etc.

These CT data are shown in Table 8.10 where the more intense absorption under columns B and C are ligand to Fe(III) LMCT [179].

Lower intensity transitions below 25,000 cm^{-1} are d-d involving forbidden states of the high spin, but exchange coupled, Fe(III) ions. The unusually high intensity is due to this exchange coupling which engenders a relaxation of the spin selection rules. However recent studies [215,216-218] argue against this hypothesis. In parallel with spin coupled Cr(III) data discussed in Chapter 7, a reduction in temperature depopulates higher spin states and increases population of the ground (total) S = 0 state. This would result in a decrease in intensity since transitions from S = 0 to excited S = 1,2,3,4 are forbidden. Such a reduction of intensity is observed with the binuclear $(enH)_2[(EDTAFe)_2O].6H_2O$ [217]. However with met-chlorohemerythrin there is an increase in intensity in the 12,500 - 16,700 cm^{-1} d-d region upon cooling the crystal. This increase in intensity is attributed to low symmetry components of the enzyme crystal field rather than exchange coupling [217].



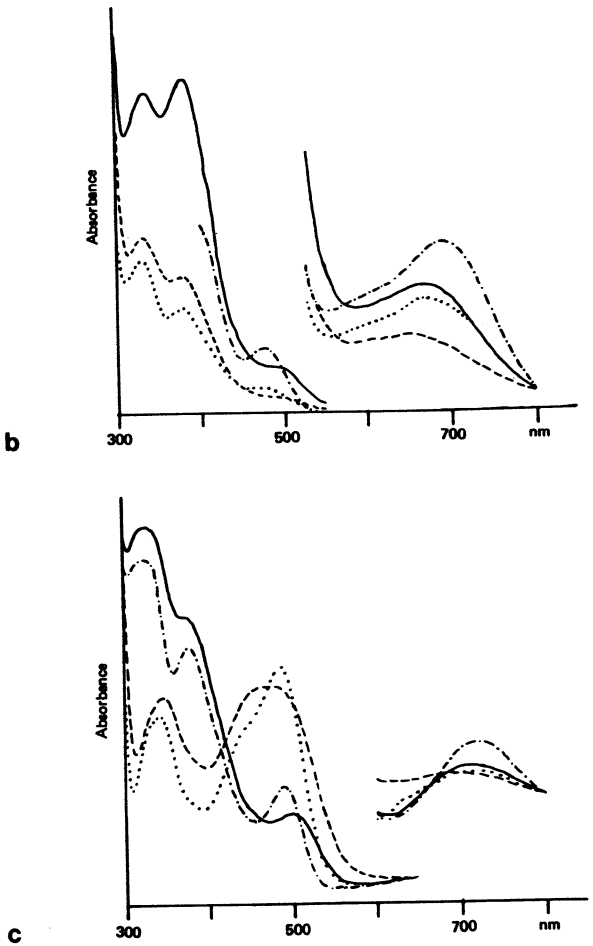


Fig.8.13 The polarised single crystal spectra of a) (previous page) oxyhemerythrin, b) chloromethemerythrin and c) azidomethemerythrin [217]. Polarisations with respect to the needle axis, and temperature are indicated as follows:- 273 K parallel (solid line), 77 K parallel (dash-dotted line), 273 K perpendicular (---), and 77 K perpendicular (...).

Some of the low energy d-d bands can be resolved into two components (Table 8.8) [184] perhaps due to the two different types of Fe(III) present.

It has been suggested [177] that some of the transitions above

25,000 cm^{-1}, of intensity 1-2000 l mol^{-1} cm^{-1}, are also 'spin forbidden' d-d transitions, of the simultaneous pair electronic excitation (spe) type (see Chapter 7).

The polarised single crystal spectra of chloromethemerythrin (Fig.8.13) show bands, in the region 20,000 - 33,000 cm^{-1} strongly polarised parallel to the Fe-O-Fe axis. Resonance Raman excitation into these bands shows enhancement of the Fe-O-Fe vibration but not Fe-imidazole or Fe-carboxylate vibrations. This is not to be expected for an SPE type excitation which should show distortion along all coordinates but would be consistent of an oxide---Fe(III) LMCT transition [218].

Gay and Solomon [187] have also reported single crystal polarised data for oxyhemerythrin, fluoro- and azidomethemeryhrin. A weak parallel (to the Fe-Fe axis) polarised band at 20,200 cm^{-1} is seen in the fluoride spectrum and is evidently a d-d band. A stronger perpendicular polarised band observed in the oxy derivative at 19,700 cm^{-1} is the π-oxygen to Fe(III) LMCT. Using transition moment theory, Gay and Solomon showed that the polarisation behaviour can only be understood in terms of a peroxide bound to one Fe(III) atom and with the Fe-O projection on the Fe-Fe axis being small [218].

Azidomethemerythrin shows at least three azide---Fe(III) LMCT bands (Table 8.10, Fig.8.13). Transitions from the two highest π^* levels of azide into the half-full d^5 Fe(III) shell could give rise to ten azide---Fe(III) LMCT bands, eight of which are allowed [217]. Some assignments have been proposed [217]. The polarisation ratio of the primarily z polarised transition near 20,500 cm^{-1}, assigned by analogy with oxyhemerythrin and oxyhemocyanin as π_v^*---$t_{2g}(b_2)$ requires the projection of the Fe-N(N_3) vector upon the Fe-Fe axis to be small. Combining this datum with resonance Raman data argues for a non-bridging structure [217]. Note also the similarity (Table 8.10) in the positions of the CT transitions from the necessarily terminal azide group in the semi-azido-methemerythrin and the azidomethemerthyrin discussed here.

The transitions listed in Table 8.10 as LMCT have energies fully consistent with the predictions of the optical electronegativity model (Chapter 5).

Ribonucleotide reductase (Table 8.10) and the acid phosphatases also contain binuclear Fe(III) sites [218].

8.3 vi) Ferritin and Phosphitin [193-197]

While the transferrins have the property of transporting iron around and organism, Ferritin is a red storage protein in which iron is stored in a water soluble fashion. It consists of a protein surrounding a core of what is essentially iron oxide. In a packed oxygen environment both octahedral FeO_6 and tetrahedral FeO_4 sites might be expected. Recalling (Chapter 6) that an octahedral high spin d^5 species will absorb at lower frequency than a tetrahedral high spin d^5 species of the same ligand, it is evident that a band near 11,000 cm^{-1} in Ferritin can be taken as evidence for

octahedral FeO_6. Higher energy d-d bands which arise from tetrahedral FeO_4 sites are obscured by CT absorption (oxygen to Fe(III)) [193,194,197].

Phosphitin, on the other hand, is green [195], and has a spectrum characteristic of a tetrahedral FeO_4 chromophore with transitions from the $^6A_1$ ground term to:

$$22{,}400(5.5)\ ^4T_1 \quad 23{,}500(5)\ ^4T_2 \quad 25{,}000\ cm^{-1}\ ^4E, ^4A_1$$

8.4 The Zinc Proteins

8.4 i) Carboxypeptidase A (CPA)

Carboxypeptidase A, a zinc enzyme of molecular weight 36,400 Daltons, catalyses the hydrolysis of the C-terminal amino acid of a polypeptide chain. Esterase activity is also observed. The active site consists of two histidine (imidazole) nitrogen atoms, a carboxylic acid link from glutamic acid, and a water molecule in a distorted tetrahedral stereochemistry about the zinc atom.

In trying to understand the mechanism of activity of this enzyme, and the way in which inhibitors function, various metal ions have been substituted for zinc. These include Mn(II), Co(II), Ni(II), Cu(II) and OV(IV). Only Co(II) CPA and Ni(II) CPA retain full peptidase activity.

Nickel(II) CPA:- Magnetic and spectroscopic data are consistent with a $^3A_{2g}$ ground term in a pseudo octahedral environment. The anticipated three low intensity spin-allowed transitions are shown in Table 8.9, together with corresponding data for carbonic anhydrase (CA), phosphoglutamase (PGM), and some model systems [44,45].

The band positions (DQ and B) seem most consistent with a N_2O_4 chromophore (compare Tables 6.23, 6.24 and 8.9) suggesting that in utilising the same site as the zinc atom, two additional water molecules are added, i.e. three water molecules in all, plus two histidine N and a glutamic O. However an NO_5 stoichiometry cannot be unequivocally excluded on this basis.

Addition of the inhibitor sodium β-phenylpropionate dramatically alters the spectrum with the near infrared ^{1} transition being lost and a new band appearing near 7000 cm^{-1}. The simultaneous presence of both low and high energy d-d features (assuming they arise from the same chromophore) eliminates both octahedral and tetrahedral species with N,O donor sets; however a five coordinate, square pyramidal stereochemistry, with N_2O_3 donor set is possible. In particular the spectrum is similar to that of the known five coordinate $[Ni(dacoda)H_2O]^-$ ion (dacoda is 1,5-diazacyclooctane-N,N'-diacetate)[46,47] (see Chapter 6, Fig. 6.58).

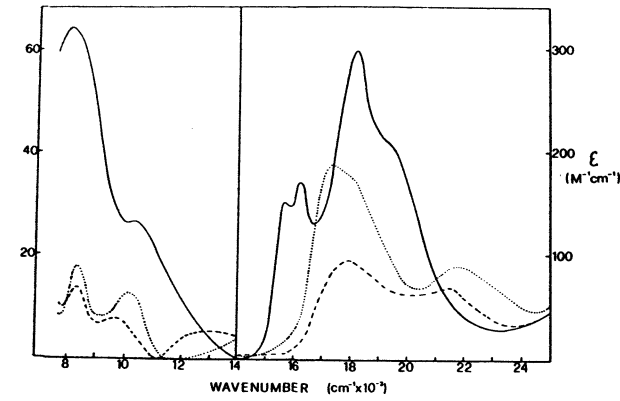


Fig.8.14 Absorption spectra of cobalt(II) bovine carbonic anhydrase (BCAB) at pH 6.1 (solid line) and of its adducts with dicyanoaurate (---) and dicyanoargentate ions (...) [62].

Cobalt(II) CPA:- Broad absorption between 17,480 and 18,020(150) cm^{-1} in Co(II) CPA, together with absorption at 8616 cm^{-1} is more consistent with a distorted five coordinate environment (see Table 6.21) than a T_d environment which would be expected to yield a more intense spectrum and somewhat lower magnetic moment [44,45] (Table 8.9).

Cobalt(III) CPA:- The cobalt(III) substituted CPA shows a d-d band at 19,880(500) cm^{-1} [70].

Copper(II) CPA:- A band at 12,580(124) cm^{-1} shifts to 11,400(180) cm^{-1} upon addition of the inhibitor sodium β-phenylpropionate [48]. In addition there is a weak feature in the near infrared at 11,000 cm^{-1} and another weak feature near 19,000 cm^{-1}. The energy and intensity (and esr spectrum) are consistent with a pseudo tetrahedral stereochemistry.

8.4 ii) Carbonic Anhydrase CA

Carbonic anhydrase catalyses the reversible hydration of carbon dioxide, the hydrolysis of some esters, and the hydration of aldehydes. It is zinc enzyme with a distorted tetrahedral site consisting of three histidine N atoms and a water oxygen. The zinc has been replaced by a range of different metal ions [54].

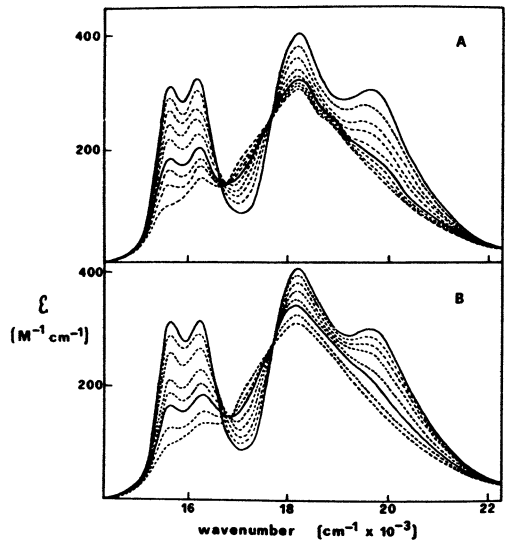


Fig.8.15 Absorption spectra of cobalt(II) bovine carbonic anhydrase (BCAB) as a function of pH. A:Unbuffered solutions at 5.8, 6.0, 6.7, 7.3, 7.7, 7.9, 8.2 and 8.8 in increasing order of intensity at 640nm. B:0.1 M HEPES solutions at pH 5.9, 6.2, 6.5, 7.0, 7.3, 7.8, 8.4, and 9.1 in increasing order of intensity at 640nm. The full lines represent the spectra at the middle and end of each titration [64].

Cobalt(II) CA: The cobalt-substituted enzyme has been extensively studied [52-57,76,77]. The spectra are markedly dependent upon pH, temperature, and the presence of exogenous ligands such as halides, carboxylic acids, and various enzyme inhibitors. The similarity of the electronic spectra of these species to those of Co(II) CPA, and to model tetrahedral systems, led earlier investigators to conclude that the species were all distorted tetrahedral, e.g. an MCD study [57]. However, as shown by Bertini and co-workers [58-61,76,77], species generated by addition of such reagents as acetate, iodide and thiocyanate ions, show an additional band near 13,000-14,000 cm^{-1} and reduced intensity absorption in the visible region (Table 8.9) and are best characterised as five coordinate. Indeed these workers demonstrated that there can be several tetrahedral species in equilibrium with each other, and with the five coordinate species, depending upon added reagent and pH. Of especial interest are the five coordinate complexes obtained by addition of dicyanoargentate and dicyanoaurate ions [62] (Fig.8.14). There is a marked dependence of the spectrum upon pH, evidence that proton dissociation, indeed two proton dissociations, are involved [63,64] (Fig.8.15). The cobalt[tris(3,5-dimethyl-1-pyrazolylmethyl)amine]

H_2O dication is an interesting model for Co(II) CA showing very similar spectroscopic characteristics [68, also see 79]. The similarity of the spectra of Co(II) CA and of Co(II) alkaline phosphatase is remarkable. Co(III) CA has also been studied [65].

Copper(II) CA: This species has a rather simple spectrum (Table 8.9), and, in contrast to cobalt(II) CA, the electronic spectrum shows little dependence upon pH. The spectrum could be consistent with either distorted tetrahedral or five coordinate stereochemistry (see Chapter 6). Nmr relaxation studies [67] lead to the conclusion that added anions such as acetate, nitrite etc, do not displace the coordinated water. Assuming they also do not displace coordinated histidine, such species are probably five coordinate. The oxalate ion does displace water, also leading perhaps to a bidentate bound oxalate five coordinate derivative (N_3O_2 ?). Azide and iodide also bind providing CT absorption [55,67].

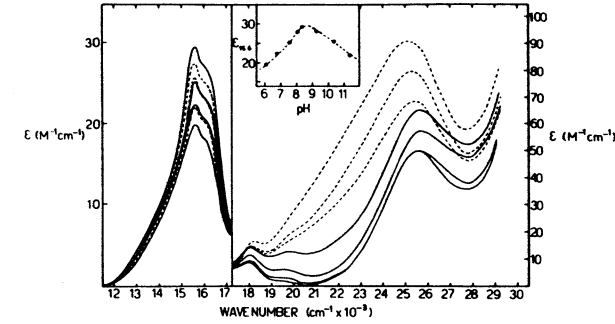


Fig.8.16 Electronic spectra of nickel(II) bovine carbonic anhydrase B; spectra at pH 6.1, 6.8, 7.6 and 8.4 are shown by solid lines in order of increasing absorbance. Spectra recorded at pH 9.3, 10.4 and 11.4 are dashed lines in order of decreasing absorbance at 15,600 cm^{-1} and increasing absorbance near 25,000 cm^{-1}. Inset shows variation of absorbance at 15,600 cm^{-1} as a function of pH [78].

Oxyvanadium(IV) CA: Nmr relaxation techniques reveal that the vanadyl ion bound in the CA site also binds an extra water molecule. The electronic spectrum (Table 8.9) is largely unaffected by the addition of exogenous anions, though the higher energy band does become better resolved [69].

Nickel(II) CA: The electronic spectrum of this species appeared in an early study [78] and appears typical of octahedral coordination (Table 8.9). The spectra are pH dependent, indicative of several proton equilibria [78] (Fig.8.16). However a more recent and

detailed absorption and CD study [81] revealed additional CD bands which were present only as ill-defined shoulders in the absorption spectrum. The presence of this additional structure implies the presence of a symmetry lower than octahedral and a five coordinate stereochemistry is tentatively proposed [81].

8.4 iii) Horse Liver Alcohol Dehydrogenase (HLAD)

Alcohol dehydrogenase is an interesting zinc enzyme in which there are two pairs of zinc atoms per sub-unit. One pair is catalytically active (labelled (c)), and the other pair, about 2.06 nm away, is inactive (labelled (n)). The former is known to involve 2 cysteine S, a histidine N and a water molecule, and the latter, which may only serve to provide structural stability, involves 4 cysteine S [172].

Cobalt HLAD: Several groups [71-72] have studied three different Co HLAD preparations namely [$Co(II)_2(c)Zn_2(n)$ HLAD], [$Zn_2(c)Co_2(n)$ HLAD] and [$Co(II)_2(c)Co_2(n)$ HLAD]. The [$Co(II)_2(c)Zn_2(n)$ HLAD] (so-called 'blue hydrid') has absorption at 15,385(1200) and 19,230(250) cm^{-1}, undoubtedly d-d in origin and suggestive of tetrahedral or perhaps five coordinate stereochemistry. A band at 28,570(7000) cm^{-1} is evidently Co-S LMCT similar to that observed in Stellacyanin etc (see above). The [$Zn_2(c)Co(II)_2(n)$ HLAD] ('green hybrid') absorbs at 15,240(1000) and 13,500(750) cm^{-1}, also d-d, and has a Co-S LMCT band at 29,240(6500) cm^{-1}. The fully substituted species has a spectrum which is a superimposition of the 'blue' and 'green' hybrids. It also exhibits structured absorption between 5500 - 10,000(270-540) cm^{-1} [73]. The difference in band energies for the Co(II) in the two isomers clearly indicates the difference in site microstoichiometry ((c) S_2NO, (n) S_4) (Table 8.9). Addition of HS^- or N_3^- to the blue hybrid leads to some additional structure in the visible band, and possible replacement of cysteine protein sites by these anions [72].

Recent studies show the formation of ternary complexes with various substrates, which modify the cobalt(II) absorption spectrum [74,75]. The pH dependence of the Co(II) HLAD spectrum also permits evaluation of pK_a values pertinent to the protein activity [75].

Copper HLAD: Treatment of catalytic site zinc-deficient enzyme with copper(II), yielded an essentially inactive mixed complex [$Cu(II)_{0.6}(c)Zn_2(n)$], where the deficiency in copper is due to difficulty in inserting the metal into the site [15]. This blue complex has an electronic (and esr) spectrum similar to the type 1 blue copper proteins (Table 8.9). The species is readily reduced to Cu(I) and it is probable that the copper(II) site is close to tetrahedral. Reaction of the Cu(II) HLAD enzyme with pyrazole shifts the d-d band to the blue (Table 8.9) and changes the esr spectrum to become more characteristic of type 2 copper. The site probably becomes more tetragonal [15]. The spectrum shows a

remarkable temperature sensitivity. The blue enzyme becomes pink upon cooling to 77 K and some bands shift by as much as 40 nm to the blue (Table 8.9).

Data for various ternary complexes with NAD^+ and pyrazole were also reported. The Cu(I) species has an LMCT spectrum similar to the metallothioneins (below) [15].

8.4 iv) Metallothioneins (Mtn)

Metallothionein is protein extraordinarily rich in cysteine, first isolated from equine renal cortex [149]. All mammalian species of Mtn studied to date contain a single polypeptide chain with 20 cysteinyl residues in a total of 61 amino acid residues, none of which are aromatic [150]. The apoprotein absorbs at 52,500 cm^{-1} and essentially no absorption occurs below 41,600 cm^{-1}. The zinc atoms may be replaced by a range of M(II) ions especially Cd(II), Hg(II), Co(II) etc. Spectroscopic studies have been undertaken especially by Vasak and co-workers [150-152] (and including MCD [213]), and typically show rich S---M LMCT transitions. The Zn(II), Cd(II) and Hg(II) proteins peak at 45,500, 40,800 and 33,000 cm^{-1} respectively. Analysis of difference spectra (metal enzyme minus apoprotein) suggests that these bands can each be deconvoluted into at least three transitions, viz:

Zn Mtn	43,300	45,700	48,700
Cd Mtn	40,000	43,300	49,700
Hg Mtn	33,000	34,000	36,000

(8.4)

Similar spectra were obtained for model complexes of these metal ions with 2-mercaptoethanol. Both sets of data were analysed by the conventional optical electronegativity procedure, using chi = 2.6 for the thiolate optical electronegativity [150]. The multiple absorption is associated with the several $\pi S(t_2)$---$M(a_1)$ and $\sigma S(t_2)$---$M(a_1)$ transitions which are possible (see Chapter 5).

The Co(II) substituted species is also clearly tetrahedral on the basis of electronic spectroscopy (Table 8.9) and shows the expected S---Co(II) LMCT. The narrowness of the visible d-d absorption leads to the conclusion that the several MS_4 sites must be very similar [151,152]. Data for Ni(II) Mtn is less clear but may also indicate tetrahedral NiS_4 units [151] (Table 8.9).

8.4 v) Thermolysin

Thermolysin is a protease of molecular weight 34,600 possessing one active zinc atom and four structural calcium atoms. The zinc ion has been replaced by copper(II) [87], cobalt(II), manganese(II) and iron(II) [88].

The zinc active site is known [89] to possess two histidyl nitrogen atoms, a glutamate residue and a water molecule. The copper(II) complex, presumably in this site, displays an 'ordinary' weak single absorption band at 13,700(90) cm^{-1}, shifting slightly

to the blue when treated with L-phenylalanine.

The Co(II) system shows weak structured absorption (CD, MCD, vis [88]) centred at 18,200 cm^{-1}, but with a well defined shoulder at 20,000 cm^{-1}, considered, by the authors, to be consistent with tetrahedral coordination, presumably by high field ligands such as nitrogen.

The addition of the enzyme inhibitor -mercaptoacetyl-L-phenylalanyl-L-alanine to Co(II) thermolysin causes a marked perturbation of the Co(II) d-d bands (suggestive, though not proof, of inhibitor binding to Co(II)) and the appearance of a new band (seen both in absorption and MCD) at 30,600 cm^{-1} assigned to S---Co(II) LMCT. The presence of this band does prove binding of the inhibitor to Co(II) [144]. A similar result was observed with carboxypeptidase A [144].

8.4 vi) Other Systems

Mn(III) acid phosphatase exhibits intense absorption at 19,420(2460) cm^{-1} attributed to tyrosine to Mn(III) charge transfer with possibly a contribution from sulphur to Mn(III) charge transfer [36,102]. Resonance Raman also results in excitation of tyrosine vibrational modes when irradiated into the CT band [102]. A bacterial superoxide dismutase type enzyme isolated from Escherichia coli contains only manganese(III) [104]. A band centered at 23,800 cm^{-1} shows well developed shoulders to lower energy. It is probably Mn-L charge transfer. Mn SOD species have also be isolated from other species, e.g. plants. In a plant species, a pair of shoulders is observed between 25,000 - 27,000 cm^{-1} and 16,500 - 17,500 cm^{-1} [200]. Similar shoulders are seen in a Mn substituted Fe SOD [201], and in various other Mn SOD species [202-209]. The lower energy band is probably d-d in origin.

The alkaline phosphatases contain 4Zn-2Mg and all six main group elements may be replaced by Co(II) [144-147]. The first two cobalt atoms occupy the catalytic sites and develop a spectrum whose absorption and MCD spectra contain bands at 15,625(260), 16,500(220) 18,000(260) and 19,600(335) cm^{-1}, sufficiently intense and of appropriate energy for tetrahedral Co(II) (though five coordinate low symmetry Co(II) is difficult to refute entirely) [146,147]. However further addition of cobalt (to structural sites) generates weak absorption near 20,000 cm^{-1} which is probably octahedral Co(II) [148]. The catalytic sites are believed to contain at least three histidine nitrogen atoms [147].

Various Fe SOD species have been reported [210-212]. They exhibit broad ill defined absorption between 20,000 - 35,000 cm^{-1}. A bacterial superoxide dismutase-type enzyme isolated from Escherichia coli contains only manganese as metallic content [104]. A band centred at 23,800 cm^{-1} shows well developed shoulders to lower energy. It is probably Mn(III)-L charge transfer.

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Chapter 9

Nephelauxetic and Spectrochemical Series, AOM Parameters and Chemical Information Content

9.1 Introduction

The well known Spectrochemical and Nephelauxetic Series were left for discussion until this final chapter, to allow the reader to refer back to data previously presented. Extensive data obtained by use of the AOM model will be collected and discussed as a means of emphasizing its value. The principle purpose however, of this chapter, is to answer the question -"What really is the information content of an electronic spectrum ?" To this end, various topics mentioned briefly in the earlier part of the book, will be reconsidered here in rather more depth.

9.2 Nephelauxetic Series: Variation of the Racah Parameter B

The Racah parameter B (and less well understood, the parameter C) for a specific ion is well known to vary as a function of the ligands bound to that ion. The value of this parameter is always reduced from that observed with the free spherical ion. B values for some first row transition metal ions are plotted versus oxidation state and electron configuration in Fig.9.1; free ion values of B and C were quoted in Table 3.4. It is evident that B increases with increasing oxidation state and with the number of electrons in the d shell. The primary factor is the size of the ion; the larger the ion, the larger the mean radial displacement of the d shell, and the smaller the mutual interelectronic repulsion. The size of the ion decreases as the oxidation state increases, and for a given oxidation state, with increasing number of d electrons. Second row transition elements being larger than their first row counterparts, will have lower B values.

Since the size of the ion is related to the effective nuclear charge experienced by the d electrons, it follows that B is not only a measure of size, but also reflects the effective nuclear charge (vide infra)[18,27].

The reduction of the value of B in a complex ion, is generally attributed to one or both of two mechanisms:

i) Central Field Covalency - If the lone pairs of the ligands penetrate the d shell, they will tend to screen the d electrons from the nucleus, decrease the effective charge experienced by the d electrons and expand the d shell. Thus B is expected to decrease if the metal ligand bond becomes partially covalent. By transferring some charge to the metal ion, the ligand will reduce the effective positive charge on the metal and allow for some expansion of the d shell.

ii) Symmetry Restricted Covalency - In an octahedral complex, for example, if there is some covalent bonding between metal and ligand, we know that the e_g orbitals will become slightly sigma-antibonding, while the t_{2g} orbitals may be involved in pi-bonding or anti-bonding with the ligand. In other words a small amount of sigma or pi electron density of the metal, may be transferred to the ligands. Such delocalisation will also decrease the mean distance between d electrons and hence reduce the value of B. This mechanism is directly observed via electron spin resonance, when metal electron density is seen to lie on ligands with nuclear moments. However central field covalency is probably the major contributor to the reduction of B in transition metal complexes.

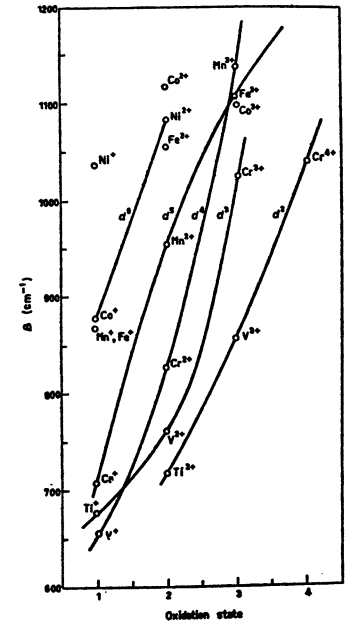

Fig.9.1 A graph of the free-ion Racah B value, in cm^{-1}, versus oxidation state. The lines connect iso-electronic ions, and illustrate the increase in B with oxidation state, for a given electronic configuration.

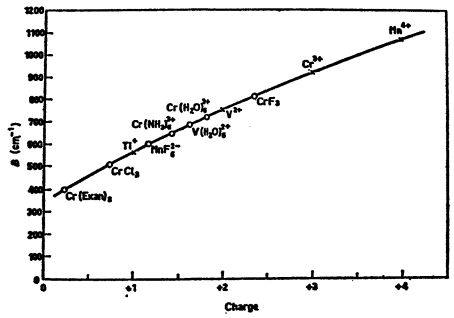

Fig.9.2 A graph depicting the variation of B in complex molecules as a function of nuclear charge. The graph is constructed by first drawing a line through the free ion values (crosses) of B for the isoelectronic d^3 Ti(I), V(II), Cr(III) and Mn(IV) ions. Various d^3 complex molecules are then plotted onto the line according to their B values. The diagram is a qualitative description of the variation in nuclear charge experienced as a function of ligand.

One may reasonably argue that the greater the reduction in B, as represented by the ratio:

$$\text{(B in complex) / (B in free ion)} = \beta \qquad (9.1)$$

the greater the covalency in the metal-ligand bond, and the smaller the effective charge experienced by the d electrons.

Figures 9.2 and 9.3 illustrate plots of interelectronic repulsion versus formal charge for d^3 and d^5 complexes. They are constructed by drawing a curve through the B values, in the case of d^3, and the ^6S-^4G term separation (directly observable and equal to 10B + 5C), in the case of d^5, for a series of isoelectronic ions. The complexes were then located onto these curves by means of their experimental B values or ^6S-^4G term separations. Such a diagram qualitatively illustrates the reduction in charge at the central ion, occurring upon complex formation.

Reduction in charge at the central ion is marked even with highly electronegative ligands such as the fluoride ion, providing support for the concept of electroneutrality as presented by Pauling. The reduction is much more marked with the more polarisable ligands and the more polarising cations, since it this combination which gives rise to the greatest degree of covalency (see Table 9.1).

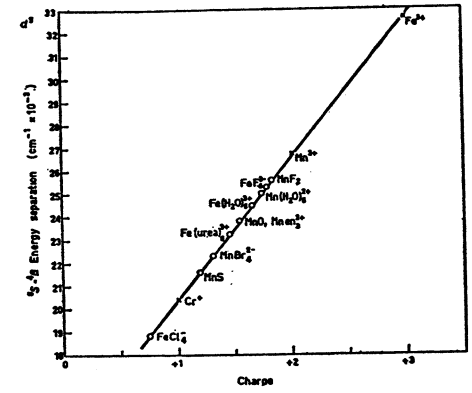

Fig.9.3 A graph depicting the variation in the ^4S--^4G energy separation in d^5 (spin-free) complex molecules as a function of nuclear charge. The graph is constructed in the same fashion as Fig.9.2, this energy separation being proportional to the interelectronic repulsion.

When, for a given ion, ligands are ordered according to the value of B in the corresponding complex, one derives the so-called **Nephelauxetic Series** (based upon Greek for 'cloud expanding'). This series is moderately independent of metal ion, with B decreasing in the sequence:

$$\text{Free ion} > F^- > H_2O > CO(NH_2)_2 > NH_3 > H_2NCH_2CH_2NH_2$$
$$= \text{oxalate} = (CO_3)^{2-} > NCS^- > Cl^- = CN^- > Br^- > N_3^- > I^-$$
$$> S^{2-} = (C_2H_5O)_2PS_2^{2-} = I^- > (C_2H_5O)_2PSe_2^{2-} > \text{diarsine} \qquad (9.2)$$

Expressed in terms of donor atoms, this series is approximately:

$$F > O > N > Cl > Br > I > S > Se > As \qquad (9.3)$$

roughly in order of the polarisability of the ligand atom. If the ligand is maintained constant a Nephelauxetic Series of metal ions can be constructed. This will vary according to the polarising power of the metal atom. B decreases in the sequence [18,27].

$$Mn(II) = V(II) > Ni(II) = Co(II) > Mo(III) > Ru(II) = Cr(III) > Fe(III) > Rh(III) = Ir(III) > Tc(IV) > Co(III) > Ag(III) > Cu(III) = Mn(IV) > Pt(IV) > Pd(IV) > Ni(IV) \qquad (9.4)$$

The value of B also tends to decrease, for a given metal and ligand, with decreasing coordination number.

Table 9.1

COLLECTION OF SPECTROCHEMICAL AND NEPHELAUXETIC PARAMETERS[a]

	Ox.No Group		10 Dq	DQ	ζ	B	C/B	Ref
Titanium								
$TiCl_6^{4-}$	2	O_h	8450	23,235		517		137
TiF_6^{3-}	3	O_h	17,500	48,120				113
$Ti(H_2O)_6^{3+}$	3	O_h	18,750	51,555				17
Vanadium								
VCl_4^{2-}	4	T_d	7900	21,720				51
VCl_6^{4-}	4	O_h	15,400	42,340				47
$V(NMe_2)_4$	4	T_d	13,300?	36,570				59
$V(OBu)_4$	4	T_d	12,415	34,135				59
VF_6^{2-}	4	O_h	20,120	55,320				47
VF_6^{3-}	3	O_h	16,100	44,300		648		520
$CsVCl_3$	2	O_h	9500	26,120		640		208
$CsVBr_3$	2	O_h	8700	23,920		870		208
$CsVI_3$	2	O_h	5900	16,200		790		208
VCl_2	2	O_h	9200	25,300		615		235
VBr_2	2	O_h	8600	23,650		530		235
VI_2	2	O_h	7900	21,720		510		235
$V(Ph_2PS_2)_3$	3	O_h	13,200	36,300		395		107
$V(CH_3SO_2)_3$	3	O_h	16,165	44,450		620		486
V(II)/CdSe	2	T_d	3400	9350		520		382
$V(bpy)_3^{2+}$	2	D_3	16,150	44,405		943		581
$V(CO)_6$	-1	O_h	25,200	69,300		430	3.02	d
Chromium								
CrO_4^{3-}	5	T_d	12,405	34,110				132
CrF_6^{2-}	4	O_h	22,000	60,500		575		520
CrF_6^{3-}	3	O_h	15,060	41,400		643		520
$Cr(en)_3^{2+}$	3	O_h	23,500	64,600		580		244
$Cr(CN)_6^{3-}$	3	O_h	26,600	73,150				
$Cr(CH_3SO_2)_3$	3	O_h	15,750	43,300		640		486
$Cr(NCSe)_6$	3	O_h	18,000	49,500		460		198
$Cr(dtc)_3$	3	O_h	15,500	42,620		440		494
$Cr(bpy)_3^{3+}$	3	D_3	23,400	64,340		495		581
$Cr(bpy)_3^{2+}$	2	D_3	17,370	47,760		604		581
$Cr(CO)_6$	0	O_h	32,200	88,550		520	3.27	d
Molybdenum								
$Mo(H_2O)_6^{3+}$	3	O_h	26,000	71,500		525		325
$Mo(CO)_6$	0	O_h	32,150	28,950		380	2.90	d
Tungsten								
$W(CO)_6$	0	O_h	32,200	88,550		390	3.33	d

cont

Table 9.1 cont

	Ox.No Group		10 Dq	DQ	ζ	B	C/B	Ref
Manganese								
MnO_4^{2-}	6	T_d	11,000	30,245				181
MnO_4^{3-}	5	T_d	10,000	27,495		800		174
MnF_6^{4-}	4	O_h	21,800	59,950		650		326
$Mn(NCSe)_6^{4-}$	2	O_h	8660	23,800		815		198
$MnCl_4^{2-}$	2	T_d	3300	9070		650	4.6	472
$MnBr_4^{2-}$	2	T_d	3100	8520		630	4.8	472
$NaMnF_3$	2	O_h	7750	21,310		845	3.6	587
MnS_b^4	2	T_d	4685	12,880		559	5.6	470
$Mn(CO)_6^{+}$	1	O_h	41,050	112,900				d
Technetium								
TcO_4^{2-}	6	T_d	17,000	46,740				181
TcF_6^{-}	5	O_h	25,000c	68,740		535		526
TcF_6^{2-}	4	O_h	28,400	78,100		530		526
Rhenium								
ReO_4^{2-}	6	T_d	19,000	52,240				180
ReF_6^{-}	6	O_h	30,000	82,485	3400			526
ReF_6^{2-}	4	O_h	32,800	90,185				526
$Re(CO)_6^{+}$	1	O_h	41,000	112,730		470	2.92	d
Iron								
FeF_6^{3-}	3	O_h	13,350	36,700		835		520
FeO_4^{2-}	6	T_d	12,600	34,640		630		274
$Fe(CN)_6^{3-}$	3	O_h	35,000	96,250				487
$Fe(ox)_3^{3-}$	3	O_h	14,140	38,900		720		176
$Fe(H_2O)_6^{3+}$	3	O_h	14,000	34,650		1010		176
$Fe(mal)_3^{3-}$	3	O_h	14,140	38,900		780		176
$Fe(urea)_6^{3+}$	3	O_h	12,950	35,600		780		176
$Fe(acac)_3$	3	O_h	14,700	40,420		536		492
$Fe(HFA)_3$	3	O_h	14,400	39,600		493		492
$FeBr_4^{-}$	3	T_d	3800	10,450	420	470	3.9	490
$Fe(o\text{-}phen)_3^{2+}$	2	O_h	13,110	36,020		602		531
$Fe(CN)_6^{4-}$	2	O_h	32,200	88,535		490		557
$Fe(CNO)_6^{4-}$	2	O_h	27,000	74,240		410		557
$Fe(HMPA)_4^{2+}$	2	T_d	6950	19,110				597
$Fe(acet\text{-}ald)_6^{2+}$	2	O_h	9960	27,400				762
$Fe(benz\text{-}ald)_6^{+}$	2	O_h	10,330	28,400				762
Ruthenium								
$Ru(en)_3^{2+}$	2	O_h	28,100	77,250		420		557
$Ru(H_2O)_6^{2+}$	2	O_h	19,800	54,440		475		556
$Ru(dien)_2^{2+}$	2	O_h	28,800	79,190		430		557
RuF_6^{-}	5	O_h	26,000	71,490	1000	425	4.75	526
RuF_6^{2-}	4	O_h	25,000	68,750		500	4.75	526
$Ru(dtc)_3$	3	O_h	19,800	54,450		660		494

cont

Table 9.1 cont

	Ox.No Group		10 Dq	DQ	ζ	B	C/B	Ref
$Ru(NH_3)_6^{3+}$	3	O_h	34,800	95,700		580		506
$Ru(H_2O)_6^{3+}$	3	O_h	28,600	78,650		640		504
$RuCl_3$/LiCl/KCl	3	O_h	18,700	51,400		490	4.67	503
$Ru(ox)_3^{3-}$	3	O_h	28,700	78,900		560		505
RuF_6^{3-}	3	O_h	23,300	64,050		737	3.78	504
RuF_6^{3-}	3	O_h	22,200	61,050	1000	550	4.75	526
$Ru(CN)_6^{4-}$	2	O_h	33,800	92,950				487
Osmium								
OsF_6	6	O_h	?		3400	310	4.75	526
OsF_6^{2-}	5	O_h	35,000	96,235	3200	410	4.75	526
OsF_6^{2-}	4	O_h	26,000	71,490	2900	500	4.75	526
$Os(CN)_6^{4-}$	2	O_h	>34,000	>93,500				
Cobalt								
CoF_6^{2-}	4	O_h	20,300	55,800		635		520
$Co(ox)_3^{3-}$	3	D_3	18,000	49,500		540		176
CoF_6^{3-}	3	O_h	13,100	36,000		787		520
$Co(CN)_6^{3-}$	3	O_h	32,200	88,535		400		487
$Co(dtp)_3$	3	D_3	14,200	39,040		400	4.05	560
$Co(S_2PF_2)_3$	3	D_3	14,240	39,155		353		843
$Co(S_2PMe_2)_3$	3	D_3	13,370	36,760		356		843
$Co(S_2PPh_2)_3$	3	D_3	13,280	36,515		355		843
$Co(S_2P(OEt)_2)_3$	3	D_3	14,220	39,100		353		843
$Co(S_2P(CF_3)_2)_3$	3	D_3	13,190	36,265		353		843
$Co(En)_3^{3+}$	3	O_h	23,160	63,680		590	4.91	200
$Co(N_3)_6^{3-}$	3	O_h	14,500	39,870		410		1099
$Co(NH_3)_6^{3+}$	3	O_h	22,870	62,880		615	5.02	200
$Co(CNO)_6^{3-}$	3	O_h	26,100	71,765		450		555
$Co(H_2O)_6^{3+}$	3	O_h	20,760	57,080		510	8.36	558
$Co(NH_3)_6^{2+}$	2	O_h	10,200	28,045		885		755
$Co(H_2O)_6^{2+}$	2	O_h	9200	25,295		850		741
$Co(PyNO)_6^{2+}$	2	O_h	8900	24,470		780		741
$Co(EtOH)_6^{2+}$	2	O_h	9050	24,885		845		741
$Co(MeOH)_6^{2+}$	2	O_h	9200	25,295		850		741
$Co(glycol)_3^{2+}$	2	O_h	8850	24,335		840		741
$Co(butanone)_6^{2+}$	2	O_h	8250	22,685		825		761
$Co(acet\text{-}ald)_6^{2+}$	2	O_h	9500	26,120		830		762
$Co(benz\text{-}ald)_6^{2+}$	2	O_h	9100	25,020		820		762
$Co(acrylamide)_6^{2+}$	2	O_h	8750	24,060				596
$Co(NNEU)_6^{2+}$	2	O_h	8700	23,920		850		808
$Co(Et_4D)_3^{2+}$	2	D_3	7900	21,720				619
$Co(benzimiz)_4^{2+}$	2	T_d	5100	14,025		716		817
$Co(imidazole)_4^{2+}$	2	T_d	5380	14,790		750		843
$Co(2MeIz)_4^{2+}$	2	T_d	5250	14,435		770		635

cont

Table 9.1 cont

	Ox.No Group		10 Dq	DQ	ζ	B	C/B	Ref
$Co(NH_3)_4^{2+}$	2	T_d	5900	16,220		710		755
$Co(MImT)_4^{2+}$	2	T_d	3845	10,570		641		739
$Co(HMPA)_4$	2	T_d	3600	9900		815		597
CoI_4^{2-}	2	T_d	2650	7285		665		723,744
$CoBr_4^{2-}$	2	T_d	2850	7835		695		723,744
$CoCl_4^{2-}$	2	T_d	3130	8605		710		723,744
$Co(su)_4^{2+}$	2	T_d	4030	11,080		575		803
$Co(Ph_3PO)_4^{2+}$	2	T_d	3680	10,120		787		723,744
$Co(Me_3PO)_4^{2+}$	2	T_d	3800	10,450		806		742
$Co(Me_3PS)_4^{2+}$	2	T_d	3890	10,690		630		742
$Co[(Me_2N)_3PS]_4^{2+}$	2	T_d	3620	9955		642		743
$Co[(Ph_2PS)_2N)]_2$	2	T_d	4025	11,065		560		547
$Co[(Me_2PS)_2N)]_2$	2	T_d	3830	10,530		632		547
$Co[(Ph_2PO)_2N)]_2$	2	T_d	3960	10,890		825		547
$Co[(Ph_2PNH)_2N)]_2$	2	T_d	4635	12,745		795		547
CdS/Co^{2+}	2	T_d	3300	9075		610		837
ZnS/Co^{2+}	2	T_d	3750	10,310		610		837
$Co(OH)_4^{2-}$	2	T_d	4230	11,630		747		845
$Co(N_3)_4^{2+}$	2	T_d	3920	10,780		658		723,744
$Co(tu)_4^{2+}$	2	T_d	4250	11,685		600		821
$Co(NCO)_4^{2-}$	2	T_d	4150	11,410		720		723,744
$Co(NCS)_4^{2-}$	2	T_d	4550	12,510		691		723,744
$Co(SAc)_4^{2-}$	2	T_d	3870	10,640		645		841
$Co(etu)_4^{2+}$	2	T_d	3780	10,395		651		839
$Co(SPh)_4^{2-}$	2	T_d	3870	10,640		600		841
$Co(DMTA)_4^{2+}$	2	T_d	3870	10,640		655		824
$Co(TCPL)_4^{2+}$	2	T_d	3910	10,750		634		840
$Co(SPhF_5)_2$	2	T_d	3660	10,065		631		841
$Co(Py2SH)_4^{2+}$	2	T_d	4500	12,375		587		701
$Co(Py4SH)_4$	2	T_d	4180	11,495		559		701
$Co(S_2PMe_2)_2$	2	T_d	4715	12,965		577		843
$Co(S_2PPh_2)_2$	2	T_d	4715	12,965		575		843
$Co(S_2P(OEt)_2)_2$	2	T_d	4945	13,595		570		843
$Co(S_2P(CF_3)_2)_2$	2	T_d	4765	13,100		580		843
$Co(S_2PF_2)_2$	2	T_d	5030	13,830		575		843
$Co(NCNPh)_4^{2-}$	2	T_d	4480	12,320		622		836
$Co(NCNPh24F)_4^{2-}$	2	T_d	4290	11,800		671		836
$Co(NCNPh2Cl)_4^{2-}$	2	T_d	3930	10,800		701		836
$Co(NCNPh35Cl)_4^{2-}$	2	T_d	4400	12,100		667		836
$Co(NCNPh246Cl)_4^{2-}$	2	T_d	4235	11,650		672		836
$Co(NCNPh246Me)_4$	2	T_d	3675	10,100		722		836
$Co(Ph_3AsO)_2Cl_2$	2	C_{2v}	3450	9485		780		845
$Co(DMF)_2Cl_2$	2	C_{2v}	3750	10,310		745		845
$Co(2\text{-}BrPy)_2Cl_2$	2	C_{2v}	4420	12,150		717		845

cont

Table 9.1 cont

	Ox.No Group		10 Dq	DQ	ζ	B	C/B	Ref
$Co(4\text{-vinPy})_2Cl_2$	2	C_{2v}	4550	12,510		670		845
$Co(Py)_2Cl_2$	2	C_{2v}	4680	12,870		705		845
$Co(3\text{-EtPy})_2Cl_2$	2	C_{2v}	4800	13,200		694		845
$Co(TDPME)Cl^{+}$	2	C_{3v}	4030	11,080		698		759
$Co(WOS_3)_2^{2-}$	2	T_d	4610	12,675		525	1049,1050	
$Co(MeTPyEA)^{2+}$	2	C_{3v}	15,000	41,245		750		1193
Rhodium								
RhF_6^{2-}	4	O_h	20,500	56,365	1000	410	4.9	526
$Rh(dtp)_3$	3	D_3	21,900	60,215		210		494
RhF_6^{3-}	3	O_h	22,300	61,300		460	4.8	526
$RhBr_6^{3-}$	3	O_h	19,300	53,050		287	5.68	507
$RhCl_6^{3-}$	3	O_h	20,700	56,900		344	5.6	507
$Rh(en)_3^{3+}$	3	D_3	34,600	95,100		422		200
$Rh(CN)_6^{3-}$	3	O_h	44,900	123,450		229		200
$Rh(ox)_3^{3-}$	3	D_3	27,300	75,050		426	5.5	504
$Rh(SCN)_6^{3-}$	3		ca.20,300	55,815				680
$Rh(NH_3)_6^{3+}$	3	O_h	34,100	93,750		428		200
$Rh(H_2O)_6^{3+}$	3	O_h	27,200	74,800		499		507
$RhCl_3$/LiCl/KCl	3	O_h	19,300	53,050		370		e
Iridium								
IrF_6	6	O_h	?			310	4.75	359
IrF_6^{-}	5	O_h	28,500	78,360	3400	360	4.9	526
IrF_6^{2-}	4	O_h	24,500		3500	641	4.9	420
IrF_6^{2-}	4	O_h	27,000	74,240	3300	510	4.9	526
$IrCl_6^{3-}$	3	O_h	25,000	68,740		300		552
$Ir(en)_3^{3+}$	3		ca.41,400	113,830				
$Ir(NH_3)_6^{3+}$	3	O_h	41,200	113,280		470		563
$Ir(NH_3)_5H_2O^{3+}$	3	O_h	40,400	111,080		550		564
$Ir(dtp)_3$	3	O_h	26,700	73,415		160		245
$Ir(dsep)_3$	3	O_h	25,000	68,740		135		245
Nickel								
NiF_6^{2-}	4	O_h	20,100			515		520
$Ni(mnt)_2^{-}$	3	D_{4h}	15,700	43,200		400		1073
NiF_6^{3-}	3	O_h	16,200	37,760	8800	703		520
$NiCl_2$	2	O_h	7700	21,170		750		961
$NiBr_2$	2	O_h	7400	20,345		650		961
NiI_2	2	O_h	7400	20,345				962
$CsNiCl_3$	2	O_h	6950	19,110		828		942
$CsNiBr_3$	2	O_h	6550	18,010		765		942
$CsNiI_3$	2	O_h	6500	17,870				963
$Ni(butanone)_6^{2+}$	2	O_h	8200	22,545		890		761
$Ni(acet\text{-}ald)_6^{2+}$	2	O_h	9450	25,985		890		762

cont

Table 9.1 cont

	Ox.No Group		10 Dq	DQ	ζ	B	C/B	Ref
$Ni(benz\text{-}ald)_6^{2+}$	2	O_h	9200	25,295		870		762
$Ni([13]\text{-}macN_4)^{2+}$	2^g	D_{4h}	21,270	58,480	C	3000[f]		1057
$Ni([12]\text{-}aneN_4)^{2+}$	2^g	D_{4h}	20,255	55,690	C	3000[f]	1058,1060	
$Ni([13]\text{-}aneN_4)^{2+}$	2^g	D_{4h}	18,460	50,755	C	3000[f]		1058
$Ni([14]\text{-}aneN_4)^{2+}$	2^g	D_{4h}	19,520	53,670	C	3000[f]		1058
$Ni([15]\text{-}aneN_4)^{2+}$	2^g	D_{4h}	18,550	51,005	C	3000[f]		1058
$Ni(2\text{-}2\text{-}2\text{-}tetN_4)^{2+}$	2^g	D_{4h}	19,520	53,670	C	3000[f]		1058
$Ni(PyNO)_6^{2+}$	2	O_h	7940	21,830				1344
$Ni(4\text{-}MePyNO)_6^{2+}$	2	O_h	7940	21,830				1344
$Ni(2\text{-}MePyNO)_6^{2+}$	2	O_h	7850	21,585				1344
$Ni(2\text{-}EtPyNO)_6^{2+}$	2	O_h	7810	21,475				1344
$Ni(2,6Me2PyNO)_6^{2+}$	2	O_h	7650	21,035				1344
$Ni(im)_6^{2+}$	2	O_h	10,030	28,320		920		913
$Ni(den)_2^{2+}$	2	O_h	11,760	32,335		784		1065
$Ni(en)_3^{2+}$	2	O_h	11,850	32,580		790		1066
$Ni(N_3)_6^{4-}$	2	O_h	9160	25,185		760	3.70	1313
$Ni(NH_3)_6^{2+}$	2	O_h	11,000	30,245		846		1066
$Ni(NCSe)_6^{3-}$	2	O_h	9100	25,020		880		198
$Ni(glycine)_3^{-}$	2	O_h	10,100	27,770				176
$Ni(\beta\text{-alanine})_3^{-}$	2	O_h	9950	27,360				176
$Ni([12]macS_4)^{2+}$	2	O_h	10,000	27,495		880		1064
$Ni([13]macS_4)^{2+}$	2	O_h	10,700	29,420		907		1064
$Ni([12]macS_3)^{2+}$	2	O_h	10,750	29,560		713		1064
$Ni(hist)_2(H_2O)_2$	2	O_h	11,250	30,930		783	4.67	1020
$Ni(dth)^{2+}$	2	O_h	10,550	29,010		747		1064
$Ni(NNEU)_3^{2+}$	2	O_h	8200	22,545		920		808
$Ni(Me_4D)_6^{2+}$	2	D_3	9000	24,745		560		619
$Ni(Et_4D)_3^{2+}$	2	D_3	8500	23,370		540		619
Ni/ZnO	2	T_d	4110	11,300		777		943
Ni/ZnS(cubic)	2	T_d	4130	11,355		592		943
Ni/ZnS(hex.)	2	T_d	4260	11,715		587		943
$Ni(CN)_4^{2-}$	2	D_{4h}	25,300	69,565	C	2800[f]		1073
$Ni(mnt)_2^{2-}$	2	D_{4h}	14,490	39,840	C	2800[f]		1073
$Ni(dtp)_2$	2	D_{4h}	17,300	47,570	C	2800[f]		1073
$Ni(dtc)_2$	2	D_{4h}	18,800	51,690	C	2800[f]		1073
NiI_4^{2-}	2	T_d	3740	10,285		631		845
$NiCl_4^{2-}$	2	T_d	4120	11,330		796		917
$NiBr_4^{2-}$	2	T_d	3830	10,530		735		917
$Ni(NCO)_4^{2-}$	2	T_d	5070	13,940		862		917
$Ni(quin)_2Br_2$	2	C_{2v}	5190	14,270		802		845
$Ni(2,3\text{-}lut)_2Cl_2$	2	C_{2v}	5560	15,290		899		917
$Ni(2,3\text{-}lut)_2Br_2$	2	C_{2v}	5560	15,290		912		917
$Ni(Ph_3PO)_2Cl_2$	2	C_{2v}	3980	10,945		856		917
$Ni(Ph_3AsO)_2Cl_2$	2	C_{2v}	4320	11,880		850		845

cont

Table 9.1 cont

	Ox.No Group		10 Dq	DQ		B	C/B	Ref
$Ni(WS_4)_2^{2-}$	2	D_{4h}	17,100	47,020				1048,1050
$Ni(WOS_3)_2^{2-}$	2	D_{4h}	17,300	47,565				1049,1050
$Ni(WO_2S_2)_2^{2-}$	2	D_{4h}	17,900	49,215				1050
Palladium								
PdF_6^{2-}	4	O_h	26,000	71,490		400	5.0	526
Platinum								
PtF_6	6	O_h	?			300	5.0	526
PtF_6^{2-}	4	O_h	30,000	82,485	2500	300	6.0	689
Copper								
CuF_6^{3-}	3	O_h	14,100	38,770		641		520,1238
$Cu([12]aneN_4)^{2+}$	2	D_{4h}	16,810	46,200				
$Cu([13]aneN_4)^{2+}$	2	D_{4h}	18,310	50,350				
$Cu([14]aneN_4)^{2+}$	2	D_{4h}	19,900	54,700				
$Cu([15]aneN_4)^{2+}$	2	D_{4h}	17,610	48,420				
Silver								
AgF_6^{3-}	3	O_h	18,400	50,600	170	470	5.0	526

a) This table should be used in conjunction with the tables in Chapter 6. The references in the final column refer to Chapter 6. b) Bis(Tetraphenyldithioimidodiphosphinato)manganese(II). c) Alternative less probable assignment yields 10 Dq = 34,600 and B = 520 cm^{-1} [527]. d) From N.A.Beach and H.B.Gray, J.Am.Chem.Soc., **90** (1968) 5713. e) J.R.Dickinson and K.E.Johnson, Can.J.Chem., **45** (1967) 2457. f) Estimated. g) Low spin.

This probably reflects a decrease in metal-ligand bond length, and a concomitant increase in covalency.

Return now to an octahedral complex, ML_6, with non-pi bonding ligands. Central field covalency would cause a general reduction in B according to the covalency in the bond. The electrons in the e_g orbitals may also suffer a decrease in mutual repulsion if they are delocalised to some degree, over the ligand L, via the sigma anti-bonding orbitals. Evidently if B were measured for the non-bonding t_{2g} electrons, and then measured for the e_g electrons, different values would be obtained. Indeed this may be the case, and reflects the phenomenon known as 'differential radial expansion'. This possibility was explored by Stephens [28] and by Jorgensen [18,27]. In octahedral chromium(III) complexes, for example, two classes of transition are expected, namely, transitions involving an electron excited from t_{2g} to e_g:

e.g. $^4T_{1g}(F) \text{---} ^4A_{2g}$ (mainly $(t_{2g})^3$ to $(t_{2g})^2(e_g)$)

and transitions within the t_{2g} set, spin flips, e.g.:

e.g. $^2E_g \text{---} ^4A_{2g}$ ($(t_{2g})^3$ S = 3/2 to $(t_{2g})^3$ S = 1/2).

The value of B obtained from analysis of the spin allowed transition will, in general, be different from that obtained from the spin forbidden transition. The two values of B are distinguished by:

$$B(\text{spin allowed})/B_o = \beta_{35}$$

$$B(\text{spin forbidden within } t_{2g} \text{ set})/B_o = \beta_{55} \qquad (9.5)$$

where the numeral subscripts refer to the Bethe nomenclature $\Gamma_3 = e_g$, $\Gamma_5 = t_{2g}$.

Similarly for d^8 complexes, a spin flip within the $(e_g)^2$ configuration:

e.g. $^1E_g \text{---} ^3A_{2g}$ ($(t_{2g})^6(e_g)^2$ S = 1 to $(t_{2g})^6(e_g)^2$ S = 0)

can provide another B value, viz:

$$B(\text{spin forbidden within } e_g)/B_o = \beta_{33} \qquad (9.6)$$

In principle, variations in β_{55} occurring within t_{2g} will reflect pi bonding or anti-bonding, while variations in β_{33} occurring with e_g will reflect sigma anti-bonding effects. Relationships between these parameters and molecular orbital coefficients for sigma and pi bonding can be developed [18,19] and Jorgensen presents tables of such parameters [19]. Certainly in complexes of highly polarisable ligands such as the sulphur donors, with polarising metals such as Rh(III) and Ir(III), very low values of these parameters are encountered (e.g. less than 0.3 in these cases, see Table 9.1), but the procedure is probably too oversimplified to be very useful. Indeed Jorgensen points out that the parameterisation discussed above can only begin to be valid in the strong field approximation when the transitions do clearly involve the configurations outlined. However the principle of more than one B value being needed to analysis an electronic spectrum is certainly valid.

For example, note the following data for fluorides [19]:

	M	β_{55}	β_{35}
CrF_6^{3-}	Cr(III)	0.93	0.89
MnF_6^{2-}	Mn(IV)	0.77	0.56
NiF_6	Ni(IV)		0.36
OsF_6	Os(VI)	0.52	
IrF_6	Ir(VI)	0.43	

(9.7)

which may well reflect increasing covalency as the metal becomes more strongly oxidising. Since the parameter β_{35} is more strongly

Table 9.2

a) SPECTROCHEMICAL f AND h FACTORS [18,27]

	f	h		f	h
Br-	0.72	2.3	$(NH_2)_2CS$	1.01	
Et_2PSe2^-	0.74		NCS$^-$	1.02	
SCN^-	0.75		NCSe$^-$	1.03	
Cl^-	0.80	2.0	$MeNH_2$	1.17	
$Et_2PS_2^-$	0.78		$NH_2CH_2CO_2^-$	1.18	
$Et_2PSe_2^-$	0.8		$MeS(CH_2)_2SMe$	1.22	
$POCl_3$	0.82		CH_3CN	1.22	
N_3^-	0.83		Py	1.23	
$(EtO)_2PS_2^-$	0.83		NCSH	1.25	
$Et_2NCSe_2^-$	0.85		$NCSHg^+$	1.25	
dtp^-	0.86	2.8	NH_3	1.25	1.4
F^-	0.9	0.8	en	1.28	1.5
$Et_2NCS_2^-$	0.90		2,2-tri	1.29	
Me_2SO	0.91	1.2	NH_2OH	1.30	
urea	0.91	1.2			
Me_2CO	0.92		SO_3^{2-}	1.3	
MeCOOH	0.94		diars	1.33	
EtOH	0.97		bpy	1.33	
Me_2NCHO	0.98		o-phen	1.34	
ox^{2-}	0.99	1.5	NO_2^-	1.4	
H_2O	1.00	1.0	CN^-	1.7	2.0
$CH_2(CO_2)_2^{2-}$	1.00		dsep$^-$		3.0

b) SPECTROCHEMICAL g AND k FACTORS [18,27]

Metal ion	g	k	Metal ion	g	k
Mn(II)	8.0		Ag(III)	20.4	
Ni(II)	8.7	0.12	Ni(IV)	22	
Co(II)	9		Mn(IV)	24	0.5
V(II)	12.0	0.08	Mo(III)	24.6	0.15
Fe(III)	14.0	0.24	Rh(III)	27.0	0.30
Cu(III)	15.7		Pd(IV)	29	
Cr(III)	15.7	0.21	Tc(IV)	31	
Co(III)	18.2	0.35	Ir(III)	32	0.3
Ru(II)	20		Re(IV)	35	0.2
			Pt(IV)	36	0.5

affected, one might conclude that sigma bonding is dominant.

As the nephelauxetic series is roughly independent of metal ion, it is possible to write a simple equation to factorise the β parameter as follows [18]:

$$(1 - \beta) = h(\text{ligands}).k(\text{central ion}) \qquad (9.8)$$

which provides a simple, but approximate, procedure for evaluating the nephelauxetic parameter. It should not be taken too quantitatively. However if one is studying a particular metal complex and the calculated β value deviates markedly from that predicted by (9.8), one should be on the alert for something unusual. Values of the h and k parameters are shown in Table 9.2.

Although one generally utilises a single (spherical) B value, there are, in an octahedral complex, ten pairs of Coulomb integrals which might realistically be expected to involve ten different interelectronic repulsion integrals. If the above treatment is used, at most three values of B can be considered. In general it is not possible to predict accurately all spin allowed and spin forbidden transitions with only one value of B, or even two. Indeed even the spin allowed bands of a 'well behaved' simple complex are not always explainable by a single value of B, e.g. consider the $[Ni(H_2O)_6]^{2+}$ ion which has absorptions at 8500 ($^3T_{2g}$), 13,800 ($^3T_{1g}(F)$) and 25,300 ($^3T_{1g}(P)$) cm^{-1}. B may be calculated in several ways (see Appendix V):

i) from ν_1 and ν_2; $\nu_2/\nu_1 = 1.624$, hence Dq/B = 1.12 and B = 759 cm^{-1}. ν_3 is predicted to lie at 23,100 cm^{-1}.

ii) from ν_1 and ν_3; $\nu_3/\nu_1 = 2.977$, hence Dq/B = 0.91 and B = 934 cm^{-1}. ν_2 is predicted to lie at 14,160 cm^{-1}.

iii) from ν_2 and ν_3; $\nu_3/\nu_2 = 1.833$, hence Dq/B = 0.86 and $E(\nu_3)/B$ = 26.4. Hence B = 958 cm^{-1}. ν_1 (10Dq) is predicted to lie at 8240 cm^{-1}.

iv) $B = (\nu_2 + \nu_3 - 3\nu_1)/15 = 907$ cm^{-1}, and ν_2 and ν_3 are predicted to lie at 14,110 and 24,990 cm^{-1} respectively.

The first calculation yields a value of ν_3 which is quite inaccurate, thus B = 759 cm^{-1} can be excluded. There is little to choose between the other calculations, but it is evident that B is not very well defined; an average value of 930 cm^{-1} might be chosen. In many cases the variation in the calculated values of B is considerably greater than the example chosen here and taking a mean value may not be very realistic. Thus B values should be discussed with some caution and their analytical significance not stretched too far.

Note that the occurrence of spin paired and spin free complexes depends upon the relative magnitudes of 10Dq and the 'pairing energy', and that the latter clearly depends upon the Racah repulsion parameters. In the absence of configurational interaction, the pairing energy is equal to:

		M(II)	M(III)
d^4	6B + 5C	23,500	28,000 cm^{-1}
d^5	7.5B + 5C	25,500	30,000
d^6	2.5B + 4C	17,600	21,000
d^7	4B + 4C	19,500	23,500

(9.9)

where typical values for M(II) and M(III) ions of the first row transition metal ions are also included. Konig and Kremer [20] have discussed 'exact pairing energies' obtained by inclusion of full configurational interaction.

One may also utilise the 'spin-pairing energy' discussed in Chapter 5 in connection with the optical electronegativity (Table 5.4, Eqn.5.9) reflecting the average spin repulsion energy of all the multiplets of a given configuration with given spin S [18,19]. As explored by Vanquickenborne and Haspeslagh [24], the difference in such average spin-pairing energy between a configuration with spin S-1 and the same configuration with spin S, is simply 2SD (cf. entries in Table 5.4, and definition of D in section 5.3). Since D is positive this provides rationalisation of Hund's first (spin) rule that a high spin configuration will have a lower energy than a low spin configuration.

However these workers [24] explore the significance of spin pairing in a range of metal ions of different configuration and derive some interesting results. They divide the total (Hartree-Fock) repulsion energy into three components, C'(dd) the intra d-d shell repulsion, C'(cd), the repulsion between the d shell and the core, and C'(c), the intra-core repulsion.

> These workers [24] parameterised the problem in terms of the change in interelectronic repulsion energy per unit change in spin, i.e. $\Delta C/2S$. In the discussion which follows, it is this term with which we deal, but for simplicity it will be referred to simply as a component of the interelectronic repulsion energy.

Firstly they observe that while spin-pairing a configuration requires additional energy, the total interelectronic repulsion energy decreases, the additional energy actually being predominantly a nuclear attraction energy. They do find that the C'(dd) term increases with decreasing spin but it is a relatively small contribution to the total repulsion energy. The term C'(c) also increases with decreasing spin caused by a concomitant decrease in core size. The major contribution to the total repulsion energy upon spin-pairing is C'(cd), the core-d shell repulsion, which strongly decreases with decreasing spin. This appears to be due to an expansion of the d shell with decreasing spin.

Across the first row transition series, from Ti(II) to Ni(II), both B and hence D increase. However this analysis reveals that the C'(dd) term actually decreases (i.e. the net change of C'(dd) upon spin pairing, decreases across the series) and once again the overall increase in D is due to a nuclear attraction term which increases across the series.

Thus although one's simple ideas about effect of spin on

repulsion energy seems to provide qualitatively the correct answer, the true situation seems, on the basis of this work [24], to be far more intricate. The possibility exists therefore, that the equally simple ideas expressed above concerning the relationship between variation of B with central field covalency and symmetry restricted covalency may only be fortuitously predictive [24,25].

There has been little attempt to rationalise the variation of the Racah parameter C, probably because it is even more difficult to quantify than B. Accurate evaluation of C would generally require a complete analysis of the spin-forbidden transitions in a molecule including a full mathematical solution to the, usually complex, matrices. However Fatta and Lintvedt did report [29] a study of a related series of tris(diketonato)iron(III) species and calculated F_2 and F_4 and, equivalently, B and C. In effect, the ratio of C/B varied from 5.4 to 7.4 in the series studied. Data were previously presented on p.453. They note that electron donating substituents, such as t-butyl, on the diketone ligand increase B, while electron withdrawing groups, such as CF_3, decrease B. Surprisingly, the reverse trends are observed with C.

9.3 Spectrochemical Series

Much of this book has emphasized the assignment of electronic transitions, methods for calculating spectra and the definition of a range of parameters which can reproduce such spectra. Clearly however the ultimate aim must be the derivation of useful chemical information from analysis of an electronic spectrum. In this chapter the kind of information which can be derived from a spectrum is summarised and discussed. In the early development of the theory, the values of Dq were compared for a wide range of ligands and metals, and the so-called 'Spectrochemical Series' was developed. The relative Dq values for a series of ligands do not depend significantly upon metal; thus one may produce, for any given metal ion, a series of ligands of increasing Dq. This may be written:

$$I^- < Br^- < O_3CrO^{2-} < dsep^- = S^{2-} = dtp^- < N_3^- < F^- < dtc^- < urea$$
$$= OH^- = IO_3^{2-} < oxalate^{2-} = malonate^{2-} = O^{2-} < H_2O < SCN^-$$
$$< pyridine = NH_3 < en = SO_3^{2-} < NO_2^- = bipy = o\text{-}phen < CH_3^- = ph^-$$
$$< CN^- < constrained\ phosphites = CO. \qquad (9.10)$$

Such an order is difficult to rationalise on the basis of crystal field theory, since ligands with little basicity (small net charge) often have high Dq values. Why should, for example, the doubly charged oxide ion exert a weaker field than the neutral water molecule ?

In terms of AOM theory, however, the series is fairly readily appreciated. Since $10Dq = 3e_\sigma(L) - 4e_\pi(L)$ it is evident that if

sigma basicity is not dominant, then the pi term must determine the position in the spectrochemical series. The $e_\sigma(L)$ value must always be positive. Those ligands which are pi acceptors ($e_\pi(L)$ is negative), lie at the high end, while ligands such as the halides with moderately large pi-donor character ($e_\pi(L)$ is positive) fall at the low end. The oxide ion with more negative charge than water, is a better pi-donor; it has therefore a larger $e_\pi(L)$ value than water.

For a given ligand, the Dq value varies approximately with metal ion, in the series:

$$\text{Mn(II)} < \text{Co(II)} = \text{Ni(II)} < \text{V(II)} < \text{Fe(III)} < \text{Cr(III)} = \text{V(III)} < \text{Co(III)} < \text{Mn(IV)} < \text{Mo(III)} < \text{Rh(III)} < \text{Ir(III)} < \text{Pt(IV)} \quad (9.11)$$

Dq increases with increasing oxidation state for a given metal ion, and increases about 25-50% for each transition series.

Various other series of groups of ligands have been discussed, e.g. the variation within a group of fairly closely related sulphur donor ligands [18,19]:

$$Cl^- < R_2PS_2^- = F_2PS_2^- < R_2NCS_2^- < RSCS_2^- < ROCS_2^- < H_2O < R_2S < NH_3 = SO_3^{2-} \quad (9.12)$$

Several series were displayed in Chapter 6, e.g. for Ti(III), p.382, for $[Co(CN)_5L]^{n-}$ species on p.477 and for $[Pd(II)S_4]$ chromophores on p.552.

Busch has published some interesting spectrochemical series involving the macrocyclic ligands. For example, the Dq values in high spin $[Ni(Mac)(NCS)_2]$ complexes decrease in the order [23,31]:

$$Me_2PydieneN_4 > Me_4[14]1,3,8,10\text{-tetraeneN}_4 > Me_2[14]1,3\text{-dieneN}_4 = Me_6[14]4,11\text{-dieneN}_4 > Me_2[14]aneN_4 > Me_6[14]aneN_4 > Me_2[14]pyaneN_4 \quad (9.13)$$

Iron(II) macrocycles follow the same sequence where data are available. Cobalt(III) macrocycles also yield a very similar sequence.

Table 9.1 contains a wide selection of spectrochemical Dq and nephelauxetic data for many metal complexes. In general the data in Table 9.1 have been culled from tables in Chapter 6 but they are by no means exhaustive.

Note that the references in Tables 9.1, 9.3-9.5 refer to Chapter 6.

However prior to such a discussion of such data, consider an early approach to rationalise Dq data.

Since essentially the same spectrochemical series is obtained for any metal ion, it is possible to factor the Dq value into a ligand factor and a metal ion factor (similar to 9.8 above), such that:

$$10Dq = f(\text{ligands}).g(\text{metal ion}) \quad (9.14)$$

This was demonstrated many years ago by Jorgensen [18] and the factors are included in Table 9.2. Once again some caution should be used with respect to this equation. If it were strictly accurate it would relegate inorganic spectroscopy to a rather well understood and somewhat mundane subject, which is not the case. Rather, as with (9.8), be on the alert if the calculated value for 10Dq in a species deviates markedly from (9.14). Both (9.8) and (9.14) are of some use when analysing data from an unknown system, such as a metalloprotein, where some clues about the metal environment are being sought, or whether an observed spectrum 'fits' a given formulation.

The above series are technically valid only for ML_6 (or ML_4) systems. Lower symmetry complexes can provide a great deal more data. However if only low resolution spectra are available, it is common to calculate an average Dq and B value for the several ligands. It is possible, under such circumstances, to talk of an average ligand field, being a weighted sum, i.e. in the case of $[ML_4Z_2]$:

$$Dq(av) = (1/6)[4Dq(L) + 2Dq(Z)] \quad (9.15)$$

One anticipates that the observed average field will be that calculated from consideration of the fields in $[ML_6]$ and $[MZ_6]$. This can be taken advantage of when considering, for example, a biological chromophore where the bound ligands are uncertain. It is also useful with ambidentate ligands (ligands which can coordinate through one of several ligating atoms, e.g. NCS^- and SCN^-, and ONO^- and O_2N^-) in order to determine which atom is ligating. However when one or both of the ligands are sterically demanding, then this approach must be used with extreme care.

Where higher resolution spectra are available (solution, if L and Z differ greatly, otherwise cryogenic single crystal or possibly mull spectra) a great deal more chemical information can be derived.

In Tables 9.3-9.5 are collected such data for a wide range of complexes, described in terms of NSH, AOM and McClure parameters.

The results, in general, support the views expressed earlier concerning the spectrochemical series and the relative order of ligands.

If more than one ligand in a low symmetry mixed ligand complex π-bonds to the metal, it is not possible to independently obtain $e_\sigma(L)$ and $e_\pi(L)$ for all ligands, from an electronic spectrum, unless at least one $e_\pi(L)$ value may be written as zero. If this is not possible, one is restricted to evaluating $d\sigma$ and $d\pi$. However, if additional information, such as anisotropic esr or magnetic data are available, a more complete analysis may be valid.

Table 9.3

NSH AND McCLURE PARAMETERS

Complex	DQ	DS	DT	dσ	dπ	Ref
V(III)						
$V(H_2O)_4Cl_2^+$	46,675	-1675	-292	-399	-305	199
Cr(II)						
$KCrF_3$	23,465	-11,970	-5965	-3390	-1465	377
Na_2CrF_4	25,525	-12,950	-5965	-3600	-1675	377
Cr(III)						
$Cr(H_2O)_4Cl_2^+$	45,380	-1675	-292	-399	-305	289
$Cr(H_2O)_5CN^{2+}$	53,075	2710	4188	1160	-192	914
$Cr(CN)_5(H_2O)^{2-}$	67,630	-2710	-4188	-1160	192	914
$c\text{-}Cr(CN)_4(H_2O)_2^-$	63,960	2710	4188	1159	-192	914
$c\text{-}Cr(CN)_2(H_2O)_4^{2+}$	54,050	-2710	-4188	-1160	192	914
$Cr(NH_3)_4F_2^+$	52,425	4460	-4324	357	1753	365
$Cr(NH_2Me)_4F_2^+$	51,680	4722	-3309	554	1622	365
$Cr(NH_2Pr)_4F_2^+$	51,480	5214	-2781	732	1630	365
$Cr(en)_2F_2^+$	53,875	5187	-4894	434	2014	269
$Cr(en)_2Cl_2^+$	52,310	-1750	-6914	-1332	900	269
$Cr(en)_2Br_2^+$	52,630	-2450	-8269	-1669	1000	269
$Cr(en)_2(OH)_2^+$	52,365	8281	-2034	1493	2149	269
$Cr(en)_2(H_2O)_2^{3+}$	56,680	2674	-4311	-24	1368	269
$Cr(NH_3)_5OH_2^{+2}$	53,430	4200	-2712	1050	2800	276
$Cr(Py)_4F_2^+$	50,065	10,990	-678	2261	2480	272
$Cr(pd)_2F_2^+$	56,400	8400	-3525	1312	2450	365
$Cr(pd)_2Cl_2^+$	52,875	-1050	-6900	-1180	1047	365
$Cr(pd)_2Br_2^+$	52,410	-3745	-9245	-2082	902	365
$Cr(pnd)_2F_2^+$	56,825	5845	-3959	705	1982	365
$Cr(chd)_2F_2^+$	56,655	5285	-4867	459	2030	365
$Cr(daud)_2F_2^+$	47,285	5670	-4162	639	1982	365
$Cr(dadd)_2F_2^+$	56,900	6090	-4013	749	2045	365
$Cr(datd)_2F_2^+$	56,780	6125	-3877	776	2027	365
Mn(III)						
$Mn(macN_4)Cl_2^+$	53,470	-8500	-16,390	-4088	1200	367
$Mn(macN_4)Br_2^+$	51,175	-7900	-19,490	-4390	1903	367
K_3MnF_6	39,500	-11,200	-7050	-3375	-1100	377
$Mn(ox)_3^{3-}$	42,345	-10,450	-8480	-3415	-675	428
$Mn(dmf)_6^{3+}$	33,405	-18,350	-13,270	-5765	-1485	430
$Mn(dmso)_6^{3+}$	31,580	-18,140	-12,050	-5555	-1665	430
$Mn(PyNO)_6$	30,550	-18,200	-11,360	-5470	-1805	430
$Mn(acac)_3$	42,725	-13,120	-5485	-3570	-1800	428
Co(II)						
$Co(H_2O)_4Cl_2$	21,790	4480[b]	-406	904	1035	702
$Co(s\text{-}Et_2en)_2(NCS)_2$	31,445	790	175	193	137	749
$Co(s\text{-}Et_2en)_2Cl_2$	27,970	-2966	-4700	-1286	231	749
$Co(s\text{-}Et_2en)_2Br_2$	25,520	-3370	-4700	-1372	144	749

cont

Table 9.3 cont

Complex	DQ	DS	DT	dσ	dπ	Ref
$Co(s\text{-}Me_2en)_2(NO_3)_2$	28,725	-4710	-3980	-1560	-275	749
$Co(3Mepyz)_4Cl_2$	25,365	-6020	-2847	-1684	-765	637
$Co(3Mepyz)_4Br_2$	24,220	-8120	-3932	-2284	-1015	637
$Co(3Mepyz)_4I_2$	22,750	-10,500	-5285	-2982	-1275	637
$Co(3Mepyz)_4(NO_3)_2$	26,260	-7770	-4065	-2228	-915	637
$Co(vinim)_4(SiF_6)_2$	26,010	-3713	-3501	-2710	-130	931
$Co(vinim)_4Cl_2$	26,240	-3713	-3501	-1280	-150	931
K_2CoF_4	22,525	-1365	-135	-311	-267	751
$Rb_2Mg(Co)F_4$	21,125	-161	-271	-382	-295	751
Rb_2CoF_4	20,000	-2345	-407	-559	-428	751
$K_2Mg(Co)F_4$	22,155	420	136	109	65	751
Ni(III)						
Rb_2KNiF_6	29,905	-8624	-6968	-2812	-563	951
Ni(II)						
$Ni(s\text{-}Et_2en)_2(NCS)_2$	32,655	-2545	-2727	-923	-42	749
$Ni(s\text{-}Et_2en)_2Cl_2$	27,560	-5072	-4394	-1695	-276	749
$Ni(s\text{-}Et_2en)_2Br_2$	27,170	-6777	-4383	-2058	-644	749
$Ni(s\text{-}Me_2en)_2(NO_3)_2$	29,395	-8116	-4414	-2350	-925	749
$Ni(thiourea)_4Cl_2$	22,125	-3393	-2912	-1130	-190	911
$Ni(Py)_4Cl_2$	27,540	-2352	-4053	-1064	243	913
$Ni(Py)_4Br_2$	26,710	-3745	-4636	-1443	52	913
$Ni(Py)_4(H_2O)_2$	28,985	-2688	-2996	-990	-23	913
$Ni(pyz)_4Br_2$	24,590	-9142	-4595	-2595	-1111	911
$Ni(2\text{-}Meim)Br^{+a}$	22,695	-9867	-7777	-3190	-680	911
$Ni(2\text{-}Meim)Cl^{+a}$	22,675	-10,895	-7560	-3380	-940	911
$Ni(imid)_4(H_2O)_2^{2+}$	28,565	-4942	-3443	-1535	-424	913
$Ni(vinim)_4(SiF_6)_2$	27,405	-5300	-5887	-1950	-50	931
$Ni(vinim)_4Cl_2$	25,275	-4680	-4895	-1680	-100	931
$Ni(5Mepyz)_4Cl_2$	26,820	-5980	-3820	-1810	-580	930
$Ni(5Mepyz)_4Br_2$	25,330	-8250	-4270	-2360	-980	930
$Ni(3Mepyz)_4Cl_2$	25,790	-6370	-3185	-1806	-788	637
$Ni(3Mepyz)_4Br_2$	24,800	-8155	-4135	-2320	-985	637
$Ni(3Mepyz)_4I_2$	23,500	-9870	-5355	-2856	-1128	637
$Ni(3Mepyz)_4(NO_3)_2$	26,190	-8505	-4474	-2442	-998	637
$Ni(en)_2(NCS)_2$	29,390	-4592	-3118	-1415	-409	928
$Ni(en)_2(NO_2)_2$	32,435	-98	922	106	-191	928
$Ni(NH_3)_4(NCS)_2$	29,815	770	217	195	25	929
$Ni(NH_3)_4(NO_2)_2$	32,255	700	1233	324	-75	929
$Ni(urea)_4Cl_2$	21,960	-2450	-3050	-947	37	1166
$Ni([14]aneN_4)Br_2$	29,900	-9688	-9285	-3360	-363	1089
$Ni([15]aneN_4)Br_2$	27,660	-6475	-6440	-2278	-200	1089
$Ni([16]aneN_4)Br_2$	24,500	-5005	-5505	-1834	-56	1089
$Ni(Me_2[14]aneN_4)Br_2$	29,820	-9625	-9353	-3356	-337	1089
$Ni(Me_6[14]aneN_4)Br_2$	29,250	-10,080	-9950	-3536	-325	1089
$Ni([14]aneN_4)Cl_2$	30,600	-6867	-8526	-2650	101	1089
$Ni([15]aneN_4)Cl_2$	28,160	-5845	-5056	-1952	-320	1089

cont

Table 9.3 cont.

Complex	DQ	DS	DT	dσ	dπ	Ref
$Ni([16]aneN_4)Cl_2$	25,120	-4641	-4704	-1645	-127	1089
$Ni([14]aneN_4)(N_3)_2$	32,545	-5831	-6073	-2090	-130	1089
$Ni([15]aneN_4)(N_3)_2$	29,485	-4200	-3592	-1397	-237	1089
$Ni([16]aneN_4)(N_3)_2$	26,870	-3864	-2548	-1180	-358	1089
$Ni([14]aneN_4)(NCS)_2$	34,015	-4340	-4202	-1511	-155	1089
$Ni([15]aneN_4)(NCS)_2$	30,355	-4865	-2277	-1358	-622	1089
$Ni([16]aneN_4)(NCS)_2$	27,685	-3010	-1559	-861	-358	1089
$Ni([14]'aneN_4)Cl_2$	29,050	-9975	-6425	-3026	-952	1197
$Ni([14]'aneN_4)(NCS)_2$	32,100	-4200	-4270	-1490	-112	1197
$Ni(TAAB)(NCS)_2$	34,200	-5761	-5138	-1945	-287	1089
Ag(II)						
AgF_6^{4-}	30,375	-11,545	-4725	-3130	-1605	1231
$AgPbF_6$	26,800	-11,740	-6588	-3426	-1300	1231

a) Recalculated. b) The data in [702] are fitted by Dq = 810, Ds = +640, Dt = 30 cm^{-1} with the sign conventions used in this book. All reference listings are those of Chapter 6.

It is dangerous to solve a spectrum by assuming that a parameter may be transferred from another complex. This may be possible for very similar complexes in which the M-L bond lengths are the same, but otherwise, while this may be accidentally accurate and may help to confirm the structural nature of a complex, it defeats the purpose of studying low symmetry systems to obtain information about how one M-L bond interacts with another.

Much of the rest of this chapter deals with interpretations of spectra in terms of the AOM model. It is surely tempting to relate the relative magnitudes of the AOM parameters to metal-ligand σ and π-bond strength. In drawing such a conclusion, two aspects should be considered. Firstly even if there were no covalent bond, but simply an electrostatic interaction, the d orbitals would be destabilised and values of the AOM parameters derived. This is certainly the situation in the more ionic complexes such as NiF_2. Thus even though the AOM model is based on metal-ligand overlap (see Chapter 1), it does not simply reflect such overlap.

Secondly, Burdett has discussed this problem in some depth [7]. Using simple perturbation theory he points out that the large destabilisation of an empty anti-bonding orbital is not always matched by a similarly large stabilisation of the corresponding bonding orbital. Some care should therefore be exercised in drawing bonding conclusions from the magnitudes of AOM parameters. The nature of the models used to describe spectroscopic data change and develop with the passing of time, but it still remains true that the total energy associated with the stabilisation or destabilisation of the d orbitals represents only a small fraction

of the total bonding energy of the molecule. It is probable that any semi-empirical model, such as the AOM, will fail if pushed too far.

We continue with some general observations concerning the data in Tables 9.1, 9.3-9.5, and subsequently discuss some aspects in more detail.

i) A specific ligand bound to a specific metal ion does possess Dq (or e_σ, or e_π) parameters which depend upon the environment and upon which other ligands are involved. The variations are often relatively small and are obviously subject to experimental error, but they are real. Thus parameters should not be transferred from one system to another without care.

ii) If one considers a series such as $[MN_4Z_2]^{n+}$, where Z is an anion, one expects that DT (or Dt) will increase as the difference between the fields of the two ligands increases. This is well known since DT reflects the difference in field strengths. However it is less well known that DS (or Ds) follows a similar series for the divalent M(II) ions of the first row transition series, thus DS becomes more negative usually in the sequence:

$$NCS^- < F^- < Cl^- < Br^- = NO_3^- < I^- \qquad (9.16)$$

This is primarily a reflection of the decrease in $e_\sigma(Z)$ in this sequence. See sequence 6.13 for another example of what has been termed the two-dimensional spectrochemical series.

iii) The magnitude of $e_\pi(L)$ is quite small in most six-coordinate first row transition divalent metal complexes. Although it is reasonably well defined by the 'goodness' of the fit to the experimental data, it is probably not meaningful in a chemical sense. For example considering two complexes with $e_\pi(L)$= 200 and $e_\pi(L')$ = -200 cm^{-1}, it would be foolhardy to conclude that L is a π-donor and L' a π-acceptor.

The e_π values for four and five coordinate complexes, however, can be significantly higher (see below).

The $e_\pi(L)$ parameter is also generally larger for trivalent ions such as Cr(III). In this case, one can reasonably conclude that some kind of π-interaction is present and its sign can provide evidence for π-donation or acceptance.

iv) The AOM approach has indicated some fairly startling interactions between various metal ions and ligands. For example the Cr(III)-F and Cr(III)-OH bonds have e_σ and e_π values which are exceptionally large (Table 9.5) suggesting a possible special bonding relationship between OH^- and F^- and Cr(III).

However Burdett [7] notes, in his simple perturbation model, the stabilisation of the bonding orbitals is greater for Cr(III)-Cl than for Cr(III)-F. Thus the large AOM values observed for Cr-F and Cr-OH, while undoubtedly telling us something about the nature of the Cr-L bond, probably do not simply reflect an increase in bond strength.

Table 9.4

AXIAL AND EQUATORIAL FIELDS

Metal ion	DQ(L)	Dq(L)	DQ(Z)	Dq(Z)	B	Ref
Vanadium						
$V(H_2O)_4Cl_2^+$	47,020	1710	46,000	1673	580[a]	
Chromium						
$KCrF_3$	30,525	1110	9350	340		377
Na_2CrF_4	32,600	1185	11,410	415		377
$Cr(CN)(H_2O)_5^{2+}$	77,870	2832	48,120	1750	560	914
$Cr(CN)_5(H_2O)^{2-}$	72,590	2640	42,840	1558	573	914
$c\text{-}Cr(CN)_4(H_2O)^-$	73,850	2686	44,130	1605	575	914
$c\text{-}Cr(CN)_2(H_2O)^+$	73,850	2686	44,130	1605	575	914
$Cr(H_2O)_4F_2^+$	47,840	1740	42,260	1537	700	269
$Cr(H_2O)_4Cl_2^+$	45,725	1663	44,690	1625	572[b]	289
$Cr(NH_3)_4F_2^+$	57,540	2092	42,190	1534		365
$Cr(NH_2Me)_4F_2^+$	55,600	2021	43,850	1595		365
$Cr(NH_2Pr)_4F_2^+$	54,770	1992	44,890	1633		365
$Cr(en)_2F_2^+$	59,670	2170	42,300	1538	692	269
$Cr(en)_2Cl_2^+$	60,490	2200	35,950	1307	670	269
$Cr(en)_2Br_2^+$	62,415	2270	33,060	1202	560	269
$Cr(en)_2(OH)_2^+$	54,770	1992	47,550	1729	778	269
$Cr(en)_2(H_2O)_2^+$	61,780	2247	46,480	1690	744	269
$Cr(py)_4F_2^+$	50,870	1850	48,460	1762		272
$Cr(pd)_4F_2^+$	60,570	2203	48,060	1748	520	365
$Cr(pd)_4Cl_2^+$	61,040	2220	36,550	1329	590	365
$Cr(pd)_4Br_2^+$	63,350	2304	30,530	1110	600	365
$Cr(pn)_2(OAc)_2$	61,920	2252	49,880	1814	625	365
$Cr(pnd)_2F_2^+$	61,510	2237	47,450	1726	566	365
$Cr(chd)_2F_2^+$	62,410	2270	45,120	1641	595	365
$Cr(daud)_2F_2^+$	62,060	2257	47,260	1719	630	365
$Cr(dadd)_2F_2^+$	61,640	2242	47,400	1724	595	365
$Cr(datd)_2F_2^+$	61,370	2232	47,590	1731	621	365
Manganese(III)						
$Mn(macN_4)Cl_2^+$	72,860	2650	14,690	534		367
$Mn(macN_4)Br_2^+$	74,235	2700	5045	183		367
K_3MnF_6	47,840	1740	22,815	830		377
$Mn(ox)_3^{3-}$	52,380	1905	22,280	810		428
$Mn(dmf)_6^{3+}$	49,110	1786	2000	73		430
$Mn(dmso)_6^{3+}$	45,837	1667	3065	111		430
$Mn(PyNO)_6^{3+}$	43,990	1600	3665	133		430
$Mn(acac)_3$	49,215	1790	29,745	1082		428
$Mn(tropol)_3^{3-}$	43,660	1588	15,340	558		428
$Mn(dtc)_3$	44,000	1600	40,140	1460		495,496

cont

Table 9.4 cont

Metal ion	DQ(L)	Dq(L)	DQ(Z)	Dq(Z)	B	Ref
Iron(II)						
$Fe(II)([13]aneN_4)(CN)_2$	60,700	2208	84,050	3057		611
$Fe(II)([14]aneN_4)(CN)_2$	55,790	2029	78,280	2847		611
$Fe(II)([15]aneN_4)(CN)_2$	50,650	1842	75,120	2732		611
$Fe(II)([15]eneN_4)(Py)_2^{2+}$	62,960	2290	27,800	1011		613
$Fe(MPZ)_4(NO_3)_2$	30,800	1120	19,700	717		637
$Fe(MPZ)_4Cl_2$	27,770	1010	18,630	678		637
$Fe(MPZ)_4Br_2$	27,910	1015	14,430	525		637
$Fe(MPZ)_4I_2$	27,910	1015	9130	332		637
Cobalt(II)						
$Co([13]aneN_4)Cl_2^+$	75,600	2750	36,980	1345		614
$Co([14]aneN_4)Cl_2^+$	70,440	2562	39,500	1437		614
$Co([15]aneN_4)Cl_2^+$	66,570	2421	40,280	1465		614
$Co([16]aneN_4)Cl_2^+$	64,370	2341	39,620	1441		614
$Co(NH_3)_5Br^{2+}$	70,940	2580	40,030	1456		638
$Co(MeNH_2)_5Br^{2+}$	64,060	2330	36,600	1332		638
$Co(EtNH_2)_5Br^{2+}$	63,380	2305	35,400	1287		638
$Co(NH_3)_5Cl^{2+}$	69,000	2510	40,250	1464	501	639
$Co(MeNH_2)_5Cl^{2+}$	65,160	2370	40,140	1460	468	639
$Co(n\text{-}BuNH_2)_5Cl^{2+}$	64,900	2360	39,500	1436	478	639
Cobalt(II)						
$Co(H_2O)_4Cl_2$	22,270	810	20,830	758		702
$Co(s\text{-}Et_2en)_2(NCS)_2$	31,235	1136	31,860	1158	780	749
$Co(s\text{-}Et_2en)_2Cl_2$	33,535	1220	16,845	613	833	749
$Co(s\text{-}Et_2en)_2Br_2$	31,080	1130	14,400	524	847	749
$Co(s\text{-}Me_2en)_2(NO_3)_2$	33,435	1216	19,305	702	875	749
$Co(3Mepyz)_4Cl_2$	28,730	1045	18,628	678	820	637
$Co(3Mepyz)_4Br_2$	28,870	1050	14,920	545	785	637
$Co(3Mepyz)_4I_2$	29,000	1055	10,240	370	775	637
$Co(3Mepyz)_4(NO_3)_2$	31,070	1130	16,635	605	820	637
$Co(vinim)_4(SiF_6)_2$	35,470	1290	7095	258	880	931
$Co(vinim)_4Cl_2$	30,380	1105	23,455	853	818	931
K_2CoF_4	22,685	825	22,215	808		751
$Rb_2Mg(Co)F_4$	21,445	780	20,485	745		751
Rb_2CoF_4	20,485	745	19,055	693		751
$K_2Mg(Co)F_4$	21,995	800	22,490	818		751
$Rh([14]aneN_4)X_2^+$		2500				
$Rh([15]aneN_4)X_2^+$		2380				
$Rh([16]aneN_4)X_2^+$		2273				
Nickel(III)						
Rb_2KNiF_6	38,150	1387	13,420	489	750	951
Nickel(II)						
$Ni(s\text{-}Et_2en)_2(NCS)_2$	35,880	1305	26,200	953	690	927
$Ni(s\text{-}Et_2en)_2Cl_2$	32,760	1191	17,165	624	864	927

cont

Table 9.4 cont

Metal ion	DQ(L)	Dq(L)	DQ(Z)	Dq(Z)	B	Ref
$Ni(s\text{-}Et_2en)_2Br_2$	32,355	1177	16,800	611	845	927
$Ni(s\text{-}Me_2en)_2(NO_3)_2$	34,615	1259	18,950	689	887	927
$Ni(thiourea)_4Cl_2$	25,570	930	15,230	554		911
$Ni(thiourea)_4Cl_2$	25,570	930	14,740	536		1166
$Ni(Py)_4Cl_2$	32,335	1176	17,955	653		828
$Ni(Py)_4Br_2$	32,200	1171	15,755	573		723
$Ni(Py)_4(H_2O)_2$	32,530	1183	21,885	796		784
$Ni(pyz)_4Br_2$	29,970	1090	13,470	490	842	911
$Ni(pyz)_4Cl_2$	31,110	1095	16,470	599	870	911
$Ni(2\text{-}Meim)Br^+$	31,895	1160	8580	312		911
$Ni(2\text{-}Meim)Cl^+$	31,620	1150	9570	348	873	911
$Ni(imid)_4(H_2O)_2^{2+}$	32,640	1187	20,400	742	770	913
$Ni(vinim)_4(SiF_6)_2$	34,370	1250	13,470	490	930	931
$Ni(vinim)_4Cl_2$	31,070	1130	13,690	498	885	931
$Ni(5Mepyz)_4Cl_2$	31,350	1140	17,770	646	822	930
$Ni(5Mepyz)_4Br_2$	30,380	1105	15,220	553	822	930
$Ni(3Mepyz)_4Cl_2$	29,560	1075	18,250	664	845	637
$Ni(3Mepyz)_4Br_2$	29,695	1080	15,020	546	840	637
$Ni(3Mepyz)_4I_2$	29,830	1085	10,825	394	825	637
$Ni(3Mepyz)_4(NO_3)_2$	31,480	1145	15,600	567	880	637
$Ni(en)_2(NCS)_2$	33,080	1203	21,995	800	883	928
$Ni(en)_2(NO_2)_2$	31,345	1140	34,645	1260		928
$Ni(NH_3)_4(NCS)_2$	29,560	1075	30,330	1103		929
$Ni(NH_3)_4(NO_2)_2$	30,795	1120	35,195	1280		929
$Ni([14]'aneN_4)Cl_2$	36,650	1425	13,850	504		1197
$Ni([14]'aneN_4)(NCS)_2$	37,145	1351	21,990	800		1197
$Ni(urea)_4Cl_2$	25,570	930	14,750	536	750	1166
$Ni([14]aneN_4)Br_2$	40,890	1487	7923	288	849	1089
$Ni([15]aneN_4)Br_2$	35,280	1283	12,425	452	818	1089
$Ni([16]aneN_4)Br_2$	31,020	1128	11,480	418	853	1089
$Ni(Me_2[14]aneN_4)Br_2$	40,890	1487	7686	279	842	1089
$Ni(Me_6[14]aneN_4)Br_2$	41,020	1492	5704	207	793	1089
$Ni([14]aneN_4]Cl_2$	40,695	1480	10,430	379	831	1089
$Ni([15]aneN_4)Cl_2$	34,150	1242	16,200	589	825	1089
$Ni([16]aneN_4)Cl_2$	30,685	1116	13,990	509	834	1089
$Ni([14]aneN_4)(N_3)_2$	39,730	1445	18,175	661	829	1089
$Ni([15]aneN_4)(N_3)_2$	33,735	1227	20,985	763	797	1089
$Ni([16]aneN_4)(N_3)_2$	29,880	1087	20,840	758	869	1089
$Ni([14]aneN_4)(NCS)_2$	38,990	1418	24,070	875	860	1089
$Ni([15]aneN_4)(NCS)_2$	33,050	1202	24,970	908	842	1089
$Ni([16]aneN_4)(NCS)_2$	29,530	1074	24,000	873	860	1089
$Ni([14]'aneN_4)Cl_2$	36,650	1333	13,845	504		1197
$Ni([14]'aneN_4)(NCS)_2$	37,150	1351	21,990	800		1197
$Ni(TAAB)(NCS)_2$	40,280	1465	22,040	801	882	1089
Ag(II)						
AgF_6^{4-}	35,965	1308	19,195	698		1231

cont

Metal ion	DQ(L)	Dq(L)	DQ(Z)	Dq(Z)	B	Ref
$AgPbF_6$	34,590	1258	11,205	408		1231

a) C = 2870 cm^{-1}. b) C = 3260 cm^{-1}. c) C = 2760 cm^{-1}.
Reference numbers refer to Chapter 6. See [1089] for additional data.

v) Ligand field strengths or AOM parameters tend to decrease with increasing metal-ligand bond length. This is most readily observed in species such as $[ML^*_4Z_2]^{n+}$ where L* is a sterically bulky ligand which inhibits the approach of the Z ligand and gives rise to a longer M-Z bond. However it is much more general than this, as discussed immediately below.

Consider now some more specific examples of information content in electronic spectra to develop some of the general comments made above.

9.3 i) Radial Distortion

With sterically bulky ligands, bond lengths may be longer than the usual sum of appropriate radii (angles may also be distorted). Since the interaction between ligand and metal is inversely proportional to a power of the metal ligand distance, such elongation will naturally result in reduced values of the various radial parameters. This fact was recognised very early in such systems as $Ni(R_2en)_2Z_2$ [1-3] where the Dq(Z) value was often much less than that expected for an analogous NiZ_6 system. However more recent work utilises data of greater resolution and attempts to quantify such variations more precisely.

There are many isolated examples where, for example, crystal field Dq values are smaller than expected because of such elongation. However the best characterised series are still tetragonal complexes of the type trans-$M(R_2en)_2Z_2$ where M is usually Co(II), Ni(II) or Cu(II) and R_2en is a symmetric or asymmetric alkyl substituted diamine (usually ethylenediamine) [4]. For example in the symmetric series with R = Et, there is steric interaction between the bulky diamine and the axial M-Z bond. As a direct consequence, the axial Dq_z values for these ligands are generally smaller than those found in analogous $[NiZ_6]^{n+}$ species (see Table 9.4). Unfortunately there are few X-ray data for this symmetric series, but there can be little doubt that some bond elongation has occurred.

Moreover there is evidently a synergistic effect in that DQ_{xy} values in such complexes are sometimes slightly larger than anticipated. Probably the weaker M-Z bond leaves a higher positive charge on the metal and results therefore in a strengthened

interaction in the xy plane. This synergism is fairly general in that weakened axial interactions often give rise to a strengthened equatorial interaction (and vice versa, though this is much less common). The effect is especially evident in Cu(II) complexes [5,6] (refer to the discussion of Tetragonality in Chapter 6) and also in an interesting series of nickel(II) macrocyclic complexes studied by Busch and co-workers [23] (Tables 9.4,9.5).

Very strong in-plane fields can give rise to weak axial fields in tetragonal nickel(II) complexes, because of steric inhibition at the axial site and/or a synergistic effect which favours the in-plane field. For example compare the axial Ni-Cl field in $Ni(s\text{-}Et_2en)_2Cl_2$ and $Ni([14]aneN_4)Cl_2$. However if a poorly fitting macrocycle is used, i.e. one with too large a central cavity (see below), the in-plane field diminishes as expected, but the axial field, subject to any steric constraints, tends to increase.

One can choose pairs of complexes with the same axial ligand displaying a small or larger field, and compare the e_σ and e_π values. In many cases, both e_σ and e_π are reduced by roughly the same percentage extent, when the overall field ($3e_\sigma\text{-}4e_\pi$) is reduced. However there are examples where $e_\sigma(L)$ is diminished but e_π is relatively large and negative. As a consequence one might conclude that the field strength was still quite reasonable but was made up primarily of a π-contribution (e.g. $[Ni(Me_6[14]aneN_4]Br_2)$. This is probably not the correct conclusion. While one may recall that the e_π values are less meaningful than e_σ for, in this case, six coordinate nickel, one might question the experiment.

This study [23] also considered unsaturated imine macrocyclic complexes and showed that, in general, Dq values increased with unsaturation. Indeed these workers were able to generate a table of structural contributions to Dq which allowed them to pre-calculate Dq on the basis of the structure of the macrocycle, e.g. number of atoms in ring, number of imine groups, nature of axial group etc.

With asymmetric diamines, rhombic complexes are obtained in which elongation along the $M\text{-}NR_2$ bond can be quite remarkable. In this case sufficient high quality electronic spectroscopic data are available, together with X-ray structural information, for NH_2, NHR and NR_2 systems, to demonstrate the relationship shown in Fig.9.4. As shown in Table 9.5, Ni-NRR' bonds have significantly different $e_\sigma(Ni\text{-}N)$ values as a consequence primarily of the variation in bond length (Fig.9.4). Quite significant reductions in $e_\sigma(Ni\text{-}N)$ are evident as the Ni-N bond becomes longer [69]. Note that these data all correspond to the same parameter $e_\sigma(Ni\text{-}N)$ but vary dramatically even when the bond distance change is minimal. Clearly AOM parameters should not be transferred from one complex to another without great care.

Other isolated examples are known where AOM parameter values appear to reflect bond length changes. A distorted five coordinate $Ni(salen\text{-}NEt_2)$ derivative shows $e_\sigma(Ni\text{-}N, imine)$ values of ca. 4000 cm^{-1} compared with an axial $e_\sigma(Ni\text{-}NEt_2)$ value of 2500 cm^{-1}; the Ni-N(amine) bond is ca. 220 pm long compared with Ni-N(imine) of 196pm [8] (data obtained from magnetic studies).

Table 9.5

AOM $e_\sigma(L,Z)$ AND $e_\pi(L,Z)$ VALUES

Complex	Point group	$e_\sigma(L)$	$e_\pi(L)$	$e_\sigma(Z)$	$e_\pi(Z)$	B	Ref
Cr(II)							
$CrCl_2$	$2\text{-}D_{\infty h}$	----	----	4500	1800	---	378
Cr(III)							
$Cr(glycinato)_3$	$6\text{-}C_3$	6700(N)	0	8800	2000	738e	1376
$Cr(en)_2F_2^+$	$6\text{-}D_{4h}$	7233	0	7811	2016	692	269
$Cr(en)_2Cl_2^+$	$6\text{-}D_{4h}$	7333	0	5558	900	670	269
$Cr(en)_2Br_2^+$	$6\text{-}D_{4h}$	7567	0	5341	1000	560	269
$Cr(en)_2I_2$+d	$6\text{-}D_{4h}$	6987	0	4292	594		365
$Cr(en)_2(OH)_2^+$	$6\text{-}D_{4h}$	6640	0	8633	2151	789	269
$Cr(en)_2(H_2O)_2^{3+}$	$6\text{-}D_{4h}$	7490	0	7459	1370	744	269
$Cr(en)_2(dmso)_2^{3+}$	$6\text{-}D_{4h}$	7534	0	6769	1653		365
$Cr(H_2O)_4F_2^+$	$6\text{-}D_{4h}$	7626	1370[b]	8510	2539	700	269
$Cr(NH_3)_4F_2^+$	$6\text{-}D_{4h}$	6967	0	7453	1751	667	269
$Cr(NH_2Me)_4F_2^+$	$6\text{-}D_{4h}$	6740	0	7479	1622		365
$Cr(NH_2Pr)_4F_2^+$	$6\text{-}D_{4h}$	6640	0	7617	1630		365
$Cr(NH_3)_5OH_2^{3+}$	$6\text{-}C_{4v}$	6866	0	7566	2800		276
$Cr(pd)_2F_2^+$	$6\text{-}D_{4h}$	7343	0	9093	2450		365
$Cr(pd)_2Cl_2^+$	$6\text{-}D_{4h}$	7400	0	5832	1052		365
$Cr(pd)_2Br_2^+$	$6\text{-}D_{4h}$	7680	0	5417	903		365
$Cr(pd)_2I_2^+$	$6\text{-}D_{4h}$	7000	0	3986	739		365
$Cr(pnd)_2F_2^+$	$6\text{-}D_{4h}$	7457	0	8394	1933		365
$Cr(chd)_2F_2^+$	$6\text{-}D_{4h}$	7567	0	8180	2030		365
$Cr(daud)_2F_2^+$	$6\text{-}D_{4h}$	7523	0	8382	1981		365
$Cr(dadd)_2F_2^+$	$6\text{-}D_{4h}$	7473	0	8472	2045		365
$Cr(datd)_2F_2^+$	$6\text{-}D_{4h}$	7440	0	8476	2027		365
Fe(II)							
$FeCl_2$	$2\text{-}D_{\infty h}$	----	----	3570	2300	---	378
Mn(III)							
$Mn(macN_4)Cl_2^+$	$6\text{-}D_{4h}$	8833	0	3380	1200		367
$Mn(macN_4)Br_2^+$	$6\text{-}D_{4h}$	9000	0	3149	1900		367
Co(III)							
$CoPy_3Cl_3$	$6\text{-}C_{2v}$	6100	-750				657
Co(II)							
$Co(s\text{-}Et_2en)_2(NCS)_2$	$6\text{-}D_{4h}$	3787	0	4044	136		749
$Co(s\text{-}Et_2en)_2Cl_2$	$6\text{-}D_{4h}$	4065	0	2350	232		749
$Co(s\text{-}Et_2en)_2Br_2$	$6\text{-}D_{4h}$	3768	0	1939	142		749
$Co(s\text{-}Me_2en)_2(NO_3)_2$	$6\text{-}D_{4h}$	4053	0	1973	-275		749
$Co(acac)_2(6\text{-}MQ)$	$6\text{-}D_{2h}$	4580	515,860	3910	675,785	800	1199
$Co(acac)_2(H_2O)_2$	$6\text{-}D_{2h}$	4500	465,795	3635	465,550	800	1199

cont

Table 9.5 cont

Complex	Point group	e_σ(L)	e_π(L)	e_σ(Z)	e_π(Z)	B	Ref
$Co(NN_3)Br$	5-TB	3800(Ax)	0	5100	2040		
		4200(Eq)	0				776
$Co(NS_3)Br$	5-TB	3100(Ax,N)	0	5000	2000		
		3540(Eq,S)	-630				776
$Co(NP_3)Br$	5-TB	900(Ax,N)	0	5400	2340		
		3680(Eq,P)	-830				776
$Co(bpc)_2$	$4\text{-}D_{2d}$	2755	-1185			780	851
$Co(L+)Br_3$	$4\text{-}C_{3v}$	4000	0	3500	1000	700	848
$Co(Quin)Br_3^-$	$4\text{-}C_{3v}$	3500	-500,0[b]	3000	450	670	850
$Co(L+)Cl_3$	$4\text{-}C_{3v}$	4250	0	3500	1100	740	848
$Co(spart)Cl_2$	$4\text{-}C_{2v}$	4200	0	3690	1080	730	862
$Co(PPh_3)_2Cl_2$	$4\text{-}C_{2v}$	4000	-1000	3500	2000	575	1043
$Co(PPh_3)_2Br_2$	$4\text{-}C_{2v}$	3500	-1000	3500	1500	575	1043
$Co(PPhEt_2)_2(mes)_2$	$4\text{-}D_{2h}$	>6000	-3400			280	1042
Ni(II)							
$Ni(L+)Br_3$	$4\text{-}C_{3v}$	5900	0	3000	850	720	848
$Ni(L+)Cl_3$	$4\text{-}C_{3v}$	6100	0	3250	1000	760	848
$Ni(Quin)Br_3^-$	$4\text{-}C_{3v}$	4000	-500,0[b]	4500	1100	670	850
$Ni(Biquin)Br_2$	$4\text{-}C_{2v}$	4200	-1000	3500	850	790	11
$Ni(Ph_3P)Br_3^-$	$4\text{-}C_{3v}$	5000	-1500	3000	700	620	935
$Ni(Ph_3P)I_3$	$4\text{-}C_{3v}$	6000	-1500	2000	600	490	935
$Ni(Ph_3P)_2Cl_2$	$4\text{-}C_{2v}$	4500	-2500	4500	2000	550	1043
$Ni(Ph_3P)_2Br_2$	$4\text{-}C_{2v}$	4000	-1500	4000	1500	550	1043
$Ni(py)_4(ClO_4)_2$[d]	$6\text{-}D_{4h}$	4300	100	1050	0	850	1342
$Ni(py)_4(SO_3F)_2$	$6\text{-}D_{4h}$	4000	50	1900	0	900	1342
$Ni(en)_2(NCS)_2$	$6\text{-}D_{4h}$	4000	0	2518	-50	890	927
$Ni(sMe_2en)_2(NCS)_2$	$6\text{-}D_{4h}$	3998	0	3952	303	862	927
$Ni(sMe_2en)_2(NO_3)_2$	$6\text{-}D_{4h}$	4197	0	1064	-925	887	927
$Ni(NH_3)_4(NCS)_2$	$6\text{-}D_{4h}$	3583	0	3829	125	851	927
$Ni(mpz)_4Cl_2$	$6\text{-}D_{4h}$	3950[b]	160	1540	-420	822	930
$Ni(mpz)_4Br_2$	$6\text{-}D_{4h}$	3950[b]	160	800	-820	822	930
$Ni(sEt_2en)_2(NCS)_2$	$6\text{-}D_{4h}$	4350	0	3120	-42	690	927
$Ni(sEt_2en)_2Cl_2$	$6\text{-}D_{4h}$	3972	0	1712	-276	864	927
$Ni(sEt_2en)_2Br_2$	$6\text{-}D_{4h}$	3923	0	1178	-644	845	927
$Ni(sEt_2en)(H_2O)Cl^+$	$6\text{-}D_{4h}$	4175	0	2200	-234	895	927
$Ni(sEt_2en)(H_2O)Br^+$	$6\text{-}D_{4h}$	4168	0	1972	-140	890	927
$Ni(sEt_2en)(H_2O)I^+$	$6\text{-}D_{4h}$	4207	0	1763	-397	882	927
$Ni(sEt_2en)_2(ONO)_2$	$6\text{-}D_{4h}$	4490	0	2550	530	850	1019
$Ni(asMe_2en)_2(TCA)_2$	$6\text{-}C_{2v}$	4545(NH_2)	0	2800	400	875	1021
		3555(NMe_2)	0				
$Ni(TMen)_2(NCS)_2$	$6\text{-}C_{2v}$	3165(NMe_2)	0	2115	-920	848	1072
$Ni(asEt_2en)_2(NCS)_2$	$6\text{-}C_{2v}$	4650(NH_2)	0	4305	530	869	1072
		2485(NEt_2)	0				

cont

Table 9.5 cont

Complex	Point group	e_σ(L)	e_π(L)	e_σ(Z)	e_π(Z)	B	Ref
$Ni(asEt_2en)_2(H_2O)_2^{2+}$	C_{2v}	4770(NH_2)	0	3050	400	880	1072
		2640(NEt_2)	0				
$Ni(asMe_2en)_2(ONO)_2$	$6\text{-}C_{2v}$	4665(NH_2)	0	2450	-510	820	f
$Ni([14]aneN_4)Br_2$	$6\text{-}D_{4h}$	4957	0	476	-364	849	1089
$Ni([15]aneN_4)Br_2$	$6\text{-}D_{4h}$	4277	0	1239	-199	818	1089
$Ni([16]aneN_4)Br_2$	$6\text{-}D_{4h}$	3760	0	1315	-57	853	1089
$Ni(Me_2[14]aneN_4)Br_2$	$6\text{-}D_{4h}$	4957	0	481	-337	842	1089
$Ni(Me_6[14]aneN_4)Br_2$	$6\text{-}D_{4h}$	4973	0	258	-324	793	1089
$Ni([16]aneN_4)Cl_2$	$6\text{-}D_{4h}$	3720	0	1527	-127	834	1089
$Ni([15]aneN_4)Cl_2$	$6\text{-}D_{4h}$	4139	0	1537	-320	825	1089
$Ni([14]aneN_4)Cl_2$	$6\text{-}D_{4h}$	4933	0	1398	101	831	1089
$Ni([14]aneN_4)(N_3)_2$	$6\text{-}D_{4h}$	4817	0	2031	-129	829	1089
$Ni([15]aneN_4)(N_3)_2$	$6\text{-}D_{4h}$	4090	0	2227	-238	797	1089
$Ni([16]aneN_4)(N_3)_2$	$6\text{-}D_{4h}$	3623	0	2049	-358	869	1089
$Ni([14]aneN_4)(NCS)_2$	$6\text{-}D_{4h}$	4726	0	2711	-155	860	1089
$Ni([14]'aneN_4)Cl_2$	$6\text{-}D_{4h}$	4443	0	408	-953		1197
$Ni([14]'aneN_4)(NCS)_2$	$6\text{-}D_{4h}$	4504	0	2517	-112		1197
$Ni(TAAB)(NCS)_2$	$6\text{-}D_{4h}$	4883	0	2290	-286	882	1089
$NiCl_2$	$2\text{-}D_{\infty h}$	----	-	4798	3098	725	378
Platinum							
$PtCl_4^{2-}$	$4\text{-}D_{4h}$	----	-	12,420	280	600	1123
$PtBr_4^{2-}$	$4\text{-}D_{4h}$	----	-	10,920	220	480	1123
$Pt(NH_3)_4^{2+}$	$4\text{-}D_{4h}$	15,620	170	---	---		1123
Palladium							
$PdCl_4^{2-}$	$4\text{-}D_{4h}$	----	-	10,150	200	540	1123
$PdBr_4^{2-}$	$4\text{-}D_{4h}$	----	-	9510	178	530	1123
Copper							
$Cu(daco)_2^{2+}$	$4\text{-}D_{4h}$	7800	1300	----	---		1046
$CuCl_4^{2-}$	$4\text{-}D_{2d}$	----	-	6800	1800		852
$CuBr_4^{2-}$	$4\text{-}D_{2d}$	----	-	5500	850		852
$CuCl_2$	$2\text{-}D_{\infty h}$	----	-	5400[c]	3300		378

a) X refers to the halogen, and L to the other ligand, except where indicated. b) Assumed. c) Extrapolated. d) Only three of the four lower tetragonal components resolved; analysis is tentative. e) C= 3020 cm^{-1}.f) From I.M.Walker, A.B.P.Lever and P.J.McCarthy, Can.J.Chem., **58** (1980) 823.

In addition see data for seven coordinate Fe(II), Co(II), Ni(II) [1162]. All references refer to listing in Chapter 6.

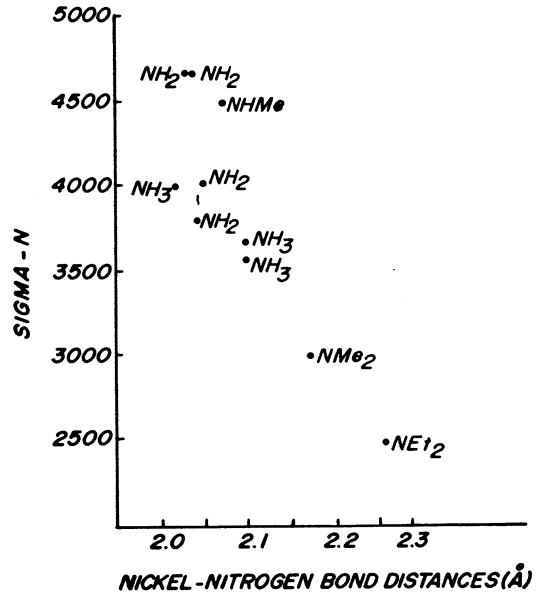

Fig.9.4 A plot of e_σ(Ni-N) values versus the Ni-N distance in a series of tetragonal and rhombic amine nickel(II) complexes [69].

In ML^+X_3 complexes with the positively charged quaternised dabco, e_σ(Ni-N) = ca. 6000 cm^{-1}, while e_σ(Co-N) = ca. 4000 cm^{-1}, attributed to a 20pm shorter bond in the nickel complexes [9]. Comparison of $[Ni(PPh_3)X_3]^-$ species shows that the e_σ(Ni-P) is greater for X = I than for X = Cl,Br, associated with a 4pm decrease in Ni-P bond length [10]. Indeed Gerloch's work on phosphine ligands is of especial interest. Significant negative pi parameters support the usual view of these ligands as pi acceptors [10,38]. Contributions from the Gerloch group usually combine both magnetic and spectroscopic data to create a more complete and

defensible picture [10,41,66].

Although some tentative conclusions may be drawn concerning the dependence of an AOM parameter on coordination number or stereochemistry, such conclusions should be treated with caution since i) the data set is not yet big enough, and ii) there can be some considerable experimental error in the determination of AOM parameter values.

Four and five coordinate cobalt(II) and nickel(II) complexes seem to have e_σ(N) values comparable to their octahedral counterparts, but the e_σ and e_π values are noticeably higher for the halogen ligands. Perhaps the increased covalency in these species allows better overlap with the polarisable halogens, while the less polarisable nitrogen ligands cannot take advantage of this.

Five coordinate complexes have been a special interest of Bertini and Gatteschi and their co-workers, who have published some interesting studies [e.g 43-45]. Some data are reported in Table 9.4 for tripod type ligands. These have long been known to introduce unusual spectrochemical effects in a trigonal bipyramidal molecule depending upon whether the axial interaction is strained or not [46]. In the examples shown, the e_σ(N) values are somewhat low probably because the axial M-N bond is lengthened. Notice, however, that the e_π halogen values are again elevated relative to simple six coordinate complexes. There may also be a synergistic effect operating here.

In these systems there is often insufficient information available in the spin-allowed spectrum to unequivocally determine all the AOM parameters. These workers choose to fix the e_π/e_σ ratio for the halogen by estimating the squared overlap integrals for the π and σ-bonding p orbitals. This seems to work reasonably well, but again must be used with caution.

Note that the sequence Br > N > S = P was obtained for both e_σ and e_π parameters in a series of the five coordinate cobalt(II) complexes.

It does appear that both the sigma and pi values for gas phase 2-coordinate species (e.g. $CuCl_2$), are greater than for the same metal and ligand in higher coordination numbers. This is eminently reasonable since the positive charge on the metal is shared with so few ligands in this case, and bond distances are probably shorter.

For example the linear series MCl_2 exhibit AOM parameters as follows (subject to some uncertainty in a few of the assignments [17]):

			dσ	dπ	
d^4	$\delta^2\pi^2\sigma^0$	$CrCl_2$	4500	1800	cm^{-1}
d^5	$\delta^2\pi^2\sigma^1$	$FeCl_2$	3570	2300	
d^7	$\delta^4\pi^2\sigma^1$	$CoCl_2$	4200	2550	
d^8	$\delta^4\pi^3\sigma^1$	$NiCl_2$	4800	3100	
d^9	$\delta^4\pi^4\sigma^1$	$CuCl_2$	ca.5400	ca.3300	

(9.17)

The decrease in sigma parameter from Cr(II) to Fe(II) probably

reflects the extra sigma anti-bonding electron. Thereafter the monotonic increase in both parameters probably reflects increasing nuclear charge on the metal ion. The pi parameters do not parallel the number of anti-bonding pi electrons; perhaps we are seeing a polarisation effect (insofar as pi bonding is concerned) rather than direct orbital overlap. Certainly this represents an interesting set of data which warrants significant expansion (other metals, other ligands).

Busch has made a study of the effect upon Dq of the hole size in a macrocycle. By comparison with Dq values obtained with analogous open chain tetradentate amines and through use of empirical force field calculations, he demonstrated [12,23,30,31] that the hole size in [15]aneN$_4$ is most appropriate for Ni(II) (high spin S = 1) while the hole size in [14]aneN$_4$ is preferred by Co(III) [12,30,31]. If the hole size is too small, the complex will either fail to form, or fold to generate a cis-species, or produce a trans-complex with excess strain energy. In this last case, Dq will generally increase because the M-L distance is reduced. If the hole size is too large, the M-L distance will be larger and Dq smaller (Table 9.4).

Fabbrizzi [52] noted that the highest Dq value for square planar S = 0 Ni(II), in a macrocycle, was observed with [13]aneN$_4$ which also had the best fit. This result suggests that M-N strain does not necessarily lead to a higher value of Dq. In support of this view, Hancock and co-workers [51] calculated the strain energy for a large number of open chain and macrocyclic ligand complexes and concluded that the maximum in-plane Dq value is obtained in a macrocycle with minimal M-N strain. Further the 'macrocyclic effect' whereby the enthalpy of formation of the M-N bond is greater than for the analogous open chain amine complex, is a consequence of this higher Dq value plus the strain in the cavity of the macrocycle. This latter effect in the gas phase, would be caused by dipole-dipole repulsion, and in solution would have a contribution from steric hindrance to solvation (see section 9.4).

Note that the Dq value clearly depends upon whether a high or low spin complex is being considered and that both high and low spin behaviour can sometimes be observed with the same ligand. For example both high and low spin d^6 Fe(II) are seen with the macrocycle Me$_6$[14]aneN$_4$ depending upon the axial ligand, in complexes Fe(mac)Z$_2$. With Z, e.g. halide, 10Dq(high spin) = 14,000 cm^{-1}, and with Z, e.g. CH_3CN, 10Dq(low spin) = 21,000 cm^{-1} [21].

It is often constructive to compare data for closely related groups of complexes, since under such circumstances, one may be able to relate trends more readily than when comparing grossly different systems. The McClure parameters for the following tetra-aquo species are telling a story (see tables for references):

	dσ	dπ
$[Ti(H_2O)_4F_2]^+$	|3040|	
$[Ti(H_2O)_4Cl_2]^+$	-1595	
$[V(H_2O)_4Cl_2]^+$	-399	-305
$[Cr(H_2O)_4F_2]^+$	662	1170
$[Cr(H_2O)_4Cl_2]^+$	-399	-305
$[Co(H_2O)_4Cl_2]$	-1017	-885 cm^{-1}

(9.18)

The change in sign for the fluoro-chromium species has been alluded to previously in that the Cr-F bond is apparently very strong. The dσ value for the Ti-Cl system is large probably because the Ti-OH$_2$ bond is very favourable, cf. data for $[TiCl_6]^{3-}$ and $[Ti(H_2O)_6]^{3+}$ in Table 6.1. The value for the Ti-F system is incredibly large given that F^- lies between Cl^- and H_2O in the spectrochemical series. It is difficult to interpret whatever its sign (which is unknown). A single crystal study would be useful.

9.3 ii) Radial distortion and the Jahn-Teller effect

ML$_6$ species exhibiting ground state Jahn-Teller distortion (high spin d^4, and d^9) (see section 4.6 ix)) are special cases where metal-ligand bond elongation (or compression) is expected to occur. Table 9.4. contains Dq$_z$ and Dq$_{xy}$ values for a range of such species. Data are also available for d^1 and high spin d^6 ML$_6$ species which exhibit excited state Jahn-Teller distortion (splitting of excited ^nE_g) and these data are included in Tables 6.1 and 6.15. Quite dramatic reductions in Dq$_z$ can be seen, relative to the in-plane Dq$_{xy}$ for the same ligand.

Where only the two transitions to the components of ^nE_g are seen, it may not be possible to determine the sense of the splitting, and hence determine whether elongation occurs along z, or in the xy plane. However most known examples occur along z, and it is assumed in the analysis in Table 9.4 that such elongation has occurred. Though the sign is uncertain, the magnitude of dσ depends solely on the splitting and is unequivocal.

Notice especially the markedly decreased values for axial M-F in such d^4 species as CrF_3, Na_2CrF_4 and K_3MnF_6, and in d^7 Rb_2KNiF_6 and for the Mn-O bond in $[Mn(dmso)_6]^{3+}$ and $[Mn(dmf)_6]^{3+}$ (Table 9.4).

It is interesting to compare ML$_6$ species with M(LL)$_3$ species. Though strictly of D$_3$ symmetry (at best), the latter systems may be tentatively assumed to parallel the ML$_6$ series. There is a general trend, though there are some exceptions, that the Jahn-Teller induced elongation in the bidentate ligand series is smaller than in the mono-dentate series. This would certainly be expected since the binding atom which moves away from the metal, is held to the

metal in the xy plane, through the bidentate framework.

9.3 iii) Angular distortion

There is no doubt that angular distortion (deviation from regular angles for point group concerned) leads to a significant variation in band energy and intensity. However systematic experimental studies of the effect of bond angle have not been carried out. Many authors have used the crystal field or AOM theories to predict the effect of changing a specific angle upon transition energies and in some cases have used this approach to fit a spectrum (see examples in Chapters 6 and 8). One may in effect use the variation in bond angle to generate distortion from regular symmetry [13-15].

The electronic spectra of cis-bidentate nitritonickel(II) complexes have been discussed [70] and the angle of bite of the nitrito ligand taken into account. A good fit was obtained between theory and experiment, assuming the crystallographically determined bite angle. A special aspect of this system is the gross difference in both transition energies and polarisation compared with trans-NiN_4O_2 analogs. The angular distortion certainly has a profound effect on the spectrum, at least in this case.

However very few studies have directly related a theoretically fitted bond angle with an X-ray structural analysis. Moreover it is not obvious that the theoretical and experimental angle should be exactly the same. In such distorted complexes, maximal bond overlap may not exactly correspond with the line joining the nuclei. In such cases the experimental and theoretical angles could differ.

A particular aspect of angular distortion is the possible rotation of 'flat' ligands such as water, pyridine etc which may only interact with a metal ion in one plane. We have met such a problem in Chapter 5 where the CT bands associated with imidazole depended upon whether the imidazole was free to rotate or not.

Schaffer and co-workers [32-34] present an interesting and important study on the interaction of pyridine as a non-linearly ligating ligand. They carry out a statistical analysis of the spectra of a wide range of $[Cr(NH_3)_4AB]^{n+}$ and $[Cr(Py)_4AB]^{n+}$ species defining the pi-interaction of the pyridine ligand in terms of an angle ψ, which the pyridine molecular plane makes with the complex molecular xy plane.

It was, unfortunately, necessary to transfer parameters between the two sets of compounds in order to complete the analysis. However, statistically, the fits to the spectra were extremely good. The final result is a negative $e_\pi(Py)$ parameter of approximately -2300 cm^{-1} (relative to $e_\pi(NH_3)$), indicative of net back-pi-bonding to pyridine, and an angle of 37.8° compared with one crystal structure analysis indicating 45°. The result has been subject to some criticism [35], but appears to be supportable in a statistical sense [33].

A similar study [34] was reported for a series of tris-(pyridine)cobalt(III) complexes where the data, though less definitive in a statistical sense, also indicated a negative $e_\pi(Py)$ parameter of $\leq$ -3000 cm^{-1}.

Little and Long [22] analysed the spectra of an extensive series of $Fe(Py)_4Z_2$ and $Fe(Py)_2Z_2$ (with Z bridging to yield a FeZ_4 plane) complexes, at ambient and cryogenic temperatures. The splitting of the excited ^5E_g level provides a measure of the McClure dσ parameter (Table 6.15) which is compared between the two series, but in this case, no angular information was derived, since no information involving dπ is obtainable from the spectrum.

9.3 iv) The AOM Model, Copper(II) Complexes and sd Mixing

The wealth of data available for low spin copper(II) complexes, primarily from the Hathaway laboratory (see Chapter 6), would seem like a gold mine for the exploitation of the AOM theory. Unfortunately the results have proved problematical. While some complexes yield AOM values which seem reasonable, others yield parameter values which appear unacceptable even though the data and assignments appear valid. Smith, Gerloch, Hathaway, and Hitchman, among others, have been interested in this field.

Evidently the AOM theory is incomplete, or is used innaccurately in dealing with copper(II) complexes. It has been supplemented by the addition of an electrostatic component [36], but this proved illusory [37]. Three factors are probably important.

Firstly, the weak field model allows only one D term for d^9 and hence all the d orbital representations possess the same interelectronic repulsion, thus the Racah B and C cancel out of the transition energies. This is probably an oversimplification. Given that there are 10 pairs of Coulomb integrals available to the d shell, it is likely that more than one value of the Racah parameter should be used, such that it does not cancel, so-called 'differential radial expansion' or 'symmetry restricted covalency' [39,40].

Secondly, spin-orbit coupling is relatively large for copper(II) and should probably be taken account of.

Thirdly the mixing in of s or p orbitals of appropriate symmetry seems a necessary pre-requisite in some copper(II) complexes. In D_{4h} systems, for example, the 4s orbital transforms in the same way as dz^2, hence sd mixing can occur. This can be a significant contribution but can be allowed for within the AOM formalism by addition of an appropriate matrix element [37,38, 41,42]. Some trigonal bipyramidal copper(II) complexes yield strange AOM parameters when solved conventionally; dp mixing may be important here [37].

Some authors maintain [37,41] that sd mixing is only important in rigidly four coordinate square planar complexes, and possibly similar systems with a vacant site (not only copper(II), but certainly other square planar systems such as Ni(II) and Co(II) e.g. [38]). When such systems are solved conventionally, they yield apparent non-zero parameter values for the vacant site. This aspect of the theory has recently been discussed by Gerloch and co-workers

[41] in terms of the potential in 'empty cells', due to the presence of atomic density above and below the molecular plane.

Vanquickenborne and co-workers [42] present a detailed theoretical discussion of sd mixing and propose that the stabilisation of the z^2 orbital, in square planar ML_4 is -4e(L) where:

$$e(L) = \langle z^2|V|s\rangle^2/[E(s) - E(z^2)]$$

The data for two square planar copper(II) complexes, were fitted as follows:

	$xy--x^2-y^2$	$xz,yz--x^2-y^2$	$z^2--x^2-y^2$	e_σ	e_π	e(L)
$[CuCl_4]^{2-}$	12,500	14,250	17,000	5333	875	1580
$[Cu(dacoda)]_2^{2+}$	18,200	20,800	23,100	7800	1300	1875

These workers [42] do extend their analysis to octahedral and six coordinate tetragonal nickel(II) complexes. The energetic effect of sd mixing is very small in octahedral nickel(II), but does slightly favour axial elongation. It is seen to be somewhat more important in tetragonal systems where it also emphasizes axial elongation. In the view of these workers it appears that sd mixing should also be considered in systems other than rigorous square planar.

9.4 Thermodynamic Correlations

9.4 i) Heats of Formation

The replacement of water in an aquo ion by stronger field ligands, such as amines, generally results in an increase in ligand field strength and an overall increase in the bond energy of the molecule. This bond energy increase can be identified via calorimetry as the heat of formation of the complex in water, ΔH_f°, e.g.:

$$[Cu(H_2O)_6]^{2+} + NH_3 \text{ ---------- } [Cu(NH_3)(H_2O)_5]^{2+} + H_2O \quad \Delta H_f^\circ \qquad (9.19)$$

This energy represents the energy of substitution of Cu-O(H_2O) by Cu-N(NH_3). In the specific case of Cu(II), the single broad d-d absorption of the aquo ion shifts to the blue upon substitution by an amine. This transition maximum corresponds to the most intense peak in the copper d-d manifold, namely $xz,yz---x^2-y^2$. A plot of this d-d band energy versus ΔH_f° (Fig.9.5) shows an extraordinarily clear linear correlation for a wide range of amines including mono-, di-, tri- and linear tetra-amines, and macrocyclic amines [47,48]. The range of variation in ΔH_f° is some 27 Kcal/M, and in wavenumbers some 6500 cm^{-1}. The origin of the line passes exactly through the wavenumber of the copper aquo ion spectrum.

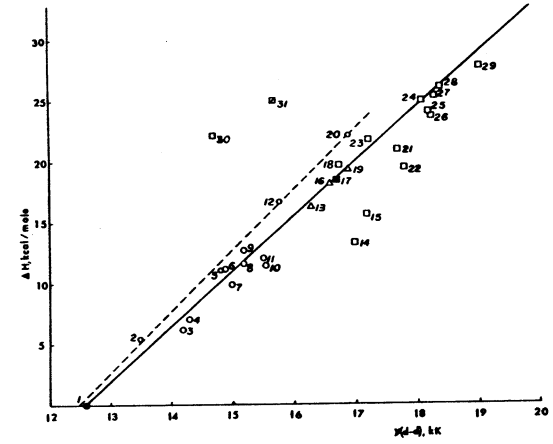

Fig.9.5 Plot of the Heat of Formation in aqueous solution versus the frequency of the principle d-d band, in a series of ammine copper(II) complexes [48]. The numbers refer to the following ligands. Water makes up the coordination sphere where necessary. A number in front of the ligand indicates how many such ligands are in the coordination sphere. If the number is absent, there are sufficient such ligands to occupy the four equatorial sites.
1. water, 2. 1 NH_3, 3. Tmen, 4. s-Et_2en, 5. 2 NH_3, 6. 1 s-Me_2en, 7. 1 as-Me_2en, 8. 1 Me-en, 9. 1 en, 10. 1 C,C-Me_2en, 11. 1 C-Me-en, 12. 3 NH_3, 13. 1,5,9-triazanonane, 14. as-Et_2en, 15. s-Et_2en, 16. 1,4,7-triazaheptane, 17. cyclen, 18. 3,3,3-tet, 19. 1,4,8-triazaoctane, 20. 4 NH_3, 21. s-Me_2en, 22. as-Me_2en, 23. 2,2,2-tet, 24. Me-en, 25. C-Me-en, 26. C,C-Me_2en, 27. en, 28. 3,2,3-tet, 29. 2,3,2-tet, 30. tren, 31. tetren.

The line for NH_3 substituents is slightly different from the general line through the other species. The equations of the lines are:

$$-\Delta H_f^\circ = 4.90\ (xz,yz---x^2-y^2) - 61.0 \quad n = 1\text{-}4,\ NH_3$$

$$-\Delta H_f^\circ = 4.33\ (xz,yz---x^2-y^2) - 54.4 \quad n = 1\text{-}4,\ \text{other N} \qquad (9.20)$$

where the d-d energy is in units of 1000 cm^{-1}.

Deviations from linearity occur when there is steric hindrance, when there is axial coordination (rather than equatorial coordination) of an amine, and when the amine cannot replace water molecules in the equatorial plane and generate a tetragonal complex, e.g. a tripod amine.

Complexes (30) and (31) in Fig.9.5 contain an axially

coordinated amine nitrogen atom and appear above the correlation line.

Various disubstituted N,N-dialkyl diamines lie below the line (e.g. (14) and (15) in Fig.9.5). It is reasonable to associate the number of Kcal/M of energy of deviation from the 'line' as an approximation to the reduction in energy due to this steric hindrance. Note however that symmetric dimethyl substituted diamine systems also lie somewhat below the line. This is believed to be due to differential desolvation. Thus ΔH_f^o may be written:

$$\Delta H_f^o = f\nu(d\text{-}d) + [\Delta H_{sol} + \Delta E] \qquad (9.21)$$

where ΔE is a spherical electrostatic term arising from the difference in the cation-anion interaction in products and reactants, and ΔH_{sol} represents the difference between the outer-sphere solvation energies of the aquo ion plus ligand, and those of the products. Hydrophobic alkyl groups may lead to greater desolvation of the products relative to the unsubstituted ligands and hence some deviation from (below) the line. Note that ΔS values for the diamine substitution increase with decreasing magnitude of $-\Delta H_f^o$ reflecting loss of water both from the inner sphere and from outer-sphere aquation [48]. Increased bond length due to steric hindrance will reduce ΔE.

Considering the energy of displacement of the water by amine, in terms of the AOM model. The energy of the acceptor $x^2\text{-}y^2$ orbital is given by [49]:

$$E(x^2\text{-}y^2) = 3e_\sigma(O) + n[(3/4)e_\sigma(N) - (3/4)e_\sigma(O)] \quad n = 1\text{-}4 \qquad (9.22)$$

where n is the number of nitrogen atoms in the equatorial plane. The reference point is the aquo ion whose $x^2\text{-}y^2$ energy is simply $3e_\sigma(O)$. Thus eqn.(9.22) may be re-written, relative to the aquo ion as:

$$\Delta H_f^o = fn[(3/4)e_\sigma(N) - (3/4)e_\sigma(O)] + [\Delta H_{sol} + \Delta E] \qquad (9.23)$$

Assuming that pi-bonding is unimportant then the energy of the xz,yz orbitals is zero in the AOM model, and the d-d transition observed in these copper(II) complexes has a energy corresponding directly with the energy of the $x^2\text{-}y^2$ orbital (9.22). Individual values of $e_\sigma(O)$ and $e_\sigma(N)$ cannot be calculated from the copper(II) data except by recourse to assumptions about the invariance of $e_\sigma(O)$ as substitution proceeds. It is more interesting to turn to nickel(II) complexes [49] where these parameters may be evaluated.

AOM parameters can be calculated from highly resolved low symmetry nickel(II) data. A plot of $(3/4)n[e_\sigma(O) - e_\sigma(N)]$ versus ΔH_f^o yields an excellent correlation for n=4 and n=6 systems. Note that in the latter case, the function was calculated assuming $3e_\sigma(O) = 8500\ cm^{-1}$ (derived from hexaquo ion assuming $e\ \pi(O) = 0$. The lines are:

$$-\Delta H_f^o = 3.6(3e_\sigma(N) - 3e_\sigma(O)) + 1.2 \quad n = 4$$

$$-\Delta H_f^o = 10.1(3e_\sigma(N) - 3e_\sigma(O)) - 4.5 \quad n = 6 \qquad (9.24)$$

where the equality assumes the $e_\sigma(L)$ values are in units of $1000\ cm^{-1}$.

The slope of 3.6, for n = 4 is directly comparable to the copper(II) data set with a slope of 4.9. The increased slope reflects the greater affinity of copper(II) for nitrogen compared with nickel(II).

These equations are expected to be valid for fairly regular substituted octahedral species (e.g. excluding tripod ligands), and in the absence of significant steric hindrance.

A first-order analysis of the energies of $[NiN_4(H_2O)_2]^{2+}$ species would reveal that the energy of ν_2 ($^3E_g(^3T_{1g}(F))\text{---}^3A_{2g}$) also corresponds with $xz,yz\text{---}x^2\text{-}y^2$ and might then be expected, by analogy with copper(II) systems, to be correlated with ΔH_f^o. Indeed this is the case for both n=4 and n=2 systems, though there is some scatter. The n=4 line intersects the wavenumber axis very close to the hexaquo ion which is satisfying. The two lines have slopes in the ratio 2.6. A ratio of 2 would be expected if Eqn.9.23 is exactly valid; the deviation is undoubtedly due to configurational interaction mixing several E_g states. The lines are:

$$-\Delta H_f^o = 4.06(^3E_g(^3T_{1g}(F))) - 55.45 \quad n = 4$$

$$-\Delta H_f^o = 1.58(^3E_g(^3T_{1g}(F))) - 18.42 \quad n = 2 \qquad (9.25)$$

The line for n = 4 is extraordinarily close to that for copper(II) (eqn.9.20). A rather scattered correlation may also be obtained by plotting ΔH_f^o against $n\nu_1$, the 10Dq transition weighted by the number of substituting nitrogen atoms.

The correlation line is:

$$-\Delta H_f^o = 0.37n\nu_1 + 0.18 \qquad (9.26)$$

but this should be used with caution due to scatter.

It is worth repeating that ligands which give rise to steric hindrance or significant distortion from a regular octahedron are not likely to obey these various correlations. Note that amino acids generally do obey these correlations because the heat of replacement of water by carboxyl is essentially zero. It is also of interest that nickel complexes of bipyridine, phenanthroline and the macrocycle [14]aneN$_4$ are well behaved.

Some data for iron(II) have been reported [49]. For the average band energy ν_a, weighted by the number of nitrogen atoms, one may write:

$$-\Delta H_f^{o} = 0.32 n\nu_a - 6.6 \qquad (9.27)$$

Only a few data points have been recorded. However the slope is essentially the same as for nickel(II).

Copper(II) amine complexes provide the opportunity to observe the σ(Cu-N)---σ^*(Cu-N) transition which should surely reflect bonding charateristics of the Cu-N bond (see section 5.12 (ix), Table 5.26 for details of spectra). Indeed the σ---σ^* transition in simple Cu(II) amines parallels the corresponding (Cu-N) stretching frequency in the infrared [50]. Not unexpectedly there is also a direct linear correlation between this frequency and the heat of formation of the complex in aqueous solution, for many, though not all amines studied [50].

Thus electronic spectroscopy may be used to estimate heats of formation in water, where a sufficient series of complexes of known heat can be accumulated to generate a data base. In some ways, the deviations from the predicted values are of more interest, since they highlight some steric or electronic problem.

These investigations should be extended into non-aqueous solvents, solubility permitting, to move toward the situation where gas phase heats might be determined from the electronic spectra of complex molecules.

9.4 ii) Relationships with Electrochemical Potentials

An electronic transition represents an excitation between a donor level and an acceptor level. In some complexes, it may be possible to record both the oxidation potential of the donor level and the reduction potential of the acceptor level. Thus the two experiments may provide complementary information.

Surprisingly little has been published to explore this possible complementarity, which may be capable of providing useful additional insights [54,55].

Certainly one does not expect that the two energy differences will be the same except under very unusual circumstances. The redox potentials are ground state processes in which the couples are at equilibrium with their solvent environment. This is not the case for the Franck-Condon type instantaneous electronic excitation. Therefore one can expect a large difference in the solvation energy contributions to the two processes. However the experiment can be used to probe such solvation energies.

In practice if one collects electrochemical data for a related range of complexes and compares them with crystal field data, there is usually no useful correlation, rather the data are widely scattered [see Fig.2 of ref. 53, for a particularly artistic scatter].

However one may anticipate better correlation between charge transfer data and differences between appropriate redox couples. For example consider a general LMCT transition:

$$M(n)L \text{----} M(n+1)L^-$$

The energy of this transition may be considered to be derivable from the two redox processes:-

$$M(n)L \text{ ----- } M(n+1)L + e^- \qquad -E[M(n)L/M(n+1)L]$$

$$e^- + M(n+1)L \text{ ---- } M(n+1)L^- \qquad E[M(n+1)L/M(n+1)L^-] \qquad (9.28\ a)$$

Thus subject to certain provisos indicated below, the LMCT energy may be related to:-

$$E[M(n+1)L/M(n+1)L^-] - E[M(n)L/M(n+1)L] \qquad (9.28\ b)$$

Such an equality may be true if i) the differences in ground state solvation energies of the various components of the redox couples are small or comparable to the difference in solvation energy of the ground and Franck-Condon excited state in the electronic transition, ii) entropy differences between the various redox components are small or zero, and iii) the optical transition is of the (0"-0') vibrational type. This appears to be the case for LMCT transitions in metallophthalocyanine derivatives [56] which can be fitted extremely well using (9.28). For the special case of the large inflexible phthalocyanine moieties in non-aqueous solvents, both requirements (i) and (ii) seem to be valid. Moreover numerical values for the required couples were available either from experiment or calculation. More usually one would have access to experimental values for $E[M(n)L/M(n+1)L]$ and $E[M(n)L/M(n)L^-]$ which is not exactly the same. See recent work in [57,71,72].

Nevertheless one may expect a correlation between an electrochemical energy difference and a CT energy in a series of related complexes where solvation effects while not zero, vary in some kind of well behaved fashion.

There are scattered reports in the literature of 'linear' correlations between CT energies as a function a redox potential (not a difference) in a series such as ML_5L' varying L'. When these are successful it is probable that the other redox process involved is not varying greatly.

Meyer and co-workers [58] reported data for a series of ruthenium(II) and ruthenium(III) ammine complexes. In addition to solvatochromic behaviour discussed previously (p.336), the relationship between the CT bands and observed redox potentials was explored.

The complex $[Ru(II)(NH_3)_5(4\text{-cyano-N-methylpyridinium})]^{3+}\ X_3$ exhibits an MLCT band near 18,000 cm^{-1} which is strongly solvatochromic. The electrochemical potentials for oxidation [Ru(III)/Ru(II)] and reduction [Ru(II)L/Ru(II)L^-], were recorded in the same solvents, and the difference, E(redox), between these two potentials shown to relate linearly with the optical MLCT transition energy. The quantity E(redox), which is not the same as that shown in (9.28), is related to the free energy change for the process:-

$$(1/2)\ [Ru(III)L] + (1/2)\ [Ru(II)L^-] \text{ ------- } [Ru(II)L] \qquad (9.29\ a)$$

or in general:

$$(1/2)\ M(n+1)L(m) + (1/2)\ M(n)L(m-1) \text{ ------ } M(n)L(m) \qquad (9.29.b)$$

where it is assumed that the redox processes involve metal oxidation and ligand reduction. Meyer proposes to use the theory developed for intervalence transitions (Chapter 7) to describe CT transitions, since the latter, involving a transfer of charge between two species is not so different from an intervalence transition in which the two valence states are separated by an energy gap.

Thus he equates:

$$E(op) = \chi_i + \chi_o + 1/(1 - \Delta\alpha)[\Delta E(redox) + (e^2/D_s d) + T\Delta S] \qquad (9.30)$$

where χ_i, χ_o and D_s were met in Chapter 7 and E(op) is the optical transition which may be E(MLCT) or E(LMCT). The factor $\Delta\alpha$ is an electron delocalisation or resonance term which reflects the variation in metal-ligand mixing before and after transition. The term $(e^2/D_s d)$ is an electrostatic term which assumes that the donor an acceptor sites may be treated as two spheres whose centres are separated by d. The two right-hand terms in (9.30) are expected to be small. A linear relationship may then be observed between E(op) and $\Delta E(redox)$ in a range of solvents, but only if χ_o is small, or is itself linearly dependent upon $\Delta E(redox)$.

The 4-cyano-N-methylpyridinium complexes show a marked dependence of the [Ru(III)/Ru(II)] couple upon solvent, but relatively little dependence in the ligand reduction couple. The interaction is believed to between the solvent and NH protons of the ammine functions, a specific interaction which invalidates the dielectric continuum model. There is therefore no fit of these data to the functions in eqn.(7.11). The straight line correlation between $\Delta E(redox)$ and E(MLCT) gave a slope of +1.71 [58].

This study is one of very few such detailed studies of CT energies and redox potentials and the marriage of electrochemistry and optical spectroscopy requires further development. Dodsworth and Lever [57] report similar data for the complex $Mo(CO)_4$(bipyrazine) in a range of solvents. In this case the potential for ligand reduction shows more dependence upon solvent than molybdenum oxidation. There is a linear relationship between $\Delta E(redox)$ (defined as 9.29) and E(MLCT) (Mo---bpz), but, remarkably, the slope is negative.

A negative slope does not readily arise in (9.30) since $\Delta\alpha$ is intrinsically positive. Most probably the negative slope is a consequence of the dominance of the χ_o term. Thus the optical transition shifts markedly to the blue with increasing polarity of the solvent because the excited state is strongly destabilised in such solvents. Possibly the molecular dipole, which lies in the molecular plane and bisects the bipyrazine ligand, reverses its direction in the excited state. In the electrochemical experiment, however, increasing polarity of solvent favours formation of bpz^- and hence decreases $\Delta E(redox)$.

These workers [57] suggest an alternate relationship between redox energy and optical transition, namely:-

$$E(op) = [\chi_i + nF\Delta E(redox) + \Delta\Delta G_s + Q] + \chi_o + \Delta(sol) \qquad (9.31)$$

where $\Delta\Delta G_s = 2\Delta G_s(ML) - [\Delta G_s(ML^+) + \Delta G_s(ML^-)]$ i.e. twice the solvation free energy of the neutral molecule less the sum of the solvation free energies of the reduced and oxidised species. $\Delta(sol)$ is the difference in solvation free energy of the equilibrated excited state less the solvation free energy of the ground state. The electrostatic term Q is the free energy of the process $ML^* + ML \text{ --- } ML^+ + ML^-$ in the gas phase. The term in square brackets in (9.31) is equivalent to $[\Delta E_g + \chi_i]$ where ΔE_g is the gas phase energy separation between ground and excited states; it is therefore solvent independent.

These two studies represent systems which behave rather differently but which, nevertheless, show correlation between electrochemical and optical data. Such a correlation allows one to probe solvent interactions and derive the magnitudes of some solvation free energy terms and reorganisation energies [71], indicated in (9.30) and (9.31) [57]. The analysis may be inverted to predict electrochemical potentials from optical energies.

Outer-sphere charge transfer between two components of an ion pair, A^+B^-, has been studied whereby the energy of the transition with given A^+, and varying B^- is seen to be a smooth function of the reduction potential of A^+, e.g. with $[Fe(CN)_5L]^{n-}$ [68].

Such correlation between optical and electrochemical data is still in its infancy but merits further investigation.

9.4 iii) Ground and Excited State Protonation Equilibria

Electronic spectroscopy has been widely used in the study of organic bases, to evaluate pK_a values. It has been less widely used in inorganic chemistry but the study of electronic spectra as a function of pH can provide useful data.

Certainly if protonation results in a (reversible) change in the electronic spectrum, then a plot of such change against pH may yield a sigmoidal curve and allow evaluation of pK_a.

It is also possible to learn something of excited state pK_a^* values. If the excited state is a stronger base than the ground state, then it will be stabilised to a greater degree by protonation than the ground state, and a red shift in the excitation may generally be expected. The reverse blue shift is expected if the excited state is a weaker base than the ground state [59]. This approach is usually explained in terms of the 'Forster Mechanism' [11].

Indeed the shift in absorption energy upon protonation can provide a measure of pK_a^* through the equation [60]:-

$$pK_a^* = pK_a + (0.625/T)(E(B) - E(BH^+)) \quad (9.32)$$

The quantities E(B) and $E(BH^+)$ are the (0"-0') transition energies (in cm^{-1}) of the unprotonated and protonated species respectively. Very frequently this does not yield a very accurate result, probably because of the difficulty in identifying the (0"-0') transition energies.

If the complex emits and the emission is quenched by protonation, then excited state pK_a^* values may be determined. A simple plot of emission against pH can yield a sigmoidal plot whose inflection point is pK_a^* but this result can be unreliable since it does not take into account the probable difference in lifetime between protonated and unprotonated species. There may not be time to establish a proper equilibrium. A more accurate evaluation is obtained from [59]:

$$pK_a^* = pH \text{ (or } H_o) + \log(\tau_a/\tau_b) \quad (9.33)$$

where pH is taken at the inflection point in the luminescence/pH plot, and the two lifetimes are for the protonated and unprotonated species respectively. If only one of the species is luminescent, this equation may not apply [61].

The $[Ru(bipyrazine)_3]^{2+}$ cation has six potentially protonatable peripheral nitrogen sites. A combined absorption and emission study [62] was able to delineate all six protonation steps and provide absorption and emission spectroscopic data for most of them. The first three protonation steps result in a red shift in both absorption and emission and are assigned to a step-wise protonation of each bipyrazine ring at sites which are more basic than the ground state. The subsequent three steps blue shift both absorption and emission and are to the remaining sites on each bipyrazine, which are less basic than the ground state. The first protonation site has a pK_a^* value some 4 units more basic than the ground state.

Petersen and Demas [63,64] report data for $[Ru(II)(CN)_2(LL)_2]$ where LL is bipyridine or o-phenanthroline. Under highly acidic solutions, aqueous solutions contain DH^+ and DH_2^{2+} species (D = complex) in equilibrium with D, protonation occurring at the cyanide sites. The excited species however only emit from D^*. This is ascribed to greatly enhanced acidity in the excited state. Note that a similar situation prevails in the $[Ru(bpz)_3]^{2+}$ case cited above, where at acidities corresponding to less than three protonation steps, emission occurs from a species more heavily protonated than the ground state, while the reverse is true for solutions at the tetra or subsequent protonation stage.

Wrighton and co-workers [65] have reported data for $[Ru(bpy)_2(4,4'-(HOOC)_2bpy)]^{2+}$ which does emit from protonated excited states. Indeed at pH = 3.5 the ground state is almost completely unprotonated but emission occurs almost wholly from the protonated form.

In the bipyridine carboxylate and bipyrazine studies, one supposes that the MLCT state, by placing extra negative charge into the heterocyclic ring, enhances its basicity. In the cyanide case, the excited state again places charge into the bpy rings, leaving a formal Ru(III) bound to cyanide. The increased charge on the ruthenium then reduces the basicity of the cyano groups.

Thus such studies can provide numerical pK_a and pK_a^* data and also insight into the nature of the excited state protonation equilibria.

Note here that the study of the emission of a species in the presence of some quencher, proton or otherwise, provides information about the chemistry of the excited state.

Such experiments, via, for example, the Stern-Volmer plot, also provide kinetic information. Thus the rate of deactivation k_q is obtained from the Stern-Volmer expression:

$$\tau/\tau_o = I/I_o = 1 - K_{sv}[Q] = 1 - k_q\tau_o[Q] \quad (9.34)$$

where [Q] is the quencher concentration, τ_o is the lifetime of the excited state in the absence of a quencher, and τ is the lifetime in the presence of the quencher. K_{sv} is known as the Stern-Volmer quenching constant. The equality between the lifetime ratio and the intensity ratio should be true for diffusional quenching, but is not true for static quenching (where the quencher forms a complex of some kind with the species being observed).

9.5 Summary of Chemical Information Content

In this section, the information which can be obtained through study of electronic spectra is briefly listed.

i) Characterisation of molecular geometry, including conformational and optical isomeric aspects.

ii) 'Fingerprinting' of a species to permit other workers to identify a species prepared elsewhere and to assess its purity.

iii) Through analysis of data, 'mapping' the energy levels in the complex.

iv) Obtaining electronic structural information concerning the metal-ligand bonding interaction, and the interaction of one metal-ligand bond with another. In combination with other techniques, such as esr, this can provide a detailed assessment of the molecular orbital description of the molecule.

v) Providing numerical estimates of AOM, NSH or other ligand field parameters to permit a comparison between the bonding in one molecule with another, and hence to discuss such variations in terms of external structural or environmental influences.

vi) Identifying the variations induced in one molecule to the next as dictated by the electron configuration of the molecule, e.g. the role of the Jahn-Teller effect in both ground and excited states.

vii) Through study of the vibrational structure in absorption spectra, obtaining additional ground state force constant data which may be unavailable from the ir or Raman data.

viii) Through study of the vibrational structure in emission spectra,

obtain excited state force constant data which may allow the evaluation of the geometric and electronic structure of the excited state.

ix) Intensive studies of high resolution spectra, especially of the heavier elements, allow testing of ever more refined theories about electronic structure which can include relativistic correction, detailed probing of the ground state wave-function etc.

x) Obtaining a wealth of thermodynamic data. Thus reaction of a species D with another species F, either associatively or substitutively can give a change in spectrum as a function of [F] leading to thermodynamic equilibrium constant data. Examples of protonation pK_a values, and of heats of formation were discussed in this chapter, but the possibilities for using electronic spectra in this fashion are endless, e.g. axial site binding in macrocyclic complexes [67].

xi) If experiments in (x) are fairly slow, then the change in spectrum can be followed as a function of time, resulting in the generation of kinetic data. Combining stop-flow and regular instrumentation can provide such data over a time frame from milliseconds to hours. Data in (x) or (xi) carried out over a temperature range will provide more fundamental enthalpy, entropy and activation data.

xii) Studies of the emission spectra provide information about the excited state electronic structure and chemistry.

xiii) Exploring solvent-solute interactions through solvatochromic behaviour and relationship with electrochemical data. Reverse the probe by observing how a given spectroscopic feature varies as a function of the mole fraction of one solvent in a mixed solvent where the two solvents differ in their interaction with the solute. Such a study reveals information about the interactions between solvents and their activities against solutes.

xiv) Investigating ion pairing and similar types of association in solution.

xv) Obtaining rates of electron transfer in, for example, mixed valence systems, and thereby developing theories of electron transfer in general. Electron transfer rates between excited states and quenchers can also be obtained.

xvi) Through combination of electrochemical and optical techniques, e.g. spectro-electrochemistry using optically thin cells, study electrochemical transients and molecules in oxidised or reduced states.

xvii) Using spectro-electrochemistry, obtain more information about the electronic structure through the accessibility of data for a range of redox levels.

etc.

9.6 The Future

This second edition differs very greatly from the first edition. Let us finish by speculating what the third edition might

contain. The great advances made in the use of optical spectroscopy in bioinorganic chemistry are likely to continue and develop. Further use will be made of CD, MCD, resonance Raman and emission spectroscopy in the bioinorganic arena. The spectroscopy of main group elements will have advanced considerably. Molecular orbital techniques which can reliably predict transition energies will exist, but will not lessen our dependence upon semi-empirical techniques, such as the AOM model. They will help us to understand more clearly the information content of these semi-empirical methods. Thus the growing collection of AOM data will continue to grow and its significance, which is still fuzzy, will clarify.

The NSH model provides a more group theoretical approach to inorganic spectroscopy and its data base will grow to provide a clearer understanding of the significance, at a molecular level, of the parameters. The ease of obtaining data from low symmetry systems, using the NSH approach, should accelerate the collectiom of such data.

The development of 'crossed beam' technology, will provide greater access to excited state spectroscopy. The next edition will contain substantial information on the absorption spectra of excited states. Indeed there will be much greater emphasis on excited state chemistry using emission, quenching of emission, ground state absorption and excited state absorption spectroscopy to provide such data. The various theoretical models will expand to include such excited states and predict their chemistry.

Both photoelectron and resonance Raman spectroscopy will continue to provide useful complementary information to identify the energy levels in a complex and will be used more often in understanding absorption spectra. The latter technique, in particular, will be used, in combination with cryogenic spectroscopy, to obtain highly detailed vibrational maps of complexes in ground and excited states. This will facilitiate the development of the theory of vibrational coupling and will lead to a much more accurate 'picture' of the molecular levels in a complex.

The rapid growth in charge transfer spectroscopy will continue, and the theories necessary for a greater understanding thereof, will develop. It is likely that this growth will provide impetus for a parallel expansion in inorganic electrochemistry. This will also stimulate the use of spectroscopy in studying solvent interactions.

There will be a growth in gas phase spectroscopy both through use of volatile complexes, and through use of inert matrices which simulate the gas phase.

Since ligands influence metal energy levels, it must be true that metals influence ligand energy levels. The variation in intra-ligand transitions, with change of metal ion, is often quite small and usually ignored. The study of intra-ligand transitions is likely to become more fruitful in the future.

There will be a much greater emphasis on the electronic spectra of carbonyl complexes, clusters and organometallic complexes in

general. The difficulties often associated with organometallic synthesis, however, make it much more difficult for a spectroscopically inclined researcher to get involved, than in conventional coordination chemistry.

May I, therefore, urge organometallic chemists to report detailed visible and ultra-violet spectra of their organometallic species even if they are unable to assign or understand them. The growth, in the literature, of a body of data (now largely absent) will provide the impetus for theoretical understanding. There can be no doubt that such electronic spectra contain information which can greatly improve our understanding of the electronic structure of organometallic species.

Electronic spectroscopy has so much to offer, that there is no doubt that it will remain a major technique, even until the fourth edition.

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Abbreviations

3-Racac	3-R-acetylacetonate
Acacen; acen	N,N'-ethylenebis(acetylacetoneiminato)
acam	acetamide
([14]'aneN$_4$)	1,4,7,11-tetraazacyclotetradecane (an isomer of-cyclam)
Et$_2$-Alanine	N,N-diethylalanine
ac-amide	acrylamide
[q]-aneN$_4$	A series of q-membered saturated cyclic amines with 4NH groups. Sometimes just written as [q] where space is restricted.
AsAs$_3$Ph	(o-Me$_3$AsPh)$_3$As
AsSb$_3$Ph	(o-Me$_3$AsPh)$_3$Sb
RAsO	Ph$_2$MeAsO
XAs$_3$	X[(CH$_2$)$_3$AsMe$_2$]$_3$ where X = P, As and Sb
Me$_6$[q]aneN$_4$	A series of q-membered saturated cyclic amines with 4NH groups and 6 methyl substituents.
Me$_p$[q]en^{2-}	unsaturated q membered macrocycles, substituted with p Me groups.
[q]aneN$_3$	A q membered cyclic amine with three nitrogen atoms
bipy; bpy	2,2'-bipyridine
bipyam	2,2'-bipyridylamine
BipyNO	2,2'-bipyridine mono-N-oxide
BipyNO$_2$	2,2'-bipyridine di-N-oxide
4,4bpy	4,4'-bipyridyl
4,4bpya	1,2-bis(4-pyridyl)ethane
4,4bpye	1,2-bis(4-pyridyl)ethene
BPZ	bipyrazine
S-bz	mercaptobenzoate
chd	cyclohexane-1,2-diamine
cod	1,5-cyclooctadiene
comNH	H$_2$NCH(CO)NCH$_2$CO$_2$
CRH	A macrocycle; so-called Curtis macrocycle.
cyclam	a 14 membered saturated macrocyclic ring with 4 N atoms
CyNP$_3$	tris(2-dicyclohexylphosphinoethyl)amine
Et$_4$D	tetraethyldithiooxamide
dan	1,8-diaminonaphthalene
dacoda	1,8-diazacyclooctane-N,N-diacetate
daud	4,8-diazaundecane-1,11-diamine
dadd	5,8-diazadodecane-1,12-diamine
datd	5,9-diazatridecane-1,13-diamine
dach	1,4-diazacycloheptane
daco	1,5-diazacyclooctane
dap	bis(3-dimethylarsinopropyl)phenylphosphine
DCA	dichloroacetate anion
den	bisdi(2-aminoethyl)amine
dma	dimethylacetamide
DMAEP	(dimethylaminoethyl)pyridine.
Et$_4$dien	1,9-N-tetraethyl-1,4,7,10-tetra-aza-decane

1,2-DiMiz	1,2-dimethylimdidazole
diphos	$(CH_3)_2P(CH_2)_2P(CH_3)_2$
dmma	N,N'-dimethyldithiomalonamide
dmso	dimethylsulfoxide
4-DMP	4-dimethylaminopyridine
dpdo	1,8-bis(diphenylphosphino)-3,6-dioxooctane
dpm	dipivaloylmethanoate
dpoe, P-P	1,2-bis(diphenylphosphino)ethane
dpo	1,8-bis(diphenylphosphino)octane
DPP	bis(2-diphenylphosphinoethyl)phenylphosphine
dppen	cis-1,2-bis(diphenylphosphino)ethene
dsep	diselenophosphate
dsp	$S_2PPh_2^-$
Dta	$CH_3CS_2^-$
dtc	diethyldithiocarbamate
dth	2,5-dithiahexane
dtaa	dithioacetylacetonate
dtma	dithiomalonamide
Dtpa	$PhCH_2CS_2^-$
dtt	1,2-dithiol-3-thione
dpma	N,N'-diphenyldithiomalonamide
$EDDHA^{4-}$	ethylenediamine di(o-hydroxyphenylacetic acid) tetr-anion
en	ethylenediamine
s-Et_2en	sym-N,N'-diethylethylenediamine
as-Et_2en	asym-N,N-diethylethylenediamine
s-Me_2en	sym-N,N'-dimethylethylenediamine
as-Me_2en	asym-N,N-dimethylethylenediamine
P3ethane	1,1,1-tris(diphenylphosphinomethyl)ethane
esu	ethylselenourea
etu	ethylthiourea
ETS	$(EtO)_2PS_2^-$
ETSe	$(EtO)_2PSe_2^-$
HDPA	2,2-dipyridylamine
6-NH2hex	6-aminohexanoic acid
Hchp	6-chloro-2-hydroxypyridine
hfacac; Hfac	hexafluoroacetylacetonate
L-hist	L-histidine
Hmhp	6-methyl-2-hydroxypyridine
imid; Im	imidazole
$imidazN_4$	2,2-bis(5-phenyl-2-imidazolyl)propane
$imidazN_2S_2$	1,3-bis(5-phenyl-2-imidazolyl)-2-thiopropane
IQ	isoquinoline
isn	isonicotinamide
2MeIm	2-methylimidazole
MImT	1-methylimidazoline-2(3H)-thione (S donor)
L^+	monoquaternised 1,4-diazabicyclo[2,2,2]octane
$[q]macS_n$	a q membered macrocycle with n sulphur atoms
mcN_4	C-meso-5,5,7,12,12,14-hexamethyl-1,4,8,11-tetraza-cyclotetradecane
meoxyac	methoxyacetate

n-Mesalim	N-Methylsalicylaldiminato
mnt	maleonitriledithiolate
MORPO	morpholine
Me4pn	N,N,N',N'-tetramethylpropanediamine
2,5-Me_2pz	2,5-dimethylpyrazine
MPA	3,3'-dimethyl-2,2'-pyridylamine anion
x-MQ	x-Methylquinoline
MQN	2-(N-morpholinomethyl)-3-quinuclidinone
1,2-mpn	1,2-diamino-2-methylpropane
p-q-pN4	$H_2N(CH_2)_pNH(CH_2)_qNH(CH_2)_pNH2$
Me_4tn; TMen	N,N,N',N'-tetramethyl-trimethylenediamine
MeTPyEA	Tris((3,5-dimethyl-1-pyrazolyl)ethyl)amine
MeTPyA	Tris((3,5-dimethyl-1-pyrazolyl)methyl)amine
MPz	methylpyrazole
3,5MPz	dimethylpyrazole
p,qN_3 or	$H_2N(CH_2)_pNH(CH_2)qNH_2$
NP_3	tris(2-diphenylphosphinoethyl)amine
N_3P	bis(2-diethylaminoethyl)-2-diphenylphosphino-ethylamine
NS_3	tris(2-tert-butylthioethyl)amine
NMeim	N-methylimidazole
NNEU	N,N'-diethylurea
NTB	tris(2-benzimidazolylmethyl)amine
opd	o-phenylenediamine
OMPA	octamethylpyrophosphoramide
OSO-NH_2	O bonded $OS(O)CH_2CH_2NH_2$
1,3pd; pd	1,3-propanediamine
o-phen	1,10-phenanthroline
2=phos	$Ph_2PC_6H_4PPh_2$
N-n-Prsalim	N-n-Propylsalicylaldiminato
n pic (n-MePy)	n-methylpyridine
P_4	Hexaphenyl-1,4,7,10,-tetraphosphadecane
PL	$CH_3(CH_2)_4C(CH_2O)_3P$ cage phosphite
pnd	1,2-propanediamine
ppp	bis(2-(diphenylphosphino)ethyl)phenylphosphine
Py	pyridine
PyO; PyNO	pyridine N-oxide
n-RPyNO	n-alkyl substituted pyridine N-oxide
pz	pyrazole
pyz	pyrazine
R-P-P-R	$R_2PCH_2CH_2PR_2$
R-Py	substituted pyridine
QAS	$(o\text{-}Ph_2As.C_6H_4)_3P$
Qn	quinuclidine
QP	$(o\text{-}Ph_2P.C_6H_4)_3P$
Qx	quinoxaline
sacsac	Dithioacetylacetone
salen	Schiff base between salicylaldehyde and en.
R-S-S-R	$RSCH_2CH_2SR$
Cl-salEt	5-Cl-sal-en-NEt_2
su	seleno-urea

TAAB	A macrocycle
tart	tartrate
TCA	trichloroacetate
tetren	tetraethylenepentammine
thz	thiazole
tmd	1,3-diaminopropane
trien	triethylenetetrammine
ttp	bis(3-diphenylphosphinopropyl)phenylphosphine
p,q,p-tet	$NH_2(CH_2)_pNH(CH_2)_qNH(CH_2)_pNH_2$
tu	thiourea
vinim	vinylimidazole

Appendix I

Character Tables for Chemically Important Symmetry Groups

1. The nonaxial groups

C_1	E
A	1

C_S	E	σ_h		
A'	1	1	x, y, R_z	x^2, y^2, z^2, xy
A''	1	−1	z, R_x, R_y	yz, xz

C_i	E	i		
A_g	1	1	R_x, R_y, R_z	$x^2, y^2, z^2, xy, xz, yz$
A_u	1	−1	x, y, z	

2. The C_n groups

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	−1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	C_3^2		$\varepsilon = \exp(2\pi i/3)$
A	1	1	1	z, R_z	$x^2 + y^2, z^2$
E	1	ε	ε^*	$(x, y)(R_x, R_y)$	$(x^2 - y^2, xy)(yz, xz)$
	1	ε^*	ε		

C_4	E	C_4	C_2	C_4^3		
A	1	1	1	1	z, R_z	$x^2 + y^2, z^2$
B	1	−1	1	−1		$x^2 - y^2, xy$
E	1	i	−1	$-i$	$(x, y)(R_x, R_y)$	(yz, xz)
	1	$-i$	−1	i		

C_5	E	C_5	C_5^2	C_5^3	C_5^4		$\varepsilon = \exp(2\pi i/5)$
A	1	1	1	1	1	z, R_z	x^2+y^2, z^2
E_1	1	ε	ε^2	ε^{2*}	ε^*	$(x, y)(R_x, R_y)$	(yz, xz)
	1	ε^*	ε^{2*}	ε^2	ε		
E_2	1	ε^2	ε^*	ε	ε^{2*}		(x^2-y^2, xy)
	1	ε^{2*}	ε	ε^*	ε^2		

C_6	E	C_6	C_3	C_2	C_3^2	C_6^5		$\varepsilon = \exp(2\pi i/6)$
A	1	1	1	1	1	1	z, R_z	x^2+y^2, z^2
B	1	−1	1	−1	1	−1		
E_1	1	ε	$-\varepsilon^*$	−1	$-\varepsilon$	ε^*	(x, y)	
	1	ε^*	$-\varepsilon$	−1	$-\varepsilon^*$	ε	(R_x, R_y)	(xz, yz)
E_2	1	$-\varepsilon^*$	$-\varepsilon$	1	$-\varepsilon^*$	$-\varepsilon$		
	1	$-\varepsilon$	$-\varepsilon^*$	1	$-\varepsilon$	$-\varepsilon^*$		(x^2-y^2, xy)

C_7	E	C_7	C_7^2	C_7^3	C_7^4	C_7^5	C_7^6		$\varepsilon = \exp(2\pi i/7)$
A	1	1	1	1	1	1	1	z, R_z	x^2+y^2, z^2
E_1	1	ε	ε^2	ε^3	ε^{3*}	ε^{2*}	ε^*	(x, y)	
	1	ε^*	ε^{2*}	ε^{3*}	ε^3	ε^2	ε	(R_x, R_y)	(xz, yz)
E_2	1	ε^2	ε^{3*}	ε^*	ε	ε^3	ε^{2*}		
	1	ε^{2*}	ε^3	ε	ε^*	ε^{3*}	ε^2		(x^2-y^2, xy)
E_3	1	ε^3	ε^*	ε^2	ε^{2*}	ε	ε^{3*}		
	1	ε^{3*}	ε	ε^{2*}	ε^2	ε^*	ε^3		

C_8	E	C_8	C_4	C_2	C_4^3	C_8^3	C_8^5	C_8^7		$\varepsilon = \exp(2\pi i/8)$
A	1	1	1	1	1	1	1	1	z, R_z	x^2+y^2, z^2
B	1	−1	1	1	1	−1	−1	−1		
E_1	1	ε	i	−1	$-i$	$-\varepsilon^*$	$-\varepsilon$	ε^*	(x, y)	
	1	ε^*	$-i$	−1	i	$-\varepsilon$	$-\varepsilon^*$	ε	(R_x, R_y)	(xz, yz)
E_2	1	i	−1	1	−1	$-i$	i	$-i$		
	1	$-i$	−1	1	−1	i	$-i$	i		(x^2-y^2, xy)
E_3	1	$-\varepsilon$	i	−1	$-i$	ε^*	ε	$-\varepsilon^*$		
	1	$-\varepsilon^*$	$-i$	−1	i	ε	ε^*	$-\varepsilon$		

3. The D_n groups

D_2	E	$C_2(z)$	$C_2(y)$	$C_2(x)$		
A	1	1	1	1		x^2, y^2, z^2
B_1	1	1	−1	−1	z, R_z	xy
B_2	1	−1	1	−1	y, R_y	xz
B_3	1	−1	−1	1	x, R_x	yz

D_3	E	$2C_3$	$3C_2$		
A_1	1	1	1		x^2+y^2, z^2
A_2	1	1	−1	z, R_z	
E	2	−1	0	$(x, y)(R_x, R_y)$	$(x^2-y^2, xy)(xz, yz)$

D_4	E	$2C_4$	$C_2(=C_4^2)$	$2C_2'$	$2C_2''$		
A_1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	−1	−1	z, R_z	
B_1	1	−1	1	1	−1		x^2-y^2
B_2	1	−1	1	−1	1		xy
E	2	0	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)

D_5	E	$2C_5$	$2C_5^2$	$5C_2$		
A_1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	−1	z, R_z	
E_1	2	$2\cos 72°$	$2\cos 144°$	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	$2\cos 144°$	$2\cos 72°$	0		(x^2-y^2, xy)

D_6	E	$2C_6$	$2C_3$	C_2	$3C_2'$	$3C_2''$		
A_1	1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	1	−1	−1	z, R_z	
B_1	1	−1	1	−1	1	−1		
B_2	1	−1	1	−1	−1	1		
E_1	2	1	−1	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	−1	−1	2	0	0		(x^2-y^2, xy)

4. The C_{nv} groups

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	−1	−1	R_z	xy
B_1	1	−1	1	−1	x, R_y	xz
B_2	1	−1	−1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	−1	R_z	
E	2	−1	0	$(x, y)(R_x, R_y)$	$(x^2-y^2, xy)(xz, yz)$

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	−1	−1	R_z	
B_1	1	−1	1	1	−1		x^2-y^2
B_2	1	−1	1	−1	1		xy
E	2	0	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)

C_{5v}	E	$2C_5$	$2C_5^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	−1	R_z	
E_1	2	$2\cos 72°$	$2\cos 144°$	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	$2\cos 144°$	$2\cos 72°$	0		(x^2-y^2, xy)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	1	−1	−1	R_z	
B_1	1	−1	1	−1	1	−1		
B_2	1	−1	1	−1	−1	1		
E_1	2	1	−1	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	−1	−1	2	0	0		(x^2-y^2, xy)

5. The C_{nh} groups

C_{2h}	E	C_2	i	σ_h		
A_g	1	1	1	1	R_z	x^2, y^2, z^2, xy
B_g	1	−1	1	−1	R_x, R_y	xz, yz
A_u	1	1	−1	−1	z	
B_u	1	−1	−1	1	x, y	

C_{3h}	E	C_3	C_3^2	σ_h	S_3	S_3^5		$\varepsilon = \exp(2\pi i/3)$
A'	1	1	1	1	1	1	R_z	x^2+y^2, z^2
E'	1	ε	ε^*	1	ε	ε^*	(x, y)	(x^2-y^2, xy)
	1	ε^*	ε	1	ε^*	ε		
A''	1	1	1	−1	−1	−1	z	
E''	1	ε	ε^*	−1	$-\varepsilon$	$-\varepsilon^*$	(R_x, R_y)	(xz, yz)
	1	ε^*	ε	−1	$-\varepsilon^*$	$-\varepsilon$		

C_{4h}	E	C_4	C_2	C_4^3	i	S_4^3	σ_h	S_4		
A_g	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
B_g	1	−1	1	−1	1	−1	1	−1		x^2-y^2, xy
E_g	1	i	−1	$-i$	1	i	−1	$-i$	(R_x, R_y)	(xz, yz)
	1	$-i$	−1	i	1	$-i$	−1	i		
A_u	1	1	1	1	−1	−1	−1	−1	z	
B_u	1	−1	1	−1	−1	1	−1	1		
E_u	1	i	−1	$-i$	−1	$-i$	1	i	(x, y)	
	1	$-i$	−1	i	−1	i	1	$-i$		

C_{5h}	E	C_5	C_5^2	C_5^3	C_5^4	σ_h	S_5	S_5^7	S_5^3	S_5^9		$\varepsilon = \exp(2\pi i/5)$
A'	1	1	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
E_1'	1	ε	ε^2	ε^{2*}	ε^*	1	ε	ε^2	ε^{2*}	ε^*	(x, y)	
	1	ε^*	ε^{2*}	ε^2	ε	1	ε^*	ε^{2*}	ε^2	ε		
E_2'	1	ε^2	ε^*	ε	ε^{2*}	1	ε^2	ε^*	ε	ε^{2*}		(x^2-y^2, xy)
	1	ε^{2*}	ε	ε^*	ε^2	1	ε^{2*}	ε	ε^*	ε^2		
A''	1	1	1	1	1	-1	-1	-1	-1	-1	z	
E_1''	1	ε	ε^2	ε^{2*}	ε^*	-1	$-\varepsilon$	$-\varepsilon^2$	$-\varepsilon^{2*}$	$-\varepsilon^*$		
	1	ε^*	ε^{2*}	ε^2	ε	-1	$-\varepsilon^*$	$-\varepsilon^{2*}$	$-\varepsilon^2$	$-\varepsilon$		
E_2''	1	ε^2	ε^*	ε	ε^{2*}	-1	$-\varepsilon^2$	$-\varepsilon^*$	$-\varepsilon$	$-\varepsilon^{2*}$	(R_x, R_y)	(xz, yz)
	1	ε^{2*}	ε	ε^*	ε^2	-1	$-\varepsilon^{2*}$	$-\varepsilon$	$-\varepsilon^*$	$-\varepsilon^2$		

C_{6h}	E	C_6	C_3	C_2	C_3^2	C_6^5	i	S_3^5	S_6^5	σ_h	S_6	S_3		$\varepsilon = \exp(2\pi i/6)$
A_g	1	1	1	1	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
B_g	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1		
E_{1g}	1	ε	$-\varepsilon^*$	-1	$-\varepsilon$	ε^*	1	ε	$-\varepsilon^*$	-1	$-\varepsilon$	ε^*	(R_x, R_y)	(xz, yz)
	1	ε^*	$-\varepsilon$	-1	$-\varepsilon^*$	ε	1	ε^*	$-\varepsilon$	-1	$-\varepsilon^*$	ε		
E_{2g}	1	$-\varepsilon^*$	$-\varepsilon$	1	$-\varepsilon^*$	$-\varepsilon$	1	$-\varepsilon^*$	$-\varepsilon$	1	$-\varepsilon^*$	$-\varepsilon$	z	(x^2-y^2, xy)
	1	$-\varepsilon$	$-\varepsilon^*$	1	$-\varepsilon$	$-\varepsilon^*$	1	$-\varepsilon$	$-\varepsilon^*$	1	$-\varepsilon$	$-\varepsilon^*$		
A_u	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1		
B_u	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1		
E_{1u}	1	ε	$-\varepsilon^*$	-1	$-\varepsilon$	ε^*	-1	$-\varepsilon$	ε^*	-1	ε	$-\varepsilon^*$	(x, y)	
	1	ε^*	$-\varepsilon$	-1	$-\varepsilon^*$	ε	-1	$-\varepsilon^*$	ε	-1	ε^*	$-\varepsilon$		
E_{2u}	1	$-\varepsilon^*$	$-\varepsilon$	1	$-\varepsilon^*$	$-\varepsilon$	-1	ε	$-\varepsilon^*$	-1	ε	$-\varepsilon^*$		
	1	$-\varepsilon$	$-\varepsilon^*$	1	$-\varepsilon$	$-\varepsilon^*$	-1	ε^*	$-\varepsilon$	-1	ε^*	$-\varepsilon$		

6. The D_{nh} groups

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	-1	-1	-1		
A_2''	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	R_z	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(R_x, R_y)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

D_{5h}	E	$2C_5$	$2C_5^2$	$5C_2$	σ_h	$2S_5$	$2S_5^3$	$5\sigma_v$		
A_1'	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	1	−1	1	1	1	−1	R_z	
E_1'	2	2 cos 72°	2 cos 144°	0	2	2 cos 72°	2 cos 144°	0	(x, y)	
E_2'	2	2 cos 144°	2 cos 72°	0	2	2 cos 144°	2 cos 72°	0		(x^2-y^2, xy)
A_1''	1	1	1	1	−1	−1	−1	−1		
A_2''	1	1	1	−1	−1	−1	−1	1	z	
E_1''	2	2 cos 72°	2 cos 144°	0	−2	−2 cos 72°	−2 cos 144°	0	(R_x, R_y)	(xz, yz)
E_2''	2	2 cos 144°	2 cos 72°	0	−2	−2 cos 144°	−2 cos 72°	0		

D_{6h}	E	$2C_6$	$2C_3$	C_2	$3C_2'$	$3C_2''$	i	$2S_3$	$2S_6$	σ_h	$3\sigma_d$	$3\sigma_v$		
A_{1g}	1	1	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	1	−1	−1	1	1	1	1	−1	−1	R_z	
B_{1g}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1		
B_{2g}	1	−1	1	−1	−1	1	1	−1	1	−1	−1	1		
E_{1g}	2	1	−1	−2	0	0	2	1	−1	−2	0	0	(R_x, R_y)	(xz, yz)
E_{2g}	2	−1	−1	2	0	0	2	−1	−1	2	0	0		(x^2-y^2, xy)
A_{1u}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1		
A_{2u}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1	z	
B_{1u}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1		
B_{2u}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1		
E_{1u}	2	1	−1	−2	0	0	−2	−1	1	2	0	0	(x, y)	
E_{2u}	2	−1	−1	2	0	0	−2	1	1	−2	0	0		

7. The D_{nd} groups

D_{2d}	E	$2S_4$	C_2	$2C_2'$	$2\sigma_d$		
A_1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	−1	−1	R_z	
B_1	1	−1	1	1	−1		x^2-y^2
B_2	1	−1	1	−1	1	z	xy
E	2	0	−2	0	0	(x, y); (R_x, R_y)	(xz, yz)

D_{3d}	E	$2C_3$	$3C_2$	i	$2S_6$	$3\sigma_d$		
A_{1g}	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	−1	1	1	−1	R_z	
E_g	2	−1	0	2	−1	0	(R_x, R_y)	(x^2-y^2, xy) (xz, yz)
A_{1u}	1	1	1	−1	−1	−1		
A_{2u}	1	1	−1	−1	−1	1	z	
E_u	2	−1	0	−2	1	0	(x, y)	

D_{4d}	E	$2S_8$	$2C_4$	$2S_8^3$	C_2	$4C_2'$	$4\sigma_d$		
A_1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	1	1	−1	−1	R_z	
B_1	1	−1	1	−1	1	1	−1		(x^2-y^2, xy) (xz, yz)
B_2	1	−1	1	−1	1	−1	1	z	
E_1	2	$\sqrt{2}$	0	$-\sqrt{2}$	−2	0	0	(x, y)	
E_2	2	0	−2	0	2	0	0		
E_3	2	$-\sqrt{2}$	0	$\sqrt{2}$	−2	0	0	(R_x, R_y)	

D_{5d}	E	$2C_5$	$2C_5^2$	$5C_2$	i	$2S_{10}^3$	$2S_{10}$	$5\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	−1	1	1	1	−1	R_z	
E_{1g}	2	$2\cos 72°$	$2\cos 144°$	0	2	$2\cos 72°$	$2\cos 144°$	0	(R_x, R_y)	(xz, yz)
E_{2g}	2	$2\cos 144°$	$2\cos 72°$	0	2	$2\cos 144°$	$2\cos 72°$	0		(x^2-y^2, xy)
A_{1u}	1	1	1	1	−1	−1	−1	−1		
A_{2u}	1	1	1	−1	−1	−1	−1	1	z	
E_{1u}	2	$2\cos 72°$	$2\cos 144°$	0	−2	$-2\cos 72°$	$-2\cos 144°$	0	(x, y)	
E_{2u}	2	$2\cos 144°$	$2\cos 72°$	0	−2	$-2\cos 144°$	$-2\cos 72°$	0		

D_{6d}	E	$2S_{12}$	$2C_6$	$2S_4$	$2C_3$	$2S_{12}^5$	C_2	$6C_2'$	$6\sigma_d$		
A_1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	1	1	1	1	−1	−1	R_z	
B_1	1	−1	1	−1	1	−1	1	1	−1		
B_2	1	−1	1	−1	1	−1	1	−1	1	z	
E_1	2	$\sqrt{3}$	1	0	−1	$-\sqrt{3}$	−2	0	0	(x, y)	
E_2	2	1	−1	−2	−1	1	2	0	0		(x^2-y^2, xy)
E_3	2	0	−2	0	2	0	−2	0	0		
E_4	2	−1	−1	2	−1	−1	2	0	0	(R_x, R_y)	
E_5	2	$-\sqrt{3}$	1	0	−1	$\sqrt{3}$	−2	0	0		(xz, yz)

8. The S_n groups

S_4	E	S_4	C_2	S_4^3		
A	1	1	1	1	R_z	x^2+y^2, z^2
B	1	−1	1	−1	z	x^2-y^2, xy
E	1	i	−1	$-i$	$(x, y); (R_x, R_y)$	(xz, yz)
	1	$-i$	−1	i		

S_6	E	C_3	C_3^2	i	S_6^5	S_6		$\varepsilon = \exp(2\pi i/3)$
A_g	1	1	1	1	1	1	R_z	x^2+y^2, z^2
E_g	1	ε	ε^*	1	ε	ε^*	(R_x, R_y)	$(x^2-y^2, xy);$
	1	ε^*	ε	1	ε^*	ε		(xz, yz)
A_u	1	1	1	−1	−1	−1	z	
E_u	1	ε	ε^*	−1	$-\varepsilon$	$-\varepsilon^*$	(x, y)	
	1	ε^*	ε	−1	$-\varepsilon^*$	$-\varepsilon$		

S_8	E	S_8	C_4	S_8^3	C_2	S_8^5	S_4^3	S_8^7		$\varepsilon=\exp(2\pi i/8)$
A	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
B	1	−1	1	−1	1	−1	1	−1	z	
E_1	1	ε	i	$-\varepsilon^*$	−1	$-\varepsilon$	$-i$	ε^*	$(x, y);$	
	1	ε^*	$-i$	$-\varepsilon$	−1	$-\varepsilon^*$	i	ε	(R_x, R_y)	
E_2	1	i	−1	$-i$	1	i	−1	$-i$		
	1	$-i$	−1	i	1	$-i$	−1	i		(x^2-y^2, xy)
E_3	1	$-\varepsilon^*$	$-i$	ε	−1	ε^*	i	$-\varepsilon$		
	1	$-\varepsilon$	i	ε^*	−1	ε	$-i$	$-\varepsilon^*$		(xz, yz)

9. The cubic groups

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	−1	−1		
E	2	−1	2	0	0		$(2z^2-x^2-y^2,$ $x^2-y^2)$
T_1	3	0	−1	1	−1	(R_x, R_y, R_z)	
T_2	3	0	−1	−1	1	(x, y, z)	(xy, xz, yz)

O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2$ $(=C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		$x^2+y^2+z^2$
A_{2g}	1	1	−1	−1	1	1	−1	1	1	−1		
E_g	2	−1	0	0	2	2	0	−1	2	0		$(2z^2-x^2-y^2,$ $x^2-y^2)$
T_{1g}	3	0	−1	1	−1	3	1	0	−1	−1	(R_x, R_y, R_z)	
T_{2g}	3	0	1	−1	−1	3	−1	0	−1	1		(xz, yz, xy)
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1		
A_{2u}	1	1	−1	−1	1	−1	1	−1	−1	1		
E_u	2	−1	0	0	2	−2	0	1	−2	0		
T_{1u}	3	0	−1	1	−1	−3	−1	0	1	1	(x, y, z)	
T_{2u}	3	0	1	−1	−1	−3	1	0	1	−1		

10. The groups $C_{\infty v}$ and $D_{\infty h}$ for linear molecules

$C_{\infty v}$	E	$2C_\infty^\Phi$	...	$\infty\sigma_v$		
$A_1 \equiv \Sigma^+$	1	1	...	1	z	x^2+y^2, z^2
$A_2 \equiv \Sigma^-$	1	1	...	−1	R_z	
$E_1 \equiv \Pi$	2	$2\cos\Phi$	...	0	$(x, y); (R_x, R_y)$	(xz, yz)
$E_2 \equiv \Delta$	2	$2\cos 2\Phi$	...	0		(x^2-y^2, xy)
$E_3 \equiv \Phi$	2	$2\cos 3\Phi$	...	0		
...	...	...	...	...		

$D_{\infty h}$	E	$2C_\infty^\Phi$	...	$\infty\sigma_v$	i	$2S_\infty^\Phi$	...	∞C_2		
Σ_g^+	1	1	...	1	1	1	...	1		x^2+y^2, z^2
Σ_g^-	1	1	...	−1	1	1	...	−1	R_z	
Π_g	2	$2\cos\Phi$	...	0	2	$-2\cos\Phi$	...	0	(R_x, R_y)	(xz, yz)
Δ_g	2	$2\cos 2\Phi$	...	0	2	$2\cos 2\Phi$	...	0		(x^2-y^2, xy)
...	...	...	...	...	...	...	...	...		
Σ_u^+	1	1	...	1	−1	−1	...	−1	z	
Σ_u^-	1	1	...	−1	−1	−1	...	1		
Π_u	2	$2\cos\Phi$	...	0	−2	$2\cos\Phi$	...	0	(x, y)	
Δ_u	2	$2\cos 2\Phi$	...	0	−2	$-2\cos 2\Phi$	...	0		

11. The icosahedral group

I_h	E	$12C_5$	$12C_5^2$	$20C_3$	$15C_2$	i	$12S_{10}$	$12S_{10}^3$	$20S_6$	15σ		
A_g	1	1	1	1	1	1	1	1	1	1		$x^2+y^2+z^2$
T_{1g}	3	$\frac{1}{2}(1+\sqrt{5})$	$\frac{1}{2}(1-\sqrt{5})$	0	−1	3	$\frac{1}{2}(1-\sqrt{5})$	$\frac{1}{2}(1+\sqrt{5})$	0	−1	(R_x, R_y, R_z)	
T_{2g}	3	$\frac{1}{2}(1-\sqrt{5})$	$\frac{1}{2}(1+\sqrt{5})$	0	−1	3	$\frac{1}{2}(1+\sqrt{5})$	$\frac{1}{2}(1-\sqrt{5})$	0	−1		$(2z^2-x^2-y^2,$ $x^2-y^2,$ $xy, yz, xz)$
G_g	4	−1	−1	1	0	4	−1	−1	1	0		
H_g	5	0	0	−1	1	5	0	0	−1	1		
A_u	1	1	1	1	1	−1	−1	−1	−1	−1		
T_{1u}	3	$\frac{1}{2}(1+\sqrt{5})$	$\frac{1}{2}(1-\sqrt{5})$	0	−1	−3	$-\frac{1}{2}(1-\sqrt{5})$	$-\frac{1}{2}(1+\sqrt{5})$	0	1	(x, y, z)	
T_{2u}	3	$\frac{1}{2}(1-\sqrt{5})$	$\frac{1}{2}(1+\sqrt{5})$	0	−1	−3	$-\frac{1}{2}(1+\sqrt{5})$	$-\frac{1}{2}(1-\sqrt{5})$	0	1		
G_u	4	−1	−1	1	0	−4	1	1	−1	0		
H_u	5	0	0	−1	1	−5	0	0	1	−1		

12. Character table for the double group O′

	E	R	$4C_3$ $4C_3^2R$	$4C_3R$ $4C_3^2$	$3C_2$ $3C_2R$	$3C_4$ $3C_4^3R$	$3C_4R$ $3C_4^3$	$6C'$ $6C_2R$	*Bethe's nomenclature*
A_1'	1	1	1	1	1	1	1	1	Γ_1
A_2'	1	1	1	1	1	−1	−1	−1	Γ_2
E_1'	2	2	−1	−1	2	0	0	0	Γ_3
T_1'	3	3	0	0	−1	1	1	−1	Γ_4
T_2'	3	3	0	0	−1	−1	−1	1	Γ_5
E_2'	2	−2	1	−1	0	$\sqrt{2}$	$-\sqrt{2}$	0	Γ_6
E_3'	2	−2	1	−1	0	$-\sqrt{2}$	$\sqrt{2}$	0	Γ_7
G'	4	−4	−1	1	0	0	0	0	Γ_8

13. Multiplication of Γ_n (single group) and Γ_J (double group) in octahedral (O_h) and tetrahedral (T_d) symmetry. The commutative law $\Gamma_a \times \Gamma_b = \Gamma_b \times \Gamma_a$ is operative

Part a: $\Gamma_n \times \Gamma_n$	*Part b:* $\Gamma_n \times \Gamma_J$	*Part c:* $\Gamma_J \times \Gamma_J$
$\Gamma_1 \times \Gamma_n = \Gamma_n$	$\Gamma_1 \times \Gamma_6 = \Gamma_2 \times \Gamma_7 = \Gamma_6$	$\Gamma_6 \times \Gamma_6 = \Gamma_7 \times \Gamma_7 = \Gamma_1 + \Gamma_4$
$\Gamma_2 \times \Gamma_2 = \Gamma_1$	$\Gamma_2 \times \Gamma_6 = \Gamma_1 \times \Gamma_7 = \Gamma_7$	$\Gamma_6 \times \Gamma_7 = \Gamma_2 + \Gamma_5$
$\Gamma_2 \times \Gamma_3 = \Gamma_3$	$\Gamma_3 \times \Gamma_6 = \Gamma_3 \times \Gamma_7 = \Gamma_8$	$\Gamma_6 \times \Gamma_8 = \Gamma_7 \times \Gamma_8 = \Gamma_3 + \Gamma_4 + \Gamma_5$
$\Gamma_2 \times \Gamma_4 = \Gamma_5$	$\Gamma_3 \times \Gamma_8 = \Gamma_6 + \Gamma_7 + \Gamma_8$	$\Gamma_8 \times \Gamma_8 = \Gamma_1 + \Gamma_2 + \Gamma_3 + 2\Gamma_4 + 2\Gamma_5$
$\Gamma_2 \times \Gamma_5 = \Gamma_4$	$\Gamma_4 \times \Gamma_6 = \Gamma_5 \times \Gamma_7 = \Gamma_6 + \Gamma_8$	
$\Gamma_3 \times \Gamma_3 = \Gamma_1 + \Gamma_2 + \Gamma_3$	$\Gamma_5 \times \Gamma_6 = \Gamma_4 \times \Gamma_7 = \Gamma_7 + \Gamma_8$	
$\Gamma_3 \times \Gamma_4 = \Gamma_4 + \Gamma_5$	$\Gamma_4 \times \Gamma_8 = \Gamma_5 \times \Gamma_8 = \Gamma_6 + \Gamma_7 + 2\Gamma_8$	
$\Gamma_3 \times \Gamma_5 = \Gamma_4 + \Gamma_5$		
$\Gamma_4 \times \Gamma_4 = \Gamma_1 + \Gamma_3 + \Gamma_4 + \Gamma_5$		
$\Gamma_4 \times \Gamma_5 = \Gamma_2 + \Gamma_3 + \Gamma_4 + \Gamma_5$		
$\Gamma_5 \times \Gamma_5 = \Gamma_1 + \Gamma_3 + \Gamma_4 + \Gamma_5$		

Appendix II

Calculation of Crystal Field Splitting Energies

There are 3 simple operations involved in the determination of crystal field energies using the ionic model:

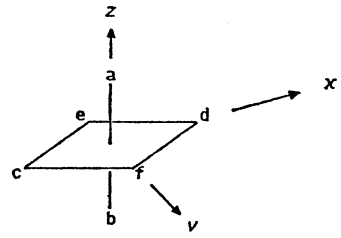
(*a*) writing the ligand (charge) coordinates in terms of the angles θ_i and ϕ_i (Fig. 4.1),

(*b*) using those values of θ_i and ϕ_i to calculate the ligand position functions D_{lm} and G_{lm} summing over all the ligands,

(*c*) using the derived D_{lm} and G_{lm} functions to calculate the H_{pq} integrals. If the off-diagonal elements ($p \neq q$) are zero, then these integrals correspond directly with the energies of the real *d* orbitals. If the off-diagonal elements are non-zero, then the secular determinant must be solved for its roots which correspond to the energy levels required. In such cases one or more of these energy levels will correspond with linear combinations of the real *d* orbitals.

OCTAHEDRAL FIELD

Assume six ligands (labelled a–f) of charge Ze are arranged at the apices of an octahedron, all equidistant from the origin



The ligand coordinates are defined by the angles θ_i and ϕ_i (see Fig. 4.1) which are as follows:

Ligand	*Coordinates*	
	θ_i	ϕ_i
a	0	–
b	180	–
c	90	0
d	90	180
e	90	270
f	90	90

Inserting these values into the ligand position functions D_{lm} and G_{lm} (Table AII, 1) and summing over all the ligands, we have:
6 ligands with $D^i_{00} = \alpha^i_0$, hence $D^i_{00} = 6\alpha^i_0$
the D^i_{40} functions are

ligand	D^i_{40}
a	$\frac{8}{3}\alpha_4$
b	$\frac{8}{3}\alpha_4$
c	α_4
d	α_4
e	α_4
f	α_4

Hence summed over all ligands, $D^i_{40} = \frac{28}{3}\alpha_4$. Similarly D^i_{44} summed over all the ligands is $4\alpha_4$. All other D_{lm} and G_{lm} position functions are zero.

These values may now be inserted into the equations given in Table AII.2 to yield the required matrix elements. Thus

$$H_{11} = 6\alpha_0 + \alpha_4$$
$$H_{22} = 6\alpha_0 - \tfrac{2}{3}\alpha_4$$
$$H_{33} = 6\alpha_0 + \alpha_4$$
$$H_{44} = 6\alpha_0 - \tfrac{2}{3}\alpha_4$$
$$H_{55} = 6\alpha_0 - \tfrac{2}{3}\alpha_4$$

TABLE AII. 1

THE LIGAND POSITION FUNCTIONS D_{lm} AND G_{lm}

$$D^i_{00} = \alpha^i_0$$
$$D^i_{20} = \alpha^i_2 (3\cos^2\theta_i - 1)$$
$$D^i_{40} = \alpha^i_4 (35/3\cos^4\theta_i - 10\cos^2\theta_i + 1)$$
$$D^i_{21} = \alpha^i_2 \sin\theta_i \cos\theta_i \cos\phi_i$$
$$D^i_{22} = \alpha^i_2 \sin^2\theta_i \cos 2\phi_i$$
$$D^i_{41} = \alpha^i_4 \sin\theta_i \cos\theta_i (7/3\cos^2\theta_i - 1)\cos\phi_i$$
$$D^i_{42} = \alpha^i_4 \sin^2\theta_i (7\cos^2\theta_i - 1)\cos 2\phi_i$$
$$D^i_{43} = \alpha^i_4 \sin^3\theta_i \cos\theta_i \cos 3\phi_i$$
$$D^i_{44} = \alpha^i_4 \sin^4\theta_i \cos 4\phi_i$$
$$G^i_{21} = \alpha^i_2 \sin\theta_i \cos\theta_i \sin\phi_i$$
$$G^i_{22} = \alpha^i_2 \sin^2\theta_i \sin 2\phi_i$$
$$G^i_{41} = \alpha^i_4 \sin\theta_i \cos\theta_i (7/3\cos^2\theta_i - 1)\sin\phi_i$$
$$G^i_{42} = \alpha^i_4 \sin^2\theta_i (7\cos^2\theta_i - 1)\sin 2\phi_i$$
$$G^i_{43} = \alpha^i_4 \sin^3\theta_i \cos\theta_i \sin 3\phi_i$$
$$G^i_{44} = \alpha^i_4 \sin^4\theta_i \sin 4\phi_i$$

TABLE AII. 2

H_{ab} INTEGRALS IN TERMS OF D_{lm} AND G_{lm}

$$H_{11} = D_{00} - 1/7\, D_{20} + 1/56\, D_{40} + 5/24\, D_{44}$$
$$H_{22} = D_{00} + 1/14\, D_{20} - 1/14\, D_{40} + 3/14\, D_{22} + 5/42\, D_{42}$$
$$H_{33} = D_{00} + 1/7\, D_{20} + 3/28\, D_{40}$$
$$H_{44} = D_{00} + 1/14\, D_{20} - 1/14\, D_{40} - 3/14\, D_{22} - 5/42\, D_{42}$$
$$H_{55} = D_{00} - 1/7\, D_{20} + 1/56\, D_{40} - 5/24\, D_{44}$$
$$H_{12} = 3/7\, D_{21} - 5/28\, D_{41} + 5/12\, D_{43}$$
$$H_{13} = -\sqrt{3}/7\, D_{22} + 5\sqrt{3}/84\, D_{42}$$
$$H_{14} = -3/7\, G_{21} + 5/28\, G_{41} + 5/12\, G_{43}$$
$$H_{15} = 5/24\, G_{44}$$
$$H_{23} = \sqrt{3}/7\, D_{21} + 5\sqrt{3}/14\, D_{41}$$
$$H_{24} = 3/14\, G_{22} + 5/42\, G_{42}$$
$$H_{25} = 3/7\, G_{21} - 5/28\, G_{41} + 5/12\, G_{43}$$
$$H_{34} = \sqrt{3}/7\, G_{21} + 5\sqrt{3}/14\, G_{41}$$
$$H_{35} = -\sqrt{3}/7\, G_{22} + 5\sqrt{3}/84\, G_{42}$$
$$H_{45} = 3/7\, D_{21} - 5/28\, D_{41} - 5/12\, D_{43}$$

All other matrix elements (the off-diagonals) are zero. Therefore these correspond directly to the roots of the secular equation (4.3) and are simply the energies of the real d orbitals in an octahedral environment:

$$\left.\begin{aligned} \mathrm{E}(d_{x^2-y^2}) &= 6\alpha_0 + \alpha_4 \\ \mathrm{E}(d_{z^2}) &= 6\alpha_0 + \alpha_4 \end{aligned}\right\} e_g$$

$$\left.\begin{aligned} \mathrm{E}(d_{xz}) &= 6\alpha_0 - \tfrac{2}{3}\alpha_4 \\ \mathrm{E}(d_{yz}) &= 6\alpha_0 - \tfrac{2}{3}\alpha_4 \\ \mathrm{E}(d_{xy}) &= 6\alpha_0 - \tfrac{2}{3}\alpha_4 \end{aligned}\right\} t_{2g}$$

The common $6\alpha_0$ term represents the repulsion of the d electrons in a spherical field environment. The splitting is seen to be $\frac{5}{3}\alpha_4$.

Other problems may be tackled in a similar manner. (See references 4 and 7 in chapter 4).

Appendix III

AOM Calculations

We are interested in deriving the energy levels of a somewhat distorted molecule. Consider, for example, a complex $[ML_4(ONO)]^{n+}$ where the nitrite group occupies a pair of adjacent sites in the pseudo-octahedron and the bite of the nitrite ligand is less than 90^o. The other ligands are assumed to be mutually perpendicular and to lie on the xy and z axes. The d^1 energy levels can be obtained by inserting the ligand position coordinates into the matrix in Table 1.12. Consider the general case where the bite of the nitrite is assumed to be $(90-2\psi)^o$ [AIII.1].

Ligands L(1) and L(2) lie in the xy plane, L(3) and L(4) along the z axis, and the nitrite is located symmetrically between the positive x and y axes. The position coordinates are:

	θ	φ
L(1)	180	90
L(2)	270	90
L(3)	0	0
L(4)	0	180
O(1)	ψ	90
O(2)	90-ψ	90

(III.1)

Insertion of these coordinates into Table 1.12 according to the procedure developed in Chapter 1, yields the following diagonal matrix elements:

$$\langle z^2|H|z^2\rangle = 2.5e_\sigma(L) + 0.5e_\sigma(O)$$

$$\langle xz|H|zx\rangle = \langle yz|H|yz\rangle = 3e_\pi(L) + e_\pi(O)$$

$$\langle xy|H|xz\rangle = 2e_\pi(L)\ L+ 1.5\sin^2 2\psi e_\sigma(O) + 2\cos^2 2\psi e_\pi(O)$$

$$\langle x^2-y^2|H|x^2-y^2\rangle = 1.5e_\sigma(L) + 1.5\cos^2 2\psi e_\sigma(O) + 2\sin^2 2\psi e_\pi(O)$$

(III.2)

Since the coordinate frame no longer sits along all the symmetry axes of the molecule, there will be some off-diagonal elements. In particular, the xz and yz orbitals do not lie along symmetry axes, and, from the viewpoint of the principle axis (bisecting the nitrite group), the z^2 and xy orbitals both transform as a_1. Thus there will be two off-diagonal elements. They are found to be (see Eqn.1.70):

$$\langle xz|H|yz\rangle = \sin 2\psi e_\pi(O)$$

$$\langle z^2|H|xy\rangle = -\sqrt{2}/2\sin 2\psi e_\sigma(O)$$

(III.3)

We should therefore preferably consider the two mixed functions:

$\sqrt{(1/2)}[xz + yz]$ whose energy is given by:

$$\sqrt{(1/2)}\sqrt{(1/2)}\langle xz + yz|H|xz + yz\rangle = 0.5[\langle xz|Hxz\rangle + \langle yz|H|yz\rangle + 2\langle xz|H|yz\rangle]$$

$$= 3e_\pi(L) + (1 + \sin 2\psi)e_\pi(O)$$

and similarly:

$\sqrt{(1/2)}[xz - yz]$ with energy $3e_\pi(L) + (1 - \sin 2\psi)e_\pi(O)$.

Since the xz and yz orbitals have equal diagonal elements, we know that they will be equally mixed in the two wavefunctions; this is not the case for the z^2 and xy orbitals. Their energies are derived from a 2x2 secular determinant with the diagonal and off-diagonal values shown above.

If the species concerned is d^1, or high spin d^4, d^6 or d^9, then the matrix elements in (III.2) and (III.3) provide the energies to fit the spectrum. If the magnitude of ψ is known, then the spectrum can be predicted as a function of the AOM parameters. If it is not known, then the value of ψ itself can be a variable parameter though there may not be enough spectroscopic data to fix it unambiguously.

If the complex concerned is high spin d^2, d^3, d^7 or d^8, then it is necessary to derive the two-electron functions, if the spin-allowed spectra are to be obtained. One might use Table 3.12 but take care to note the conditions for validity noted in the footnotes. The relationships shown between one-electron and two-electron elements in Table 3.12 are only true for the conditions noted. In lower symmetries there may be additional non-zero one-electron elements contributing to the two-electron elements.

If the complex belongs to C_{2v} or higher symmetry, it is not necessary to use Table 3.12, since the solutions for high spin d^2, d^3, d^7 and d^8 have already been presented in the NSH formalism, in Tables 3.9 - 3.11. Thus the spectra may be solved to derive the NSH parameters. Using Table 1.9 (or 1.10) one can then determine the one-electron d orbital energies in the NSH formalism. Since the one-electron energies in the AOM formalism are available (in this case, III.2 and III.3 above), it is possible to map the two together to obtain the AOM parameters.

This is achieved by setting up a series of simultaneous equations. For example the energy difference between xz and x^2-y^2 orbitals in the NSH and AOM formalisms may be equated, to provide one of these equations. Note also that the off-diagonal elements in the two formalisms may be directly equated, but be careful of phase.

If there is no symmetry, then the NSH approach has too many

parameters and cannot be usefully employed. It is then necessary to set up the complete ^{n}F and ^{n}P matrices (for the cases discussed here) within the AOM formalism and diagonalise them.

Reference

AIII.I This model was initially developed by I.M.Walker

Appendix IV

The C^{q}_{k} Matrices for d^{1} Basis

This appendix provides the C^{q}_{k} matrices which are solutions to Eqn.(3.48) in the l = 2 basis. Subsequently, the four-fold DQ and DT matrices for an l = 2 basis are provided, being the solutions to equations such as (3.55) and (3.56). The DS matrix is the same as C^{2}_{0}, while the DU and DV matrices are obtained simply by multiplying the C^{2}_{2} and C^{4}_{2} matrices, respectively, by $(1/\sqrt{2})$.

THE C20 MATRIX FOR D1

	<22	<21	<20	<2-1	<2-2
<22	-.28571				
<21		.14286			
<20			.28571		
<2-1				.14286	
<2-2					-.28571

THE C22 MATRIX FOR D1

	<22	<21	<20	<2-1	<2-2
<22			-.28571		
<21				-.34993	
<20	-.28571				-.28571
<2-1		-.34993			
<2-2			-.28571		

THE C40 MATRIX FOR D1

	<22	<21	<20	<2-1	<2-2
<22	.04762				
<21		-.19048			
<20			.28571		
<2-1				-.19048	
<2-2					.04762

THE C42 MATRIX FOR D1

	<22	<21	<20	<2-1	<2-2
<22			.18443		
<21				-.30117	
<20	.18443				.18443
<2-1		-.30117			
<2-2			.18443		

THE C44 MATRIX FOR D1

	<22	<21	<20	<2-1	<2-2
<22					.39841
<21					
<20					
<2-1					
<2-2	.39841				

THE C43 MATRIX FOR D1

	<22	<21	<20	<2-1	<2-2
<22				-.28172	
<21					.28172
<20					
<2-1	.28172				
<2-2		-.28172			

THE DQ MATRIX FOR D1 ALONG C2(Z)

	<22	<21	<20	<2-1	<2-2
<22	.03637				.18185
<21		-.14548			
<20			.21822		
<2-1				-.14548	
<2-2	.18185				.03637

THE DT MATRIX FOR D1 ALONG C2(Z)

	<22	<21	<20	<2-1	<2-2
<22	.03074				-.21517
<21		-.12295			
<20			.18443		
<2-1				-.12295	
<2-2	-.21517				.03074

Appendix V

Transition Energy Ratios for Cubic Complexes

This appendix provides solutions to the energies of cubic (octahedral, tetrahedral, cubic) complexes of high spin d^2,d^3,d^7 and d^8 transition metal ions. The data are tabulated and readers may extrapolate linearly between the presented data to evaluate their own Dq and B values. Alternatively, large scale graphs may be drawn (e.g. Figs. AI, AII) from these data and the solutions for specific complexes mapped thereon. This latter approach is especially useful as an undergraduate exercise.

There are two tables, AV1 for A_2 ground terms, and AV2 for T_1 ground terms. For example the $[VF_6]^{3-}$ ion shows transitions at 14,800 and 23,000 cm^{-1} attributed to the transitions $^3T_{2g}$---$^3T_{1g}$ (ν_1) and $^3T_{1g}(P)$---$^3T_{1g}$ (ν_3). Thus Table AV2 is necessary. The ratio (ν_3/ν_1) = 1.554. This value corresponds with Dq/B = 2.55 and ν_3/B = 36.53. Thus Dq = 1610 and B = 630 cm^{-1}. For further examples see A.B.P.Lever, J.Chem.Ed., **45** (1968) 711.

Table AV1

TRANSITION ENERGY RATIOS FOR A_2 GROUND TERM IONS

Dq/B	ν_3/B	ν_2/ν_1	ν_3/ν_1	ν_3/ν_2
.05	15.603	1.795	31.205	17.389
.1	16.211	1.789	16.211	9.062
.15	16.825	1.783	11.217	6.291
.2	17.446	1.777	8.723	4.909
.25	18.074	1.771	7.229	4.083
.3	18.708	1.764	6.236	3.535
.35	19.35	1.757	5.529	3.146
.4	20	1.75	5	2.857
.45	20.658	1.743	4.591	2.634
.5	21.325	1.735	4.265	2.458
.55	22	1.727	4	2.316
.6	22.685	1.719	3.781	2.199
.65	23.379	1.711	3.597	2.102
.7	24.083	1.702	3.44	2.021
.75	24.797	1.694	3.306	1.952
.8	25.521	1.685	3.19	1.893
.85	26.255	1.676	3.089	1.843
.9	27	1.667	3	1.8
.95	27.755	1.657	2.922	1.763
1	28.521	1.648	2.852	1.731

Table AV1 cont

Dq/B	ν_3/B	ν_2/ν_1	ν_3/ν_1	ν_3/ν_2
1.05	29.297	1.638	2.79	1.703
1.1	30.083	1.629	2.735	1.679
1.15	30.879	1.619	2.685	1.658
1.2	31.685	1.61	2.64	1.64
1.25	32.5	1.6	2.6	1.625
1.3	33.325	1.59	2.563	1.612
1.35	34.158	1.581	2.53	1.601
1.4	35	1.571	2.5	1.591
1.45	35.85	1.562	2.472	1.583
1.5	36.708	1.553	2.447	1.576
1.55	37.574	1.544	2.424	1.57
1.6	38.446	1.535	2.403	1.566
1.65	39.325	1.526	2.383	1.562
1.7	40.211	1.517	2.365	1.559
1.75	41.103	1.508	2.349	1.557
1.8	42	1.5	2.333	1.556
1.85	42.903	1.492	2.319	1.555
1.9	43.81	1.484	2.306	1.554
1.95	44.723	1.476	2.293	1.554
2	45.639	1.468	2.282	1.554
2.05	46.56	1.46	2.271	1.555
2.1	47.485	1.453	2.261	1.556
2.15	48.414	1.446	2.252	1.557
2.2	49.346	1.439	2.243	1.559
2.25	50.281	1.432	2.235	1.561
2.3	51.22	1.425	2.227	1.563
2.35	52.161	1.419	2.22	1.565
2.4	53.105	1.412	2.213	1.567
2.45	54.051	1.406	2.206	1.569
2.5	55	1.4	2.2	1.571
2.55	55.951	1.394	2.194	1.574
2.6	56.904	1.388	2.189	1.576
2.65	57.86	1.383	2.183	1.579
2.7	58.817	1.377	2.178	1.582
2.75	59.776	1.372	2.174	1.585
2.8	60.736	1.367	2.169	1.587
2.85	61.698	1.361	2.165	1.59
2.9	62.662	1.356	2.161	1.593
2.95	63.627	1.352	2.157	1.596
3	64.593	1.347	2.153	1.599
3.05	65.561	1.342	2.15	1.601
3.1	66.53	1.338	2.146	1.604
3.15	67.5	1.333	2.143	1.607
3.2	68.471	1.329	2.14	1.61
3.25	69.443	1.325	2.137	1.613
3.3	70.416	1.321	2.134	1.616
3.35	71.39	1.317	2.131	1.618
3.4	72.365	1.313	2.128	1.621

Table AV1 cont

Dq/B	ν_3/B	ν_2/ν_1	ν_3/ν_1	ν_3/ν_2
3.45	73.341	1.309	2.126	1.624
3.5	74.318	1.305	2.123	1.627
3.55	75.295	1.302	2.121	1.63
3.6	76.273	1.298	2.119	1.632
3.65	77.252	1.294	2.116	1.635
3.7	78.232	1.291	2.114	1.638
3.75	79.212	1.288	2.112	1.64
3.8	80.192	1.284	2.11	1.643
3.85	81.174	1.281	2.108	1.646
3.9	82.155	1.278	2.107	1.648
3.95	83.138	1.275	2.105	1.651
4	84.121	1.272	2.103	1.653
4.05	85.104	1.269	2.101	1.656
4.1	86.088	1.266	2.1	1.658
4.15	87.072	1.263	2.098	1.661
4.2	88.057	1.261	2.097	1.663
4.25	89.042	1.258	2.095	1.666
4.3	90.028	1.255	2.094	1.668
4.35	91.014	1.253	2.092	1.67
4.4	92	1.25	2.091	1.673
4.45	92.987	1.247	2.09	1.675
4.5	93.974	1.245	2.088	1.677
4.55	94.961	1.243	2.087	1.68
4.6	95.949	1.24	2.086	1.682
4.65	96.937	1.238	2.085	1.684
4.7	97.925	1.236	2.084	1.686
4.75	98.913	1.233	2.082	1.688
4.8	99.902	1.231	2.081	1.69
4.85	100.891	1.229	2.08	1.693
4.9	101.881	1.227	2.079	1.695
4.95	102.87	1.225	2.078	1.697
5	103.86	1.223	2.077	1.699

Table AV2

TRANSITION ENERGY RATIOS FOR T_1 GROUND TERM IONS

Dq/B	ν_3/B	ν_2/ν_1	ν_3/ν_1	ν_3/ν_2
.05	15.305	2.242	38.015	16.957
.1	15.62	2.234	19.279	8.629
.15	15.945	2.227	13.042	5.857
.2	16.279	2.22	9.93	4.473
.25	16.621	2.213	8.067	3.645
.3	16.971	2.207	6.828	3.094
.35	17.328	2.201	5.947	2.702

Table AV2 cont

Dq/B	ν_3/B	ν_2/ν_1	ν_3/ν_1	ν_3/ν_2
.4	17.692	2.195	5.288	2.408
.45	18.062	2.19	4.777	2.181
.5	18.439	2.185	4.37	2
.55	18.822	2.18	4.038	1.852
.6	19.209	2.175	3.763	1.73
.65	19.602	2.171	3.531	1.627
.7	20	2.167	3.333	1.538
.75	20.402	2.163	3.163	1.462
.8	20.809	2.159	3.014	1.396
.85	21.219	2.155	2.883	1.338
.9	21.633	2.151	2.768	1.286
.95	22.051	2.148	2.665	1.241
1	22.472	2.145	2.572	1.199
1.05	22.897	2.142	2.489	1.162
1.1	23.324	2.138	2.414	1.129
1.15	23.754	2.136	2.346	1.098
1.2	24.187	2.133	2.283	1.071
1.25	24.622	2.13	2.226	1.045
1.3	25.06	2.127	2.173	1.022
1.35	25.5	2.125	2.125	1
1.4	25.942	2.123	2.08	.98
1.45	26.387	2.12	2.039	.961
1.5	26.833	2.118	2	.944
1.55	27.281	2.116	1.964	.928
1.6	27.731	2.114	1.93	.913
1.65	28.182	2.112	1.899	.899
1.7	28.636	2.11	1.869	.886
1.75	29.09	2.108	1.842	.874
1.8	29.547	2.106	1.816	.862
1.85	30.004	2.104	1.791	.851
1.9	30.463	2.103	1.768	.841
1.95	30.923	2.101	1.746	.831
2	31.385	2.099	1.725	.822
2.05	31.847	2.098	1.705	.813
2.1	32.311	2.096	1.687	.805
2.15	32.776	2.095	1.669	.797
2.2	33.242	2.093	1.652	.789
2.25	33.708	2.092	1.636	.782
2.3	34.176	2.091	1.621	.775
2.35	34.645	2.089	1.606	.769
2.4	35.114	2.088	1.592	.762
2.45	35.584	2.087	1.579	.756
2.5	36.056	2.086	1.566	.751
2.55	36.527	2.084	1.553	.745
2.6	37	2.083	1.542	.74
2.65	37.473	2.082	1.53	.735
2.7	37.947	2.081	1.519	.73
2.75	38.422	2.08	1.509	.725

Table AV2 cont

Dq/B	ν_3/B	ν_2/ν_1	ν_3/ν_1	ν_3/ν_2
2.8	38.897	2.079	1.499	.721
2.85	39.373	2.078	1.489	.717
2.9	39.85	2.077	1.48	.713
2.95	40.327	2.076	1.471	.709
3	40.804	2.075	1.462	.705
3.05	41.283	2.074	1.454	.701
3.1	41.761	2.073	1.446	.697
3.15	42.24	2.073	1.438	.694
3.2	42.72	2.072	1.431	.691
3.25	43.2	2.071	1.423	.687
3.3	43.681	2.07	1.416	.684
3.35	44.162	2.069	1.41	.681
3.4	44.643	2.068	1.403	.678
3.45	45.125	2.068	1.397	.675
3.5	45.607	2.067	1.39	.673
3.55	46.09	2.066	1.384	.67
3.6	46.573	2.066	1.378	.667
3.65	47.056	2.065	1.373	.665
3.7	47.539	2.064	1.367	.662
3.75	48.023	2.063	1.362	.66
3.8	48.508	2.063	1.357	.658
3.85	48.992	2.062	1.352	.655
3.9	49.477	2.062	1.347	.653
3.95	49.962	2.061	1.342	.651
4	50.448	2.06	1.337	.649
4.05	50.934	2.06	1.333	.647
4.1	51.42	2.059	1.328	.645
4.15	51.906	2.059	1.324	.643
4.2	52.393	2.058	1.32	.641
4.25	52.88	2.057	1.316	.639
4.3	53.367	2.057	1.312	.638
4.35	53.854	2.056	1.308	.636
4.4	54.342	2.056	1.304	.634
4.45	54.829	2.055	1.3	.633
4.5	55.317	2.055	1.297	.631
4.55	55.805	2.054	1.293	.629
4.6	56.294	2.054	1.29	.628
4.65	56.782	2.053	1.286	.626
4.7	57.271	2.053	1.283	.625
4.75	57.76	2.053	1.28	.624
4.8	58.249	2.052	1.277	.622
4.85	58.739	2.052	1.274	.621
4.9	59.228	2.051	1.271	.619
4.95	59.718	2.051	1.268	.618
5	60.208	2.05	1.265	.617

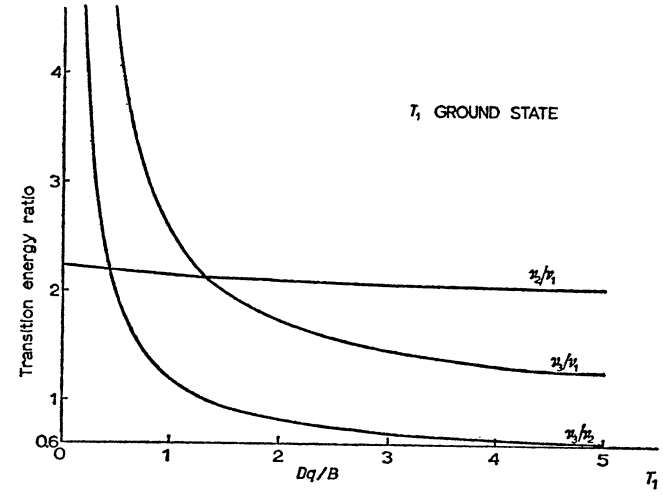


Fig.AV.1 T_1 Ground term, transition energy ratios versus Dq/B.

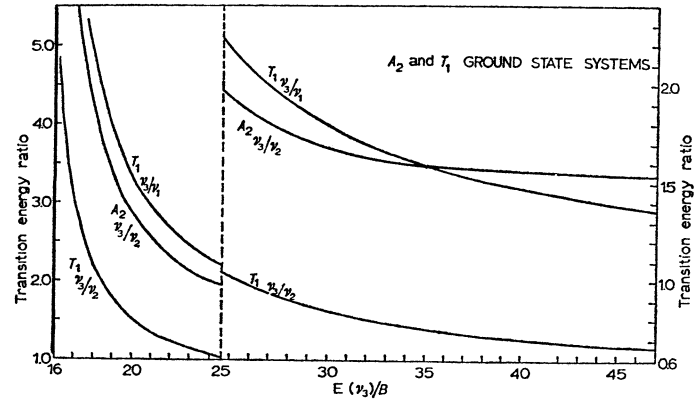


Fig.AV.2 A_2 and T_1 ground terms, transition energy ratios versus $E(\nu_3)/B$. Note that the left-hand ordinate refers to $E(\nu_3)/B$ (16-24.6) and the right-hand ordinate to $E(\nu_3)/B$ (24.6-47).

Appendix VI

Energy Levels in Tetragonal High Spin d^2,d^3,d^7 and d^8 Complexes and Some Thoughts on Fitting Rhombic Spectra using the Trace Method

This appendix provides additional assistance for solving the spectra of tetragonal complexes of high spin d^2,d^3,d^7 and d^8 transition metal ions.

Note that the d^1 matrix elements are given in Eqns.1.29 and 1.53. The matrices for d^2,d^3,d^7 and d^8 in the NSH formalism are in Tables 3.9 and 3.10 for two different bases. For regular tetragonal complexes of the type ML_4Z_2, one would commonly use Table 3.9.

The energies in the classical crystal field model are to be found in Eqns.3.15, 3.17, and 3.18, applicable directly to d^2 and d^7. They may be used for d^3 and d^8 after multiplying Dq, Ds and Dt (but not B) by -1.

The matrices may be programmed on a micro-computer and an iterative program written to compare experimental transition energies against calculated energies. This involves solving a cubic equation obtained by expansion of the 3x3 E secular determinant. For convenience, a procedure for solving a cubic equation is outlined here.

A 3x3 secular determinant may be expanded to yield a general cubic expression:

$$y^3 + py^2 + qy + r = 0$$

which may be reduced to the form:

$$x^3 + ax + b = 0$$

by substituting y by the value (x-(p/3)), and where:

$$a = (1/3)(3q - p^2),\ b = (1/27)(2p^3 - 9pq + 27r)$$

We expect three real roots in the case of the E secular determinant; this will be true if $(b^2/4) + (a^3/27) < 0$.

In this case we can define an angle ϕ as follows:

$$\cos\phi = (-b/2)/[\sqrt{(-a^3/27)}]$$

Then the three roots are, in terms of x:

$$x(1) = 2[\sqrt{(-a/3)}]\cos(\phi/3)$$
$$x(2) = 2[\sqrt{(-a/3)}]\cos(\phi/3 + 120)$$
$$x(3) = 2[\sqrt{(-a/3)}]\cos(\phi/3 + 240)$$

Unfortunately no simple transition energy ratio calculations have yet appeared to be used to solve the tetragonal spectra rapidly. A number of approaches have, however, been presented to alleviate the problem.

The first order weak field tetragonal energies are, relative to their octahedral parents as zero, for d^3 and d^8 (multiply by -1 for d^2 and d^7):

A_{2g}	B_{1g}	7Dt	
T_{2g}	B_{2g}	7Dt	separation (-35/4)Dt
	E_g^a	-(7/4)Dt	
$T_{1g}(F)$	A_{2g}	-4Ds + 2Dt	
			separation 6Ds - (5/4)Dt
	E_g^b	2Ds + (3/4)Dt	
$T_{1g}(P)$	A_{2g}	2Ds - 8Dt	
			separation -3Ds + 5Dt
	E_g^c	-Ds - 3Dt	

(VI.1)

Usually the sign of Dt is known from the known relative fields of the two ligands, thus the relative order of the splitting of T_{2g} can be guessed.

To first order, the energy separation between the E_g^a and B_{2g} components of T_{2g} is simply (35/4)Dt. However the equations in (VI.1) neglect configurational interaction. Allowing for CI, the mixing of this E_g level with the other two E_g levels by configurational interaction, causes the separation of the E_g^a and B_{2g} levels to deviate from this simple expression. If Dt is evaluated with the first order expression, it will not be accurate. Busch and co-workers [AVI.1] report a polynomial expansion which, they claim, yields Dt to within 0.5 , namely:

$$Dt(corr) = 0.989Dt - (3.80 \times 10^{-4})Dt^2 + (3.53 \times 10^{-7})Dt^3 - (1.5 \times 10^{-10})Dt^4 \quad \text{(VI.2)}$$

where Dt is the apparent value from the first order equation, and Dt(corr) is the 'true' value.

Note that the separation between the B_{1g} and B_{2g} states is always exactly 10Dq inclusive of configurational interaction. Thus the energy of the $^nB_{2g}$---$^nB_{1g}$ transition in d^3 and d^8 species provides 10Dq. In d^2 and d^7 species, 10Dq is the difference between the transitions to $^nB_{1g}$ and $^nB_{2g}$ from the ground state.

Bartecki published some expressions for d^3 and d^8 tetragonal complexes [AVI.2], which were later developed for d^2 and d^7 [AVI.3].

They rely on the ability to observe all six possible spin-allowed transitions in the spectrum of a tetragonal species.

a) d^3 and d^8 $A_{2g}(F)$ ground term in O_h and $B_{1g}(F)$ in D_{4h}.

The following transition labels are assumed. Their energy order is irrelevant:

$$\nu_1=E^a_g \leftarrow B_{1g};\quad \nu_2=B_{2g} \leftarrow B_{1g};\quad \nu_3=A^a_{2g} \leftarrow B_{1g};\quad \nu_4=E^b_g \leftarrow B_{1g}$$
$$\nu_5=A^b_{2g} \leftarrow B_{1g};\quad \nu_6=E^c_g \leftarrow B_{1g}$$

$$DS= -7Ds=\tfrac{1}{2}(\nu_2+5(\nu_3+\nu_5) - 4(\nu_1+\nu_4+\nu_6) - 15B) \qquad (VI.3)$$

$$DT= -(7\sqrt{15}Dt)/2=(\sqrt{15}/20)(11\nu_2-(\nu_3+\nu_5)-2(\nu_1+\nu_4+\nu_6) + 45B) \qquad (VI.4)$$

$$DQ= (0.6\sqrt{21})\nu_2+(\sqrt{7}/\sqrt{5})DT \qquad (VI.5)$$

(N.B.; 10Dq = ν_2)

$$B= 1/990[-(36\nu_1-54\nu_3+36\nu_4-54\nu_5+36\nu_6)$$
$$\pm[(36\nu_1-54\nu_3+36\nu_4-54\nu_5+36\nu_6)^2 - 1980(\nu_2^2+3(\nu_3+\nu_5)^2$$
$$+4(-\nu_1\nu_3+\nu_1\nu_4-\nu_1\nu_5+\nu_1\nu_6-\nu_3\nu_4$$
$$-\nu_3\nu_6-\nu_4\nu_5+\nu_4\nu_6-\nu_5\nu_6))]^{\frac{1}{2}}] \qquad (VI.6)$$

b) d^2 and d^7 $T_{1g}(F)$ ground term in O_h with two possible ground terms in D_{4h}.

i) A_2 ground term in D_{4h}. The following transition labels are assumed:

$$\nu_1=E^a_g \leftarrow A^a_{2g};\quad \nu_2=E^b_g \leftarrow A^a_{2g};\quad \nu_3=B_{2g} \leftarrow A^a_{2g};\quad \nu_4=B_{1g} \leftarrow A^a_{2g}$$
$$\nu_5=A^b_{2g} \leftarrow A^a_{2g};\quad \nu_6=E^c_g \leftarrow A^a_{2g}$$

The relevant equations are

$$DS= -7Ds= -\tfrac{1}{2}((\nu_3+\nu_4)+5\nu_5-4(\nu_1+\nu_2+\nu_6)-15B) \qquad (VI.7)$$

$$DT= -7\sqrt{15}\ Dt/2= (\sqrt{15}/20)(11\nu_3-3\nu_4-\nu_5 \qquad (VI.8)$$

$$-2(\nu_1+\nu_2+\nu_6) + 45B) \qquad (VI.9)$$

$$DQ= (0.6\sqrt{21})(\nu_4-\nu_3)+(\sqrt{7}/\sqrt{5})DT \quad (N.B.;\ 10Dq = \nu_4-\nu_3$$

ii) E ground term in D_{4h}. The following transition labels are assumed:

$$\nu_1=A^a_{2g} \leftarrow E^a_g;\quad \nu_2=E^b_g \leftarrow E^a_g;\quad \nu_3=B_{2g} \leftarrow E^a_g;\quad \nu_4=B_{1g} \leftarrow E^a_g$$
$$\nu_5=A^b_{2g} \leftarrow E^a_g;\quad \nu_6=E^c_g \leftarrow E^a_g$$

$$DS = -7Ds = 0.5[5(\nu_1 + \nu_5) - 4(\nu_2 + \nu_6) + (\nu_3 + \nu_4) - 15B] \qquad (VI.10)$$

$$DT = -(7\sqrt{15}Dt)/2 = (\sqrt{15}/20)[11\nu_3 - 3\nu_4 - (\nu_1 + \nu_5) - 2(\nu_2 + \nu_6) + 45B] \qquad (VI.11)$$

$$DQ = (0.6\sqrt{21})(\nu_4 - \nu_3) + \sqrt{(7/5)}DT \quad 10Dq = \nu_4 - \nu_3 \qquad (VI.12)$$

The energy sequence is again irrelevant. Expressions for B may be derived by appropriate substitution and expansion of expression (VI.6) but this is most readily accomplished within the computer program. Thus Eqn.(VI.6) is programmed for d^3 (d^8) and the following substitution made within the program for d^2 (d^7). The superscript N identifies the transition energy in Eqn.(VI.6).

A_2 ground term.

$$\nu_1^N=\nu_1-\nu_4;\quad \nu_2^N=\nu_3-\nu_4;\quad \nu_3^N=-\nu_4;\quad \nu_4^N=\nu_2-\nu_4$$
$$\nu_5^N=\nu_5-\nu_4;\quad \nu_6^N=\nu_6-\nu_4$$

E ground term

$$\nu_1^N=-\nu_4;\quad \nu_2^N=\nu_3-\nu_4;\quad \nu_3^N=\nu_1-\nu_4;\quad \nu_4^N=\nu_2-\nu_4$$
$$\nu_5^N=\nu_5-\nu_4;\quad \nu_6^N=\nu_6-\nu_4$$

Two values of B will arise from solution of (VI.6). Generally the lower in magnitude (often negative) is incorrect.

Indeed these latter substitutions may be made directly into the programmed expressions for the d^8 DQ, DS and DT parameters so that it is not necessary to program both sets [AVI.4].

There are some difficulties with this approach. It is based upon fits to the sums (traces) of the A_2 and E matrices. There are redundancies in that six observables are used to fit three parameters.

If the data set fit the matrices well, it will generate a set of parameters essentially identical to that obtained by a direct iterative fit of the matrices. If only five observables are available, a program can iterate over the sixth. However it is imperative to compare the calculated set against the experimental set, whether all six, or only five, observables are used. If the data set do not fit the matrices too well, one finds a good fit to the sums of the transition energies, but a poor fit to individual transition energies. In such a case, the parameter values may be significantly in error.

It is an attractive approach, but must be used with caution.

Rhombic Spectra and the Trace Method

Finally some comments on the use of matrix traces as a means of fitting spectra.

In low symmetry complexes, there are generally quite a few

parameters. An iterative program may be written to vary these parameters to fit an observed spectrum. The time necessary to obtain a fit can be greatly reduced by inclusion of boundary requirements based on the traces of the secular determinants.

For example, consider a C_{2v}(I) complex of Ni(II). The matrix elements in Table 3.9 are relevant (multiplied by -1). There are six parameters, DQ, DS, DT, DU, DV and B. Suppose all three B_1 transitions can be observed. Their transition energies are given by the difference between the roots of the B_1 secular determinant and the ground state energy. We do not know the individual roots to the B_1 secular determinant but the trace (sum of the diagonal elements) is equal to the sum of the three roots, thus:

Sum of observed B_1 transition energies = Trace of B_1 determinant - 3 x Ground state energy

Using Table 3.9, we can then write:

Sum B_1 transition energies = [0.14548DQ - 0.14286DS + 0.12296DT

- 0.24744DU - 0.21296DV +15B] - 3 x -0.43644DQ

=1.4548DQ - 0.14286DS + 0.12296DT - 0.24744DU - 0.21296DV + 15B

Similarly:

Sum B_2 transition energies = [0.14548DQ - 0.14286DS + 0.12296DT

+ 0.24744DU + 0.21296DV + 15B] - 3 x -0.43644DQ

=1.4548DQ - 0.14286DS + 0.12296DT + 0.24744DU + 0.21296DV + 15B

Sum A_2 transition energies = [0.14548DQ + 0.28571DS - 0.2459DT

+ 15B] - 3 x -0.43644DQ

= 1.4548DQ + 0.28571DS - 0.2459DT + 15B

(VI.13)

These equations provide three boundary conditions which greatly reduce the complexity of the problem.

Note the octahedral sum relationship (DS = DT = DU = DV = 0) that the sum of the three B_1 (or B_2) equals:-

1.4548DQ + 15B = 40Dq + 15B

as expected for a sum to T_2, T_1(F) and T_1(P) in O_h.

The tetragonal sums are (DU = DV = 0):

Sum of E transition energies (B_1,B_2) = 1.4548DQ - 0.14286DS

+ 0.12296DT + 15B

= 40Dq + Ds - 25Dt + 15B

Sum of A_2 transition energies (A_2) = 1.4548DQ + 0.28571DS

- 0.2459DT + 15B

= 40Dq - 2Ds - 20Dt + 15B (This includes transition to B_{2g} in D_{4h})

Recall relationships in Eqns.1.83.

References

AVI.1 L.Y.Martin, C.R.Sperati and D.H.Busch, J.Am.Chem.Soc., **99** (1977) 2968.

AVI.2 A.Bartecki and K.Kurzak, Bull.Acad.Pol.Sci., **XXVI** (1978) 385.

AVI.3 a) A.B.P.Lever, Coord.Chem.Rev., **48** (1982) 63;b) **53** (1984) 296.

AVI.4 Unfortunately there was a consistent sign error in ref.AVI.3. As shown in (VI.8) and (VI.11) above, a negative sign should be removed from Eqn.A6 [AVI.3 a], and from the DT expression in [AVI.3 b].

Appendix VII

Some Additional Wavefunctions

1. NSH Hamiltonian for f Orbitals

The NSH Hamiltonians must be extended to l = 6 for use with an f orbital basis. The operator eigenstates which follow are written as the set up to l = 4, shown as [H(4)], plus the additional components to l = 6.

Quantized along C_4^z.

$|A_{1g}(0)|$ [H(4)] + √(1/8)C^6_0 - √(7/16)[C^6_4 + C^6_{-4}]

$|E_g(0)|$ [H(4)] + √(7/8)C^6_0 + √(1/16)[C^6_4 + C^6_{-4}]

$|E_g(2+)|$ [H(4)] + √(5/32)[C^6_2 + C^6_{-2}] + √(11/32)[C^6_6 + C^6_{-6}]

$|T_{2g}(xy)|$ [H(4)] - √(1/2)[C^6_6 + C^6_{-6}]

See [AVII.1], Table VII for further data.

2. d^3 Wavefunctions

The weak field ^4P and ^4F functions are:

$\|L,M_L,S,M_S\rangle$	Determinantal Wavefunctions
$\|1,1,3/2,3/2\rangle$	√(3/5)($2^+,1^+,-2^+$) - √(2/5)($2^+,0^+,-1^+$)
$\|1,0,3/2,3/2\rangle$	-√(1/5)($2^+,0^+,-2^+$) + √(4/5)($1^+,0^+,-1^+$)
$\|1,-1,3/2,3/2\rangle$	-√(3/5)($2^+,-1^+,-2^+$) + √(2/5)($1^+,0^+,-2^+$)
$\|3,3,3/2,3/2\rangle$	-($2^+,1^+,0^+$)
$\|3,2,3/2,3/2\rangle$	($2^+,1^+,-1^+$)
$\|3,1,3/2,3/2\rangle$	-√(2/5)($2^+,1^+,-2^+$) - √(3/5)($2^+,0^+,-1^+$)
$\|3,0,3/2,3/2\rangle$	√(4/5)($2^+,0^+,-2^+$) + √(1/5)($1^+,0^+,-1^+$)
$\|3,-1,3/2,3/2\rangle$	√(2/5)($2^+,-1^+,-2^+$) + √(3/5)($1^+,0^+,-2^+$)
$\|3,-2,3/2,3/2\rangle$	($1^+,-1^+,-2^+$)
$\|3,-3,3/2,3/2\rangle$	($0^+,-1^+,-2^+$)

Reference

AVII.1 J.C.Donini, B.R.Hollebone and A.B.P.Lever, Prog.Inorg.Chem., **22** (1977) 225. Note that due to printing difficulties with this book, the number of typographical errors is greater than could be wished. There are a few errors in Table VII. They can usually be detected by noting that a) the total (squared) contribution of any $|J,M_J\rangle$ function, to all the representations, must add up to unity, and b) that the squares of the functions in the normalised function subduction coefficients must also add up to unity.

Appendix VIII

1. DETERMINANTAL WAVE FUNCTIONS

Consider a two-electron system in which electron (1) resides in the orbital a_1 and electron (2) resides in orbital a_2. Each orbital a_i defines both space and spin coordinates, and thus from the Pauli Exclusion Principle, can only take one electron. Such orbitals are therefore called one-electron orbitals or spin-orbitals. We may write the overall wave function:

$$\psi = a_1(1)a_2(2) \qquad \text{(VIII.1)}$$

There is no reason however why we should not put electron (1) into a_2 and electron (2) into a_1, to obtain:

$$\psi' = a_1(2)a_2(1) \qquad \text{(VIII.2)}$$

Since both (IV.1) and (IV.2) are as good as each other, we must combine them (form a linear combination), to obtain the complete wave function.

We have two ways of combining them:

$$\psi_{\text{sym}} = \psi + \psi' = a_1(1)a_2(2) + a_1(2)a_2(1) \qquad \text{(VIII.3)}$$

and

$$\psi_{\text{asym}} = \psi - \psi' = a_1(1)a_2(2) - a_1(2)a_2(1) \qquad \text{(VIII.4)}$$

If we now interchange electrons (1) and (2), the symmetrical wave function ψ_{sym} is unaltered. This implies that both electrons are in identical orbitals, and therefore contravenes the Pauli Exclusion Principle. A similar interchange in the asymmetric function ψ_{asym} causes a change in sign of the wave function; therefore ψ_{asym} does not contravene the Pauli principle. In general all wave functions must be antisymmetric with respect to the interchange of two electrons.

The asymmetric wave function may be written in the form of a determinant:

$$\psi_{\text{asym}} = \begin{pmatrix} a_1(1) & a_1(2) \\ a_2(1) & a_2(2) \end{pmatrix} \qquad \text{(VIII.5)}$$

often called 'Slater' determinants. It is the property of a determinant that interchanging a row or column changes its sign. Such a manoeuvre is the equivalent of interchanging electrons, and therefore the determinant has the required antisymmetric nature. Also note that should two electrons have the same set of quantum numbers (be in the same one-electron orbitals) the determinant, and the wave function, would be zero as required. The determinant (VIII.5) may be written $[a_1(1)\, a_2(2)]$ for short, or in general for an n electron system:

$$[a_1(1)\, a_2(2)\, a_3(3) \ldots a_n(n)] = \begin{vmatrix} a_1(1) & a_1(2) & a_1(3) & \ldots & a_1(n) \\ a_2(2) & a_2(2) & a_2(3) & \ldots & a_2(n) \\ . & . & . & \ldots & \\ . & . & . & \ldots & \\ a_n(1) & a_n(2) & a_n(3) & \ldots & a_n(n) \end{vmatrix} \qquad \text{(VIII.6)}$$

where in the short hand form we have written the principal diagonal elements. If each individual a_i is normalised, then the normalisation constant for the determinant is $1/(n!)^{\frac{1}{2}}$.

$[a_1(1)\, a_2(2) \ldots a_n(n)]$ therefore represents an $n \times n$ determinant in which we have *all possible permutations of the n electrons in the n orbitals.* It is called a determinantal wave function. The terms of the expanded determinant may be written in the form:

$$\psi = 1/(n!)^{\frac{1}{2}} \sum_x (-1)^x p^x \, [a_1(1)\, a_2(2) \ldots a_n(n)] \qquad \text{(VIII.7)}$$

We may illustrate this by expanding a 3×3 determinant, *viz.*

$$\begin{vmatrix} a_1(1) & a_1(2) & a_1(3) \\ a_2(1) & a_2(2) & a_2(3) \\ a_3(1) & a_3(2) & a_3(3) \end{vmatrix}$$

$$= a_1(1) \begin{vmatrix} a_2(2) & a_2(3) \\ a_3(2) & a_3(3) \end{vmatrix} - a_1(2) \begin{vmatrix} a_2(1) & a_2(3) \\ a_3(1) & a_3(3) \end{vmatrix}$$

$$+ a_1(3) \begin{vmatrix} a_2(1) & a_2(2) \\ a_3(1) & a_3(2) \end{vmatrix}$$

$$= a_1(1)\, a_2(2)\, a_3(3) - a_1(1)\, a_2(3)\, a_3(2) - a_1(2)\, a_2(1)\, a_3(3)$$

$$+ a_1(2)\, a_2(3)\, a_3(1) + a_1(3)\, a_2(1)\, a_3(2) - a_1(3)\, a_2(2)\, a_3(1).$$

$$= \Sigma (-1)^x p^x \, [a_1(1)\, a_2(2)\, a_3(3)]$$

Each of the factors is obtained by interchanging (permuting) the electron numbers in $a_1(1)\, a_2(2)\, a_3(3)$. The factor is negative if an odd number, x, of permutations, P, are required to obtain it, and even if x is even.

We assume that the orbitals a_i are orthogonal to one another, *i.e.*:

$$\int a_i^* a_j \, d\tau = 0 \text{ if } i \neq j$$
$$= 1 \text{ if } i = j$$
$$= d_{ij}$$

where d_{ij} is the Kronecker Delta which is zero if $i \neq j$ and one if $i = j$.

Thus for example, the microstate $(1^+, 2^+)$ represents the normalised wave function:

$$\tfrac{1}{\sqrt{2}} \, [1^+(1), 2^+(2) - 1^+(2), 2^+(1)]$$

2. ONE-ELECTRON OPERATORS

In the discussion which follows, the function a_i refers to a spin-orbital which will accept only one electron. The operators under discussion are assumed to operate only upon the orbital function of the spin-orbital so that the spin may be integrated straight out. In other words, all the matrix elements will be multiplied by $\int s_{a_i}^* \, s_{a_j} \, ds$ which, because of orthogonality will be unity if the spins of a_i and a_j are the same but zero otherwise.

We are interested in the matrix elements of $\int \psi_{\mathrm{I}}^* \hat{F} \psi_{\mathrm{II}} \, d\tau$ where ψ_{I} and ψ_{II} are antisymmetric determinantal wave functions and F is a one-electron operator, being a sum of one-electron operators $F(1)$, $F(2)$, etc., *i.e.* it operates on one electron at a time. Consider the determination of the matrix elements of $\int (1^+, 2^+)^* \hat{F} (1^+, 2^+) \, d\tau$ (*i.e.* in this case $\psi_{\mathrm{I}} = \psi_{\mathrm{II}}$).

Expanding the determinants we obtain

$$\int (1^+, 2^+)^* \, \hat{F} (1^+, 2^+) \, d\tau = \int \tfrac{1}{\sqrt{2}} \, [1^+(1)\, 2^+(2) - 1^+(2)\, 2^+(1)]^* \hat{F}$$
$$\tfrac{1}{\sqrt{2}} \, [1^+(1)\, 2^+(2) - 1^+(2)\, 2^+(1)] \, d\tau$$

$$= \tfrac{1}{2} \int 1^+(1)^*\, 2^+(2)^* \, \hat{F} \, 1^+(1)\, 2^+(2) \, d\tau$$
$$\text{A}$$
$$+ \tfrac{1}{2} \int 1^+(2)^*\, 2^+(1)^* \, \hat{F} \, 1^+(2)\, 2^+(1) \, d\tau$$
$$\text{B}$$
$$- \tfrac{1}{2} \int 1^+(1)^*\, 2^+(2)^* \, \hat{F} \, 1^+(2)\, 2^+(1) \, d\tau$$
$$\text{C}$$
$$- \tfrac{1}{2} \int 1^+(2)^*\, 2^+(1)^* \, \hat{F} \, 1^+(1)\, 2^+(2) \, d\tau \qquad \text{(VIII.8)}$$
$$\text{D}$$

Consider integral A, this may be further expanded:

$$\text{A} = \tfrac{1}{2} \int 1^+(1)^* \, \hat{F} \, (1) \, 1^+(1) \, d\tau \int 2^+(2)^* \, 2^+(2) \, d\tau$$
$$+ \tfrac{1}{2} \int 2^+(2)^* \, \hat{F} \, (2) \, 2^+(2) \, d\tau \int 1^+(1)^* \, 1^+(1) \, d\tau$$
$$= \tfrac{1}{2} \langle 1^+ | \hat{F} | \, 1^+ \rangle + \tfrac{1}{2} \langle 2^+ | \hat{F} | \, 2^+ \rangle$$

since for orthogonal d orbitals the second integral in each product equals unity. Similarly

$$\text{B} = \tfrac{1}{2} \langle 1^+ | \hat{F} | \, 1^+ \rangle + \tfrac{1}{2} \langle 2^+ | \hat{F} | \, 2^+ \rangle$$

Now consider C,

$$-\text{C} = \tfrac{1}{2} \int 1^+(1)^* \, \hat{F}(1) \, 2^+(1) \, d\tau \int 2^+(2) \, 1^+(2) \, d\tau$$
$$+ \tfrac{1}{2} \int 2^+(2)^* \, \hat{F}(2) \, 1^+(2) \, d\tau \int 1^+(1)^* \, 2^+(1) \, d\tau$$

For reasons of orthogonality the second integral in each product

is zero; therefore C and D are zero.

$$\therefore \int (1^+, 2^+)^* \hat{F}(1^+, 2^+)\, d\tau = A + B + C + D$$
$$= \langle 1^+ |\hat{F}| 1^+\rangle + \langle 2^+ |\hat{F}| 2^+\rangle$$

It is clear from this result that the matrix elements of an integral involving a one-electron operator may be derived quite readily without necessarily expanding the determinantal wave functions.

Consider now the situation when ψ_I differs from ψ_{II};

$$\text{let } \psi_I = (a_1, a_2)$$
$$\psi_{II} = (a_1, b_2)$$

then $\int (a_1, a_2)^* \hat{F}(a_1, b_2)\, d\tau$

$$= \tfrac{1}{2} \int a_1(1)^* \hat{F}(1)\, a_1(1)\, d\tau \int a_2(2)^*\, b_2(2)\, d\tau$$
$$+ \tfrac{1}{2} \int a_2(2)^* \hat{F}(2)\, b_2(2)\, d\tau \int a_1(1)^*\, a_1(1)\, d\tau$$
$$+ \tfrac{1}{2} \int a_2(1)^* \hat{F}(1)\, b_2(1)\, d\tau \int a_1(2)^*\, a_1(2)\, d\tau$$
$$+ \tfrac{1}{2} \int a_1(2)^* \hat{F}(2)\, a_1(2)\, d\tau \int a_2(1)\, b_2(1)\, d\tau$$
$$+ C + D$$

Products C and D, as before will be zero but so will the first and fourth of the above products since the second integral in each will be zero (non-orthogonality).

Hence $\int (a_1, a_2)^* \hat{F}(a_1, b_2)\, d\tau = \langle a_2 |\hat{F}| b_2\rangle$

It follows logically that if ψ_I and ψ_{II} differ in more than one constituent spin-orbital, all the matrix elements will be zero.

3. TWO-ELECTRON OPERATORS

Two-electron operators operate on two electrons simultaneously; it follows from the preceding discussion that for at least one of the matrix elements to be non-zero, ψ_I and ψ_{II} should not differ in more than two constituent spin-orbitals.

Case I. $\psi_I = \psi_{II} = (a_1, a_2)$

Then $\iint \psi_I^* \hat{H}_{12}\, \psi_{II}\, d\tau_1\, d\tau_2 = \iint \frac{1}{\sqrt{2}} [a_1(1)^*\, a_2(2)^* - a_1(2)\, a_2(1)]$

$$\hat{H}_{12} \tfrac{1}{\sqrt{2}} [a_1(1)\, a_2(2) - a_1(2)\, a_2(1)]$$
$$= \tfrac{1}{2} \iint a_1(1)^*\, a_2(2)^*\, \hat{H}_{12}\, a_1(1)\, a_2(2)\, d\tau_1\, d\tau_2$$
$$\text{A}$$
$$+ \tfrac{1}{2} \iint a_1(2)^*\, a_2(1)^*\, \hat{H}_{12}\, a_1(2)\, a_2(1)\, d\tau_1\, d\tau_2$$
$$\text{B}$$
$$- \tfrac{1}{2} \iint a_1(1)^*\, a_2(2)^*\, \hat{H}_{12}\, a_1(2)\, a_2(1)\, d\tau_1\, d\tau_2$$
$$\text{C}$$
$$- \tfrac{1}{2} \iint a_1(2)^*\, a_2(1)^*\, \hat{H}_{12}\, a_1(1)\, a_2(2)\, d\tau_1\, d\tau_2$$
$$\text{D}$$

Since A = B and C = D, we find

$$\iint (a_1, a_2)^*\, \hat{H}_{12}\, (a_1, a_2)\, d\tau_1\, d\tau_2$$
$$= \pm [\iint a_1(1)^*\, a_2(2)^*\, \hat{H}_{12}\, a_1(1)\, a_2(2)\, d\tau_1\, d\tau_2$$
$$- \iint a_1(1)^*\, a_2(2)^*\, \hat{H}_{12}\, a_1(2)\, a_2(1)\, d\tau_1\, d\tau_2$$
$$= [\langle a_1\, a_2 |\hat{H}_{12}| a_1\, a_2\rangle - \langle a_1\, a_2 |\hat{H}_{12}| a_2\, a_1\rangle] \qquad \text{(VIII.9)}$$

If ψ_I and ψ_{II} contain more than two spin-orbitals, then the expression (VIII.9) is summed over all pairs of electrons.

That is, if $\psi_I = \psi_{II} = (a_1, a_2, \ldots a_n)$

$$\langle \psi_I |\hat{H}_{12}| \psi_{II}\rangle = \sum_{i>j=1}^{n} [\iint a_i(1)^*\, a_j(2)^*\, \hat{H}_{12}\, a_i(1)\, a_j(2)\, d\tau_1\, d\tau_2 -$$
$$- \iint a_i(1)^*\, a_j(2)^*\, \hat{H}_{12}\, a_i(2)\, a_j(1)\, d\tau_1\, d\tau_2]$$

Case II. ψ_I and ψ_{II} differ in one constituent spin-orbital—assume

$$\psi_I = (a_1, a_2, \ldots a_j \ldots a_n)$$
$$\psi_{II} = (a_1, a_2, \ldots b_j \ldots a_n) \qquad \text{(VIII.10)}$$

In deriving the matrix elements of $\langle \psi_I |\hat{H}_{12}| \psi_{II}\rangle$, it is first necessary to order ψ_I and ψ_{II} so that similar spin-orbitals lie in the same sequence. Thus the columns of ψ_I, for example, are

permuted, so that they correspond exactly with those of ψ_{II}. If an odd number of permutations are required, the negative sign of expression (IV.11) below is required. Then

$$\langle \psi_I | \hat{H}_{12} | \psi_{II} \rangle$$
$$= \pm \sum [\int\int a_i(1)^* a_j(2)^* \hat{H}_{12} a_i(1) b_j(2) \, d\tau_1 \, d\tau_2$$
$$- \int\int a_i(1)^* a_j(2)^* \hat{H}_{12} a_i(2) b_j(1) \, d\tau_1 \, d\tau_2 \quad \text{(VIII.11)}$$

Where l runs over the $n-1$ orbitals common to ψ_I and ψ_{II}.

Recall that each of the products in expression (VIII.11)' is also multiplied by $\int a_1^* a_1 \, d\tau \int a_2^* a_2 \, d\tau \ldots$ and also by the spin integrals, all of which are unity. Since the integral $\int a_j^* b_j \, d\tau$ is zero, it follows that any element involving terms such as

$$\int\int a_m(1)^* a_n(2)^* \hat{H}_{12} a_m(1 \text{ or } 2) a_n(2 \text{ or } 1) \, d\tau_1 \, d\tau_2$$

will be zero since they are multiplied by $\int a_j^* b_j \, d\tau$; thus as indicated in expression (IV.11), all non-zero matrix elements of $\langle \psi_I | \hat{H}_{12} | \psi_{II} \rangle$ must involve products containing a_j and b_j.

Case III. ψ_I and ψ_{II} differ in two constituent spin-orbitals—assume

$$\psi_I = (a_1, a_2 \ldots a_j, a_k \ldots a_n)$$

and

$$\psi_{II} = (a_1, a_2 \ldots b_j, b_k \ldots a_n)$$

Following from the previous argument all the matrix elements of $\langle \psi_I | \hat{H}_{12} | \psi_{II} \rangle$ will now be zero unless the jth and kth orbitals are being operated on by H_{12}.

Thus
$$\langle \psi_I | \hat{H}_{12} | \psi_{II} \rangle$$
$$= \pm [\int\int a_j(1)^* a_k(2)^* \hat{H}_{12} b_j(1) b_k(2) \, d\tau_1 \, d\tau_2$$
$$- \int\int a_j(1)^* a_k(2)^* \hat{H}_{12} b_j(2) b_k(1) \, d\tau_1 \, d\tau_2]$$

The negative sign refers to the situation where an odd number of permutations is required to order ψ_I as ψ_{II}.

As an example we derive the interelectronic repulsion energy of the wave function

$$\tfrac{1}{\sqrt{2}} [\overset{+}{(xz}, \overset{-}{yz}) - \overset{-}{(xz}, \overset{+}{yz})] \quad \text{(VIII.12)}$$

It is not necessary to expand the individual determinantal wave functions since we would simply double the number of matrix elements and multiply the result by $\frac{1}{2}$.

The interelectronic repulsion operator, e^2/r_{12}, is a two-electron operator. The interelectronic repulsion energy may be determined by evaluating

$$\int \tfrac{1}{\sqrt{2}} [\overset{+}{(xz}, \overset{-}{yz}) - \overset{-}{(xz}, \overset{+}{yz})] \frac{e^2}{r_{12}} \tfrac{1}{\sqrt{2}} [\overset{+}{(xz}, \overset{-}{yz}) - \overset{-}{(xz}, \overset{+}{yz})] \, d\tau_1 \, d\tau_2$$

$$= \tfrac{1}{2} \int\int \overset{+}{(xz}, \overset{-}{yz}) \frac{e^2}{r_{12}} \overset{+}{(xz}, \overset{-}{yz}) \, d\tau_1 \, d\tau_2 \qquad \text{A}$$

$$+ \tfrac{1}{2} \int\int \overset{-}{(xz}, \overset{+}{yz}) \frac{e^2}{r_{12}} \overset{-}{(xz}, \overset{+}{yz}) \, d\tau_1 \, d\tau_2 \qquad \text{B}$$

$$- \tfrac{1}{2} \int\int \overset{-}{(xz}, \overset{+}{yz}) \frac{e^2}{r_{12}} \overset{+}{(xz}, \overset{-}{yz}) \, d\tau_1 \, d\tau_2 \qquad \text{C}$$

$$- \tfrac{1}{2} \int\int \overset{+}{(xz}, \overset{-}{yz}) \frac{e^2}{r_{12}} \overset{-}{(xz}, \overset{+}{yz}) \, d\tau_1 \, d\tau_2 \qquad \text{D}$$

A and B equal $J(xz, yz)$. The spins of the spin-orbitals in C and D are not ordered correctly to be integrated out—thus it is necessary to interchange the columns of one of the determinants in each integral. One interchange (permutation) requires that the resulting integral be multiplied by -1. The resulting integral is

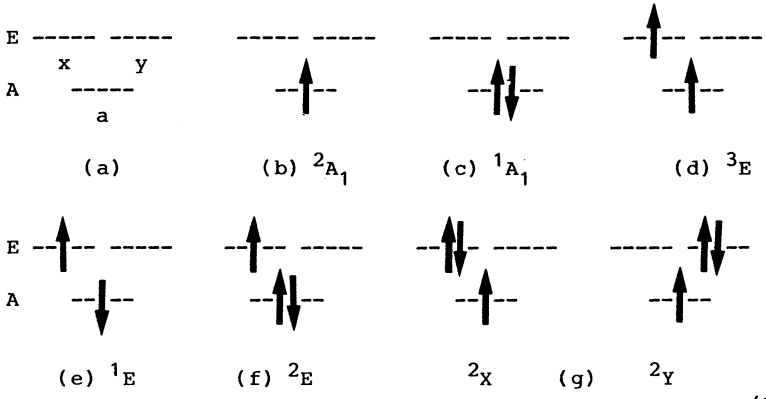
$$\int\int \overset{-}{(xz}, \overset{+}{yz}) \frac{e^2}{r_{12}} \overset{-}{(yz}, \overset{+}{xz}) \, d\tau_1 \, d\tau_2$$

which is equal to $K(xz, yz)$. Therefore, the interelectronic repulsion energy of the wave function (VIII.12) is

$$\tfrac{1}{2} J(xz, yz) + \tfrac{1}{2} J(xz, yz) - (-\tfrac{1}{2} K(xz, yz)) - (-\tfrac{1}{2} K(xz, yz)) =$$
$$= J(xz, yz) + K(xz, yz).$$

Let us continue with a rather more difficult example which raises some additional facets.

Consider a set of levels as shown in (a) with examples of various configurations, and where the components of the E level (in say C_{4v}) are labelled x,y and the unidimensional A state is labelled a.



(VIII.13)

The energies of configurations (b)-(e) follow readily from the discussion in 3.2, and using W(A) and W(E) as the one-electron energies, namely:

(b) W(A); (c) 2W(A) + J(a,a); (d) W(A) + W(E) + J(a,x) - K(a,x)
(e) W(A) + W(E) + J(a,x) + K(a,x)

Configuration (f) may be written as the determinantal function $(a^+a^-x^+)$ and its energy may be written $\langle(a^+a^-x^+)|H|(a^+a^-x^+)\rangle$. The coulombic and exchange terms are obtained by expanding the determinant as follows:

$$\langle(a^+a^-x^+)|e^2/r_{12}|(a^+a^-x^+)\rangle = \langle a^+(1)1/\sqrt{2}[a^-(2)x^+(3) - x^+(2)a^-(3)]$$

$$- a^-(1)1/\sqrt{2}[a^+(2)x^+(3) - x^+(2)a^+(3)] + x^+(1)1/\sqrt{2}[a^+(2)a^-(3) - a^-(2)a^+(3)]$$

$$|e^2/r_{12}|a^+(1)1/\sqrt{2}[a^-(2)x^+(3) - x^+(2)a^-(3)] - a^-(1)1/\sqrt{2}[a^+(2)x^+(3)$$

$$- x^+(2)a^+(3)] + x^+(1)1/\sqrt{2}[a^+(2)a^-(3) - a^-(2)a^+(3)]\rangle$$

(VIII.14)

Labelling left-hand first, second and third square bracketed functions as (A), (B) and (C), and the right-hand analogs as (A'), (B') and (C'), this integral may be expanded further.

Thus the first cross product (AA') is:

$$(AA') = \langle a^+(1)a^+(1)\rangle.1/2.[\langle a^-(2)x^+(3)|e^2/r_{12}|a^-(2)x^+(3)\rangle$$

$$+ \langle x^+(2)a^-(3)|e^2/r_{12}|x^+(2)a^-(3)\rangle$$

$$- 2\langle a^-(2)x^+(3)|e^2/r_{12}|x^+(2)a^-(3)\rangle]$$

$$= 1.1/2.[J(a,x) + J(a,x) - 0\ ]$$
$$= J(a,x)$$

(VIII.15)

The third term is zero because of spin orthogonalisation, i.e. note that integration out of the spin yields the spin integrals $\langle -(2)+(2)\rangle\langle +(3)-(3)\rangle$ which are zero. The product (BB') expands as:-

$$(BB') = \langle a^-(1)a^-(1)\rangle.1/2.[\langle a^+(2)x^+(3)|e^2/r_{12}|a^+(2)x^+(3)\rangle$$

$$+ \langle x^+(2)a^+(3)|e^2/r_{12}|x^+(2)a^+(3)\rangle$$

$$- 2\langle a^+(2)x^+(3)|e^2/r_{12}|x^+(2)a^+(3)\rangle]$$

$$= 1.1/2.[J(a,x) + J(a,x) - 2K(a,x)]$$
$$= J(a,x) - K(a,x)$$

(VIII.16)

and similar to (AA'), (CC') yields J(a,a).

The cross terms (AB'), (BC') and (AC') are all zero because they are multiplied by the integrals $\langle a^+(1)a^-(1)\rangle$, $\langle a^-(1)x^+(1)\rangle$ and $\langle a^+(1)x^+(1)\rangle$ respectively, all of which are zero. The total energy of (f), ^2E is:

$$(f)\ ^2E = 2W(A) + W(E) + 2J(a,x) + J(a,a) - K(a,x)$$

(VIII.17)

By inspection of (f), this result is consistent with the two pairs of (a,x) interactions (2J(a,x)), one (a,a) pair (J(a,a)) and a parallel pair of (a,x) (-K(a,x)). There is also an anti-parallel (a,x) pair, but this is seen not to contribute +K(a,x) to the energy when there is simultaneously a parallel pair contributing -K(a,x). ^2E comprises the denegerate pair $(a^+a^-x^+)$ and $(a^+a^-y^+)$.

Excitation of an A_1 electron to E(x) or E(y) provides (g), an excited version of (f). These states, ^2X and ^2Y are unidimensional, and are not components of an E term.

This fact can be ascertained via the direct products. Hence, for (d), the direct product $a_1 \times e = e$ - therefore the state is ^3E. For (g), the direct product is $a_1 \times e \times e = a_1 + a_2 + b_1 + b_2$, where, from earlier considerations, the quartet state is $^4A_2$ and the remaining states are all spin singlets.

Thus ^2X and ^2Y must be mixed in and out-of-phase to generate two unidimensional representations:

$$^2A_1 = 1/\sqrt{2}\ [^2X + {}^2Y] \qquad ^2B_1 = 1/\sqrt{2}\ [^2X - {}^2Y]$$

whose energies are given by:

$$^2A_1 = 1/2[\langle ^2X|H|^2X\rangle + \langle ^2Y|H|^2Y\rangle + 2\langle ^2X|H|^2Y\rangle]$$

$$^2B_1 = 1/2[\langle ^2X|H|^2X\rangle + \langle ^2Y|H|^2Y\rangle - 2\langle ^2X|H|^2Y\rangle]$$

(VIII.18)

where

$$\langle ^2X|H|^2X\rangle = \langle (a^+x^+x^-)|H|(a^+x^+x^-)\rangle$$
$$= W(A) + 2W(E) + 2J(a,x) + J(x,x) - K(a,x)$$

$$\langle ^2Y|H|^2Y\rangle = \langle (a^+y^+y^-)|H|(a^+y^+y^-)\rangle$$
$$= W(A) + 2W(E) + 2J(a,y) + J(y,y) - K(a,y)$$

(VIII.19)

and J(a,x)=J(a,y), J(x,x)=J(y,y), K(a,x)=K(a,y), and these results are obtained by inspection of (g) using the arguments previously learned, or by full expansion as described above.

The off-diagonal element is expanded as follows:

$$\langle ^2X|H|^2Y\rangle = \langle (a^+x^+x^-)|e^2/r_{12}|(a^+y^+y^-)\rangle \text{ (W terms are zero)}$$

$$= \langle a^+(1)1/\sqrt{2}[x^+(2)x^-(3) - x^-(2)x^+(3)] - x^+(1)1/\sqrt{2}[a^+(2)x^-(3)$$
$$- x^-(2)a^+(3)] + x^-(1)1/\sqrt{2}[a^+(2)x^+(3) - x^+(2)a^+(3)]$$
$$|e^2/r_{12}|a^+(1)1/\sqrt{2}[y^+(2)y^-(3) - y^-(2)y^+(3)] - y^+(1)1/\sqrt{2}[a^+(2)y^-(3)$$
$$- y^-(2)a^+(3)] + y^-(1)1/\sqrt{2}[a^+(2)y^+(3) - y^+(2)a^+(3)]\rangle$$

(VIII.20)

The first term product (AA') is:

$$(AA') = 1/2\ \langle a^+(1)a^+(1)\rangle[\langle x^+(2)x^-(3)|e^2/r_{12}|y^+(2)y^-(3)\rangle$$
$$+ \langle x^-(2)x^+(3)|e^2/r_{12}|y^-(2)y^+(3)\rangle$$
$$- \langle x^+(2)x^-(3)|e^2/r_{12}|y^-(2)y^+(3)\rangle$$
$$- \langle x^-(2)x^+(3)|e^2/r_{12}|y^+(2)y^-(3)\rangle]$$

(VIII.21)

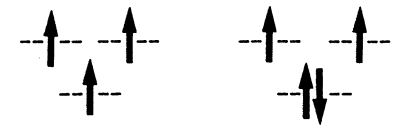
The last two integrals are zero because of spin orthogonality. The first integral is re-arranged to $\langle x^+(2)y^-(3)|e^2/r_{12}|y^+(2)x^-(3)\rangle$ = K(x,y) (see Footnote to Table 3.2); similarly the second integral is equal to K(x,y), hence (AA') product is K(x,y). All other terms in the expansion of the off-diagonal element (VIII.20) are zero due to orthogonality in the product of their labelled (1) electron, e.g. (BB') is multiplied by $\langle x^+(1)y^+(1)\rangle$ which is zero, etc. Thus $\langle ^2X|H|^2Y\rangle$ = K(x,y) and the total energies of the $^2A_1$ and $^2B_1$ states are:

$$^2A_1 = W(A) + 2W(E) + 2J(a,x) + J(x,x,) - K(a,x) + K(x,y)$$

$$^2B_1 = W(A) + 2W(E) + 2J(a,x) + J(x,x,) - K(a,x) - K(x,y)$$

(VIII.22)

As a final two examples note:



(h) $^4A_2$ (i) $^3A_2$

whose energies by inspection are:-

(h) $^4A_2 = W(A) + 2W(E) + 2J(a,x) + J(x,y) - 2K(a,x) - K(x,y)$

(i) $^3A_2 = 2W(A) + 2W(E) + J(a,a) + 4J(a,x) + J(x,y) - 2K(a,x)$
$- K(x,y)$

(VIII.23)

4 Ladder Operators

Ladder, or step-up and step-down operators were met in section 3.7. They are a most useful way to relate wavefunctions and we provide a further example here.

The step-up operator was defined in Eqn.(3.79) and can be used for either orbit or spin. The corresponding step-down operator is defined:

$$L_-|L,M_L\rangle = h[(L - M_L + 1)(L + M_L)]^{1/2}|L,M_L - 1\rangle$$

(VIII.24)

where for the spin analog replace L by S, and M_L by M_S. Suppose we wish to know what 2-electron kets ($|m_1,m_1'|$) contribute to the ^1G function $|L,M_L,S,M_S\rangle = |4,3,0,0\rangle$. We begin with a relationship we know and use the ladder operators to reach the desired information, by a series of equalities. Thus we know by inspection that:

$$|4,4,0,0\rangle = |2^+,2^-|$$

We can write the equation:

$$L_-|4,4,0,0\rangle = L_-|2^+,2^-|$$

from which:

$$\sqrt{8}|4,3,0,0\rangle = 2|1^+,2^-| + 2\ |2^+,1^-|$$

(VIII.25)

where on the left-hand side L = 4, M_L = 4, and, on the right, l = 2 and m_1 = 2, and where each d electron is acted upon independently. Therefore:-

$$|4,3,0,0\rangle = (1/\sqrt{2})[|2^+,1^-| - |2^-,1^+|]$$

Similarly we might seek the ^3f function $|3,2,1,0\rangle$. By inspection, we can write:

$$|3,2,1,1\rangle = |2^+,0^+|$$

and then equate:

$$S_-|3,2,1,1\rangle = S_-|2^+,0^+|$$

$$\sqrt{2}|3,2,1,0\rangle = 1|2^-,0^+| + 1|2^+,0^-|$$

hence:

$$|3,2,1,0\rangle = (1/\sqrt{2})[|2^-0^+| + |2^+,0^-|]$$

where on the left $S = M_S = 1$, and on the right $s = m_s = 1/2$.

Appendix IX

A Selection of 3j Symbols

The 3j symbols were introduced in section 3.4. Explicit formulae for many 3-j symbols are presented in [AIX.1,AIX.2] and an extensive listing in [AIX.2].

Here the symbols are listed as $(j_1j_2j_3/m_1m_2m_3)$. If a specific symbol within this set cannot be found, try permutation using the phase rules provided in section 3.4.

$j_1j_2j_3/m_1m_2m_3$	$j_1j_2j_3/m_1m_2m_3$
110/000 = $-\sqrt{(1/3)}$	111/-101 = $\sqrt{(1/6)}$
110/1-10 = $\sqrt{(1/3)}$	211/-101 = $-\sqrt{(1/10)}$
111/000 = 0	211/0-11 = $\sqrt{(1/30)}$
211/000 = $\sqrt{(2/15)}$	211/1-10 = $-\sqrt{(1/10)}$
211/2-1-1 = $\sqrt{(1/5)}$	123/10-1 = $\sqrt{(2/35)}$
123/1-10 = $-\sqrt{(1/35)}$	123/000 = $-\sqrt{(3/35)}$
222/20-2 = $\sqrt{(2/35)}$	222/10-1 = $\sqrt{(1/70)}$
222/000 = $-\sqrt{(2/35)}$	242/-13-2 = $-\sqrt{(1/18)}$
222/-121 = $-\sqrt{(3/35)}$	242/20-2 = $\sqrt{(1/630)}$
242/10-1 = $\sqrt{(8/315)}$	242/000 = $\sqrt{(2/35)}$
242/-220 = $\sqrt{(1/42)}$	242/-121 = $\sqrt{(4/63)}$
242/24-2 = 1/3	
132/12-3 = $\sqrt{(1/7)}$	132/02-2 = $-\sqrt{(1/21)}$
132/-12-1 = $\sqrt{(1/105)}$	323/-12-1 = $\sqrt{(2/35)}$
323/-321 = $\sqrt{(1/42)}$	323/-220 = $-\sqrt{(1/21)}$
330/000 = $-\sqrt{(1/7)}$	332/3-30 = $\sqrt{(5/84)}$
332/2-20 = 0	323/10-1 = $-\sqrt{(3/140)}$
323/02-2 = $-\sqrt{(1/21)}$	323/000 = $\sqrt{(4/105)}$
321/000 = $-\sqrt{(3/35)}$	321/10-1 = $\sqrt{(2/35)}$
330/1-10 = $\sqrt{(1/7)}$	
330/2-20 = $-\sqrt{(1/7)}$	331/-101 = $\sqrt{(1/14)}$
330/3-30 = $\sqrt{(1/7)}$	331/000 = 0
331/1-21 = $\sqrt{(5/84)}$	331/1-10 = $\sqrt{(1/84)}$
331/2-31 = $-\sqrt{(1/28)}$	331/2-20 = $-\sqrt{(1/21)}$
331/2-1-1 = $\sqrt{(1/84)}$	331/3-30 = $\sqrt{(3/28)}$
332/-202 = $-\sqrt{(1/21)}$	332/-1-12 = $\sqrt{(2/35)}$
332/-101 = $-\sqrt{(1/210)}$	332/000 = $\sqrt{(4/105)}$
332/1-32 = $\sqrt{(1/42)}$	332/1-21 = $\sqrt{(1/28)}$
332/1-10 = $-\sqrt{(3/140)}$	332/2-31 = $-\sqrt{(5/84)}$
332/2-20 = 0	332/3-30 = $\sqrt{(5/84)}$
333/-303 = $\sqrt{(1/42)}$	333/-2-13 = $-\sqrt{(1/21)}$
333/-202 = $\sqrt{(1/42)}$	333/-1-12 = 0
333/-101 = $-\sqrt{(1/42)}$	333/000 = 0

$j_1 j_2 j_3 / m_1 m_2 m_3$	$j_1 j_2 j_3 / m_1 m_2 m_3$
341/-34-1 = 1/3	343/-34-1 = √(1/33)
343/-24-2 = - √(5/99)	341/-321 = √(1/252)
341/-101 = √(1/42)	341/-121 = √(5/84)
341/000 = √(4/63)	341/-220 = √(1/21)
343/-303 = √(1/154)	343/2-20 = √(1/462)
343/-202 = √(7/198)	343/-101 = √(1/1386)
343/1-21 = -√(20/693)	343/000 = - √(2/77)
343/12-3 = √(3/77)	432/000 = 0
432/-202 = √(1/21)	432/-1-12 = -√(2/63)
432/-101 = -√(1/42)	432/0-22 = √(1/63)
432/0-11 = √(5/126)	432/1-32 = -√(1/210)
432/1-21 = -√(7/180)	432/1-10 = -√(1/84)
432/2-31 = √(3/140)	432/2-20 = √(4/105)
432/2-1-1 = -√(1/252)	432/3-30 = -√(1/20)
432/3-2-1 = -√(1/140)	432/3-1-2 = √(1/18)
432/4-31 = √(1/15)	432/4-2-2 = -√(2/45)

References

AIX.1 D.M.Brink and G.R.Satchler, "Angular Momentum", OUP, Oxford, 1968, Table 3.

AIX.2 M.Rotenberg, R.Bivins, N.Metropolis and J.K.Wooten, Jr., "The 3-j and 6-j Symbols", Technology Press, MIT, Cambridge, Mass. 1959.

Appendix X

Correlation Tables

Some point group correlation information is contained in (1.37), (1.44) and Tables 1.5 - 1.7. Additional information, selected from the detailed presentation in [AX.1] is shown here.

To carry out a correlation from a high group to a low group which is not directly indicated, carry out the correlation in steps, e.g. O_h---D_{3d}---C_{3v}.

O_h	T_d	D_{4h}	D_{3d}
A_{1g}	A_1	A_{1g}	A_{1g}
A_{2g}	A_2	B_{1g}	A_{2g}
E_g	E	$A_{1g} + B_{1g}$	E_g
T_{1g}	T_1	$A_{2g} + E_g$	$A_{2g} + E_g$
T_{2g}	T_2	$B_{2g} + E_g$	$A_{1g} + E_g$
A_{1u}	A_2	A_{1u}	A_{1u}
A_{2u}	A_1	B_{1u}	A_{2u}
E_u	E	$A_{1u} + B_{1u}$	E_u
T_{1u}	T_2	$A_{2u} + E_u$	$A_{2u} + E_u$
T_{2u}	T_1	$B_{2u} + E_u$	$A_{1u} + E_u$

T_d	D_{2d}	C_{3v}	C_{2v}
A_1	A_1	A_1	A_1
A_2	B_1	A_2	A_2
E	$A_1 + B_1$	E	$A_1 + A_2$
T_1	$A_2 + E$	$A_2 + E$	$A_2 + B_1 + B_2$
T_2	$B_2 + E$	$A_1 + E$	$B_1 + B_2 + A_1$

D_{3d}	C_{3v}	C_{2h}	D_3
A_{1g}	A_1	A_g	A_1
A_{2g}	A_2	B_g	A_2
E_g	E	$A_g + B_g$	E
A_{1u}	A_2	A_u	A_1
A_{2u}	A_1	B_u	A_2
E_u	E	$A_u + B_u$	E

D_{2d}	$D_2(c_2-c_2^z)$	C_{2v}
A_1	A	A_1
A_2	B_1	A_2
B_1	A	A_2
B_2	B_1	A_1
E	$B_2 + B_3$	$B_1 + B_2$

Reference

AX.1 E.B.Wilson,Jr., J.C.Decius and P.C.Cross, "Molecular Vibrations", McGraw Hill, New York, 1955, Table X-14.

Appendix XI

The C^q_k Matrices for d^2 Basis

This appendix provides the C^q_k matrices being the solutions to Eqn.(3.50) for d^2 in the L = 1 and L = 3 basis.

C 20 MATRIX

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11	.2					-.27994				
<10		-.4					-.34286			
<1-1			.2					-.27994		
<33				-.14286						
<32										
<31	-.27994					.08571				
<30		-.34286					.11429			
<3-1			-.27994					.08571		
<3-2										
<3-3										-.14286

C 22 MATRIX

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11			.4899	-.44263				-.11429		
<10					-.25555				-.25555	
<1-1	.4899					-.11429				-.44263
<33	-.44263					-.09035				
<32		-.25555					-.12778			
<31			-.11429	-.09035				-.13997		
<30					-.12778				-.12778	
<3-1	-.11429					-.13997				-.09035
<3-2		-.25555					-.12778			
<3-3			-.44263					-.09035		

C40 MATRIX

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11						-.11664				
<10							.19048			
<1-1								-.11664		
<33				-.14286						
<32					.33333					
<31	-.11664					-.04762				
<30		.19048					-.28571			
<3-1			-.11664					-.04762		
<3-2									.33333	
<3-3										-.14286

C42 MATRIX

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11				-.04762				-.18443		
<10					.16496				.16496	
<1-1						-.18443				-.04762
<33	-.04762					-.34993				
<32		.16496					.08248			
<31			-.18443	-.34993				.30117		
<30					.08248				.08248	
<3-1	-.18443					.30117				-.34993
<3-2		.16496					.08248			
<3-3			-.04762					-.34993		

C44 MATRIX

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11										-.25198
<10										
<1-1				-.25198						
<33			-.25198					-.30861		
<32									-.39841	
<31										-.30861
<30										
<3-1				-.30861						
<3-2					-.39841					
<3-3	-.25198					-.30861				

The DQ and DT Matrices for a d^2 Basis

This appendix provides the DQ and DT matrices for d^2 L = 2 and L = 3, (see 3.55, 3.56).

THIS IS THE DQ MATRIX-C2(Z) AXIS

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11						-.08909				-.11501
<10							.14548			
<1-1				-.11501				-.08909		
<33			-.11501	-.10911				-.14086		
<32					.25459				-.18185	
<31	-.08909					-.03637				-.14086
<30		.14548					-.21822			
<3-1			-.08909	-.14086				-.03637		
<3-2					-.18185				.25459	
<3-3	-.11501					-.14086				-.10911

THIS IS THE DT MATRIX-C2(Z) AXIS

	<11	<10	<1-1	<33	<32	<31	<30	<3-1	<3-2	<3-3
<11						-.07529				.13608
<10							.12295			
<1-1				.13608				-.07529		
<33			.13608	-.09221				.16667		
<32					.21517				.21517	
<31	-.07529					-.03074				.16667
<30		.12295					-.18443			
<3-1			-.07529	.16667				-.03074		
<3-2					.21517				.21517	
<3-3	.13608					.16667				-.09221

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